

Antonio Di Ieva *Editor*

The Fractal Geometry of the Brain

Second Edition

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Antonio Di Ieva
Editor

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Editor

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*Infinity in every line, a never-ending climb
The beauty of fractals, they'll leave you blind.
ChatGPT*

Foreword to the 2nd Edition

Prof. Klaus Schwab has described the 4th Industrial Revolution as “a culmination of emerging technologies fusing the physical and biological worlds the likes of which has not been seen before” – we are all living through this era of change and disruption. Rarely a day goes by without hearing of new advancements in machine learning, or advances made by researchers working on modern quantum computers, and many other areas that fall under this umbrella. Domains of applications of these new scientific and technological discoveries include the pharmaceutical industry, education, precision medicine, engineering, and so on.

Nevertheless, it is all too easy to overlook the advances made by researchers working in many other exciting fields who are discovering new ideas and techniques in science and mathematics. A plethora of new domains have emerged as a result, which have benefit from these efforts. One such example is the field of Complexity Science. An area of study which helps us better understand the natural world, and utilize our new found knowledge in creating novel solutions to the problems that humanity currently faces.

Stephen Hawking was once asked what he believed would be the field most important in the twenty-first century, he replied, “I think the next century will be the century of complexity.” Hawking’s co-author, the cosmologist Professor George Ellis, has also been active within the field of complex systems. Promoting ideas related to causality, both classical bottom-up causality familiar to physicists and top-down causality that often appears as a characteristic of living systems.

“Fractal Geometry” is a field of study within the realm of complexity, first discovered by the mathematician Benoît Mandelbrot, approximately 50 years ago. Mandelbrot initially began studying a family of complex behaviours and phenomena, starting with notions of self-similarity. Nevertheless, he soon realized what he was attempting to capture was the notion of “roughness,” and would eventually settle on the following definition, “a fractal is by definition a set for which the Hausdorff-Besicovitch dimension strictly exceeds the topological dimension.” While this is an elegant definition mathematically, it is quite unwieldy in practical terms, and various techniques now exist to perform the practical calculations (such as “Box Counting”).

In a 2010 TED talk called, “Fractals and the art of roughness,” Mandelbrot’s work is said to have “led the world to a deeper understanding of fractals, a broad and powerful tool in the study of roughness, both in nature and in humanity’s works.” In the presentation, Mandelbrot explains how the arrival of the computer provided the means for him to explore new mathematical worlds, which first burst onto the scientific stage in the 1970s with the discovery of the “Mandelbrot Set” – a beautiful image created by endlessly repeating a simple rule in mathematics.

As a mathematician, I find the work presented in this book particularly pleasing, it not only introduces the reader to new areas but also examines the use of such theories to explore new technologies that enables the clinician to study complexities of nature, in particular complexities found in neurosciences.

Professor Antonio Di Ieva, an internationally recognized expert in the domain of using fractal geometry within the field of neuroscience, has not only made a significant contribution to this book himself, but has included contributions from other professionals, thereby creating a truly valuable resource.

This new edition of *The Fractal Geometry of the Brain* surpasses the comprehensive collection that appeared in the first edition. The current edition contains four parts.

Part I covers introductory aspects of Fractal Geometry and its applications to neurosciences. For the clinician, it includes introductory chapters on fractal and multi-fractal analysis including the important technique of “box counting.” Box Counting provides the practitioner with a practical technique of using fractals in real-world situations.

Part II contains chapters covering fractals in neuroanatomy; for instance, it includes a chapter on “The morphology of the brain neurons”.

Part III is a substantial body of knowledge covering some 25 chapters. On a spectrum of topics ranging from “Fractal analysis in neurodegenerative diseases” to “Fractals in neuropsychology and cognitive neuroscience.”

Part IV addresses topics related to the computational and translational aspects of fractals in neurosciences. Including a chapter on “Fractal analysis in MATLAB: A tutorial for neuroscientists”

A refreshing, well-organized, and timely book that should be of interest to anyone working in neuroscience, or anyone interested in learning more about the emerging field of Fractal Geometry and its applications.

Former CTO and Professor of Informatics,
Warwick, UK
September, 2023

Felix Hovsepiyan, PhD, FIMA

Foreword to the 1st Edition

The *Fractal Geometry of the Brain* adopts an encyclopedic view of the multiple ways the notion of fractal enters into the clinical, as well as scientific, understanding of the human brain, with multiple papers addressing the brain's form, function, and pathologies. This ambitious task is undertaken from the perspective of making available to the biomedical community a measure of complexity that can characterize both disease and wellness, a measure, the fractal dimension, which is fully consistent with the complexity of biomedical phenomena and with phenomena that have historically resisted being characterized by one or a few scales.

A fractal is a geometric concept used to describe the scaling of physical objects in space, which cannot be characterized by one or a few scales. This is not an esoteric mathematical concept, but is a concrete practical measure of variability in complex phenomena. Thus, the folds of the brain's surface, the regions of space that confine the flow of material through the brain, as well as tumor growth, are not described by smooth integer dimension paths, or surfaces, but have dimensions between the integers. These fractal dimensions entail a scaling across multiple scales, with the fractal dimension determining how the scales are tied together. This scaling gives rise to a self-similarity of the object in space, which like nested Russian dolls repeat the same structure at ever smaller scales. Multiple examples of such space-filling biomedical phenomena are presented throughout the book and used to explain how such scaling is related to robustness and its loss to disease.

A fractal can also describe a dynamic process, as observed in physiologic time series. Here again the variability cannot be determined by a given rate or any other characteristic scale determined by an average. The dynamic changes in physiologic time series are coupled across a wide range of time scales, with no one scale being dominant. A fractal dimension again determines how the slowest and the fastest changes are coupled together, even though one may be on the order of fractions of a second and the other on the order of days or even weeks. Such coupling gives rise to a self-similarity of the dynamic variability in time. The scale at which a physician ought to intervene, in order to control the dynamics of the process, is often the crucial question. Here again a number of dynamic phenomena and their disruptions are described throughout the book.

Finally, there is the statistical fractal, where the self-similarity does not reside in the space or time variation of the variable of interest, but rather this fractal is contained in the scaling of the distribution describing the random fluctuations in physiological variables. But here, as in the cases of the two other kinds of fractals, it is not the mathematics that is emphasized; although the multiple ways the fractal dimension can be estimated from data is discussed, it is the biomedical implications of the self-similar statistics that is of most concern. How the scaling opens up a new way of understanding, of both the normal behavior and the multiple ways normality can be subverted, is of most importance to clinical medicine.

I have taken the space to distinguish the three manifestations of fractal used in the book for good reason. Each fractal form assists in understanding a different aspect of the brain. The phrase fractal geometry in the book's title is not geometry in the classical mathematical sense, but that should not be a surprise. Mathematicians have been arguing over the meaning (and existence) of fractals and fractal geometry, since Benoit Mandelbrot introduced his idiosyncratic definitions into the scientific lexicon, over 40 years ago. Mandelbrot was more interested in understanding the complexity of natural and social phenomena, using relatively simple mathematical models, than he was in satisfying the scientific criteria in any given discipline, including those laid down by mathematicians.

The brain, with its rich complexity of neuronal interconnections, the broad spectrum of time scales in its dynamics, and its statistical variability in the performance of tasks, is a natural candidate for fractal modeling. The brain's complexity is defined by a balance between regularity and randomness as captured by its various fractal dimensions. The fractal dimension is ubiquitous in the various complex processes within the brain because fractal processes have been proven to have an evolutionary advantage over those that do not.

Mandelbrot recognized a number of ways that his fractal concept could be used to gain a deeper understanding of physiologic complexity,¹ using what he candidly admitted were toy models. The handbook, *Fractal Physiology*,² extending his applications of the fractal concept to real-world physiological data, was introduced in 1994, but it took another 20 years for a journal of the same name to become available. Ewald Weibel, in the memorial volume dedicated to Mandelbrot,³ entitled his contribution: "How Benoit Mandelbrot Changed My Thinking About Biological Form." This remembrance details his personal, but not atypical, journey a number of biomedical scientists have taken, from the traditional perspective of biology and physiology to one encompassing the rich variability observed in the data and

¹ B.B. Mandelbrot, *Fractals, Form, Chance and Dimension*, W.H. Freeman and Co., San Francisco, 1977.

² J.B. Bassingthwaighe, L.S. Liebovitch, and B.J. West, *Fractal Physiology*, Oxford University Press, NY, 1994.

³ M. Frame and N. Cohen, Eds., *Benoit Mandelbrot; A Life in Many Dimensions*, World Scientific, NJ, 2015.

captured by the concept of fractal dimension. This book is in this tradition, with a focus on the human brain, and is remarkably successful in its undertaking.

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March 2016

Bruce J. West, PhD

Preface to the 2nd Edition

Despite the fascination related to the application of fractal geometry to the neurosciences, due to its inherent beauty related to the ability to describe self-similar patterns of complex networks of interconnected cells, scaling properties within neural systems, and firing patterns of individual neurons at the micro-scale and whole brain at the macro-scale, fractal analysis is still a niche of research for very few dedicated scholars.

Fractal geometry has had its Spring a few decades ago, although nothing comparable to the Spring currently occurring to the use of artificial intelligence, which nowadays is applied to any social and scientific field, with great alternating hypes and hopes, dystopian fears and utopian illusions, epic failures, and remarkable progresses.

In spite of such a niche, fractal geometry has had and still has the capability to gain attention in multiple domains of analysis, with a myriad of cross-translational applications in different fields, including the brain sciences.

When I published the first edition of *The Fractal Geometry of the Brain* in 2016, I was not expecting to reach a great audience of readers, but eventually I was wrong. There have been about 100,000 electronic accesses to the book, a few thousand copies sold (plus many more on illegal circuits to download PDFs for free), and several hundreds of citations, not to mention the hundreds of requests for collaboration that I and many of the book's authors have received over the last few years, including the thrilling offer by Springer Publisher to work on a second edition of the book.

In February 2016, I concluded the preface of the first edition of *The Fractal Geometry of the Brain* with the hope to ignite new ideas in the open minds of out-of-the-box thinkers who are interested in the study and treatment of the brain from a novel and progressive perspective. With my great joy and surprise, over the last 7 years, I have found many students and scholars around the world extremely keen and thrilled to learn the principles of fractal geometry and to apply it to the most translational and sensational cross-disciplinary projects within medicine and the neurosciences in general.

In 2018, thanks to some major research funds, I was able to concretize an idea that I have had for many years earlier, i.e., the foundation of the Computational NeuroSurgery (CNS) Lab (at Macquarie University, Sydney, Australia), where diseases of neurosurgical interest are studied by means of computational modelling and AI. Useless to say, a big slice of the computational modelling cake has been taken by fractal geometry, and computational fractal-based analysis has become the workhorse of several novel research I and my team have been working on, including fractal-based radiomics of brain tumors, fractal analysis of eye tracking, fractal analysis in connectomics and neuroimaging, with some sporadic translational applications in different fields, like ecology and neuroaesthetics, for example.

After carefully considering the updates in the field over the last few years and the increasingly limited time for leading a major editorial collaboration, most of the last year has been dedicated to update the first edition of the book. In this second edition, more than 90% of the chapters have been updated by myself and the other authors, who have been very happy (and probably surprised by the invitation) to do so. Many further authors have been invited to join the new edition, and 16 new chapters have been added into the current edition, clearly showing an extension of the fractal horizon in the neuroscience landscape. In the second edition, five chapters from the first edition have been removed, including one on the use of GPU for fractal inference (as its knowledge can be considered obsolete nowadays – the use of GPU is currently state-of-the-art) and another excellent chapter on the application of fractal analysis to electroencephalography (EEG) time series by Prof. Włodzimierz Klonowski, who unfortunately passed away on the 4th of April 2020. Prof. Klonowski was a Polish biomedical physicist and a member of the Polish Academy of Science with an outstanding expertise in the fields of biomedical engineering, awarded the Golden Cross of merit by the President of the Republic of Poland in 2004. Klonowski was an enthusiastic expert of fractal geometry, and amongst his many contributions, he successfully applied chaos theory and fractal analysis to signal and image analysis, above all, to EEG signal dynamics. I met last Włodzimierz in Los Angeles in April 2018, at the 15th Annual World Congress of the Society for Brain Mapping and Therapeutics, when I organized and chaired a session on “Fractal geometry and Computational clinical neurosciences.” In such a session, I invited Włodzimierz to share his deep knowledge on fractals in the neurosciences, and I was very humbled that he kindly accepted to fly to California to deliver a talk on his favorite topic, the “Application of Higuchi fractal dimension method in neuroscience and in clinical diagnosis.” It was a great talk and we had a very inspiring brainstorming, and I still remember his gentle sense of humor. It’s hard to think that it would have been the last time that we would have met each other, and that two years later we would have suddenly and tragically stopped our email correspondence on the use of fractals in neurophysiological time series analysis. Although his chapter is obviously missing in the new edition, I warmly suggest the scholars interested in the field to read his outstanding chapter in the first edition of the book.

After years, I still consider the fractal geometry of the brain “brain food” for curious minds. From a private correspondence with Professor Dusan Ristanovic

(who unfortunately also passed away – in 2017), it is well known that Mandelbrot considered his *Fractal Geometry of Nature* book an essay (surely a very successful one, I would say), since a real fractal geometry of nature does not exist as a true geometry. It is known that every geometry must start with basic notions and axioms. Euclidean geometry, advanced by David Hilbert, starts with three basic notions (i.e., point, straight line, and plane) and 25 axioms. Riemann and Lobachevsky geometries use the same notions and axioms except one. No basic notion and axioms exist in fractal geometry! So, fractal geometry does not exist as a true geometry, if not the analysis related to it and its parameters, such as the fractal dimension, for example. Having said that, it is hard to remove concepts with such a powerful root and meaning from the common language, in spite of their mathematical inaccuracy.

With the exponential evolution of AI and with the more and more limited free time on the horizon, it might be difficult to deliver a third edition of this book, but science is often and unpredictably recurrent (self-similar?), and I wouldn't be very surprised if fractals get a new Spring and see new sun rises in the next years.

Sydney, Australia
May 2023

Antonio Di Ieva

Preface to the 1st Edition

This book's title, *The Fractal Geometry of the Brain*, recalls Benoit Mandelbrot's famous book in which he described *the fractal geometry of nature* more than 30 years ago. Since then fractals have found application in many fields, especially in the biomedical sciences and, in particular, the neurosciences. In light of the hodological interconnection of ideas regarding the study of the amazing and complex system called the brain, it seems natural that neuroscientists also peered through the lens of fractal analysis to view the brain as fractal, self-similar in both neuroanatomical structures and neurophysiological time series. Nonetheless, in practice fractals – often deemed too exotic, esoteric, misunderstood, or simply not understood – do not yet have clearly defined roles in biomedical research and clinical neurosciences. With an aim of defining those roles, the scope of this book summarizes the state of the art of fractal analysis in neuroscience.

This book reviews the most intriguing applications of fractal analysis to the neurosciences with a focus on current applications and future perspectives, limits, advantages, and disadvantages. It is essentially for scientists, scholars, physicians, and students of different disciplines interested in applying fractal-based computational models to the study of the brain. This book is mainly clinically oriented, offering suggestions for applying fractals “from bench to bedside,” but also covers numerous research aspects. Thanks to the valuable inputs of 58 experts worldwide (physicians, basic and clinical neuroscientists, computer scientists, mathematicians, psychologists, biologists, physicists, and engineers, among others) from 19 different countries, who have made personal contributions in the field, this book is a holistic body of work on fractals and the brain that describes how fractal analysis can help analyze the marvelous and complex system called the brain in its entire physiopathological spectrum.

The book is organized into four parts, each introduced by a short chapter that summarizes the topics of its section:

- Part I “Introduction to Fractal Geometry and Its Applications to the Neurosciences” (Chaps. 1, 2, 3, and 4)¹ summarizes the “nuts and bolts” of fractal geometry and its applications to basic and clinical neurosciences.
- Part II “Fractals in Neuroanatomy and Basic Neurosciences” (Chaps. 5, 6, 7, 8, 9, 10, and 11) investigates the brain as a fractal object in a hierarchical, multilevel analysis from neurons and microglia to the whole brain itself, with all its structural and functional complexity. A specific chapter is dedicated to the fractal geometry of cranial sutures, and the concluding chapter explores an evolutionary perspective on the fractal geometry of the human brain.
- Part III “Fractals in Clinical Neurosciences” (Chaps. 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, and 30) summarizes applications of computational fractal-based analysis to the clinical neurosciences, moving sequentially through the different disciplines. Chapters 13, 14, and 15 summarize applications of fractal analysis to neurological diseases and in particular neurodegenerative diseases. Chapters 16, 17, and 18 discuss cerebrovascular-related topics, and Chaps. 19 and 20 focus on neuroimaging. Chapter 21 bridges neuroimaging to the field of neuro-oncology, and Chaps. 22, 23, and 24 investigate brain tumors from neuropathological, neuroradiological, and oncological points of view. Next, Chaps 25, 26, and 27 cover fractal-based analysis of brain time series (i.e., EEG, EMG, and biosignals in Parkinsonian diseases). Chapter 28 focuses on applications of fractals to the sensory extension of the brain, the eye. Chapter 29 covers cognitive neurosciences and psychiatric diseases, and lastly in the section, Chap. 30 focuses on the relationships between fractals and visual experiences, including in the visual arts.
- Part IV “Computational Fractal-Based Neurosciences” (Chaps. 31, 32, 33, 34, 35, and 36) illustrates applications of fractal analysis to neuroscience by means of software and computer science tools. Chapters 35 and 36 introduce some bioengineering-related aspects of fractal geometry, from fractal neuronal interfaces to future perspectives in artificial intelligence.

I hope that this book provides the stimulus to ignite new ideas in the open minds of out-of-the-box thinkers who are interested in the study and treatment of the brain from a novel and progressive perspective.

Sydney, Australia
February 2016

Antonio Di Ieva

¹ These chapters numbers refer to the first edition of the book, whereby the new edition has different chapters numbers.

Acknowledgments to the 2nd Edition

As usual, I would like to acknowledge all of the contributors, who have been able to dedicate some spare time to update and/or write from scratch the chapters included in the second edition of *The Fractal Geometry of the Brain*.

And, again, I would like to thank Jennilee Davidson for her lovable, continuous support in proofreading and contributing to some book chapters. Since the first edition of the book, I have been lucky enough to continue to get Jennilee's scientific (and more) support, also considering that in the meantime she has also become my loveable wife. What a fantastic partner in science and life!

Furthermore, without any big changes from the first edition, a special thanks goes to Carlo Russo – again! The indefatigable friendship and support by Carlo are even witnessed by his incredible effort to join me in Sydney for two years (2019–2021) to help to set up the Computational NeuroSurgery (CNS) Lab. In spite of the pandemic-related restrictions of that strange time, our collaboration has continued to produce great outcomes, and I feel extremely lucky to continue to enjoy his friendship and to benefit from his outstanding expertise. Indeed, Carlo is still an outstanding member of the CNS Lab in Sydney, without any temporal and spatial limitations between our different jobs, different countries (i.e., Australia and Italy), and very different time zones. As already written in the acknowledgments of the first edition, it is still a collaboration (and friendship) more unique than rare!

Last but not least, my great thanks were, are, and always be dedicated to my parents, Giuseppe and Tina Di Ieva, who have helped me to track my pathways and always supported my turbulent decisions.

Antonio Di Ieva

Acknowledgments to the 1st Edition

First of all, I would like to acknowledge all of the contributors, without whose expertise and enthusiasm this comprehensive book would not have seen the light of day. Among them, a special thanks goes to Audrey Karperien, as she has not only written three extraordinary chapters but, despite her own duties, has also enthusiastically agreed to review chapters written by others (Preface, Chaps. 1, 3, and 31).

I would like to thank Jennilee Davidson for her lovable, continuous support in proofreading a great part of the book. Moreover, I'd like to acknowledge Timothy Lam for his help to draft Chap. 20² and Joseph Di Michele for Chap. 24. The great enthusiasm I have found in several of my students at the University of Toronto in performing fractal-based research in clinical neurosciences has been a source of great satisfaction, and I also thank Dr. Michael Cusimano for the great neurosurgery-based and scientific network we've shared.

In addition, I would like to thank the professors (who have become friends) who supervised my Ph.D. in clinical neurosciences at the Medical University of Vienna in the period of 2009–2011, supporting my research on the applications of computational fractal-based analyses into the fields of neurosurgery and neuroanatomy, neuropathology, and neuroradiology: Manfred Tschabitscher, Christian Matula, and Johannes Hainfellner.

Last but not least, a special thanks goes to two old friends of mine, Carlo Russo and Fabio Grizzi. Although not present among the contributors of this book, their presence is shadowed in every page of mine. I will not forget the endless discussions we have had (and still have) on the applications of fractals in biomedicine, in the lab of the Humanitas Institute in Milan and in Vienna and in Toronto and in Sydney as well as online almost on a daily basis, without any spatiotemporal barriers. Also unforgettable are the endless hours spent to run experiments during my on call nights, when, between an emergency in the operating theater and a case to manage

² These chapters numbers refer to the first edition of the book, whereby the new edition has different chapters numbers.

in the ER, several computers were computing our calculations in parallel over nights and weekends. The indefatigable support by Carlo Russo, who has programmed the requested software applications, and who has always been very keen to brainstorm “in real time,” is something more unique than rare!

A unified fractal model of the brain would not have been possible without this self-similar network of great friends and precious colleagues from all around the world.

Antonio Di Ieva

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Part I
Introduction to Fractal Geometry and Its
Applications to Neurosciences

Chapter 1

The Fractal Geometry of the Brain: An Overview



Antonio Di Ieva

Abstract The first chapter of this book introduces some history, philosophy, and basic concepts of fractal geometry and discusses how the neurosciences can benefit from applying computational fractal-based analysis. Further, it compares fractal with Euclidean approaches to analyzing and quantifying the brain in its entire physiopathological spectrum and presents an overview of the first section of this book as well.

Keywords Brain · Fractals · Fractal geometry · Fractal analysis · Multifractal · Neurosciences · Fractal dimension · Lacunarity · Self-similarity · Scaling

You cannot explain the third dimension to ones living in a 2D world.
Neither can you explain the fractal dimension to some counting only in integer numbers.
You cannot describe the light to people living in the darkness.
Or ask people to think out of the box, if they don't know anything else than the box itself.
Sharing knowledge is the final aim, teaching is an illusion.
You can just hope to ignite stimuli in the right minds. (A.D.I., Caserta, Italy, March 2015)

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1.1 From *The Fractal Geometry of Nature* to Fractal Analysis in Biomedicine

Among the cycling trends characterizing how humans have built knowledge over time, relatively recent applications of the reductionist approach have allowed for great discoveries and paradigm shifts in several modern scientific disciplines. Even given the limits of reductionism, this approach has favored development of very specific individual topics and has allowed for important advances as diverse as space exploration and the discovery of DNA, amongst many other discoveries. Ironically, though, the reductionist approach, encroaching on its own heels, has also ushered in a more holistic view of nature, while several disciplines focusing on the study of very specific and restricted topics have reemerged in the “omics” (a suffix referring to the concept of wholeness or “completion”): genomics, metabolomics, proteomics, transcriptomics, connectomics, etc. Genomics, for instance, in attempting to define and explain individual pathologies by analyzing the genome, spurred further need to analyze each gene product (i.e., the proteins), thus spawning the age of proteomics and transcriptomics. Fortunately, the even more detailed view of the nature of things has still left space for the general idea that everything is interconnected.

Fractal geometry, even though it developed from a holistic view of nature, as theorized by the fractal pioneer Benoit Mandelbrot (1924–2010), has had a similar destiny. As a field, it underwent rapid development of methods and ideas, culminating in the discipline of “fractalomics” [1], a view wherein everything is fractal, where *Fractals Everywhere* (to use the title of Michael Barnsley’s book [2]), where everything can be explained or at least described by means of fractal geometry. This view, however, is inadequate. As emphasized by Mandelbrot in his seminal *The Fractal Geometry of Nature* [3], fractals are not everywhere and are not really a panacea [4]. As happens with so many theories, the initial enthusiasm and explosion of scientific publications, conferences, and ideas has been scaled down, owing at least in part to intrinsic limitations of the methods and pitfalls described by both “fans” and critics of the latest fashion, with the typical seasonal interchange.

The application of fractalomics to biomedical science has essentially followed within this path. In the beginning, fractal analysis was applied to the study of histological and/or microscopic images [5–7], planting roots in several branches of biology and medicine. Yet, substantial amounts of research and numerous publications regarding fractals in biomedicine notwithstanding, fractal geometry was not fully introduced into the lexicon of physicians, generally considered reluctant to accept paradigm shifts. In neuroscience in particular, as widely described in this book, several applications of “neuro-fractalomics” have been used to explain and quantify physiological and pathological phenomena, yet a common protocol, which might give fractal analysis the dignity of a consolidated scientific methodology, is still lacking.

One reason for the slow transition to clinical applications in biomedicine and neuroscience is practicality. The substantial work on fractals is not readily accessible to the clinical world. This is evident when one searches the keywords “fractal”

with “brain” on PubMed, which provides more than 1000 articles, or even more on Google Scholar, which yields a few thousand, neither of which gives sufficient merit that fractal analysis has had in the field of neuroscience and its potential for further investigation [8, 9]. Fractal geometry, above all in medicine, is a relatively new paradigm and, like many new paradigms, has required, and still requires, a long time to be “metabolized,” to become accepted by the general audiences of physicians, scientists, and laypeople. On one hand, it is true that fractal geometry has had a spring and a winter (and multiple repetitions of such seasons, in a cyclic and recursive manner); on the other hand, it should be said that it is time to accept fractal analysis in a less enthusiastic but more mature way.

In such a perspective, this book links together several experts in the field to spread fractal knowledge and to ignite new ideas and potential collaborations for a new perspective on the study of the human brain in its entire physiopathological spectrum. This first chapter introduces fractal geometry and fractal analysis and briefly outlines the contents of this book’s first section on the rudiments of fractal analysis in neuroscience, also including practical applications of fractal modelling.

1.2 From Euclid to Fractal Metrology

Fractal geometry is a mathematical and philosophical model with both strengths and weaknesses. Mandelbrot coined the word “fractal” from the Latin adjective *fractus* (meaning broken or fractured), with the corresponding Latin verb *frangere* signifying “to break” or to create irregular fragments [3]. As is well known to the readers of Mandelbrot’s works, fractal geometry was initially developed on the basis of what were considered “monsters” of Euclidean geometry: mathematical or natural phenomena not fully explicable or quantifiable by means of ideal, platonic, and/or Euclidean approaches. Mandelbrot “visited” galleries of such monsters, populated by bizarre mathematical creatures described by Cantor, Peano, Hausdorff, Sierpinski, and Urysohn, among several other mathematicians, and drafted a manifesto of fractal geometry based upon a body of well-defined laws and coherent principles, including those derived from chaos theory [10] “to bear resemblance to pathologic entities or mathematical monsters, [...] despite, or owing to, their beauty, richness and fascinating shapes” [11].

Similar to the way that music can be considered a universal language [12], fractal geometry can be seen as a universal language by which nature can be explained or at least described and quantified. Expanding the music analogy, the “notes” of fractal geometry are fractal dimensions (denoted as FDs, D_{FS} , or simply D in some papers) arrived at through fractal analysis. FD can be considered a dimension, a quantifier, a parameter, an estimator, and so on but is essentially a metric that quantifies something. This “something” can be the “geometrical complexity” or the “space-filling property” of an object or a pattern, the roughness of a surface, or the variation of a time series, for instance, where the more irregular a phenomenon or the more an object or pattern fills the space in which it is embedded (e.g., the

microvascular network in a tissue), the higher the FD [3]. FD is also a continuous variable, easy to use and compare by means of mathematical and statistical methods, and suited for describing the physiological gradualism of natural phenomena in a “non-saltatory” manner. There are different methods to compute FD, with the box-counting algorithm being the most basic and frequently used one (see Chaps. 2, 3, 4, and 5, among others).

In his milestone paper on Britain’s coastline, published in the journal *Science* in 1967 [13], Mandelbrot defined the concepts of fractal dimension and self-similarity. According to Mandelbrot’s definition, the topological dimension (D_T), which can be informally understood as the commonly known dimensions of typical (“Euclidean”) geometry, is always an integer, but FD need not be an integer. Expanding further, a fractal is a set for which FD (or more specifically, a measure known as the Hausdorff–Besicovitch dimension) strictly exceeds the topological dimension [3].

The value of FD arrived at for a particular phenomenon is an intrinsic and unique value of that phenomenon itself but represents an “averaging” of variation in data [14]. The fact that it is strictly linked to the geometrical complexity of an object does not mean that it is the object. Different objects can share the same FD value, and a series of objects with the same FD do not necessarily belong to the same class [15]. It should also be emphasized that owing to logarithmic calculations involved in determining an FD value (see Chaps. 2 and 3 for details), small changes in FDs can correspond to big differences in the shape of an object. In sum, FD describes the overall structure and complexity of an object but does not tell the shape of the object (neither can a Euclidean dimension). In principle, this is similar to other metrics, such as the unit meter, for instance, which can quantify the height of objects or persons, where a chair and table can have the same height in meters but are not the same object, just as people are unique but may have the same height and weight values. These general principles apply to FDs determined for other phenomena such as time series as well as objects.

The universality of fractal geometry is shown by the fact that the same concepts or methods may be applied to completely different disciplines, from meteorology to medicine and from physics to biology. As an example, the box-counting method, despite having several intrinsic limitations, can be used to analyze the geometrical complexity of a microvascular network of a tumor and the trees’ pattern in a forest or the surface roughness of a cell and the coastline of an island.

Of course, as in every model, there are limitations. Despite being a “universal language,” fractal geometry has not proposed or found any Holy Grail nor, as alluded to above, is it a cosmological constant that is able to describe and/or explain everything. The search for cosmological constants (e.g., Albert Einstein) and theories of everything (e.g., Stephen Hawking) have opened extraordinary streams of discoveries but eventually have failed to find a scientific Holy Grail valid for every theory and application. The concepts and methods of fractal analysis applied to “biomedical fractalomics,” for example, have neither explained nor defeated cancer, and “neuro-fractalomics” has not unveiled the great mysteries of the brain. A positivist view of fractal geometry might lead to its own failure as a model—bringing to a new endless winter of fractal geometry! The same applies to artificial

intelligence, which has also seen oscillations of springs and winters over the last decades, with the most recent successful peak of enthusiasm and applications in any modern sector and scientific field.

Moreover, mathematicians may argue that the existence of a fractal pattern is the condition sine qua non for applying fractal analysis, and, from a strictly mathematical point of view, this is correct. Mathematical fractals are invariant under certain transformations of scale (self-similar) infinitely, whereas natural “fractals” are self-similar within limits [13]. To elaborate, it is not appropriate to extrapolate for biological systems (e.g., the brain) or features (e.g., cell morphology or event frequency) beyond certain limits of investigation; rather, natural fractal or fractal-like patterns can be considered statistically self-similar only within the upper and lower limits (the scaling window), in which the relationship between the scale of observation and measured size or length can be established [16]. Indeed, it has been proposed that biological fractality can only be considered to be supported by the evidence when the experimental scaling range covers at least two orders of magnitude [1, 17].

Despite the reality of limited or statistical self-similarity, however, fractal analysis of biological phenomena is viable. The box-counting method, for example, offers a quantitative parameter, the box-counting dimension, which is related to the space-filling property (or the “level of occupancy”) of an analyzed object regardless of whether the object is fractal, pseudo-fractal, quasi-fractal, semi-fractal, super-fractal, or any of the other definitions that have been proposed. In this respect, it cannot be emphasized enough that fractal geometry is a model that, like other models, can be an approximation. It is reasonable to use fractals as models of natural objects in the same way that a perfect circle is often used to represent the cross section of an artery, for example [18].

Focusing the interest on biomedicine, fractal geometry has dual usefulness: (1) providing a unique model of analysis to attempt to explain the peculiarity and singularity of phenomena and (2) proposing a system of universal quantification, a simple tool used to describe natural things in an appropriate and more realistic way.

These two points require some clarification. Every phenomenon has its peculiar features, its own “fingerprint” of features, which makes it unique or different from others. As described in Chap. 27, the pattern of distribution of microvessels in tumors, for example, is different from the one found in normal tissues. In the same way, neoplastic cells have different shapes and features in comparison to normal cells. Similarly, electroencephalographic records of healthy individuals have their own specific and unique features that are different from those of patients affected by epilepsy. In general, each phenomenon has its spatiotemporal peculiarity, which may be very difficult to describe in terms of Euclidean geometry or nonlinear mathematics but more readily described in terms of fractal geometry.

In such a view, fractal geometry indeed plays a double role: on one hand, it aims to outline the “cutoff,” which separates different phases of transition, from physiological to pathological states, for example, with an increase or decrease

in geometrical complexity or space-filling properties. On the other hand, fractal analysis provides valid, objective, and more realistic tools of measurement.

The objectiveness of the parameters is of course related to the standardization of the methods of data acquisition (e.g., image resolution or scale window limits); otherwise, comparing outputs computed by means of a heterogeneous dataset may lead to various (and wrong) results (see the “black box” concept in Chap. 13). This has been (and still is) one of the most notorious pitfalls encountered in the clinical applications of fractal analysis. But this problem is also quite “ancient”; indeed, in the history of measurement, samples have always been collected or standardized in order to be used as a reference, such as the international prototype kilogram, deposited at the International Bureau of Weights and Measures in Paris, or, the second, defined as the second division of the hour by 60 (the first division being the minute) or, more precisely, the duration of 9,192,631,770 cycles of radiation corresponding to the transition between two energy levels of the cesium-133 atom at rest at a temperature of 0 K.

1.3 The Fractal Geometry of the Brain

The title of this book implicitly assumes that the brain is a fractal structure and that the fractal geometry of the brain exists. This might be wrong, or at least misleading, as several mathematicians do not consider fractal geometry as a formal geometry in the traditional sense. In general, geometry was developed for the measurement of the Earth, as the Greek etymology of the word suggests, and is the branch of mathematics related to the analysis of shape, size, relative positions, and properties of objects. Although developed for practical purposes, traditional geometry is not sufficient to fully describe the complexity of physical phenomena. Regardless of the existing controversies on the nature of fractal geometry and the distinctions between fractal and formal Euclidean geometry, both serve practical purposes.

Whereas the concepts and rules of Euclidean geometry may be considered adequate for an ideal world [19], fractal geometry provides tools of analysis more appropriate for the real world. It searches for the generators, or common rules, that nature, for entropic economy, adopts for phenomena which can be otherwise apparently completely different and through fractal analysis provides valid, objective, and more realistic tools like the fractal dimension to analyze the shapes and properties of natural objects. As stated by Mandelbrot: “Many patterns of Nature are so irregular and fragmented, that, compared with Euclid—a term used in this work to denote all of standard geometry—Nature exhibits not simply just a higher degree but altogether a different level of complexity. The existence of these patterns challenges us to study those forms that Euclid leaves aside as being *formless*, to investigate the morphology of the *amorphous*” [3].

According to Mandelbrot, the fractal dimension has been “injected” into several different scientific fields [20], from physics to geomorphology and from economics to biology. Among a myriad of examples, fractals have been used to analyze Internet

traffic [21], econometrics time series [22], DNA base sequences [23] and packing of chromatin in the human genome [24], recursive genome function [25], social networks [26], ecosystems [27], metabolite exchange [28], automatic keyword extraction in text [29], human gait [30], advanced neuroimaging [31], computer graphics, eye tracking [32], forest fragmentation in ecology [33, 34], aesthetics in artworks and psychology [35], and even the architecture of Gothic cathedrals and the neurobiological mechanisms of human creativity [36]. The application of fractal geometry to the neurosciences has been the consequence of a new trend of research focused on the analysis of the intrinsic complexity of biological systems, which have, according to Ludwig von Bertalanffy's *weltanschauung*, a tendency to form multilevel structures of "systems within systems," each of which forms a "whole in relation to its parts and is simultaneously part of a larger whole" [37]. Fractal geometry represents a holistic framework to explain biological ontogenesis involving a fractal hierarchy, from a single cell to the function of the human brain and its intrinsic and extrinsic interactions [38, 39].

With the intention to avoid philosophical or mathematical diatribe on the meaning of fractal geometry or any speculations on the real fractal geometry of nature and of the brain, this book is essentially focused on the translational practical applications of fractal analysis to the neurosciences, specifically, brain sciences. Computational neuroscience is interdisciplinary, where different fields, such as cognitive science and computer science, mathematics and physics, psychology, and neurobiology, converge in the study of the nervous system. Fractal analysis can be seen as a tool, or a methodology, of computational neuroscience. Computational fractal-based analysis is most likely the most correct term, considering that in this perspective fractals "simply" become a tool in the hands of the scientists dealing with biomedical sciences. Putting fractal analysis under the same umbrella as all the computational methods used to analyze the brain can be considered wrong, and the use of FD as a universal quantifier to describe every natural object can be misleading. But it is true that fractal geometry offers powerful tools for an objective and more realistic description of complex systems, such as the brain, and that some estimators, such as the box-counting dimension and lacunarity, are robust quantitative parameters. Despite continuous debates about the use of the term fractal to describe natural objects, fractal geometry is still a useful tool to describe and quantify the inherent complexity of objects [3], including neurons, astrocytes, and the brain itself [40]. A tool is a tool, and as famously stated by Lars Leksell, the Swedish neurosurgeon who pioneered stereotaxis and radiosurgery, "A fool with a tool is still a fool." It rests upon the operator how to use such a powerful tool to explain phenomena and quantify natural things.

Of course, fractal analysis does not "simply" offer some quantitative parameters. FDs and related parameters obtained through fractal analysis, for example, with all their pros and cons, are powerful geometrical tools. But fractal geometry has something more than a normal geometry. The concept of self-similarity, the repetition of iterations generating natural patterns, the scale invariance, the multifractal scaling, the strange attractors, and the chaotic patterns are all concepts with much more intriguing roots. The elegant beauty of the Koch curve (the "snowflake") and the

sublime fascination of the Mandelbrot set evoke something more than mere rules and equations of other descriptive geometries. It is not by coincidence that fractal patterns have been found in music and visual arts and computer graphics has offered a true exploitation of fractal art.

I have personally always been fascinated by Hofman’s concept that “the macroscopic organization of the brain in mammals is governed by a few simple generative rules and that these internal factors of brain design, bearing no relation to the selective reasons of initial enlargement, may be the primary determinants directing the evolution of the brain” [41] (see Chap. 12). The repetition of patterns to microscopic, mesoscopic, and macroscopic levels is a peculiarity of mathematical fractals, but natural objects own such a peculiarity in a finite range, and this cannot be emphasized enough. This resolution scaling effect applies to cell morphometry (including neuron and astrocyte morphometry) and overall brain morphometry. Convolved brain tissues, tortuous brain vessels, ramified neural cells, and jagged electroencephalographic records, for instance, all resist description by Euclidean and other more traditional geometries yet, within their respective limits, yield gracefully to the universal language of fractal analysis.

As the old adage goes, “not everything that matters can be measured and not everything that can be measured matters.” But it is also true that if it is worth doing, it is worth measuring. Fractal geometry offers a novel perspective on the description and measurement of nature. Nature shows complexity, roughness, apparent chaos and mystery, and beauty—the sublime. And what is there in nature more complex, apparently chaotic and mysterious, beautiful, and sublime than the human brain?

1.4 Fractal Dimensions and Neurosciences

At this point, it is clear that natural, especially biological, morphologies are irregular, and when we can assess them (using computers) as distributions of points in space and time, we can often describe and quantitate them rather compactly by the methods of fractal analysis [42, 43].

But how is fractal analysis performed? As expressed in the preface to this book, the first section offers one answer to this question by outlining the key tenets of fractal analysis. In particular, Chaps. 2, 3, 4, and 5 explain the basic concepts of fractal geometry (e.g., fractal dimension, self-similarity, and scaling) and introduce specific methods and examples of the practical applications of fractal analysis to neuroscience.

The fractal dimension is a powerful tool for characterizing features, but its meaning has to be fully understood in order to avoid misuse of fractal analysis in biomedicine [44]. Several important limitations, caveats, and recent methodological considerations have therefore also been discussed in those chapters. In addition, supplementary methods of fractal analysis are discussed. In Chap. 2, for instance, Audrey Karperien and Herbert Jelinek discuss lacunarity, a measure that complements FD. In box counting, FD measures how much the object (or data) fills the

space [45], but lacunarity measures how the data fill the space [3, 46]. Other fundamental parameters and concepts of fractal analysis, such as the Hurst coefficient (H), the scaling exponent (α), the power exponent (β), the fractional differencing parameter, and fractal noise, are discussed in Chap. 3 by Tatjana Stadnitski. While Chap. 2 focuses mainly on applications of fractal analysis to images, Chap. 3 is essentially dedicated to time series analysis. Chap. 4, by Renaud Lopes, discusses limitations of the “monofractal” approach, which describes phenomena by means of a single D value, introducing multifractal behavior measured by a spectrum of D values, with special emphasis on the applications in neuroimaging. Finally, in Chap. 5, Camillo Porcaro, Sadaf Moaveninejad, Valentina D’Onofrio, and Antonio Di Ieva summarize again the terms, formalism, and meaning of the fractal estimation of time series, with particular emphasis on the performance of the fractal parameters. In conclusion, section I offers the nuts and bolts of fractal analysis, which is very useful to understand the other sections of the series and to learn the methodologies to translate fractal analysis to the reader’s own field of research.

The reader who is already an expert in fractal analysis, familiar with basic concepts, and up to date on methodological developments may jump directly to section II of this book, but less versed readers can read the next three chapters in order to comprehensively understand the rest of this book wherein specific applications of fractals in basic and clinical neurosciences are discussed.

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Chapter 2

Box-Counting Fractal Analysis: A Primer for the Clinician



Audrey L. Karperien and Herbert F. Jelinek

Abstract This chapter lays out the elementary principles of fractal geometry underpinning much of the rest of this book. It assumes a minimal mathematical background, defines the key principles and terms in context, and outlines the basics of a fractal analysis method known as box counting and how it is used to perform fractal, lacunarity, and multifractal analyses. As a standalone reference, this chapter grounds the reader to be able to understand, evaluate, and apply essential methods to appreciate and heal the exquisitely detailed fractal geometry of the brain.

Keywords Box counting · Fractal · Fractal analysis · Lacunarity · Multifractal · FracLac · Theoretical models

2.1 Fractal Analysis: Measuring Self-Similar Details

Fractal analysis is a way of measuring phenomena when the details of design are as important as gross morphology [1]. It has been used to solve nebulous problems having complex details in diverse fields, including finance [2, 3], materials technology [4–6], geology [7, 8], epidemiology [9–12], search and rescue [13, 14], space science [15, 16], and neuroscience, such as in signal processing (electroencephalogram/electrocardiogram (EEG/ECG)) [11, 17, 18], diagnostic imaging [19], histology [19–21], tumor morphology [22, 23], vasculature [24, 25], brain structure and function [22, 26–28], consciousness [29], and mental fatigue [30]. It has illuminated intersections amongst complex details in areas such as where dance meets posture [31], music meets breathing [32, 33], rivers flow [7], cities grow [34–37], mud cracks [38], and clinical trials tackle macular degeneration [39].

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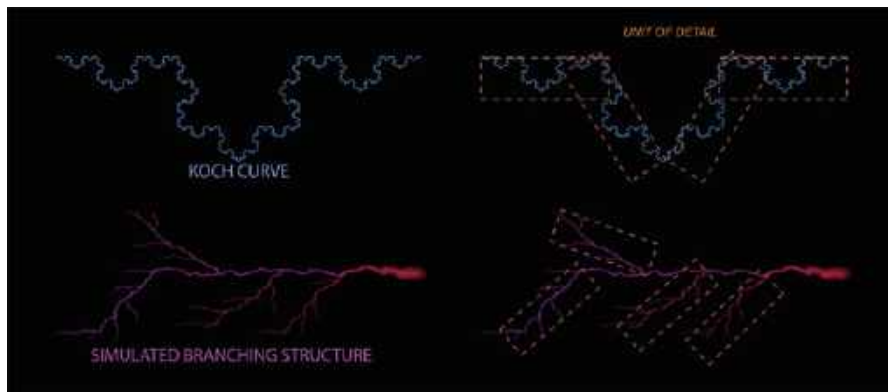


Fig. 2.1 Two patterns illustrating identical (Koch) vs. approximate (branch) self-similar repetition. Both patterns are constructed of four repeating subunits (shown in the boxed “units of detail”) that are, respectively, identical to or are approximately like the whole structure, just $1/3$ smaller. (Koch fractals generated with the ImageJ Fractal Generator [47]; branching fractals generated with MicroMod [48, 49])

It enlightens us when we need to compare or assign a number to peculiarly detailed phenomena that are difficult to describe and measure, except to say that they repeat themselves, or, in the parlance of fractal analysis, are “self-similar” [40, 41].

Self-Similar

Self-similar means repeating a part of a greater identity; *similar* connotes either “being identical to” or “resembling” an overall pattern, as illustrated in Fig. 2.1.

The two self-similar phenomena illustrated in Fig. 2.1 will be referred to several times in this chapter. Both depict physical phenomena, but, as the introductory paragraph might suggest, the term “self-similar” can describe many types of features in time and space, or both, and might, but need not necessarily, reflect a single underlying physical structure. A self-similar signal, such as from a heartbeat or from flow integrated across any network, for example, can emerge from several inputs rather than a singular physical fractal structure [18, 42–45]. Indeed, one might expect, within today’s trends in big data, artificial intelligence, sensors, and the Internet of things, to find self-similarity being discovered in many complex system views in neuroscience (e.g., see Lin et al.’s work [46] linking fractal features of cellular phenomena to fractal features of ambient words and other sounds, possibly mediated through effects on cytoskeletal proteins) [17, 19, 26, 44].

Self-similarity is the basis for fractal analysis, meaning quantitating the peculiarly difficult-to-describe phenomena listed above using a metric called the *fractal dimension* (D_F). Whereas the D_F applies to all temporal and spatial realms, this chapter, a primer for the rest of this book, applies it chiefly to patterns in two dimensions (2D), self-similar phenomena for which the complexity can be measured in a process analogous to zooming in on the detail on a slide under a microscope or

a digital image on a computer screen. In this context, a working definition of a D_F for this chapter follows:

Fractal Dimensions D_F s)

A fractal dimension can be defined as a measure of a pattern's *complexity*; a unitless number describing self-similar scaling as an index of change in detail with change in resolution [50]; strictly speaking, the number conveys that a pattern scales as a fractal only if the pattern both has *meaningful detail* and repeats that detail irrespective of the magnification or resolution at which we inspect it.

In this definition, the phrase “meaningful detail” is critical. To understand it, imagine inspecting two types of patterns. Imagine first zooming in on a simple line and always seeing the same basic pattern magnified—this is trivial (not fractal) geometrical scaling. Now, imagine zooming in on the Koch fractal from Fig. 2.1 but always being startled to see the same unit of detail reproduced at every level of resolution—this is fractal scaling [40, 41, 51]. Figure 2.2 illustrates this distinction as well as the differences between theoretical and practical self-similar scaling that are essential for understanding the rest of this chapter.

The series in column (a) in Fig. 2.2 tells the story of theoretical fractal curves, which are scale-independent, infinitely reproducing their unit of detail regardless of magnification or resolution. The column represents what we would see if we could inspect a fractal pattern through an infinite magnifying glass or what we can simulate in dynamic online computer graphics. Note that the bottom row in this column emphasizes that the unit of detail captured in each box is reproduced four times for each 1/3 scaling, rather than three times for each 1/3 scaling, thus highlighting an unintuitive feature of fractals: that the unit of detail is generally reproduced a certain number of times that does not simply recapitulate the scaling factor used.

The story in column (b) contrasts with the theoretical story. It shows a practical constraint to understanding the meaningful detail in theoretically infinite scaling that comes into play when a fixed digital representation of a theoretical fractal pattern is zoomed in on. Unlike a theoretical fractal, a fixed digital version of one has a limit that may no longer respect the underlying unit of detail. Thus, irrespective of the resolution used to create a static image, as we zoom in, the detail eventually diminishes to a point where the diameter of the line used to render the image is merely magnified. The bottom frame in column (b) stops at a point that reproduces the unit of detail, but this is only because of the authors' arbitrary choice; no amount of magnification would make a truly infinitely scaling theoretical fractal resolve into its basic unit of detail.

The story in column c is also essential to understanding “meaningful detail,” but, in this case, the story is about how trivial scaling differs from meaningful fractal scaling. That is, column c shows what “meaningful detail” is not. To elaborate, a simple line, either theoretical or graphical, may scale self-similarly and, for a theoretical one, infinitely, but the scaling in both cases is trivial, ever resolving into virtually straight segments and always lacking detail.

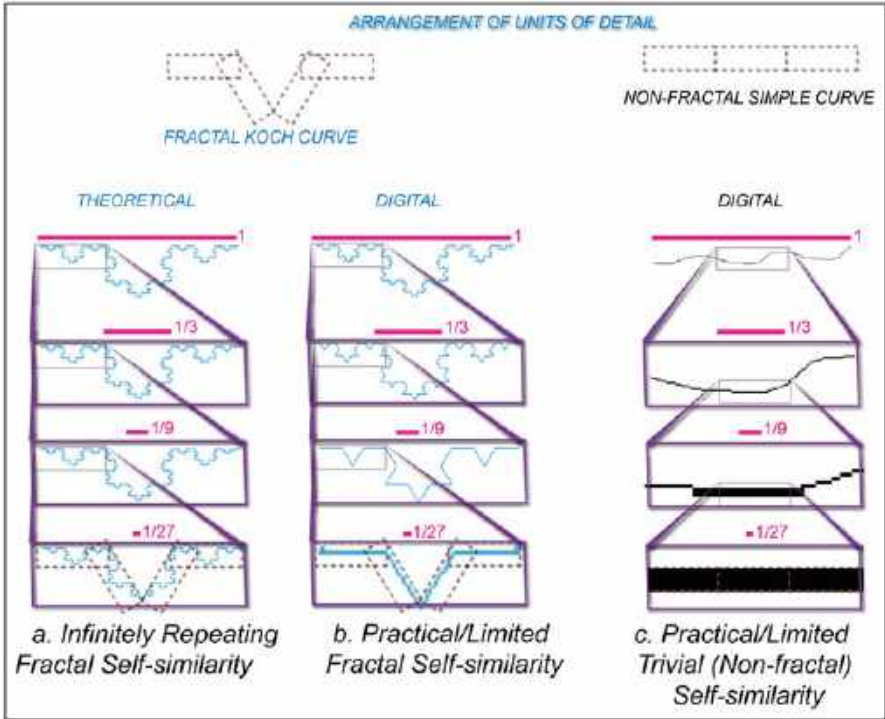


Fig. 2.2 Infinite, practical, and trivial self-similar scaling for fractal and non-fractal curves. This figure illustrates what happens when we zoom in on one unit of detail theoretically rather than practically. The top of this figure shows units of detail for the Koch and simple curve, respectively. These units show how each pattern can be broken down/built up so that it repeats a fundamental element of self at every level of magnification. Columns (a–c) show the scaling process. Each step in a series shows what happens as we zoom in on $1/3$ of the pattern above it. The top of each column shows the starting pattern and a bar for the net length of the pattern at the first level of resolution, assigned an arbitrary length of 1. Moving vertically down the columns, each successive row shows $1/3$ of the row above magnified by 3 (with the smaller boxes magnified to the larger boxes); to add perspective, each bar shows the new net length relative to the net length of the starting row for each column. (Koch fractals generated with the ImageJ Fractal Generator [47])

One further story to know in order to understand *meaningful detail* for this chapter involves digital imaging fidelity. It is important to understand the limits of imaging. Fixed digital images are rendered from arrays of fixed pixels so that, at the finest resolutions, lines or edges, including arcs, are approximated by pixel-sized steps (see Fig. 2.3) [52]. Although we can either imagine meaningful detail infinitely scaling as in column (a) in Fig. 2.2 or mimic it infinitely scaling in dynamic computer graphics zooming in on online games and simulators, we can only approximate it in fixed pixel-based digital and printed images. This is because they repeat only to the practical minimum element size used to capture or render

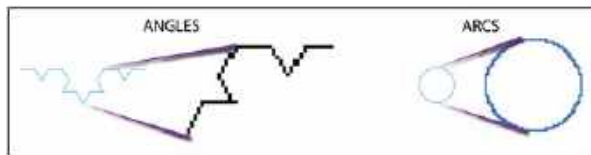


Fig. 2.3 Lack of fidelity when emulating angles and arcs using pixels. This figure illustrates what happens when one zooms in on digital images. Some lines may appear smooth and uniform (the smaller figures) but are actually discontinuous approximations. Digital images are rendered on arrays of fixed and discrete pixel points that emulate but cannot perfectly represent lines in all possible orientations. As a result, at the finest resolutions, edges that are not vertical or horizontal to the array (e.g., as shown here) can only be approximated. (Koch fractals generated with the ImageJ Fractal Generator [47])

them (e.g., pixels or printer resolution). This general practical limitation, which is that the meaningful detail is not infinitely continuous and is likely to be distorted at some point, applies to whatever optics or method of imaging is used to inspect a pattern acquired for analysis from a fixed image.

2.1.1 Supplementary Measures in Fractal Analysis

In addition to a D_F , peculiar phenomena with difficult-to-describe details can be typically quantified with complementary metrics, such as *lacunarity* and *multifractality*. At the other end of the spectrum to self-similarity, lacunarity measures heterogeneity (Sect. 2.4), while multifractality, at a place that introspects fractality, identifies patterns that are better probed for a series of D_F s (Sect. 2.7). The rest of this chapter explains a way to calculate and critically evaluate all three in neuroscience.

2.2 How Is D_F Calculated?

One of the simplest ways to calculate a D_F appeals to the basic working definition that was provided above: It is the logarithmic ratio of the *change in detail*(N) to the *change in scale*(S) for a pattern (Eq. 2.1):

$$D_F = \frac{\log N}{\log S} \quad (2.1)$$

The relationship between detail and scale is the key to understanding what a D_F conveys about complexity [40, 50, 53]. A D_F elegantly relates the changes in

detail and scale to each other somewhat like “narrowness” relates length to width, “circularity” relates circumference to diameter, and “density” relates content to container. But whereas people can decide how relatively narrow, circular, or dense something is by looking at it, most of us cannot so easily gauge how relatively complex something is by looking at it (albeit as fractal theory and analytical methods are taught and used more and more in math, computing, biology, health, and social realms, it seems that may be changing).

It may seem unintuitive, but the relationship between detail and scale is explainable in a straightforward way. Imagine, for example, zooming in under a microscope on the contour of a tumor or on the circumference of a crenelated plasma membrane, or similarly, zooming in using any device screen on the length of a winding DNA strand or tortuous blood vessel in a digital image. Depending on our reason for measuring any of these structures, we might be interested in different attributes, like the fully unraveled length or, alternatively, the net distance covered from one end to the other with all of the twists and kinks in place. Likewise, depending on how we expect to use our measurement of the Koch fractal from Fig. 2.1, we might measure it as either as long as the sum of all the steps along its winding path or as long as the box enclosing it. For Eq. 2.1, the change in detail is based on the first measure, which is found by unraveling the twists or, more generally, counting the units of detail, and the change in scale is based on the second measure, i.e., the net length:

Change in Detail N in Eq. 2.1)

Change in detail is the number of times a unit of detail repeats at each scaling, on the basis of which the number of new parts in each scaled piece is used to calculate a D_F . In terms of the Koch fractal in Fig. 2.2, for instance, the unit of detail has four new parts at each scaling. Although the arrangement of the new parts in the unit of detail is certainly extremely important and will be discussed in depth later, the number of new parts is sufficient for calculating a D_F (see Sects. 2.3.1 and 2.4.2.1 for a further discussion of the pattern itself).

Change in Scale S in Eq. 2.1)

Change in scale can be defined as the relative size of the unit of detail compared to the parent, on the basis of which the change in magnification or resolution in each scaled piece is used to calculate a D_F and the reciprocal of the ratio is multiplied by the original size to acquire a new size. In terms of the Koch fractal in Fig. 2.2, the enclosing box’s net length is multiplied by $1/3$ in each successive row so that $S = 3$. The numerator in this example is 1 but that is not a restriction—for scaling a geometric pattern into, for example, 4 pieces, each $3/8$ the original size, $S = 8/3$.

Thus, using these definitions and substituting the values for N and S into Eq. 2.1 for both the Koch fractal and the branching structure in Fig. 2.1 yields a $D_F = \log 4 / \log 3 = 1.26$. A deeper discussion of mathematics is beyond the scope of this chapter (please consult References, such as Falconer [40], for a deeper understanding).

2.2.1 Practical Points Relevant to the D_F

Before going further, there are some key practical points that a fractal analyst should know about the D_F and Eq. 2.1.

2.2.1.1 Not Unique

First, it is important to realize that a D_F is not a unique descriptor. As an example, Table 2.1 illustrates several combinations of N and S that lead to a D_F of 1.26, the value for the theoretical Koch fractal. In the same vein, it does not speak directly about the form of the self-similarity. One cannot work backward from a D_F or even N and S to determine the arrangement or form of the unit of detail.

2.2.1.2 Not Necessarily Exact, Infinite, and Known

Second, it is important to know some key points about the basic D_F equation. The reader might have noticed with trepidation that Eq. 2.1 asks for singular N and S values, for instance. This suggests that fractal scaling must be exact, infinite, and known in terms of N and S to determine a D_F using the formula, yet that would exclude finite images, patterns like the branching structure from Fig. 2.1,

Table 2.1 Examples of ratios that have a D_F of 1.26

N Change in detail	S Change in scale	D_F LOG N /LOG S
4	3	1.261859507
5	3.58030993	1.261859507
6	4.136854782	1.261859507
7	4.67438362	1.261859507
8	5.196152423	1.261859507
9	5.704522495	1.261859507
10	6.201278705	1.261859507
11	6.687813864	1.261859507
12	7.165242665	1.261859507
13	7.634475763	1.261859507
14	8.096269924	1.261859507
15	8.551263155	1.261859507
16	9	1.261859507
17	9.442950209	1.261859507
18	9.880522794	1.261859507
19	10.31307682	1.261859507
20	10.74092979	1.261859507

and virtually all of the peculiar phenomena one might wish to quantify with a D_F in neuroscience, which tend to scale less reliably than does the Koch fractal.

Fortunately, the paradox of needing to know in order to know is not borne out. Unlike the exactly scaling Koch fractal, for instance, even some infinite theoretical fractals may scale like the branching structure in Fig. 2.1, which is only statistically self-similar, as defined below [40, 41, 50, 51].

Statistical Self-Similarity

Statistical self-similarity can be defined as follows: self-similar scaling that is not necessarily infinite or not necessarily exact (or both), scale invariance, the type of scaling typically seen in nature, and a statistically self-similar pattern or process, for example, which may be composed of units of detail similar, but not necessarily identical, to the whole and that may repeat in a regular but not necessarily strictly defined pattern, as shown in Fig. 2.1 and explored in Fig. 2.4.

In addition, it is possible to use the equation to estimate a D_F by setting a practical limit to the scaling and then sampling a pattern over a practical range of N and S (to be discussed in detail in Sect. 2.3.2) without knowing anything upfront about either. These points might be liberating to the concerned researcher, who can now feel free to investigate fractal scaling in any phenomenon without having to first determine whether the phenomenon itself is a “legitimate” fractal for Eq. 2.1.

2.2.1.3 Density Vs. Dimension

Finally, a third practical point to know to understand Eq. 2.1 relates to density. To understand how density affects the D_F , consider how the Koch curve compares to a line and a plane. A straight line, when scaled by $1/3$, resolves into three new parts, each part $1/3$ the length of the original. The line, therefore, has a $D_F = \log 3 / \log 3 = 1.00$. Similarly, a filled plane scaled by $1/3$ would have nine new parts and a $D_F = \log 9 / \log 3 = 2.00$. Now, revisit the Koch fractal from Figs. 2.1 and 2.2, which when scaled by $1/3$ resolves into four new parts. It has a D_F of 1.26, lying between the limits of one and two dimensions (i.e., the values for the line and filled plane, respectively).

Going further, now consider Fig. 2.5, which compares the Koch fractal and its unit of detail to a 32-segment quadric fractal and its unit of detail. The quadric fractal pattern’s unit of detail has 32 new parts scaled to $1/8$ and arranged as shown, giving it a greater $D_F = 1.67$, meaning that it is more complex than the Koch fractal.

Taken together, these comparisons suggest that the D_F approaches the dimension of the space that a pattern is embedded in and that the closer a pattern comes to filling the space it is embedded in, the higher the D_F . But the reader should note that a D_F in any domain is not just a proxy for density, and all D_F s do not necessarily predictably correlate with it. All the patterns in Fig. 2.4, for instance, were assigned a theoretical D_F of 1.41, strictly based on branch length and number. A D_F that accounts for the changing mass and tortuosity with resulting overlap in the extracted pattern, however, might be different (see Sect. 2.4.2.1) [40, 41, 50, 54].



Fig. 2.4 Exactly and statistically self-similar branching. All six images represent the same branching pattern with a unit of detail that generates four new branches, with each 3/8 the length of the parent at $\pm 45^\circ$ to the parent, for a theoretical $D_F = \log 4 / \log(8/3) = 1.41$. (a) A skeletonized version and a filled silhouette of the pattern both generated using “exact self-similarity,” whereby the pattern reproduces itself by generating four scaled branches always at the same relative size, location, and orientation relative to a parent branch. (b) The rest of the versions, all generated using statistical self-similarity. These patterns reproduce themselves by generating not exactly but, on average, four new branches that are 3/8 the length of the parent, based on a given probability that four scaled branches will occur at the same size and orientation relative to their parent branches. For the four fixed angle images, all branches continued growing at $\pm 45^\circ$ relative to the parent; in contrast, for the probabilistic angle and tortuosity images, the orientation changed after sprouting at $\pm 45^\circ$ relative to the parent. After that point, branches grew according to a given probability of the orientation changing with each increment. It should be noted that as the manifested structures become more subject to random influences, the underlying pattern does not change, yet it becomes increasingly difficult to discern; at the same time, the images start to resemble the natural phenomena more. (Images generated using MicroMod for ImageJ [48, 49])

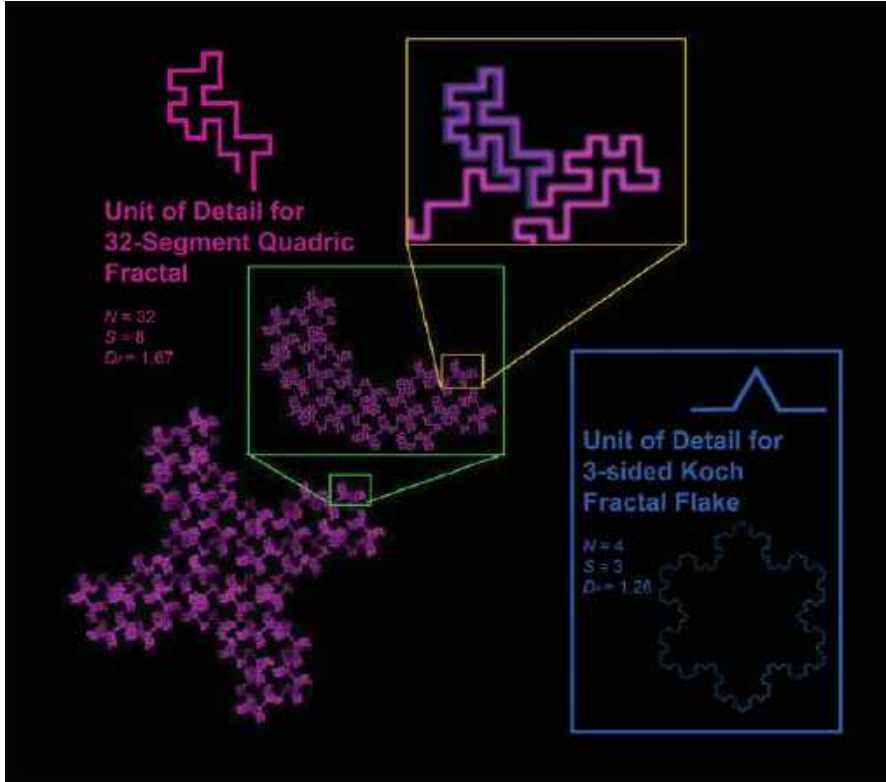


Fig. 2.5 Scaling and units of detail for a 32-segment quadric fractal vs. a Koch curve arranged as a 3-sided flake

2.2.2 D_F s in Neuroscience

The complexity that a D_F captures seems to run rampant in the peculiar phenomena of neuroscience. Gazing through the structural lens that this chapter adopts, one might find the complexity in the contour of the Koch fractal reminiscent of the complexity in the borders of tumor masses, the plasma membranes of individual tumor cells, and the membranes of cell parts such as the nuclei within a tumor cell on histopathology slides. Similarly, the complex branching shown in Fig. 2.2 suggests multifarious networks of scaling, branching, and tortuous nervous system structures such as tracts, vessels, neurons, and glia.

Indeed, the D_F has been applied to topics ranging from gross overall nervous system structure and function [17, 19, 22, 26–28, 45, 55] to tracts, circuitry [17, 56], and vasculature, including the most tortuous of vessels, great and small [12, 25, 57–59], to tumor morphology and growth [22, 23, 60, 61], and further to cells

and groups of cells and on to the finest levels within cells and the extracellular milieu [62–65], all represented by a variety of imaging modalities, including retinal photography, radiography, histopathology, electron microscopy, etc. [20, 66–70].

Exploring D_F s across these levels from gross systems to the tissue level to cellular levels and further to the subcellular and molecular levels has shown that, generally, all ways of inspecting a phenomenon may or may not necessarily converge on a single D_F . Whether or not this happens depends on the particular method of extracting a pattern from a brain structure or a process, where several pertinent D_F s may exist. Popovic et al., for example, found preliminary evidence that the D_F for retinal vasculature depends on the region of the vasculature being investigated [71]. Other researchers investigating a different topic, namely, the D_F s for cultured rat oligodendrocytes, found them to be consistent over a ten-fold range of magnification when comparing portions of cells, whole cells, and entire fields of cells [71]. Still others found that the D_F s for different layers of newborn rat retina have distinct values where altered oxygen exposure leads to histoarchitectural changes characterized in some layers by increased but in others decreased D_F s [58, 73]. Thus, it is important to be aware that different patterns can be extracted from the same image or underlying phenomenon and may or may not generate different D_F s (see also Sect. 2.4.2.1).

How were all of these D_F s calculated for these complex and detailed phenomena? Although we have discussed how to find one when both S and N are known, in practice, units of detail in complex neurobiological phenomena are not readily identifiable. Going back to the patterns in Fig. 2.4, for instance, the multifarious branching patterns found in the brain resemble more the statistically self-similar pattern with probabilistic angles and tortuosity than the more deterministic examples in that figure. Emphasizing this point, Fig. 2.6 illustrates scale invariance in membrane detail and branching patterns for two glial cells abundant in the brain.

As might be inferred from Fig. 2.6, one reason natural fractal scaling is typically not easily measured is sheer practicality. When fractal analysis first started being applied in neuroscience in the 1970s and 1980s, patterns were laboriously extracted [79], people were manually tracing, unraveling, determining angles, and measuring lengths and widths of parts of patterns, then identifying and counting them as units of detail, and then further analyzing the measurements to find scaling trends. Now, it would be prohibitively tedious and time-consuming to do this, especially for large datasets of images, electroencephalogram (EEG) signals, etc.

Fortunately, the tedium of finding units of detail and scaling rules can be left to software and the joy of discovery to the analyst. Without manually measuring any part of the cells in Fig. 2.6, one can estimate their D_F s using any of several accessible software packages. From the freeware FracLac for ImageJ [49], for instance, the measured D_F of the astrocyte in Fig. 2.6 was found to be 1.58 and the microglial cell 1.59, both calculated using default values and a quick click. These values are not theoretically but empirically determined D_F s called box-counting dimensions (D_{BS}).

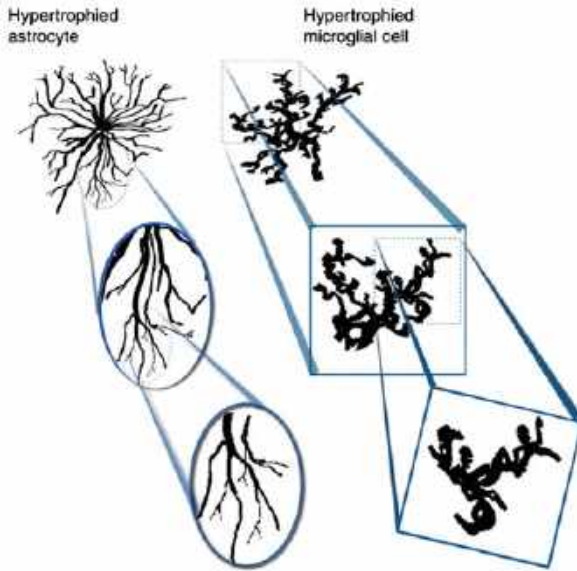


Fig. 2.6 Silhouettes of two types of brain cells from histopathology slides (shown at different scales) characterizing statistical self-similarity as is typically found in nature. The insets highlight the units of detail roughly repeated at each scaling as revealed by the respective contrast method used to image the cells. “Hypertrophied” refers to morphological changes after brain trauma compared to in normal tissue. These cell morphologies were deliberately selected to illustrate fractal scaling; in reality, however, both astrocytes and microglia comprise morphologically and functionally heterogeneous groups of which the cells depicted are but one type. These particular instances of the types were selected because they are useful examples, whereas, in practical situations, quantifiable self-similar units of detail are not always readily identifiable in these cells [74–78]

Box-Counting Dimension D_B)

A box-counting dimension is a type of empirical D_F typically calculated using software implementing methods that measure proxies for N and S for Eq. 2.1; a D_B does not identify or depend on identifying the actual unit of detail and its scaling ratio but is instead usually inferred from sampling data.

One can analyze the fractal geometry of the brain using any of several D_F s (e.g., dilation, mass vs. radius, or local connected) [80–85], but a D_B stands out as an elegant, robust, and available measure suitable for learning about, understanding, and performing fractal analysis at an expert level [75, 77, 78, 86–93].

2.3 Box Counting

What exactly does the D_B measure for the cells in Fig. 2.6? It measures the usual ratio of change in detail with change in scale from Eq. 2.1, with a slight twist to

measure detail indirectly. The basic method examines images at increasing levels of resolution using a series of sampling elements decreasing in size and then determines the scaling ratio from the data gathered, based on the box-counting S (S_B) and N (N_B).

This is similar to how the Koch fractal was scaled and magnified in Fig. 2.2, except that in box-counting algorithms, S_B and the unit of detail are not always known upfront, so an image is sampled using arbitrary values of S_B , an arbitrary sampling element, and an arbitrary pattern of placing the element (grid orientation or G). The unit of detail is not directly identified, but its existence is inferred after the fact. Figure 2.7 illustrates the essential process for binary images (defined below) of the structures introduced in Fig. 2.1.

Binary Images

Binary, in the context of fractal analysis usually refers to digital images having only two possible pixel values (usually black and white), one of which is deemed *foreground* and the other *background*. Binary images are the usual input for binary box-counting fractal analysis. They are usually extracted from original color or grayscale (Sect. 2.5) digital images as three basic types: *binary silhouettes* are filled patterns, *binary contours* are unfilled outlines, and *binary skeletons* are tracings of a pattern without regard for diameter (such as the branching pattern in Fig. 2.4). Importantly, all three types can generally be extracted from the same original image (see Sect. 2.4.2.1 for a discussion of the pattern type and D_B).

Note that for a and b in the figure, N and N_B are identical. This is only because b was initially set up using $S_B =$ the Koch fractal's known S and then, for illustration, the G deemed visually closest to ideal was selected out of many random trials. But all this downplays perhaps the most important potential pitfall of box counting—that for any S_B or series, a different G can yield a noticeably different N and a correspondingly different D_B . So, N_B corresponds to N only in an average and contrived sense in the illustration.

The figure also shows the differences in the sampling element shape and orientation for the theoretical vs. empirical methods. Owing again to the point that the unit of detail is generally not known in practice, one must choose a sampling element. The elements are usually conceptually boxes, meaning square arrays of pixel points, but other sampling elements can also be used (e.g., rectangles, triangles, hexagons, ovals, complex shapes, or even virtual volumes [56] instead of areas). Whatever the shape, elements are characteristically laid down in a pattern that does not overlap them, in a “tight” or “fixed-grid” pattern to sample images exhaustively. However, again, this is not the only possibility, and as was shown when the unit of detail was “boxed in” for the Koch fractal in the first part of this chapter (e.g., see Fig. 2.1), actual units of detail can indeed overlap when counted using boxes. Alternative algorithms address this by various methods such as oversampling (overlapping sampling elements) or, conversely, undersampling (sampling non-exhaustively), as in, respectively, sliding-box and random sampling algorithms [37, 52, 81, 95–98].

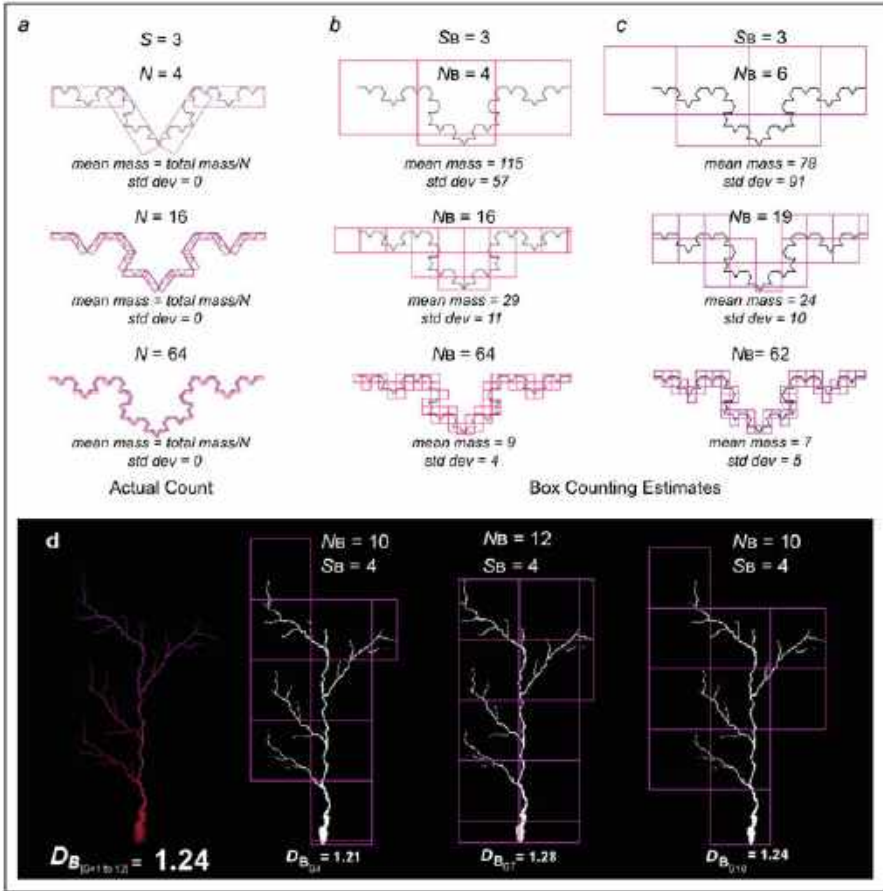


Fig. 2.7 Measuring the change in detail with change in scale. N (or N_B), the number of boxes containing patterns (or foreground pixels); mean mass for N (or N_B), the mean amount of patterns (or number of foreground pixels) per box; std. dev, standard deviation for the mean mass; G , grid orientation; G_X , the X^{th} G in a set of 12 arbitrary G s; $D_{B[G]}$, the D_B from a series of box sizes in a pattern over a given G . **(a)** Measuring a Koch curve precisely using overlapping boxes based on its known unit of detail and S (see Fig. 2.2). **(b)** Optimized box counting, approximating the process in **a** based on a priori knowledge of S_B and the unit of detail. **(c)** Unoptimized box counting, approximating **a** based on a priori knowledge of S_B and a random guess at G . For the fixed-grid method in **b** and **c**, images are broken into evenly spaced boxes despite the fact that some edge blocks in this figure are truncated for display; these are counted as an entire box if the mass > 0 . **(d)** Optimized box counting over a simulated branching structure with known theoretical $D_F = \log(4)/\log(3) = 1.26$, using known $S_B = 3$, a series of 75 box sizes, and 12 random guesses at G . This figure lists the average empirical D_B over all 12 G s ($D_{B[G=1-12]} = 1.24$) and an intermediate D_B for 3 selected G s. It shows the actual orientation for those G s, comparing the largest size only, to emphasize the profound effect of G on N_B and therefore the D_B (Koch fractals generated with the ImageJ Fractal Generator [47]; branching fractals generated with MicroMod [48, 49]; **b**, **c**, and **d** generated in FracLac using the “draw grids” option [94])

2.3.1 Converting S_B and N_B to a D_B

Table 2.2, in columns i–iii, shows that in box counting, S_B and N_B for Eq. 2.1 each comprise a series of values, with S_B at any size ($S_{B[S]}$) and N_B at any size ($N_{B[S]}$). $S_{B[S]}$ relates to the sampling element's size and $N_{B[S]}$ to its contents at each size. For 2D images, $S_{B[S]}$ approximates some measure of the sampling unit's length, such as one side or diagonal of a box, the diameter of a hexagon, or an axis or root of the area of an ellipse or oval. From the example in Fig. 2.7a using rectangular sampling elements and the length as the measure, if the top image is 1 arbitrary unit, at a 1/3 scale, then the relative box "size" would be $1 \times 1/3 = 1/3$ with $S_{B[S]}$ (its reciprocal) = 3.

$N_{B[S]}$ is a little more complicated, depending on the type of analysis being performed:

- For **fixed-grid binary box counting**, $N_{B[S]}$ is the number of boxes containing any foreground pixels at that size, as illustrated in Fig. 2.7.
- For **binary mass box counting**, $N_{B[S]}$ becomes $N_{M[S]}$, based on the average foreground pixels per box at a size [84, 96, 99] and is used to determine the fractal metrics other than the D_B , including mass dimensions, mass lacunarity (Sect. 2.4), and mass multifractal spectra (Sect. 2.7) [50, 95, 96]. It should be noted that the masses in b and c in Fig. 2.7 were rounded, so the total mass accounted for differs across the methods despite that the same image was used for all.
- For **3D and grayscale box counting**, $N_{B[S]}$ is based on an average projected volume per box size (Sect. 2.5) [40, 84, 99].

2.3.1.1 Regression Lines

Table 2.2, row a, column iv, typifies the case for data gathered with a priori knowledge. The results in this case are generally consistent for any element size in a series and so are amenable to being plugged directly into Eq. 2.1. In contrast, rows b and c, column iv show that this is not true for the typical data obtained without such knowledge. Rather, in this case, the D_B is inferred through regression analysis as the limit of the ratio for Eq. 2.1 as the size gets smaller and smaller, calculated as the slope of the regression line for $\log N_B$ vs. $\log S_B$. (Note that because the starting image size is the same for all sample sizes, S_B can be determined from the reciprocal of each box size.) [40, 50, 94, 95].

2.3.2 Methodological Issues in Box Counting

Knowing that one must decide how to sample, one might wonder just how to make that decision and then how to squeeze the sampled data into Eq. 2.1. As noted earlier,

Table 2.2 Theoretical, idealized vs. empirical methods for calculating the D_B for the Koch fractal

Sampling method	i. Box size (one side)	ii. $S_B[S]$	iii. $N_B[S]$	iv. $\log N/\log S$	vs. D_B as log or reg	vi. r^2
(a) G for Fig. 2.7a, idealized, a priori knowledge	Relative ^a					
	$(1/3) \times 1 = 1/3$	3	4	1.26	Log = 1.26	1.0000
	$(1/3)^2 \times 1 = 1/9$	9	16	1.26	Log = 1.26	
	$(1/3)^3 \times 1 = 1/27$	27	64	1.26	Log = 1.26	
(b) G for Fig. 2.7b, box counting, optimized, a priori knowledge	Pixels					
	$(1/3) \times 81 = 27$	1/27	4	0.42	Reg = 1.26	1.0000
	$(1/3)^2 \times 81 = 9$	1/9	16	1.26		
	$(1/3)^3 \times 81 = 3$	1/3	64	3.79		
(c) G for Fig. 2.7c, box counting, not optimized, no a priori knowledge, the same sizes, different G than Fig. 2.7b	Pixels					
	27	1/27	6	0.54	Reg = 1.06	0.9999
	9	1/9	19	1.34		
	3	1/3	62	3.76		
(d) G not shown, box counting, not optimized, no a priori knowledge, different sizes and G than Fig. 2.7b, c	Pixels					
	40	1/40	5	0.44	Reg = 1.22	0.9897
	20	1/20	15	0.90		
	5	1/5	67	2.61		

Pixels, length in pixels of a side of the square used for sampling, $\log \log(N)/\log(S)$ from Eq. 2.1; Reg, regression line slope from Eq. 2.1 = slope of $\log(N_B)$ vs. $\log(S_B)$; r^2 , Pearson's correlation coefficient for the slope of $\log(N_B)$ vs. $\log(S_B)$

^aLength of the enclosing rectangle relative to the length of the original image (81 pixels)

software removes the tedium and preserves the delight of fractal analysis—it can make the sampling decisions, sample the images, and calculate the D_B s. However, the default values for the decisions in software may not always be relevant to a given research question. In general, prudent application requires some preliminary work to establish suitable methods for specific research questions [100, 101].

This is especially important because box-counting algorithms and software have become commonplace and readily available in the rapidly changing face of digital connectivity, placing some burden on the fractal analyst about what method to use. As for any tool in computational neuroscience and science in general, using a tool requires some fundamental background knowledge about how to ensure that the controls are in place in order to avoid seriously threatening pitfalls [40, 51, 100–103]. Accordingly, this section discusses several methodological issues faced in box counting and offers ways for the analyst to validate results in the light of those issues. Before getting into the details, perhaps the most overriding methodological issue for the savvy analyst to consider is transparency. The reader is most strongly advised to ensure that the box-counting implementation they use provides full access to the methodology (e.g., the source code for software) to ensure that all of the issues noted below are adequately addressed.

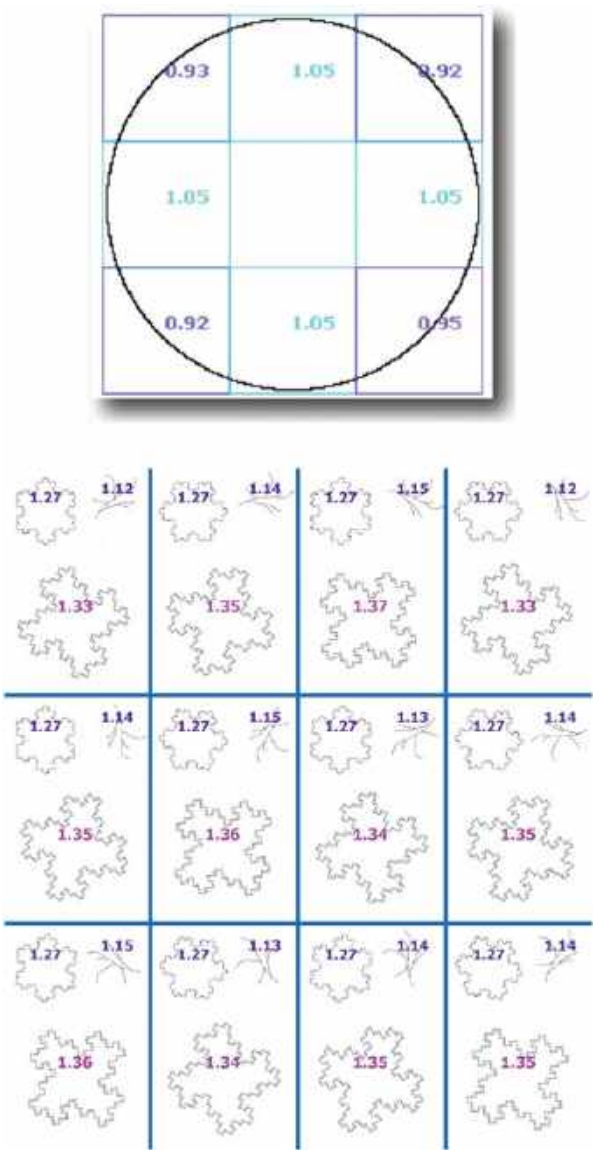
2.3.2.1 High Correlation Coefficients: Necessary but Not Sufficient

One criterion for validating a D_B is a high correlation coefficient for the data (r^2 in Table 2.2). By the properties of regression lines, if the relationship between $\log N$ and $\log S$ is linear for a large proportion of the sizes, then the correlation coefficient will be high. However, a valid linear relationship, i.e., a high r^2 , does not necessarily validate a D_B . Indeed, several regression lines with r^2 greater than 0.99 (with 1.00 being the perfect correlation) but widely differing slopes (D_B s) can be found for the same control image with a known D_F by manipulating the scaling between and the range of sizes as well as how boxes are positioned [40, 49, 63, 100]. In other words, a low r^2 invalidates a D_B and a high r^2 is necessary for but does not guarantee high validity.

2.3.2.2 Sampling Size, Location, and Orientation/Rotation Biases

Thus, to meaningfully validate the regression line slope, box-counting implementations must also address $S_{B[S]}$ (e.g., the smallest and largest valid sizes and the method of scaling sizes between them) and G and make available to the analyst all assumptions about them. How profoundly the D_B depends on G is shown in Fig. 2.7, wherein images were sampled using different G s, resulting in the ambiguous situation of there being many D_B s and mass-related measures for any pattern assessed, even when using the same $S_{B[S]}$ (e.g., compare the D_B s from Table 2.2 in rows a–d) [1, 100, 104, 105]. Fortunately, robust ways to address these have been established and are discussed in the section *Box-Counting Solutions* before going

Fig. 2.8 Differences in the D_B with different orientations. The figure shows the D_B over 9 arcs of a circle as well as for 2 control fractals and a simple skeletonized structure rotated 12 times. All images were rendered at low resolutions. (Images generated using, respectively, the “subscanning” and “rotation” functions in FracLac2020 [94])



there, though, it is wise to discuss another key issue underpinning G -related bias: rotation [63, 100].

As illustrated in Fig. 2.8, the rotational orientation (R) of a pattern or, from an alternative perspective, of the sampling grid, can affect the D_B [95, 100]. It is known, for instance, that R affects the digital images of non-stellate more than stellate neurons [106]. Recapitulating the basic message in Fig. 2.3 about rendering arcs and angles in digital images, the R at which a phenomenon is captured in the

Table 2.3 Effect on the D_B of rotating 800×800 pixel digital images 12 times by 30° increments (created using the “rotate” function in FracLac) and then sampling each over 12 G per rotation (for a total of 144 samples per image)

Image	$\mu(R)$	$D_{BR} = \Sigma D_{B[G = 1-12] R[0-360]}/12$				$D_{BT} = \Sigma D_{B[G = 1-12; R = 0-360]}/144$		
		Min(R)	Max(R)	$\sigma(R)$		Min	Max	σ
b-Koch curve-o	1.25	1.24	1.26	0.005		1.22	1.29	0.022
b-Koch flake-o	1.27	1.22	1.29	0.001		1.21	1.30	0.020
b-rectangle-o	1.02	1.02	1.02	0.001		0.99	1.04	0.019
b-rectangle-f	1.85	1.74	1.85	0.048		1.72	1.91	0.055
b-mic-o	1.52	1.52	1.52	0.002		1.50	1.57	0.022
b-mic-f	1.59	1.58	1.60	0.003		1.57	1.63	0.021
b-Ast-o	1.59	1.57	1.59	0.009		1.55	1.62	0.021
b-Ast-o	1.58	1.58	1.60	0.008		1.56	1.61	0.020
g-TEM-s	1.61	1.58	1.67	0.046		1.56	1.72	0.051
g-TEM-c	1.62	1.61	1.62	0.000		1.60	1.66	0.021
g-retina-s	1.25	1.21	1.31	0.043		1.18	1.34	0.045
g-retina-c	1.28	1.28	1.28	0.002		1.26	1.30	0.009

D_{BR} , the box-counting dimension averaged over 12 rotated images, each itself first averaged over 12 G ; D_{BT} , the box-counting dimension averaged over all R s and G s; $\mu(R)$, mean D_{BR} = mean D_{BT} ; $\text{Min}(R)$, D_{BR} for rotation image with the lowest D_{BR} ; $\text{Max}(R)$, D_{BR} for rotation image with the highest D_{BR} ; $\sigma(R)$, standard deviation for D_{BR} ; Min and Max , lowest and highest D_{BT} , respectively; σ standard deviation for D_{BT} ; b binary, g grayscale, o outlined, f filled, s square sample, c circular sample, *Mic* microglia (Fig. 2.6); *Ast* astrocyte (Fig. 2.6); *TEM* transmission electron micrograph of the mouse cerebellum (Fig. 2.14); Retina, retinal fundus photograph

first place and, similarly, rotating an image through an arc digitally require making approximations that can change the primary data—in sum, create a new pattern [1, 100]. (Note that this is a digital image-rendering phenomenon and is not the same as the feature known as *rotational invariance* measured by lacunarity and discussed in Sect. 2.4.) [59].

Table 2.3 quantitates the effects for rotating various control images through 360° in 30° increments from two perspectives, looking at the D_B for both each R and the overall set of R s. The same data were used for both, just analyzed in different ways, gathered based on $G = 12$ and the default series of box sizes provided in FracLac [49, 95]. The first part of the table calculates a D_B for each individual R then averages those, and the second takes the average of the full 144 D_B s as one group [49, 94].

The table highlights more trends to bear in mind when applying box counting in neuroscience. Although the overall mean is the same for each image from both approaches, the ranges and variations are not, and different patterns are affected to different degrees. The standard deviations for *b-Koch Flake-q* for instance, show a case where the range over R is broad yet the variability is not correspondingly high. In this case, a small part of the arc, through which the image was rotated, yielded a markedly different value compared to most of the arc. The table also shows that the D_B for the filled rectangle was considerably more affected by R than the outlined and

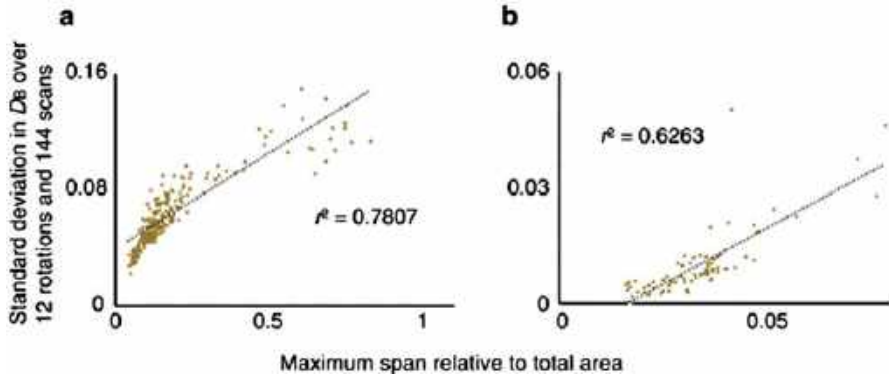


Fig. 2.9 Positive correlation between variation in the D_B with digital rotation and the ratio of maximum span to total area. **(a)** 310 binary outlines of astrocytoma nuclei. **(b)** 79 binary outlines of microglia. Both samples were also positively correlated with pixel density (not shown: astrocytoma $r^2 = 0.7119$ and microglia $r^2 = 0.5831$) but not circularity (astrocytoma $r^2 = 0.0427$ and microglia $r^2 = 0.0253$)

that the circular texture samples depended less on R (i.e., they have lower standard deviations) compared to square samples from the same areas on the same images.

In general, related to the point about stellate neurons noted above, the variation with digital rotation also depends on the overall shape of and pixel density in a pattern. Figure 2.9 illustrates this using the ratio of the maximum span across a pattern to the total area of the pattern for two samples (binary contours: one from the nuclei only in astrocytoma and one from whole microglial cells).

2.3.2.3 Box-Counting Solutions

All these methodological issues revisit the paradox we discounted in Sect. 2.1—that accurate and valid results require knowledge beforehand of the very features we hope to quantify. If we knew a priori about the unit of detail and the optimal series of sizes, orientation, and box placement for each image, then it might not add much value to perform an analysis other than to replicate results or validate controls. Those are essential tasks but are unlikely to enlighten us about the peculiar phenomena that we cannot yet characterize. Providentially, though, as was already mentioned when the basic paradox was brought up, manifold solutions are available in box-counting software to address these tremendously important problems. These include solutions for sampling over multiple G s and R s, controlling for density differences, empirically determining optimal samplings such as minimum and maximum covers, etc. [100, 104, 105, 107].

The method to select depends on the research needs. For instance, to generate a series of sizes when the scaling is known ahead of time (such as to validate software settings using control images of known D_{FS} as in Fig. 2.7), one might choose an

implementation that allows a custom series to be calculated. For more general cases when the scaling is not known, one would want software that offers a variety of arbitrary designs, such as a simple linear decrease (e.g., subtracting 1 from each size), rational scaling (e.g., scaling by $1/2$ or $8/3$), or power increments (e.g., going upward, $\text{size}_{(n+1)} = (\text{size}_n)^x$), and also that performs post-processing to choose an optimally covering series.

It is also important to be able to control or at least be aware of the minimum and maximum sizes, specified in pixels or as a percentage of image size. An important guideline in this respect is to be aware that, for control images at least, results tend to deviate from theoretical as box size exceeds roughly 45% of the pattern size (generally the smaller dimension if it is not symmetrical) [100].

To account for the profound effects of G that were discussed earlier here, one strategy is to run multiple scans and select data from the covering with the smallest number of boxes, highest correlation coefficients, least variation, etc. FracLac, for example, oversamples an image by selecting different G s and R s and then extracts an optimized G series using methods to select minimum covers and the smoothest scaling patterns, including methods for automatically optimizing minimum and maximum box sizes (see also Chap. 40). Under optimal conditions, these methods typically yield accuracies of 0.00%–3.00% from theoretical for fractals, non-fractals, and in silico biological models [94, 100].

2.4 Lacunarity

The second type of analysis promised to be covered here is lacunarity analysis. Lacunarity, complementary to the D_B , is based on the mass data that can normally be gathered during box counting [81, 84, 96, 99, 108].

Lacunarity

Lacunarity can be defined as a measure of heterogeneity in patterns, variously characterized as “gappiness,” rotational invariance, and visual impressions of patchiness in texture, all of which are typically calculated during box counting based on mass measures (i.e., the relative content of each sampling unit of a particular size) (see “mass” in Fig. 2.7).

2.4.1 Calculating and Applying Lacunarity

Like the D_B , lacunarity comes in various types that depend on the box-counting sampling method (e.g., “sliding-box” and “fixed-box” lacunarities) [99, 109]. Lacunarities calculated during box counting are often based on the mean mass per box at each sampling size, which results in a series of lacunarity values, one for each size. Various strategies reduce this monstrous amount of data to a single measure

describing an entire pattern, such as taking the mean of the mean for all sizes, and further selecting a representative value for an image if multiple G s are used [84, 100]. The lacunarity type we discuss here, i.e., fixed-grid, mass-based lacunarity (Λ), is based on the mean of all squared *coefficients of variation* (standard deviation divided by the mean) [40, 50, 84, 99]. Some ways to calculate lacunarity are redundant with the calculated D_F because both depend on the same data; however, this is not the case with Λ and the D_B because the mass from which Λ is calculated is independent of the count from which the D_B is calculated [84, 99, 100].

Lacunarity has been applied in several contexts in neuroscience [26, 44, 84]. As one example, Λ based on binary patterns extracted from dynamic contrast-enhanced computed tomography of head and neck tumors, images that reflect both tumor blood supply and capillary permeability, has been used to answer questions about how blood flow patterns in tumors relate to patient outcomes, showing that lower Λ of the overall vascularization pattern is associated with better outcomes, suggesting that a more heterogeneous tumor blood supply predicts worse outcomes [110].

2.4.2 Understanding the D_B and Λ

The D_B and Λ work together to objectively quantify the subtle differences in various phenomena, where patterns indistinguishable by their D_B are often distinguishable by their mass-related Λ , and vice versa [71, 74, 84, 99, 111–113]. Both, for instance, are sensitive to subtle changes in branching brain cell morphology that are not visually detectable yet have significant functional implications, but, whereas the D_B is more sensitive to branching, Λ is superior for particular features such as soma size relative to process length [78, 100, 109].

Section 2.4.1 outlines the computational basis of this complementary relationship; Fig. 2.10 illustrates it. In the figure, the D_B measures increase from left to right and Λ from bottom to top. To understand Λ , consider the column near 1.26, which has two images with similar calculated D_B s but drastically differing Λ s. Rotating Koch flake (KF), which has low Λ , affects it little; but rotating Henon map (HM), a canonical multifractal with a higher Λ , affects that image markedly—it has a distinct orientation relative to its surroundings. In terms of gaps, KF has one large gap, but HM has many irregular gaps. Similarly, for quadric cross (QC) and microglia (ramified, filled; Mrf) in the column from 1.50 to 1.51, moving up (i.e., as Λ increases), Mrf depends more on orientation and is more heterogeneous in its gaps [84, 99].

2.4.2.1 Pattern Idiosyncrasies

Another important point to be aware of is that the type of pattern selected and how it is extracted can affect the results. This includes factors like the thresholding method used to convert an image to a binary format [115] as well as any processing after

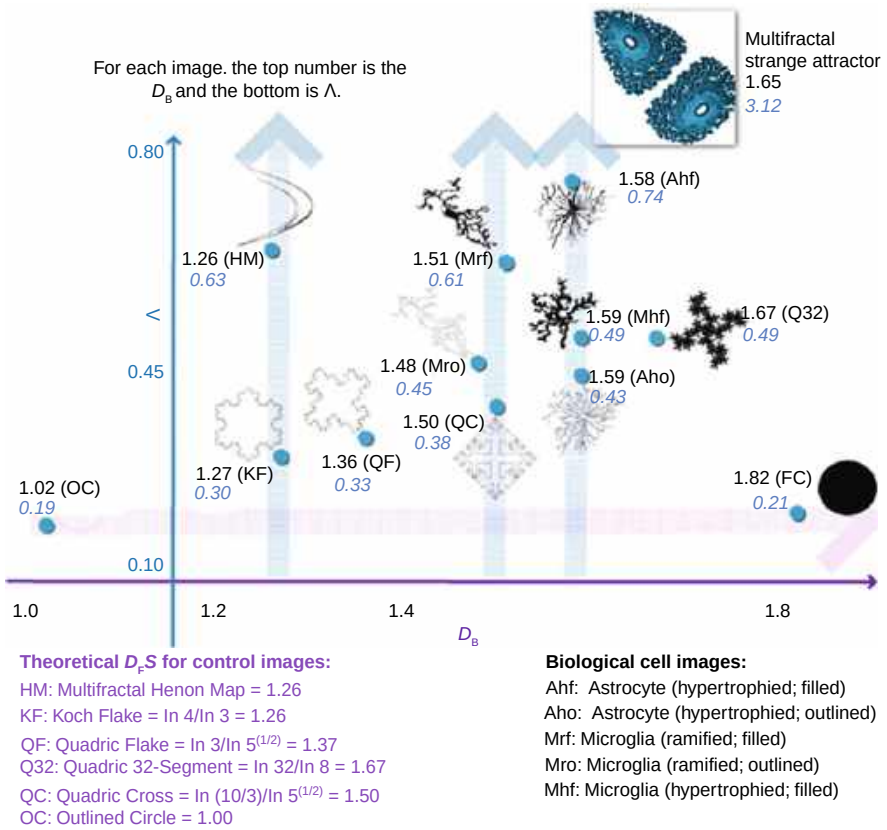


Fig. 2.10 Λ vs. D_B for selected binary images of theoretical fractals and brain cells The vertical axis's range is a limitation of the dataset, not of Λ per se, as indicated by the much higher Λ of the strange attractor at the top of the figure; this image is colored to emphasize the pattern, but the D_B and Λ are for its binary silhouette. The biological cells are drawn at roughly the same size because D_F s are scale-independent; however, the original astrocyte (Ahf/Aho) spans a broader area than do both the original microglia and the ramified microglia (Mrf/Mro) spans a broader area than does the hypertrophied microglia (Mhf) (D_B and Λ calculated using FracLac [49]; QF generated with the ImageJ Fractal Generator [47]; multifractals generated with fractal growth models [114])

that. Binary silhouettes typically have higher D_B s and Λ s than do either outlined or skeletonized counterparts of the same image, for instance. For a typical branching structure, filled silhouettes and outlined contours bear information about changing branch diameter, but skeletonized patterns, because they generally use a consistent (usually a single pixel) diameter line, tend to lose that information. The higher D_B s of filled patterns can be understood as the filled state resembling a filled plane (which has a theoretical $D_F = 2$, as was noted in Sect. 2.2.1.2) more than do its outlined or skeletonized states. The higher Λ s can be understood by considering silhouettes, contours, and skeletons as the same collection of containers in different

states (e.g., the outlined circle in the lower left of Fig. 2.10 has one container and it is empty) where the outlined and skeletonized states have only empty or mostly identical containers but the silhouette has some filled and some empty, thus making it more heterogeneous. In this regard, similar to the D_B (see Sect. 2.2.1.2), Λ is moderately correlated, but not redundant, with density [58, 116].

These are not strict rules. The astrocyte in Fig. 2.10, represented as an outlined and as a filled pattern (Aho/Ahf), for instance, defies the trend: Λ is lower in Aho as expected, but the D_B is essentially the same in both. This can be understood as the containers (branches in this case) being narrow so that filling them does not substantially change the number of groups of pixels but does change the distribution, the variation in gap size. The circle in Fig. 2.10 also challenges the trend; FC has a much higher calculated D_B than does the outlined circle (OC) but barely higher Λ . This is understandable when one considers that both are round, rotationally invariant containers, each with only one container state and, therefore, similar Λ s, but the filled state is markedly closer to filling the plane and, therefore, has a higher D_B [40, 79].

Related to this is the degree of tortuosity in a structure, a notion explored previously in Fig. 2.4 and which, especially for patterns of living things, can influence the digital pattern extracted. The images in this chapter were selected with a bias to visually convey a unit of detail, but this selection bias does not reflect real neuroanatomy, where cells and cell parts overlap extensively and dynamically. Overall, the fractal analyst evaluating results or deciding what type of pattern to assess needs to be aware that these trends exist but cannot assume they are or are not pertinent without investigating to what extent they affect the results [78, 100, 106].

2.4.2.2 Applying Lacunarity and D_B in Complement

In terms of practical applications, the complementary relationship between the D_B and Λ has been used in several areas. The D_B and Λ were used in tandem to characterize optic nerve astrocyte morphology as related to where the cells were located, for instance [117]. Similarly, Barreto et al. [118] measured binary outlines of astrocytes to study how complexity and heterogeneity changed over the hours and weeks after ischemia induced by surgical middle cerebral artery occlusion in mice. Astrocyte complexity increased soon after injury, then slowly dropped back, whereas Λ generally decreased, but along a different schedule than the increase. Moreover, a neuroprotective (heat shock protein 72) genotype decreased the effect on complexity but not Λ [109, 118]. Barreto et al. attributed the difference in D_B at least in part to redistribution of normally varied morphologies that astrocytes adopt even within one region, with a shift away from cells with lower complexity [112, 118]. They assumed that Λ initially quantified large, irregular vacant spaces between processes (higher Λ), becoming filled more homogeneously with increasing numbers of radial processes (lower Λ).

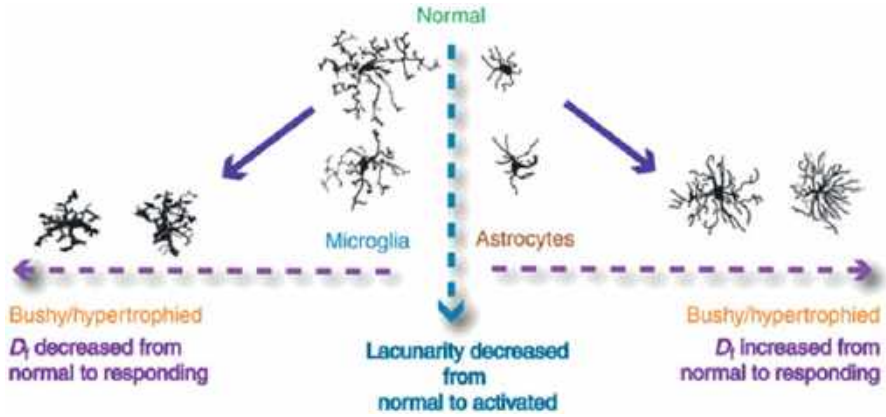


Fig. 2.11 As microglia hypertrophy, they typically decrease in the overall complexity, after potentially a brief increase (not shown); astrocytes, in contrast, increase. Both generally decrease in lacunarity [76, 77, 86, 113, 117–121]. (See Chap. 6 for a more detailed discussion of the nuances in the trends in microglial morphology)

These trends for astrocytes are especially intriguing when compared to what is known for microglia, which are pertinent to compare to astrocytes, with both being peppered throughout the brain, sharing the same basic microenvironment in time and space, thus potentially relevant to fractal analysis of that environment as a whole. Figure 2.11 compares the trends for astrocytes and microglia as they change their activation state (the illustration does not compare them along the same time scale).

2.5 Grayscale Volumes and Box Counting

In addition to binary images, box counting can also be performed on grayscale images, but there are some differences. Grayscale analysis is generally pertinent when the questions are about texture. It has been used to investigate the fractal nature of whole brain structures, electron microscopy images [75], whole histological images where the environment cells are embedded in is important, and intracellular structures, including the nuclei. Both texture and structure can be analyzed by either binary [27, 75] or grayscale methods, but, as illustrated in Fig. 2.12, each method is more suited to certain types of questions [23].

The grayscale D_B (D_G) is related to neither count nor mass but to pixel intensity. This is because rather than being just a binary choice, grayscale pixel values cover a range (e.g., from 0 to 255 in 8-bit grayscale images). Box counting for grayscale images can be performed by projecting the image into volumes based on this range, after partitioning the image into the standard sampling elements as for binary box counting. The areas are extruded to 3D columns having height as the difference,

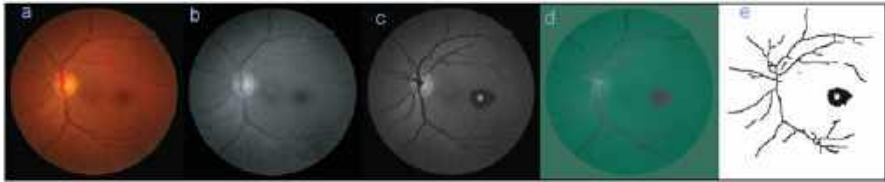


Fig. 2.12 Different pattern extraction methods for box-counting analysis of retinal fundus photographs. (a) The original retinal fundus photograph. (b) The image converted to grayscale for a full texture analysis. (c) The image processed using automated methods to isolate vessels and a region of interest (ROI) for the macular area (d). The image from b prepared for grayscale box counting by masking everything but the pixels isolated in c (the green will be ignored). (e) The image from c prepared for binary box counting by automatically thresholding the mask into black pixels and the rest into white pixels. (Preprocessing done with ImageJ [122])

δ , between the maximum and minimum pixel intensity within the area (A) of the sampling unit (see Fig. 2.13). The grayscale N (N_G) is then based on the mean volume, which, depending on the method being used, is usually calculated as one of:

- (i) $\delta \times A$
- (ii) $(\delta + 1) \times A$ or
- (iii) $(\delta + 1)$

Then, as usual, the D_G is the slope of the regression line for the log–log ratio of N_G vs. S_B [1, 49].

2.6 Subscanning

An application of box counting that a prudent interpreter of box-counting fractal analysis should understand is subscanning analysis, used to investigate variation over space. Subscanning is simply analyzing several domains that may or may not overlap in one image independently of other areas and then presenting the results as a whole so that the variation in the local D_B can be compared and assessed [65, 94, 107]. As an example, this technique was used earlier in this chapter to assess the process of box counting itself to generate the rotation images in Fig. 2.8. Figure 2.14 illustrates how subscanning using color coding can be used in general to investigate variation over tissue samples in the local D_B .

In one study of astrocytes, subscanning showed that the D_B was consistent (around 1.7) from the perisomatic area to the most peripheral extent of a branch's territory faithfully revealed by light microscopy but rapidly dropped off at the tips of the processes. The variation, however, was probably attributable to the inability of light microscopy to resolve the finest processes at the ends because when astrocytes were reconstructed via 3D modeling, the branching pattern at the ends was the same as it was everywhere else [65].

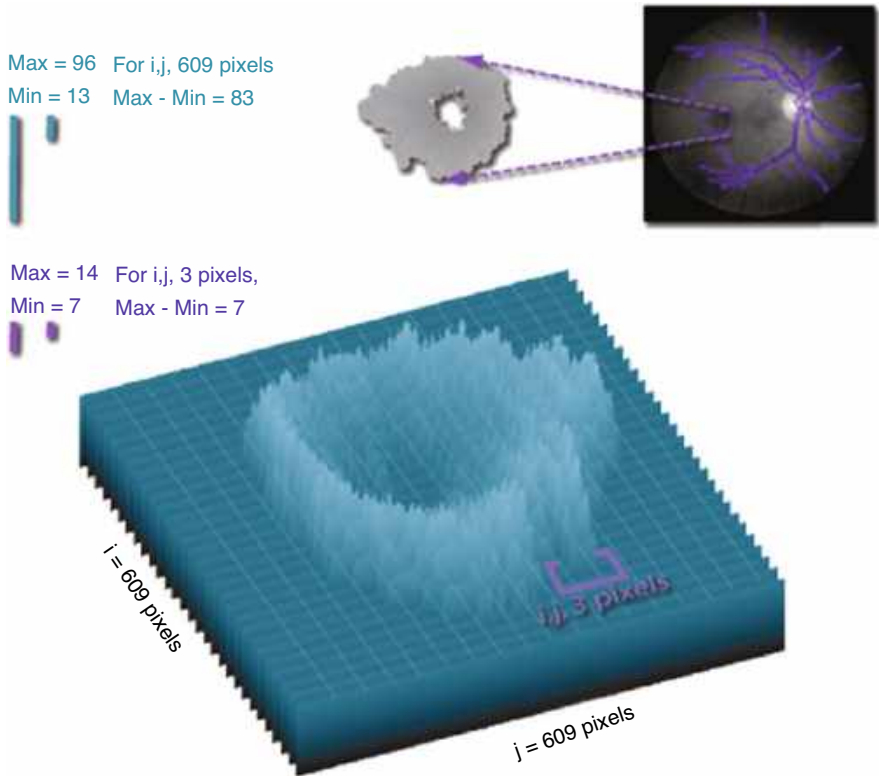


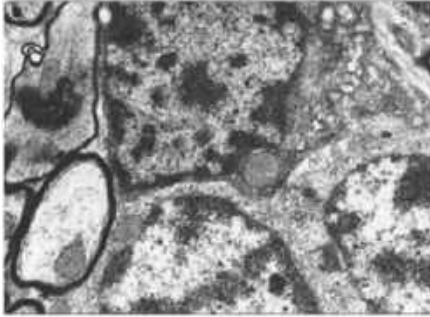
Fig. 2.13 A grayscale surface plot illustrating the general idea of converting 2D grayscale images into 3D information. This figure shows that $\delta = 83$ over the full image and $\delta = 7$ for one sample taken at coordinates i and j at the area within the block marked off using a sampling element with a side length of three pixels (i.e., $A = 3^2 = 9$). The inset shows the source image, the macular area from a retinal fundus photograph (surface plots generated in Fiji [123])

2.7 Multifractal Analysis

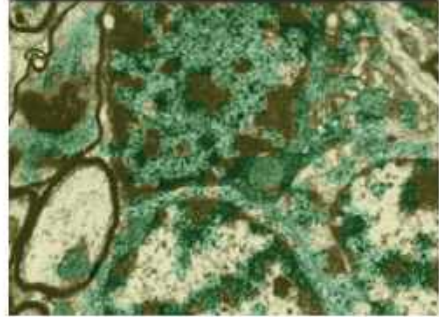
The third type of box-counting analysis promised to be covered is multifractal analysis. The box-counting analysis described thus far is sometimes called “monofractal analysis,” to distinguish it from “multifractal analysis,” which identifies patterns characterized better by a spectrum of D_F s than a single D_F [50, 124, 125]. The multifractal analysis process is analogous to applying warping filters to an image to exaggerate features that might otherwise be unnoticeable with standard box counting. Mono- and non-fractals are little or not affected by the distortions, but multifractals are affected in characteristic ways [124].

The warp filters are a set of arbitrary exponents traditionally denoted by the symbol Q , which are applied to the mass captured in a box (mass is illustrated in Fig. 2.7). In fractal analysis software, the Q set is something that the user typically

a. Digital Transmission Electron Micrograph of Mouse Cerebellum



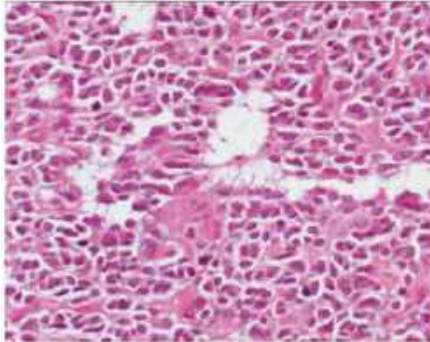
Post fixed in osmium tetroxide,
stained with lead and uranyl acetate
Original magnification 12,000x



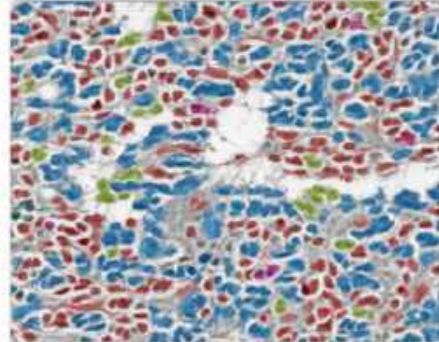
Subcanned as grayscale in FracLac:

- 1) FracLac>Sub>ImageType>Gray1
- 2) Method>Rectangles
- 3) Subscan size = 4%>OK
- 4) FracLac>Scan

b. Digital Histology Image of Human Astrocytoma



- Stained with H&E
- Original magnification 40x



*Converted to grayscale and nuclei
autosegmented in ImageJ then
subcanned as grayscale in FracLac:*

- 1) Process>Binary>MakeBinary
- 2) Analyze>Analyze Particles >Add to Manager
- 3) File>Revert (back to grayscale)
- 4) FracLac>BC>ImageType>Gray1>OK
- 5) FracLac>ROIs

Fig. 2.14 Subscanning identifies local variation in the D_B using color coding. (a) Structures within an area of mouse cerebellum differentiated by differences in the local D_B (green vs. yellow areas). (b) Variation in the local D_B among several of the same structure (i.e., different colors of the nuclei in a histopathology slide of astrocytoma)

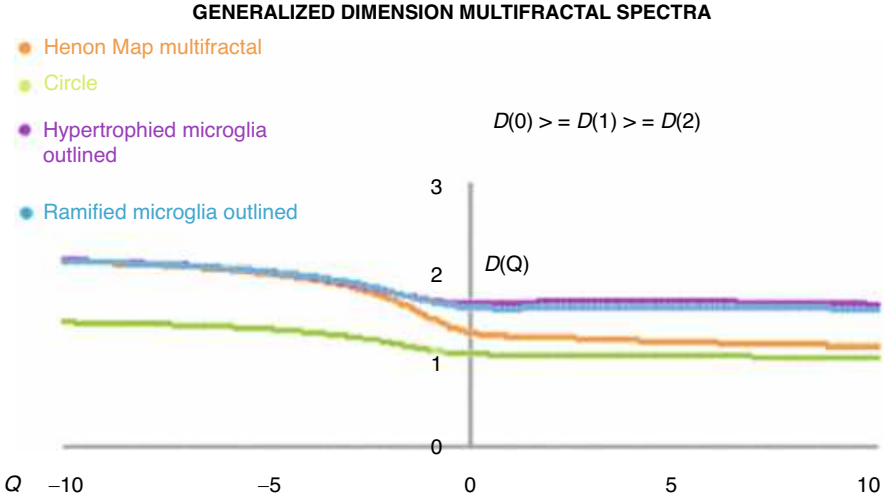


Fig. 2.15 The generalized dimension graph frequently used for multifractal analysis (generated for images in Fig. 2.10). The horizontal axis, Q , is an exponent, and the vertical is a probability-based D_F that varies with Q . Being exponents applied to probabilities (i.e., between 0 and 1), negative Q s amplify and positive Q s shrink the value on the vertical axis, as is most clearly seen in the curve for the multifractal. Multifractals are characterized by generally steeper slopes around $Q = 0$, whereas the flatter mono- and non-fractal curves stay close to one value for all Q [40, 41, 50, 54]. (Data generated using FracLac [49, 124])

manipulates from a starting default set (e.g., bracketing 0 symmetrically such as from -5 to 5). A generalized dimension (D_Q) is determined for each Q , based on the mass dimension (D_M). N for the D_M (N_M) is the mean of the probability distribution of all masses at one size, reflecting the density over the image. For the D_Q , each mass is distorted by being raised to Q , and then N_Q is the mean for this distorted density distribution for a size. Graphs based on these data have predictable features that distinguish amongst non-, mono-, and multi-fractal scaling and quantify multifractal features.

Figure 2.15 shows one such graph, the graph of D_Q vs. Q that is commonly used in multifractal analysis [40, 50, 126]. See also Chap. 4.

2.7.1 Reading the D_Q s. Q Curve

The generalized dimension curve (D_Q vs. Q) is typically decreasing, sigmoidally around $Q = 0$ for multifractals, but much flatter for mono- and non-fractals. Indeed, if distortion over all scales reveals noticeable differences (i.e., the pattern has a set of unique D_Q s), then the scaling is considered multifractal, but if a pattern has essentially one value for all Q s, then the pattern is considered not multifractal (i.e.,

mono- or non-fractal). As explained in Fig. 2.15, at any scale, negative Q s amplify and positive Q s shrink the pattern, thus affecting different densities nonlinearly.

The curve can be assessed objectively using a characteristic known as “dimensional ordering,” whereby $D_{(0)} \geq D_{(1)} \geq D_{(2)}$, which defines an interval used to assess the spectrum’s amplitude [40, 50, 100, 124, 126, 127]. Thus, as was mentioned, software may let the researcher select values for Q , but, in order to assess the amplitude, it should also ensure that the Q set includes integer values for 0, 1, and 2.

There are some points to be aware of when interpreting dimensional ordering values. To elaborate, in a multifractal analysis, $D_{(0)}$, $D_{(1)}$, and $D_{(2)}$ generally correspond to quantities known as, respectively, the “capacity dimension” (closely related to the D_B), the “information dimension” (analogous to a quantity known as the Shannon entropy), and the “correlation dimension,” which can be calculated for different types of fractal patterns (i.e., including temporal) using different methods of fractal analysis. However, there are practical issues affecting the actual values obtained for these by box counting. For example, in practice, the D_B obtained from optimized box counting (Sect. 2.3.2.3) differs from the $D_{(0)}$ obtained from optimized multifractal analysis, owing to the differences in optimizing methods. Similarly, the Shannon entropy for $D_{(1)}$ cannot be determined directly using standard box counting [7].

Applying the basic rules for reading the D_Q curve to Fig. 2.15, one can see that the curve for the Henon multifractal is a typical multifractal curve showing considerable heterogeneity in scaling, in contrast with the curve for the circle, which is flat, as is typical of homogeneous non-fractals. In keeping with the trends shown earlier in Fig. 2.10, increased heterogeneity with multifractal scaling is complementary to the heterogeneity of different types of lacunarity.

Similarly, one can see differences for microglia at different activation levels. The figure shows results for two outlined patterns extracted from, respectively, the inactivated microglial cell with $D_B = 1.48$ (Mro) and the more activated microglial cell (Mhf) from Fig. 2.10 (for the outlined version of Mhf, not shown in Fig. 2.10, the calculated $D_B = 1.54$). That the hypertrophied cell had a higher D_B is evident from the portion of its curve slightly above that for the ramified cell. Their curves are both flat, neither suggesting multifractal scaling. These curves are similar to curves from images of comparable monofractals. The curve (not shown) for the 32-segment quadric fractal from Fig. 2.10 (Q32), for instance, has a similar slope near $Q = 0$ but lies slightly above the cells (i.e., it is a monofractal with a higher D_B , i.e., $D_B = 1.67$) [40].

2.7.2 Reading the $f(\alpha)$ Curve

The second commonly used graph discussed here, the $f(\alpha)$ spectrum, is a typically convex curve opening downward, as shown in Fig. 2.16. Again, the shape distin-

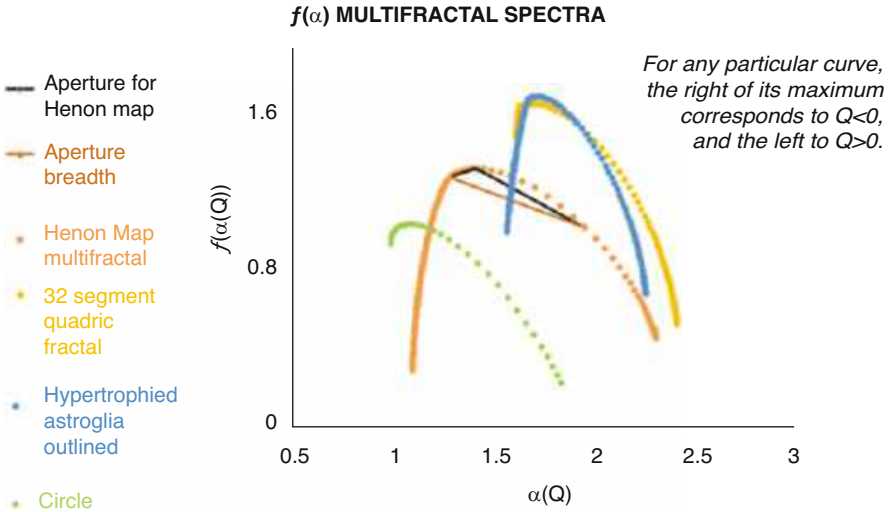


Fig. 2.16 A graph of the $f(\alpha)$ spectra used for multifractal analysis (generated for images in Fig. 2.10). The horizontal axis, $\alpha(Q)$, is a D_F based on Q , and the vertical is a D_F dependent on $\alpha(Q)$ and is further modified by Q . Compared to mono- and non-fractals, multifractals are characterized by broader $f(\alpha)$ spectra, which can be measured by their apertures (see legend) [40, 41, 50, 54]. (Data generated using FracLac [49, 124])

guishes multifractals from other patterns, being (1) broader and (2) less convergent, particularly on the left for multifractals, than for mono- and non-fractals.

One feature used to quantify the $f(\alpha)$ spectra is the *aperture* (illustrated for the Henon map in Fig. 2.16), defined by lines from $Q = 1$ to 0 to -1 and the total breadth from $Q = 1$ to -1 . The slopes and lengths of the aperture lines quantify the scaling regardless of any other values of Q used. Defining the aperture interval as from 1 to -1 seems reversed but can be understood knowing that the variable along the horizontal axis, $\alpha(Q)$, itself decreases with Q [40, 124].

The G -related bias discussed in Sect. 2.3.2.2 can invalidate multifractal analysis results. A common problem in multifractal analysis is anomalous spectra created from inappropriate sampling. Signs of this include that the left ($Q > 0$) and right ($Q < 0$) portions of the $f(\alpha)$ spectra (Fig. 2.16) fail to meet in a continuous curve, flip upward, or cross over. This often means that an image has a density distribution with very small probabilities taking on inappropriately greater significance in some, but not all, G .

Values of $Q < 0$ are most relevant to this problem, so sometimes using a different range of Q s can fix the problem; similarly, sampling only above a certain density or limiting the smallest size can also sometimes fix this problem. If different settings are used for different samples within a dataset, then these arbitrary solutions can alter the results in unexpected ways or make comparisons difficult, so a preferred alternative is to use the same density, Q , etc. but increase G and R until normal

results are obtained. As a practical illustration, the multifractal spectra in Fig. 2.14 were generated for the set of images from Fig. 2.10, using FracLac [49] to implement the solutions suggested above, including multiple G s. For most of the control images, normal multifractal curves appeared by 12 runs, but, for the astrocytes and microglia, the results could be read only after running 500 different sampling locations (taking seconds more per image, where that value would generally vary with the image size and equipment used to run the analysis) [1, 125, 128].

This has practical implications for fractal analysis in neuroscience. As an example, microglia are known to scale as multifractals, but relatively rarely, less than 10% of cells, when in a hyper-ramified or transitional state between their normal/ramified and pathological/hypertrophied states, and more often, for example, in the elderly human brain [78, 100, 113]. An investigator formulating a question about hyper-ramification and multifractality and then analyzing a sample using too low a value for G might see only unreadable multifractal spectra and consequently abandon their questions about multifractal scaling in these cells. Alternatively, being aware of what inadequate sampling looks like and that a small amount of extra time processing might solve the problem could inspire further curiosity.

2.7.2.1 Applying Multifractal Analysis

Multifractal analysis has been applied in novel ways in neuroscience. It has been used in hemodynamics, for example, showing in arteriovenous malformations correlations of size and flow with $D_{(0)}$, $D_{(1)}$, and $D_{(2)}$ (these correlations are similar to, although weaker than, correlations of size and flow with the D_B) [68, 70, 91, 129]. Multifractal analysis has also revealed postmortem capillary distribution differences in Alzheimer's disease, small vessel disease dementia, and control from three different brain regions [55]. Box-counting multifractal and lacunarity differences together have also been used to distinguish between embryos from control vs. folate-deficient mice in the cornea, lens, retina, and vitreous, with multifractal analysis detecting even slight differences in tissue textures, which the study authors speculate are attributable to altered extracellular matrix molecules [98]. Multifractal analysis has further been used to differentiate normal from pathological retina, with D_Q curves for binary vessel tracings extracted from normal retinal fundus photographs showing smaller amplitudes than those from pathological tissues (diabetic macular edema) and the $f(\alpha)$ spectra showing distinctively different shapes and asymmetry. Also of general import for the multifractal analyst, similar to the trends noted in Sect. 2.4.2.1, it has been shown that skeletonized retinal patterns have lower D_Q spectra than the filled ones [68, 70, 91, 116, 129].

2.8 The Validity of 2D Patterns from Four-Dimensional (4D) Reality

An overarching issue anyone using or interpreting fractal analysis should become familiar with is the validity of the inputs themselves. The source images we have been discussing were 2D. Yet, biology is not. Microglia, for instance, play active roles in brain structure and function, in development, normal cognition, alcoholism, autism, chronic stress, pain mediation, Alzheimer's disease, diabetes, aging, etc. [74, 76, 113, 119, 130–135]. They are dynamic in space and time. In living, normal brains, they are found waving and winding their processes around neurons at speeds of roughly $1.5 \mu\text{m}/\text{min}$; in pathological or developing tissue, in vivo and in vitro, they are found migrating and phagocytosing rapidly or sometimes more slowly [113, 130, 133, 136]. When examining microglia intertwined and interacting with neurons within the central nervous system (CNS) or even alone in a dish, the microgliologist realizes these are 3D biological phenomena, indeed, 4D phenomena changing over time, as are virtually all phenomena in neuroscience. Nonetheless, here, we have discussed the fractal analysis of 2D, still images. A question often asked is: how to resolve the discrepancy in dimensions?

One way is by appealing to an analogy in facial recognition software. To elaborate, given an image, facial recognition software uses many features to identify a face within the image. Intriguingly, one such feature is the simple isosceles triangle that can be drawn on an image across the eyebrows with its vertex at the mouth [137]. Like cell profiles or black and white images of any structure, such drawings are 2D. That is, a 2D pattern (the triangle) extracted from an image of a face serves to meaningfully identify an originally 3D, biological structure (the actual face) that changes in space and time.

A similar principle applies to fractal analysis of 2D patterns extracted from images representing 3D structures such as dynamic cells existing in the 3D space of the brain or cell culture over the fourth dimension of time. In the same way that the triangle on the face is not the face, and that the face is not a triangle, yet still the triangle is vital for the software to recognize the face, and the patterns extracted from images of cells are not the cells but hold the descriptors we are concerned with in fractal analysis. Indeed, in some cases, 2D representations convey 3D information that may be integral to a pattern, such as when vessel diameter or length is smaller for vessels farther away from the image plane, and this relative distance is itself what is being investigated. As such, a discerning approach acknowledges that 2D images can be reasonable proxies for 3D phenomena, subject to the limits that they are investigated within.

A further implication of this is that the D_B calculated for a digital image is not necessarily the D_F of the phenomenon as a whole; rather, it is always the D_B of the pattern that the researcher is interested in. Choosing a pattern type (e.g., grayscale texture, binary silhouette, contour, or skeleton) and a method of extracting it depends on several factors, with one of the foremost being the research question itself.

2.8.1 Control and Calibration

Another issue to be aware of in box-counting fractal analysis is verifying the results. When possible, fractal analysis software should be calibrated using controls, such as the control images from Fig. 2.10 and modeled structures from Fig. 2.4. It is important to match benchmark images to the images being investigated (e.g., in size and number of foreground pixels) and to the goals of the research (e.g., mono- vs. multifractal images or in silico cell models created using known scaling parameters) and to consider deviations from theoretical and potential influences like R (Sect. 2.3.2.2) [1, 48, 91, 100]. In multifractal analysis, control images are particularly useful for establishing parameters for dimensional ordering and the $f(\alpha)$ aperture (Sect. 2.7) [87, 124, 125].

2.9 Conclusions

In summary, this chapter has laid out the elementary principles of fractal geometry and explained fractal analysis as a way of measuring phenomena when the details of design are as important as gross morphology. This chapter outlined how the details change with scale in theoretical and natural fractal patterns. It also grounded the reader in the basics of binary and grayscale box counting and explained trends and complementary relationships among the D_B , Λ , and multifractal spectra with application to topics in neuroscience. It illustrated that although the tedium of fractal analysis can be left to the click of a button, prudent and defensible interpretation of that analysis needs some regard for common pitfalls [40, 100–103]. Accordingly, it reviewed several methodological considerations, such as different types of pattern idiosyncrasies and sampling issues, and discussed the practical methods of addressing them. Overall, this chapter prepares the fractal analyst to conscientiously and confidently use the elegant, robust techniques of box counting to measure and interpret the D_B , lacunarity, and multifractality of the exquisitely detailed fractal geometry of the brain.

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Chapter 3

Tenets and Methods of Fractal Analysis (1/f Noise)



Tatjana Stadnitski

Abstract This chapter deals with the methodical challenges confronting researchers of the fractal phenomenon known as pink or 1/f noise. This chapter introduces concepts and statistical techniques for identifying fractal patterns in empirical time series. It defines some basic statistical terms, describes two essential characteristics of pink noise (self-similarity and long memory), and outlines four parameters representing the theoretical properties of fractal processes: the Hurst coefficient (H), the scaling exponent (α), the power exponent (β), and the fractional differencing parameter (d) of the ARFIMA (autoregressive fractionally integrated moving average) method. Then, it compares and evaluates different approaches to estimating fractal parameters from observed data and outlines the advantages, disadvantages, and constraints of some popular estimators. The final section of this chapter answers the questions: Which strategy is appropriate for the identification of fractal noise in empirical settings and how can it be applied to the data?

Keywords 1/f noise · Fractal parameters · ARFIMA · Long memory

3.1 Tenets and Methods of Fractal Analysis (1/f Noise)

Pink or 1/f noise is a well-studied fractal (i.e., self-similar) phenomenon. As will be explained here, the word “pink” in the name arises from its characteristically long “memory” being intermediate between that of *white noise* (no memory) and *red* or *brown noise* (infinite memory). Pink noise serves as an adequate model for many biological processes because it exhibits both stability and adaptability, two properties typical of healthy complex systems [4]. Highly correlated fluctuations of pink noise are typical for neural dynamics. For instance, cognitive reactions generated in psychological experiments, including mental rotation, lexical decision,

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shape and color discrimination, and word naming demonstrated a $1/f$ noise character [17, 41, 42]. Pink noise was also detected in visual perception [3], in rhythmic tasks requiring coordination of motor and cognitive activities [8, 10], and in brain dynamics of electroencephalogram records [1, 2, 7, 24]. The oscillatory neural synchronization [29], the spontaneous phase-transition process of many interacting neurons [5], and the fit between the phase-transition conjecture and the criticality condition [43, 44] have all been discussed as the cause for these observations in the recent neurophysiological literature. Some researchers view $1/f$ noise as a communication channel for complex networks through which the brain influences complex processes and, in turn, is influenced by them [1]. Generally, it is believed that $1/f$ noise reflects the coordinative, metastable basis of cognition [22]. Brain dynamics have been statistically investigated either by monofractal methods or by studying autocorrelation functions (ACFs) of the related time series. Another alternative for the statistical analysis of brain dynamics is to examine their spectral density functions and periodograms obtained by the fast Fourier transform algorithm. Different methodological approaches to detecting $1/f$ noise and the interrelations between them will be discussed in the following subchapters.

3.2 Statistical Terms: Parameter, Estimator, and Estimate

To understand how pink noise is evaluated in fractal analysis, one should first know the differences between the following statistical concepts: “parameter,” “estimator,” and “estimate.” A *parameter* is a quantity that defines a particular system, such as the mean of the normal distribution. Strictly speaking, to obtain a population parameter, one must measure an entire population, which is mostly infeasible, so, instead, one generally uses *estimators* (rules or formulae) to infer population parameters from observed samples. For any parameter, there are usually multiple estimators with diverse statistical properties. As an example, suppose we have n observations of some phenomenon X ; then, we can estimate the population mean (μ) using two well-known estimators, the sample mean $\hat{\mu}_1 = \frac{1}{n} \sum_{i=1}^n X_i$ and the median $\hat{\mu}_2 = X_{0.5}$. In contrast to parameters, estimators are not numbers but are functions characterized by their distributions, expectancy values, and variances, where the quantity to be estimated generally differs from its estimation. Thus, because every estimation method involves some estimation error, the investigator must choose which method best suits the task at hand. Based on the above example, for instance, μ may differ from both $\hat{\mu}_1$ and $\hat{\mu}_2$, and an investigator might choose the former as the better estimator of μ if the phenomenon being investigated is better understood using a smaller variance and narrower confidence intervals. In fractal analysis, one way to decide which estimator to use is by means of *Monte Carlo simulations*. For instance, computational algorithms can generate a population with a known parameter value, and repeated samples of the same size can be drawn from this population, e.g., 1000 time series with $T = 500$, and then different estimators can

be applied to the series, yielding 1000 estimates of the parameter per method. An *estimate* is a particular numerical value obtained by applying an estimator. Good estimators are unbiased, i.e., their means equal the true parameter value, and have small variability, i.e., their estimates do not differ strongly. Considering that just one estimate per method is available in a typical research situation, an estimator with a narrow range is usually better than one with a broad range.

3.3 Properties of 1/f Noise: Self-Similarity and Long Memory

Fractals are *self-similar structures or processes* where the whole has the same shape as its parts. The concept of self-similarity, as well as examples such as the Koch snowflake or the Sierpiński triangle, is covered in Chap. 2. To summarize here, there are two types of self-similarities: strict and statistical. Strictly self-similar geometrical objects like the Koch snowflake are composed of exact copies of the whole object. Statistical fractals like Romanesco broccoli consist of sub-pieces that closely replicate the whole structure. Pink noise is statistically self-similar, meaning its smaller segments look like its larger segments and all exhibit the same statistical properties as the overall process. It is also *scale-invariant*, meaning its characteristics remain similar when viewed at different scales of time or space.

These properties, self-similarity and scale invariance, can be mathematically modeled by *power laws*. A power law is a functional relationship between two quantities, where one quantity varies as a power of another. For instance, the relation between external stimuli and internal sensation can be described by a power law proposed by the American psychologist Stevens: $S = cI^b$, where S is the sensation magnitude, I is the stimulus intensity, c is an arbitrary constant to determine the scale unit, and b is the power exponent, which depends on the type of stimulation. This *exponential function* states that the perceived strength of subjective sensation increases proportionally with the intensity of the physical stimulus. Assuming c is positive, if the exponent b is zero, the graph of the function is a horizontal line; if b is one, the graph is a line of slope c ; if b is greater than one, the line curves upward; and if b is between 1 and 0, it curves downward. Most relevant here, when the log is taken of both sides, power functions have the convenient feature of becoming linear, with a slope equal to the value of the exponent. For instance, the logarithmic form of Stevens' power law is $\log(S) = \log(c) + b\log(I)$, i.e., a straight line with slope of b (Fig. 3.1). In general, multiplying the argument I of the Stevens' or similar functions by a constant k causes *proportional scaling* of the function itself: $f(kI) = c(kI)^b = k^b f(I) \propto f(I)$. Therefore, the slope of the straight line on the log-log plot can be viewed as the signature of a power law or, in fractal analysis parlance, a measure of fractality.

Power laws express self-similarity in fractal processes by demonstrating that the same principles hold for all segments of the process and on all scales. For instance,

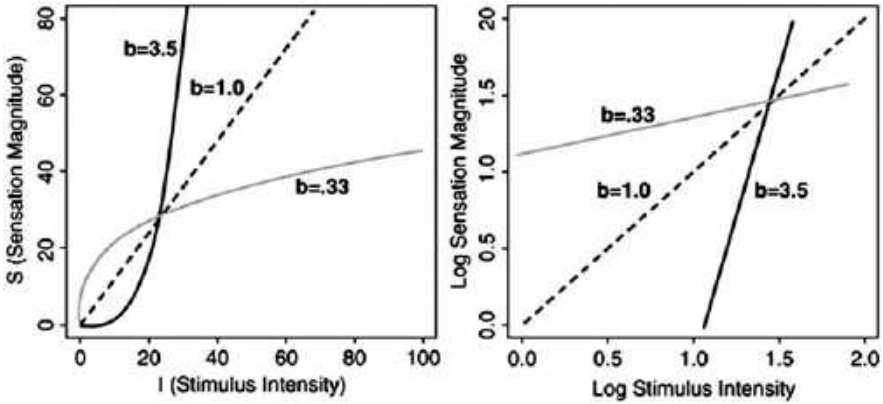


Fig. 3.1 Stevens' power-law functions

fractals are often described by a power law stating that the frequency of occurrence (F) of an element of a given size is inversely proportional to some power of its size ($F_s \propto 1/\text{size}^{\text{power}}$). Dividing a process into intervals of equal length n allows viewing it on different scales. Fluctuations (F) of fractal processes are proportional to n , i.e., they increase with growing interval length ($F_n \propto n^{\text{power}}$). Another power law assumes that the variance of fractal processes (S) is inversely proportional to their frequency (f) ($S_f \propto 1/f^{\text{power}}$).

3.3.1 Memory

Pink noise is generally measured as a time series (i.e., as successive measurements made over a time interval). Self-similarity within a time series can be modeled by the *autocorrelation function* (ACF). The autocorrelation function describes the correlation of a signal with itself at different lags. In other words, it reflects the similarity between observations in reference to the amount of time between them. Informally, this relates to the notion of the “memory” of a process, which can range from no memory to infinite memory. To elaborate, white noise is a sequence of time-ordered, uncorrelated, random variables, sometimes called random shocks or innovations, and is therefore considered to have “no memory.” A process that can be predicted by its immediate past, although the autocorrelations decay quickly as the number of intervening observations increases, is considered to have a “short memory.” On the other end of the spectrum, red noise results from integrating white noise and so can be represented as the sum of random shocks such that the impact of a particular innovation never dissipates, which implies no decay in autocorrelation and infinite memory.

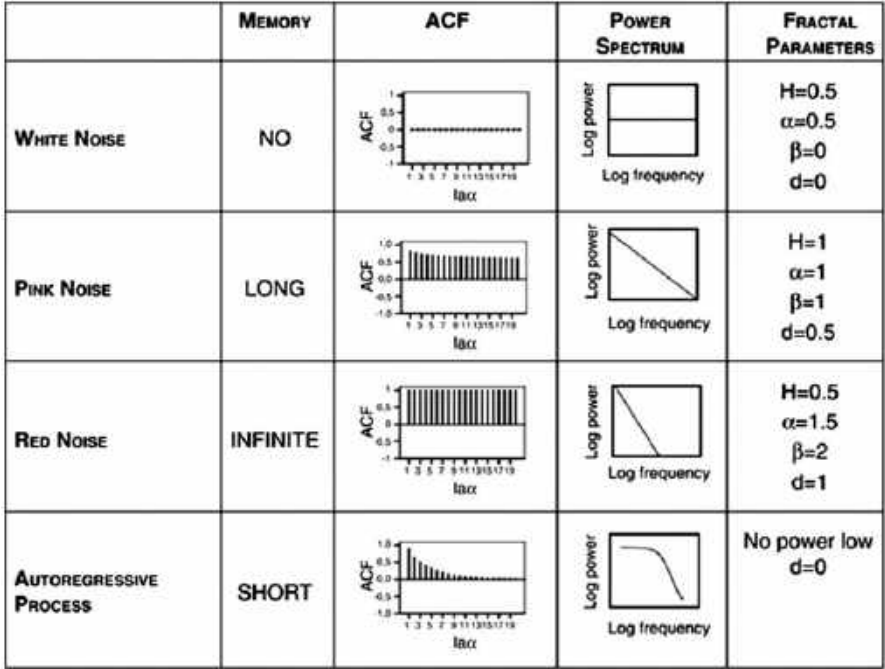


Fig. 3.2 Theoretical autocorrelation functions, logarithmic power spectra, and parameter values of pink noise compared with non-fractal patterns. (Note: white, pink, and red noises are self-similar processes.) Pink noise is an intermediate between white and red. Autoregressive processes are not self-similar

As mentioned earlier, pink noise is an intermediate between white and red. The self-similarity of pink noise manifests in a power-like decaying autocorrelation function. Pink noise possesses very slowly decaying autocorrelations, reflecting statistical dependence between observations separated by a large number of time units. This explains why pink noise can be viewed as an intermediate between white noise with no correlation in time and red noise with no correlation between increments. Due to persistence of its autocorrelations, $1/f$ noise is also called a *long-memory process* or *long-range-dependent process*. The ACF of processes with different memory characteristics is compared in Fig. 3.2.

3.3.2 Stationarity

One of the most important concepts of time series analysis is stationarity. A process is said to be stationary if its mean, variance, and autocorrelations do not change over time; otherwise, it is nonstationary. Nonstationary behaviors can evolve from cycles and stochastic or deterministic trends. Changing data, as a rule, are unpredictable

and cannot be modeled or forecasted unless stabilized; therefore, nonstationary time series are usually transformed to make them stationary. White noise, also called *Gaussian noise*, is stationary with constant mean and variance, whereas red noise, also called *Brownian motion*, is nonstationary. Because summing Gaussian noise produces Brownian motion, differencing Brownian motion creates Gaussian noise, which means that increments of red noise are stationary. For this reason, red noise is sometimes termed *difference stationary*. As an intermediate between white and red noise, pink noise is both stationary and nonstationary, with slow fluctuations of its parameters. Due to this property, the $1/f$ process becomes both stable and adaptive.

Generally, there are two different classes of long-memory processes in fractal analysis: *nonstationary fractional Brownian motions* (fBms) and *stationary fractional Gaussian noises* (fGns) [26, 27]. Mandelbrot and van Ness [26] introduced fBm as Brownian motion with correlated increments. A positive correlation implies that an increasing trend in the past is expected to be followed by an increasing trend in the future. Therefore, fBms with positive correlations of increments are called *persistent*. Fractional Brownian motions with negative correlations of increments are called *anti-persistent* because an increasing trend here is likely to be followed by a decrease in the future. Mandelbrot and Wallis [27] defined fractional Gaussian noise as a series of successive increments in an fBm. This means that each fBm is related to a specific fGn and both are characterized by the same fractal dimension [25].

A *fractal dimension* (D) is an index of complexity of self-similar objects describing how details in a pattern change with the scale at which it is measured. Euclidean geometry is three-dimensional: The dimension of a line is 1, of a square is 2, and of a cube is 3. According to the following power law $NR \propto R^D$, the number of self-similar pieces into which the object may be broken (N) is proportional to some scaling factor (R). Hence, the fractal dimension can be calculated as $D = (\log N)/(\log R)$. For example, a line is one-dimensional because we can break it into N self-similar pieces, with each $1/N$ the length of the original (i.e., $N = R$). A square is two-dimensional because it consists of N^2 self-similar copies of itself, each of which must be magnified by a factor of N to yield the original figure (i.e., $N = R^2$). For instance, we can decompose a square into 4 self-similar sub-squares with the scaling factor 2, 9 self-similar pieces with the scaling factor 3, or 16 self-similar pieces with the scaling factor 4. For description of fractal images like the Koch snowflake, the Sierpiński triangle, or a piece of broccoli, the spaces of integer dimension are not enough. Fig. 3.3 visualizes that the Koch snowflake is constructed by iteratively scaling a starting segment into four new pieces with the scaling factor 3: All original line segments are replaced with four, each a self-similar copy that is $1/3$ the length of the original (i.e., $N = 4$, $R = 3$). In contrast to a line or a square, the Koch snowflake has a non-integer dimension: $D = (\log 4)/(\log 3) \approx 1.26$. Similarly, the Sierpiński triangle breaks into three self-similar pieces with the scaling factor 2, resulting in the non-integer dimension $D = (\log 3)/(\log 2) \approx 1.58$. Fractal dimension can also be understood as a measure of the space-filling capacity of an object. Hence, a square is larger than a line, while the Koch snowflake and the Sierpiński triangle take a position between these two sets: The Koch snowflake is closer to a line

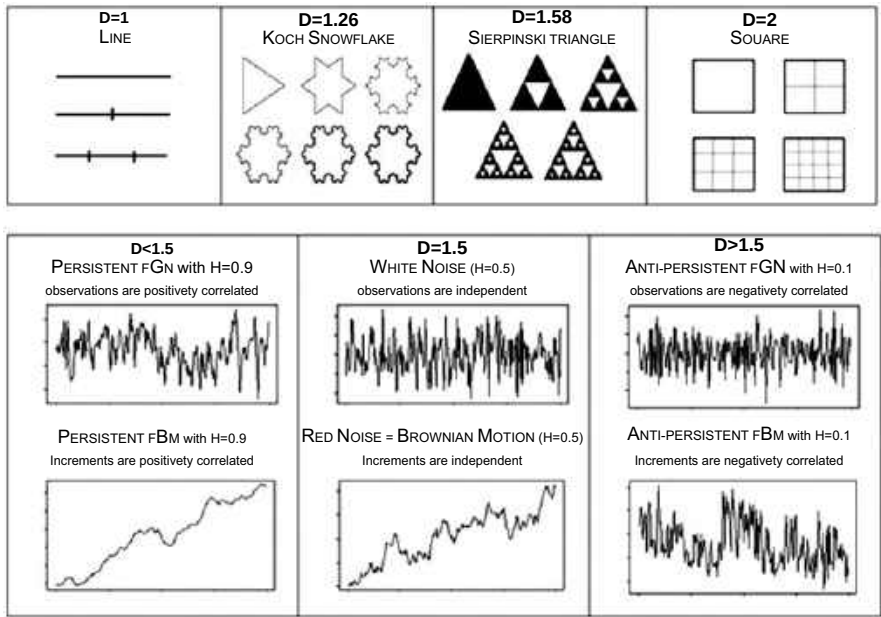


Fig. 3.3 Geometric objects and time series with different dimensions

and the Sierpiński triangle is closer to a square. For the same reason, jagged and irregular patterns of white and red noise yield a fractal dimension of 1.5. Successive observations of white noise and increments of red noise are independent of each other, so $D = 1.5$ describes a random independent system. By contrast, fGn and fBm are correlated noises. The course of persistent series is smoother and closer to a line, implying lower fractal dimensions than that of uncorrelated noises. Being more jagged, anti-persistent series occupy more space and have therefore higher fractal dimensions closer to $D = 2$ (Fig. 3.3). Stationary fractional Gaussian noises and nonstationary fractional Brownian motions represent a continuum; pink noise marks the boundary between fGn and fBm. Successive observations of pink noise time series are positively correlated, resulting in fractal dimensions near $D = 1$.

3.4 Fractal Parameters

The extent of self-similarity and the memory properties of a process are often described using four fractal parameters: the Hurst coefficient (H), the scaling exponent (α), the power exponent (β), and the fractional differencing parameter (d).

3.4.1 Hurst Coefficient

H expresses the probability that an event in a process is followed by a similar event. Investigating the behavior of the Nile river, Hurst [21] observed that wet years were often followed by wet years resulting in wet periods and dry years by dryness-producing droughts. This tendency to self-similarity, which is typical of many climatic processes, is known as the Hurst phenomenon or Joseph effect [28]. As a probability measure, the Hurst coefficient ranges between 0 and 1, which corresponds to a fractal dimension between 1 and 2 ($D = 2 - H$). $H = 0.5$ for both white and red noise because white noise is a sequence of independent innovations and red noise consists of uncorrelated increments. The Hurst coefficient of fractal processes deviates from 0.5, whereby $H < 0.5$ denotes anti-persistent and $H > 0.5$ positively persistent processes. Anti-persistence implies negative correlations between successive increments. Pink noise is characterized by $H = 1$. Recall that differencing Brownian motion creates Gaussian noise and summing Gaussian noise produces Brownian motion. Thus, related fGn and fBm are both characterized by the same Hurst exponent (Fig. 3.3).

3.4.2 Scaling Exponent (α)

As described above, self-similarity can be modeled mathematically by power laws. If we divide a process into segments of equal length, n , we can effectively view it on different scales. The detrended fluctuation analysis (DFA) proposed by Peng et al. [30] demonstrates that fluctuations (F) of fractal signals are related to the interval size (n) by the following power law:

$$F(n) \propto n^\alpha$$

For white noise, $\alpha = 0.5$, and for red, $\alpha = 1.5$ (Fig. 3.2). Fluctuations of pink noise are proportional to n , i.e., they increase with growing interval length, which implies that $\alpha = 1$. For stationary fractional Gaussian noises, $\alpha < 1$, and for nonstationary fractional Brownian motions, $\alpha > 1$.

3.4.3 Power Spectra

For studying discrete-time processes, there are two major approaches: *time domain* and *frequency domain analysis*. Although time and frequency domains are mathematically equivalent, they examine data from different perspectives and pursue different goals. In time domain analysis, the central concept is the memory of the process: To what extent is its present state predictable from its past? Therefore,

statistical assessments are mostly based on autocorrelations. The main goal of frequency domain analysis is to detect cycles in the data by identifying frequencies that explain variance in a time series. Here, the power spectrum and spectral density function serve as the most important statistical tools.

The power spectrum determines how much power (i.e., variance or amplitude) is accounted for by each frequency in a series. The term “frequency” describes how rapidly things repeat themselves. Thus, there exist fast and slow frequencies. For instance, a time series with $T = 100$ observations can be reconstructed by 50 periodic or cyclic components ($T/1, T/2, T/3, \dots, 2$). The frequency is the reciprocal of the period and can be expressed in terms of the number of cycles per observation. $f = 0$ implies no repetition, $f = 1/T$ the slowest, and $f = 0.5$ the fastest frequency. The *spectral density function* gives the amount of variance accounted for by each frequency measured. Spectral analysis represents a special type of analysis of variance (ANOVA) where the overall process variance is divided into components of independent cycles with different lengths. For nonperiodic processes like white noise, the variance is equally distributed across all possible frequencies. For cyclic processes, a few so-called major frequencies explain a great amount of the series’ variance (i.e., all of the series’ power is concentrated at one or a few frequencies).

3.4.4 Power Exponent

β Quantifies the extent of self-similarity. Fractal processes have self-similar power spectra with a spectral density that is proportional to the reciprocal of the frequency and can be expressed by the following power law:

$$S(f) \propto 1/f^\beta.$$

For white noise, the variance is equally distributed over the frequencies; thus $\beta = 0$ and the logarithmic power spectrum has a slope of 0. For red noise, power falls off rapidly with increasing frequency, i.e., low-frequency components predominate, $\beta = 2$, and the power spectrum has a slope of -2 . For pink noise, $\beta = 1$, implying that its variance is inversely proportional to its frequency and that the power spectrum is a straight line with slope -1 . This property becomes more vivid in the logarithmic expression: $\log(S) = -\beta \log(f)$. The power exponent of stationary fractional Gaussian noises can be any real value in the range $[-1; 1]$ with $-1 < \beta < 0$ for anti-persistent and $0 < \beta < 1$ for positively persistent signals. For nonstationary fractional Brownian motions, $1 < \beta < 2$ means anti-persistent and $2 < \beta < 3$ positively persistent series. These power laws do not hold for short-memory autoregressive processes, the spectra of which are not straight lines. The linear relation between the power and the frequency of such processes breaks down at the low frequencies where random variation appears. As a result, a flat plateau (the zero slope of white noise) dominates low frequencies in spectral plots. Figure 3.2 shows the power spectra of the discussed processes.

3.4.5 Differencing Parameter (d)

As described previously, one of the typical characteristics of pink noise is a power-like decaying autocorrelation function with a long memory. Granger and Joyeux [18] and Hosking [20] demonstrated that the memory properties of fractal time series can be parsimoniously modeled by the *differencing parameter* d of the Box–Jenkins ARIMA (autoregressive integrated moving average) methodology [6], allowing it to take on continuous values.

In the time domain, each time series can be described by three types of mathematical models: autoregressive (AR), moving average (MA), and integrated (I). In an AR model, the value of the current observation depends on the values of the previous observations,

$$Y_t = \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + u_t,$$

where ϕ quantifies the magnitude of the dependence, p specifies the order of the dependence, and u_t is a sequence of purely independent and identically distributed innovations or white noise.

An MA process is described by:

$$Y_t = u_t - \theta_1 u_{t-1} - \cdots - \theta_q u_{t-q}.$$

Here, performance at time t depends on both the current and past error terms. A process containing both autoregressive and moving-average components is called mixed. An integrated process is represented by the equation

$$Y_t = Y_{t-1} + a_t,$$

where the random part a_t can be generated by any ARMA (autoregressive moving average) process. Red noise is an integrated process with $a_t = u_t$; therefore, it can be expressed as the sum of white noise. The term “integrated” implies that the impact of the random component in the series does not dissipate over time. As a result, integrated processes are nonstationary and show instability in level. That is why red noise is also called *random walk*. Differencing red noise produces stationary white noise:

$$\Delta Y_t = Y_t - Y_{t-1} = u_t.$$

The equation demonstrates that increments of red noise, ΔY_t , are stationary. Integrated models are also known in the time series literature as *unit root processes*. A situation of nonstationarity is called the unit root problem when the regression coefficient of a first-order autoregressive model equals one. The name unit root is due to the fact that $\phi = 1$. In this case, the autoregressive model can be written as $(1 - L)Y_t = u_t$, where L is the lag operator: $LY_t = Y_{t-1}$, $L^2 Y_t = Y_{t-2}$ and so

on. The term *unit root* refers to the root of the polynomial in the lag operator. If $(1 - L) = 0$, we obtain $L = 1$.

The autoregressive integrated moving average (ARIMA) method describes processes through the three parameters p , d , and q . For example, the following process

$$Y_t = \phi_1 Y_{t-1} + u_t + \theta_1 u_{t-1}$$

is called ARMA (1, 1) because it contains one autoregressive (Y_{t-1}) and one moving-average term (u_{t-1}). Therefore, the value of the autoregressive parameter p reflects how many preceding observations influence the current observation. The value of the moving-average term q describes how many previous random shocks must be taken into account when describing the dependency present in the time series. Within the Box–Jenkins ARIMA framework, d is a whole number and refers to the order of differencing that is necessary to make a process stationary ($d = 0$). Thus, the ARMA (1, 1) process can also be written as ARIMA (1, 0, 1). White noise is ARIMA (0, 0, 0). Models with $d = 1$ are called integrated of order 1. Red noise is ARIMA (0, 1, 0). Autoregressive fractionally integrated moving average (ARFIMA) modeling extends the traditional Box–Jenkins approach by allowing the differencing parameter d to take on non-integer values. This enables ARFIMA models to give parsimonious descriptions of any long-range dependencies in time series. Pink noise has a d of 0.5. Stationary fractal noises can be modeled with $-0.5 < d < 0.5$. Values of d in the interval $[-0.5; 0]$ suggest anti-persistence. Processes with $d > 0$ are long-term persistent. Therefore, finite long memory can be modeled with $0 < d < 0.5$. For $0.5 \leq d \leq 1.5$, the process is nonstationary.

To summarize, there are different mathematical methods to express self-similar power-law organization of fractal structures. Since fractal signals can be analyzed in both time and frequency domains, there are different fractal parameters like the Hurst coefficient, the scaling exponent α , the power exponent β of spectral analysis, or the differencing statistic d of the ARFIMA framework. It is crucial to understand that these parameters express exactly the same characteristics, which implies that each quantity can be converted to the other. For instance, interrelations between the parameters for stationary processes are $H = (\beta + 1)/2$, $\alpha = H = d + 0.5$, and $\beta = 2d$. The expected theoretical values for 1/f noise are $H = 1$, $\alpha = 1$, $\beta = 1$, and $d = 0.5$. Table 3.1 outlines the relationship between the parameters and demonstrates how one quantity can be transformed into the other (Table 3.1).

3.5 Estimators of Fractal Parameters

As previously noted, parameters are descriptive measures of a population that are usually unknown and therefore estimated from a sample. A sample is a limited subset of a population, which can present a distorted image of the whole. Thus, the quality of inference depends on the properties of the observed sample. Fractal parameters are inferred from empirical time series, their autocorrelations, and esti-

Table 3.1 Fractal parameters and the relations between them

	H	α	β	d
White noise	0.5	0.5	0	0
Pink noise	1	1	1	0.5
Red noise	0.5	1.5	2	1
Stationary fGn	[0; 1]	[0; 1]	[-1; 1]	[-0.5; 0.5]
Anti-persistent	<0.5	<0.5	<0	<0
Positively persistent	>0.5	>0.5	>0	>0
Nonstationary fBm	[0; 1]	[1; 2]	[1; 3]	[0.5; 1.5]
Anti-persistent	<0.5	<1.5	<2	<1
Positively persistent	>0.5	<1.5	>2	>1
Relation to other parameters for fGn	$H = \alpha$	$\alpha = H$	$\beta = 2H - 1$	$d = H - 0.5$
	$H = (\beta + 1)/2$	$\alpha = (\beta + 1)/2$	$\beta = 2\alpha - 1$	$d = \alpha - 0.5$
	$H = d + 0.5$	$\alpha = d + 0.5$	$\beta = 2d$	$d = \beta/2$
Relation to other parameters for fBm	$H = \alpha - 1$	$\alpha = H + 1$	$\beta = 2H + 1$	$d = H + 0.5$
	$H = (\beta - 1)/2$	$\alpha = (\beta - 1)/2 + 1$	$\beta = 2(\alpha - 1) + 1$	$d = \alpha - 0.5$
	$H = d - 0.5$	$\alpha = d + 0.5$	$\beta = 2d$	$d = \beta/2$

mates of the spectral density function like a *periodogram*. Figure 3.4 demonstrates that independent samples (three time series with $T = 200$ observations) derived from the same pink noise population can produce rather different ACFs and periodograms (Fig. 3.4). Furthermore, it illustrates that empirical power spectra and estimated autocorrelation functions can substantially deviate from the theoretical pattern of the generating process depicted in Fig. 3.2.

Slow-decaying autocorrelations and the straight line on the log–log periodogram plot are necessary but not sufficient conditions for fractal time series. In fact, there are many ways to generate finite amounts of data that mimic the pink noise behavior. For instance, it is known that a combination of non-fractal signals like white noise and random walk can produce patterns similar to $1/f$ noise. Figure 3.5 shows that autoregressive ($p \neq 0, d = 0, q = 0$) and integrated ($p \neq 0, d = 1, q \neq 0$) ARIMA models can imitate fractal structures. Thus, accurately inferring fractal parameters from empirical data is a challenging statistical issue. Estimating methods must be able to discern fractal behavior in the presence of non-fractal noise and to reliably distinguish between genuine fractals and fractal-like signals (Fig. 3.5).

Numerous procedures for measuring the fractal parameters β , α , H , and d have been developed in recent years. Generally, the estimation methods can be assigned to three categories: (1) exact or approximate maximum likelihood or conditional sum of squares ARFIMA estimation of d like the exact method proposed by Sowell [32], the approximate algorithm (fracdiff) of Haslett and Raftery [19], and the conditional sum of squares approach introduced by Chung [9]; (2) fractal methods predicated on

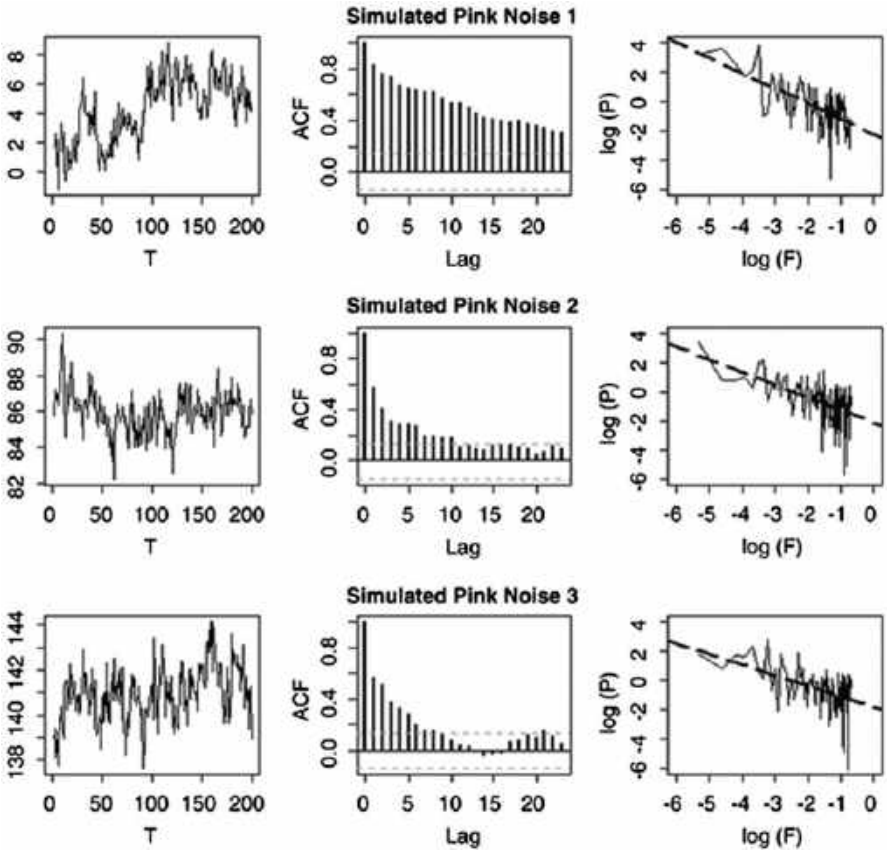


Fig. 3.4 Simulated empirical pink noise time series with their ACFs and log–log plots

the relationship $F(n) \propto n^\alpha$ such as detrended fluctuation analysis (DFA) proposed by Peng et al. [30] and signal summation Conversion (SSC) developed by Eke et al. [14]; and (3) periodogram-based procedures like power spectral density (PSD) method (Eke et al. [15]), Whittle (FDWhittle) [39], Sperio (fdSperio) [31], and Geweke/Porter-Hudak (fdGPH) [16]. Recall that a periodogram is an estimate of the spectral density function, and the following power-law relation is expected for fractal processes: $S(f) \propto 1/f^\beta$. Table 3.2 contains estimators of the fractal parameters from the open-source *statistical software* R (www.r-project.org). Because of its availability, R is increasingly used in the neurosciences. As compared with other software tools, R has many further advantages: It provides a variety of statistical and graphical techniques and is highly extensible, allowing users to develop new computational routines by combining the existing functions. Various packages are under constant development for R and cover a wide range of modern statistics (Table 3.2).

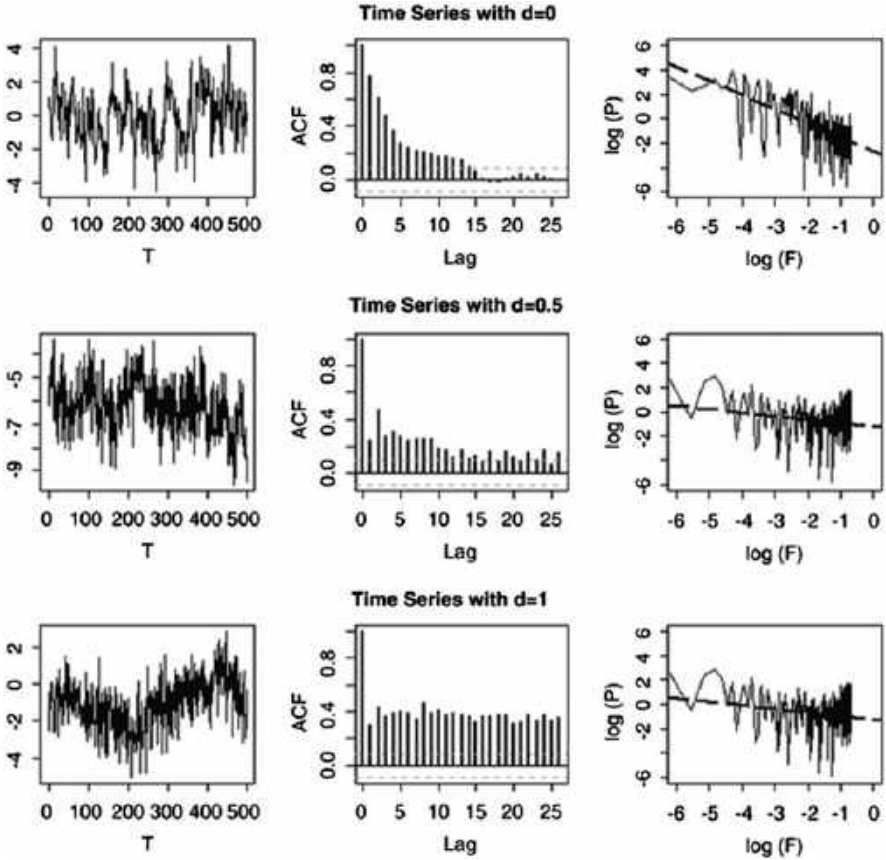


Fig. 3.5 Simulated empirical fractal and non-fractal time series with their ACFs and log–log pots

Table 3.2 Estimators of fractal parameters available in the software package R

Procedure	Outputs estimates of	Can be applied to	Available
DFA	α	fGn and fBm	Library <i>fractal</i>
SSC	α	fGn and fBm	[21] ^a
PSD	β	fGn and fBm	[22]
^{low} PSD _{we}	β	fGn and fBm	[22] ^a
HurstSpec	α	fGn and fBm	Library <i>fractal</i>
fdGPH	d	fGn and fBm	Library <i>fracdiff</i>
fdSperio	d	fGn and fBm	Library <i>fracdiff</i>
FDWhittle	d	fGn and fBm	Library <i>fractal</i>
fracdiff approximate ML	d	Only fGn	Library <i>fracdiff</i>

^aR codes for SSC and ^{low}PSD_{we} estimations are available at http://www.psychologie.uni-heidelberg.de/projekte/zeitreihen/R_Code_Data_Files.html

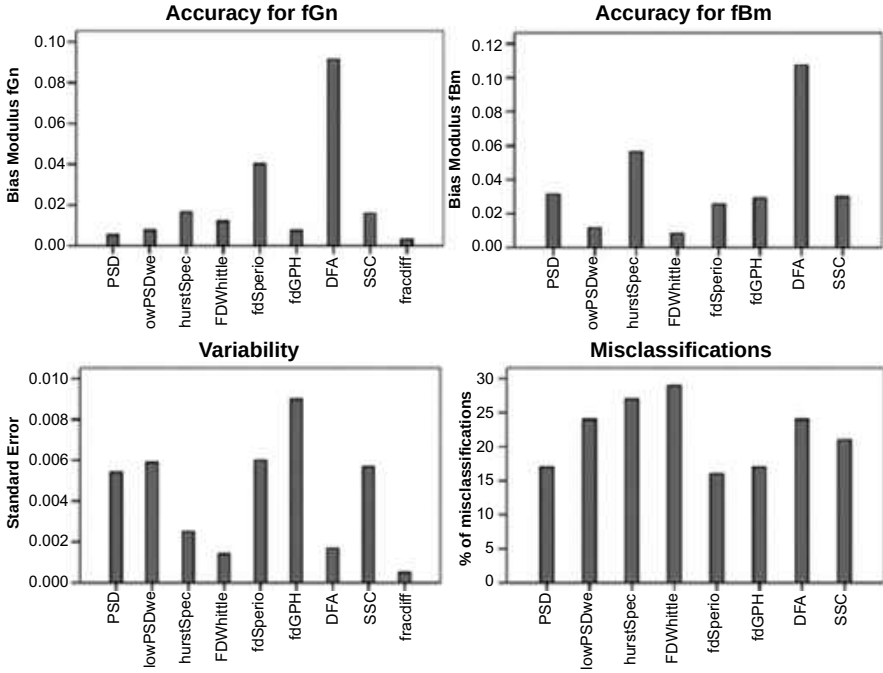


Fig. 3.6 Evaluation of the fractal methods implemented in R by means of a Monte Carlo method

Due to the complexity of the fractal methods, it is not possible to compare them mathematically. Therefore, their properties (e.g., accuracy, variability) are examined by means of a Monte Carlo method. Figure 3.6 demonstrates findings obtained in one simulation study. The estimators are compared and evaluated using bias moduli ($|\text{mean of estimates} - \text{true parameter}|$), standard errors of estimates, and the percentage of signal misclassifications employing the following rules: $\beta < 1 \rightarrow \text{fGn}$ and $\beta > 1 \rightarrow \text{fBm}$, $\alpha < 1 \rightarrow \text{fGn}$ and $\alpha > 1 \rightarrow \text{fBm}$, and $d < 0.5 \rightarrow \text{fGn}$ and $d > 0.5 \rightarrow \text{fBm}$. Estimations are averaged for different sample sizes (128, 512, 1024, 2048) and parameterizations (e.g., H variations from 0.1 to 0.9) (Fig. 3.6).

ARFIMA algorithms were thoroughly evaluated by Stadnytska and Werner [37] and Torre et al. [40]. The main advantage of the ARFIMA methods is the possibility of the joint estimation of the short- and long-memory parameters. This solves a potential finite sample problem of biased overestimation of fractality in time series that contain both long- and short-range components. Moreover, goodness-of-fit statistics based on the likelihood function, like the Akaike information criterion (AIC) or the Bayesian information criterion (BIC), allow determining the amount of “short-term contamination” and enabling a reliable discrimination between short- and long-memory processes. The greatest problem with ARFIMA estimators is

that they only work for stationary series because their range is confined to $[0; 0.5]$. As a result, nonstationary fractional Brownian motions ($0.5 \leq d \leq 1.5$) tend to receive estimates of d close to 0.5. Consequently, such empirical series can easily be misclassified as $1/f$ noise. Hence, checking for stationarity is a necessary precondition for ARFIMA estimation, which can be done using special procedures called *unit root tests* (see [35], for a comprehensive overview). The augmented Dickey–Fuller test (ADF) [12], the most popular method, checks the null hypothesis “ H_0 : $d = 1$, series has a unit root””; thus, the rejection of H_0 implies that the series under study is probably stationary. Dolado et al. [13] developed a procedure that extends the Dickey–Fuller approach to fractionally integrated processes: the fractional Dickey–Fuller (FDF) test. In contrast, the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test [23] assumes that processes are stationary (H_0 : $d = 0$). Therefore, the properties of the series under study can be determined using a combination of both procedures: (1) if the ADF is significant and the KPSS is not, then the data are probably stationary with $d \in [0; 0.5]$; (2) in the Brown noise case, insignificant ADF and significant KPSS results are expected; (3) $d \in [0; 1]$, if both tests are significant.

Eke et al. [14], Delignières et al. [11], Stroe-Kunold et al. [38], Stadnytska et al. [36], and Stadnitski [33] systematically analyzed different power law-based methods. In contrast to ARFIMA procedures, these estimators can be directly applied to different classes of time series. Consequently, they represent adequate tools for distinguishing fGn and fBm signals. However, due to considerably larger biases and more pronounced standard errors, the precision of fractal and periodogram-based methods is distinctly inferior to that of the ARFIMA approaches. Moreover, algorithms like PSD, DFA, or SSC use different data transformations like detrending or filtering. As a result, the performance of estimators strongly depends on the manipulations employed. For instance, numerous modifications have been suggested to improve the PSD estimation. The method $^{\text{low}}\text{PSD}_{\text{we}}$ consists of the following operations: (1) subtracting the mean of the series from each value, (2) applying a parabolic window to the data $[w]$, (3) performing a bridge detrending $[e]$, and (4) estimating β excluding 7/8 of high-frequency power estimates $[low]$. The estimator PSD is constructed without transformations. Simulation studies demonstrated that PSDs were more accurate for fGn noises, whereas $^{\text{low}}\text{PSD}_{\text{we}}$ were accurate for fBm signals. The greatest disadvantage of fractal and periodogram-based methods is their poor performance for complex processes that combine long- and short-term components. For example, simulation studies demonstrated that FDWhittle, the best procedure for pure noises, showed the worst accuracy in complex cases.

Table 3.3 Point estimates of the fractal parameters obtained from different fractal methods

Procedure	Autoregressive process ($d_{\text{true}} = 0$)	Pink noise process ($d_{\text{true}} = 0.5$)	Integrated process ($d_{\text{true}} = 1$)
DFA	0.65	0.43	0.33
SSC	0.40	0.36	0.61
PSD	0.64	0.21	0.05
HurstSpec	0.66	0.44	0.24
fdGPH	0.37	0.36	0.77
fdSperio	0.24	0.37	0.80
FDWhittle	0.71	0.24	0.24
fracdiff	0.50	0.24	0.23
Approximate ML	(0) ^a	(0.46) ^a	(0.46) ^a

To make the comparisons more convenient, all measures are transformed to d

^aJoint estimation of long- and short-memory components

3.6 Identification of Fractal Noise in Empirical Settings

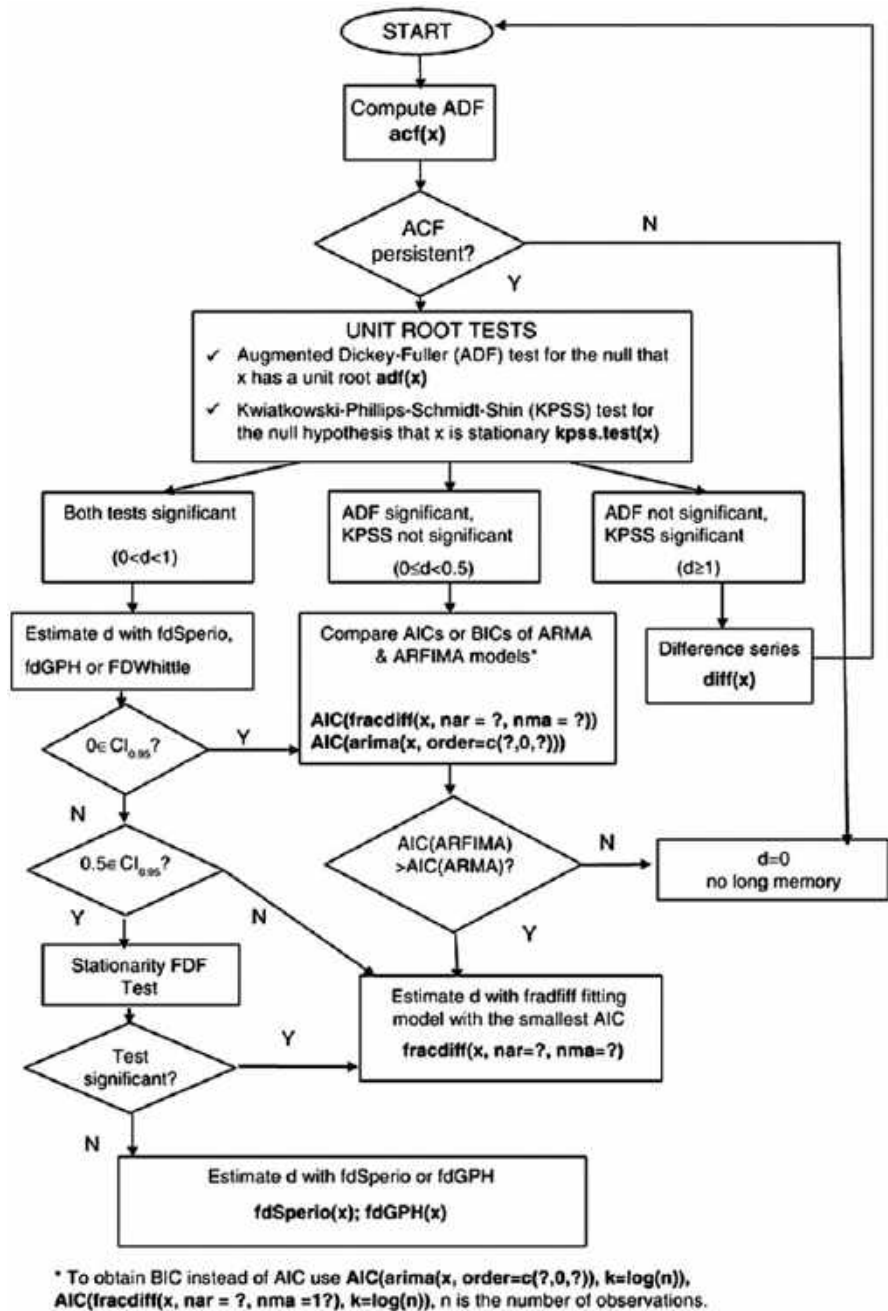
The most important findings from the reported evaluations for researchers studying fractality can be summarized as follows. First, all estimators require at least 500 observations for acceptable measurement accuracy. Further, it is essential to know that there exist different procedures with diverging characteristics. The central difficulty is that there is no clear winner among them: None of the procedures is superior to the other. As a result, the choice of the measurement method determines the outcome of fractal analyses. The performance of the methods strongly depends on aspects like the properties of the underlying process (e.g., stationary vs. non-stationary) or empirical context. For instance, maximum likelihood algorithms, the most accurate estimation techniques of the time domain, can only handle stationary data. Most estimators from the frequency domain can be applied to stationary and nonstationary time series directly; however, they are less precise than ARFIMA methods. Moreover, they tend to fail in empirical series that combine fractal and non-fractal dynamics. Hence, the key finding from simulation studies is that fractal estimators can produce erroneous results under disadvantageous conditions. Further, estimates from the same time series obtained from different methods can vary considerably. Consequently, comparisons of results from studies in which fractality was determined with different measurement instruments are problematic and must be interpreted cautiously.

To emphasize the importance of the last conclusion, fractal parameters of the simulated time series presented in Fig. 3.5 are estimated with different methods. Table 3.3 shows the obtained point estimates. To make the comparisons more convenient, all measures are converted to d . The demonstration aims to point out that non-fractal series could be easily misclassified as fractal noises. It is even possible to erroneously attribute the $1/f$ properties to autoregressive or integrated processes (Table 3.3).

Depending of the hypotheses of the research, diverse objectives of fractal time series analyses can be distinguished: to discriminate between fractal and non-fractal patterns for diagnostic purposes, to test for the effective presence of genuine persistent correlations in the series, to provide an accurate estimation of the strength of these long-range dependencies, or to identify the short-term process that accompanies a fractal pattern. Simulation studies demonstrated that the performance of fractal methods strongly depends on aspects like the complexity of the underlying process or parameterizations. As a result, strategic approaches are necessary when analyzing fractals. Moreover, different objectives require distinct strategies. An example of such a strategic approach is described by Stadnytska et al. [36]. The algorithm presented in Fig. 3.7 allows an accurate estimate of the fractional differencing parameter d by combining different tests and statistical techniques. Furthermore, it distinguishes between fractal and non-fractal empirical time series. For instance, the procedure will probably debunk autoregressive or integrated time series from Fig. 3.5 as non-fractal. Stadnitski [34] demonstrated the application of this strategy to an empirical time series using R (Fig. 3.7).

3.7 Summary

Self-similarity and long memory are essential characteristics of a fractal pattern. Slowly decaying autocorrelations and power laws reflect these properties in the parameters H , α , β , and d . These fractal parameters express exactly the same statistical characteristics; thus, each quantity can be converted to the other. The expected theoretical values of pink noise are $H = 1$, $\alpha = 1$, $\beta = 1$, and $d = 0.5$. There are two major types of estimators for these parameters: ARFIMA algorithms and procedures searching for power laws. The former are very accurate methods capable of measuring both long- and short-term dependencies, but they can only handle stationary processes. The latter are adequate for stationary and nonstationary data; their precision, however, is distinctly inferior to that of the ARFIMA methods and they tend to overrate fractal parameters in series containing short-memory components. The greatest problem is that no estimator is superior for a majority of theoretical series. Moreover, in a typical research situation, it is usually unclear what kind of process generated empirical data. Consequently, the estimation of fractality requires elaborate strategies. The estimation method proposed by Stadnytska et al. [36] represents an example of such a strategic approach.

Fig. 3.7 Strategy for estimating d proposed by Stadnytska et al. [36]

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Chapter 4

Multifractal Analysis in Neuroimaging



Renaud Lopes

Abstract The characteristics of biomedical signals are not captured by conventional measures like the average amplitude of the signal. The methodologies derived from fractal geometry have been a very useful approach to study the degree of irregularity of a signal. The monofractal analysis of a signal is defined by a single power-law exponent in assuming a scale invariance in time and space. However, temporal and spatial variation in the scale-invariant structure of the biomedical signal often appears. In this case, multifractal analysis is well-suited because it is defined by a multifractal spectrum of power-law exponents. There are several approaches to the implementation of this analysis, and there are numerous ways to present these.

In this chapter, we review the use of multifractal analysis for the purpose of characterizing signals in neuroimaging. After describing the tenets of multifractal analysis, we present several approaches to estimating the multifractal spectrum. Finally, we describe the applications of this spectrum on biomedical signals in the characterization of several diseases in neurosciences.

Keywords Multifractal analysis · Multifractal spectrum · Electroencephalogram · Brain imaging · Neurological diseases

4.1 Introduction

The methodologies derived from fractal geometry have been a very useful approach to study the degree of fragmentation (or irregularity) in natural, artificial, and statistical structures or processes [1]. Fractal structures are characterized by self-

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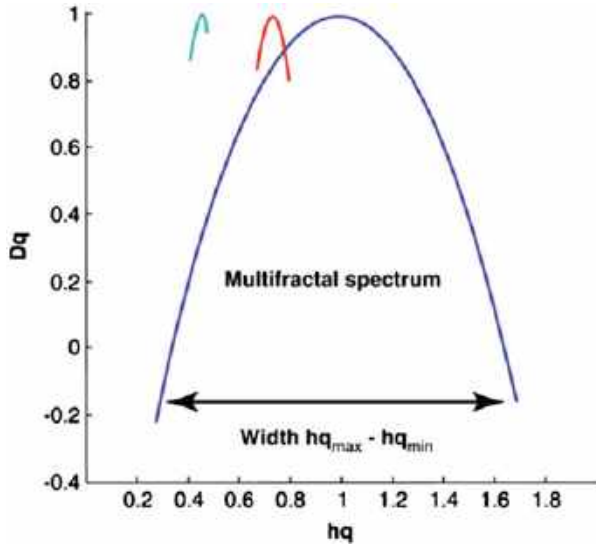


Fig. 4.1 An example of multifractal spectra for monofractal, multifractal, and white noise signals

similarity, scaling independence, and a fractal dimension, an exponent obtained from a power or scaling law.

However, due to the complexity of some processes, one exponent may not be enough to characterize a complex phenomenon. Multifractal formalism allows using more exponents. In this case, the object of analysis is divided into several fractal sets, each generating a fractal dimension that is then translated into a continuous spectrum of exponents (the so-called spectrum of singularities or multifractal spectrum). Figure 4.1 shows an example of multifractal spectra for monofractal, multifractal, and white noise signals. The multifractality degree (MD) obtained from this continuous spectrum allows measuring the information content. Multifractal systems are common in nature, especially in geophysics. They include fully developed turbulence, stock market time series, heartbeat dynamics, human gait, and natural luminosity time series, among others.

In neuroimaging, multifractal analysis has been a very useful approach to studying problems related to neurodegenerative diseases, epilepsy, schizophrenia, and brain tumors, among others [2].

This chapter reports the tenets, methods, and applications of multifractal analysis in neuroimaging. The purpose of this chapter is to provide a survey of multifractal analysis approaches and to discuss the main results in neurosciences. It is organized as follows: in the next section, we introduce more formally the tenets of multifractal analysis; Sect. 4.3 gives the survey of multifractal analysis approaches, their principles, and limitations. Section 4.4 discusses the main applications of multifractals in the neurosciences and the methods used.

4.2 Tenets of Multifractal Analysis

Many signals, recorded from real data, exhibit a very irregular behavior (see, e.g., Gallos et al.; Mandelbrot and Van Ness; Shivamoggi; and Meyer [3–6]). They vary rapidly in small intervals and display oscillations of different sizes. For instance, they can be time series of prices of financial assets, they can also be heartbeat time series, and so on. In the sequel, we denote by f one of these signals. Roughly speaking, the local roughness of f at some place, in other words, the “typical” size of its oscillations in a neighborhood of some fixed point t_0 , provides relevant information on the signal f at this place. Also, it is crucial to wonder whether local roughness of f changes from place to place and in that situation to have a global and synthetic description of all of its values. This is the keystone of multifractal analysis (see, e.g., Gallos et al. and Meyer [3, 6]).

The main goal of this section is to explain some important notions, which formalize the ideas presented in the first paragraph.

First, we define the notion of a *pointwise Hölder exponent*, which provides a precise measure of the local roughness of a signal f at an arbitrary fixed point t_0 . Let α be any given nonnegative number; one says that f satisfies a *Hölder condition of order α at t_0* , whenever there exists a polynomial P of degree less α , and there is a constant c , such that, for any t in the neighborhood of t_0 (in other words, $|t - t_0|$, the distance between t and t_0 is small), one has

$$|f(t) - P(t - t_0)| \leq c|t - t_0|^\alpha \quad (4.1)$$

Observe that P and c depend a priori on t_0 ; in other words, one should expect that they change if t_0 is replaced by another point t_1 . Also, observe that when α lies between 0 and 1, then P is always the constant polynomial equal to $f(t_0)$; thus, in this case, the inequality (4.1) reduces to $|f(t) - f(t_0)| \leq c|t - t_0|^\alpha$. The pointwise Hölder exponent of f at t_0 is denoted by $\alpha_f(t_0)$ and defined as the supremum of the numbers α satisfying (4.1).

The well-known *Brownian motion* and its most natural generalization called the *fractional Brownian motion* (fBm) have traditionally been used in order to model rough signals. Maybe one can say that the Brownian motion is the keystone of classical mathematical finance; among many other things, the Brownian motion plays a key role in the very famous Black–Scholes model for pricing options on risky financial assets [7]. With regard to the fractional Brownian motion, this self-similar Gaussian process with stationary increments was first introduced in 1941 by Kolmogorov [8] in order to model the velocity of turbulence at intermediate scales (the inertial zone). Yet, its denomination “fractional Brownian motion” (fBm) only goes back to the late 1960s and was first used in the seminal article [4] by Mandelbrot and Van Ness. We mention that the fBm has been made to be known and adopted by a large audience, thanks to this seminal article and to several other works by Mandelbrot, which have followed it (see Mandelbrot and Wheeler and Mandelbrot [1, 9]). The fBm is denoted by B_H , since it mainly depends on the *Hurst*

parameter H , which belongs to the interval $(0,1)$. Up to a renormalization factor, B_H can be defined as the centered Gaussian stochastic process with covariance satisfying, for all t and s ,

$$E [B_H(t), B_H(s)] = \frac{1}{2} \left(|t|^{2H} + |s|^{2H} - |s - t|^{2H} \right)$$

Observe that the fBm reduces to the Brownian motion in the particular case where $H = 1/2$. In all the other cases (that is $H \neq 1/2$), the increments of B_H are correlated, and they even display long-range dependence when $H > 1/2$. Surely, the additional degree of freedom provided by the Hurst parameter makes the fBm a considerably more flexible model than the Brownian motion. Yet, the fBm has to be abandoned as a model for the velocity of turbulence. It is important for us to precisely explain the reasons for this abandonment. Indeed, in this way, we can clarify the differences between the so-called monofractal and multifractal signals.

Assume that f is an arbitrary rough signal. The so-called “structure function of f ” is denoted by I_f and is defined, for all h and positive p , as

$$I_f(h, p) = \int |f(t+h) - f(t)|^p dt \quad (4.2)$$

We mention, in passing, that some important variants of the structure function can be defined in terms of wavelet coefficients, continuous wavelet transforms, and the so-called wavelet leaders (see, e.g., Gallos et al. and Mandelbrot [3, 9]). Generally speaking (except in pathological cases), if $|h|$ is small, then the quantity $I_f(h, p)$ behaves approximately as

$$I_f(h, p) \sim |h|^{z_f(p)} \quad (4.3)$$

When f is a signal issued from the velocity of a turbulent flow, various experiments, based on wind tunnels measurements, have shown that “there is no linear dependence between p and the exponent” $\zeta_f(p)$. The situation is very different in the case where f is the fBm B_H because it can then be shown that the exponent $\zeta_{B_H}(p)$ satisfies $\zeta_{B_H}(p) = Hp$, which clearly means that $\zeta_{B_H}(p)$ linearly depends on p . This linear dependence can be partially explained by the fact that the pointwise Hölder exponent of the fBm B_H remains the same all along its sample path; more precisely, for all t , one has $\alpha_{B_H}(t) = H$. In sharp contrast, for a signal f issued from the velocity of a turbulent flow, the pointwise Hölder exponent $\alpha_f(t)$ keeps changing from point to point in an unpredictable and chaotic way. What are then the values that $\alpha_f(t)$ is most likely to have? In order to answer this question, it is useful to introduce the notions of “iso-Hölder sets” and of “spectrum of singularities.” To each fixed nonnegative number, ρ is associated with an iso-Hölder set $S_f(\rho)$, defined as the set of all the points t such that $\alpha_f(t) = \rho$. The Hausdorff dimension of $S_f(\rho)$ is denoted by $D_f(\rho)$. The spectrum of singularities of f is defined as the whole collection of the values taken by $D_f(\rho)$, when ρ runs through the set of all the

nonnegative numbers; therefore, this spectrum is denoted by D_f . Observe that in the case of the fBm B_H , the spectrum D_{B_H} is *trivial*, since it only consists of two values. It is for this reason that the fBm is called a *monofractal signal*. In sharp contrast, when a spectrum D_f contains infinitely many different values (i.e., in practice, a great number of different values), then the corresponding f is called a *multifractal signal*. Notice that models for such signals can be obtained through multiplicative cascades (see, e.g., Gallos et al. and Meyer [3, 6]). We end this paragraph by emphasizing the fact that the spectrum of singularities D_f provides a global and synthetic geometrical description of all the values of the pointwise Hölder exponent of a multifractal signal f . Thus, D_f contains relevant and useful information related to the roughness of f to which it is important to access. This leads us to *the natural question on how the spectrum of singularities D_f can be explicitly obtained and determined in practice*.

Trying to compute D_f using its very definition is definitively an impossible task, even to the most powerful computer. The first obstruction is that this task requires the calculation of the pointwise Hölder exponents of f at infinitely many points. The second obstruction is that calculating the Hausdorff dimension of a set, by using the very definition of this dimension, can very hardly be done in practice. Thus, it is a challenging issue to provide satisfactory answers to the fundamental question raised at the end of the last paragraph. In order to deal with this issue, multifractal analysis, more particularly the so-called multifractal formalism, was introduced by the physicians Parisi and Frisch in [10]. Since almost three decades, multifractal analysis is a very active interdisciplinary research area where mathematicians, physicians, and other scientists interact. It is worth mentioning that the main idea of the multifractal formalism, due to Parisi and Frisch, is that the exponent $\zeta_f(p)$, governing the local asymptotic behavior of the structure function (see (4.3) and (4.2)), can be obtained *as the Legendre transform of the spectrum of singularities D_f* . That is, one has

$$\zeta_f(p) = \inf_{\rho} \{p\rho + 1 - D_f(\rho)\} \quad (4.4)$$

As a consequence, $D_f(\rho)$ can be recovered through *the Legendre inversion formula*:

$$D_f(\rho) = \inf_p \{p\rho + 1 - \zeta_f(p)\} \quad (4.5)$$

We conclude this section by stressing that the two crucial formulas (4.4) and (4.5) are, among other things, very useful for clinicians who want to obtain multifractal spectra through computations. Yet, these two formulas are not always satisfied. The main preoccupation of multifractal analysis is to determine the weakest possible conditions under which these are valid (see, e.g., Gallos et al. and Meyer [3, 6]).

4.3 Methods of Multifractal Analysis

The multifractal spectrum (also called the spectrum of singularities) D_f of a signal f can be estimated by two families of methods. The first one uses the “box-counting” approach, which is only based on time (for 1D signal) and space (for an n -Dimensional signal, $n > 1$) domain. The term “box counting” is used because the signal f is divided into boxes or segments in the computation of D_f . The procedure is repeated with different sizes of boxes. The second one uses time-frequency decomposition of the signal f for the estimation of D_f . The wavelet theory is commonly used for the decomposition. In this case, the scale of the wavelet decomposition can be compared to the size of boxes in the first family of methods. The methods and their benefits and drawbacks are summarized in Table 4.1.

4.3.1 Time Domain Methods

4.3.1.1 Generalized Fractal Dimensions and the Multifractal Spectrum

Standard box-counting techniques are used to analyze point's sets. Each set is described by an infinite number of generalized dimensions, D_q , also called “Renyi's dimension,” and by the multifractal spectrum (or spectrum of singularities) D_f . The generalized dimensions D_q are computed as a function of the order of the probability moment q and then the multifractal spectrum can be obtained by the Legendre transform.

The Legendre transform could lead to some errors [11]. In 1989, Chhabra and Jensen defined a method for the direct estimation of the multifractal spectrum [12]. This method is widely used [13, 14] but has the drawbacks of the box-counting methods.

Besides problems arising when the boxes contain few points, the algorithms are characterized by low statistics, emphasized by negative exponents ($q < 0$); this, in turn, makes the measure diverge exponentially [15].

4.3.1.2 The “Sandbox” or Cumulative Mass Method

The sandbox method, developed by Vicsek [16], is useful for the assessment of the generalized fractal dimensions for both positive and negative moment orders, q , permitting the reconstruction of the complete multifractal spectrum.

This method consists of randomly selecting N points belonging to the structure and then counting, for each point i , the number of pixels $M_i(r)$ that belong to the structure inside a disk of diameter r centered at this point. The generalized dimensions D_q are obtained using the mean of $M_i(r)$ for various rs .

The advantage of this method is that the boxes are centered on the structure, so there are no boxes with too few elements (i.e., pixels) inside. Indeed, for $q < 0$,

Table 4.1 Review of approaches to estimating the multifractal spectrum

Name	Pros	Cons
Time domain methods		
Generalized fractal dimensions and the multifractal spectrum	Easy implementation	Highly dependent on the box sizes
	Development into 1D, 2D, or 3D signals	Problems arising when the boxes contain few points
Sandbox method	Easy implementation	Dependent on the box sizes
	Development into 1D, 2D, or 3D signals	Only for binarized signals in the case of 2D or 3D signals
	Boxes centered on the structure	
	Not always a concave spectrum	High computational cost
The large deviation multifractal spectrum		Difficult to apply in 2D and 3D signals
	Easy implementation	Dependent on the box sizes
Multifractal detrended fluctuation analysis or multifractal detrended moving average	Development into 1D, 2D, or 3D signals	Only for binarized signals in the case of 2D or 3D signals
	Use of the “profile” of the signal	
	Robust even for signals of small lengths	
	Time–frequency domain methods	
Wavelet transform modulus maxima	Less dependent on the wavelet and number of scales than “box-counting” methods	High computational cost
	Development into 1D, 2D, or 3D signals	The continuous wavelet transform introduced some discontinuities at the end points of the signal
Wavelet leaders-based multifractal analysis		No mathematical background
	Less dependent on the wavelet and number of scales than “box-counting” methods	Need large signals
	Development into 1D, 2D, or 3D signals	
	Mathematical background	
Multifractal Brownian motion	Robust to the estimation of the right part of the spectrum	
	Estimation of the Hölder exponent to each point of the signal	Very few applications
	Development into 1D, 2D, or 3D signals	Need large signals

boxes that contain a small number of elements (because they barely overlap with the cluster) give anomalously large contributions.

The sandbox procedure represents a solution to the border effect problem (i.e., the presence of almost empty cells containing few points not centered in them), permitting the reconstruction of the multifractal spectrum, which is also for negative q [17].

Finally, we can quote that this method has just been developed for binarized signals.

4.3.1.3 Large Deviation Multifractal Spectrum

When the multifractal spectrum is estimated using the abovementioned methods, its shape is always concave. The advantage of the large deviation multifractal spectrum is that it will not always be concave, and so, less information loss will occur. However, much more numerical computation is required and the method becomes difficult to apply in two and three dimensions. Indeed, the algorithm necessitates the calculation of two limits, instead of just one as for the two previous methods.

In view of the high computational cost, this method is primarily used for 1D signals [18, 19]. This spectrum can also be applied to image segmentation [20].

4.3.1.4 Multifractal Detrended Fluctuation Analysis (MDFA)

The detrended fluctuation analysis (DFA) method is a widely used technique for the quantification of monofractal properties. This method has the advantage of avoiding spurious detection of correlations that are artifacts of non-stationarities in the time series and has successfully been applied in several fields such as neuron spiking [21] and DNA sequences [22]. In 2002, Kantelhardt et al. [23] proposed a generalization of the DFA method for the multifractal characterization of nonstationary time series and called it the multifractal detrended fluctuation analysis (MDFA) procedure. It consists of three main steps. Briefly, the first one is to construct the “profile” of the signal. The second one is to divide the profile into nonoverlapping segments of equal length, and the third one is to calculate the local trend for each segment by a least-square fit of the data and calculate the variance. For more details of the MDFA method, see in Kantelhardt et al. [23].

This method is sensitive to the length of segments.

4.3.1.5 Multifractal Detrended Moving Average (MDMA)

In 2010, Gu and Zhou [24] developed an approach similar to the MDFA procedure based on the moving-average technique. They called it the multifractal detrended moving average (MDMA). Both methods rely on a measure of fluctuations but differ in calculating the moments of the fluctuation function. Indeed, in the MDMA

approach, the signal is divided into segments after the calculation of the profile and the detrending step. For more details of the MDMA method, see Gu and Zhou [24].

It has been shown that MDMA outperformed MDFA in the estimation of scaling exponents [24].

4.3.2 Time-Frequency Domain Methods

The second class of methods is defined in the time-frequency domain. All of the next approaches for the estimation of the multifractal spectrum are based on the wavelet transform. The wavelet transform of a signal is used like an “oscillating” box to represent its components. Some methods use the discrete wavelet transform, whereas others are based on the continuous wavelet transform.

4.3.2.1 Wavelet Transform Modulus Maxima (WTMM)

The wavelet transform modulus maxima (WTMM) method, originally developed by Muzy et al. [25], is based on the concept of wavelets in general and the use of the continuous wavelet transform in particular. It has been developed and used in one and two dimensions in several fields [26–28]. In the study by Kestener and Arneodo [29], the method was applied in 3D and the authors showed its robustness using simulated 3D multifractal models. In the WTMM method, the wavelet transform maxima are used to define a partition function whose power-law behavior is used to estimate the local exponents and to assess the dimension of the detected exponents. A matching set of fractal exponents and dimensions is used to obtain the multifractal spectrum.

The method presents some limits. It is more difficult to implement than the previous methods. It has some freedom degrees, like the wavelet choice and the scales number. Also, the continuous wavelet transform introduced some discontinuities at the end points of the signal, due to zero padding. It is necessary to define a cone of influence for the analysis, and it is defined as the region of the wavelet spectrum, in which the decrease of the wavelet amplitude near the edges becomes significant. However, Kestener and Arneodo [29] showed that it is more effective than the “box-counting” methods.

4.3.2.2 Wavelet Leaders-Based Multifractal Analysis (WLMA)

The wavelet leaders-based multifractal analysis (WLMA) computes the scaling exponents by measuring how the absolute value of the wavelet coefficients changes as a function of scale. The data are transformed to the wavelet domain with a discrete wavelet transform, and the wavelet leaders are calculated by finding the maximum wavelet coefficient among the adjacent wavelet coefficients for the current scale and

all smaller scales. The estimated multifractal formalism has a mathematical validity and the right part of spectrum is valid ($q < 0$), meaning that the partition function (obtained by wavelet leaders) is also valid when $q < 0$ [30]. For more details of the WLMA, see Wendt et al. [30].

4.3.2.3 Multifractal Brownian Motion (mBm)

Based on multifractal modeling, Lopes et al. [31] introduced an extension of the fBm, which overcomes some of its limitations. The proposed model is based on multifractional Brownian motion (mBm). The intuitive idea that leads to mBm consists of replacing the fBm Hurst parameter by a smooth function depending on the space variable t . Briefly, a signal is modeled by an mBm. The estimation procedure of the Hölder exponent at each t was inspired by both the variance method and the localized generalized quadratic variation method [32]. The novelty of the procedure is that the estimator was obtained through its wavelet coefficients. The advantage in using wavelet coefficients is that they offer more numerical stability than do sampled values; wavelet coefficients are less sensitive to perturbations by white noise than sampled values. For more details of the mBm procedure, see the study by Lopes et al. [31].

4.4 Applications of Multifractal Analysis

Over the last decade, multifractal analysis has been applied in four main fields of neuroimaging: electroencephalogram (EEG) signal analysis, blood oxygen level-dependent (BOLD) signal analysis, magnetic resonance imaging (MRI) brain image analysis, and brain network analysis. In this section, the methods and applications are reviewed and summarized in Table 4.2.

Table 4.2 The methods for the estimation of the multifractal spectrum and their applications in neuroimaging

Methods	Applications			
	EEG	fMRI	Brain imaging	Brain networks
Fractal dimensions			[33, 34]	[35, 36]
Sandbox				
Large deviation spectrum				
MDFA or MDMA	[37–39]	[40]		
WTMM				
WLMA	[41]	[42]		[43]
mBm				

4.4.1 *Electroencephalogram (EEG) Signals*

An electroencephalogram (EEG) is a recording of the electrical activity of the brain from the scalp. The recorded waveforms reflect the cortical electrical activity. Many studies in neuroscience have converged to raise the hypothesis that the underlying pattern of neuronal activation from the EEG signals is nonlinear, with self-affinitive dynamics, while EEG signals themselves are nonstationary. Therefore, the fractal analysis of EEG signals has shown scaling behaviors that may not be consistent with the monofractal process. These signals may be better modeled as an underlying multifractal process. Several studies have confirmed this hypothesis and are summarized below.

Multifractal analysis may outperform traditional EEG signal analysis, such as spectral analysis, in detecting the degree of cognitive impairment in brain diseases. The MDFA approach predicted cognitive function scores in an out-of-sample EEG dataset of patients with Alzheimer's disease, mild cognitive impaired, and age-matched controls [37]. The MDFA model also improved accuracy in correlating with cognitive performance as compared to spectral analysis in traumatic brain injury [38].

Interestingly, the study of multifractal behavior provided meaningful insights into the study of functional brain–heart interplay. The synchronized co-occurrence of multifractal changes in brain and heartbeat dynamics, measured by the WLMA method, highlights the complexity of the brain–heart interplay phenomenon, especially with synchronous changes at higher EEG frequency bands and through nonlinear cardiovascular control [41].

Multifractal-derived features could be used as signal features for machine learning in clinical applications. However, complex multifractal analysis may not offer a significant advance over a simpler, already established method. One of the main reasons would be that several aspects need to be considered when analyzing the multifractal properties of EEG signals, such as the choice of the appropriate estimation method, estimation parameters, and the influence of the time series variance on signal features [39].

4.4.2 *Brain Imaging*

Functional MRI (fMRI) measures brain activity by detecting associated changes in blood flow through the BOLD signal. BOLD signals are known to have a colored noise structure mainly occurring at a low frequency. Resting-state motion-corrected BOLD signals have long memory in time or $1/f$ spectral properties. The heartbeat or breathing cycle may also demonstrate scale invariance in temporal dynamics (“scale-free dynamics”). Several studies of the temporal scale-free or fractal property in fMRI have demonstrated that this constitutes an intrinsic feature of ongoing brain activity [40, 42]. It was shown that the Hurst coefficient

decreases during task in activating and deactivating brain regions. However, using one scaling exponent assumes Gaussianity and self-similarity, while fMRI signals may significantly depart from those either of those two assumptions [42]. To address this issue, the scaling properties of fMRI signals can be investigated using multifractal analysis, thus measuring a collection of scaling exponents. Evidence for the presence of multifractality in both resting-state and task-related fMRI signals has been demonstrated in almost the whole brain.

There are fewer published works on multifractal analysis in other brain imaging modalities. Takahashi et al. [33] used the multifractal spectrum to quantitatively evaluate white matter hyperintensity on MR images. Lopes et al. [34] used a 3D adaptation of the multifractal spectrum to characterize the local changes in homogeneity in brain single-photon emission computed tomography (SPECT) images. There was a statistically significant difference between the control group and the pathological group.

4.4.3 Brain Network Analysis

Functional brain imaging has been widely used to study the neural basis during a task. However, the brain maintains a level of spontaneous activity at rest that can be captured by functional connectivity approaches in fMRI. Distributed sets of brain regions, whose spontaneous activity exhibited a large degree of temporal coherence, can be identified through the computation of functional connectivity maps, to extract the brain resting-state networks. Resting-state networks can also be identified using EEG. Although functional connectivity has expanded our knowledge on how functional brain networks are organized in space and time, the study of fluctuations of the network topology and those of individual connections between brain regions, known as dynamic functional connectivity, has expressed multifractal scaling [35]. The dynamic nature of global network metrics and the local individual connections in the networks have been characterized by multifractal time series analysis using EEG and fMRI [36, 43].

The spatiotemporal organization of brain resting-state networks has been demonstrated on healthy subjects using the multifractality of dynamic functional connectivity in EEG [35]. The results showed that the multifractal analysis framework may provide potential biomarkers for clinical research. For example, combined with traditional resting-state features, multifractal analysis contributed to distinguish Alzheimer's disease in healthy subjects [44].

4.5 Conclusions

A limitation of fractal analysis is its inability to describe objects by a single fractal value, thus exhibiting a multifractal behavior. Multifractal analysis is a response to this limit. It allows computing a spectrum or a set of fractal dimensions.

Many algorithms exist to evaluate the spectrum of singularities D_f (the so-called multifractal spectrum) of a signal f , and numerical differences between the methods appear. The choice of the method is not a trivial task because no “gold-standard” approach exists. However, the advantages and limitations of each method have to be known in relation to input data.

About time domain approaches, MDFA and MDMA methods are computationally more stable than are others methods in terms of box (or segment) size. Moreover, they were the most used in the previous studies. The choice between MDFA and MDMA is not trivial. Eke et al. [45] showed that MDFA is applicable to more types of signals than is MDMA.

About time-frequency domain approaches, the WTMM method is the most used in the literature. However, WLMA overcomes major difficulties partially solved by WTMM such as no mathematical results are expected to hold for the WTMM method, the high computational cost of WTMM, and the easy theoretically and practical generalization to higher dimensions of WLMA [30]. For the mBm method, its low number of studies does not allow to draw conclusions about its robustness.

There is no consensus on the choice between time or time-frequency domains approaches. The only strong result is that MDFA is more consistent than WTMM and WLMA methods for shorter signals (less than 7500 time points) [44, 46].

Most of the applications of multifractal analysis in neuroimaging are interested in 1D signals, such as EEG and functional MRI signals. Evidence for the presence of multifractality in both EEG and fMRI signals has been largely demonstrated. Multifractal analysis is also able to discriminate two or more states in a signal or to characterize patients from healthy volunteers. Few studies are interested in the use of multifractal analysis in 2D or 3D brain images. The initial works were interesting, but they used “box-counting” methods; it will be interesting to see the influence of more robust methods, such as wavelet-based methods.

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Chapter 5

Fractal Time Series: Background, Estimation Methods, and Performances



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Abstract Over the past 40 years, from its classical application in the characterization of geometrical objects, fractal analysis has been progressively applied to study time series in several different disciplines. In neuroscience, starting from identifying the fractal properties of neuronal and brain architecture, attention has shifted to evaluating brain signals in the time domain. Classical linear methods applied to analyzing neurophysiological signals can lead to classifying irregular components as noise, with a potential loss of information. Thus, characterizing fractal properties, namely, self-similarity, scale invariance, and fractal dimension (FD), can provide relevant information on these signals in physiological and pathological conditions. Several methods have been proposed to estimate the fractal properties of these neurophysiological signals. However, the effects of signal characteristics (e.g., its stationarity) and other signal parameters, such as sampling frequency, amplitude, and noise level, have partially been tested. In this chapter, we first outline the main properties of fractals in the domain of space (fractal geometry) and time (fractal time series). Then, after providing an overview of the available methods to estimate the FD, we test them on synthetic time series (STS) with different sampling

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frequencies, signal amplitudes, and noise levels. Finally, we describe and discuss the performances of each method and the effect of signal parameters on the accuracy of FD estimation.

Keywords Fractal dimension · Time series · Neurophysiology · Hurst exponent · Detrended fluctuation analysis · Slope of Power Spectral Density · Higuch’s fractal dimension · Katz’s fractal dimension

Abbreviations

BC	Box counting
DFA	Detrended fluctuation analysis
fBm	Fractional Brownian motion
FD	Fractal dimension
fGn	Fractional Gaussian noise
f_s	Sampling frequency
GHE	Generalized Hurst exponent
H	Hurst exponent
HFD	Higuchi’s fractal dimension
KFD	Katz’s fractal dimension
SNR	Signal-to-noise ratio
sPSD	Slope of power spectral density
STS	Synthetic time series
TLF	Takagi–Landsberg function
WCF	Weierstrass cosine function

5.1 Introduction

Over the past 40 years, fractal analysis has become a central and emerging concept in most natural sciences, including geography [30], financial forecasting [65], meteorology [69], human anatomy [22, 24], genetic variability in human phenotypes [4, 34], and brain sciences [12].

In neuroscience, fractal analysis has progressively emerged as a method to describe brain anatomy at both the microscopic and macroscopic levels [25, 39, 71] as well as for exploring brain functions in both physiological [8, 46, 58] and pathological conditions [15, 57, 59, 61, 72, 73]. Brain signals, such as potentials recorded through the scalp in electroencephalography (EEG) and magnetoencephalography (MEG) or metabolic activity detected with functional imaging techniques, offer a fundamental tool for understanding the brain activity underlying many physiological and pathological processes. Fractal analysis has been increasingly applied to describe those dynamic signals and, therefore, to characterize brain activity [8, 46, 48, 76] and neurological diseases [14, 56, 60, 61, 72, 75].

To date, most of the knowledge in neurophysiology has been acquired by applying the theory of linear systems. Accordingly, neurophysiological signals have been widely analyzed using linear methods, such as Fourier transform (FT) and wavelet transformation (WT), whose application requires the preliminary assumption of several characteristics of the signal, e.g., their stationarity. FT allows decomposing a signal into a set of elementary sinusoids that are then represented in the frequency domain and described by their power spectral density (PSD). Thus, in an FT signal, power distribution is described across frequencies, and the PSD of frequency components has been widely used as a quantitative measure of brain activity.

However, although these methods are well-suited for stationary time series, most neurophysiological signals are nonstationary and nonlinear. Moreover, inferring their properties a priori could represent a bias in their analysis, and therefore, signals displaying an irregular behavior could be wrongly considered as noise.

Conversely, nonlinear methods have been increasingly applied in the last few decades to analyze these signals in their complexity and to detect information potentially hidden in physiological time series.

Although fractal analysis has been traditionally applied to measure geometrical complexity (i.e., in the domain of space), some fractal properties can be demonstrated in the time domain and can be thus characterized for time series. Fractal properties have been described in several neurophysiological time series at different levels, from neuronal spiking activity to EEG, MEG, and electromyography (EMG) signals [49, 67, 74]. These properties of neurophysiological time series can be characterized and quantified with specific parameters, such as the fractal dimension (FD) and the Hurst exponent (H), which will be extensively described in the following sections.

In this chapter, we first present the main properties and characteristics of fractal objects in the domain of space, providing some basic concepts of fractal geometry. We then discuss the characterization of the fractal processes in the time domain. Finally, we will report several methods to evaluate the fractal properties of time series, and we will test them on synthetic data to evaluate their performances.

5.2 Fractal Geometry

In Euclidean geometry, axioms and rules can describe objects with discrete dimensions, such as a point (nondimensional, 0D), a line (monodimensional, 1D), a triangle (bidimensional, 2D), or a parallelepiped (tridimensional, 3D).

If, on one hand, geometrical figures and Platonic solids can be represented by means of Euclidean geometry, natural objects show a geometrical complexity that can be represented by these integer dimensions only in a simplistic and reductionistic way. The geometry of some complex objects can be generated using recursive algorithms [18, 45]. These complex objects, whose dimensions lie in

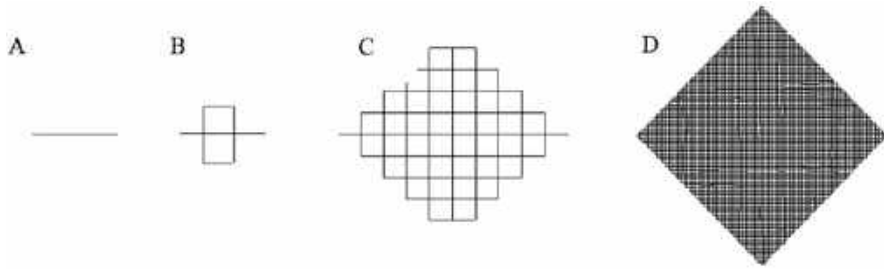


Fig. 5.1 Peano's curve (D) is generated by dividing a line segment (A) into three parts and placing two squares symmetrically above and below the middle part (B). Further iteration of this process, where the pattern is repeated at a reduced scale for each segment of the line and the squares (C). The iteration of this process leads to a filled square (D), a space-filling curve with fractal properties

between Euclidean integer dimensions (between D and 2D in two-dimensional objects and between 2D and 3D in three-dimensional objects), have been named “fractals” by Mandelbrot [41], who coined the term from the Latin word *fractus* (meaning “broken” or “fragmented”), to emphasize their irregular characteristics. Unlike Euclidean geometry, fractal geometry is not defined by axioms but rather by a set of properties that indicate whether the analyzed object is fractal [18, 45] or, at least, can be quantified by means of fractal geometry principles and tools. For example, the continuous curve that passes through every point in a unit square, described *ante litteram* by Giuseppe Peano (1890) [53], is a space-filling curve with fractal properties (Fig. 5.1).

The most important properties of fractal geometry are (i) self-similarity, i.e., the resemblance of the parts of an object to the whole object, (ii) scale invariance, which means that the measured properties do not change with the scale at which they are measured, and (iii) fractal dimension, which provides a quantitative measure of self-similarity and scaling [14, 45].

5.2.1 Self-Similarity

An object is defined as geometrically self-similar when its parts are identical to the whole object [45], i.e., irrespective of whether each part can be considered a reduced-scale copy of the whole [25]. This definition refers to exact (i.e., mathematical) fractal objects, such as those shown in Fig. 5.2. The self-similarity of these abstract objects results from the application of a recursive algorithm, which iterates the same rule (ideally with infinite iterations) on the object itself (see Sect. 5.2.3).

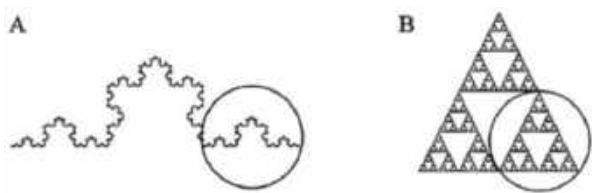


Fig. 5.2 The Koch curve (A, also called the Koch snowflake) and the Sierpiński triangle (B) are exact mathematical fractal objects displaying self-similarity, i.e., their parts (encircled in the figure) are identical rescaled copies of the whole object. The Koch curve has a fractal dimension (FD) of 1.261, whereas the FD of the Sierpiński triangle is 1.585

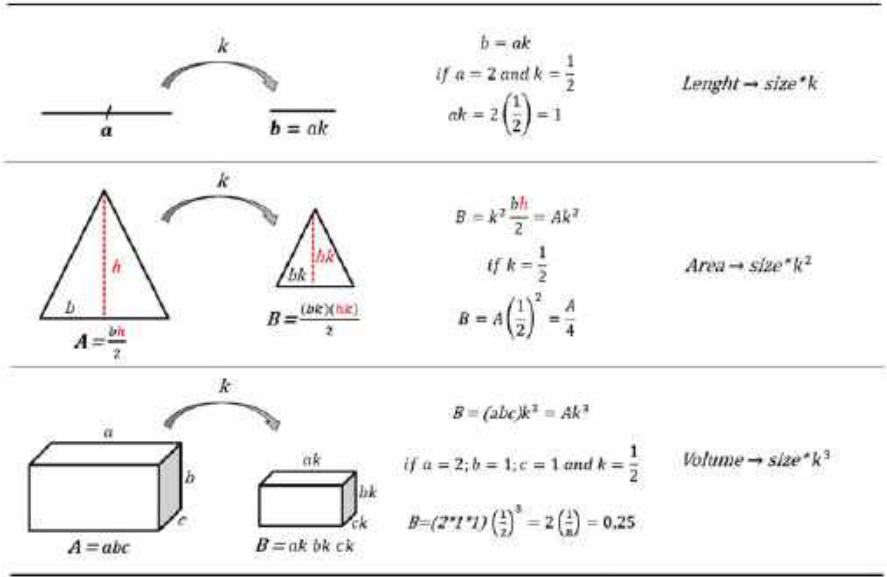


Fig. 5.3 Scaling relationships between monodimensional (straight line), bidimensional (triangle), and tridimensional (parallelepiped) objects

5.2.2 Scaling

To better understand the differences between Euclidean and fractal geometries, we should use scaling and self-similarity to measure an object. As an example, let us consider Fig. 5.3: we can say that segment b has half the length of segment a . Thus, $b = k * a$, with $k = \frac{1}{2}$. So, if $a = 2$, then we can say that $b = (\frac{1}{2}) * 2 = 1$. Therefore, we can generalize that the dimension of a segment or a line is: length = size $* k^1$.

In the case of a triangle, its area is $A = (bh)/2$, where b is the base of the triangle and h is its height, according to Euclidean geometry. If we want to know the area

B of a second smaller triangle at a scale factor $k = 1/2$, using the scaling rule on a segment, we can multiply the base and the height of the triangle, i.e., $b * k$ and $h * k$, thus obtaining for the area of triangle B as:

$$B = \frac{kb * kh}{2} = \frac{k^2(bh)}{2} = Ak^2$$

In this case, we can generalize the rules again as $\text{area} = \text{size} * k^2$. Finally, for a parallelepiped, its volume according to Euclidean geometry is $A = a b c$, where a , b , and c are the sides of the parallelepiped. As previously done for the triangle, by applying the scaling rule to the sides of the parallelepiped, we can calculate the volume B of a second parallelepiped with a scale factor k of $1/2$ so that $B = (ka)(kb)(kc) = k^3(abc) = k^3A$. Generalizing again, we have: $\text{volume} = \text{size} * k^3$.

Thus, by considering the scaling factor k , we can define some general rules regarding the scaling of monodimensional, bidimensional, and tridimensional objects (a line, a triangle, and a parallelepiped, respectively), which can be summarized as follows:

- Length = $\text{size} * k^1$
- Area = $\text{size} * k^2$
- Volume = $\text{size} * k^3$

where the k exponent is the dimension of the object under investigation.

5.2.3 From Euclidean Dimension to Fractal Dimension

Before moving on to the formal definition of fractal dimension (FD), let's explain it intuitively with an example. A fractal object can be generated starting from simple objects with definite dimensions such as a line, a triangle, and a parallelepiped (initiators) by applying transformation rules (generators), with ideally infinite iterations. To clarify this process, let's create a fractal object from the black triangle (see Fig. 5.4). In this case, the generator eliminates the central white triangle, and the iteration applies infinite times the generator to each smaller black triangle obtained from the generator itself. This is the method to create the Sierpiński triangle, a mathematically generated self-similar structure.

Since calculating the area of the Sierpiński triangle is quite complex (if not impossible) using only the rules of Euclidean geometry, how can we calculate it? Moreover, what is its dimension?

Let's consider the scaling and self-similarity rules applied in Sect. 5.2.3, particularly the scaling rule applied for calculating the area of the triangle.

Based on the rule $\text{area} = k^2 * \text{size}$, we can consider that the area B of the smaller Sierpiński triangle S_2 in Fig. 5.5 can be calculated as:

$$B = k^D A \tag{5.1}$$

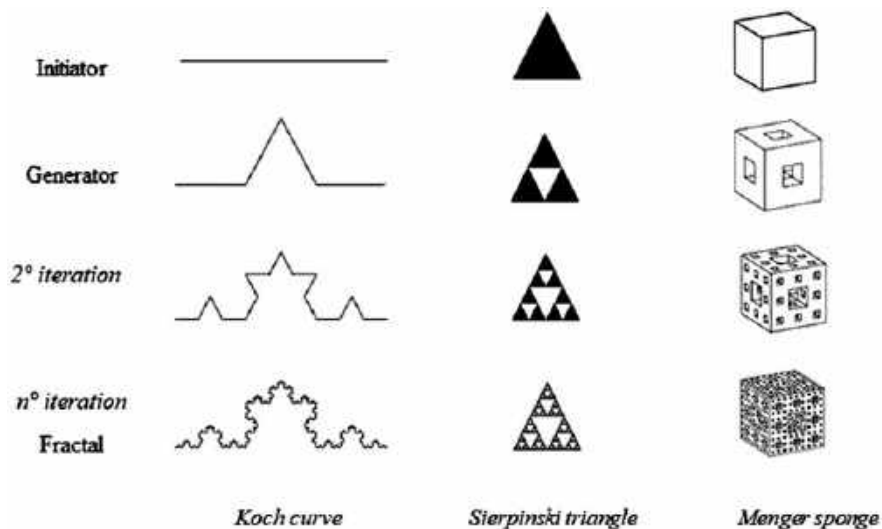


Fig. 5.4 How to create fractal objects. The Koch curve (left column) can be built by repetitively applying a generator (a central break) to a monodimensional straight line (initiator) with ideally infinite iterations. Starting with a two-dimensional black triangle (initiator) and applying a white central triangle (generator), it is possible to create a Sierpiński triangle (middle column). Finally, a Menger sponge can be generated starting from a cube (initiator) by recursively removing a cross-shaped pattern of smaller cubes (generator) from each face and the interior, leading to a three-dimensional fractal with infinitely repeating complexity

where D is the dimension of the triangle S_2 and A is the area of the bigger triangle S_1 . If triangle S_1 is three times the size of triangle S_2 , then $A = 3B = 3k^D A$. Dividing the two terms by A , we obtain $1 = 3k^D$. Since the scaling factor k between the triangles is $k = 1/2$, we have that:

$$1 = 3 \left(\frac{1}{2} \right)^D.$$

To find D , we take the logarithm of both sides:

$$\log(1) = \log \left(3 \left(\frac{1}{2} \right)^D \right)$$

Since $\log(1) = 0$, we can simplify to:

$$0 = \log(3) + D \log \left(\frac{1}{2} \right).$$

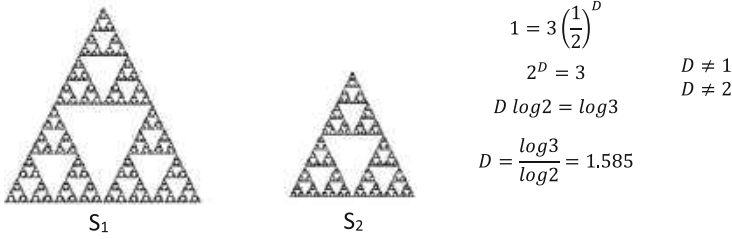


Fig. 5.5 Calculation of the dimension of the rescaled Sierpiński triangle S_2

Solving for D gives us:

$$D = -\frac{\log(3)}{\log\left(\frac{1}{2}\right)} = \frac{\log(3)}{\log(2)} = 1.585.$$

Thus, the dimension D of the Sierpiński triangle S_2 is a noninteger value falling between 1 and 2. This value represents its fractal (non-integer) dimension.

5.2.4 Fractal Dimension

Formally, self-similarity and scaling properties can be quantitatively evaluated using the change in detail with the change in resolution [41]. This quantity represents the FD, an adimensional parameter ranging from 1 to 3 (or from 1 to 2 in a bidimensional plane). The name FD derives from the characteristic of non-integer values between Euclidean dimensions. FD can be defined in many ways. One of the most intuitive definitions could be: FD is a unitless number that measures a scaling factor as a change in detail with the change in scale. Strictly speaking, the value of FD indicates that a pattern resizes itself as a fractal if it has significant detail and repeats that detail at all levels of magnification or resolution. In other words, when a geometrically self-similar object is examined with a finer resolution, further small replicas of the entire object are resolved. The size of self-similarity describes how many new pieces that are geometrically similar to the whole object are observed as the resolution is made finer. If we change the scale by a factor k and find L pieces identical to the original one, then self-similarity (which we call D : fractal dimension) will be:

$$B = k^D A$$

$$\frac{B}{A} = k^D$$

$$\begin{aligned} \text{Setting } \frac{B}{A} &= L & L &= k^D \\ \log(L) &= D \log(k) \\ D &= \frac{\log(L)}{\log(k)} \end{aligned} \tag{5.2}$$

5.3 Fractality in Natural Objects

Natural objects are rarely self-similar, and their parts are not identical reduced copies of the whole object. Because of physical constraints, parts are only approximately similar to the whole. These objects can be defined as statistically self-similar if their parts display statistical properties (i.e., probability distribution and descriptive statistical measures) proportional to those of the whole object [21].

5.3.1 Mathematical Formalization of Self-Similarity and Scaling

As explained earlier, an object is defined as self-similar when a property of the object measured at a higher resolution is proportional to the same property measured on the whole object at a lower resolution. Let L be a measured property of the object (i.e., length, area, volume) and r the resolution at which the property $L(r)$ of the object is measured. The value $L(ar)$, where $a < 1$, is the property measured at a finer resolution ar . Self-similarity means that $L(r)$ is proportional to $L(ar)$, that is:

$$L(ar) = kL(r) \tag{5.3}$$

where k is the constant of proportionality (or scaling factor), which may depend on a . The graphical representation of this relationship appears as a straight line whose slope corresponds to k (Fig. 5.6).

Statistical self-similarity can be formally described using the probability density function (pdf), defining the number of times that L assumes a specific value. Thus, an object is statistically self-similar if the pdf of the property L , measured at a finer scale ar , equals the pdf of k times L , measured at the lower scale r .

$$\text{pdf } [L(ar)] = \text{pdf } [kL(r)] \tag{5.4}$$

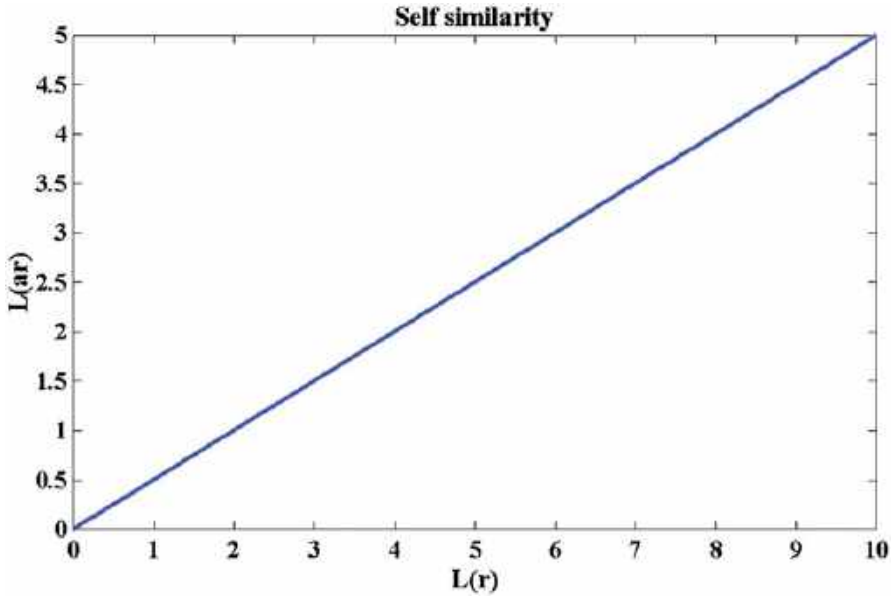


Fig. 5.6 Graphical representation of the relationship between $L(ar)$ (y-axis) and $L(r)$ (x-axis) of an ideal self-similar object. In self-similar objects, there is a linear relationship of a property L measured at different resolutions (ar and r), which is graphically represented as a straight line. The scaling factor k is set at 0.5 and represents the slope of the line

5.3.2 Self-Affinity

Another important concept that needs to be distinguished from self-similarity is self-affinity [21]. Self-similar objects are isotropic, i.e., their properties remain unchanged with scaling in all directions. Therefore, to demonstrate the self-similarity of an object, its parts should be uniformly enlarged in all directions.

Conversely, self-affinitive objects show an anisotropic scaling, i.e., the proportions between the enlarged pieces in one direction are different from those in another direction. This distinction is often blurred, and self-similarity is used to indicate self-affinity [40]. However, self-affinity represents an important concept when considering the time domain, and, formally, physiological time series are self-affinitive temporal structures [18].

5.3.3 Power Law

In statistics, a functional relationship between two quantities is defined as power law. According to power law, a [relative change](#) in one quantity results in a proportional relative change in the other quantity, independent of the initial size of those

quantities: one quantity varies as a **power** of another:

$$L(r) = A r^\alpha \quad (5.5)$$

One attribute of power law is to be scale invariant (or scale-free). The power-law scaling is a result of self-similarity where α and A are constant values:

$$L(ar) = A(ar)^\alpha = a^\alpha L(r) = k L(r) \quad (5.6)$$

where $k = a^\alpha$.

Applying logarithms to Eq. (5.5) on both sides, we obtain:

$$\log[L(r)] = \log(Ar^\alpha)$$

$$\log[L(r)] = \log(A) + \log(r^\alpha)$$

$$\log[L(r)] = \log(A) + \alpha \log(r)$$

$$\log[L(r)] = b + \alpha \log(r) \quad (5.7)$$

where $b = \log(A)$.

Equation (5.7) is similar to the linear equation $y = \alpha x + b$, describing a straight line with slope α and intercept b . In our case, the power-law scale Eq. (5.7) reveals itself as a straight line when the logarithm of the measured property $L(r)$ is plotted against the logarithm of the scale r in which it is measured (Fig. 5.7).

Objects that follow the power law and are represented by straight lines on log-log graphs are fractal objects. Although not all power-law relationships are due to fractals, the existence of such a relationship should alert the observer to consider if the system is self-similar.

This relationship is extremely important for time series recorded in neurophysiology, and the power-law test is the first to be performed to understand if the neurophysiological time series has fractal characteristics (see the following paragraphs).

5.3.4 The Hurst Exponent

The Hurst exponent (H) can be generically defined as a method to evaluate the long memory of time series or the amount of deviation of the series from a random walk. Initially introduced by Harold Hurst in 1951 as a hydrological measure to characterize the Nile river's fluctuations over time [29], the use of H has been extended to measure the tendency (or memory) of time series data to regress around the mean or gravitate toward a particular direction [36]. H ranges between 0 and

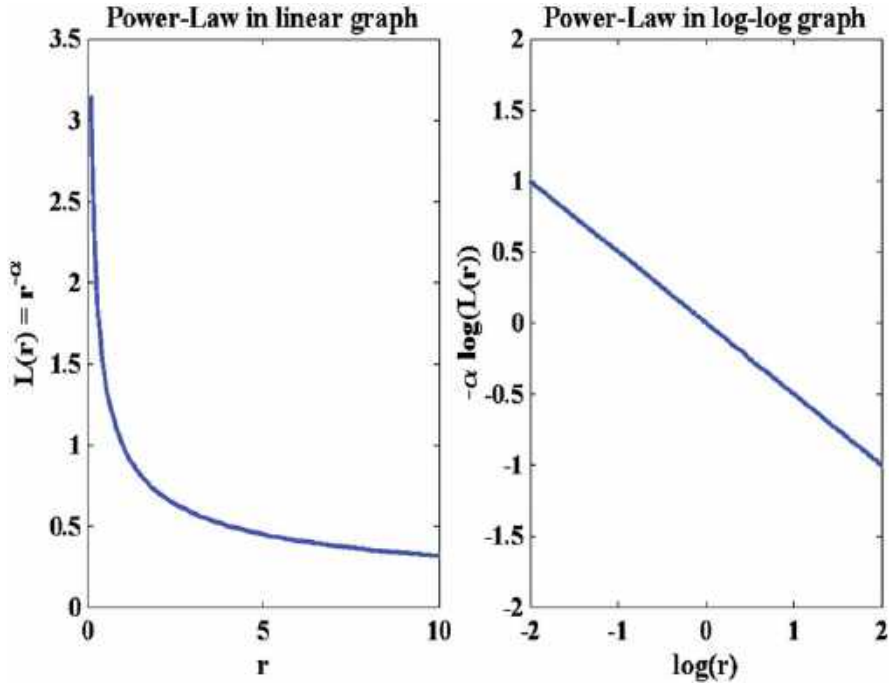


Fig. 5.7 Graphical representation of the relationship between the property $L(r)$ and the scale r (left panel) and the logarithm of both measures (right panel). The relationship is represented as a straight line in the log-log graph (right panel)

1, where values $0.0 < H < 0.5$ indicate that the data contain alternating high and low values over a majority of the time series and values $0.5 < H < 1.0$ reflect that the data within the time series contain autocorrelation. A H value of 0.5 means that the data may start correlated but will decay rapidly over the time series to zero. This corresponds to a geometrical random walk and is characteristic of completely random processes, such as Brownian motion. In other words, values of H close to 1 describe smoother signals, while values approaching 0 describe more jagged-appearing signals (i.e., high and low) [44]. Importantly, this method implies assumptions about the time series' stationarity.

The Hurst exponent and the FD are related to each other through a simple relationship:

$$D = 2 - H \quad (5.8)$$

In simple terms, the relation between H and FD can help characterize the level of randomness of the data within a time series, which is very useful in neurosciences.

Several methods have been developed to calculate H and will be discussed in the following sections.

5.4 Fractal Analysis of Time Series

The main concepts described for fractal geometry can be translated into the time domain, thus defining fractal time series. The fractal analysis of time series aims to identify the presence of one or more of the following properties: self-similarity, scale invariance, and power-law scale.

An important feature of the time series is their stationarity. A time series is defined as stationary if its probability distribution does not change over time, displaying time-independent descriptive statistical measures, such as mean and variance (Fig. 5.8). Moreover, stationary time series also display invariant auto-correlation functions under time translation. Different methods applied to extract information from time series can assume or not signal stationarity. Despite their frequent non-stationarity, neurophysiological signals are generally analyzed with methods assuming stationarity, such as FT.

Behind the complex and apparently random fluctuations of neurophysiological signals, a time-invariant mechanism can be hidden, which can be described by fractal analysis over time. The analysis of fractal time series, therefore, could be important to understand the underlying neurophysiological processes overlooked by traditional and linear methods.

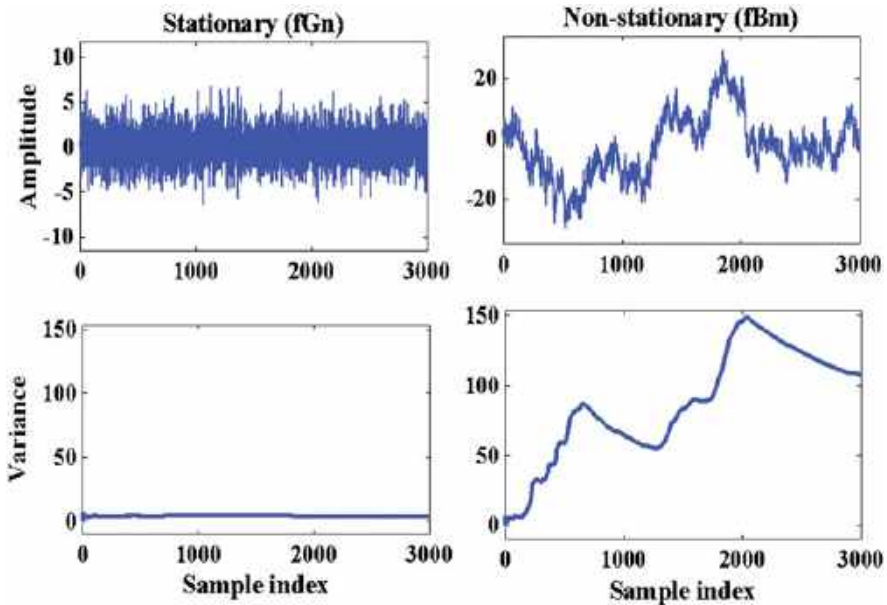


Fig. 5.8 Stationary and nonstationary stochastic time series. On the left side, a stationary signal fractional Gaussian noise (fGn) displaying constant variance over time (left lower panel). On the right side, a nonstationary signal fractional Brownian motion (fBm), obtained from the cumulatively summed series of fGn. Unlike the stationary signal, the variance of the nonstationary signal varies with time (right lower panel)

In the time domain, an FD represents the amplitude of the signal on a plane as a function of time. Because of this planar geometry, FD values range between 1 and 2, since a line has a Euclidean dimension equal to 1, while a plane has a Euclidean dimension equal to 2. FD equals or tends to 2 in time series characterized by high irregularity and no memory, such as a Brownian motion, while it equals or tends to 1 in very regular time series such as a ramp line, which is characterized by full memory (i.e., the overall signal can be predicted using a portion of the signal itself).

5.4.1 Overview of Methods to Generate and Estimate FDs in Synthetic Time Series

Many methods for the estimation of FDs in the time domain have been proposed for the analysis of neurophysiological signals, including: (1) slope of power spectral density (sPSD) [20, 50], (2) detrended fluctuation analysis (DFA) [5, 26], (3) generalized Hurst exponent (GHE) [7, 16, 68], (4) Higuchi's fractal dimension (HFD) [27], (5) Katz's fractal dimension (KFD) [31], and (6) box-counting (BC) method [45]. All these algorithms will be described in more detail in Sect. 5.4.3.

Although many comparative studies reported a greater accuracy in FD estimation with Higuchi's method [2, 51, 63], some others suggested that different methods may have different performances depending on the characteristics of the analyzed signals [19]. However, all the reported methods have been applied for the estimation of FDs, especially for EEG recordings [35] and functional magnetic resonance imaging (fMRI) data [16, 23, 57, 64, 73].

However, insufficient attention has been paid to the possible influence of several signal parameters on FD estimation, particularly: (i) sampling frequency (f_s), (ii) signal amplitude (Amp), and (iii) noise level. All are fundamental parameters of neurophysiological signals.

With respect to point (i), neurophysiological signals can be recorded using a broad spectrum of f_s , with higher f_s , leading to an increase in the temporal resolution of the recordings. For example, EEG signals can be acquired with sampling frequencies ranging from 100 Hz to 4096 Hz. Therefore, it is essential to understand whether the FD estimates are affected by changes in this parameter.

Here, we aim to characterize the performances of the different available methods in estimating the properties of the fractal time series and to describe their sensitivity to different signal parameters. To do so, we will first apply these methods on synthetic time series (STS) with a known FD (FD_{th}), created using well-known functions, namely, (1) Weierstrass cosine function (WCF), (2) Takagi–Landsberg function (TLF), (3) fractional Brownian motion (fBm), and (4) fractional Gaussian noise (fGn). By using these functions, STS with different sampling frequencies (f_s), amplitude (Amp), and noise will be generated. Then, the performances of several algorithms will be compared in terms of accuracy in estimating the FD (FD_e) and sensitivity to the amplitude, f_s , and noise of the STS.

We believe that this is a crucial preliminary step to define the main advantages and drawbacks of every method and to indicate the more appropriate tests for studying signals with different characteristics.

5.4.2 Generating Synthetic Fractal Time Series

Fractal time series can be generated using several mathematical functions and setting FD as a parameter of the function. These synthetic signals with a known FD can be used to test the performance of different methods in the estimation of FD. In this work, we used two deterministic functions (WCF and TLF) and two stochastic functions (fGn and fBm) to generate STS with different FD (Fig. 5.9). In the following sections, these functions will be briefly described.

5.4.2.1 Weierstrass Cosine Function (WCF)

The WCF is a deterministic, continuous but nowhere differentiable function and depends on four parameters, namely, N , λ , m , and H . N is the number of

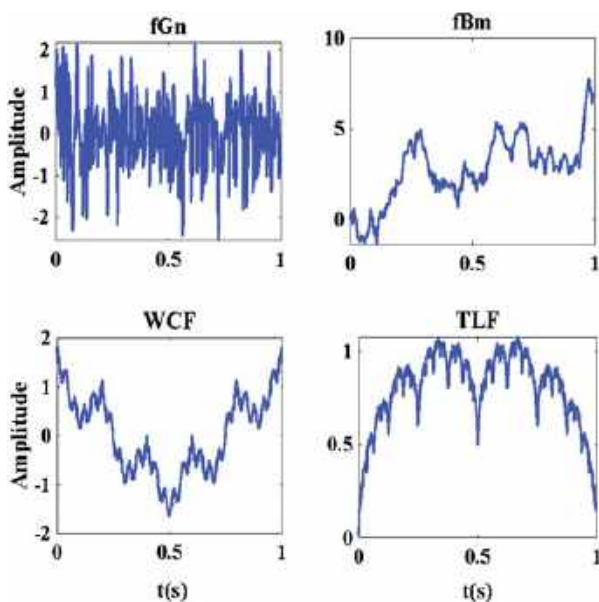


Fig. 5.9 Synthetic time series with $FD_{th} = 1.5$ and $f_s = 256$ Hz. Upper panels: fractional Gaussian noise (fGn) and fractional Brownian motion (fBm). Lower panels: Weierstrass cosine function (WCF) and Takagi-Landsberg function (TLF). All functions are displayed in a time interval t of 1 s (x -axis)

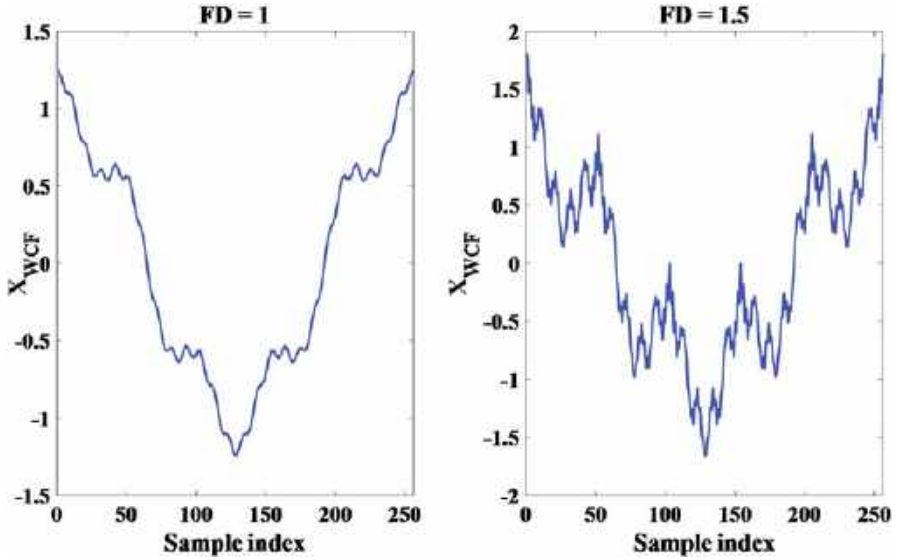


Fig. 5.10 Graphical representation of WCF with the same amplitude and sample index but different FDs: FD = 1 (on the left) and FD = 1.5 (on the right)

samples of the time series, H is the Hurst exponent (see Sect. 5.3.4), and λ is an integer. The function is periodic with period 1. Accordingly, for each point (i), $i \in (0, 1, \dots, N - 1)$, the sample $X_{WCF}(i)$ is obtained as follows:

$$X_{WCF}(i) = \sum_{m=0}^{\infty} \lambda^{-(mH)} \cos(2\pi \lambda^m t(i)), 0 < H < 1, \lambda > 1 \quad (5.9)$$

To generate time series with a duration of 1 s and FDs equal to 1 and 1.5, H is set to 1 and 0.5 (as explained in Sect. 5.4.2), respectively. Another input parameter in the WCF is λ , which is set to the fixed value of 5. Finally, the sum is calculated for a finite duration of $m = 0-26$ (Fig. 5.10).

5.4.2.2 Takagi–Landsberg Function

Following Weierstrass' discovery of an everywhere non-differentiable function published in 1875, another example of such a function was introduced by Teiji Takagi in 1903 and later extensively revised [9]. The TLF is a curve constructed by positive midpoint displacements of segments of a straight line. It is defined in the unit interval by:

$$X_{TLF}(i) = \sum_{n=0}^{\infty} a^n \Delta(2^n i) \quad (5.10)$$

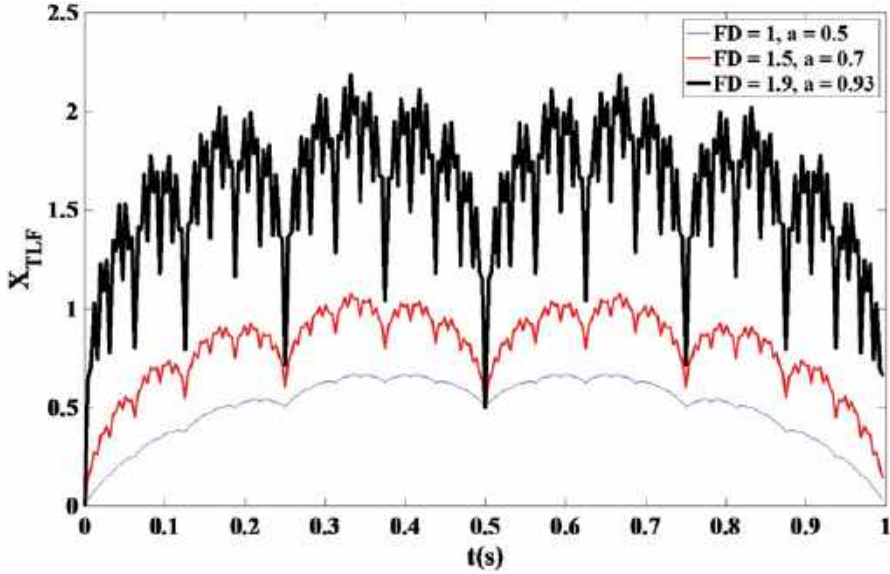


Fig. 5.11 TLFs with different values of FDs and of a , sampled at the same frequency ($f_s = 256$ Hz) in a time interval of 1 s

where a is a real number $\in [0,1]$ and $\Delta(i)$ is the distance of i to the nearest integer such that $\Delta(i) = \text{dist}(i, \mathbb{Z})$. The graph in Fig. 5.11 shows that the function is centered at $t = 1/2$. Importantly, TLF displays self-affinity in the interval $0 < t < 1$ [33].

An immediate generalization of the Takagi function is to set for integer $b \geq 2$ [55]:

$$X(i) = \sum_{n=0}^{\infty} a^n \Delta(b^n i) \quad (5.11)$$

The function is everywhere continuous but nowhere differentiable if $ab \geq 1$ [17]. According to previous works [63], by setting $b = 2$, the FD of TLF will be:

$$FD_{\text{TLF}} = \frac{\log(4a)}{\log(2)} \quad (5.12)$$

Following the relationship in Eq. (5.8) between the Hurst exponent (H) and FD, the Hurst exponent for TLF (H_{TLF}) is:

$$H_{\text{TLF}} = -\log_2^a \quad (5.13)$$

As previously done in Sect. 5.4.2.1 for WCF, we want to generate TLFs with FDs of 1 and 1.5 and a f_s of 256 Hz. To this aim, the parameters a and i must be set in

the TLF Eq. (5.11). The parameter i refers to the distance between two successive points in the interval $[0, 1]$ and is defined by $\frac{1}{f_i}$.

Regarding parameter a , the steps are as follows:

$$FD_{\text{TLF}} = 1$$

$$H_{\text{TLF}} = 2 - FD_{\text{TLF}} = 1$$

$$a = 2^{-H_{\text{TLF}}} = 2^{-1} = 0.5$$

5.4.2.3 Fractional Brownian Motion (fBm)

Fractional Brownian motion (fBm) was first introduced by Mandelbrot and van Ness (1968) [42] to describe a group of nonstationary stochastic processes where successive increments are stationary and correlated [11]. fBm can be considered a generalization of the standard Brownian motion (Bm), a stochastic process that can be represented as the random movement of a single particle. However, in standard Bm, successive increments in position are not correlated. In fBm, the correlation between consecutive points of the series exists and defines its persistency. In a persistent series, there is a positive correlation between points and an increasing trend in the past will likely be followed by an increasing trend in the future. Conversely, series displaying negative correlations are defined as anti-persistent and show a decreasing trend. Accordingly, fBm expresses non-stationarity processes, and, thus, its variance (as well as other statistical properties) is time-dependent [10] (Fig. 5.8). Specifically, the variance of fBm follows the rule:

$$\text{Var}(\Delta x^2) \propto \Delta t^{2H} \quad (5.14)$$

Meaning that the variance of a displacement Δx is a power function of the time interval Δt where it is observed. The scaling factor H is the Hurst exponent (see Sect. 5.3.4), a real number in the range $0 < H < 1$. By definition, the H of a standard Bm equals 0.5, while in fBm, $H < 0.5$ corresponds to persistent series, while $H > 0.5$ characterizes anti-persistent series.

fBm displays self-similarity and long-range dependence and has been used as a function to model many stochastic processes. Although fBm shows nonstationary features, the successive increments are stationary.

The fBm time series were generated according to the wavelet-based synthesis method proposed by Abry and Sellan [1]. In the dedicated MATLAB toolbox, four input parameters can be set to generate fBm: N , H , σ , and μ . N is the length of the output sequence, H is the Hurst exponent, σ is the standard deviation of the noise, and μ is the mean of the noise (set to 0) (Fig. 5.12).

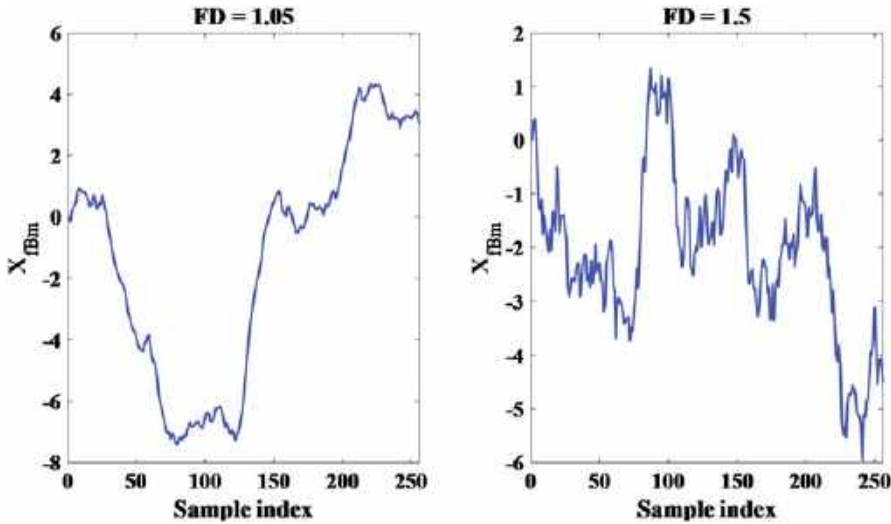
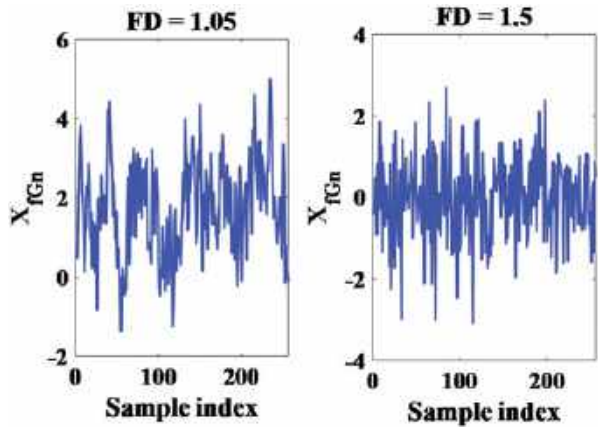


Fig. 5.12 Fractal Brownian motion signal with different FD values (FD = 1.05 on the left and FD = 1.5 on the right) and the same sampling frequency $f_s = 256$ Hz in a time interval of 1 s

Fig. 5.13 fGn signal with different FD values (FD = 1.05 on the left, FD = 1.5 on the right) and the same $f_s = 256$ Hz in a time interval of 1 s



5.4.2.4 Fractional Gaussian Noise

Fractional Gaussian noise (fGn) represents a group of self-similar stationary functions [18], thus displaying constant mean and variance over time (Fig. 5.8).

fGn and fBm are related to each other. Notably, each fBm corresponds to a specific fGn, and both are characterized by the same H . fGn can be obtained by differentiating the corresponding fBm [11], i.e., by considering the differences between the neighboring elements of a fBm. Conversely, the integration of an fGn corresponds to fBm, i.e., fBm can be obtained by summing an fGn (Fig. 5.13) [10, 18].

5.4.3 Methods to Estimate Fractal Dimensions

In the following sections, we will briefly introduce different algorithms for the estimation of fractal analysis of time series.

5.4.3.1 Slope of Power Spectral Density

An important class of signals with scale-free dynamics have autocorrelation or power spectral density (PSD) that follows a power law. A power-law process has a PSD of the form $C|\omega|^{-\alpha}$ for some positive constant C and some exponent α . In some instances, the signal of interest exhibits a power-law PSD. In other cases, the signal of interest is corrupted by noise with a power-law PSD. These noises are often referred to as colored (e.g., white or pink noises).

Keeping this in mind, the fast Fourier transform (FFT) of a signal leads to a power-law relationship between the power spectrum and the frequency:

$$|A(f)|^2 \propto cf^{-\beta} \quad (5.15)$$

where c is a constant, A is the amplitude of the FFT at a given frequency f , and β is the spectral index. In a log–log plot, this relationship can be presented as a straight line with slope $-\beta$.

The index β has the following relationship with FD [18]:

$$FD = \frac{5 - \beta}{2} \quad (5.16)$$

5.4.3.2 Detrended Fluctuation Analysis

This method, introduced by Peng et al. (1994) [54], was developed to improve the analysis of nonstationary time series characterized by long-range temporal correlations. DFA can be used to evaluate the self-affinity of time series, by calculating a scaling exponent α , which is related to FD as follows:

$$D = 3 - \alpha \quad (5.17)$$

For a given signal y , DFA can be performed by defining the suite x_n of the cumulative series of original signal y_i fluctuations about its mean:

$$x_n = \sum_{i=1}^n (y_i - \bar{y})$$

And then divide x_n into boxes of equal length τ . In each box, fit the local trend of x_n by a polynomial $P(\tau, n)$ that represents the local trend of the box. For the given τ

box size, compute the root-mean-squared detrended fluctuation of the signal x_n as:

$$F(\tau, n) = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - P(\tau, n))^2}$$

The last step is repeated for each of the available τ scales (box size) to obtain the empirical relationship between the overall fluctuation $F(\tau, n)$ and the box size τ :

$$F(\tau, n) \propto \tau^\alpha$$

The scaling exponent α quantifies the empirical strength of the long-range power-law correlations of the signal. Long-range correlation features can be found when the signal is not random.

DFA has been largely applied to EEG and MEG signals [47] to study both physiological processes (i.e., perception and sensorimotor transformations) and characterize many neuropsychiatric conditions, such as depression, Alzheimer's disease (AD), and spastic diplegia [26, 35].

5.4.3.3 Generalized Hurst and Range-Scale-Based Exponent

Several methods have been developed to calculate the Hurst exponent (H). One of the most frequently used methods is the generalized Hurst exponent (GHE), first introduced by Barabási and Vicsek [3]. GHE is a generalization of the classical method provided by Hurst [29] and is related to the scaling behavior of some statistical properties of time series. Given a time series $X(t)$ of length T , the scaling is characterized based on the statistic $K_q(\tau)$, defined as:

$$K_q(\tau) = \frac{\langle |X(t + \tau) - X(t)|^q \rangle}{\langle |X(t)|^q \rangle} \quad (5.18)$$

The statistic scales are defined as:

$$K_q(\tau) \approx c\tau^{qH}$$

The evaluation of Hurst exponent H following the methodology of Mandelbrot and Wallis [43] (called range-scale (R/S) analysis) uses the range of the partial sums of deviations of the signal y from its mean, rescaled by its standard deviation. The R/S statistic is given by:

$$(R/S)_n = \frac{1}{\sigma_n} \left[\max_{1 \leq k \leq n} \sum_{j=1}^k (y_j - \bar{y}_n) - \min_{1 \leq k \leq n} \sum_{j=1}^k (y_k - \bar{y}_n) \right]$$

where n is the length of the signal, $\bar{y}$ is its mean, and σ is its standard deviation given by:

$$\sigma_n = \left[\frac{1}{n} \sum_{j=1}^n (y_j - \bar{y}_n)^2 \right]^{1/2}$$

In this work, to estimate H , we used the GHE by applying the Eq. (5.18).

5.4.3.4 Higuchi's Fractal Dimension

Higuchi's method (HFD), first proposed by Tomoyoki Higuchi (1990) [28], is an iterative method designed to analyze fractal properties in the time domain.

In HFD, in order to describe and quantify the presence of similar features at different time scales, the original signal $X(t)$ with N total samples is downsampled every k samples. Therefore, a new self-similar time series X_m^k can be calculated as:

$$X_m^k : x(m), x(m+k), x(m+2k), \dots, x\left(m + \text{int}\left(\frac{N-m}{k}\right)k\right) \\ \text{for } m = 1, 2, \dots, k$$

where m is the first sample (initial time); k is the time interval, ranging from 1 to $k_{\max}$; $k_{\max}$ is a free parameter; and $\text{int}(r)$ is the integer part of the number r .

The length $L_m(k)$ of each curve X_m^k is calculated as:

$$L_m(k) = \frac{1}{k} \left[\sum_{i=1, \text{int}\left(\frac{N-m}{k}\right)} |X(m+ik) - X(m+(i-1)k)| \cdot \frac{N-1}{k \text{int}\left(\frac{N-m}{k}\right)} \right]$$

where N is the length of the original time series $X(t)$ and $(N-1)/\{\text{int}[(N-m)/k]\}$ is a normalization factor. $L_m(k)$ is then averaged for all m , forming the mean value of the curve length $L(k)$ for each $k = 1, 2, \dots, k_{\max}$ as:

$$L(k) = \frac{\sum_{m=1}^k L_m(k)}{k}$$

An array of mean values $L(k)$ is obtained, and the FD (D_H) can be estimated as:

$$D_H = \frac{\log(L(k))}{\log\left(\frac{1}{k}\right)} \text{ for } k = 1, 2, \dots, k_{\max}$$

The choice of the free parameter k is crucial in FD estimation [57]. Conversely, window length and overlapping windows, do not affect FD estimation [32, 58].

5.4.3.5 Katz's Fractal Dimension

In Katz's method (KFD), the FD (D_k) of a time series is defined as:

$$D_k = \frac{\log\left(\frac{L}{a}\right)}{\log\left(\frac{d}{a}\right)} = \frac{\log(n)}{\log(n) + \log\left(\frac{d}{L}\right)} \quad (5.19)$$

where L is the length of the time series (i.e., the sum of the Euclidean distances between successive points of the sample), a is the average distance between successive points, d is the planar extent of the time series (i.e., the maximum distance between the first point of the series and any other point), and $n = L/a$. [31].

Previous works comparing HFD and KFD performances described a lower accuracy of KFD, in terms of underestimation of the FD of a system [19, 70]. However, the observed differences were higher for longer recording lengths, while the accuracy was similar for short recordings [19]. However, some studies also suggested that KFD is less sensitive to noise and can better discriminate changes in FD (e.g., those observed between different brain states) [19, 70].

5.4.3.6 Box-Counting Method

The box-counting (BC) method has been traditionally applied to describe fractal structures in the space domain, with the aim of describing how objects fill the space [45]. Despite being less suited for the analysis of fractal characteristics of time series, BC has been applied to estimate their FDs (covering dimension or box-counting dimension). The BC dimension is generally deduced by means of the Minkowski–Bouligand method, and it is therefore also called the Minkowski–Bouligand dimension.

The method consists of covering the waveform with small square boxes of size $\varepsilon \times \varepsilon$ and counting the number $N(\varepsilon)$ of boxes required to cover the waveform. Then, the covering procedure is repeated using boxes of different sizes and the value of $N(\varepsilon)$ is determined again. The box-counting dimension D_B is defined as:

$$D_B = \frac{\log N(\varepsilon)}{\log \frac{1}{\varepsilon}} \quad (5.20)$$

When plotting $\log N(\varepsilon)$ against $\log 1/\varepsilon$, a straight line will be suggestive of self-similarity, and FD_B represents the slope of the regression line. In natural objects (which are not self-similar over an infinite range of magnitudes), the BC requires a fractal (scaling) window, defined by a superior limit (ε_{max} , also known as Kolmogorov dimension or, more confusedly, entropy dimension) and a lower dimension (ε_{min} , also known as Minkowski dimension). A restricted fractal window defines the local box-counting dimension D_B , which is pragmatically useful to quantify the space-filling properties of objects embedded in their two- and/or three-

dimensional space (e.g., the microvasculature of a brain tumor in a bidimensional histological slide, quantified by means of the local microvascular fractal dimension, introduced by Di Ieva et al. [13]. As mathematical self-similarity is not a typical property of natural objects, biological objects and generally defined *random fractals* retain their fractal self-similarity in a defined scaling window of at least two orders of magnitude [37, 38, 52, 66]. The fractal window should always be reported when fractal analysis is used to quantify natural objects and time series.

Many methods have been proposed to overcome some of the pitfalls of BC applied in time series, such as the multiresolution BC [62], and other modified versions [6].

5.4.4 Results and Discussion on the Methods of Estimating Fractal Dimensions on Synthetic Time Series

In this section, we report the results of the application of six different methods (sPSD, DFA, GHE, HFD, KFD, and BC) on four STS with known FDs (WCF, TLF, fBm, and fGn). We first tested the different algorithms on all STS with different FDs, to evaluate their performances on each function. Then, to test the sensitivity of each method to different signal parameters, we generated STS with different sampling frequencies (f_s), signal amplitudes, and noise levels. All the tested STS were generated in a time interval of 1 s.

5.4.4.1 Performances of Methods for Estimating Fractal Dimensions

To evaluate the performances of each method in estimating the FDs of different functions, we generated several STS with different FD_{th} ranging from 1.05 to 1.95 with steps of 0.1. For all the tested STS, the f_s was fixed at 256 Hz, a time interval of 1 s was considered, and no noise was added. Then, the FD_e provided by each method on individual functions was compared with the FD_{th} (Fig. 5.14), and the corresponding mean squared errors (MSEs) were calculated (Table 5.1).

The FD_{th} of fGn tested with DFA and sPSD was calculated using different approaches with respect to the other methods. For DFA, after determining the parameter α (corresponding to the slope of the signal in a log–log space), the FD_{th} of fGn was calculated using $D = 2 - \alpha$. Conversely, for the other functions tested with DFA, the Eq. (5.17) ($D = 3 - \alpha$) was used.

For sPSD, the index β was first determined as the slope of the PSD in the log–log space, and, then, the FD of fGn was calculated using the equation $D = \frac{3-\beta}{2}$. For the other STS, the relationship of Eq. (5.16) ($D = \frac{5-\beta}{2}$) was applied in FD estimation.

Our results indicate that DFA and PSD have the highest accuracy in FD estimation on fGn signals, while they show poor performances when applied to TLF. In particular, sPSD provides an underestimation of FD values of TLF, while DFA

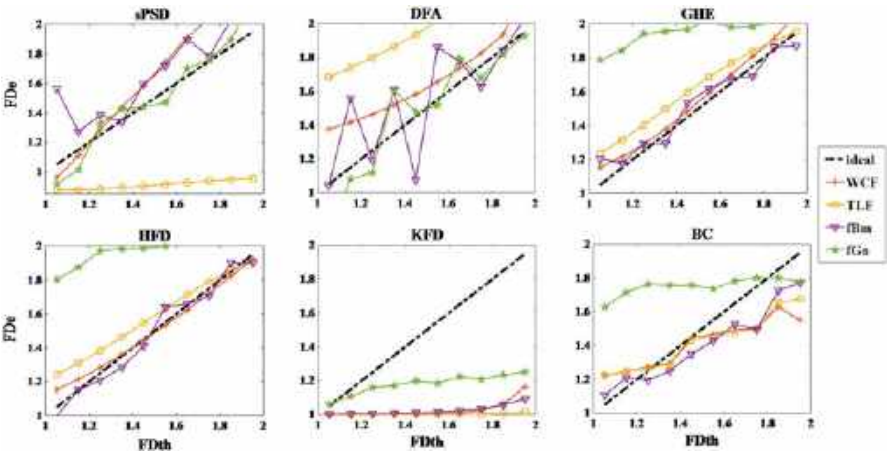


Fig. 5.14 Comparison between methods for estimating the FDs of different STS. The black dashed lines represent the ideal performance of each method, with the estimated FD (FD_e) corresponding to the FD (FD_{th}) set in each STS. FD_e values above 2 are not displayed

Table 5.1 Mean squared errors (MSEs) between the theoretical FD (FD_{th}) and the estimated FD (FD_e) for all the methods tested (DFA, sPSD, GHE, HFD, KFD, and BC) on the STS (WCF, TLF, fBm, and fGn). MSE values below 0.15 are displayed in bold for clarity

	sPSD	DFA	GHE	HFD	KFD	BC
WCF	0.19	0.19	0.07	0.04	0.54	0.19
TLF	0.64	0.50	0.13	0.11	0.57	0.16
fBm	0.13	0.36	0.09	0.09	0.54	0.15
fGn	0.09	0.15	0.5	0.52	0.39	0.4

overestimates them. Furthermore, both methods show good performances when applied to WCF, and both methods do not appear less accurate for fBm than GHE and HFD (Fig. 5.14). Moreover, although the MSEs correspond to those by BC, the estimations appear more variable across different FDs. The MSEs of the distances between the FD_{th} and the FD_e are reported in Table 5.1.

Conversely, HFD and GHE show the highest accuracy in predicting the FDs of TLF, WCF, and fBm, with HFD providing a slightly better performance on WCF and TLF with respect to GHE. However, both methods also provide the poorest estimation of fGn, largely overestimating the FD values.

Similarly, BC provides an inaccurate estimation of FD when applied on fGn while providing more accurate results for the other STS (Table 5.1). However, despite a good performance of BC for lower FD values, at higher FD values, the method becomes less reliable (Fig. 5.14).

Among the six methods evaluated, KFD is the only one providing a poor performance in FD estimation for all the considered STS, by underestimating FD values of all the functions tested.

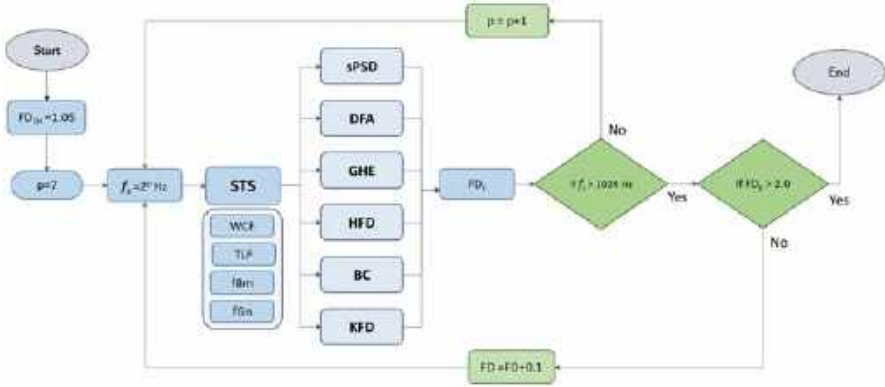


Fig. 5.15 Flowchart for testing the effect of f_s on FD estimation for different methods

Overall, our results suggest that HFD is the most accurate method for calculating the FDs of all the tested STS (WCF, TLF, and fBm), except for the stationary stochastic function (fGn), where DFA and sPSD provide the best performance in estimating FDs. These results are in line with previous works [19, 62]. GHE provides a good estimation of the FDs of WCF, TLF, and fBm, while it overestimates the FD of fGn.

5.4.4.2 Effect of Sampling Frequency

The effect of f_s on FD estimation was tested on all the algorithms (sPSD, DFA, GHE, HFD, KFD, and BC) by generating STS with f_s of 128 Hz, 256 Hz, 512 Hz, and 1024 Hz and different FDs ranging from 1.05 to 1.95 with 0.1 steps (Fig. 5.15). Regarding the two stochastic signals (fGn and fBm), the FD estimation was repeated for 10 iterations, and the estimated FD corresponds to the average over 10 iterations. All the STS were generated in the same time duration of 1 s and no noise was added to the signals.

sPSD does not show significant sensitivity to f_s when applied to stochastic signals (fBm and fGn). For all the tested frequencies, the prediction is less accurate for lower FD values for fBm (Fig. 5.16, left panels). When tested on TLF, the method shows a slight change in FD estimation with f_s , with better performances for lower frequencies (Fig. 5.16, Table 5.2). Although the performance of sPSD on WCF is poor at the f_s of 128 Hz, there is no clear effect of f_s .

Concerning the effect of f_s on DFA estimation accuracy, there is no clear effect for stochastic functions (fGn and fBm) (Fig. 5.16, right panels), while for both WCF and TLF, higher f_s are associated with a slight increase in the accuracy of FD prediction (Fig. 5.16 and Table 5.2).

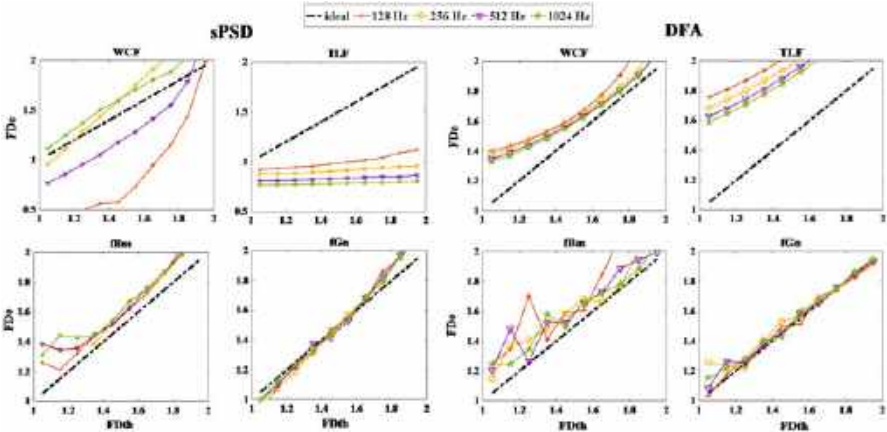


Fig. 5.16 Comparison of the sensitivity to f_s in FD estimation in four different STS for sPSD (left panels) and DFA (right panels). For sPSD applied to WCF and TLF, the lower y-axis limits were set at 0.5 for clarity. FD_e values above 2 are not displayed

Table 5.2 Mean squared errors (MSEs) in computing the fractal dimension (FD) at different sampling frequencies (f_s) for the slope of power spectral density (sPSD, on the left) and detrended fluctuation analysis (on the right). MSE values below 0.15 are displayed in bold for clarity

sPSD					DFA				
	128 Hz	256 Hz	512 Hz	1024 Hz		128 Hz	256 Hz	512 Hz	1024 Hz
WCF	0.68	0.19	0.26	0.14	WCF	0.24	0.19	0.15	0.14
TLF	0.55	0.64	0.72	0.77	TLF	0.57	0.5	0.45	0.41
fBm	0.13	0.14	0.16	0.16	fBm	0.17	0.16	0.17	0.14
fGn	0.08	0.07	0.08	0.08	fGn	0.06	0.05	0.05	0.06

Although sPSD and DFA provide the best FD estimate for fGn at all the considered f_s , the accuracy decreases at extreme FD values for sPSD and, to a lesser extent, at low FD values for DFA.

Both GHE and HFD are not sensitive to the f_s when estimating the FDs of WCF, fBm, and fGn (Fig. 5.17 and Table 5.3). A slight and progressive improvement of estimation accuracy was observed for both methods applied to TLF at higher f_s . Despite the low accuracy of the FD values for fGn, both GHE and HFD provide the best FD estimations for the other STS at all the f_s tested (Table 5.3). FD_e values above 2 are not displayed.

Our results suggest that KFD does not perform well in estimating the FDs of all the STS at all the tested f_s . However, among all the tested methods, the performance of KFD is less affected by f_s , with the exception of fGn, where lower f_s are associated with a higher accuracy in FD prediction.

Concerning BC, the resolution (i.e., the size of the boxes) was adapted to each f_s tested by calculating the maximum number of boxes covering the waveform. Figure

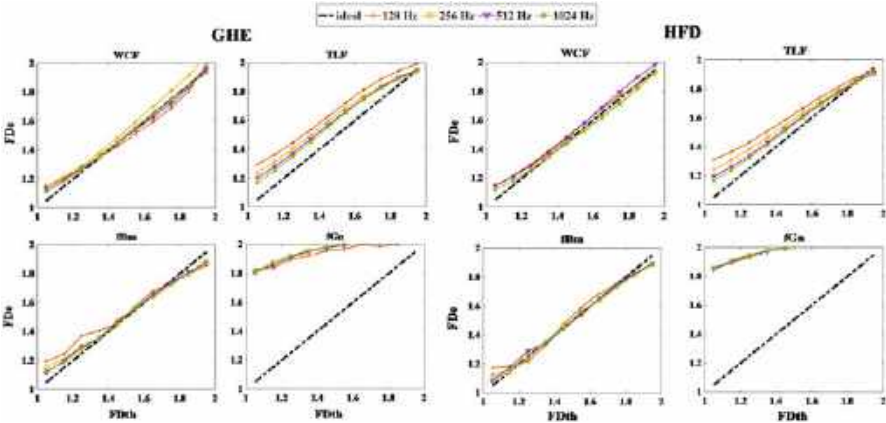


Fig. 5.17 Plots representing the estimated FD (FDe) versus the theoretical FD (FDth) of synthetic signals (WCF, TLF, fBm, and fGn) with different sampling frequencies (f_s) for generalized Hurst exponent (GHE, left panels) and Higuchi's method (HFD, right panels)

Table 5.3 Mean squared errors (MSEs) at different sampling frequencies (f_s) for the generalized Hurst exponent (GHE, left table) and Higuchi's method (HFD, right table)

GHE					HFD				
	128 Hz	256 Hz	512 Hz	1024 Hz		128 Hz	256 Hz	512 Hz	1024 Hz
WCF	0.05	0.07	0.03	0.02	WCF	0.05	0.04	0.05	0.03
TLF	0.17	0.13	0.11	0.09	TLF	0.15	0.11	0.08	0.06
fBm	0.07	0.05	0.05	0.04	fBm	0.03	0.03	0.03	0.03
fGn	0.5	0.51	0.52	0.52	fGn	0.54	0.54	0.53	0.54

5.18 shows that the method is sensitive to f_s in FD estimation for the STS, achieving the best prediction accuracy for intermediate FD values (Table 5.4).

Overall, our results show that HFD and GHE are the methods that provide the best accuracy for FD estimation on WCF, TLF, and fBm at all the tested f_s and that the performance is not markedly affected by f_s . Concerning the FD prediction for fGn, sPSD and DFA are the most suited methods at all the considered f_s and the FD estimation of these algorithms is not affected by f_s . Despite providing the worst FD estimation for WCF, TLF, and fBm, KFD is the method showing the lowest sensitivity to f_s in the prediction of FD in these functions.

5.4.4.3 Effect of Signal Amplitude

To test the effect of waveform amplitude on the FD estimation provided by different methods, the amplitude of each STS was increased by changing the root mean

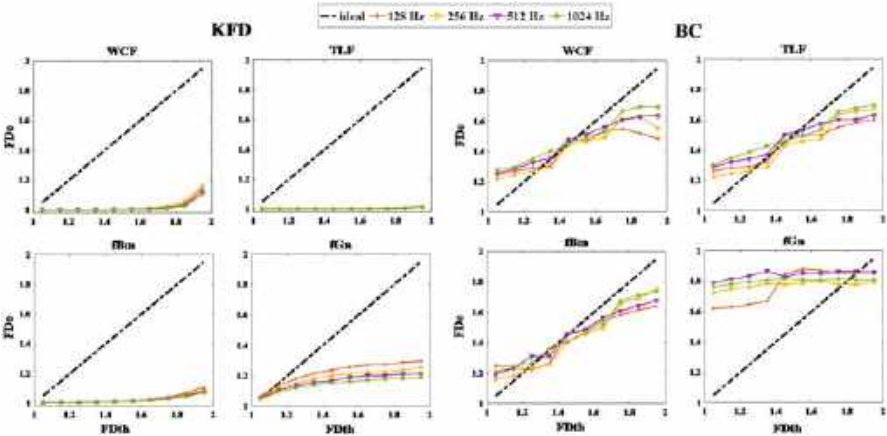


Fig. 5.18 Plots representing the estimated FD (FDe) versus the theoretical FD (FDth) of four synthetic signals (STS) with different sampling frequencies (f_s) for the box-counting method (BC, left panels) and Katz’s method (KFD, right panels)

Table 5.4 Mean squared errors (MSEs) at different sampling frequencies (f_s) for Katz’s method (KFD, left table) and box-counting method (BC, right table)

KFD					BC				
	128 Hz	256 Hz	512 Hz	1024 Hz		128 Hz	256 Hz	512 Hz	1024 Hz
WCF	0.54	0.54	0.55	0.55	WCF	0.21	0.17	0.16	0.14
TLF	0.57	0.57	0.57	0.57	TLF	0.18	0.14	0.17	0.16
fBm	0.54	0.54	0.55	0.55	fBm	0.16	0.12	0.13	0.12
fGn	0.3	0.39	0.41	0.43	fGn	0.34	0.39	0.43	0.41

square (RMS) over all the waveform samples, and the FD was then computed. Five different signal amplitudes were considered, corresponding to different added RMSs: 1, 5, 10, 20, 40, and 60.

Figure 5.19 shows that FD estimations provided by sPSD, DFA, GHE, HFD, and BC are not affected by signal amplitude, and the curves representing the performances of each method are perfectly superimposed. Therefore, changing the amplitude of the signal does not affect the accuracy of these methods in calculating the FD.

As previously demonstrated by Raghavendra and Dutt [63], our results show that the FD estimation provided by the KFD is affected by the amplitude of the signal. In particular, an increase in signal amplitude is associated with an increase of the predicted FD values. Overall, this leads to an improvement of the KFD performance only at lower and intermediate FD values (Table 5.5 and Fig. 5.20).

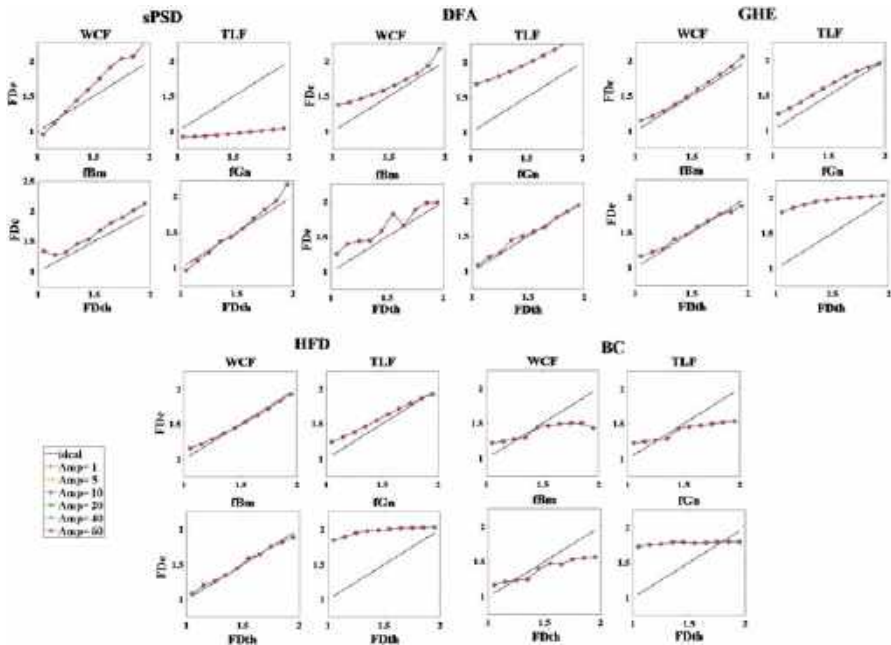


Fig. 5.19 Plots of the FD estimated by different methods (sPSD, DFA, GHE, HFD, and BC) versus the theoretical FD (FDth) of four synthetic signals (STs) with different amplitudes (1, 5, 10, 20, 40, 60)

Table 5.5 Mean squared errors (MSEs) between the performance of Katz’s method (KFD) on the STS of different amplitudes and the ideal performance (FDe corresponding to FDth)

	KFD				
	5	10	20	40	60
WCF	0.42	0.29	0.12	0.23	0.32
TLF	0.39	0.23	0.08	0.34	0.45
fBm	0.44	0.32	0.15	0.24	0.44
fGn	0.22	0.30	0.74	1.5	2.2

5.4.4.4 Effect of the Noise Level

The effect of the noise level on the FD estimation was assessed in all methods by generating STS with different signal-to-noise ratios (SNRs), as in previous studies [19]. An SNR was defined as the difference between the signal amplitude in decibels (dB) and the amplitude of the noise (dB). Additive white Gaussian noise (AWGN) was selected as the noise to be added to the STS.

Therefore, when an SNR equals 0 dB, both signal and noise have the same amplitude, while positive SNR values indicate that signal is higher than noise and negative values reflect a higher noise amplitude with respect to the signal.

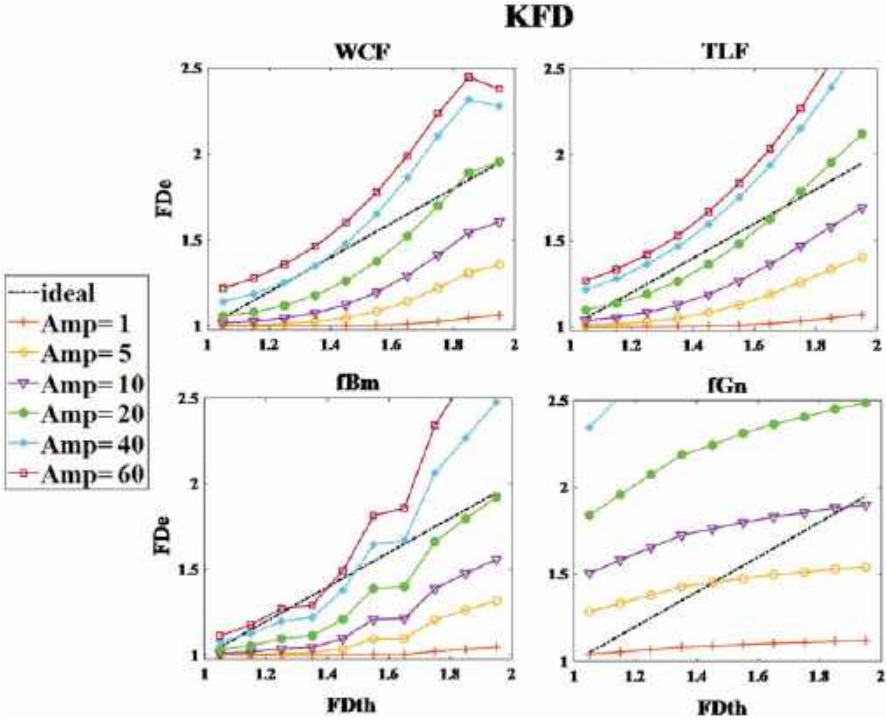


Fig. 5.20 Plots representing the FD estimated (FDe) for different STS of different amplitudes with Katz's method (KFD) versus the theoretical FD (FDth) of the signals

STS were generated setting five separate SNRs: 0 dB, 5 dB, 10 dB, 20 dB, and 40 dB. Then, their FDs were estimated using different methods. The performances of each method were evaluated by calculating the MSEs from the signal without noise (SWoN) and the ideal performance. For each method, the MSEs calculated between functions with different SNRs and the corresponding signals without noise provide information on the sensitivity of the method to noise level and allow to indicate putative thresholds of SNR where the noise can be considered to have a negligible effect on the FD estimation. On the other hand, the comparison with the ideal performances provides information regarding the accuracy of FD estimation at a particular SNR.

For all the considered methods applied on all STS, lower SNRs (corresponding to higher noise levels) were associated with higher values of the estimated FD. In fact, considering Fig. 5.21, we can observe how introducing noise determines an increase of the space filled by the curve, therefore resulting in higher FD values (i.e., closer to 2) with respect to the corresponding signals without noise.

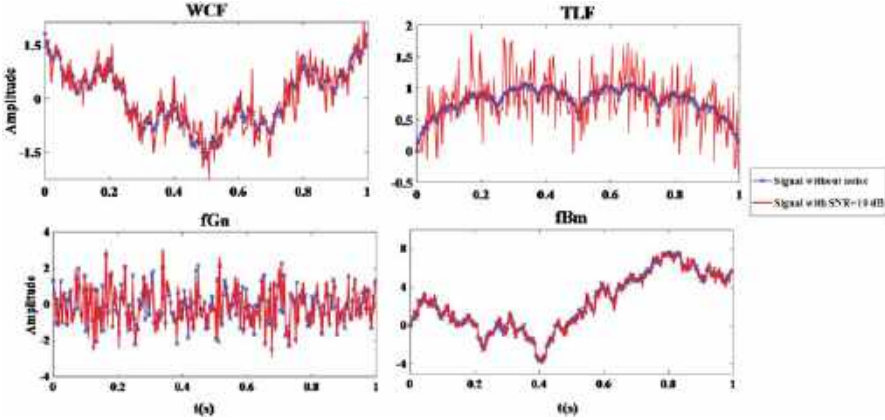


Fig. 5.21 STS with $\text{SNR} = 10$ dB, signal amplitude = 1 dB, $\text{FD}_{\text{th}} = 1.5$, and $f_s = 256$ Hz. The blue traces represent signals without noise, while the red traces correspond to signals where additional white Gaussian noise (AWGN) was added, setting an SNR at 10 dB. Upper panels: fractional Gaussian noise (fGn) and fractional Brownian motion (fBm). Lower panels: Weierstrass cosine function (WCF) and Takagi-Landsberg function (TLF). All signals are displayed in a time interval t of 1 s

Our results show that sPSD is markedly affected by the noise levels when estimating the FDs of WCF and TLF, as shown by the comparison of the MSEs between signals with different SNRs and the corresponding functions without noise (Fig. 5.22 and Table 5.6). Moreover, sPSD provides a poor estimation of FDs of these functions even for the signal displaying low noise, i.e., with an SNR of 40 dB.

Similarly, FD calculation for fBm by sPSD is affected by noise level, but the accuracy in the prediction increases for signals with lower noise levels (i.e., SNRs of 20 and 40 dB). Conversely, sPSD retains a good estimation of FDs for fGn at all the SNRs tested, although providing more variable estimations at extreme FD values and at lower SNRs. For WCF and TLF, the FD estimation is affected by noise, particularly at low FD values, but the effect of noise has lower relevance for $\text{SNRs} > 20$ dB.

DFA is affected by noise levels when applied to the stationary stochastic signal (fGn) only at higher FD values (Fig. 5.23 and Table 5.7). Regarding fBm, DFA is not markedly affected by noise levels, as shown by the low MSE values emerging from the comparison with the performances on the signal without noise. Moreover, when comparing, fBms display different SNRs with the ideal performance, and there are minor changes in the estimation, further suggesting that the method is little affected by the noise level for this signal.

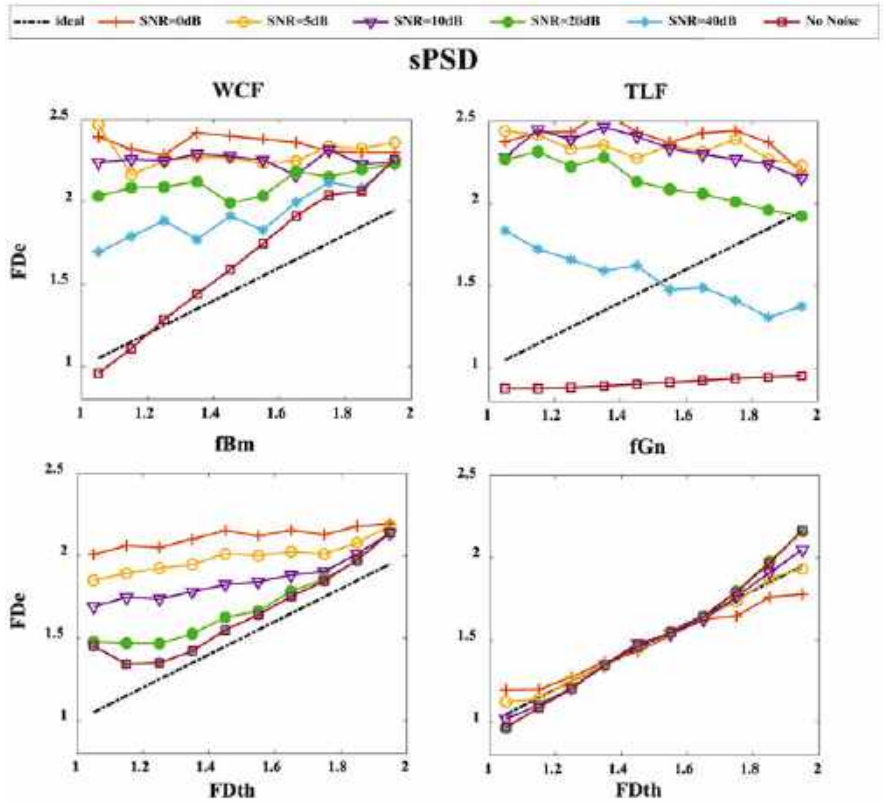


Fig. 5.22 FD estimated using sPSD (FDe) on different STS versus the theoretical FD (FDth) set for each STS with different signal-to-noise ratios (SNRs). The black dashed lines represent the ideal performances of each method, with the estimated FD (FDe) corresponding to the FD (FDth) of the STS. The magenta lines represent the performances of the sPSD applied on signals without noise (SWoN)

Table 5.6 Mean squared errors (MSEs) between the performance of sPSD on STS with different SNRs (0, 5, 10, 20, and 40 dB) and the performances on signals without noise (SWoN) and with the ideal performance (“ideal,” highlighted in gray). MSE values below 0.15 are displayed in bold for clarity

sPSD		0 dB	5 dB	10 dB	20 dB	40 dB
WCF	SWoN	0.83	0.77	0.74	0.6	0.4
	Ideal	0.9	0.84	0.8	0.66	0.46
TLF	SWoN	1.5	1.4	1.42	1.22	0.66
	Ideal	1	0.9	0.9	0.75	0.44
fBm	SWoN	0.52	0.4	0.25	0.07	0.001
	Ideal	0.66	0.53	0.4	0.22	0.17
fGn	SWoN	0.17	0.1	0.05	0.005	0.0007
	Ideal	0.09	0.03	0.04	0.09	0.09

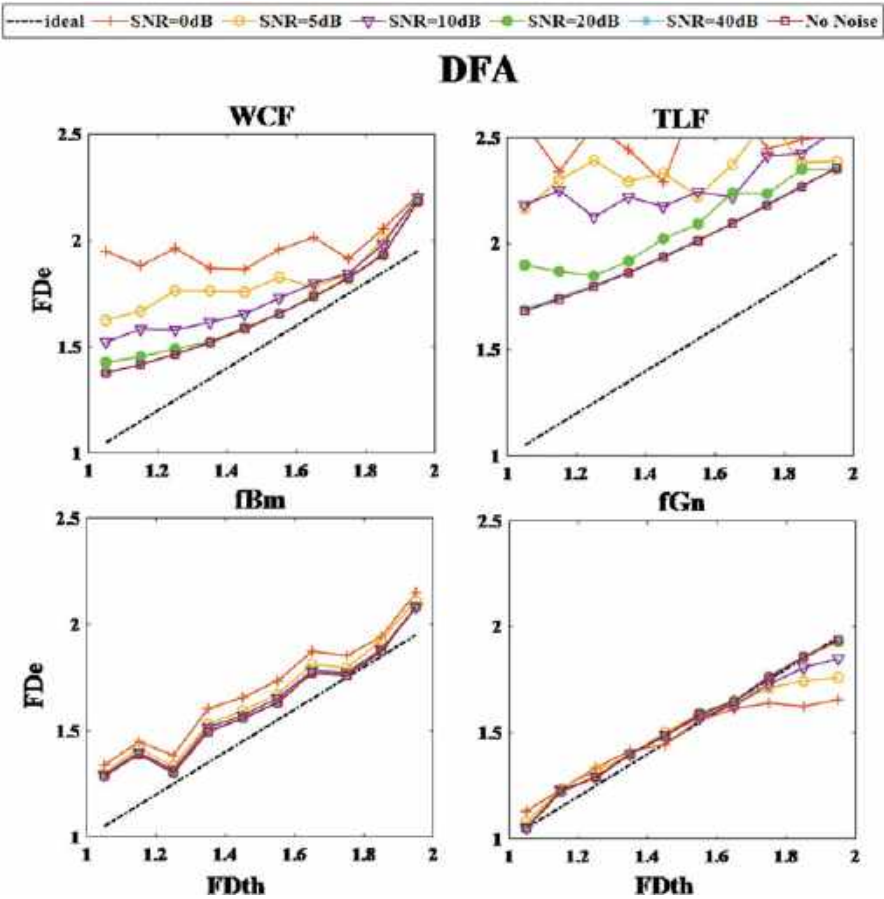


Fig. 5.23 As for Fig. 5.22, performances of detrended fluctuation analysis (DFA) applied on different STS with different SNRs

Table 5.7 Mean squared errors (MSEs) calculated between the performance of DFA on each STS and the performance of DFA on the corresponding signal without noise (SWoN) and with the ideal performance (“ideal”)

DFA		0 dB	5 dB	10 dB	20 dB	40 dB
WCF	SWoN	0.34	0.18	0.095	0.02	0.001
	Ideal	0.52	0.36	0.28	0.2	0.19
TLF	SWoN	0.56	0.4	0.3	0.1	0.006
	Ideal	1.05	0.88	0.8	0.6	0.5
fBm	SWoN	0.09	0.04	0.01	0.002	0.0001
	Ideal	0.2	0.16	0.15	0.14	0.14
fGn	SWoN	0.13	0.07	0.03	0.004	0.0004
	Ideal	0.13	0.08	0.05	0.04	0.04

Our results show that GHE and HFD are sensitive to noise levels when estimating the FDs of WCF, TLF, and fBm, achieving more accurate predictions only at high SNR (40 dB), showing extremely low MSEs with respect to the performances of the corresponding signal without noise. Conversely, both methods overestimate FD values of fGn at all the considered noise levels, and the SNR has a minimal influence on the estimation (especially at lower FDs) (Tables 5.8 and 5.9; Figs. 5.24 and 5.25).

Overall, despite its poor performance in estimating the FDs of all the tested STS, KFD is the method less affected by noise levels, followed by BC, in line with the results of the study by Esteller et al. [19] (Tables 5.10 and 5.11; Figs. 5.26 and 5.27).

Table 5.8 Mean squared errors (MSEs) calculated between the prediction of GHE on each STS and the corresponding signal without noise (SWoN) and with the ideal performance (“ideal”)

GHE		0 dB	5 dB	10 dB	20 dB	40 dB
WCF	SWoN	0.52	0.51	0.46	0.27	0.009
	Ideal	0.57	0.55	0.5	0.32	0.07
TLF	SWoN	0.44	0.42	0.44	0.39	0.08
	Ideal	0.57	0.55	0.56	0.51	0.2
fBm	SWoN	0.43	0.33	0.2	0.04	0.0008
	Ideal	0.47	0.38	0.25	0.09	0.05
fGn	SWoN	0.03	0.02	0.008	0.001	0.0001
	Ideal	0.53	0.52	0.51	0.51	0.51

Table 5.9 Mean squared errors (MSEs) calculated between the prediction of HFD on each STS and the corresponding signal without noise (SWoN) and with the ideal performance (“ideal”)

HFD		0 dB	5 dB	10 dB	20 dB	40 dB
WCF	SWoN	0.53	0.49	0.41	0.2	0.005
	Ideal	0.55	0.51	0.44	0.23	0.05
TLF	SWoN	0.47	0.46	0.45	0.37	0.05
	Ideal	0.6	0.6	0.6	0.5	0.15
fBm	SWoN	0.34	0.22	0.12	0.02	0.0003
	Ideal	0.35	0.23	0.13	0.04	0.03
fGn	SWoN	0.03	0.02	0.006	0.0009	0.00006
	Ideal	0.55	0.54	0.54	0.54	0.53

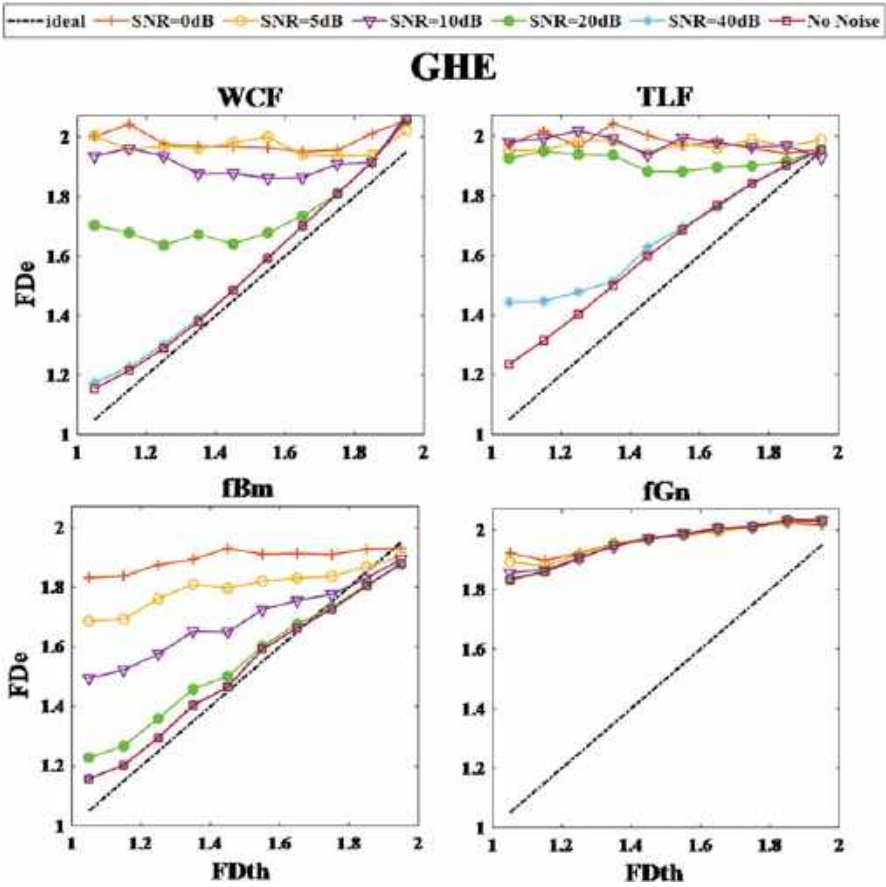


Fig. 5.24 As for Fig. 5.22, performances of the generalized Hurst exponent (GHE) in estimating the FDs of four separate STS with different SNRs

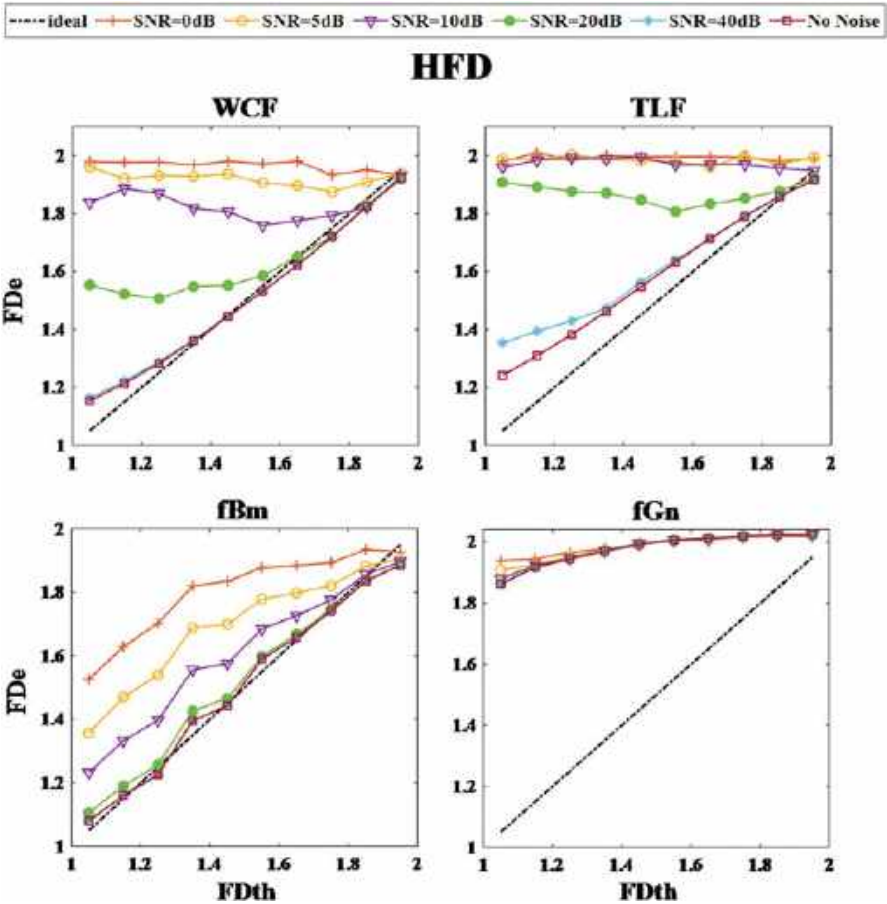


Fig. 5.25 Performances of Higuchi’s method (HFD) in FD estimation of STS with different SNRs

Table 5.10 Mean squared errors (MSEs) calculated between the FD estimated with Katz’s method (KFD) for each STS and the corresponding signal without noise (SWoN) and with the ideal performance (“ideal”)

KFD		0 dB	5 dB	10 dB	20 dB	40 dB
WCF	SWoN	0.1	0.05	0.02	0.002	0.00004
	Ideal	0.46	0.5	0.52	0.54	0.54
TLF	SWoN	0.1	0.05	0.02	0.002	0.00004
	Ideal	0.49	0.53	0.56	0.57	0.57
fBm	SWoN	0.09	0.04	0.02	0.002	0.00004
	Ideal	0.47	0.51	0.53	0.54	0.54
fGn	SWoN	0.04	0.02	0.005	0.0006	0.00003
	Ideal	0.37	0.38	0.38	0.39	0.39

Table 5.11 Mean squared errors (MSEs) calculated between the FD estimated for each STS with the DFA and the corresponding signal without noise (SWoN) and with the ideal performance (“ideal”)

BC		0 dB	5 dB	10 dB	20 dB	40 dB
WCF	SWoN	0.1	0.09	0.07	0.04	0.005
	Ideal	0.2	0.2	0.2	0.22	0.2
TLF	SWoN	0.12	0.11	0.1	0.08	0.01
	Ideal	0.19	0.19	0.19	0.19	0.19
fBm	SWoN	0.09	0.06	0.03	0.006	0.001
	Ideal	0.09	0.06	0.03	0.006	0.001
fGn	SWoN	0.01	0.008	0.007	0.006	0.003
	Ideal	0.39	0.38	0.38	0.38	0.38

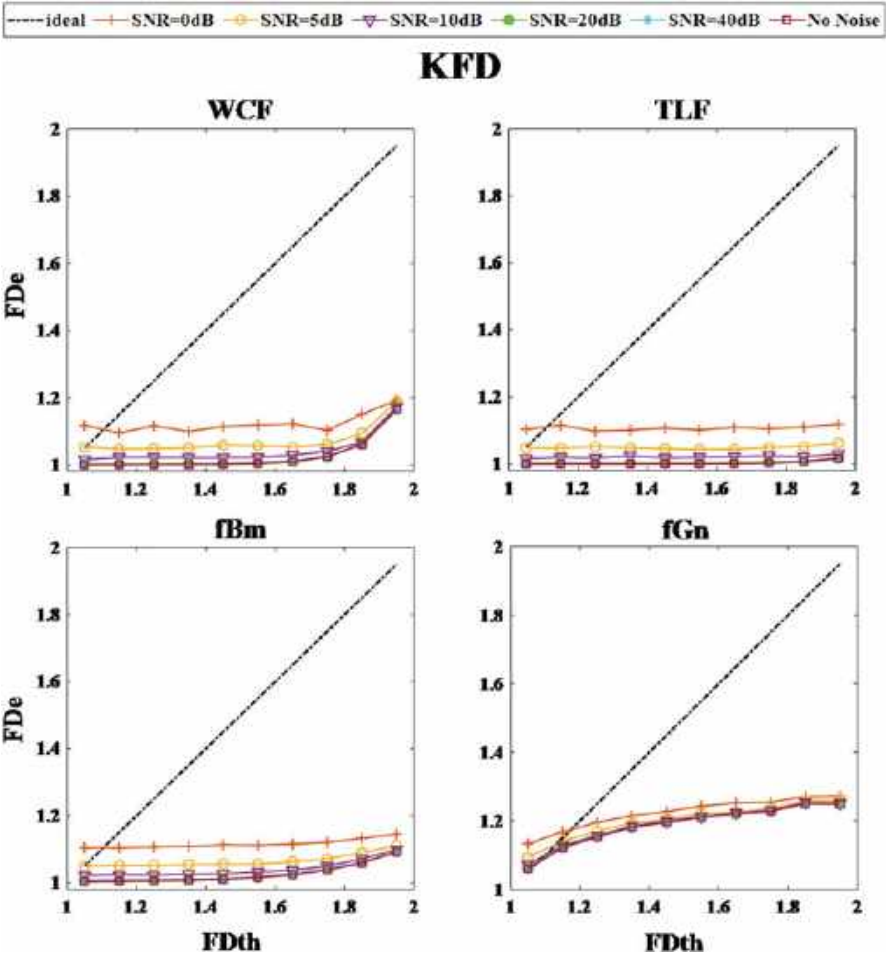


Fig. 5.26 Performances of Katz’s method (KFD) applied on different STS with different SNRs

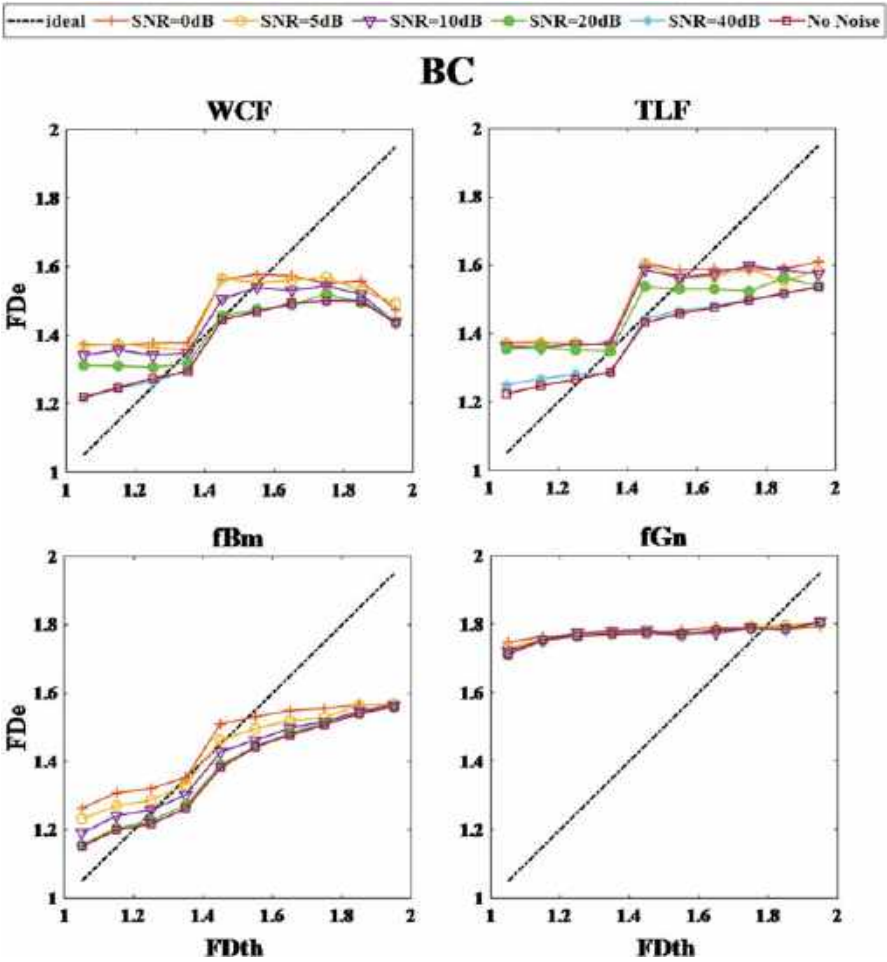


Fig. 5.27 Performances of the box-counting (BC) method applied on different STS with different SNRs

5.5 Conclusions

Fractal analysis represents an emerging and valuable approach in the time series study, providing a useful tool for analyzing neurophysiological signals. Several methods have been proposed and applied to estimate the FDs of different neurophysiological signals, but the importance of several signal characteristics in their performances has never been systematically assessed. Using different methods on synthetic time series with a known FD, we showed that different signal and recording features could influence the accuracy in FD estimation. These results offer the theoretical basis to guide the choice of the more appropriate method to

compute FDs. Particularly, the stationarity and the SNR (representing the noise level) of the signals should be preliminarily considered. HFD and GHE represent the methods displaying better performances on the synthetic stochastic stationary signal (fBm). Conversely, for the synthetic stationary signal (fGn), DFA and sPSD have provided the most accurate FD estimations, with a minor influence on both f_s and noise levels. Interestingly, all the tested methods, with the exception of KFD, are not sensitive to changes in signal amplitude. Considering the poor performances of all the STS tested with respect to the other methods, and the marked sensitivity to signal amplitude, KFD does not appear to be the best choice to estimate FDs in fractal time series.

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Part II
Fractals in Neuroanatomy and Basic
Neurosciences

Chapter 6

Fractals in Neuroanatomy and Basic Neurosciences: An Overview



Antonio Di Ieva

Abstract The introduction of fractal geometry to the neurosciences has been a major paradigm shift over the last decades as it has helped overcome approximations and limitations that occur when Euclidean and reductionist approaches are used to analyze neurons or the entire brain. Fractal geometry allows for quantitative analysis and description of the geometric complexity of the brain, from its single units to the neuronal networks.

As illustrated in the second section of this book, fractal analysis provides a quantitative tool for the study of the morphology of brain cells (i.e., neurons and microglia) and its components (e.g., dendritic trees, synapses), as well as the brain structure itself (cortex, functional modules, neuronal networks). The self-similar logic which generates and shapes the different hierarchical systems of the brain and even some structures related to its “container,” that is, the cranial sutures on the skull, is widely discussed in the following chapters, with a link between the applications of fractal analysis to the neuroanatomy and basic neurosciences to the clinical applications discussed in the third section.

Keywords Brain · Microglia · Neuroanatomy · Neuron · Fractal · Self-similarity · Neuronal networks

By echoing some concepts expressed by the seventeenth-century English scholar Richard Bentley, Benoit Mandelbrot expressed that “Clouds are not spheres, mountains are not cones, coastlines are not circles, and bark is not smooth, nor does lightning travel in a straight line” [1].

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6.1 What About the Brain?

Although Mandelbrot's fractal geometry has often been seen as “mathematical esoterica,” candidly confessed by himself, the new model has been applied in the biomedical domain, including the neurosciences [2–4]. The natural complexity of the brain, its hierarchical structure, and the sophisticated topological architecture of the neurons organized into micro- and macro-networks can only roughly be described by means of Euclidean geometry and linear dynamics approaches [3].

Regardless of the mathematical meaning of fractal geometry, it should be emphasized that in the context of neuroscience (and all the biomedical sciences), fractal geometry should not really be considered as a model or a theory but rather as a framework. It is a useful framework to describe the complexity of the brain functional networks, in both its physiological and pathological states, in all its hierarchical structure, from protein/synapse networks to macroscopic organizations of neurons and other cells, as confirmed by the small-world topologies seen in cellular and large-scale neuronal networks [5, 6].

6.2 Fractals, Neurons, and Microglia

The adult human brain contains about 86 billion neurons and 85 billion nonneuronal cells (an adult male with an average brain of 1.5 kg) [7]. It is now known that the complexity of brain function is not strictly related to such a high number of cells but rather to its network connectivity.

Fractal analysis provides a quantitative tool for the study of the morphology of brain cells and its components (i.e., dendritic trees and synapses), as well as the brain structure itself, as summarized below and in the entire second section of this book.

Fractal analysis has been proven to be useful in quantifying the cytoneuroarchitecture, by providing a global estimate of axonal and dendritic branching, in discriminating between certain functionally different neuronal types that are visually similar [8–16], as well as in following the change of dendritic complexity induced by drugs [17]. The fractal dimension (FD) of a neuron increases with the ruggedness of the cellular border, the degree and pattern of branching, and the space-filling capacity. The fractal-based neuronal and nonneuronal morphometry even allows the characterization of cells in pathological conditions, as demonstrated, for example, by a lower complexity of the astrocytic cells' borders in the brain cortex of babies, which becomes higher in healthy adults, and then again smaller in aged subjects, and is significantly reduced in subjects affected by dementia [18]. Inversely to the FD, lacunarity has been shown to increase with age, and the relationship between FD and lacunarity can be used for a holistic and dynamic representation of the neuroglial network over time, from the process of differentiation to the senescence of cells. As elucidated in Chap. 7 by Audrey Karperien and Herbert

Jelinek, together microglia and neurons contribute to the fractal topology of the brain that defines its computational capabilities. The methods, aims, and results to compute the fractal dimension of neurons are presented in Chap. 7, as well as in Chap. 8 by Nebojša Milošević. The fractal dynamics of the synapse formation process are illustrated by Małgorzata Kołodziej and Przemysław Waliszewski in Chap. 9.

6.3 The Brains and Trees

It is not necessary to be a botanist to be able to distinguish a weeping willow from a pine. How? Both have a trunk, roots, and branches, but nonetheless, they look very different. They share many specific qualities of trees; however, among other characteristics, they each have a different geometrical shape, which is one reason why it is not difficult to distinguish them: the romantic weeping willow, with its crying leaves vs. the austere shape of a Scots pine! These two trees have different geometrical shapes, or better, they each fill the space differently. Considering that they could have the same Euclidean measures (e.g., height, width, and number of branches), they each own a distinctive fractal dimension for their peculiar geometrical complexity in the three-dimensional space.

A systematic and exhaustive research about the specific fractal geometry of all trees has not been published yet, in spite of this simple and intuitive idea.

However, the literature is full of evidence about the existence of a geometrical fingerprint found in many biological systems: bronchial trees, vascular trees, neuronal trees, and so on. Clearly, nature follows specific rules, which are common for several distinct systems, and some morphological *generators* generate different hierarchical levels of anatomical systems. The use and abuse of the word “tree” in biomedical sciences are not a coincidence. Botanic and vascular trees have branches, bifurcations, and trunks, something that, in short, can only be roughly estimated by means of Euclidean approaches. Are trees fractal structures? The answer is controversial; however, regardless of the fractal or quasi-fractal or semi-fractal nature of the trees, what is sure is that fractal analysis can help describe and quantify their geometrical complexity. The brain also shows many fractal tree-like structures. Not by coincidence, the old anatomical books mention the “*arbor vitae*” (Latin word meaning “tree of life”), for example, in reference to the white matter of the cerebellum; some researchers have confirmed its fractal nature or at least used fractal analysis to differentiate some brain and cerebellar diseases. Fractal analysis has been used to successfully distinguish between the shapes of different classes of neurons with their specific dendritic arborizations, as previously written. Moreover, peculiar vascular and microvascular networks characterize the brain and its pathological counterparts (e.g., brain tumors or cerebrovascular malformations, as discussed in the third section of this book).

6.4 Increase of the Fractal Dimension from “Too Smooth to Too Folded” Human Brains

In 1991, Hofman showed that the whole brain cortex has a fractal structure [19], as confirmed in certain studies [20, 21], although, from a mathematical point of view, the brain is not strictly self-similar [22]. The fractality of the convoluted brain is based on the hypothesis that the convolutions are a result of fractal folding and compartmentalization of neurons into modular circuits, governed by simple generative rules to generate the brain design [19]. The cerebral cortex is characterized “by the hierarchical organization of groups of neurons or, in terms of fractal geometry, infinitely nesting clusters of neurons” [19]. Comparing several species and performing allometric analysis, Hofman showed that among mammals, there are two major groups: (1) species with smooth, non-convoluted brains, where the cortical surface area increases as the two-third power of brain volume, and (2) species with convoluted brains, where the cortical morphology is described by more complex mathematical models, with a cutoff from the transition of a lissencephalic brain to a gyrencephalic one, as illustrated below. The FD of the human brain was computed to be 2.70 ± 0.07 , and the folding of the cortical surface to such a value and the compartmentalization of neurons into modular circuits were suggested to be the architectural reason for the high degree of parallel processing occurring in the cerebral cortex to maximize the processing of transfer of information [19]. The relationship between the cortical surface and the square root of cortical thickness has implications regarding the origin of folding from an evolutionary and developmental perspective, as well as in pathologies such as human lissencephaly, as recently reconfirmed [23]. The calculated ~ 2.5 fractal dimension value of gyrencephalic brains has intriguingly been found remarkably close to the value of crumpled sheets of paper [23].

In human pathology, the term *lissencephaly* refers to “smooth brain” phenotypes present in a wide spectrum of malformations of the normal cortical folding, caused by dysfunction in the migration of neurons during the process of corticogenesis. The terms used for a complete lack of the gyri (agyria) or broad gyri (pachygyria) add a qualitative element to the description of the lissencephalic patterns, although no objective quantitative indexes are in clinical use. On the other side of the spectrum of the abnormal cerebral gyrogenesis (the process of formation of the gyri) and sulcogenesis (formation of the grooves) are lesions characterized by the brain cortex being “hyperfolded,” and/or more gyri are present (polymicrogyria).

Computational fractal-based analyses have proposed FD and further complementary indexes as objective parameters to estimate the topological complexity of natural objects, such as the brain cortex, as well as imaging markers of normal vs. diseased human brains in a very wide range of neurological diseases (see the third section). In an unpublished exploratory analysis, we computed the folding of a few human lissencephalic vs. polymicrogyric brains, finding an average FD value of 2.15 vs. 2.85, respectively. It is interesting to note that these values are

at the extreme wings of a normal distribution of the patterns according to the increasing complexity of the brain folding. Computational fractal-based analyses can add a computer-aided quantitative parameter to the qualitative (and often subjective) description of the brain structure performed by morphologists (e.g., neuroradiologists or neuroanatomists/pathologists).

6.5 Neuronal Networks

The functional modularity of the brain shows self-similar features at each level of analysis, although such self-similarity is statistical rather than exact, like in many other biological systems. The fractal dimension has been shown as a valid parameter to characterize the topological structure and space-filling properties of networks [24], including the neuronal ones [5].

The brain and neuronal networks' "self-similarity logic" is intriguingly introduced by Diego Guidolin, Cinzia Tortorella, Raffaele De Caro, and Luigi Agnati in Chap. 10. Interestingly enough, such logic can also be applied to the analysis of cranial sutures, as covered in Chap. 11 by Takashi Miura. By means of a multifractal approach, it has been shown that the brain maintains a state of self-organized criticality [25]. Neural networks have been shown to use a self-similar branching structure of dendrites and downstream networks in interconnected reciprocal feedback loops: output from the outer branch nodes of the network tree enters the inner branch nodes of the dendritic tree in single neurons [26]. The recurrent fractal neural network theory has been suggested to provide a strategy to understand the exchange of global and local information processing, in a holistic view of the translation from neuronal computation into conscious experience [26]. The topological concept of self-similar networks of the brain seems to reconcile the reductionistic approach to dissect the brain into single units (neurons, columns, modules, nuclei, and so on) in a broad, highly connected network, which gives the mind its unity. Small-world networks and scale-free topologies, observed on subcellular and cellular levels and in large-scale brain networks (as seen in functional magnetic resonance imaging (fMRI)) [5], may be the anatomical and functional basis of the high efficiency of information flow in the brain and of the emergence of consciousness.

In conclusion, the analysis of self-similar patterns can suggest some operational modes and principles on which the brain architecture is shaped, from its anatomical (i.e., brain design) to functional (i.e., information processing) spectrum. In this context, an evolutionary and comparative anatomy perspective is intriguingly and brightly presented by Michel Hofman in Chap. 12. This holistic chapter on the fractal geometry of the brain links the second section to the next more clinically oriented section (i.e., section III) of this book.

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Chapter 7

Morphology and Fractal-Based Classifications of Neurons and Microglia in Two and Three Dimensions



Audrey L. Karperien and Herbert F. Jelinek

Abstract Microglia and neurons live physically intertwined, intimately related structurally and functionally in a dynamic relationship in which microglia change continuously over a much shorter timescale than do neurons. Although microglia may unwind and depart from the neurons they attend under certain circumstances, in general, together both contribute to the fractal topology of the brain that defines its computational capabilities. Both neuronal and microglial morphologies are well-described using fractal analysis complementary to more traditional measures. For neurons, the fractal dimension has proved valuable for classifying dendritic branching and other neuronal features relevant to pathology and development. For microglia, fractal geometry has substantially contributed to classifying functional categories, where, in general, the more pathological the biological status, the lower the fractal dimension for individual cells, with some exceptions, including hyper-ramification. This chapter provides a review of the intimate relationships between neurons and microglia, by introducing 2D and 3D fractal analysis methodology and its applications in neuron–microglia function in health and disease.

Keywords Fractals · Fractal analysis · Microglia · Neuron · Theoretical models · Box counting · Dendritic spines

7.1 A Brief Introduction to Neurons and Microglia

One can tell much from observing the general profile of a cell under a microscope. Students of neuroanatomy and neuropathology, for instance, are taught to predict the function, age, and location of a cell based on general morphological characteristics

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[1–3]. Indeed, cell shape is a well-known feature upon which to make diagnostic and clinical decisions [4–7]. Sometimes criteria that decisions can be based on are readily apparent and easily quantified, such as when the three-dimensional (3D) radius or two-dimensional (2D) diameter is used to classify red blood cells, but, other times, objectively measurable criteria are elusive. This is largely the case with the two brain cells that this chapter focuses on, namely, neurons, the relatively stable structural and functional units of physiological signaling, and neuroinflammatory microglia, the more dynamic attendants of those units [8, 9]. The last decade has seen advancements not only in mathematical analysis methods but also in histology, computing, and modeling.

7.1.1 Neuronal and Microglial Morphology in Context

Microglia live closely intertwined with neurons in the complex 3D environment of the central nervous system (CNS). Both have characteristically radial structures with multiple ramified processes. In both cells, the processes are usually relatively long compared to their essentially central cell bodies but come in varieties having specialized form and function [10–13].

Neuronal design includes centralizing somata of varying sizes and shapes, richly branching, spine-encrusted dendritic arbors that bring in and integrate signals, and generally less branched and sometimes much longer axons that send signals along to the target tissues. Depending on a neuron's location and function, its fundamental features are formed and arranged differently. Indeed, morphology such as dendritic spine density and cell body size are often all the information required to classify neurons and even deduce their location and function [14–16]. However, these features may, in some cases, not be sufficient as is the case for retinal ganglion cells and pyramidal neurons of the visual pathways [17, 18].

Similar to neurons, microglia normally have a central, although often elongated, soma, highly ramified processes specialized in such tasks as tending to dendritic spines, and (at least in mice) a dedicated process that extends along the initial axon segment [19] (a neuronal part responsible for action potential initiation) [20–28]. One key difference between the morphologies of these interlinked cells is that in normal adult humans, microglia are usually considerably smaller than the neurons they attend, spanning an area comparable to that of a neuron's soma. They are motile and unevenly distributed throughout the CNS, hovering nearer the somata more in the gray than in the white matter, generally inclined to stay near neuronal processes, and constantly and rapidly moving and waving their own processes in close proximity to the neuronal processes they attend [21–28]. Another important morphological difference is that microglia periodically adopt a wholly unbranched morphology to perform some of their functions. Whereas they are capable of and perform phagocytosis in all their guises (e.g., in spine pruning), for mass phagocytosis and waste disposal, they can fully wind in their processes to adopt an amoeboid form that is able to migrate to the sites of trauma [21–23, 25–33].

Like neuronal morphology, microglial morphology tells much about what a cell does, and many of the same factors that influence the changes in neuronal morphology also affect microglial morphology at the same time that microglia influence neuronal morphology [19, 21–23, 26, 28]. An important difference, however, is that microglia change both form and position much more quickly and dramatically in real time with important consequences. In a normal adult brain, the microglial soma is essentially stationary, but processes move to contact both axonal boutons and dendritic spines in patterns that include distinct contact and rest intervals [34]. In a traumatized brain, the microglial process that contacts the initial axon segment, for instance, loses contact when the microglia retract the processes and adopt their amoeboid form [19, 21–28].

Being able to quantitatively assess these physically and functionally interlinked brain cell types is vital to many of the goals of neuroscience. But just what should one measure about a neuron with a dense dendritic arbor on one end and a relatively sparse but extremely much longer axon on the other? And how might one characterize and compare the multiple personalities of an individual microglial cell? Further, how might one quantitate the 3D structural attributes and connectivity between neurons and microglia? Neuroanatomical examples are many wherein traditional measures such as total cell area, branch length, number of branch points, and angle of branching could not differentiate between visually subtle but important differences in the populations of both cell types, yet introducing fractal measures, alone or in conjunction with other measures, made it possible to draw distinctions [35–39].

Thus, this chapter broadly surveys fractal approaches to classifying neurons and microglia and discusses interpretations and implications of such work for understanding the fractal geometry of the brain. This chapter reviews these topics separately, reflecting the current dearth of studies comparing and assimilating results of fractal analysis of both, and introduces new studies in 3D fractal analysis and modeling. This chapter draws on the key definitions, terminologies, and methodological considerations explained in the introductory chapters of this book.

7.2 Fractal Analysis of Neurons

Table 7.1 lists a sample (from studies cited in the US National Institutes of Health's PubMed database) of the methods and approaches that have contributed to the state of the art of neuronal classification by fractal measures. To sum up the field, the fractal dimension (D_F) characterizes very well the neuronal structure in terms of the dendritic arbor (see Sect. 7.2.1) and offers a basis for objectively distinguishing among several accepted neuroanatomical and functional categories of neurons. As Table 7.1 indicates, the D_F has further been used to draw distinctions between neurons in pathological and nonpathological scenarios and has shown promise for gauging developmental stages. The use of fractal analysis has also now extended, as expected, to include D_F in neuron and glia 2D and 3D modeling as well as

to applying machine learning and deep learning for classification tasks. Current research has been placing more emphasis on utilizing accepted fractal analysis methods rather than identifying novel scaling features. Whereas there is in place a solid basis for objectively quantifying several neuron types to sort and compare them using the D_F , direct correlations between the D_F and neurophysiological events or overall function remain in the theoretical domain [50, 51].

7.2.1 Fractal Analysis of Dendritic Arbors

The 3D topology of the CNS underlies its computational capabilities, and evidence indicates that its functioning has fractal components [36, 50, 52–64]. Dendritic arbors are chiefly the local signal-integrating wiring of that topology, and much of the work published classifying neurons using fractal methods focuses on their D_F (i.e., a D_F for which somata and axons are ignored). This is not to say that the evidence for fractal scaling implicates the dendritic arbor to the exclusion of a neuron's other parts or collections of other parts or that there is reason to reject the possibility of fractal scaling in the gross or fine structure of axons and somata [65, 66]. Indeed, there are specific reasons to explore those topics such as the intimate physical arrangement of axonal terminals relative to dendrites, spontaneous neurotransmitter release that has been shown to occur on a fractal schedule [58], and fractal features of axonal networks [56] and gross brain matter in 2D and 3D image representations [57, 67–70]. Of interest is that fractal analysis is increasingly playing a greater role in neuropathology for structural and functional assessments [71–73].

The reasons for classifying neurons based on their dendritic arbors are practical. At the tissue level, the different parts of a neuron often comprise seemingly distinct compartments. For example, axons form tracts and local collections of somata or dendritic arbors form defining layers in various brain structures. Soma shape and branching within the dendritic arbor, in particular, are usually seen more completely under a microscope than are typically longer axons and are readily captured in the visual field of imaging systems. On the basis of individual cells, a richly branching, signal-integrating dendritic arbor is visually suggestive of fractal scaling and itself comes in many morphological arrangements that have been used to characterize neurons descriptively and functionally [5, 55, 58, 74]. As Table 7.1 illustrates, characterizing neurons by their D_F for dendritic branching alone has demonstrated practical utility in objectively quantitating several heuristically defined functional and morphological categories of neurons.

This success notwithstanding, there are some key issues to note with respect to classifying neurons based on the D_F of the dendritic arbor. One general point is that there are differences among species in the D_F for dendritic branching in at least some categories of neurons [75, 76]. This is especially important to bear in mind when applying fractal results based on animal models of disease such as the studies listed in Table 7.1 under *Pathology*.

Table 7.1 The selected sample of fractal analysis approaches to neuronal morphology cited in PubMed (August 2015; using the search terms “neuron” and “fractal”)

Topic and summary	Model	References	Analysis method
<i>Neuroanatomy</i>			
(a) Higher D_F for apical dendritic branching pattern in superficial rather than deep pyramidal neurons	Rat	[40]	Modified Richardson's (1.030 ± 0.004 vs. 1.010 ± 0.005 ; $p < 0.008$)
	Cerebral cortex		
	2D binary skeletonized images		Box counting (1.129 ± 0.014 vs. 1.085 ± 0.012 ; $p < 0.029$)
	Multiplane microphotography		10 vs. 10 cells
(b) Deep neural networks of diverse cortical and subcortical neurons and glia	Superficial pyramidal = soma in laminae II–III; apical dendrites terminating in layer I; deep = soma in laminae V–VI, apical dendrites terminating in layers III–IV	[41]	Box counting (1.33 ± 0.06 vs. 1.24 ± 0.04 ; $p < 0.001$)
	Rat histological and simulated examples	[42]	15 vs. 15 cells Fractal analysis unspecified – part of simulation feature input NeuroMorpho.org Hippocampome.org Chinese academy of sciences (CASIA) rat brain dataset
(c) Different D_B s between the dendritic arbors of inner and outer sublaminae as four classes: simple inner and outer and complex inner and outer	Adult rat; both genders	[43]	Box counting
	Retinal ganglion cells		
	2D binary skeletons		Simple $< 1.39 < \text{complex}$
	Scanned camera lucida		27 vs. 32 cells

(continued)

Table 7.1 (continued)

Topic and summary	Model	References	Analysis method
<i>Pathology</i>			
(d) Lower D_F of Purkinje cell dendritic arbors in a pathological model	Mouse (reeler mutation)	[44]	3D box counting and Sholl analysis
	Cerebellar cortex		
	3D images; synchrotron X-ray microscopy of Golgi staining		
	Gait, posture, balance, tremor disorders		
(e) Lower D_F of corneal nerve fibers (also decreased length and increased tortuosity) in corneal small fiber sensory neuropathy	In vivo observation with corneal confocal microscopy	[45]	Box counting
	Amyotrophic lateral sclerosis (ALS)		
(f) Anterior thalamic nucleus lesion, sham operation, and enriched environment-affected D_A fractal analysis	Rat CA1 hippocampal neurons Anterior thalamic nucleus lesion Enriched environment Sham operation	[46]	D_A fractal analysis: box counting, mass-radius method 3D-reconstructed CA1 hippocampal cells
(g) Higher D_F following Bacillus Calmette-Guerin (BCG) immunization Alzheimer's disease	APP/PS1 mice BCG		
(h) D_F ($P < 0.0001$) reduced for diabetic sensorimotor polyneuropathy (DSPN) compared without DSPN	Human diabetes corneal confocal microscopy diabetic sensorimotor polyneuropathy (DSPN)	[48]	Box counting Corneal nerve fiber fractal dimension (CNFrd)
<i>Development</i>			
(i) D_F positively correlated with neuronal size and developmental changes in a mathematical model of human fetal development	Human	[49]	Box counting and Sholl analysis
	Fetal dentate nucleus		
	Scanned camera lucida drawings of Golgi-impregnated neurons		

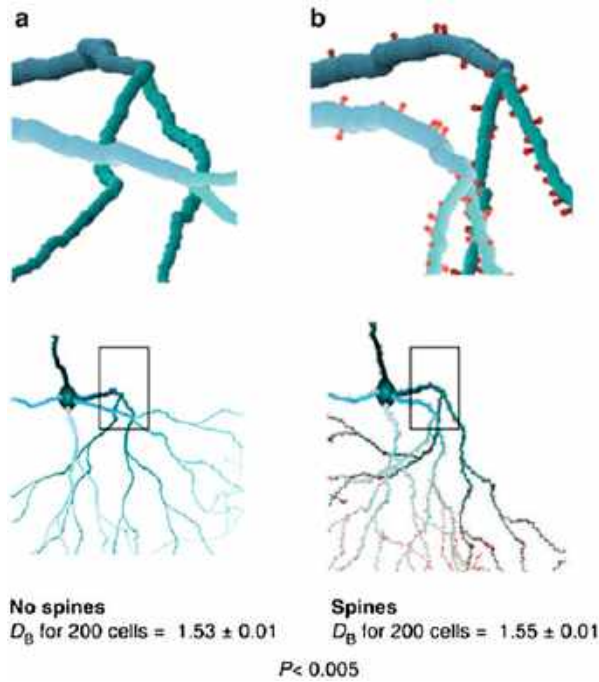


Fig. 7.1 Greater D_B s for simulated planar stellate dendritic arbor neurons with spines (200 cells modeled with and 200 without dendritic spines). **(a)** An example of the original model with no spines. **(b)** An example of the original model with spines modeled as “mushroom”-shaped at a rate of 0.014, length = $2 \times$ terminal diameter, spine diameter = $0.5 \times$ terminal diameter, head diameter = $3 \times$ spine diameter, spine angle = 90° . The examples are representative of each set, both modeled using one set of base parameters with random variation to simulate natural variation (binary patterns generated in MicroMod2015 [78] based on a built-in stellate model)

Another point to be aware of is what is meant by the dendritic arbor per se. Many studies calculate a D_F for binary skeletonized, single pixel-wide patterns extracted from digital images, but this is not the only pattern that could be assessed; binary silhouettes, for instance, can also be extracted and assessed and, for some fractal analysis methods, silhouettes tend to have higher D_F s than do skeletons (see Chaps. 2 and 32). Moreover, the level of resolution and the extent to which dendritic branching is captured can influence the result, reflecting not differences in fundamental branching per se but methodological matters [77].

Furthermore, the general dendritic branching pattern investigated by fractal analysis often discounts dendritic spines, as was done in the work listed under items (a) and (b) in Table 7.1, for instance, which used binary skeletons devoid of spines and reduced to single pixel-wide strands. In this respect, as explained in Fig. 7.1, for in silico stellate neuron models, binary silhouettes extracted from models with spines have a higher average D_F than those without. The D_F used for measuring the modeled spines was the box-counting dimension (D_B), demonstrated to be robust

and sensitive to neuronal and other cell morphologies, including microglial (see Sect. 6.3) [5, 43, 79, 80].

The overall branching pattern defines one feature of the dendritic arbor, but spines are a morphological and functional feature that defines another element of neurophysiological significance. Spines play important roles in both increasing the available surface area for and integrating information in neuron signaling. Moreover, from a structural perspective, they change over time. In addition to being modified with learning and experience in general, they have been shown to retract and swell with experimental temperature manipulation and in rodent pyramidal cells to be lost toward the distal aspects of the apical dendrite with chronic stress [14, 81–86].

7.2.2 *Methodological Issues*

That spines affect the D_B for models is just one part of the general consideration that method matters in the fractal classification of neurons. Methodological considerations include, in addition to the choice of which parts of a neuron to measure, the D_F type and technique of determining it. Table 7.1 lists a representative but not comprehensive selection of approaches and methods that have been used (e.g., other methods not listed include multifractal [5], local connected fractal dimension [87], and wavelet-based [88–90] analysis).

That each method has its strengths and weaknesses can be appreciated by considering the studies listed under item (a) in Table 7.1. Those studies compared cortical pyramidal neurons (a broad category having distinct apical and basal dendrites and branching axons) based on binary, skeletonized patterns derived from digital images of only the signature apical dendrites. Both studies showed that two subtypes, superficial and deep pyramidal cells, differed significantly in the D_F , yet from one study to the next, the actual values differed substantially for the same neuroanatomical subcategory of neurons. The important point to discern here is that this reflects methodological differences, and the result to focus on is the relative difference when the same method is applied [77].

With neurons in particular, the method matters. The first study under item (a) in Table 7.1 used two types of D_F s, one from a modified Richardson's method (D_R) and one a type of D_B . As noted, both successfully differentiated between superficial and deep pyramidal cells, but the actual values obtained were notably higher for the D_B .

This is probably attributable to the fundamental differences between these methods that can affect the results for neurons. In essence, the D_R method uses a caliper to directly measure branching in radial patterns and, as such, would be expected to be highly sensitive to the patterns assessed in the study. In contrast, box counting uses grid-based sampling to estimate scaling and does not necessarily measure the same feature the D_R measures, even from the same pattern. The more general D_B is not restricted to nor specialized for skeletonized radial patterns in the way that the modified D_R is, but the D_B has been used to successfully classify

such patterns (e.g., see $p < 0.001$ under item (a) in Table 7.1.) as well as space-filling patterns such as detailed membrane features and mass distributions, including branching cell processes where intracellular contents are relevant (e.g., the spines in the models in Fig. 7.1) [14, 16, 91].

Even within a method, issues particularly relevant to neurons may be important. Digital rotation, for instance, is a known issue that affects box counting and is especially pertinent if not addressed for patterns with the generally elongated shape spanned by the apical dendrite of pyramidal cells (reviewed in Chap. 32) [16]. One strategy (e.g., under item (a) in Table 7.1) partially addresses digital rotation effects by aligning the main trunk of each dendrite in the same orientation for all images being compared, but this strategy leaves the part of the pattern representing all subbranches uncompensated for. This is because subbranches in the dendritic arbor are not oriented in space in the same way as the main branch, so neither is their pixel-related orientation in a digital image. As such, this type of correction renders comparisons of the main trunk, but not of the subbranches, more valid from one image to the next, leaving open the possibility of considerable inherent variation not being addressed if assessed using only one orientation. This sort of correction also ignores the broader issue of sampling location bias in general that affects box counting as explained in Chaps. 2 and 32.

An alternative that addresses both digital rotation and sampling location bias more comprehensively is to sample a broad range of digital rotations using multiple sampling locations. Figure 7.2 illustrates the effect of digital rotation for a set of 200 branching structures modeled on manual measurements from the digital images of pyramidal cells. The set models statistically equivalent versions of an apical dendrite based on the fractal parameters listed in Fig. 7.2b. The figure suggests that variation in the D_B by digital rotation can be notable for structures such as apical dendrites but can be controlled by appropriate sampling [92]. In general, the choice of method and the details of its implementation are relevant not only when interpreting D_F s but also when planning large-scale studies in particular (e.g., calculating a D_R is tedious, whereas validated, free, open-source box-counting software that automatically deals with digital rotation and sampling bias is currently available) [93].

7.2.2.1 Complementary Methods

Fractal analysis can also be a key part of broader characterizations of neuronal morphology. The work on human fetuses listed in Table 7.1f emphasizes this. The researchers in that study combined fractal analysis with other measures, including another type of analysis called Sholl analysis. They measured several parameters and found that three were positively correlated with the growing size of neurons but at different rates over the gestational period in humans. The parameters were the global D_F and radius of the arbor and another measure of branching, the number of dendritic intersections. The researchers combined these parameters to develop a mathematical model describing prenatal development of the dendritic

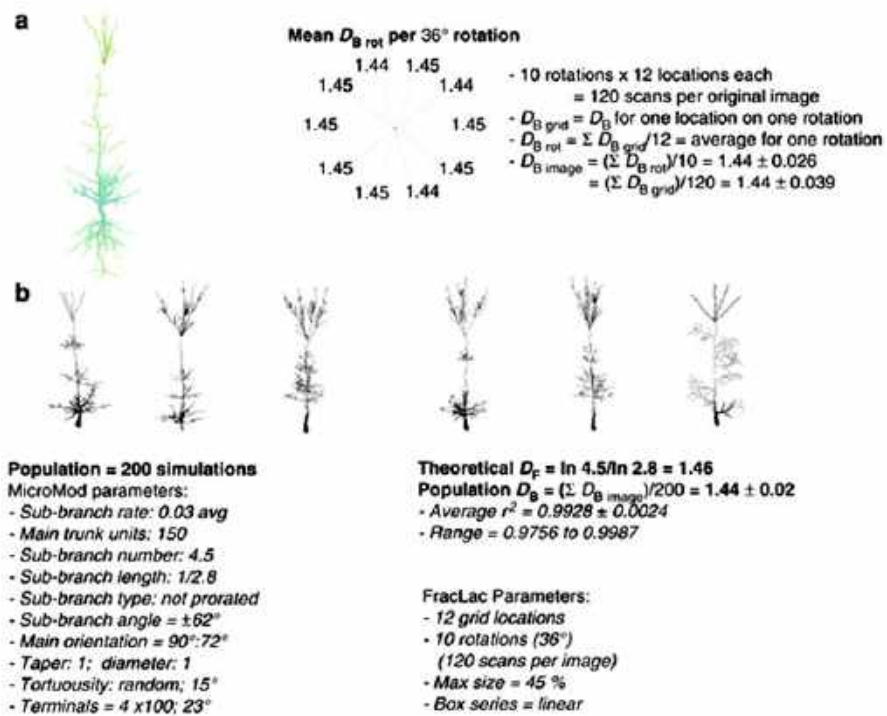


Fig. 7.2 Box-counting dimension for a simulated branching pattern for pyramidal cell apical dendrites. **(a)** The clocklike figure shows the variation in the D_B with digital rotation for one sample model rotated in 36° increments (rotated each time from the starting orientation rather than by accumulating distortions) and how the D_B was calculated for each model, using 12 grid locations and 10 rotations for each. **(b)** A sample of binary contours from 200 simulations made using one set of parameters with average branching and random variation. Similar results were obtained for binary skeletons (not shown) (simulated using MicroMod2015 for ImageJ; D_B determined using FracLac2015 [78])

arbor of neurons in the dentate nucleus, with the potential for developing methods of monitoring developmental abnormalities [49, 94].

7.2.2.2 3D Analysis

In many neurons, the dendritic arbor is relatively symmetrical or essentially “flat” (contained within a relatively planar volume within the CNS) as found with retinal ganglion cells, for instance, which is related to laminar organization at the tissue level and makes it possible to capture relevant details in digital images from histological preparations. In the study listed under item (d) in Table 7.1,

researchers found that although the dendritic arbors of cerebellar Purkinje neurons in normal mice were essentially planar and arranged parallel to each other, arbors in the mutated model depended on the visualizing orientation (i.e., were larger perpendicular to the sagittal plane and smaller in the same plane) [44]. The differences presumably relate to disrupted laminar organization seen at the tissue level in the mutation mouse model. Whether the D_F differed by plane was not specified.

This approach is applicable to the assumption of 2D analysis that 2D representations adequately represent 3D structures. This does not mean that a 2D D_F is equivalent to a 3D D_F , similar to the way an area is not equivalent to a volume, but that it will consistently describe the part of the scaling in the underlying 3D structure that is being observed. The assumption is explained in detail in Chaps. 2 and 32. The crux of the matter is that it is assumed that the salient scaling features of a dendrite are consistent and will thus be captured in unbiased samples. The assumption is borne out for neurons in a general sense by findings in the literature of reasonably small standard deviations in the D_F within populations of neurons, suggesting that, insofar as the plane for imaging has been varied in the published literature, the plane does not affect the fractal dimension [5]. Such circumstantial evidence notwithstanding, it is important to be aware that D_F s calculated for dendritic arbors might differ by plane, especially in pathological circumstances, and may provide a means to quantify features of structural abnormalities. Early work on 3D images using fractal analysis was reported by Schierwagen who investigated the multiscale fractal dimension of mouse pyramidal neurons. However, the analysis was based on thresholding the 3D images to represent the neuron shape as a set to 1 voxel [95]. Rowland et al. discuss connectivity between dendrites and other neurons in terms of optimizing function to the cost of building the connections [46]. From an electrotonic perspective, dendritic arbors depend on their mass distribution, dendritic branching, and taper that can be characterized by a 3D fractal analysis [36, 96] 3D imaging and 3D fractal analysis have also been applied in several projects, including evaluation of the structural brain complexity from magnetic resonance imaging (MRI) images from the MassIVE (Mass Spectrometry Interactive Virtual Environment) dataset and applying the 3D box-counting algorithm [72]. Similar studies were reported for autism spectrum disorder, subcortical gray matter structures in schizophrenia, and changes in brain structure associated with extended memory tasks [66, 97, 98]. Earlier work by Zhang et al. analyzed white matter structure in the human brain and degeneration of white matter in normal aging [99, 100]. Neuropathology was investigated using 2D and 3D fractal dimension analyses as well as lacunarity for predicting the meningioma grade [101]. This is by no means an extensive list of current work in 3D fractal analysis but aims to provide an overview of how 3D analysis has progressed.

7.3 Microglia

As with neurons, for microglia, work has been published using fractal analysis to study and classify microglial morphology in neuroanatomy, pathology, and development. Table 7.2 lists samples of that work in normal and pathological models, in 2D and 3D, using individual cellular branching patterns and other features.

One difference between the two fields of fractal analysis of neurons and of microglia is that the D_F for microglia is more clearly correlated with function. Using box counting, it has been shown that, basically, microglia in a normal, healthy brain are highly ramified with a relatively high D_B and that in response to certain stimuli such as chronic stress [103], they may hyper-ramify to a state with a slightly higher D_B , but, when responding to fully noxious pathological events such as brain trauma, they enter a cycle of de-ramification with concomitantly decreasing D_B , to a rounded, amoeboid form with a very low D_B , and then return through a cycle of increasing re-ramification and increasing D_B when resuming normal activity (see Fig. 7.3) [28, 39, 106, 109–115]. Different stages within the cycle shown in Fig. 6.3 are often classified using categories that describe both form and function, and the D_F has been shown to objectively quantify these categories [39].

Much of the fractal analysis work published for microglia has used box counting based on binary silhouettes or contours, with a strong point being that this methodology is robust for all of the general morphological types, branched and unbranched, facilitating comparisons such as in Fig. 7.3.

Similar to what has been seen for neurons, the D_F is typically positively correlated with the increasing size of the span covered by a cell but measures features in addition to size. It has been shown to delineate microglia based on other categories, such as age (see Table 7.2f, g). Also similar to what has been found for neurons, the D_F for microglia objectively distinguishes cells in different pathological states (e.g., Alzheimer's) or diffuse brain injury [116], healthy and diseased retina [117], and multiple sclerosis [118]. One strength of this approach is that it differentiates differences that can reflect important pathological events or other activities that are subtle or undetectable visually [39, 105, 114]. In the work listed in Table 7.2d, for instance, the D_F was useful for quantitating not only gross pathological responses but also less easily observable responses over different brain regions after transient global ischemia [104, 105].

This general capacity for quantifying subtleties of microglial morphology has important implications because microglia are not only immunoinflammatory cells but also much more, playing multiple roles in maintaining normal brain structure and function and interacting with all parts of the neuron [26, 28, 115]. They are involved in pruning spines, facilitating synaptic transmission, interacting with the extracellular matrix, and in axonal excitability [19, 25]. They secrete growth factors and inflammatory mediators and play major roles in dealing with stress at all levels from sleep deprivation [119] to subtle cognitive experience [37] to pain modulation [120] and severe trauma [24, 29, 121–124]. In this respect, many roles

Table 7.2 Selected samples of fractal analysis approaches to microglial morphology cited in PubMed (August 2015; search terms “microglia” and “fractal”)

Topic and summary of classification criterion	Model	References	Analysis method
<i>Neuroanatomy</i>			
(a) Consistent D_B by location but substantial variability in hull size correlated with process size and complexity	Healthy adult rat	[102]	Box counting and Sholl analysis
	Prefrontal cortex		Convex hull
<i>Pathology</i>			
(b) D_B defines an activation-based cycle with two sides (higher $D_B =$ lower activation, except transient peak of hyper-ramification)	Multiple models	[39]	Box counting, lacunarity, and multifractal
(c) Hyper-ramification in chronic stress increases D_B (peak of the cycle before decreasing to full activation); minocycline decreases D_B ; larger cells are affected more than the smaller ones	Adult male rats	[103]	Box counting, Sholl analysis, and other morphometrics
	Chronic stress induces hyper-ramification coincident with an increase in depression-like behaviors		
	Attenuated by minocycline		
(d) D_F correlates with the course and intensity of pyramidal cell degeneration but with different dynamics in different brain regions	Adult (3-month-old) female rats	[104, 105]	Mass radius, dilation, and other morphometrics

(continued)

Table 7.2 (continued)

Topic and summary of classification criterion	Model	References	Analysis method
(e) Decreased D_B (higher activation) with pathology and increased D_F (lower activation) with neuroprotectin D1	Global ischemia (cardiac arrest)	[106]	2D and 3D box counting
	Upper layer cerebral cortex, stratum radiatum of CA1, and hilus of dentate; no visible neurodegeneration but microglial reaction in the upper layer of the cerebral cortex, weak relative to hippocampal areas		
	Mice		
	Laser-induced choroidal neovascularization Attenuated by neuroprotectin D1 eye drops		
<i>Development</i>			
(f) Higher D_B (less activated) in young adult microglia compared to middle-aged individuals	Rats 3–5 and 15–19 months	[107]	Box counting
	Dorsal root ganglia and lumbar spinal cord		
(g) D_F increased over time but with different dynamics by brain region	Rat	[108]	Dilation
	Developing cerebral hemisphere		

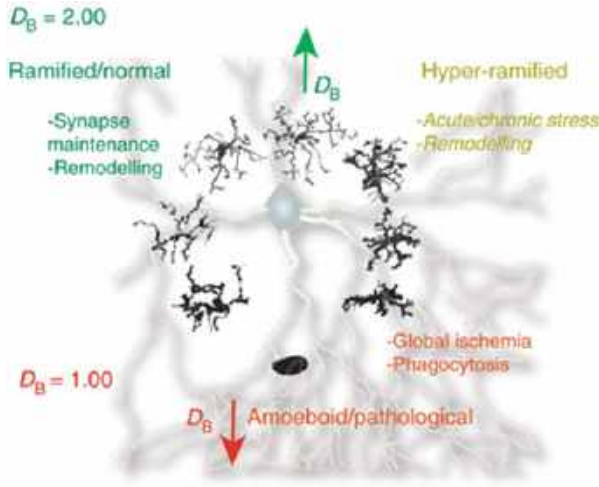


Fig. 7.3 A schematic showing the cycle of microglial morphology, activation level, and D_B . Microglial complexity is the highest in the ramified (top cells) and the lowest in the amoeboid (bottom cell) forms responding to severe compromise, where the D_B is close to 1.00 (bottom red arrow). The D_B generally corresponds to the level of re-ramification or de-ramification between the peak values. The D_B objectively differentiates not only between the gross differences separating morphological/functional categories but also between the subtle differences within the cycle to quantitate subtle differences in activity. Using cell span, lacunarity, and multifractality (not shown) further differentiates between cells with similar D_B s

of microglia in normal brain function and subtler dysfunction are currently being investigated, suggesting that the D_F of microglia will have potential application in many fields (e.g., drug monitoring, learning [21], stress [103], alcoholism [125], and schizophrenia [126]; see Table 7.2c, e).

Along with the D_F , lacunarity and multifractal analysis have been used to classify microglial morphology. In particular, lacunarity has been used to objectively discern among highly similar-looking microglial morphologies within an activation category and having the same D_B (e.g., more “rodlike” have higher Λ than more radial cells; rodlike forms appear when reacting microglia fuse but could also reflect physical confines or active migration) [24, 39, 105, 127]. In silico modeling has shown that the D_B is more sensitive to branching structures, but lacunarity is for particular features that are not always visually recognized such as soma size relative to process length [39, 111, 112, 128]. Multifractal analysis has shown promise for identifying microglia, particularly in transitional states [5, 39, 113, 129].

7.4 Future Directions

Whereas neurons are the critical structural and functional basis of signaling, microglia are their dynamic, constant attendants, physicians, and undertakers. Being able to quantitatively assess both of these brain cell types is vital to the overall science of understanding, monitoring, and modifying nervous system function. This chapter has outlined ways that the D_F has been used to describe and categorize these two cell types in normal and pathological models, in two and three dimensions, using not only individual cellular branching patterns [5, 87, 130, 131] but also other features [5, 89, 132]. This chapter has also explained that there are a variety of approaches, each with its own strengths and weaknesses, making some preferred over others depending on the application. Box counting of silhouettes holds particular promise for comparisons of neurons and microglia, in being able to capture features relevant to studies of both cell types in all potential forms.

The detailed designs of neurons and microglia are inherently related. The topological structure of neurons dictates the brain's computational capacities and also interacts reciprocally with the topological structure of microglia, in not only normal but also pathological states. Moreover, the fundamental features of neuronal morphology change. Being part of a living system, neuronal morphology is reciprocally and profoundly influenced by the system it provides central communication and control for, and microglia are a major part of that plasticity [21–28, 133]. In both cells, branching and other features of morphology change with experience and, in the long term over time, with age, alcohol exposure, etc. [21, 84, 85, 134–136].

It is intriguing in this regard that despite such lifelong plasticity, the D_F s for neurons based on dendritic arbors and for microglia are stable for different neuroanatomical/functional categories. This brings up questions about the normal variety in the D_F that should be expected within a category and how that variety itself might be related to cognitive capacities etc. In particular, it brings up questions about the D_F of neurons where spines are included and how studying microglial and neuronal function together, including this feature, might bring about valuable insights.

In conclusion, the stage is set to draw these two fields closer, to use fractal measures of neuronal and microglial categories to investigate complex features and interactions such as signal integration, spine morphology, chemical mediators, network effects, and 3D orientation [56, 69, 74, 137, 138]. Classifying tissues using this type of approach offers a perspective on connectivity and synchronicity, complementing the perspective afforded by studying dendritic branching or microglial morphology alone [137, 139]. Fractal properties of three-dimensional representations of cortical and subcortical structure–function relationships and association with pathology, including improvements in fractal analysis methodology, have been a major focus for nonlinear, statistically self-similar images [140].

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Chapter 8

The Morphology of Brain Neurons: The Box-Counting Method in the Quantitative Analysis of 2D Images



Nebojša Milošević

Abstract This chapter begins by showing the difference between fractal geometry and fractal analysis. The text shows the difference between mathematical and natural fractals and how they are best defined by explaining the concept of fractal analysis. Furthermore, the text presents the most famous technique of fractal analysis: the box-counting method. Defining this method and showing the methodology that leads to the precise value of the fractal (i.e., the box) dimension is done by demonstrating the images of human dentate neurons. A more detailed explanation of the methodology was presented in the previous version of this chapter.

This version promotes the notion of monofractal analysis and shows how three types of the same neuronal images can quantify four image properties. The results showed that monofractal parameters successfully quantified four image properties in three nuclei of the cerebellum. Finally, the author discusses the results of this chapter and previously published conclusions. The results show how the monofractal parameters discriminate images of neurons from the three nuclei of the human cerebrum. These outcomes are discussed along with the results of previous studies.

Keywords Box-counting method · Fractal geometry · Human · Image preprocessing · Monofractal analysis · Neuronal morphology · Two-dimensional images

8.1 Introduction

Although standard quantitative methods in science are based on classical Euclidean geometry [21], fractal geometry has been developed as a new geometry of nature [63]. The most common biological patterns can be characterized by fractal geometry [9], and, thus, this type of geometry is being used in diverse research areas [9,

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43]. Particularly, this is an increasingly useful tool in quantitative image analysis in medical science [28].

Fractal analysis is derived from fractal geometry [40] as a modern mathematical method of measuring the complexity of patterns in geometry and nature [43]. This type of analysis uses a simple parameter [21]: the fractal dimension (FD). The FD is more prominent if the object's border is rugged, the branching pattern is abundant, and the lines are more irregular and twisted [9, 20].

The first version of this chapter presented one fractal analysis method (i.e., the box-counting method) and studied its application to a 2D neuron image [28]. Images of the dentate nucleus of an adult human were used for this [27]. However, the previous version was more concerned with the correct methodology of preparing the image to calculate the box dimension's (BD's) precise value. The text repeats the conclusions of the previous version and primarily explores how the box-counting method can quantify the four properties of the neuronal image. This chapter shows the research results of the three nuclei of the human cerebrum. Finally, the results were compared with the previously published results of computational parameters.

8.1.1 From Fractal Geometry Toward Fractal Analysis

The basic idea (i.e., the starting point) of fractal geometry is the term “fractal” [40], which can be comprehended as a theoretical abstraction that cannot be defined analytically [21]. In 2D space, a mathematical (geometrical) fractal is a mathematical construction defined by never-ending cascades of similar structural details [21]. Thus, each mathematical fractal has four features [40]: (a) starting object (initiator), (b) recursion algorithm, (c) geometrical self-similarity, and (d) fractal dimension (FD). Thus, every mathematical fractal starts with an initiator [9, 40] and ends with a limit fractal [36]. Incomplete fractal constructs are called prefractals [9, 36]. All of them represent a geometrical fractal set [40].

Fractals, in general, possess two essential properties: self-similarity and scaling [21]. As reported previously [13, 24], there is a substantial equivalence between these two properties for mathematical fractals. The pieces are an exact duplication of the whole object; therefore, it is commonly said that a mathematical fractal is geometrically self-similar [21].

On the other hand, this definition has to be corrected for natural objects [28]. Their pieces are rarely identical copies of the whole object, and rather than being geometrically self-similar, they are statistically self-similar [40]. They are self-limited in their scaling due to their functional requirements [28]. Therefore, self-similarity implies that the measured property depends on the scale's size (or resolution) used [2], i.e., the property scales with the measure.

Finally, the self-similarity and scaling of natural objects usually estimate the fractal dimension (FD). This measure describes the complexity of an object's form and space-filling property. The scaling relationship is the rule that shows

how the measured value depends on the resolution of the measurement. The most straightforward scaling relationship is provided by power-law scaling [2].

8.1.2 *Fractal Analysis*

Consequent to the fractal theory, fractal analysis could be considered an experimental technique that uses the methodology of Mandelbrot's fractal geometry and Euclid's traditional geometry to investigate several properties of natural objects [26, 40]. Previous work on fractal analysis traditionally represented a 2D analysis [4] that mainly focused on analyzing the border's outlines of natural objects [20]. These papers mainly investigate the object's border, its complexity (or simply the object's structure), and function [42]. The problem is guided to calculate the fractal dimension and analyze the meaning of the parameter value to the considered the object [40].

By analogy with the investigation of a geometrical fractal set, the use of fractal analysis postulates the following attributes: (a) the shape of the starting object (which corresponds to the initiator in fractal geometry), (b) an algorithm (enabling its successive application to the starting object), (c) scaling (or statistical self-similarity), and (d) fractal dimension. The limit generator is the same as the starting object, but it was not changed during repeated algorithm application [9].

In fractal analysis, there are two basic approaches to measuring the fractal dimension of objects in a plane [43]. The first and most commonly used is the length-related method comprising the classical Richardson's coastline method [27], box-counting method [42, 43], and dilation method [2]. The second method is a mass-related method [43]. Using different fractal analysis methods has led to difficulties in comparing the results because each method of determination of the fractal dimension produces slightly different results when analyzing the same structure [2, 9]. It should be noted that the coastline method is equivalent to the box-counting method only when one analyzes the border of a system [9]. There is still a problem in presenting, analyzing, and comparing different methods of measuring the fractal dimension [40].

8.2 The Box-Counting Method

The segment-counting method [33], a version of Richardson's method for measuring the length of any unformed line [34], is robust but simultaneously tedious and time-consuming [40]. Thus, the need for more handsome methods emerges, and conventional box counting [42] appears to be the method that suitably measures the fractal dimensions of natural objects [3, 9, 11, 20, 29, 40, 43, 48]. This method is either equal to or similar to the idea from traditional calculus when the area of

a plane region within any closed unformed boundaries has to be measured [34]. Briefly, this area is superimposed onto a net of equivalent squares [46].

Conventional box counting covers the object (in the plane) with a set of rectangular coordinate boxes [28]. The number of boxes (N) intersected by the pattern and within it is counted. Each set is characterized by the square side (r). Finally, N is presented as a function of r . Fractal dimension (in further text, box dimension or D_B) is determined as the slope of the log–log relationship between N and r . The reader can see a more detailed mathematical explanation for determining D_B in the following publications [8, 28, 40].

8.2.1 Software for Box Counting

When writing the second edition (August 2022), a search on “fractal analysis software” throughout the World Wide Web yielded more than 2,190,000 results. Among them, several main fractal programs use box counting and other techniques. In addition, there are appropriate plugins in the main programs for image and signal analysis. Our further investigation will be restricted to “ImageJ” software (www.imagej.nih.gov/ij/) and the appropriate plugin (Analyze:Tools:Fractal Box Count), which emerge in “FracLac” (www.imagej.nih.gov/ij/plugins/fractalac/fractalac.html), developed by Audrey Karperien [16].

8.2.2 Application on 2D Digital Images

When the box-counting method is applied to digitized images, it covers the image with a grid of square cells (with cell size r). For binary images, the cell size is expressed as the number of pixels. The number of squares $N(r)$ needed to cover the image is given by a power law, and the box dimension (BD in further text) is obtained as an absolute value of the slope of the log–log relationship between $N(r)$ and r . Figure 8.1 shows this procedure on a 2D neuronal image from the monkey dentate nucleus [25].

The box size is progressively smaller (i.e., using geometric progression) where the sequence of box sizes is reduced by a factor of $1/2^n$ ($n = 1, 2, 3 \dots$) from one grid to the next. Here, it is important to state that the choice of scaling factor can dramatically impact the results. Thus, a scaling factor of 2 is appropriate for the box-counting procedure (Fig. 8.1). Previous research [24] has promoted a hypothesis that 2D neuronal images could be considered fractal over several decades of scale if the box sizes are scaled as a power of 2. This sequence represents a finite increasing geometric progression $\{r_n\}$ and, by definition, $r_{n+1}/r_n = q$ for every n , where q is the constant common ratio [24].

As previous investigations have concluded, choosing the size of boxes as a finite increasing geometric progression represents a better solution to the fitting problem

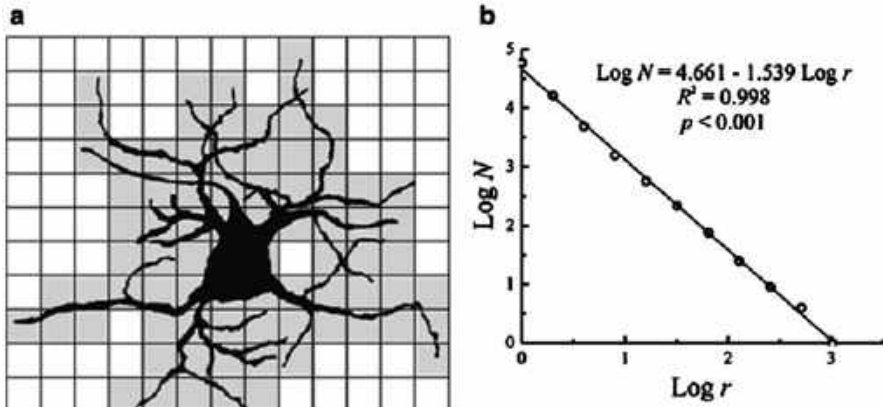


Fig. 8.1 Application of the box-counting method to a “black-and-white” image of a neuron from the monkey dentate nucleus: **(a)** the whole image is covered with a set of squares and squares in which cover dendrites (light gray color) are counted. **(b)** A log–log plot between the number of squares (N) and square size (r) is fitted by a straight line. R is the corresponding correlation coefficient. The fractal dimension D is the negative value of the slope of the fitted line (in this case, 1.539)

[40]. Finally, in performing the box-counting method, the box sizes should be taken from 2^0 to 2^k pixels, where k is the value for which N is equal to one. In this case, the relationship between $\log N$ and $\log r$ has been linear for more than two decades of the range [28].

8.2.3 The Methodology

As the biological image (i.e., an image of a neuron) does not reflect fractal in the theoretical or ideal sense, interpreting the image using the BD is meaningless [13]. Nevertheless, the BD may still be valuable as a quantitative parameter of neuronal images since it indicates complexity or the scale dependence of a pattern [1]. In addition, fractal analysis of biological images (particularly images of a neuron) is not intended to indicate that the object is fractal [9, 40]. Results have suggested that variations in sampling and preparing images for analysis, as well as the analysis itself, can have nontrivial effects on the estimation and interpretation of the BD [13].

8.2.3.1 Image Size and Resolution

Theoretically, images of identical objects of different sizes and the exact resolution should not influence the magnitude of fractal dimension [13]. Following this idea, we previously analyzed the group of neurons at three resolutions [33]. The image

size was changed, while the resolution was kept the same. The results of this analysis suggested that for this sample of images (mainly images with the same width and height), there was an increase in BDs with size in a solid mathematical sense. In an actual situation, it could be stated that the BD remains the same, while the image size increases. In addition, it must be noted that the effect on the FD associated with the resizing may not be related to the size per se but to the computer processing [13].

Scanning the same cells at different resolutions returned different FD values, even when all other parameters were kept constant [9]. In their previous work, Jelinek et al. concluded that cells scanned at low resolution had higher values of FDs than did those obtained at high resolution [13]. In order to test this hypothesis, we experimented. Our sample of cells was printed on A4 paper, keeping their size to the A5 format. Each cell was scanned from a resolution of 100 dpi to 1100 dpi, as the final resolution was restricted by a scanner (Mustek 1200, Mustek Systems Inc., Taiwan). The BD was calculated for each group of neurons, and their mean values were plotted against the resolution [33]. This study concluded that the most accurate BD values were obtained when all cells were scanned at the exact resolution. Examining the effect of scan resolution on BD accuracy is not meaningful.

8.2.3.2 Image Rotation

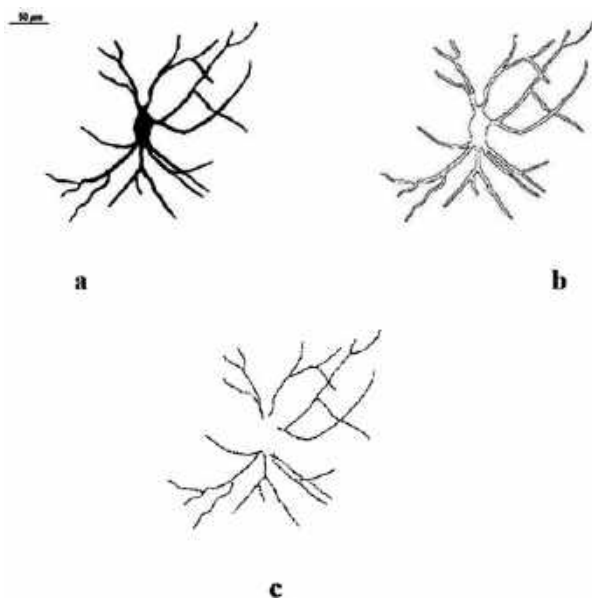
It is well-known that a picture exhibits rotational symmetry if rotation by a specific angle around some central axis point can return the picture to its original configuration [41]. For instance, squares and hexagonal snowflakes are typical examples of “ideal symmetry.” It is reasonable to imagine that the BDs of such a picture in these positions have the same value. Previously, we conducted an experiment wherein we explored how the BD changes in the course of continuous rotation of non-stellate neurons [33].

Our sample of 17 images was continuously rotated ($0\text{--}360^\circ$), increasing the angle by 15° . The conclusion indicated that image rotation influenced the value of the BD, mainly if the so-called non-stellate neuron was analyzed [34]. In this case, the axis should be constructed, four angles should rotate the image ($45^\circ + k\pi/2$, where $k = 0, 1, 2$, and 3), and the apparent BD should be recorded. An accurate BD will be the mean of these values [34]. However, this conclusion has a significant limitation: the correct value of the BD implies a change in the position of the recorded neuron. It indeed introduces a perturbation to the results.

8.2.3.3 Image Representation

When the box-counting method is applied to a 2D image of the object, the FD depends on image presentation [9]. Thus, many researchers use different processed images [31], primarily binary (or black and white) [43] and outline (or a binary

Fig. 8.2 A representative digital image of a group IV aspiny neuron from the human neostriatum. **(a)** Binary, **(b)** binary-outline, and **(c)** binary-skeleton image. (The original image can be found in the study by Vranes et al. [47] and was obtained with permission)



silhouette) [13, 28]. There is another type – the skeletonized image – mainly when the image of a neuron is quantified [9]. It is created when a binary image is converted into a one-pixel-thick curved line [28]. Previous research has shown that such an image simultaneously quantifies dendrite tortuosity and dendritic tree complexity [33]. However, recent research has shown that this type best quantifies the complexity of the dendritic field if the neuron body is first removed from the binary image and such an image is skeletonized [47].

All three forms of the image (binary, outline, and skeletonized) were produced with the ImageJ program, using the appropriate commands (Process–Binary–Make binary, Process–Binary–Outline, and Process–Binary–Skeletonize). A BD is calculated for each type of image (D_{bin} , D_{out} , and D_{skel}), and these values are used in further calculations. Figure 8.2 illustrates the three types using an example of an image of a multipolar neuron from the human neostriatum [47].

8.3 Materials and Methods

8.3.1 Samples from the Human Cerebellum

Material from the three nuclei of an adult human (dentate nucleus, neostriatum, and principal olivary nucleus) was collected between 2009 and 2018, from either the Center for Forensic Medicine or the histology collection of the Institute of Anatomy

(University of Novi Sad, Serbia). All materials were obtained with the approval of the Ethics Committee (Faculty of Medicine, University of Novi Sad, Serbia). The research was performed following the ethical standards defined by the 1964 Declaration of Helsinki and all subsequent revisions. In some cases, the experiments were undertaken with the written informed consent of the families of the deceased (whose anonymity was preserved).

The samples were collected from 10 (dentate nucleus), 29 (neostriatum), or 30 (principal olivary nucleus) brains, of both sexes, without visible disturbances in the central nervous system. These samples were cut into blocks whose dimensions corresponded to the size of the nucleus being investigated. Detailed information on the histological procedures can be found in previous publications [19, 22, 32, 37, 38]. The next subchapters will discuss recording, acquisition, and image quantification.

8.3.2 Recording of Samples and Formation of an Image

The samples were analyzed by the optical microscope “Leica DC 100” (Leica Microsystems, Wetzlar, Germany) using a system magnification of 400x [22]. Adequate impregnated neurons were fully visible in a series of optical sections, i.e., horizontal cross-sectional images [17]. Any problematic cells (i.e., broken or poorly impregnated images) were excluded from the sample. Conversely, only those impregnated with dense and homogeneous precipitates were recorded [32]. All identified neurons were transformed into digital images using the digital camera “Leica DC 100” with the accompanying software program “Digital Camera System” (Leica Microsystems, Heerbrugg, Switzerland). Any whole neuron was recovered by analysis of several images that ranged from 3 to 10 [22], 20 [17], or 12 [32] in each set. This range is correlated with the body size of a neuron. Each cross-sectional image was taken as color images (red–green–blue (RGB)) and saved as a TIFF (Tag Image File Format).

Then, all horizontal sections of each neuron were loaded into ImageJ (<http://rsbweb.nih.gov/ij>, National Institute of Health, Bethesda, MD). The image of the entire neuron was reconstructed by the subcommand “Z Project” (all sections for a given cell merged into a single image). The RGB image was converted into grayscale [17, 22]. In a further step, each grayscale image was converted into a binary format by the subcommand of ImageJ (“Threshold” by adjusting the contrast). Figure 8.3 shows examples of the binary images of neurons from the three nuclei of the human brain.

All following steps in editing were made using various subcommands of ImageJ by an anatomist/specialist, who first reconstructed the trajectory of the soma and each dendrite (or traced the dendritic axis). Then, the axon and other artifacts were removed [17, 22, 32].

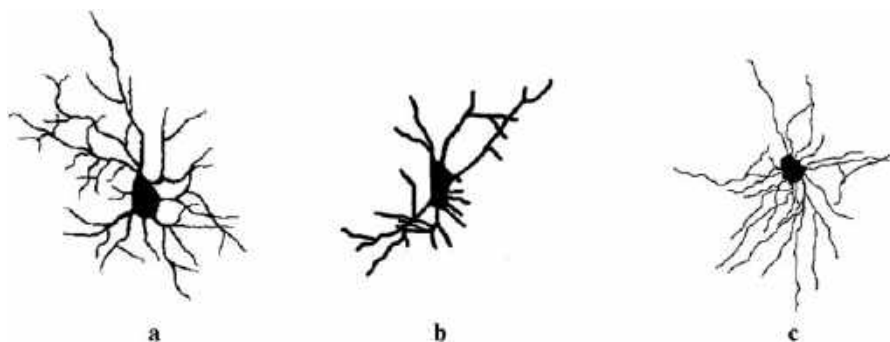


Fig. 8.3 Examples from the dentate nucleus (a), neostriatum (b), and principal olivary nucleus (c) in the adult human. (With permission, images were taken from previous publications [18, 22, 32])

8.3.3 Quantifying the Image

The binary image of a neuron is the easiest to quantify with various computational parameters. The same can be done with fractal analysis on the three different types of neurons: binary, outline, and skeleton images [28]. Monofractal analysis can be considered a form of fractal analysis in which the object is characterized by a unique fractal dimension value [33]. In this study, monofractal analysis was performed using the box-counting method incorporated into ImageJ, either with the standard command [9] or with the appropriate plugin [16]. Depending on how the neuronal image is displayed, monofractal analysis quantifies it through three fractal dimensions and lacunarity. These parameters describe four properties of the image: size, shape, complexity, and homogeneity.

Neuron size and complexity were quantified on the binary image (Fig. 8.2a), using two parameters: the fractal dimension of the binary image (D_{bin}) and lacunarity (Λ). The D_{bin} indirectly estimates the size of the neuron by measuring the space-filling property concerning the square constructed around the neuron [28]. The value of the fractal dimension is between one (straight line) and two, where a value closer to two is linearly related to the size of the square [34]. The Λ evaluates the gaps in the neuron image and the rotational invariance [30], calculating the number of pixels per all possible sizes of boxes [41]. The heterogeneity of the image was estimated during its simultaneous translation and rotation. A higher Λ means that the image has larger or more numerous gaps and larger rotational variance [47].

The neuron's shape is most efficiently quantified by the neural field [33], i.e., by constructing a polygon around the dendritic ends [18]. After filling the polygon with black pixels, the newly created image is converted into an outline image, whose border is marked with a one-pixel-wide line [28]. By calculating the D_{out} , the deviation of the border from the circle ($D_{\text{out}} = 1$) is estimated.

The complexity is quantified by assessment of the dendritic tree. The body is removed from the binary image of the neuron, and the rest is converted into a

skeleton image [9]. More precisely, the command “Skeletonize” removes pixels from the edges of an object until they are reduced to a single-pixel-wide shape [47]. The calculated parameter value (D_{ske}) simultaneously evaluates the number of primary dendrites and the degree of dendritic tree density [34].

8.3.4 Statistics

Distribution of the data was tested using four criteria: coefficient of variation (CV), skewness (m_3), kurtosis (m_4), and the p -value of the Kolmogorov–Smirnov/Shapiro–Wilk test. The estimation of values for all parameters, between the two groups, was carried out by nonparametric statistics. More specifically, the mean values were tested between two independent samples of different variances, while the p -value was read from the table for the two-tailed t -test [35]. A value of $p < 0.05$ was considered statistically significant. The SPSS software package v26 (IBM SPSS Statistics, Chicago, IL, USA, demo version) was employed for statistical analysis.

8.4 Results

The results presented here began with examining the character of the distribution of four parameters on samples of images from the mentioned nuclei. This was done because the analyzed images were only part of the total sample and to see how the obtained results would be processed further. The distribution was analyzed by calculating the coefficient of variation (CV), skewness (m_3), kurtosis (m_4), and p -value of the Kolmogorov–Smirnov or Shapiro–Wilk test, which depended on the number of images (less than or more than 100). Table 8.1 shows the calculated values of the mentioned parameters for all the three nuclei.

The table clearly shows that only the p -value indicated that the distribution of Λ does not conform to a normal distribution for images of the dentate nucleus and the neostriatum. As for the neostriatum, an aberration from the normal distribution was also observed in m_4 . On the other hand, in the images of the olivary nucleus, the aberration from the normal distribution was registered only in D_{bin} . Although only one parameter deviated from the normal distribution in all nuclei, further analysis was based on nonparametric statistics, i.e., with more rigorous techniques.

8.4.1 The Dentate Nucleus

From the total sample, previous research [10] classified 269 multipolar neurons according to the topology: into central (153) and border (116). Here we choose

Table 8.1 The coefficient of variation (CV), skewness (m_3), kurtosis (m_4), and p -value of the Kolmogorov–Smirnov or Shapiro–Wilk test in the three brain nuclei

Parameters	CV (%) ^a	m_3 ^a	m_4 ^a	p ^a
Dentate nucleus (78 cells)				
D_{bin}	3.0	−0.344	−0.388	0.200 ^b
D_{out}	3.3	−0.031	−0.630	
D_{skel}	3.6	0.177	−0.459	
Λ	12.1	0.174	−0.810	0.003*
Neostriatum (82 cells)				
D_{bin}	3.7	−0.186	1.252	0.057
D_{out}	3.4	−0.212	−0.452	0.200 ^b
D_{skel}	4.2	0.035	−0.283	
Λ	24.8	0.286	−1.072*	0.026*
Olivary nucleus (124 cells)				
D_{bin}	1.9	−0.157	0.583	0.001* ^c
D_{out}	1.8	0.142	−0.512	0.195 ^c
D_{skel}	4.0	−0.405	0.336	0.051 ^c
Λ	17.8	−0.038	0.164	0.509 ^c

^a $m_3 \in (-1,1)$ symmetric distribution, $m_4 \in (-1,1)$ light tailed to the normal distribution, and $p > 0.05$ normal distribution
^bThe Lilliefors significance correction
^cValues from the Shapiro–Wilk test

78 neurons (44 central and 34 borders) to test their morphology using 4 monofractal parameters (D_{bin} , D_{out} , D_{skel} , and Λ). The sample was selected according to the computational parameters of body size ($15,000 \text{ pix}^2 < A_s < 25,000 \text{ pix}^2$) and dendritic field complexity ($N_m > 10$).

The border neurons had higher values for D_{bin} , D_{out} , and D_{skel} than did the central ones, whereas the mean Λ was higher in the central ones (Table 8.2). Moreover, a significant difference between the mean values existed only in Λ . The displayed values of the standard deviations (SDs) were intended to represent the variability of the calculated mean values of the monofractal parameters of the two groups of neurons.

8.4.2 The Neostriatum

From the total number of neurons of the neostriatum (552), which were previously quantitatively processed [18], for this study, neurons with a large body and high complexity of the dendritic tree were chosen, i.e., group IV [17]. This study analyzed 82 aspiny cells from the 2 nuclei of the neostriatum: 42 from the putamen and 40 from the caudate nucleus. This research aimed to examine the possibility

Table 8.2 The mean values, standard deviations (SDs), and standard errors (SEs) of the monofractal parameters of four image features (D_{bin} , D_{out} , D_{skel} , and Λ) of nucleus dentate neurons. This table also shows the p -values from the table for the two-tailed t -test

Parameters	Statistics	Central (44 cells)	Border (34 cells)	p
D_{bin}				0.383
	Mean	1.545	1.555	
	SD	0.048	0.048	
	SE	0.007	0.008	
D_{out}				0.183
	Mean	1.375	1.388	
	SD	0.046	0.042	
	SE	0.007	0.007	
D_{skel}				0.125
	Mean	1.262	1.278	
	SD	0.044	0.045	
	SE	0.007	0.008	
Λ				0.001
	Mean	0.472	0.431	
	SD	0.056	0.045	
	SE	0.008	0.008	

of distinguishing the same group of aspiny neurons between two nuclei using monofractal parameters that quantify four image properties.

Table 8.3 shows the means, standard deviations, and standard errors (SEs) of the four monofractal parameters (D_{bin} , D_{out} , D_{skel} , and Λ) in two nuclei. The table also shows p -values that show the statistical evaluation of means, read from the table for the two-tailed t -test. Standard deviations in the table were listed to easily estimate the parameters’ variability. The mean values of D_{bin} , D_{out} , and D_{skel} were higher in the putamen than in the caudate nucleus, whereas the opposite case was spotted for the Λ . In addition, values of standard deviations show small variability of means (values below 20%, Table 8.3). Finally, examining the means of all parameters between nuclei showed that there was no significant difference only between D_{out} .

8.4.3 The Olivary Nucleus

The previous investigation [44] classified neurons into three groups according to the size of the body (i.e., to the A_s). However, this section examines only two groups (small and large), quantifying four image features with monofractal parameters (D_{bin} , D_{out} , D_{skel} , and Λ).

Table 8.4 shows the mean values, standard deviations (SDs), and standard errors (SE) of the four parameters in two groups of neurons. The symbol * indicates a significant difference in the mean values between the groups, and the displayed

Table 8.3 The four parameters (D_{bin} , D_{out} , D_{skel} , and Λ) from the two nuclei of the neostriatum (caudate and putamen) are shown by the mean values, standard deviations (SDs), and standard errors (SEs). The last column shows the statistical evaluation of means between nuclei read from the table for the two-tailed t -test

Parameters	Statistics	Caudate (40 cells)	Putamen (42 cells)	p
D_{bin}	Mean	1.506	1.551	< 0.001
	SD	0.052	0.050	
	SE	0.008	0.008	
D_{out}	Mean	1.244	1.259	0.094
	SD	0.040	0.043	
	SE	0.006	0.007	
D_{skel}	Mean	1.137	1.162	0.022
	SD	0.042	0.051	
	SE	0.007	0.008	
Λ	Mean	0.49	0.39	< 0.001
	SD	0.11	0.10	
	SE	0.02	0.02	

$p < 0.05$ indicates a significant difference

values of SDs indicate a small variability of the means (less than 20%). Although the small group had higher means for D_{bin} and D_{out} , only the D_{bin} was significantly different. On the other hand, the large group had higher means of D_{skel} and Λ , significantly different from the mean values of the small group (Table 8.4).

8.5 Discussion

Extracting the concepts of fractal geometry toward the biomedical sciences has led to significant progress in understanding complex functional properties and structural features [2]. Generally, various systems in nature, if they accomplish four properties (Sect. 8.1.2) during their construction or formation, can be considered ideal or natural fractal objects [40]. On the other hand, fractal analysis is the estimation of the fractal characteristics of data [9, 43]. It is derived from fractal geometry [9], and it is used to describe the organization of objects found in nature [13], quantifying their complexity with a value of the FD [28]. The measurement of the FD is usually not intended to indicate whether an object is a fractal or not but rather to provide a measure of the complexity of scaling inherent in the object [9].

Once fractal geometry was formulated, many scientists adopted fractal analysis as an appropriate method for objective quantitative analysis [28]. It is necessary to mention that the box-counting method is perhaps the most reliable and popular fractal analysis technique. It is used in various scientific fields [49] where it is necessary to analyze an image. In the case of 2D neuron imaging, this technique has been used for more than 20 years in various fields of medical science [5–7, 13, 30, 45].

Table 8.4 Parameters of the human olivary nucleus’s small and large neurons (D_{bin} , D_{out} , D_{skel} , and Λ) are shown as means, standard deviations, and standard errors. The last column shows the statistical evaluation of means between nuclei read from the table for the two-tailed t -test. The symbol * indicates a statistical difference in the mean values between groups

Parameters	Statistics	Small neurons (65)	Large neurons (59)
D_{bin}			
	Mean	1.861	1.830 [***small]
	SD	0.033	0.030
	SE	0.004	0.004
D_{bin}			
	Mean	1.069	1.067
	SD	0.019	0.019
	SE	0.002	0.002
D_{bin}			
	Mean	1.134	1.158 [**small]
	SD	0.045	0.044
	SE	0.006	0.005
Λ			
	Mean	0.78	0.90 [***small]
	SD	0.14	0.13
	SE	0.02	0.02

** $p < 0.01$ and *** $p < 0.001$

By looking at an image of a neuron, the use of the box-counting method allows for differentiation between neurons [10, 12, 14, 39], quantifying the size and complexity of the dendritic tree [25, 36]. The previous edition paid attention to the methodology of the box-counting method in order to obtain the correct BD value. Besides preprocessing tasks, such as image size, resolution, and rotation (Sects. 8.2.1 and 8.2.2), this text shows the significance of the BD calculated for different presentations of the same neuronal image.

This version of this chapter postulates the term “monofractal analysis.” Moreover, the text presents how the box-counting method can quantify four properties of a neuronal image: size, shape, complexity, and homogeneity. Quantification was performed on samples of three nuclei of the human cerebellum (Sect. 8.4), examining previously published results [15, 23, 44, 47]. In this regard, the results of this study either confirm or enhance the previous ones.

8.6 Conclusions

In conclusion, to calculate the FD of an object, it is not necessary to investigate its main fractal properties, as these properties are the main subject of fractal geometry. The text shows the box-counting method and its methodology in neuron

image analysis. Finally, it demonstrated how box dimension and lacunarity can successfully quantify four features of an image.

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Chapter 9

Neuronal Fractal Dynamics



Małgorzata Kołodziej and Przemysław Waliszewski

Abstract Synapse formation is a unique biological phenomenon. The molecular biological perspective of this phenomenon is different from the fractal geometrical one. However, these perspectives are not mutually exclusive and supplement each other. The cornerstone of the first one is a chain of biochemical reactions with the Markov property, that is, a deterministic, conditional, memoryless process ordered in time and in space, in which the consecutive stages are determined by the expression of some regulatory proteins. The coordination of molecular and cellular events leading to synapse formation occurs in fractal time space, that is, the space that is not only the arena of events but also actively influences those events. This time space emerges owing to coupling of time and space through nonlinear dynamics. The process of synapse formation possesses fractal dynamics with non-Gaussian distribution of probability and a reduced number of molecular Markov chains ready for transfer of biologically relevant information.

Keywords Neurons · Synapse · Differentiation · Dynamics · Gompertz growth · Fractal · Fractal time space · Persistent homology · Complexity

9.1 Synapse Formation from the Perspective of Molecular and Cellular Biology

The word synapse originates from the Greek words $\sigma\upsilon\nu$ (together) and $\acute{\alpha}\pi\tau\epsilon\iota\nu$ (to fasten). Indeed, a synapse enables transfer of electrical or chemical impulses between neurons within the nervous system or from neurons to target cells, such as muscle cells. A synapse is a complex molecular structure composed of an axon

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and the dendrites of an adjacent neuron. It possesses both a presynaptic and a postsynaptic membrane. There is a synaptic cleft between them. In fast electrical synapses, the presynaptic and postsynaptic membranes are connected by special molecular channels called gap junctions. These channels enable a direct flow of electric current from the presynaptic membrane to the postsynaptic one. The slower chemical synapses use a number of neurotransmitters to regulate the function of the postsynaptic neurons. These relatively simple chemical compounds are released from the axon membrane and bind to the specialized receptors in the postsynaptic membrane. In consequence, the postsynaptic neuron may change immediately its potential and initiate an electrical response. It may also activate some secondary molecular signaling pathways that inhibit or excite it.

A neuromuscular junction is an example of a specialized synapse that not only transfers electrochemical impulses but also activates some molecular events, leading to muscle contraction. A neuromuscular junction consists of a motor neuron, a myofiber, and a Schwann cell. In this kind of synapse, the impulse depolarizes the motor neuron by releasing the neurotransmitter, acetylcholine. Acetylcholine voyages crossways through the synaptic cleft and docks to its receptors in the sarcolemma. The muscle contraction begins after opening of the receptor ion channels. The entire synapse is covered by a myelin sheath provided by the Schwann cell that protects the junction [14].

The formation of synapses is a complex process that starts during embryogenesis and continues in a lifetime. Cell types involved in the synapse formation originate from different regions of the growing embryo. The myoblasts originating from the mesoderm form a multinucleated myotube. During the myotube formation, motoneurons from the neural tube initiate preliminary contacts with the myotube. The Schwann cells that originate from the neural apex are directed by the axons to their targets. After contact between the Schwann cell and axon is established, both cells create a loose, unmyelinated covering over the axons [17].

In the central nervous system, astrocytes are nonneural cells that regulate a number of activities in the brain, such as synaptic transmission, neurometabolism, plasticity of synapses, and cerebrovascular tone [17]. In particular, astrocytes secrete two protein factors that regulate the process of synapse formation [7]. The first one is known as thrombospondin (TSP). This factor determines the formation of morphologically normal but functionally silent synapses. The second factor, which has not been isolated yet, converts those silent synapses into fully functional ones [7]. In addition, the functional maturation of synapses is regulated directly by glia as “support cells.” TSP is also expressed in glia as an extracellular matrix protein that binds heparin. It was also demonstrated that TSP affects the relocation of synaptic proteins to new synapses rather than inducing their expression over there [7].

The specific patterning of synapse development at the neuromuscular junction shows that the majority of muscles are innervated at their centers. Although it may seem that the axons specifically target the midpoint of the myotube, a number of observations suggest that this is not a valid supposition. After the initial axonal contact, the newly formed myotube gates grow symmetrically from the point of

innervation. In combination with the fact that receptor mass is the result of axonal contact at its place of origin, the structural patterns of muscle fibers can be classified as both myotatic development and axonal innervation [17].

The first contact between a motoneuron and a myotube generates synaptic transmission, but the signal is very weak. One week later, a fully functional synapse is formed. This synapse plays a role in the development of new axons [9]. To form a presynaptic terminal, molecular signaling must go through a number of subcellular sections. In the soma, presynaptic proteins need to be synthesized, packaged together, and attached to microtubule motors for delivery through the axon. Within the axon, transport of presynaptic packages is controlled to confirm that developing synapses achieve a sufficient volume of factors. At individual axonal sites, extracellular interactions must be translated into intracellular signals that can incorporate mobile transport vesicles into the nascent presynaptic terminal. Once the initial recruitment process is complete, the components and subsequent functionality of the presynaptic terminals are constantly altered. For example, the axon of a motoneuron releases arginine that binds to a muscle-specific kinase receptor in the postsynaptic membrane. This binding induces the downstream activation of the cytoplasmatic protein rapsyn. Rapsyn contains a domain that is responsible for clustering of acetylcholine receptors in the postsynaptic membrane [14]. The coordination of all those processes in time and in space is the most mysterious part of the process of synapse formation [4].

The competition between axons is an activity-dependent process. It is based on synaptic pruning or synaptic elimination. It has been suggested that a strong synapse, which produces enough action potential or exhibits strong activity to depolarize the postsynaptic membrane, wards off the weaker axons [14]. The most important consequence of synaptogenesis is the ability of the motoneuron to discriminate between fast- and slow-twitch muscle fibers; fast-twitch muscle fibers are innervated by “fast” motoneurons, and slow-twitch muscle fibers are innervated by “slow” motoneurons. Here, we suppose that nonselective pathways indicate that the axons are led to their targets by the matrix through which they travel. Finally, the axons may innervate muscle fibers nonspecifically and cause the muscles to gain the characteristics of the axon that innervates them. In this path, a “fast” motoneuron can convert any muscle fiber into a fast-twitch muscle fiber [14]. These changes in receptor subunits lead to the growth of the postsynaptic cluster size and cluster development [2].

The main transmitter in the peripheral synapses is acetylcholine. Its counterpart in the central nervous system is glutamate and its receptors, especially the NMDA (*N*-methyl-D-aspartate) receptor. The activation of the NMDA receptor induces synaptogenesis over the activation of downstream products. The most important messenger in the brain and spinal cord is glycine, which docks on glycine receptors. Glycine receptors play an important role in the excitability of spinal cord and brain stem neurons [2]. Because of embryo development, some of these receptors' attributes undergo important modifications, resulting in main variation of their physiological functions such as switching of receptor structures. Previous data have

shown that development is associated with the conversion of the monomeric alpha or heteromeric alpha-2 beta receptors in immature neurons to the alpha-1 beta receptors in mature neurons. Together with these changes, the apparent receptor correspondence between glycine and strychnine, as well as that between $\text{Zn}(2+)$ and ethanol, increases during the development process. The established receptor shows a slow desensitizing current and a high sensitivity to modulation by protein kinase C. The high level of glycinergic transmission in juvenile spinal neurons modulates neuronal excitability, thus initiating membrane depolarization and alterations in intracellular calcium, which leads to chronic inhibition of glycinergic transmission, and initiates neurite outgrowth and changes in the level of synaptic transmission induced by GABA(A) and AMPA receptors. The plasticity of juvenile glycine receptors is associated with cytoskeleton dynamics [2].

The environment in which the neurons are located plays an important role in synaptogenesis. Brain-derived neurotrophic factor (BDNF), which is produced by the brain, regulates the process of synapse development, including synthesis of a transmitter, concentration of the transmitter in vesicles, and cholesterol biosynthesis, which is important for synaptogenesis. Indeed, BDNF-null mutants show a defect in neuronal growth and formation of synapses [15]. The second important molecule is neuroligin, which plays a role in cell adhesion in the postsynaptic membrane. A defect in genes encoding neuroligin leads to mental retardation and autism [22]. However, many of the signaling processes regulating synaptogenesis in the central nervous system can also be regulated by matrix metalloproteinases (MMPs) [6]. An important component of the controlled actin assembly, Abelson interacting protein-1 (Abi-1), was identified as a binding partner for the postsynaptic density (PSD) protein ProSAP2/Shank3. The dynamic and locally specific regulation of actin in the cytoskeleton is critical for the emergence of synaptic plasticity [13]. It has been demonstrated that the downregulation of Abi-1 by small-interfering RNA leads to excessive dendrite branching and a reduction in a number of synapses. The overexpression of Abi-1 has the opposite consequence. Abi-1 can play a role as a specific synapto-nuclear messenger and is involved in dendrite and synapse formation [13]. Furthermore, filopodia (dendritic spine morphology) play a role in synaptogenesis by initiating contact between the axons of other neurons. It has been reported that new neurons are contacted by axosomatic, axodendritic, and axospinous synapses. Dendritic spines synapse primarily on multiple synapse boutons that are already present in the synapse. The connectivity of new neurons continues to change until at least 2 months, long after the formation of the first dendritic protrusions [16]. The brain stimulation following birth, after weaning, or during adulthood leads to formation of numerous synapses [3]. It was also demonstrated that the subcellular anatomical characteristics of astrocytes in the brain enabled them with both identification and a response to changes in neuronal and synaptic activities [11].

9.2 Fractal Time Space in the Dynamic Process of Synapse Formation

The explanation how all the above-outlined molecular and cellular phenomena spanning different levels of the complex, hierarchical, dynamic cellular system are coordinated in time and in space is one of the greatest challenges of modern biological sciences. Here, the process of synapse formation is an interesting experimental model that may also stimulate theoretical research [7]. Molecular biological models depict synapse formation as a kind of Markov process, that is, a deterministic, conditional, memoryless process ordered in time and in space, in which the consecutive stages occur owing to a transition rule that depends only on the current state. Although these models are static, they suggest that some important molecular events occur in neurons in a collective manner, that is, a manner marked by similarity among or with the neurons of a given anatomical region.

The application of fractal calculus and the principles of fractal geometry brings about an utterly different perspective. Cells represent supramolecular dynamic systems that grow and interact in both space and time [18, 19, 20, 21]. Neuroanatomists have found that morphological irregularities in both neuron and neural network structures, which result from such interactions, can be measured by the spatial fractal dimension [12]. The existence of the spatial fractal dimension indicates the existence of some underlying emergent molecular and electrochemical processes in the neural tissue. In addition, experimental data obtained from the study of the cellular system P19/RAC 65 show that the dynamics of synapse formation *in vitro* can be described by a family of sigmoidal curves, such as the Gompertz one that is one of the solutions of Eq. 9.2 (see Appendix) [21]. This equation suggests that the process of synapse formation is a cooperative phenomenon, that is, the events occur not only in conjunction with others but also facilitate further evolution of the process. In this equation, the dynamics of synapse formation, the function of probability distribution, and the anharmonic potential are all coupled to each other. In the background, there is coupling of two fundamental dimensions that define time and space in which all events occur (see Appendix, Eqs. 9.1 and 9.6). It is worth noticing that the emerging time space is not just an arena for events and processes. This time space influences them actively. This fact is reflected by the presence of both temporal and spatial fractal dimensions in Eq. 9.6 (see Appendix).

Indeed, synapse formation requires long-range interactions and diffusion of some important gene products between neural cells. The interacting cells first create a dynamic system with its own attractor (i.e., a fragment of time and space where the dynamic processes occur and where no further evolution of the system is possible at all owing to the action of the intrasystemic forces (drift) unless some extrasystemic forces (diffusion) act upon it). The extrasystemic impulses and their interaction with a network of neurons assume further control of the process. In addition, cells differentiate into neurons and create synapses with a conjugated probability and non-Gaussian distribution rather than with a classical probability and Gaussian distribution (see Appendix, Eqs. 9.11–9.15). As a consequence of the Markovian

nature of the molecular circuits, entropy of such a chain of coupled molecular reactions is always lower than that of a set of random and independent biochemical reactions (see Appendix, Eqs. 9.7–9.10). Moreover, a number of molecular circuits that can actively transfer biologically relevant information in a given time is much lower than the total number of such circuits (see Appendix, Eq. 9.16).

Benoit Mandelbrot was one of the first researchers who noticed that the random walk model can describe a wide range of neuronal activities in terms of two parameters [8]. Furthermore, the analysis of the inter-spike intervals of the firing cortex neurons reveals different non-Gaussian probability distributions, such as exponential, gamma, log-normal, and power-law. The coexistence of different probability distributions unveils different stochastic processes, including molecular ones that stand behind those distributions. Fractal scaling in the dynamics of those intervals is correlated with the coupling between neuron firing and movement of muscle groups. These neuron firings reveal long-range temporal correlations. The correlations are missing if the firings are not associated with some movements [10]. Fractal time space is a prerequisite condition for the phenomenon of a dynamic coalition of neurons. This coalition is based on the collective behavior of neurons that participate in generating some muscular movements by entering the coalition and forming a network of firing neurons or leaving it for some time. In other words, the existence of fractal time space enables both plasticity and adaptation of the neuronal network to different situations.

The process of synapse formation can also be visualized at the macroscale as a geometric simplicial complex and analyzed by persistent homology. This approach has the ability to identify important topological features in data across multiple scales [5]. Recently, the persistence of homology fractal dimension has been defined using the sum of the lengths of the intervals in the i -dimensional persistent homology of the filtered simplicial complex [1]. The probabilistic fractal dimension and the metric space global capacity fractal dimension complement each other and combine both topological and geometric perspectives for the comprehensive analysis of synapse formation.

A.1 Appendix

If one assumes that the spatial variable x and the temporal variable t are coupled to each other in a linear manner into a single, complex spatiotemporal variable θ such that

$$\theta = \mu x + t \quad (9.1)$$

then the Gompertz function, the function of probability distribution $P(\theta)$, the anharmonic potential $U(\theta)$, and the diffusion coefficient D are all related to each other through the one-dimensional differential operator (9.2). This operator contains the function of probability distribution [21]:

$$-\frac{1}{D} \frac{\partial^2 P(\theta)}{\partial \theta^2} + \frac{D}{4} P(\theta) + U(\theta) P(\theta) = 0 \quad (9.2)$$

This linear coupling of variables can also be defined as a function with both spatial and temporal fractal dimensions. It is known from experimental data from the in vitro cellular system of P19/RAC 65 that the number of cells (or their volume) also changes in time t according to the Gompertzian function $f(t)$ [19]. The volume of the spheroid V is given by the equation:

$$V = V_k F(t_0) e^{a(1-e^{-bt})} = V_k f(t_0) a t^{b_t} = V_0 a t^{b_t} \quad (9.3)$$

where V_k is the mean volume of a single cell, n stands for the number of cells in the spheroid, and the Gompertzian function can be fitted with the fractal function $f(t) = a t^{b_t}$ with extremely high accuracy; for a coefficient of the nonlinear regression $R \gg 0.95$ for $n \geq 100$ pairs of coordinates), a stands for the scaling coefficient, b_t is the temporal fractal dimension, and t is scalar time.

The volume V of the spheroid can also be expressed as a function of the scalar geometrical variable x (i.e., the radius of a family of the concentric spheres covering the entire spheroid) by Eq. (9.4):

$$V = a_1 x^{b_s} \quad (9.4)$$

where a_1 stands for the scaling coefficient, b_s is the spatial fractal dimension after scalar time t_1 , and x is the scalar, geometrical variable, which locates an effect in space.

Hence, we get Eq. (9.5):

$$V = a_1 x^{b_s} = V_0 a t^{b_t} = a_0 x^{b_{s0}} a t^{b_t} \quad (9.5)$$

where a , a_0 , and a_1 stand for the scaling coefficients, b_t is the temporal fractal dimension, b_{s0} and b_s are the spatial fractal dimensions after time t_0 and t , respectively, and x is the geometrical variable.

Finally, Eq. (9.6) relates space and time in which proliferation, differentiation, and synapse formation occur. This equation defines the geometrical variable x as a function of the scalar time t and both temporal and spatial fractal dimensions

$$\ln x = \frac{1}{b_s - b_{s0}} \ln \frac{a_0 a}{a_1} + \frac{b_t}{b_s - b_{s0}} \ln t \quad (9.6)$$

where t stands for scalar time, x is the geometrical variable, b_s is the spatial fractal dimension, and b_t is the temporal fractal dimension.

A.1.1 Entropy and Dynamics of Synapse Formation in Fractal Time Space

It is worth noticing that the assumed Markov model of molecular interactions within differentiating neurons implies at least three important consequences. First, the entropy (i.e., missing information) H_M of the Markov chain of the coupled molecular reactions is always lower than the entropy of a set of random and independent biochemical reactions H_R . Indeed, entropy is defined as the expected value of missing information H_p :

$$H_P = H(X) = - \sum_{j=1}^N p_j \log p_j \quad (9.7)$$

where $p = (p_1, p_2, \dots, p_j)$, $j \in N$, is the probability density function over a generic variable X , and if $p_j = 0$, then $H_p = 0$ and $\log$ is a natural logarithm providing a unit of measure.

Hence, the conditional entropy $H(X_k | Y_{k-1})$ of the X_k reaction stands for which conditional information is determined when the state $Y_{k-1} = i$, is given by the following equation:

$$H(X_k | Y_{k-1} = i) = - \sum_j p_{ij} \log p_{ij} \quad (9.8)$$

The conditional entropy of the Markov chain H_C is given by (9.9):

$$H_C = H(X_k | X_{k-1}) = - \sum_i p_i \sum_j p_{ij} \log p_{ij} \quad (9.9)$$

Finally, we get Eq. (9.10) for the n first steps of the Markov chain $X_1, X_2, \dots, X_n$ from (9.7), (9.8), and (9.9), the principle of additivity of independent random events, and from the analog principle for the conditional probabilities, we get:

$$\begin{aligned} H_M &= H(X_1) + \sum_{k=2}^{k=n} H(X_k | X_{k-1}) = - \sum p_j \log p_j + (n-1) H_C < n H_P \\ &= - \sum_j p_j \log p_j = H_R \end{aligned} \quad (9.10)$$

Second, the Gompertzian dynamics of molecular cellular growth can be normalized, i.e., the growth dynamics of various tissue systems can be described by a single normalized Gompertz function $f_N(t)$ (9.11). In fact, this normalized Gompertz

function is both a dynamics function $f_N(t)$ and a probability function $p_N(t)$ (for details, see Waliszewski [19]).

$$f_N(t) = e^{-e^{(-bt)}} = p_N(t) \quad (9.11)$$

Consider a coupling of probability function $p_N(t)$ and anti-probability function $-\log p_N(t)$, where $r = b$:

$$\frac{dp_N(t)}{dt} = -rp_N(t) \log p_N(t) \quad (9.12)$$

This equation defines the relationship between entropy $H(t)$ and the normalized Gompertzian dynamics of growth $p_N(t)$:

$$p_N(t) = \int \frac{\partial p_N(t)}{\partial t} dt = -r \int p_N(t) \log p_N(t) dt = rH(t) \quad (9.13)$$

Finally, from (9.11) and (9.12), we get (9.14):

$$H(t)_{\text{Gompertz}} = \frac{1}{b} e^{-e^{(-bt)}} \quad (9.14)$$

According to the Shannon theorem, of all the continuous distribution densities for which the standard deviation exists and is fixed, the Gaussian, (i.e., normal) distribution has the maximum value of entropy H :

$$H_{\text{Gauss}} = - \int_{-\infty}^{\infty} \frac{e^{-\frac{t^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \log \frac{e^{-\frac{t^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} dt = \frac{1}{2} \log(2\pi e\sigma^2) \quad (9.15)$$

In the case of growing supramolecular cellular system such as neurons, entropy or missing information $H(t)$ is a function of time related to the dynamic function of growth in fractal space time. For $b = 1$, both the normalized Gompertz function (9.11) and the entropy function (9.14) overlap each other. However, $b \ll 1$ for the majority of cellular systems. The distribution of probability is in such cases is non-Gaussian.

Third, there is a relationship between the number of elements in the Markov chain and entropy. If $M_p(n)$ stands for a number of Markov chains of length n with the total probability p , $0 < p < 1$, then there exists the same limit for each probability p that equals entropy H :

$$\lim_{n \rightarrow \infty} \frac{\log M_p(n)}{n} = H \quad (9.16)$$

If the total number of states of the supramolecular cellular system equals 2^m , then the number of molecular reactions interconnected in the Markov chains of length n is 2^{nm} . It is clear from (9.16) that only 2^{nH} molecular Markov chains with probability $1-\varepsilon$, $\varepsilon > 0$ will be involved in the transfer of biologically relevant information.

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Chapter 10

A Self-Similarity Logic May Shape the Organization of the Nervous System



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Abstract From the morphological point of view, the nervous system exhibits a fractal, self-similar geometry at various levels of observations, from single cells up to cell networks. From the functional point of view, it is characterized by a hierarchical organization in which self-similar structures (networks) of different miniaturizations are nested within each other. In particular, neuronal networks, interconnected to form neuronal systems, are formed by neurons, which operate thanks to their molecular networks, mainly having proteins as components that via protein–protein interactions can be assembled in multimeric complexes working as micro-devices. On this basis, the term “self-similarity logic” was introduced to describe a nested organization where, at the various levels, almost the same rules (logic) to perform operations are used. Self-similarity and self-similarity logic both appear to be intimately linked to the biophysical evidence for the nervous system being a pattern-forming system that can flexibly switch from one coherent state to another. Thus, they can represent the key concepts to describe its complexity and its concerted, holistic behavior.

Keywords Fractal geometry · Cell networks · Molecular networks · Neuroanatomy · Neurodynamics

10.1 Introduction

According to Jacob [73], from a structural standpoint, living systems are characterized by a hierarchical pattern of organization in which structures are nested within one another. It has also been pointed out that this conceptual proposal by

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Jacob can be applied not only to living individual but also, in a complex organism, to its main organs [2, 5]. In fact, as stated by Grizzi and Chiriva-Internati [53], all anatomical systems exhibit the pivotal property to form multilevel structures, each of which forms “a whole in relation to its parts and is simultaneously part of a larger whole” [53]. A key morphological consequence of this feature is that the irregular and complex shapes in which the structural detail increases (at least for quite a wide range of scales) with increasing magnification or resolution [18]) characterize biological structures (but, more in general, the majority of natural objects). Thus, the conventional (Euclidean) view that natural structures have a well-defined form, which can be determined at some characteristic scale, appears to be just an approximation of the reality. The two related problems of defining the concept of a “form” of a particular biological structure and that of providing its description in quantitative terms have been deeply debated among morphologists since the classic work of Thompson [118] and are still under careful scrutiny. A breakthrough in this field occurred with the development of “fractal geometry” by Benoit Mandelbrot [91, 92], based on the previous mathematical achievements of Poincaré and Cantor. From a mathematical point of view, fractal objects are the result of iterative processes, a complex irregular shape characterizes them, and their measured properties depend on the scale at which they are measured. The most important and characterizing feature of fractal objects, however, is “self-similarity,” a term indicating that they are composed of parts that are smaller, exact duplications of the whole object over infinity of length scales. In other words, geometrical configurations characterizing a fractal object can be found again and again as far as magnification is increased (Fig. 10.1a). For what it concerns biological morphology, this mathematical framework is of key importance for two reasons. On one side, it provides a conceptual support to describe the complexity of the form exhibited by biological objects. Of course, natural objects can at best be a quasi-fractal, exhibiting self-similarity only within a limited scale range and only statistically, since small details of a given anatomical system are rarely identical copies of the whole system. Nevertheless, they are often characterized by the same architectural scheme with the same structural complexity (Fig. 10.1b). Examples include the bronchial tree, the ductal system of glands, the biliary tree of the liver, and the vascular system [33, 34, 60, 88]. On the other side, fractal geometry has also led to the definition of parameters, thus allowing a quantitative characterization of complex shapes (see [42, 67, 108]). “Fractal dimension” (D), for instance, measures the rate of addition of structural details with increasing magnification, scale, or resolution [37]. Thus, it represents an estimator of morphological complexity: the more irregular an object, the higher its D value [92]. It can be quite easily evaluated by a variety of methods, with the “box-counting method” (see [17, 108]) being the most widely used in biological applications. The information provided by D is often complemented by the measurement of another fractal parameter, called “lacunarity.” In a general sense, this parameter can be considered as a measure of the nonuniformity (heterogeneity) of a structure, potentially useful to characterize how

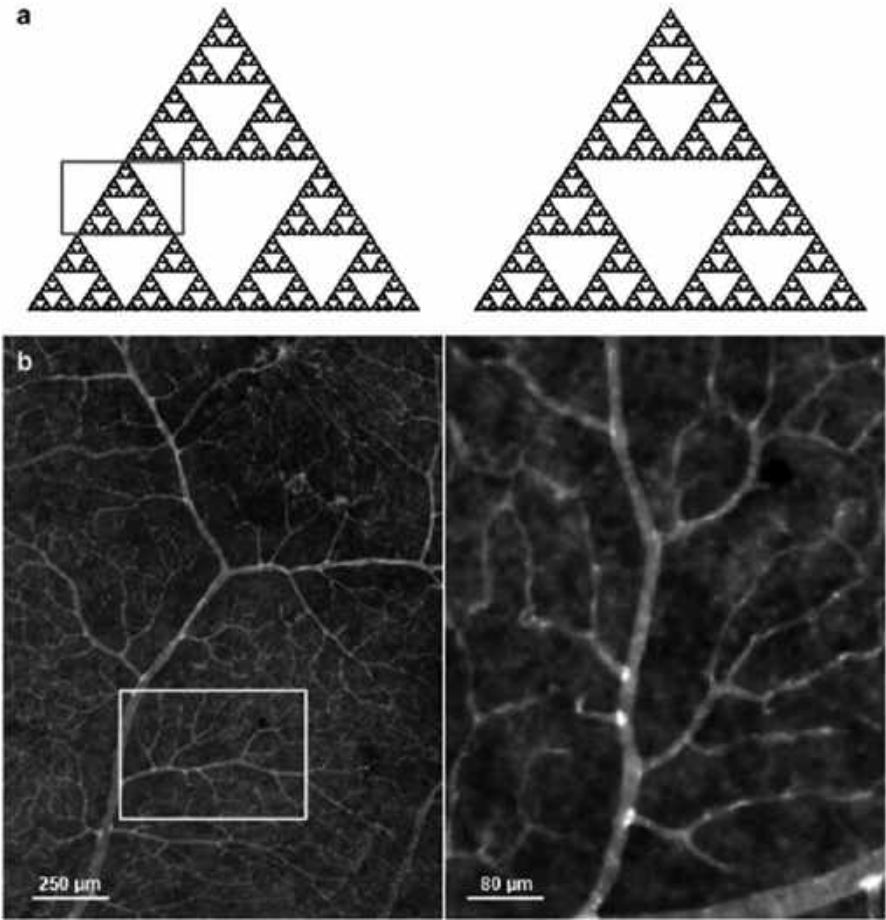


Fig. 10.1 (a) A classical mathematical fractal, the Sierpiński triangle. As illustrated, each part is the exact duplication of the whole object over infinity of length scales. (b) A natural structure, the vascular network of the retina, exhibiting self-similarity. It can be considered a quasi-fractal: self-similarity only occurs within a limited scale range and only statistically since structural details are not identical copies of the whole system. Nevertheless, they are characterized by the same architectural scheme

it fills the available space [108]. Roughly speaking, lacunarity exhibits higher values when the analyzed structure has large gaps. Several further methods and parameters based on fractal geometry (such as Hurst coefficient and detrended fluctuation analysis) have been proposed for the analysis of natural objects (see [46, 115]), and in the last decades, the use of fractal geometry-based tools has significantly expanded to describe the morphology of biological structures and the changes they

undergo under physiological or pathological processes. Examples include, among others, the application of fractal principles to measure the irregular and complex membrane ultrastructure of cells at specific functional and pathological stages [87, 89], changes in the self-organization of endothelial cells *in vitro* induced by anti- or pro-angiogenic factors [56], characterization of alveolar shape in a normal lung and in emphysema [14], processes of vascular remodeling [55, 60], changes in the morphology and spatial distribution of the connective tissue in the human carotid body during aging [61], chromatin architecture [85], and consequences of its changes in terms of chromosomal alterations [47].

The morphology of living organisms, however, follows a nested organization, since a hierarchical scheme can also be recognized at the functional level, with each physiological function being the combined result of processes occurring at the system, tissue, cell, and molecular levels. In this respect, it has also been proposed that some auto-similarity could prevail at each nested level [4], meaning that the different involved elements can interact, irrespective of their degree of miniaturization, according to similar rules. A well-known example is the feedback mechanism found in metabolic molecular networks where the amount of the final product inhibits its further production by acting directly upon a certain rate-limiting step. Feedback, however, is also found at the organ and system levels, e.g., in the circuits that maintain muscle length, or in arterial blood pressure control. For this reason, the term “self-similarity logic” has been suggested to convey the concept of a multilevel structural organization in which highly similar functional rules (logic) apply at each level [8]. This view is in agreement with the statement of Russell and Aloy [105] that a key concept in biological system design is modularity: Nature duplicates and reuses existing parts and design principles again and again.

This essay illustrates the hypothesis that both the structural (geometrical) “self-similarity” concept and the broader “self-similarity logic” principle are particularly apparent in the nervous system, likely providing a unitary conceptual scheme to describe many aspects of its morphological and functional organization.

10.2 Structural Self-Similarity of the Nervous System

As pointed out by Di Ieva et al. [39], the application of Euclidean geometry to neuroanatomy is strongly limited by several factors, such as the natural complexity of the brain, its hierarchical structure, and the sophisticated topological architecture of the neurons organized in micro- and macro-networks. For this reason, the introduction of new geometrical frameworks, such as fractal geometry, and the tools they provide to quantitatively characterize the shape of the nervous structures, have led to a rapid increase in the amount of data supporting the prevalence of properties such as self-similarity in the brain at various levels of observation. They will be briefly reviewed in the sections that follow.

10.2.1 Cell Level: Complex Geometry of Neurons and Glial Cells

As illustrated in Fig. 10.2a, neurons are characterized by quite a complex morphology. In his foundational work, *The Fractal Geometry of Nature*, Mandelbrot [92] specifically mentioned the Purkinje neurons of the cerebellum and wrote, “it would be nice if neurons turned out to be fractal.” Studies undertaken to test this hypothesis were in general supportive (see [39, 126] for more comprehensive reviews as well as Chaps. 7 and 8). Fractal shapes in the branching patterns of dendritic trees were identified in retinal neurons [26] and in thalamic neurons [79]. Fractal analysis allowed a distinct differentiation of neuron types in the different laminae [96] of the dorsal horn of the spinal cord, and variations in the fractal structure of the pyramidal neuron dendrite branching were associated with differences in the functional capacity among the regions of the visual cortex [131]. Wen et al. [124] provided a clear-cut indication of self-similarity in neurons by examining the connectivity repertoire of basal dendrite arbors of more than 2000 pyramidal neurons. This study identified a universal statistical process underlying the construction of dendrite arbors, which led to structures in which fragments of the arbor were statistically similar to the entire arbor, thus displaying self-similarity. Additional evidence came from image analysis, in particular from 3D reconstruction techniques using scanning electron microscopy or confocal microscopy [106].

More recently, a specific analysis has been reported on the origin of neuron fractality and on its impact on functionality [109]. The results indicated that the fractal character of dendrites is determined not only by the range of branch lengths (as in conventional trees) but also by the ways in which they fork and weave through space. The combination of these features allows to generate a fractal-like structure, thus achieving maximal connectivity with minimal costs associated with building and operating the neuron’s branches.

A morphology characterized by self-similarity properties, however, is not limited to neurons. Various mammalian astroglial cell types have been classified by means of fractal analysis [100], and lacunarity and fractal dimension have been used to show the physiological changes of neuronal and astroglial structures in the process of aging, whereby the fractal dimension decreases and lacunarity increases in neurons, while the contrary occurs for astrocytes [58, 116]. As far as other glial cell types are concerned, the sequence of oligodendrocyte developmental stages parallels changes in fractal dimension [20], and fractal analysis of cell ramification and space-filling patterns differentiate microglial cells into two categories, depending on whether their fractal dimension did or did not increase after brain injury [110].

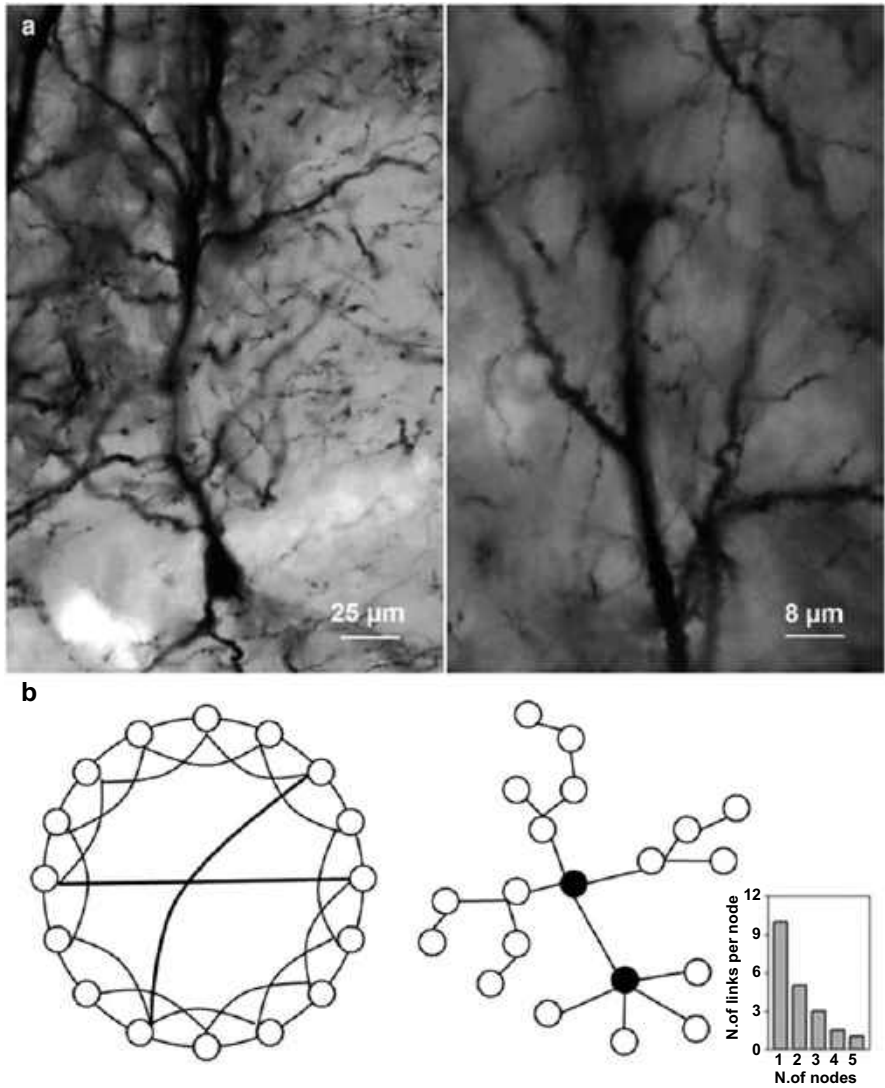


Fig. 10.2 (a) A Golgi-stained hippocampal pyramidal neuron. As illustrated in the right panel, fragments of the dendritic arbor are statistically similar to the entire arbor, thus displaying self-similarity. (b) A schematic representation of the two network architectures identified in the CNS (see text). The left panel illustrates a “small-world network” and the right panel a “scale-free network.” In the latter, some nodes (hubs of connectivity) have a high number of connections to other nodes, whereas most nodes have just a few connections. More specifically, the distribution of the links among the nodes of the network follows a power law (shown in the inset)

10.2.2 *Tissue Level*

10.2.2.1 Central Nervous System

The central nervous system (CNS) can be described as a huge network of cells interacting through communication pathways. It has been proposed that two main modes (see [9, 63] for reviews) of communication (wiring transmission (WT) and volume transmission (VT)) could be distinguished in the CNS. The first is characterized by a physical “wire” connecting the nodes of the network (as in the synaptic transmission between neurons), while the second concerns the three-dimensional diffusion of signals in the extracellular space for a distance greater than the synaptic cleft. Thus, this communication mode uses the several often spatially divergent tortuous channels made by the clefts (about 20 nm in diameter; [30]) between cells and filled with extracellular fluid and the extracellular matrix. In the context of this discussion, quite recent data suggesting that the geometry exhibited by the CNS extracellular space is characterized by fractal features [95, 112] are of substantial interest. VT is characterized by a very high divergence, since one source usually sends signals to a great many targets, including not only neurons but also other types of cells in the CNS, such as astrocytes [117] and microglial cells [43]. This leads to the formation of “complex cellular networks,” exchanging signals in a certain volume of brain tissue and, due to this cross talk, integrating their activity [3]. In this context, the relationship between neurons and astrocytes is the best-studied (see Fellin and Carmignoto [45]), and recent findings have highlighted the involvement of “neuron–astroglial interactions” in the higher brain functions. As a matter of fact, the concept of “tripartite synapse” has been introduced, since, in most glutamatergic central synapses, the extremity of a protoplasmic astrocyte process wraps the synaptic cleft. It should be noted that astrocytes express membrane receptors to neurotransmitters and can release their own chemical messengers (gliotransmitters). Thus, they establish a cross talk with both pre- and postsynaptic neurons. Several astrocytes participate in this functional organization, coupled to each other by gap junctions (see [25]) and communicating by “calcium waves” [98]. Thus, neuro-astroglial networks do exist, which not only control dynamic glucose delivery [103] but also participate in cognitive functions [102].

The architecture of the CNS extends over a range of up to five orders of magnitude of scales: from microns for cell structures at one end to centimeters for inter-areal neuronal connections at the other. From the morphological point of view, the CNS appears to be strictly self-similar (i.e., a quasi-fractal) at a low scale, as shown by studies on the shape of neuronal and glial elements (see Sect. 10.2.1). As far as the neuronal networks are concerned, the connections between different cortical areas were shown to possess an organization in the form of “small-world networks” [122], forming clusters of nearby cortical areas with short links, which, in turn, have long-range connections to other clusters [113, 114]. Within clusters, functional magnetic imaging identified a network topology of the type

called “scale-free” [40], in which some nodes (hubs of connectivity) have a high number of connections to other nodes, whereas most nodes have just a handful. Using the Human Connectome Project dataset (www.humanconnectome.org) [120], these nontrivial architectures have been recently confirmed to characterize the human CNS at the individual level [84].

Both the abovementioned types of network organizations (see Fig. 10.2b) display quite a high level of geometrical self-similarity [111], suggesting that the CNS (at least for its neural component) can be considered self-similar over all its length scales [130].

10.2.2.2 Peripheral Nervous System

No studies specifically aimed at analyzing the overall architecture of the peripheral nervous system (PNS) in terms of self-similarity and fractal properties are currently available. It should be outlined, however, that quite recent studies have highlighted a close structural and developmental relationship between blood vessels and nerves [24, 41, 74, 83]. This association seems to be of particular relevance in patterning the peripheral nervous system. The molecular mechanisms regulating the common wiring of nerves and blood vessels have attracted considerable interest over the past decade. Some common morphogenic signals and mechanisms have been recognized that direct formation of either structures (see [101]). Indeed, axon guidance molecules, such as semaphorins, ephrins, slits, and netrins, also function as angiogenic and vessel-guiding factors. Angiogenic growth factors, such as vascular endothelial growth factor (VEGF), express neurogenic potential as well. This assumption may explain why in peripheral tissues, blood vessels are often aligned with nerves and display a similar branching pattern [97]. On the one hand, endothelial cells produce signals, such as endothelin-3, artemin, and neurotrophin-3, which guide axons to track alongside developing vessels [71, 81]. On the other hand, nerves may also produce signals such as VEGF-A to attract blood vessels and stimulate the surviving endothelial cells.

This mutual structural relationship between the two systems is of particular importance for our discussion. In fact, the question about whether vascular networks could be considered strictly self-similar structures was addressed by quite a large number of studies (see [60]). Lorthois and Cassot [86] provided a discussion of the published results, together with a new multiscale analysis performed on high-resolution image datasets of vascular networks [27]. Their work demonstrated that healthy vascular structures are a superposition of two components: at low scale, a non-fractal “mesh-like” capillary network, which becomes homogeneous and space-filling over a cutoff length of the order of its characteristic length (25–75 μm), and a “treelike” distribution network, which can be demonstrated to be quasi-fractal (i.e., strictly self-similar) for quite a wide range of length scales. Thus, the observed structural association between peripheral nerves and the “treelike” component of the vascular system legitimates the hypothesis of a self-similar architecture for the PNS as well.

10.3 A Self-Similarity Logic Drives the Functional Features of the CNS

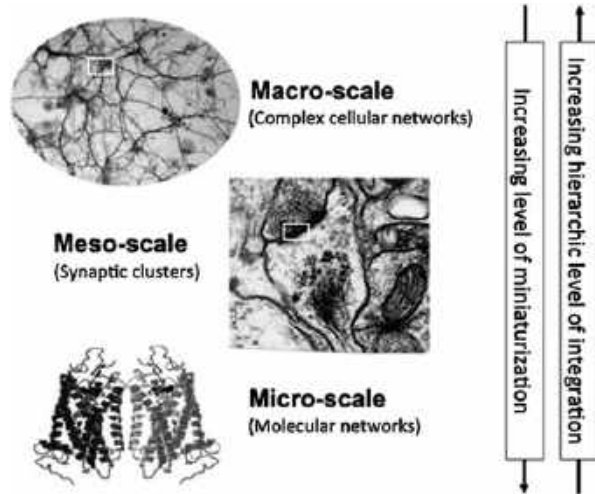
From a functional point of view, it has been observed that the CNS parenchyma self-organizes in functional units. The best-known examples are the cortical columns (see [99]) consisting of an array of cooperating neuronal groups extending radially across the cortical cellular layers and representing units of operation. As far as the anatomical boundaries of these units are concerned, they appear loosely delimited in morphological terms, since they are dynamic entities changing according to functional needs. A role in delimiting them is probably played by astrocytes, since the astroglial cells, especially in mammalian brains, define the microarchitecture of the parenchyma by dividing the gray matter into relatively independent structural units through the process known as “tiling” [23].

A further characteristic of these basic units, important for our discussion, is suggested by a very simple observation. CNS functions (as, for instance, motor control) can indeed be studied either as the output of the entire system (e.g., by means of studies on reflex responses) or of some of its regions (e.g., by extracellular electrophysiological recordings from the motor areas) or even at the level of single neurons (e.g., by intracellular electrophysiological recordings from spinal motor neurons). These different approaches are not simply a consequence of the increased analytical power of the experimental tools employed, but rather they depend on an intrinsic characteristic of the system. In fact, the CNS is the holistic assembly of a multilevel functional organization. In other words, a hierarchical criterion holds meaning that system components of different miniaturizations can not only be arranged to work together either in parallel or in series but also nested within each other [35]. For this reason, a structure of multiple layers encased one within the other like “Russian Matryoshka dolls” has been argued for the basic functional units, and the term “functional module” (FM) has been proposed to indicate them [8].

In particular, it has been suggested [4, 121] that in each FM, at least at the macro-, meso-, and microscale levels, could be recognized (see Fig. 10.3).

- The macroscale level is a level in which it is possible to recognize neuronal networks and complex cellular networks [3].
- The mesoscale level is the level of single neurons and, in particular, of the so-called synaptic clusters (SCs), in which multiple synapses act cooperatively to modulate their strength [52, 107]. SCs are often organized around the dendritic spines and partially isolated from the surrounding environment by glial cells [36, 52].
- At the microscale level are the molecular networks, made by molecules that function as a metabolic and/or regulatory signaling pathway in a cell [21]. Of particular interest are the “receptor mosaics,” i.e., macromolecular complexes formed at the membrane level by G protein-coupled receptors [48] as a conse-

Fig. 10.3 A schematic representation of the hierarchical organization of the nervous system. The three main miniaturization levels, i.e., the macro-, meso-, and microscale, are illustrated



quence of direct allosteric receptor–receptor interactions (see Guidolin et al. [62] for a review).

As pointed out by some authors, the same principles could be applied across the different levels of organization to account for the flux of dynamically emergent macrostate brain configurations [8, 125]. More specifically, processes resulting in a “network dynamics” [75, 93] among interacting elements operate at each level of organization, leading to the emergence of “self-similarity logic” as a characteristic functional feature of the CNS. Furthermore, it is noteworthy that a common rule also seems to exist at all levels for what it concerns the remodeling processes in the CNS. This appears to be mainly based on the self-organization of the existing components to form a new complex capable of new “emergent” properties. These aspects will be discussed in the sections that follow.

10.3.1 *Interaction-Dominant Dynamics in the CNS*

From a general point of view, the types of dynamics that a biological system exhibits can be classified into two broad classes [16]. In component-dominant dynamics, behavior is the product of a rigidly delineated architecture of modules, each with predetermined functions. On the contrary, interaction-dominant dynamics is crucially based on the plasticity of the system components and on the network of communication processes among these components. The nervous system can be better described as an interaction-dominant dynamics system where interaction processes alter the integrative action of the single components and where it is difficult, and sometimes impossible, to assign tightly defined and unique roles to each specific component.

For what it concerns such a complex dynamics of the CNS, the best-investigated level is the neural network level. Information handling by neural networks has a long story behind it. In this regard, it is enough to cite the classical article by McCulloch and Pitts [93] and the further development that followed (see [68] for a historical review), which established that from networks of abstract models of neurons as switching devices, quite complex behavior can emerge.

In a network, processing is carried out by a usually large number of simple processing elements (called nodes), having a nonlinear response function [72]. Each node receives input (excitatory or inhibitory) from some number of other nodes, responds to that input according to its response function, and in turn excites or inhibits other nodes to which it is connected. Thus, every input to the network will result in a flow of collective configurations of the nodes, often converging to some stable pattern (also called the “attractor”) that represents the response of the network. There are some key characteristics worth noting. First, what such a system “knows” is essentially captured by the pattern of connections and the efficiency associated with each of them [80]. Second, rather than using symbolic representations, the “vocabulary” of such a system consists of patterns of activation across different units. Furthermore, what made this type of dynamics so intriguing was the possibility of “learning procedures” [104], by which the network would adjust the connection efficiencies in small incremental steps (for instance, based on examples of a target behavior) in such a way that over time the network’s response accuracy would improve. Thus, the behavior emerging in a network relies on a distributed rather than a centralized control.

It has to be pointed out that the identification of different classes of VT signals (see [9] for a review), including chemicals (such as neurotransmitters, ions, gases, and enzymes) and physical signals (such as electrotonic currents, temperature gradients, and pressure waves generated by arterial pulses), opens the possibility that the above-described dynamics could not be limited to neural networks but could also drive the behavior of the “complex cellular networks” (see Sect. 10.2.2.1), in particular, of the neuro-astroglial networks [59]. In this context, an organizational aspect deserving consideration concerns the possibility (see [3]) that the communication pathway between a signal source and its targets could be modulated by a “modifier” responding to teaching signals that could originate, for instance, from environmental inputs. Such a modifier could be spatially situated along a VT diffusion pathway and/or an astrocyte network [51].

A network dynamics, however, also appears to characterize the meso- and microscale levels of brain organization. At the mesoscale level, the abovementioned synaptic clusters represent the functional bridge between the cellular networks and the molecular level. They have been shown to be very plastic entities from both structural [70] and functional [123] points of view. Notably, it has been reported that plastic changes induced by long-term potentiation (LTP) at one synaptic contact lower the threshold for the induction of LTP at neighboring synapses at a stimulation strength that did not cause any plastic changes under control conditions [52]. Gally et al. [49] described a simple mechanism allowing such a temporal correlation between synaptic sites (see also [57]). It was based on a simple network dynamics in

which the release from the depolarized postsynaptic spine of a diffusible substance (such as NO) was capable of acting as a signal returning from postsynaptic to both presynaptic and postsynaptic sites within some volume of the tissue. The production of the diffusible substance at a particular time was a function of its activity at that time, and the movement of the substance by diffusion provided the possibility of establishing temporal correlations between the activities of the neighboring sites. For what it concerns the microscale level, in recent years, increasing experimental evidence has supported the hypothesis that G protein-coupled receptors can form high-order receptor oligomers at the cell membrane (see [62, 63] for reviews). The existence of these supramolecular complexes is considered of particular importance because it allows the emergence of integrative functions performed by a protein aggregate as a whole. In fact, owing to receptor–receptor interactions in a suitable network dynamics, a configuration change of a given receptor will transform the probability of changing the configuration for the adjacent receptors, and the effect will propagate throughout the cluster, leading to a complex collective behavior and to an integrated regulation of multiple effectors [6]. It was suggested that this molecular mechanism might also lead to a transient and/or permanent change of the synaptic efficacy and then contribute to memory storage and engram formation (see [57] for a review).

Thus, network dynamics appears a characterizing feature of almost all the levels of the functional organization of the nervous system, providing the first key element of the “self-similarity logic” that shapes its design from the functional point of view (see [82]). This point will be further illustrated in the next sections by taking into consideration some basic operations performed by CNS networks according to Charles Sherrington’s studies (see [7] for a review) and that can be realized at different miniaturization levels.

10.3.1.1 The Concept of “Fringe”

A fringe is an area of overlap between two networks and can lead to facilitation. As shown in Fig. 10.4a, if neuron B has a low discharge rate, neuron B1 does not fire, but if neuron A fires at high frequency, neuron B1’s excitability increases and it is easier for the activity of neuron B to excite B1. Neuron B1 is said to be in the subliminal facilitation fringe of A.

As illustrated in Fig. 10.4b for receptor complexes, the concept of fringe (as other classical types of controls such as feedback or feedforward and spatial or temporal summation) can also be used to describe some significant behavior of molecular networks. The concept of fringe is of particular importance since it can suggest mechanisms that at the molecular network level regulate the cross talk between pathways allowing the production of complex cellular responses [3].

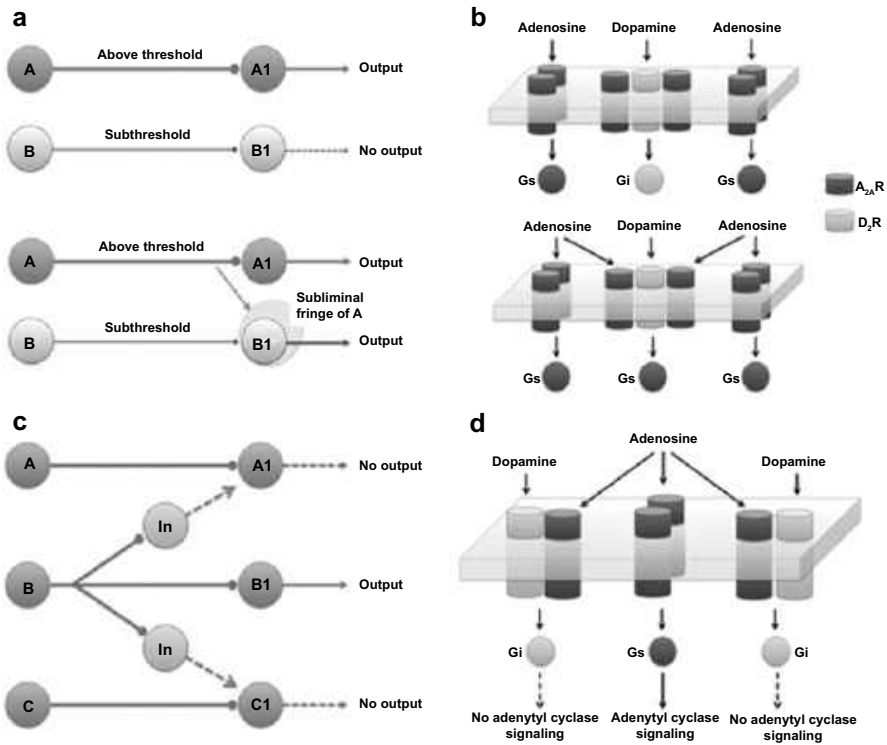


Fig. 10.4 (a) A schematic representation of the concept of fringe. Two input neurons, A and B, directly acting on two output neurons (A1 and B1, respectively) are considered. It is surmised that only A provides an above-threshold input to its target neuron (upper panel). However, if it fires at a high frequency, then it can lower the threshold for the firing of B1 (lower panel). (b) The same basic scheme is applied at the microscale to receptor mosaics formed by A₂A and D₂ receptors with reciprocal inhibitory activity (see Guidolin et al. [62]). A low concentration of adenosine can activate only Gs-mediated signaling from A₂A homodimers (upper panel). However, high concentrations of adenosine can induce, via receptor–receptor interactions, an inhibition of the D₂ receptor in the heterotrimer and its shift from Gi- to Gs-mediated signaling (lower panel). (c) Lateral inhibition mediated by interneurons (In) leads to focus the activity of a neural circuit in selected groups of neurons. (d) A possible lateral inhibition process at the microscale level in a system involving A₂A/D₂ heterodimers and A₂A homodimers. Activation of D₂ receptors by dopamine leads to a reduction in the affinity of A₂A receptors in the heterodimers and, hence, in a sharpening of the Gs-mediated signaling

10.3.1.2 The Concept of “Lateral Inhibition”

Figure 10.4c illustrates a classical representation of lateral inhibition for neural circuits (see Bear et al. [19]) that can be transferred (Fig. 10.4d) to molecular networks. Thus, besides “physical compartmentalization” that allows a very efficient spatial segregation of signaling pathways, a functional segregation, via lateral inhibition, can also produce a similar effect.

The fact that a similar organization of the circuit, as in the case of fringe, can be recognized for lateral inhibition at both the neural network and at molecular network levels underlines once again the potential heuristic value of “self-similarity logic.”

10.3.2 Remodeling Processes in the Nervous System

The experience-based reshaping of the brain structures [94] is another general feature that deeply characterizes the nervous system, and a “self-similarity logic” principle appears as a useful descriptive concept for this aspect as well.

Cell networks can be redeployed (or reused) for multiple functions, as shown by Anderson [15], following a careful survey of neuroimaging data. In this context, it was pointed out, for instance, that the Broca area is involved in multiple functional cognitive tasks besides containing an FM involved in the control of speech [54]. Changes in connectivity between neurons can be obtained not only by establishing new connections (see, for instance, the neural circuitry responsible for seasonal breeding in several species [1]) but also by reactivating “silent” synapses when needed [77]. At the molecular level, the existence of receptor–receptor interactions [10] can lead to the assembly of oligomeric receptor complexes with different properties even if formed by the same types of monomers [11].

It is noteworthy that a common rule seems to exist at all levels. It is mainly based on the self-organization of existing components to form a new complex capable of “emergent” properties. “Proliferation” (production, generation) of new system components seems to play a minor role in the remodeling processes occurring in the CNS. In this respect, the “self-similarity logic” driving the remodeling of the CNS appears to significantly differ from that operating in other plastic systems, such as, for instance, the vascular system (see [60]). To illustrate this concept, the sprouting phenomenon (occurring in both systems) can be considered. Sprouting of capillaries involves the recruitment of many endothelial cells and the production of specialized endothelial cells called tip cells, whose extending and retracting filopodia define the direction in which the new vascular sprout grows. Thus, these cells perform functions analogous to the growth cones of neurons [50]. The latter, however, are dynamic, actin-supported extensions of single developing axons.

Using a metaphor originally proposed by Agnati et al. [8], the “self-similarity logic” emerging in the CNS remodeling processes can be illustrated with the term “mosaic” as it is defined in figurative art, i.e., as the process of making pictures by inlaying small bits of colored stones (“tesserae”). Thus, the term indicates how from a given set of elements it is possible to achieve different patterns endowed with different emergent properties.

10.4 Fractal Features and Pathological Processes in the Nervous System

An intriguing aspect deserving consideration features the fractal architecture that the CNS may express under pathological conditions. In recent years, a number of studies have provided the first pieces evidence of this issue from both a structural and a functional perspective.

The fractal organization of the brain networks involved in pain perception has been characterized by a study combining intracranial electroencephalogram (EEG) and functional magnetic resonance imaging (fMRI) data [44]. Brain connectivity during pain was found to be organized in four neocortical modules, interrelated via “connector hubs.” The results indicated that a pain stimulation enhanced the modularity and the small-world (see Fig. 10.2b) organization of the network as compared to other stimuli, providing evidence for pain-dependent changes in network organization and communication.

On the contrary, self-similarity measures of functional brain networks were found to be diminished in patients diagnosed with disorders of consciousness arising from severe brain injury [90]. Using diffusion MRI tractography, the authors demonstrated a quantitative association between fractal dimension and chronic loss of consciousness, highlighting the close connection between structural complexity of the brain networks and ability to support cognitive function.

Structural changes following local injuries were explored by magnetic resonance imaging in patients with focal brain tumors [65]. Fractal complexity was computed with a wavelet-based estimator of the Hurst exponent [22] and was found to be reduced in the peritumoral region. As distance from the tumor increases, however, this initial suppression was balanced by a subsequent increase before finally normalizing. Thus, the results indicated that focal tumors affected the brain globally, a long-range effect traditionally not considered by neurosurgical management paradigms. On this basis, the possibilities offered by analysis methods aimed at characterizing the fractal features of brain networks to optimize neurosurgical strategies have been quite recently proposed and discussed [66].

Studies using blood oxygenation-level-dependent fMRI (BOLD fMRI) provided data using the fractal logic of many brain functional processes. In a study on brain dynamics during recovery from sport-related concussion [32], the temporal dynamics of BOLD signals were examined throughout concussion recovery using fractal analysis, providing evidence of a BOLD scale-free logic driving long-term recovery after the trauma.

BOLD signals have also been used to study impaired attention and working memory following stressful life events [31]. The results indicated that the distress associated with the stressful events altered the scale-free brain dynamics and that fractal measures of signal complexity can be sensitive measures of the interaction between psychological and physical distress.

10.5 Concluding Remarks: A Place for Self-Similarity in a Global Model of the Nervous System?

A quite large body of presently available data identified self-similar structures at all levels of both morphological and functional organization of the nervous system. They appear to be almost embedded within one another, recalling the image of the Russian Matryoshka dolls [4, 8], with a “self-similarity logic” (i.e., a set of almost identical rules) operating at the successive levels of such a nested system. As pointed out by Werner [126], however, these observations remained largely islands in the otherwise rapidly advancing theoretical neuroscience with different priorities. Thus, a better integration of self-similarity with other insights into brain organization and complexity would be needed.

In this respect, of substantial interest is the biophysical evidence for the brain being a complex system in a regime of criticality, as understood in statistical physics (see Chialvo [29] and Werner [126] for reviews). In statistical physics, systems operating at the critical point of transition between ordered and random behaviors are metastable with respect to a set of control parameters and are capable of rapid qualitative change in response to fluctuations of external input. The important feature of system organization following a critical transition is the formation of new objects with distinct properties. Typical physical examples are phase transitions, as, for instance, that from water to ice or from ferro- to paramagnetism.

For what it concerns the nervous system, criticality has a distinguished history. As early as 1950, Turing [119] postulated that the brain as a dynamical system would operate near a critical state as the prerequisite for an instantaneous reaction to novelty. A quite large body of experimental observations of sudden transitions between stable states was subsequently accumulated. Illustrative examples include the motor behavior [64, 76] and the visual system [69]. They were considered evidence for the brain being a pattern-forming system that can flexibly switch from one coherent state to another.

As far as the significance of criticality is concerned, three attributes deserve consideration [126]. The first is the just-mentioned characteristic of critical transitions, making them a universal mechanism for rapid switching between different cooperative patterns of neuronal behavior. Furthermore, excitable systems at criticality were shown to exhibit an optimal dynamical range for information processing [78]. Finally, a model that reproduces the typical features of systems at a critical point can learn and remember complex logical rules [38]. According to Werner [126], these intriguing features of critical systems are at the basis of their current appeal as a general theoretical framework to describe the nervous system.

An important question for our discussion can now be raised. Is there any relation between the just briefly summarized biophysical features of the nervous system and the self-similar organization it exhibits? Also documented by Aguirre et al. [13], the intimate relation between self-similar structures and nonlinear dynamics in dissipative systems is evident and well-substantiated, with self-similarity and self-similarity logic appearing as the basic design principles for complex systems. Two

aspects, in particular, can be emphasized here. The first point is of morphological nature. A self-similar structure, such as a network (see Sect. 10.2.2), is likely the most economic architecture supporting dynamic transitions associated with the formation and dissolution of attractors (see [129]). In addition, the fractal morphology of the elements forming the cell networks in the nervous system is consistent with the constraint that the time for propagating signals and the length of distribution pathways should be minimized [127]. The second point concerns dynamics. In fact, at critical transitions, a reordering process of a system is needed in which events at many scales make contributions of equal importance [128]. A hierarchical nesting of self-similar structures operating according to similar rules (“self-similarity logic”) could, therefore, well support a process of this type, representing a compact format to store a range of temporal and/or spatial stimulus parameters: the metaphor for a set of strings resonating at specific frequencies has been proposed [126] to illustrate this concept. Also from the point of view of the computational tasks that the nervous system can perform [59], it is possible to appreciate the relevance of this “vertical” morphofunctional organization, since it allows an enormous number of possible configurations for each FM, providing it with an extraordinary potential capability to process and store information [12].

In conclusion, although a great many questions concerning the dynamical properties of the nervous system as a hierarchical system of networks are still open, the self-similarity characterizing its structural and functional organization appears not only an efficient design principle but also a unifying concept to describe its complexity and its concerted, holistic behavior. In this respect, the concept of a “collectome” (replacing that of “connectome”) has been recently proposed [28] to emphasize the characteristics of the neural framework that has a bicontinuous style of work from micro- to macroscale based on fractal rules.

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Chapter 11

Fractality of Cranial Sutures



Takashi Miura

Abstract It has long been known that skull suture has a typical fractal structure. Although the fractal dimension has been utilized to assess morphology, the mechanism of the fractal structure formation remains to be elucidated. Recent advances in the mathematical modeling of biological pattern formation provided useful frameworks for understanding this mechanism. This chapter describes how various proposed mechanisms tried to explain the formation of fractal structures in cranial sutures.

11.1 Biology of Skull Suture Development

Skull sutures are thin, soft tissues between the skull bones that work as growth centers of the skull. At birth, sutures are relatively wide and straight (Fig. 11.1a, [1]). During development, the skull grows by brain expansion, and new bones are added to the bone edges at the suture region. As a result, the skull can grow while protecting the brain. In addition, the suture tissues gradually wind and develop interdigitating patterns without changing their width (Fig. 11.1b, [2]). This winding structure sometimes becomes very complex and exhibits self-similarity (Fig. 11.1c, [3–9]). After adolescence, the suture tissue gradually disappears, and the skull bones become fused [10].

Skull suture biology is an interdisciplinary field that includes medicine, developmental biology, anthropology, and applied mathematics [10, 11]. In medicine, the timing of skull fusion is clinically important because premature fusion of the skull bones causes a pathological state called craniosynostosis [12], which leads to a deformed skull. Several candidate genes for craniosynostosis, such as fibroblast growth factor receptors (FGFR) [13], transcription factor twist [14], and ephrins [15], have been cloned from craniosynostosis patients [10].

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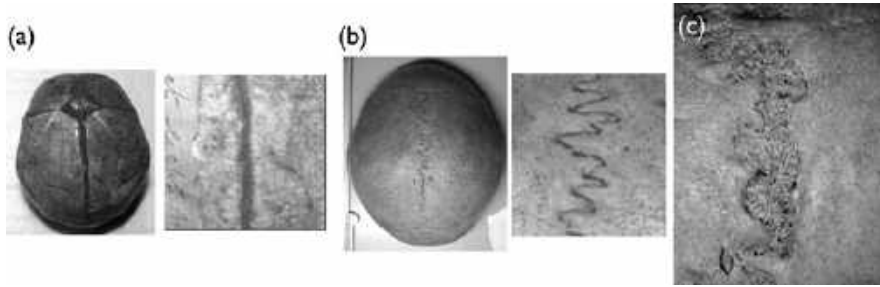


Fig. 11.1 Suture development. (a) The newborn suture is straight and relatively wide. (b) The adult suture is thin and interdigitated. (c) A lambda suture of the human skull. The suture can develop complex fractal patterns. Modified from [16]

Developmental biology of the suture has been classically studied by morphometrics and surgical operations [17–23]. Recent advances in transgenic technology enable us to assay various genes' functions in suture development. Suture development involves several transcription factors (TWIST [24], MSX2 [25], Alx4 [26], Foxc1 [27], RUNX2 [28], Dlx5 [29]), extracellular signaling molecules (TGF β s [30], FGFs and their receptors [31, 32], BMPs and hedgehog family [33]), and extracellular matrices [34].

Anthropologists are more interested in the mechanical aspects of suture patterns. Suture interdigitation is thought to strengthen the mechanical connection between bones; this idea has been tested by direct measurement [35]. The effect of external forces has also been investigated in artificially deformed skulls [36].

Since the emergence of fractal geometry in the 1980s, it has been speculated that skull sutures are good examples of fractal structures. Long et al. [3] first mentioned the fracture of the skull suture. They described the similarity between the skull suture and the Koch curve (Fig. 11.2a) and measured the fractal dimension of the skull suture. Since then, numerous measurement work using the box count method (Fig. 11.2b) has been undertaken [5–9, 37, 38]. However, how the fractal pattern is generated is relatively less understood.

11.2 Definition of Fractal Patterns

The fractal pattern can be defined as an irregular geometric object with nesting of structure in various spatial scales [39]. The pattern is described by several keywords in introductory literature (self-similarity, non-integer dimension, and scaling) [39].

Here we consider a classic example of a fractal structure, the Koch curve (Fig. 11.2a). This structure is generated by repeated application of a specific rule to line segments at different spatial scales. Initially, the line segment is divided into three sub-segments, and the middle segment is reformed into an equilateral triangle.

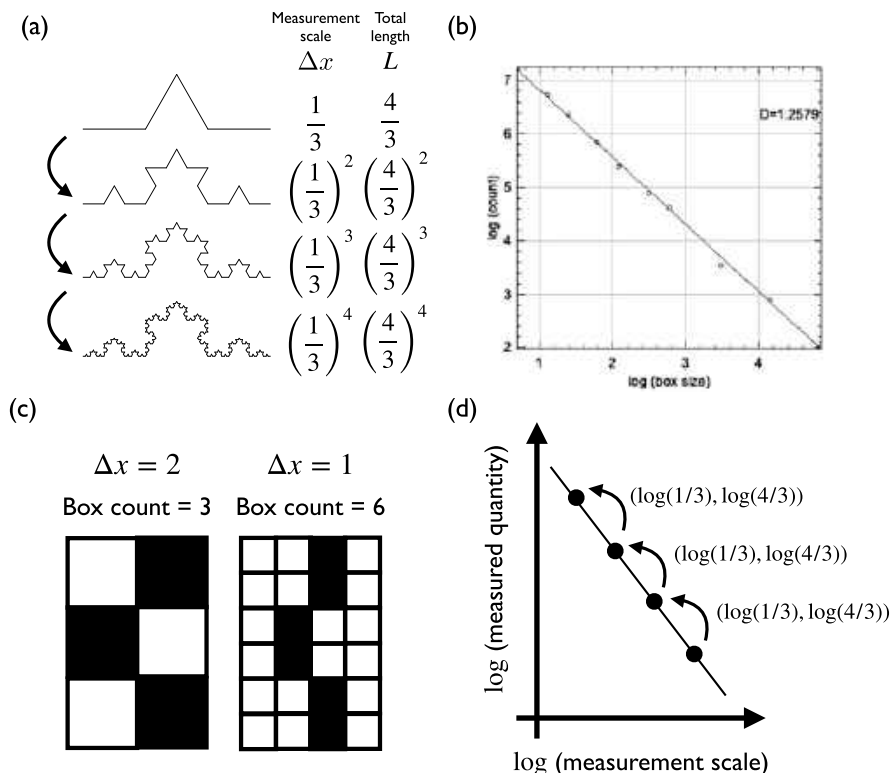


Fig. 11.2 Fractal dimension and scaling. (a) Definition of Koch curve. (b) A result of the box count method to measure the fractal dimension of a Koch curve. (c) Definition of box count method. (d) Relationship between length (count) and measurement scale Δx in Koch curve

Repetition of this process results in the Koch curve, a complex structure that is clearly self-similar.

The easiest way to determine the fractality of the pattern is via measurement of the fractal dimensions using the box count method by ImageJ/Fiji software [40] as follows:

1. Prepare an image that has a complex structure.
2. Convert the image to 8-bit image ("Image > Type > 8-bit").
3. Binarize the image ("Process > Binary > Make Binary").
4. Calculate the fractal dimension ("Analyze > Tools > Fractal Box Count...").

Then you obtain linear distribution in log-log plot and Fractal dimension (Fig. 11.2b).

Box count method uses lattices (boxes) of a certain size (Δx) which contain the target structure (black in Fig. 11.2c). Then we vary the box size Δx , count the number of boxes that covers the target structure, and plot $\log(\text{size of the box})$ versus

$\log(\text{number of boxes})$ (Fig. 11.2b). The fractal dimension is calculated by the slope of this line.

Intuitively how the linear distribution in the log–log plot appears by the box count of the fractal structure is as follows: one process of segment elongation corresponds to the reduction of box size Δx by one-third. Then the measured quantity, total length L of the structure (which corresponds to the number of counts), is increased by $4/3$ (Fig. 11.2a). Then, by one process, the point $(\log \Delta x, \log L)$ moves at a constant direction and distance $(\log(1/3), \log(4/3))$ on the log–log plot, results in linear distribution (Fig. 11.2d).

11.3 Models to Generate Fractal Pattern

There are several classes of models that can generate fractal patterns. In this section, we briefly review three of them, which have been utilized to understand the skull suture pattern formation.

11.3.1 Geometric Model

This model class uses the same rule repetitively to generate complex geometric patterns. The most famous one is the Koch curve, which was introduced in the previous subsection (Fig. 11.2a) and was referred to in the first report on the fractal dimension of cranial sutures [3]. This model class applies the same rule to the different spatial scales, resulting in a self-similar pattern. This type of model is generally deterministic—the same model always generates the same pattern.

11.3.2 Eden Front

Eden front model was first proposed as a general growth model of cell colonies [41] (Fig. 11.3a–c). The model begins with an array of occupied (black) and unoccupied (white) sites on a rectangular lattice. In each time step, a site adjacent to the preexisting occupied site (gray) is randomly chosen and converted to an occupied site (Fig. 11.3a). Repetition of this process results establishes the characteristic ragged surface (Fig. 11.3b). The structure's perimeter is proportional to the cluster size on a log–log plot (Fig. 11.3c).

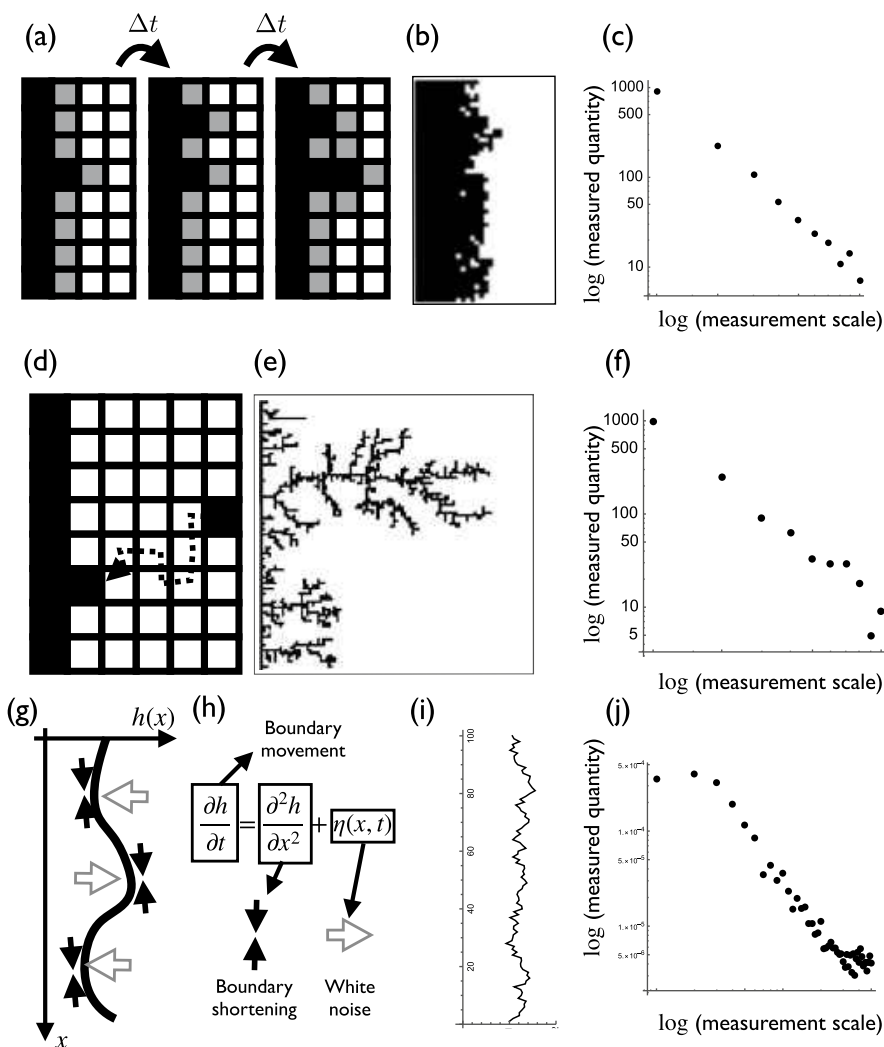


Fig. 11.3 Models that generate fractal structure. (a–c) Eden front. (a) We define occupied sites (black) and unoccupied sites (gray and white) on a rectangular lattice. On the smaller spatial scale at $\Delta t = 1$, an interface (gray) site is randomly chosen at each time step Δt and turned black. (b) Eden model generates a ragged interface. (c) Log–log plot of box count method of Eden pattern shows linearity. (d–f) Diffusion-limited aggregation (DLA). (d) Definition of the model. (e) A structure generated by the DLA model. (f) Log–log plot of box count method of DLA pattern shows linearity. (g–j) Edwards-Wilkinson model. (g) Intuitive explanation of Edwards-Wilkinson model. (h) Governing equation. (i) Example of the pattern generated by the EW model. (j) Power spectrum shows linearity on a log–log plot

11.3.3 Diffusion-Limited Aggregation

Diffusion-limited aggregation (DLA) is another class of models that generates fractal structures (Fig. 11.3d–f, [42]). In DLA, the original shape of the occupied region is defined on a square lattice. A particle is then released far from the edge of the region and randomly moves until it hits the occupied region. Here, we suppose that the occupied region is a bone structure. When the free particle encounters the bone, it is converted to bone, and another particle is released far from the bone-mesenchyme boundary (Fig. 11.3d). Eventually, this process forms a ragged surface (Fig. 11.3e). The log–log plot shows linearity (Fig. 11.3f).

11.3.4 Stochastic Differential Equation (SDE) Models

This class of model utilizes differential equations with the stochastic term, which defines the change of a certain value (in this case, interface coordinate $h(x, t)$ at location x and time t , Fig. 11.3g) per unit time. This is similar to the reaction-diffusion equation [43], but the model contains the stochastic term $\eta(x, t)$.

The simplest example of the SDE model that generates fractal structure is the Edwards-Wilkinson (EW) model [44], which consists of white noise term $\eta(x, t)$ and boundary shortening term $\partial^2 h / \partial x^2$ (Fig. 11.3g, h). In this model, the pattern is generated by the white noise, but the effect of boundary shortening term smoothes the pattern (Fig. 11.3g). As a result, we could observe stochastic fluctuation pattern (Fig. 11.3i) and linear distribution in the log–log plot of power spectrum of the pattern ($|\hat{h}(k, t)|^2$) (Fig. 11.3j).

11.4 Models of Skull Suture Fractal Pattern Formation

This section describes the models to generate the fractal structure of the skull sutures, which apply the models described above.

11.4.1 Eden Collision Model

To our knowledge, the first attempt to explain fractality in skull sutures was the Eden collision model [45, 46]. This model regarded the suture pattern as two confronting Eden fronts (Fig. 11.4). These fronts proceed and collide, forming the ragged interfaces of two bones.

The advantage of this model is that it can reproduce the initial phase of suture formation before the two confronting osteogenic front collides. Another advantage is that the authors made a three-dimensional model [46] due to its simplicity.

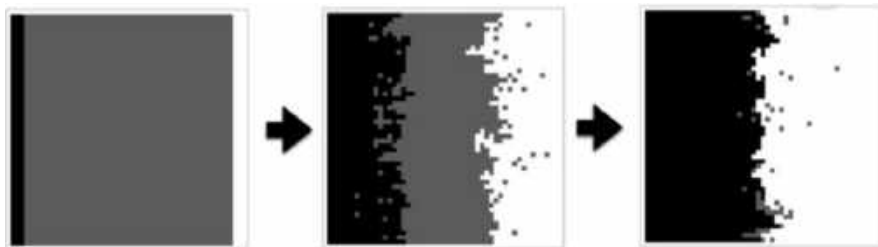


Fig. 11.4 Edén collision model of suture development [45, 46]. The model consists of two confronting Edén fronts. The originally straight interface roughens to form the Edén front, and two interfaces collide to form the interdigitated boundary of the bones

Although the model is a pioneering work on fractal suture patterns, the model has several shortcomings. First, the time course of suture interdigitation development appears to differ from experimental observation. Real sutures become interdigitated *after* the approach of two osteogenic fronts (Fig. 11.1). In the Edén collision model, the two osteogenic fronts are ragged from the beginning, and the fractal dimension reduces after the collision. Second, the model cannot reproduce the maintenance of thin narrow suture tissue. This characteristic is clinically important for elucidating the mechanism of craniosynostosis. Third, known molecular interactions are difficult to incorporate into this system.

11.4.2 *Partial Differential Equation (PDE)-Based Model and the Koch Curve*

Another class of models are based on partial differential equations (PDEs) [16, 47, 48]. The model proposes two variables: the tissue differentiation state ($u(x, y, t)$) and distribution of the substrate factor molecules ($v(x, y, t)$) (Fig. 11.5a). The tissue differentiation state represents each spatial coordinate $((x, y))$ at a certain time (t) as bone ($u = 1$) or mesenchyme ($u = -1$). The tissue differentiation state influences the generation of substrate factors, which are diffusible signaling molecules (such as FGF18 and BMP4) produced by the mesenchyme. The substrate molecule v promotes osteogenic differentiation (Fig. 11.5a).

By combining these two effects, this model reproduces the maintenance of narrow suture tissue and the formation of interdigitation (Fig. 11.5a, b).

These behaviors are intuitively explained as follows: by default, the osteogenic front proceeds under the effect of substrate factors from the suture tissue. When two osteogenic fronts approach, the concentrations of the substrate factors decrease because their source (the mesenchyme tissue) is narrowed. Eventually, the osteogenic front progression is halted. If the suture tissue becomes too narrow, the substrate factor becomes so low that the osteogenic fronts retract. Consequently, this mechanism establishes an optimal width, maintaining a band-like region of

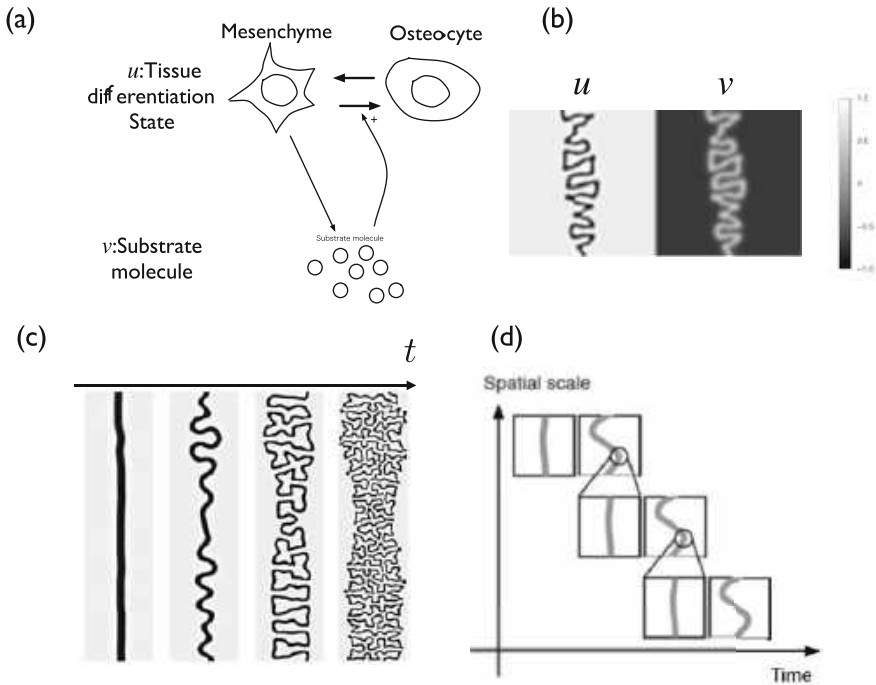


Fig. 11.5 PDE-based model of fractal structure generation [16]. **(a)** Model definition. The model has two variables: the tissue differentiation state $u(x, y, t)$ and the substrate molecule distribution $v(x, y, t)$. u has two stable states, bone ($u = 1$) and mesenchyme ($u = -1$). Substrate molecules passively diffuse from the mesenchyme tissue, promoting osteogenic differentiation. **(b, c)** Numerical simulation results of the model. Although interdigitation develops in **(b)**, we observe only two characteristic lengths (the width and curvature of the suture). Fractal structures are established in **(c)**. **(d)** Intuitive explanation of fractal pattern formation. The pattern is formed similarly to the Koch curve. Modified from [16]

mesenchyme tissue. The suture tissue's optimal width depends on the substrate molecules' diffusion length.

We now consider the two-dimensional situation. When the band-like solution is slightly winding, the osteogenic front becomes slightly convex because the surrounding region contains more mesenchyme tissue, and the local concentration of substrate molecules is high. Therefore, any initial slight perturbation is amplified, causing the narrow band-like structure interdigitation. The wavelength of the interdigitation was derived by a mathematical analysis [49].

The above-described system has only two fixed characteristic lengths and cannot replicate the scaling behavior observed in fractal structures. However, the system is well understood, and the width of the suture tissue and characteristic length of the interdigitation can be mathematically determined [49]. To obtain fractal structures, we need an additional mechanism. We assume that the spatial effect (governed by the diffusion coefficients) in the system is gradually decreased [16]. After

implementing the time-dependent effect (Fig. 11.5c), the system tends to generate smaller structures at later time points. This process is qualitatively similar to the procedure that generates the Koch curve and exhibits similar scaling behavior under an appropriately chosen growth function.

The PDE model has two significant advantages. First, it includes a known molecular network; therefore, we can predict the effects of modifying a specific gene. Second, the model incorporates the maintenance of tissue width and thus handles the distinction between normal and craniosynostosis sutures. There are several known shortcomings of this model. First, the model does not seem to replicate the observed suture development, in which the rapid generation of larger structures is followed by the emergence of smaller structures [16]. Second, the model is currently only confined to two-dimensional due to the complexity of the model.

11.4.3 Mechanics-Based Model Using DLA

Another model [50] generates suture fractal patterns by mechanical principles (Fig. 11.6). Zollikofer's model first defines the boundary of two bones, then calculates the mechanical stress distribution φ inside the bone, assuming that this stress is balanced and static. Under such conditions, φ satisfies Laplace's equation $\Delta\varphi = 0$. Next, the boundary is iteratively grown by a stochastic process. The probability of growth depends on φ . In addition, the boundary is smoothed by a surface tension term. This model reproduces the formation of complex interdigitated patterns (Fig. 11.6b).

One significant advantage of this model is that it can generate a very complex structure with a high fractal dimension. Zollikofer's model directly incorporates

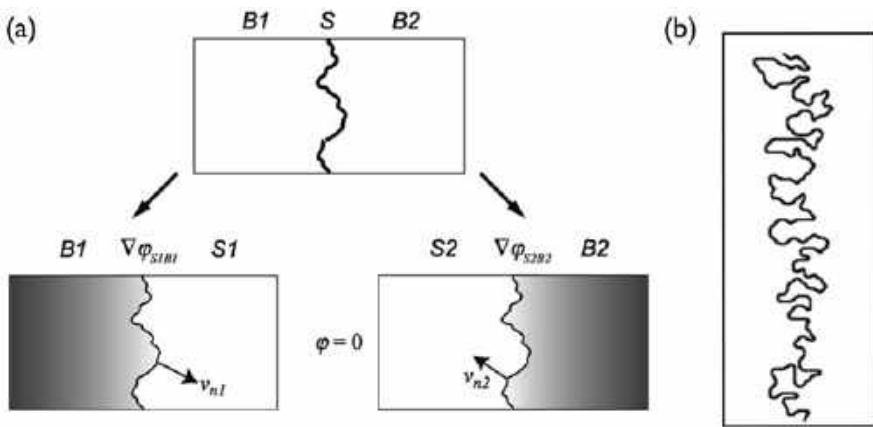


Fig. 11.6 Mechanics-based model of suture interdigitation formation [50]. (a) The model defines two bones, B1 and B2, with boundary S. The interface iteratively grows from both sides according to the stress distribution φ . (b) Suture pattern generated by the model

the mechanical aspect of bone growth. Mechanics is an important component of anthropological research, and this model predicts the enhanced interdigitated structure under increased mechanical load [51]. The shortcoming of the model is twofold: the nature of “particle” is not clear under their formulation. There is a more rigorous model of interface instability induced by mechanical stimuli [52]. In addition, suture width is not implemented by their model.

11.4.4 SDE-Based Model at the Anterior Part of the Sagittal Suture

A recent study used SDE to generate specific aspects of the suture pattern [53]. It is known that the sagittal suture has different shape characteristics—at the anterior part of the suture, the suture pattern is stochastic, while in the posterior part, the pattern is more prominent and seems to have some characteristic length (Fig. 11.7a).

The model is based on the PDE model described above [16, 47] and incorporates stochastic noise. The parameter set of the PDE part is chosen so that interface

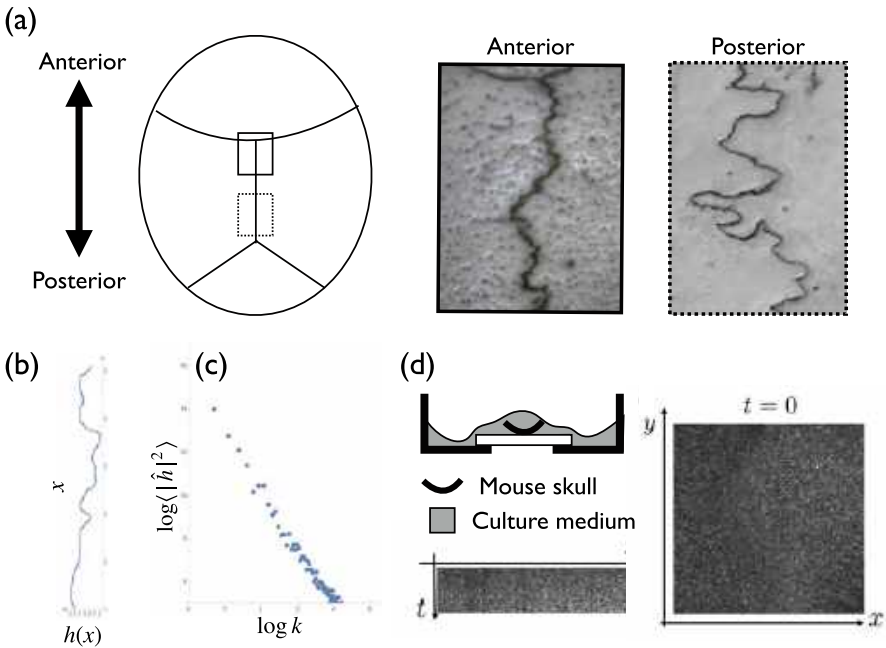


Fig. 11.7 SDE model of suture pattern formation. (a) Difference between the anterior and posterior part of the sagittal suture. (b) The digitized shape of the anterior part of the sagittal suture. (c) Log-log plot of the power spectrum of (b). (d) Experimental observation of noise term ($\eta(x, t)$). Organ culture of transgenic mice having FRET sensor to monitor the phosphorylation state of ERK was used. Modified from [53]

instability is not observed—in other words, the model does not generate a pattern without noise term. Then the noise term generates the irregular pattern, which is smoothened due to the osteogenesis dynamics. Smoothing means reducing the high-frequency component in the power spectrum. As a result, the power spectrum shows the linear distribution in a log–log plot, which can be interpreted as a fractal structure (Fig. 11.7b, c). The existence of spatiotemporal noise is also assayed using transgenic mice visualizing FGF signal input (Fig. 11.7d).

This model has several advantages—the model can reproduce a specific part of the suture while reproducing suture width maintenance and incorporating the known molecular interactions. Mathematical analysis is more accessible in this model than in the discrete systems described above. However, the model only reproduces a specific type of pattern—a more complex pattern with overhung is difficult to be handled with this model.

11.5 Future Directions

11.5.1 *Experimental Verification of Theoretical Models*

These theoretical models described above must be verified in carefully designed experiments. In the model of [50], mechanical stress plays a similar role to the diffusible signaling molecules in [16]. The stress distributions in an elastic body and the distribution of diffusible substances in an equilibrated solution both obey Laplace's equation ($\Delta u = 0$); therefore, we cannot distinguish these two mechanisms by their model behaviors alone. In addition, the chemical and mechanical factors are not independent in actual suture; for instance, mechanical stress can alter the gene expression in suture tissue [54], and ongoing differentiation obviously affects the mechanical property of bone [51].

The measurement of model parameters remains technically challenging. The diffusion coefficients of specific morphogens in the PDE model can be measured by imaging techniques such as fluorescence recovery after photobleaching (FRAP). Unfortunately, we lack consensus on the most critical molecule in this system, so it is not easy to select a target molecule. The mechanics model [50] requires the mechanical properties of suture tissue, which may be measurable using atomic force microscopy. At present, the mechanism of fractal pattern formation remains unclear and must be elucidated in the further experimental study.

Recent advances in single-cell RNA-Seq provided numerous gene expression data [55–57]. Currently, we do not have a good methodology to utilize the data to validate the models. Incorporating this information into a modeling study seems to be a challenging but promising area of research.

11.5.2 *Application of Fractal Aspect of Skull Suture*

The relationship between fractal dimension and craniosynostosis can be one promising application of these models. In general, lowering the effect of the signaling molecule v promotes curvature in the PDE model. Conversely, increasing the effect of the signaling molecule in the PDE model should induce craniosynostosis. Therefore, craniosynostosis development may be negatively correlated with the degree of curvature. The degree of interdigitation can be estimated from the amplitude of the winding pattern. However, an easier approach is to measure the fractal dimension using the box count method. Therefore, the fractal dimension provides a simple diagnostic tool for craniosynostosis.

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Chapter 12

The Fractal Geometry of the Human Brain: An Evolutionary Perspective



Michel A. Hofman

Abstract The evolution of the brain in mammals is characterized by changes in size, architecture, and internal organization. Consequently, the geometry of the brain, and especially the size and shape of the cerebral cortex, has changed notably during evolution. Comparative studies of the cerebral cortex suggest that there are general architectural principles governing its growth and evolutionary development. In this chapter, some of the design principles and operational modes that underlie the fractal geometry and information processing capacity of the cerebral cortex in primates, including humans, will be explored. It is shown that the development of the cortex coordinates folding with connectivity in a way that produces smaller and faster brains.

Keywords Brain design · Brain evolution · Cerebral cortex · Cortical folding · Fractal geometry · Human brain · Information processing · Primate neocortex · Neural wiring · Neural networks · Primates · Scaling

12.1 Introduction

During the past decades, considerable progress has been made in explaining the evolution of brain size in mammals in terms of physical and adaptive principles (see, e.g., [36, 47, 51, 61, 82]). In addition, a quantitative approach to the comparative morphology of the brain has made it possible to identify and formalize empirical regularities in the diversity of brain design, especially in the geometry of the cerebral cortex (e.g., [12, 13, 32, 35]). Although many aspects of brain evolution still remain unexplained, these comparative investigations, using scaling methods and mathematical models, have provided new insights into the evolutionary dynamics of the brain and its morphological constraints. The object of this chapter is to

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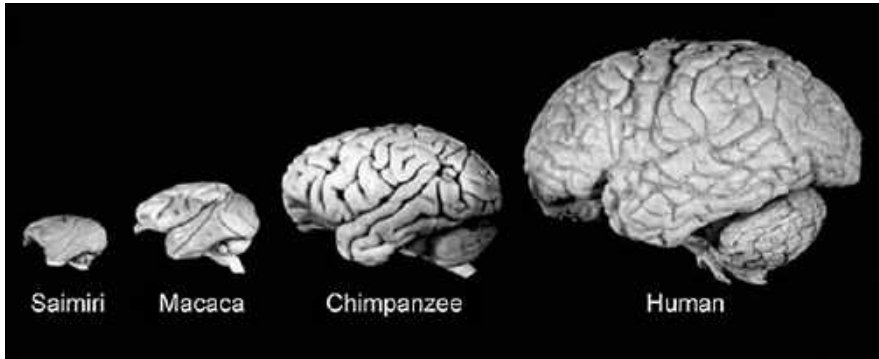


Fig. 12.1 Lateral views of the brains of some anthropoid primates showing the evolutionary expansion of the neocortex. The diverse configurations and gyral and sulcal patterns should be noted. *Saimiri sciureus*: $E = 22$ g; *Macaca mulatta*: $E = 95$ g; *Pan troglodytes*: $E = 420$ g; *Homo sapiens*: $E = 1350$ g. (Reproduced with permission from Hofman [36])

examine some of the design principles and operational modes that underlie the fractal geometry of the cerebral cortex in primates, including humans. It is shown that the development of the cortex coordinates folding with connectivity in a way that produces smaller and faster brains, and it will be argued that there are limits to information processing in a neuron-based system.

12.2 Principles of Brain Evolution

The evolution of the brain in mammals has been accompanied by an internal reorganization as a result of differential growth of its major parts. Consequently, the geometry of the brain, and especially the size and shape of the cerebral cortex, has changed notably since the late Cretaceous [40]. The evolutionary expansion of the cerebral cortex, indeed, is among the most distinctive morphological features of mammalian brains. Particularly in species with large brains, and most notably in great apes and marine mammals, the brain becomes disproportionately composed of the neocortex [1, 66, 67, 73] (see Fig. 12.1). The volume of the cortical gray matter, for example, expressed as a percentage of the total brain volume increases from about 25% for insectivores to 50% for humans [23, 31], whereas the relative size of the entire cerebral cortex (including the white matter) goes from 40% in mice to about 80% in humans [2, 25, 26, 31]. The relative size of the cerebellum, on the other hand, remains constant across phylogenetic groups, occupying about 10–15% of the total brain mass in different orders [29, 50, 71].

12.2.1 *Evolution of the Cerebral Cortex*

It is now well-established that the cerebral cortex forms as a smooth sheet populated by neurons that proliferate on the ventricular surface and migrate outward along radial glial fibers (for reviews, see [16, 57]). Differences in the duration of neurogenesis, which increases more rapidly with brain size for the cerebral cortex than for subcortical areas [15, 22, 48], lead to a systematic increase in the ratio of the cortical to subcortical regions. Whereas in small-brained species, the cortical volume expands by virtue of a combined increase in surface area and cortical thickness, the increase of the cortical volume in species with a brain size of more than $3\text{--}4\text{ cm}^3$ is almost entirely due to a disproportionate expansion of the cortical surface area [32]. It is the increase of the cortical surface area beyond that expected for geometrically similar objects of different volumes that creates the need for cortical folding [32, 35].

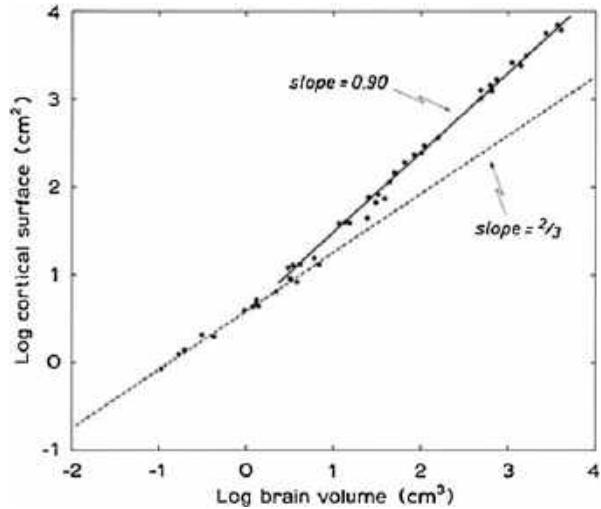
Consequently, the brains of larger species, like primates, are not well-described by the ideal constructs of Euclidean geometry. Mandelbrot [52] coined the word “fractal” to identify complex geometrical forms and developed the concept of fractal scaling to describe their organized variability. An important feature of fractal objects is that they are invariant, in a statistical sense, over a wide range of scales, a property that is known as scaling (for a review, see [35]).

In mammals with convoluted brains, among which are almost all primates, the cortical surface area, rather than being proportional to the $2/3$ power of geometric similarity, is nearly a linear function of brain volume [30, 31] (Fig. 12.2). It means that if a mouse brain (volume = 0.5 cm^3) were scaled up as the $2/3$ power to the size of the human brain (volume = 1400 cm^3), it would have a cortical surface of only about 480 cm^2 . The actual surface area of the human cortex, however, is about 2000 cm^2 , which is more than four times larger than would be predicted assuming geometric similarity, indicating that mammalian brains change their shape by becoming folded as they increase in size.

12.2.2 *Mechanisms of Cortical Folding*

Bok [5] was among the first comparative neuroanatomists who carefully observed and measured the folding of the human cortex. He cataloged possible geometrical deformations that could emerge in the folding of a layered system, having a radial structure which retained a constant volume during the folding process (Fig. 12.3). Since then, several hypotheses about cortical folding have been developed, mainly emphasizing mechanisms *intrinsic* to the cortical gray matter (for reviews, see [3, 32]). Later studies suggest that *extrinsic* factors are more important and that tension along axons in the white matter is the primary driving force for cortical folding [76]. By keeping the aggregate length of axonal and dendritic wiring low, tension should contribute to the compactness of neural circuitry throughout the cortex.

Fig. 12.2 Total cortical surface area as a function of brain volume in terrestrial mammals. Logarithmic scale. The slope of the regression line is 0.90 ± 0.012 , representing the surface–volume relationship for convoluted brains. Note that the cortical surface area of the species with convoluted brains (area $> 10 \text{ cm}^2$) is nearly a linear function of brain volume, rather than being proportional to the $2/3$ power of geometric similarity. (Modified with permission from Hofman [33])



Herculano-Houzel et al. have found that connectivity and cortical folding are directly related across species and that a simple model based on a white matter-based mechanism may account for increased cortical folding in the primate cerebral cortex [27, 28, 53, 58]. They argue that the mechanical tension generated by the pattern of connectivity of fiber bundles traveling through the white matter may account for the observed pattern of cortical surface convolutions. The authors propose that the degree of tension, taken as directly proportional to the morphological characteristics of the fiber bundle (i.e., axonal length and average cross-sectional area and the proportion of efferent neurons), determines how much the cortical surface folds inward.

This model is used to explain how surface convolutions vary with brain size and how gray matter thickness varies. Thus, the local wiring and cortical folding is a simple strategy that helps fit the large sheetlike cortex into a compact space and keeps cortical connections short. An important evolutionary advantage of this design principle is that it enables brains to be more compact and faster with increasing size [24, 42].

Although the buckling and folding of curved multilayered surfaces, such as the cortical surface in mammals, has been characterized experimentally, the transitions in the folding pattern cannot be reliably predicted by the current theoretical models, owing to the nonlinearity of the underlying stretching and bending forces. Recently, Stoop and colleagues [72] presented a theory that builds on general differential geometry principles, which describes the wrinkling morphology and pattern selection on curved multilayered surfaces. The framework developed in their study enables a systematic classification of wrinkling phenomena in complex geometries, by examining the symmetry properties of effective higher-order differential operators built from the surface metric and film-substrate coupling forces. A comparison to earlier experiments suggests that the theory is universally applicable

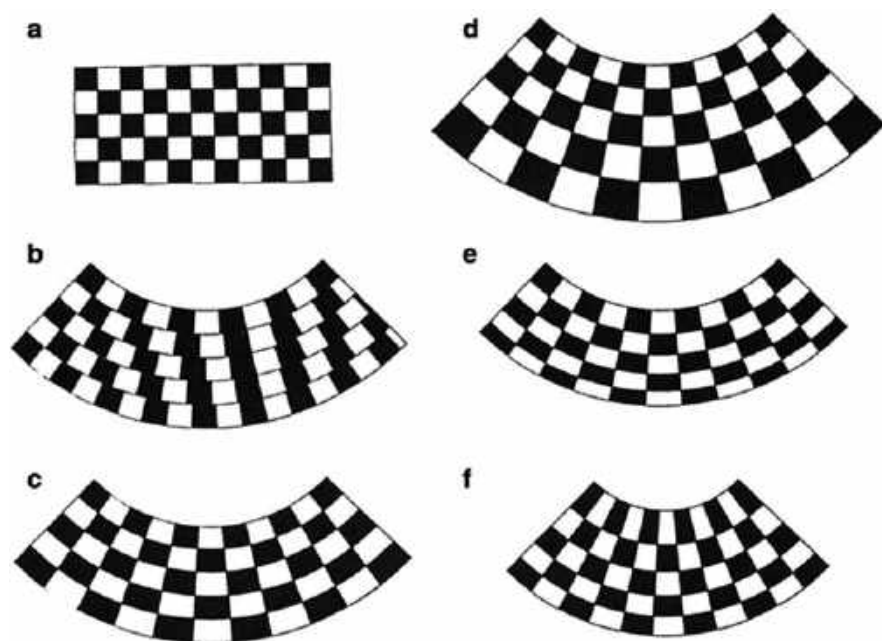


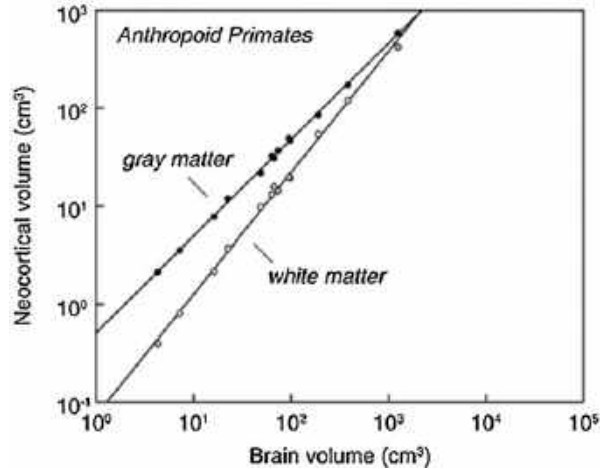
Fig. 12.3 Several ways for a simple pattern to change its intrinsic curvature, in which in each case different properties are maintained (after Bok 1929). In (b), the elements have kept their original form and surface area; in (c), their original arrangement and height; in (d), their original arrangement and form; and in (e), their original arrangement and surface area. In (f), here added to Bok's original series, the elements have kept their original arrangement and surface area, as in panel (e), but here, the elements of the center row, instead of those of the inner row, have kept their original form. Although none of the patterns are found in the mammalian cortex *sensu stricto*, the pattern of (f) corresponds most with the transformations found in the geometry of the cerebral cortex during folding. (Reproduced with permission from Hofman [32])

to microscopic and macroscopic systems, and, it, therefore, may help contribute to a better understanding of tissue mechanics and morphogenesis of multilayered tissues, including the development of brain convolutions.

12.2.3 *Scaling of the Primate Neocortex*

Analysis of the cerebral cortex in anthropoid primates revealed that the volume of the neocortex is highly predictable from absolute brain size [32, 35, 83]. The volume of the cortical gray matter, containing local networks of neurons that are wired by dendrites and mostly nonmyelinated axons, is basically a linear function of brain volume, whereas the mass of long-range axons, forming the underlying white matter volume, increases disproportionately with brain size (Fig. 12.4). As a result, the

Fig. 12.4 Volumes of the cerebral gray and white matter as a function of brain volume in anthropoid primates, including humans. Logarithmic scale. The slopes of the regression lines are 0.985 ± 0.009 (gray matter) and 1.241 ± 0.020 (white matter). Note the difference in the rate of change between the gray matter (neural elements) and white matter (neural connections) as brain size increases. (Reproduced with permission from Hofman [34])



volume of gray matter expressed as a percentage of the total brain volume is about the same for all anthropoid primates. The relative white matter volume, on the other hand, increases with brain size, from 9% in pygmy marmosets (*Cebuella pygmaea*) to about 35% in humans, the highest value in primates [32, 36].

Volumetric measurements of the gray and white matter in the neocortex of anthropoid primates, furthermore, have shown that the “universal scaling law” of neocortical gray to white matter applies separately for frontal and non-frontal lobes and that changes in the frontal (but not non-frontal) white matter volume are associated with changes in other parts of the brain, including the basal ganglia, a group of subcortical nuclei functionally linked to executive control [65, 68]. These comparative analyses indicate that the evolutionary process of neocorticalization in primates is mainly due to the progressive expansion of the axonal mass that implement global communication, rather than the increase in the number of cortical neurons and the importance of high neural connectivity in the evolution of brain size in anthropoid primates.

12.3 The Fractal Geometry of Convolved Brains

The principal idea underlying scaling is that, although biological systems may evolve by rules distinct from those governing the development of a physical system, they cannot violate the basic physical principles. One of the standard problems of classical scaling is that for any series of similar objects, the surface area is proportional to the square of a length dimension, whereas the volume is proportional to the cube. According to this geometric principle, also known as Galileo’s principle of “similitude,” the surface area is proportional to the $2/3$ power of volume or in its generalized form

$$\text{Area} = k (\text{Volume})^{D/3} \tag{12.1}$$

where k is the scaling constant and $D = 2$, the topological or Euclidean dimension of geometric similarity.

To determine the surface dimension D for mammalian brains, the empirical exponents are related to the generalized surface–volume relation (Eq. 12.1). The allometric equations are given in Table 12.1. The surface dimension of non-convoluted brains coincides with the Euclidean dimension [30], indicating that mammals with a smooth cerebral cortex satisfy the geometric scaling model, highly similar to the spherical surface area of an Euclidean hemisphere, where $\text{area} = 3.84 (\text{volume})^{2/3}$. Convoluted brains, on the other hand, with their surface dimension of $D = 2.70 \pm 0.07$, have a fractal dimension far above the expected value of standard dimensional analysis (Table 12.2; see also [19, 20, 33, 35]).

Table 12.1 Allometric scaling of the cerebral cortex against brain volume in mammals with convoluted brains

Covariate	(Sub)orders	Species	Intercept, b	Standard major axis, α ($\pm$ SD)	r
Outer cortical surface area (cm ²)	7	18	4.47	0.726 $\pm$ 0.021	0.994
Total cortical surface area (cm ²)	9	23	3.77	0.901 $\pm$ 0.022	0.994
Mean cortical thickness (cm)	7	16	0.106	0.129 $\pm$ 0.019	0.832
Cortical volume (gray matter, cm ³)	7	16	0.520	0.982 $\pm$ 0.017	0.998
Cortical volume (white matter, cm ³)	7	16	0.046	1.280 $\pm$ 0.045	0.991
Cortical volume (total, cm ³)	9	21	0.396	1.099 $\pm$ 0.028	0.994

Reproduced with permission from Hofman [33]

The model is $\log y = \log b + \alpha \log x$, where x is the variate (i.e., brain volume in cubic centimeters), y is the covariate in centimeter units, $\log b$ is the log- y intercept at $\log x = 0$, and α is the slope of the standard major axis. The strength of the relationship is reflected by Pearson’s correlation coefficient, r

Table 12.2 The fractal geometry of convoluted brains

Cortical parameter	Topological dimension	Empirical dimension	Difference (p -value)
Cortical thickness	1	0.39 $\pm$ 0.057	<0.001
Outer cortical surface	2	2.18 $\pm$ 0.062	<0.02
Total cortical surface	2	2.70 $\pm$ 0.065	<0.001
Cortical gray matter	3	2.95 $\pm$ 0.050	NS
Cortical white matter	3	3.84 $\pm$ 0.134	<0.001
Total cortical volume	3	3.30 $\pm$ 0.084	<0.01

Reproduced with permission from Hofman [33]

The model is based on the equation: $y = b x^{D/3}$, where y is the cortical structure, x is the brain volume, b is the allometric constant, and D is the topological (Euclidean), respectively, empirical (fractal) dimension of the structure. The fractal dimensions are given as mean $\pm$ standard deviation

NS not significant

Most studies of the fractal properties of the cerebral cortex have focused on computing whole-brain measures. Recently, King [43] has been able to compute the local fractal dimension of the human cortex as extracted from high-resolution magnetic resonance imaging (MRI) scans. It turns out that the local fractal dimension values of the cerebral cortex vary roughly between 2.5 and 3.0, depending on the size of the region over which the counting is performed.

By Mandelbrot's definition, a fractal is any object or process of which the dimension, given by the equation

$$D = \log N / \log (1/r) \quad (12.2)$$

strictly exceeds its topological or Euclidean dimension. It means that every set with a non-integer D , as in convoluted brains, is a fractal. To derive this formula for dimension, consider a straight line of unit length divided into N segments of length r (r = ratio vector). By definition, $N \times r = 1$ and so $r = 1/N$. Analogously, a unit square contains N pieces of edge length r in an area of 1, so $r = 1/(N^{1/2})$ and a unit cube N pieces of edge length r in a volume of 1, so $r = 1/(N^{1/3})$. Denoting the dimension by D , the relationship can be generalized to $r = 1/(N^{1/D})$, and the equation can be solved for the fractal dimension D . Dimension D is called the fractal dimension because it is not necessarily an integer.

In general, D is the number that tells us something about the overall structure and complexity of an object. The empirical area–volume relation, for example, found for convoluted brains indicates that the cortical surface is partly space-filling and that its surface area fractally evolves into a volume or that its volume, by fractal folding, attains the properties of an area [33].

12.3.1 Principles of Scaling

The central idea of the scaling principle is that many fractal forms can be generated by a procedure that replicates the same geometric configuration on every hierarchical level, a property that is known as self-similarity. In order for the exponent of self-similarity to have formal meaning, the sole requirement is that the shape be self-similar, i.e., the whole may be split up into N parts, obtainable from it by a similarity of ratio r (cf. Eq. 12.2).

A number of linear fractals can be represented by an initial polygon (the initiator) and a polygon (the generator), which replaces the sides of the initiator. Thus, a fractal structure could be generated, having the form and fractal dimension of a convoluted brain, by dividing the sides of a regular hexagon into $N = 3$ segments of length $r = 4/9$ (Fig. 12.5). The resulting similarity dimension, applying Eq. 12.2, yields $D = \log 3 / \log (4/9)^{-1}$ or a number equal to 1.3548. A similar procedure in 3D space, with a decahedron as the initiator, yields a fractal dimension of the surface area of 2.7095, a value that is almost identical to the empirical dimension, $D = 2.70$, for convoluted brains (see also [19, 32, 44]).

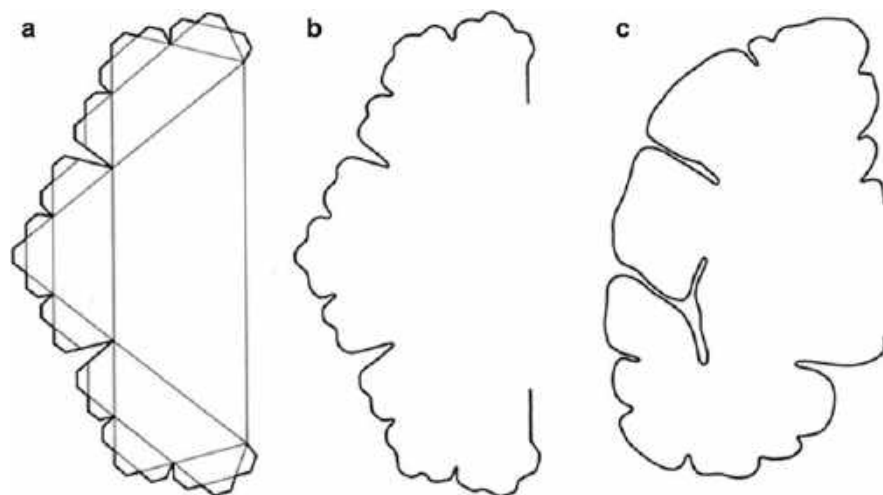


Fig. 12.5 A fractal model of the mammalian cerebral cortex. (a) The form and fractal dimension of a convoluted brain can be generated by dividing the sides of a regular hexagon (the initiator) into three segments of length $r = 4/9$. Continuing this procedure ad infinitum, it is obvious that the perimeter grows without any bound. The resulting similarity dimension is not an integer but is a number equal to 1.3548. For further details, see text. (b) The ultimate form cannot, of course, be depicted, but the smoothed third or fourth stage is a good approximation of the form of a highly convoluted brain with its rich pattern of gyri and sulci. (c) The Cortical surface of a human hemisphere as seen in the frontal section at the level of the anterior commissure. (Reproduced with permission from Hofman [33])

It is clear that specific geometric production rules may lead to the generation of “objects,” which show a high resemblance to natural phenomena. However, one cannot conclude that a series of objects with the same fractal dimension belongs to the same class, since several sets of instructions can lead to the same numerical value. In other words, many different objects may have an equal D but have a completely different appearance. In the next section, some organizing principles that may account for the fractal geometry of the mammalian brain will be presented.

12.3.2 *Fractal Scaling of the Neocortex*

From anatomical studies, we know that the primate neocortex is made up of distinct neural networks, which are organized in columnar arrays stretched out through the depth of the cortex [8, 11, 18, 38, 54, 55]. These modules, in turn, are linked together into larger neural networks by coalescing adjacent sets of columnar units. These networks, which also have a columnar arrangement, are found through the cerebral cortex [48, 57] so that we may consider columnar arrangements of neural elements as a general organizational framework of the primate cortex.

The arrangements of these clusters of neurons in vertical columns perpendicular to the pial surface, however, leads to a geometric dilemma with the evolutionary expansion of the brain. It is this dilemma – the requirement of the cortical surface area to keep abreast with the volumetric growth of the brain itself – that creates the need for cortical folding. Studies in mammals have shown that in species with convoluted brains, the mass of interconnective nerve fibers, forming the underlying white matter, is proportional to the 1.28 power of brain volume [31, 33] (see also Table 12.1), meaning that the cortical white matter is a fractal system. As a result, the total cortical surface area (S_{cortex}), including all gyri and sulci, scales approximately as the 2/3 power of the white matter volume (V_{white}) or

$$S_{\text{cortex}} = 36.4 (V_{\text{white}})^{2/3} \quad (12.3)$$

In other words, the surface area of the cerebral cortex, and with that the total number of neuronal columns, is geometrically similar to the amount of white matter, i.e., with the number and length of the interconnective nerve fibers. In small species with non-convoluted brains, a similar relationship was found between the cortical surface area and the mass of myelinated nerve fibers [33].

Apparently, the fractal geometry of the mammalian brain is a consequence of the design of the cerebral cortex, in which each cortical module, containing a large number of neurons, is connected to its environment by a specific number of axons. Here, we have an analogy to Rent's rule for computer geometry [45], which says that the number of components (C) in each module of complex computer circuits is related to the number of terminals (T) according to $T \sim C^{D/D_T}$, where D is the fractal dimension and D_T is the Euclidean dimension. In a spatial circuit, where the components are organized in columnar units, while they are in contact with the outside by their surfaces, as in the cortex, $D_T = 3$ and D is somewhere between 2 and 3. The ratio D/D_T increases with the degree of parallelism that is present in the design [45, 52].

Therefore, a fractal dimension of $D = 2.70$, as found for convoluted brains, suggests a high degree of parallel processing to take place in the cerebral cortex, especially in highly corticalized mammals, such as monkeys and apes. To reach the state of integral parallelism in which each neural component has its own terminal, the length and number of the interconnective axons must be reduced in order to set limits to the axonal mass.

12.4 Fractal Principles of Neural Wiring

The most obvious problem imposed by large brains is increasing distances among the neuronal somata of functionally related regions and the inevitable lengthening of their essential communication lines, the axons. Importantly, the axonal length and volume increase much more rapidly than the number of neurons. Furthermore, a proportional increase of neurons and connections would inevitably lead to a rapid

increase of synaptic path length, defined as the average number of monosynaptic connections in the shortest path between two neurons [9, 69, 79]. To limit the path length with increasing brain size, shortcut connections can be inserted, resulting in small-world- and scale-free-type networks [7, 70].

Although such a solution can effectively decrease path length within the neocortex, the increased lengths of the axons and the associated increased travel time of the action potentials still pose serious problems. As compensation for these excessive delays, axon caliber and myelination should be increased [39]. An indication that larger brains deploy both more shortcuts (long-range connections) and larger-caliber axons is that the volume of the white matter increased at $4/3$ power of the volume of gray matter during the course of evolution.

Wen and Chklovskii [81] have shown that the competing requirements for high connectivity and short conduction delay may naturally lead to the observed architecture of the mammalian neocortex. Obviously, the brain functionally benefits from high synaptic connectivity and short conduction delays. A magnetic resonance imaging study, furthermore, focusing specifically on the prefrontal cortex, has shown that the volume of the white matter underlying the prefrontal areas is disproportionately larger in humans than in other primates [63]. It suggests that the connectional elaboration of the prefrontal cortex, which mediates such important behavioral domains as planning, aspects of language, attention, and social and temporal information processing, has played a key role in human brain evolution.

Although the frontal lobe as a whole has not been differentially enlarged throughout human evolution [64], there is increasing evidence for its reorganization, as some regions with known functional correlates are either bigger or smaller in the human brain than expected when compared with the same region in great apes. Comparative studies, for example, suggest that the human prefrontal cortex differs from that of closely related primate species less in relative size than it does in organization (for a review, see [74]).

Specific reorganizational events in neural circuitry may have taken place either as a consequence of adjusting to increases in size or as adaptive responses to specific selection pressures. It appears that the evolution of the human brain was accompanied by discrete modifications in local circuitry and interconnectivity of selected parts of the brain and that these species-specific adaptations may affect these parts differently [36, 37, 62, 73]. But the similarity in brain design among primates, including humans, indicates that brain systems among related species are internally constrained and that the primate brain could only evolve within the context of a limited number of potential forms.

12.4.1 Neocortical Wiring

In the neocortex, billions of neurons are interconnected via a massive yet highly organized network of axonal and dendritic wiring. This wiring enables both near and distant neurons to coordinate their responses to external stimulation. Specific

patterns of cortical activity generated within this network have been found to correlate with cognitive and perceptual functions [77, 78]. Understanding the organizing principles of cortical wiring, therefore, represents a central goal toward explaining human cognition and perception in health and disease. Despite more than a century of endeavors, however, the organizing principles and function of cortical connectivity are still not well-understood (see e.g., [4, 6, 21]).

Recent network studies, using diffusion tensor imaging (DTI), have demonstrated that not only are the neurons in the neocortex structurally and functionally highly organized but that it also holds for the wiring of the entire brain [75, 80]. The interconnecting white matter axonal pathways are not a mass of tangled wires, as thought for a long time, but they form a rectilinear three-dimensional grid continuous with the three principal axes of development. The topology of the brain's long-range communication network looks like a 3D chessboard with a number of highly connected neocortical and subcortical hub regions. Structural connectivity networks, as defined by DTI, have identified a common hub in the medial parietal cortex of humans, chimpanzees, and macaque monkeys. However, the apparent lack of medial prefrontal hubs in humans that are present in chimpanzees and macaque monkeys, coupled with evidence of increased gyrification in the human prefrontal cortex, suggests important evolutionary changes in the connectivity of the human prefrontal cortex ([49, 56]; for a review, see [59]).

The competing requirements for high connectivity and short conduction delay may lead naturally to the observed architecture of the human neocortex. Obviously, the brain functionally benefits from high synaptic connectivity and short conduction delays. The design of the primate brain is such that it may perform a great number of complex functions with a minimum expenditure of energy and material both in the performance of the functions and in the construction of the system. In general, there will be a number of adequate designs for an object, which, for practical purposes, will all be equivalent.

12.4.2 *Neural Network Communication*

We have shown that in species with convoluted brains, the fraction of mass devoted to wiring seems to increase much slower than that needed to maintain a high degree of connectivity between the neural networks [35]. These findings are in line with a model of neuronal connectivity, which says that as brain size increases, there must be a corresponding fall in the fraction of neurons with which any neuron communicates directly. The reason for this is that if a fixed percentage of interconnections is maintained in the face of increased neuron number, then a large fraction of any brain size increase would be spent maintaining such degree of wiring, while the increasing axon length would reduce neural computational speed [60]. The human brain, for example, has an estimated interconnectivity of the order of 10^3 , based on data about the number of modular units and myelinated nerve fibers [35]. This implies that, on average, each cortical module is connected to a thousand

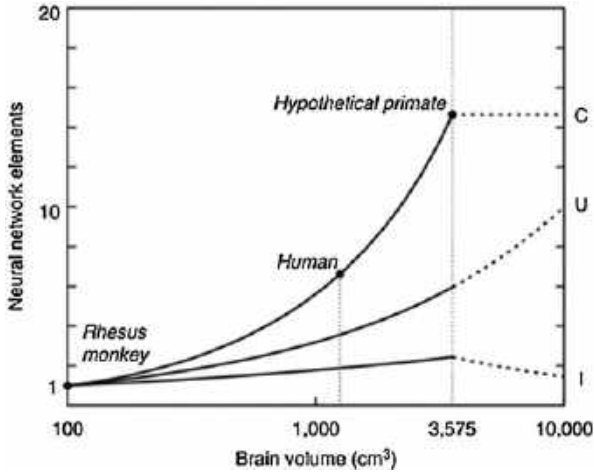


Fig. 12.6 The number of connections (C), cortical processing units (U), and level of interconnectivity (I) in the primate neocortex as a function of brain size. Semilogarithmic scale. Values are normalized to one at a brain volume of 100 cm^3 , the size of a monkey brain. Note that the number of myelinated axons increases much faster than the number of cortical processing units. The human cerebrum, for example, contains six times more myelinated axons than that of a rhesus monkey, whereas the number of cortical processing units is only three times larger. Dashed lines show the potential evolutionary pathway of these neural network elements in primates with very large brains, i.e., beyond the hypothetical upper limit of the brain's processing power (see text and Hofman [34, 35]). Note that a further exponential growth in the number of cortical processing units, without an increase in the number of connections, will lead to a decrease in connectivity between these units and thus to more local wiring. (Reproduced with permission from Hofman [35])

other modules and that the mean number of processing steps, or synapses, in the path interconnecting these modules, is about two.

In a binary system, with an interconnectivity of $I = 1$, the growth of connections (C) is a linear function of the number of units (U), and, in a fully connected network, where $I = U$, both variables are related according to the equation: $C = U(U - 1)$, which is nearly equivalent to $C = U^2$. In such a system, the number of connections increases much faster than the number of units (Fig. 12.6). Generally, the growth of connections to units is a factorial function of the number of units in a fully connected network and a linear function of the number of units in a minimally connected network. The human cerebral cortex, with an estimated interconnectivity of about 10^3 , lies somewhere between these extremes, and close to the line for $I = U^{0.5}$. It implies that in humans, the number of myelinated axons scale to the 1.5 power of the number of modular processing units, that is, a fourfold increase in modular units requires an eightfold increase in the number of nerve fibers [34, 35].

Herculano-Houzel et al. [28], furthermore, have shown that in primates, the mass of the white matter scales linearly across species with its number of nonneuronal cells, which is expected to be proportional to the total length of myelinated axons in the white matter. Decreased connectivity in the brain is compatible with previous

suggestions that neurons in the cerebral cortex are connected as small-world and scale-free networks, which slow down the increase in global conduction delay in cortices with larger numbers of neurons [4, 69, 70, 78].

12.4.3 *Limits to Information Processing*

The primate cortex, as we have seen, has evolved from a set of underlying structures that constrain its size and the amount of information it can store and process. Therefore, the functional capacity of a neuronal structure is inherently limited by its neural architecture and signal processing time (see e.g., [14, 35, 46]).

The processing or transfer of information across cortical regions, rather than within regions, in large-brained species can be only achieved by reducing the length and number of the interconnective axons in order to set limits to the axonal mass. The “number” of fibers can be reduced by compartmentalization of neurons into modular circuits in which each module, containing a large number of neurons, is connected to its neural environment by a small number of axons. The “length” of the fibers can be reduced by folding the cortical surface and thus shortening the radial and tangential distances between brain regions. Local wiring – preferential connectivity between the nearby areas of the cortex – is a simple strategy that helps keep cortical connections short. In principle, efficient cortical folding could further reduce connection length, in turn, reducing white matter volume and conduction times [9, 10, 17, 35]. Thus, the development of the cortex does seem to coordinate folding with connectivity in a way that could produce smaller and faster brains.

Once the brain has grown to a point where the bulk of its mass is in the form of connections, then further increases (as long as the same ratio in interconnectivity is maintained) will be unproductive. Increases in the number of units will be balanced by decreased performance of those units due to the increased conduction time. Model studies of the neocortex growth in primates at different brain sizes, using a conservative scenario, revealed that with a brain size of about 3500 cm³, the total volume of the subcortical areas (i.e., cerebellum, brain stem, diencephalon, etc.) reaches a maximum value [34, 36]. Increasing the size of the brain beyond that point, following the same design principle, would lead to a further increase in the size of the neocortex, but to a reduction of the subcortical volume. Consequently, primates with very large brains (e.g., more than 5 kg) may have a declining capability for neuronal integration despite their larger number of cortical neurons.

This architectural dilemma explains why large-brained primates, in order to maintain processing capacity, tend to show more specialization in the neocortex. Indeed, an increase in the number of distinct cortical areas with increasing brain size has been reported [41, 73]. It may also explain why large-brained species may develop some degree of brain lateralization as a direct consequence of size. If there is evolutionary pressure on certain functions that require a high degree of local processing and sequential control, such as linguistic communication in human brains, these will have a strong tendency to develop in one hemisphere.

12.5 Concluding Remarks

The evolution of the cerebral cortex in mammals is mainly characterized by the development and multiplication of clusters of neurons, which are strongly interconnected and in physical proximity. Since these clusters of neurons are organized in vertical columns, an increase in the number and complexity of these neuronal networks will be reflected by an expansion of the cortical surface area beyond that expected for geometric similar brains. As a result, the cortical surface area fractally evolves into a volume with increasing brain size.

It appears that the potential for brain evolution results from the self-similar compartmentalization and hierarchical organization of neural circuits and from the invention of fractal folding, which reduces the interconnective axonal distances. The competing requirements for high connectivity and short conduction delay may naturally lead to the observed architecture of the brain. The similarity in brain design among mammals, furthermore, indicates that brain systems among related species are internally constrained and that the brain could only evolve within the context of a limited number of potential forms.

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Part III
Fractals in Clinical Neurosciences

Chapter 13

Fractal Analysis in Clinical Neurosciences: An Overview



Antonio Di Ieva

Abstract Over the last years, fractals have entered into the realms of clinical neurosciences. The whole brain and its components (i.e., neurons and astrocytes) have been studied as fractal objects, and even more relevant, the fractal-based quantification of the geometrical complexity of histopathological and neuroradiological images as well as neurophysiopathological time series has suggested the existence of a gradient in the pattern representation of neurological diseases. Computational fractal-based parameters have been suggested as potential diagnostic and prognostic biomarkers in different brain diseases, including brain tumors, neurodegeneration, epilepsy, demyelinating diseases, cerebrovascular malformations, and psychiatric disorders as well. This chapter and the entire third section of this book are focused on practical applications of computational fractal-based analysis into the clinical neurosciences, namely, neurology and neuropsychiatry, neuroradiology and neurosurgery, neuropathology, neuro-oncology and neurorehabilitation, neuro-ophthalmology, and cognitive neurosciences, with special emphasis on the translation of the fractal dimension and other fractal parameters as clinical biomarkers useful from bench to bedside.

Keywords Biomarker · Classifier · Fractal analysis · Fractal dimension · Brain tumor · Epilepsy · Neurodegeneration · Aneurysm · Arteriovenous malformation · Electroencephalography · Neuroimaging · Neurology · Neuro-ophthalmology · Cognitive neurosciences · Psychiatry

To see is to believe

Data on the specific fractal characteristics of distinct biological systems (in both physiological and pathological states) accumulate daily. The researches in the field have two specific aims: (1) finding the geometrical fingerprint that characterizes

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each physical and biological system and (2) looking for the quantitative parameters that could be helpful to distinguish different physiopathological states (e.g., normal tissue versus cancer).

The greatest promise of fractal analysis is that it offers objective measures of seemingly random structures and it is a tool for examining the mechanistic origins of pathological forms [1].

Since the first applications of fractal analysis to biological images [5, 6], it has been clear that such a computational-based method can help recognize and quantify patterns to suggest potential diagnostic and prognostic biomarkers. Fractal analysis has been shown to be able to characterize the complexity of images or “more precisely their texture composition” [19]. The quantification of the geometrical complexity of histopathological and neuroradiological images as well as neurophysiopathological time series suggests the existence of a gradient in the pattern’s representation and networks from those in the normal brain to those affected by pathologies (e.g., brain tumors, epilepsy, and dementia). For such a reason, fractal analysis has found large applications in the clinical neurosciences, as briefly summarized in this chapter and illustrated in the following chapters of the third section of this book. Limitations of the translational applications of fractals into clinical practice are discussed at the end of this chapter.

13.1 Fractals in Neurosurgery and Neural Systems

Neurosurgery is a broad and complex field solidly based on the multidisciplinary background from different fields of expertise, including neuropathology, neuro-radiology, neurology, and many others, not to mention the skills derived from different complementary surgical fields. Computational modeling, fractal analysis, and artificial intelligence (AI) are the backbone of the new field of computational neurosurgery, as proposed and summarized by *Antonio Di Ieva* in Chap. 14, where the analysis of patterns and fractal dynamics emerge and merge together. In Chap. 14, the author also presents a personal journey in the fractal neurosciences, spanning many topics of research performed in different countries and laboratories.

The fractal dimension (FD) and other complementary parameters have been investigated to characterize the topological complexity of the brain in its entire physiopathological spectrum and to characterize systems’ dynamics, as exploited by *Elzbieta Olejarczyk, Milena Cukic, Camillo Porcaro, Filippo Zappasodi, and Franca Tecchio* in their chapter on the Clinical Sensitivity of Fractal Neurodynamics (Chap. 15).

13.2 Clinical Neurology and Cerebrovascular System

In Chap. 16, *Leticia Diaz Beltrán, Christopher Madan, Carsten Finke, Stephan Krohn, Antonio Di Ieva, and Francisco Esteban* review the main applications of the computational fractal-based analysis for the study of several neurological conditions, with special emphasis on the diagnostic precision of the FD.

The fractal dimension has been proposed as a potential surrogate biomarker of the degree of brain damage in neurodegenerative diseases, and even in the normal cerebral aging process [7], as widely illustrated from the neuroimaging point of view by *Jennilee Davidson, Luduan Zhang, Guang Yue, and Antonio Di Ieva* in Chap. 17 and more from the neuropathology perspective by *Daniel Pirici, Laurentiu Mogoanta, Daniela Ion, and Samir Kumar-Singh* in Chap. 18.

The vascular tree of the brain has also been characterized by means of its fractal features, not only from its angiostructural point of view but also in its functionality. The spatiotemporal complexity of cerebral hemodynamics has in fact been studied by means of fractal characterization [4, 15, 20, 26, 28, 34]. Although there is speculation about the fractality, pseudo-fractality, or multifractality of vascular trees [31, 35–37], the fractal-based morphometrics of the vascular and microvascular networks of the brain has been used to find morphometric image markers. The fractal-based analyses of the cerebrovascular system physiopathology are discussed by *Martin Soehle* in Chap. 19, with special emphasis on the subarachnoid hemorrhage. With regard to cerebrovascular pathologies, *Gábor Závodszy, Dániel Gyürki, György Károlyi, István Szikora, and György Paál* illustrate their intriguing research on the fractality and chaotic patterns in the hemodynamics of intracranial aneurysms in Chap. 20, while *Antonio Di Ieva* and *Gernot Reishofer* apply computational fractal-based modeling to the analysis of the brain arteriovenous malformations in Chap. 21, proposing a new image and prognostic biomarkers of such pathologies.

13.3 Neuroimaging

Brain imaging was first revolutionized by the clinical application of angiograms, ultrasound, and computed tomography (CT) followed by magnetic resonance imaging (MRI), positron emission tomography (PET), and single-photon emission computed tomography (SPECT) later on. MRI is the most widely used technique in neuroimaging to characterize the structure (structural imaging) and even the function (functional imaging, by means of functional MRI, fMRI) of the nervous system, which completely revolutionized the clinical neurosciences (e.g., neurology, psychiatry, and neurosurgery) for patient management. Different MR images can be obtained by selecting the pulse sequences (e.g., T1, T2, and proton density images), used to differentiate gray matter, white matter, and cerebrospinal fluid, as well as the different pathological conditions. Fractal analysis has been shown to be appropriate

for the analysis of MR images, including fMRI biosignals. Functional MRI allows for the investigation of the amplitude of activation in the neural networks of the brain, and the fractal-based time-series analysis is successfully used to identify the process of dysregulation of dynamic interactions between different brain regions in healthy adults versus pathological states [3, 25].

In Chap. 22, *Salim Lahmiri, Mounir Boukadoum, and Antonio Di Ieva* offer an excursus on the use of the fractal dimension and the related Hurst exponent in the characterization of several pathologies in neuroimaging, with special emphasis on the MR image classification. The application of a relatively novel MR imaging sequence, the susceptibility-weighted imaging (SWI) into neuro-oncology (also with some perspectives in neurotraumatology), is then presented by *Antonio Di Ieva* in Chap. 23. The chapters regarding neuroimaging bridge the discussion toward the application of fractals to the analysis of brain tumors, as illustrated by *Syed Reza, Atiq Islam, and Khan Iftekharuddin* in Chap. 24, in which the fractal-based classification and texture estimation for MRI brain tumor features are presented.

13.4 Neurohistology, Neuropathology, and Neuro-oncology

As stated by Baish and Jain in 2000, “By focusing on the irregularity of tumor growth rather than on a single measure of size such as diameter or volume, fractal geometry is well suited to quantify those morphological characteristics that pathologists have long used in a qualitative sense to describe malignancies – their ragged border with the host tissue and their seemingly random patterns of vascular growth” [1]. Cancer has been often described as a complex, chaotic, and heterogeneous system, emerging from the disruption of tissue architecture, which begs the question “If heterogeneous, how much?”; fractal analysis responds by offering some parameters for the patterns quantification.

The existence of a gradient in the geometrical complexity of histological texture or microvascular networks from those in the normal brain to those in malignant brain tumors has been speculated upon. The complex network dynamics and fractality of brain oncogenesis (i.e., development of brain tumors) are discussed by *Jackson Sánchez and Miguel Martín-Landrove* in Chap. 25. The texture analysis of histological features of brain tumors (mainly meningiomas) is the topic of Chap. 26 by *Omar Al-Kadi and Antonio Di Ieva*, while in Chap. 27, the same authors present their findings on the analysis of the geometrical complexity of brain tumors’ (mainly gliomas) microvascular networks. Although the translation of the concepts of “fractal tumors” or “fractal microvasculature” into clinical practice (“from bench to bedside”) still has some difficulties to overcome, future research in neuro-oncology should focus on finding new and reliable parameters correlated with biological and epidemiological characteristics. From this perspective, fractals and multiparameter morphometric computer-aided analyses could be of benefit in translational research and in the medical armamentarium as a conventional procedure to improve diagnostic capacity and accuracy.

Experimental studies have not only focused on the geometrical complexity of the neoplastic cells but also on the fractal structure of the environment surrounding such cells [33]. In Chap. 28, *Lee Curtin* explores the use of radiomics and fractal metrics to quantify glioblastomas' features in neuroimaging for diagnostic and prognostic purposes, whilst *Alessandra Lumini, Guilherme Freire Roberto, Leandro Alves Neves, Alessandro Santana Martins, and Marcelo Zanchetta do Nascimento* reach a similar diagnostic aim by applying fractal geometry to generate “percolation” images.

In summary, in histology and pathology, fractal analysis is used to quantitatively and objectively characterize cancer tissues in order to overcome the operator-dependent bias of classification and categorization [1, 6, 9, 13, 16, 18, 21, 29]. Moreover, the fractal dimension has also been shown useful to quantify the response for cancer treatment, in animal models [32], as well as in neuroimaging of patients affected by brain tumors [8].

Following the new paradigm which has shifted the reductionist approach to a systems biology approach [12, 24], cancer has been interpreted in its multilevel hierarchical system. Using an “ecological approach” [23], it has been shown that the spread of cancer cells (human glioma cells), like the spread of a specific tree species in a forest, have a specific “spatiotemporal signature,” characterized by a particular fractal geometry and scale invariance of the boundaries of the patterns generated in the different physiopathological states [22]. Fractal analysis offers a common language for the identification of such spatiotemporal fingerprinting in different natural systems.

13.5 Fractal-Based Time-Series Analysis in Neurosciences

As discussed in the first section of this book, fractal and multifractal analysis can be applied to the study of the time-series biosignals. Examples of neurophysiological time series are the electroencephalographic (EEG), magnetoencephalographic (MEG), or electromyographic (EMG) records, and intracranial pressure (ICP) curves as well [11, 17].

According to the latest discoveries regarding the network connectivity of the human brain, it is very intriguing to note that brain function depends upon adaptive self-organization of large neural assemblies and that the brain functional networks are characterized by small-world properties, implying a scale-invariant or fractal small-world topological organization [2]. The fractal small-world connectivity of the brain, based on the connectivity of hubs and topologically pivotal nodes, is the anatomical and functional background of its architecture, in physiological as well as pathological states [2, 27, 30]. EEG, MEG, fMRI, and other methods offer valid tools of analysis of such networks.

Time-series analysis is discussed by *Lorenzo Livi*, who in Chap. 30 discusses the multifractal detrended fluctuation analysis and multiscaling features found in patients affected by Parkinson's disease.

13.6 Neuro-ophthalmology, Pain Medicine, Cognitive Sciences, Neuropsychology, and Psychiatry

In consideration of the close relationship of the eye to the brain and neurological diseases, a specific chapter is focused on the application of fractal analysis into neuro-ophthalmology and eye research (Chap. 31 by *Marina V Zueva, Natalia Neroeva, Anastasia Zhuravleva, Anna Bogolepova, Vladislav Kotelin, Denis Fadeev, and Irina Tsapenko*).

As mentioned above, fractal analysis can be used to characterize the selfsimilarity in neural networks, as done and summarized in Chap. 32 by Camille Fauchon, H       Bastuji, Roland Peyron, Luis Garcia-Larrea, who have applied fractal analysis to the study of brain networks related to pain.

Abnormal fractal dynamics have been observed in several behavioral and psychological conditions as well as neuropsychiatric diseases, opening the venues to the applications of computational fractal-based analysis in the cognitive neurosciences, neuropsychology, and psychiatry. A fractal-based approach to analysis electrophysiological signals to detect and monitor depression is exploited by *Milena   uki  , Elzbieta Olejarczyk, and Maie Bachmann* in Chap. 34, whilst in Chap. 35, Mohammed Khan and Herbert Jelinek introduce a new approach (the Point-of-Care Testing) to correlate EEG with psychopathology. In Chap. 36, *Sergio Iglesias-Parro, Maria Felipa Soriano, and Antonio Ib    ez-Molina* provide a panoramic view on the application of nonlinear approaches to the study of mental disorders, including affective and anxiety disorders. Complementary approaches on the applications of fractal analysis in neuropsychology are illustrated by *Stephen Wolfson, Ian Kirk, Karen Waldie, and Chris King* in Chap. 37 and *Antonio Cerasa* in Chap. 38.

13.7 Limitations of Application of Fractal Analysis into Clinical Neurosciences

The translation of fractal analysis into clinical practice still has some obstacles.

The fractal dimension is a geometrical feature of irregularly shaped objects, but it is not a magic number. It is a dimension, and as two different objects can have the same size (the same three different Euclidean dimensions of length, width, and height), two very different patterns can share the same fractal dimension, as discussed in Chap. 1. This means that FD should be considered as a complementary method along with other parameters.

The fractal-based morphometric approach does not necessarily constitute a superior tool in comparison to other approaches. Different types of nonlinear analyses can be performed along and integrated into other approaches in order to achieve a holistic view of the same problem.

Despite the difficulty to discriminate between fractality and “apparent fractality originating from underlying randomness” [14], fractal analysis does not offer a

mere mathematical theoretical model because in clinical medicine it is aimed at finding reliable parameters such as diagnostic, prognostic, and therapeutic biomarkers. Despite the universality of the fractal language, fractal geometry cannot be considered as a Rosetta stone for the interpretation of every phenomenon; this has to be clear to the clinician and neuroscientist applying computational models for the translation of data into clinical applications.

13.8 The “Black Box”

Clinicians are not required to fully understand how fractal analysis and other computational methods work, just like a person using a computer for various tasks does not necessarily have to understand how the computer works. Clinicians should rather understand the proper way to use computational methods, address a problem, understand the inputs that could be analyzed by means of different and less conventional approaches, and correlate such findings with clinical findings, in order to make them useful in a clinical perspective.

As discussed in several chapters, the computation and use of FD have pros and cons and it should be understood that different values may be the result of “true differences in image texture or a result of certain arbitrary decisions made during the estimation process” [19]. The standardization of the technique as well as the delineation of the outcome assessment has to be clearly defined in order to guarantee an internal and external validation of the method and to avoid the bias of operator-dependent variability.

By using a concept from the artificial intelligence language, the concept of the black box will be used to explain this [9] (Fig. 13.1). The black box is substantially the machine, the computing algorithm. Although its way to function may not be obscure to computer scientists and engineers, it may be “black” to clinicians, who eventually do not understand exactly what is happening inside of the machine. For example, clinicians and sometimes even radiologists do not fully understand the physics background of computed tomography and magnetic resonance imaging, but when specific algorithms transform the raw data into images, they can interpret the meaning of such images very well. In such an example, the hardware and software transforming the raw data into images are the “black box.” Different algorithms may run in the black box, e.g., the Fourier transform of the raw magnetic resonance signals, or the box-counting method for computing the fractal dimension of an image like a histological slide, the detrended fluctuation analysis of a time series, etc. A specific task of the black box can be requested by a clinician for a specific application, but the competence to work “inside the black box” is still left to other kinds of specialists and technicians.

The role of the clinician is then to understand how that black box can be used, selecting the proper inputs to be analyzed. Examples of inputs could be microvessels or other features in histological samples or specific radiological sequences. The clinician, whom we define here as the “operator,” has to choose

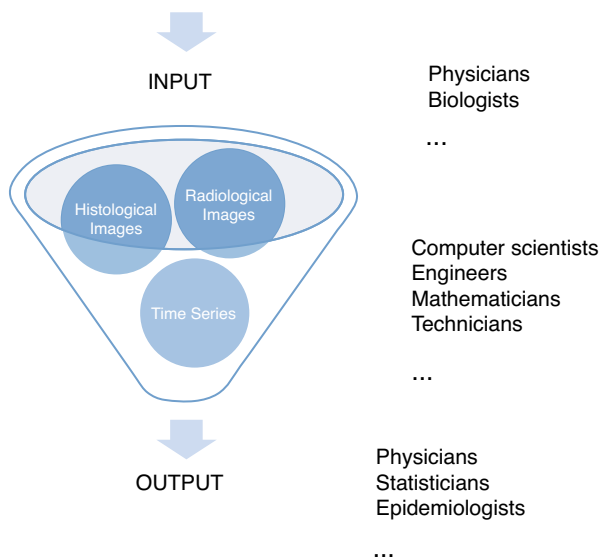


Fig. 13.1 The diagram shows the concept of the “black box” [38]. The black box here is represented by a funnel, where the data processing, by means of fractal analysis or other computational methods, occurs. The spheres are the data input to processing, which can be histological images (e.g., stained brain tumors, immunostained microvessels), radiological images (e.g., region of interests of brain tumors in MR or CT imaging), or time series (e.g., EEG, MEG, or intracranial pressure signals). The data are processed in the black box by means of standardized methods (e.g., box-counting method). The outputs are the results of the analysis, which are values useful for statistical analyses and epidemiological correlations, for example, or that can be validated as potential morphometric markers and/or diagnostic/prognostic biomarkers. The different professional figures involved in the process may be the ones quoted on the right of the figure

not only the inputs but also the methods to identify them (e.g., which kind of immunohistochemistry to detect endothelium, the threshold to extract binarized images, the selection of region of interests in histological or radiological images, etc.). This is an essential task because the lack of standardization of such a step would introduce a heterogeneous cohort of data into the black box, and the outputs, although computed in the same way, will be very heterogeneous as well (and therefore, useless). Therefore, the operator provides the inputs to the black box, but he/she is still required for the interpretation of the outputs. The outputs are the data generated by the computational method working in the black box and can be expressed in terms of morphometric parameters of angioarchitecture, for example, or the index of complexity of a radiological image. These data have to be analyzed by the clinician again to be correlated with epidemiological findings, for example, or clinical features, in order to be demonstrated as potential biomarkers, which might be useful in clinical practice. Although the different steps can be performed by different operators, the entire procedure should be

carefully orchestrated; otherwise, the translation of results into clinical practice might be misleading. The literature is full of controversial results on the same topics, e.g., researches finding higher or lower values of FD in the same tissues. The heterogeneity of the results has likely been the main reason for the failure of the widespread use of fractal geometry into biomedicine. For example, in a review, we compared the results of the morphometrics of microvascularization in different subtypes of pituitary adenomas, finding completely different and noncomparable results [10]. By analyzing the reason of such discrepancies, it turned out that all of the authors chose different analytic methods and different antigens, which gave rise to a plethora of incomparable results due to their heterogeneity. The different inputs and outputs in the “black box” inevitably outputted completely different results, giving wrong and naïve conclusions on the role of the microvascularization of the pituitary tumors. The comparison of oranges and apples may lead to wrong conclusions. Every computational method applied into the biomedical sciences, and therefore into the clinical neurosciences, should be standardized and verified in order to offer meaningful and useful results. This is essential to avoid the rejection a priori of a useful model!

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Chapter 14

Fractals, Pattern Recognition, Memetics, and AI: A Personal Journal in the Computational Neurosurgery



Antonio Di Ieva

Abstract In this chapter, the personal journey of the author in many countries, including Italy, Germany, Austria, the United Kingdom, Switzerland, the United States, Canada, and Australia, is summarized, aimed to merge different translational fields (such as neurosurgery and the clinical neurosciences in general, biomedical engineering, mathematics, computer science, and cognitive sciences) and lay the foundations of a new field defined computational neurosurgery, with fractals, pattern recognition, memetics, and artificial intelligence as the common key words of the journey.

Keywords Fractal analysis · Patterns · Artificial intelligence · Neurosurgery

Neurosurgery is a broad and complex field solidly based on the multidisciplinary background from different fields of expertise, including neuropathology, neuroradiology, neurology, and many others, not to mention the skills derived from different complementary surgical fields. Beyond the obvious manual skills that are required to perform neurosurgical operations, a neurosurgeon is challenged daily to recognize patients' signs and symptoms as well as findings from the different ancillary tests that make a disease worthy of surgical or other kinds of treatments. To frame the problem with a modern perspective, the neurosurgeon's inference is based on extracting features from patients' information and clinical data, images, and texts, with the aim to recognize patterns, i.e., diseases, and how they affect patients. This diagnostic step is then followed by the search for the best treatment according to the patient's prognosis. All these steps require a long training curve, made of theoretical

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and practical learning, successes and mistakes, knowledge acquired “osmotically” from colleagues, and actively improved thanks to the experience developed year after year. In 2008, I proposed that this “osmotic process” of learning is based on memetic principles, i.e., imitation, as many other cultural phenomena: children learn to speak by imitating their parents’ and relatives’ language; musicians learn to play by studying and imitating their teachers; and surgeons generally learn to operate by imitating experienced surgeons [1]. The appendix of this chapter includes a reproduction of that paper, in which Richard Dawkins’ meme was applied to neurosurgery and neuroscience in general.

Successful experience and mistakes build the neurosurgeon’s backbone by means of what could be defined “reinforced learning.”

The terminology used here, i.e., features extraction, pattern recognition, and reinforced learning, is obviously not casual, as there is a consolidated tradition to compare computers to the human brain, or to forge a machine’s artificial intelligence on the human intelligence template. Or to explain the complexity of human intelligence by means of computer science-based examples.

The inception of artificial intelligence (AI) has been a combination of science and technology, science fiction, hypes and hopes, and often more fears than real paradigm shifts. Following periods of ups and downs, where AI has gone through an alternance of a few winters followed by blooming springs, the last decade has witnessed a new enthusiastic surge of work which has put AI at the center of attention in any domain, from industry to economy, from science to health, and of course medicine and neurosurgery have not been passive in front of this cultural revolution.

Taking the field of neuro-oncology as an example, as surgical neuro-oncology is my main area of interest and expertise, brain tumors are exemplary of this cultural (r)evolution over time. The evolution of brain tumor diagnosis spans the past century beginning with neurological examination to crudely localize the tumor for surgical resection, progressing to the introduction of air into the ventricles during X-ray studies to indirectly confirm mass lesions (i.e., pneumoencephalography) [2], and arriving at the present day where we rely heavily on the latest noninvasive neuroimaging methods such as magnetic resonance imaging (MRI). Harvey Cushing, the father of modern neurosurgery, operated on patients supposedly affected by brain tumors with only the guidance of neurological examination and surely would have been astonished by MRI, and likely could never have imagined that AI would perform better than doctors in detecting the presence of brain diseases, as, for example, demonstrated at the world’s first “Human vs AI” neuroimaging diagnostic contest organized by *BioMind* in China in 2018 [3]. A similar pathway has been followed with regard to many other diseases, including cerebrovascular pathologies.

Recently, computational modeling and AI methodologies have been introduced for feature extraction and pattern recognition in many different domains (e.g., face and speech recognition, language), including biomedical imaging, with the aim of automating recognition of relevant target features or structures (e.g., automatic segmentation of brain tumors for volumetric analysis, automatic diagnosis, and even patient’s prognostic stratification). The use of AI in medicine is skyrocketing

[4–8], although the field is still challenged by concerns like the lack of validation against experts' opinion, issues related to models' generalizability and algorithms' explainability, as well as the lack of a standardized methodology of analysis.

14.1 From Fractals to Artificial Intelligence

When I started to operate pituitary tumors in Italy in the 2000s, the fact that pituitary adenomas were less vascularized than a normal pituitary gland was well known, but when I reviewed the literature on the morphometry of the microvessels of the pituitary gland and adenomas [9], I realized that there were too many controversial findings, due to high heterogeneity of analytical methods and lack of standardized tools to describe and quantify the microvascular networks in biological tissues. Aware of the principles of fractal geometry, which deals with the geometry of nature, my group and I, therefore, applied the fractal geometry Mandelbrot's box-counting algorithm, initially used to characterize the coastlines (in a seminal paper published in *Science* in 1967) [10], to characterize the microvascular patterns of the pituitary gland for the first time, in its entire physiopathological spectrum (from the normal gland to its tumorous counterpart). The study, published in 2007, provided evidence that the switch from the physiological gland to the pathological one is related to a microvascular disarrangement and, in addition, that the computational fractal-based estimation of the vascular patterns is an objective and reliable tool to be used in quantitative medicine [11]. The fractal dimension (FD) and other fractal-based parameters were valid estimators of natural patterns, to be explored for brain tumor characterization [12, 13]. By means of histopathological and neuroradiological computer-aided analyses, the extension of the preliminary methods from my and other research groups showed that unique microvascular profiles, i.e., microvascular fingerprints, exist in other brain tumors, including gliomas [14–17] and meningiomas [18]. These findings encompassed various researches (including the one within my PhD at the Medical University of Vienna, Austria, in the period 2008–2011, under the formal supervision of Prof. Manfred Tschabitscher, neuroanatomist, and other scientists from different fields) performed in multiple countries (e.g., Austria, the United States, Switzerland, and Canada) in collaboration with scientists from around the world. Indeed, a further work published in 2012 showed that a computer can successfully recognize the histopathological grade of gliomas (grade 2 vs. 3 vs. 4) by automatically categorizing the microvascular disarrangement quantified by means of some computational fractal-based parameters, including the novel microvascular fractal dimension [14]. Since the beginning of my research in fractal analysis, which always runs in parallel to my clinical and surgical routine as a practicing neurosurgeon, I realized that scientists with interest and expertise in fractals were rare to be found. For this reason, during the first year of my Ph.D. in Vienna (2008), in collaboration with Drs. Carlo Russo and Fabio Grizzi, we founded the Virtual Fractal Lab (vFL, www.fractal-lab.org), an Internet community joined by several hundreds of scholars around the world, aimed to collaborate on

the application of computational fractal-based analysis to the biomedical sciences. This “laboratory” acted as a proper research center, although only in a “virtual” way, as the collaboration happened mostly via emails and meetings, in an era in which teleconferences were not used as much as nowadays. This “virtual” community also attempted to publish a new journal, the Fractal Lab Journal, although unfortunately the initiative was aborted for lack of time of the editors and collaborators. In any case, the seed of the vFL allowed a variety of translational research to bloom, spanning from the clinical neurosciences to ecology and mathematics, and editorial collaborations, including two seminal reviews, published in *The Neuroscientist* [19, 20], and the first and second editions of the book *The Fractal Geometry of the Brain* (Springer Publisher).

Already in that multifold translational research and those papers, many machine learning methodologies, which are much more popular nowadays, were used, including Principal Component Analysis (PCA), Independent Component Analysis (ICA), Support Vector Machine (SVM), etc., thanks to the tight collaboration with mathematicians and computer scientists, first among all, Dr. Carlo Russo.

The translation of the method into neuroimaging Aware of the limitations of the analysis on neuropathology slides (including the obvious need to have brain tissue sampled in neurosurgical operations), the fractal-based computational methodology was applied to neuroimaging, including nuclear medicine position emission tomography (PET) [21] and MRI [15, 16]. Indeed, by bringing together a multidisciplinary group of computer scientists, engineers, neurosurgeons, and neuroradiologists, we were able to translate the fractal-based pattern recognition of brain tumors’ features into 3D magnetic resonance images [22], using such a technique to differentiate gliomas subtypes on ultra-high field (7 Tesla) MRI [15], and then gliomas subtypes and different types of tumors (such as metastases and lymphomas) on 3 Tesla MRI [16]. Once in Australia (although this kind of research actually goes back to my period as a neurosurgeon in Italy already in the 2005–2008 period), a further extension of the idea was the application of advanced imaging and machine learning techniques to recognize features in the spectroscopic fingerprinting of brain tumors by means of magnetic resonance spectroscopy, toward the concept of “spectrobiopsy,” which is the diagnosis based on the nature and concentration of the metabolites as seen and quantified on MRI [23, 24]. Complementary approaches were also used to characterize other diseases and different problems, for example, the microangioarchitectural complexity of arteriovenous malformations’ nidus in relation to the obliteration response following gamma knife radiosurgery treatment [25]. Further applications, developed during my Australian (and current) period, include the use of fractal analysis to compare MRIs and to check the accuracy of synthetic MR images [26] or to use the fractal radiomics parameters to characterize brain tumors [27, 28].

Regardless of the specific application, it was clear that computational fractal-based analysis was aimed to extract features (e.g., the architectural complexity of the disease) and recognize patterns (i.e., predict the nature of the disease, or even its evolution over time, related to patient’s prognosis). The multidimensional

complexity of the analysis was also approachable by means of machine learning techniques, aimed to mine data, cluster patterns, and stratify and predict patients' outcomes.

Computational modeling, computerized techniques, fractal geometry, machine learning, and more recently deep learning are all aimed toward data mining, features extraction, and recognition of any kind of pattern. From such a perspective, fractal analysis was the computational model which allowed me to join the wonderland of computational modeling and AI in the clinical neurosciences. Further applications in a very assorted plethora of diseases, fields, and sub-specialties have been the natural and physiological consequence of merging the different computational techniques, including fractal analysis and AI, and different conditions of neurosurgical interest. This pathway led to the foundation of the Computational NeuroSurgery (CNS) Lab at Macquarie University, Sydney, Australia, in 2018, thanks to the initial grant and research support from Macquarie University and the Royal Australasian College of Surgeons (RACS) when I was awarded the John Mitchell Crouch Fellowship, the most prestigious research award from the RACS. The CNS Lab activity aimed (and aims) to find diagnostic, prognostic, and therapeutic markers of disease of neurosurgical interest by means of computational modeling and AI, and where different projects, several of which under the broad umbrella of machine vision research, run in parallel thanks to the tight collaborations between national and international neurosurgeons, clinical neuroscientists, computer scientists, and machine learners, engineers, and statisticians. The laboratory has generated multiple papers, collaborations, and research grants in the order of several millions of dollars in the last years. The creation of a novel and hybrid Computational Neurosurgery Fellowship, where a neurosurgeon is merging their neurosurgical expertise with research in computational modeling and AI, is also an innovative aspect of this personal journey, and my personal legacy to the new field of computational neurosurgery.

This journey reflects the author's search for a structural and functional fingerprint within different structures and domains of applications, as summarized in Fig. 14.1.

14.2 Fractals, Artificial Intelligence, and Cognitive Neurosciences

Understanding the relationship between vision and cognitive perceptions (or how an image is interpreted) is one of the greatest challenges in neuroscience. By training many students, residents, registrars, and researchers in the field of the clinical neurosciences (including neuroimaging and neuropathology) over the last 15 years, I noticed how "random" the eye movements of unexperienced beginners can be vs. the ones of experienced specialists' when they are looking at medical images and searching for pathologies and features related to any kind of clinical clue. This is very well known in the field of cognitive neurosciences [29–32], but I was

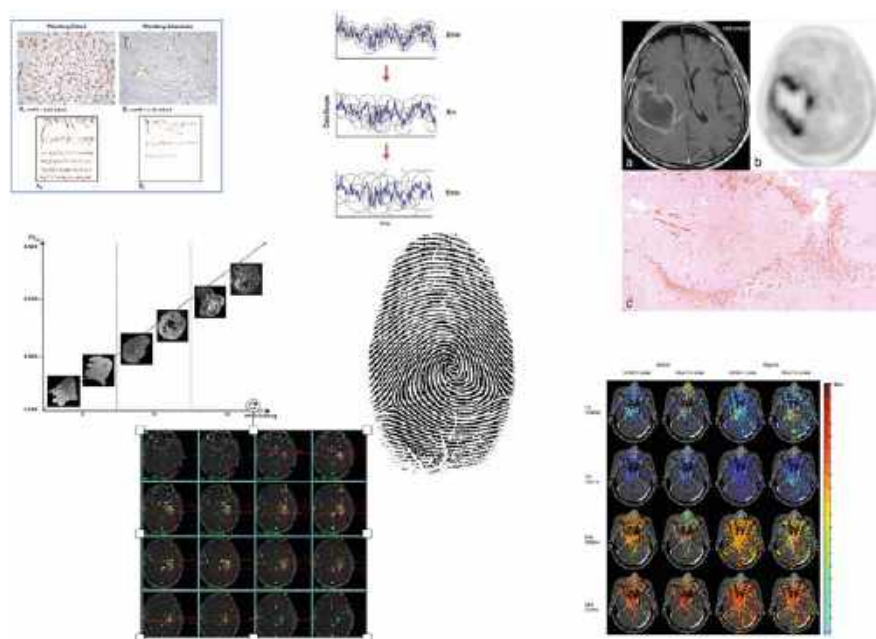


Fig. 14.1 Some examples of fractal analysis' application to characterize the fingerprinting of different structures within different applications, including the microvascularization and other features of brain tumors in histopathology and neuroimaging, the angioarchitecture of arteriovenous malformations in radiosurgery, the intracranial pressure time series, the eye tracking of experts looking at neuroradiology images, etc.

looking for a computational fractal-based model to characterize eye tracking in order to compare them, stratify experts from beginners and naïve, and then implement AI-based models by “boosting” and reinforcing the predictive models by means of specialists’ cognitive expertise. In this project, I considered eye tracking, also called “scanpaths,” [33] as a proxy, i.e., surrogate biomarkers, of higher cognitive function. In summary, I aimed to use computational fractal-based analysis to create a fractal map of images and of the eye movements required to track the images themselves. Such a fractal map could be used for a cognitive neuroscience purpose in an efficient way (i.e., the way experts focus their attention on relevant features in an image while excluding the presence of “noise” (e.g., non-diagnostic and non-relevant features) and also to stratify or quantify the learning process of beginners through their pathway toward expertise. More specifically, I was looking for a novel computational method based on the fractal maps of eye tracking that could quantify the cognitive fingerprint for analyzing complex images. It was also important for this method to longitudinally score the acquired visual efficiency of novices during their training period, looking for a potential computational marker of cognitive efficiency. In one of my successful grants (an Australian Research Council Future Fellowship, 2019–2023), I proposed to use fractals as the Rosetta Stone for the translation of

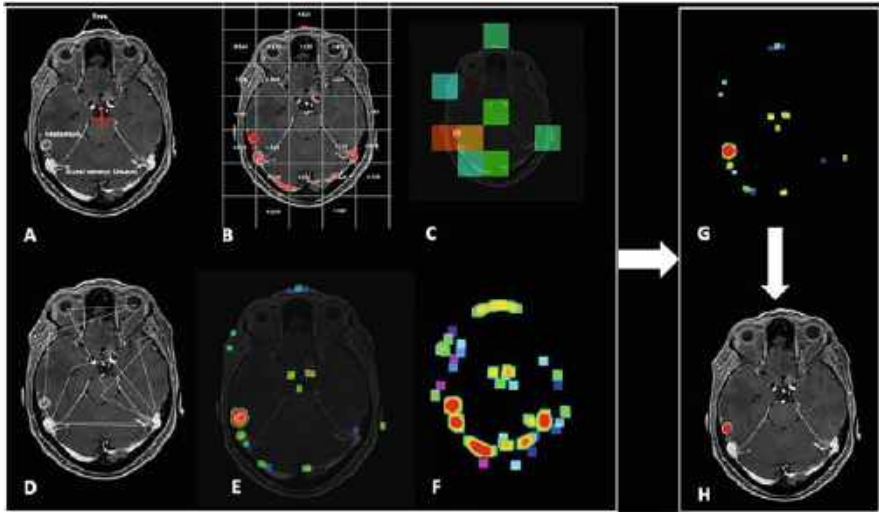


Fig. 14.2 (a) An axial slice of a brain MRI (T1 post-gadolinium) shows a brain metastasis. (b) The fractal map computed on the image and (c) its heat map based on the regional fractal dimension value (related to the space-filling properties of the object). (d) Example of a participant’s eye-tracking scanning at the image; (e) visual heat map of an expert looking at the image, vs. (f) a naïve looking at the same image. (g) Generation of a saliency map targeted on the main region of interest by merging the fractal map (b) with the expert eye tracking (e) “denoised” from the nonexpert visual tracking (f). (h) Final result of the computed most salient region of interest in the image

experts’ decision-making and cognitive efficiency into a computerized method to recognize visual patterns, with one of the final aims to automatically recognize relevant features in images, paving the way for a new generation of computer tools that enable human brain-like detection of salient features in any kind of images. Figure 14.2 summarizes the pipeline.

At my laboratory, we used the fractal dimension of the scanpath to measure the geometric complexity of the eye movements of experts vs. nonexpert radiologists looking at brain MRIs and we introduced the Hurst exponent to analyze them as time series, adding the temporal analysis to the geometrical features [34]. Chapter 42 summarizes the main finding of the research, with further works in press and in progress using the methodology to differentiate the scanpaths of neurosurgeons looking at brain MRIs, ophthalmologists and otorhinolaryngologists looking at images within their domain of expertise, medical students looking at medical images and re-scanned longitudinally during their training period, to show their potential “cognitive efficiency” gained over time, musicians reading classic music sheets, artists looking at images, layperson looking at general images, etc.

14.3 Toward the Augmented Decision-Making of the Future

While it seems clear that AI could assist the natural intelligence (NI) of experts to improve diagnosis and treatment of several diseases, it is not yet clear how to integrate AI to improve diagnostic decision-making. The underlying fear of a dystopic challenge where AI is pitted against NI can be overcome by viewing AI as a means to create an *augmented physician*, wherein experts' NI is augmented by AI (which recursively learns from NI's experience and expertise), shifting from a human-vs-machine to a human-AND-machine paradigm. In this model, *I foresee that the machine will not replace physicians but that physicians using AI will soon replace those not using it*, as their expertise will be enhanced by the fast and reliable computational power of AI, as proposed in my letter published in The Lancet [35]. The same applies to multidisciplinary teams (MDTs) whereby AI will help future interdisciplinary meetings to become more objective and reliable (Fig. 14.3), as currently, MDTs are the gold standard for the best decision-making in clinical medicine, although susceptible to many biases, which makes them not infallible.

A point of convergence shared by AI and NI is the analysis of patterns: a machine can extract features to recognize a pattern in a different but complementary way to a human expert who recognizes and differentiates aspects of an image to deduce the diagnosis, or at least to reach a differential diagnosis. In both ways, the specific peculiarity of the pattern (e.g., of a brain tumor) can be summarized by its intrinsic fingerprint. Such a fingerprint can be analyzed and quantified by means of computational fractal-based modeling, as shown in my previous research.

Fig. 14.3 The augmented multidisciplinary team, where a team of different specialists aimed to characterize patients' diseases and propose the best treatment, can be enhanced by the power of AI

Augmented Multidisciplinary Team



14.4 Limitations and Future Perspectives

Despite the advances of fractal analysis and AI in the last years, there is still a lack of standardization and routine use of the capacity of fractal-based computational modeling and AI to be synergistically integrated into healthcare. Indeed, many publications on the application of AI to neuroimaging have been made possible by the mono-disciplinary work of computer scientists without involving clinical neuroscientists, with scarce relevance and translational application in the clinical scenario. For example, since the introduction of the Brain Tumour Segmentation (BraTS) challenge in 2012 [36], several AI-based models have been introduced to automatically segment the edges of brain tumors on MRI. Despite the great achievement of the proposed models, the definitions introduced by the BraTS committee as well as the results of the winners of the challenge every year have not been translated into the clinical application and modified any decision-making related to patients. The main reason for this is related to the involvement of many computer scientists, very few neuroradiologists, and no other clinical neuroscientist (e.g., neurosurgeons, radiation oncologists, neurologists, etc.) in the challenge. Aware of such a limitation, by merging the translational expertise of different physicians and scientists, we successfully introduced a heuristic approach to bring deep learning-based automatic segmentation of brain tumors on MRI into the clinical scenario [37]. By using deep learning and fractal-based radiomics of brain tumors and other diseases of neurological and neurosurgical interest (including concussion, traumatic brain injury, cerebrovascular diseases, chronic pain, etc.), we aimed to implement doctors' armamentarium with objective and fast computational tools. Further research is in progress to create an ethical framework for introducing computational fractal-based analysis and AI in the clinical scenario, in order to avoid breaching of data security and patients' privacy and rights.

In order to achieve these aims, new translational professional figures are required, where a neurosurgeon, for example, needs some expertise in computational modeling, and where a computer scientist or biomedical engineer needs to be involved more in the clinical habitat, in order to better understand the nature of the problem and how they can help. In such a perspective, some private funds secured allowed me to create a new hybrid fellowship, the Computational Neurosurgery Fellowship, in which a neurosurgeon (consultant or senior trainee) can learn the basics of computational modeling, fractal analysis, and AI and apply such powerful tools to different diseases of neurosurgical interest.

The applications in the field are endless, and it is a matter of time and energy to realize aims and projects, hoping to fulfill the expectations.

As the adagio would say, the sky is the limit.

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Chapter 15

Clinical Sensitivity of Fractal Neurodynamics



Elzbieta Olejarczyk, Milena Cukic, Camillo Porcaro, Filippo Zappasodi, and Franca Tecchio

Abstract Among the significant advances in the understanding of the organization of the neuronal networks that coordinate the body and brain, their complex nature is increasingly important, resulting from the interaction between the very large number of constituents strongly organized hierarchically and at the same time with “self-emerging.” This awareness drives us to identify the measures that best quantify the “complexity” that accompanies the continuous evolutionary dynamics of the brain. In this chapter, after an introductory section (Sect. 15.1), we examine how the Higuchi fractal dimension is able to perceive physiological processes (15.2), neurological (15.3) and psychiatric (15.4) disorders, and neuromodulation effects (15.5), giving a mention of other methods of measuring neuronal electrical activity in addition to electroencephalography, such as magnetoencephalography and functional magnetic resonance. Conscious that further progress will support a deeper understanding of the temporal course of neuronal activity because of continuous interaction with the environment, we conclude confident that the fractal dimension has begun to uncover important features of the physiology of brain activity and its alterations.

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15.1 Introduction

One of the most prominent features of the brain is its fractal nature expressed through self-similarity of patterns occurring in its anatomical and functional organization at different spatiotemporal scales [1–4]. This property depends on brain state, and it changes in different physiological and pathological conditions. Thus, the brain is a complex system, which can be examined by measures derived from nonlinear dynamics. In the second half of the 1980s and the beginning of the 1990s, there was a growing interest in application of fractal dimension in analysis of EEG signals [5]. Many different measures were defined. They based mainly on the phase space reconstruction which required a time-consuming calculations. Contrary to these measures, the Higuchi fractal dimension (HFD) [6] is particularly useful in real-time clinical applications due to its high computational efficiency because this measure is defined in time domain and can be calculated for a short segment of length about 100 samples. HFD has been widely used mainly in the analysis of electrophysiological signals (EEG, EKG, EOG, and EMG) but has also proved useful in the analysis of signals from other modalities such as fMRI.

In this chapter, we discuss the changes in the EEG complexity, evaluated by means of HFD, related to various physiological processes such as sleep, anesthesia, and healthy aging as well as related to different neurological and psychiatric disorders such as epilepsy, stroke, Alzheimer's disease, depression, and schizophrenia. Moreover, the EEG is sensitive to the influence of an external stimulation by electromagnetic field (photostimulation, microwaves, transcranial electrical, and magnetic stimulation). A systematic review of the results of HFD applied as a single measure or, more broadly, as an element of a feature set together with other measures is provided.

15.2 Physiological Processes

15.2.1 Sleep

In 1994, HFD was for the first time used in clinical studies of sleep to evaluate changes in the EEG complexity of alpha waves from wakefulness to sleepiness [7]. At the beginning of 2000s, Klonowski and Olejarczyk continued this research. They used HFD method to analyze EEG signals during the entire course of sleep, and they noticed that HFD allows for the differentiation of individual stages of sleep, which made it possible to apply this method to construct a hypnogram useful in the diagnosis of sleep disorders. In 2003, the first HFD-based sleep stager was

developed [8–12]. The 4-second EEG segments were analyzed in each of the sleep stage in 15 healthy persons and 15 patients suffering with insomnia. The EEG signals were recorded in First Psychiatric Clinic of Medical Academy in Warsaw using the standard 10/20 electrode system with a sampling frequency of 128 Hz, 12-bits analog-to-digital converter, and with the current source density (CSD)—an approximation of the surface Laplacian, as the reference electrode. Then, HFD was averaged over all segments in a given sleep stage and over all patients. The results were compared with six vigilance stages (wake, N1, N2, N3 and N4, and REM) assessed by a specialist according to the criteria of Rechtschaffen and Kales (1968) [13], valid until 2007, as they were updated by American Academy of Sleep Medicine (AASM 2007) by unification of stages N3 and N4 to only one stage N3. The mean HFD for each sleep stage was as follows: 1.58 ± 0.04 (wake), 1.49 ± 0.06 (N1), 1.49 ± 0.06 (N1), 1.38 ± 0.04 (N2), 1.30 ± 0.04 (N3), and 1.48 ± 0.09 (REM). The parameter k_{max} was equal to 8. The advantage of HFD algorithm is that it is fast and therefore it can be used for online monitoring of changes in EEG signal complexity during sleep as well as for monitoring the drowsiness in drivers. Thus, Klonowski and Olejarczyk patented and implemented this algorithm in the platform of TECMIC—Tecnologias de Microelectronica SA (programs FRAST™ for automatic sleep staging written by E. Olejarczyk) in the SENSATION project realized from 2004 to 2006 (www.sensation-eu.org). A typical HFD time course and corresponding automatic hypnogram are shown in Fig. 15.1. HFD of whole-night sleep of a healthy person reveals the characteristic quasi-periodic dynamics of sleep stages. The highest HFD is during wakefulness. HFD decreases when sleep becomes deeper. HFD increases again during REM stage to a lower level than in the wake stage. During the whole night, there are from 3 to 5 such cycles.

No differences in HFD of individual sleep stages between healthy controls and patients with insomnia were found. The difference between both groups consisted only in much more irregular dynamics of sleep stages in insomniacs. Moreover, a comparison between 44 healthy controls and 35 chronic insomniacs showed differences in the cortical reactivity of both groups to eye closing and to fatigue related to the performance of a task [14]. In the occipital lobe, FD decreased significantly more in chronic insomniacs than in controls which can be related to an increased alpha power, mainly in the right hemisphere, in patients with insomnia.

A logistic model with four parameters was proposed to study the frequency-dependent topographic differences between two states: wake and drowsiness [15]. The model confirmed that the main difference between these states relies on the reduction of alpha activity. The transition from wake to drowsiness during the resting state with eyes closed caused a decrease in the number of clusters from four to two which can be explained by an alpha activity gradient with the highest value in the occipital lobe and the lowest in the frontal part of the brain, with exclusion of the temporal regions because of the unchanged presence of auditory cortex activity.

HFD calculation does not require a large number of samples (about 100). This feature makes HFD useful for the study of sleep spindles as well. HFD together with other nonlinear measures, such as Sevcik's fractal dimension [16], for example, allows to improve the classification of sleep spindles due to the fact that these

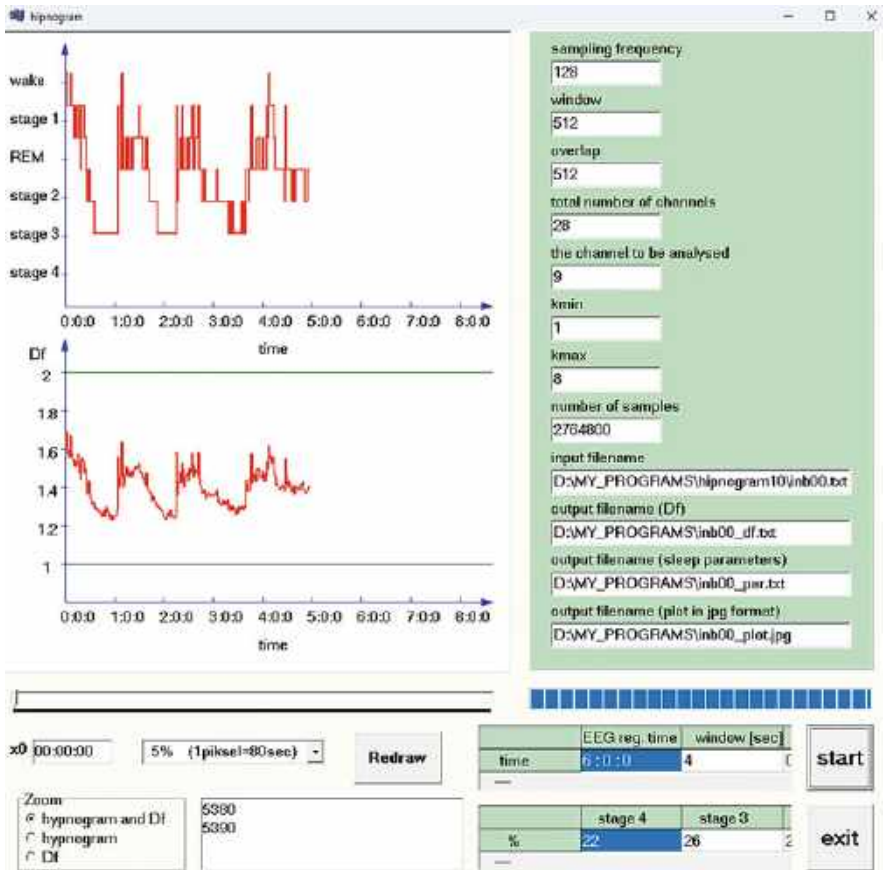


Fig. 15.1 The graphical interface of FRAST™ program showing a typical HFD time course marked with red line in the lower window and the corresponding automatic hypnogram in the upper window. HFD of whole-night sleep of a healthy person reveals the characteristic quasi-periodic dynamics of sleep stages. The highest HFD is during wakefulness. HFD decreases when sleep becomes deeper. HFD increases again during the REM stage to a lower level than in the wake stage. During the whole night, there are from 3 to 5 such cycles

features are separated well in the feature space [17]. The authors showed that the accuracy of the multilayer perceptron (MLP) classifier was bigger in comparison to the k -NN classifier to reach this objective.

HFD depends on changes in circadian rhythms, which is associated with changes in alertness and performance [18]. The 10-minute EEG data were recorded in 21 healthy young males four times during the day (T1: 8.00 a.m., T2: 11.30 a.m., T3: 2.30 p.m., and T4: 7.00 p.m.) in two conditions: eyes open and eyes closed. HFD decreased during the whole day in frontal, temporo-parietal, and occipital regions in condition with eyes open, while in condition with eyes closed, HFD increased at T2 in central and frontal regions. The authors suggested that multifractal

topographic analysis of sleep EEG might be more adequate to distinguish among different sleep stages. Moreover, the results of regression analysis, performed to test the relationship between clinical tests and two classes of independent measures—linear (power spectra in different frequency bands) and nonlinear (HFD), showed that up to 20% of the variance was due to HFD, which testifies the presence of nonlinear phenomena in sleep EEG.

Recently, Jain et al. [19] proposed to combine 23 different measures (temporal, spectral, time frequency, nonlinear, and statistical) as features in the sleep stages classification. Among the nonlinear features, HFD, Hurst exponent, Lempel–Ziv complexity, sample entropy, and detrended variation were relevant for classification.

The availability of a unique intracranial EEG (iEEG) database of 1772 electrode locations, provided publicly by the Montreal Neurological Institute and Hospital, the Centre Hospitalier de l'Université de Montreal, and Grenoble-Alpes University Hospital (<https://mni-open-ieegatlas.research.mcgill.ca/>), allowed for topographical analysis of the iEEG complexity using HFD and for the creation of a complete functional brain atlas for 38 cortical regions across the various stages of vigilance [20]. As we expected, HFD decreased with the sleep deepening, but in a region-specific manner (see Fig. 15.2). HFD had the highest values during wakefulness in the frontal lobe ($\text{HFD} > 1.7 \pm 0.1$) and the lowest values during N3 in the parahippocampal gyrus and fusiform gyrus ($\text{HFD} < 1.35 \pm 0.1$). The only area in which the HFD did not differ between any two vigilance stages was the transverse temporal gyrus. The region-specific signatures of transition between the successive vigilance stages were: (1) no change from wake to N2 in the fusiform and parahippocampal gyri; (2) REM and wakefulness differed mainly in the precentral gyrus; and (3) REM and N2 were similar only for the postcentral gyrus. Generally, the fronto-central regions allowed for clear differentiation between the vigilance stages, in contrast to the temporo-occipital regions that slept differently. Moreover, the intraindividual analysis of the same iEEG data showed that the relationship between HFD of the primary motor (M1), somatosensory (S1), and auditory (A1) cortices ($\text{M1} > \text{S1} > \text{A1}$) was preserved in all sleep stages [21].

15.2.2 Anesthesia

The results of the previous studies on the EEG complexity during the whole sleep showed that HFD is useful in differentiation of the individual vigilance stages. Anesthesiologists from the University of Silesia in Katowice hypothesized that HFD may also be useful in evaluation of the depth of anesthesia. The analysis of EEG signals recorded in 58 patients during general anesthesia, induced with fentanyl, midazolam, and propofol, showed high correlations ($r > 0.7$) between HFD and BIS, which is a standard used parameter in clinical practice. However, the change in HFD was better expressed and occurred earlier than the change in BIS, which was especially observed at the time of anesthetic induction at the beginning of the anesthesia and at the time of intermittent boluses of propofol during the course

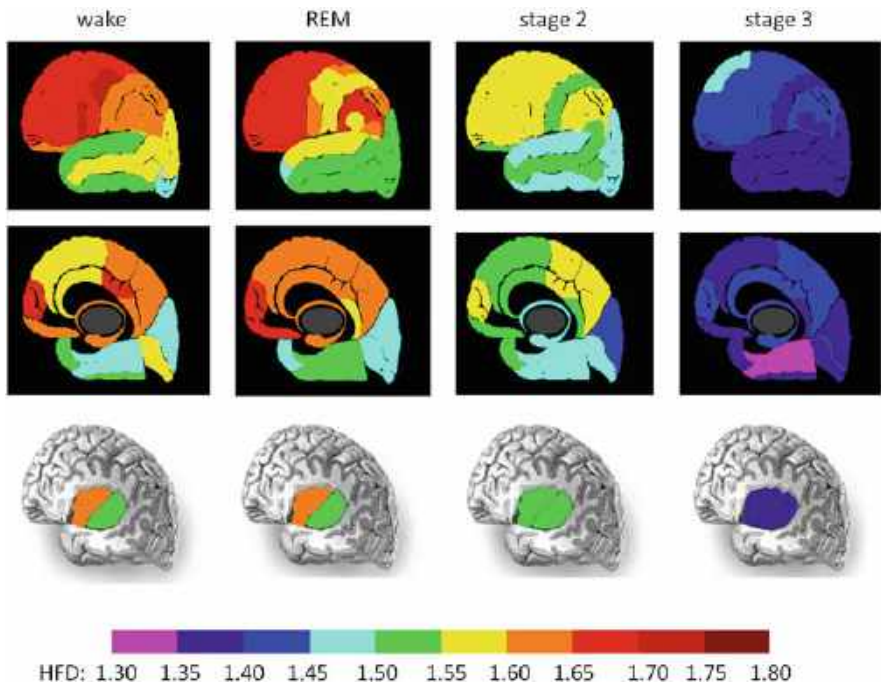


Fig. 15.2 HFD of different brain areas in four sleep stages (wake, R, N2, and N3). The results are shown on lateral, medial, and insular views (from the upper to the lower row). HFD decreases from wake to N3. The highest HFD is observed in the frontal lobe during the awake state and stage R (red areas). The lowest HFD was in parahippocampal and fusiform gyri during stage N3 (lilac color). (The figure is reproduced from Ref. [20] under a CC-BY license)

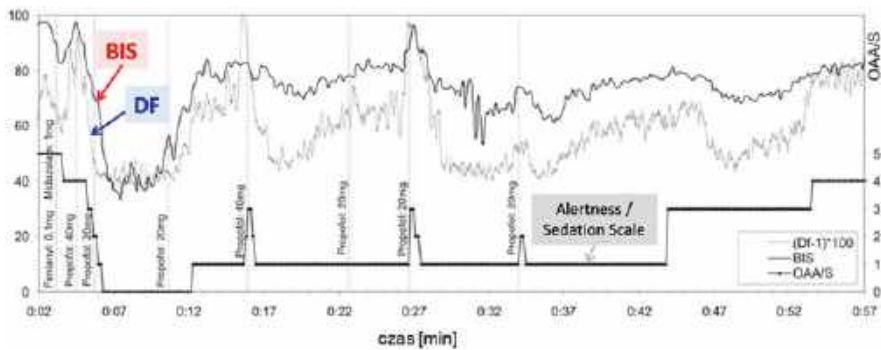


Fig. 15.3 Comparison of HFD and BIS parameters. HFD was more pronounced and occurred earlier than the change in BIS, which is especially visible at the time of induction at the beginning of anesthesia, and at the time of anesthetic administration during anesthesia. (The figure is a modification of Case 4 from Ref. [23])

of anesthesia (Fig. 15.3). HFD momentary rapidly increased at the time of the induction and then decreased to the level indicating the depth of anesthesia; deeper anesthesia corresponded to lower HFD values [11, 12, 22–24]. Since one of the advantages of the HFD algorithm is its high computational efficiency, it can be used online to monitor the depth of anesthesia. This algorithm was implemented in 2006 to the platform of TECMIC—Tecnologías de Microelectronica SA (program FRANES™ for monitoring the depth of anesthesia written by E. Olejarczyk).

HFD depends strongly on the frequency band and on the dose of anesthetic. For higher frequencies and lower anesthetic concentration, HFD takes higher values. HFD decreased from 0.90 to 0.55 when the frequency band changed from 6–47 Hz to 0.5–19 Hz, and from 0.90 to 0.83 when the remifentanyl concentration was increased from 0 to 4 ng/ml [25].

It is worth noting that during volatile anesthesia (isoflurane, sevoflurane, or desflurane), a characteristic epileptiform patterns, i.e., spikes, sharp waves, discharges, appear, which may affect the incorrect assessment of the depth of anesthesia [26–34]. Therefore, several algorithms were developed for the automatic detection of these patterns to eliminate them from the EEG recording before it is analyzed using anesthetic depth indices such as HFD and BIS [34–37].

Topographic HFD changes induced by propofol anesthesia were observed mainly in the frontal lobe [38]. HFD decreased with the deepening of anesthesia, which can be related to a cessation of the higher-level information processing in the cortex.

A comparison of the mean HFD of EEG recorded in rats anesthetized with ketamine and nembutal showed the differences between the action of both anesthetics in different brain structures [39]. In particular, HFD was higher in the cortex and in the right hippocampus, which can be related to lower activity of delta and higher activity of beta and gamma during anesthesia with ketamine than during anesthesia with nembutal, because HFD decreases for lower frequencies (delta) and increases for higher frequencies (beta and gamma). Moreover, the effect of nembutal anesthesia on the EEG complexity was layer-dependent [40]. The intracerebral EEG was recorded in rats from different layers of the cerebellar cortex: molecular layer (ML), Purkinje cell layer (PCL), granular layer (GL), and the white matter (WM) in the vermal part of cerebellar lobule V. HFD increased from ML to WM. HFD showed a negative correlation with theta activity in ML but a positive correlation with gamma activity in ML and GL.

15.2.3 *Maturation and Healthy Aging*

HFD has also been successfully applied to assess brain maturity of 36 preterm infants [41]. Twelve features were considered: HFD, spectral entropy (SpEn), spectral edge frequency, relative delta power, mean, variance, skewness, kurtosis, median of the absolute values of minima and maxima, Hilbert transform amplitude, inter-hemispheric symmetry, and inter-burst interval length. The calculations were performed in 10-minute EEG segments without artifacts, as well as separately

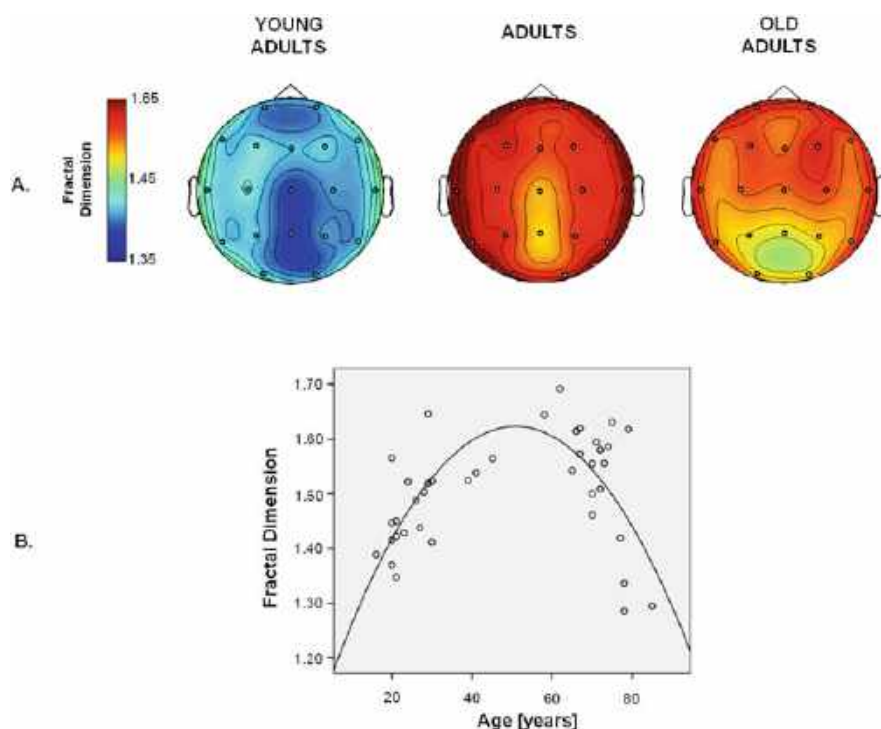


Fig. 15.4 (a) Topographies of HFD in the 3 groups: young adults (<25 years), adults (25–66 years), and old adults (>66 years) (b) HFD in function of age. The curve describing changes in HFD values over the full age range (from 16 to 85 years) is a parabola whose vertex corresponds to the age of about fifty years. (Figure reproduced from Ref. [42] under a CC-BY license)

during burst and inter-burst periods identified in these segments independently by three observers. The highest Spearman correlations were during the burst periods for HFD (0.644, 0.618, and 0.441) and for SpEn (0.688, 0.453, and 0.425), while during the inter-burst periods for spectral edge frequency (0.550, 0.547, and 0.536).

The aim of the study of Zappasodi et al. [42] was to investigate the usefulness of HFD in studying changes in the complexity of EEG during the resting state in 40 healthy people depending on age. The results showed that the curve describing changes in HFD values over the full age range (from 16 to 85 years) is a parabola whose vertex corresponds to the age of about 50 years (Fig. 15.4). HFD increased from 16 to 50 years and then decreased. These changes were topographically specific. In the elderly, the changes in the inter-hemispheric asymmetry were found in the frontal and central areas.

15.3 Neurological Disorders

15.3.1 Epilepsy

The EEG signal during epileptic seizure is characterized by an increase of amplitude and a decrease of frequency, which corresponds to a decrease of HFD [11, 12]. It has also been noticed that HFD of the ECG signals decreases during an epileptic seizure [43].

In many studies of seizure prediction based on HFD and other nonlinear measures [11, 12, 44–49], the publicly available dataset provided by the Department of Epileptology at the University of Bonn (<http://epileptologie-bonn.de>) [50] has been explored. The database contains five sets of 100 single-channel EEG segments lasting 23.6 s and recorded with a sampling rate of 173.61 Hz. There are two sets with surface EEG data recorded during the resting state in healthy controls with eyes open (set A) and with eyes closed (set B) and three sets with the intracranial EEG acquired in patients with epilepsy, including one set with seizure activity (set E) and two sets with interictal activity, one from the epileptogenic zone (set D) and other from the hippocampal formation of the opposite hemisphere of the brain (set C). The following values of mean HFD were received for these five sets: set A: $D_f = 1.43 \pm 0.08$; set B: $D_f = 1.35 \pm 0.09$; set C: $D_f = 1.27 \pm 0.08$; set D: $D_f = 1.23 \pm 0.06$; and set E: $D_f = 1.25 \pm 0.08$ [11, 12]. HFD of EEG during the resting state with eyes closed was lower than that with eyes open, which was caused by the presence of alpha rhythm in condition with eyes closed. Moreover, HFD of epileptic EEG was significantly lower than HFD of normal EEG. Although HFD differentiated well between focal and nonfocal areas of brain during seizure-free intervals (p -level = 0.00162), no difference was found in HFD of interictal and ictal activity.

Another data were recorded at the Johns Hopkins Epilepsy Center in 45 patients with partial epilepsy, including 17 patients with mesial temporal lobe epilepsy, 5 with neocortical temporal lobe onset epilepsy, 15 with extratemporal lobe epilepsy, and 8 patients unclassified due to multifoci onset seizures or rapid bilateral spread. HFD allowed to detect only onsets of neocortical extratemporal seizures, and not mesial ones, which may allow for their differentiation [51].

In 2013, Hu et al. [52, 53] proposed a monitoring system, based on a combination of several measures such as power spectrum, Petrosian fractal dimension, HFD, and Hjorth parameters, for seizure detection and focus localization, with 98.3% and 75.5% accuracy, respectively.

Next, an approach based on a combination of nonlinear measures and machine learning initiated a series of studies aimed at developing an optimal system for seizure prediction.

HFD was used as a feature in classification of preictal and ictal intracranial EEG activity using Bayesian linear discriminant (BLD) classifier. The results of classification were smoothed by Kalman filter. The sensitivity of this algorithm, evaluated using 21 intracranial EEG recordings from the Freiburg database, was

86.95% and 89.33% for the seizure occurrence period of 30-minute and 50-minute, respectively. This algorithm may be suitable for real-time seizure prediction due to the high computational efficiency of both methods, HFD and BLD classifier [54].

Another study consisted in classification of five nonlinear features such as Hurst exponent, HFD, largest Lyapunov exponent, and sample entropy, using three different classifiers: support vector machine (SVM) with different kernel functions, decision tree (DT), and k-nearest neighbor (KNN) [55]. The best classifier was SVM with radial basis function (RBF) kernel, which provided the accuracy, sensitivity, and specificity of 98%, 99.5%, and 100%, respectively.

Other set of five features, i.e., Shannon entropy, collision entropy, distance entropy, Hjorth parameters, and HFD, were used in classification performed by means of hierarchy hidden Markov model in three groups: healthy controls (G1), patients during interictal (G2), and ictal period (G3) [56]. The accuracy of classification for G1 vs. G2, G1 vs. G3, and G2 vs. G3 was 95.62%, 96.67%, and 95.00%, respectively.

Furthermore, the support vector machine (SVM) with HFD and Katz fractal dimension (KFD) in five frequency bands (delta, theta, alpha, beta, and gamma) as features, allowed for classification with the accuracy of 100% for normal vs. ictal state, and 99.5% for normal vs. interictal state [57], whereas SVM with statistical features of discrete wavelet transform (DWT) and nonlinear features, such as entropies and fractal dimension, allowed for classification with an accuracy of 99.7% [58]. Lu et al. [59] obtained the accuracy of SVM classification of two features: HFD and sample entropy equal to 89.8%. Both measures had lower values in epileptic than in non-epileptic EEG. The combination of two features gave better results of classification than only one of them.

From 2019, the Bonn database was extensively explored by various research groups, which allows for comparison of the results obtained using different algorithms.

The novelty of the algorithm proposed by Jindal et al. [44] consisted in the decomposition of the EEG signal into an optimal number of time–frequency subbands using Tunable Q-wavelet Transform (TQWT). An advantage of TQWT in comparison to other transforms is its capability to tune Q-factor, which defines the number of oscillations of given wavelet. Next, three nonlinear features, i.e., approximate entropy, HFD, and KFD, were estimated from decomposed subbands. Two classifiers were used in this study: support vector machine (SVM) and random forest tree classifier. SVM classifier was performed for different kernel functions (PUK kernel, poly kernel, normalized poly kernel, and RBF kernel). The classification was performed separately for individual features and for feature vector combined from three features. The highest accuracy of 100% was obtained for SVM classifier with PUK kernel for combined feature vector and for Q-factor equal to 5.

Abdulhay et al. [45] noticed that the performance of the classification depends on the choice of an appropriate classifier for a specific feature class. They considered three classes of features: (1) wavelet-based entropy (approximation entropy, sample entropy, and permutation entropy); (2) nonlinear features (Hurst exponent, HFD), and (3) higher-order spectra (mean, normalized entropy-1, and normalized entropy-

2), and three classifiers (k -Nearest Neighbor (k -NN), support vector machine (SVM), and Naive Bayes). The highest performance was obtained in the following cases: (1) classification of normal and interictal activity using k -NN classifier for entropy features; (2) classification of ictal and interictal activity using Naive Bayes for nonlinear features; and (3) classification of normal and ictal activity using SVM classifier with radial basis function kernel, for higher-order spectra features.

Brari et al. [46] showed that derivatives of EEG signals contain important information for epileptic seizure classification. Five features were constructed: HFD of EEG signal and HFD of the first, the second, and the third derivative of EEG for parameter $k_{\max} = 18$, and HFD of the first derivative of EEG for $k_{\max} = 6$. The k -NN classifier reached the highest accuracy for the combination of the last three features.

The k -NN classification of ten features: mean absolute value, standard deviation, average power of both measures, HFD and sample entropy, and the expected activity measurement coefficient calculated for four sub-bands, gave better accuracy (98%) than artificial neural network classifier (94%) [49].

The combination of Higuchi and Katz's fractal dimensions with multidistance fluctuation-based dispersion entropy, based on permutation and Shannon entropies, improved the performance of SVM classifier in seizure detection (normal vs. ictal) [48].

Torabi et al. [47] proposed four nonlinear measures of EEG, such as HFD and KFD, Hurst exponent, and Lempel–Ziv complexity, calculated in whole frequency range as well as in individual frequency bands (delta, theta, alpha, and beta) as a set of features for classification. The features selection was performed using ReliefF algorithm, and three different classifiers (multilayer perceptron neural networks (MLPNN), linear support vector machine (SVM), and RBF SVM) were applied for the selected features. The best feature was KFD for theta and beta bands. The MLPNN classifier reached the highest accuracy equal to 99.91% for ABCD/E, 98.19% for AB/CD/E, 98.5% for A/D/E, 100% for A/E, and 99.84% for D/E.

15.3.2 Stroke

Although HFD is a very promising marker in stroke research, it has not been widely explored in this application. The studies carried out so far have been based on the data collected in Ospedale San Giovanni Calibita Fatebenefretelli in Rome.

The first dataset contained the EEG signals recorded during the resting state in 36 patients during the period from 4 to 10 days after a unilateral ischemic stroke in the area of the middle cerebral artery and in 19 healthy controls [60]. A subject of particular interest in acute stroke research is the degree of clinical impairment and the prognosis of recovery. The results showed that HFD was lower in patients ($HFD = 1.45 \pm 0.09$) than in healthy controls ($HFD = 1.53 \pm 0.11$), and its decrease was associated with a worse clinical state, whereas a greater inter-hemispheric asymmetry indicated a worse prognosis for recovery after six months.

In the second study of Zappasodi et al. [61], the EEG signals were acquired in 24 patients in two phases: acute (2–10 days after a unilateral ischemic stroke) and sub-acute (after 2.5 months) in both hemispheres, ipsilateral and contralateral to the lesion. HFD was lower in acute than in sub-acute phase which indicates an improvement in the patients' condition. In both phases, HFD was lower in ipsilateral than in contralateral hemisphere.

15.3.3 *Alzheimer's Disease*

Metal intoxication, in particularly with aluminum occurring commonly in water, food, and medicines, may be one of the reasons for the emergence of Alzheimer's disease (AD) risk. The study of aluminum-treated rats showed a lower complexity of electrocortical activity in the intoxicated animals than in the controls. The HFD decreased mainly in the parietal cerebral cortex, but it was not changed in the paravermal area of cerebellum [62].

Nonlinear analysis of EEG signals in a fairly large number of 79 patients with AD and 82 controls was performed using five measures: HFD, SpEn, spectral centroid (SC), spectral roll-off (SR), and zero-crossing rate (ZCR) [63]. Two classifiers were compared: multilayer perceptron (MLP) and support vector machine (SVM). The highest performance of MLP classification for individual features was obtained for HFD, and it was equal to $66.84 \pm 1.84\%$. All features were lower in AD than in controls. HFD was lower mainly in the parietal and temporal parts of the brain. The concatenation of five features improved the MLP accuracy to $72.65 \pm 1.88\%$. The application of SVM classifier allowed to obtain even higher performance (from $73.29 \pm 1.52\%$ to $77.61 \pm 1.69\%$ in the range of sigma from 10 to 40). However, a disadvantage of this study is reuse of EEG data recorded previously to another objective, i.e., to study event-related potentials generated using an auditory oddball paradigm. Later, a dedicated study was performed in 67 AD patients and 41 healthy controls during resting state with eyes closed [64]. The results of this study showed a region-specific reduction of HFD in AD compared to the elderly controls mainly in occipital and temporal regions, while HFD was lower in the elderly compared to the younger controls in central and parietal areas. Moreover, HFD was reduced even more in AD patients with higher levels of non-ceruloplasmin-bound copper (NCC) but only in the parietal regions, whereas no changes in hemispheric asymmetry were found in AD patients compared to elderly healthy persons. Nevertheless, the subsequent studies Al-Nuaimi et al. [65] showed that in the parietal regions, the changes caused by AD appear first, and then the brain complexity gradually decreases over time from the posterior part to other brain regions. The k-NN classification with HFD from five EEG channels (Fz, Cz, Pz, P3, and T6), selected as the best features, allowed for discrimination between AD and norm with sensitivity and specificity of 100% and 80%, respectively. The combination of three measures: HFD, Tsallis entropy (TsEn), and Lempel–Ziv complexity (LZC) derived only from the selected frequency bands and EEG channels allowed to improve the

classification performance reaching the values of sensitivity and specificity greater than 90% [66]. It has already been noted earlier that the calculation of HFD and KFD in the frequency range narrowed to the beta band allows to increase the classification efficiency to 99.3% with a sensitivity of 100% and a specificity of 97.8% [67]. Further, these results were confirmed by another research group [68, 69]. They found that the temporal-scale-dependent measure like HFD correlates better with cognitive decline in AD patients in higher frequency range (beta and gamma bands) even for short 5-second EEG recordings.

The combination of the multiple signal classification and empirical wavelet transform (MUSIC-EWT) with nonlinear features (box dimension, HFD, KFD, and the Hurst exponent), and the enhanced probabilistic neural network (EPNN) classifier allowed for differentiation between mild cognitive impairment (MCI) and AD patients with an accuracy of 90.3% [70].

Recently, a new approach based on the Kuramoto mean-field model was proposed to study the relation between functional connectivity and complexity of simulated signals [71]. The functional connectivity was evaluated using the weighted phase lag index (wPLI). Three measures of complexity were considered in this study, i.e., HFD, Lempel–Ziv complexity (LZC), and fluctuation-based dispersion entropy (FDispEn). The simulated signals were generated using the connectivity matrix of wPLI obtained from EEG data recorded in 13 mild cognitive-impaired AD (MCI-AD) patients and in 15 healthy controls. A comparison of real EEG signals in both groups showed a reduced complexity in MCI-AD patients, but the topography of these changes depended on the complexity measure. The lower values were obtained in the anterior, posterior, and central regions for LZC, in the anterior and central regions for FDispEn, and only in the central part of the brain for HFD. However, the comparison between simulated EEG signals of patients and controls gave more restrictive results; the group differences were found only for LZC in the anterior and central regions, whereas the comparison between simulated EEG signals in lesion-induced controls and controls indicated the lower values of LZC in the anterior and central regions, FDispEn in the posterior and central regions, and HFD in the central region. Interestingly, both simulations revealed a positive correlation between the relative changes in functional connectivity and complexity, i.e., the reduction of connectivity in MCI-AD patients caused a lower complexity in this group in relation to the controls.

15.4 Psychiatric Disorders

15.4.1 Depression

Nonlinear methods in the analysis of EEG signals in depression were used for the first time in 2012 [72]. The EEG data were recorded in two groups, in 12 patients with major depressive disorder (MDD) and in 12 controls, during the resting state

with eyes closed using the standard 10–20 system of electrode placement. Only seven frontal channels were considered in this study. The nonlinear analysis of EEG was performed using two methods, HFD and KFD, in whole frequency range from 1 to 70 Hz and in five bands (delta, theta, alpha, beta, and gamma) obtained using wavelet analysis with a Daubechies wavelet of order 44. HFD in the range of higher frequencies (beta and gamma bands) was higher in patients with MDD than in healthy controls, and this difference is more significant in the left than in the right hemisphere. No group differences were in KFD. The classification accuracy of 91.3% was obtained using an enhanced probabilistic neural network as a classifier.

Following this lead, other investigators made a detailed comparative analysis based on four nonlinear measures (HFD, detrended fluctuation analysis (DFA), correlation dimension (D2), Lyapunov exponent (L1)) and linear measures (power of four frequency bands) of 19-channel EEG recorded in 45 untreated patients with MDD and 45 healthy persons [73]. Three classifiers: k-nearest neighbors (k-NN), linear discriminant analysis (LDA), and logistic regression (LR), were applied for patient classification with the abovementioned measures as features. Moreover, the genetic algorithm (GA) was used for feature selection. The authors stated that this technique gives a better classification accuracy than other methods like principal component analysis (PCA). The performance of classifiers was determined separately for every feature as well as for their combinations composed of linear or nonlinear features. Considering the linear features, the highest accuracy (73.3%) was obtained for alpha power, especially in the left hemisphere. The combination of all power bands did not improve the accuracy of classification. Among nonlinear measures, the correlation dimension using LR classifier provided the highest performance of 83.3%. Moreover, the combination of all nonlinear features allowed to improve accuracy up to 90%. Thus, these results showed that the combination of nonlinear features allows for higher classification accuracy of MDD patients, in comparison to the accuracy obtained by linear features.

The previous two studies were focused mainly on optimization of the classification. However, these studies did not provide complete information on the values of the considered measures and on the size and direction of their changes. In particular, Ahmadi et al. [72] did not consider the parietal and occipital EEG channels which are the most important in the patient classification [74]. A comparison of two measures, HFD and spectral asymmetry index (SASI), which is based on the evaluation of the relative difference of powers in two EEG frequency bands, i.e., bands of higher and lower frequencies than the maximum of alpha band spectrum, was performed [74]. The results showed that both measures allowed for differentiation between MDD patients and healthy controls basing on a single EEG channel in the parietal lobe. Although HFD provided a better performance, SASI allowed for a clearer differentiation due to the occurrence of an opposite polarity in both groups of participants (positive in depression and negative in norm), whereas HFD averaged over all EEG channels was higher in MDD patients than in controls (1.71 vs. 1.66).

Further, the concept described by Hosseinifard et al. [73] has been expanded by putting the emphasis on the importance of the appropriate choice of EEG channels

in group classification [75]. For this purpose, the analysis of 30-channel EEG was performed separately for each channel using six measures, including three linear measures (SASI, alpha power variability (APV), and relative gamma power (RGP)) and three nonlinear measures (HFD, DFA, and Lempel–Ziv complexity (LZC)) in a group of 13 MDD patients and 13 healthy controls. According to the results of Hosseinifard et al. [73], LR classifier with leave-one-out cross-validation was applied for group classification. All measures were higher in depression than in norm. A comparison of single measures revealed that linear measures (APV and RGP) had the highest accuracy of 81% in central, temporal, and parietal regions, while nonlinear measures (HFD and DFA) reached an accuracy of 77% in central and parietal regions. A combination of three linear and three nonlinear measures calculated in a single channel of the central region gave better result of classification than a combination of four nonlinear measures calculated from 19 channels (accuracy of 92% vs. 90%).

Automatic classification of MDD patients is more efficient than the classification performed by two standard psychological tests [76]. The classification efficiencies amounted to 64.8% for SASI and 55.2% for HFD, while for two psychological tests, HAM-D and self-rated EST-Q-D, were 44.8% and 28.8%, respectively. Moreover, the combination of SASI and HFD allowed to improve the classification up to 78.4%. The calculation of both measures was performed in group of 125 patients using 7-minute EEG segments recorded at only one channel (Pz).

Individual variability evaluated by six measures: SASI, APV, RGP, HFD, DFA, and LZC, was lower than the differences between MDD patients and healthy controls [77]. The EEG analysis was based on recordings acquired in 15 sessions during a period of 14 months.

Another extension of Ahmadlou et al. [72] study was proposed using the same methods not only to frontal but also to parietal regions [78]. The complexity evaluated by HFD and KFD was higher in depression than in norm in both brain regions in full frequency range and in higher-frequency bands (beta and gamma). Next, these researchers compared different nonlinear measures (KFD, HFD, LZC, and Kolmogorov complexity (KC)), and they found that LZC and KC outperform other measures in patient classification [79]. Moreover, they examined the impact of emotional stimulation on EEG complexity in both groups using a battery of five different nonlinear measures such as KFD, HFD, LZC, KC, and Shannon entropy (ShEn) [80]. The negative and positive emotions have been induced by noise and music, respectively. In healthy persons, the EEG complexity evaluated by KFD, HFD, and LZC decreased during musical stimulation but increased during noise stimulation in frontal and parietal regions. Whereas, in MDD patients, both stimulations elicit an increase of EEG complexity in the frontal region, though it was higher during the negative stimulation.

In contrast to the musical stimulation, repetitive Transcranial Magnetic Stimulation (rTMS) has a positive effect in depression [81]. HFD decreased in MDD patients after 20 sessions of rTMS. Another method allowing to reduce EEG complexity is biofeedback realized by means of diaphragmatic paced breathing (6 breaths/min) or a balloon game [82]. The EEG analysis was performed using

three measures: HFD and SampEn obtained directly from EEG, and HFD obtained after applying Hilbert–Huang Transform (HHT). Only the last method allowed for differentiation between all conditions.

Both nonlinear measures, HFD and sample entropy (SampEn) are powerful enough to discriminate between different phases of depression such as recurrent depressive episode, and remission as well as MDD patients and healthy controls independently on classification method [83, 84]. Seven classifiers were compared: naive Bayes classifier (NB), multilayer perceptron, decision tree (D3), logistic regression (LR), random forest, and support vector machine (SVM) with the linear and polynomial kernel. The normalized values of HFD and SampEn, obtained from 19-channel EEG, were used as features. PCA was applied for feature selection. It was shown that an appropriate choice of features (and feature selection method) is the decisive step in obtaining high-performance classification. The highest accuracy of 95.12% was provided by NB classifier using only the first-principal component. The performance of other classifiers can be improved to the same level by increasing the number of principal components to 10.

The study of Cukic et al. [83, 84] was performed in a rather small group of 21 MDD patients and 20 age-matched controls. Shortly thereafter, another systematic comparison of features and classification methods, but using a larger dataset (200 patients with MDD and 200 age- and gender-matched healthy controls), was published [85]. However, another set of features and classifiers were analyzed. Four measures (band power, coherence, HFD, and KFD) and four classifiers (k-NN, LDA, SVM, and SVM with conformal kernel transformation (CK-SVM)) were considered. The conformal kernel was introduced to improve the performance of SVM deteriorated due to large data variability. The total number of features was very high (1859) because four measures were calculated for all 26 electrodes and five frequency bands. Moreover, coherence was calculated for all combinations of inter- and intra-hemispheric regions as well as for every pair of electrodes inside each hemisphere for four frequency bands. In this study, a sequential backward selection (SBS) method was used for feature selection. This approach does not seem correct because of the too big and unbalanced number of features, coming mainly from coherence. Moreover, as the authors themselves stated, the SBS method did not consider again features that were early rejected. This can explain the result that the coherence was found to be the best feature. For the same reason, CK-SVM was better than other classifiers because SVM (or CK-SVM) performs better for unimodal distributions. Other nonstandard conditions and parameters used in this study also raise doubts: (1) EEG was recorded during the resting state with eyes open; (2) the occipital channels were excluded from this analysis; and (3) three arbitrary chosen parameters were used in calculation of HFD ($k_{max} = 50, 100, \text{ and } 150$).

In the next study was proposed an approach consisting in combining of multiple classifiers, which allowed to obtain better performance than that obtained from any single classifier [86]. Accuracies of five classifiers (k-NN, LDA, SVM with the radial basis function kernel (RBF-SVM), NB, and D3) and their combinations were compared. Ensemble classifiers accuracies were evaluated using three methods:

majority voting, weighted voting, and AdaBoost. As features, linear (SASI, APV, and relative band power (RBP) for theta, alpha, beta, and gamma bands) and nonlinear (HFD, LZC, and DFA) measures and their combinations were used. All these 9 measures were calculated for each of 18 EEG channels. The best sets of channels for every measure were selected according to the ranking derived from F-tests and from ReliefF algorithm. For example, the best channels for relative beta power were the occipital channels (O1 and O2). Firstly, the performances of all classifiers were calculated for every measure separately using all EEG channels. The highest accuracies ($>80\%$) were obtained for HFD using D3 (82.52%) and for LZC using RBF-SVM (80.70). Next, the accuracies were evaluated for all measures only using the best channels. The highest accuracies were obtained for features selected by ReliefF algorithm, i.e., for relative alpha and beta bands (90%) and for nonlinear measures (HFD: 81%; LZC: 81.95%; and DFA: 86%), using k-NN classifier. Then, the performance was evaluated for ensemble classifiers composed of nine classifiers of single measures calculated for the best channels. The highest accuracy of 93.30% was obtained using NB classifier with majority voting and features selected by ReliefF algorithm.

Finally, the classification was performed for a combination of all selected features derived from nine measures together. The highest classification accuracy of 95% was received using k-NN or D3 with only the best three features selected by ReliefF algorithm, i.e., relative power of alpha band at O1 and O2, and relative power of beta at O2. Thus, the occipital channels, excluded by Wu et al. [85], proved to be the most important features in the classification of MDD patients. Moreover, in this study, other two conditions were fulfilled. EEG was recorded during the resting state with eyes closed, and HFD was calculated for an optimal k_{max} value equal to 8. Nevertheless, the analysis was performed for a small group of 10 MDD patients and 10 controls.

15.4.2 Schizophrenia

The idea of using the fractal dimension of EEG as a feature vector in classification was applied for the first time by Nouranian et al. [87] to classify normal EEG and EEG under music stimulation using a HFD vector obtained from 19-channel EEG in a feed-forward artificial neural network with back propagation learning algorithm as a classifier.

This approach was then extensively explored in the diagnosis of patients with schizophrenia by an Iranian research group using EEG data recorded in 20 patients with negative symptoms of schizophrenia and in 20 healthy controls recruited at the Center for Clinical Research in Neuropsychiatry, Perth, Australia. Different sets of EEG measures were used as a feature vector in various classifiers to discriminate between patients with schizophrenia and healthy controls. First, the feature vector composed of autoregressive (AR) model parameters, band power, and HFD was applied in fuzzy classifier with an accuracy of over 80% [88]. Then,

the set of features has been extended with wavelet energy as an additional feature [89]. Moreover, the more informative channel was selected using two methods, bidirectional search and plus-L minus-R (LRS) techniques. The classification was performed independently by two classifiers: linear discriminant analysis (LDA) and support vector machine (SVM). Their accuracies depended on the method of channel selection (for bidirectional search: 84.62% and 99.38% for LDA and SVM, respectively; for LRS technique: 88.23% and 99.54% for LDA and SVM, respectively). In the next study by Boostani et al. [90], the set of features was composed of AR model parameters, band power, HFD, KFD, and Petrosian fractal dimension. It was shown that the accuracy of boosted version of direct linear discriminant analysis (BDLDA) is higher (87.51%) than other classifiers such as standard LDA (85.36%), Adaboost (85.41%), support vector machine (SVM), and fuzzy SVM (<50%). In another study of this research group [91], five entropy and complexity-based features: Shannon entropy (ShEn), spectral entropy (SpEn), approximate entropy (ApEn), Lempel–Ziv complexity (LZC), and Higuchi fractal dimension (HFD), were classified by two classifiers, linear discriminant analysis (LDA) and adaptive boosting (Adaboost) with accuracies of 86% and 90%. All analyzed features were lower in schizophrenia than in healthy controls. The best set of features, selected using a genetic programming, were: ApEn, LZC, and HFD from temporal and frontal channels, corresponding to the limbic area, which is involved in schizophrenia. The accuracy of classification increased to 89% (LDA) and 91% (Adaboost) for the set of selected features. In 2011, Sabeti et al. [92] proposed one more new approach consisted of two-stage feature selection. In the first step, the mutual information was applied to select the best channels. Next, the best features were chosen using genetic programming. This approach gave better results than the method of feature selection based on principal component analysis (PCA). In this study, only three features were considered: AR model parameters, band power, and HFD in classification. The classification accuracy using ten-fold cross-validation was 85.90% and 91.94% for LDA and Adaboost, respectively.

Another research group analyzed HFD of EEG during the resting state with eyes closed recorded in 33 males, including 18 neuroleptic-naïve patients with recent-onset schizophrenia and 15 healthy controls [93]. HFD was higher in patients with positive symptoms of schizophrenia, but lower in patients with negative symptoms of schizophrenia in comparison to HFD in healthy controls.

In the study of Thilakvathi et al. [94], EEG data were recorded in 55 patients with schizophrenia and 23 healthy controls in three conditions, during the resting state with eyes closed and during mental activity stimulated by two modified odd-ball paradigms. Six features, i.e., HFD, Kolmogorov complexity, Shannon entropy, spectral entropy, information entropy, and approximate entropy, were analyzed in both conditions. HFD was higher in patients with schizophrenia than in the healthy controls for HFD, Kolmogorov complexity, and approximate entropy. The highest accuracy of classification (88.5%) was obtained during the stimulation.

Later, Alimardani and Boostani [95] recorded EEG data in 26 patients with schizophrenia and 27 patients with bipolar disorder at the Clinic of Psychiatry of Hafez Hospital and Shiraz University of Medical Sciences in Iran to differentiate

both groups. They calculated 12 features, i.e., AR model parameters, kurtosis, HFD, KFD, and Lyapunov exponent, for 19-channel EEG in two conditions, during the resting state with eyes open and with eyes closed. In this work, the authors presented interesting methodological considerations regarding the importance of choosing an appropriate classifier and the method of features selection. They underlined that classifiers such as support vector machine (SVM) or Bayesian classifier should be used for unimodal distributions, whereas k -NN classifier is better for multimodal distributions [96]. Moreover, they proved in this study that the accuracy of k -NN classifier can be improved by using of a combination of Manhattan, Euclidean, and Minkowski distances in the voting formula [96]. Finally, the application of an efficient feature selection algorithm, called Davies–Bould [97], in fast feature reduction allowed to improve the accuracy even more (up to 87.51%). The accuracies of k -NN classification, determined for each feature separately, were 81%, 79%, 73%, 72%, and 71% for AR model parameters, kurtosis, Lyapunov exponent, HFD, and KFD, respectively. However, the highest accuracy was obtained when all features together were considered. Group comparison results showed that 54% of all HFD values were higher for patients with schizophrenia patients than in patients with bipolar disorder.

Sharma et al. [98] reported that the classification of patients with schizophrenia can reach an accuracy of 100% by using a combination of two time domain features (HFD and correlation between each pair of EEG channels) and two statistical features (mean and variance), and by the selection of the best EEG channels.

In the next study, Goshvarpour and Goshvarpour [99] proposed an original strategy of feature combination. A publicly available 19-channel EEG data recorded in two groups, 14 patients with paranoid schizophrenia and 14 healthy controls, were used for this analysis [100]. Three nonlinear measures, including HFD, Lempel–Ziv complexity, and Lyapunov exponents, were considered. The fusion strategies were applied to three features vectors with 19 components, corresponding to three nonlinear measures of 19-channel EEG. The following combinations were considered: sum, product, division, and two weighted sums with F -values or Shannon entropy-based information gain ratios, as weights. Then, the patient classification was performed for different fusion strategies using the probabilistic neural network, tested by different cross-validation methods such as k -fold cross-validation (for $k = 2, 5$, and 10) and a holdout method (for $p = 0.15, 0.2$, and 0.25). The highest accuracy of 100% was obtained for the holdout method, independently on parameter p , using only division rule as the fusion strategy, while for ten-fold cross-validation, four out of five rules were necessary to reach this performance. The analysis of individual nonlinear measures revealed that the dependence of differences between groups on nonlinear measure and on topography. Unlike the other two measures, HFD was lower in schizophrenia than in norm. These changes were region-specific. HFD and Lempel–Ziv complexity were higher in the frontal lobe as opposed to the Lyapunov exponent which was higher in the central and occipital regions. These differences between groups were better expressed for HFD than for Lempel–Ziv complexity.

15.5 Neuromodulation

15.5.1 Photostimulation and Microwaves Stimulation

The effect of photostimulation on the EEG complexity of healthy persons depends on the stimulation frequency [11, 12, 101]. The study protocol consisted of a sequence of 5-second stimuli with frequencies increased from 3 Hz to 27 Hz and then decreased, every 3 Hz. There were one-second pauses between the individual stimuli. Interestingly, HFD increased for higher frequencies, reaching the maximum value at 18 Hz. This effect was observed mainly in the occipital part of brain.

An increase of HFD of about 1.7% was also observed in 14 healthy persons exposed to microwave radiation with a frequency of 450 MHz modulated at 40 Hz, even at nonthermal level of microwave field [102]. The microwave power was 1 W, and the field power density near the head was 0.16 mW/cm².

15.5.2 Electrical and Magnetic Stimulation

A new approach consisted in a combination of HFD with ICA was proposed in the study of stimulation of a single snail BR-type neuron by static magnetic field of 2.7 mT [103]. As a result of magnetic stimulation, HFD of the neuronal activity decreased. Interestingly, two components were found in the complexity of signal induced by the field, namely, plasticity and elasticity. Although these are not clinical studies, the concept of applying ICA to HFD is interesting and it could be applied to such studies as well.

Another study showed that HFD allows for functional differentiation of spatially close areas of the brain like primary somatosensory (S1) and motor cortices (M1) independently on brain state [104]. For this purpose, EEG was recorded in a group of 20 healthy persons in four conditions: during the resting state with eyes closed and with eyes open, during motion consisting in squeezing a bulb with constant pressure for about 200 ms, and during galvanic electrical stimulation of the median nerve at the wrist. The activities of S1 and M1 were extracted from EEG using the functional source separation (FSS) method. The EEG complexity increased after motor or sensory stimulation in comparison to the resting state. HFD was higher in M1 than in S1 in every person, and this relation did not depend on condition which suggests that the HFD could be a useful measure for functional mapping of the whole brain.

Classification of event-related potentials evoked in EEG by transcutaneous electrical stimulation to the middle finger allowed to assess the degree of pain perception in healthy persons [105]. Two nonlinear measures, HFD and Grassberger–Procaccia (GP) correlation dimension, without or with moving variance or with autocorrelation function, were calculated for ERPs in all 16 EEG channels or in five channels selected using Fisher score. As a result, twelve feature vectors were used

in the classification, six for 16-channel EEG and six for 5-channel EEG. Four pain perception levels were defined: control (C), sensation (S), pain (P), and maximum pain (MP). The highest accuracy of 100% was obtained for the combination of HFD and GP with a moving variance and without channel selection for 3-level classification (C, P, and MP), while the accuracy of 4-level classification was 87.5%. At least 20 trials were required to receive a satisfactory result.

15.5.3 Transcranial Electrical Stimulation (tES)

A transcranial direct current stimulation (tDCS) through using an electrode of a personalized shape covering the primary somatosensory cortex (S1) allows to reduce multiple sclerosis (MS) fatigue [106]. This effect was seen only in the left hemisphere which was impaired in MS patients compared to healthy controls before stimulation. Moreover, the relationship between HFD from S1 and from the primary motor cortex (M1) was disturbed before stimulation, and it was restored after the tDCS.

15.5.4 Repetitive Transcranial Magnetic Stimulation (rTMS)

The repetitive transcranial magnetic stimulation (rTMS) can be successfully used in therapy of drug-resistant patients with depression [81]. The results of this study showed a significant decrease of HFD after 20th session of rTMS in MDD patients. As mentioned earlier in Sect. 15.4.1, HFD in depression was elevated in comparison to HFD in controls. Therefore, a reduction of EEG complexity is a desired effect of rTMS therapy.

15.6 Other Modalities than EEG

15.6.1 MEG

Similar to the results of EEG analysis, the HFD of spontaneous magnetoencephalographic (MEG) signals was lower in Alzheimer's disease patients than in healthy controls for all channels ($\text{HFD} = 1.81 \pm 0.07$ vs. 1.89 ± 0.03 for $k_{\text{max}} = 48$) [107]. Comparison between AD patients and elderly controls showed the differences between HFD in both groups only in 48% of 148 MEG channels, located mainly in the temporal region. The accuracy, sensitivity, and specificity were 87.8%, 80%, and 95.24%, respectively.

15.6.2 *fMRI*

For the first time, the usefulness of the HFD in the analysis of short time series obtained from fMRI was demonstrated in studies of limbic dysregulation in anxiety [108]. It was shown that the parameter k_{max} of 14 was appropriate for the calculation of HFD of just 33-sample-long signal. The complexity analysis showed the importance of dorsolateral and ventromedial prefrontal cortices, amygdala, and hippocampus in modulation of emotional response. Further studies analyzed extensively the influence of various factors on the results of HFD and other measures evaluated in fMRI time series [109]. They underlined the importance of an adequate filtering of these short, sparse, and noisy signals.

The fMRI study of the thalamo-cortical network activity during the resting state in patients with chronic migraine showed that HFD of BOLD signal was higher in the dorsal attention network (DAN) and in the anterior part of default-mode network (DMN), but lower in the posterior DMN compared to healthy controls [110]. Additionally, HFD of BOLD activity in DAN correlated with right fractional anisotropy and mean diffusivity, although both measures showed no changes in chronic migraine. In this study, HFD was calculated in 150-second windows in the k_{max} range from 2 to 25, corresponding to half of the total number of fMRI volumes.

The study of complexity of BOLD activity in consciousness disorders at different levels of severity (norm, minimally conscious state, and vegetative state) showed that the lower the HFD, the worse the state of consciousness [111].

The analysis of data from simultaneous EEG-fMRI in 25 healthy persons during the resting state showed a positive correlation between BOLD signal and HFD of EEG signal in the frequency range from 2 to 20 Hz in several cortical areas of the brain (sensory, motor, language, and occipital) and in subcortical limbic system [112].

15.7 Conclusions

Analysis of EEG complexity in healthy individuals showed that the highest HFD is in wakefulness, mainly during tasks that require increased brain activity [7–40]. On the other hand, HFD decreases in states with reduced brain activity or alertness [7–40]. These changes are age-dependent [41, 42]. The complexity of pathological EEG signals changes, but this change may not always mean a decrease in neuronal activity and thus a decrease in HFD. The reduction of HFD may be related to the impairment of brain function in various neurological disorders (epilepsy [11, 12, 43–59], stroke [60, 61], Alzheimer’s disease [62–71]), but in the case of psychiatric diseases, such as depression [72–86] and schizophrenia [87–100], we observe increased EEG complexity in certain areas of the brain. Proper

neuromodulation, such as photostimulation [11, 12, 101] and transcranial electrical or magnetic stimulation [81, 106], can improve the brain function.

The only parameter in the HFD method is k_{max} , but its value depends on various factors, including signal acquisition modality and related signal preprocessing [6, 7, 11, 12, 107–109]. The tendency of complexity evaluated by HFD does not depend on the modality; for example, HFD decreased for both EEG and MEG in Alzheimer's disease [62–71, 107], or for EEG and fMRI in the state of lower consciousness [7–40, 111]. Moreover, HFD plays an important role in any set of features used for patient classification allowing to improve its performance.

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Chapter 16

Fractal Dimension Analysis in Neurological Disorders: An Overview



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Abstract Fractal analysis has emerged as a powerful tool for characterizing irregular and complex patterns found in the nervous system. This characterization is typically applied by estimating the fractal dimension (FD), a scalar index that describes the topological complexity of the irregular components of the nervous system, both at the macroscopic and microscopic levels, that may be viewed as geometric fractals. Moreover, temporal properties of neurophysiological signals can also be interpreted as dynamic fractals. Given its sensitivity for detecting changes in brain morphology, FD has been explored as a clinically relevant marker of brain damage in several neuropsychiatric conditions as well as in normal and pathological cerebral aging. In this sense, evidence is accumulating for decreases in FD in Alzheimer's disease, frontotemporal dementia, Parkinson's disease, multiple sclerosis, and many other neurological disorders. In addition, it is becoming increasingly clear that fractal analysis in the field of clinical neurology opens the possibility of detecting structural alterations in the early stages of the disease, which highlights FD as a potential diagnostic and prognostic tool in clinical practice.

Keywords Brain tumors · Complex systems · Cognition · Clinical neurology · Fractal dimension

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16.1 Introduction

Fractals are geometric objects that are self-similar at distinct scales and can be found widespread in nature, appearing in many research domains, including dynamical systems, physics, biology, and behavioral sciences [61]. In recent years, fractal analysis has emerged as a powerful tool for characterizing irregular and complex patterns found in the nervous system. Specifically, the concept of fractal geometry objectively analyzes and quantifies the intricate structures conforming the topological and functional patterns of the human brain, elucidating how it works at a system's level. Fractal analysis has thus become a promising method to aid physicians and scientists in the diagnosis of certain neurological conditions, prediction of clinical outcomes, classification between pathological states, and detection of disease progression in the brain, with the ultimate goal of aiding novel therapeutic strategies and preventative medicine.

In the field of neuroscience, fractal analysis is used to measure the scaling properties inherent to neurological systems (e.g., neuroanatomical structures at the micro, meso-, and macroscale) and is typically applied by estimating the fractal dimension (FD), a scalar index that describes the topological complexity of the object under investigation. A wide range of neural structures—from neurons to complex networks—can be characterized as structural or dynamical fractals to quantify their intrinsic complexity. In this sense, the spatial properties of irregular components of the nervous system, both at the macroscopic and microscopic levels, may be viewed as geometric fractals, while temporal properties of neurophysiological signals should be interpreted as dynamic fractals [14, 15]. This chapter provides an updated overview of the applications of fractal analysis in human neuroimaging and neurodiagnostics, with a particular focus on clinical neurology.

16.2 Geometric Fractal Analysis Applied to Neuroscience

The convoluted structure of the cerebral cortex poses the problem of how to characterize its morphological complexity quantitatively [19]. Several methods and tools are available for the quantitative measurement of brain volume [34, 62, 66]. However, different factors including the type of MRI machine, brain volume artifacts, clinicopathological conditions, and variability among the standardized protocols may yield important variations in volume estimates [3]. As the field of neuroscience evolves, significant efforts have been made to develop new techniques to analyze the brain and to complement the existing ones, providing crucial knowledge on functionality as well as information surrounding the structural alterations, like those seen in neurological diseases. In this sense, one of the most promising indices is the fractal dimension (FD), a measure that has long been applied to estimate the topological complexity of natural structures [42] and that has been employed to evaluate brain complexity across multiple scales [17, 25, 45].

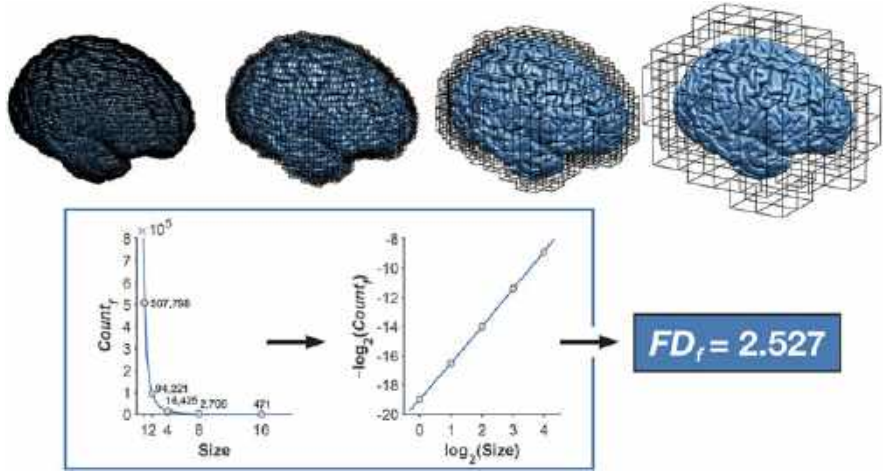


Fig. 16.1 Illustration of the box-counting method of calculating fractal dimensionality. Top: visualization of the number of boxes necessary to contain the brain structure across different box sizes. Bottom: Plot of the power-law relationship between the number of boxes (count) and box size, followed by a conversion to log–log space. The slope of line for this count–size relationship in log–log space represents the fractal dimensionality estimate of the structure. (Figure reprinted from Madan [36] with permission)

Given its sensitivity for detecting changes in brain morphology, FD has been explored as a clinically relevant marker of brain damage in several neuropsychiatric conditions as well as in normal and pathological cerebral aging [40, 55]. There are several different methodologies to compute the FD index empirically from structural MRI data. One of the most commonly applied methods is the box-counting algorithm, which is suitable to study the cerebral cortex [25, 38] and is illustrated in Fig. 16.1. Nonetheless, other approaches to measure cortical complexity through FD include techniques based on the Fourier transform, surface dilation, and spherical harmonic reconstructions [6, 18, 39, 46, 65]. In this sense, significant efforts to elucidate how much FD is related to other indices such as thickness and gyrification are of special interest to characterize the potential value of this measure for new insights into cortical architectural features [38, 53]. It is worth highlighting that several studies reported FD estimates as sensitive to multiple aspects related to data acquisition and processing, which may cause these research outcomes to be heterogeneous and therefore not directly comparable [45]. Nonetheless, formal evaluations have provided evidence that FD has higher test–retest reliability than conventional measures of brain structure [39], as well as is more robust to head motion-related artifacts [35]. This chapter reviews and summarizes recent literature on the application of FD to describe cortical brain complexity in healthy and clinical individuals, as well as to outline future lines of research in this field.

16.3 Relationship Between Fractal Dimension and Healthy Subjects

16.3.1 *Aging and Development*

A wide range of computational methods for calculating FD have been used to perform different experiments involving age, development stage, and clinical condition, among others. Generally, these research analyses focus on how cortical complexity varies during development and aging in nonpathological situations. For example, in addition to the studies described in Esteban et al. [17], Reishofer et al. [55] concluded that alterations of WM complexity can be reliably measured with fractal analysis in a large cohort of healthy aging subjects. This outcome suggests that FD may be considered a valuable biomarker for identifying WM structural changes in the human brain. More recently, Marzi et al. [43] conducted machine learning models for individual age prediction in two large datasets of healthy individuals. Their findings showed that the selections of the interval of spatial scales were relevant for a more suitable characterization of fractal properties of the cerebral cortex, as well as for a more accurate estimation of FD. Moreover, this work reported a monotonic reduction in structural complexity of the cortex with age during almost all the lifetime. Furthermore, an overall reduction in FD with age is reported in several studies that analyzed this index in healthy adults and elderly subjects [22, 28, 31, 32, 37, 38, 40, 43, 44]. Interestingly, a negative association between age and FD has been described at different spatial levels, including the whole brain [38, 40, 43, 44], the left dorsolateral prefrontal cortex [32], and in the posterior wall of the right central sulcus [28]. A comparative FD analysis was performed by [22] in different age groups, where altered values of FD were observed between young adults (<45 years), middle adults (45–60 years), and old adults (>65 years); more precisely, FD decrease was reported from young to middle age in the bilateral frontal, left frontal, and right limbic lobes, while from middle to old age, the FD reduction was observed in the bilateral temporal, parietal, and left limbic lobes, respectively. Nevertheless, an increase in FD estimates of the left middle orbitofrontal cortex region was highlighted from young to middle age and of the occipital lobe area from middle to old age groups.

Furthermore, two longitudinal studies yielded similar results, observing a negative relationship between FD and age ([31, 37]; reviewed in [45]). In comparison to other cortical measures, such as cortical thickness or gyrification index, FD appears to be more sensitive to age-related differences. Last, regarding gender differences, two studies reported that the reduction in FD was faster and more significant in males than in females [22, 28]. Similarly, one study in a large sample from the UK Biobank observed differential patterns of age-related changes in cortical FD according to sex, with relatively more age-affected regions in males [48].

16.3.2 Cognition

With respect to cognitive functioning, as reported by Meregalli et al. [45], several studies [20, 24, 31, 44, 47] have established that higher FD values were related to better cognitive performance, except for Lu et al. [32], who did not find any relationship between FD and scores derived from the Mini-Mental State Examination (MMSE). Im et al. [20] reported a positive relation between FD of the whole brain and years of education, as well as between FD of the right hemisphere and IQ in a group of healthy young individuals. More recently, Liu et al. [31] analyzed the association of FD values and cognition in subjects ranging from 70 to 90 years old and reported significant correlations between FD and global cognition in distinct brain areas, including the bilateral temporal lobe, left occipital lobe, and several subcortical structures, in line with the results previously obtained by Mustafa et al. [47] for whole-brain fractal analysis. Additionally, Kinno et al. [24] observed positive correlations between FD of several frontotemporal brain areas and memory function in a cohort of older subjects (over 75 years old), assessed by using the scores obtained in the Wechsler Memory Scale-Revised (WMS-R). On the other hand, McDonough and Madan [44] evaluated the association among FD, brain activity, and cognitive functioning in middle-aged and older subjects who were at risk of dementia. Their findings yielded that decreased FD was linked to higher brain activity during memory retrieval in several posterior brain areas and that this pattern was related to cognitive dysfunction.

Finally, Aguiard et al. [1] investigated patterns that emerged in brain signals in response to external stimulating image regimes recorded in a sample of young adults (18–22 years) that were analyzed using nonlinear techniques. Their results showed that subjects who were visually stimulated by a series of mixed images (a randomized set of neutrally or negatively arousing images) exhibited a significantly higher fractal dimension compared to subjects visually triggered by pure images (an organized set of either all neutral or all negatively arousing images). In addition, their findings also revealed that subjects who performed better on memory recall showed higher fractal dimension computed from the electroencephalogram.

16.4 Fractal Analysis and Neurological Disorders

Recent studies have started to compare differences in FD values between healthy control participants and patients with neurological or psychiatric conditions. Here, we summarize some relevant researches carried out in this field, in addition to the investigations outlined in Esteban et al. [17]. Overall, only a few have been conducted so far in these conditions and, therefore, the results obtained are limited and heterogeneous. However, the great majority reported the presence of global or regional FD changes in the clinical population, highlighting the sensitivity of fractal analysis across a wide spectrum of neuropsychiatric disorders [45].

16.4.1 Alzheimer's Disease (AD)

Different studies showed a general decrease in FD in patients with AD in comparison to healthy individuals. In particular, Villamizar et al. [63] performed a comparative analysis between the FD of neuroimages from healthy control subjects and AD patients. Their results showed lower general FD values in AD patients related to neurodegeneration. Also, Nobukawa et al. [50] applied Higuchi's fractal algorithm to evaluate specific temporal fractal properties. Their work established a comparison between the values of the temporal scale-specific FD of AD patient and healthy control groups. Their findings demonstrated that the AD group displayed reduced fractality at both slow and fast temporal scales; furthermore, they were able to correlate fractality at fast temporal scale with cognitive decline, which may be used as a basis for a useful methodology to characterize temporal neural dynamics in AD or other neurodegenerative diseases.

On the other hand, Nicastro et al. [49] and Ruiz de Miras et al. [57] characterized FD using spherical harmonic reconstruction, both reporting a decrease in whole-brain FD in AD patients. Regarding regional specificity, both studies showed reduced FD values in AD patients when compared to healthy subjects, particularly in areas embracing the insula, the temporal medial lobe, and the posterior cingulate cortex. In this line, Ruiz de Miras et al. [57] also observed differences in this index in the temporal pole, whereas Nicastro et al. [49] identified FD values variances between AD patients and controls in the precentral and postcentral gyri.

Moreover, Ruiz de Miras et al. [57] also assessed FD in patients presenting mild cognitive impairment (MCI), detecting that subjects who would develop manifest AD in the following 4 years exhibited lower white matter FD values than non-converter MCI subjects, particularly in the medial frontal lobe. By contrast, McDonough and Madan [44] calculated FD in subjects at risk for dementia without associated cognitive impairment and reported no relation between FD values and AD risk.

Finally, Nicastro et al. [49] and Ruiz de Miras et al. [57] also revealed positive correlation analyses regarding FD and cognitive performance, such that higher cognitive impairments were associated with lower FD estimations.

16.4.2 Frontotemporal Dementia

One of the main studies assessing FD in patients with frontotemporal dementia (FTD) was performed by Nicastro et al. [49], showing an FD decrease in patients when compared to healthy controls, both on the whole-brain level and in areas such as the insula, posterior cingulate cortex, and precentral and post-central gyri. They also conducted correlation analyses, detecting an association between memory, language, and fluency impairment, and decreased FD in the left insula, inferior temporal, and medial orbitofrontal gyri. Likewise, Sheelakumari et al. [59]

investigated FD in two subtypes of FTD: behavioral variant FTD (bvFTD) and primary progressive aphasia (PPA). An overall reduction in FD was observed in both subtypes when compared with healthy individuals. In addition, whereas PPA appeared to be characterized by low FD in the left hemisphere of the brain, bvFTD was associated with an FD decrease in the right hemisphere, potentially linked to the behavioral manifestation of FTD. This hypothesis was supported by the identification of a negative correlation between the FD of the right hemisphere of bvFTD patients and their score on the frontal system behavior scale, which showed that less complexity was related to worse clinical status. Finally, Ziukelis et al. [68] provide a systematic review of fractal analysis in neurodegenerative diseases and dementia.

16.4.3 Multiple Sclerosis

Apart from the studies highlighted in Esteban et al. [17], research by Roura et al. [56] applied cortical FD as calculated by the box-counting method for the prediction of disability worsening in a cohort 146 of subjects suffering from relapsing-remitting multiple sclerosis (RRMS) with up to 5 years of clinical follow-up. Fractal analysis of the brain structures was conducted to evaluate whether significant differences exist between RRMS patients and healthy controls. Their findings unveiled that whole brain, cortical and deep grey matter FD, and whole brain, cortical and white matter FD lacunarity differed between the two cohorts. Furthermore, a longitudinal analysis assessing the 5-year follow-up revealed a significant reduction in fractal dimension and an increase in lacunarity. The authors applied linear mixed models that fitted the distribution of the annual variations in the FD and lacunarity of the brain areas under study in order to evaluate the changes that may appear in brain fractal geometry in relation to the duration of the disease. The models exhibited significant changes in the cortical and white matter FD and for cortical, grey matter, and white matter lacunarity in association with the disease duration. In particular, a decrease in the cortical FD, white matter FD, and grey matter FD was found. Last, a multivariate analysis was performed considering the relationship between brain volume and the presence of lesions with disability worsening, and results indicated a higher risk for disability worsening for cortical FD when adjusting for the presence of lesions and a trend for higher risk of disability worsening when adjusting for grey matter volume. To sum up, their results showed that cortical FD identified a subset of patients with brain damage who were at a higher risk of disease progression in short- to mid-term. The comparison of the risk based on cortical and grey matter fractal geometry yielded cutoffs associated with a higher risk of disability accumulation on several disability scales. Fractal geometry was thus pictured as a complementary measure that may represent a biomarker for disease monitoring and prognosis in patients with multiple sclerosis, even though multicenter validation studies in prospective cohorts will be required to confirm these findings.

16.4.4 *Parkinson's Disease (PD)*

As outlined in Meregalli et al. [45], two main studies are worth noticing regarding the application of fractal analysis in PD. First, Li et al. [29] described decreased values of FD in patients suffering from the disease in different brain areas, such as left precentral and postcentral cortex, left superior frontal cortex, left caudal middle frontal cortex, bilateral superior parietal cortex, and right superior temporal cortex. Additionally, a lower FD value in the left postcentral cortex was associated with a longer duration of the disease under study. By contrast, Kubera et al. [26] did not discover changes in FD in association with PD, despite the significant thinning of the cortex. However, it should be considered that the number of samples analyzed in the latter work was low, what may have limited statistical power.

16.4.5 *Epilepsy*

In addition to the research described in Meregalli et al. [45], two recent studies have extended this field of research. Lu et al. [33] used two indicators, Sample Entropy (SampEn) and Higuchi's Fractal Dimension (HFD) to identify epileptic signals characterized as being chaotic and nonlinear. By applying a support vector machine (SVM) classifier to these features, they were able to automatically classify the most EEG signal segments with an accuracy of 89.8%. Although preliminary, their results may provide theoretical guidance for the recognition or prediction of epileptic EEG signals in clinical practice.

Moreover, Malekzadeh et al. [41] proposed a new method for early identification of epileptic seizures in EEG signals by applying nonlinear features based on FD and a deep learning model and using Bonn and Freiburg datasets to perform empirical analyses. Their findings yielded that the proposed methodology achieved an accuracy of 99.736% and 99.176% in the automatic diagnosis of epileptic seizures in EEG signals for the Bonn and Freiburg datasets, respectively.

16.4.6 *Autism Spectrum Disorder (ASD)*

In addition to the articles described in Esteban et al. [17], two recent studies have addressed the study of FD in autistic individuals. Zhao et al. [67] measured FD values of the right and left cerebellar cortex in male children with autism spectrum disorder (ASD) and typically developing peers, identifying a decrease in FD estimates in the right cerebellar cortex of autistic children. Moreover, they detected a significant positive correlation in the ASD group regarding the FD values and the difference between performance and verbal IQ. On the other hand, Radhakrishnan et al. [54] investigated EEG signals of ASD using Higuchi Fractal Dimension

(HFD). These authors analyzed brain responses for auditory/visual stimuli in typically developing children and children with autism through EEG, attempting to characterize brain dynamics with HFD. Since one of the key parameters implicated in the computation of HFD is the time interval parameter “ k ,” most researchers arbitrarily fix the value of “ k ” in the range of all channels to estimate the degree of nonlinearity in an EEG signal. As a result of their research, they ultimately proposed an algorithm to estimate the optimal value of the time parameter for each channel. Their statistical analysis revealed that a difference of 30% was observed between autistic and TD children.

16.4.7 Attention Deficit Hyperactivity Disorder (ADHD)

Avelar et al. [4] hypothesized that the dynamics of exploratory movements may predict differences in the use of information by children with ADHD when compared to their typical development (TD) peers. Their findings supported this hypothesis, suggesting that ADHD patients exhibit deficits in dynamic touch that may be crucial for the organization of effective, task, and context-sensitive movement patterns, which manifested as reduced sensitivity to mechanical information supporting perception of object properties by wielding. Importantly, they reported significant differences between children with and without ADHD related to temporal correlations (degree of fractality), particularly how the fractality of exploratory movements of the rod and hand moderated children’s emphasis on the first principal moment of inertia (I1). In TD children, lower hand fractality and greater rod fractality were related to increased emphasis on I1. By contrast, children with ADHD increased emphasis on I1 when the exploratory movement was characterized by greater hand fractality and lower rod fractality. Generally, their study suggested that this result may be taken to reflect differences in the coordination of multiscale processes involved in information sampling and use, although further research on this topic is required. On the other hand, Wang and Kamata [64] proposed a novel classification approach using 3D fractal dimension complexity map (FDCM) for ADHD automatic diagnosis. By calculating the Hausdorff fractal dimension of grey matter density data extracted from structural MRI information, they designed a 3-dimensional convolutional neural network that allowed to estimate features from FDCM and discerning between ADHD children and typically developing peers. Additionally, Dawi et al. [7] analyzed the fractal dimension and sample entropy of EEG signals to investigate the changes in EEG signals’ complexity, concretely the decrease in attention and memory of patients with ADHD with respect to normal subjects while playing a serious game. Their findings exhibited a reduction in FD and sample entropy of EEG signals for ADHD individuals, which may reflect their lower attention. They concluded that the decrease in attention and memory in ADHD patients could be mapped on the reduction of complexity and memory of their EEG signals. Last, Sho’ouri [60] looked for differences in eye movements in patients with ADHD and healthy people while performing an attention-related task.

The two groups were classified using the vector derived from the calculation of two features, approximate entropy (ApEn), and Petrosian's fractal dimension (Pet's FD) of electrooculogram (EOG) signals, along with support vector machine (SVM) and neural gas (NG) as another two classifiers. Their findings described that the values of both features were significantly lower in the ADHD group compared to the control group. Furthermore, the SVM classifier was more successful in separating the two groups than the NG. In conclusion, the reduction in ApEn and Pet's values in the EOG signals of the ADHD patients group explained that their eye movements were slower than the control group and this difference was due to their attention deficit.

16.4.8 Brain Tumors

Di Ieva et al. [8, 9, 10, 11, 12, 16] showed that the fractal dimension is an objective parameter to characterize brain tumors on neuroimaging and histopathology, offering a new prognostic, diagnostic, and therapeutic biomarker.

Moreover, fractal parameters can be used as morphometric and radiomic parameters to differentiate different types of brain tumors, as demonstrated by Jang et al. [21], Karsy et al. [23] and Petrujkic et al. [52].

Benson et al. [5] proposed a robust method for MR brain image classification based on FD. Their methodology assumed that brain tumors may increase the intracranial pressure (ICP), thus changing its normal structure and that those alterations could be effectively characterized by using fractal analysis. Their findings showed that the presence of tumors can cause significant changes in the fractal dimension of grey matter and white matter and used this variation in FD for the classification of MR brain images into two groups: with and without tumor. More recently, Sánchez and Martín [58] also explored morphological and fractal properties of brain tumors; they hypothesized that parameters extracted from the tumor interface fluctuations would allow for the characterization of the particular growth model, which could be relevant for an appropriate diagnosis and the correspondent therapeutic strategy. In this sense, they analyzed these fluctuations to determine generalized FD. Their results showed that tumor-interface FD, along with other parameters, may help to discriminate between gliomas and meningiomas or schwannomas.

16.4.9 Vascular Disease

A recent study conducted by Aminuddin et al. [2] explored the potential use of FD as a marker in cerebral small vessel disease (CSVD) and correlation with clinical sequelae of CSVD. Their findings revealed that asymptomatic CSVD subjects (from the incidental MRI finding of the presences of low white matter hyperintensity (WMH) burden) had significantly lower FD in comparison to the control subjects,

suggesting that FD may serve as a promising vascular neuroimaging biomarker for asymptomatic CSVD subjects from the presence of low WMH burden, since FD is an index that measures the complexity of a self-similar and irregular structure, such as circle of Willis and its tributaries. Previously, Pantoni et al. [51] evaluated changes in FD between healthy subjects and patients with small vessel disease (SMD) and mild cognitive impairment (MCI). By calculating white and grey matter FD changes, the authors observed a decrease in whole-brain white matter FD in individuals suffering from SVD. Moreover, their findings also reported a positive correlation between FD values and cognitive performance, such that a reduction in white matter FD was related to worse cognitive outcomes.

On the other hand, Lemmens et al. [27] conducted a systematic review of FD in the retinal vasculature to explore its role in neurodegeneration and stroke. The literature suggested that, overall, the central pathology is related to a reduction in FD as a measure of microvascular network complexity and considered this index as a promising noninvasive and cost-effective biomarker for the diagnosis of neurodegenerative and cerebrovascular disease. Similarly, Liew et al. [30] applied fractal analysis to estimate the branching complexity and density of the retinal vessels and its relationship with long-term stroke mortality. They reported that a decrease in retinal vascular FD, that is, a sparser vascular network, may be associated with cerebral tissue hypoxia and an increased risk of long-term stroke mortality. Nevertheless, further analyses should be conducted to validate FD performance as a neurodegeneration biomarker in clinical practice.

Di Ieva et al. [13, 14] have introduced a novel methodology to characterize the geometrical complexity of the arteriovenous malformations' (AVM) vascular nidus, proposing the nidus' FD as a promising angioarchitectural parameter predicting the response to gamma knife radiosurgery.

16.5 Conclusion

Overall, evidence is accumulating for decreases in FD across a variety of neuropsychiatric disorders, including AD, FTD, PD, and MS, and many other neurological diseases. It is becoming increasingly clear that fractal analysis in the field of neurodegenerative disorders opens the possibility of detecting structural alterations in the early stages of the disease, which highlights FD as a potential diagnostic and prognostic tool in clinical practice. In addition, FD alterations were also found to be linked to clinical worsening in different neurological conditions. Thus, further research is warranted to evaluate the association between topological complexity and specific clinical parameters and to assess the reliability of the different methodologies employed to calculate FD, with the objective of specifying the most appropriate techniques for the respective research questions. In conclusion, this overview shows that fractal analysis represents a valuable tool to describe brain complexity in healthy populations and to identify morphological alterations related to developmental or pathological conditions. Nonetheless, the present chapter also

revealed a lack of longitudinal studies in the existing FD literature, which will be of great relevance to better understand the prognostic role of fractal analysis in different neurological conditions and disease stages.

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Chapter 17

Fractal Dimension Studies of the Brain Shape in Aging and Neurodegenerative Diseases



Jennilee M. Davidson, Luduan Zhang, Guang H. Yue, and Antonio Di Ieva

Abstract The fractal dimension is a morphometric measure that has been used to investigate the changes of brain shape complexity in aging and neurodegenerative diseases. This chapter reviews fractal dimension studies in aging and neurodegenerative disorders in the literature. Research has shown that the fractal dimension of the left cerebral hemisphere increases until adolescence and then decreases with aging, while the fractal dimension of the right hemisphere continues to increase until adulthood. Studies in neurodegenerative diseases demonstrated a decline in the fractal dimension of the gray matter and white matter in Alzheimer's disease, amyotrophic lateral sclerosis, and spinocerebellar ataxia. In multiple sclerosis, the white matter fractal dimension decreases, but conversely, the fractal dimension of the gray matter increases at specific stages of disease. There is also a decline in the gray matter fractal dimension in frontotemporal dementia and multiple system atrophy of the cerebellar type and in the white matter fractal dimension in epilepsy and stroke. Region-specific changes in fractal dimension have also been found in

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Huntington's disease and Parkinson's disease. Associations were found between the fractal dimension and clinical scores, showing the potential of the fractal dimension as a marker to monitor brain shape changes in normal or pathological processes and predict cognitive or motor function.

Keywords Alzheimer's disease · Amyotrophic lateral sclerosis · Aging · Epilepsy · Fractal dimension (FD) · Frontotemporal dementia · Huntington's disease · Multiple sclerosis · Multiple system atrophy · Neurodegenerative disease · Parkinson's disease · Shape complexity · Spinocerebellar ataxia · Stroke

17.1 Introduction

Fractal analysis has been applied to the characterization of the nervous system, from simple cells to complex brain structures [10]. Studies have shown that fractal geometry can be used to quantify shape complexity changes of the brain with age [4, 15, 28, 37, 46, 61, 62, 69, 81, 86] and neurodegenerative diseases such as multiple sclerosis [12, 13], Alzheimer's [30, 31], amyotrophic lateral sclerosis [57], frontotemporal dementia [66], multiple system atrophy [82], stroke [87], epilepsy [7, 17, 40], spinocerebellar ataxia [21, 24, 44, 78, 79, 83], Huntington's disease [34], and Parkinson's disease [35, 39]. The term "fractal" was introduced by Mandelbrot [43] to describe the irregular but self-similar shapes of natural objects. It summarizes the structural details of the object in a range of spatial scales into a numerical value (fractal dimension [FD]), which can serve as an index of morphometric variability and complexity of the object. The more complex the object (e.g., a more convoluted cortical surface, a white matter [WM] structure with a more complicated branched pattern), the higher its FD value (Fig. 17.1). The FD is based on a logarithmic scale, and small changes in the FD correspond to large changes in complexity [70]. In addition, neurodegenerative-associated brain atrophy can be detected by the pattern of change of FD. The fractal geometry of the brain was presented by Hofman using gross specimen data [20] and confirmed by magnetic resonance imaging (MRI) studies [22, 33, 41, 85]. MRI has been frequently used to study brain shape. Besides a few specimen studies [20, 75], most fractal research has analyzed MRI data to evaluate brain shape complexity at macroscopic and microscopic scales using T1- [4, 7, 12, 13, 15, 17, 22, 28, 30, 31, 33, 37, 38, 40, 41, 46, 57, 61, 62, 82, 85–87] and T2-weighted magnetic resonance (MR) images [36, 72], respectively.

This chapter reviews MRI-based fractal studies of the brain shape in aging and neurodegenerative diseases.

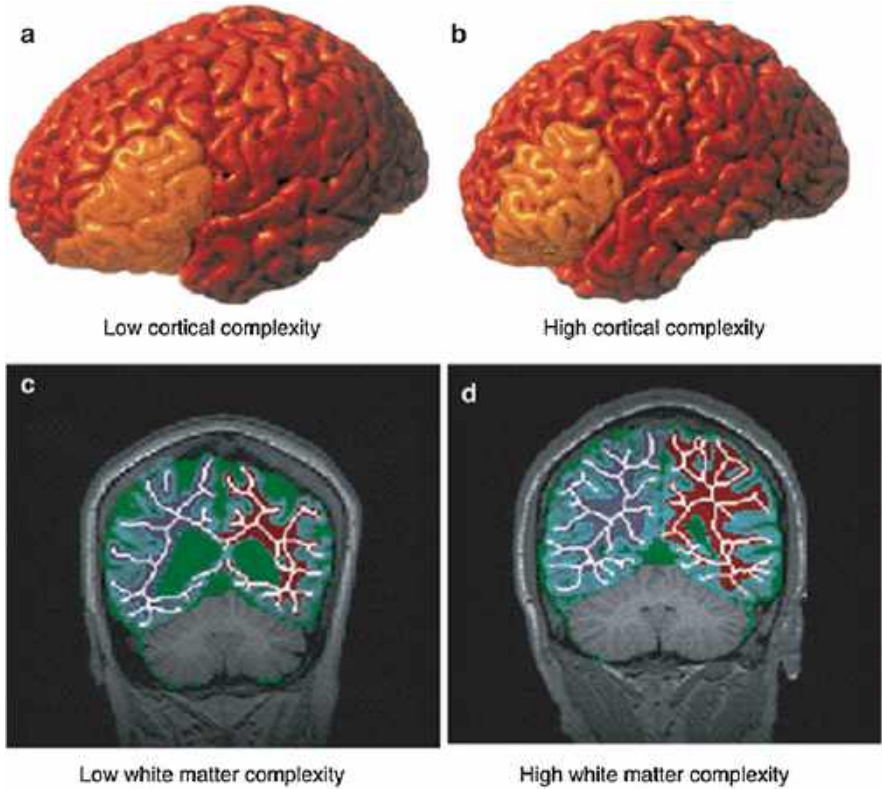


Fig. 17.1 Representative examples of brain shape complexity. Low and high cortical complexity in inferior frontal regions is shown in a 6-year-old female (a) and a 16-year-old female (b), separately. Low and high white matter complexity is demonstrated on a MR slice from an old subject (c) and a young subject (d). Low complexity values indicate a flat structure area with few secondary branches, while higher complexity values indicate a greater number of convolutions and more higher-order branches ((a, b) Reprinted from Blanton et al. [4], images courtesy of Arthur W. Toga, PhD, USC Laboratory of Neuro Imaging, and (c, d) reprinted from Zhang et al. [86], with permission from the publishers)

17.1.1 Anatomical Shape Features of Interest

Depending on the pathophysiological processes, FD values were computed on different shape features of the brain WM and gray matter (GM) to evaluate shape complexity changes with age or neurological diseases. Table 17.1 summarizes the shape features studied in the current literature. In general, three shape features were investigated in the WM fractal studies—interior (skeleton), surface (gray–white matter interface), and general structures (Fig. 17.1) [7, 12, 15, 17, 31, 41, 44, 46, 57, 59, 62, 82, 85–87]. The skeleton describes a thin line of an object that summarizes

Table 17.1 Anatomical shape features of interest and fractal dimension methods in brain shape studies in normal structure, aging, and neurodegenerative diseases

Pathology	Shape features of interest	MR	Fractal dimension methods
Normal structure	3D gray matter (cortical ribbon) [33]	T1	Fast Fourier transform-based method [33]
	2D skeleton of the cerebellar white matter [41]	T1	Box counting on 2D images [41]
	3D external cortical surface (pial surface) [22]	T1	Box counting on a 3D tessellated surface [22]
	3D white matter skeleton, surface, and general structure [85]	T1	Box counting on 3D images [85]
Aging	<i>White matter</i>		
	3D skeleton, surface, and general structure [15, 86]	T1	Box counting on 3D images [85]
	3D general structure [46, 62]	T1	Box counting on 3D images [46, 60, 62]
	Microstructure of deep white matter [72]	T2	Multifractal measure [72]
	<i>Gray matter</i>		
Alzheimer's disease	3D external cortical surface [4]	T1	Surface-based [75]
	3D external cortical surface [69], 81]	T2	Box counting on 3D images [69, 81]
	Pseudo 3D-skeletonized cerebral surface [37]	T1	Box counting on 3D images [37]
	2D-skeletonized cerebral cortex [28]	T1	Box counting on 2D images [28]
	Surface of central sulcus [38]	T1	Surface-based [38, 75]
	3D cortical ribbon [61]	T1	Box counting on 3D images [60]
	3D cerebral cortex	T1	3D box counting [25]
	2D coronal and axial profiles of the cerebral cortical ribbon [30]	T1	Box counting on 2D cortical ribbon images [30]
	3D cortical ribbon, 3D gray–white matter surface [31]	T1	Box counting on a 3D tessellated surface [22, 26, 31]
	3D average and probabilistic maps of primary deep sulci (including supracallosal, cingulate, marginal, parieto-occipital, anterior and posterior calcarine sulci, and sylvian fissures) [74]	T1	Box counting on 3D images [74]
	FD of edges in the Hilbert domain [36]	T2	Box-counting method [36]
	3D FD of the entire volume of the brain and of 3D skeleton [27]	T1	Box counting on a 3D volume generation from 2D images [27]

	3D FD of cortical surfaces [29]	T1	Box counting on 3D images Cube counting (3D cube counting algorithm) [29]
	Cortical FD and cortical thickness [32]	T1 3D whole brain	Surface-based morphometry (SBM) using CAT12 Toolbox. Spherical harmonic reconstructions [32]
	Global FD, regional WM FD (insula, temporal pole, medial temporal lobe, posterior cingulate cortex) [59]	T1	Spherical harmonic reconstructions [59]
	Regional FD, cortical thickness, and central surface of left and right hemispheres [53]	T1	Spherical harmonic reconstruction method [53]
	WM, whole brain stem [54]	T1	Higuchi's and detrending moving average (DMA) method [54]
	FD and cortical thickness [65]	T1	Method based on spherical harmonics [65]
Amyotrophic lateral sclerosis	3D white matter skeleton, surface, and general structure [57]	T1	Box counting on 3D images [85]
	Cortical surface [6]	T1	Spherical harmonic reconstructions [6]
Frontotemporal dementia	3D GM skeleton, surface, and general structure FD [66]	T1	3D box counting [66]
Epilepsy	2D white matter contour (gray–white matter interface) [7]	T1, proton density	Box counting on 2D images [7]
	3D white matter surface [17]	T1	Surface-based [17]
	3D external cortical surface [40]	T1	Surface-based [75, 76]
Multiple sclerosis	2D white matter border, skeleton [12]	T1	Box counting on 2D images (HarFA software, http://www.fch.vutbr.cz/lectures/imagesci/ ; [85])
	3D gray matter (cortical ribbon) [13]	T1	Box counting on 3D images [13]
	3D FD entire volume of the brain and also of its 3D skeleton [27]	T1	3D box counting, web platform (http://3dfd.ujaen.es) [27]

(continued)

Table 17.1 (continued)

Pathology	Shape features of interest	MR	Fractal dimension methods
Multiple system atrophy of the cerebellar type	3D cerebellar white matter, cerebellar gray matter [82]	T1	Box counting on 3D images [82]; extended from HarFA 2D box counting: [85])
	3D FD in supratentorial regions and focal region GM [79]	T1	3D box counting [79]
	3D FD cortical folding [78]	T1	Box counting [21, 78]
	3D FD of cerebral and cerebellar cortices [83]	T1	Box counting [83]
	3D FD-based network [21, 24, 83]	T1	Box counting [21, 24, 83]
Spinocerebellar ataxia (SCA)	3D FD of GM anatomical regions from entire brain [78]	T1	Box counting [78]
	Cortical GM and WM of the cerebellum and cerebrum [44]	T1	3D box counting [44]
	2D white matter skeleton [87]	T1	Box counting on 2D images [85, 87]
	Cortical thickness and complexity (FD) of cortical surface folding [35]	T1	Spherical harmonic reconstructions 3D images [35]
	Cortical thickness and complexity (FD) of cortical surface folding [34]	T1	Spherical harmonic reconstructions 3D images [34]
Stroke	Cortical local FD and local gyrification index [39]	T1	Spherical harmonic reconstructions [39]
Huntington's disease			
Parkinson's disease			

Abbreviations: 2D 2 dimensional, 3D 3 dimensional, ALS amyotrophic lateral sclerosis, ALSFRS-R ALS Functional Rating Scale Revised, AD Alzheimer's disease, FvSBe Frontal Systems Behaviour Scale, FTD frontotemporal dementia, bvFTD behavioral variant FTD, GM grey matter, HD Huntington's disease, MMSE Mini-Mental State Examination, MS multiple sclerosis, MSA-C multiple systems atrophy—cerebellar type, PD Parkinson's disease, PPA primary progressive aphasia, SARA Scale for the Assessment and Rating of Ataxia, SCA-3 spinocerebellar ataxia type 3, SCA-2 spinocerebellar ataxia type 2, TPISA triangular prism surface area, UHDRS Unified HD Rating Scale, WM white matter, WMS-R Wechsler Memory Scale Revised

its shape, size, orientation, and connectivity [8]. It represents the interior shape and is the essential structure of the WM. The surface is a set of boundary voxels in WM images. The general structure is the whole set of WM voxels. Besides the macroscopic structures described above, a study examined the microstructure of the WM using T2-weighted MR images [72].

In the GM fractal studies, the cortical surface was investigated because of the obvious fractal properties of the sulci-gyri convolution patterns. Specifically, pial surface (external cortical surface), gray-white matter interface (internal cortical surface), and cortical ribbon (the cortical surfaces and the structure between them) structures were examined to assess cortical complexity changes (Fig. 17.2) [4, 13, 22, 27, 28, 30, 31, 33, 37, 38, 40, 61, 66, 69, 81, 82]. The gray-white matter interface may be considered as a shape feature of either the WM or the GM as it may reflect the changes of either or both of the structures.

Given that fractal properties arise secondary to gyrification [20], the FD of the cortical surface measures the folding pattern. Two studies reported a strong association between the FD of the pial surface and shape measures, such as folding area, sulcal depth, and curvature, and a weak association between the FD of the pial surface and cortical thickness [22, 26]. A recent study found strong positive correlations between the FD of the cortical ribbon and cortical thickness and between the FD of the cortical ribbon and gyrification index [30]. Thus, in the neurodegenerative process, the effect of the decrease in cortical thickness is complementary to the impact of the reduction in the gyrification index. Shape measurements such as thickness, sulcal depth, and curvature represent local features of the cortex. The gyrification index [88], a global descriptor of cortical complexity, is sensitive to the direction of the imaging slices [76]. These findings suggest that the FD is a superior global measurement for the characterization of the folding pattern of the entire cortex. Figure 17.3 demonstrates the effects of the cortical thickness and gyrification index on the FD.

17.1.2 Fractal Dimension Methods

Several FD methods were used in the quantification of brain shape complexity (Table 17.1), including box counting [7, 22, 26, 28, 30, 31, 37, 41, 46, 60, 62, 69, 81, 82, 85–87], surface-based [4, 17, 38, 75, 76], and fast Fourier transform-based methods [33]. Among them, box counting is the most widely used technique because it can evaluate a fractal structure with or without strict self-similarity [85].

In general, the box-counting method works by repeatedly covering fractal images using meshes with different-sized boxes (r) and then counting the number of boxes (N) occupied by the studied structure [43]. The process results in a logarithmic function, $\ln(N) = D \cdot \ln(1/r) + \ln(k)$. The slope of the function (D) is referred to as the FD.

The surface-based algorithm was used to evaluate cortical surface complexity. It estimates the FD using the slope of a regression line between the logarithmic

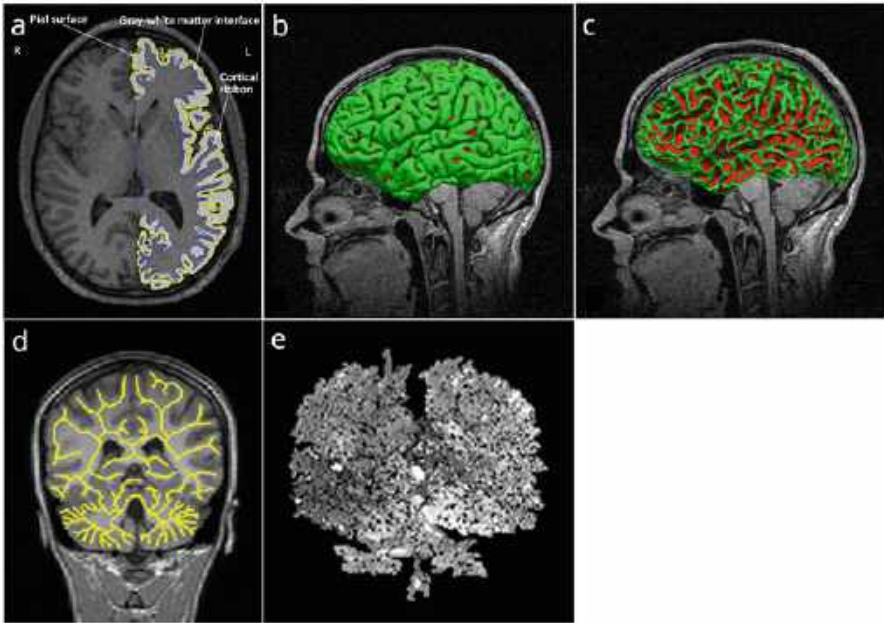


Fig. 17.2 Brain shape features on magnetic resonance imaging (MRI). (a) Illustration of the gray matter shape, including the pial surface, the gray–white matter interface, and the cortical ribbon. (b) Lateral visualization of the pial surface. (c) Lateral visualization of the gray–white matter interface (white matter surface) of the same subject demonstrated in (b). (d) 2D white matter skeleton of an MRI slice. (e) 3D white matter skeleton extracted from the same image shown in (d) ((a–c) Modified from Sandu et al. [61], images courtesy of Anca-Larisa Sandu, PhD, Aberdeen Biomedical Imaging Center, University of Aberdeen, UK, and (d, e) reprinted from Zhang et al. [85], with permission from the publishers)

least squares of the surface area and the logarithmic spatial frequency of the surface mesh that models the cortex [4]. A box-counting algorithm has been used recently to compute the FD of tessellated cortical surfaces extracted from MR images to evaluate cortical complexity [22, 26, 31]. The Fourier-based method was proposed to examine cortical ribbon complexity, and the FD is dependent on the Fourier function of the cortical shape [33].

The quantification of the local FD with spherical harmonization (SHFD) [67] has high test reliability [42] and has two main advantages compared to the box-counting method [84]: (1) high-resolution results since SHFD does not need to down-sample the cortical surface and (2) FD estimations are independent of the orientation of the surface, which is a caveat of the box-counting method. The SHFD method calculates FD complexity maps from the cortex surface at different scales: global, regional, and local. Global scales include a single value for the whole brain hemisphere; regional scales are a set of values for regions of interest; and local scales have a value for each surface vertex [59].

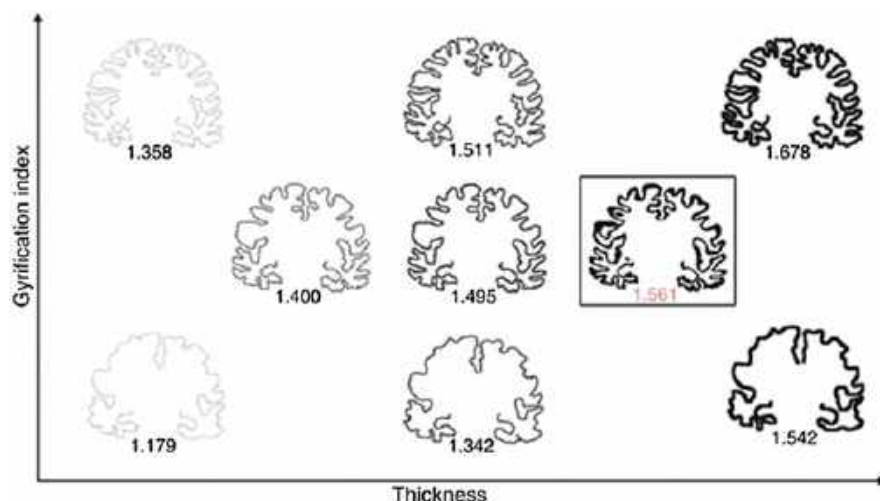


Fig. 17.3 The effects of cortical thickness and gyrification index on measured fractal dimensionality. A coronal slice from a control subject and its fractal dimension are seen in the box. The remaining cortical ribbons are artificial data demonstrating fractal dimension changes with variation in cortical thickness, gyrification index, and the combination of the two. The fractal dimension of each slice is indicated by the number below the slice. Changes in the cortical thickness are seen on the horizontal axis with increasing thickness toward the right. Changes in the gyrification index are seen on the vertical axis with values increasing upward. Thinning of the cortical ribbon and lowering the gyrification index both decrease fractal dimensionality. (Reprinted from King et al. [30], with permission from the publisher)

A multifractal analysis has been used to evaluate microstructural changes in the WM on T2-weighted MR images [72]. It divides an image into squares, and the probability density of each square box is defined. The multifractal measure is computed from a series of equations derived from the slope of a log-log plot of the probability and the size of the square [72]. Different from the conventional FD measure that evaluates macroscopic fractal structure, the multifractal index measures local microscopic shape changes by quantifying intensity fluctuation.

17.2 Fractal Dimension Studies of the Brain Shape

17.2.1 Aging

The changes of brain shape complexity occur throughout the life span with age, involving both GM and WM [4, 15, 25, 28, 37, 46, 61, 62, 69, 72, 81, 86].

The FD studies of the GM structure have reported that cortical complexity increases in early fetal life [69, 81], from infancy (<1 year old) until puberty (15 years old) for the left hemisphere and until adulthood (>25 years old) for the

right hemisphere [4, 28], and then decreases with aging [37, 61]. Specifically, the FD of the cortical surface was found to be relatively stable before 28 weeks of gestational age and increases rapidly afterward [81]. Kalmanti and Maris measured the FD of the 2D-skeletonized cerebral cortex in 93 healthy individuals (3 months–78 years; mean $\pm$ SD: 16.6 ± 16.6 years old) that were divided into four groups: infants (<1 year old), children (1–10 years old), prepubertals (10–15 years old), and adults (>25 years old) [28]. They found that basal ganglia development, mainly in the left hemisphere, was heavily dependent upon age until adolescence (15 years old). Both left and right cortical complexities were equally developed between the ages of 1 and 15 years, while the right cortex continued to develop in adulthood. A smaller sample-size study ($n = 24$) compared the FD scores of the external cortical surface between 24 normal children (6–11 years old [$n = 13$]; mean $\pm$ SD: 9.23 ± 1.71 years old) and adolescents (12–16 years old [$n = 11$]; mean $\pm$ SD: 13.25 ± 1.14 years old) and found that cortical complexity increased in the left and right inferior frontal and the left superior frontal regions [4]. Sandu et al. [61] investigated postadolescent developmental changes in cortical complexity by comparing the FD values of adolescents ($n = 17$; 13.3–14.4 years old; mean $\pm$ SD: 14.1 ± 0.27 years old) with those of adults ($n = 14$; 21–30.1 years old; mean $\pm$ SD: 24.24 ± 2.76 years old). They found significant reductions in the FD of the cortical ribbons for the whole brain, left and right hemispheres, and frontal and parietal lobes for both genders and only for males in the left temporal lobe. Similarly, Jao et al. [25] reported decreased FD in the lateral frontal lobes, which spread to the temporal and parietal lobes in middle-aged participants ($n = 98$; 53.74 ± 4.32 years old). However, males experienced earlier decline in FD in the right frontal and left parietal lobes ($n = 57$; 53.61 ± 4.25 years old). Interestingly, males exhibited more decreased FD in the subregions on the left, whereas females ($n = 41$; 53.94 ± 4.44 years old) exhibited more decreased FD in the lateral subregions [25]. Lee et al. [37] compared the FD values of skeletonized cortical ribbons between healthy young ($n = 31$; mean $\pm$ SD: 24.48 ± 5.02 years old) and old subjects ($n = 31$; mean $\pm$ SD: 63.23 ± 9.42 years old) and found decreased FD scores in the older group. Besides the cortical studies, one group investigated age-related changing patterns in the shape complexity of the anterior and posterior walls of the central sulcus (CS) in 295 healthy subjects (18–94 years old) [38]. The authors found significant reductions in the FD of the posterior wall of the right CS and a trend of FD reduction in the anterior walls of the bilateral CS during normal aging. No age-related changes in the FD were observed for the posterior wall of the left CS. Research has shown that the complexity of the cortical surface has a significant association with intelligence and education [22], which suggests that cortical complexity may serve as a biological marker of cognitive development.

Age-related WM FD changes throughout adulthood (20–80 years old) demonstrated an inverse U-shape pattern, with a gradual increase from young to mid-age and a sharp decrease to the old ($n = 209$) [15]. Specifically, the FD values of the WM general structure for the whole brain and each hemisphere increased until the late 40 s and then decreased with age. The FD values of the WM skeletons for the whole brain and left hemisphere declined from the 20 s to the age of 30, then

rose until the age of 50, declined again to the age of 70, and then rose again. In contrast, the FD of the skeletons for the right hemisphere increased slightly until the 40 s and reduced sharply afterward. The FD reduction in aging was also observed in a smaller sample-size study (24 young subjects, 17–35 years old; mean $\pm$ SD, 27.7 ± 4.4 years old; 12 old subjects, 72–80 years old; mean $\pm$ SD, 74.8 ± 2.6 years old) that reported significant reductions in the FD of the WM skeleton of the whole brain and left hemisphere and in that of the WM general structure of the whole brain [86]. This study also observed a nonuniform distributed pattern in the degeneration of WM complexity between the genders and across brain hemispheres, with a decreased WM interior shape complexity found in the left hemisphere in old men but in the right hemisphere in old women. Neither of the above two studies observed FD changes in the gray–white matter surface. A longitudinal study measured the FD of the WM general structure on 148 participants at ages 68 years and followed them until 73 years. The authors reported significant FD reductions in the WM over the 5-year period in late life [62]. These FD decreases were associated with cognitive function decline, specifically processing speed (Digital Symbol score), verbal memory (Auditory Verbal Learning Test), reasoning (Raven's Standard Progressive Matrices), and general cognitive ability after adjustment for childhood mental ability. This is consistent with a study that reported significant associations between the FD of brain WM structure and cognitive changes over the life course from age 11 to 68 years [46]. Specifically, the FD was correlated positively with lifelong fluid change, estimated by a standardized difference between Moray House Test (MHT) and Raven's Standard Progressive Matrices test (RPM), and negatively with a decline in cognition in late life, estimated by a standardized difference between the National Adult Reading Test (NART) and RPM. No significant correlations were found between fractal measures and cognitive maturation from 11 years to adulthood, measured by the standardized difference between the MHT and NART. These results suggest that those with a greater brain WM structural FD had greater than expected fluid abilities at age 68 years than that predicted by their childhood intelligence and less cognitive decline at age 68 years. Takahashi et al. [72] performed a multifractal study to examine age-related microstructural changes in the deep WM on T2-weighted MR images without visible abnormal intensities. They reported an increased multifractal measure in the frontal region, not in the parieto-occipital region, and a higher executive dysfunction score in healthy elderly subjects compared to young healthy controls. The executive dysfunction score was found to be positively associated with the multifractal measure in the frontal region, but not in the parieto-occipital region. These findings suggest that microstructural changes in the WM preferentially occur in the frontal region during normal aging, and these changes are correlated with executive cognitive decline reflective of frontal-subcortical dysfunction [72]. The above studies suggest that WM complexity may predict retention of cognitive ability in late life.

17.2.2 *Alzheimer's Disease*

Alzheimer's disease (AD) is an irreversible, progressive brain disorder in which amyloid plaques and tau tangles present throughout the brain and neurons degenerate and die, causing problems with memory, thinking, and behavior [51]. Although an early study did not find differences in the FD of the sulcal surface between patients with AD and a control group [74], FD studies since then have demonstrated a decrease in the shape complexity of the cerebral cortex and WM in AD on structural MR images [30, 31]. Differences in the FD analysis method can affect the FD outcome of AD MR images. An analysis of the whole brainstem FD using Higuchi's method resulted in no differences between the AD and the control group, whereas when the investigators used the detrending moving average (DMA), an increased FD of AD patients was found compared to control and mild cognitive impairment subjects ($n = 232$ MCI, 200 control, 200 AD) [54].

A decrease in the FD of the 2D cortical ribbon was found on the anterior tip of the temporal lobe, the mammillary bodies, the superior colliculus, the most posterior edge of the corpus callosum, the inferior colliculus, and the massa intermedia of the thalamus in patients with AD [30]. A further study by the same group demonstrated that the FD values of the 3D cortical ribbon and the gray–white matter surface were significantly lower in patients with mild AD than controls, with highly significant differences detected by the FD of the cortical ribbon [31]. Significant correlations were found between the cortical gyrification index and the FD scores of both the cortical ribbon and the gray–white matter surface. However, only the FD of the cortical ribbon had a significant association with cortical thickness and the most commonly used neuropsychiatric assessment battery in AD clinical trials, AD Assessment Scale cognitive (ADAS-cog) score. This study showed that the FD of the cortical ribbon had the highest discrimination power in separating controls and mild AD patients among the studied matrices, including cortical thickness, the gyrification index, and the FD measures of the gray–white matter surface and the pial surface. The FD of the pial surface was not recommended by the authors, since atrophic changes on the pial surface could either decrease or increase FD, depending on how the atrophy occurs. For example, two types of atrophic changes in the pial surface were observed in this study: decreased folding area causing decreased complexity and increased sulcal depth leading to increased complexity.

Another group recommended that spherical harmonic-based FD may be a complement to thickness loss analysis in monitoring longitudinal changes of AD progression. This group investigated cortical surface abnormalities in AD and found decreased thickness, local gyrification index (LGI) average scores, and spherical harmonic-based FD values, as well as local WM FD [59]. In this study, the decreased global FD and local WM FD were associated with the conversion from mild cognitive impairment to AD within the 4-year follow-up period and the local WM FD positively correlated with scores from the widely used mini-mental state examination (MMSE) for cognitive impairment. This is consistent with the observed decrease in global GM 3D FD of 63 subjects with mild to moderate AD compared

to healthy controls [27]. When analyzing the FD in AD, it is important to note that older age is associated with a decrease in FD and that specific cutoffs need to be considered when comparing healthy controls over 60 years old and AD patients [27].

Several studies have noted differences in regional FD in AD. One subject with AD had a decreased FD predominantly in the medial temporal lobes and bilateral parietal lobes [29]. A larger study of AD, amnesic MCI due to AD (aMCI), and healthy subjects ($n = 110, 104$, and 64 , respectively) compared FD to cortical architecture, which was assessed as a function of cognitive reserve (years of education) [65]. They found that cognitive reserve modulates the cortical architecture only at the pre-dementia stage. The aMCI subjects with high compared to low cognitive reserve showed a significant decrease of cortical thickness in the right temporal and left prefrontal lobe, but increased FD in the right temporal and left temporo-parietal lobes. It was suggested that the aMCI subjects had atrophy of the brain due to neurodegeneration, but the higher cognitive reserve mechanisms reflected the richer brain folding in temporo-parietal areas. However, AD patients with higher cognitive reserve had reduced cortical thickness as well as decreased FD compared to low cognitive reserve AD subjects. Thus, these results confirmed the existence of a limited time window to modulate cognitive reserve at the aMCI stage. Other regional areas that have reduced FD in AD are the insula, posterior cingulate, pre- and post-central gyri, and bilateral parahippocampal gyri ($n = 32$ AD and 28 healthy controls) (Fig. 17.4) [53]. Interestingly, specific neuroanatomical regions in that study were associated with specific subscores of Addenbrooke's cognitive examination-revised (ACE-R) battery test of cognitive domains. The memory subscore of the ACE-R was positively correlated with the FD of the bilateral insula, left superior temporal cortex, and isthmus cingulate. The memory, language, and fluency subscores of the ACE-R were positively correlated with the FD of the left insula, inferior temporal cortex, and medial orbitofrontal gyri. Together, these results suggested that the FD of these regions was vulnerable to memory and language impairments. Compared to participants with frontotemporal dementia (FTD), no significant reduction in FD was observed for AD. Relative to AD, FTD patients showed decreased FD bilaterally in the orbitofrontal cortex, left anterior insula, and paracentral gyrus (Fig. 17.5).

Memory scores have also been correlated with the FD of the frontotemporal regions of elderly patients aged 75 years and older, including patients with AD. The FD in the bilateral frontotemporal regions was positively correlated with recognition memory, paired associate memory, attention, and visual-and-working memory scores on the Wechsler memory scale-revised (WMS-R) neuropsychological memory test [32]. Specifically, the visual-and-working memory correlated with the cortical FD in the right posterior cingulate cortex as well as the lateral occipital cortex, and the attention score correlated with the right precuneus cortex, right pericalcarine cortex, and cuneus cortex.

Together, these studies demonstrate the potential clinical application of the FD as a quantitative marker of the cortical structure in AD.

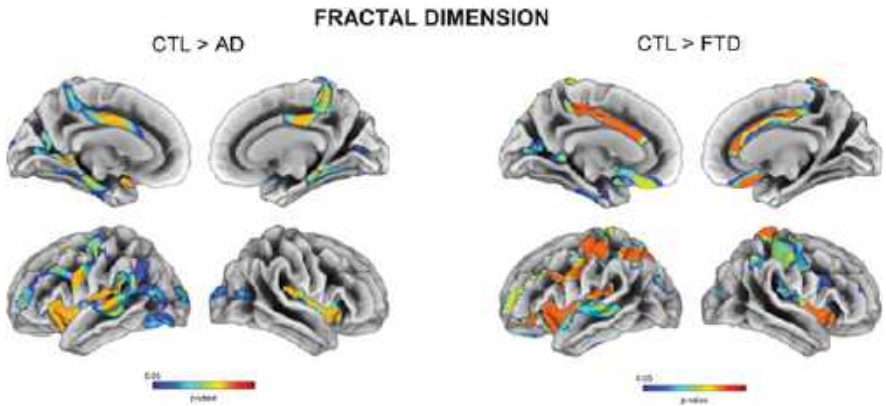


Fig. 17.4 Vertex-wise FD group comparisons between controls, AD, and FTD (FDR-corrected $p < 0.05$). (Reprinted from Nicastro et al. [53], with permission from IOS Press)

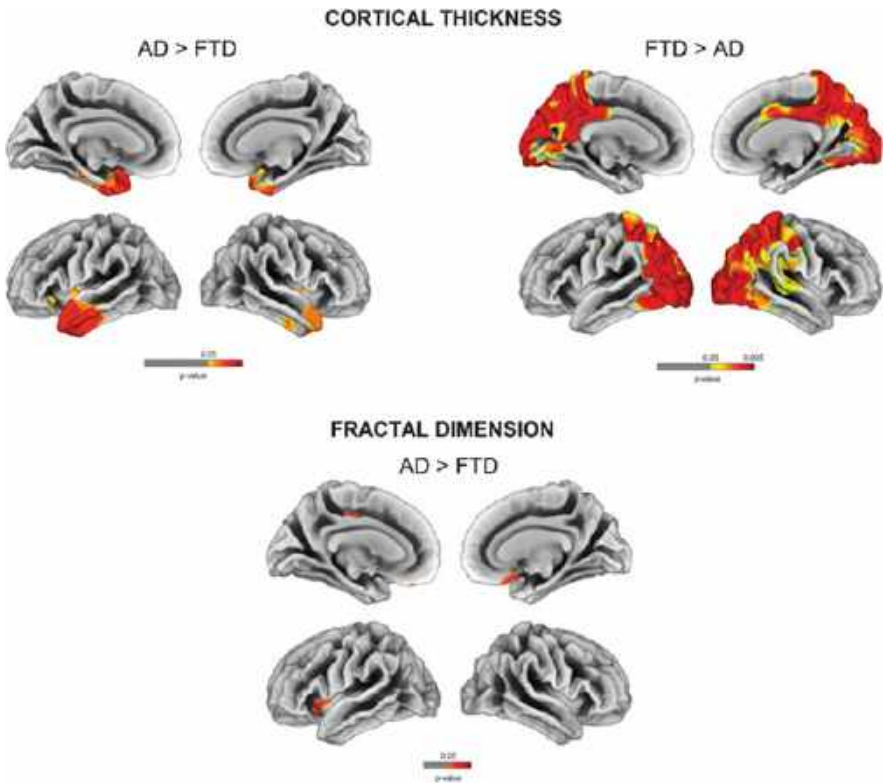


Fig. 17.5 Vertex-wise cortical thickness and FD comparisons between AD and FTD (FDR-corrected $p < 0.05$). (Reprinted from Nicastro et al. [53], with permission from IOS Press)

17.2.3 *Amyotrophic Lateral Sclerosis*

Amyotrophic lateral sclerosis (ALS) is a rapid, progressive neurodegenerative disease that is the most common form of motor neuron disease (MND) [48]. ALS is characterized by the degeneration of upper and lower motor neurons in the brain and spinal cord and is genetically, pathologically, and clinically related to a form of dementia called frontotemporal dementia (FTD). In fact, patients can have both, ALS/FTD [58].

Rajagopalan and colleagues investigated WM shape changes using FD analysis in ALS patients with different clinical phenotypes (ALS patients: $n = 89$; controls: $n = 11$) [57]. In their study, ALS patients were assigned into 4 subgroups: ALS-FTD, upper motor neuron (UMN)-predominant ALS patients with corticospinal tract (CST) hyperintensity on T2/PD-weighted images (ALS-CST+), UMN-predominant ALS patients without CST hyperintensity identified on T2/PD-weighted images (ALS-CST-), and ALS-classic patients (ALS-CI). Among these subgroups, they found the greatest degeneration in WM shape complexity in the ALS-FTD patients. Specifically, the FD values of the whole-brain WM skeletons, right-hemisphere skeletons, and whole-brain general structure were significantly reduced in the ALS-FTD patients when compared with the ALS-CST+ patients (Fig. 17.6). No significant voxel-based morphometry changes were found between the controls and the ALS patients and among the ALS subgroups. These findings may suggest that the FD is a more sensitive index of brain WM integrity than volumetric measurement in ALS population. A recent study with 22 patients diagnosed with ALS and 20 healthy subjects evaluated GM FD in 102 neuroanatomical regions of interest [6]. Decreased FD values were found in the left precentral gyrus and central sulcus, left circular sulcus of insula (superior segment), left cingulate gyrus and sulcus (middle posterior part), right precentral gyrus, and right postcentral gyrus of ALS patients. These results are consistent with the disease pathogenesis which affects the motor cortex or precentral gyrus region. Moreover, the FD values in the right precentral gyrus and left circular sulcus of the insula (superior segment) were negatively correlated with disease duration.

ALS functional rating scale (ALSFRS-R) is a measure of daily functional activity impairment (e.g., speech, swallowing, and walking) and used for evaluating the functional status of patients with ALS. The significant association between the FD and ALSFRS-R indicates that the shape complexity of the WM [57] and the GM of the right precentral gyrus [6] reflects the functional status of ALS patients, and the FD could potentially serve as a biomarker of ALS pathophysiology.

17.2.4 *Frontotemporal Dementia*

Frontotemporal dementia (FTD) is caused by the progressive degeneration of neurons mainly in the frontal and temporal lobes of the brain and is associated

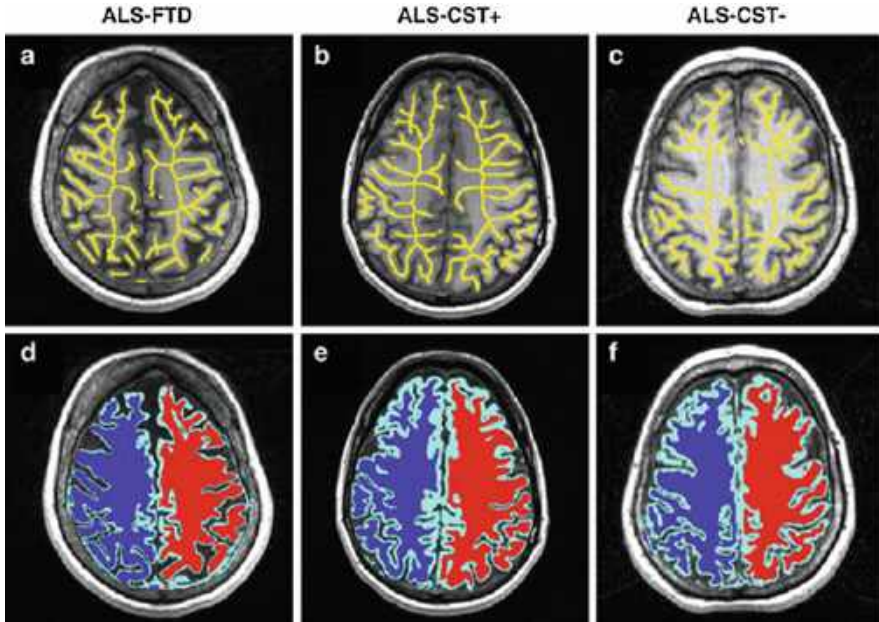


Fig. 17.6 2D illustration of reduced complexity in the white matter skeleton and general structure on anatomical MR images. The complexity levels of the WM skeleton and general structure were reduced in an ALS-CST- patient (**c, f**) and an ALS-FTD patient (**a, d**) compared to an ALS-CST+ patient (**b, e**). ALS-FTD: ALS patients with frontotemporal dementia. ALS-CST+: Upper motor neuron (UMN)-predominant ALS patients with corticospinal tract (CST) hyperintensity on T2-/PD-weighted images. ALS-CST-: UMN-predominant ALS patients without CST hyperintensity identified on T2-/PD-weighted images. (Reprinted from Rajagopalan et al. [57], with permission from the publisher)

with ALS. Alterations in behavior, speech, or language are often the first noticeable symptoms. FD can also be used as a biomarker for structural changes in FTD. Patients with the behavioral variant of FTD (bvFTD) showed significant reductions in FD values of GM whole brain, right hemisphere and left hemisphere skeletons, and general structures compared to controls ($n = 27, 20$, respectively) [66]. Comparison between bvFTD and primary progress aphasia subjects ($n = 12$) revealed smaller FD values in the whole-brain skeleton and general structure of bvFTD patients. There was also a significant negative correlation between the FD of the right hemisphere skeleton GM and the frontal systems behavior scale (FrSBe) total score in the bvFTD group [66]. These results indicate that variants of FTD are associated with disease-specific morphometric complexity patterns.

17.2.5 *Epilepsy*

Epilepsy is a neurological disorder in which nerve cell activity in the brain becomes disturbed, causing seizures and sometimes loss of consciousness [50]. Fractal studies have shown decreased cortical and WM complexity in epilepsy [7, 17, 40]. In the cortical complexity study, a decreased FD of the external cortical surface was found in all lobar regions except the right frontal region in patients with left mesial temporal lobe epilepsy [40]. In patients with right mesial temporal lobe epilepsy, decreased complexity was observed in the bilateral temporal, occipital, and left parietal regions. Cook, Free et al. investigated brain WM shape complexity in epilepsy by measuring the FD values of the 2D WM border and 3D WM surface on conventional MR images [7, 17]. FD reductions in the WM border (gray–white matter interface) were found in more than half of the patients with frontal lobe epilepsy without identifiable lesions on structural MR images [7]. In the 3D study, abnormal FD scores of the WM surface were found in 8 out of 16 patients with cryptogenic epilepsy and a gyral abnormality on MR images and in 9 out of 23 patients with epilepsy and normal MR images [17]. These findings suggest that the FD can be used to measure subtle cortical changes, and the disruption of the cortical ribbon appeared in many patients with epilepsy.

17.2.6 *Multiple Sclerosis*

Multiple sclerosis (MS) is a chronic inflammatory and neurodegenerative disease of the central nervous system in which axons have been denuded of the myelin sheaths that protect them and allow the conduction of nerve signals [23]. WM lesions are well recognized in MS [2], and extensive damage of GM has also been reported in recent studies [56, 89].

Esteban et al. reported decreased WM FD values and increased GM FD measures in patients with MS [12, 13]. Such different abnormality patterns of the GM and WM indicate that different pathological processes occurred in each structure. Specifically, the FD measurements of the gray–white matter interface (WM borders) and WM internal structures (WM skeletons) decreased significantly in patients with MS compared with healthy controls. The same FD decrease was found for WM borders in all MS subtypes, including clinically isolated syndrome (CIS), relapsing–remitting MS (RRMS), secondary progressive MS (SPMS), and primary progressive MS (PPMS), and in all subtypes except for PPMS for WM skeletons. The decrease of the FD values in the WM borders was affected not only by the presence of MS plaques but also by changes in the non-visible WM abnormalities such as normal-appearing WM. For the WM skeletons, the decrease was only observed in MR slices with MS lesions. The FD of the WM skeletons was found to be correlated with both T1 and T2 lesion load. No associations were found between the WM FD and MS-related clinical disability scales such as Expanded Disability Status Scale (EDSS)

and the MS Functional Composite (MSFC) score [12]. Jimenez et al. [27] analyzed the images from this study to create a web-based platform employing the 3D box-counting computation and the skeleton generation algorithm. The group focused on the interactive 3D visualization of the volumetric data involved in the 3D FD analysis with WebGL, a technology with several advantages for representing 3D models on the web.

The GM study focused on MS patients at the early to intermediate phases of the disease and found increased FD values in either first attacks of MS (FAMS) or relapsing-remitting MS (RRMS) compared with healthy subjects [13]. The FD of the RRMS was significantly higher than FAMS. These findings suggest that GM shape complexity is abnormal in patients with MS and the FD changes occur in the early stage of the disease and increase with disease progression. However, the authors stated that the FD of the GM in the progressive patients was difficult to estimate since GM atrophy was highly prominent in this subgroup. For the brain-lesion load, the FD of the GM was found to be correlated with T1 and T2 WM lesion load, but not with the GM atrophy or the disability scores such as EDSS and MSFC. Several explanations were provided in the study: (1) GM atrophy is less pronounced in the early stages of the disease, and there may be lack of statistical power to detect an association. (2) GM atrophy might preclude GM tissue damage that has been associated with the FD changes. Another group also reported significantly decreased FD values using the triangular prism surface area (TPSA) method in MR images of four MS patients compared to 12 normal controls [73]. It should be noted that this study assessed significance according to 95% confidence intervals; however, no statistical analysis was shown and the specific brain structures measured were not stated.

17.2.7 Multiple System Atrophy

Multiple system atrophy (MSA) is a progressive neurodegenerative disease in which different types of nerve cells in the brain and spinal cord lose function and die, affecting both the autonomic nervous system and movement [49]. Research has shown that the FD of the cerebellar WM and GM decreased significantly in patients with MSA of the cerebellar type compared to controls. Such cerebellar degeneration was dominated by morphological changes in the cerebellar WM [82]. Although the FD was related to volumetric measurement in this study, it was suggested as a better index than the conventional volumetric measurement in assessing the shape changes of the WM and GM, since it generated smaller variances and less gender effects.

17.2.8 *Spinocerebellar Atrophy*

Unstable polyglutamine expansions cause the neurodegenerative disease spinocerebellar ataxia (SCA), which is traditionally associated with pontocerebellar degeneration [14, 44, 55]. Magnetic resonance imaging shows loss of cerebellar GM and WM and loss of volume in the brainstem and cerebellar peduncles [45]. More recently, damage of the cerebral hemispheres has also been implicated. Clinically, SCA is characterized by slow, progressive clumsiness in the arms and legs, a staggering gait, lower limb spasticity, difficulty with speech and swallowing, and impaired eye movements [55]. The most prominent form is SCA type 3 (SCA3), also known as Machado–Joseph disease.

Studies have combined the clinical features of SCA and FD methods to distinguish whether cortical complexity is associated with clinical symptoms and whether there is a correlation between progression of cerebellar and cerebral dysfunction. The polyglutamine (cytosine, adenine, guanine [CAG]) repeat length was correlated with SCA disease onset. Patients with SCA3 with longer CAG repeat length demonstrate earlier disease onset; however, the CAG repeat length does not significantly correlate with cerebellar 3D FD values or SARA scores [21]. Moreover, the duration of disease was not correlated with any 3D FD values. Interestingly though, patients with SCA3 show a significant reduction in 3D FD values compared to age-matched healthy controls in the right anterior lobe, left anterior lobe, right posterior upper lobe, left posterior lower lobe, and whole cerebellum, suggesting that the cerebrum in addition to the cerebellum is affected by SCA neuropathogenesis [21].

Wang et al. [78] evaluated the MR imaging of 48 patients with SCA3 and 50 healthy controls using a 3D FD approach. Interestingly, cortical involvement was more extensive in specific supratentorial regions than involvement of the traditional olivopontocerebellar region and the corticocerebellar system in SCA3 participants. In contrast to the previous study [21], there was a negative correlation with SARA scores and the 3D FD value of the cerebellar cortex for SCA3 patients. There was also a significant correlation between decreased 3D FD values and the disease duration, which was suggested as an indication of atrophy of the cerebellar and cerebral cortexes in SCA3. Furthermore, cerebral abnormality was more apparent in the later stages of the disease with lower 3D FD values, which was consistent with the presence of dementia at the later stage of SCA3. This study demonstrated that 3D FD is effective to quantify cortical changes and evaluate the association between dysfunction of the cerebellar cortex and cortical complexity.

The investigation by Wang et al. [79] consistently reported that atrophy in SCA3 is not limited to the infratentorial regions and showed extensive involvement of extracerebellar regions. Participants with SCA3 had lateralized atrophy predominantly in the left hemisphere. Specifically, decreased 3D FD values were only found in the left hemisphere occipital and temporal lobes. In the frontal cortex, the premotor cortex (left precentral gyrus, bilateral superior frontal gyrus, and left-middle frontal gyrus), supplementary motor cortex (left supplementary motor

area), and primary motor cortex (left precentral gyrus) were significantly atrophied (decreased 3D FD values).

Brain structural networks are disrupted in SCA3 (based on a correlation matrix of 3D FD values from MR images), which may reflect the clinical characteristics of SCA3. The cortico-cerebral circuits are disrupted and segregated into occipital-parietal and frontal-pre-frontal clusters. The cerebellum of SCA3 participants clustered in a frontal-temporal cluster and is segregated from cerebellum-temporal-frontal circuits [83]. The brain network in SCA3 was also explored by Jao et al. [24] by employing a 3D FD-based network to detect network differences between SCA3 and healthy control groups ($n = 48$ and $n = 48$). Consistent with previous studies, patients with SCA3 exhibited lateralized atrophy (lower 3D FD values) in the left supratentorial regions and lower modularity values. Interestingly, the cerebellar regions were found to be dissociated from higher-level brain networks. There was decreased intramodular connectivity in all lobes but increased intermodular connectivity in the frontal and parietal lobes of patients with SCA3. Huang et al. [21] have cautioned investigators of interpreting the results of structural networks constructed according to morphological covariance, including 3D FD values as the structural covariance may arise from genetic influence, normal development, experience-related plasticity, aging, and mutual trophic influences [1].

In spinocerebellar ataxia type 2 (SCA2), pontocerebellar degeneration has been detected through the application of FD to cerebellar changes; however, the concurrent cerebral changes and progression were not clear. Longitudinal brain MRI of nine patients with SCA2 and 16 age-matched healthy controls was investigated twice (3.6 ± 0.7 and 3.3 ± 1.0 years apart, respectively) on the same 1.5 T MR scanner [44]. For participants with SCA2, the cerebellar GM and WM FD values were significantly lower, whereas the cerebral WM FD was not different from healthy controls. The simplified tissue architecture and volume loss associated with demyelination and axonal loss at neuropathological examination of the cerebellar WM likely explain the decreased FD in the cerebellum of SCA2 participants as proposed by Marzi et al. Moreover, the cerebral GM was significantly lower in SCA2 participants. However, there were no differences in the rate of change in FD between groups. It was suggested by the investigators that the relatively low sensitivity of the fractal analysis of the GM and WM structural complexity progression in this study is reflective of the small sample size.

17.2.9 Stroke

A stroke occurs when the blood supply to an area of the brain is interrupted or when a blood vessel in the brain bursts, spilling blood into the spaces surrounding brain cells [47]. Brain cells die when the brain is deprived of oxygen and nutrients or there is bleeding into or around the brain.

One study has shown a decreased FD in brain WM following stroke (Fig. 17.7) [87]. The FD was lower in the stroke-affected hemisphere, and the greater FD of

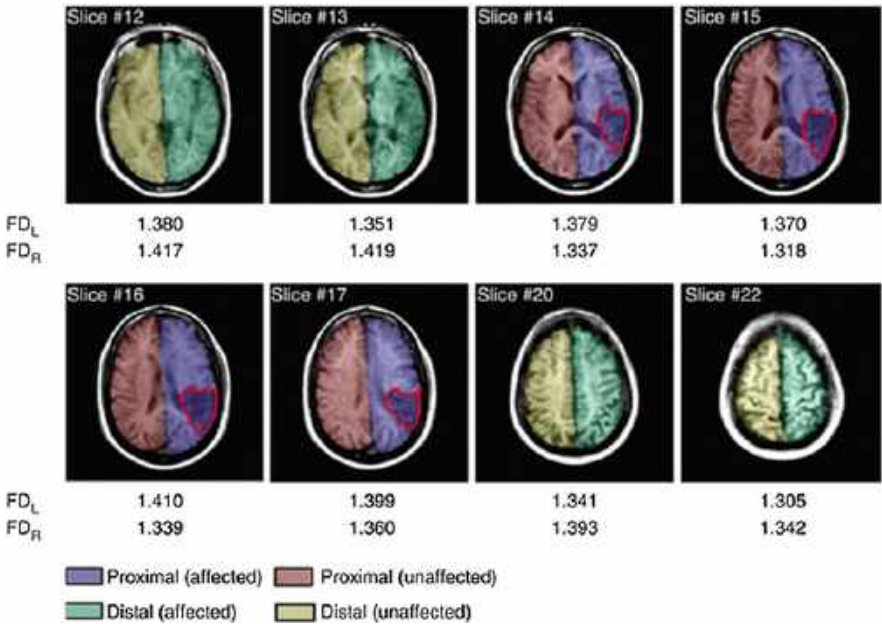


Fig. 17.7 Illustration of the fractal dimension (FD) values of the left and right white matter (WM) in eight horizontal slices from one subject with a right-hemisphere lesion (outlined in red). The FD values of the left hemisphere were smaller than those of the right hemisphere in slices without apparent lesions (slices #12, #13, #20, and #22). The FD values of the left hemisphere were greater than those of the right hemisphere in slices with a lesion (slices #14–#17). Images are displayed in neurological convention, with the left side corresponding to the left hemisphere. FD_L, the FD of the left WM skeleton; FD_R, the FD of the right WM skeleton. The colors were designated to illustrate regions of interest. Blue, proximal region on the affected side; red, proximal region on the unaffected side; green, distal region on the affected side; yellow, distal region on the unaffected side. Note that the proximal region indicates slices with lesion and distal region specifies slices without lesion. (Reprint from Zhang et al. [87], with permission from the publisher)

the residual WM was associated with less impaired upper extremity motor function in patients with left-subcortical or right-cortical lesions. This correlation was more robust in patients with right-hemisphere lesions. No significant associations were found between lesion volume and motor function, suggesting that, at least in this study, the FD was more sensitive in predicting upper extremity motor function than volumetric measurements post stroke. The FD may potentially serve as a useful clinical index to evaluate the level of residual WM structural changes following stroke or any accompanying brain plastic changes to guide future therapeutic interventions.

17.2.10 *Huntington's Disease*

Huntington's disease is caused by an expanded CAG (cytosine, adenine, guanine) trinucleotide repeat in the huntingtin gene (*HTT*) leading to a neurodegenerative mixed movement disorder with psychiatric manifestations and cognitive decline [71]. Striatal volume loss likely begins many years before the unequivocal signs of Huntington's neurodegeneration [11, 80].

The complexity of cortical folding, which was used as a measure of cortical neurodevelopment, differs between preHD individuals ($n = 57$) and healthy controls ($n = 57$) in motor and visual system areas as well as in temporal cortical areas [34]. In preHD, the left hemisphere has a lower FD in a cluster including the supramarginal and postcentral cortex, precentral cortex, and the lingual gyrus. However, the middle and superior temporal cortex have a greater FD. In the right hemisphere, there is a lower FD in the precuneus, posterior and middle cingulate gyrus, and lingual gyrus. Furthermore, preHD participants have lower GM volume in the putamen bilaterally, whereas no regions had a higher GM volume. The investigators used putamen volume as an indicator of the extent of HD-induced neurodegeneration and used a regression model to predict mean putamen GM volume with age, CAG repeat length, and neuroanatomical FD as independent variables. They found that age, CAG repeat length, and FD of parietal regions (left supramarginal gyrus and right precuneus) were significant predictors for mean putamen GM volume. Mean putamen GM volume was a significant predictor of the Unified Huntington's Disease Rating Scale (UHDRS) motor score. However, there was no association between loss of putamen volume and FD of motor regions. This suggests that the volume and cortical folding of regions associated with voluntary movement may not be related. Overall, the outcomes of the study suggested that *HTT* CAG repeat length may influence cortical folding, but there was no evidence that this leads to developmental pathology or was clinically meaningful. The authors proposed that the *HTT* CAG repeat expansion may influence the processes governing cortical neurodevelopment but that influence appears to be independent of the events leading to neurodegeneration [34].

17.2.11 *Parkinson's Disease*

Parkinson's disease (PD) is characterized by a progressive decline in motor, cognitive, and behavioral symptoms. Neurobehavioral complications in PD such as impulsivity have been investigated as they often lead to psychosocial impairment [77].

Most structural and functional studies in PD focus on impulse control disorders (ICDs), disregarding populations of patients with PD that potentially have increased levels of trait impulsivity [35]. Hence, Kubera et al. [35] investigated whether

impulsivity is related to cortical thickness and the complexity of cortical folding (based on FD) in non-demented PD individuals with no history of ICD compared to healthy controls. Although both parameters are related to early brain development, the complexity of cortical folding is assumed to be more stable during life span than cortical thickness [5, 52]. The MR images of 22 PD patients and 18 healthy participants matched for age, gender, and education (mean years of age 64.59 ± 2.204 and 62.67 ± 2.332 , respectively) were compared with an expectation of differences between neuroanatomical areas. A spherical harmonic reconstruction approach, which reflects the frequency of sulcal and gyral convolutions, was used for analyzing the complexity of cortical folding. Although there were differences between the groups for cortical thickness, no significant differences were found in the complexity of cortical folding. The authors suggested that these findings support the complexity of cortical folding as a stable neurodevelopmental marker that is not influenced by cortical thickness changes in PD. Since the PD participants included in the study had good response to dopaminergic treatment, it was also suggested that clarification was required to determine whether the complexity of cortical folding predates symptom onset in drug-naïve patients.

A larger follow-up study investigated the cortical complexity and gyrification patterns using local FD and local gyrification index (LGI) [39]. Brain structural MRI data of PD participants ($n = 60$) were compared to healthy controls ($n = 56$). These groups were similar in age, sex, education, and mini-mental state examination (MMSE) (mean years of age 61.60 ± 6.93 and 63.07 ± 5.54 , respectively). The PD participants had widespread lower local FD values in the left pre- and post-central cortex, left superior frontal cortex, left caudal middle frontal cortex, bilateral superior parietal cortex, and right superior temporal cortex compared to healthy controls. The PD group also had a significant negative correlation between the local FD of the postcentral cortex and duration of illness, which may explain the progressive loss of balance and gait impairments in PD. However, no significant correlations were found between local FD changes of the left postcentral cortex and Unified Parkinson's Disease Rating Scale (UPDRS)-III scores or Hoehn and Yahr (H&Y) stages. This was attributed to the fact that these assessments were conducted while PD participants were on medication, thus likely reducing the possibility of detecting a significant relationship with local FD of this region. Additionally, there were no significant differences observed for local gyrification index. In contrast to the aforementioned study, these results indicate that the local FD may provide a more sensitive diagnostic biomarker and the investigators suggested that the negative correlation between the local FD of the left postcentral cortex and duration of illness may be a prognostic imaging biomarker for PD. They further suggested that the decreased local FD of the cerebral cortex may reflect the pathophysiology of the cortico-striatal-thalamic-cortical circuit. The authors also proposed that microstructural degradation in the subcortical white matter and basal ganglia may also contribute to the decrease in local FD.

17.3 Discussion

FD studies have demonstrated that brain shape complexity changes with age throughout life and in neurodegenerative diseases (Table 17.2). In general, the FD of the GM increases in early fetal life [69, 81] and continues to develop equally in the left and right hemispheres between 1 and 15 years old. After adolescence, the FD of the left cortex decreases, while the FD of the right cortex continues to increase until adulthood and then decreases with aging [4, 28, 37, 61]. The inverse-U pattern of the FD changes of the left cortex agrees with the changing pattern of a metabolism measure, entropy production, over the same time frames [3]. The FD of the WM increases from young until mid-age and then decreases with aging [15, 62, 86]. During middle age, there is a decrease in FD that spreads from the lateral frontal lobes to the temporal and parietal lobes. Sex differences are also apparent with a reduction in FD mainly in the right frontal and left parietal lobes earlier for males. Males also exhibit a lower FD in subregions in the left hemisphere than the right hemisphere, whilst females exhibit a lower FD in lateral subregions [25]. The reduction in the FD of the WM and GM in late life with aging parallels with the decrease in glucose metabolism [68]. In the fractal studies of neurodegenerative diseases, the FD was found to decline in the GM in AD [30, 31] and MSA of the cerebellar type [82], and in the WM in ALS [57], epilepsy [7, 17, 40], MS [12], MSA of the cerebellar type [82], and stroke [87]. Such a decreased pattern is consistent with the metabolic studies demonstrating a decline in glucose metabolism with diseases [63]. An increased FD was found in the GM in MS [13], in which a rise in glucose metabolism was observed [63]. These findings may suggest a link between FD-measured shape complexity and metabolism, supporting the association between brain shape and function.

Besides the potential link between the FD and metabolism, the FD of brain shape has been found to be significantly associated with common clinical functional scores, such as AD Assessment Scale Cognitive Score [31], attention, and memory (Wechsler Memory Scale Revised) [32], Mini-Mental State Examination [59], and the revised Addenbrooke's Cognitive Examination memory, language and fluency subscores [53] in AD, the ALS Functional Rating Scale [57], the Frontal Systems Behavior Scale in behavioral variant FTD [66], the Scale for the Assessment and Rating of Ataxia in SCA3 [79], and upper extremity motor function scores (Wolf Motor Function Test and Fugl-Meyer Motor Assessment Score) in stroke [87]. In addition, the changes in the WM FD with age have been shown to match with the changing pattern of cognitive abilities throughout life [9, 46].

These studies suggest that the FD may be used as a quantitative index in the evaluation of the brain shape changes in aging, AD, ALS, bvFTD, SCA3, and stroke. No associations were observed between the FD and MS-related clinical disability scales (EDSS and MSFC) [12, 13]. This might be due to the interaction between the FD abnormality in the GM and WM. In the two MS fractal studies, the association of the FD with functional scores was investigated in the GM and WM separately. Further study may be needed to examine the relationship between

Table 17.2 Fractal dimension changes of the brain shape with age and neurodegenerative diseases and their association with clinical scores and other morphometric measures

Pathology	Fractal dimension (FD)	Associations
Aging	↑FD of the external cortical surface in early fetal life [69, 81]; ↑FD of the left cerebral cortex from infancy till adolescence (1–15 years old) and then ↓ with aging; ↑FD of the right cortex till adulthood and then ↓ with aging [4, 28, 37, 61]	FD of the cortical surface ↔ cortical thickness, folding area in both hemispheres and several lobe regions, sulcal depth only in the left temporal region, IQ, and number of years of education [22]
	Inverse-U pattern on the FD of the white matter: ↑FD from young till mid-age then ↓ to the old [15, 62, 86]	FD of the white matter ↔ information processing speed (digital symbol score), verbal memory (auditory verbal learning test), reasoning (Raven's standard progressive matrices), general cognitive ability in specific models (with adjustment for childhood mental ability), lifelong fluid change (positive), and cognitive decline in late life (negative) [46, 62]
	↓ FD in the lateral frontal lobes, which then spreads to the temporal and parietal lobes in the middle-age group; ↓ FD, mainly in the right frontal and left parietal lobes, earlier for males; males exhibited more of a ↓ FD in the subregions on the left than those in the right, whereas females exhibited more of a ↓ FD in the lateral subregions [25]	
Alzheimer's disease (AD)	↓ FDs of the cortical ribbon and gray–white matter surface [30, 31]	FD of the cortical ribbon ↔ AD assessment scale cognitive (ADAS-cog) score [31]
		FDs of the cortical ribbon and gray–white matter surface ↔ cortical gyrification index, cortical thickness [30, 31]

(continued)

Table 17.2 (continued)

Pathology	Fractal dimension (FD)	Associations
	No significant difference in FD of the sulcal surface between groups, or between hemispheres within groups [74]	
		FD in bilateral frontotemporal regions ↔ positively correlated with recognition memory, paired associate memory, attention, and visual-and-working memory scores on WMS-R FD in right precuneus cortex, right pericalcarine cortex, and cuneus cortex ↔ attention FD in left lateral occipital cortex ↔ visual and working memory [32]
	Incorporating Hilbert Domain Edge FD with spectral energy signature into a diagnostic algorithm yields a sensitivity and specificity of 100% and CCR of 96% for AD [36]	
	↓ 3DFD of the GM [27]	3DFD of the GM ↔ age [27]
	↓ FD with a predominance noted in the medial temporal lobes and bilateral parietal lobes (of 1 subject) [29]	
	↓ global FD and local WM FD; ↓ SHFD global values, thickness, and local gyrification index (LGI) average scores [59]	Global FD and WM FD ↔ conversion from MCI to AD within 4 years follow-up period [59] Local WM FD ↔ positively correlated with MMSE score [59]
	↓ Regional FD (insula, posterior cingulate, pre- and post-central gyri, bilateral parahippocampal gyri, and right post-central gyrus) [53]	FD of bilateral insula, left superior temporal cortex, and isthmus cingulate ↔ positively correlated with memory subscore on ACE-R [53] FD of left insula, inferior temporal cortex, and medial orbitofrontal gyri ↔ positively correlated with ACE-R memory, language, and fluency subscores [53]
	DMA method: ↑ FD of whole brainstem in AD relative to control and MCI subjects (no statistical analyses shown) [54] Higuchi's method: FD of whole brainstem similar across groups [54]	DMA > Higuchi's method in categorizing EMCI stage [54]

	↑ FD in the right temporal and in the left temporo-parietal lobes ↓ FD in the left temporal cortices in participants with AD with high cognitive reserve compared to low cognitive reserve [65]	FD of the left middle temporal gyrus ↔ positively associated with aMCI-High Cognitive Reserve group Digit span backward scores [65] FD of the left middle temporal gyrus ↔ positively associated with aMCI-Low Cognitive Reserve group 15-Word List scores [65] FD of the white matter ↔ ALS functional rating scale [57]
Amnestic lateral sclerosis (ALS)	↓ FD of the white matter in patients with different clinical signs; greatest reductions in ALS patients with frontotemporal dementia [57] ↓ FD GM ↓ regional FD (left precentral gyrus and central sulcus, left circular sulcus of insula (superior segment), left cingulate gyrus and sulcus (middle-posterior part), right precentral gyrus, and right postcentral gyrus) [6]	FD values in the right precentral gyrus ↔ positively correlated with ALSFRS-R scores [6] FD values in the right precentral gyrus, the left circular sulcus of the insula (superior segment) ↔ negatively correlated with disease duration [6] FD of right hemisphere GM skeleton ↔ FrSBe score in bvFTD subjects [66]
Frontotemporal dementia	↓ FD of GM bvFTD [66]	
Epilepsy	↓ FD of the external cortical surface in all lobar regions except the right frontal regions in patients with right mesial temporal lobe epilepsy; ↓ FD of the external cortical surface in the bilateral temporal occipital, and left parietal regions in patients with right mesial temporal lobe epilepsy [40] ↓ FD of the 2D WM contour in more than half of patients with frontal lobe epilepsy [7] Abnormal FDs of WM surface in half of the patients with cryptogenic epilepsy and a gyral abnormality on MR images and 9/23 patients with epilepsy and normal MR images [17]	
Multiple sclerosis (MS)	↓ FD of the white matter border, skeleton in all MS and subtypes [12] ↑ FD of the GM at the early to intermediate phases of disease [13]	No associations between FDs of the white matter and gray matter and MS-related clinical disability scales (expanded disability status scale and MS functional composite score) [12, 13] FDs of the white matter skeletons and GM ↔ T1 and T2 WM lesion load [12, 13]

(continued)

Table 17.2 (continued)

Pathology	Fractal dimension (FD)	Associations
	↓ FD (no statistical comparison results shown) [73] ↑ FD of GM ↑ FD in GM in RRMS relative to CIS [27]	
Multiple system atrophy of the cerebellar type	↓ FD of 3D cerebellar WM, cerebellar GM [82]	FD ↔ volume [82]
Spinocerebellar ataxia type 3 (SCA-3)	↓ regional FD of cerebral lobes, subcortex, and cerebellum (difference most marked in cerebellum) ↓ FD of cerebrum and cerebellum in SCA-3 subjects with DD > 8 years 3D-FD values of the bilateral thalamus not significantly between MS and control subjects [78]	FD of cerebellum negatively correlated with SARA score ↓ 3DFD ↔ disease duration [78]
	↓ regional FD (all cerebral lobes and cerebellum) [79] ↓ FD of the cerebellum [21]	Cerebellar cortex 3D FD negatively correlated with SARA score [79] 3D FD ↔ cortical structure [21]
	FD correlation maps constructed to model a structural network; network properties indicated loss of small world topology, regional reductions in nodal betweenness and greater network vulnerability [83]	
	FD correlation matrix constructed to model a structural network; ↓ intramodular connectivity in all lobes, ↑ intermodular connectivity in frontal and parietal lobes, and fewer links between cerebellum and cerebrum; ↓ FD lateralized in the left supratentorial regions and lower modularity values [24]	
Spinocerebellar ataxia type 2 (SCA-2)	↓ FD of the cerebellar and cerebral GM; ↓ FD of the cerebellar WM [44]	

Stroke	↓ FD of the white matter in the stroke-affected hemisphere [87]	FD of the residual white matter ↔ upper extremity motor function (Wolf motor function test, Fugl-Meyer motor assessment score) [87]
Huntington's disease	↓ FD in the areas of motor and visual cortex In pre-HD ↑ FD in the temporal cortex areas in pre-HD [34] Regional cortical FD not significantly different [35]	Left supramarginal gyrus and right precuneus FD ↔ putamen volume but did not predict UHDRS motor score Motor region FD not a predictor of putamen volume [34]
Parkinson's disease	↓ Cortical FD in the left pre- and postcentral cortex, left superior frontal cortex, left caudal middle frontal cortex, bilaterally superior parietal cortex, and the right superior temporal cortex [39]	Left postcentral cortex FD ↔ negatively correlated with duration of illness [39]

↓ decreased, ↑ increased, ↔ is associated with
Abbreviations: 2D 2-dimensional, 3D 3-dimensional, ACE-R revised Addenbrooke's Cognitive Examination, AD Alzheimer's disease, ALS amyotrophic lateral sclerosis, ALSFRS-R ALS Functional Rating Scale Revised, aMCI amnesic mild cognitive impairment, CJS clinically isolated syndrome, EMCI early mild cognitive impairment, FrSBe Frontal Systems Behavior Scale, FTD frontotemporal dementia, bvFTD Behavioral Variant FTD, GM grey matter, HD Huntington's disease, MMSE Mini-Mental State Examination, MS multiple sclerosis, MSA-C multiple systems atrophy—cerebellar type, PPA primary progressive aphasia, RRMS relapsing-remitting MS, SARA Scale for the Assessment and Rating of Ataxia, SCA-3 Spinocerebellar Ataxia Type 3, SCA-2 Spinocerebellar Ataxia Type 2, PD Parkinson's disease, UHDRS Unified HD Rating Scale, WM white matter, WMS-R Wechsler Memory Scale Revised

the clinical scores and the combination of the FD measurements of the GM and WM to assess the usefulness of FD in MS. The FD of the left supramarginal gyrus and right precuneus did not predict UHDRS motor score in pre-HD [34]. This may be due to the fact that UHDRS was not developed for clinical investigation of pre-HD and the participants were not clinically unwell, so the lack of association between neurodevelopmental measures with motor ratings was not surprising to the investigators. Cortical complexity (FD) did not contribute significantly to the prediction of BIS attention score in PD [35]. The fractal studies in epilepsy [7, 17, 40] and MSA of the cerebellar type [82] did not address the associations between the FD and clinical scores, which are needed in the future to investigate the potential applications of FD analysis in these clinical settings.

To the best of our knowledge, limited studies have systematically examined the pathophysiological mechanism of the FD changes with age in neurodegenerative disease. There is a significant decrease in 3D FD of the WM in both AD and normal aging. Patients younger than 60 years old with AD showed 3D FD values significantly different from healthy controls, while patients older than 60 years with AD only showed significant differences from healthy controls with a chosen significance cut-off [27]. A longitudinal study between patients with SCA2 and age-matched healthy controls (3.6 and 3.3 years between scans, respectively) did not find differences in the rate of change in FD between groups [44]. However, the low sensitivity of the FD analysis in this study was suggested to be reflective of the small sample size.

It is hypothesized that FD reductions in the brain WM could be due to axon loss, increased water content, decreased myelin content, and other inflammatory events that may lead to a more amorphous tissue. The decrease of the WM border can be due to both juxtacortical WM lesions and GM abnormalities [12]. The FD increase may be due to inflammatory component (i.e., microglial activation) and cellular changes (synapse pruning, demyelination, blood–brain barrier changes, etc.) [13].

Shape analysis has been suggested to provide new information that is not accessible by volumetric measurements [18, 35]. Thus, the FD can be considered as an independent morphometric measure, although some studies showed that the FD was associated with volumetric measurements [12, 13, 34, 59, 82]. Moreover, research has shown that the FD is a more sensitive morphometric measure than volumetric measures to detect subtle structural changes in the brain WM [57, 82, 86]. Besides FD analysis, different approaches were used for quantifying brain topological changes in aging and neurodegenerative disorders, such as magnetization transfer ratio (MTR) [16, 64] or diffusion tensor imaging (DTI) [19]. These techniques may measure different morphological features due to the different physical principles of each technique. No FD studies to date have examined the relationship between the FD and MTR, DTI, or magnetic resonance spectroscopy, which may be necessary in the future to identify the biological usefulness of fractal analysis in aging and neurological disorders. The incorporation of spectral energy signature with the Hilbert Domain Edge FD into a diagnostic algorithm yielded sensitivity and specificity of 100% and CCR of 96% for AD [36]. The FD measures of 2D and 3D shape features have been investigated in aging and diseases (Table 17.1). Although

3D measures are generally superior to 2D metrics, sometimes it is useful to evaluate local changes using 2D features [30, 87]. Studies may be needed in the future to extract local 3D shape features to assess local shape complexity changes.

In conclusion, several studies have demonstrated the potential of FD as a clinical measure for assessing brain shape changes in aging and neurodegenerative diseases. However, in order for the FD technique to be more widely adopted in clinical settings and to aid clinicians in making more objective and accurate diagnoses of brain disease and injury, further studies are needed to investigate the associations between the FD measure and pathophysiological mechanism. Additionally, it is important to identify local shape features to assess changes in specific regions to improve the sensitivity and accuracy of the measurement.

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Chapter 18

Fractal Analysis in Neurodegenerative Diseases



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and Samir Kumar-Singh

Abstract Neurodegenerative diseases are defined by progressive nervous system dysfunction and death of neurons. The abnormal conformation and assembly of proteins is suggested to be the most probable cause for many of these neurodegenerative disorders, leading to the accumulation of abnormally aggregated proteins, for example, amyloid β ($A\beta$) (Alzheimer's disease and vascular dementia), tau protein (Alzheimer's disease and frontotemporal lobar degeneration), α -synuclein (Parkinson's disease and Lewy body dementia), polyglutamine expansion diseases (Huntington disease), or prion proteins (Creutzfeldt–Jakob disease). An aberrant gain-of-function mechanism toward excessive intraparenchymal accumulation thus represents a common pathogenic denominator in all these proteinopathies. Moreover, depending upon the predominant brain area involvement, these different neurodegenerative diseases lead to either movement disorders or dementia syndromes, although the underlying mechanism(s) can sometimes be very similar, and on other occasions, clinically similar syndromes can have quite distinct pathologies. Non-Euclidean image analysis approaches such as fractal dimension (FD) analysis have been applied extensively in quantifying highly variable morphopathological patterns, as well as many other connected biological processes; however, their application to understand and link abnormal proteinaceous depositions to other clinical and pathological features composing these syndromes is yet to be clarified. Thus, this short review aims to present the most important applications of FD in investigating the clinical-pathological spectrum of neurodegenerative diseases.

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Keywords Fractal dimension · Amyloid plaques · Cortical atrophy · Cerebral blood flow · Electroencephalogram · Gate analysis · Demyelination

18.1 Alzheimer's Disease and Vascular Dementia

Alzheimer's disease (AD) is the most frequent form of neurodegenerative brain disease and leads to a progressive loss of memory and cognitive abilities. It is responsible for cognitive decline in up to three-quarters of all patients with dementia [45]. Deposition of amyloid β ($A\beta$) plaques and tau neurofibrillary tangles (NFTs) in affected brain regions are the two pathological hallmarks of AD [2, 93].

Amyloid plaques are extracellular insoluble protein aggregates composed of $A\beta$, a 40–43 amino acid peptide that results from abnormal processing of the amyloid precursor protein (APP) (Fig. 18.1). APP is a single transmembrane protein and due to alternative splicing is expressed as 695, 751, and 770 amino acid long isoforms with APP695 being most abundantly expressed in neurons [44]. Constitutively, APP is processed by a series of proteases called α and γ secretases that cut the protein to generate extracellular soluble fragment (soluble-APP α) and an APP intracellular domain (AICD) fragment that bear multiple signaling roles, as well as N-truncated $A\beta$ fragment called p3 [16, 85]. In the $A\beta$ secretion pathway, however, APP is cleaved by β and γ secretases, leading to the secretion of the APP ectodomain (soluble-APP β), AICD, and $A\beta$ fragments of 40 ($A\beta$ 40) or 42 ($A\beta$ 42) amino acids [85]. It is speculated that any remaining uncleaved APP holoprotein at the cell surface can be re-internalized via clathrin-coated pits and subsequently recycled through the endosomal system [72]. Some of those proteins will be recycled to the cell surface, while others will enter the endosomes and lysosomes and will be degraded. $A\beta$ generated within the cells of the brain parenchyma (mostly, but not exclusively, neurons) is the probable source for the most $A\beta$ in extracellular amyloid plaques [11]. A complete image of the cellular $A\beta$ metabolism is still elusive; however, based on a number of in vitro experiments, it is believed that during the endosomal recycling, $A\beta$ 40 can be generated and released to the surface of the cells. Endoplasmic reticulum (ER) shows considerable amount of $A\beta$ 42, whereas Golgi vesicles contain both $A\beta$ 40 and $A\beta$ 42 [85]. Mutations in the gene encoding for APP and the core γ -secretase component presenilins (*PSEN1* and 2) constitute an important cause of familial forms of AD, and the subsequent resulting phenotypes manifested as pure AD or associated with a predominant vascular pathology [32, 34, 95].

As already stated, amyloid depositions are basically extracellular aggregates, extremely complex in their composition, morphology, and localization. They can be present almost anywhere in the brain parenchyma, ranging from a diffuse to a more dense texture, as well as in the walls of blood vessels, a feature known as cerebral amyloid angiopathy (CAA) [50, 92, 97]. Although numerous studies describe the overall diversity of all these depositions and their relation with the clinical evolution, there is also increasing evidence toward a close physiopathological connection

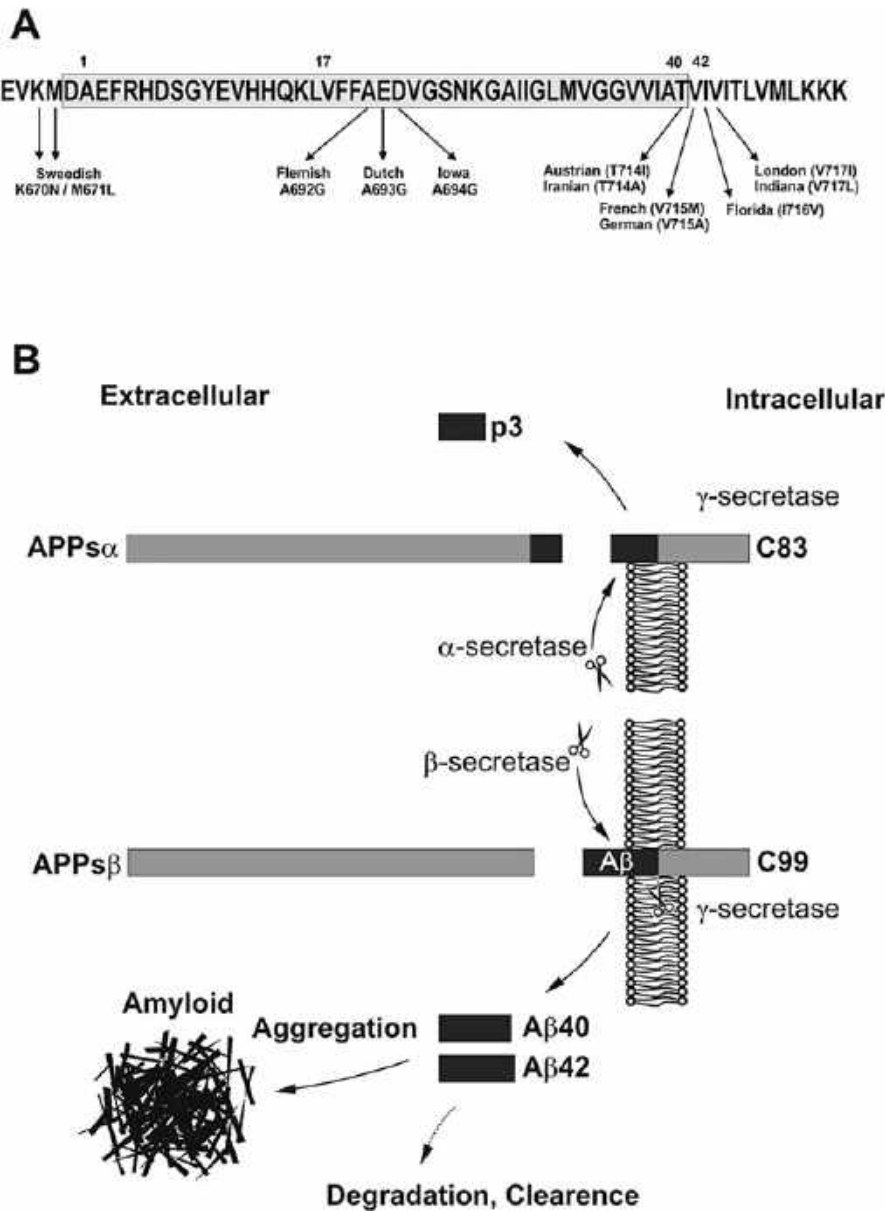


Fig. 18.1 Schematic diagram of the amyloid precursor protein (APP) with pathogenic mutations and metabolic processing. (a) The aminoacid sequence of A β fragment exhibits common pathogenic aminoacid substitutions. Sites of secretase cleavage points are indicated by arrows (position 1, 17 and 40–42 for β -, α -, and γ -secretases). (b) The constitutive proteolytic cleavage by α - and γ -secretases leads to the formation of the short p3 peptide, and the alternative pathway leads to the formation of A β peptide; in both cases with a consecutive release of a C-terminal APP intracytoplasmic domain (C83 and C99) and APPs soluble fractions. A β 42 and A β 40 diffuse in the parenchyma and precipitate as fibrillar aggregates or “plaques”

between CAA, dense types of depositions, and the brain vasculature [52, 54]. Many lines of transgenic mice overexpressing mutant forms of APP and PSEN have been developed in an attempt to mimic the amyloid pathology observed in the brains of human AD patients [12, 29].

In both human pathology and mouse models, it has been shown that this diversity of plaques exhibits different affinities for fibril-binding dyes such as Congo red or Thioflavin-S (ThS), the more positive deposits being thus considered more “compact” or “dense,” compared to the “diffuse” protein deposits negative for such fibril-binding dyes [50, 92, 97]. Moreover, relative proportions of A β 40 and A β 42 have also been found to differ between these types of plaques, for instance, diffuse deposits are shown to be composed predominantly of A β 42, while the compact plaques contain both A β 42 and A β 40 [21, 40]. Diffuse plaques are also considered to mature into compact plaques by co-depositing A β 40; however, recent data suggest that diffuse and dense plaques might in fact develop in distinct pathways [50, 92].

Subjective morphological classifiers such as “dense,” “diffuse,” or “reticular,” as well as objective Euclidean morphometric features like areas, diameters, or densities, have had limited utility in describing and classifying the highly variable amyloid patterns [50, 92, 97]. Complex and nonregular objects can be complementarily described by utilizing more objective, scale-invariant parameters, which are generally studied under the concept of fractal dimension (FD). FD of an object is a measure of the morphological complexity of that object, the inner self-similarity as measured at different scales, or simply put, the efficiency with which the object is filling the space it occupies [56]. As already described before, this concept is now widely used in pathology to describe many highly irregular normal and pathologically occurring patterns and processes like changes in cerebral blood flow, tumor angiogenesis, and chromatin distribution in malignant cells [13, 55, 63]. Relatively recently, FD analysis has been introduced in studying AD for both pathology and brain imaging studies.

18.1.1 Fractal Dimension—A Classifier for the AD Pathology

FD analysis has been utilized as a morphological classifier to study amyloid plaques in human, monkey, camel, dog, and felines, and although such an initial study has been based on a low number of analyzed amyloid deposits, it showed that diffuse plaques had significantly lower FD values compared to mature (dense or dense-core) plaques even when considering this across different species, a fact that showed support to the upcoming idea that these two types of deposits might form in different ways [60, 66]. Indeed, while diffuse plaques seem most probably to originate in the neuropil following an imbalance between the diffusion of A β oligomers and fibrils with different precipitation properties (mainly dictated by the A β 40/A β 42 ratios) (reviewed in [50, 51]) and the catabolic properties of a plethora of enzymes and cellular key players, for dense plaques, there have been postulated theories

that place them both as the developmental successors of diffuse deposits and as separate entities occurring at sites of perivascular amyloid drainage failure along the Virchow-Robin spaces [17, 54, 92, 100].

A fractal analysis study assessing the three most prominent types of plaques in human AD (diffuse, compact without a core, and dense-core plaques), based on the box-counting algorithm applied on binarized images grabbed on slides after A β immunohistochemistry, showed important differences in the FD of these types of plaques [70]. This study involved studying more than 6000 binarized images of amyloid plaques stained with A β 40, A β 42, and total A β immunohistochemistry (IHC) (Fig. 18.2) [70]. Presuming that the complexity of plaques increases over time, the diffuse type of depositions seems to be the earliest plaques and parallel studies showing that diffuse type of depositions is the first type of plaques to appear in AD [92]. However, the different degrees of complexity also sustain an alternative hypothesis that the diffuse and compact plaques have different patterns of formation and evolution [66]. The evolutionary link between the coreless compact and dense-core plaques is even more elusive. It is very much possible that the central halo observed in IHC for amyloid core is a method dependent artifact, where anti-A β antibodies are unable to bind to the most compact amyloid and remains unstained central region. On the other hand, knowing the close pathological association of the dense-core plaques with the vessels, it could also be that all three types of plaques follow a distinct pattern of evolution.

Assessing the FD of A β 40 component, the more soluble A β fraction responsible for the compactness of the plaques, could, first of all, differentiate between depositions from AD patients and sporadic plaques found in aged individuals (Fig. 18.3). Moreover, A β 40 FD values could also differentiate between the phenotypes induced by mutations that lead to different A β 40/A β 42 ratios and that for both human pathology and mouse models bearing these mutations [36, 53, 70]. In this line, a similar pattern of mostly dense-core plaques had been documented for patients bearing the APP Flemish (A692G) mutation and Tg2576 mouse line bearing the APP Swedish mutation (APPK670N/M671) [52, 54]. Although the sizes of the plaques themselves differ grossly, they both showed a similar A β 40 FD profile, with data above the 95th percentile revealing almost identical values [70].

It has been suggested that increasing the relative levels of secreted A β 42 drives increasing burdens of parenchymal diffuse plaques [35, 54]. The A β 42 fraction is less soluble compared to A β 40, and this translates into less complex plaques' architecture (lower values for A β 42 FD) for PSEN1 mutation carriers and the corresponding mouse models [70], and other studies of amyloid deposition in human and animal aging conditions also revealed higher FD values for A β 40/dense plaques compared to A β 42/diffuse depositions [60, 66]. The higher diffusibility of A β 40 and thus its longer availability to form more complex patterns are again in perfect agreement with the observed FD differences.

Compact plaques could be readily distinguished from diffuse plaques for all A β species and for both human patients and mouse models utilized [70]. Although it can be conceived that dense-core plaques represent advanced stages of development

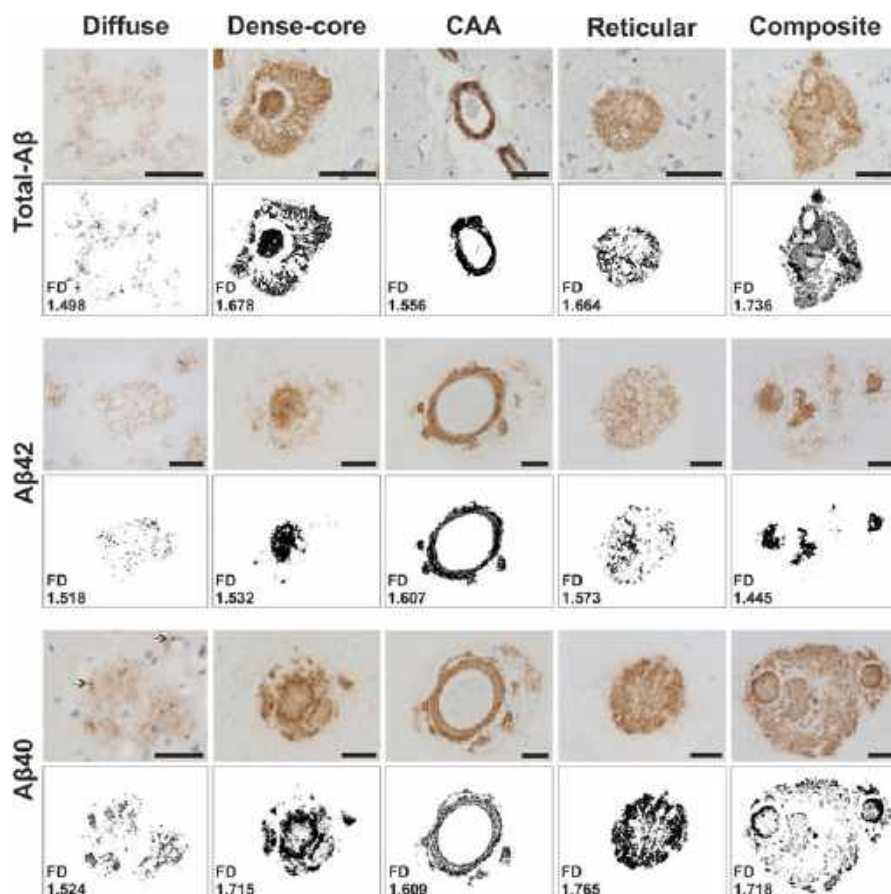


Fig. 18.2 Fractal analysis of the most prominent types of amyloid plaques in AD patients. Original immunohistochemistry images reflect representative types of plaques as detected by anti-total A β , A β 42 and A β 40 antibodies, and the binary images below represent corresponding segmented data for FD analysis. Respective FD values are illustrated in each binarised image; scale bar represents 25 μ m. (Reprinted from Pirici D. et al. [70], with permission of Elsevier)

and compaction starting from diffuse plaques, or post-production modulation, as microglia associate only with dense plaques and can phagocytize A β [18], it still remains to be shown that higher FD values truly reflect increased histopathological modeling of the plaque.

Regarding the perivascular amyloid drainage failure hypothesis, a complex “glymphatic” pathway has been described to drive the parenchymal interstitial fluid to flow out of the CNS [38]. Thus, the CSF flows following arterial pulsations in spaces between the basement membranes of the vascular smooth muscle cells and pia mater and is transported across the perivascular astrocytic end-feet to enter the interstitial fluid of the perivascular space, also known as Virchow-Robin space.

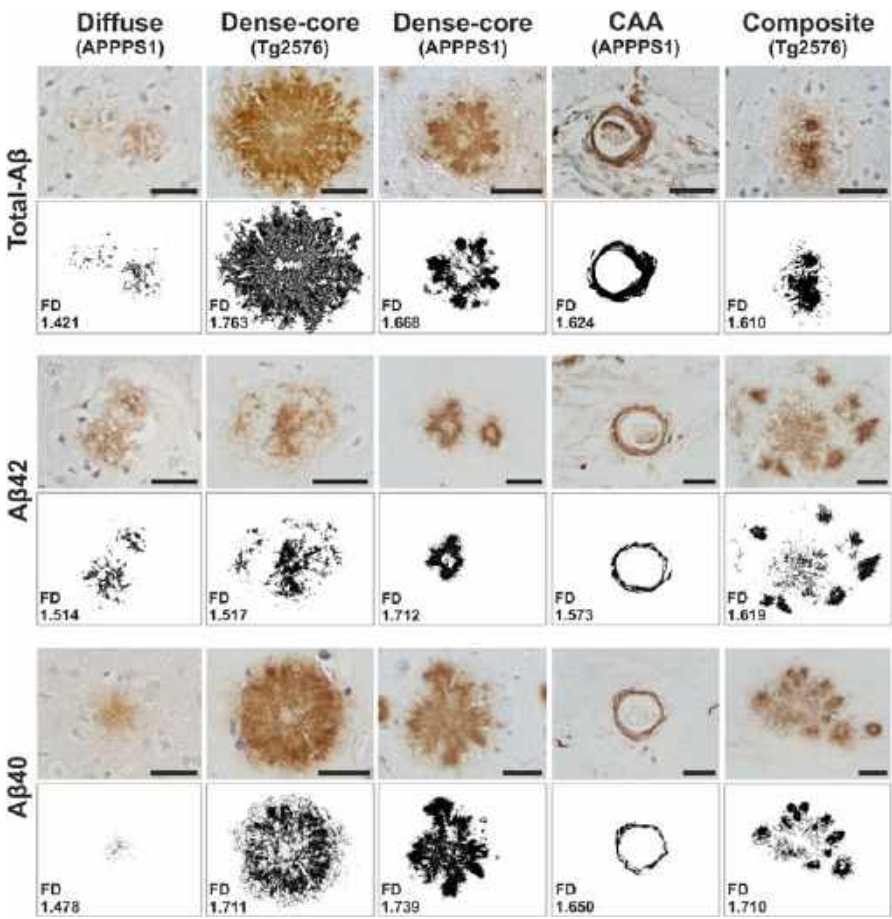


Fig. 18.3 Fractal analysis of the most prominent types of amyloid plaques in mouse models of amyloidosis. Original immunohistochemistry images reflect representative types of plaques as detected by anti-total Aβ, Aβ42 and Aβ40 antibodies, and the binary images below represent corresponding segmented data for FD analysis. Respective FD values are illustrated in each binarised image; scale bar represents 25 μm. (Reprinted from Pirici D. et al. [70], with permission of Elsevier)

Perivascular interstitial fluid then drains back into the CSF, either along the direct perivascular space or along the vascular basement membranes between the smooth muscle cells and the endothelial cells, but in both cases, it must cross through the astrocytic end-feet barrier [62]. In the astrocyte end-feet, aquaporin AQP4 is the main water channel that drives the bidirectional fluid flux around the blood vessels [74]. Based on this hypothesis, a two-photon microscopy *in vivo* study showed that treating wildtype C57BL/6 mice with the TGN-020 AQP4 inhibitor leads to a delay of soluble fluorescent Aβ40 clearance around blood vessels resulting in perivascular accumulation of Aβ40 [79]. Moreover, a bidirectional modulation of the astrocyte

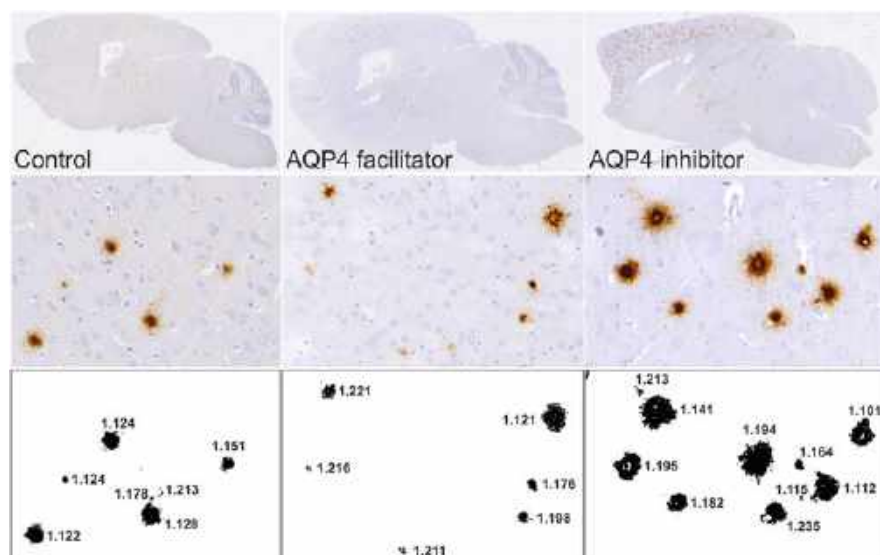


Fig. 18.4 Aquaporin 4 facilitation decreases the total amyloid load in PSAPP mice and the plaques' FD, while its inhibition increases amyloid burden and the overall complexity of the plaques. Immunohistochemistry for total A β is showed on whole brains in the first row of images, and on selected exemplificative areas below. The last row illustrates corresponding binarized images with equivalent FD values for each deposit. (Pirici et al., unpublished data)

water channels with the TGN-020 AQP4 inhibitor and the TG-73 facilitator on double transgenic PSAPP mice (APP Swedish KM670/671NL mutation and PS1 L166P mutation) showed that the inhibitor increases total A β burden as well as the fractal dimension of the deposits, while the facilitator decreases the total amyloid load and simplifies the morphology of the plaques, i.e., lower FD values (Fig. 18.4). Altogether, the AQP4 facilitator shifts plaque morphology from dense to more diffuse type of deposits, and with further characterization underway, AQP4 modulation pathway might be a valuable avenue for preventing A β accumulation in the aging brain. Neurofibrillary tangles, the other pathological denominator of AD and other neurodegenerative diseases, have not been evaluated yet from a non-Euclidian morphological point of view, but since cognitive deficit is more strongly linked to NFT rather than amyloid plaques [9], this might be of interest in the future.

FD algorithms have also been used as a measure of complexity for dendritic neuronal arborization or glial cell meshwork [5, 59, 77, 82, 88]. The most abundant cellular component of the central nervous system (CNS), astrocytes, reacts to chronic/acute brain injury, becomes activated, and undergoes a series of morphological and functional changes. Acutely reactive astrocytes become swollen, eosinophilic, with blunt cytoplasmic processes, and with enlarged and eccentrically located nuclei. In time, these acutely reactive astrocytes exhibit long cytoplasmatic extensions that finally intercalate as a dense network of cytoplasmic processes,

a state called astrogliosis. The abundance of specific cytoskeleton protein, glial fibrillary acidic protein (GFAP), makes them highly distinguishable by immuno-histochemistry. Although their morphology evidently depends on the subsequent pathology and evolutionary stages, a study analyzing fibrous, protoplasmatic, and activated astrocytes pooled together from mild dementia and from patients with recent as well as lacunar/chronic ischemia revealed that these cell types share common features among these pathologies [69]. Thus, after calculating individual FD values for binarized silhouettes and outline masks, protoplasmatic astrocytes had the smallest FD values (1.43 ± 0.04 for silhouettes/ 1.29 ± 0.03 for outline masks) for both stroke and dementia patients, which was significantly different from both fibrous ($1.46 \pm 0.06/1.34 \pm 0.02$) and activated astrocytes ($1.50 \pm 0.05/1.35 \pm 0.03$) (Fig. 18.5) [69]. Although there was enough variability among astrocytes pooled from frontal, temporal, and striatal areas that did not allow any clear-cut differentiation between these areas for the same pathological subgroup, sporadic AD cases showed the smallest calculated FD values between all pathological instances only in the temporal lobe areas [69]. Silhouette binarizations were, as expected, more

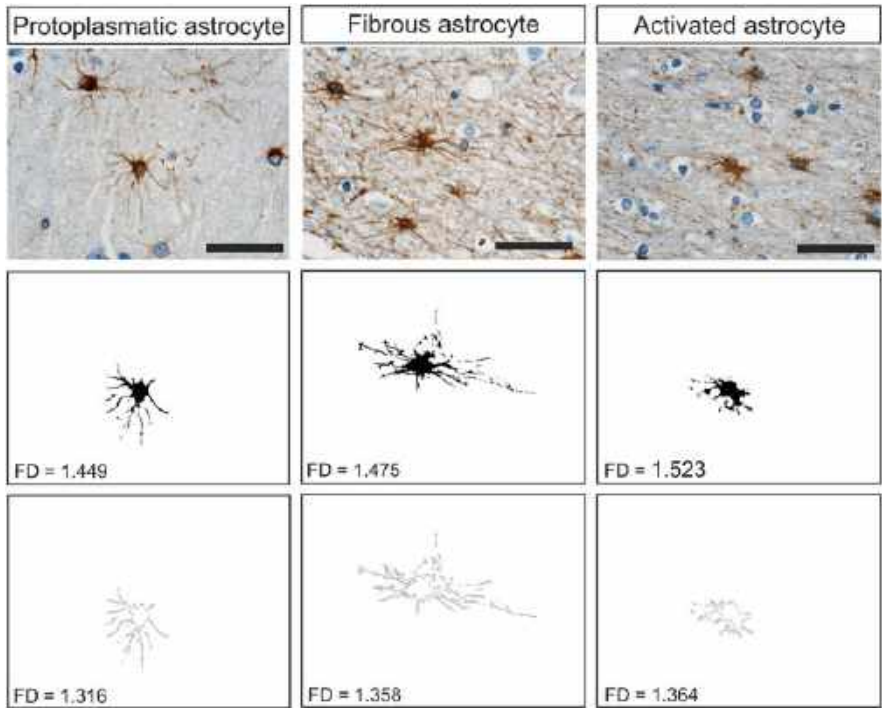


Fig. 18.5 Fractal analysis of three different types of astrocytes. The first row shows the original immunohistochemistry images, the second row shows the binary silhouette of the cells, and the last row shows the outline masks, each binarized image included the corresponding FD values. (Reprinted from Pirici D. et al. [69], with permission of the Romanian Society of Morphology)

informative than outline profiles, probably due to the loss of lacunarity data for the second type of image processing.

Astrocytic FD analysis could also combine functional data, as it has been recently shown that, for example, water and glutamate buffering is done mostly by the same astrocytes showing a high degree of colocalization for Aquaporin 4 (AQP4) and Glutamate transporter 1 (GLT-1), the two main water and glutamate gates in the CNS [61]. Plausibly, FD analysis of astrocytes based on the immunohistochemistry expression patterns of these markers will shed more light on the behavior of different astrocyte classes and their interplay with the neuropil and the blood vessels.

18.1.2 Imaging and Fractal Analysis in AD

Quantitative brain MRI studies have greatly contributed to the knowledge of in vivo anatomical changes related to aging, and volumetric analysis can now reliably differentiate patients with AD from controls [15, 20, 22, 49, 67]. Moreover, the complexity and folding of the cerebral cortex, as well as the complicated organization of the white matter fibers, have demonstrated that macroscopically the human brain exhibits fractal properties [23, 39, 48, 102]. In AD, loss of neurons and cerebral atrophy with widening of sulci and thinning and decreased folding of the cortical ribbon lead to a decreased complexity of the affected cortices [3].

Segmentation of two-dimensional cortical ribbons from control subjects and patients with mild to moderate Alzheimer's disease, followed by computation of the FD on the full thickness of the cortices using the box-counting algorithm revealed lower values for AD patients compared to age-matched controls (Fig. 18.6) [46]. This finding was evident especially in the medial temporal lobe, superior and inferior colliculus, mammillary bodies, corpus callosum, and thalamus [46]. Although assessing the complete thickness of the ribbons was superior as discriminating value to the analysis of only the outline of the pial surface or of the grey/white matter junction, it was limited to the bidimensional sectioning of the cortices, not being able to take into account the entire cortical surface [39, 42, 46]. A more recent imaging study from the same group utilized a three-dimensional approach to analyze the ruggedness of the cortical ribbon layer based on the data of 70 patients (35 control individuals and 35 patients with mild AD) imaged with a high-contrast magnetic resonance imaging technique (three-dimensional magnetization prepared rapid acquisition 3D inversion recovery; or MP-RAGE) [47]. The pial surface, the grey/white matter boundary, and the total thickness of the cortex were segmented and analyzed based on a custom 3-dimensional cube-counting fractal algorithm. The fractal dimensions of the total cortical ribbons were significantly higher for control compared to AD subjects, the grey/white matter boundary fractal dimension showed smaller but significant differences between the two pathologies, while the analysis of the pial surface revealed no significant difference between the two groups [47]. Only the total cortical thickness FD showed a significant correlation with cortical thickness itself and with the cognitive decline.

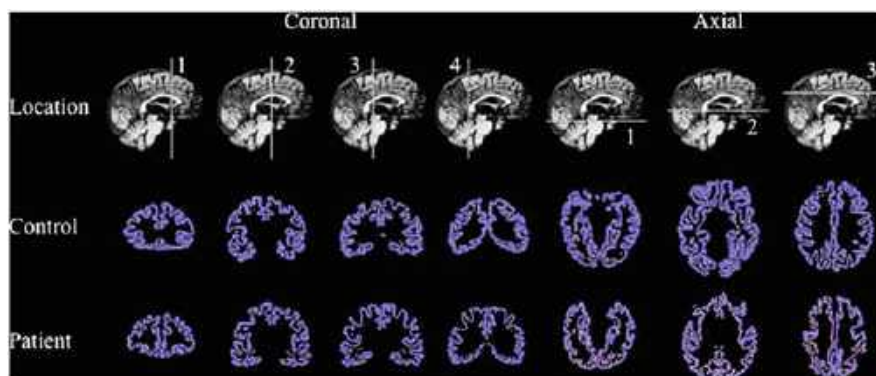


Fig. 18.6 Example of two-dimensional segmentation for fractal analysis of cortical ribbons from a control case and a patient with Alzheimer's disease. The upper row of images identifies the sectioning planes for seven different anatomical regions. The middle and lower rows display the cortical ribbons segmented from the brains of a control individual and from Alzheimer's disease patient brain, respectively, and extracted for analysis. Non-cortical areas (cerebellum, white matter, basal nuclei, and brain stem) were not considered for this analysis. The narrowing of the cortical ribbon and widened sulci are evidently leading to a less complex morphology and lower FD values for the FD patient. (Reprinted from King R.D. et al. [46], with permission of Springer)

Single photon emission CT (SPECT) of the cerebral blood flow (CBF) has been used to evaluate and compare the dynamics of cerebral circulation and brain metabolism in dementia [26, 58]. In AD and vascular dementia (VaD), blood flow is reduced in temporoparietal areas even before cerebral atrophy is noticeable, and with the progress of dementia, the metabolic deficit extends toward the anterior areas of the brain [19, 25, 27, 41, 90]. Data suggest that reduced CBF in temporoparietal areas together with the atrophy of the medial temporal lobe is a feature of AD and might be linked to the loss of entorhinal cortical neurons projecting to the dentate gyrus of the hippocampus [43].

To objectivize evaluation of tomograms, heterogeneity of CBF was assessed utilizing intensity-driven cutoff algorithm to threshold and automatically compute 3-dimensional FD on SPECT images that demonstrated that higher FD values are associated with uneven CBF [63, 65, 101]. Applied to the whole brain, this approach revealed significantly lower values for the control group compared to AD and VaD and with no differences between AD and VaD [101]. When the authors looked individually at the anterior and posterior brain regions, for the AD patients, FD values were significantly higher for posterior areas, while VaD patients had higher FD values for anterior areas [101]. Moreover, posterior CBF heterogeneity was well correlated with the cognitive impairment in AD, being able to distinguish between mild-stage AD and elderly controls [63, 65].

Aside functional MRI, the analysis of the **electroencephalogram (EEG)** is currently used as a diagnostic tool in patients with cognitive dysfunction involving either a general or a localized decline of the brain functions. In AD patients, this

results in a shift of the EEG spectrum toward lower frequencies and decreased coherence between interhemispheric signal sources [91]. Given the nonlinear nature of neuronal activity, EEG pattern analysis seemed a good candidate for FD analysis. For multiple recording locations, full spectrum EEG revealed significantly lower FD values for AD compared to age controls [6, 99]. In discriminating AD patients from age controls, limited EEG in the β -band of the eyes-closed condition revealed an accuracy of 99.3%, with a sensitivity of 100% and a specificity of 97.8%, [1]. It has also been shown that magnetoencephalography recordings on 20 AD patients were significantly less complex and more regular for 71 out of 148 recording channels, compared to similar recordings from 21 age-matched control individuals [31]. In a more recent study, resting-state eyes-closed EEG recordings from 41 healthy controls and 67 AD patients revealed that not only the signal FD values were reduced for AD patients compared to education-matched healthy controls, especially for temporal and occipital areas, but FD values also correlated with the decline in the cognitive status assessed on a mini-mental state examination [89].

18.2 Other Neurodegenerative Diseases

Frontotemporal lobar degeneration (FTLD) consists of a spectrum of clinical syndromes exhibiting progressive degenerative changes in behavior, language, personality, cognitive skills, and motor function and are pathologically characterized by selective frontal and temporal lobe atrophy. Although FDLT is not a very common cause of dementia, it is especially prevalent in the age group of 45–65 years [68]. FTLD has varied presentations and also have overlaps with other neurodegenerative diseases such as AD (i.e. frontal presentation of AD), it can take years for an accurate diagnosis and because of its lower onset of age, it has a greater impact on social life, work, and economic costs compared to AD [28]. Besides neuronal loss in the affected areas, there are five recognized FTLD pathological subtypes based on the protein profiles of a large spectrum of intraneuronal or cytoplasmic inclusions: (i) FTLD with tau-positive inclusions (tauopathies or, classically, Pick's disease); (ii) FTLD with TDP-43 positive intraneuronal inclusions; (iii) FTLD with protein aggregates consisting of the RNA-binding protein fused in sarcoma (FUS); and (iv) FTLD with no inclusions. Available data estimate that up to 43% of FTLD patients have a positive family history, and for 10.2–27% of the patients, there is an autosomal dominant pattern of inheritance [73, 76, 78, 83, 98]. Relative recent advances identified culprit mutations in the genes encoding microtubule-associated protein tau (*MAPT*) on chromosome 17q21–22 [37], progranulin (*P GRN*) on chromosome 17q21–22 [4, 14], valosin-containing protein (*VCP*) on chromosome 9p21-p12 (*VCP*) [96], TAR DNA-binding protein TDP-43 (*TARDBP*) on chromosome 1p36 [8], or charged multivesicular body protein 2B gene (*CHMP2B*) on chromosome 3p13 [87].

SPECT imaging studies assessing the heterogeneity of CBF have also been performed for FTLD patients compared to AD and control patients. In a study

involving 21 FTLN patients and 21 AD patients in the early stages of dementia, the authors utilized a derived FD function to assess technetium-99 m hexamethylpropylene amine oxime SPECT scannings [64]. For heterogeneity analysis, SPECT imaging data had a threshold with a 35% cutoff and a 50% cutoff of the maximal voxel radioactivity and then FD values were calculated as the logarithms of the threshold values and the remaining numbers of voxels for both anterior and posterior brain regions. While posterior FD values could not differentiate between FTLN and AD, anterior and anterior-to-posterior FD ratios showed significantly higher values for FTLN cases compared to AD patients [64]. Increased heterogeneity and reduced CBF in frontal, anterior cingulate, temporal, orbitofrontal, and ventrolateral prefrontal cortices had been already proved before in FTLN compared to AD, where this heterogeneity was more pronounced in parietal, superior occipital, and temporo-occipital cortices [94].

Morphological changes such as gray matter atrophy together with white matter degeneration had been documented by MRI in the frontal and anterior temporal lobes for FTLN patients. Quantitative voxel-based 3D morphometry is currently utilized to assess the cortical and subcortical changes in FTLN; however, this does not offer information on the complexity and minute architecture of these changes. A recent study segmented the gray matter from weighted MR whole-brain images and ran an FD analysis on the skeleton, surface profiles, and general architecture pattern of the cortex in an attempt to find and assess subtle differences between 39 FTLN patients and 20 control individuals [86]. The authors reported a significant reduction of the FD for the skeleton profile and for the general pattern of the cortex for FTLN patients, which also correlated with the score of the Frontal Systems Behavior Scale.

Parkinson's disease (PD) is another common type of dementia/dementia syndrome clinically presenting with a combination of clinical rigidity, bradykinesia, and resting tremor, and in later stages dementia. This neurodegenerative disease results from the death of dopamine-secreting cells in the substantia nigra, with dopamine synthesis balance being important, among others, for lowering the inhibitory effect of basal ganglia over the motor system that control voluntary movements. This results in the overall reduction of the motor output or bradykinesia [7]. As there are many causes and patterns of manifestation for movement disorders, these apparently chaotic changes have been ideally investigated through nonlinear algorithms like FD analysis. Dynamic recording while walking using a triaxial accelerometry technique, a study on 11 PD patients and 10 healthy elderly controls, found out that fractal patterns of the movements were higher in PD patients for all three orthogonal axes compared to controls [84]. Moreover, it has been iterated that even normal gait has a stochastic fractal pattern which seems to be altered in PD to a point where it can be separated by machine learning and neural network classifiers, although long-time recordings from gait signal are still necessary to verify this hypothesis [33, 57, 81]. Nevertheless, FC applied to individual human pallidal neuronal spike trains recorded from patients with PD during ablative surgery of the globus pallidus, led to the conclusion that fractal dynamics may characterize the activity of single neurons in the CNS and act as an indicator of a dysfunctional CNS network [75].

Multiple sclerosis (MS) is the most common demyelinating disease of the human central nervous system leading initially to reversible neurological deficits and in time, to permanent disability [10]. Although the etiology and pathogenesis of MS are poorly understood, with both genetic and environmental factors being incriminated, damage to the white matter is mediated by autoimmune mechanisms, including T-cell-mediated and antibody-mediated injuries [30]. On gross examination of the brain sections, areas of demyelination can be observed as sharply demarcated zones, or “plaques,” distributed predominantly in the white matter but may involve gray matter as well. On microscopic examination, plaques are composed of perivascular mononuclear cell inflammatory infiltrates, moderate gliosis, activated microglia, myelin loss, and axonal swelling [71].

A study comparing control subjects with patients with MS on classical MR imaging revealed that skeletonized white matter had lower FD values for MS patients, for both visible and normal-appearing white matter, thus being able to detect changes even at an early phase of the disease [23]. Moreover, the same group showed that the FD of the segmented gray matter is increased in these patients compared to healthy controls, and this did not correlate with gray matter atrophy or neurological disability, suggesting that these changes might appear very early in the course of the disease [24]. More recently, a study on 146 patients with relapsing-remitting MS followed up for up to 5 years with clinical and brain MRI recordings revealed a significant decrease of the FD, with an increased lacunarity of the cortex and white matter for different brain regions, and this algorithm was successfully utilized to identify patients at risk for impending relapsing phase [80].

18.3 Conclusion

The data presented in this review strongly suggests that FD analysis is a valid tool not only in differentiating and grading pathological features based on histological texture analysis of specific proteinaceous deposits in different types of neurodegenerative diseases but it can also lead to genotype–phenotype correlations as is the case with familial forms of Alzheimer’s disease. These data therefore recommend FD analysis to be potentially utilized as a diagnostic aid in neurological diseases and also as a tool for further in-depth investigation of the dynamic phenomena that leads to these distinct but overlapping syndromes.

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Chapter 19

Fractal Analysis of the Cerebrovascular System Pathophysiology



Martin Soehle

Abstract The cerebrovascular system is characterized by parameters such as arterial blood pressure (ABP), cerebral perfusion pressure (CPP), and cerebral blood flow velocity (CBFV). These are regulated by interconnected feedback loops resulting in a fluctuating and complex time course. They exhibit fractal characteristics such as (statistical) self-similarity and scale invariance which could be quantified by fractal measures. These include the coefficient of variation, the Hurst coefficient H , or the spectral exponent α in the time domain, as well as the spectral index β in the frequency domain. Prior to quantification, the time series has to be classified as either stationary or nonstationary, which determines the appropriate fractal analysis and measure for a given signal class. CBFV was characterized as a nonstationary (fractal Brownian motion) signal with spectral index β between 2.0 and 2.3. In the high-frequency range (>0.15 Hz), CBFV variability is mainly determined by the periodic ABP variability induced by heartbeat and respiration. However, most of the spectral power of CBFV is contained in the low-frequency range (<0.15 Hz), where cerebral autoregulation acts as a low-pass filter and where the fractal properties are found. Cerebral vasospasm, which is a complication of subarachnoid hemorrhage (SAH), is associated with an increase in β denoting a less complex time course. A reduced fractal dimension of the retinal microvasculature has been observed in neurodegenerative disease and in stroke. According to the decomplexification theory of illness, such a diminished complexity could be explained by a restriction or even dropout of feedback loops caused by disease.

Keywords Cerebral blood flow velocity · Hurst coefficient · Spectral exponent · Spectral index · Subarachnoid hemorrhage · Decomplexification · Fractal Brownian motion

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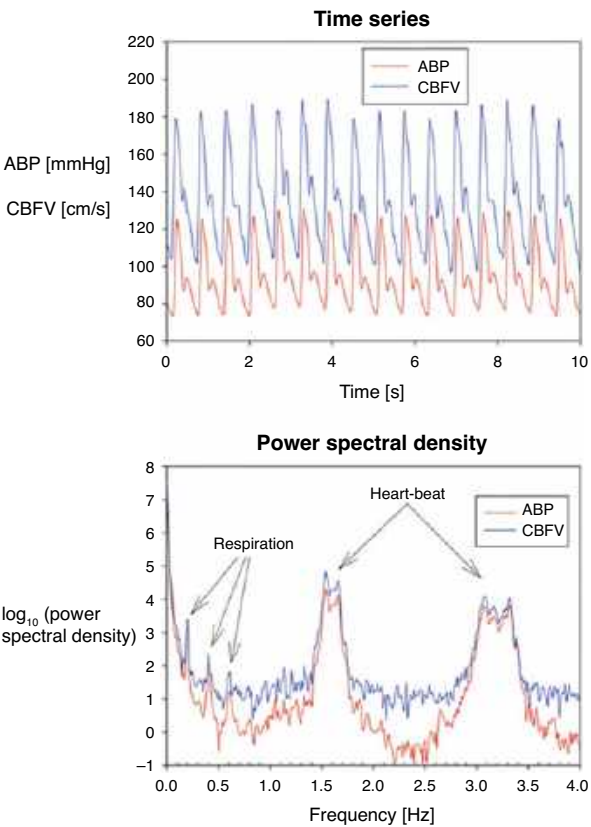
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19.1 Introduction

Physiologic parameters such as arterial blood pressure (ABP) or cerebral blood flow (CBF) are not constant over time but fluctuate (Fig. 19.1). These fluctuations are either periodic, driven by an oscillatory process, such as the heartbeat, or they are nonperiodic, for example, random or nonrandom. These nonperiodic fluctuations have been shown to contain valuable information recently [16]. A time series is regarded as fractal time series if it possesses the characteristics of self-similarity and scale invariance [12]. Fractal analysis allows us to identify and quantify these characteristics.

Fig. 19.1 *Upper part:* Time series of arterial blood pressure (ABP, shown in red) and cerebral blood flow velocity (CBFV, illustrated in blue) in a 48-year-old woman suffering from subarachnoid hemorrhage. The time course of CBFV follows that of its major determinant ABP. *Lower part:* The power spectral density of the ABP and CBFV time series shown above is presented at different frequencies. The peaks at 1.6 Hz (=96/min) and 3.2 Hz are caused by the fundamental frequency related to heartbeat (96/min) and its second harmonic, respectively. Likewise, peaks at 0.2 Hz (=12/min), 0.4 Hz, and 0.6 Hz are related to respiration (respiratory rate = 12/min) and its higher harmonics. The highest power spectral density is obtained for frequencies between 0 and 0.15 Hz, a range where fractal characteristics were observed for both ABP and FV [34]



19.2 Cerebral Autoregulation as a Feedback Loop

Cerebral perfusion pressure (CPP), which is calculated as the difference between mean arterial blood pressure (MAP) and intracranial pressure (ICP), is a major determinant of cerebral blood flow (CBF) [11]. CBF would decrease whenever CPP declines, which is prevented by the mechanism of cerebral pressure autoregulation [20]: As CPP drops, cerebral arterioles dilate until CBF is restored. In reverse, cerebral arterioles constrict when CPP increases. In terms of control theory, cerebral pressure autoregulation is a feedback loop with CPP as input and CBF as output. In addition, CBF is determined by arterial carbon dioxide partial pressure ($p_a\text{CO}_2$) and cerebral metabolism as well [17, 28], which form two additional feedback loops that control CBF. In control theory, feedback loops are mathematically described by differential equations [9] which inherently exhibit a nonlinear behavior. This explains why the output parameter CBF shows nonlinear properties as will be discussed below. Many physiologic parameters, among others ABP, body temperature, arterial pH, and electrolyte concentrations, are controlled by feedback loops as well which keep them in a physiologic range. Thus, the organism maintains its interior milieu by regulatory mechanisms, a concept known as homeostasis [7].

19.3 Variability and Complexity

Despite cerebral autoregulation, CBF varies notably: Oscillations induced by heartbeat are observed, which are seen in ABP likewise (Fig. 19.1). These periodic changes occur in a frequency range above $0.5 \text{ Hz} = 30/\text{min}$, which is too fast for autoregulation to react [10]. In the frequency range below, CBF variations of 5–10% (coefficient of variation = standard deviation/mean) are observed despite autoregulation [26, 34], which are highly irregular and complex. A loss of fluctuation might be of clinical relevance as it could be associated with increased morbidity and mortality. This has been extensively investigated for a different parameter, namely, heart rate variability (HRV), where a reduction of HRV is associated with poorer prognosis and/or increased mortality in patients suffering from coronary artery disease [32], dilated cardiomyopathy [37], congestive heart failure [30], and postinfarction patients [18]. CBF variability has been investigated in patients suffering from subarachnoid hemorrhage (SAH), which is caused by a rupture of an aneurysm of a cerebral artery. SAH is associated with high morbidity and mortality [8] not only due to the initial arterial hemorrhage but also due to the complication of cerebral vasospasm, which is characterized by a narrowing of cerebral arteries leading to cerebral ischemia [2].

19.4 Methodology of Variation and Fractal Analysis

Continuous CBF measurements are difficult in the clinical setting; therefore, CBF velocity (CBFV) is assessed instead, which is a valid replacement as long as vessel diameter does not change during the measurement period [36]. To do so, transcranial Doppler sonography (TCD) is performed which enables to record time series of CBFV with high temporal resolution and long duration [1]. Variation of a time series can be quantified by calculating its standard deviation (SD):

$$SD = \sqrt{\frac{\sum (x - m)^2}{n - 1}}$$

where x denotes the data, m the mean, and n the number of samples. As the duration of a time series increases, its SD might either remain constant or vary which is denoted as *stationary* or *nonstationary*, respectively. The relative standard deviation or coefficient of variation (CV) is calculated as $CV = SD/m$.

For fractal analysis, the time series is investigated to possess fractal characteristics, one of which is *self-similarity*: Segments of a fractal are similar to larger segments or the whole [13]. This could refer to geometrical self-similarity as known from the Mandelbrot set or to statistical self-similarity as for the case of fractal time series. Here, the segments of the time series have similar statistical properties (e.g., SD) as the entire time series. This statistical similarity is independent of the time scale chosen, a property known as *scale invariance*. Different parameters exist to quantify a fractal time series, one of which is the Hurst coefficient H .

19.5 Hurst Coefficient H_{bdSWV}

H_{bdSWV} is calculated in the time domain using the bridge-detrended scaled windowed variance (bdSWV) method: First, a given fractal time series is bridge-detrended to remove linear trends [6]. Then, the time series is divided into (nonoverlapping) windows of length τ . The SD is calculated for each window and averaged yielding $mSD(\tau)$. This computation is repeated for a wide range of window sizes τ starting with a reference window size τ_0 . Finally, $mSD(\tau)/mSD(\tau_0)$ is plotted versus τ/τ_0 on a double-logarithmic scale, and the Hurst coefficient H_{bdSWV} is estimated as the slope of the regression line [6, 23] (Fig. 19.2):

$$H_{bdSWV} = \frac{\log(mSD(\tau) / mSD(\tau_0))}{\log(\tau / \tau_0)}$$

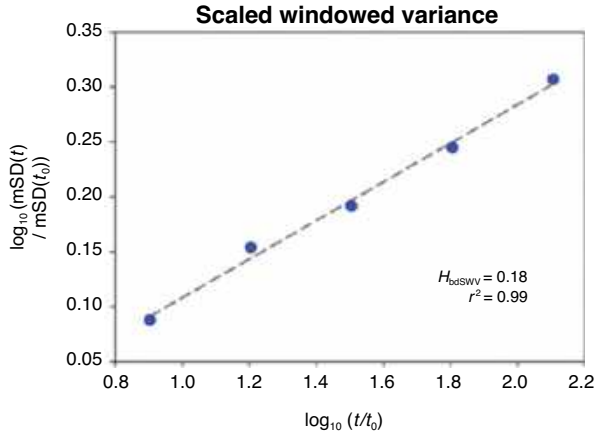


Fig. 19.2 Fractal analysis of cerebral blood flow velocity by means of bridge-detrended scaled windowed variance (bdSWV) [6]. The variance, or mean standard deviation (mSD) to be more precise, is calculated (and shown as *blue circles*) for five different signal lengths t , starting with reference length t_0 . Both the variance mSD (ordinate) and the signal length t (abscissa) are displayed on a logarithmic scale. The slope of the linear regression line (shown as *gray dashed line*) is denoted as the Hurst coefficient H_{bdSWV} . The correlation coefficient r^2 indicates the strength of the relationship between $\log$ (variance) and $\log$ (time scale)

H_{bdSWV} is applicable to nonstationary time series only [12]. Therefore, the parameter spectral index β is calculated in the frequency domain, which differentiates between stationary and nonstationary time series.

19.6 Spectral Index β

As a general principle, a Fourier analysis is performed to obtain the power spectral amplitude $|A(f)|^2$ at frequency f [13]. According to the “1/f power law,” $|A(f)|^2$ decreases exponentially as frequency f increases in fractal time series [13] (Fig. 19.3):

$$|A(f)|^2 \propto cf^{-\beta}$$

where c is a constant and β is the spectral index, which is calculated as the negative slope of the linear regression line of $\log(|A(f)|^2)$ on $\log(f)$. Fractal time series with $1 < \beta < 3$ are characterized as nonstationary fractal Brownian motion (fBm) time series, whereas stationary fractal Gaussian noise (fGn) time series are having $-1 < \beta < 1$ [12].

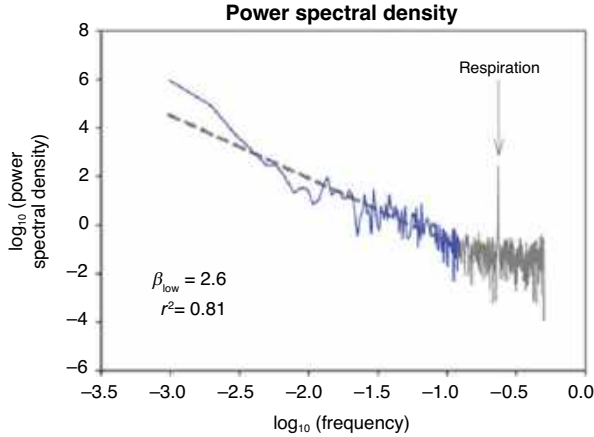


Fig. 19.3 Fractal analysis of cerebral blood flow velocity using the power spectral density method [13]. Both the power spectral density and the frequency are shown on a logarithmic scale. The peak at -0.63 ($=\log_{10}(0.23 \text{ Hz})$) is caused by respiration. Only frequencies below -0.9 ($=\log_{10}(0.125 \text{ Hz})$) are considered for further analysis (shown in *blue*): The negative slope of the linear regression line (shown as *gray dashed line*) of $\log(\text{power spectral density})$ on $\log(\text{frequency})$ is denoted as spectral index β_{low} . The strength of the relationship is displayed as correlation coefficient r^2

19.7 Spectral Exponent

The spectral exponent α can be calculated in the time domain by detrended fluctuation analysis (DFA) [29]: A given time series $y(k)$ of length N is detrended by subtracting the local trend $y_n(k)$, and its root mean square fluctuation $F(n)$ is determined as:

$$F(n) = \sqrt{\frac{1}{N} \sum_{k=1}^N [y(k) - y_n(k)]^2}$$

This computation is repeated for different time scales n , providing a relationship between $F(n)$ and n . The slope of the regression line of $\log[F(n)]$ and $\log(n)$ is denoted as the scaling exponent α , with higher α values indicating a smoother time series.

19.8 Fractal Analysis of Human CBF

So far, fractal analysis of human CBFV has been performed in a few patient studies only (see Table 19.1). These include studies in which the cerebrovascular

system is either directly altered by an intracranial disease such as subarachnoid hemorrhage [34] or indirectly affected by causes such as cardiogenic shock [4] or autonomic failure [5]. In addition, fractal parameters of CBFV have been analyzed in healthy volunteers [33, 38, 39]. All studies identified fractal properties in their CBFV time series, such as scale invariance and self-similarity. They all classified CBFV as fractal Brownian motion (fBm), which is defined as having a spectral index $1 < \beta < 3$ [12]. Hence, CBFV is nonstationary which means that its observed variance increases with the length of the observation period [12]. Eke et al. [12] have stressed the importance of classifying fractal time series according to the dichotomous fGn/fBm model, as the signal class determines the appropriate method of quantifying its fractal properties. Values between 2.0 and 2.3 were observed for the fractal parameter β both in patients [5, 34] and healthy volunteers [39] using the power spectral density (PSD) method. This PSD technique has been described as being appropriate for both fGn and fBm signals. In contrast, Rossitti and Stephensen [33] as well as West et al. [38] observed higher β -values of 2.5 and 2.7, respectively, but they applied dispersional analysis, a method which should be used for fGn time series only [12]. Much lower β -values between 0.7 and 0.9 were described by Bellapart et al. [4] in healthy volunteers and cardiogenic shock patients. They applied DFA, a method that allows signal classification as fGn/fBm either but leaves some uncertainty around α -values of 1.0. They reported an α between 0.86 and 0.97 which is in the area of uncertainty. Moreover, for DFA $2^{15} = 32,768$, samples are required to achieve a 95% precision [13]; however, 6000 samples were analyzed only. Zhang et al. [39] and Reynolds et al. [31] observed a kink in their PSD data and consequently calculated different β -values for different frequency ranges: They reported low β -values of 0.3 and 0.7 for the very low-frequency range 0.002–0.02 Hz [39] and 0.005–0.06 Hz [31], respectively. Many times higher β -values of 2.3 and 1.6 were calculated for the frequency range 0.02–0.5 Hz [39] and 0.06–0.5 Hz [31], respectively. Hence, CBFV might not only exhibit a monofractal, but probably a bifractal or even multifractal behavior.

Fractal parameters such as α , β , and H give more precise estimates, that is, less mean squared error ($MSE = \text{variance} + \text{bias}^2$), with increasing length of the time signal n . To obtain detailed information in the low-frequency range (LF, <0.15 Hz), low sampling frequencies (<1 Hz) are desired, which results in long recording times. However, in clinical practice, long recording times (>30 min) are hardly feasible in TCD, which is susceptible to probe dislocation, and will be tolerated neither by patients nor nursing staff for long periods. Therefore, using the analysis techniques with the highest precision at a given signal length is preferred. The spectral index β obtained by power spectral density analysis reliably classifies a fractal time series as fGn or fBm but gives imprecise estimates. Therefore, the Hurst exponent H is calculated as a fractal parameter, which gives two magnitude higher precision estimates than PSD [12].

Depending on the fGn or fBm signal class, H is estimated using the dispersional or the bdSWV method, respectively [12]. Failure to match the fGn/fBm signal class with the appropriate method (dispersional/bdSWV method) results in a serious error in estimating H [12]. A Hurst coefficient of 0.26 ± 0.13 was obtained for CBFV

Table 19.1 Fractal analysis studies of cerebral blood flow velocity (CBFV) obtained in healthy volunteers (upper part) or patients (lower part)

Reference	Subjects	Time series duration at $f_{\text{(samples)}}$, method	Freq. range	Spectral index β	Comments
Healthy volunteers					
Rossitti and Stephensen [33]	$n = 4$	512 s at 1 Hz		(2.52 ± 0.18)	$D = 1.24 \pm 0.09$
		($n = 512$), RD			
West et al. [38]	$n = 6$	2 h, beat-to-beat, RD		(2.70 ± 0.08)	$D = 1.15 \pm 0.04$
Zhang et al. [39]	$n = 6$	512 s at 1 Hz,	0.002–0.02 Hz	0.34 ± 0.09	
		($n = 512$), PSD	0.02–0.5 Hz	2.31 ± 0.08	
Patients					
Reynolds et al. [31]	$n = 106$	3.4 min at 5 Hz	0.005–0.06 Hz	0.670	
	Premature Neonates	($n = 1024$), PSD	0.06–0.5 Hz	1.592	
Blaber et al. [5]	$n = 2 \times 8$				
	Control vs. autonomic failure	$n = 256$ beats, CGSA, PSD		2.0 ± 0.15	60° tilt: $\beta = 1.6 \pm 0.25$
Bellapart et al. [4]				2.2 ± 0.2	60° tilt: $\beta = 1.45 \pm 0.2$ ($p < 0.05$)
	$n = 3 \times 5$	5 min at 20 Hz	0.04–0.15 Hz		
	Healthy	($n = 6000$), DFA		(0.72 ± 0.32)	$\alpha = 0.86 \pm 0.16$
	Card. Shock without VAD			(0.94 ± 0.54)	$\alpha = 0.97 \pm 0.27$
Soehle et al. [34]	Card. Shock with VAD			(0.90 ± 0.40)	$\alpha = 0.95 \pm 0.20$
	$n = 31$ SAH	17 min at 1 Hz	0.001–0.125 Hz		
		($n = 1024$), PSD			
	Before vasospasm			1.97 ± 0.35	During vasospasm: $\beta = 2.23 \pm 0.43$ ($p < 0.05$)
	Contralateral side of vasospasm			2.09 ± 0.56	Side of vasospasm: $\beta = 2.29 \pm 0.61$ ($p < 0.05$)

$f_{\text{(samples)}}$, method sampling frequency, β spectral index as determined based on the power spectral density, n number of patients or number of samples, D fractal dimension as calculated by the relative dispersion (RD) method, PSD power spectral density, CGSA coarse-graining spectral analysis, α spectral exponent as determined by detrended fluctuation analysis (DFA), VAD ventricular assist device, SAH subarachnoid hemorrhage. Values of β shown in brackets are calculated as $\beta = 5-2D$ or $\beta = 2\alpha-1$. Values = mean ± standard deviation

in SAH patients [34], which is similar to the value of 0.24 ± 0.02 obtained in the rat cerebral microcirculation [12]. In both studies, the Hurst coefficient was unequal to 0.5 which means that the variations in CBF are not random. Moreover, the H value <0.5 indicates anticorrelation within the CBF time series, that is, increases of the signal in the past influence decreases at present. Hence, a long-term memory [3, 13] was present in the CBF signal. In general, variability and fractal analysis provide complementary information: Whereas variation analysis investigates standard deviation on a single time scale only, bdSWV analyzes standard deviation over a wide range of time scales, and H quantifies the roughness of the time series on a time-invariant scale [3].

19.9 Decomplexification

Cerebral vasospasm, a complication of SAH that is characterized by a narrowing of cerebral arteries, is associated with a significant increase in the spectral index β (Table 19.1) [34]. Likewise, SAH patients exhibit a significant higher β on the side of vasospasm as compared to the contralateral non-vasospastic side [34]. Such an increase in β is equivalent to a less complex time course or decomplexification of CBFV, which could be explained by the decomplexification theory of illness [15]. Parameters such as CBF are regulated by multiple interconnected feedback loops resulting in a highly complex time course. In illnesses such as cerebral vasospasm, these feedback loops are either restricted in their range of action or even out of order which causes a less complex time course. Decomplexification has been shown in Cheyne–Stokes respiration, parkinsonian gait, neutrophil count in leukemia, and fever in Hodgkin’s disease [15] as well as during intracranial hypertension following traumatic brain injury [35].

19.10 Frequency-Dependent CBF Variability

CBFV analysis in the frequency domain exhibits peaks in power spectral density which are related to heartbeat ($\sim 80\text{--}120 \text{ min}^{-1} = 1.3\text{--}2 \text{ Hz}$) and respiration ($\sim 8\text{--}20 \text{ min}^{-1} = 0.13\text{--}0.3 \text{ Hz}$) [34] (Fig. 19.1). Transfer function analysis reveals a high coherence between ABP and CFBV in the higher frequency range (HF $> 0.15 \text{ Hz}$), which indicates that a high proportion of CBF variability is explained by likewise heartbeat and respiration-induced fluctuations in ABP [27]. One might assume that the fractal behavior of CBFV might be simply explained by the fractal properties of ABP, a theory that is refuted by cerebral autoregulation (CA). CA behaves like a high-pass filter, which freely passes fast ABP changes ($>0.15 \text{ Hz}$) but effectively attenuates low-frequency (LF, $<0.15 \text{ Hz}$) oscillations [5, 14]. In fact, it has been shown that changes in cerebrovascular resistance due to CA contribute significantly to CBF variability in the LF range [26]. Moreover, the LF range contains most of the

spectral power (or variability) of CBF [26], and hence patient studies have focused on the LF range [4, 34]. The feedback loops of CA with its inherent nonlinear behavior are active in the LF range and could explain the fractal properties of CBF. However, the currently available techniques of fractal analysis quantify fractal properties but cannot prove a causal relationship. Therefore, it has not been proven that the fractal behavior of CBF is due to the fractal behavior of CA.

19.11 Fractal Analysis of the Cerebral Microvasculature

Fractal properties are not only observed in the blood flow signal of the cerebral vasculature, but in its vessel architecture as well. Although the bony skull prevents a direct view of the cerebral vascular architecture, it is possible to photograph the brain surface during a neurosurgical operation. For example, Komatsu et al. [19] analyzed the box-counting fractal dimension D of cerebral microvessels based on a 1×1 cm area of the cortical surface. They observed a significantly higher fractal dimension in patients suffering from moyamoya disease (median $D = 1.69$ [IQR, 0.08]) as compared to patients with atherosclerotic disease ($D = 1.64$ [0.04]) or nonischemic disease (e.g., tumors, $D = 1.61$ [0.03]) of the brain. Moyamoya disease is a cerebrovascular disease characterized, among other things, by a greater meandering or divergence of cortical pial arteries, which explains the increased anatomical complexity of the cerebral microvasculature.

Another possibility is to investigate the retina, which shares anatomic similarities with and functional characteristics of the brain because of the common embryological origin. Moreover, the retina is regarded as an integral part of the brain; therefore, the retina offers a “window to the brain” which is easily observable through the transparent eye [21]. A reduced retinal FD was associated with increased age (correlation coefficient $R^2 = 0.18$) as well as with arterial hypertension [22]. McGrory et al. [24] reported a reduced FD of the retinal microvasculature to be related to small vessel disease of the brain, even though the effect size was small. In addition, a positive association between retinal vascular FD and CBF was observed by Nadal et al. [25], this association being most pronounced ($R^2 = 0.35$) between the FD of the retinal venular vessels and the CBF of the anterior brain. In a systematic review, Lemmens et al. [21] found a significant association between decreased retinal vasculature FD and various degrees and etiologies of neurodegenerative disease, including Alzheimer’s disease and vascular dementia. In addition, a decreased retinal FD seemed to be associated with stroke, although differences in FD between subtypes of ischemic stroke remain unclear [21]. These findings are more straightforward in neurodegenerative disease than in stroke, where more conflicting results have been reported. The reduced retinal vasculature FD means a loss of complexity in age and disease which supports the abovementioned decomplexification theory of illness. Nonetheless, many of the observations described above need to be confirmed by larger studies.

19.12 Conclusions

Time series of CBF (and its surrogate CBFV) as well as images of the cerebral microvasculature possess fractal properties such as scale invariance and self-similarity. Fractal analysis allows to quantify these properties and give insights into the system of cerebrovascular system pathophysiology in general and of cerebral autoregulation in particular. At a given signal length, the Hurst coefficient provides a more precise estimation than other fractal parameters; however, different methods exist to calculate it based on the signal classification as fGn or fBm. Therefore, a correct signal classification is required, which is based on the robust method of estimating the spectral index β by power spectral density analysis. Few studies have performed fractal analysis in patients with cerebrovascular system pathophysiology so far. These reported a reduced complexity in CBFV during vasospasm following SAH as well as in the retinal microvasculature in both neurodegenerative disease and stroke. Further studies are required to investigate the relationship between fractal analysis parameters and patient outcome.

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Chapter 20

Fractals and Chaos in the Hemodynamics of Intracranial Aneurysms



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Abstract Computing the emerging flow in blood vessel sections by means of computational fluid dynamics is an often applied practice in hemodynamics research. One particular area for such investigations is related to the cerebral aneurysms, since their formation, pathogenesis, and the risk of a potential rupture may be flow-related. We present a study on the behavior of small advected particles in cerebral vessel sections in the presence of aneurysmal malformations. These malformations cause strong flow disturbances driving the system toward chaotic behavior. Within these flows, the particle trajectories can form a fractal structure, the properties of which are measurable by quantitative techniques. The measurable quantities are well established chaotic properties, such as the Lyapunov exponent, escape rate, and information dimension. Based on these findings, we propose that chaotic flow within blood vessels in the vicinity of the aneurysm might be relevant for the pathogenesis and development of this malformation.

Keywords Cerebral aneurysm · Computational fluid dynamics · Chaotic advection · Passive tracers · Fractals · Information dimension

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20.1 Introduction

Intracranial aneurysms are focal dilations of branches of the circle of Willis, most frequently involving bifurcation points. Intracranial aneurysms are mostly of the *saccular* type (“berry” aneurysms) and affect about 2–5% of the population. The aneurysm rupture incidence is 6–10 per 100,000 inhabitants per year. About 45% of the ruptures may lead to death or permanent disability during the first year [12, 16, 32, 33]. Aneurysms are often incidentally found on imaging studies performed for different reasons. Neurosurgeons and neuro-interventionists often face the dilemma whether or not to treat unruptured aneurysms. Rupture can be catastrophic, but the treatment risk itself is also non-negligible [12, 33].

The pathogenesis, growth, and rupture of intracranial aneurysms are still not fully understood. It is likely a multifactorial disease, with genetic and hemodynamic factors, as well as lifestyle and coexisting diseases playing a role. Recent research demonstrates a strong impact of hemodynamics on aneurysm initiation. Local changes of hemodynamics, particularly that of wall shear stress, initiate focal vessel dilatation (vascular remodeling), which may result in unbalanced flow and further dilatation eventually resulting in aneurysm formation (see reviews and [13, 21]).

The treatment of intracranial aneurysms has undergone significant changes over the last few decades. Initially, open surgery was dominant during which a metallic clip was placed over the neck of the aneurysmal sac, thereby separating it from the blood flow in the parent artery. With the advent of less-invasive therapies, more and more aneurysms, particularly ruptured ones are treated by endovascular methods. In all these therapies, the common goal is to reduce the blood flow in the aneurysmal sac and induce intrasaccular thrombosis leading to aneurysm occlusion. From a mechanical point of view, the aneurysm occlusion reduces the pressure and wall shear stress load on the aneurysm walls, slowing down or stopping the degeneration process. One of these methods, coil embolization, is based on filling the sac with microcoils, thereby increasing the flow resistance within the aneurysm sac [6]. Another attempt was to fill the sac with polymer foam, but it did not gain wide acceptance. Nowadays, by far, the most popular method is to use flow diverter (FD) devices, which are also called stents [22]. They are similar in appearance to the more well-known coronary stents, but their function is different. They are implanted within the parent artery in front of the neck of the aneurysm, and the struts of the stents provide flow resistance. This flow resistance reduces the communication between the flow in the parent artery and the aneurysm and thus leads to the reduction of flow velocity in the sac.

As described above, the analysis of the flow parameters (e.g., wall shear stress, pressure, and oscillatory shear index) may play a critical role in understanding the formation, growth, and rupture of aneurysms. However, in vivo flow measurements using different imaging techniques do not provide the required spatial and temporal resolution. As a result, computational fluid dynamics (CFD) has been introduced into aneurysm research at an early stage. Using CFD, ex vivo numerical flow

simulation is feasible. An overview of the generally applied experimental and computational methods is given in [24].

Initially, idealized models were used in simulations (such as a sphere attached to a cylindrical tube), later real geometries were simulated [15]. The images are extracted from medical imaging studies such as 3D rotational subtractural angiography. Depending on the CFD method, the geometry is smoothed and remeshed or left in the original DICOM format.

Most of the commercial CFD codes use the so-called finite volume (FV) method, in which the investigated space domain is subdivided into small subvolumes, and for each volume, the discretized conservation equations are solved. This method needs smooth walls, and intermediate steps are required. An emerging other method to simulate flows is the so-called lattice Boltzmann method that uses simplified molecular-level collision statistics, which, on the macroscopic level, leads to the same conservation laws. This method needs a voxel mesh. Medical imaging using the DICOM standard produces exactly that file format.

We note that since the image recording takes several seconds to complete, the geometry used in the simulations effectively represents a time-averaged surface during several cardiac cycles.

The flow simulation needs a lot of simplifying assumptions. First, that blood is assumed to be a Newtonian fluid. It is common consensus that the non-Newtonian properties of blood exert their effect only in small blood vessels, in the sub-millimeter range. As arteries of the circle of Willis are well above this size, the assumption is justified. Detailed discussion of the importance of non-Newtonian effects can be found in reviews [2, 3]. The second assumption is that the flow is laminar, incompressible. Considering the flow parameters, this is also reasonable. Third, the vessel walls are assumed to be rigid. Since the artery walls are flexible with a complicated material behavior, it is difficult to model them and compromises are necessary. In addition, little is known about the surrounding tissues that support the artery from the outside. It seems that if we compare the elastic wall to the rigid wall case, in spite of some quantitative differences, the qualitative behavior of the flow is similar. Since brain arteries have little deformation during the cardiac cycle, for our purposes the rigid wall model is sufficient [14, 29].

A crucial point is the choice of the proper boundary conditions for the simulations. Very often CFD-based researchers do not reveal any details about this. Sticking with the assumption of rigid walls, the inlet and outlet boundary conditions have to be determined at the beginning and the end cross sections of the investigated artery segment. It has been shown that the inlet boundary condition has a much larger influence on the flow in the interesting region than the outlet one [29]. The inlet boundary condition is usually given by a time-dependent flow rate or velocity; the outlet one is by pressure. It would be desirable if the flow rate or velocity curve was patient-specific but it is rarely possible. Therefore, general flow rate curves are often used [5, 15]. If velocity is given at the inlet, a spatial distribution has to be given that can be uniform or parabolic, or even better, the spatio-temporal Womersley distribution [29]. The pressure at the outlet is usually assumed to be

constant. This is justified by the fact that at the peripheries, on the level of capillaries, the cyclic variation of the flow is negligible.

A large amount of studies have been published in connection with flow simulation of intracranial aneurysms [3–5, 15]. Most of them aim at determining the temporal or spatial distribution of wall shear stress or identify typical flow patterns [23]. It has also become possible to simulate treatments, in particular microcoils [30] or FD stents [31]. The development of CFD for intracranial aneurysms is progressing toward a semiautomatic easy-to-use CFD software with a few minutes of runtime for clinical applications.

For reasons provided later, the accuracy of the flow simulation is absolutely essential in this work. For this reason, we use an in-house lattice Boltzmann code. The detailed description of the code and the parameters can be found in [35]. In that work, the code was validated against detailed velocity measurements in an aneurysm model.

In this chapter, the 2D geometries are idealized, artificial geometries, while the 3D geometry is extracted from CT-angiography images. The 3D geometry is the same one validated in [35]. The inlet boundary condition is a laminar paraboloid inflow profile whereby the peak of the profile changes temporally according to a synthetic volumetric flow rate curve which bears the main properties of a real heartbeat volume flow curve. This curve was used in several works, for example [15, 35]. The outlet boundary condition is constant pressure.

The accurate flow simulations provide a spatially and temporally well-resolved velocity field in and around the aneurysm. For the rest of this chapter, this velocity field will be considered as given and in this field, a large number of individual particle paths are calculated. We describe how the time-varying flow field in and around an intracranial aneurysm results in the chaotic motion of particles transported by blood. We also show the fractal patterns traced out by these particles. For computational convenience, we use both a simple two-dimensional (2D) model for parametric studies and also illustrate the generality of the phenomenon by using real, three-dimensional (3D) blood vessel geometries.

20.2 Fractal Patterns in Time-Dependent Flows

The motion of particles transported by moving fluid is known to be chaotic in many cases, a phenomenon called *chaotic advection* [1, 10]. Chaotic particle motion is associated with a sensitive dependence of particle paths on their initial position [27]. This means that particles that start near each other typically deviate rapidly from each other, the distance between the initially close particles typically diverges exponentially with time. This very strong separation of particles leads to the observation that a patch of particles, like a blob of dye, rapidly forms long and curving filaments. These filaments, formed by the stretching and folding action associated with chaos, form fractal structures that are hence traced out by ensembles of particles transported in the fluid.

In particular, motion of particles within blood vessels has been investigated from this point of view [17–19]. It has been found that a disruption of the flow, like a branching, a narrowing, or a bulge of the blood vessel can result in the chaotic motion of particles transported in the blood.

It has been speculated that the formation of fractal patterns by platelets has an important effect on platelet activation [17]. Similar effects were found previously in many situations ranging from microfluids through plankton blooms to environmental processes [26]. In cases when filamentary fractal patterns are generated in fluid flows, the underlying biological or chemical processes are enhanced by the large perimeter where the species can interact and by the strong chaotic mixing. This enhancement of processes manifests itself in a nontrivial, singular term in the chemical rate equations [9, 11, 28] or in the population dynamical equations [8, 20]. Hence, the formation of fractal patterns in fluid flows is expected to have very important consequences on the underlying biochemical processes that take place in the blood as well [17, 34].

The simplest model to follow the motion of particles and to investigate their chaotic properties in a known flow field is to assume that the particles are very small and lack inertia. In this simplest approximation, the velocity of the particle coincides with the instantaneous velocity of the fluid at the location of the particle. For a steady (time-independent) flow, this would imply that the particle follows the streamlines of the flow. If, however, the flow is non-steady, like in case of blood pulsating periodically, the particle trajectories deviate from the streamlines and can become very complex, chaotic, even if the flow itself is simple [1, 10].

When the size and inertia of particles are neglected, the equation of motion for a particle can be written as $d\mathbf{r}/dt = \mathbf{v}(\mathbf{r}, t)$, where $\mathbf{r}(t)$ is the position of the particle at time t and $\mathbf{v}(\mathbf{r}, t)$ is the speed of the fluid at position $\mathbf{r}$ at time t . As discussed above, the right-hand side of this equation is known from numerical simulation. Hence, this is a first-order, explicit differential equation for the particle position $\mathbf{r}(t)$. We solve this equation using the fourth-order Runge–Kutta method [34]. Given an arbitrary position of a particle, we can follow its path along the blood vessel.

A few example particle paths are shown in Fig. 20.1 for a 3D aneurysm. Three particles were started from different but extremely close (within 1 μm) positions at the inlet of the shown segment, and their paths were traced numerically. For a while, the three particles moved along very similar paths, until they entered the aneurysm sac. There they spent very different times before leaving it, and both within the sac and after leaving it, they followed completely different paths. This picture illustrates the very sensitive dependence of the particle paths on their initial position, a unique sign of chaotic motion.

20.3 Basic Concepts Demonstrated on a Simplified 2D Case

The geometry of the surface of a real aneurysmal vessel section is rather intricate. Its shape is influenced by an immense number of factors. Thus, in order to be able

Fig. 20.1 Trajectories of three particles (*red, green, and blue*) traveling through the transient flow field of a real aneurysm. Despite the small initial distance between the particles (within $1\text{ }\mu\text{m}$), they travel along substantially different paths. The inlet is toward the bottom of the image; the three outlets are on the top and on the left side of the figure

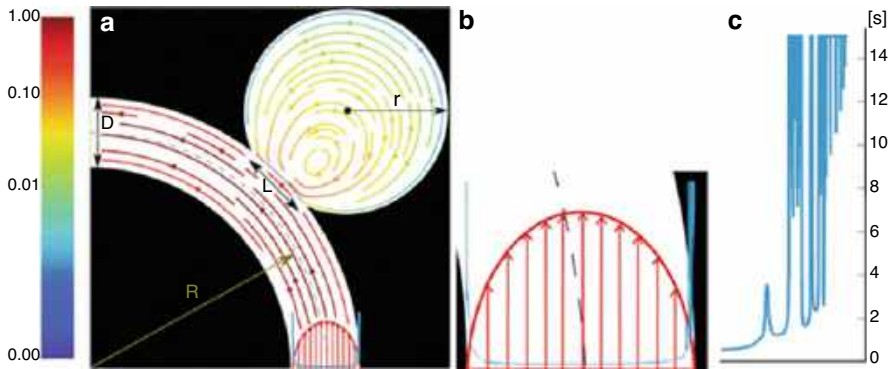
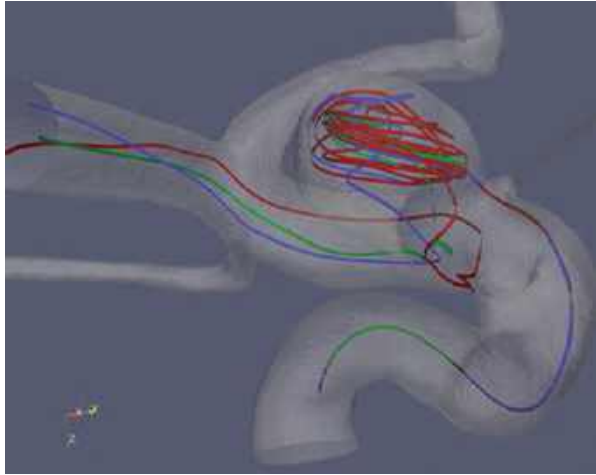


Fig. 20.2 (a) Synthetic aneurysm geometry automatically generated from the following parameters: aneurysmal sac radius (r), aneurysmal neck arc length (L), parent artery diameter (D), and parent artery curvature defined by its radius (R). The typical velocity streamline arrangement is also shown in the figure colored by a normalized logarithmic scale. (b) The parabolic inlet velocity profile (*red*) is also shown together with the residence time (*light blue*) as a function of the initial position. (c) The enlarged residence time function

to gain some general insight into the ongoing dynamics and the possible parameters driving it, a heavily simplified model was investigated first. This 2D model, shown in Fig. 20.2a, represents a synthetic cerebral aneurysmal case generated automatically from a few well selected parameters.

The reduced number of parameters and degrees of freedom lead to a higher computational efficiency and make parametric studies more feasible. Hence, a simple study that might indicate local tendencies around a physiologically correct general set of parameters becomes possible. Apart from the geometric properties, this set includes several quantities associated with the flow condition such as the average flow velocity or the heartbeat frequency. In our case, the former is expressed

as the nondimensional Reynolds number (Re) using the inlet vessel diameter and the average inlet velocity at the systolic peak, the latter is defined through the Womersley number (Wo) with the same inlet diameter and the frequency of the heart beats. Applying the aforementioned numerical methods, this model yields a high-resolution transient velocity field. The emergent flow field is smooth and laminar. Figure 20.2a displays a typical arrangement of streamlines within the computational domain. The qualitative picture of the flow remains the same under the effects of the different parameters. However, investigating the flow field from the viewpoint of tiny immersed particles, the picture varies substantially.

As we have seen in the previous section in the 3D case, the time it takes the particles to enter and leave the aneurysm sac may vary greatly as we change slightly their initial positions. The same behavior can be observed in the case of our simplified 2D model, as shown in Fig. 20.3a–d. Here, three particle paths are shown with a very small difference in their initial position (approximately $6\text{ }\mu\text{m}$). The particles follow the same path until they reach the aneurysm, they penetrate it, and start to whirl around inside the sack. It is the particle with the green path that leaves the sac first, after about 11 s, that is, 11 heartbeat periods. Later, after about 22 s, the particle with blue path exits. However, the particle with the red path is still within the aneurysm after 60 s (that is, roughly after 60 heartbeats). This indicates that the residence time can be a quite irregular function of the initial conditions, as also shown in Fig. 20.2b, c. Here, as a function of the initial position, we can see in light blue the time it takes the particle to pass along the vessel segment. For particles in the center of the blood vessel, the residence time is short and is a regular function of the initial position: these particles do not enter the aneurysm; they just follow the undisturbed part of the flow. However, particles at the edge have a chance to enter the sac and to get trapped there for a long time. We see here that the residence time depends sensitively on the initial position of these particles. Particles initially close to each other can have very different residence times. The sensitive dependence on the initial position is, again, a unique sign of chaotic particle motion.

Residence times shown in Fig. 20.2b, c feature a large number of peaks, that is, many particles have anomalously long residence times (up to 15 heartbeats in this case). In fact, in such flows, it is known that there is an infinity of initial positions with infinitely long residence times [10]. These initial positions form a fractal set, and they all belong to particle paths that are trapped forever in the vessel segment shown in the figure.

To better understand what happens to those particles that are trapped within the aneurysm, we follow many particles from the vicinity of the wall initiated from a straight line. Figure 20.3e shows where these particles are to be found after 15 s. We see that the particles, originally along a straight line, either leave the vicinity of the aneurysm or spread along an intricate curve within the aneurysm. The particles that have not left the sack trace out a filamentary fractal pattern.

This intricate pattern is caused by the existence of a mathematical object called *chaotic set* within the aneurysm. The chaotic set is a union of all particle paths that eternally stay within the aneurysm. Neither in the future nor in the past they are to be found outside the aneurysm. Examples for such particles can be periodic

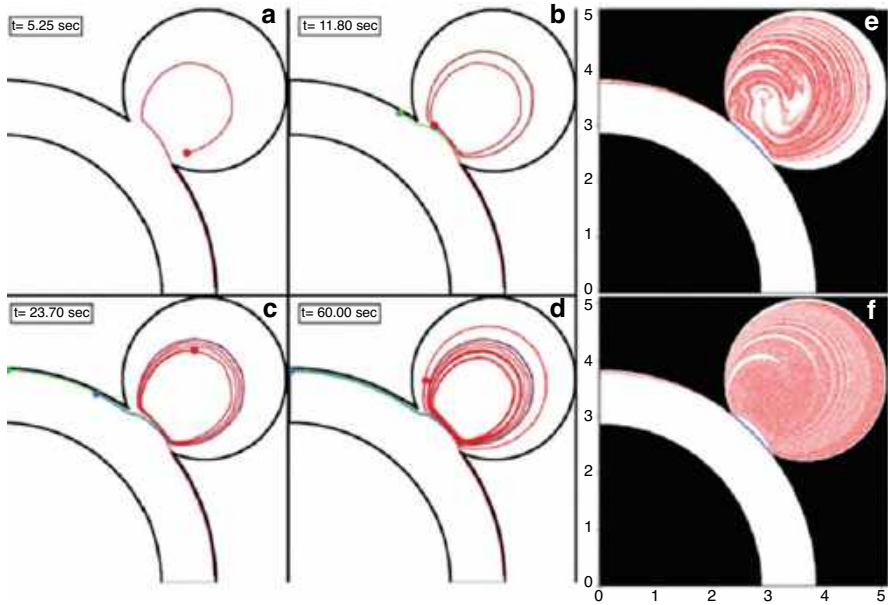


Fig. 20.3 (a–d) Sample of strongly deviating trajectories in the 2D flow. The three particles (*red, green, and blue*) were started close to each other (a few μm); however, they spend significantly different amounts of time in the central vortex. (e, f) Positions of one million tracer particles after 15 heartbeat cycle (15 s) (e) initiated from a line segment across the inlet close to the wall and (f) initiated from within the aneurysmal sac using uniform spatial distribution

particle paths that, after one or more periods of the flow, return to their previous position and then follow the same path again and again. In typical chaotic systems, the number of such periodic particle paths increases *exponentially* with the length of the time period [10, 27]. That is, the number of periodic particle paths increases very strongly with the length of the period, the longer we wait, the more periodic paths reveal themselves to an observer.

It is important to note that the chaotic set consists of infinitely many particle paths, but it is still exceptional that a particle path belongs to the chaotic set. In mathematical terms, the chaotic set is a set of measure zero among all possible particle paths, typically all particles leave the aneurysm sooner or later. Also, the particle paths that belong to the chaotic set are unstable in the sense that any small deviation from the particle path will result in a completely different path, leaving the aneurysm after some time.

Nevertheless, these trapped paths have a great effect on other particle paths. Whenever a particle gets close to a particle path that belongs to the chaotic set, like those in Fig. 20.3e with long residence time, for a while it follows segments of particle paths belonging to the set: The closer it gets to the chaotic set, the longer it takes to leave its vicinity. This effect generates the peaks on the residence time plot shown in Fig. 20.2b, c: the high peaks correspond to particles that spend a long

time wandering chaotically in the vicinity of the chaotic set. This implies that the particles trapped for a long time close to the chaotic set will eventually form the image shown in Fig. 20.3f.

The property that an unstable chaotic set governs the other particle paths forming a fractal pattern as shown in Fig. 20.3e is quite robust. For example, changing fine details of the shape of the aneurysm does not result in a qualitative change in the fractal pattern of particles.

20.4 Measuring Chaotic Quantities from Residence Times

A relatively simple technique to measure important quantities that characterize chaotic motion is the free energy formalism [25]. We initiate a large number of particles along a line segment of length L so that this segment contains at least some high peaks of the residence time, that is, it contains particle initial positions that approach the chaotic set. Then we measure the residence time for each particle, see Fig. 20.4a for an example, and count the number $N(t)$ of intervals along L with residence time longer than t .

Let $l_i(t)$, $i = 1 \dots N(t)$ denote the length of these intervals along L , see Fig. 20.4b. For a large enough t , the free energy function $F(b)$ can be defined as in [25]:

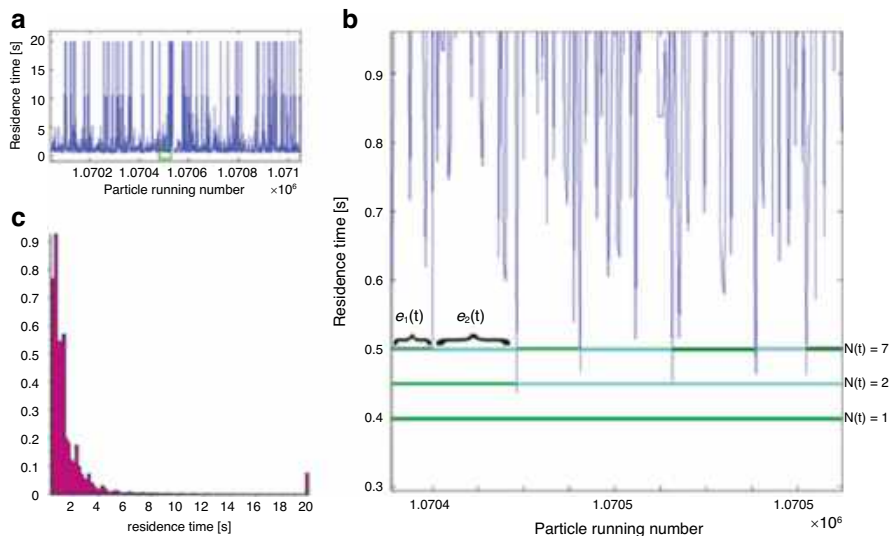


Fig. 20.4 (a) A typical residence time plot. (b) Enlarged region from the residence time plot from the green rectangle with the explanation of quantities for the computation of the $bF(b)$ function. (c) Histogram of residence times, note the final peak which represents the particles still trapped after 20 s

Table 20.1 The default set of parameter values containing both the geometric and the flow parameters

Parameter	Value
R	1.5 [mm]
R	3.5 [mm]
L	1 [mm]
D	1 [mm]
Re	100 [–]
Wo	1.36 [–]

$$e^{-bF(b)} \sim \sum l_i(t)^b.$$

The important quantities describing chaotic advection can be derived from this function [25]. For instance, the fractal dimension D_0 of the set of infinitely high peaks in Fig. 20.4a can be found as the b value at which $bF(b)$ equals zero. In this case, it is more robust [34] to obtain the information dimension D_1 of the same set, which usually is not far from D_0 .

Another important quantity is the average Lyapunov exponent, which is the average rate of exponential separation of initially close particles. This can also be obtained from $F(b)$ as the slope of the $bF(b)$ function at $b = 1$.

Also, the escape rate can be obtained from $F(b)$. The escape rate is the exponential rate of decay of the number of particles still staying within the investigated segment of the blood vessel. It can be considered as the reciprocal of the average lifetime of chaotic motion within the vessel segment [27]. It is computed simply as the value of $F(b)$ at $b = 1$ [25].

The study used a default parameter set (see Table 20.1 for details) as a basis for comparison. These parameters reflect a general realistic physiological case. In every additional sample, one parameter was either increased or decreased significantly relative to these values. This change results in a modified flow field and consequently in modified paths for the tracer particles. The major question here is how the changes in the parameters affect the chaotic behavior of the particles.

The answer can be extracted from the results of the numerical simulations. We compute the flow field and then follow many particle trajectories to generate the residence times in each parameter setup. Using the residence times, we derive chaotic quantities. The change in the obtained chaotic characteristic numbers, listed in Table 20.2, indicates the effects of local deviations from the default parameter set and might give additional insight on what to expect in the 3D realistic cases.

Table 20.2 shows the change in the major measurable chaotic properties as a function of the change in the default parameter set. The table contains several intuitive and a few counter-intuitive results. Regarding the first group: one major separation point for the advected particles is the distal part of the neck of the aneurysm. In a geometry with decreased bulge radius and maintained velocity, the overturn time of the particles trapped in the vortex inside the aneurysm is shorter, thus the particles revisit this region of separation more often. This should result in an increased Lyapunov exponent, as well as an increased escape rate. Since the

Table 20.2 Results of the parametric study of the most important quantities characterizing the chaotic particle motion

Changed parameter and the amount of change	D_1 (information dimension)	κ (escape rate)	λ (Lyapunov exponent)
Default set of parameters	0,922	0,019	0,243
Parent artery radius (R) +2,5 [mm]	0,927	0,024	0,330 ↑
Parent artery radius (R) -2,5 [mm]	0,937 ↑	0,016	0,257
Parent artery diameter (D) +0,5 [mm]	0,919	0,024	0,301
Parent artery diameter (D) -0,5 [mm]	0,911	0,009 ↓	0,100 ↓
Sac neck width (L) +1,0 [mm]	0,853 ↓↓	0,056 ↑	0,383 ↑
Sac neck width (L) -0,5 [mm]	0,923	0,017	0,219
Sac radius (r) +2,5 [mm]	0,902	0,005 ↓↓↓	0,046 ↓↓↓
Sac radius (r) -0,7 [mm]	0,923	0,064 ↑↑↑	0,826 ↑↑↑
Flow velocity (Re) +50	0,926	0,053 ↑	0,712 ↑↑↑
Flow velocity (Re) -90	0,916	0,018	0,208 ↓
Heart beat freq. (Wo) +0,56 (0,5 s cycle)	0,879 ↓	0,040 ↑	0,332 ↑
Heart beat freq. (Wo) -0,4 (2 s cycle)	0,935 ↑	0,003 ↓↓↓	0,049 ↓↓↓

The significant deviations from the default values are denoted by colored arrows

overturn frequency strongly influences the stretching effect of the flow as well, the increase in the Lyapunov exponent is expected to be more dominant. Increasing the radius should produce the opposite results. This is well reflected in Table 20.2. The change in the flow velocity has a similar effect. Its increase will yield higher escape rates, since the average velocity increases throughout the domain and also causes a higher vortex velocity, thus shorter overturn times and an increased Lyapunov exponent. For the second group, where Table 20.2 gives less plausible results, an interesting example is the outcome of changing the heartbeat frequency. The shorter cycle time increased, and the longer cycle time strongly decreased the escape rate. This might not be directly obvious, since the average flow velocity remained the same. However, the increased frequency provides an increased excitation for the flow possibly driving it toward stronger chaotic behavior. Another example is the change in the width of the neck of the aneurysm. On the one hand, changing it modifies the escape rate significantly. The wider the neck is, the lower the escape rate becomes. On the other hand, it has only a moderate effect on the Lyapunov exponent. As a result, in accordance with the Kantz–Grassberger relation [7], it modifies the fractal dimension of the particle trajectories. Increasing the neck width decreases the fractal dimension. Fractal dimension can be thought of as a measure of how densely the chaotic trajectories are packed in space. Therefore, a higher D_1 implies a stronger overall effect of the chaotic behavior on the flow.

20.5 Appearance of Chaotic Flow Inside Intracranial Aneurysms

The previous part of this chapter introduced some concepts on what might be expected in simplified cases based on some basic geometric and fluid flow properties. Comparing cases of real geometries, however, seems to be a more challenging task. Since every geometry is unique, the combined effects of the different parameters (which are naturally immensely more numerous than in our simplified case) yield a single result. This renders carrying out a parametric study nearly impossible. Nevertheless, we might showcase the effects of changing the Reynolds number with relative ease by changing the inlet flow velocity. Figure 20.5a, b shows the inlet cross section of the vessel geometry shown in Fig. 20.1; it is the same geometry used in [35] for code validation and in [34] for the exploration of fractal properties. The color of every point on the surface represents the residence time of a particle initiated from that particular point. Similar to the simplified case, there are initial locations from where the particles take much more time to leave the investigated domain than one would expect based on the average flow velocity, for some examples see Fig. 20.1. Particles traveling these paths are usually trapped in some vortex residing mainly inside the aneurysmal sac or in some cases near vessel bifurcation points. On several occasions, the aneurysm displays a very intricate flow profile possibly incorporating multiple vortices enabling for even more intertwined particle paths. In the current case, the aneurysm has a single vortex core under the investigated boundary conditions. In Fig. 20.5a, we present the residence times on the inlet cross section at a Reynolds number of 50. In Fig. 20.5b, we present the same cross section with a different boundary condition: The Reynolds number was increased to 450. The difference is clearly visible. Increasing the flow velocity resulted in a richer, more complex pattern that displays the signs of the well-known “stretching and folding” actions related to chaos. In our experience, based on ten different geometries, this phenomenon seems to be general, though different models are affected to a different extent.

As shown in Fig. 20.5, most of the particles leave the investigated domain under 1 s resulting in low residence times. As a comparison, a typical heartbeat cycle is around 1 s, so it takes a particle less than a heartbeat to get advected out of the observed zone. The choice of particle release time within this 1 s can have a significant effect, since the flow velocities change greatly during the cycle. As a consequence, the different release times can yield vastly different residence time patterns.

Suh et al. [36] investigated particle residence times in abdominal aneurysms. They took into account the pulsatility of blood flow by dividing the heart cycle into 10 points and calculated the particle residence times using the average of these points. Following similar logic, the heart cycle in this work is also decomposed into 10 instants (shown in Fig. 20.6) and at each of these instants, one million particles are released from the same initial locations. Figure 20.6 shows the inlet plane of the vessel geometry colored according to the particle residence times at

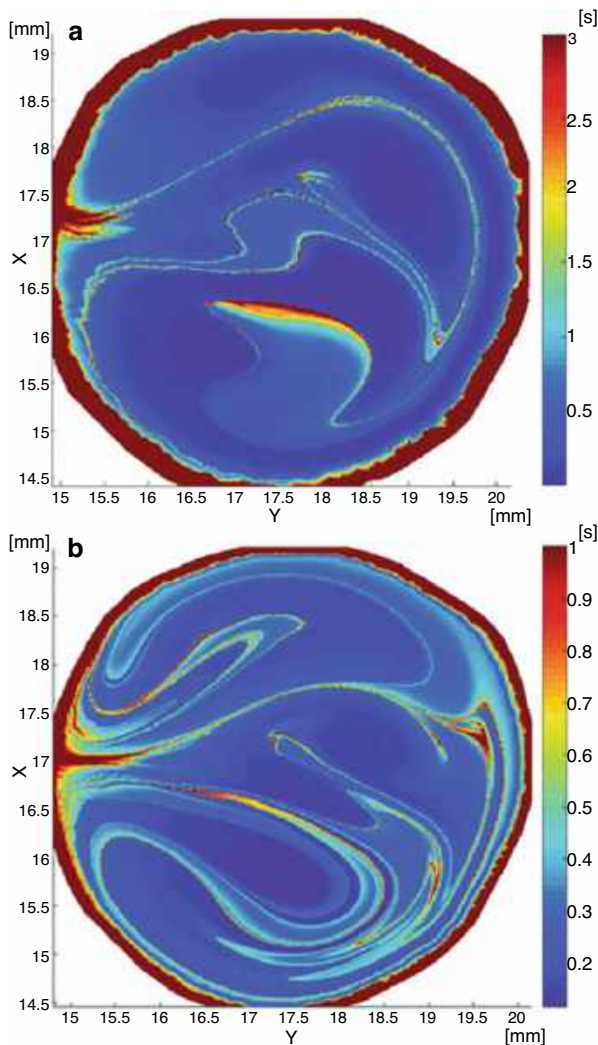


Fig. 20.5 Residence times of the particles initiated from the inlet cross section of the parent artery in case of (a) $Re = 50$ and (b) $Re = 450$. Note that the color scales are different due to the different average velocity

different releasing instants. The folded, filamentary structures differ during the peak and the following decelerating phase of the heart cycle (Fig. 20.6d–f). The reason for this is the generally higher velocities during this part of the cycle. Similar phenomena occur in other real-life aneurysm geometries too, but again, the extent of the effect of the releasing times differs from geometry to geometry. Therefore, when investigating strongly advective transient flow domains, where the residence

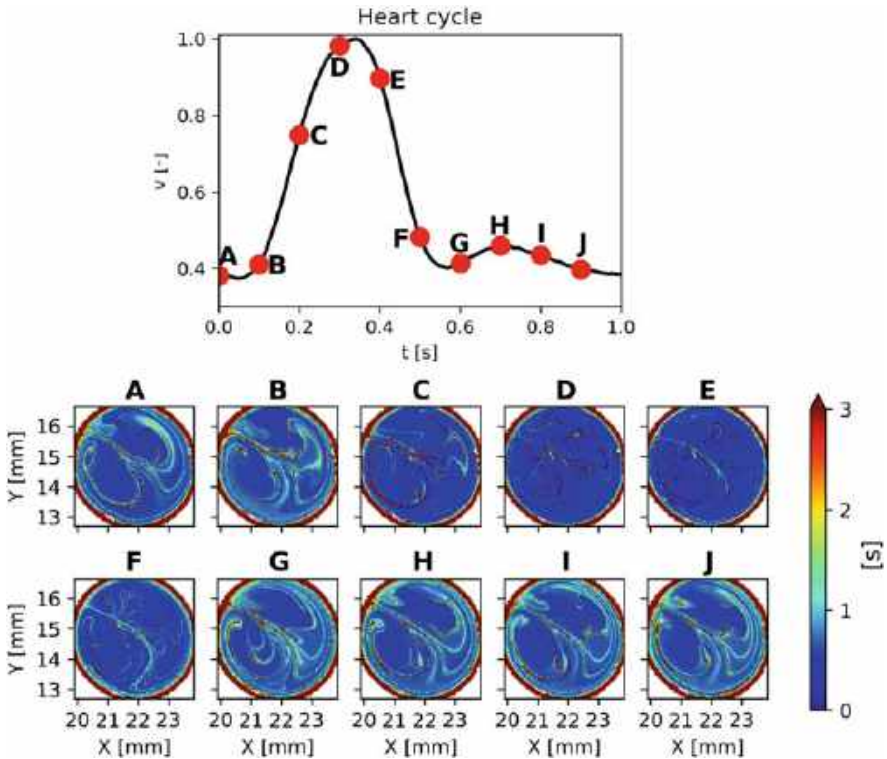


Fig. 20.6 Top: the heart cycle used in the simulations, with the particle releasing times noted. Bottom: residence times of the particles initiated from the inlet cross section of the parent artery released at different time instances during the heart cycle (a–j)

time falls on the same time scale as the transient perturbation, the effects of the initial time point have to be considered.

20.6 Concluding Remarks

Some researchers believe that the rupture risk is associated with the irregularity of the flow pattern around the region of the aneurysm [5]. According to our findings, the emergent flow properties are in strong connection with the chaotic behavior of the advection in it. Thus, the measured quantities might be used as an indicator of the flow pattern complexity. Quantitative measures of the chaotic and fractal behavior for the same geometry, similar to those for the 2D simplified geometry here, were computed in [34]. It was also discussed in [17] that chaotic advection is capable of influencing the ongoing biochemical reactions significantly. It can be speculated that this might induce a cascade of far-ranging changes in the ongoing processes

from endothelial functions to nutrition mechanisms leading to altered behavior in the wall tissues. For example, the walls of these arteries are dependent on oxygenated blood for their metabolic needs. The presence of chaotic structures might alter the residence time and thus the level of deoxyhemoglobin present inside the aneurysm.

Therefore, using this study about the very fabric of the emergent flow structures, the chaotic quantities might turn out to be useful for characterizing cerebral aneurysms as they might predict the biochemical behavior of the concerned vessel section.

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Chapter 21

Fractal-Based Analysis of Arteriovenous Malformations (AVMs)



Antonio Di Ieva and Gernot Reishofer

Abstract Arteriovenous malformations (AVMs) are cerebrovascular lesions consisting of a pathologic tangle of the vessels characterized by a core termed the nidus, which is the “nest” where the fistulous connections occur. AVMs can cause headache, stroke, and/or seizures. Their treatment can be challenging requiring surgery, endovascular embolization, and/or radiosurgery as well. AVMs’ morphology varies greatly among patients, and there is still a lack of standardization of angioarchitectural parameters, which can be used as morphometric parameters as well as potential clinical biomarkers (e.g., related to prognosis).

In search of new diagnostic and prognostic neuroimaging biomarkers of AVMs, computational fractal-based models have been proposed for describing and quantifying the angioarchitecture of the nidus. In fact, the fractal dimension (FD) can be used to quantify AVMs’ branching pattern. Higher FD values are related to AVMs characterized by an increased number and tortuosity of the intranidal vessels or to an increasing angioarchitectural complexity as a whole. Moreover, FD has been investigated in relation to the outcome after Gamma Knife radiosurgery, and an inverse relationship between FD and AVM obliteration was found.

Taken altogether, FD is able to quantify in a single and objective value what neuroradiologists describe in qualitative and/or semiquantitative way, thus confirming FD as a reliable morphometric neuroimaging biomarker of AVMs and as a potential surrogate imaging biomarker. Moreover, computational fractal-based techniques are under investigation for the automatic segmentation and extraction of the edges of the nidus in neuroimaging, which can be relevant for surgery and/or radiosurgery planning.

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21.1 Introduction

Arteriovenous malformations (AVMs) are vascular lesions of the brain and, more rarely, of the spinal cord, occurring with an incidence of 1/100,000/year and prevalence of 18/100,000/year [1]. In about 70% of cases, AVMs present with hemorrhage, accounting for 2% of all strokes, while in 20% of patients, seizures are the first presentation. In some cases, AVMs can be asymptomatic or can cause headache, focal neurological deficits, cognitive dysfunction, and pulsatile tinnitus. The combined rate of mortality and major morbidity related to untreated AVMs is 2.7% per year [22].

Morphologically, AVMs consist of a pathologic tangle of vessels, i.e., fistulous connections of afferent arteries draining blood directly into dilated efferent draining veins without the interposition of the brain parenchyma (Fig. 21.1). The core of the AVMs is defined as the nidus, which is the nest where the pathologic connections occur. Their treatment can be challenging requiring surgery (by means of microneurosurgery techniques), endovascular embolization or radiosurgery (e.g., Gamma Knife), or a combination of such modalities. The aim of the treatment is the removal and/or obliteration of the nidus in order to lower the risk of bleeding and/or seizures.

Several systems have been proposed to classify AVMs, with the most used and clinically relevant being the Spetzler–Martin classification system [28] and its subsequent modification, the Spetzler–Ponce classification system [27]. Both systems quantify specific features in a final score, i.e., (a) the size of the nidus, (b) the functional eloquence of the brain region where the AVM is located, and (c) the type of venous drainage. On one hand, these systems are very relevant because they are related to outcome (prognostication) and therapy modality, but on the other hand, no information is added with regard to the angioarchitecture. AVMs with the same score can show completely different shapes, different number of vessels forming the nidus, peculiar branching patterns, and distinct geometrical complexity. To overcome this limitation, several methods have been proposed for the morphometric analysis of AVMs, including a computational fractal-based approach, as discussed in this chapter.

21.2 Neuroimaging of AVMs

The diagnosis of AVMs is based on multimodal imaging techniques, which encompass various magnetic resonance imaging (MRI) techniques, computed tomography (CT), and catheter biplane digital subtraction angiography (DSA), which is still con-

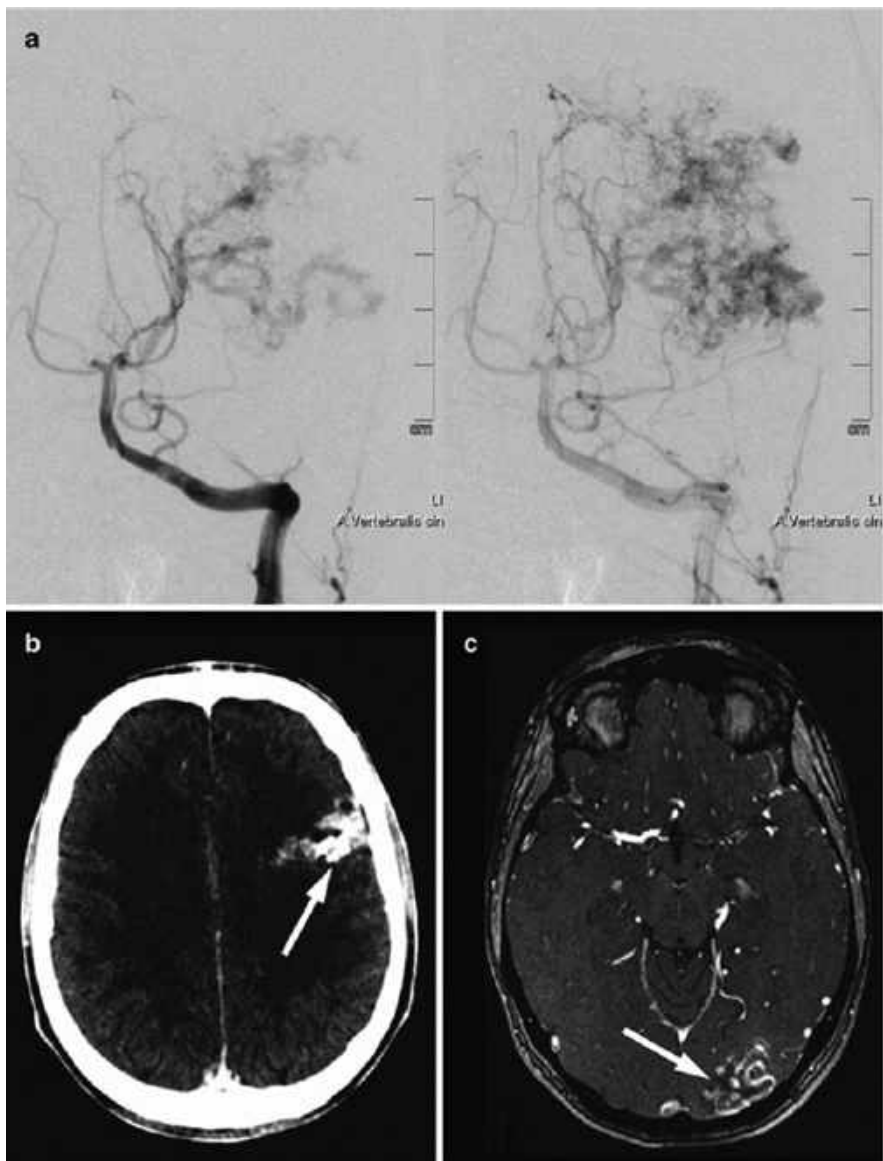


Fig. 21.1 CTA and MRA for different patients with AVM. (a) DSA of the vertebral artery imaged in an earlier phase (*left*) giving more weight to the feeding arteries and imaged in a later phase (*right*) giving a better contrast to the nidus. (b) CTA and (c) MRA of a unilateral AVM of two different patients marked with a *white arrow*

sidered the gold-standard technique [21] (Fig. 21.1). The high temporal resolution of DSA allows for the subsequent subtraction of a baseline image from contrast-enhanced images. Different states of contrast media uptake provide an excellent differentiation between the arteries, the veins, and the nidus. However, since this technique only generates 2D information from two orthogonal directions, the entire geometrical complexity of the nidus and the vascular system is not fully represented. Magnetic resonance angiography (MRA) and computed tomography angiography (CTA) overcome this limitation and provide a 3D representation of the vasculature. In MRA, techniques with contrast media (CE-MRA) or without contrast media (TOF-MRA) can be used to visualize the vascular tree. A disadvantage of all standard 3D techniques utilizing contrast media (MRA and CTA) is given by the fact that generated images are snapshots of the contrast media flowing through the vascular system. This means that imaging an early phase put stronger weights on the arterial system, while imaging a later phase puts more weights on the venous system. The time between bolus injection and data acquisition is therefore crucial for the image quality and a challenging task. 4D techniques that are available for both MR and CT are able to measure sequential angiograms and allow analyzing the dynamics of the flow within the AVM. Dynamic CT angiography (dCTA) provides good spatial and temporal resolution but exposes the patient to a high dose of ionizing radiation [20]. In MRI, dynamic contrast-enhanced sequences (DCE-MRI) based on T1-weighted MR scans are used to provide an overview of the AVM flow dynamics (Fig. 21.2), but this technique is limited in spatial and temporal resolution. Recent developments in MR imaging utilize advanced imaging reconstruction techniques such as highly constrained back projection with phase contrast as a constrain (HYPR flow) to improve spatial and temporal resolution [29].

For the optimal representation of the acquired data, several techniques exist for image fusion in multimodal imaging and 3D rendering of the AVM [29].

21.3 AVMs' Angioarchitecture Morphometrics

AVMs vary greatly among patients, and experts have attempted to classify them based on several angioarchitectural parameters, by looking for similar clinical presentation and posttreatment outcome. However, these parameters have not been standardized.

An objective morphostructural analysis of AVMs should provide useful information for classification, prognostication, therapeutic decisions, and follow-up.

Several grading systems have been proposed, which generally take into account the number of feeding arteries from the different vascular territories, the size and location of the AVM, and other features, with the Spetzler–Martin (5-tiered score) and Spetzler–Ponce (3-tiered score) as the most clinically relevant and used ones, as previously quoted. By the way, such classification systems do not take into account the morphological complexity of the nidus, referring to its geometry only in terms of the size. Of course, AVMs having the same Euclidean parameters (i.e., maximal diameter and volume) may have a very different pattern of distribution

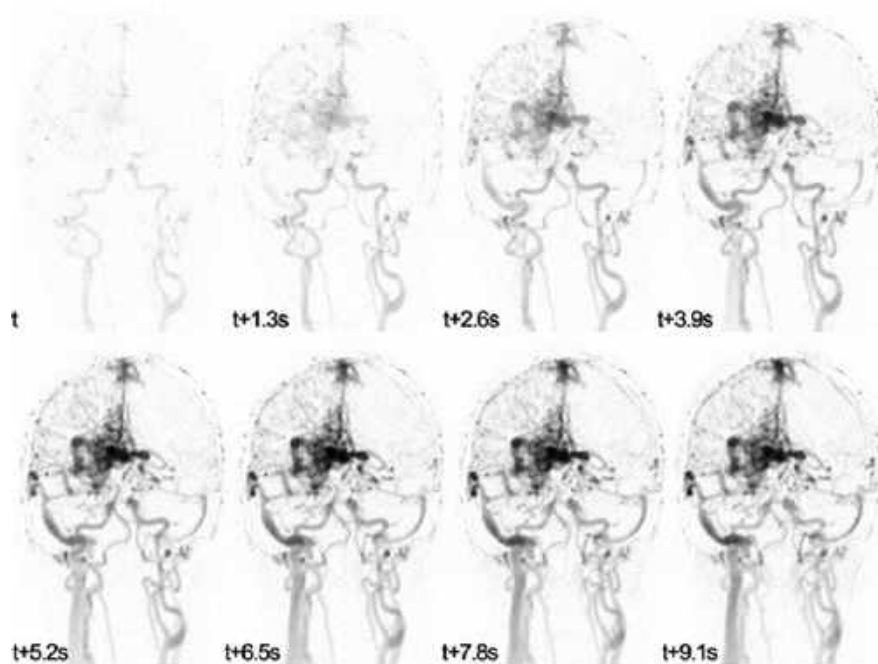


Fig. 21.2 Maximum intensity projection of a 4D dynamic MRA with a temporal resolution of 1.3 s. (Images are displayed in inverted view)

of the vessels forming the nidus (e.g., different density, compactness, tortuosity of the vessels, in one word, different angioarchitecture). Moreover, the distribution of the vessels within the nidus can leave different grades of “gaps” among the vessels themselves. According to the nonlinearity of the AVM’s shape, “compactness” and “diffuseness” of the nidus have been investigated as a better mean of quantification; these parameters have been shown to be important factors in predicting the difficulty to resect an AVM [11].

Fractal analysis is a novel computer-aided mathematical model, which has also been shown to have several applications in clinical neurosciences, including neuroimaging [7]. Fractal dimension and lacunarity seem to offer valid tools to quantify AVM’s geometrical complexity, as presented hereinafter.

21.4 Computational Fractal-Based Analyses of AVMs

Fractal analysis is a mathematical model that, among other parameters, offers the fractal dimension (FD) and lacunarity as measures of the roughness and/or geometrical complexity of natural objects, including physiological and pathologic

brain structures in neuroimaging. As several natural objects characterized by a branching pattern, AVMs are suitable for fractal analysis as well. Computing FD of the nidus by means of the box-counting method, for example, adds an objective morphometric parameter to quantify the space-filling properties of the tangle itself, thus offering a potential morphometric biomarker, which could be added to the existing classification systems.

21.4.1 AVMs' Fractal Dimension

The first study that investigated AVMs by means of FD was published by Reishofer et al. in 2012 [25]. Given the fact that the direct observation of the nidus' vasculature using MRI angiography is not an easy task due to limited spatial resolution, the architecture of surrounding arteries was analyzed by means of different measures of FD. The nidus can be supplied by one or more feeding arteries that are directly connected to draining veins leading to a vascular system with a higher geometrical complexity compared to a normal vascular system. The aim of this study was to investigate whether FD, obtained from 3D-time-of-flight (TOF) MR images, is a suitable biomarker to represent changes in the vascular geometry and, furthermore, whether FD is related to pathologic and physiological parameters such as the nidus size and vascular flow.

Ten patients with unilateral supratentorial AVM and ten healthy controls participated in this study. The patients and controls underwent a standard MR imaging protocol including T₁- and T₂-weighted sequences, 3D-TOF, and dynamic contrast-enhanced (DCE) MR imaging. 3D-TOF MR angiography is an imaging technique that allows the visualization of the cerebral arterial system with high accuracy up to small vessel diameters. The following parameters were used to visualize cerebral arteries covering the circle of Willis and vertebral-basilar arteries: TR = 22 ms, TE = 3.68 ms, flip angle = 18°, FOV = 200 mm, phase FOV = 75%, image matrix = 384 × 288, number of slabs = 3, slices/slab = 52, and slice thickness = 0.65 mm. DCE-MRI data were acquired to image the vascular flow using a 3D-FLASH sequence with the following parameters: TR = 2.67 ms, TE = 1.05 ms, flip angle = 16°, FOV = 230 mm, image matrix = 320 × 320, number of slabs = 1, slices/slab = 12, and slice thickness = 6 mm. A dose of 0.2 ml/kg body weight contrast agent (ProHance®, Bracco Diagnostics, Inc., Princeton, NJ, USA) was injected intravenously via a power injector (Spectris; Medrad Inc., Indianola, PA, USA) at a flow rate of 3 ml/s. All measurements were carried out on a 3 T Tim Trio system (Siemens Medical Systems, Erlangen, Germany) using a 12-channel head coil. The following image processing steps have been applied in order to prepare 3D-TOF data for fractal analysis (Fig. 21.3):

- Normalization to the standard MNI space (Montreal Neurological Institute) by means of linear co-registration using FLIRT [16] from the FMRIB Software Library (FMRIB Centre, University of Oxford, UK; <http://www.fmrib.ox.ac.uk/>)

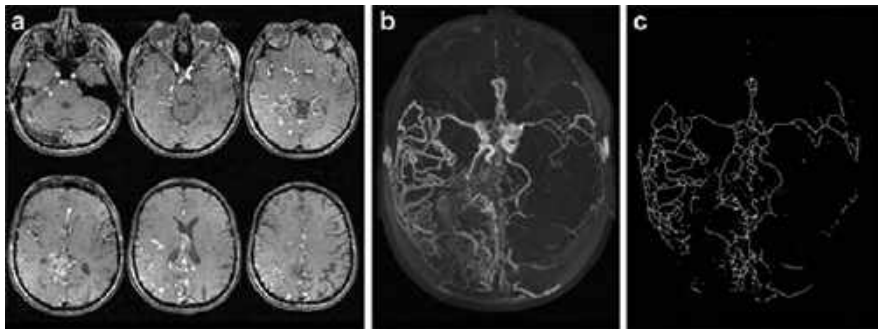


Fig. 21.3 Normalized 3D-TOF images (a) provide the basis for MIP images (b). FD evaluated from MIP images after the image processing step segmentation, binarization, and skeletonization (c). (From Reishofer et al. [25])

fsl/fsl/downloading.html) to account for intersubject variability regarding head size and FOV positioning.

- Maximum intensity projection (MIP) was performed to generate a two-dimensional representation of the 3D-TOF data using MRIcro software (Chris Rorden, University of Nottingham, UK; <http://www.nitrc.org/frs/download.php/414/mrzip.zip>).
- The background was separated from the vascular tree using a k-means clustering algorithm.
- Images were converted into binary images and skeletonized using the ImageJ software v.1.45 (Wayne Rasband, National Institutes of Health, USA; <http://rsbweb.nih.gov/ij/download.html>).
- To enable a statistical comparison between the hemisphere with AVM and the hemisphere without AVM for both patients and controls, the images ($364 \times \text{pixel}$) were split into two halves ($182 \times 436 \text{ pixel}$) and analyzed separately.

The fractal dimension was approximated using the box-counting dimension (D_b) and the Minkowski dimension (D_m) [17]. All of the fractal measurements were evaluated for the whole brain in patients (P_{total}) and healthy controls (HC_{total}) and also for both hemispheres separately in patients (P_{AVM} , $P_{\text{no AVM}}$) and controls (HC_{left} , HC_{right}). A 2×2 mixed-design ANOVA test with a within-subject factor of hemisphere (for patients, AVM, no AVM; for healthy controls, left, right) and a between-subject factor of group (P , HC) was applied to test for significant differences.

Patients showed significantly higher FD values for both methods in hemispheres with AVM compared with the hemispheres without AVM (D_b , $p = 0.002$; D_m , $p = 0.002$). Controls had similar values for FD in both hemispheres with no significant differences in FD (D_b , $p = 0.982$; D_m , $p = 0.892$). No significant differences were observed comparing the non-affected hemisphere of patients with healthy controls (HC_{left} , D_b , $p = 0.574$; D_m , $p = 0.918$; HC_{right} , D_b , $p = 0.691$; D_m , $p = 0.872$) but significant differences when comparing the patient's hemisphere

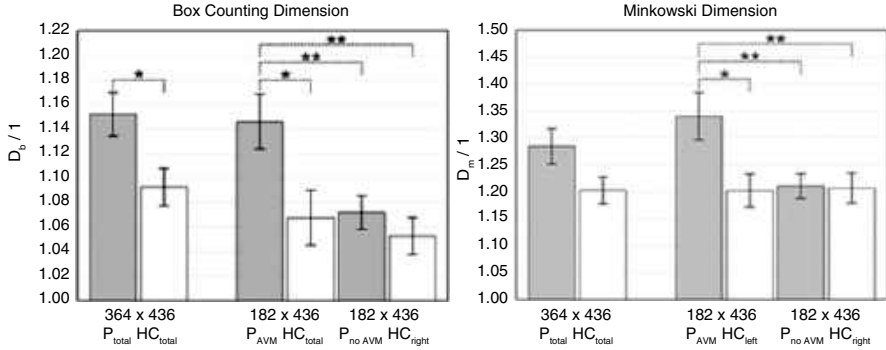


Fig. 21.4 Mean values of D_b (left side) and D_m (right side) comparing patients (P_{total}) with healthy controls (HC_{total}) for the whole image, hemispheres of patients with AVM (P_{AVM}) compared to the hemisphere without AVM ($P_{no AVM}$), and comparison of the left and right hemispheres of healthy controls (HC_{left} , HC_{right}). Asterisks indicate significant differences in group comparisons: * $p < 0.05$, ** $p < 0.01$

with AVM with the hemispheres of healthy controls (HC_{left} , D_b , $p = 0.015$; D_m , $p = 0.020$; HC_{right} , D_b , $p = 0.010$; D_m , $p = 0.024$) (Fig. 21.4).

These results suggest that FD is a sensitive parameter able to detect changes in the geometrical complexity of the cerebral arterial system with high accuracy.

Values of FD are strongly correlated with physiological and pathologic parameters. This has been demonstrated by correlating FD with the maximum slope of contrast media transit obtained from dynamic contrast-enhanced (DCE) MRI data. A linear regression analysis showed a strong and positive linear correlation between FD of the affected hemisphere and the maximum slope of contrast media transit for D_b and D_m (D_b , $r = 0.913$; $p < 0.0001$; D_m , $r = 0.926$; $p < 0.0001$) (Fig. 21.5). Given the fact that the amount of arterial inflow is related to the number of vessels feeding the nidus, this result was not unexpected. A strong correlation was also found between FD and the nidus size (D_b , $r = 0.944$; $p < 0.0001$; D_m , $r = 0.963$; $p < 0.0001$). The nidus size was estimated from DSA data. This strong correlation is not that obvious but suggests that there is a relation between vascular complexity and nidus size.

It has to be noted that absolute values obtained from the box-counting method are dependent on the image matrix size [24]. Given the fact that the box size is doubled at each iteration step starting with the size of 1 pixel, a complete covering of the image is only possible if the dimensions of the image to be analyzed are given as a power of two. If this is not the case, FD is dependent on the initial location of the grid. To overcome these limitations in the box-counting method, extensions of this technique have been proposed [19] such as the sliding box-counting dimension. When using this technique, each box is slid over the image overlapping the previous box making this method independent from the initial state but at the cost of computational time. The assessment of FD by evaluating the Minkowski dimension uses a different concept in which the structure itself is covered through geometrical

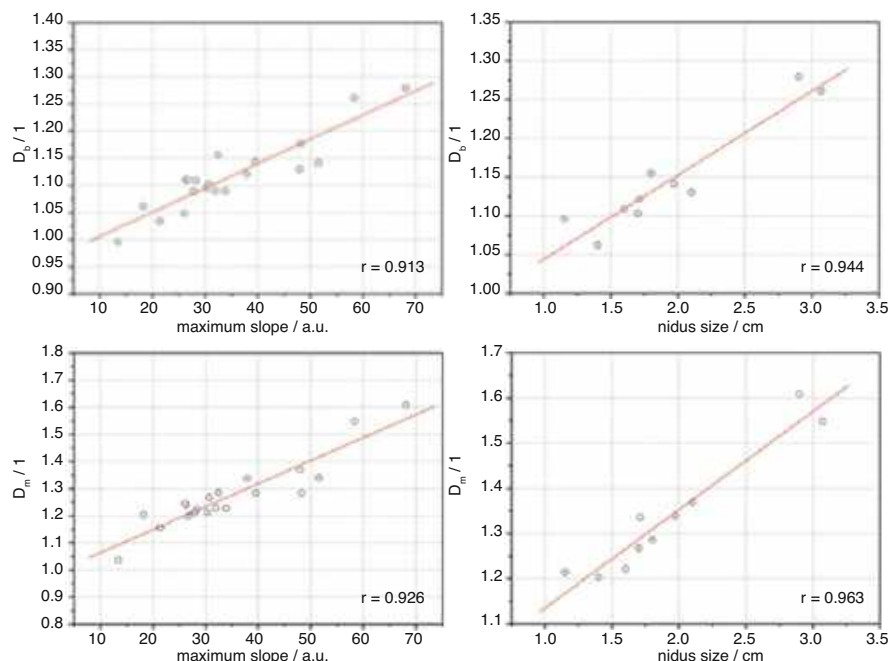


Fig. 21.5 Correlation between FD values (D_b , D_m) and the maximum slope of contrast media transit obtained from DCE-MRI data (*first column*) and correlation between FD values (D_b , D_m) and the nidus size evaluated from X-ray angiography data (*second column*)

objects such as circles, triangles, or squares making this method independent from different image matrix sizes. Another limitation is given by the fact that an MIP of 3D-TOF MR images is only an approximate representation of the vascular system. Superimposed vessels cannot be discriminated for a fixed projection plane, but since the projection of a fractal from an m -dimensional space into a $(m-1)$ dimensional subspace is well defined [12], FD from MIP images is an acceptable approximation. However, FD analysis of the vascular cerebral system is a simple and robust technique that yields an objective measure and may assist neuroradiologists in the diagnosis of complex cerebrovascular diseases.

21.4.2 Fractal Dimension of the Nidus and Its Relevance in Radiosurgery

Di Ieva et al. investigated the applicability of FD as a potential neuroimaging biomarker of the AVMs' angioarchitecture, especially in relation to the patients' outcome and follow-up in radiosurgery [9]. The Gamma Knife (GK) radiosurgery treatment, as well as other radiosurgical modalities, is aimed to obliterate the

fistulous connections within the nidus, lowering the risk of bleeding. By means of specialized equipment, the stereotactic GK focuses multiple beams of radiation on the target (e.g., a brain tumor or AVM). The radiosurgical effect of obliteration of the vessels occurs over time, in general in 2–3 years; the 3-year obliteration rates following stereotactic radiosurgery range from 60 to 86.6% [3, 13, 23]. The box-counting method was used for the FD computation in a retrospective case series of 54 patients affected by brain AVMs who underwent GK radiosurgery at the University of Toronto (Ontario, Canada) [9]. In this study, almost 54% of patients presented with hemorrhage, 13% only with headache, 9% with seizures, while other patients were incidentally diagnosed or reported minor focal neurological deficits. All of the patients underwent multimodal imaging, including CTA, MRA, and angiogram as well, but in order to standardize the image and the following fractal analysis, only MRA images performed on the same scanner with the same parameters were considered in the analysis. By means of such selection in the inputs (see the “black box” concept in Chap. 12), it was possible to obtain a homogeneous dataset for analysis. 3T GE Medical System Signa HDxt with an eight-channel head coil was used for all patients, with the following MRI parameters: axial 2D FRSE sequence with a rectangular matrix of 320, an echo train length of 15, a TR of 5500 ms, a TE of 91 ms, a slice thickness of 2 mm without interslice gap, and with a rectangular field of view of 20 cm resulting in a plane resolution of 0.625 mm square. Images were transferred to the radiosurgery software, which is a modification of CMI software (Montreal Stereotactic Planning System; CMI Services, Montreal, QC, Canada) [2] (see Methods in [9]). A 4C GK radiosurgery system (Leksell Gamma Knife® Perfexion™, Sweden) was used for the stereotactic radiosurgery treatment. A standard dose of 25 Gray (Gy) for AVM volumes $<4\text{ cm}^3$ and 20 Gy for volumes $>4\text{ cm}^3$ was administered (limiting to 15 Gy the dose to the nidus whether near to eloquent brain areas). Checking the imaging 3 years after the treatment, the outcome of radiosurgery was dichotomized as complete obliteration, defined as the total disappearance of the nidus on the MRA, no abnormal flow voids on MRI and no early draining vein on the angiogram.

The image analysis was based on the following steps (Fig. 21.6):

- Selection of the MR sequences (FRFSE T2 post-gadolinium) and images containing the nidus.
- Selection of the nidus itself (the “region of interest” (ROI)), according to the outline of the target zone for the radiosurgical procedure, as chosen by consensus of six experts (including neurosurgeons, neuroradiologist, radiation oncologist, and physicist).
- Normalization of the intensity of the pixels within the ROI, according to the Brightness Progressive Normalization (BPN) algorithm, introduced by C. Russo [26], which allows to use a single gray-scale threshold for the automatic segmentation of the nidal vessels [10].
- Segmentation of the nidal vessels.
- Application of the box-counting method. A fractal window in a two-order scale was chosen, starting at the highest resolution of the MR images

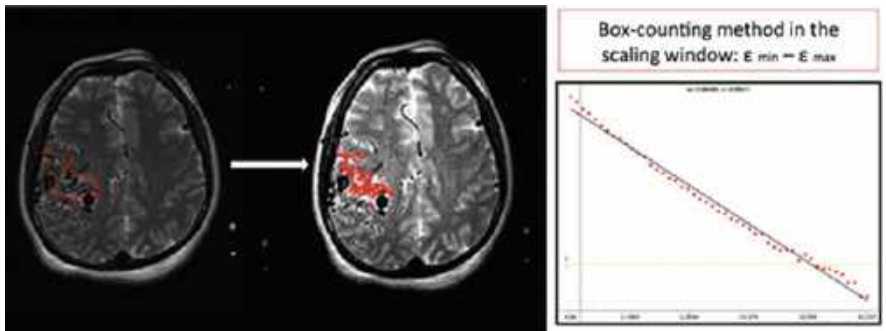


Fig. 21.6 Computation of FD on MR images of AVMs. The region of interest of the AVM is manually segmented on each slice, and an RGB threshold is used for the computer-aided segmentation of the vessels forming the nidus. The box-counting method is then applied in the fractal window, and the results are plotted on a graph, the slope of which represents FD. (From Di Ieva et al. [8], reprinted with Permission of the Publisher)

Table 21.1
Angioarchitectural
parameters associated with
fractal dimension in brain
AVMs

Angioarchitectural parameter
Arterial enlargement presence
Presence of non-sprouting angiogenesis
Presence of sprouting angiogenesis
Flow pattern
Venous ectasia
Presence of venous rerouting
Presence of pseudophlebitic pattern
Size

($\epsilon_{\min} = 0.86 \text{ mm}$ – $\epsilon_{\max} = 86 \text{ mm}$). Monofractal behavior was found in such range.

The obtained FD values were statistically compared to (a) several angioarchitectural parameters, commonly used by the neuroradiologist for the qualitative description of the nidus (e.g., arterial enlargement, flow-related or intranidal aneurysms, flow pattern, neoangiogenesis, venous ectasia, nidus size, number of draining veins, venous drainage type, presence of pseudophlebitic pattern, etc.) and (b) the radiosurgical outcome, in terms of obliteration, 3 years after the treatment.

With regard to the angioarchitectural parameters, FD was strongly associated with volume and size of the nidus and significant associations were also found with several other indices (Table 21.1). These results suggested that FD is able to quantify in a single and objective value what neuroradiologists otherwise describe in qualitative and/or semiquantitative way, confirming FD as a reliable morphometric neuroimaging biomarker of AVMs. Higher FD values may signify that the AVM has more tortuous and compact vessels within the nidus, for example, with a higher grade of roughness.

Regarding the outcome, only a few morphometric parameters were significantly associated with it, including the presence of sprouting angiogenesis, moderate-to-high flow pattern, and the presence of a venous pouch. Size and venous drainage were both strongly related to outcome, signifying that bigger AVMs (>3 cm of diameter) and/or with deep venous drainage are more related to incomplete obliteration, with a lower chance to be completely cured by means of GK radiosurgery alone. Although no statistical significance was found, interestingly enough the AVMs with higher FD values had a lower chance of complete obliteration at the follow-up (i.e., lower response to GK) (Fig. 21.7). Every increase in FD value by 0.1 was associated with a 1.21 increase of odds to get an incomplete obliteration. This inverse relationship between FD and AVM obliteration should be further investigated, but at least it would suggest that the fractal analysis may add some parameters to the morphometric of AVMs to predict whether a patient will respond or not to radiosurgery alone and whether he/she will require alternative treatment or a combination of therapeutic modalities (e.g., embolization and/or surgery for “pruning” the nidus, followed by radiosurgery to obliterate the remaining fistulous vessels). The potential of FD as a clinical surrogate biomarker for prognosis should be further investigated in larger prospective studies.

Moreover, the technique could be extended to other neuroimaging modalities (e.g., DSAs), as well as to the same sequence imaged over time, in order to make a longitudinal analysis of FD during the patients’ follow-up. The longitudinal analysis could offer some indices for prognostication too. For example, in a case in which an AVM is radiologically followed over time (e.g., every 6 months), if the FD values remain constant, it may signify that the patient most likely is a nonresponder to radiosurgery and may require further treatment (e.g., multimodal treatment with endovascular embolization and/or microneurosurgery). More studies are warranted to validate such computational methods for the assessment of the AVM angioarchitecture in relation to the outcome.

21.5 Limitations

As it may occur in several other neuroimaging analyses, the major limitation of the application of fractal analysis to AVMs’ morphometrics is related to the image analysis itself, i.e., to the choice of the ROI (the edges of the nidus). This procedure is in fact generally performed on a consensus of experts, who may introduce some intra- and interobserver variability; moreover, it is time-consuming too. The imaging parameters, as well as the fractal analysis ones, have to be standardized in order to obtain a homogeneous dataset to be analyzed and a homogeneous series of output data, which can correctly be related to some clinical information.

Several techniques are under investigation for the automatic extraction of the ROI as well as for the standardization of the methods of segmentation of the intranidal vessels, as summarized in the following paragraph.

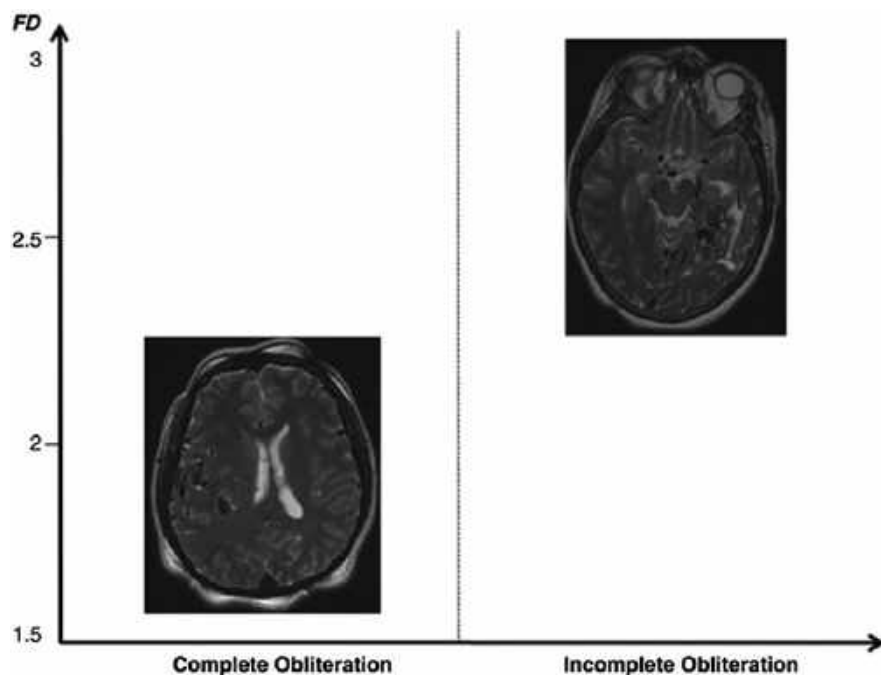


Fig. 21.7 Relating between AVMs' angioarchitecture (quantified by means of FD) and radiosurgical outcome: the nidus with lower geometrical complexity (lower computed fractal dimension, on the *left*) might have higher chance to be completely obliterated after Gamma Knife radiosurgery treatment, while an incomplete obliteration could occur in a more geometrically complex nidus (higher FD, on the *right*). (From Di Ieva et al. [8], reprinted with Permission of the Publisher)

21.6 Computational Techniques for the Automatic Nidus Identification

Several computational techniques are under investigation for the automatic extraction of the nidus' edge from the neuroradiological images. The pixel profiling technique is one of the most promising methods to classify and automatically extract the nidus' edges, by means of a computerized analysis of the pixel intensity distribution inhomogeneity [5]. Threshold-based segmentation of the AVM's vascular tree or voxel-wise support vector machine (SVM) and fuzzy logic have also been proposed as a novel and promising means for fast image extraction of the nidus and therapy planning [4, 14, 15].

Computational fractal-based methods have been investigated as well for the computer-aided identification of the nidus [8]. Lahmiri et al. recently computed FD of cerebral hemispheres via the scaling exponent of detrended fluctuation analysis (DFA), using it in an SVM to differentiate the nonpathologic cerebral hemisphere from the one affected by an AVM, obtaining a perfect classification accuracy [18]

(see also Chap. 19). A further method for the automatic detection of AVMs in MR images was developed by computing several fractal-based parameters (including the Hurst exponent) into a supervised machine-learning algorithm, resulting in a computational method for the ROI detection as well as a further confirmation of the fractal properties and scale-invariant structure of the AVMs (see Methods in [18]). Moreover, assuming that the different pixel distribution within the nidus in MR imaging may affect the variance (or clustering volatility) of the general pixel distribution in images of normal brains, the generalized autoregressive conditional heteroskedasticity (GARCH) technique, initially proposed for clustering financial time series data [6], was applied for quantifying such differences [8]. The GARCH technique was demonstrated to be potentially helpful in automatically detecting or characterizing the AVMs' nidus.

These computational methods seem to be promising for the automatic segmentation and features extraction of the AVMs but should be tested on different images (different MRI sequences and other modalities, including CTAs and DSAs) and on larger patients' series.

21.7 Conclusion

Fractal analysis is a novel approach to quantify the vascular complexity of the arteriovenous malformations. Higher FD values are related to AVMs characterized by an increased number and tortuosity of the intranidal vessels, or to an increasing angioarchitectural complexity as a whole, confirming FD as a reliable computational morphometric neuroimaging biomarker. Moreover, FD might be a potential surrogate biomarker for treatment response as well, relevant for prognostication. Fractal analysis can add some parameters, which could be added into a multiparametric morphometric analysis of the angioarchitecture of the nidus itself in relation to some relevant clinical information. The use of lacunarity should also be investigated as an angiostructural morphometric parameter. Further studies are required to extend the application of fractal analysis for the study of AVMs.

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Chapter 22

Fractals in Neuroimaging



Salim Lahmiri, Mounir Boukadoum, and Antonio Di Ieva

Abstract Several natural phenomena can be described by studying their statistical scaling patterns, hence leading to simple geometrical interpretation. In this regard, fractal geometry is a powerful tool to describe the irregular or fragmented shape of natural features, using spatial or time-domain statistical scaling laws (power-law behavior) to characterize real-world physical systems. This chapter presents some works on the usefulness of fractal features, mainly the fractal dimension and the related Hurst exponent, in the characterization and identification of pathologies and radiological features in neuroimaging, mainly, magnetic resonance imaging.

Keywords Computed tomography · Detrended fluctuation analysis · Fractal dimension · Hurst exponent · Magnetic resonance imaging · Neuroimaging · Classification · Statistical tests

22.1 Introduction

Fractal analysis is becoming an attractive method in biomedical engineering to describe and quantify the morphological complexity in computed tomography (CT) and magnetic resonance imaging (MRI). In recent years, it has been applied in neuroimaging for automatic classification purposes, in the context of medical

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diagnosis and prognostication [24–27, 29]. Indeed, fractal features such as the fractal dimension have been found to be effective in describing the complex morphology of the cortex and hence suitable to distinguish between different physiopathological states in brain MR imaging [6].

Fractal analysis has been adopted in several studies on brain magnetic resonance image classification, thanks to the ability it provides to evaluate the self-affinity at different scales and the long-range correlations of an image [11, 39] and to use the results for discrimination. Recent works in the field have assumed high self-affinity and long-range correlation in normal brain MRI in comparison to those of some neurological diseases, based on the hypotheses that brain morphology is more regular in healthy subjects and that differences can be detected and quantified by means of fractal analysis [6, 24–27, 29].

This chapter reviews recent works related to the usefulness of fractal analysis to characterize MRI features, especially by means of the fractal dimension (FD) and Hurst exponent (HE), and therefore help detect neurological pathologies through the morphological changes that bring to a abnormal brain image. This chapter is organized as follows: Sect. 22.2 presents recent works on the classification of brain magnetic resonance images by using fractal metrics as the main features; Sect. 22.3 reviews other works with different applications of fractals in neuroimaging. In Sect. 22.4, some conclusions and future perspectives are summarized. A technical description of fractal analyses techniques used in brain magnetic resonance image classification is given in the appendix, including classic range-scale analysis, detrended fluctuation analysis (DFA), and generalized Hurst exponent.

22.2 Fractals in Brain Magnetic Resonance Image Classification

A new MRI classification approach based on fractal geometry and the spectral energy distribution of edges in brain magnetic resonance images was proposed by Lahmiri and Boukadoum [25] to identify patients affected by Alzheimer's disease (AD), brain tumors (e.g., gliomas), brain infections, as well as demyelinating diseases such as multiple sclerosis (MS). Edges represent abrupt changes in image intensity, their distributions in normal and abnormal brain MR images can be different, and thus the authors used morphological properties for distinguishing normal anatomical silhouettes from abnormal ones. The considered features were the fractal dimension of edges after expressing the initial image as an analytic object and the skewness (used to measure symmetry/asymmetry in the data distribution) and kurtosis (used to measure the degree of tailedness in the data distribution) of its energy distribution at different frequencies. More specifically, the Hilbert transform [14] was applied to the brain MR images in the first step, and the Sobel filter [38] was applied to the Hilbert-transformed image to detect strong edge strength at the brain boundaries affected by pathologies. Figure 22.1 shows

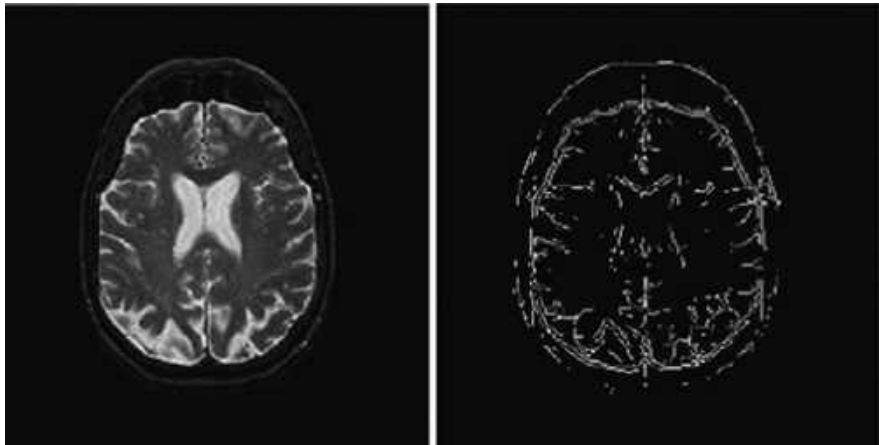
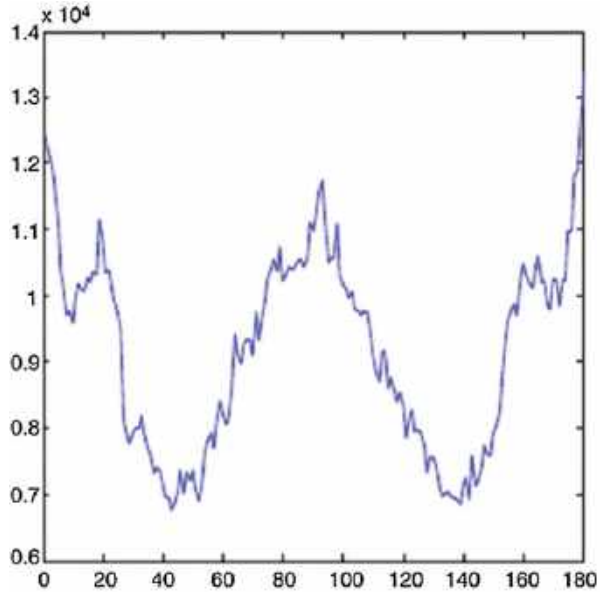


Fig. 22.1 Normal MRI (*left*) and its Sobel-filtered image in Hilbert domain (*right*)

an example of a normal MRI and its Sobel-filtered image in Hilbert domain. The extracted edges are analyzed for feature extraction by the fractal-based box-counting method [11, 15] and the Fourier power spectrum [12]. A feature vector was built with the obtained image fractal dimension and the skewness and kurtosis of the spectral energy distribution of the Sobel-processed image. Figure 22.2 provides an example of spectral energy distribution from which skewness and kurtosis were computed. Finally, the feature vector was fed to a backpropagation neural network [13] trained by the scaled conjugate numerical algorithm [13, 33] to perform classification. The proposed fractal dimension and spectral energy (FDSE) distribution system outperformed the popular discrete wavelet transform (DWT) and principal component analysis (PCA)-based means of analysis [21]. Indeed, the approach based on the FD, skewness, and kurtosis of the edge spectrum energy distribution achieved $91.78 \pm 0.0148\%$ correct classification rate of the normal against images in different pathological conditions (e.g., Alzheimer's disease, gliomas and metastases, herpetic infections, and MS as well), while the conventional DWT-PCA-based approach achieved $82.69 \pm 0.0796\%$ correct classification rate.

In a subsequent study, Lahmiri and Boukadoum [27] described a modification of the work in [25] that is suitable for pathologies such as gliomas and brain metastases. The brain MR images were converted to a one-dimensional signal by row concatenation to speed up feature extraction, and the skewness and kurtosis were replaced by additional geometrical features of the image. Detrended fluctuation analysis was adopted as an alternative method to determine the local trends in a biological signal and measure its level of persistence. It differs from the box-counting method to evaluate the fractal nature of an image by evaluating its stability with regard to trend removal. This in turn leads to a more robust estimate of the scaling properties in a nonstationary environment. In both cases, the fractal aspect of the processed image is evaluated with the Hurst exponent (see Appendix). In the new work, the

Fig. 22.2 Spectral energy distribution of the normal Sobel-filtered image in Hilbert domain



feature vectors were composed of the box counting-based Hurst exponent, the DFA-based Hurst exponent, and the energy of the detrended fluctuations of the derived bio-signal. Figure 22.3 plots in log–log coordinates of the relationship between the detrended fluctuation function and the granularity of analysis (number of bins per signal; see Eq. 22.7 in Appendix) for signals of healthy patients vs. AD ones, where the red curve is the empirical DFA and the blue line is the theoretical DFA for a random walk signal, for which the DFA is a straight line with a 45° slope. The support vector machine (SVM) with quadratic kernel was chosen as the main classifier, and it achieved 100% detection accuracy of normal brain image against glioma and metastatic bronchogenic carcinoma, regardless of whether the processed 1D signal was obtained by concatenating the rows or columns of the MR image.

A potential improvement upon the basic Hurst exponent analysis to characterize the fractal feature of brain magnetic resonance images is multiscale analysis (MSA) [8]. Multiscale analysis uses the generalized Hurst exponent method, which determines the scaling properties of a signal by computing the q th-order moments of the distribution of the signal's increments [8]. MSA can detect many types of signal dependencies with a high computational efficiency. Since each order q allows the estimation of a different Hurst exponent, varying this value allows for obtaining the multifractal spectrum of the signal in a straightforward way. Furthermore, by using MSA as an alternative for the box counting-based Hurst exponent and DFA-based exponent estimation, the feature extraction time is reduced considerably. In [27], MSA was investigated for the classification of normal against AD images. Figures 22.4 and 22.5 plot in log–log coordinates of the obtained relationship between the generalized Hurst exponent $H(q)$ and the time scale of analysis for different orders

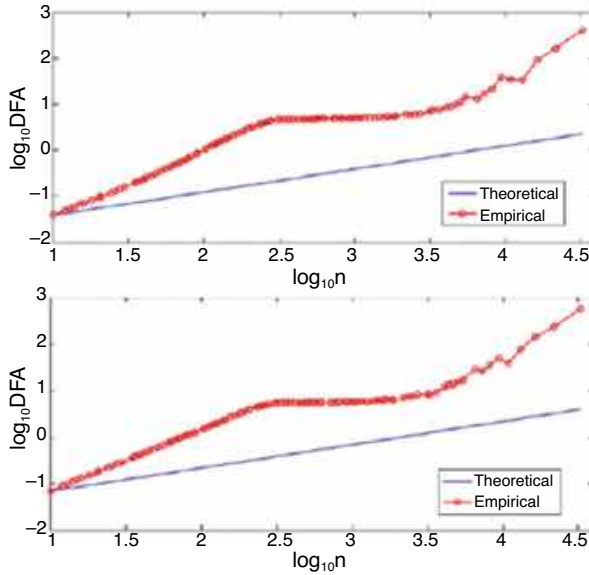
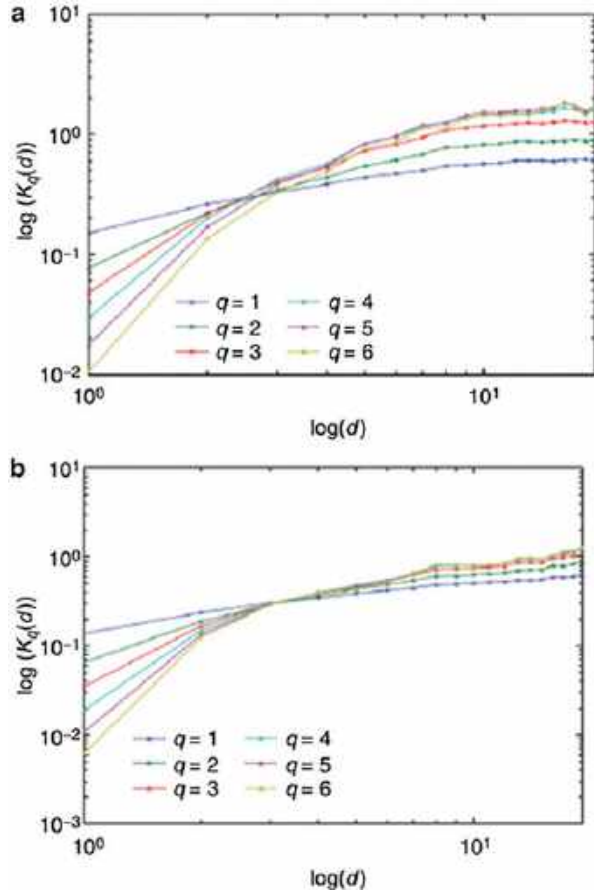


Fig. 22.3 DFA ($F(n,N)$) plot of a normal (*top*) and AD signal (*bottom*) as a function of binning

q (see Eq. 22.9 in [Appendix](#)) of normal brain images versus patients affected by Alzheimer's disease. The extracted fractals were used as features to differentiate healthy brain MRI from those of AD by SVM classifier. The results indicated a $99.18 \pm 0.01\%$ classification accuracy, 100% sensitivity, and $98.20 \pm 0.02\%$ specificity. These results showed the potential of using multiscale fractal analysis to differentiate healthy brain images from ones affected by Alzheimer's disease in near real time as the MSA algorithm took only 5.64 s to analyze a brain MRI.

In a further work, Lahmiri and Boukadoum [28] extended their work in [27] to a multiclassification problem. They explored the effectiveness of classifying brain images of healthy subjects against those with AD or mild cognitive impairment (MCI) by using support vector machines. Three experiments were conducted: In the first one, the SVM was trained to classify AD against normal images. In the second experiment, the SVM was trained to classify AD against MCI, and, in the third experiment, a multiclass SVM was trained to simultaneously classify all three types of images. In order to statistically evaluate the discriminative power of the Hurst exponent, two statistical tests were performed. The first one tested the null hypothesis of equal means for the HE values of healthy MCI and AD data based on the F-statistic of an analysis of variance (ANOVA) test. The second test checked the null hypothesis of equal distribution variances based on the Brown–Forsythe statistic. The results from these statistical tests indicated rejection of the two hypotheses at the 5% level of significance for all scales of analysis (the obtained p-values were all <0.05). Therefore, it was concluded that the distributions of HEs between images of normal brain and images of patients

Fig. 22.4 Multiscale analysis results on log–log scale of healthy image (a) and AD (b)



affected by MCI and AD are statistically different. Thus, these fractal measures are potentially discriminative to distinguish normal MCI and AD images. Indeed, the classification results indicated that the SVM achieved $97.08 \pm 0.05\%$ correct classification rate, $98.09 \pm 0.04\%$ sensitivity, and $96.07 \pm 0.07\%$ specificity for the classification of healthy against MCI images, thus outperforming recent works found in the literature. For the classification of MCI against AD, the SVM achieved $97.5 \pm 0.04\%$ correct classification rate, 100% sensitivity, and $94.93 \pm 0.08\%$ specificity. The third experiment also showed that the multiclass SVM provided highly accurate classification results. The processing time for a given image was 25 s. From these findings, the authors concluded that this approach is efficient and may be promising for clinical applications.

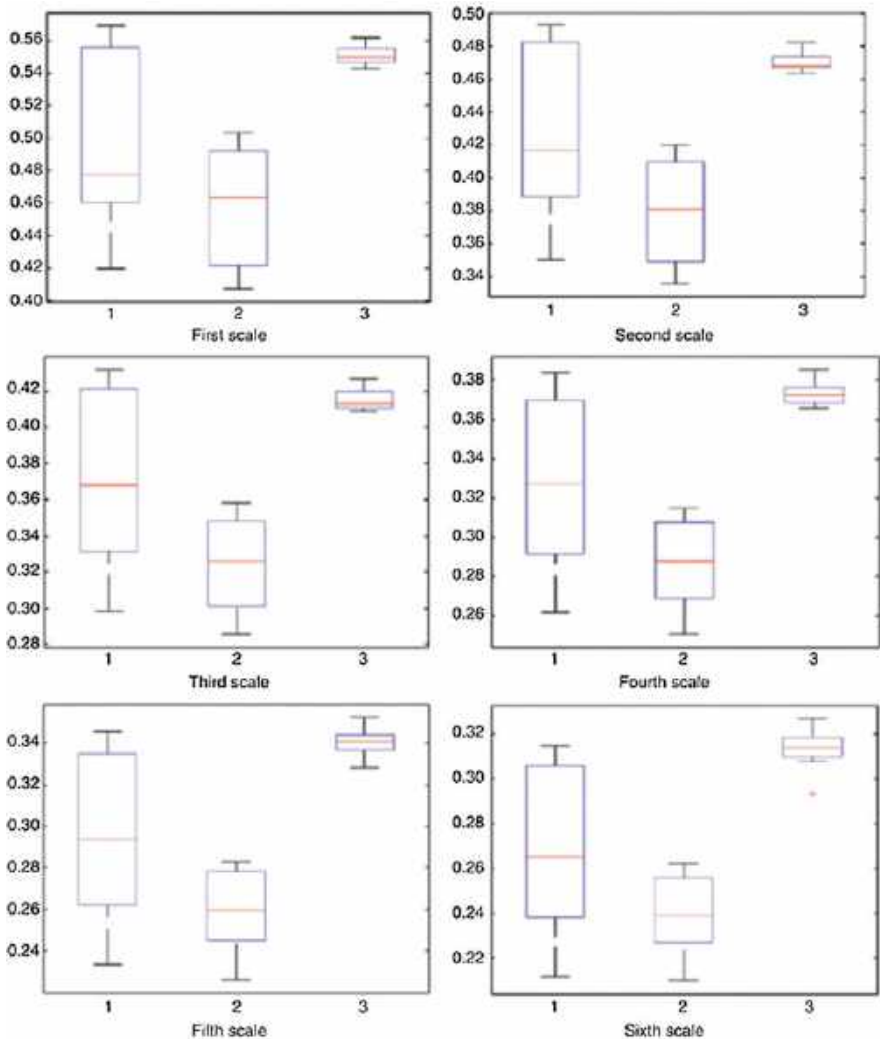


Fig. 22.5 Multiscale analysis-based boxplots in brain imaging of patients affected by MCI (1), AD (2), vs. images of disease-free brains (3)

22.3 Other Applications of Fractal Analysis in Neuroimaging

In the context of brain tumor detection, the fractal dimension was found to be effective in distinguishing between tumor and nontumor regions in various imaging modalities. For instance, Zook and Iftekharuddin [42] used the analysis of variance (ANOVA) to test whether fractal dimension statistics in healthy subjects are significantly different from those affected by brain tumors. They found that FD

might be exploited successfully to determine the possible presence and location of brain tumors in MR and CT images. Wardlaw et al. [41] quantified the blood oxygen level-dependent (BOLD) signals' temporal complexity using the fractal dimension with the purpose to allow maps to be generated that are physiologically distinctive in nature and allow potential insight into tumor microvasculature. In other words, they used fractal dimension mapping to identify physiological regions that are dominated by low-frequency vascular components or by high-frequency components. They concluded that low fractal dimension regions are significant as they highlight probable regions of regularized tumor metabolism and microvascular patterns. Iftekharruddin et al. [16] investigated the effectiveness of fusing two texture features along with intensity in multimodal magnetic resonance images for pediatric brain tumor segmentation and classification. The two texture features were obtained with the piecewise-triangular-prism-surface-area (PTPSA) algorithm for fractal feature extraction and the fractional Brownian motion (fBm) framework, which combines both fractal and wavelet analyses, for fractal wavelet feature extraction. The ensuing tests were applied to MRI T1, T2, and fluid-attenuated inversion recovery (FLAIR) sequences, and the obtained experimental results showed that the fusion of fractal, fractal wavelet, and intensity features in multimodality MR images offers better tumor segmentation results when compared to that of just fractal and intensity features in single modality MR images. Indeed, the segmentation accuracy was 100%. In addition, the classification results obtained with a multilayer feedforward neural network with automated Bayesian regularization to classify the tumor regions from nontumor regions indicated that the true-positive fraction (TPF) values range from 75% to 100% for different patients, with an average value of 90%. Di Ieva et al. used fractal analysis to characterize MR susceptibility-weighted images and to differentiate types and grades of brain tumors [4, 7] (see Chap. 20). Moreover, the fractal dimension, lacunarity, and other fractal indexes can be used as radiomics parameters to characterize the shape and features of brain tumors on neuroimaging, mainly, MRI [17, 20]. Indeed, fractal-based radiomics features can be mined through computational analysis to produce quantitative imaging biomarkers useful in the clinical setting [17]. For example, Petrujkić et al. used computational quantitative MR image features, including fractal radiomics parameters, to differentiate glioblastoma from solitary brain metastases on multimodal MR imaging [35].

Other studies [3, 4] have also focused on the role of fractal dimension in the characterization of brain images. For example, Akar et al. [1] estimated the fractal dimension of the cerebellum in order to characterize Chiari malformation type I (CM-I). They calculated the areas of white matter (WM) and gray matter (GM) and computed the corresponding fractal dimension values by using the 2D box-counting method in healthy and CM-I patients' magnetic resonance images. The obtained results indicated that CM-I patients had significantly higher fractal dimension values of GM and WM tissues compared to healthy patients. In addition, they found that the fractal dimension and area values of GM tissues in the patients group were correlated. They concluded that the fractal dimension values of the cerebellum may be a useful marker for investigation of CM-I patients. The findings were extended

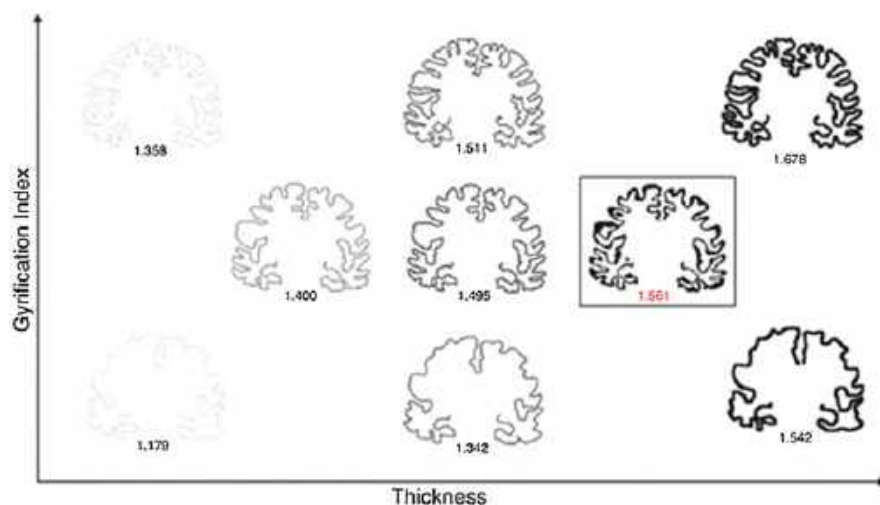


Fig. 22.6 Changes in cortical thickness (*horizontal axis*) and gyrification index (*vertical axis*) of the brain (coronal slice from a control healthy subject seen in the box) affect the fractal dimension (indicated by the number below the slice) of the cortical ribbons. (Reprinted from King et al. [22], with permission from Springer)

at Di Ieva's lab by using deep learning on MRIs of patients affected by Chiari malformation type I [40]. Sandu et al. [37] used fractal dimension values to reveal brain structure irregularities in patients with schizophrenia as they allow quantifying the shape complexity of cortical folding of the human brain. They segmented magnetic resonance images and the calculated fractal dimension values for the gray/white matter boundary for the whole brain and the hemispheres separately by using the box-counting and Minkowski–Bouligand methods. The authors found that the patients affected by schizophrenia showed higher FD values than the healthy group, for the whole brain volume and right hemisphere. King et al. [23] computed the fractal dimension of the cortical ribbon to distinguish between normal controls and mild Alzheimer's disease patients using a custom cube-counting triangle-intersection algorithm (Fig. 22.6). They found that the fractal dimension of the cortical ribbon showed statistically high differences between the control and Alzheimer's disease subjects. It was concluded that the fractal dimension of the cerebral cortical ribbon is a useful quantitative marker of the cerebral cortex structure in mild Alzheimer's disease.

The fractal analysis has also been shown to be more accurate than other methods for the detection of changes in the brain white matter in diseases such as multiple sclerosis and amyotrophic lateral sclerosis (ALS) [9, 10, 36]. Esteban et al. [9, 10] investigated whether fractal dimension estimates of gray matter in T1 magnetic resonance imaging sequences can identify gray matter abnormalities in patients with multiple sclerosis in the early phase of the disease. First, the image was segmented by a voxel-based morphometric approach optimized for multiple sclerosis. Then,

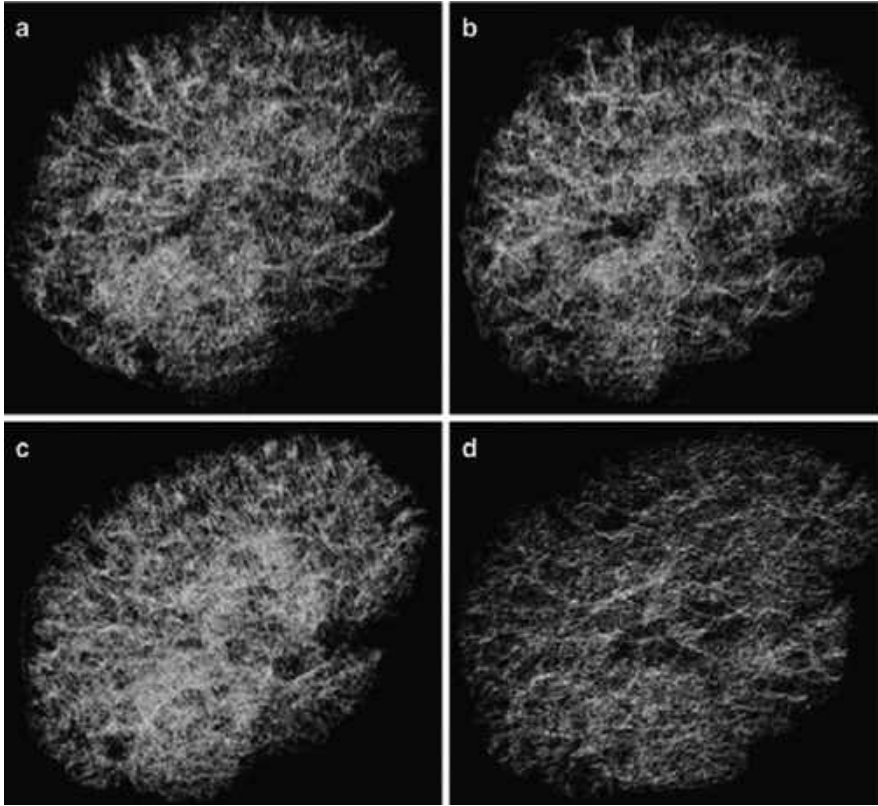


Fig. 22.7 3D rendering of white matter skeleton image in a typical (a) control subject, (b) amyotrophic lateral sclerosis (ALS) patient with dementia, (c) ALS patient with corticospinal tract hyperintensity, and (d) ALS patient without corticospinal tract hyperintensity. The FD has been shown to be more sensitive than volumetric voxel-based morphometry in detecting changes of white matter. (Reprinted from Rajagopalan et al. [36], with permission from the Publisher)

they estimated the three-dimensional FD of the gray matter in multiple sclerosis patients and healthy controls. They found that patients with multiple sclerosis had a significant increase in the fractal dimension of the gray matter in comparison to the images of the healthy group. Figure 22.7 shows examples of white matter skeleton images in healthy subjects vs. some pathological conditions.

The FD measures were also used in various other neuroimaging applications. Jayasuriya et al. [18] used FD estimates of the textures present on MR images to detect the midsagittal plane (MSP) of the brain. They found that fractal dimension measures are robust in identifying this symmetry plane in three-dimensional brain magnetic resonance images. In addition, they were found to be robust with respect to strong noise and low resolution. An optimized vascular fractal tree model using the level set distance function was developed in [2] to construct a region of human cerebral vasculature, which is enclosed in a realistic brain envelope. The

optimized vascular fractal tree model was found to be successful in producing a geometrically optimized model of the cerebral vasculature with the preferential distribution of large arteries on the brain cortex. Computational fractal-based models have also been applied for the characterization of water molecule diffusion in vivo on diffusion tensor MR imaging [19]. Fractal analysis has also been used to investigate the accuracy of synthetic MR images generated by novel imaging modalities, as introduced by Liu et al. [30].

A comprehensive survey on fractals in neuroimaging can be found in [6], as well as in Chaps. 13, 14, 18, 20, and 21.

22.4 Conclusion and Future Perspective

The efficacy of fractal-based features for the classification of brain magnetic resonance images is demonstrated through many recent studies in this area of biomedical engineering. Range-scale analysis, detrended fluctuation analysis, and multiscale analysis based on the generalized Hurst exponent were adopted to estimate fractal features; artificial neural networks and support vector machines were employed to perform classification. The performance of fractal-based features was compared to conventional methods. The results so far show the effectiveness of fractal features in the classification of healthy against unhealthy images both in binary and multiclassification schemes.

For future works, several directions have been proposed in the literature. For instance, previous works [24, 26, 29] can be extended by focusing on the appropriate selection of scale-based features and applying the fractal-based features to other brain imaging modalities such as functional magnetic resonance imaging, CT, and positron emission tomography (PET). Zook and Iftekharuddin [42] suggested including examining the effects of tumor type and size as well as the effect of noise in fractal dimension analysis. Wardlaw et al. [41] recommended to effectively filter the cardiac and respiratory frequencies from the BOLD signal prior to generating fractal dimension maps so as to improve identification of regions of active metabolism. Iftekharuddin et al. [16] suggested to direct future works to discriminate multiple types of brain tissues such as the white matter, gray matter, cerebrospinal fluid, and skull from solid tumor and edema. Akar et al. [1] recommended comparing the effectiveness of fractal dimension measures in detecting Chiari malformation type I anomaly with other disorders such as MS, fibromyalgia, migraine, spinal cord tumors, and Chiari Malformation type II. Jayasuriya et al. [18] suggested using the location of midsagittal plane for asymmetry analysis of pathological brains. The use of fractal analysis in aneurysms and arteriovenous malformations, aimed at identifying new diagnostic and prognostic neuroimaging biomarkers of cerebrovascular diseases [5], are discussed in Chaps. 17 and 18, respectively.

In the connectomic era, new insights will be offered by the application of computational fractal-based analyses into the studies of diffusion tensor imaging.

Moreover, the application of fractal analysis to nuclear medicine tools [single-photon emission computed tomography (SPECT) and positron emission tomography (PET)] may offer new biomarkers useful in the clinical setting [32].

As can be seen from these potential perspectives, the field is still open for discovery and improvement.

Appendix A: Appendix: Fractal Analysis Techniques

In this appendix, we summarize fractal analyses techniques used in recent studies related to brain magnetic resonance image classification, including classic range-scale analysis, detrended fluctuation analysis, and generalized Hurst exponent.

Range-Scale-Based Hurst Exponent

The Hurst exponent H [15] characterizes the scaling behavior of the range of cumulative deviations of a given signal y from its mean. The evaluation of Hurst exponent H following the methodology of Mandelbrot and Wallis [31] and called range-scale (R/S) analysis that uses the range of the partial sums of deviations of the signal y from its mean, rescaled by its standard deviation. The R/S statistic is given by:

$$(R/S)_n = \frac{1}{\sigma_n} \left[\max_{1 \leq k \leq n} \sum_{j=1}^k (y_j - \bar{y}_n) - \min_{1 \leq k \leq n} \sum_{j=1}^k (y_k - \bar{y}_n) \right] \quad (22.1)$$

where n is the length of the signal, $\bar{y}$ is its mean, and σ is its standard deviation given by:

$$\sigma_n = \left[\frac{1}{n} \sum_{j=1}^n (y_j - \bar{y}_n)^2 \right]^{1/2} \quad (22.2)$$

Hurst [15] found that many natural signals can be empirically described by the following relation:

$$(R/S)_n = (n/2)^H \quad (22.3)$$

The exponent H is estimated for the whole signal length n . Mandelbrot and Wallis [31] incorporated the ordinary least square regression to estimate it:

$$\log [(R/S)_n] = \alpha + H \log(n) + \varepsilon \quad (22.4)$$

where $\log (\cdot)$ is the logarithmic operator, α is the constant term of the regression, and ε is the error term.

There exists a simple relationship between Hurst exponent and the fractal dimension D of the original signal. For instance, if the signal is self-similar and self-affine, the relationship is $D = 2 - H$ [11]. Thus, Hurst exponent also measures the global fractal dimension or roughness of a one-dimensional signal.

Detrended Fluctuation Analysis

Peng et al. [34] developed the detrended fluctuation analysis (DFA) to appropriately handle nonstationary data. Its main advantage is avoiding the spurious detection of long-range dependence due to nonstationary data. For a given signal y , the algorithm is described as follows:

- (a) Define the suite x_n of the cumulative series of original signal y_i fluctuations about its mean:

$$x_n = \sum_{i=1}^n (y_i - \bar{y}) \quad (22.5)$$

x_n divided into boxes of equal length τ .

- (b) In each box, fit the local trend of x_n by a polynomial $P(\tau, n)$ that represents the local trend of the box.
- (c) For the given τ box size, compute the root-mean-squared detrended fluctuation of the signal x_n as:

$$F(\tau, n) = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - P(\tau, n))^2} \quad (22.6)$$

The last step is repeated for each of the available τ scales (box size) to obtain the empirical relationship between the overall fluctuation $F(\tau, n)$ and the box size τ :

$$F(\tau, n) \propto \tau^\alpha \quad (22.7)$$

The scaling exponent α quantifies the empirical strength of the long-range power-law correlations of the signal. If the signal is not random, it is characterized by long-range correlation features.

Generalized Hurst Exponent

Consider a signal $s(t)$ defined at discrete time intervals $t = v, 2v - T$ over a period T that is an integer multiple of v . Then the q th-order moments of the distribution that characterize the statistical evolution of $s(t)$ are defined as follows [8]:

$$K_q(d) = \frac{\langle \|s(t+d) - s(t)\|^q \rangle}{\langle |s(t)|^q \rangle} \quad (22.8)$$

where $d \in [v, d_{\max}]$ is a time interval and $d_{\max}$ is its predetermined upper limit. The generalized Hurst exponent $H(q)$ is defined from the scaling behavior of $K_q(d)$ according to the following empirical relation [8]:

$$K_q(d) \propto \left(\frac{d}{v} \right)^{qH(q)} \quad (22.9)$$

If $K_q(d)$ and d satisfy a linear relationship in log–log scale for a given order q , the Hurst exponent $H(q)$ can be estimated by running a linear regression of $\log(K_q(d))$ versus $\log(d)$. The generalized Hurst exponent $H(q)$ describes the long-memory dependence or persistence in the signal $s(t)$. The multiscaling structure of signal $s(t)$ is related to the different orders q of $H(q)$.

In general, when $H(q) > 0.5$, the signal fluctuations related to order q are persistent. When $H(q) < 0.5$, the signal fluctuations related to order q are anti-persistent. Finally, the signal fluctuations are those of a random walk if $H(q) = 0.5$ [8]. Notice that $H(q = 2)$ corresponds to the classic Hurst exponent.

In order to determine Hurst exponent, the original MRI is transformed first into a one-dimensional signal by row concatenation [24, 25, 27, 28]. Then, classic range-scale analysis, detrended fluctuation analysis, and generalized Hurst exponent are applied to the obtained one-dimensional signal.

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Chapter 23

Computational Fractal-Based Analysis of MR Susceptibility-Weighted Imaging (SWI) in Neuro-Oncology and Neurotraumatology



Antonio Di Ieva

Abstract Susceptibility-weighted imaging (SWI) is a magnetic resonance imaging (MRI) technique able to depict the magnetic susceptibility produced by different substances, such as deoxyhemoglobin, calcium, and iron. The main application of SWI in clinical neuroimaging is detecting microbleedings and venous vasculature. Quantitative analyses of SWI have been developed over the last few years, aimed to offer new parameters, which could be used as neuroimaging biomarkers. Each technique has shown pros and cons, but no gold standard exists yet. The fractal dimension (FD) has been investigated as a novel potential objective parameter for monitoring intratumoral space-filling properties of SWI patterns. We showed that SWI patterns found in different tumors or different glioma grades can be represented by a gradient in the fractal dimension, thereby enabling each tumor to be assigned a specific SWI fingerprint. Such results were especially relevant in the differentiation of low-grade versus high-grade gliomas, as well as from high-grade gliomas versus lymphomas.

Therefore, FD has been suggested as a potential image biomarker to analyze intrinsic neoplastic architecture in order to improve the differential diagnosis within clinical neuroimaging, determine appropriate therapy, and improve outcome in patients.

These promising preliminary findings could be extended into the field of neurotraumatology, by means of the application of computational fractal-based analysis for the qualitative and quantitative imaging of microbleedings in traumatic brain injury patients. In consideration of some evidences showing that SWI signals are correlated with trauma clinical severity, FD might offer some objective prognostic biomarkers.

In conclusion, fractal-based morphometrics of SWI could be further investigated to be used in a complementary way with other techniques, in order to form a

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holistic understanding of the temporal evolution of brain tumors and follow-up response to treatment, with several further applications in other fields, such as neurotraumatology and cerebrovascular neurosurgery as well.

Keywords Brain tumors · Fractal dimension · SWI · Susceptibility-weighted imaging · Traumatic brain injury

23.1 Introduction

Susceptibility-weighted imaging (SWI) is a magnetic resonance imaging (MRI) technique, introduced in the mid-1990s [34, 35]. SWI is able to depict the magnetic susceptibility produced by different substances, such as deoxyhemoglobin, calcium, and iron, with a high sensitivity, relative to other MRI sequences. SWI is mainly used to show vasculature (e.g., intratumoral microvasculature in brain tumors) and microhemorrhages, offering a valid tool to investigate several neurological diseases (e.g., multiple sclerosis) along with neurotrauma in traumatic brain injury (TBI) patients and grade tumors, and assist in determining treatment or prognosis. Several applications have been reviewed in the different fields of clinical neurosciences, namely, neurology, neuro-oncology, cerebrovascular diseases, neurotraumatology, and functional neurosurgery (see [24]), summarized in Table 23.1.

Table 23.1 Synopsis of the applications of SWI in different fields of clinical neurosciences [24]

Applications of SWI in the clinical neurosciences	
Neurology	Multiple sclerosis, amyotrophic lateral sclerosis
	Iron deposits
	Brain calcifications
	Stroke
Neuro-oncology	Brain tumor architecture, vasculature, bleedings, necrosis
	Brain tumor differential diagnosis (grading and histotypes)
	Differential diagnosis between brain abscess and tumor
	Brain tumor follow-up and/or response to treatment
Cerebrovascular	Thrombosis detection
	Stroke
	Vascular malformations
Neurotraumatology	Hemorrhage/microhemorrhage detection in TBI
	Dichotomize hemorrhagic from nonhemorrhagic diffuse axonal injury (DAI)
Functional neurosurgery	Anatomical localization of veins and deep gray nuclei for targeting in deep brain stimulation and Gamma Knife Radiosurgery

23.2 Technical Aspects of SW Imaging

The signal intensity on the MR image is determined by three basic parameters: (1) proton density (PD), (2) T1 relaxation time, and (3) T2 relaxation time. The T1 (longitudinal) and T2 (transverse) relaxation times define the way protons revert back to their resting states after the initial radio-frequency (RF) pulse. The transverse magnetization starts decreasing in magnitude immediately as protons start going out of the phase. This process of dephasing and reduction in the amount of transverse magnetization is called transverse relaxation. Spin–spin relaxation (T2) and local field inhomogeneities (T2*) are the two mechanisms leading to dephasing and form the basis of transverse relaxation in gradient recalled echo (GRE) sequences [13]. Substances causing local field inhomogeneities, and therefore T2* shortening, include deoxyhemoglobin, iron, calcium, metallic implants, paramagnetic contrast, and blood products (iron). As a result of T2* shortening, these substances cause signal loss on T2*-GRE images and produce areas of hypointensity, which can be detected.

A major limitation with GRE is the fact that T2* is a 2D sequence that produces thicker image slices if a long echo time (TE) is used [35], which causes smaller hemorrhages to not be depicted in T2* images [59]. The ability to employ a long TE while still preserving image quality can only be accomplished with 3D GRE [35]. This idea, initially developed by Reichenbach et al. [84], nonetheless exploits the blood–oxygen level-dependent (BOLD) effect and uses refined phase data for contrast enhancement in MR images [35].

The SW sequence is a high-spatial resolution T2*-weighted sequence with additional post-processing using both the magnitude and phase information. To create SW images, a high-pass filter is used to develop a phase mask from MR phase images [35]. The filtered phase images are then multiplied with the magnitude images to enhance the conspicuity of smaller veins and other substances causing susceptibility effects [13]. By performing the aforementioned steps, SW phase images can be used for the differentiation between diamagnetic (i.e., calcium) and paramagnetic (i.e., deoxyhemoglobin, hemosiderin, and ferritin) substances [101].

SW images can be further analyzed by determining the maximum intensity projection (MIP) and minimum intensity projection (mIP), which provide information on the voxels presenting with the maximum and minimum intensities, respectively. This information on MIP and mIP found via SW images is especially useful to visualize the geometry of cerebral veins and deep gray matter structures, among other components in the brain.

SWI has been used in MR machines at different magnetic fields, including 1.5 Tesla (T), 3 T (high magnetic field), and 7 T (ultrahigh magnetic field). The higher the magnetic field, the higher the signal-to-noise (SNR) ratio, thereby offering higher resolution of fine details in the brain structures, such as submillimeter-sized veins which are not possible to see by means of other neuroimaging techniques [21]. Very high field strengths may cause distortion of some structures characterized

by high iron or calcium concentration [65] and create susceptibility artifacts, especially in proximity of the bone.

23.3 SWI in Neuro-Oncology

Several studies have shown higher sensitivity of SWI in relation to other MRI techniques (such as T1, T2, contrast-enhanced T1, and FLAIR) to detect microvasculature and intratumoral microhemorrhages in brain tumors. Table 23.2 offers a synopsis of the literature on SWI in neuro-oncology. As illustrated in Chap. 27, microvasculature might be used to differentiate brain tumors of different histological grades, such as low-grade gliomas from higher-grade gliomas. Glioblastoma (grade IV) is among the most vascularized human cancer [62] also showing a very high microvascular heterogeneity [20]. Moreover, its architectural heterogeneity is also shown by the presence of necrotic and/or hemorrhagic areas. Indeed, the previous classification systems defined it *multiforme*, from Latin, meaning “multishapes.” Importantly, these features can be clearly depicted in SW imaging (Fig. 23.1).

23.3.1 Morphometrics and Fractal-Based Analysis of SWI in Brain Tumors

Several morphometric techniques have been suggested to quantify the different patterns shown in SW images in order to differentiate between tumor subtypes (Table 23.3). These techniques demonstrate the potential validity of SWI as an image biomarker useful in the differential diagnosis in neuroimaging. For example, hypointense spots on SWI images have been quantified by the frequency and maximum diameter [103]. Furthermore, the hypointense dot-like or linear structures shown in SW imaging of brain tumors have been quantified in terms of intratumoral susceptibility signals (ITSSs), by means of a semiquantitative method, which has been shown to be useful in differentiating cerebral lymphomas from high-grade gliomas. By means of this technique, GBM has been shown to have higher ITSS scores compared to lymphomas. More specifically, GBMs typically elicit a much higher intratumoral SWI signal relative to lymphomas, which show very little, if any, intratumoral SWI signal [77, 78]. However, a major limitation of this technique pertains to the subjectivity of the measure since the computation of ITSSs is mainly operator-dependent. Hori et al. introduced a grading system based on the ratio of SWI hypointensity relative to the size of the tumor, which led to a more accurate correlation between score and tumoral grade being found when compared to grading [41]. The susceptibility grading system ranges from 0 (no SWI hypointensity detectable) to 3 (presence of vascular structures alone), with grade I signifying hypointensity in less than half of the tumor (and, according

Table 23.2 Literature synopsis of applications on SWI in neuro-oncology

Author, year of publication, Ref	Main findings
Baldawa et al., 2012 [5]	Blooming effects shown in SWI indicating intratumoral hemorrhage
Bian et al., 2014 [10]	7 T SWI is more effective in detecting the number of CMBs from radiation injury involving the frontal lobe, parietal lobe, cerebellum, or brainstem; 7 T and 3 T SWI are equally sensitive in detecting the amount of CMBs if radiotherapy applied to the temporal lobe or basal ganglia
Chen et al., 2020 [14]	ITSS is useful to differentiate intracranial solitary fibrous tumors from meningiomas
Deguchi et al., 2019 [17]	Pretreatment intratumoral SWI signals correlate with response to high-dose methotrexate and progression free-survival in primary brain tumors
Ding et al., 2014 [26]	PCNSL has fewer intralesional vessels found on SWI relative to high-grade gliomas and brain metastases
Deistung et al., 2013 [18]	QSM, which uses SWI data, can differentiate intratumoral hemorrhagic components from calcification in glioblastoma
Di Ieva et al., 2012 [22]	Fractal analysis applied in SW images effectively monitored changes in intratumoral microvasculature in patients undergoing antiangiogenic therapy
Di Ieva et al., 2013, 2016 [23, 25]	Fractal analysis applied in SW images objectively determined tumor grades in gliomas and differentiated different brain tumor histotypes
Fellah et al., 2011 [27]	SWI can monitor changes in intratumoral microvasculature over time in patients undergoing antiangiogenic medication
Furter et al., 2014 [29]	PCNSL can be differentiated from GBM based on ITSSs. A threshold of 1.5 ITSSs or higher is indicative of GBM, while an ITSS value less than 1.5 is suggestive of PCNSL
Gaudino et al., 2020 [30]	ITSS is useful for typing and grading in pediatric brain tumors at 1.5 T
Grabner et al., 2012 [32]	SWI at 7 T depicts intratumoral microvasculature and microhemorrhages in patients taking antiangiogenic medication; SWI findings correlate with histopathology
Gramsch et al., 2013 [33]	SWI lesions did not correlate with metastases transformation; abnormalities on contrast-enhanced T1 can detect metastases
Hori et al., 2010 [41]	Ratio of hypointensity from SW images within tumor boundaries is the most objective measure in grading
Jurkiewicz et al., 2010 [49]	SWI detected hemosiderosis after tumor resection
Kang et al., 2021 [50]	SWI can be performed after gadolinium injection without information loss or signal change and is useful in evaluating brain tumors

(continued)

Table 23.2 (continued)

Author, year of publication, Ref	Main findings
Kim et al., 2009 [53]	ITSSs observed in SWI can differentiate high- and low-grade tumors
Khosasani et al., 2023 [52]	SWI is useful for glioma grading
Li et al., 2010 [55]	SWI can clearly depict intratumoral hemorrhage and microvasculature; image findings correlated with pathology
Li et al., 2015 [56]	Low-grade gliomas have less ITSSs compared to high-grade gliomas
Löbel et al., 2010 [59]	SWI detected more hemorrhages over time than T2*
Lou et al., 2009 [60]	SWI can clearly detect early BGG; tumor visibility on SWI poorer in late BGG
Lou et al., 2012 [61]	SWI can differentiate between intracranial ectopic germinoma from subacute lacunar infarct using hypointensity differences
Lupo et al., 2012 [63]	More CMBs discovered in the previous tumor region after radiation therapy using 7 T SWI
Lupo et al., 2013 [64]	Higher percent volume of hypointensity on SW image of a T1-weighted contrast-enhancing lesion best predicted patient response to concomitant therapy
Menon et al., 2009 [67]	SWI could detect calcification in subfrontal gangliocytoma
Moenninghoff et al., 2010 [68, 69]	7 T SWI allowed intratumoral vessel branches 250 µm in size to be depicted
Natsumeda et al., 2022 [71]	7 T SWI can be used to differential astrocytomas from oligodendrogliomas
Nishiguchi et al., 2013 [72]	SW-PRESTO can assess ischemia before and after tumor embolization surgery
Nossek et al., 2009 [75]	SWI helped to determine tumor malignancy based on microvascular assessment
Park et al., 2009 [76]	Use ITSSs in SW images to objectively grade tumors; ITSSs correlate with rCBVmax
Park et al., 2010 [77]	GBM distinguished from metastatic tumors using ITSSs in SW images
Peters et al., 2012 [78]	Glioblastoma correctly differentiated from lymphomas using ITSSs in SWI
Peters et al., 2013 [79]	SWI can detect lesions in medulloblastoma patients who underwent radiation therapy
Petrujčić et al., 2019 [80]	Texture features of multimodal MR imaging (including SWI) are a useful tool in differentiating glioblastoma from solitary brain metastases
Qin et al., 2022 [82]	SWI can distinguish radionecrosis from recurrence in gliomas but not in brain metastases after radiotherapy
Radbruch et al., 2012 [83]	PQ of ITSSs can distinguish between cerebral metastases originating from different sources (e.g., breast carcinoma and malignant melanoma); however, PQ could not differentiate between all metastase types (e.g., breast carcinoma and bronchial carcinoma)

Sehgal et al., 2006 [87]	SWI is more sensitive than contrast-enhanced T1 and T2 in depicting tumor boundaries, intratumoral microvasculature, and internal architecture; vessel findings in SWI correlate with histopathology
Tanji et al., 2022 [91]	High intratumoral susceptibility signade on SWI is a risk factor for post-stereotactic biopsy hemorrhage
Toh et al., 2012 [92]	SWI can distinguish glioblastoma from brain abscess, unlike in T2
Vossough et al., 2012 [96]	Red nucleus degeneration seen in SWI after tumor resection in the posterior fossa
Wieczorek-Pastusiak et al., 2013 [98]	“Vessel view” computer counting program found more vessels in SWI than contrast-enhanced T1
Wu et al., 2009 [99]	SWI found more (smaller) calcified lesions than CT
Zeng et al., 2011 [102]	SWI can assess the extent of radiation injury based on the number of hemorrhagic lesions; findings matched histopathology
Zhang et al., 2010 [103]	Small brain metastases less than 1 cm and CMBs found in SWI, but not seen in contrast-enhanced T1 and T2*
Zulfiqar et al., 2012 [106]	Greater number of calcium deposits found intratumorally when using SWI and conventional MRI together, compared to conventional MRI alone
CMBs cerebral microbleeds, PCMSL primary central nervous system lymphomas, QSM quantitative susceptibility mapping, ITSSs intratumoral susceptibility signals, BGG basal ganglia germinoma, SW-PRESTO SW principles of echo shifting with a train of observations, rCBV _{max} maximum relative cerebral blood volume, GBM glioblastoma, PQ percentagewise quantification	

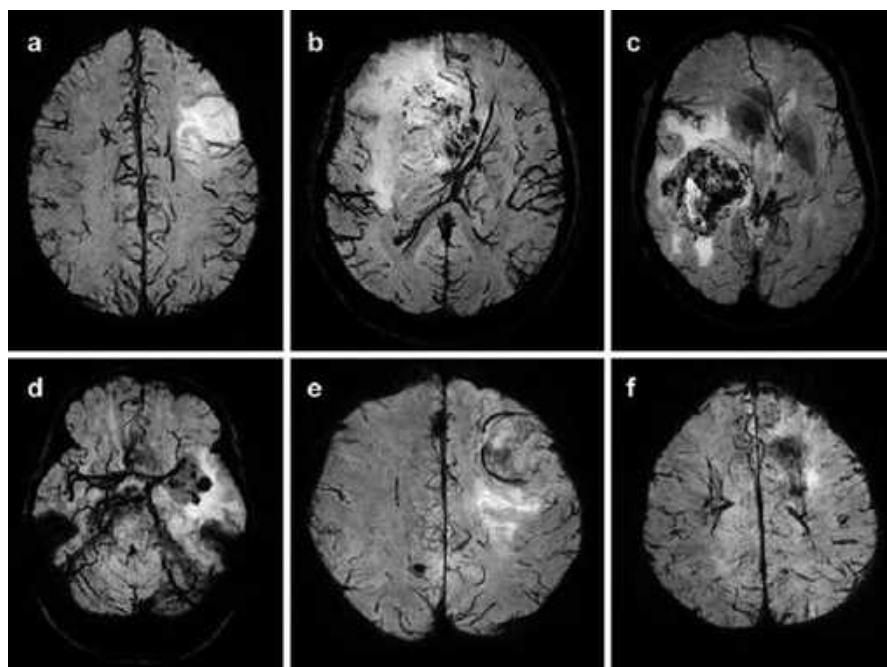


Fig. 23.1 Examples of different tumors showing distinct SWI signal patterns, shown in minimal intensity projection. (a) Left frontal grade II glioma ($FD_{SWI} = 1.25$). (b) Right fronto-callosal grade III glioma ($FD = 2.02$). (c) Right temporal grade IV glioma ($FD_{SWI} = 2.40$). (d) Left temporal metastasis ($FD_{SWI} = 2.10$). (e) Left frontal meningioma ($FD_{SWI} = 1.88$). (f) Left frontal lymphoma ($FD_{SWI} = 1.87$). (Reproduced from Di Ieva et al. [25] with permission of the Publisher)

to the authors, related to bleeding), while the minimum requirement for grade II is characterized by hypointensity found in more than half of the tumor in one or more SW images depicting linear and/or tortuous shapes (which is associated with bleeding and/or vascular structures). This technique offered advancement in the quantification of the intratumoral SWI patterns, although still limited by some intra- and interobserver variability [23]. In order to overcome the subjectivity of the scoring systems, percentagewise quantification (PQ) of the ITSSs has been suggested to be useful in characterizing brain metastases, although the technique focuses on quantitative aspects while neglecting the distribution and morphology of ITSS [83].

Quantitative susceptibility mapping (QSM) offers a very promising mean for susceptibility quantification of different brain structures [18, 36]. Specifically, QSM has the potential to produce lesion-specific susceptibility maps (derived from GRE phase data), which can provide more information on blood products and calcification [16, 86]. The ability to detect calcification in tumors can be useful for tumor grading and in deciding patient prognosis as calcification in tumors has

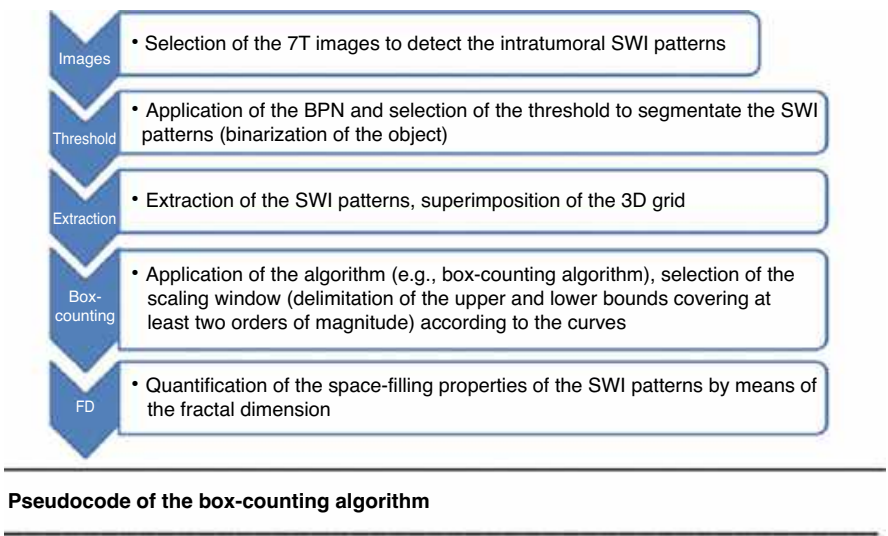
Table 23.3 Morphometric analyses used to quantify SWI in brain tumors

Technique	Description	Pros	Cons
Frequency and maximum diameter	Computer-aided or manual count and measure of the hypointensity spots	Easy to perform	It does not consider the morphology and distribution of the spots
ITSSs	Semiquantitative method to assess the morphology of the SWI signals	Useful for comparison among different tumors	Biased by operator-dependent variability
SWI/volume	Grading system based on the ratio of SWI hypointensity relative to the size of the tumor	Useful for comparison among different tumors	Limited by some intra- and interobserver variability
PQ	Quantification of the percentage of ITSSs	Very useful for comparison and follow-up	Strictly quantitative with no consideration on the shape and distribution of ITSSs
QSM	Quantification in ppm of the susceptibility of distinct anatomical and pathological regions of the brain	Very promising for precise susceptibility quantification of different structures	Requires optimal post-processing
FD _{SWI}	Fractal dimension computed by means of the box-counting method expressing the space-filling properties of the SWI pattern within the tumor	Single value able to give in a snapshot information on density, roughness, and geometrical complexity of the SWI pattern	Requires optimal computational processing. Not able to give information on the density alone, for example, but all the morphometric findings as a whole

FD_{SWI} SWI fractal dimension, *ITSSs* intratumoral susceptibility signals, *ppm* part per million, *PQ* percentagewise quantification, *QSM* quantitative susceptibility mapping

been found to be associated with longer patient survival compared to tumors without calcification [48, 87].

Finally, a computational fractal-based method to quantify the intratumoral SWI patterns was introduced by Di Ieva et al. in 2012 [22, 23]. The geometrical complexity of the intratumoral SWI-related architecture, resulting from microbleedings, necrotic areas, and/or neoplastic microvasculature, was quantified by means of the fractal dimension (FD). Most importantly, a relationship between FD of the SWI patterns (FD_{SWI}) and glioma grades was found [23]. FD was computed by means of the box-counting method. According to the principles of fractal geometry, higher values of FD signify higher space-filling properties of the intratumoral SWI patterns, meaning a higher density and geometrical complexity of the microvasculature and/or microbleedings within the tumor. It should be emphasized that apparently small changes of FD correspond to a high increase of the space-filling properties of the SWI patterns, when considering a linear approximation in a small interval of the



```
For z=1 to (number of images of the 7T-MR stack)
    Array[Z]=Binarize (image[z])

L= $\epsilon_{min}$ 
While L< $\epsilon_{max}$ 
    n[L]=count boxes on 3D grid of size L on Array
    L=L*1.1

FD=-[Slope of log(L) vs -log(n) graph]
```

Fig. 23.2 Algorithm and pseudo-code used to measure the 3D fractal dimension of the SWI patterns on 7 T MR images (BPN, brightness progressive normalization [85]). (Reproduced from Di Ieva et al. [22] with permission of the Publisher. The same algorithm has also been used on 3 T MR images [25])

nonlinear FD parameter. For example, an increase of the FD value from 2.07 to 2.23 corresponds to an increase of about 20% of the geometrical complexity of the object. Computing the FD on the stacks of MR images (i.e., on the 3D reconstructions of the tumor) gives rise to fractal values, whose noninteger and nonlinear values are included in a 3D space, between 0 and 3. Figs. 23.2 and 23.3 show the algorithm and technique for the computation of FD of SWI patterns.

When analyzing the SWI 7 T MR images of 36 patients affected by grades II–IV gliomas, we found a trend of higher FD values in higher-grade gliomas [23]. The mean FD value was 2.086 ± 0.413 , with the FD values ranging from a mean value of 1.682 ± 0.278 in grade II to 2.247 ± 0.358 in grade IV gliomas ($p < 0.05$) (Fig.

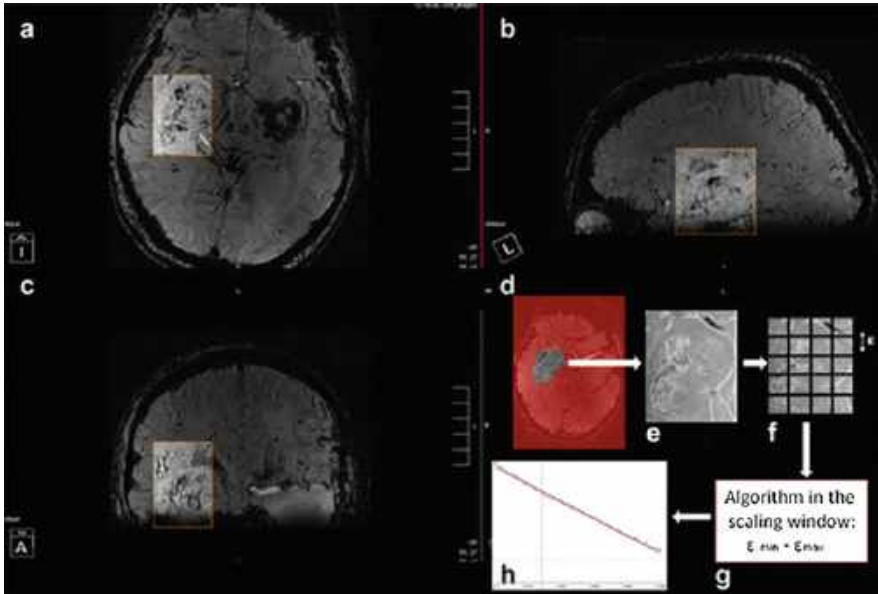


Fig. 23.3 Method used for fractal analysis of 3 and 7 T MR-SW images. In the example, (a) axial, (b) sagittal, and (c) coronal sections of 7 T MR images (Magnetom 7 T, Siemens Healthcare, Erlangen, Germany, performed at the Medical University of Vienna, Austria) centered on a recurrence of GBM (in the insets). In (d), the region of interest was extracted (i.e., the tumor), and in (e) there were segmentation and subsequent binarization of the SWI patterns (in the figure, the signal has been inverted, i.e., the SWI black hypointensities appear here as white). (f) Superimposition of the grid of boxes of ε size. The Z-axis sequential sections for the 3D reconstruction are shown according to the algorithm in Fig. 23.2. (g) Repetition of the algorithm with ε of different sizes in the scaling window ranging from $\varepsilon_{\min} = 0.23$ to $\varepsilon_{\max} = 23$ mm (10% increment of the box size at each step until to reach $\varepsilon_{\max}$). (h) Results plotted on a log-log graph (x-axis, ε , the side length of the box; y-axis, $N(\varepsilon)$, on the grid, the number of boxes of ε side which cover completely the SWI patterns). The slope of the graph of $\log [N(\varepsilon)]$ against $\log (1/\varepsilon)$ represents the fractal dimension. The software for the analysis was developed in visual C++ language by Carlo Russo. (Reproduced from Di Ieva et al. [22] with permission of the Publisher)

23.4). Figure 23.5 illustrates some examples of different intratumoral SWI patterns, which were found in the study.

To this first demonstration that fractal analysis in SW imaging can offer reliable and useful morphometric image biomarkers to distinguish between low- and high-grade gliomas, the application of the methods was extended further to other kinds of brain tumors, such as brain metastases, meningiomas, and lymphomas as well [25]. Moreover, considering that 7 T MR scanners are very expensive, not easily available, and are mostly used for research purposes, the study was performed at 3 T to show the feasibility of this computational neuroimaging technique on more commonly available scanners. In this study, we considered 78 patients affected by brain tumors of different histopathology (13 grade II gliomas, 11 grade III, and 25 grade IV gliomas, 15 metastases from different sources, 8 grade I meningiomas; and

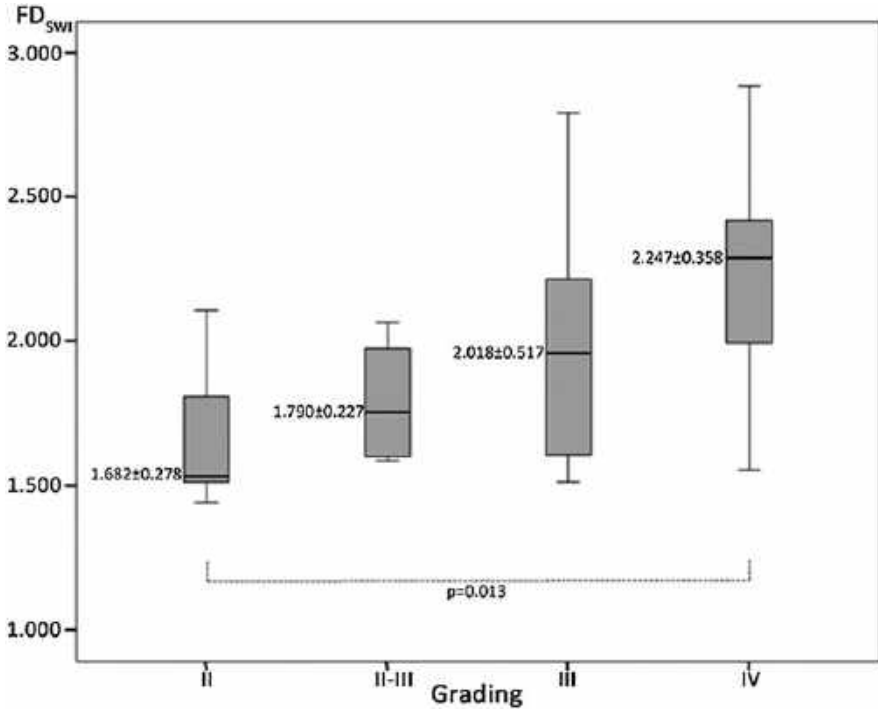


Fig. 23.4 Box plots showing the FD_{SWI} values for each histopathological group of gliomas. Four cases were histopathologically considered “ambiguous” and were put in the group “II–III” grades. The overall significance was statistically significant, $p < 0.05$, with the strongest significance in grade II versus grade IV gliomas ($p = 0.013$). (Reproduced from Di Ieva et al. [23] with permission of the Publisher)

6 primary B-cell lymphomas), computing the FD_{SWI} as well as the volume fraction of the SWI signal within the tumors. Both values were able to differentiate low grade from grades III and IV gliomas, as well as metastases and meningiomas, although FD alone was statistically more significant, even to differentiate lymphomas (mean $FD_{SWI} = 1.69 \pm 0.59$) from GBM (mean $FD_{SWI} = 2.03 \pm 0.45$, $p < 0.05$) (Fig. 23.1) [25]. These results confirmed the findings of the previous study performed at 7 T. More importantly, these findings demonstrate that the computational fractal-based analysis of the SWI patterns can help differentiate contrast-enhancing lesions, which often poses problems in the differential diagnosis for neuroimaging, such as between lymphomas and malignant gliomas.

Taken together, our findings showed that lower FD_{SWI} values are associated with tumoral vasculature, while higher values (signifying more space-filling objects) associated with intratumoral bleedings and/or necrotic areas, as previously indicated in qualitative, semiquantitative, and quantitative studies [19, 32, 41]. FD_{swi} appears to be a reliable morphometric parameter of the tumoral architecture, showing a gradient of complexity among tumors of different histotype and demonstrating

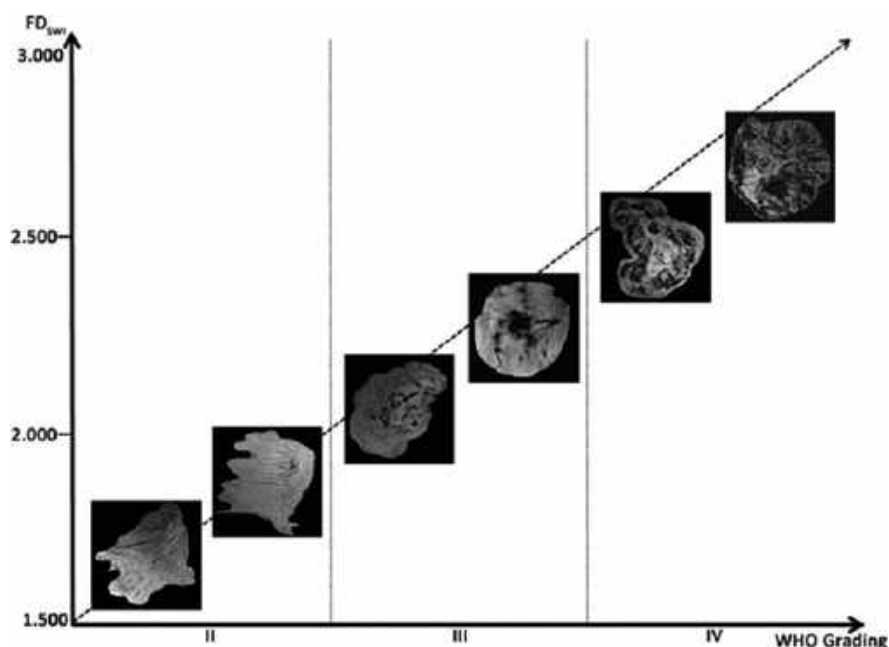


Fig. 23.5 Schema showing the correlation between the histological glioma grade and the FD_{SWI} values. The intratumoral SWI patterns consist of vascular structures (linear structures in grade II tumors, increasing in tortuosity and density in higher-grade gliomas) and conglomerated dot shapes, which most likely indicate intralesional bleeding and/or necrotic areas (as evident in grades III and IV gliomas). It should be noted that the image and fractal analysis were performed on the 3D volume of the reconstruction of the entire tumor in MRI, while the images shown here are selections of the axial sections taken from the whole stack. (Reproduced from Di Ieva et al. [23] with permission of the Publisher)

the ability to quantify the “architectural fingerprinting,” which has also been shown in the histological features and microvasculature of brain tumors as seen in histopathology [19] (see Chaps. 23 and 24).

Besides the grading and differentiation of brain tumors (*diagnostic aim*), SWI has also been applied longitudinally in follow-up analyses of patients, in order to track tumor evolution and/or treatment response (*follow-up aim*). Different percentages of SWI hypointensity in brain tumors have in fact been correlated with different response to therapy in newly diagnosed GBM [64]. Antiangiogenic drugs (e.g., bevacizumab, an antibody targeted against the vascular endothelial growth factor (VEGF)) can be used in recurrences of malignant gliomas, having three fundamental effects: (1) inhibition of the neoplastic angiogenesis, (2) normalization of the angioarchitecture (resulting in a more efficient delivery of drugs and oxygen to the cancer cells) [47], and (3) reduction of the peritumoral edema [2, 28, 74]. Intratumoral and/or brain bleedings are rare but known side effects of the treatment. SWI can be potentially used to assess whether antiangiogenic therapy is producing

favorable effects (like decreased intratumoral microvasculature) or unfavorable results (increased intratumoral microvasculature in nonresponder patients and/or microbleedings).

We conducted a pilot study on the application of computational fractal-based analysis to quantify the changes of the intratumoral SWI patterns, visualized by means of ultrahigh field (7 T) MRI, in four patients affected by malignant glioma recurrences undergoing second-line treatment with bevacizumab, and followed longitudinally [22]. The imaging was in fact performed at time T_0 (before beginning the antiangiogenic treatment) and repeated 2 weeks (T_1) and 4 weeks (T_2) after therapy was initiated. In two patients (like in the case illustrated in Fig. 23.6), it was shown that the microvascular proliferation and eventually the intratumoral bleedings were increasing over time, as confirmed by the increase of the value of FD from T_0 to T_1 . This means that, despite the antiangiogenic treatment, over a 1-month longitudinal follow-up, there was an increase in the intratumoral microvasculature and/or intratumoral bleedings, thereby suggesting that the patient is nonresponsive and/or experiencing side effects of the antiangiogenic treatment itself (like intratumoral bleeding). These morphological changes, qualitatively described by the neuroradiologists and quantitatively shown by the morphometric fractal analysis, were associated with a progressive neurological decline of the patients, who died within 10 weeks. A third patient was shown to improve neurologically after the antiangiogenic treatment. Although in this case, the neuroradiologist reported only a reduction of the peritumoral edema and no differences regarding the tumor itself, the computational analysis revealed a minimal decrease of FD over time ($FD_{SWI} = 2.34$ in T_0 , decreasing to 2.26 at T_1 and T_2). The fourth patient showed an improvement in the peritumoral edema along with improvement in clinical and neurological status (albeit temporarily), without any differences in FD_{SWI} values (however, this is because the tumor mainly presented a large necrotic area, visualized in forms of a huge black spot in SWI).

Although this was a limited study performed on a very small cohort of cases, it was the first to introduce fractal analysis as a quantitative method for objectively comparing sequential images taken during the neuroradiological follow-up of patients affected by brain cancer. Moreover, although such computational analysis can be performed on lower magnetic fields, it has been shown that 7 T MRI is also the best means to visualize the presumed intratumoral vascularity in gliomas [69, 71].

Nonetheless, it has to be emphasized that it is anatomically incorrect to speak about “microvasculature.” In fact, the mean radius of capillaries is 5–10 μm , while even by means of 7 T MRI, the smallest detectable vessel diameter is about 0.3 mm. Therefore, by means of neuroradiological imaging, it is not possible to visualize tiny angiogenic sprouts, for example, as it happens in histology. This is precisely the reason why we introduced the “microvascular fractal dimension” in histopathology, but in neuroradiology such term was replaced with “FD of the SWI pattern” [19, 23].

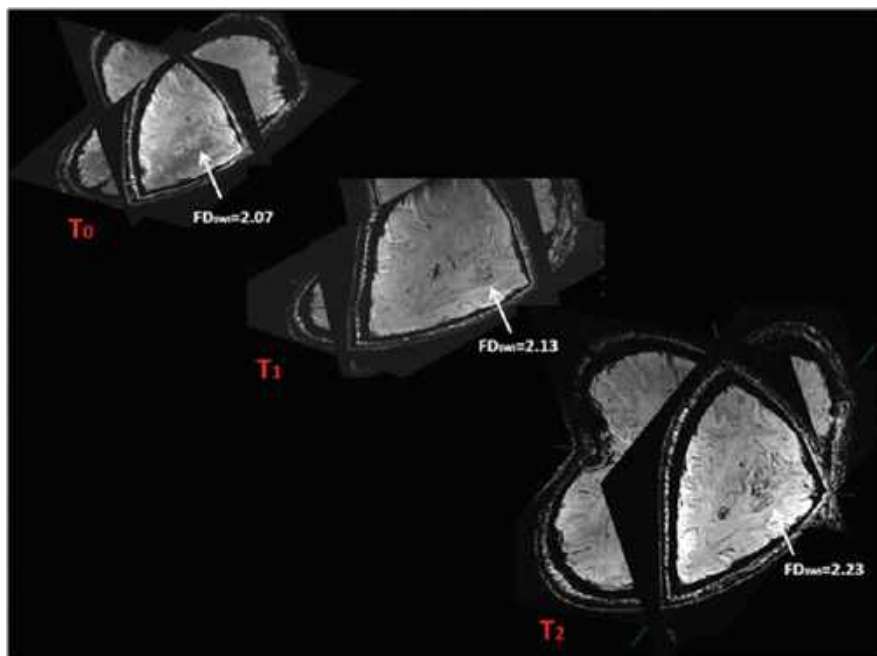


Fig. 23.6 Multiplanar reconstructions of the 7 T SWI MR images in time T_0 , which were repeated serially after 2 (T_1) and 4 weeks (T_2) since the beginning of antiangiogenic treatment. The *white arrows* show the recurrence of the GBM, and from T_0 to T_2 , it is evident that there is an increase in the signal of the SWI patterns. The value of the fractal dimension of the intratumoral SWI patterns (FD_{SWI}), reported for each temporal sequence, increased over time (see text). (Reproduced from Di Ieva et al. [22] with permission of the Publisher)

23.4 Future Perspective of SWI in Neurotraumatology

Traumatic brain injury is a leading cause of death and disability worldwide. Efforts to develop better biomarkers of brain injury are important to improve diagnosis, treatment, and prognosis of these patients. Newer MRI techniques hold promise in providing such a biomarker. SWI is an effective imaging technique to detect microbleedings in the white matter of patients after TBI [35]. Specifically, SWI is more sensitive in comparison to other techniques to detect very small cerebral microbleedings in the brainstem and corpus callosum. This neuroradiological technique has been shown to be useful in dichotomizing diffuse axonal injury (DAI) patients as hemorrhagic and nonhemorrhagic, which is very useful in terms of a clinical perspective since treatment, outcome, and prognosis between these two groups of patients are quite different [95]. Table 23.4 summarizes selected literature on the application of SWI to TBI patients.

Besides from differentiating between the subtypes of DAI, SWI can provide details on the clinical status of TBI patients. For example, the number and volume

Table 23.4 Literature synopsis of applications of SWI in neurotraumatology

Author, year of publication, Ref	Main findings
<i>Studies supporting SWI usage</i>	
Akiyama et al., 2009 [1]	SWI detected three times more lesions than T2*
Babikian et al., 2005 [4]	Lesions detected on the brainstem, thalamus, and basal ganglia offered the best prediction of neuropsychological outcome
Beauchamp et al., 2011 [7]	SWI found more lesions than CT and T2 MRI, even if MRI not performed until 2–8 weeks after TBI onset
Beauchamp et al., 2013 [8]	More lesions seen on SWI correlated with lower intellectual ability
Chastain et al., 2009 [12]	Median lesion number or volume on SW image could not predict good and poor outcome
Choi et al., 2014 [15]	Additional lesions found with SWI, but not FLAIR, were a good predictor of poor clinical outcome
Colbert et al., 2010 [16]	Presence of microhemorrhage was the best predictor of outcome in children with nonaccidental trauma
Geurts et al., 2012 [31]	SWI detected most lesions, especially in the brainstem and corpus callosum, compared with T2, T2*, and FLAIR
Hageman et al., 2022 [37]	SWI can visualize microbleeds and predict cognitive outcome and persistent posttraumatic complaints in patients with mTBI
Hasiloglu et al., 2011 [38]	SWI found more lesions than T2 and T2* in amateur boxers
Helmer et al., 2014 [39]	SWI can detect changes in hypointensity of smaller (<5 mm) cerebral microbleeds over different time points in patients who had concussions
Henry et al., 2015 [40]	SWI could detect shearing of axons in the corpus callosum and prefrontal cortex
Huang et al., 2013 [42]	SWI lesions in the brainstem could differentiate DAI from cerebral fat embolism
Huang et al., 2015 [43]	The presence of cerebral microbleeds detected in SW images of mTBI patients was related to deficits in short-term memory, as assessed by the digit span test
Keightley et al., 2012 [51]	SWI could not detect structural abnormalities after child concussion or mTBI
Liu et al., 2015 [57]	Greater number of cerebral microbleeds was detected on SWI relative to T2* imaging in patients with mTBI. The number of microbleeds found on SW images could predict the incidence of post-concussive syndrome, whereby an increase in every microbleed would increase the risk of post-concussive syndrome by 1.5 times
Maugans et al., 2012 [66]	Structural damage not observed in children with sports-related concussion
Moenninghoff et al., 2015 [70]	7 T SWI can detect greater numbers of cerebral microbleeds at a similar and higher spatial resolution compared to 3 T SWI
Niwa et al., 2011 [73]	Hypointensity changes in SW images indicate hemorrhage progression over long term
Park et al., 2009 [76]	CMBs in the frontal, occipital lobes, and brainstem seen only in TBI patients

(continued)

Table 23.4 (continued)

Author, year of publication, Ref	Main findings
Sigmund et al., 2007 [88]	SWI detects more lesions than T2 and FLAIR
Spitz et al., 2013 [90]	SWI is more sensitive than FLAIR to detect lesions from TBI, more lesions indicative of poorer neuropsychological outcome
Tong et al., 2003 [93]	SWI detected more lesions than T2-GRE, especially in the brainstem, corpus callosum, and cerebellum
Tong et al., 2004 [94]	Lesions found on SW images in 7 of 9 brain regions indicate higher likelihood of poor outcome
Toth et al., 2013 [95]	SWI can distinguish between mTBI with and without hemorrhage
Wang et al., 2011 [97]	SWI can detect hemorrhage in patients with acute cervical spinal cord injury
Wu et al., 2010 [100]	SWI effective in detecting intraventricular hemorrhage

TBI traumatic brain injury, *mTBI* mild traumatic brain injury

of SWI hypointensities in TBI patients have been correlated to the scores of the most used scale in neurotraumatology, the Glasgow Coma Scale (GCS) [3, 9, 16, 31, 45, 58, 76, 88, 90]. This means that SWI hypointensities are correlated with trauma clinical severity, and its quantification can offer some prognostic indicators (*prognostic surrogate biomarkers*). Furthermore, microbleeds can be quantified with the use of susceptibility maps [36]. Finally, the computational fractal-based analysis of SWI patterns holds promise in the quantification and description of brain lesions in TBI patients, with higher FD_{SWI} associated with worse prognosis, especially with higher FD_{SWI} values in the brainstem [Di Ieva, unpublished data].

23.5 Limitations

The methodological limitations pertaining to the application of fractal analysis to MR-SW imaging are related to the image resolution and intensity of the SWI signal, as already discussed above. Moreover, SWI requires post-processing computation, while the parameters as well as the number of multiplied phase masks should be standardized, in order to make the input data homogeneous (see Chap. 12). The choice of the threshold to segment and automatically extract the SWI hypointensities can be affected by an operator-dependent variability, and this can affect the fractal dimension computation as well. In order to solve the problem of different luminosities being found in different MR images, we introduced an algorithm for the progressive normalization of brightness, which normalizes the luminosity intensity, thereby making it possible to use the same threshold to extract the SWI hypointensities from each image [22, 85].

As summarized in [22], further studies should be focused on the following objectives:

1. Decrease the operator-dependent variability in selecting the edges of the tumor by moving toward automatic detection of the tumor outline using fractal-based algorithms [44, 54, 105] (see also Chap. 19) or neural networks [6, 46, 104].
2. Improve the imaging and post-processing parameters to standardize the methods and to reduce technical variability.
3. Compare 3 and 7 T MR images to the histological findings from brain tumors to definitively assess the significance of SWI patterns, similar to studies that were performed with 3 T MRI [81].
4. Combine the fractal analysis of SWI patterns at 7 T with some quantitative measurements of vessel morphology, as defined by magnetic resonance angiography [11] or perfusion parameters [89] to find valid and objective indexes for monitoring patients affected by brain tumors.
5. Associate the fractal quantification of the SWI patterns with other neuroradiological parameters, including (but not limited to) the volume of the tumor, peritumoral edema, or the histogram analysis of diffusion tensor imaging-derived maps.
6. Look for any robust correlation of the neuroradiological morphometric findings with clinical parameters, and subsequently test them as reliable predictive or prognostic biomarkers.

Due to high translationality of this field of research, the solution of these problems requires a multidisciplinary collaboration of distinct specialists working together in different fields: Points 1 and 2 should be investigated by computer scientists, technicians, and engineers, for example; point 3 by neuroradiologists and neuropathologists; and points 4–6 by a large group of different biomedical specialists, including neurosurgeons, biologists, epidemiologists, oncologists, and radiotherapists.

23.6 Conclusion

SWI has been shown to offer some diagnostic and prognostic indexes, but it is of paramount importance to underline that it should be used in a complementary way with other techniques, such as perfusion and diffusion imaging, in order to get a holistic knowledge of the temporal evolution of brain tumors as well as to follow up response to treatment. SWI can still be developed, and several issues on processing procedures or correlation with neuroradiological results, for example, have yet to be explored and addressed, but here we focus on the fact that SWI research should also be aimed to decrease the reliance on intra- and interobserver variability in order to enhance the quantification and objectivity of SWI. Fractal analysis can help in such aim, offering computational fractal-based parameters (e.g., FD of the SWI pattern) as potential neuroimaging biomarkers in neuro-oncology as well as in neurotraumatology, although the field could be expanded into other applications as well (e.g., follow-up of arteriovenous malformations or cavernomas).

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Chapter 24

Texture Estimation for Abnormal Tissue Segmentation in Brain MRI



Syed M. S. Reza, Atiq Islam, and Khan M. Iftekharuddin

Abstract This chapter discusses multifractal texture estimation and characterization of brain lesions (necrosis, edema, enhanced tumor, nonenhanced tumor, etc.) in magnetic resonance (MR) images. This work formulates the complex texture of tumor in MR images using a stochastic model known as multifractional Brownian motion (mBm). Mathematical derivations of the mBm model and corresponding algorithm to extract the spatially varying multifractal texture feature are discussed. Extracted multifractal texture feature is fused with other effective features to enhance the tissue characteristics. Segmentation of the tissues is performed using a feature-based classification method. The efficacy of the mBm texture feature in segmenting different abnormal tissues is demonstrated using a large-scale publicly available clinical dataset. Experimental results and performance of the methods confirm the efficacy of the proposed technique in an automatic segmentation of abnormal tissues in multimodal (T_1 , T_2 , Flair, and $T_{1\text{contrast}}$) brain MRIs.

Keywords Brain tumor · Segmentation · Texture feature · Lesion · Classification · Multi-modal MR

24.1 Introduction

Brain tumor segmentation in MR image is clinically significant for tumor diagnosis and grading, therapy, medication dose selection, surgery, and patient follow-up. Manual delineation of lesions by radiologists for large volumes is tedious, error-prone, and often suffers from variability. Therefore, robust tumor volume

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segmentation as well as classification of different tumor tissues such as active tumor (enhanced and nonenhanced), necrosis, and edema is required. However, automatic segmentation of brain tumor and other abnormal tissues is a challenging task in medical image analysis due to the unpredictable appearance of these different tissues and their abnormal infiltrations among healthy tissues, and variations in intensity, shape, size, and location in the MRI. The problem is further complicated by the MR artifacts, for example, intensity inhomogeneity, bias in the magnetic field, and noise in the signals.

Many different automatic brain tumor segmentation techniques proposed in the literature are based on either features or atlas-dependent registration or a combination of both. Since our developed texture-based tumor segmentation method is a feature-based method [24], we briefly provide a literature review on recent feature-based techniques. Among the recent feature-based techniques, Lee et al. [11] propose support vector machine (SVM)-based discriminative random fields (DRFs) with a set of multiscale image and spatial alignment-driven features, spatial probabilities of normal tissues (white matter, gray matter, and cerebrospinal fluid), spatial expected intensity maps, and left-to-right symmetry characteristics. However, the method does not allow training and interpatient testing across the different patients. Corso et al. [5] use the conditional random field (CRF) to model cascade of boosted discriminative classifier. Popuri et al. [22] study Dirichlet-priors, Gabor-like texton [12], and the level set features to build statistical models for brain tissues. However, level set techniques are very sensitive to initialization and tend to suffer from boundary leaking problems. Zikic et al. [27] use spatial nonlocal features and initial probabilities with the classification forest (CF) to segment the brain tumor. Bauer et al. [3] integrate random forest (RF) classification with hierarchical CRF regularization in an energy minimization scheme. In ref. [7], Geremia et al. introduce a symmetry feature and use discriminative random decision forest for voxel-wise classification of tumor voxels. Tustison et al. [26] use the Gaussian mixture model (GMM) and maximum a priori estimation with Markov random fields (MAP-MRF) to generate connected component-based geometric features for brain tumor classification. Meier et al. in [17] extend the discriminative model [3] to a generative-discriminative hybrid model which generates initial tissue probabilities for enhancing the classification and spatial regularization. The authors extract 44 features including first-order texture, gradient information, symmetry features, and seven voxel-wise tissue probabilities for brain tumor segmentation. Other important texture-based techniques are also found in the literature. Ghoneim et al. [15] have proposed a 3D co-occurrence matrix-based texture analysis to classify gliomas. However, they use a manually segmented volume of the region of interest. Pachai et al. [19] have shown a multiresolution pyramid algorithm to segment multiple sclerosis lesions in the brain MR image. Pitiot et al. [21] have presented a texture-based MR image segmentation approach with a novel combination of a two-stage hybrid neural classifier.

However, none of the features used in the above techniques capture the multiresolution spatially varying properties of the brain tissues. We argue that the complex structure of brain tumor in MRI may be more amenable to multiscale spatially

variable texture analysis such as piecewise triangular prism surface area (PTPSA) [24] and mBm [24]. The detailed multiresolution analysis of fractal features such as mBm [9] enables capturing the randomly varying spatial tumor texture at different scales. In our prior works [2, 9, 24], we study the effectiveness of these features and feature selection methods for segmenting posterior fossa tumor. In [9], we show that the fractal feature-based technique is also efficient in segmenting different types of tumor, such as medulloblastoma, astrocytoma, and low-grade gliomas. In [23, 24], we use these sophisticated texture features, fractal PTPSA and mBm along with intensity and intensity differences among the modalities to analyze multimodal MRI for characterizing the different abnormal and normal brain tissues.

24.2 Background Review

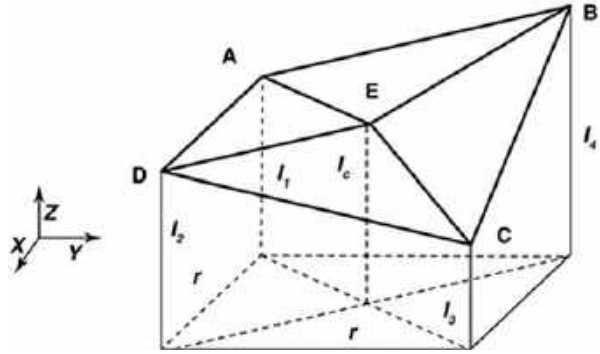
Different techniques mentioned in the previous section clearly indicate that extracting effective features is one of the key factors for successful segmentation. Fractal-based spatially varying texture can be one of the effective features for the segmentation of abnormal brain tissues in MRI. This section briefly reviews the relevant background of our proposed fractal-based texture fractal-PTPSA [8] and multiresolution mBm texture modeling [9] and feature extraction. The detailed mathematical derivations of mBm can be found in [8, 9].

24.2.1 Fractal (PTPSA) Texture Feature Extraction

A fractal is an irregular geometric object with infinite nesting of a self-similar structure at multiple scales. The concept of the fractal is first proposed by Mandelbrot [16] to describe the geometry of the natural objects. The fractal dimension (FD) feature is a noninteger real number that characterizes the texture of objects. In a prior work [8], we investigate statistically the effectiveness of PTPSA feature for brain tumor detection. In PTPSA, the image is divided into several equal sub-images. For each sub-image (Fig. 24.1), the intensities of the four corners (I_1 , I_2 , I_3 , and I_4) and their average (I_c) at the center pixel form four triangles (ABE , ADE , DCE , and BCE).

The FD is calculated by the slope of the log–log plot of the total surface areas of the four triangles vs. sub-image size.

Fig. 24.1 Illustration of fractal PTPSA feature extraction for a sub-image [1]



24.2.2 Multifractal Brownian Motion (mBm) Process and Feature Extraction

The mBm is a nonstationary zero-mean Gaussian random process that corresponds to the generalization of fractional Brownian motion (fBm). The fBm is a part of the set $1/f$ process and considered to be homogeneous or monofractal. The Hurst index (Holder exponent), H in fBm process, is same at all-time instances. The value of H ($0 < H < 1$) determines the randomness of the fBm process, for example, if $H = 0.01$, the signal is very rough, while for $H = 0.99$, the signal is very smooth. Figure 24.2 shows an example of a simulated 1D fBm process vs. time plots for different H values. The figure confirms variation of signal roughness with the variation of H values. However, like many other natural signals, the roughness of the tumor texture varies in space; therefore, tumor texture is more amenable to the multifractal structure. Consequently, this work attempts to estimate tumor texture using the *mBm* process. In the mBm process, the Hurst index, H , is a time-varying parameter, which effectively captures the spatially varying heterogeneous texture of brain tissues.

The mBm process is defined as $x(at) = a^{H(t)}x(t)$, where $x(t)$ is the mBm process with a scaling factor, a , and the time-varying Hurst index, $H(t)$. The covariance function for a 2D mBm process is given as

$$\begin{aligned} \gamma_Z(\vec{u}, \vec{v}) &= E(z(\vec{u})z(\vec{v})) \\ &= \frac{\sigma_u^2}{2} \left[|\vec{u}|^{2H(\vec{u})} + |\vec{v}|^{2H(\vec{v})} - |\vec{u} - \vec{v}|^{2H(\vec{u})} \right] \end{aligned} \quad (24.1)$$

where $z(\vec{u})$ is the mBm process, $\vec{u}$ denotes the vector position (u_x, u_y) of a point in the process, σ_u^2 is the variance, and $H(\vec{u})$ is the Hurst index for a 2D signal. In order to estimate H from multiscales, multiresolution wavelet decomposition is used. After a series of mathematical derivations, the expectation of the squared magnitude of wavelet coefficients $E \left\{ \left| W_Z(\vec{b}, a) \right|^2 \right\}$ is given by

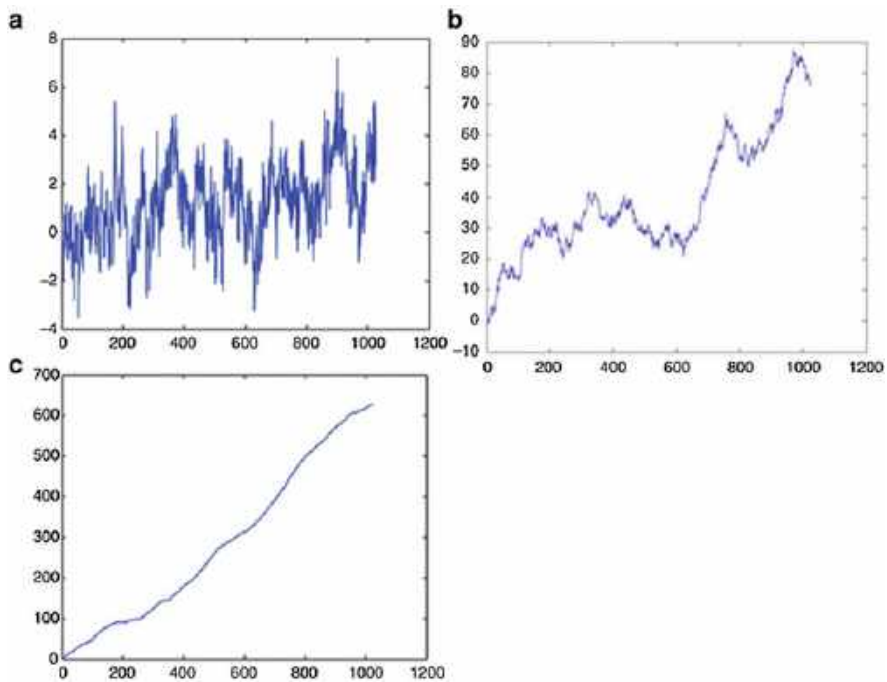


Fig. 24.2 Simulated 1D fBm process with different H values: (a) $H = 0.01$; (b) $H = 0.5$; and (c) $H = 0.99$ [9]

$$E \left\{ \left| W_Z \left(\vec{b}, a \right) \right|^2 \right\} = \frac{1}{M+N} \sum_{x=0}^{N-1} \sum_{y=0}^{M-1} \left| W_Z \left(\vec{b}, a \right) \right|^2 \quad (24.2)$$

where a and b are the scaling factor and 2D translation vector of the wavelet basis, respectively. Here, N and M are the dimensions of the image. The Hurst index for 2D image is calculated as follows:

$$2H(\vec{u}) = \lim_{a \rightarrow 0^+} \frac{\log \left(\left(\frac{1}{M+N} \right) \sum_{x=0}^{N-1} \sum_{y=0}^{M-1} |W_x(b, a)|^2 \right)}{\log a} \quad (24.3)$$

Finally, FD is obtained as

$$FD = E + 1 - H(u), \quad (24.4)$$

where E is the Euclidean dimension ($E = 2$ for 2D images).

For a given image, we first divide the image into nonoverlapping blocks or sub-images and calculate the FD for that corresponding sub-image. The process

Algorithm 1: The mBm process computation for tissue texture extraction

MultiFD(*image*, *block_size*, *wavelet*, *level*)

image: $N \times M$ brain MRI

wavelet: Wavelet filter that verifies the usual admissibility criterion

level: The number of wavelet levels used to compute FD ($\text{level} > 1$)

1. For each block of size *block_size* from image do
 - a. For *a* in 1 to *level*
 - i. Compute the wavelet coefficients at scale *a*.
 - ii. Compute $E\{|W_2(\vec{b}, a)|^2\}$ as shown in (21.2)
 - b. Compute $H(\vec{u})$ from the linear regression of $\log(E\{|W_2(\vec{b}, a)|^2\})$ versus $\log(a)$ as shown in (21.3).
 - c. Compute FD in each image block as shown in (21.4):

$$FD = 2 + 1 - H(\vec{u})$$
 2. end
-

Fig. 24.3 Algorithm for mBm texture feature extraction [9]

is repeated for all the sub-images. The simplified algorithm for mBm texture feature extraction is shown in Fig. 24.3.

The detailed mathematical derivations of mBm and the description of the above algorithm shown in Fig. 24.3 for the mBm texture feature can be found in [9].

24.3 Methodology

In this section, we describe our developed pipeline for abnormal brain tissue segmentation from the 3D MRI volume images (T1, T2, T1c, and Flair). Figure 24.4 shows the steps in the pipeline, which is the proposed method in our previous work [24]. The process starts with linear co-registration among the modalities, which reduces the alignment, rotation, and scaling mismatches. Then 2D MRI slices are obtained from 3D volume for subsequent processing. We briefly discuss each of the steps in Fig. 24.4.

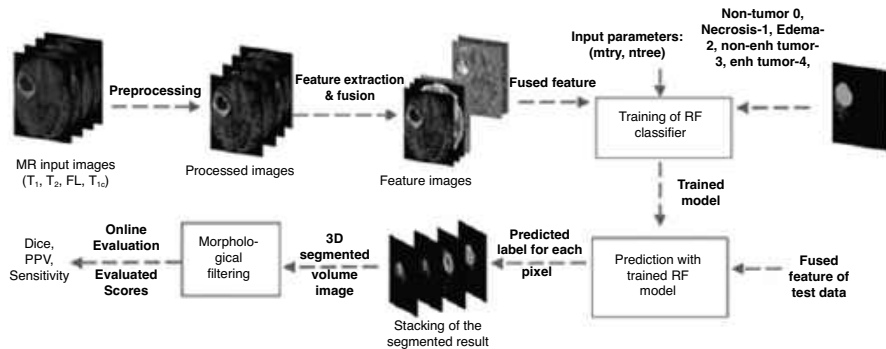


Fig. 24.4 Generic flow diagram of the proposed method [24]

24.3.1 Preprocessing

The preprocessing step involves aligning and co-registration among the channels, resampling, skull stripping, MR bias field, and intensity inhomogeneity correction. Co-registration is performed to correct the misalignment among the modalities and can be done with several tools such as SPM, ITK, and Slicer3D. The MR bias field signal is a low-frequency and very smooth signal that blurs the image and reduces the high-frequency components from the images. Therefore, it is very important to reduce the bias field corrected, and several techniques are found in the literature. In this work, we perform MRI bias correction with N4ITK [25] MRI bias correction tool of Slicer3D. In order to minimize the intensity inhomogeneity of the MR image, intensity normalization is performed. The intensity normalization for MR images is very important because the intensity of the same tissue type can vary from patient to patient and even slice to slice of the same patient. Several MR intensity inhomogeneity correction techniques [10, 18] are available in the literature. The intensity inhomogeneity correction method in [18] is a two-step normalization method [18], where the image histograms are modified such that the histograms match a mean histogram obtained using the training data. After inhomogeneity correction, the intensity values for the same tissue in different MR images fall into a single range of the scale. The method in [10] also comprises two steps. In the first step, 10-point histogram matching is performed, where the reference images of four modalities are an arbitrary set from a single patient data. Intensity values below the mean of the input volume are considered as the background pixel and excluded from the histogram matching process. The next step is normalizing all the intensity values around the mean intensity value of cerebrospinal fluid (CSF). In [10], two class classification (CSF vs. rest) is performed using the random forest to separate the CSF from the other tissues. We notice that thresholding the intensity differences among the modalities gives a fair output to get the CSF mask. We take the histogram-matched images, inhomogeneity-corrected images, and this CSF

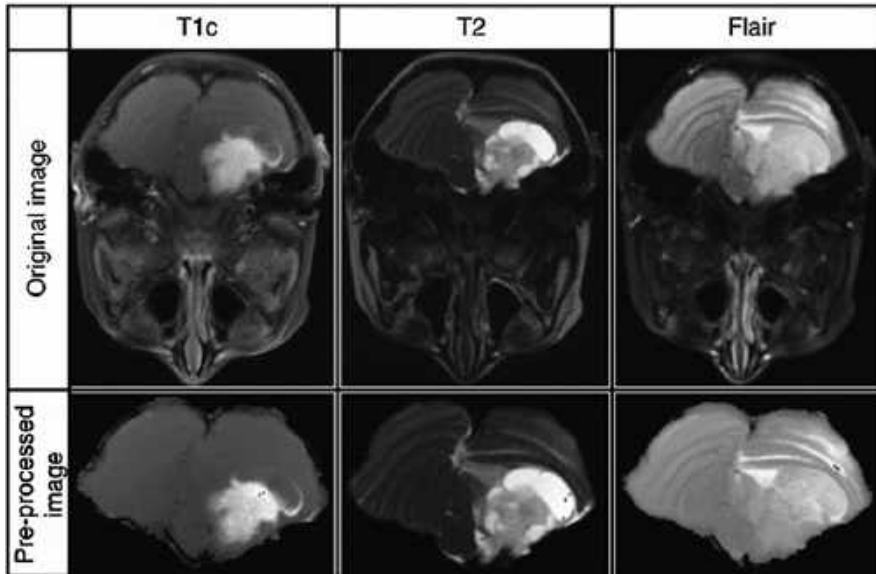


Fig. 24.5 Example images of original (*top row*) and preprocessed images (*bottom row*)

mask as features. Figure 24.5 shows example images of original and preprocessed images.

24.3.2 Feature Extraction, Fusion, Ranking, and Selection

For each slice of the input images, we extract two primary sets of features such as nonlocal and spatial/texture:

- *Feature type 1 (nonlocal feature)*: The nonlocal features are the pixel intensities of four MRI modalities and the corresponding differences in pixel intensities among the modalities. These features do not depend on the local texture patterns, and, thus, we named them as nonlocal features. We use the intensities (I_{T1} , I_{T2} , I_{F1} , and I_{T1c}) and the intensity differences ($d_1 = I_{T1} - I_{T2}$, $d_2 = I_{T2} - I_{F1}$, and $d_3 = I_{F1} - I_{T1c}$) to capture the global characteristics of different tissues. These different features (d_1 , d_2 , and d_3) describe the amount of intensity variation at each pixel among the MR modalities. It is instinctive that different tissues (WM, GM, CSF, tumor core, necrosis, and edema) may display a different amount of intensity variation among the modalities. A similar type of difference feature is also used in [27].
- *Feature type 2 (spatial/texture feature)*: The spatial or texture features are those features that depend on the local texture pattern of an area. As the tumor is the

uncontrolled growth of the tissues, it infiltrates into the surroundings and takes a different texture pattern which is expected to be different from the nontumor region. In order to characterize the tumor surface variation, we employ our novel fractal texture features such as fractal PTPSA, mBm [1, 2, 9]. More details on our fractal texture features can be found in ref. [2, 9]. We also use regular Gabor-like texton [12] since these features have an important association for image segmentation. Texton features are useful to decompose an image into its constituent components and reduce redundant information. In general, total 48 filters, including 3-scales, 6-orientations, 2-phases, and 8-center-surround derivatives, are used in texton feature extraction process. Instead of using all 48 texton filters' outputs, we manually selected a few filters that which show visually significant differences among the abnormal and normal tissue regions.

- *Feature fusion:* After feature extraction from each corresponding slice, we perform feature domain fusion. Each row of resulting fused feature matrix represents all the feature values for a corresponding pixel location of the brain. To illustrate this fusion, let us consider each of the input image volumes is of size $V (= X \times Y \times Z)$, where X , Y , and Z indicate the number of rows, columns, and slices of an input image, respectively. For N_f number of features extracted, the 3D fused feature matrix will be of size $(N_r \times N_f \times Z)$, where $N_r (= X \times Y)$ is the number of rows in the feature matrix, N_f is the number of columns, and Z is the number of slices as usual.
- *Feature ranking and selection:* We know that all the features are not equally important and redundancy among the features degrades the classifier's performance. To identify the most useful features from the whole feature set, different feature ranking and selection methods are used. From the recent techniques, we use mutual information-based implementation of a minimum redundancy maximum relevance (mRMR) [20] feature ranking technique. The method works in two steps. In the first step, it uses mutual information to search the maximum relevance between the individual feature and the class label. However, the features selected with the maximum relevance could have high redundancy among them, and it is intuitive that the combination of these redundant features may have poor class-discriminating power. Therefore, in the second step, the method uses minimal redundancy to select the mutually exclusive features. In this work, we perform the feature ranking on the whole feature set and then heuristically select 19 top-ranked features out of 38 features. Among all the features, we noticed that the intensity, intensity differences, and the mBm features are in the top-ranking list. This confirms the effectiveness of our novel texture features (mBm and PTPSA) in segmenting posterior fossa brain tumor in our previous work [2, 9].

24.3.3 Classification with Random Forest

Random forest for classification which is also known as classification forest (CF) has a very fast and efficient multiclass handling capability. In this work, we use CF [4] for the tissue classification. The classification forests are ensembles of binary classification trees which are formed using randomly drawn data samples and features/predictors. This random selection of data samples and also the subset of feature/predictors help CF to perform robust classification compared to other classifiers such as support vector machines and neural networks. In the CF classification method, each tree offers a classification/vote, and for prediction, the class with the maximum votes is assigned as the final label. Instead of voting, class probability may also be used. In that case, at each node n , the classification tree randomly takes a subset of training samples X_n and predicts a class probability, $p_i^n(\Omega_i|x)$, where p_i is the probability of the sample x in class Ω_i . Depending on the feature dimension, CF randomly resamples the training dataset at every node and assigns the partitions X_L and X_R to the left/right nodes. Tree growing continues up to a certain tree depth (D_T). In the testing phase, the overall probability is calculated as the average of all tree probabilities. Finally, the class with the highest probability is assigned as the actual class. More details about CFs can be found in ref. [6].

In general, the classical random forests algorithm for both classification and regression starts with drawing the N_{tree} bootstrap samples from the training dataset which is the number of trees necessary for good predictions. The next step is to grow the classification trees for each bootstrap data sample where at each node, randomly sampled m_{try} predictors are chosen and the best split searched from the chosen predictors. Finally, for a new data sample, the predictions of all N_{tree} classification trees are collected and averaged for the final output. Usually, the parameters N_{tree} and m_{try} are fixed heuristically for the best performance. In our case, we gradually increase N_{tree} and m_{try} each at a time and observed the overall classification performance. After extensive investigation, we set $N_{\text{tree}} = 100$ and $m_{\text{try}} = 4$ in this work. The detailed description of the RF classifier's training and testing used in this work can be found in ref. [13].

24.4 Results and Discussion

This section reports the segmentation results obtained from the predicted pixel labels of the RF classifier. These 2D abnormal tissue segments are then stacked to generate volume image. Figs. 24.6 and 24.7 show example tissue segments using three slices taken from randomly chosen patients of training and test cases, respectively.

Evaluation

The following similarity coefficients are exploited to evaluate any of our segmentation performance:

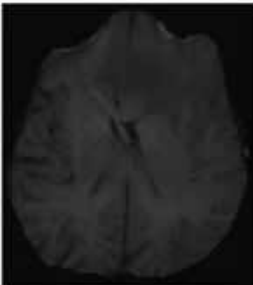
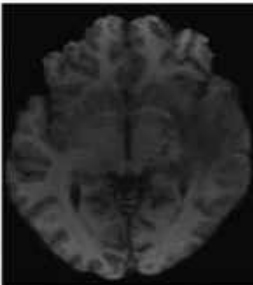
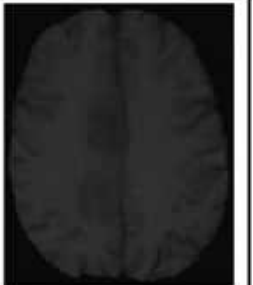
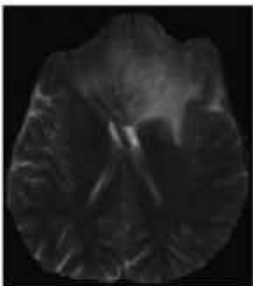
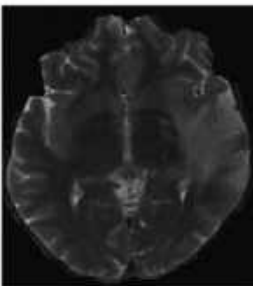
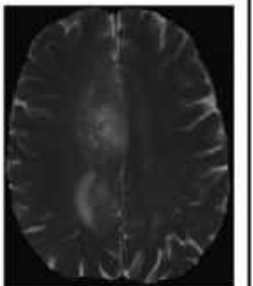
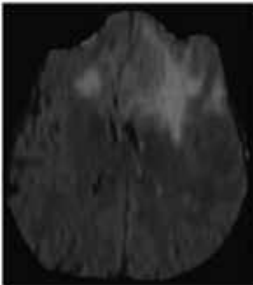
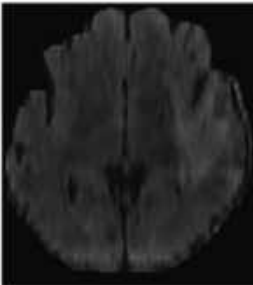
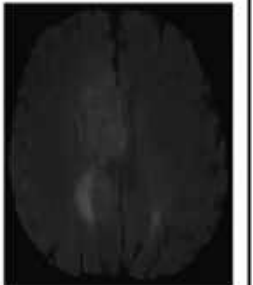
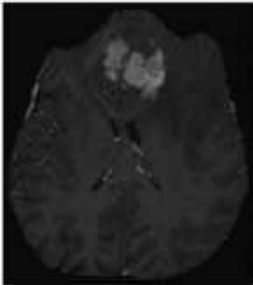
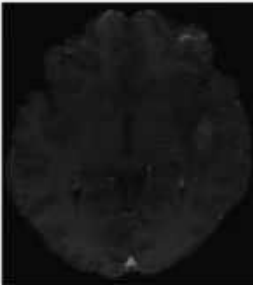
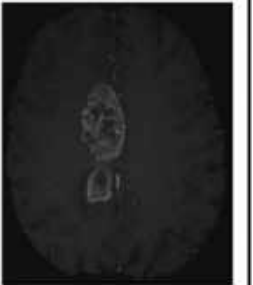
	Example image 1	Example image 2	Example image 3
T1			
T2			
Flair			
T1c			

Fig. 24.6 Segmented tissues with corresponding input and ground-truth images of three training patients. Each column represents an example set of multimodality MRI slices. Input: T1, T2, Flair, T1contrast. Output: segmented image and ground truth. Labels in the ground-truth: 1 – necrosis, 2 – edema, 3 – nonenhancing tumor, 4 – enhancing tumor, and 0 – everything else

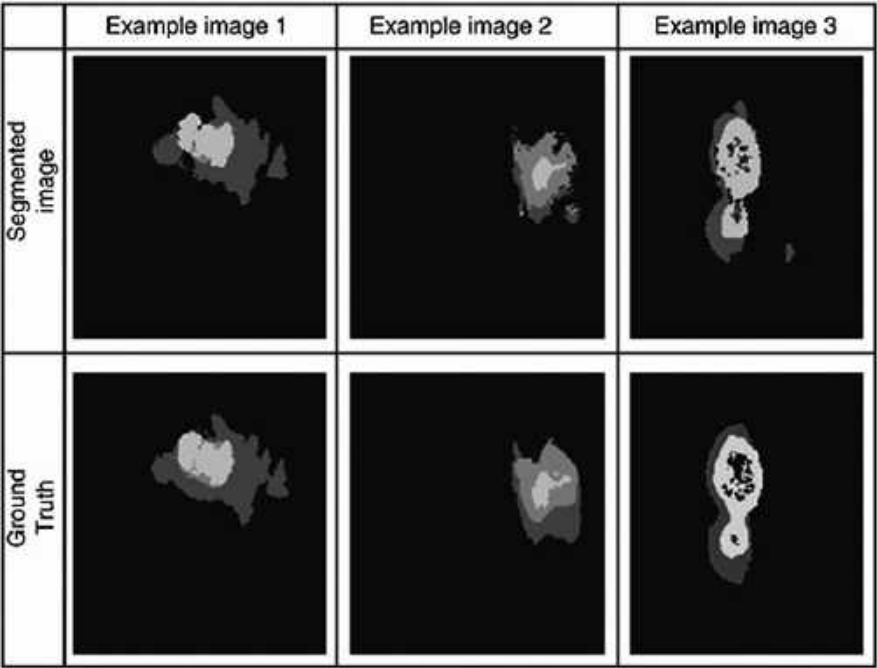


Fig. 24.6 (continued)

- 1. Dice Coefficient = $2TP / (2TP + (FP + FN))$
- 2. Sensitivity = $\frac{TP}{TP + FN}$.

where TP = true positive, FP = false positive, and FN = false negative. Three different categories, which are complete tumor, tumor core, and other tissues, are considered for the evaluation. The details on these three categories are as follows: complete tumor (1 – necrosis, 2 – edema, 3 – nonenhancing tumor, and 4 – enhanced tumor), tumor core (3 – nonenhanced tumor and 4 – enhanced tumor), and individual tissues.

Quantitative Evaluation

We evaluate our preliminary abnormal tissue segmentation results using the BRATS-2013 [13] and BRATS-2014 [14] clinical dataset. BRATS-2013 dataset consists of multicontrast MR scans from 65 glioma patients with low-grade (astrocytoma or oligoastrocytomas) and high-grade (anaplastic astrocytoma and glioblastoma multiforme) tumors. Similarly, BRATS-2014 dataset consists of 400 patients’ images. We perform threefold within patient cross-validation on 213 training patients of BRATS-2014 dataset; the average scores using the proposed method are shown in Table 24.1. Our segmentation rate using threefold cross-validation varies 73–87% using dice overlap metric for tumor core and complete

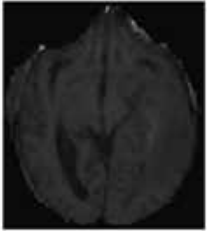
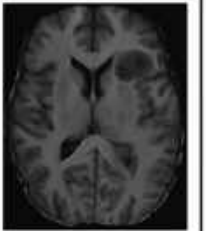
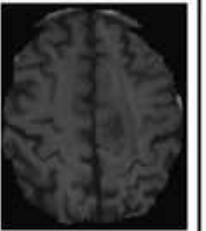
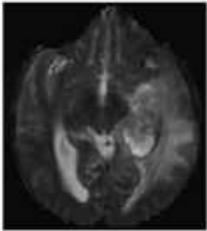
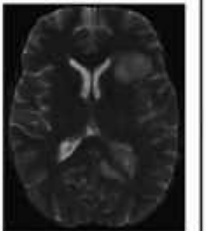
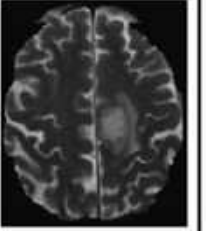
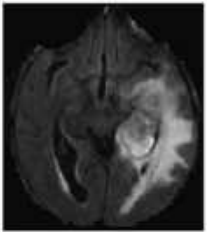
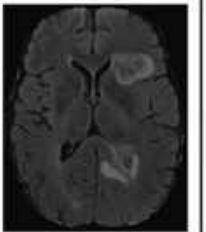
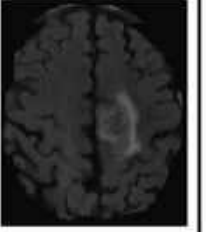
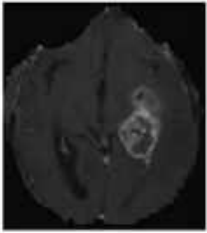
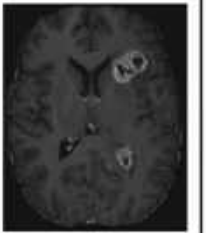
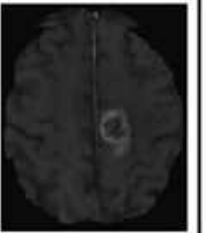
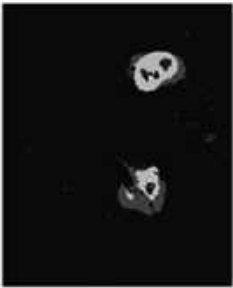
	Example image 1	Example image 2	Example image 3
T1			
T2			
Flair			
T1c			
	Example image 1	Example image 2	Example image 3
Segmented image			

Fig. 24.7 Segmented tissues with corresponding input and output images of three test patients. Each column represents an example set of multimodality MRI slices. Input: T1, T2, flair, T1 contrast. Output: segmented image. Labels in the ground-truth: 1 – necrosis, 2 – edema, 3 – nonenhancing tumor, 4 – enhancing tumor, and 0 – everything else

Table 24.1 Average tumor segmentation (threefold within patient cross-validation) results of 213 patients (BRATS-2014) using the proposed method

	Dice						Sensitivity					
	Necrosis	Edema	Non-enh	Enhance	Core	Complete	Necrosis	Edema	Non-enh	Enhance	Core	Complete
Mean	0.42	0.76	0.13	0.74	0.73	0.87	0.72	0.80	0.09	0.84	0.72	0.89
Std.	0.30	0.13	0.13	0.23	0.24	0.08	0.34	0.15	0.11	0.21	0.26	0.09

tumor. In low standard deviation, 8–24% indicates that the proposed method offers consistent results for different abnormal tissue segmentation.

The patient-wise cross-validation results using our algorithm in Table 24.1 suggest that one may obtain reasonably good results for any representative patient dataset. In order to measure the robustness of the method, we use the trained RF classifier with BRATS-2013 data and test on BRATS-2014 dataset. Quantitative scores of 213 training patients of BRATS-2014 with the proposed method (Fig. 24.4) are shown in Table 24.2.

Results in Table 24.2 shows that the dice score varies from 63% to 76% using the proposed method, which is very promising. From the tissue-wise results, we notice that our texture-based method performs comparatively better for larger lesion size (edema and enhanced tumor), while for smaller lesions, the performance is compromised. We realize that the interpolation method in the texture feature extraction process penalizes the detail of the smaller lesions, and thus the classification performance is less than satisfactory. Again from the patient-wise results, we notice that the proposed algorithm usually performs better for high-grade (HG) tumors than low-grade (LG) tumors. Therefore, we observe that the MRI containing HG tumor surface may contain higher randomness in texture. In summary, the results in these tables show that our proposed method is effective in segmenting different grades of glioma tumors.

24.5 Conclusion and Future Work

In this chapter, we propose a novel texture feature estimation technique using the fractal nature of the brain tissues. The extracted texture features in the MRIs are shown to be effective in discriminating the abnormal and normal brain tissues. The developed feature-based segmentation method is evaluated using a large-scale publicly available clinical dataset known as BRATS. Experimental results obtained with LG and HG patient data confirm the efficacy of our method for multiclass abnormal brain tissue segmentation. Our results also offer comparable brain tumor segmentation performance when compared to the other state-of-the-art works posted on the BRATS website [13]. The method offers comparatively higher dice overlap score for larger size tumors in comparison with smaller ones. Further, substantial false positives in our detections and presence of anomaly in the segmented tumor core area compromise the overall performance. More investigation for lesions with smaller size is required. Our future works include study of more effective features and sophisticated feature selection technique for improved brain tumor segmentation.

Table 24.2 Average tumor segmentation results of 213 patients of BRATS-2014 using the proposed method. RF classifier is trained with 20 HG patients of BRATS-2013

	Dice					Sensitivity				
	Necrosis	Edema	Non-enh	Enhance	Core	Complete	Necrosis	Edema	Non-enh	Enhance
Mean	0.31	0.62	0.03	0.63	0.63	0.76	0.49	0.59	0.02	0.85
Std.	0.27	0.17	0.05	0.28	0.22	0.15	0.31	0.18	0.03	0.18

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Chapter 25

Multifractal Analysis of Brain Tumor Interface in Glioblastoma



Jackson Sánchez and Miguel Martín-Landrove

Abstract The dynamics of tumor growth is a very complex process, generally accompanied by numerous chromosomal aberrations that determine its genetic and dynamical heterogeneity. Consequently, the tumor interface exhibits a non-regular and heterogeneous behavior often described by a single fractal dimension. A more suitable approach is to consider the tumor interface as a multifractal object that can be described by a set of generalized fractal dimensions. In the present work, detrended fluctuation and multifractal analysis are used to characterize the complexity of glioblastoma.

Keywords Multifractal · Glioblastoma · Fractal dynamics · Complexity

25.1 Introduction

Multifractal and detrended fluctuation analysis has been used for the evaluation of the texture in medical image analysis [1–3]. In particular, it has been used for brain tumor classification and grading [4–7], brain tumor segmentation through the characterization of image texture [8–14], and as a radiomic feature [15–20]. Fractal dimension has been used as a prime feature associated with image intensity to detect and classify brain tumors [21, 22], characterize brain tumor microvasculature [23, 24] and in IoT developments [25]. In all of the previously mentioned approaches, the derived fractal quantities give information on the bulk tumor growth dynamics and heterogeneity. A different approach to studying the

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dynamics of tumor growth comes from the fractal and scaling analysis of the tumor interface as originally proposed by Brú et al. [26–29] *in vitro* and *ex vivo* tumor samples. Application of fractal and scaling analysis to the tumor interface for *in vivo* contrast-enhanced T_1 weighted MRI of brain tumors was first done by Martín-Landrove and Pereira [30]. The analysis was further extended to three-dimensional contrast-enhanced T_1 -weighted MRI of brain images [31–33]. In this chapter, a general approach based on multifractal and detrended fluctuations analysis is used to characterize fluctuations of quantities supported on the tumor interface, such as the radius, interface width, and contrast-enhanced image intensity.

25.2 Image Selection and Segmentation

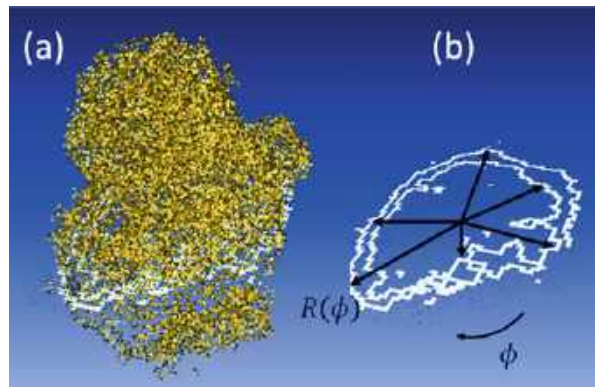
Images for glioblastoma were extracted from collections in The Cancer Imaging Archive [34, 35], the TCGA-GBM collection [36] and from the RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) Challenge 2021 [37–39], which includes a classification according to O(6)-methylguanine-DNA methyltransferase (MGMT) promoter methylation [40, 41]. Among these collections, T_1 -weighted images, either contrast-enhanced or not, were selected and further reviewed. Tumor lesions selected for image processing were identified as such and separated from anatomical structures.

Images were segmented according to a dynamical quantum clustering algorithm [42, 43] applied to medical image data [44–46].

25.3 Multifractal Analysis

To determine the multifractal scaling exponents for the one-dimensional ordered series extracted from the tumor interface, as seen in Fig. 25.1, a general procedure of fluctuation analysis is used [47, 48]. In the following, the method is derived for the ordered series corresponding to the radii of the interface points but it can be

Fig. 25.1 (a) Tumor interface point cloud as extracted by the segmentation method previously described; white interface points correspond to a particular slice. (b) Construction of the ordered series from points extracted from the tumor interface, in this case, the radius $R(\phi)$



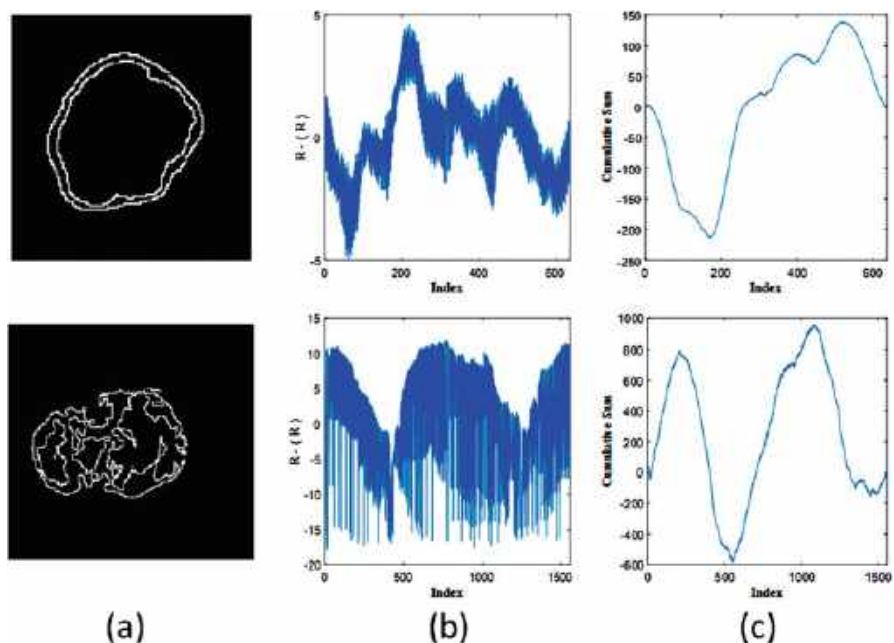


Fig. 25.2 Some examples of extracted ordered series. (a) Tumor interface point cloud corresponding to a slice. (b) Ordered series of the fluctuations of the radii of the tumor interface and (c) the cumulative sum of the ordered series

easily extended to any variable associated with these points such as the average interface width or its contrast-enhanced intensity. First, the profile of the ordered series is determined by a cumulative sum,

$$R(i) \equiv \sum_{k=1}^i (r_k - \langle r \rangle) \quad (25.1)$$

where $\langle r \rangle$ represents the mean radius of the tumor interface (see Fig. 25.2). The profile series is then partitioned into $N_s \equiv \text{int}(N/s)$ segments of equal length s , the box probability $p_s(v)$, which is the sum of the values r_k within each segment v of size s , is defined as

$$p_s(v) \equiv R(vs) - R((v-1)s) \quad (25.2)$$

The scaling properties and exponents can be obtained through the partition function

$$Z_q(s) \equiv \sum_{v=1}^{N_s} |p_s(v)|^q \quad (25.3)$$

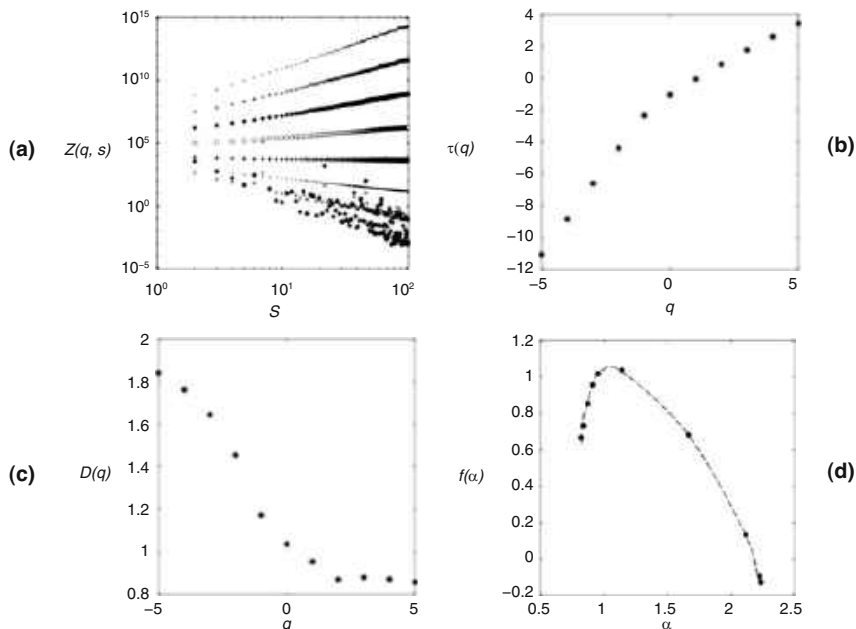


Fig. 25.3 Different steps in multifractal analysis for one-dimensional ordered series extracted from the tumor interface. **(a)** Partition functions $Z_q(s)$. **(b)** Large scale exponents $\tau_1(q)$. **(c)** Generalized fractal dimensions, $D_1(q)$, and **(d)** Singularity spectrum $f_1(\alpha)$

For large values of s , a power law behavior is obtained for $Z_q(s)$ (Fig. 25.3a), allowing for a definition of the scaling exponent $\tau_1(q)$ that characterizes the one-dimensional fluctuations,

$$Z_q(s) \sim s^{\tau_1(q)} \quad (25.4)$$

as seen in Fig. 25.3b.

The generalized fractal dimensions (Fig. 25.3c) are then defined as

$$D_1(q) \equiv \frac{\tau_1(q)}{q-1} \quad (25.5)$$

and the generalized one-dimensional Hurst exponents can be obtained from the relation

$$\tau_1(q) = qh_1(q) - 1 \quad (25.6)$$

A complementary way to characterize the multifractal tumor interface is by determining the singularity spectrum $f_1(\alpha_1)$ (Fig. 25.3d), which is related to $\tau_1(q)$

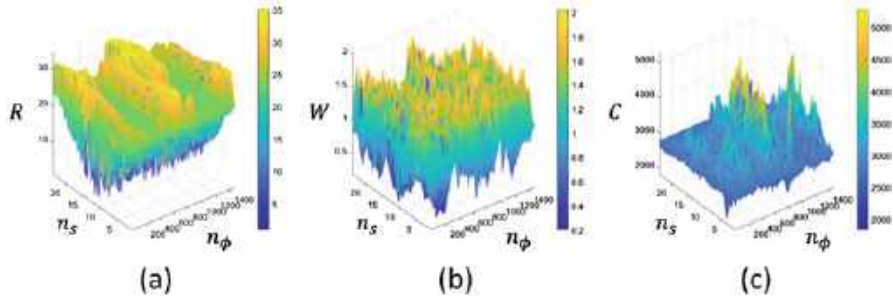


Fig. 25.4 Two-dimensional arrays from the tumor interface. (a) Radii extracted from the tumor interface point cloud. (b) Fluctuations of the radii for a fixed scale and (c) contrast-enhanced image intensity at the tumor interface

by a Legendre transform

$$\alpha_1 = \frac{d\tau_1}{dq},$$

$$f_1(\alpha_1) = q\alpha_1 - \tau_1(q) \quad (25.7)$$

The ordered series that can be extracted from any slice represents a one-dimensional sampling of the tumor interface and as a consequence an incomplete picture of the tumor interface fluctuations. A more general approach is possible if a two-dimensional detrended fluctuation analysis [49] is performed. In this case, the tumor interface is parameterized as a two-dimensional array with elements $r(n_\phi, n_z)$, the radii of the tumor interface, or any other quantity associated with it, as shown in Fig. 25.4, and is partitioned in two-dimensional segments of size s .

The detrended fluctuation function in each segment is given by

$$F^2(v, w, s) = \frac{1}{s^2} \sum_{i=1}^s \sum_{j=1}^s \epsilon_{v,w}^2(i, j) \quad (25.8)$$

where $\epsilon_{v,w}(i, j)$ is the difference between the cumulative sum of $r(i, j)$ and its trend over the segment (v, w) . The average of the detrended fluctuation over all the segments is

$$F_q(s) = \left\{ \frac{1}{M_s N_s} \sum_{v=1}^{M_s} \sum_{w=1}^{N_s} [F(v, w, s)]^q \right\}^{\frac{1}{q}} \quad (25.9)$$

for $q \neq 0$ and for $q = 0$,

$$F_0(s) = \exp \left\{ \frac{1}{M_s N_s} \sum_{v=1}^{M_s} \sum_{w=1}^{N_s} \ln[F(v, w, s)] \right\} \quad (25.10)$$

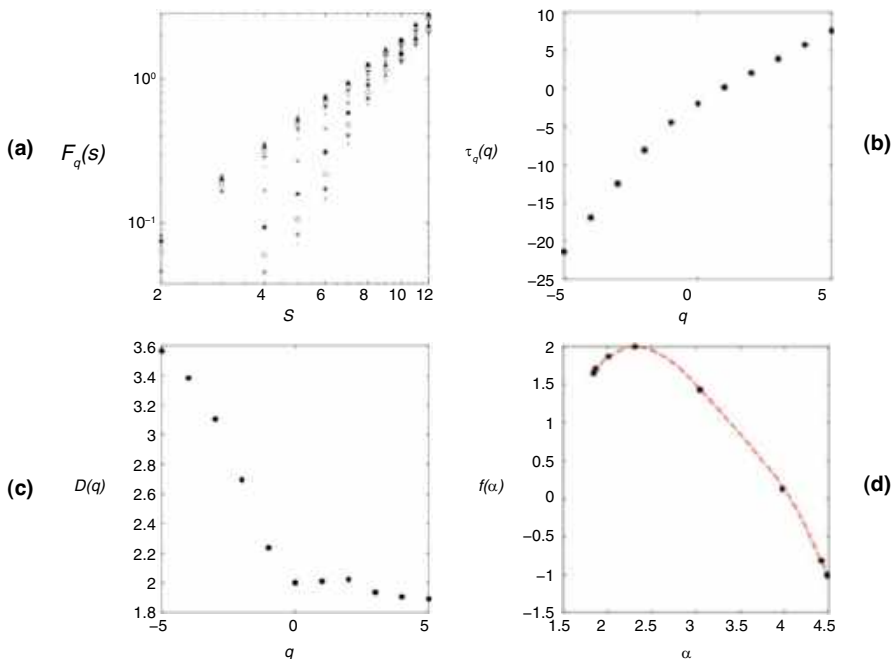


Fig. 25.5 Different steps in detrended fluctuation analysis for two-dimensional arrays extracted from the tumor interface. **(a)** Detrended fluctuation functions $F_q(s)$. **(b)** Large scale exponents $\tau_2(q)$. **(c)** Generalized fractal dimensions, $D_2(q)$, and **(d)** Singularity spectrum $f_2(\alpha)$

For large values of s , F_q behaves as a power law,

$$F_q(s) \sim s^{h_2(q)} \quad (25.11)$$

The multifractal nature of the fluctuation is characterized by the scaling exponents $\tau_2(q)$ and related to the generalized two-dimensional Hurst exponents $h_2(q)$ by

$$\tau_2(q) = qh_2(q) - 2 \quad (25.12)$$

and Eq. 25.5 holds for the generalized fractal dimensions $D_2(q)$. In a similar way, as in Eq. 25.7, the singularity spectrum $f_2(\alpha)$ can be determined. The general scheme is presented in Fig. 25.5.

25.4 Results and Discussion

25.4.1 *Multifractal Analysis of One-Dimensional Ordered Series*

The results corresponding to the multifractal analysis of the one-dimensional ordered series are summarized in Table 25.1 showing the average value of the information dimension $D_1(1)$, the correlation dimension $D_1(2)$, and the Hurst exponent, $h_1(2)$.

Also from Table 25.1, the generalized fractal dimension $D_1(0)$, which corresponds to the Euclidean dimension, is very close to 1, as it should be since the ordered series is one-dimensional. Comparison of the values for $D_1(1)$, $D_1(2)$, and $h_1(2)$ among the MGMTp(0) (unmethylated) and MGMTp(1) (methylated) groups yields to the result that both groups are very similar in these quantities, with an increase in favor of the MGMTp(0) group. Also, the Gbm group, which belongs to the TCGA-GBM collection, exhibits results that are very close to the other groups, possibly because this group represents an unknown mixture of unmethylated and methylated MGMT promoters. The general behavior can be seen in Fig. 25.6.

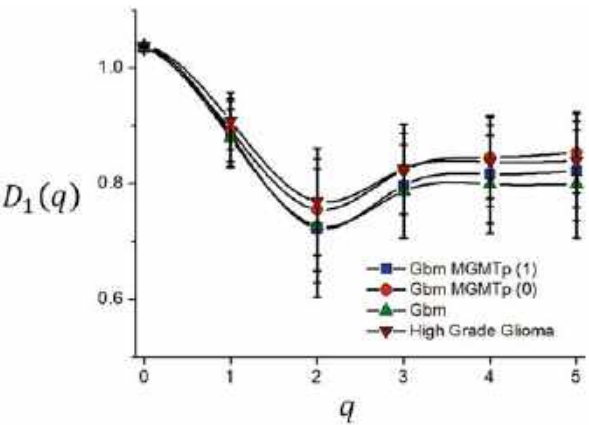
Table 25.1 Generalized fractal dimensions D_1 and Hurst exponent, $h_1(2)$ for the one-dimensional ordered series of the radii extracted from the tumor interface

Data set	$D_1(0)$	$D_1(1)$	$D_1(2)$	$h_1(2)$
MGMTp (0) ^a	1.037 ± 0.002	0.90 ± 0.05	0.76 ± 0.11	0.88 ± 0.05
MGMTp (1) ^a	1.036 ± 0.002	0.89 ± 0.06	0.73 ± 0.12	0.86 ± 0.06
Gbm ^b	1.037 ± 0.002	0.88 ± 0.05	0.73 ± 0.10	0.87 ± 0.05

^a RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) Challenge 2021. (0) Unmethylated MGMT promoter, (1) methylated MGMT promoter

^b The Cancer Imaging Archive TCGA-GBM collection

Fig. 25.6 Generalized fractal dimensions $D_1(q)$, associated with one-dimensional ordered series for data sets of glioblastoma and high-grade gliomas



25.4.2 Detrended Fluctuation Analysis of Two-Dimensional Tumor Interface Data

Different data were extracted from the tumor interface: the set of radii of the interface points, R , the set of the interface points fluctuations for a fixed scale, that is, a width evaluated on each interface point, W , and the set of contrast-enhanced image intensities evaluated at each interface point, I . The results for the average value of $D_2(1)$, $D_2(2)$, and $h_2(2)$ are summarized in Table 25.2. Notice in general that all the quantities decrease as the data set is changed from $R \rightarrow W \rightarrow I$ and an increase in favor of the methylated group MGMTp(1) for all the data sets.

The relationship between the generalized fractal dimensions is shown in Fig. 25.7.

25.4.3 Generalized Hurst Exponents and Singularity Spectra

A complementary view of the multifractal nature of the tumor interface can be obtained by the evaluation of the generalized Hurst exponents $h(q)$ and the singularity spectrum $f(\alpha)$. In Tables 25.1 and 25.2, the quantity known as the Hurst exponent, that is, $h(2)$ is evaluated, indicating how correlated the fluctuations in the tumor interface are, but as seen in Fig. 25.8, the dependence on q of the generalized Hurst exponents is more pronounced for $q < 0$ (more sensitive to small fluctuations). This behavior is observed both for the multifractal one-dimensional analysis and for the two-dimensional detrended fluctuation analysis with the possibility to discriminate between the glioblastoma groups.

Table 25.2 Generalized fractal dimensions D_2 and Hurst exponent, $h_2(2)$ for two-dimensional arrays extracted from the tumor interface. R interface radius, W interface width, I contrast image intensity at the interface

Data set	$D_2(1)$	$D_2(2)$	$h_2(2)$
R MGMTp (0) ^a	1.91 ± 0.18	1.82 ± 0.35	1.92 ± 0.17
W MGMTp (0) ^a	1.87 ± 0.21	1.73 ± 0.43	1.87 ± 0.21
I MGMTp (0) ^a	1.84 ± 0.24	1.69 ± 0.48	1.84 ± 0.24
R MGMTp (1) ^a	1.95 ± 0.17	1.90 ± 0.34	1.95 ± 0.17
W MGMTp (1) ^a	1.90 ± 0.17	1.80 ± 0.35	1.90 ± 0.17
I MGMTp (1) ^a	1.88 ± 0.18	1.75 ± 0.36	1.88 ± 0.18
R Gbm ^b	1.93 ± 0.15	1.87 ± 0.30	1.93 ± 0.15
W Gbm ^b	1.88 ± 0.14	1.76 ± 0.28	1.88 ± 0.14
I Gbm ^b	1.86 ± 0.13	1.72 ± 0.26	1.86 ± 0.13

^a RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) Challenge 2021. (0) Unmethylated MGMT promoter, (1) methylated MGMT promoter

^b The Cancer Imaging Archive TCGA-GBM collection

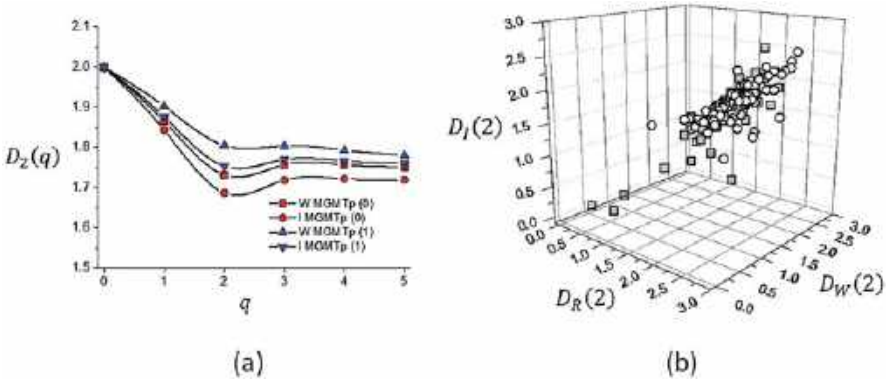


Fig. 25.7 (a) Generalized fractal dimensions $D_2(q)$, associated with two-dimensional data arrays extracted from the tumor interface and corresponding to the width of the interface at a certain scale, W , and the contrast-enhanced image intensity, I .; (0) corresponds to unmethylated MGMT promoter and (1) to methylated MGMT promoter. (b) Relationship between the correlation dimensions $D_R(2)$, $D_W(2)$ and $D_I(2)$ with MGMTp(0) in square symbols and MGMTp(1) in circles

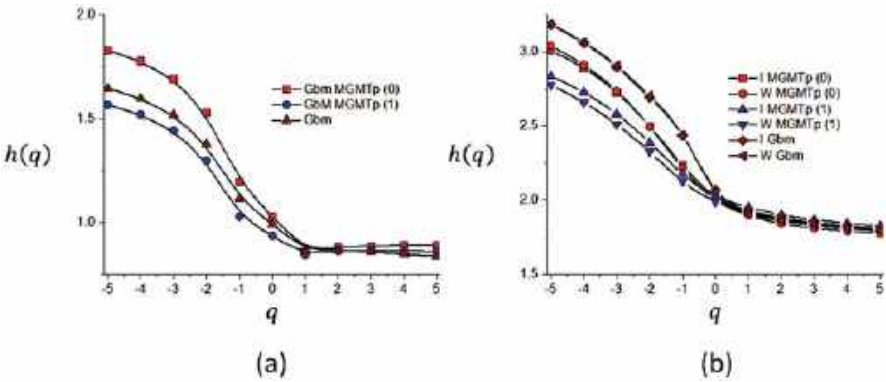


Fig. 25.8 Generalized Hurst exponents, $h(q)$. (a) Results obtained by multifractal analysis applied upon one-dimensional ordered series extracted from the tumor interface. (b) Results obtained by two-dimensional detrended fluctuation analysis applied upon two-dimensional arrays of interface data; I corresponds to contrast-enhanced image intensity while W is the width of the tumor interface at a fixed scale. (0) corresponds to unmethylated MGMT promoter, (1) to methylated MGMT promoter

Figure 25.9 shows the corresponding singularity spectra obtained by multifractal analysis on one-dimensional ordered series, Fig. 25.9a and two-dimensional detrended fluctuation analysis, Fig. 25.9b.

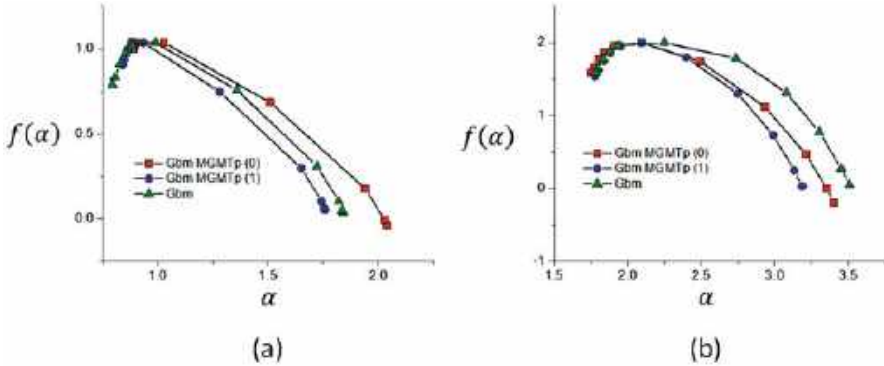


Fig. 25.9 Singularity spectrum $f(\alpha)$. **(a)** Results obtained by multifractal analysis applied upon one-dimensional ordered series of the radii extracted from the tumor interface. **(b)** Results obtained by two-dimensional detrended fluctuation analysis applied upon the two-dimensional array of the radii extracted from the tumor interface. (0) corresponds to unmethylated MGMT promoter, (1) to methylated MGMT promoter

25.5 Conclusions

Multifractal and detrended fluctuation analysis are powerful tools to characterize the heterogeneity and complexity of the tumor interface. The determination of generalized fractal dimensions, generalized Hurst exponents, and related quantities such as singularity spectra besides the commonly used fractal dimension could greatly improve the description of the tumor growth dynamics. From a practical point of view, these quantities add up to the standard set of radiomic features that can be used as discriminators or image biomarkers for tumor detection and classification.

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Chapter 26

Fractal-Based Analysis of Histological Features of Brain Tumors



Omar S. Al-Kadi and Antonio Di Ieva

Abstract The structural complexity of brain tumor tissue represents a major challenge for effective histopathological diagnosis. Tumor vasculature is known to be heterogeneous, and mixtures of patterns are usually present. Therefore, extracting key descriptive features for accurate quantification is not a straightforward task. Several steps are involved in the texture analysis process where tissue heterogeneity contributes to the variability of the results. One of the interesting aspects of the brain lies in its fractal nature. Many regions within the brain tissue yield similar statistical properties at different scales of magnification. Fractal-based analysis of the histological features of brain tumors can reveal the underlying complexity of tissue structure and angiostructure, also providing an indication of tissue abnormality development. It can further be used to quantify the chaotic signature of disease to distinguish between different temporal tumor stages and histopathological grades.

Brain meningioma subtype classifications' improvement from histopathological images is the main focus of this chapter. Meningioma tissue texture exhibits a wide range of histological patterns whereby a single slide may show a combination of multiple patterns. Distinctive fractal patterns quantified in a multiresolution manner would be for better spatial relationship representation. Fractal features extracted from textural tissue patterns can be useful in characterizing meningioma tumors in terms of subtype classification, a challenging problem compared to histological grading, and furthermore can provide an objective measure for quantifying subtle features within subtypes that are hard to discriminate.

Keywords Fractal dimension · Texture analysis · Brain histopathology · Meningioma · Tissue characterization · Pattern classification

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26.1 Introduction

Diagnosis of brain tumors is mainly based on visual examination of cell morphology and tissue distribution. The accurate diagnosis of histopathological brain tissue is essential for selecting the appropriate treatment procedure. However, the various cellular arrangements and complex tissue vasculature render the brain one of the most challenging and complex organs in the human body. Additionally, tumor angiogenesis further increases the heterogeneity of the complex cellular- and tissue-level spatial relationships.

The unique folding of brain tissue in a *fractal-like* geometry is an interesting characteristic that is worthy of attention. The development of consistent fractal patterns in normal brain tissue acts as a sign of order and represents how well-structured the patterns can be. The fractal characteristics of the brain reflect its complexity.

Fractal analysis in tissue characterization has been deployed in many useful applications in histopathological imaging [29]. The development of a computer-aided diagnostic system based on fractal-model design to assist in histological diagnosis of brain tumors is significant for effective management of disease. This could improve pathologist's ability to accurately detect and characterize subtle tissue abnormalities and hence lead toward a more reliable diagnosis and promoting better patient outcome.

Quantifying the fractal properties of pathological tissue and cellular properties by means of its textural patterns resembles how pathologists perceive and identify distinctive features of disease. A computer-aided diagnostic system can play the role of a visual perception function through mimicking the behavior of pathologists: first, by perceiving the different unique patterns in an image (e.g., extracting fractal features) and then by using past experience and experimental knowledge to link the perceived patterns and appropriate diagnosis (e.g., decision-making algorithms). This has an advantage over the classical visual diagnosis approach. Limitations due to subjective factors in the classical visual diagnosis approach like preconception, expectations, relying on diligence, and fatigue contribute to differences in image perception, which leads to inter- and intraobserver variability assessments. In contrast, an automated decision system can overcome such bias in diagnosis by maintaining consistent perceived patterns. It would also enhance the ability to discriminate complex texture patterns that would otherwise be difficult via ordinary human vision. Computer-assisted tools can therefore help in offering objective diagnostic and prognostic image biomarkers to improve patients' management.

26.2 Fractal Morphometry of Tissue Complexity

The concept of *fractality* is used to: (a) analyze complex patterns that look the same across different scales of magnification; (b) describe the self-similarity of irregular

and self-similar objects in nature; and (c) quantify the surface roughness and geometrical complexity of objects [39]. Various researchers have been investigating the application of fractal analysis to imagery in a number of diverse fields including, but not limited to, physics [12], acoustic, geology [1], ecology [26], radiology [2], and fluid mechanics [41]. In histopathology, fractal geometry has been used as a measure of morphological complexity of cells and tissues providing compact description of the textural characteristics for both normal and pathological cases [29].

Histopathological assessment is mainly based on the study of underlying biological features of disease for clinical diagnosis. Certain descriptive features can reveal more details when examined at increasing resolution. Many tissue patterns tend to exhibit statistically self-similar features at different levels and in different sizes. The assessment of the pattern irregularity at different scales could assist in better understanding tissue pathology and thus assist pathologists in identifying type and grade of disease. However, fractality in textural patterns of histopathological tissue is considered *quasi-fractal*, as they scale in a statistical fashion. Biological tissues, like several natural objects, are fractal in a limited range of magnification (in the so-called “fractal window”). The resemblance between shapes seen at different scales is usually approximate and is considered to be random rather than strictly self-similar. The exhibited geometrical variability mainly refers to deviations in cell structures and distributions over the tissue sample.

The application of the fractal approach in histopathology also has a potential role in characterizing the morphological information for deeper insight and understanding the biology of tumor tissues. Tumor vasculature is spatially heterogeneous and does not have a clear hierarchical organization consisting of irregularly formed vessels that are both leaky and often permeable. This heterogeneity in tissue texture calls for a method that can improve the capability of discriminating higher-order statistical textural information, which would otherwise be difficult using classical examination approaches. The size and geometrical parameters of irregular and complex structures in tumor tissue differ when examined at increasing resolution. Hence, fractal geometry, which best characterizes non-Euclidean structures, can be used to measure topological and geometrical properties of irregular and complex structures in tumor tissue.

26.2.1 Fractal Dimension Estimation

Fractal pattern analysis has been applied to many of the physiological processes in biomedical research. Unlike Euclidean geometry, which is best developed to measure regular structures, complex biological structures, which are known to be irregular, could be best quantified based on the principles of fractal geometry. The fractal dimension (FD) and other related dimensional parameters provide a compact representation of the complexity of the biological patterns. Nevertheless, the

accuracy of FD estimation depends on the precision of the input shape description, and therefore several estimation methods have been developed for this purpose.

FD is widely used as a tool to quantify the geometrical complexity of a structure by means of its space-filling properties. There are several methods used to inspect self-similarity and dedicated to providing a reliable estimate of the FD. The most well-known methods are briefly discussed below:

- *Fractional Brownian motion (fBm)*: The fBm is a nonstationary model known for its capability to describe random phenomena [42]. It is a generalization of the Brownian motion where the increments of the process are normally distributed but not independent [39]. The correlation of a random process $B(t)$ can be represented as the expected value $E(B)$ of the product of non-overlapping increments of the fBm process:

$$E((B(t) - B(0)) \cdot (B(t+h) - B(t))) = \left((t+h)^{2H} - t^{2H} - h^{2H} \right) / 2 \quad (26.1)$$

where the Hurst index, H , is a real number in the range $(0, 1)$ and describes the roughness of the motion, with a higher value leading to a smoother motion [38]. The value of H determines the process of the fBm, such that when $H > 1/2$ indicates a positive correlation between the increments. A Brownian motion is achieved for the special case $H = 1/2$, and the increments are negatively correlated when $H < 1/2$. For an image, H is practically estimated by plotting the mean absolute difference of pixel pairs as a function of scale on a log-log scale (see Fig. 26.1), where H will represent the slope of the curve that is used to estimate the FD as: $FD = 3 - H$.

- *Box counting*: This is a recursive method that involves covering the contour lines with a grid of n -dimensional boxes having a side length δ and counting the number of non-empty boxes $N(\delta)$. An overlaying grid approximates each contour line, and then the number of boxes or cell intersections is counted. The smaller the size of the boxes becomes, the larger the length estimate as finer details are captured. The technique is performed recursively with different box sizes covering the contour lines.

For a one-dimensional curve having a length L , the relation can be expressed as [39, 56]: $N(\delta) \approx \frac{L}{\delta}$. Then, the generalized form of the slope of the logarithmic plot of the number of boxes against their size would represent the FD (see Fig. 26.2)

$$N(\delta) \propto \frac{1}{\delta^{FD}} \quad (26.2)$$

$$FD = \lim_{\delta \rightarrow 0} - \frac{\log N(\delta)}{\log(\delta)}. \quad (26.3)$$

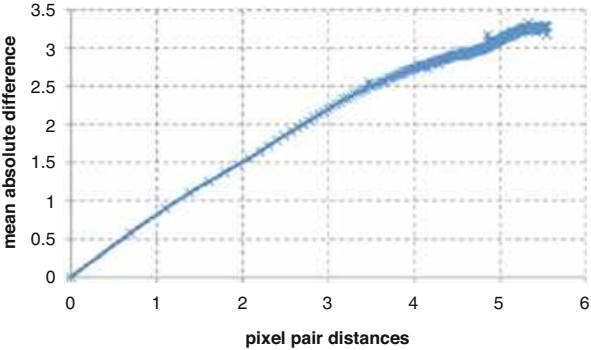


Fig. 26.1 Mean absolute difference versus pixel pair distances in log–log coordinates for a transitional meningioma histopathological image in Fig. 26.4

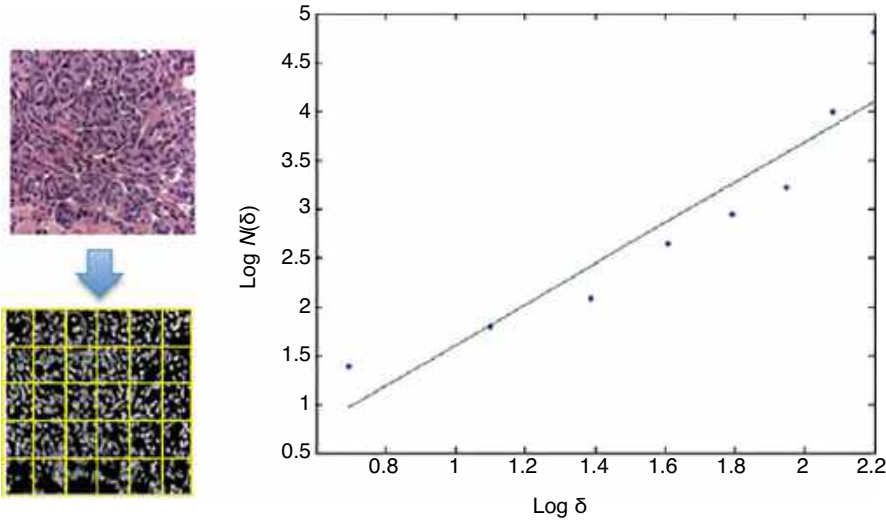


Fig. 26.2 Application of the box-counting method to a transitional meningioma subtype image (FD = 1.81). [left] Histopathological image and corresponding segmented cell nuclei image covered with a set of squares, where the squares covering the cell nuclei are iteratively counted; [right] Estimating the FD from the slope of the linear regression line of the log–log plot between the number of squares $N(\delta)$ and square size δ^2

Also based on the same principle but with a different shape, *prism counting* was introduced [16], where the combined area of four-sided triangular prisms defined by the corner points is computed and summed over the surface. Similarly, the iterative *Epsilon-Blanket* method computes the FD of a scale-varying curve (or surface) based on the corresponding area (or volume) [44]. The set of points that lie at a certain distance ε from the curve are used in forming a blanket having a strip of width of 2ε covering the curve and for different values of ε . The FD for a curve

can be calculated using the relation $L(\varepsilon) \propto \varepsilon^{1-FD}$, where $L(\varepsilon)$ is the curve length defined from the blanket area $A(\varepsilon)$ as $L(\varepsilon) = A(\varepsilon)/2\varepsilon$. For the case of a surface, the blanket is formed from a set of points in the 3-dimensional space, which are also ε far from the surface and having a width of 2ε . The surface area is defined as $A(\varepsilon) = V(\varepsilon)/2\varepsilon$, and the FD is estimated from the relation $A(\varepsilon) = \varepsilon^{2-FD}$.

- *Walking Divider*: This method uses a ruler or a chord length acting as a *step size* and then measures the number of steps or *length* required to cover the fractal curve, which can be represented as [39]: $L(\delta) = K\delta^{1-FD}$, where $L(\delta)$ is the length of trail, δ is the step size, and K is a constant. The method operates by iteratively walking the divider along the curve and then counting the number of steps required to cover the curve for each case. This process is repeated at different resolutions, and the FD, which is the relation between the step size and chord length, can be estimated.
- *Power spectrum*: Given a signal $f_i = 1, 2, \dots, N$ (where N has the power of 2), the Fourier power spectrum can be used to estimate the FD as: $\hat{P}_i = c|k_i|^{-\beta}$, where c is a constant and the spectral exponent β , which relates to the FD, can be estimated by fitting the least squares error line to the data. Also, since the Fourier transform of the *autocorrelation function* is the power spectrum of the function [45], the autocorrelation function ρ as a function of the shifting distance a expressed as $\rho(a) \propto a^{-2FD}$ can also be used to estimate the FD.
- *Area/perimeter relationship*: In this approach, the FD is estimated based on the power-law relationship, where the areas and the respective perimeters of the different curves of a group are measured [57]. The boundary perimeter P is related to the enclosed area (A) by $A = kP^{FD/2}$ for $1 < FD < 2$, where k is a constant for a given shape (e.g., 4 for square and $2\sqrt{\pi}$ for a circle, etc.)
- *Pixel Dilation*: This method relates the FD to the curve border length. The dilation operation is performed by convolving various filters having different diameters with the curve border. The computed area would be divided by the size of the kernel, and a log-log plot would be used to determine the slope of the linear regression line S ; and hence the $FD = 1 - S$.

26.2.2 Related Work

A significant progress has been made to understand how irregular shapes and structures relate to the analysis of the morphological complexity of tumor cells and tissues. A number of research studies have been applied to histopathological images for different tumors in an attempt to automate the diagnosis procedure. Some relied on a single texture measure for feature extraction, such as extraction of wavelet-based features [33, 46, 52], or employing other measures like the gray-level co-occurrence matrix or color space features for classification [11, 51]. Multiple measures for classification have been applied as well, such as using spatial and frequency texture features for classification by regression tree analysis [58].

Topological properties of the brain histopathological tissue were also extracted by a graph-based representation through probabilistically assigning a link between a pair of cells [19]. Morphological characteristics for feature extraction were applied in [54, 60], while others focused more on classifier improvement [24, 50]. However, the complexity and hierarchical structures of the vascular histopathological patterns in the brain are technically challenging to characterize [36]. The spatial resolution is also not sufficient to resolve the complex structure of capillary vessels [27] and to reliably collect morphometric data from histopathological tissue at high resolutions [15, 30]. Therefore, the fractal characteristics of the pathologic vascular networks were exploited to resolve structure complexity and to derive meaningful features that could assist in the diagnosis and/or staging of tumors. Fractal geometry was introduced as a better means of quantifying the microvasculature of normal pituitary glands and pituitary adenomas, and it was found that the use of the surface FD is more appropriate than microvessel density estimates [20]. Also in another work, statistical analysis was performed to study the fractal microvasculature and provides a quantitative assessment of grade complexity for WHO grade II and III gliomas [23]. The angioarchitecture heterogeneity of glioblastoma was shown to be able to distinguish the tissue variability between the different grades and can be used to differentiate low-grade from malignant tumors in histological specimens [21, 22].

With regard to brain meningiomas, Table 26.1 lists work recently performed for the purpose of automated histopathological subtype classification. All approaches developed in Table 26.1 were applied to the same meningioma dataset, which facilitates the comparison between the different classification results. This thereby avoided possible deformation in tissue appearance due to variation in tissue sample preparation/protocol or inconsistency attributed to stain reactivity from different manufacturers or using different microscopic scanners. This would be useful in preventing such variation effects on subsequent quantitative analysis. Moreover, Table 26.1 shows that the highest classification accuracy was achieved once the fractal analysis was performed in a multiresolution manner. The FD measure can breakdown the complexity of tumor heterogeneity and hence better discern the pathological patterns between the different subtypes.

The way the FD meaningfully relates to underlying physiology, in terms of quantifying tumor texture heterogeneity—as the rougher the surface, the more chaotic the tissue structure [5, 6, 8] explains the better performance of the fractal-based method compared to other texture analysis methods, hence providing a reliable characterization for diagnostic interpretation of histopathological images. The general design of a computer-aided diagnosis system and the effectiveness of histological fractal-based analysis in quantifying spatial heterogeneity in meningioma brain tumor subtypes are discussed next.

Table 26.1 Different texture analysis methods for histopathological brain meningioma classification^a

Reference	Features	Classification	Cross-validation	Accuracy
Lessmann et al. [33]	Color transforms and wavelet transform	Self-organizing map	Unsupervised (clustering)	79.00%
Wirjadi et al. [59]	Grey-level and color features	Classification and regression trees	Hold-out	83.40%
Qureshi et al. [48]	Wavelet packet and local binary patterns	Support vector machine	5-fold	80.00%
Qureshi et al. [47]	Wavelet packet and grey-level co-occurrence	Support vector machine	5-fold	82.10%
Fatima et al. [25]	Morphological and grey level co-occurrence features	Multilayer perceptron neural network	Leave-one-patient-out	92.50%
Zeng et al. [61]	Morphological and color features	<i>K</i> -Nearest Neighbor	Leave-one-out	85.00%
Strange and Zwiggelaar [53]	Morphological features	Random forest	10-fold	91.25%
Al-Kadi [5, 6]	Fractal dimension and wavelet packets	Bayesian	Hold-out	92.50%
Al-Kadi [8]	Fractal dimension, lacunarity, and wavelet packets	Support vector machine	Leave-one-patient-out	94.12%

^aMeningioma dataset provided by the Bethel Department of Neurosurgery in Bielefeld, Germany

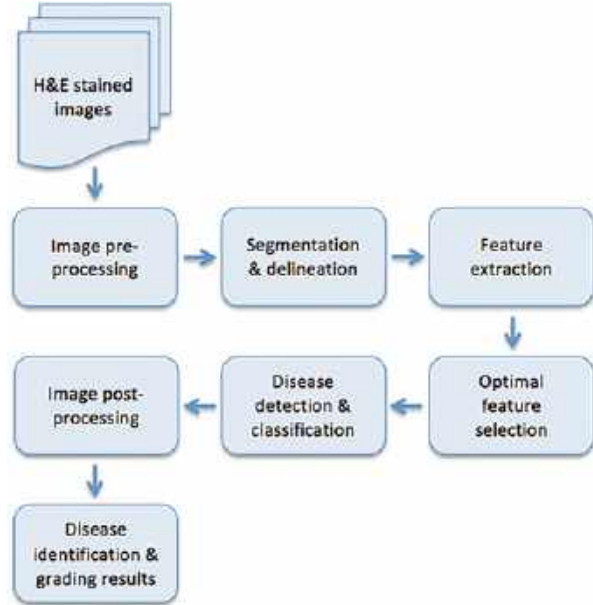
26.3 Automated Histopathological Image Analysis

General model design for automated histopathological tissue characterization and classification, which is also applicable to brain tumor subtype classification, is briefly discussed below:

26.3.1 Image Preparation

A tissue specimen runs through several basic preparation steps before it is ready for image texture analysis. This includes chemical fixation for cell/tissue preserving, tissue dehydration processing for replacing with an embedding medium, wax embedding, sectioning of samples so the microstructure can be clearly observed, and Hematoxylin and Eosin (H&E) staining—or using any other suitable stains—to increase tissue contrast and highlight cell nuclei features. Once the tissue samples are treated with the appropriate histology stain, images are acquired at a certain magnification level via microscopic imaging and made ready for histopathological

Fig. 26.3 A typical computer-aided diagnosis system for classification and grading of histopathological images



analysis. Figure 26.3 illustrates a typical computer-aided diagnosis system, which can be used to implement histopathological diagnosis.

26.3.2 Pre-processing and Focal Regions Segmentation

Pre-processing is associated with improving the features of interest that assist in determining the pathology. Particularly, image structures are enhanced for simultaneous visualization of the feature and image space, providing a method for exploration of correlations of texture patterns in the feature space to histological image characteristics.

Image regions having certain textural properties considered essential in the diagnosis process can be highlighted by different pre-processing approaches such as color normalization and separation to overcome stain artifacts; subsampling while preserving the minimum sampling rate (i.e., Nyquist rate); smoothing and denoising by simple Gaussian/median filtering or by employing an adaptive filtering technique that could reduce local tissue deformation while preserving edges and fine structures; contrast stretching and histogram equalization; or by other image enhancement techniques (e.g., dynamic range compression operations). Segmentation and delineation techniques such as nonlinear thresholding, edge detection, active contours, labeling, or clustering can follow the pre-processing stage for separating the foreground focal regions from the image background.

26.3.3 Feature Extraction and Classification

This stage is considered the heart of the computer-aided diagnosis system. Tissue patterns' characteristics are quantitatively captured for classification based on textural, spectral, morphometric, or color space features and could be performed at different resolution levels. This is followed by feature reduction, providing an effective way to select the most relevant features for alleviating the *curse of dimensionality* and achieving more accurate classification. The features can be refined using linear methods such as principal component analysis, linear discriminant analysis, or by nonlinear methods via mapping to higher dimensional feature space. The cost of prediction errors can be considered in the following stage of disease detection and classification by means of machine learning (i.e., supervised classification and similarity function) or statistical tests. Then, efficiency is assessed by the accuracy of the quantitative representation to classify patterns. Gaining significant acceptance by pathologists usually relies on achieving effective diagnostic performance while maintaining a certain degree of simplicity, i.e., avoiding behaving like a black box. Accuracy can be improved either by developing an effective feature extraction algorithm, which can provide high-quality features, or by designing a sophisticated classifier that can deal with nonlinear and high dimensional features; the main focus of this chapter will be on the former problem.

26.3.4 Qualitative Enhancement and Grading Results

Pathologists may choose to add complementary pathological-guided post-processing operations (e.g., morphological operations, shape extraction, annotation, etc.) to further “qualitatively” refine the results to be optimized for visual appearance. The system terminates with the disease identification and grading results stage, providing a complementary “second opinion” to pathologists in the diagnosis process.

26.4 Characterizing Tissue via Fractal Properties

Large numbers of cell nuclei are expected to set the structure of the tissue patterns at low magnification levels and few cell nuclei at medium and high magnifications. However, images of tissue texture usually tend to have a composite of multipatterns. The different patches encountered within these patterns can be exploited to analyze the fractality or statistical scale invariance of tissue, and hence the possibility of characterizing tumor features, e.g., its microvascularization [31]. The appearance of tissue patterns can reflect its abnormality in a sense that more chaotic vasculature means a “rougher” texture surface and hence a higher FD [4]. The development

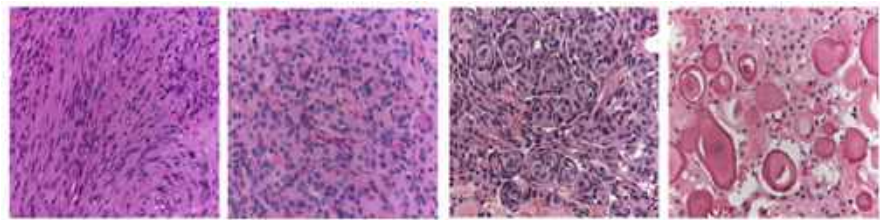


Fig. 26.4 Four types of grade I meningioma, from left to right (fibroblastic, meningothelial, transitional, and psammomatous)

Table 26.2 Main histological textural features for the four grade I meningioma subtype images shown in Fig. 26.4

Subtype	Characteristics
Fibroblastic (M_F)	Spindle-shaped cells resembling fibroblasts in appearance, with abundant amounts of pericellular collagen
Meningothelial (M_M)	Broad sheets or lobules of fairly uniform cells with round or oval nuclei
Psammomatous (M_P)	A variant of transitional meningiomas with abundant psammoma bodies and many cystic spaces
Transitional (M_T)	Contains whorls, few psammoma bodies, and cells having some fibroblastic features (i.e., spindle-shaped cells)

of a tumor alters the structure of the tissue in a disorganized fashion; thus, tissue patterns are not identical in all image samples and a certain degree of variation exists within and between image samples. Tissue heterogeneity places tissue patterns commonly in the category of stochastic textures that can exhibit quasi of *fractal-like* characteristics over several scales of magnification.

Pathologists usually look for several discriminative features that are indicative of tumor cells, e.g., abnormal size, shape, color, and arrangement. As an example of histological brain tumors, Fig. 26.4 shows four subtypes of grade I meningioma tissue biopsies distinguished according to the World Health Organization grading system [37]. Meningiomas exhibit a wide range of histological patterns and a single meningioma may show a combination of patterns. Patterns of tissue textures ranging from fine (micro-scale) to coarse (macro-scale) are encountered therein. An example of the former is the arrangement of fibroblastic and transitional cells in a concentric circle or whirlpool shape, while the large round nuclei in the meningothelial and the many cystic spaces in the psammomatous bodies contribute to the coarseness of the latter texture pattern. The histopathological features as an indication of tumor subtype are summarized in Table 26.2.

From a texture analysis perspective, the descriptive fractal features should characterize the qualitative characteristics used in clinical practice. This would assist in discarding irrelevant details and improve tissue classification of the different subtypes. More precisely, the subtle tissue distribution and the prominent cell nuclei arrangements in the coarse texture can be simultaneously localized and hence used for tumor subtypes differentiation. As for the tissue stochastic texture where it does

not usually have a repetitive pattern but homogeneous statistical properties, the directionality and density of the discontinuities in the texture's surface help locate changes in the pathological state of tissue. Moreover, tissue with a large amount of edges is considered rich with features, which will make the feature extraction algorithm more capable of delivering a better characterization.

26.5 Quasi-Fractal Texture Representation

Tumors tend to introduce irregularity into tissue pattern structure and cause deviations in cell nuclei distribution. This disorder increases the surface roughness of the examined texture pattern and affects pattern fractality. Regarding histopathological patterns, not all parts show self-similarity at different scales due to the inherent variability and effect of disease. An FD estimation method that can give an accurate estimation of the quasi-fractal texture patterns would be required. The box-counting and fBm algorithms are widely used for FD estimation as they can be applied to patterns that are not self-similar (nonlinear fractal images) and are relatively easy for empirical estimation [43]. Therefore, a comparison is made between the two methods when applied to histopathological images of brain meningioma (see Fig. 26.5). Examining the output FD texture images, the fBm generates more visible discontinuities compared to the box-counting method. More discontinuities in the absence of significant tissue deformation would mean a texture with richer information and hence a more reliable estimation of the roughness of the image surface. Thus, we can say that the fBm is better suited to represent textures with textures having fine structures, as the case with histopathological images.

26.6 Multifractality Analysis

Histopathological tissue properties of brain tumors can be investigated at different levels of resolution. This is essential to determine the grading of tumor progression, as fine fractal structures within the tumor tissue could be subtle or not easy to discern. The cell nuclei texture, shape, orientation, intensity, and denseness are different from one subtype to another and sometimes within the same subtype. The fine fractal structures having low contrast would be best characterized at higher resolution levels, and vice versa. Therefore, an integrated approach that incorporates both low and high levels of resolution can better account for morphological variability.

Advanced fractal analysis of tissue texture could be performed by estimating the FD and corresponding lacunarity measures at different levels of resolutions. The employment of fractal analysis in a multiresolution approach facilitates revealing many of the fine cellular and tissue details, and in a sense assessing tissue irregularities at multiple scales of different resolutions. A wavelet packet—which

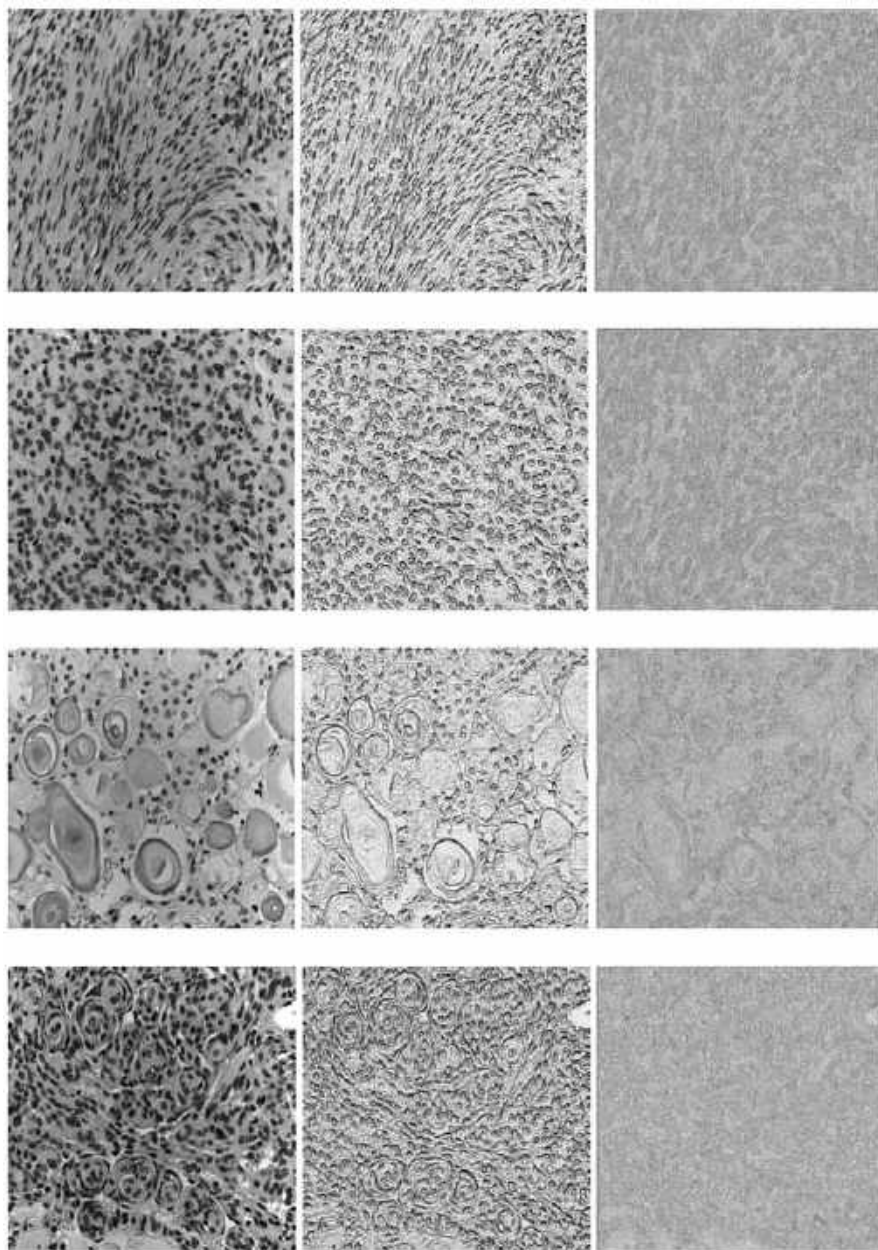


Fig. 26.5 First column represents the blue channel image for meningioma fibroblastic, meningothelial, psammomatous, and transitional subtypes, whereas the second and third columns are their corresponding fractal dimension images computed using the fractional Brownian motion and the differential box-counting algorithm, respectively (Image reproduced from [4])

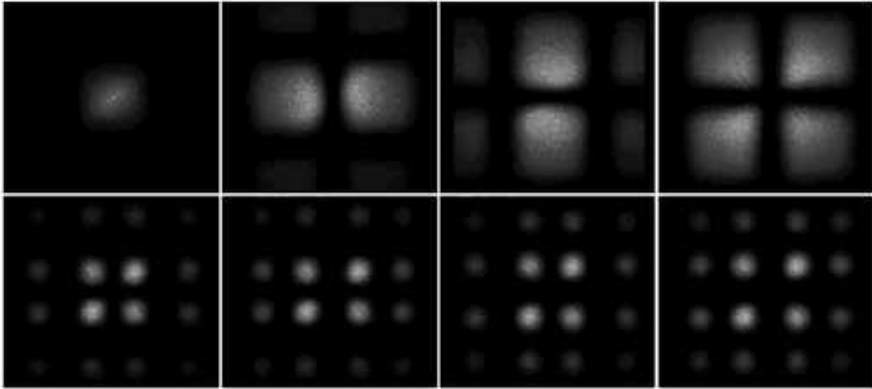


Fig. 26.6 Example of fibroblastic meningioma power spectrum of low-low, low-high, high-low, and high-high wavelet packet subbands for the 1st (upper row) and 3rd (lower row) levels of resolution (Image reproduced from [8])

is a generalized multiresolution approach—can decompose the complexity of the texture pattern into different frequency subbands, giving the opportunity to characterize the texture structure at the appropriate frequency channel. A Daubechies wavelet, which is widely used in characterizing signals exhibiting fractal patterns [17], can be used to obtain the wavelet packet subbands. Each subband is analyzed using its associated FD instead of energy, which has the advantage of being less sensitive to linear transformations of image intensity and abrupt changes in tissue texture [8, 9]. Then, the subband with the highest FD would be chosen for further decomposition in a tree-structured approach [5]. This facilitates the investigation of the possibility of higher frequency channels to provide significant information on tumor vascularization. Figure 26.6 supports this trend where the middle and high wavelet packet subbands for the first level of decomposition had a stronger power spectrum—the distribution of image power at a certain spatial frequency—as compared to the low-frequency channels. This indicates that the multiresolution fractal analysis can better detect the fractal characteristics inherent in meningioma tissue texture.

26.6.1 Assessing Fractal Texture Heterogeneity

In order to differentiate between two textures if they have identical FD values, even though the two textures might not look similar, the lacunarity—which is a measure of deviation from scale translation invariance of the FD texture—can assist in quantifying aspects of patterns that exhibit scale-dependent changes in structure [28, 35, 39]. Namely, lacunarity measures the “lumpiness” (or “gappiness”) of the fractal data, providing meta-information about the dimension of the fractal texture.

Table 26.3 Fractal dimension (FD) and its corresponding lacunarity (L) for each of the histopathological patterns/shown in Fig. 26.3

Fibroblastic (M_F)			Meningothelial (M_M)			Transitional (M_T)			Psammomatous (M_P)		
I	FD	L	I	FD	L	I	FD	L	I	FD	L
1	1.73	0.034	1	1.75	0.036	1	1.80	0.025	1	1.70	0.042
2	1.77	0.029	2	1.76	0.033	2	1.81	0.025	2	1.67	0.046
3	1.79	0.031	3	1.76	0.032	3	1.80	0.024	3	1.68	0.041
4	1.74	0.033	4	1.77	0.034	4	1.75	0.031	4	1.69	0.042

The lower the lacunarity value, the more heterogeneous the examined fractal area, and vice versa. Given a fractal image I_{FD} , the lacunarity (L) can be defined in terms of the ratio of the variance over the mean value as in (26.4), where M and N represent the size of I_{FD} [45].

$$L = \frac{\frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} I_{FD}(x, y)^2}{\left(\frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} I_{FD}(x, y) \right)^2} - 1 \quad (26.4)$$

To practically represent how lacunarity can further differentiate between two or more texture patterns that may exhibit similar dimensions, different texture images representing meningioma grade I patterns were used. After generating pixel-by-pixel FD images from each of the meningioma patterns, some of the textural patterns, although different, had their FD values similar as shown in Table 26.3. Therefore, computing corresponding lacunarity can add another dimension for discrimination between *FD-alike* textures. The spatial scale invariance that the FD exhibits can be complemented with the property of translation invariance of the *lacunarity* measure, hence, providing a rich description of pattern texture. For example in Table 26.3, an FD value of 1.77 & 1.75 was recorded for image textures 2 & 4 and 1 & 4 corresponding to the fibroblastic & meningothelial and meningothelial & transitional tumor subtypes of grade I meningiomas, respectively. This also reflects on the complementary relation between FD and lacunarity, where the many cystic spaces represented as large gaps between the psammoma bodies in the psammomatous subtype contribute to a higher lacunarity value. At the same time, this means less tissue heterogeneity, i.e., low FD value, as the associated *structure* sparsity can cause fewer occurrences of sudden perturbations within the fractal texture. Thus, the size distribution of the gaps within the fractal texture can act as another dimension to quantify the spatial heterogeneity and further differentiate between fractal textures having similar dimensions, but with a different appearance.

26.6.2 Performance Under Tissue Distribution Variation

Varying amounts and types of noise are inevitable during histological processing that affect the accuracy of the estimated texture measures. Natural variation in cell structures and distribution of cells in histological tissue contributes to texture complexity, while induced noise (i.e., color variations in tissue appearance) would increase disorder and reduce the quality of the extracted features. An intuitive approach would be to mitigate the impact of noise before reaching the feature extraction stage and then leaving the task of quantifying texture complexity to the feature extraction methods to deal with.

Variations in the staining conditions of histopathological samples may arise due to different factors. Examples of major source of noise that can affect tissue biopsies, but not limited to, include: uneven distribution of staining, poor contrast between H&E stains, nuclear staining errors, appearance of reddish-brown nuclei, dark precipitate on slides, and white patches on slides after deparaffinization step. A number of color normalization methods have been developed to deal with the issue of color variation in histopathology images caused by inconsistent biopsy staining [13, 32, 34, 40, 62]. However, the capability of the texture analysis method to accurately characterize tumor heterogeneity relies on how well it will perform under inter- and intra-subtype variability. Distinctive textural characteristics related to each subtype could be obscured and would vary from one sample to another when extracted from a single patient (see Fig. 26.7), also samples extracted from different patients within a subtype would vary as well (see Fig. 26.8). This renders the characterization of the different discriminating textural histological features even more challenging, given that this inherent variation would reduce the separability between the subtype classes as illustrated in Fig. 26.8. For instance, the fibroblastic subtype is harder to differentiate, as it can be easily misclassified to transitional or meningothelial subtypes due to the heterogeneity of the tissue cell nuclei [55].

Practically, different images for a certain meningioma subtype do not necessarily have an identical structure or the same number of cells. The diverse texture measures attempt to effectively quantify the structural complexity despite the uncertainty associated with the image texture surface. Therefore, the performance of statistical and model-based texture analysis methods was investigated under different cell tissue density conditions in ref. [6]. It was found that texture analysis performance based on the FD method was more stable in medium cell density and was less susceptible to tissue heterogeneity in sparse image structures (i.e., tissue with low cell nuclei density). Overall, the histological fractal-based features showed significantly improved classification performance as compared to other well-known statistical and model-based texture analysis methods [8].

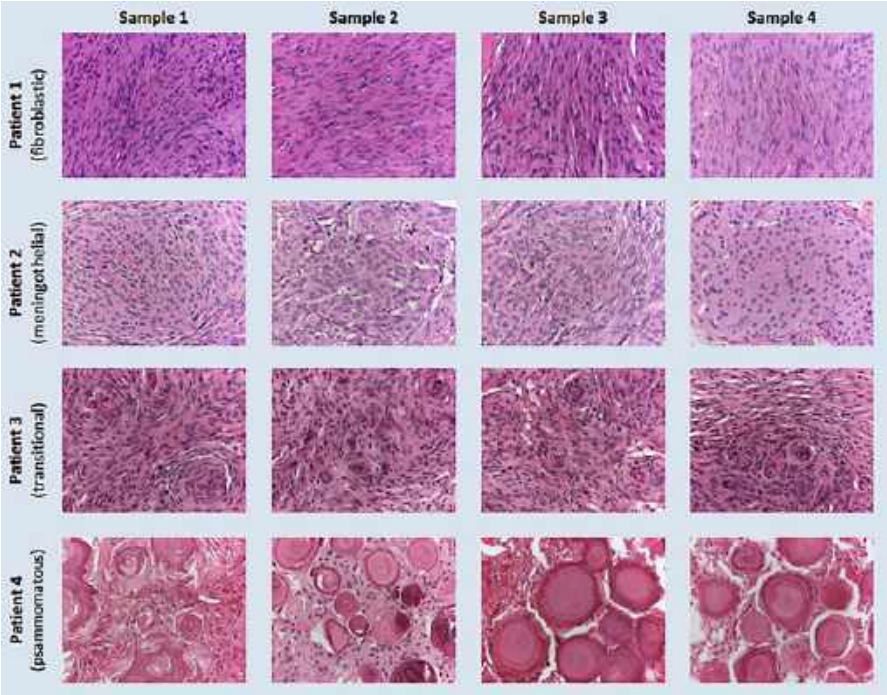


Fig. 26.7 Four different histopathological samples from four different patients showing the meningioma image texture behavior heterogeneity within the same tissue case (patient)

26.6.3 Automated Classification Using Multiresolution Fractal Features

Employing a computerized feature extraction approach at a single resolution might be limiting. The accurate delineation of the interlaced histopathological patterns is generally challenging—specifically in subtype classification. Characterizing texture at different resolution levels has the ability to filter out noise while highlighting dominant features that contribute better to subtype distinction [3, 7]. Gabor filters is an example of a multiresolution approach that can break down image texture morphological complexity. Also its high sensitivity to local features facilitates for pre-attentive or subtle texture discrimination [57].

Gabor filters are designed to resemble the performance of the mammalian visual cortical cells, in a sense of extracting features at different orientations and scales [18]. This sensitivity at various resolutions is advantageous for extracting meaningful features that can characterize underlying physiology. Features are derived from Gabor coefficients, as they cannot be used explicitly for texture analysis due to their high variability [14]. Texture signatures such as entropy or local energy [49] were used to characterize the different spatial-frequency decompositions to provide

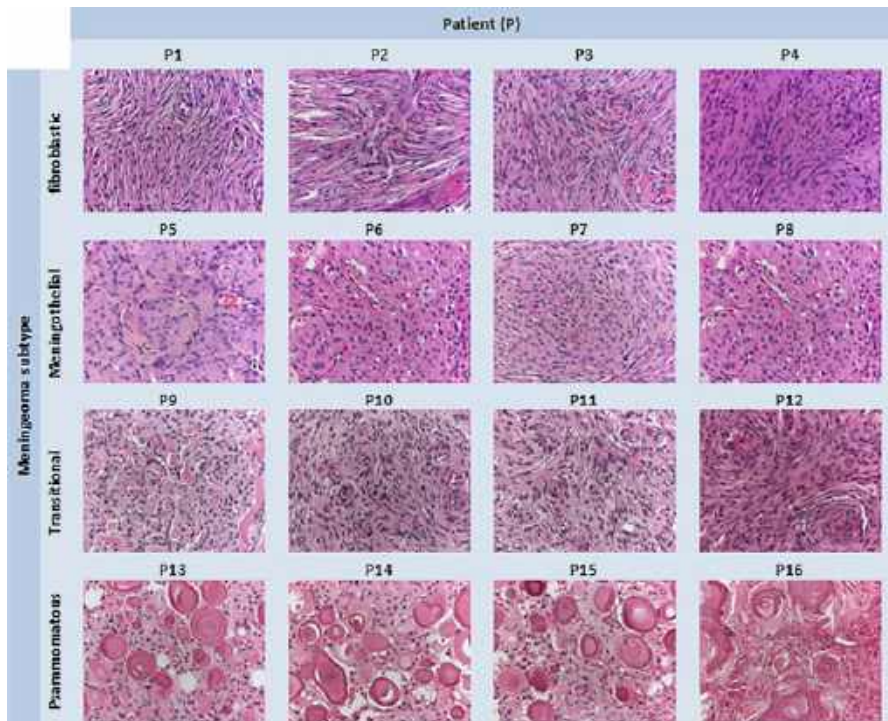


Fig. 26.8 Histopathological image samples extracted from 16 different patients referring to 4 different subtypes. The first row shows 4 different samples from 4 patients having a meningioma fibroblastic subtype, similarly for the rest of the cases. The different samples show that the simple observation of separation between the different subtypes is generally difficult to capture in an analytic solution

better spatial-frequency localization. Herein, the fractal dimension (FD) is used as a supporting feature (signature) to complement the texture characterization capability of the Gabor filter energy output [57].

Image texture can be analyzed in a multiresolution representation using Gabor filters. It can be defined as a Gaussian-modulated sinusoid with a capability of multiresolution decomposition due to its localization both in spatial and spatial-frequency domain. The real impulse response of a 2-D sinusoidal plane wave with orientation θ and radial center frequency f_o modulated by a Gaussian envelope with standard deviations σ_x and σ_y , respectively, along the x and y axes is given by [10]

$$h(x, y) = \frac{1}{2\pi\sigma_x\sigma_y} \exp \left\{ -\frac{1}{2} \left[\frac{x^2}{\sigma_x^2} + \frac{y^2}{\sigma_y^2} \right] \right\} \cos(2\pi f_o x) \quad (26.5)$$

where

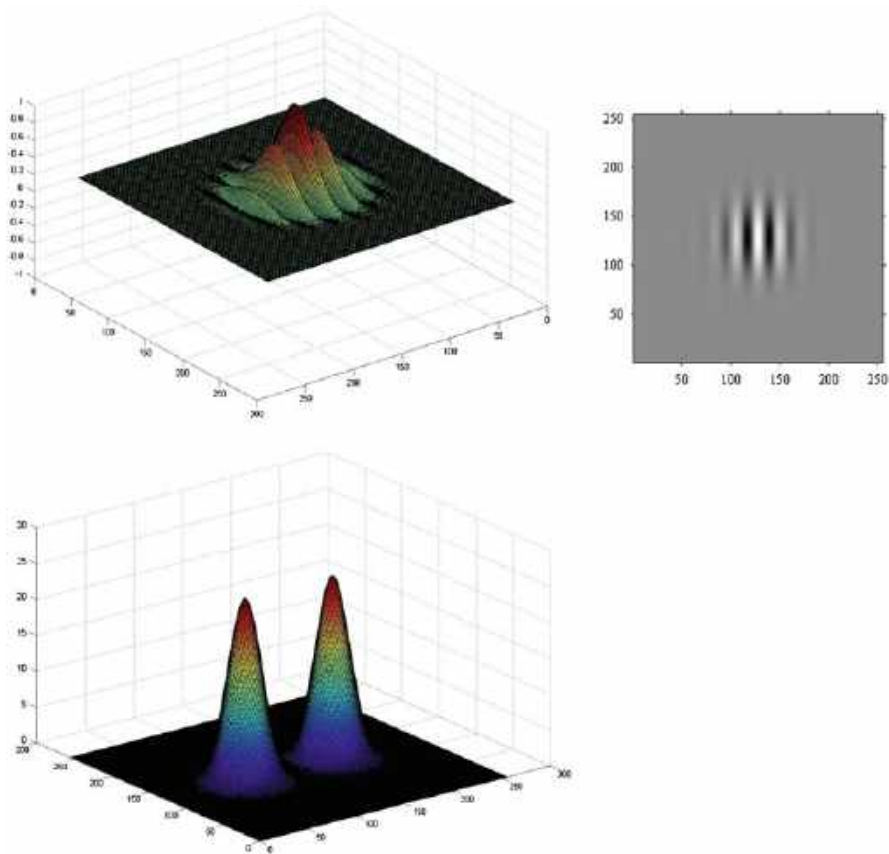


Fig. 26.9 Gabor filter having 0° orientation and 32 cycles/image width for a 256×256 size image

$$x = x \cos \theta + y \sin \theta \quad y = -x \sin \theta + y \cos \theta.$$

The Gabor filter in the corresponding spatial-frequency domain would be represented as two symmetrically spaced Gaussians as follows

$$H(u, v) = \exp \left\{ -2\pi^2 \left[(u - f_0)^2 \sigma_x^2 + v^2 \sigma_y^2 \right] \right\} + \exp \left\{ -2\pi^2 \left[(u + f_0)^2 \sigma_x^2 + v^2 \sigma_y^2 \right] \right\} \quad (26.6)$$

and spatial and corresponding spatial-frequency responses are graphically shown in Fig. 26.9. Multiple filters covering the spatial-frequency domain can be generated by varying the filters' center location by tuning the frequency f_0 with a specific angle θ .

The energy signature of each filter output can be estimated and used as a feature vector for classification. The Gabor filter energy $G_f(E)$ and fractal dimension (FD)-

Table 26.4 Classification accuracy comparison of Gabor filtered meningioma subtype images (G_f) after extraction of energy (E) and in combination with fractal dimension (FD) texture measures

Feature Extraction	Meningioma subtype				Total Accuracy
	M_F	M_M	M_P	M_T	
$G_f(E)$	83.75	86.25	90.00	73.75	83.44%
$G_f(E \& FD)$	100	87.50	96.25	96.25	95.00%

fused signatures are extracted from the meningioma dataset in Table 26.1 and used subsequently for automated classification using a Bayesian classifier and cross-validated by leave-one-out approach. It can be observed from Table 26.4 that an improved classification accuracy of 95% was achieved when combining the image texture Gabor filters’ energy and the fractal characteristics $G_f(E \& FD)$ as a feature vector—without performing any prior cell nuclei segmentation.

The meningioma textural properties are characterized at various scales and orientations for improved separability between different extracted features. Classification results gave an improved accuracy for all subtypes when the multiresolution Gabor filter method was combined with the image texture fractal dimension. This indicates that characterizing the cell-nuclei self-similarity properties via Gabor filters can assist in achieving an improved meningioma subtype classification.

As the cell nuclei size varies between the different subtypes, the multiresolution approach can better identify these differences and represent them by a measurable quantity (i.e., Gabor filter banks energy signature) exemplifying the amount of information at different scales. Yet relying on the energy of the filter outputs alone is deemed insufficient (c.f. Table 26.4); thus, analyzing the fractal features of texture could contribute toward highlighting aspects overlooked by the Gabor filter banks. The integration of the FD signature in the multiresolution texture analysis process provides a means to check for irregularity of salient meningioma features described in Table 26.2; namely, coarse image surface texture attenuations of cell nuclei local shape and spatial distribution would result in higher FD estimates, and vice versa. Since FD is considered a morphological parameter that can characterize the complexity of the shape and structure of an object, it could provide more stable parameter estimation amidst the intravariability of the number of cell nuclei in the same subtype.

26.7 Diagnostic Challenges and Future Perspectives

Thus far, histological fractal-based analyses of brain tumors have shown that they can effectively describe tissue complexity and irregularity. Nevertheless, time has come to assess how reliable the developed algorithms and techniques are under different variability conditions. An extensive application of these fractal analysis techniques to a wider patient population will assist in gaining wider acceptance

in the clinical practice as an essential tool for diagnosis that complements the pathologists' perceptive abilities. This includes having a large unified and publicly available dataset of brain histopathological images for various grades and subtypes. The database should simulate the diverse tissue variation and artifacts encountered in pathological diagnosis, whether it was due to human interaction (i.e., user or protocol variation), stain reactivity from different providers, or use of microscopic scanners from different manufacturers with different parameters of digitalization. Moreover, to assure clinical significance, other implementation issues related to how the developed methods are relatively easy for empirical estimation, and how computationally complex with respect to running time and memory usage should be evaluated as well. Hence, the robustness of the methods developed by investigators can be justifiably compared.

26.8 Conclusion

The aim of this chapter was to provide insight into the usefulness of the FD for classifying brain histopathology images. In particular, the employment of fractal analysis in a multiresolution approach for an improved tissue texture characterization. Combining the FD with the lacunarity measure can provide a good synergy for enhancing the capability for recognizing textural patterns exhibiting scale- and translation-invariant properties (i.e., characterizing the complexity and homogeneity, respectively).

Computer-aided diagnosis for histopathological image analysis has become essential in tumor grading and classification. This is attributed to the advances of algorithms and methodologies in tumor tissue characterization that allow for faster and more accurate clinical decision-making. However, due to the large variations and complexity in tumor tissue, differences among subtypes could be subtle, as brain meningiomas exhibit a wide range of histological patterns and a single meningioma may show a composite of multipatterns. A wise approach is to relate the extracted features to the geometrical properties of irregular and complex structures in tumor tissue, hence exploiting the fractality characteristics of brain tissue can better represent tissue complexity and heterogeneity of textural patterns. Currently, histological fractal-based methods are expected to play a promising role as tumor-grading tools that are minimally operator dependent to reduce variability associated with the manual grading system, thereby reducing uncertainty that may impact patient outcome.

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Chapter 27

Computational Fractal-Based Analysis of Brain Tumor Microvascular Networks



Antonio Di Ieva and Omar S. Al-Kadi

Abstract Brain parenchyma microvasculature is set in disarray in the presence of tumors, and malignant brain tumors are among the most vascularized neoplasms in humans. As microvessels can be easily identified in histologic specimens, quantification of microvasculature can be used alone or in combination with other histological features to increase the understanding of the dynamic behavior, diagnosis, and prognosis of brain tumors. Different brain tumors, and even subtypes of the same tumor, show specific microvascular patterns, as a kind of “microvascular fingerprint,” which is particular to each histotype. Reliable morphometric parameters are required for the qualitative and quantitative characterization of the neoplastic angioarchitecture, although the lack of standardization of a technique able to quantify the microvascular patterns in an objective way has limited the “morphometric approach” in neuro-oncology.

In this chapter, we focus on the importance of computational-based morphometrics, for the objective description of tumoral microvascular fingerprinting. By also introducing the concept of “angio-space,” which is the tumoral space occupied by the microvessels, we here present fractal analysis as the most reliable computational tool able to offer objective parameters for the description of the microvascular networks.

The spectrum of different angioarchitectural configurations can be quantified by means of Euclidean and fractal-based parameters in a multiparametric analysis, aimed to offer surrogate biomarkers of cancer. Such parameters are here described from the methodological point of view (i.e., feature extraction) as well as from the clinical perspective (i.e., relation to underlying physiology), in order to offer

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new computational parameters to the clinicians with the final goal of improving diagnostic and prognostic power of patients affected by brain tumors.

Keywords Angioarchitecture · Brain tumor · Fractal dimension · Fractal analysis · Glioblastoma · Microvasculature

27.1 Introduction

The brain is among the most perfused organs of the human body, and in order to meet its high metabolic demand, it has one of the most complex vascular and microvascular systems. Different regions of the brain are characterized by specific microvascular networks [8, 23–25, 39] whose level of complexity most likely reflects the function related to that specific area [15]. Since tumors are supported by different patterns of vascularization [35], it is possible to speculate that tumorigenesis (i.e., the sequential accumulation of mutations within tissue cells) is related to the disarray of the physiological microvascular bed or even related to a programmed rearrangement of a new and specific microvascular network. The study of the neoplastic angioarchitecture seems to be of paramount importance not only for research purposes but for clinical applications as well since it has been shown that the “normalization” of the abnormal microvascular architecture of tumors by means of anti-angiogenic agents, for example, can reduce regional hypoxia and eventually increase the efficacy of therapies [33, 34]. According to this model, the imbalance between pro- and anti-angiogenic factors in a tumor disorganizes the angioarchitecture, which might be brought back from its irregular pathological state to a physiologic state and then “normalized.” Fractal analysis offers several tools for the quantification and temporal follow-up of such angiostructural changes [26, 46].

27.2 Brain Tumors and Vascularization

The World Health Organization grading system classifies brain tumors according to histological features in a sort of benign-to-malignant gradient (from grade 1 to grade 4, with grade 4 being the most malignant) [43], although in the most recent classification systems, the genetic profile of tumors has added further diagnostic and prognostic information of paramount importance to make the classification more accurate [52]. The grading has diagnostic, prognostic, and therapeutic value, considering that grade 1 lesions have low proliferative and infiltrative potential and can be virtually cured by means of surgery and/or radiosurgery alone, while grade 4 cancer requires multimodal approaches (i.e., surgery followed by radiochemotherapy or more recent treatments) and has fatal prognosis.

Malignant brain tumors are among the most vascularized tumors in humans [43]. As microvessels can be easily identified in biopsy specimens, it has been suggested

Table 27.1 Examples of brain tumors showing peculiar microvasculatures

Different brain tumors	Same brain tumors of different grades
Gliomas, gliosarcomas, metastases, lymphomas	Gliomas: grades 1 to 4
Meningiomas vs. solitary fibrous tumors	Meningiomas: grades 1 to 3
Pituitary neuroendocrine tumors vs. normal pituitary gland	

that quantification of microvascularity might be used alone or in combination with other histological features to increase the understanding of the dynamic behavior and prognosis of brain tumors, as well as for diagnostic purposes. However, microvascularity is not always considered a feature related to brain tumor grading. Microvascular proliferation is considered a histopathological hallmark of glioblastoma [43], which is the most frequent and malignant glioma (WHO grade 4), but the differences in microvascularity between grade 2 and 3 gliomas, for example, are less clear and not used for grading. Also, meningiomas of different grades, as well as meningiomas in comparison to solitary fibrous tumors, clearly display different vascular patterns (see Table 27.1), but other features are still used to differentiate the different grades and/or tumors [2]. This is also related to the lack of standardization of a technique able to quantify the microvascular patterns in an objective way.

The vascularization of malignant brain tumors is the target for anti-angiogenic treatment (such as bevacizumab), providing reasoning on the importance of the different methods in quantifying and assessing the changes in tumor microvascularity. On the one hand, it is true that different brain tumors show different angiogenic patterns, but on the other, it seems that single histotypes, such as glioblastoma, show a very highly heterogeneous microangioarchitecture [5, 19, 20, 47]. We here summarize the concepts and methods related to microvascular detection and image analysis for the following morphometric analyses.

27.2.1 Immunohistochemistry

In histological specimens, microvessels can be detected by means of immunohistochemistry (IHC). Several antibodies have been described for the immunodetection of the endothelium, each one with its own pros and cons, summarized in Table 27.2. The most commonly used are CD34 and CD31 (CD refers to “cluster of differentiation”), which are considered pan-endothelial markers, while CD105 and Endocan are more commonly used for newly formed vessels (i.e., in the process of angiogenesis). Factor VIII and Ulex europaeus agglutinin I (UEAI) have almost fallen into disuse due to their multiple limitations. The ratios of CD34:CD105, CD34:Endocan, or CD31:Endocan immunoreactive vessels can provide the “angiogenic fraction” of the tumor [Di Ieva, unpublished results].

Table 27.2 Vascular markers commonly used in immunohistochemistry

Antibody	description	Pros	Cons
CD31	Also known as platelet-endothelial cell adhesion molecule (PECAM-1); membrane-bound glycoprotein and member of the immunoglobulin family	Stains vessels in both neoplastic and healthy tissue; sufficient sensitivity for blood vessels of all sizes	Frequent antigen loss during retrieval
CD34	Endothelial membrane-bound glycoprotein	Stains blood vessels of all sizes	Pan-endothelial: stains newly formed as well as preexisting vessels, making it ineffective for visualizing only angiogenesis. May lack sensitivity in some tissues
CD105	Endothelial membrane-bound glycoprotein; essential for angiogenesis. Also called endoglin	Stains only newly formed blood vessels during tumorigenesis; good for visualizing angiogenesis	Ineffective as pan-endothelial marker; weak or no staining in normal tissue
Factor VIII	Glycoprotein polymer involved in platelet aggregation; comprised von Willebrand factor and smaller antihemophilic factor	Effective for larger vessels staining	Lacks sensitivity for smaller vessels in comparison to other antibodies
Endocan	Cell-specific-molecule-1, proteoglycan secreted by endothelial cells; regulated by vascular endothelial growth factor and related to angiogenesis	Marker of small and large neoangiogenic vessels	Scattered immunostaining of vessels; generally not expressed in non-tumoral tissues

See review in Di Ieva et al. [22]

27.3 Morphometrics of Microvasculature

Qualitative rather than quantitative analyses have generally been used to describe the differences in the vascularization of the normal brain vs. brain tumors.

It has been suggested that vessel quantification can be added into the tumor classification system for grading as well as for prognostication. If the aim is to provide a morphometric biomarker (i.e., predicting patient prognosis, treatment response, etc.), the limitations are due to the lack of standardization of the proposed parameters. Morphometrics aims to reduce shapes to numbers, offering a reductionist approach to describe the geometrical complexity of natural objects. In pathology, morphometric analyses add a quantitative element to the qualitative description of the tissue [15], generating continuous variables, which can be used for statistical comparison.

Table 27.3 Morphometric parameters used to quantify the angio-space

Euclidean-based parameters	Fractal-based parameters
Microvessel density (MVD)	Fractal dimension (FD), microvascular fractal dimension (mvFD)
Density/number of vessels	Local box-counting dimension
Total vascular area (TVA)	Tortuosity
Local vascular area (in the hot spot)	Lacunarity
Perimeter	Representative elementary volume (REV) length scale
Mean diameter of the microvessels	Hurst exponent
Branching count	Other non-Euclidean parameters
Major and minor axis length	
Pathologist-based analysis	
Semiquantitative scoring systems, that is, 0 (no clusters), 1 (clusters), and 2 (very prominent clusters)	
	Compactness
	Shape factor
	Distance maps
	Microvascular structural entropy

Defining the “tumor-space” as the neoplastic volume, the “angio-space” refers to the network of microvessels filling such volume, not just in terms of quantity but also in terms of size, shape, and pattern of distribution [15].

Pathologists recognize distinct vascular angiogenic subtypes, described as palisades, glomeruloid vascular proliferation, vascular garlands, vascular clusters, and microvascular sprouting, and may also use semiquantitative scores to quantify the level of “clusterization” of the microvessels [21] (see Table 27.3). This seems to be relevant, especially when considering the relationship between vascular patterns and clinical outcomes, as different angiogenic protein expressions associated with different angioarchitecture can respond in different ways to chemotherapy [5]. These descriptions lack standardization, and some intra- and interobserver variability may be present.

Since the introduction of the use of the vessel density in 1972 [6], several other parameters have been proposed as potential “quantitators” of the microvascular system, used to quantify the angio-space (Tables 27.3 and 27.4).

We here classify these morphometric parameters in “Euclidean-based” and “fractal-based” parameters.

27.3.1 Euclidean-Based Parameters

One of the most used parameters is the microvessel density (MVD), based on the measure of the number of immunostained microvessels per square millimeter

Table 27.4 Morphometric parameters used to quantify the microvascular pattern identified in histological specimens of grade 2 and 3 gliomas

Angioarchitectural morphometric parameters	
A_{Tum}	Area covered by the specimen of brain tumor on the histological section
TVA	Total vascular area, the area of the microvessels of the whole specimen
A%	Ratio between TVA and A_{Tum}
MVs/mm ²	Microvascular density expressed as the mean number of microvessels per mm ²
Hot spot	1 mm ² area of the specimen having the highest local A(%)
Local A(%)	The % vascular area in the hot spot
mvFD	Microvascular fractal dimension: monofractal dimension of the microvascular pattern of the whole specimen (in a pre-determined scaling window)
Loc mvFD	Local mvFD: monofractal dimension of the microvascular pattern expressed in the hot spot area (scaling window, $\varepsilon_{min} = 1 \mu m$; $\varepsilon_{max} = 100 \mu m$)
Loc bcD	Local box-counting dimension: Value of fractal dimension estimated by the box-counting method in a non-monofractal range (scaling window, $\varepsilon_{min} = 1 \mu m$; $\varepsilon_{max} = 1000 \mu m$)

From Di Ieva et al. [21]

in the most densely vascularized area of the histological specimen (termed “hot spot”). Different techniques have been proposed for the MVD calculation, and several seminal papers have investigated its role in every human tumor (see review in [12]), including the brain and pituitary tumors (reviews in [12, 17]), but in conclusion, it has been shown that it is not a valid measure for guiding and/or evaluating anti-angiogenic treatment [32], nor for distinguishing different grades of brain tumors. All the histological analyses are limited by the sampling, especially in glioblastoma, which shows a very high histologic heterogeneity. The greatest limitation of the use of the MVD is the choice of the hot spot, due to (a) high inter- and intraobserver variability in its selection, (b) potential lack of representativity of the whole specimen, and (c) potential lack of representation of a complex 3D structure, such as the microvascular tree, in a 2D histological area [14, 20, 45].

Other Euclidean parameters can be computed on the histological specimens of brain tumors (see Table 27.3). Multiparametric analyses of combinations of such indices have shown microvascular morphometrics as a valid tool to differentiate different grades of brain tumors [20, 38, 47].

The greatest limitation of the described parameters is that Euclidean geometry can only quantify regular and smooth objects and therefore can only offer approximations of the roughness expressed by natural objects, such as microvascular trees. The complexity of vascular systems depends on (a) the number of vessels, (b) their size and shape, and (c) the pattern of the vessel distribution and the nonlinear, temporal, and spatial advance of the promotion, progress, mediation, and inhibition of angiogenesis generating the complex ramified network irregularly filling the environment surrounding the tumor, which can be described by Euclidean geometry only with great approximation [29].

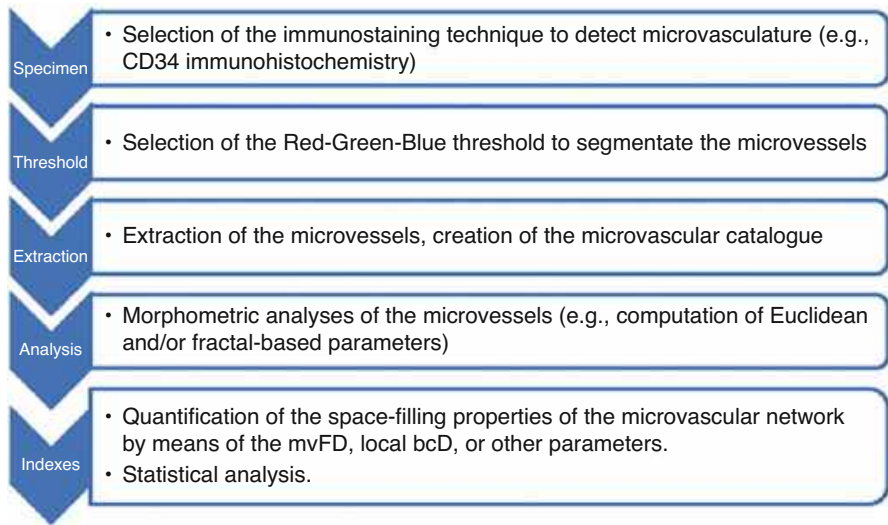


Fig. 27.1 Flowchart used for image and morphometric analyses of microvasculature in histological specimens

Fractal geometry overcomes the limits of Euclidean parameters. The fractal-based parameters are described next.

27.3.2 Image Analysis

The first step to apply any morphometric analyses on the neoplastic angio-space is the choice of the technique to visualize the microvessels and the following image analysis, which should be standardized (Fig. 27.1).

When the target of analysis is the microvascular network, several issues have to be considered, for example, whether the analysis is performed on the basis of the space-filling area of the vessels or on skeletonized representations of the microvessels.

In the image analysis, eventual artifacts related to the segmentation of immunoreactive vessels have to be considered. For example, nonspecific immunoreactivity can limit the automatic detection of the vessels, as in the case when antigens diffuse to other compartments (e.g., CD34 antigens which immunostain cell nuclei or necrotic areas rather than endothelial cells). In such cases, a pathologist is still required to identify any artifacts, which should be eliminated from the analysis.

Other digital image acquisition calibration parameters should be standardized as well, in terms of regulating the light intensity of the microscope, maintaining the same resolution over the whole series, and using an identical threshold to extract the immunoreactive vessels. A virtual microscope can be used to digitize the entire

Fig. 27.2 Example of a vascular catalog (in this case, of a glioblastoma, with magnification of the vessels in the *inset*). The catalogs give information at a glance on the number and shapes of all the microvessels extracted from the entire specimen



specimen rather than a few areas selected manually. The vessels can be extracted as a whole from the specimen and organized in a vascular catalog, giving a snapshot on the number and shape of the microvessels within the specimen (Fig. 27.2).

27.4 Fractal-Based Morphometric Analyses of Microvessels

Fractal geometry provides a computer-aided method to describe and quantify the roughness and geometrical complexity of the microvessels distributed within the tumor, as it has been shown in a vast variety of neoplasia, including prostate, kidney, lung, colon, skin, and brain tumors as well. By concentrating on the irregularity of tumor growth rather than individual measure of size (such as diameter and volume or other Euclidean parameters), fractal geometry is well suited to quantify the morphological features that pathologists have long used to describe malignancies in a qualitative way (i.e., wrinkled borders within the host tissue,

random microvascular patterns, etc.) [3]. Several researchers have demonstrated the fractality (or “semi-fractality”) and multifractality of the vascular and microvascular trees of many tissues and organs, in physiologic as well as pathologic states, including the human brain [1–3, 8, 10, 11, 27, 28, 30, 36, 42, 46, 49].

The main aim of computational fractal-based analysis of the microvascular patterns is the quantification of parameters, which can be used as surrogate biomarkers.

27.4.1 *Microvascular Fractal Dimension (mvFD)*

The most widely used parameter in fractal geometry is the fractal dimension (FD or D_F), which describes the space-filling properties of irregularly shaped objects [4, 10, 11, 27]. FD has been shown to be the most robust estimator of vascular networks [37, 51]. Extending the use of FD to the quantification of microvessels, the microvascular fractal dimension (mvFD) quantifies the space-filling properties of the microvessels within a tissue; in the case of a tumor, mvFD quantifies the level of geometrical complexity of the microvessels embedded within the tumor. The mvFD adds a qualitative component to the parameterization of the angio-space, trying to answer the following question: assuming an equal number of vessels, what can be said about their shape, size, and pattern of distribution? This means that histological specimens of similar or different tissues can show the same value of MVD or total vascular area, for example, but a very different pattern of distribution, with different values of mvFD. Of course, it can also happen that histological tissues with the same mvFD values differ for the Euclidean parameters, making the approaches complementary methods of quantification. Therefore, mvFD offers useful information about the tumoral space filled by microvessels. The parameter is computer-aided, and its objective measure makes it a comparable parameter among different operators. If it is important to analyze the distribution of the vessels within the tissue, by means of FD, for example, it is equally important to analyze complementary parameters, the “vascular gappiness,” that is, the avascular spaces, by means of the lacunarity or other methods, like the distance maps [46].

In a 2D histological specimen, mvFD ranges from 0 to 2. For example, a tumor with a hypothetical mvFD value of 1.95 owns a microvascular network almost filling its area, like the Peano’s curve, with a very complex geometric and highly space-filling microvascular pattern (Fig. 27.3). This microvascular pattern virtually fills the whole tissue in which it is embedded, reducing the avascular spaces to the minimum (therefore, increasing the lacunarity) [4] (Fig. 27.4). In the 3D space, such as in electron microscopy reconstructions of the microvascular trees in the 3D volume, FD trends toward a maximal value of 3 [46, 49].

In several kinds of tissues, it has been shown that FD and lacunarity values differ in the physiological and pathological vascular networks [4, 13, 26]. This means that the disarray of the microvascular network, which extends from the normal tissue (e.g., the brain) to the tumor, can be quantified by means of fractal analysis, in a kind

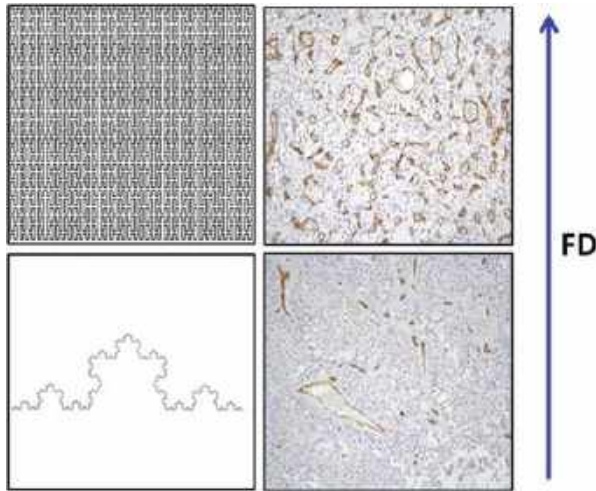


Fig. 27.3 Values of the fractal dimension (FD) on a 2D plane, comparing two fractal objects with the microvasculature (endothelial CD34 immunostaining) of a pituitary neuroendocrine tumor (*lower image*) with the specimen of a normal pituitary gland. The *lower image on the left* shows the Koch's curve ("snowflake"), with $FD = 1.26$. FD increases in more space-filling objects, namely, in the Peano's curve (*higher image on the left*, with $FD \approx 2$) and in the microvasculature of the normal pituitary gland, which is more homogeneously distributed within the parenchyma of the gland

of gradient of morphometric differences between tissues. Based on the evidence that the neoplastic microvascular network is abnormal, it has been experimentally shown that the "normalization" of such disarray can be used for therapeutic purposes, as previously quoted, and the FD can be used as a surrogate biomarker for diagnosis and/or treatment follow-up [4, 26]. By means of 2D and 3D analyses, FD values of normal brain microvasculature have been found to be higher than that of neoplastic regions, meaning that the normal brain is better suited to more space-occupying microvasculature [4, 26, 49]. However, due to the use of different methods and techniques, these results are controversial. For example, in one experimental study, the vascular FD values of primate and rat brain tumors have been found to be higher than those of normal brain parenchyma, thus indicating a greater metabolic supply and bigger surface area for gas exchange within the tumors [46]. In numerical terms, fractal-based demonstration that tumoral and normal vascular networks are intrinsically different; Baish et al. introduced a complementary metric to the FD, the tortuosity, defined as the ratio between the minimum vascular path joining two points and their geometric distance from each other [4]. A greater tortuosity has been correlated with a worse response to drugs, confirming again the importance of the analysis of the angiostructure from the therapeutic perspective.

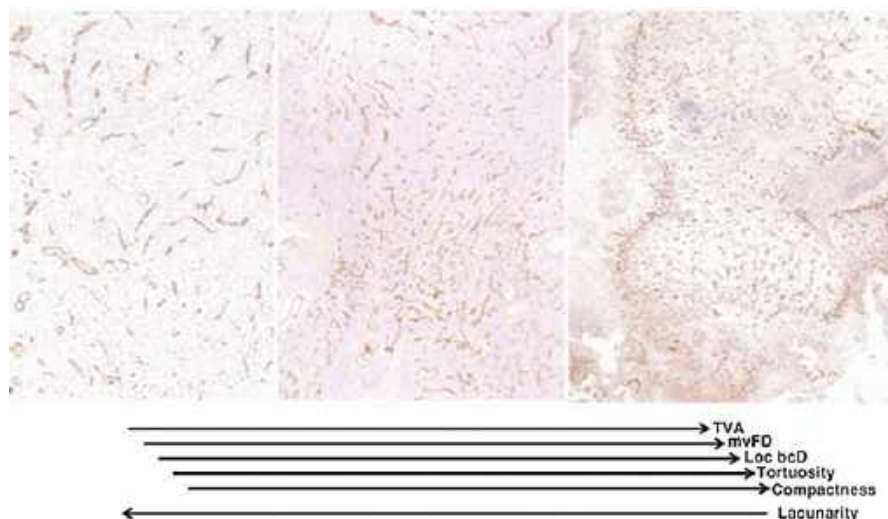


Fig. 27.4 Specimens of three cases of GBM, showing different microvasculature (microvessels detected by using antibodies raised against CD34 with brown coloration). The angio-space of each tumor can be described by means of several parameters (*TVA* total vascular area, *mvFD* microvascular fractal dimension, *loc bcd* local box-counting dimension. See Table 27.4)

Among several reasons, including their relative simplicity and the speed to be computed, the most used methods to compute *mvFD* are the box-counting and the sandbox methods (see Chap. 2). These methods have been applied to whole histological specimens of brain tumors as well as on selected spots. The choice of the “hot spot” can be performed manually (i.e., the pathologist chooses the area subjectively considered to be the most vascularized and representative of the microvascular network of the tumor) or by computer-aided methods (e.g., the spot with the highest vascular area, as calculated by the computer by means of different methodologies, including artificial intelligence-based methods). The first method is biased by the introduction of the operator-dependent choice of the hot spot, while the second method can be biased by the operator-dependent choice of the threshold to automatically extract the immunoreactive vessels from the specimen or from manually erasing artifacts outside the region of interest (or by overfitting in AI models). An ideal method should be objective and reproducible, without any operator-dependent biases. Moreover, as it often happens in pathology, the analysis can be biased by the choice of the specimen itself (the problem of sampling, which could not be representative of the whole tumor, like a sample resembling a WHO grade 2 glioma within a grade 3 tumor). To limit this problem, other parameters have been introduced.

27.4.2 Local Fractal Dimension and Local Box-Counting Dimension

In order to avoid the sampling of nonrepresentative areas of the tumors such as necrotic areas, the local fractal dimension considers FD in the hot spot. The subjective selection of the hot spot would limit the objective reproducibility of such parameters in the same way as with the MVD. In order to avoid such limitations, the hot spot can be automatically chosen, for example, by computing the vascular area for each 1 mm^2 area on a grid drawn on the specimen (Fig. 27.5) and extracting the spot with the highest ratio (immunoreactive surface of vessels/tumoral area). Once selected, the 1 mm^2 spot with the highest vascular area becomes the hot spot for the following analysis (Figs. 27.5 and 27.6). On such a spot, the local mvFD can be computed. A previous analysis showed that a monofractal behavior was found considering a two orders of magnitude window, that is, between 1 and $100 \text{ }\mu\text{m}$ [21]. Assuming an average microvessel diameter between 5 and $10 \text{ }\mu\text{m}$, such a fractal window considers the smallest features of the microvessels (i.e., the roughness of the outline), as well as the agglomerate of microclusters of vessels (at the magnitude of $100 \text{ }\mu\text{m}$). As shown, this was considered insufficient to cover the entirety of the histopathological features of the microvessels, such as the macro-agglomerates in large clusters (like the garland-like vascular structures). A fractal window ranging between $1 \text{ }\mu\text{m}$ and 1 mm , that is, covering three orders of magnitude, was considered significant to cover all of the mean histopathological characteristics of the microvessels, including the intrinsic morphologic features and their clustering properties. Applying the box-counting method in such a range, a monofractal behavior was not confirmed (see Fig. 27.6), although a similar slope of the curve was found across all the analyzed specimens. The several power laws showed that glioma microvasculature clearly demonstrates a multifractal distribution of geometrical complexity of the glioma's microvessels when observed at various scales of magnification. The slope of a straight line interpolating the points on the log-log graphs was considered as the mean value expressing the space-filling properties of the microvessels in the hot spot, and the value was indicated as the local box-counting dimension (local bcD) [21]. Both mvFD and local bcD express, in a single variable, the ability of the microvessels to fill the space in which they are embedded (i.e., the tumor), which is a characteristic intrinsically correlated to with their number, size, shape, and distribution pattern.

27.5 Fractal-Based Analysis of the Angio-Space in Brain Pathology

We define the concept of “angio-space” as the set of parameters quantifying the microvasculature within a tumor or the tumoral space occupied by its own angioarchitecture (Di Ieva [13] and unpublished data). According to the speculation

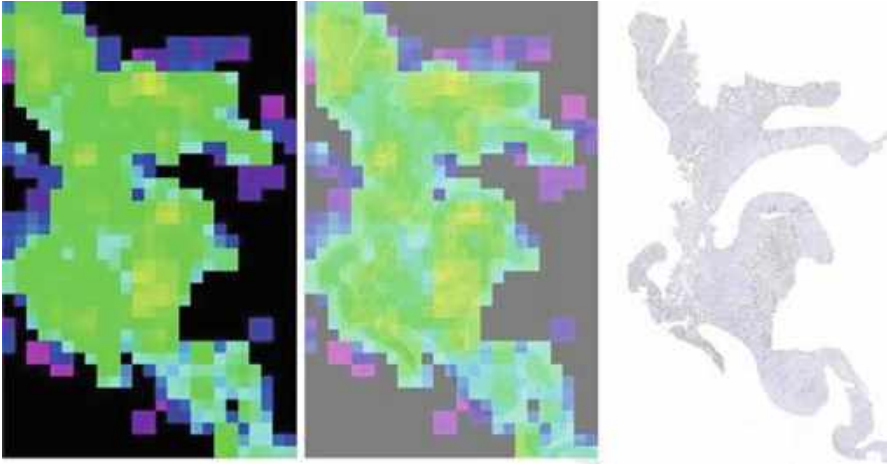


Fig. 27.5 Microvascular map for the automatic selection of the hot spot. Over-imposing a grid of 1 mm^2 boxes on the entire specimen, the vascular area can be computed on each box and a color map can be associated with the different ratios. In this example, the yellow boxes represent the regions of the specimen with the highest vascular area. Raw data also show the value of the vascular area for each box, and the one with the highest value is automatically chosen as hot spot for the following computational analyses. (Software by Carlo Russo)

that different tumors, or even different subtypes of the same tumor, have a specific “microvascular fingerprint” [13], it has been speculated that a gradient of geometrical representation (and then of fractal angioarchitectural parameters, including FD) exists between physiological and pathological tissues, as well as different states of the same tissues. The pituitary gland is an exemplary model: Euclidean-based approaches (i.e., vascular area and MVD) and fractal-based analyses have clearly shown that it is not only more vascularized but also exhibits a higher microvascular heterogeneity and geometrical complexity in the comparison of the pituitary neuroendocrine tumors [17, 18, 22]. Regarding the different subtypes of pituitary tumor, analyses of the MVD have shown very discordant results [17], with some preliminary fractal-based analyses not showing any statistically significant differences in the angioarchitecture of micro- vs. macro-adenomas, for example [Di Ieva, unpublished results]. The pituitary gland is a simple model from which to extend the fractal analysis of the microvasculature to the study of brain tumors, by analyzing histological specimens treated with antibodies raised against the endothelium (in the whole specimen and/or hot spots), or volumetric reconstructions of histological tissues and/or radiological images (see Chaps. 19, 20, 21, 22, and 23). If the pituitary gland shows a relatively low variability of the microvascular patterns (meaning that the vascular network is quite homogeneous) [16], malignant brain tumors show very high microvascular heterogeneity. By analyzing 114 glioblastoma specimens (regardless of the IDH or other genes status, as not known at the time of that study), a mean mvFD value of about 1.44 (ranging 1.06–1.87) was found,

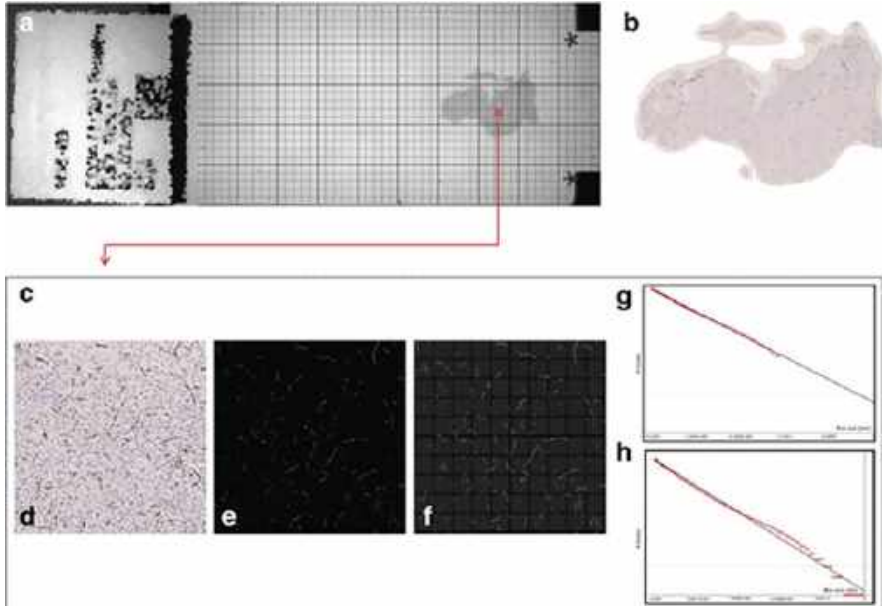


Fig. 27.6 Computer-aided technique for fractal analysis of the microvasculature in histological specimens. Estimation of the morphometric fractal parameter in the whole section and in the hot spot region of specimens of brain gliomas; detection of microvessels using antibodies raised against CD34. The extraction of the whole specimen and the segmentation of the immunoreactive vessels (**a**, **b**) allow the automatic measurement of the morphometric parameters listed in Table 27.4. Moreover, a grid of boxes measuring 1 mm to a side length is superimposed on the whole histological section. For each box, the ratio between the local vascular area and the local neoplastic area is calculated [*local A*(%)]; the box with the highest *local A*(%) is automatically selected and extracted to be used as the “hot spot” representative of the whole specimen (**c**). (**d**) Hot spot of the specimen automatically extracted. (**e**) Selection of the RGB threshold to automatically extract the CD34+ vessels. (**f**) Box-counting method: superimposition on the image of grids with boxes of different lengths, from a minimum size ($\varepsilon_{\min}$) to a maximum size ($\varepsilon_{\max}$), in the preselected scaling window. (**g**) A monofractal behavior was found by limiting the scaling window to a range of two orders of magnitude ($\varepsilon_{\min} = 1 \mu\text{m} - \varepsilon_{\max} = 100 \mu\text{m}$). The slope of the obtained curve is the microvascular fractal dimension (mvFD) [Axes of the log–log graph (box-counting method): *X*-axis = Box size (mm); *Y*-axis = *N* Boxes]. (**h**) Considering that a maximum size of the box of $100 \mu\text{m}$ does not cover the complete hot spot, a wider scaling window was also applied (Reproduced from Ref. Di Ieva et al. [21])

with a coefficient of variation (CV, standard deviation divided by the mean) of 44% [20]. For comparison, the CV of pituitary tumors was found less than 10% [16]. This very high variability in the space-filling properties of microvasculature of glioblastoma reflects the very high angioarchitectural heterogeneity shown by this malignant tumor. Glioblastomas’ microvascular networks can be represented in a very broad and continuous spectrum of possible geometric configurations; this was the first mathematical demonstration that glioblastoma is *multiforme* (from the Latin

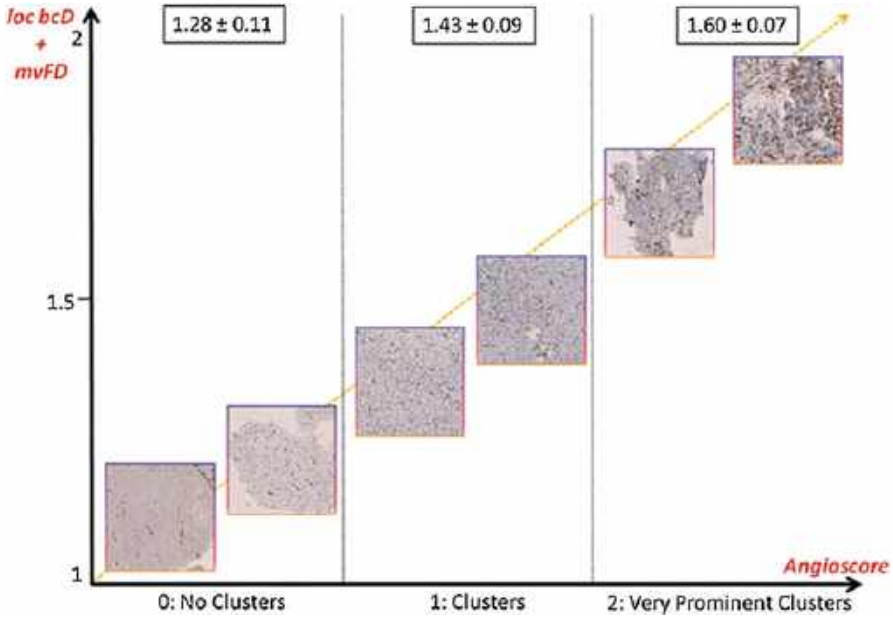


Fig. 27.7 Correlation between the fractal angiormorphometric parameters and the level of clustering of the microvessels. Schema showing the correlation between the angioscore assigned by the neuropathologist to each specimen and the fractal-based variables [local box-counting dimension (loc bcD) and microvascular fractal dimension (mvFD)]. The value reported for each angioarchitectural group is the mean value of loc bcD $\pm$ standard deviation. The yellow line indicates the increasing grade of malignancy of the tumor. The fractal parameters assigned nearly 75% of the gliomas to the correct histological grade and angioscore. (Reproduced from Ref. Di Ieva et al. [21])

meaning “multi-shapes,” according to its previous definition) also in relation to its microvasculature [20].

Morphometric and computational fractal-based analyses also showed that grade 2 gliomas have different angio-spaces in comparison to grade 3 gliomas (Fig. 27.7), although this might be less relevant in the new genetic profile era. Among several morphometric parameters, the local mvFD and especially the local bcD were shown to be the most reliable quantitative indicators of the neoplastic microvasculature, making them potential surrogate biomarkers [21]. The histopathological grading of gliomas can be challenging in some cases, especially in grade 2 vs. grade 3 gliomas, and the erroneous diagnosis can result in patients’ over- or undertreatment [50]. For this reason, computational methods, which can help in differential diagnosis, are more than welcome to be added in the pathologists’ and clinicians’ armamentarium.

In a study on 43 patients affected by glioblastoma, the microvascular fractal dimension and lacunarity have been shown to be statistically correlated with the expression of genes related to neoplastic cytoarchitecture and angiogenesis [40], eventually offering also the background for the association between genetic

profile and microvessels' morphology. Moreover, a study on 207 glioma patients, including 94 glioblastoma, confirmed Di Ieva's findings [13] that the microvascular fractal dimension is a better parameter than microvascular density to quantify the microvasculature complexity and determined that the microvascular fractal dimension can be a potential prognostic factor as related to patients' overall survival, progression-free survival, and response to chemotherapy [9].

27.6 Limitations

If fractal-based angioarchitectural classification of brain tumors can be useful for diagnostic and prognostic purposes, there are still some limitations to acknowledge. For a description of the limitation of image and fractal analysis, see Chaps. 1 and 13. Below, we summarize some main points. First of all, according to the basic principles of fractal geometry, it does not mean that objects owning the same mvFD, for example, show the same microvascular pattern. As previously stated, this means that fractal indexes should be analyzed as complementary to other morphometric parameters in a multiparametric analysis.

When the analysis is performed on histological specimens, the sampling problem can limit its benefit. The analysis of the whole specimen or of the computationally chosen most representative spot of the specimen tends to decrease the limitations of other analyses, like the MVD-based ones, but a histological sample still often does not represent anything more than a very small fraction of the whole tumor, and this should be taken into consideration. The estimation of a 2D representation of the microvascular tree could not reflect the real 3D complexity of the microvascular network because 2D methods performed on histological specimens are length related rather than mass related, and this can underestimate the true 3D complexity of the tree in space [7, 44]. This issue can be limited by using volumetric analyses and 3D reconstructions of tissues. The effect of geometrical deformation (e.g., cracks) during the process of sample preparations, and other artifacts such as nonhomogeneity of staining, may also have some impact on the accuracy of the fractal analysis. For this reason, also the sampling, immunostaining, and preparation of the histological specimen should be standardized.

Moreover, the histological section represents a "snapshot" in time, a "frozen photo" of the tumor at the moment of the surgical operation, and each quantification refers to that specific moment, not taking in account the dynamic process on the temporal trend of the tumor. For this reason, the temporal changes of tumors should also be considered by means of neuroradiological and/or nuclear medicine findings over time (such as perfusion parameters or methionine or other aminoacids' uptake in PET, positron emission tomography) [19].

As previously stated, the value of FD and other fractal-based parameters is highly dependent on the image analysis parameters (and "pixel related," according to the resolution). Several research inquiries have considered different tissues (brain cortex, tumors) in different models (humans, rats), visualized in different ways after

undergoing different treatments. When comparing the several values found in the published research, a plethora of results show contrasting findings and this could be avoided only by a standardization of techniques, materials, and methods.

The standardization ought to be done in the analytical analysis of the fractal parameters. It is in fact known that biological fractals, such as microvascular networks, are statistically self-similar only within a specific scaling window (i.e., covering at least two orders of magnitude) [42]. The choice of different windows may give different results, and such choice should be standardized as well or at least justified: if a window ranging 1–100 μm , for example, considers an FD that reflects the organization of the vessels in microclusters and their roughness; on the other hand, a range of 100–10,000 μm does not take into account the roughness of the single vessels but rather the distribution of vascular clusters on a larger scale [12]. The lack of standardization of all these elements could produce results no more comparable between different laboratories. Last but not least, microvessels cannot be real fractals, as pointed out, and the “fractalization” of such natural objects can be a mathematical idealization limited by computational manipulation. In order to avoid the several definitions expressed in the very thick related literature on fractality, quasi-fractality, pseudo-fractality, or multifractality of microvessels [28, 41, 48, 53, 54], it is here enough to say that fractal analysis, far from being an infallible and universal method, is still able to offer several reliable parameters for an objective, reproducible, and realistic description and quantification of the microvasculature in normal as well as pathological tissues.

27.7 Future Perspectives and Conclusion

As the architectural pattern of each tumor seems to be its specific microvascular fingerprinting, objective and reproducible morphometric biomarkers are required to describe and quantify the so-called angio-space. Such indexes should be (a) able to quantify what pathologists describe in a qualitative way on the histological specimens of tumors, (b) objective, that is, not affected by intra- and interobserver variability, and (c) clinically meaningful, having prognostic and/or predictive value [21]. Fractal analysis offers several parameters that are promising candidates for such purposes.

In addition, the methodological phase of testing morphometric parameters should be followed by clinically oriented research; to be tested as a potential biomarker, in fact, the morphometric fractal-based parameters should be analyzed in the same way that new oncologic therapies are introduced into clinical practice, in step-by-step successive phases of study [31]. Although the translation of the concept of “fractal microvasculature” into clinical practice (i.e., from “bench to bedside”) is still difficult to be realized [15], it should be emphasized that the research in the field of microvessel morphometrics should run alongside research focused on tumoral angiogenesis (research on molecular biomarkers, such as vascular endothelial growth factor, hypoxia-induced factor, etc.), in order to offer a holistic

view on the effects of structure and function on the environment and viability of brain tumors. This means that the morphometric analyses should run alongside the biological approach, and eventually be implemented by multiomics approaches (e.g., proteomics, genomics, etc.). Moreover, morphometric analyses should be integrated into *in vivo* techniques (neuroimaging and nuclear medicine methods) to follow-up brain tumors and response to treatment. AI methods can also be used to implement features' extraction from tumors' microvascular patterns. For example, deep learning methods could be used to characterize the angio-space, whilst machine learning methods could be used to mine data from multi-parameter analyses of the neoplastic microangioarchitecture.

In light of the speculations and findings that have been illustrated in this chapter, computational fractal-based and multiparameter morphometric analyses will ultimately find their place in the oncologist's toolbox as a conventional diagnostic procedure.

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Chapter 28

Fractal-Based Morphometrics of Glioblastoma



Lee Curtin

Abstract Morphometrics have been able to distinguish important features of glioblastoma from magnetic resonance imaging (MRI). Using morphometrics computed on segmentations of various imaging abnormalities, we show that the average and range of lacunarity and fractal dimension values across MRI slices can be prognostic for survival. We look at the repeatability of these metrics to multiple segmentations and how they are impacted by image resolution. We speak to the challenges to overcome before these metrics are included in clinical care, and the insight that they may provide.

Keywords Glioblastoma · Morphology · Mathematical oncology · Imaging biomarkers

28.1 Introduction

Glioma is the most common primary brain tumor with around 13,000 estimated to be diagnosed in the USA in 2021 alone [13]. Glioblastoma, the most aggressive form of glioma, has a short median survival from diagnosis of 14–16 months with the current standard of care treatment [16–18]. This short median survival is accompanied by a small 5-year survival of only 7.2% in the USA [13]. During the course of care, the clinical team considers a multitude of factors in treatment planning but do not typically consider many tumor imaging features in a quantitative fashion. An exception to this is in the Response Assessment in Neuro-Oncology (RANO) criteria for assessing tumor progression, which relies on changes to the largest diameter that can be drawn on a 2D image slice [21].

Historically, pretreatment magnetic resonance imaging (MRI) has been used to identify the existence of these tumors and their location but segmentations of

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abnormalities were not part of a standard clinical workflow. Given the availability of MRI for patients with glioma and the relative inaccessibility of tissue samples, quantitative MRI analyses have been a vast resource for research. Two of the most common MRI sequences are T1-weighted with gadolinium contrast (T1Gd), which highlights blood-brain-barrier breakdown and necrotic regions in the brain, and T2-weighted (T2), which highlights edematous regions of the brain. There is also fluid-attenuated inversion recovery (FLAIR), which is similar to T2 in that it highlights edema but is able to suppress normal bright fluid-associated signal. Segmentations of these pathologies have shown prognostic impact in a variety of studies, typically through their T1Gd volumes [1, 5, 8, 22]. Some segmentation shape metrics such as surface regularity have also shown prognostic impact in glioblastoma [14]. With regard to lacunarity and fractal dimension, these have shown prognostic implications and differences between glioma grades, suggesting that they may provide insight into the aggressiveness of the disease [4]. There is also a study showing differences in fractal dimension of differing MRI abnormalities between Verhaak glioblastoma subtypes [20, 23].

In this chapter, we focus on the prognostic implications of 2D fractal dimension and lacunarity values for patients with glioblastoma and how these results differ for different imaging regions. We then speak to the clinical, imaging and segmentation variables that can play a role in these morphometrics, some of which will need to be standardized if we are to include lacunarity and/or fractal dimension as prognostic indicators in clinical care.

28.2 Fractal Analysis of Pretreatment MRI Provides Prognostic Information for Glioblastoma

28.2.1 Thresholding Morphometric Values to Discriminate Survival

A paper by Liu et al. found the fractal analysis of necrotic regions related to patient survival and RNA-Seq in a cohort of 95 patients with glioblastoma [11]. By thresholding, they found that higher values of fractal dimension (≥ 1.56) and lower values of lacunarity (≤ 0.46) were associated with longer overall survival and progression free survival [11]. They carried out a gene ontology analysis on a subset of patients with available RNA-seq ($n = 32$) to determine which biological processes and pathways were associated with lacunarity. They found that inflammatory response, negative regulation of apoptosis, and immune response were positively associated with lacunarity. This suggested that higher lacunarity values are a byproduct of increased inflammation, immune activity, and a lack of regulated cell death.

We aimed to validate the association between necrosis and survival on a larger dataset of patients diagnosed with glioblastoma (402 patients) and extend this concept to other abnormalities present on T1Gd MRI and T2/FLAIR MRI [3]. These regions are: necrosis (present on T1Gd MRI), enhancement with necrosis

(present on T1Gd MRI), and edema (present on T2 and FLAIR MRIs). Patients with cystic glioblastoma (determined through imaging) were excluded from this study. Cysts are typically round and present in addition to or instead of necrosis and likely would have thrown off any true signals of necrosis [2]. Using the same methodology as Liu et al., we computed 2D lacunarity and fractal dimension values using the FracLac ImageJ [9] plugin for each slice. We computed the mean, median, and range of lacunarity and fractal dimension values across slices [3]. Unlike Liu et al, who presented the mean values across slices, we primarily presented median values and found significant associations with survival. In that work, we used the median values as we believed them to be more robust to outliers, which we had seen particularly on the most superior and inferior abnormality-associated MRI slices. We allowed the threshold for binning lacunarity and fractal dimension values to vary and implemented a p-value correction for multiple comparisons within the maxstat package in R [7]. We present an overview in Fig. 28.1.

We found that lower values of lacunarity in T2/FLAIR (edema) and T1Gd abnormalities (enhancement with necrosis) and higher fractal dimension values of T2/FLAIR regions were associated with longer overall survival across the whole cohort. We found that higher values of lacunarity were associated with progression free survival for necrotic regions. In a subcohort of patients who were known to receive the standard of care therapy, we found that lower fractal dimension scores were associated with longer overall and progression free survival, and that higher lacunarity was associated with longer overall survival.

The thresholds that optimally distinguished our cohort were not the same as thresholds presented by Liu et al., which may be owing to differences in cohort, image resolution, treatment received, imaging protocols, or segmentation methodology. We have tested the thresholds of Liu et al. on mean values computed on our cohort and did not see significance, even when subsetting to images with the same resolution as their cohort. Our mean lacunarity values were slightly lower than those in the paper of Liu et al. and our mean fractal dimension values were higher.

This lack of generalizability for lacunarity and fractal dimension thresholds does raise concern for the ability to use these thresholded results in the clinic. We

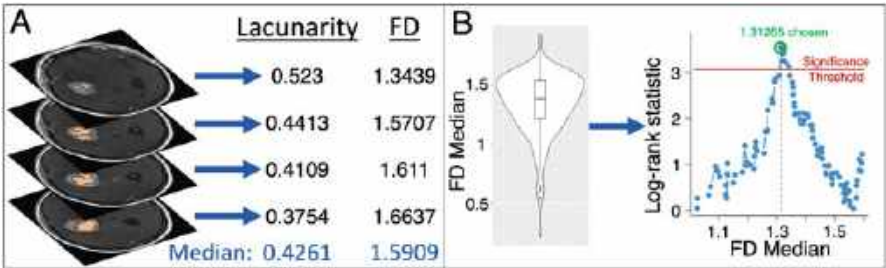


Fig. 28.1 (a) Morphometrics are calculated on each slice, and their median is selected. (b) The medians across patients are tested for their log-rank statistic, and the maximum is chosen. Significance is adjusted using maxstat in R. Figure from [3]

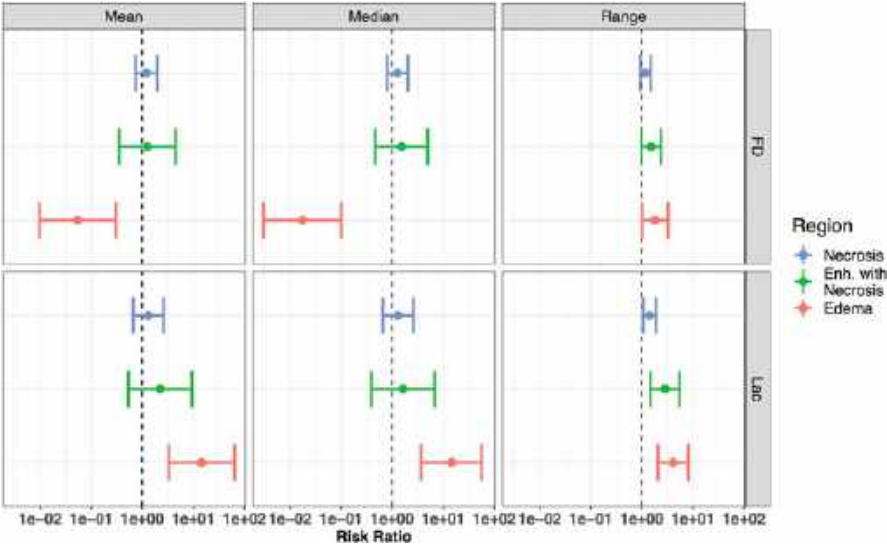


Fig. 28.2 Univariate Cox proportional hazard models for each variable within each imaging region tested. Fractal dimension and lacunarity of edema were both significantly associated with overall survival for median and mean values. Interestingly, the range of morphology values was significant in all tests except for fractal dimension on necrotic regions

wanted to test the more robust use of lacunarity and fractal dimension as continuous variables, which we present in the next section.

28.2.2 Morphometrics as Continuous Prognostic Variables

We were interested in testing fractal dimension and lacunarity values as continuous prognostic variables, especially given the differences in Liu’s results and our own. We presented univariate and multivariate Cox proportional hazard models of median morphometric values in previous work [3]. Here, we extend on that work by also presenting the mean and range of lacunarity and fractal dimension values. We do this across all three imaging regions. The results of univariate Cox proportional hazard models are presented in Fig. 28.2. Interestingly, the range of lacunarity values provides more significant univariate results than the median or the mean. We see that the lacunarity range is significantly prognostic for overall survival for all three imaging regions tested (necrosis $p = 0.021$ HR = 1.4, enhancement with necrosis $p = 0.0016$ HR = 2.8, edema $p < 0.0001$ HR = 4.1), while fractal dimension is for edema ($p = 0.045$, HR = 1.8). We see that the median fractal dimension and lacunarity edema values are significant ($p < 0.0001$ HR = 0.018, and $p < 0.0001$ HR = 14.4, respectively), as are the mean fractal dimension and lacunarity edema values ($p = 0.00090$ HR = 0.054, and $p = 0.00040$ HR = 14.5, respectively).

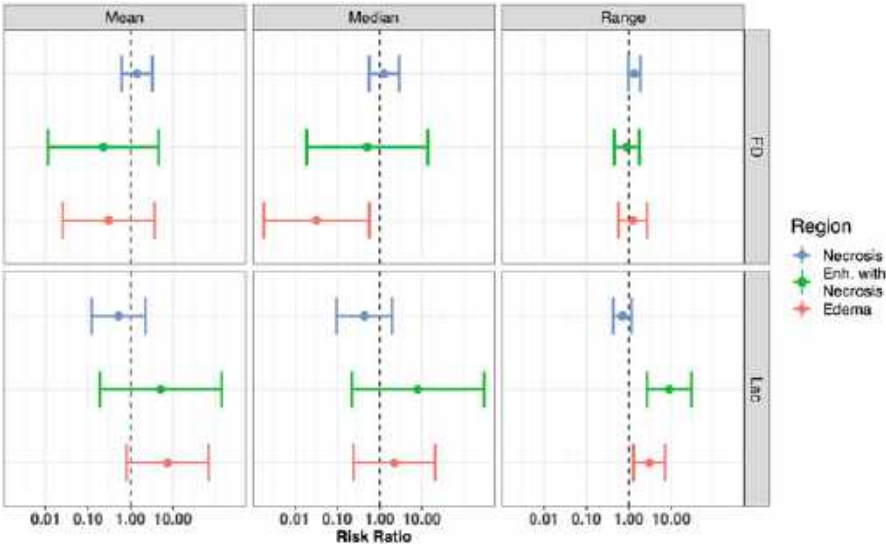


Fig. 28.3 Multivariate Cox proportional hazard models across each statistic (mean, median, and range) alongside age at diagnosis and abnormality volume (not shown). The median value of fractal dimension was significant for edema regions. The range of lacunarity values were significant for edema and enhancement with necrosis

In multivariate analyses with age at diagnosis and abnormality volume, we see that lacunarity and fractal dimension can still provide significant information. Due to the negative correlation between lacunarity and fractal dimension, we did not expect them to both provide prognostic information when compared together. Indeed, we see that fractal dimension provided significant results for mean and median values, while lacunarity showed significance for range. We present multivariate adjusted Cox proportional hazard results in Fig. 28.3 across 250 patients with all three imaging abnormalities available. A total of three tests were implemented, one for each statistic (mean, median, range) across all imaging regions. The median fractal dimension of edema remained prognostic ($p = 0.019$, $HR = 0.03$) against all other median values as well as imaging abnormality volumes and patient age. The range across slices of lacunarity values remained prognostic for both enhancement with necrosis ($p = 0.00036$, $HR = 8.9$) and edema ($p = 0.011$, $HR = 3.0$) in this multivariate setting. These Cox proportional hazard models show that lacunarity and fractal dimension are able to provide prognostic information for patients with glioblastoma as continuous variables.

In all significant results, we see that the range of lacunarity and fractal dimension values are detrimental to survival the larger they are. Consistency of fractal dimension and lacunarity values, and therefore a low range across slices, is generally better for the patient. Lower average fractal dimension values of edema are beneficial for survival, showing that less complexity in the shape of edema is beneficial. Higher average lacunarity values of edema are associated with worse prognosis, which points to inhomogeneous edema as detrimental.

28.3 Reliability of Lacunarity and Fractal Dimension

Although we found lacunarity and fractal dimension could have significant associations with patient survival across the cohort, there are multiple challenges to overcome before quantitative MRI-derived features are used in the clinic on a patient level. MRI was not developed for consistent quantitative imaging on a universal scale, rather with a focus on showing clinicians what they needed to see. Thus, different MRI scanners, acquisition parameters, and protocols can lead to different imaging features, including segmentation volume [15]. Segmentation techniques also lead to some variability [6, 19]. In addition, different people segmenting the same image with the same software may include/exclude different voxels based on variables such as their own judgement and the intensity profile at which they are viewing the image. We present two segmentations carried out on a T2 MRI slice by different individuals in Fig. 28.4, chosen as an extreme example. These individuals underwent the same training in how to segment. In one case the threshold for capturing signal was lower, leading to more signal being captured. These naturally led to different lacunarity and fractal dimension values.

28.3.1 Intraclass Coefficients of Morphometrics on Repeated Segmentations

Considering this source of variation, we have tested the reliability of lacunarity and fractal dimension across multiple segmentations of the same images on a subcohort of the data presented in the previous section. We have 211, 201, and 136

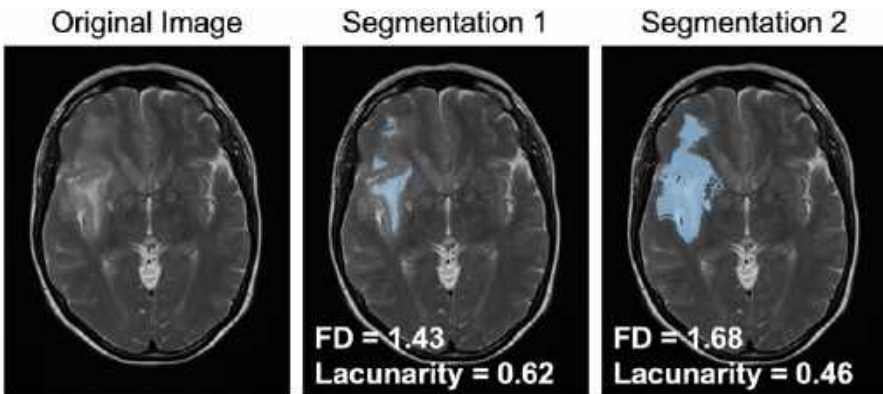


Fig. 28.4 A T2 MRI slice showing how thresholding can affect signal captured and subsequently impact morphometrics. The higher signal threshold (labeled Segmentation 1) led to a smaller segmentation with a lower fractal dimension and higher lacunarity value. This example was chosen as an extreme case

Table 28.1 Intraclass coefficients for different imaging regions and morphometrics for images with multiple segmentations available. Highest ICC values are shown in bold for each imaging abnormality and morphometric. Significant metrics of multivariate Cox proportional hazard models (see Fig. 28.3) are shown in italics. For fractal dimension, median values consistently showed the most reliable signal. For lacunarity, mean values were slightly more reliable in T1Gd MRI and median values were slightly more reliable for T2/FLAIR MRI. In general, lacunarity values were more reliable than fractal dimension values

		T1Gd MRI		T2/FLAIR MRI
		Necrosis (<i>n</i> = 201)	Enhancement with necrosis (<i>n</i> = 211)	Edema (<i>n</i> = 136)
Median	Lacunarity	0.937 [0.918, 0.952]	0.898 [0.867, 0.922]	0.942 [0.920, 0.958]
	Fractal dimension	0.898 [0.866, 0.922]	0.915 [0.890, 0.935]	<i>0.916 [0.882, 0.940]</i>
Mean	Lacunarity	0.948 [0.932, 0.960]	0.905 [0.876, 0.927]	0.939 [0.915, 0.956]
	Fractal dimension	0.877 [0.838, 0.906]	0.843 [0.796, 0.879]	0.798 [0.719, 0.855]
Range	Lacunarity	0.892 [0.858, 0.918]	<i>0.792 [0.731, 0.840]</i>	<i>0.842 [0.780, 0.887]</i>
	Fractal dimension	0.686 [0.588, 0.761]	0.586 [0.464, 0.682]	0.459 [0.247, 0.612]

images with two or more segmentations of enhancement with necrosis, necrosis, and edema, respectively. We used intraclass coefficients (ICCs) for lacunarity, fractal dimension, and segmentation volume, across the three different abnormality regions and statistics presented in this chapter. We find that the median 2D lacunarity and fractal dimension values across slices are all reasonably consistent (all ICCs > 0.89), as are the means (all ICCs > 0.79). The range of values were less consistent, particularly for fractal dimension; this is likely owing to the impact of peripheral slices. See Table 28.1 for a summary of these ICC values, including confidence intervals with the most reliable lacunarity and fractal dimension values for each imaging abnormality shown in bold. ICCs above 0.9 are generally considered as excellent reliability [10]. Significant morphometric values from the multivariate Cox proportional hazard models showed high ICC values which were all equal to or greater than 0.79 (shown in italics in Table 28.1).

28.3.2 Impact of MRI Resolution on Morphometrics

Another source of variability is in imaging resolution. Over time, MRI scanners have become able to provide higher resolution images in standard clinical care, Fig. 28.5. Image resolution naturally impacts the detail at which tumor borders can be delineated, which in turn impacts the fractal dimension and lacunarity values produced. We see that increased image resolution is associated with an increased value of both fractal dimension and lacunarity (Fig. 28.5). Although in our cohort of mixed resolutions we did see significant survival outcomes, lacunarity, and fractal dimension thresholds that significantly distinguished survival in one cohort may not be reproducible within cohorts of differing image resolutions and MR sequence

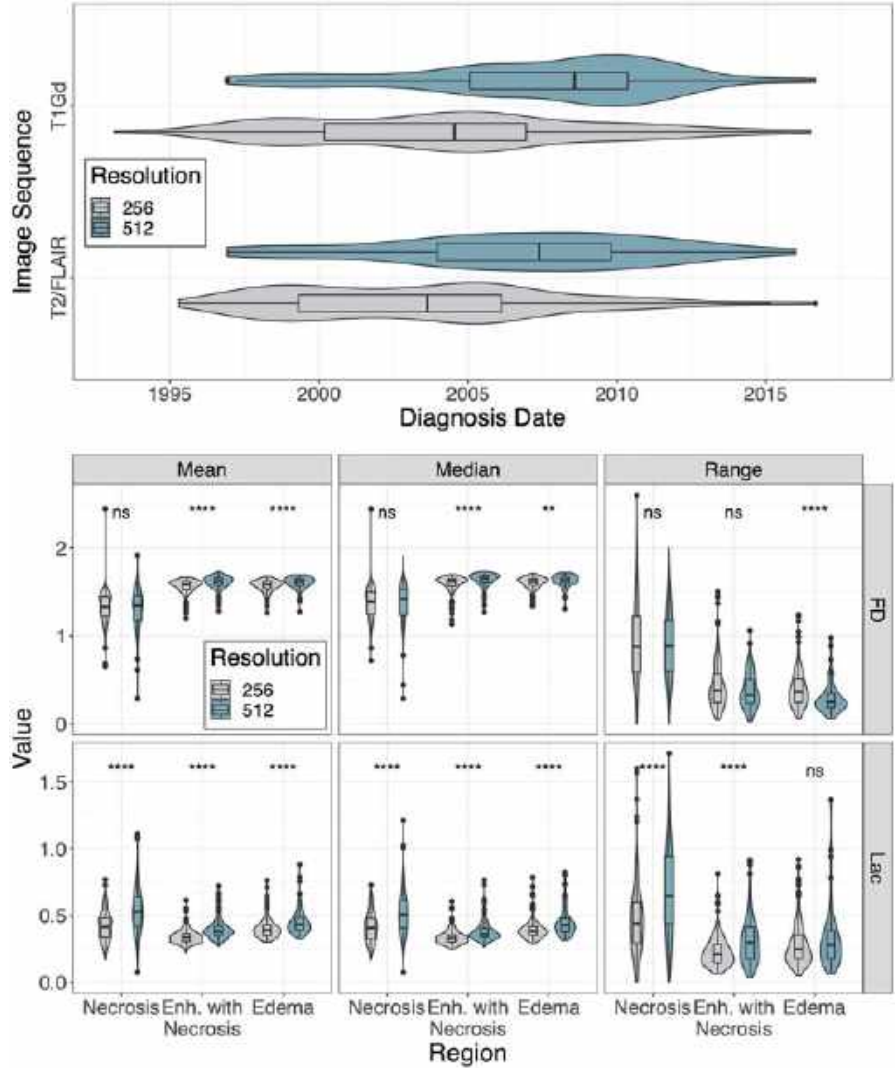


Fig. 28.5 (Top) Diagnosis date shown against image resolution for the two most common voxel resolutions in our dataset, 256 and 512. As clinically standard MRI resolution improves over time, we should expect to see changes in lacunarity and fractal dimension values that are produced. (Bottom) Indeed, lacunarity and fractal dimension values are significantly impacted by image resolution. We see this consistently across mean, median, and range values. In general, lacunarity and fractal dimension are higher in the higher resolution cohort. The only comparison to significantly drop with a higher resolution was the fractal dimension range of edema

protocols. This is naturally a concern for applicability of these methods to patient outcome. Rather than thresholding, continuous variables are less affected by this issue. In previous work, we showed the inclusion of image resolution in Cox proportional hazard models did not dilute the significance of lacunarity or fractal dimension in distinguishing overall survival as continuous variables [3].

28.4 Conclusion

Fractal analysis can provide meaningful insights into clinical data that are not obvious to the naked eye. In this chapter and previous work, we have found that fractal dimension and lacunarity scores of different MRI abnormalities associated with glioblastoma are related to patient survival. Although these results are promising, more work is required before these results can be included to regularly inform clinical care on a patient level. In particular, a morphometric survival study with standardized MRI protocols of patients with glioblastoma who received similar treatment would be a worthwhile undertaking. It is important to note the 2021 World Health Organization changes to the brain tumor grading system that no longer consider IDH1 mutant tumors as glioblastoma [12]. Unfortunately, IDH1 status is not retrospectively available for the majority of patients presented here. Recreating these works within large cohorts of patients with IDH1 wild-type glioblastoma and more consistent imaging will be an important step before any clinical application is possible. We chose to carry out these analyses on 2D slices, in part due to software availability but also to avoid the influence of slice thickness on lacunarity and fractal dimension values, which also varies. Future work could also compute 3D fractal dimension and lacunarity values which would in a sense combine the average and range of values presented here into a single value. Although there is more work to be done, fractal analysis has been shown as a promising method for extracting prognostic information from clinically standard imaging for patients with glioblastoma.

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Chapter 29

Percolation Images: Fractal Geometry Features for Brain Tumor Classification



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Abstract Brain tumor detection is crucial for clinical diagnosis and efficient therapy. In this work, we propose a hybrid approach for brain tumor classification based on both fractal geometry features and deep learning. In our proposed framework, we adopt the concept of fractal geometry to generate a “percolation” image with the aim of highlighting important spatial properties in brain images. Then both the original and the percolation images are provided as input to a convolutional neural network to detect the tumor. Extensive experiments, carried out on a well-known benchmark dataset, indicate that using percolation images can help the system perform better.

Keywords Deep learning · Fractal features · Classification ensemble · Brain tumors · Feature representations

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29.1 Introduction

The diagnosis, prognosis, and therapy of tumors all benefit from the automatic segmentation and categorization of medical pictures. Since brain tumor cancer is one of the fatal and most growing diseases, early detection of brain tumors allows a quicker response to therapy, which raises the survival percentage of patients. According to the WHO Classification of Tumors of the Central Nervous System (CNS) [1], CNS tumors can be classified into biologically and molecularly defined groups with characterize natural histories and distinguish tumor types and subtypes, with the goal of a better understanding of prognosis and optimal therapy for patients with specific CNS tumors. The most diffused types of brain tumor according to [2] are three: meningiomas, gliomas, and pituitary. Meningiomas are benign tumors that originate from the thin membranes that surround the brain and spinal cord, specifically, from the cap cells of the arachnoid layer; anyway, grade III meningiomas, albeit rare, can be very aggressive, behaving like cancer. Gliomas are diffuse tumors that grow within the substance of the brain; they are quite aggressive, but if taken in the grade I can be cured by means of surgery. Pituitary adenomas arise from the pituitary gland, a gland in the skull base, which sometimes can involve the brain. The classification of brain tumors is carried out using several medical imaging techniques, anyway magnetic resonance imaging (MRI) is the most used technique to acquire information about tumors (tumor type, shape, size, location, etc.), necessary for their diagnosis. In MRI, pituitary adenomas and meningiomas appear as uniform intensities, while glioma tumor spreads over the glial cells of the brain and exhibits varying intensities. The location, shape, and size of these tumors, among other characteristics, differ from one tumor to another. Since manual segmentation and classification of brain tumors in a huge collection of medical images takes a lot of time and effort due to the large amount of information contained in each image, it is desired and useful to implement an automated detection and classification method for this task [3]. Two different problems can be addressed: (i) brain tumor segmentation (BTS) consists in differentiating tumor tissues from healthy ones, i.e., as a 2-class classification problem at a pixel-level, (ii) brain tumor classification (BTC) consists in classifying a whole image according to the above cited taxonomy. Computer vision-based biomedical imaging technology has attracted more and more interest because it provides radiologists with recognition information for more effective treatment: the recent developments in deep neural network modeling have provided novel methods for the study, segmentation, and classification of brain malignancies [4, 5].

In recent years, numerous methods have been suggested for classifying or segmenting brain tumors in MRI. Published studies have evolved from using classical machine learning [6–9], where the classification pipeline includes a pre-processing stage designed for feature extraction and a classification stage, to the latest deep learning techniques, designed to enable computers to discover the features that best represent the data [10–20].

Among classical machine learning approaches, Sachdeva et al.[6] proposed a semi-automatic segmentation system which extracts 71 features using the intensity profile, the co-occurrence matrix, and Gabor functions; such features are coupled with standard classifiers: support vector machines (SVM) and artificial neural networks (ANN). Analogously, in [7] an approach for classifying brain tumors that utilized ten characteristics and a back propagation network as the classifier is proposed, and in [8] fractal wavelet features are used with a self-organizing map (SOM) classifier. Cheng et al. [9] extracted intensity histogram, gray-level co-occurrence matrix (GLCM), and bag-of-words (BoW) model-based features coupled with SVM classifier.

The advent of deep learning (DL) had profound influence on the design of medical imaging systems, and most recent research on brain tumors detection and classification is based on convolutional neural networks (CNN) and fully convolutional networks (FCN). DL methods have gained increasing amounts of attention due to their outstanding performance in many image classification tasks. However, using CNNs in the realm of medical imaging is difficult, due to the limited amount of data available for training which may lead the system to overfitting. A feasible solution consists in transfer learning (TL), i.e., using DL models pre-trained on a large dataset and transfer learned knowledge to the target (small) dataset.

Several works in the recent literature have proposed transfer learning for brain tumor classification: the authors of [10] carried out extensive experiments comparing nine pre-trained DL models (InceptionResnetv2, Inceptionv3, Xception, Resnet18, Resnet50, Resnet101, Shufflenet, Densenet201, and Mobilenetv2) finding that InceptionResnetv2 outperforms the other DL models on Figshare [21] and Kaggle [22] datasets. While other models have been shown to work better in a clinic scenario [19]. Similarly, other authors [4, 11] designed approaches for brain tumor classification using TL. Badža and Barjaktarovi [12] introduced a novel CNN architecture based on the modification of an existing pre-trained network: the proposed model showed a good generalization capability and good execution speed. Mzough et al. [13] proposed a 3D CNN model for glioma brain tumor categorization into two classes (low-grade and high-grade glioma): they suggested the use of a pre-processing strategy based on intensity normalization and adaptive contrast enhancement to overcome the data heterogeneity and studied the impact of the pre-processing and data augmentation on classification accuracy. Díaz-Pernas et al. [14] presented a fully automatic system for both tumor segmentation and classification: their model, a Deep Convolutional Neural Network that includes a multiscale approach, processed the images in three spatial scales along different processing pathways. An ad-hoc model for three-class brain tumor classification is proposed in [15], where a deep dense inception residual network is designed by customizing the output layer of InceptionResnetv2 with a deep dense network and a softmax layer. Another ad-hoc architecture is presented in [5] where a variant of EfficientNet is proposed with dense and drop-out layers added. Moreover data augmentation and min-max normalization is adopted to increase the contrast of tumor cells. The authors of [16] presented a multi-level attention mechanism network (MANet) which incorporates both spatial and cross-channel attention. A

hybrid deep convolution neural network and deep watershed auto-encoder (CNN-DWA) model is proposed in [17] for tumor detection and segmentation. One problem of existing approaches is adaptability to different MRI machine: in order to improve flexibility, Alanazi et al. [18] suggested using a transfer-learning model strategy to categorize brain MRI data first in two classes (tumor/no tumor) then into the three tumor subclasses. Russo et al. [20] proposed a method based on a pre-processing phase for dealing with the problem of the lack of data standardization to adjust for different CNN models and MR machines. They dealt with tumor segmentation problem and showed the advantage of transforming input data using 3D spherical coordinates over the Cartesian-input trained model. In summary, as observed from the above research studies, the classification performance with DL approaches for brain MRI classification is significantly higher than with traditional ML techniques. However, traditional handcrafted features, which do not require a massive amount of data for training, can help solve this hard classification problem. In this paper we propose a method that mixes fractal geometry features and DL methods for brain MRI classification. The classification system proposed in this study is a hybrid solution that exploits handcrafted fractal feature (i.e., percolation) [23] to generate a “percolation image” that can highlight important spatial properties in brain MRI, in the next step both the original image and the percolation image are used to identify the tumors using a CNN. Our experiments, conducted on the well-known Figshare dataset [21], show that the use of percolation images can help improve the performance of the system.

29.2 Percolation

Handcrafted fractal features have been used for the classification of medical images since before most of the current state-of-the-art CNN models became available. These features still are a viable alternative for the quantification of images from datasets with a low number of samples. While Fractal Dimension (FD) and Lacunarity (LAC) have been applied in several studies as the main approaches for measuring fractal properties in images, percolation has recently shown to be another suitable way for obtaining features from medical images. Percolation is a physical concept regarding the transport of fluids through porous media. This concept has recently been applied to image feature extraction, where properties such as number of clusters and percolation occurrence can be measured.

One of the main studies on the application of percolation theory for tumor detection in medical images was published in [24], where the authors proposed that vascular growth of tumors can be described using percolation models. Recently, handcrafted percolation features have been used to quantify different types of histology images [23, 25]; however, the application of similar approaches to other types of medical images is limited. In [26], percolation theory was applied to evaluate topological images of the cortical activity, obtained from electroencephalograms; however, we found no recent applications of similar approaches for BTC in MRI.

In addition, an alternative approach for handling the obtained percolation features has been proposed. In [27], handcrafted fractal features are rearranged to form an RGB image, which is given as input to a CNN. However, the images generated from this approach do not retain spatial information. In this paper, we propose a technique to generate an image from the application of percolation concepts. The generated percolation image should be able to highlight important spatial properties in brain tumor's MRI, while also preserving spatial information.

29.3 Materials and Methods

29.3.1 Brain Tumors Datasets

Clinical situations typically allow the acquisition and availability of a limited number of brain MRI slices, each having a significant slice gap. With such a limited amount of data, creating a 3D model is challenging. Most of existing datasets for brain tumor detection are therefore based on 2D slices. In this study we perform a set of experiments on a publicly available brain MRI dataset (Figshare) [21]. Figshare is a collection of data from 233 patients, acquired in China from 2005 to 2010; it consists of 3064 images acquired in the three common views (sagittal, coronal, and axial) and split into three tumor classes: meningiomas (708 images), gliomas (1426 images), and pituitary tumors (930 images). Figure 29.1 shows examples of these three types of tumors. The images have a resolution of 512×512 pixels and have a segmentation mask, in which tumor pixels are labeled as 1 and healthy pixels as 0. The dataset does not include view labels, which were added manually for the purposes of this work. This dataset also provides a testing protocol based on fivefold cross-validation. Fivefold cross-validation (CV) is a testing protocol in which all data are randomly split into fivefolds, then the model is trained using fourfolds, while onefold is left for testing. This procedure is repeated five times, changing the test folder each time, and averaging the classification performance. In our case the split process is not random since the indices are given by the authors [9].

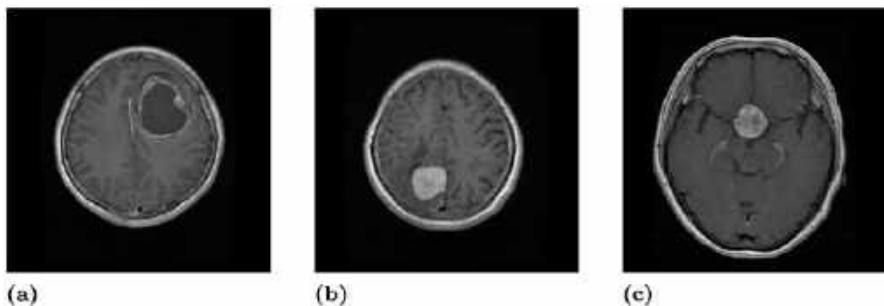


Fig. 29.1 Samples of images from the Figshare dataset: (a) Glioma; (b) Meningioma; (c) Pituitary

29.3.2 Dataset Pre-processing

All the MRI images were pre-processed as proposed by Kang et al. [28], in order to remove noise and unwanted areas that negatively affect classification performance and cropped to a fixed size as required by DL approaches. The pre-processing procedure involves the following steps: (a) conversion to grayscale and slightly blurring, (b) thresholding the image, then perform a series of morphological erosions and dilations to remove any small regions of noise, (c) locating contours in thresholded image, then determine the most extreme points along the contour, (d) cropping the image using the most extreme points, pad to square (to avoid image stretching) and resizing to 352×352 . Figure 29.2 shows an example of the pre-processing steps.

29.3.3 Method Overview

The proposed method consists mainly on applying some of the percolation concepts defined in [27]. First, the gliding-box algorithm is applied to the input image. It involves placing a box of size $L \times L$ on the upper-left corner of the image and then moving it one pixel at a time until it reaches the bottom-right corner. At each iteration, a pixel similarity analysis is performed on the area covered by the box. The analysis is performed by comparing the intensity level of the central pixel F_c with that of each pixel F_i in the box using the chessboard distance Δ , given by Eq. 29.1:

$$\Delta = \max(|F_i - F_c|). \quad (29.1)$$

Then, the obtained Δ is compared to the value of L . If $\Delta \leq L$, then the box is labeled as 1, otherwise, it is labeled as 0. This procedure generates a binary matrix for each box, wherein the pixels labeled as 1 are considered pores. Following this step, we check which boxes match the percolation threshold criteria. This is done by evaluating the ratio of the number of pixels labeled as pores, given by Ω , to the total

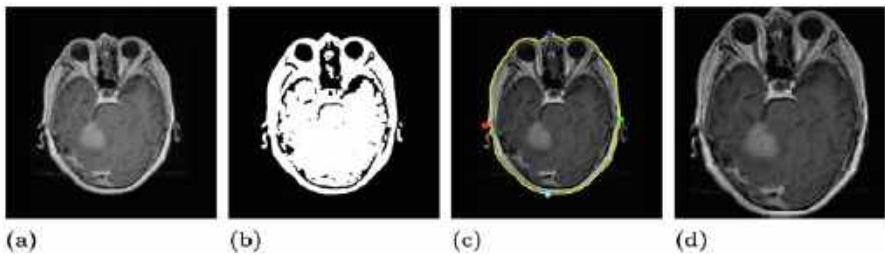


Fig. 29.2 Pre-processing steps: (a) Blurred image; (b) Thresholded image; (c) Extreme points; (d) Cropped image

number of pixels in the box, given by L^2 . If this value is greater than or equal to the threshold for square matrices (0.59275, [27, 29]), the value 1 is assigned to the box and 0 otherwise. These values are then used to generate a binary mask representing the percolation properties of the input image at the L scale. Since the input image has a resolution of $H \times W$, the mask size S for a scale L is given by Eq. 29.2:

$$S(L) = \left(H - \frac{L-1}{2}\right) \times \left(W - \frac{L-1}{2}\right). \quad (29.2)$$

Then, the input image is cropped to match the resolution of the mask. Next, an element-wise multiplication is performed between the input image and the complement image of the mask. This results in a percolation image. In Fig. 29.3, some examples of the obtained masks and their respective percolation images are shown, considering $L = 3$. The next step of the proposed method consists in classifying the input and percolation images with a CNN.

29.3.4 Classification

The images are classified through a ResNet-50 CNN. We have chosen this model as it has shown the best balance between performance and computing cost in preliminary tests. In this stage, both the input and percolation images are given as input to a pre-trained ResNet-50 model. The classifications are done separately, which means that results obtained for each image are independent. After performing classifications, the scores (i.e., class probability distributions) obtained from the softmax layers of each CNN are combined by the sum rule. After this summation, the class with the highest score is considered the predicted class. An overview of the proposed method is shown in Fig. 29.4.

29.3.5 Performance Evaluation

The performance of the proposed method was investigated through the following experiments: First, different sets of percolation images are generated with varying L values. These sets are given as input to the ResNet-50 and the obtained results are evaluated, aiming to verify how the choice of different scales for the gliding-box algorithm affects the performance. Then, the original image is also classified, and resulting scores are combined with the ones generated by the classification of the percolation image sets. Next, the results are compared to the ones obtained by traditional feature extraction and classification approaches.

We have also evaluated the classification performance of each of three available perspectives separately. The dataset was split into three subgroups: sagittal, coronal, and axial. The procedures applied for training and classification of the full dataset

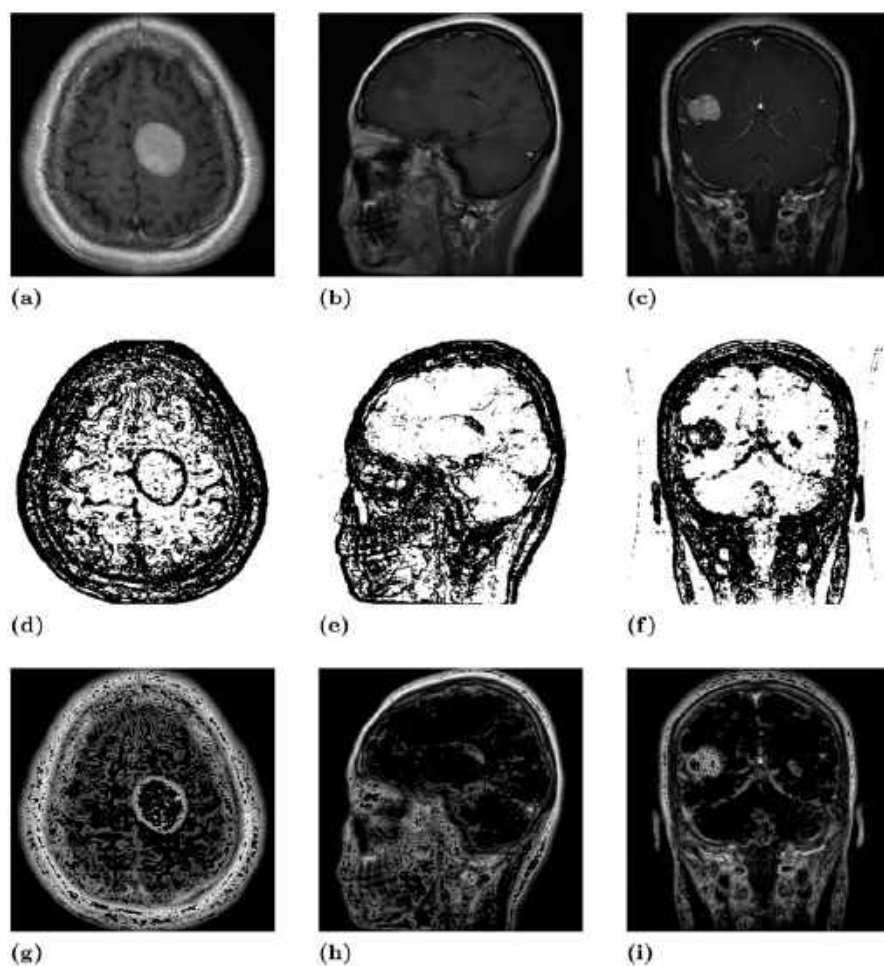


Fig. 29.3 (a), (b), and (c): Original images; (d), (e), and (f): masks obtained from applying the proposed method on the images (a), (b), and (c), respectively, with a scale of $L = 3$; (g), (h), and (i): percolation images obtained from the multiplication of the masks with their respective input images

were also applied for evaluating the subgroups. In Fig. 29.5, an example from each point-of-view available is shown.

The extraction of the percolation features, the generation of the masks and percolation images, as well as the CNN classifications were all performed in Matlab R2019b. All tests were run on a *Intel Xeon Silver 4116 CP@2.10GHz*, 128 GB RAM and a *NVIDIA GeForce RTX 2080Ti* card. The fivefold cross-validation technique was adopted in all test steps to avoid problems such as *overfitting*. For training the CNN models, the default values of hyper-parameters available from the

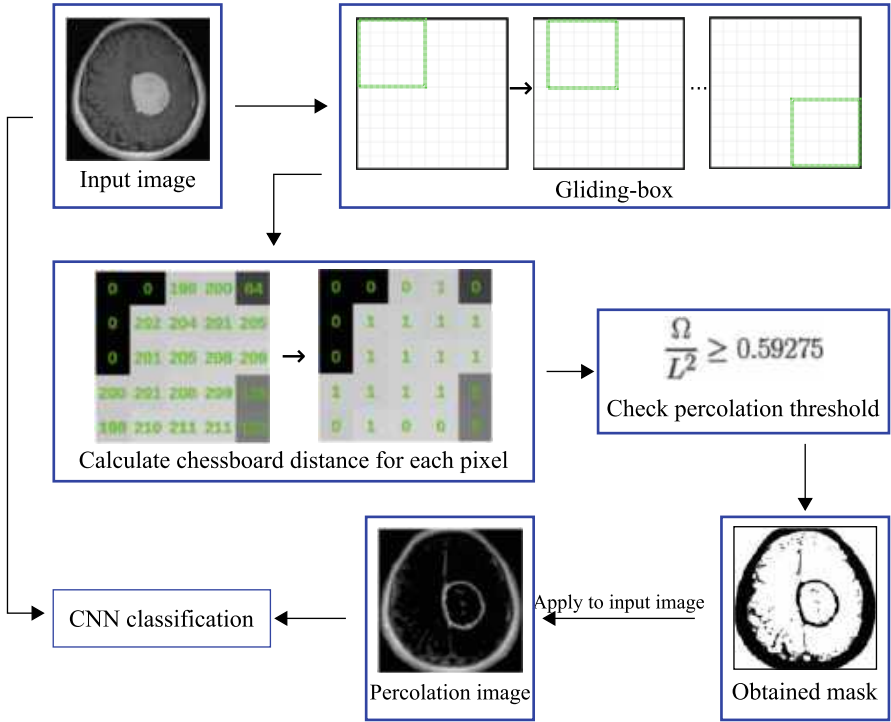


Fig. 29.4 Overview of the proposed method

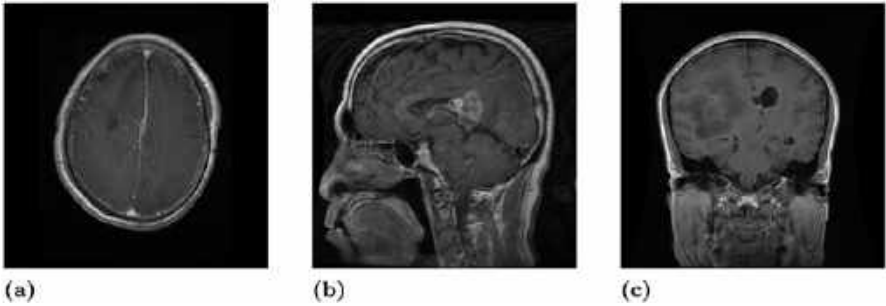


Fig. 29.5 Examples of the three available views: (a) axial, (b) sagittal, (c) coronal

Deep Learning Toolbox of the software *MATLAB R2019b* were used, which are an initial learning rate of 0.01; a learning rate drop period of 2 epochs; a learning rate drop factor of 0.75; a mini-batch size of 32; an L2 regularization of 0.0001 and using solver SGDM with ten training epochs.

29.4 Results and Discussion

The first experiments consisted on evaluating the performance of the generated percolation images across different scales. The values chosen for evaluation were $L = 3, 7, 11, 15, 19, 23, 27$. The results obtained for all evaluated scenarios are shown in Table 29.1.

When the full dataset is considered, the best performance was obtained with a scale value of $L = 3$, with an accuracy value of 91.45%. With an exception for the scale $L = 11$, the classification performance gradually deteriorates as the scale increases. This trend is also observed for the classification of subgroups representing axial and sagittal views, where the lowest accuracy values occurred at the highest L values. The best accuracy value for these two subgroups was obtained at $L = 7$. For the coronal subgroup, larger L scales perform better than for the other subgroups. We believe this may indicate that the images obtained from this viewpoint do not contain relevant information at the borders, because the larger the scale, the more border pixels are cropped from the input image, to fit the mask resolution. The second stage of experiments was to evaluate the CNN ensembles considering the classification of the original and percolation images, combined by the sum rule. The results are shown in Table 29.2.

For the classification of the entire dataset, the best result was an accuracy rate of 94.22%, which was obtained in the ensemble composed by the original images and the percolation images generated at a scale of $L = 3$. This represents an

Table 29.1 Performance of the percolation images with different scales

	Full dataset	Axial	Sagittal	Coronal
$L = 3$	91.45%	89.22%	86.43%	88.33%
$L = 7$	91.02%	90.13%	89.18%	89.56%
$L = 11$	91.22%	88.72%	88.50%	91.18%
$L = 15$	90.67%	88.52%	88.30%	90.42%
$L = 19$	89.52%	85.30%	87.02%	88.90%
$L = 23$	89.36%	85.90%	86.33%	90.42%
$L = 27$	88.54%	84.79%	85.25%	90.32%

Table 29.2 Performance of the ensemble composed by the classification of the input and percolation images

	Full dataset	Axial	Sagittal	Coronal
Original image	93.83%	92.25%	90.86%	92.79%
Original + Percolation image ($L = 3$)	94.42%	92.04%	91.15%	92.22%
Original + Percolation image ($L = 7$)	94.22%	93.05%	92.43%	93.64%
Original + Percolation image ($L = 11$)	94.06%	91.44%	92.43%	93.07%
Original + Percolation image ($L = 15$)	93.77%	91.24%	91.84%	93.45%
Original + Percolation image ($L = 19$)	93.28%	91.14%	91.64%	92.60%
Original + Percolation image ($L = 23$)	93.64%	91.84%	90.86%	92.60%
Original + Percolation image ($L = 27$)	93.31%	89.93%	90.86%	93.17%

improvement of 0.59% over the classification accuracy of the original images alone. For the classification of the three subgroups represented by each of the different perspectives, the best results were obtained in the ensemble composed by the percolation images generated with a scale of $L = 7$. In these four scenarios, the inclusion of a CNN for classifying the percolation images improved the accuracy obtained. The best improvement was observed in the classification of the sagittal subgroup, where accuracy increased by 1.57%. We also analyzed the preformance of the individual classes using other metrics besides accuracy, such as sensitivity (true positive rate), specificity (true negative rate), and F1-score. The results are shown in Tables 29.3, 29.4, 29.5 and 29.6. The confusion matrix for the full dataset classification is shown in Table 29.7. The pituitary class performed best in the classifications using the full dataset and in the scenarios related to the sagittal and coronal view classification. For the axial view, the best performance was obtained by the glioma class. The meningioma class provided the lowest metrics in all scenarios, indicating greater difficulty in performing the classification.

To test how our approach compares with other traditional feature extraction techniques in computer vision, we obtained LBP, Haralick, and fractal descriptors from the original images and gave them as input to a Random Forest classifier,

Table 29.3 Performance measures for the full dataset using the proposed ensemble with $L = 7$

Class	Accuracy	Sensitivity	Specificity	F1-score
Meningioma	94.75%	0.872	0.970	0.885
Glioma	96.31%	0.960	0.966	0.960
Pituitary	97.39%	0.969	0.976	0.956

Table 29.4 Performance measures for the axial images using the proposed ensemble with $L = 7$

Class	Accuracy	Sensitivity	Specificity	F1-score
Meningioma	94.26%	0.841	0.969	0.860
Glioma	96.17%	0.964	0.960	0.962
Pituitary	95.67%	0.938	0.964	0.927

Table 29.5 Performance measures for the Sagittal images using the proposed ensemble with $L = 7$

Class	Accuracy	Sensitivity	Specificity	F1-score
Meningioma	94.00%	0.899	0.955	0.888
Glioma	94.30%	0.919	0.961	0.932
Pituitary	96.56%	0.953	0.971	0.945

Table 29.6 Performance measures for the Coronal using the proposed ensemble with $L = 7$

Class	Accuracy	Sensitivity	Specificity	F1-score
Meningioma	94.12%	0.879	0.959	0.868
Glioma	95.26%	0.942	0.962	0.950
Pituitary	97.91%	0.969	0.984	0.966

Table 29.7 Confusion matrix for the full dataset using the proposed ensemble with $L = 7$

		Actual class		
		Meningioma	Glioma	Pituitary
Predicted as	Meningioma	617	53	17
	Glioma	44	1369	12
	Pituitary	47	4	901

Table 29.8 Accuracy values obtained by different computer vision methods

	Full dataset	Axial	Sagittal	Coronal
LBP	68.70%	72.41%	60.28%	72.39%
Haralick	74.45%	75.53%	73.75%	77.89%
Fractals (FD, LAC, and percolation)	72.55%	75.73%	72.47%	75.14%
ResNet-50 (original image)	93.83%	92.25%	90.86%	92.79%
ResNet-50 (percolation image, $L = 7$)	91.02%	90.13%	89.18%	89.56%
Proposed (ensemble, $L = 7$)	94.22%	93.05%	92.43%	93.64%

using the same fivefold partitioning used in previous experiments. The comparison is shown in Table 29.8. It can be noted that the proposed method significantly outperforms the other techniques, even when only the percolation images are given as input to the CNN. This indicates that our proposal is a promising approach, and we believe that it may be able to fill the gap related to the classification of MRI datasets using percolation concepts.

29.5 Conclusion

In this paper, we proposed an approach to extract percolation features from brain MRI images and then generate a mask, in order to apply it to the original images and obtain a percolation image. Both the original image and percolation image were given as input to a CNN ensemble and accuracy values of up to 94.42% were obtained. The percolation image can be obtained using different scales. We evaluated seven possible scale values and observed that smaller values tend to provide slightly better results. We also evaluated the classification of subgroups composed by images obtained only from a given perspective: axial, sagittal, or coronal. Our proposed ensemble outperformed the standard CNN classification and other computer vision approaches evaluated in all scenarios tested using a scale of $L = 7$.

We believe this is a promising approach, which can be further explored to provide contributions to research regarding classification of MRI datasets using percolation concepts. For future work, we propose the evaluation of different datasets and CNN models, in addition to the Resnet50. We also consider the inclusion of other percolation properties, such as number of percolating clusters per box and size of the largest cluster, in the percolation image generation process.

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Conflict of Interest The authors declare that they have no conflict of interest.

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Chapter 30

On Multiscaling of Parkinsonian Rest Tremor Signals and Their Classification



Lorenzo Livi

Abstract Self-similar stochastic processes and broad probability distributions are ubiquitous in nature and in many man-made systems. The brain is a particularly interesting example of (natural) complex system where those features play a pivotal role. In fact, the controversial yet experimentally validated “criticality hypothesis” explaining the functioning of the brain implies the presence of scaling laws for correlations. Recently, we have analyzed a collection of rest tremor velocity signals recorded from patients affected by Parkinson’s disease, with the aim of determining and hence exploiting the presence of scaling laws. Our results show that multiple scaling laws are required in order to describe the dynamics of such signals, stressing the complexity of the underlying generating mechanism. We successively extracted numeric features by using the multifractal detrended fluctuation analysis procedure. We found that such features can be effective for discriminating classes of signals recorded in different experimental conditions. Notably, we show that the use of medication (L-DOPA) can be recognized with high accuracy.

Keywords Scaling · Fluctuation analysis · Parkinsonian rest tremors · Deep brain stimulation · Classification

30.1 Introduction

Memory is one of the most interesting aspects characterizing many natural, social, and financial processes [6, 17, 23]. Memory can be quantified in different ways, depending on the particular features and effects that one wants to highlight. One of the most common approaches in the study of real-valued time series is the analysis of the autocorrelation function. However, autocorrelation captures only linear contributions of the past on future events. A more general setting is offered by

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measures based on mutual information, such as excess entropy and related quantities [9, 13, 56]. In the linear setting provided by the study of the autocorrelation function, the memory extent can be quantified as the characteristic time scales defining the decay of the autocorrelation. A time series possesses short memory when such decay is consistent with an exponential functional form, meaning that the influence of the past on the current state is limited in time. On the other hand, when the decay follows a power law, the time series is said to manifest *long-term correlations* (LTC), and the strength of such correlations is referred to as the degree of *persistence* of the generating stochastic process. Persistence of a (stationary) stochastic process [46] is quantified by the self-similarity coefficient, called the Hurst exponent $H \in [0, 1]$. In practice, when the process corresponds to uncorrelated noise, we obtain $H = 0.5$; if the process is persistent (correlated) or anti-persistent (anti-correlated), H will be greater than and less than 0.5, respectively.

Several methods have been proposed to extract and analyze the fluctuations of the stationary component of a time series [2, 22, 45, 54], including the well-known detrended fluctuation analysis (DFA). DFA has been shown to be successful in a broad range of applications [30, 34, 49]. The DFA procedure has been generalized [5, 12, 31, 41] in order to account for multiscaling, implying the use of several scaling exponents each tailored for a different magnification level of the fluctuations. A time series is called fractal or multifractal whether a single or multiple scaling exponents are necessary in order to characterize the fluctuations. Multifractal time series are also characterized by a nontrivial *multifractal spectrum*, also called a singularity spectrum, which provides a synthetic description of the process complexity. Multifractality could also emerge as a by-product of an underlying heavy-tailed distribution of the data [18]. Two prototypical examples of self-similar processes are fractional Brownian motion and fractional Lévy stable motion [23]. Both families of processes are characterized by stationary increments. In the first case, the increments follow a Gaussian distribution, while in the latter increments are distributed according to an α -stable distribution, accounting thus also for heavy tails – the latter is in fact a generalization of the former since a Gaussian distribution is a particular example of α -stable distribution.

Fractal and multifractal analysis of time series plays a pivotal role in many scientific contexts, including neuroscience and medicine in general [15, 16, 44, 55]. Typical examples of applications are human gait analysis [19], background neuronal noise-like activity in the human and mouse hippocampus [48], analysis of cervical tissue samples [21], MRIs for tumor characterization [29], EEG signals [33], protein contact networks [37], and electromyograms for the diagnosis of neuromuscular diseases [52].

Parkinson's disease (PD) is a neurodegenerative disorder that targets the central nervous system. Such a disease is characterized by the progressive loss of dopaminergic neurons in the substantia nigra of the midbrain. The most evident symptoms associated with PD are tremors, bradykinesia, rigidity, and postural instability, while in more advanced stages of the disease, other factors might be present, such as different types of cognitive impairments (e.g., dementia) and changes in behavior and/or emotional states [8, 25]. The causes of PD are, however, still largely

unknown. This has led to multidisciplinary research involving, for instance, the use of artificial neural networks for the purpose of prediction of related signals [20, 57] and mutual information-based methods for detecting upper limb motor dysfunction [14]. Deep brain stimulation [38, 39, 42, 51] is a neurosurgical procedure that involves a surgical intervention to implant electrodes in brain areas suitable for receiving electrical impulses [3]. DBS proved to be effective in the treatment of PD and other diseases, such as obsessive-compulsive disorders [10].

In our recent study [35], we have analyzed 48 signals recorded from 12 distinct subjects affected by PD [8, 24, 53]. Tremor signals were recorded by means of a velocity laser pointed to the patients' index finger. For each subject, signal recording was performed in four different settings, given by the combination of the use of DBS and L-DOPA medication. Using the same data, Yulmetyev et al. [58] performed a comprehensive analysis by using the statistical theory of non-Markov processes and flicker-noise spectroscopy. In addition, the attenuation effects of DBS on locomotion and tremor over different time scales were further investigated by Beuter and Modolo [7], who developed a computational model of biological neural networks.

In our work [35], we have analyzed such data by focusing on LTC and multifractal signatures, both used for the purpose of representation (e.g., feature extraction) and further discriminative analyses. In order to obtain a baseline for comparison, the developed feature-based representation (FBR) was compared with the one offered by the power spectra. Our results showed that such rest tremor signals present a clear multifractal signature and different forms of LTC. Notably, the effect of medication was clearly recognizable in the signals, suggesting also qualitative change in LTC from antipersistent to persistent. We successively tested several instances of the two FBRs of the signals in the setting of supervised classification and (nonlinear) feature transformation. We considered three different classification problems involving the recognition of (i) the presence of medication, (ii) the use of DBS, and (iii) the high- and low-amplitude tremor groups. Classification results showed that the use of medication can be discriminated with higher accuracy. In particular, results with the highest accuracy were obtained with a parsimonious, two-dimensional representation encoding only the magnitude of the LTC and multifractality present in the signals.

Experimental evidence suggests that such a two-dimensional time-series representation could play an important role in clinical applications involving the study of Parkinsonian rest tremors. In addition, our results reconfirmed the (already well known) usefulness of fractal analysis in the analysis of biomedical signals.

The structure of this chapter is as follows. In Sect. 30.2, we provide the details of multifractal detrended fluctuation analysis (MFDFA), the procedure used to calculate LTC and MFS. In Sect. 30.3, we describe the results presented in Livi et al. (2016). Finally, in Sect. 30.4, we offer concluding remarks and discuss future directions.

30.2 Multifractal Detrended Fluctuation Analysis for Nonstationary Time Series

Conventional methods employed to analyze the LTC properties of a time series (e.g., fluctuation, spectra, and R/S analysis [4, 47, 49]) could be misleading when such time series are affected by non-stationarities [11, 30]. In many cases, a process is driven by an underlying trend [27], which operates at specific time scales, like seasons in the analysis of data related to a natural phenomenon and days in financial market analysis. Usually, when investigating memory properties of a process, one is interested in the fractal properties pertaining to the intrinsic fluctuations of the process. Several methods have been proposed to extract and analyze the fluctuations of the stationary component of a time series, such as DFA, detrended moving average [2], wavelet leaders [1, 54], adaptive fractal analysis [45], and the so-called geometric-based approaches [22]. The DFA procedure has been generalized in the so-called multifractal detrended fluctuation analysis (MFDFA) [5, 12, 31, 41], which accounts also for multiscaling thus implying the possibility to characterize a time series from a multilevel perspective.

In the following, we provide the essential technical details regarding MFDFA [31]. A very useful and hands-on guide discussing the details of MFDFA has been recently published by Ihlen [28], which also includes a link to a ready-to-use MATLAB[®] toolbox.

Given a time series x of length N , the following steps are performed in order to estimate the LTC and multifractal properties of x . MFDFA operates by analyzing the profile $Y(i)$, which is obtained by integrating $x(j)$, with $j = 1, 2, \dots, i$. Y is then divided into nonoverlapping segments of equal length s . Since N might not be a multiple of s , the operation is repeated by starting from the opposite end, obtaining thus a total of segments. Successively, a local detrending operation is executed by calculating a polynomial function on each of the segments. Computing a local fit helps remove non-stationarities compatible with trends that are well modeled by polynomial functions. The degree of the fitting polynomial has to be tuned according to the expected trending order of the time series. For instance, by choosing a quadratic function, it would be possible to remove quadratic trends in Y and, accordingly, linear trends in the original time series, x . However, periodic or fast-changing trends require different and more advanced methods of detrending [36].

The procedure then computes the local variance, denoted as $F^2(v, s)$, for each segment $v = 1, \dots, 2N_s$. The q th-order moment of the local variance over all segments is evaluated as

$$F_q(s) = \left\{ \frac{1}{2N_s} \sum_{v=1}^{2N_s} \left[F^2(v, s) \right]^{q/2} \right\}^{1/q}, \quad (30.1)$$

with $q \in \mathbb{R}$, the q -dependence of the fluctuation function (30.1) allows to highlight the contributions of both high and low fluctuation magnitudes. Hence, the q parameter serves as a sort of lens in order to inspect the time series at different resolutions.

Notably, for $q > 0$, large fluctuations have a higher impact in Eq. 30.1; conversely, for $q < 0$, the impact of the smaller fluctuations is enhanced. The case $q = 0$ cannot be computed by means of Eq. 30.1 and so a logarithmic form has to be used [31]. In order to check if the scaling of the variance is self-similar, the last steps are repeated for different scales s of increasing size.

The self-similarity exponent characterizing the scaling of the fluctuations can be determined by computing the slope of the doubly logarithmic plot of $F_q(s)$ versus s , computed for each value of q . If the series x is long-term correlated, then $F_q(s)$ is well approximated – for large values of s – by the power law form:

$$F_q(s) \sim s^{H(q)}. \quad (30.2)$$

The $H(q)$ exponent is the generalization of the Hurst exponent; the original DFA formulation is obtained setting $q = 2$. When $H(2) > 1/2$, the time series possesses long-term, positive correlations; for $H(2) < 1/2$, the series is anticorrelated; and $H(2) = 1/2$ indicates that the series is compatible with uncorrelated noise – e.g., white Gaussian noise. A homogeneous scaling over all fluctuation magnitudes indicates that $H(q)$ is independent of q , suggesting that the series is monofractal. For instance, monofractal time series can be obtained by exploiting the fractional Brownian motion model. On the other hand, when the small fluctuations scale differently from the large ones, then the series could be considered as multifractal. Many real-world time series possess multifractal properties. Synthetic examples can be easily obtained as well by exploiting the α -stable distributions proper of the Lévy motion.

The self-similarity exponent, $H(q)$, can be used to bridge time series with the conventional fractal and multifractal analysis of measures [26]. This is performed by exploiting the concept of generalized partition function, $Z_q(s)$. Starting from Eq. 30.1 and using Eq. 30.2, it is possible to define

$$Z_q(s) = \sum_{v=1}^{N/s} [F(v, s)]^q \sim s^{\tau(q)}, \quad (30.3)$$

where $\tau(q) = qH(q) - 1$ is the q -order mass exponent (also called Rényi scaling exponent) of the generalized partition function. The multifractal spectrum, denoted as $f(\alpha)$, provides a compact description of the multifractal character of the time series. It can be obtained via the Legendre transform of $\tau(q)$:

$$f(\alpha) = q\alpha - \tau(q), \quad (30.4)$$

where α is equal to the derivative of $\tau(q)$ with respect to q . It is worth noting that $f(\alpha)$ corresponds to the Hausdorff dimension of a subset of the input data where the Hölder exponent is equal to α . Using $\tau(q)$ it is possible to express the generalized Hurst exponent, $H(q)$, directly in terms of α and $f(\alpha)$, and vice versa:

$$\alpha = H(q) + q \frac{\partial H(q)}{\partial q}; f(\alpha) = q [\alpha - H(q)] - 1. \quad (30.5)$$

The multifractal spectrum (30.5) encodes important information regarding the degree of multifractality and the peculiar sensitivity of the time series to fluctuations with high/low magnitudes. The width of the support of $f(\alpha)$, shortened in the following as MFSW, is defined as

$$\Delta\alpha = \alpha(q_{\min}) - \alpha(q_{\max}). \quad (30.6)$$

MFSW offers an important quantitative indicator of the multifractal signature that is present in the data. The higher the width, the higher the complexity of the time series. In Eq. 30.6, $q_{\min}$ and $q_{\max}$ represent the lower and upper values adopted for the q range, respectively.

30.3 Evidence of Multiscaling in Parkinsonian Rest Tremor Velocity Signals

In the following, we use the quantities introduced above in order to characterize time series representing rest tremor velocity signals recorded from patients affected by Parkinson's disease.

Figure 30.1 offers a visual representation of the statistics for H and MFSW for the three different categorizations of the data. LTC properties of the signals yield statistically significant differences only when considering the impact of medications. It can be observed that the use of L-DOPA changes the signal LTC properties toward the positively correlated pole, while in the absence of medication, the signals are clearly anti-correlated (upper panels). In our opinion, this is an interesting aspect that stresses the need for further developments in future research studies. All signals are multifractal with a relevant multifractal signature quantified by the MFSW (middle panels). It is worth noting that the multifractal signature is sufficiently preserved after shuffling the time series (lower panels), suggesting that LTC is not the only source for the observed multifractality. In fact, shuffling the time series destroys LTC and any deterministic trend that might influence the actual degree of multifractality present in the data. As discussed in Ref. [31], multifractality could be observed as a consequence of LTC, a broad probability density function of the data, or as a combination of both cases.

Let us focus now on a specific patient. We take into account the effect of using the medication on the resulting LTC and multifractal properties. Fig. 30.2 shows four plots depicting the scaling of the fluctuations (Fig. 30.2a), the generalized Hurst exponents (Fig. 30.2b) while varying q , the q -order mass exponents (Fig. 30.2c), and finally the MFS (Fig. 30.2d). As it is possible to observe, the use of medication (indicated as med-On) yields a persistent signal ($H(2) = 0.67$), while

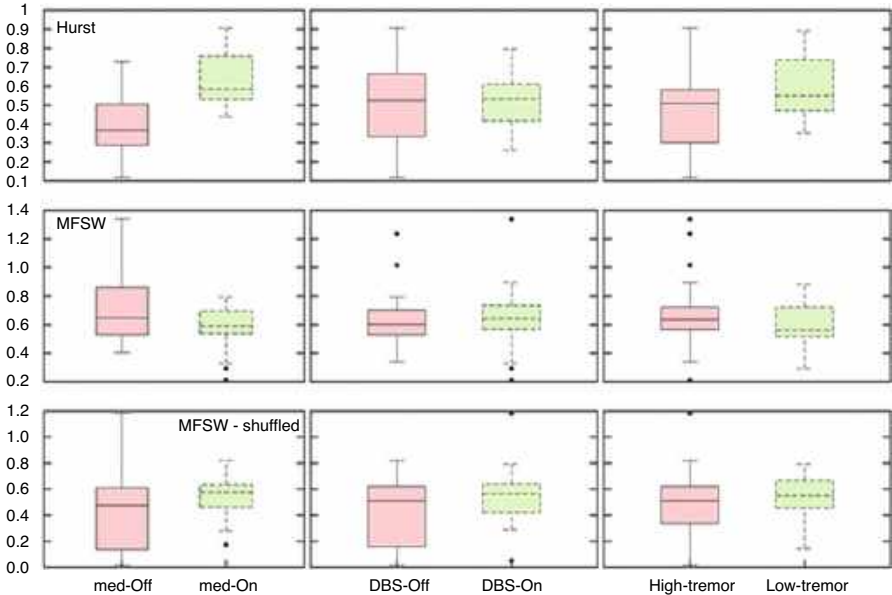


Fig. 30.1 Statistics of all Parkinsonian rest tremor signals. (Taken from Livi et al. [34])

without medication we obtain anti-persistence ($H(2) = 0.38$). This is an important qualitative difference noted in the data. In addition, we notice that the use of medication produces a signal with a narrower MFS than considering the signal recorded without the medication effect ($\Delta\alpha = 0.64$ vs. $\Delta\alpha = 0.85$). This suggests that the absence of medication produces a tremor velocity signal with a richer and hence more complex structure, i.e., more heterogeneous in terms of scaling. The negative correlation dimension ($\tau(2) = -0.26$) for the med-Off case is a direct consequence of the antipersistence. Finally, as previously mentioned, shuffling the time series does not entirely destroy the multifractality of the signals. Table 30.1 summarizes all relevant quantities.

Let us now turn our attention to the problem of classifying such signals. Automatic classification of rest tremor signals is an important and useful research objective.

A first step in this direction when processing time series consists in reducing the dimensionality of the extracted features. To this end, we have used principal component analysis (PCA) and its nonlinear version called kernel-PCA (kPCA) [50]. In order to provide evidence for the effectiveness of the developed FBR using LTC and MFSW, in the following we report the results obtained on three different classification problems conceived on the same dataset. Notably, we face the problem of discriminating between the two groups (high- and low-amplitude tremors) and recognizing the use of both medications and DBS.

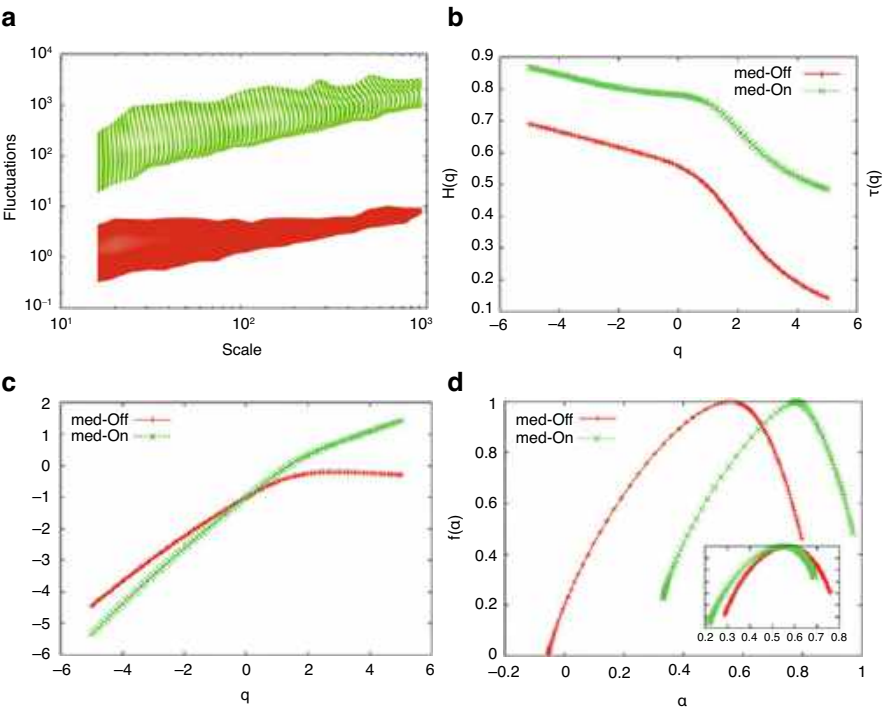


Fig. 30.2 Details of MF DFA calculations for patient “Hg2.” Scaling of fluctuations in (a) has been vertically displaced in order to improve visualization. The effect of medication produces signals with higher persistence, as shown by the generalized Hurst exponents in (b). The multifractality is evident from the nonlinear nature of the q -order mass exponent in (c) and the considerable width of the MFS in (d). *Inset* in (d) shows spectra calculated after shuffling the time series

Table 30.1 Summary of results shown in Fig. 30.2

	Med-off	Med-on
$H(2)$	0.38	0.67
$\tau(2)$	−0.26	0.35
$\Delta\alpha$	0.85	0.64
$\Delta\alpha$ (shuffled)	0.47	0.49

We have selected the well-known support vector machine (SVM) as supervised classification system [50], which is configured with a Gaussian kernel. Notably, we used a version known as C-SVM, where C is a hyperparameter controlling the complexity of the resulting model; in SVM, the structural complexity of the model is measured by considering the number of SVs computed during the training stage. SVM offers an efficient and fast solution for classification problems, even when dealing with a dataset of limited size (as in our case). Both hyper-parameters, i.e., C and the width of the Gaussian kernel, are determined by preliminary tests using a grid search scheme. Since our dataset, regardless of the adopted FBR, is limited to

48 patterns, we tested the recognition capability of C-SVM according to the leave-one-out setting: each pattern is tested on one pattern at a time by learning a C-SVM model on the remaining 47 patterns. We report the number of errors (for each class), AUC, and the average number of SVs as an indicator of C-SVM model structural complexity.

Table 30.2 summarizes the results for the three classification problems presented in [35]. For each classification problem, we report the results obtained on the five FBRs under consideration. For POWER-PCA and POWER-kPCA we use, in both cases, the first two, three, and four principal components in order to evaluate the performances by varying the dimensionality of the representation space given by the Fourier power spectrum. Results show that the effect of medication allows for a more accurate classification. We note that the best result (achieving AUC equal to 0.85) is obtained with the parsimonious, two-dimensional representation denoted as H-MFSW. The per-class errors are also more balanced with respect to the results obtained with the two-dimensional version of POWER-kPCA. In general, results for the last two problems, namely, “DBS-Off/DBS-On” and “high-tremor/low-tremor,” are not convincing – we obtain results compatible with a random classifier. This fact suggests that the effect of medication on the 48 subjects seems to be more characterizing, allowing for an effective classification regardless of the use of DBS or the membership to the high- or low-amplitude rest tremor groups.

Three different classification problems are faced by considering several feature-based, low-dimensional representations: recognition of (i) medication Off–On, (ii) DBS Off–On, and (iii) high–low amplitude tremor.

30.4 Concluding Remarks and Future Research Perspectives

Methods for analyzing the (multi)fractal properties of time series are instrumental for investigating the nonlinearity of biomedical signals recorded from the brain. Here we discussed recent results [35] obtained on a collection of signals recorded from patients affected by Parkinson’s disease. We have shown that the analysis of long-term correlations and multifractal properties allows to deduce important insights on such signals. Our results show that use of medication (L-DOPA) can be recognized with high accuracy. A result of notable interest is the fact that the use of medication implies a qualitative change in long-term correlation features with statistically significant differences. In fact, experimental conditions involving the use of medication produce signals with positive correlations; the absence of medication produces signals having negative self-similar correlations. Successively, we have exploited features of correlations and multifractality for the purpose of developing feature-based representations of the original time series. Classification results show that the use of medication can be recognized with higher accuracy. Notably, the best results have been obtained by means of a parsimonious, two-dimensional representation of the original time series. Such representation encodes

Table 30.2 Classification results

Representation	Dimension	Errors	AUC	SVs
<i>Med-off/med-on</i>				
H-MFSW	2	7 (5/24, 2/24)	0.85	35.2
MFS-PCA	4	16 (8/24, 8/24)	0.67	44.2
MFS-kPCA	4	16 (10/24, 6/24)	0.67	31.6
POWER-PCA	2	22 (22/24, 0/24)	0.54	46.0
POWER-PCA	3	22 (22/24, 0/24)	0.54	46.5
POWER-PCA	4	23 (23/24, 0/24)	0.52	46.5
POWER-kPCA	2	13 (13/24, 0/24)	0.73	31.5
POWER-kPCA	3	9 (9/24, 0/24)	0.81	31.1
POWER-kPCA	4	8 (8/24, 0/24)	0.83	31.1
<i>DBS-off/DBS-on</i>				
H-MFSW	2	48 (24/24, 24/24)	0.00	47.0
MFS-PCA	4	29 (14/24, 15/24)	0.40	47.0
MFS-kPCA	4	26 (8/24, 18/24)	0.46	44.6
POWER-PCA	2	48 (24/24, 24/24)	0.00	46.9
POWER-PCA	3	48 (24/24, 24/24)	0.00	46.9
POWER-PCA	4	48 (24/24, 24/24)	0.00	46.9
POWER-kPCA	2	23 (3/24, 21/24)	0.52	45.2
POWER-kPCA	3	23 (3/24, 21/24)	0.52	45.3
POWER-kPCA	4	23 (3/24, 21/24)	0.52	45.9
<i>High-tremor/low-tremor</i>				
H-MFSW	2	16 (16/16, 0/32)	0.50	32.7
MFS-PCA	4	21 (14/16, 7/32)	0.45	39.2
MFS-kPCA	4	15 (12/16, 3/32)	0.58	32.2
POWER-PCA	2	16 (16/16, 0/32)	0.50	31.3
POWER-PCA	3	16 (16/16, 0/32)	0.50	31.5
POWER-PCA	4	16 (16/16, 0/32)	0.50	32.2
POWER-kPCA	2	13 (12/16, 1/32)	0.61	29.2
POWER-kPCA	3	14 (13/16, 1/32)	0.58	30.2
POWER-kPCA	4	14 (13/16, 1/32)	0.58	28.9

Taken from Livi et al. [35]

only the Hurst exponent and a signature of complexity as expressed by the width of the multifractal spectrum.

Future research directions include the possibility to exploit the so-called local Hurst exponent [40] and a direct estimation of the multifractal spectrum [12], which proved to be effective also with time series of limited length. Another future direction could be the analysis of the (multi)scaling of the cross-correlation [32, 43]. In such a scenario, the idea is to perform a multilevel analysis of pairs of signals in order to characterize the complexity of their coupling over time.

We believe that the presented results could be potentially useful to neuroscientists, suggesting the relevance of using long-term correlations and multifractal

signatures for the analysis of Parkinsonian rest tremor velocity signals. The developed two-dimensional representation provides a parsimonious and cost-effective description for rest tremor signals. We suggest that it could be adopted in different real-world clinical settings, even by considering resource-constrained scenarios (e.g., on hardware having a limited computational capability).

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Chapter 31

Fractal Phototherapy in Maximizing Retina and Brain Plasticity



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Abstract The neuroplasticity potential is reduced with aging and impairs during neurodegenerative diseases and brain and visual system injuries. This limits the brain's capacity to repair the structure and dynamics of its activity after lesions. Maximization of neuroplasticity is necessary to provide the maximal CNS response to therapeutic intervention and adaptive reorganization of neuronal networks in patients with degenerative pathology and traumatic injury to restore the functional activity of the brain and retina.

Considering the fractal geometry and dynamics of the healthy brain and the loss of fractality in neurodegenerative pathology, we suggest that the application of self-similar visual signals with a fractal temporal structure in the stimulation therapy can reactivate the adaptive neuroplasticity and enhance the effectiveness of neurorehabilitation. This proposition was tested in the recent studies. Patients with glaucoma had a statistically significant positive effect of fractal photic therapy on light sensitivity and the perimetric MD index, which shows that methods of fractal stimulation can be a novel nonpharmacological approach to neuroprotective therapy and neurorehabilitation. In healthy rabbits, it was demonstrated that a long-term

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course of photostimulation with fractal signals does not harm the electroretinogram (ERG) and retina structure. Rabbits with modeled retinal atrophy showed better dynamics of the ERG restoration during daily stimulation therapy for a week in comparison with the controls. Positive changes in the retinal function can indirectly suggest the activation of its adaptive plasticity and the high potential of stimulation therapy with fractal visual stimuli in a nonpharmacological neurorehabilitation, which requires further study.

Keywords Neurodegeneration · Retinal and brain plasticity · Fractal dynamics · Fractal photostimulation · Neurorehabilitation

31.1 Introduction in the Problems of Neuroprotection and Neurorehabilitation

31.1.1 *The Retina and Brain Neurodegenerative Diseases*

Neurodegenerative diseases of the central nervous system (CNS) include a group of pathologies with certain common characteristics [200]. Some CNS disorders affecting the brain and spinal cord are characterized by ocular symptoms that often precede clinical manifestations in the diagnosis of such neurodegenerative conditions [14, 149, 179, 183, 195, 238, 304, 372, 388]. It is commonly known that Alzheimer's disease (AD) [15, 111, 288] and Parkinson's disease (PD) [138, 142, 189] are associated with specific changes at the retinal level.

On the other hand, various retinal diseases share common pathophysiological mechanisms and characteristics with the CNS neurodegenerative lesions [112, 162, 183, 200]. The effectiveness of modern therapies is mainly symptomatic, and the identification of new targets for therapeutic interventions [59] and new strategies for neurorehabilitation is required.

Glaucomatous optic neuropathy (GON), a progressive degeneration of the optic nerve that specifically characterizes primary open-angle glaucoma (POAG), is both a neurodegenerative disease of the retina and brain [52, 110, 173, 237, 374]. Interestingly, age-related macular degeneration (AMD) is also associated with a higher risk of cognitive decline and dementia compared to physiological aging [361, 390]. Neurodegenerative aspects are well documented in the cases of diabetic retinopathy (DR), which is typically characterized by the early neurodegeneration onset [151, 225, 253, 379]. Massive degeneration of photoreceptors in retinitis pigmentosa (RP) leads to progressive narrowing of the visual field and loss of visual functions [232]. During RP, aberrant spontaneous activity of retinal ganglion cells (RGCs) [104, 201, 210] and the brain activity [75] also develops.

Despite the differences in symptoms and etiologies of the CNS disorders (including the visual system), many mechanisms of neurodegeneration are similar. They include oxidative stress, mitochondrial dysfunction, excitotoxicity, inflammatory changes, iron accumulations, and protein aggregation [10, 77, 226, 230, 285].

In neurodegenerative retinal and brain diseases, not only medicinal but also nonmedicinal methods of neuroprotection and neurorehabilitation (neurorestoration) are widely used in order to protect and restore the structure and function of nervous tissue. Currently various therapeutic methods are being developed based on the plasticity – intrinsic property of the brain, its ability to change and adapt to changing the input signals of the internal or external environment [8, 22, 23, 268, 289]. Neuroprotection and neurorehabilitation refer to various aspects of disease-modifying treatment that slows down the clinical evolution of neurodegenerative disease either by suppressing primary events (neuroprotection) or by enhancing compensatory and regenerative mechanisms in the nervous tissue (neurorehabilitation or neurorestoration) [90, 97]. Both processes are based on the properties of CNS neuroplasticity and differ in the targets and their application time (the earliest and later events in the neurodegeneration development). At the same time, they use common molecular mediators, such as neurotrophic factors, neurotransmitters, and intracellular signaling cascades. At the early (preclinical) stages of neurodegenerative diseases, positive CNS changes develop characterizing the adaptive plasticity and contributing to the protection and preservation of the nervous tissue structure and function.

Neuroprotective therapy, applied at the preclinical stage, can activate adaptive plasticity and may have great potential to resist primary events in neurodegeneration, stopping or significantly slowing the disease progression [391, 392]. In more advanced stages, the task of external therapeutic influence will be to strengthen compensatory and regenerative mechanisms in the brain.

By now, in clinical trials, the effects of neuroprotective agents such as oxygen, corticosteroids, progesterone, amphetamines and other promoters of neuroaminergic neurotransmission, erythropoietin, magnesium sulfate, cerebrolysin, citicoline and other cholinergics, cyclosporine-A, rivastigmine, piracetam, and other medicines have been studied. However, despite more than 30 years of experience, a clinically effective neuroprotective agent has not yet been recommended in the treatment of TBIs, which allows achieving a favorable outcome with minimal side effects [42, 177, 213]. A recent systematic review analyzed 18 large studies of 4637 patients, which studied the use of neuroprotective medicines for the treatment of TBIs, including the Cochrane meta-analyses [213]. The authors concluded that usefulness of many neuroprotectors in brain lesions has not yet been confirmed from the standpoint of evidence-based medicine.

In ophthalmology, direct neuroprotectors aimed at various molecular mechanisms of glaucoma are rarely used and mainly in advanced stages of POAG and against the background of already existing “maximum” hypotensive therapy. Meanwhile, it is known that there is a time interval between the violation of the RGC function and degeneration. The shifted timing of structural and functional changes indicates the existence of a plastic stage, a therapeutic window during which functional disorders are reversible [219, 257, 400]. At this stage of plastic changes, neuroprotective therapy should be maximally effective [257]. Consequently, the expectation of the initial-stage POAG transition to the advanced stages (to prescribe neuroprotective therapy) is physiologically unjustified.

There are biases in the medical community toward the early use of neuroprotectors due to their possible side effects, and the ethical problem of prescribing treatment to patients in the early plastic stage, when there are still no clinical disease manifestations. Considering this fact, increasing the effectiveness of the treatment of neurodegenerative diseases of various etiologies will be determined by fundamentally new approaches to modifying therapy, including nonmedicine methods of neuroprotection and neurorehabilitation.

31.1.2 Key Problems of Axon Regeneration

As a result of neurodegenerative diseases, such as glaucoma and optical neuropathies, and after traumatic lesions to the optic nerve, the RGCs are unable to regenerate their axons, which in many cases lead to irreversible blindness.

The review by Yves Sauve and Frederic Gallard [280] discusses various aspects and problems of visual rehabilitation for injuries and diseases of the visual system. Most CNS neurons lose their ability to regenerate axons in the early period of development, before axon myelination [36, 53, 291]. In numerous experimental studies devoted to the restoration of neural circuits, a number of objective limitations of visual regeneration have been identified, in particular, related to the establishment of relevant interaction between regenerating axons or graft integration in the host retina [349]. Other challenges are to provide optimal conditions (microenvironment) for directed axon ingrowth [2, 301, 349] and the achievement by a sufficient number of axons of their targets in the brain [160, 281]. Solving the problem of early plastic destructive changes, including the loss of synapses and dendrites, disruption of neural networks, and cell death, could mitigate the manifestations of neurodegeneration or increase the efficiency of recovery [280]. Maximizing plasticity in the adult visual system as an approach to visual rehabilitation in the RGC pathology will be of clinical importance, even in the absence of RGC regeneration of axons and their functional re-integration in brain circuits, as it will maximize the use of the remaining intact RGC connections to increase functionality [62]. On the other hand, even when the RGC regeneration and their functional re-integration into the brain are achieved, the experience-dependent plasticity and activity of visual circuits, as a response to re-innervation, will be extremely important to achieve the goal of maximum recovery of all the aspects of visual functions.

Various ways of manipulating the brain plasticity window in adults, which is usually a characteristic only for the period of development, in order to restore vision are being actively studied [66, 81, 306, 321]. The visual chiasm is an obstacle blocking the successful ingrowth of axons to their targets, although some authors consider this conclusion controversial [73, 345]. On the other hand, subcortical visual areas, such as the lateral geniculate nucleus (LGN) and *superior colliculus*, can also be targets for regenerative plasticity [132]. Focusing on the state of the RGCs targets in the brain and methods to maintain them in a functional state during degenerative and regenerative processes is believed to be critical to maximize

plasticity and functional recovery [62]. Currently, significant LGN and visual cortex degeneration in patients with glaucoma and other degenerative retinal conditions is well documented [258]. The degeneration of neurons in the higher subcortical and cortical visual centers that occurs in the early stages of some diseases can contribute to the overall progression of the disease at all the levels [376]. This highlights the importance of reactivating the plasticity of the LGN and visual cortex in order to increase the success of strategies aimed at preserving and restoring vision.

In addition, with the progression of neurodegenerative retinal pathology (such as glaucoma), there is a loss of the retinal-controlled circadian function, which is also associated with mood disorders and depression in patients [175]. Therefore, achieving even a small regeneration of the disrupted RGC's circuits with the suprachiasmatic nucleus, an area of the brain that is not involved in image formation can be the patient's significant benefit [62].

Today, diverse strategies of neurorehabilitation and visual rehabilitation are being developed, including visual and cognitive training, various approaches to stimulation therapy, visual prosthetics, and pharmacological methods for stimulating neurons, activating axon growth and synapse formation, restoring or remodeling damaged neural networks. This chapter will focus on the brain and retinal plasticity and some nonmedicine strategies for visual recovery and neurorehabilitation in diseases of the visual system and neurological disorders that have the potential to improve the activity of sensory systems and cognitive functions. Special emphasis will be placed on prospects of stimulation therapy with fractal visual signals.

31.2 Retina Neuroplasticity and Aspects of Visual Rehabilitation

31.2.1 Retinal Plasticity in Degenerative Disorders

Structural retinal plasticity is marched in the development of various retinal diseases, including glaucoma, AMD, RP, DR, and others. At the level of the internal limiting membrane (ILM), diseases such as glaucoma lead to the lesions of dendrites and axons of retinal ganglion cells (RGCs). As a result, RGCs lose synaptic connections and become apoptotic. At the level of the outer limiting membrane, in AMD, the close interaction of photoreceptors with retinal pigment epithelium (RPE) is disrupted due to drusen, leading to detachment and death of RPE cells. Therefore, photoreceptors shorten the outer segments and degenerate.

Synaptic retinal connections of vertebrates are very plastic and dynamically regulated by ambient lighting and circadian rhythms by means of light-activated neuromodulators [239]. The circuitries of electrically connected neurons provide plastic reconfigurable circuits that play a key role in the transmission and processing of visual information. On the other hand, gap junctions also play a role in progressive cell death and aberrant activity observed in various pathological conditions.

Thus, gap junctions are potential targets for new methods of neuroprotective treatment of neurodegenerative retinal diseases, such as glaucoma and ischemic retinopathy [239].

Retinitis pigmentosa Nowadays, the manifestations of retinal plasticity in RP are well described. Rod bipolar cells, after deafferentation, create new synapses with photoreceptors, however, in the case of severe hereditary disease (such as RP) after photoreceptor degeneration, bipolar cells and other retinal neurons exhibit a non-adaptive neuroplastic response, leading to a change in the pattern of synaptic retinal contacts with the formation of aberrant connections [152]. Negative (nonadaptive) remodeling complicates the creation of functional synapses between photoreceptors and bipolar cells that limits the clinical effectiveness of retinal prosthetics strategies. Studies on mouse models of photoreceptor degenerations rd1 [313], rd10 [260], and *Crx*^{-/-} [252, 305] have shown that synaptic dysfunction precedes photoreceptors degeneration, and abnormal synaptic input from photoreceptors to bipolar cells causes hyperactivity of neuronal circuits in the inner retina. The loss of inputs from photoreceptors produces aberrant rhythmic hyperactivity in the inner retina, increasing the release of glutamate from bipolar cells [34, 104, 105, 121, 201, 202, 210, 335]. This pathological activity is a factor that interferes the encoding of information and visual processing.

Age-related macular degeneration It should be noted that events characterizing retinal remodeling, such as the neuronal sprouting and modification of synaptic connections, are important elements and early events of neuronal degeneration of various etiologies. Such plastic processes, in particular, are described in the development of the dry form of AMD. Robert K.P. Sullivan and coauthors compared the structure of the retina in healthy individuals and patients with dry AMD in different zones: central, middle periphery, and far periphery [315]. They found that in patients with AMD, many photoreceptor synapses in all zones of the retina were pulled into the outer nuclear layer from the OPL. These events stimulated the outgrowth of dendrites of bipolar cells postsynaptic to photoreceptors and the remodeling of the scheme of their synaptic contacts. Thus, in AMD, changes of the retina, characterizing its structural plasticity, develop at all the eccentricities, but not only in the macular region.

On the other hand, experimental studies have found that when in the rabbit retina photoreceptors are destroyed by photocoagulation, rods and cones move to the ablation zone for several weeks, reducing the blind spot [25]. The rod bipolar cells, unlike the cone bipolar cells, expand the dendritic branching, forming new synapses with healthy photoreceptors outside of the lesion zone, which helps to restore visual functions. The fact that the preserved retinal neurons of the patients with dry AMD are able to form new synapses [315] may be critically important for the justification of new methods for the restoration of the damaged retina. The current approaches to AMD treatment can only slow the geographic atrophy progression but cannot restore the damaged RPE or photoreceptors.

Diabetic retinopathy Currently, it is well documented that the progression of diabetes is characterized by classical microvascular DR, and also neuronal dysfunction and degeneration, and glial activation (cited by [316]). Apoptosis of RGCs and reactive gliosis of Müller cells are the most important histological signs of retinal neurodegeneration. Glial activation is a general retinal glial response to lesions, which is characterized by aberrant expression of glial fibrillary acid protein (GFAP) in Müller cell [20, 216]. Since Müller cells secrete factors in the retina that can modulate blood flow, vascular permeability, and survival of retinal neurons, impairment of their function plays a significant role in the pathogenesis of early microcirculation disorders [16]. In addition, reactive gliosis can contribute to the death of nerve cells, for example, by the mechanism of glutamate excitotoxicity [297]. Amacrine cells AII and A17 play a key role in the processing of scotopic visual signals. The study [47] has shown that diabetes alters the conduction of synaptic receptors of rod bipolar on A17 cells, but not on AII cells. Reduced Ca^{2+} permeability and nodular extensions on the dendrites of amacrine A17 cells in diabetic retinas of animals were noted as well. This change in the dynamics of neurons and the microcircuit of neural connections (dysfunction of the synaptic dyad of bipolar and amacrine cells) can explain the decrease in the dynamic range and sensitivity of scotopic vision in the case of diabetes [47].

Glaucoma Taking into account the etiopathogenetic mechanisms known to date, primary glaucoma can be considered a multifactorial neurodegenerative disease, which is characterized by progressive optical neuropathy, pathological changes in the field of vision, and the death of RGCs. The reduction of intraocular pressure (IOP) today is not a guarantee of stabilization of the glaucoma process, which continues to progress in most patients. Considering also that primary glaucoma has signs of neurodegenerative changes with disruption of neural circuits not only in the retina but also in the brain [44, 45, 110, 173] various strategies that prevent or slow down the degeneration of RGCs may be critically important for the preservation of visual functions in glaucoma [174]. The most studied in this regard are neuroprotectors. In modern guidelines for the treatment of glaucoma, in addition to local hypotensive therapy, treatment with neuroprotectors comes to the fore [78]. The main regulators of processes of neuroprotection and neuroplasticity are neurotrophic factors. Existing ideas about the plasticity of the RGCs in glaucoma [60] have documented the specific effect of neurotransmitters on the vertical outgrowth and development of the RGCs dendrites. The input signals from retinal neurons to the RGCs were found to be necessary for the normal maturation and stratification of dendrites and for RGC functions.

The earliest events in the preclinical stage of GON include the truncation of the RGC dendrites and the loss of synapses in the inner retina, as well as the loss of axon terminals in the LGN [44, 188, 219, 400]. The functional activity of RGCs (according to the pattern-electroretinogram) in initial glaucoma changes much earlier than measurable changes in the thickness of the retinal nerve fiber layer. This “critical period of dysfunction” preceding axon death is also called

the plastic stage, during which changes in visual functions can be reversed [257]. Nowadays, changes in the complexity of the dendritic branching of RGCs in the inner plexiform layer of the retina [155] and neurons in the LGN [188] have been demonstrated. The documented loss of synapses [69] and the formation of new synapses in the preclinical stage of GON [245] are signs of adaptive plasticity in the case of early glaucoma, which may be the target of neuroprotective therapy. In the LGN cells, excessive growth and branching of dendrites are observed at first, but then the size of the dendritic field and the complexity of branching decreased [79].

In the detailed review by Nicolás Cuenca and coauthors [64], common responses of the retina at the molecular and cellular levels for various retinal neurodegenerative diseases were analyzed. The authors noted that in retinal pathology of different nature, such as AMD and glaucoma, the neuroplastic retinal responses are largely identical and lead to changes in the retinal structure and remodeling of the retinal neuron circuits. They also assume that in various pathologies, considering commonality of their molecular mechanisms and signs of retinal plasticity, the definite common therapeutic approaches can be used in order to preserve and restore the retinal structure and neuronal activity, such as neuroprotection, gene engineering, transplantation therapy, and visual prosthetics [64].

31.2.2 Achievements and Problems of Visual Rehabilitation

Among eye diseases, the main causes of irreversible visual impairment are glaucoma, AMD, corneal opacity, etc. affect mainly the elderly people whose share is more than 80% among people aged 60 years and older [35]. Visual rehabilitation includes a wide range of therapeutic measures aimed at improving the quality of life and socialization of patients with visual impairment. The strategies of visual rehabilitation are divided into compensatory, substitution, and rehabilitation (restitution) therapies. In rehabilitation therapy, various approaches of medical and nonmedical visual rehabilitation are used. Several innovative areas are actively developing: visual prosthetics, 3D printing, cell therapy, the use of stem cells, and gene therapy (see reviews [51, 54, 58, 64, 209, 385]).

In AMD with geographic atrophy, methods of cellular therapies aim to restore RPE, which supports the metabolism and functions of photoreceptors to prevent the complete loss of photoreceptors. The potential for the treatment of glaucoma was described by the transplantation of RGCs obtained from human embryonic stem cells (hESCs) [384, 386] and from male germline stem cells [314]. The positive role of induced pluripotent stem cells for the survival of transplanted RGCs has been shown [370].

The strategy of gene editing is aimed at correcting the underlying the disease genetic. Numerous studies have shown that gene replacement therapy can slow the progression of retinal degeneration and restore visual functions in the early disease stages [50, 51, 113, 156, 271].

Another method using the achievements of genetic engineering is optogenetics – a genetic way of activation or inhibition of individual neurons by light [92, 93, 95, 164, 165, 180, 274]. A successful experience of using optogenetics to activate receptors directing axon ingrowth has been published [80, 107, 234, 247]. It is believed that in the near future, 3D printing of retinal tissue may become possible for its transplantation into the affected retina although today a serious obstacle to 3D-printing methods is ensuring cell viability and functionality of retinal tissue [172]. It has been shown [184, 185] that RGCs and glial retinal cells of adult rats can be printed by means of piezo-jet technology. In pilot studies, visual prostheses were able to partially restore visual functions in patients with irreversible vision loss, which cannot be restored in any other way. A recent review [356] provides an overview of modern optoelectronic visual prosthetics devices with clinical trial experience. These devices can convert visual signals into electrical and electrical stimuli into visual signals and are designed for visual rehabilitation of patients with severe retinal diseases with massive photoreceptor lesions, such as RP and progressive AMD with geographical atrophy [233].

Behavioral therapy, methods of visual perception training, and stimulation therapy (electro-, magneto-, and light stimulation) are also widely used in visual rehabilitation strategies.

The analysis of all the existing methods of visual rehabilitation therapy is not included in the purposes of this chapter. The authors of this study will briefly focus on nonmedical approaches to visual rehabilitation, including those related to stimulating therapy, further focusing on the prospects of using intermittent stimuli with fractal dynamics in phototherapy.

31.2.3 Methods of Visual Rehabilitation Based on Neuroplasticity

Numerous computer training programs include repetitive sessions of visual perceptual learning. The stimuli are presented in the area of the “blind” zone and at the border of the blind and intact zones, while the observer performs the task of identifying the desired signal or differences between signals and tracking moving stimuli. Compensation therapy includes repeated and prolonged (for several weeks) training of patients to effectively analyze visual information, for instance, optimizing coordinated saccadic eye movements [144, 275, 333, 334]. Computer-based training programs for visual rehabilitation often use augmented and virtual reality (AR/VR) methods. In the study of 20 patients with hemianopia due to a stroke and control groups of the appropriate age, Lorenz B. Dehn and coauthors [67] showed that cognitive training in virtual reality game (VR) improved spatial visual memory, reduced visual field defects, and depression symptoms.

The results of clinical studies suggest that it is possible to partially restore light perception after intensive visual perception training in the area of residual

vision. For the first time, Josef Zihl and D.Y. von Cramon [393] drew attention to the fact that CNS lesion does not always lead to irreversible loss of function, and it is possible to increase the sensitivity of neural networks by repetitive visual stimulations in the area of residual function.

Bernhard Sabel and Erich Kasten developed the ideas of Zihl and von Cramon [393], putting forward the theory of activation of residual vision [157, 158, 273]. These authors and their colleagues have proposed a number of computerized methods for the treatment of patients with cerebral blindness and found the improvement of visual functions and visual field defects of patients with the consequences of severe TBI and strokes [3, 157, 158]. The use of repetitive visual stimulation of the area of residual vision (a method called “Visual Restoration Therapy” – VRT) in glaucoma patients led to an increase in light sensitivity. However, a positive effect was detected only by means of a high-resolution perimeter and was absent while using standard perimeters (white-on-white and blue-on-yellow) [109, 272]. Also, the VRT training caused a reduction in reaction time and improved health-related quality of life (HRQOL), but not the visual-related quality of vision (VRQOL).

31.2.4 Biofeedback Training Systems for Visual Rehabilitation

For visual rehabilitation of patients with degenerative diseases of the macular region (progressive type of AMD with geographic atrophy), for whom there are no effective therapies today, a positive effect of perceptual learning and other methods based on neuroplasticity has been shown aimed at reorganizing neural connections in the primary visual cortex [198, 254, 322]. Other methods of visual rehabilitation based on the visual system plasticity in the pathology of the macular retinal region include two strategies with biofeedback rehabilitation: with using a microperimeter and analysis of VEP in real time. Patients suffering from macular pathology, with low visual acuity and unstable fixation use extrafoveal retinal regions – preferred retinal loci (PRL) in order to compensate for the loss of central vision. In AMD patients, while using a biofeedback simulator based on the Nidek MP-1 microperimeter (10 sessions of 10 minutes each), fixation stability, and reading ability improved [350, 351]. More recent studies have confirmed this approach potential to visual rehabilitation of patients suffering from macular pathology [208, 218, 262, 282, 352]. Another innovative method of visual training is the acoustic biofeedback method based on VEP, which promotes the relocation of the patient’s gaze fixation from the pathologically altered macular region to the PRL – the intact adjacent (paracentral) retinal region, with remodeling of the primary visual cortex [346]. The effectiveness of this method for improving visual functions has been documented in patients with amblyopia, severe TBI injury, and AMD [57, 82, 346]. Both biofeedback rehabilitation strategies are promising for their use in the clinic for visual rehabilitation therapy of patients with macular pathology and further detailed clinical trials of them on large cohorts of patients are needed [9].

31.2.5 Other Methods of Visual Recovery

Other strategies for visual recovery in cortical blindness and retinal pathology involve the use of various methods of noninvasive stimulation therapy. They include transcranial electrical stimulation (TES), as well as methods of percutaneous electro- and magneto-stimulation and phototherapy. In patients with chronic stroke, the transcranial direct current stimulation technique in combination with VRT reduced visual field defects and improved daily activity, according to the authors, due to the strengthening of innate plasticity mechanisms associated with learning [255]. Miura and co-authors have shown the improvement in BCVA and the perimeter index MD (mean deviation) using transdermal electrical stimulation with cutaneous electrodes in 10 RP patients (20 eyes) [215]. The results of this study differ from the data of other authors who previously did not find a statistically significant improvement in visual acuity and perimetry data after 1 year of transcorneal electrical stimulation in patients with RP [283, 353]. In patients who underwent a 10-day course of visual training in parallel with transcranial random noise stimulation, the improvement in recognition of complex movements in the “blind field” occurred faster than in those who underwent only visual training or training + false stimulation [130]. Thus, the results showed a significant improvement in the ability to long-term plastic changes and restoration of the networks of the visual system when neuromodulation is combined with visual training. These studies indicate a high potential of combined approaches, including noninvasive electrical stimulation (transorbital or transcranial) and visual perception training.

It is important that only a few of the listed methods of visual rehabilitation have passed randomized controlled clinical trials (RCTs). The Cochrane Systematic Review [339] evaluated the results and evidence base of 44 RCTs using various strategies of visual rehabilitation (psychological therapy, vision improvement methods, interdisciplinary rehabilitation programs, and others). The authors concluded that all visual rehabilitation techniques had no significant impact on HRQOL, with very low reliability of evidence. However, there is a reasonable need for further research with a better methodological base and the study of longer-term supportive effects of visual rehabilitation measures.

31.3 Neuroplasticity and Problems of Neurorehabilitation

31.3.1 Central Nervous System Plasticity

Neuroplasticity of the brain’s sensory and motor systems refers to the brain’s intrinsic ability to change constantly, simultaneously adapting to an ever-shifting world. Brain plasticity persists throughout an entire person’s life [83, 84, 98, 317] and plays an important role in the maturation and development of the CNS, the

acquisition of new skills [43, 246, 299, 354]. The plasticity potential is the highest during critical and sensitive periods of the early child developing, when incoming information is necessary for the correct formation of specific neural connections [26, 81, 106, 124, 128, 140, 211, 223, 310, 317, 358]. The dynamics of processes of excitation and inhibition control the time course of critical periods. In the early postnatal life, most neuronal contacts are excitatory ones. During the brain maturation process, inhibitory neurotransmission progressively increases, and when a certain threshold is reached, the critical period of uniquely high plasticity closes [128]. The adult brain is much less plastic than the developing child one; however, even in an elderly person, neural connection circuits may be reconstructed by means of experience.

The plasticity of the visual cortex is the result of a complex interaction between the human genetic background and the environment (epigenetic factors) [205]. The influence of the child's environment, favorable or unfavorable (such as neglect in childhood), interferes with all the processes of CNS development, including neurogenesis, formation of neuronal sprouting, synaptogenesis, refinement of synaptic circuits, and myelination. The development of synaptic pathways sticks to the "use it or lose it" rule [250, 293]. The high plasticity of neural networks of the mature adult brain allows adapting to quantitative and qualitative changes in sensory inputs in accordance with the experience [40, 46, 159]. Therefore, optimal structural and functional rearrangements of neural networks ensure the adaptation of the central nervous system in response to endogenous and exogenous stimuli and during the implementation of memory and learning processes. The CNS plastic changes may be caused by loss or excess of mono- and multimodal stimulation or also may be a response to brain damage. Therefore, they occur as a result of nonuse or excessive use of sensory systems, as well as a result of training, while acquiring new skills [222].

Neuroplasticity is accompanied by changes in the cytoskeleton, receptor-barrier-transport system, biopolymer synthesis system, and intracellular homeostasis system [76, 135]. Neuroplasticity also implies the plasticity of all the life support systems of a neuron, including neuroglia and the blood circulation regulation system of the brain. Thus, the neurovascular unit also takes an active part in the implementation of neuroplasticity processes. Neuroplasticity manifests itself in changes in nerve circuits and synapses that occur at various levels of the central nervous system due to various changes in the external and internal environment. They include migration of new neurons and their integration into existing neural networks, growth and breakage of neurites, axonal and dendritic sprouting, synaptogenesis, loss and modulation of mature synapses, changes in their functional activity, and other plastic events [123, 246, 354].

Neuroplasticity plays an important role in recovery after brain lesion. Reparative processes and compensation of impaired functions occur due to the reorganization of certain brain parts or the whole brain [32]. Endogenous neuroprotective reactions are triggered with each CNS lesion, after a certain latent period. It was noted that with local damage to the nervous tissue, the mechanisms of reactive and reparative neuroplasticity are activated as well. If reactive plasticity serves as a direct response

of surviving neurons to the pathological effects, then reparative neuroplasticity is characterized by the inclusion of all compensatory-restorative mechanisms [204]. Neurotrophic factors are the main regulators of neuroprotection and neuroplasticity processes. Neurotrophic growth factors refer to physiologically active polypeptides that regulate the growth and differentiation of neurons in developing systems and their functional stability. The first functional evidence of the role of neurotrophins in brain plasticity came from the visual cortex [48, 49, 380].

31.3.2 Brain Plasticity During Aging and Neurodegenerative Diseases

Nowadays, the elongation of life duration is one of the reasons for a significant increase in the average age and the general aging of the population [369]. An increase in the proportion of elderly people in the total population leads to an increase in the prevalence of age-related diseases [161], including AD, PD, and other neurodegenerative disorders [148].

It is generally accepted that normal physiological aging is accompanied by cognitive decline. Age-related morphological changes include a loss of the complexity of dendritic branching [39], a reduction in the number of dendrites, dendritic spines, and synapses and are most significant in the prefrontal cortex [241] and in the mesial temporal regions [221]. Since brain function depends on the preservation of neural networks (brain connectivity), the effects of aging are manifested at the level of synaptic plasticity, as evidenced by the age-related decrease in the number of synapses and aberrant synaptic transmission in some areas of the brain. Moreover, a sign of brain aging is an increased level of neuroinflammation generated by glial cells, which may contribute to changes in neural/synaptic function [187].

Key interrelated factors underlying plastic changes and functional decline in the aging brain include a reduction in the schedule of active brain activity (an increase in the “nonuse” period), an increase in noise power in spontaneous neuron activity (noisy processing), and weakened control by neuromodulators and “negative” learning [193, 194].

Brain strokes The recovery problem of patients who have suffered an acute cerebrovascular accident, TBI, or focal brain damage due to other causes is extremely relevant in the modern world. One of the most significant consequences of focal brain damage is the development of cognitive deficits. The frequency of post-stroke cognitive impairment is 95% [166]. 1 out of 10 patients develops dementia shortly after the first stroke, and with the development of repeated cerebral circulatory disorders, more than 30% suffer from dementia [154]. Like the other post-stroke symptoms, cognitive deficits may exhibit spontaneous recovery, but this process is largely time-limited. Despite this, cognitive deficit progressed in 23.2% of patients [166]. The state of the cognitive sphere is largely a determining factor in the functional recovery of the patient as a whole [381]. The negative impact of the post-

stroke cognitive deficit is so significant that it can be considered an independent risk factor for the development of recurrent stroke. An analysis of 12 prospective studies (a total of 82,899 participants) available in the PubMed, EMBASE, and Web of Science databases for the period from January 1, 1980, to October 1, 2013, proved that in patients with global cognitive decline, the relative risk of stroke was 1.19 (95% CI 1.12–1.27).

Traumatic brain injury Another important cause of the development of acute focal brain damage is TBI. Cognitive impairment is one of the most common problems after TBI, occurring in about 70% of people with severe injuries [24]. In addition, mild but recurrent TBI may be a risk factor for neurodegenerative diseases. Pathophysiological processes in the central nervous system, such as abnormal protein accumulation, neuroinflammation, and mitochondrial dysfunction, may cause secondary brain damage, often leading to cognitive dysfunction. Most patients with TBI have long-term memory and attention disorders, depression, fatigue, and cognitive decline [170].

Alzheimer's disease AD is a progressive neurodegenerative disease and is one of the most common causes of dementia, which leads to social, professional, and household patients' maladaptation. Due to the fact that dementia is most common in older people, the rapid increase in the proportion of elderly people in the population, which is currently observed, has led to an increase in the prevalence of AD. The methods of diagnosis and treatment of this disease are being improved with the search for the most accessible biomarkers of AD and therapeutic targets. The course of all neurodegenerative diseases is steadily progressing, and unlike patients with focal brain damage due to various causes, there is no possibility of spontaneous improvement. All the therapeutic measures are aimed at reducing the rate of progression and stabilizing the condition for a while. Among medications, acetylcholinesterase inhibitors are used that are aimed at restoring the imbalance between neurotransmitters and making up for their deficiency, as well as a reversible NMDA receptor blocker – memantine [373]. Research is being conducted related to the search for disease-modifying strategies for the treatment of AD. The main purpose of these studies is to influence the process associated with the pathological deposition of A β and hyperphosphorylated tau-protein. Another direction being developed is the use of monoclonal antibodies aimed at A β binding and removing.

Currently, the recovery of cognitive impairments is approached from the position of a complex effect combining restorative training, some approaches of motor therapy and medicine correction. The main goal of any *rehabilitation* intervention is the maximum functional recovery, independence of the patient, restoration of working capacity, and improvement of the quality of life. The main goal of *cognitive rehabilitation* is to restore and compensate for the impaired daily skills of the patient that have arisen as a result of cognitive deficits, and to increase the patient's ability to participate in activities that are limited due to disorders in one or more cognitive functions. Cognitive rehabilitation based on the principles of neuroplasticity is considered one of the promising ways to mitigate the effects of TBI [91]. The fact

that the brain has the ability to rebuild neural connections and create new connection schemes makes it urgent to search for methods that stimulate not only the restoration of broken circuits but also the creation of new ways to compensate for the existing functional deficit. Therefore, for cognitive rehabilitation, those internal strategies that are aimed at restoring not so much damaged tissue as the lost function may be the most important ways.

Two main approaches to cognitive rehabilitation are relevant today:

- Restorative treatment, the purpose of which is to improve the cognitive sphere for functioning as a whole. It is aimed at restoring the affected cognitive functions;
- Compensatory treatment that teaches solutions to specific tasks, such as using memory notebooks or learning self-learning strategies. The patient learns new strategies for solving the task with the use of preserved cognitive functions.

Until recently, more attention was paid to methods of external compensation, but advances in neuroplasticity research have again intensified the search for potentially effective internal (restorative) strategies [91, 163, 249]. The fact that the brain has the ability to rebuild neural connections and create new connection circuits makes it urgent to search for methods that stimulate not only the restoration of broken circuits but also the creation of new ways to compensate for the existing functional deficit [30].

Modern technologies, including computer stimulating programs, are increasingly used in neuropsychological correction today. An analysis of studies performed during the period between 2010 and 2017 and included in the PubMed, Scopus, Cochrane, and Web of Sciences databases has shown that virtual reality may provide an effective tool for assessing and rehabilitating cognitive and behavioral disorders in patients suffering from TBI [190]. Interactive virtual scenarios and audiovisual stimulation, creating full sensory involvement, facilitate the restoration of attention, visual-spatial, and executive dysfunction [214]. In the method of neurobiological control, EEG indicators are used as biofeedback [266]. The use of digital technologies in cognitive rehabilitation is promising because of their use with the help of telemedicine for the rehabilitation of patients in remote areas, if it is difficult or impossible for the patient to visit the appropriate specialist. The expediency of joint physical training and cognitive rehabilitation by means of using virtual reality is shown [38, 191].

Thus, the potential of neuroplasticity of adults decreases during the aging process [41, 246], and even more so in brain injuries and disorders. In this regard, in elderly people suffering from chronic TBI and neurodegenerative diseases, the ability to adapt to changing environmental conditions and restore the structure and dynamics of brain activity after acquired injuries of various etiologies is severely limited. For the same reason, it is difficult to expect a high clinical result for any neurorehabilitation methods if they are used against the background of reduced neuroplasticity, not sufficient for experience-dependent reinforcement, reconstruction, or creation of new connections.

It is important to note that even a short-term increase in the plasticity potential of the adult brain could enhance the effectiveness of neural network remodeling and

facilitate the restoration of the functions they serve using already known traditional, complementary, or alternative therapies. Therefore, the most effective ways to activate (maximize) the structural and functional plasticity of the brain in order to improve the cognitive functions and quality of life of such patients are being searched nowadays (see Sect. 31.4).

31.4 Natural and Artificial Fractals in Rehabilitation Therapy

31.4.1 Plasticity-Based Stimulation Therapy Using Natural and Artificial Fractals

Numerous studies accumulated the experience of various approaches to rehabilitation therapy that expand the knowledge on neurophysiological basis for adaptive and nonadaptive plasticity of the visual system, substrates of impact, and limitations of visual rehabilitation [23, 276, 290, 357]. The application of growth factors showed high effectiveness in experimental studies. However, their clinical application is limited by the complicated delivery of large protein molecules neurotrophins to neuronal structures. Special attention is paid to pharmaceutical drugs that exert an effect similar to growth factors. Cerebrolysin is one of such drugs that showed effectiveness in the therapy for neurodegenerative diseases and focal brain lesions [13, 33, 94, 348]. The range of rehabilitation procedures includes pharmacotherapy, cognitive and sensory training techniques, and methods of noninvasive electric, magnetic, and photo stimulation of the brain and eye (see Sects. 31.2 and 31.3).

This chapter does not pretend to be a detailed and critical analysis of the existing methods of stimulation therapy, but only includes a brief overview of them. Our task is to determine the place in these neurorehabilitation strategies of approaches that use natural and artificial fractals, as well as stimulation of the visual system and the brain with scale-free (fractal) optical signals.

Among nonpharmacological methods for enhancing neuroplasticity, physical activity has a good evidence basis. It was proved that physical activity decreased the risk of AD and dementia by 45% and 28%, respectively [116]. Physical training contributes to the survival of neurons, optimizes neuroendocrine and physiological reactions to psychosocial and physical stress, and increases the sensitivity of the vegetative nervous system and CNS by stimulating such processes as synaptic plasticity, neurogenesis, angiogenesis, and autophagy [192]. Some other methods also target the involvement of the remaining partially preserved structures and the enhancement of their plasticity [289]. Computer games, learning new languages, and playing musical instruments are used to slow down the age-related reduction of cognitive functions [108]. However, it should be mentioned that the effectiveness of the developed programs and technologies is not high and does not have a high level of verification in evidence-based medicine. For example, a positive

effect of physiotherapy and training programs involved virtual reality and target the improvement of the walking structure, postural stability, and quality of life in patients with Parkinson's disease (PD) was acknowledged as weak or very weak in a Cochrane review [74].

The studies proved the relevance of the search for new methods of neurorehabilitation and visual rehabilitation that reactivate the plasticity of the visual system and the brain. The maximization of neuroplasticity is necessary to provide the maximal adaptive CNS response to therapeutic intervention and adaptive reorganization of neural networks in patients with neurodegenerative pathology and traumatic injury that restore the functionality of the retina and the brain.

31.4.1.1 Stimulation Therapy with Sensory Stimuli

For over 70 years, the high sensitivity of the brain to external rhythmic impact and its capacity to synchronize its dominant activity with the rhythm of the external impact had been known [21, 139, 240, 320, 321, 389]. Sleep disorders and depression treatment and mood and cognitive performance improvement in patients with diseases and injuries of the brain widely include sensory rhythmic stimulations (audio, visual, and vibrating). Their effect is associated with the phenomenon of "brainwave entrainment" (BWE), in particular, with the principle of neurofeedback [168, 308, 320, 330, 389]. The influence of the methods of sensory rhythmic stimulation on cognitive oscillation and mood is associated with the fact that various cognitive conditions are characterized by various types of dominating fluctuations in the ongoing spontaneous brain activity. Intensive cognitive activity is typically characterized by the domination in the EEG of oscillations at the beta rhythm frequency (13–30 Hz), the alert state is associated with alpha rhythm (9–13 Hz), drowsiness is characterized by theta rhythm (up to 5–8 Hz), and delta rhythm oscillation (0.5–4 Hz) dominates in the condition of deep sleep [319].

Some neurologic and psychiatric diseases are characterized by the selective weakening of brain oscillation activity in specific frequency ranges. For example, clinical symptoms of PD and behavioral impairments are associated with changes in the brain activity in the beta range [63, 126, 131]. In patients with schizophrenia, the weakening of working memory and other cognitive functions is associated with anomalous oscillation rhythm in the gamma range [12, 292, 337]. The changes in the activity in the alpha range are observed after a stroke and craniocerebral injury [169, 360].

Brain rhythms weakened in separate ranges can act as a selective target for stimulation therapy. For example, in noninvasive brain stimulation (NIBS), rhythmical ranges are divided into two groups ("direct" and "indirect" modes) depending on the effect registered by EEG [295]. A "direct" frequency mode corresponds to the range of the impaired EEG rhythm, which is associated with a direct interference in the stimulation rate and generator rate [256, 331, 378]. An example of an "indirect" mode of rhythmic NIBS is continuous theta burst stimulation, which increases the

activity in the theta range and decreases the activity in the beta range, which is shown on EEG when recorded at a resting state [331].

It should be noted that in the stimulation therapy for the improvement of cognitive functions, relaxation, or meditation, simple sensory (visual and audio-visual) signals are used with one chosen frequency in the alpha, beta, gamma, or theta rhythm or binaural audio stimulation (binaural beat with illusory frequency). There are physiological grounds to believe that these methods can be useful for the rehabilitation of cognitive functions. For example, it was shown that alpha synchronization was associated with a creative (divergent) mindset [87, 88], and binaural rhythms exerted some effect on cognitive functions and mood [171]. However, during the past years, some studies appeared that showed the ambiguity of not only the power but also the sign of impact of binaural rhythms: improvement or worsening. This ambiguity was obvious when affect the cognitive functions of the subjects with initially various levels of intellect and divergent and convergent mindsets [265]. This shows that binaural rhythms, and probably other forms of sensory stimulation, based on the BWE phenomenon, require additional studies to provide fundamental grounds for specific indications for their application in neurorehabilitation.

In certain conditions, low-intensive photic stimulation can change cortical activity and improve episodic memory in young and senior adults using a certain stimulation frequency. Flickering visual stimuli (9.5–11.0 Hz) modulated cortical activity and improved episodic memory in young and senior people [364, 365]. This effect is sensitive to the frequency of stimulation. In cognitively healthy senior people, it significantly improved the recollection of words in remote periods after their memorization only after the exposure to 10.2 Hz flickers. After the exposure to flicker frequency lower than 9.0 Hz and higher than 11.0 Hz, this effect disappeared [364].

However, it should be highlighted that the rhythms of audial or photic signals with constant frequency (determined vibrations) in programs of sensory stimulation therapy can locally improve electric activity in the chosen EEG range but cannot restore the complicated dynamics of the brain activity that is characteristics of a healthy person (see Sect. 31.4.1.3 and further sections).

It should be noted that the safety of rhythmic stimulation with determined dynamics for the brain and retina is understudied. Weak interest in this subject is explained by the fact that all the studies involve low intensive and safe for the retina stimuli. These stimuli have much lower power than is required for the retina injury (photochemical or thermal). An exception is a widely studied issue on the danger of intermittent photic stimulation in photosensitive subjects who develop paroxysmal reactions registered by EEG [347]. In some patients, paroxysmal reaction develops in response to high-contrast images and stripes [363]. Flickering visual stimuli cause seizures in patients with photosensitive epilepsy. Discontinuous changes in the parameters of visual stimuli in the 3–70 Hz range impose a potential risk of seizure with its maximum in the range of 15–20 Hz [362]. The study with the combined

application of EEG and fMRI [343] revealed that the cortico-subcortical network that generates alpha oscillations at rest differs in patients with photosensitive epilepsy comparing the control subjects by a decrease in alpha-dependent inhibition of visual cortex activity and sensorimotor networks at rest. This study showed the association between neuronal processes involved in the generation of alpha rhythm and processes involved in photosensitive epilepsy.

Arnold Wilkins and coauthors highlighted that flickering did not have to be rhythmic (at one frequency) to provoke seizure [362]. This confirms the importance of comprehensive studies on the safety of intermittent visual stimulation in the diagnostic (EEG) and treatment (photic and audio stimulation) considering not only stationary parameters (frequency, power, and color) but also the dynamics of intermittent stimuli. The authors of this chapter suggest that the sign of the effect and its power can differ for regular rhythm (determined fluctuations) and nonlinear fluctuations of the stimuli parameters, including stochastic and deterministic-chaotic (fractal) behavior.

31.4.1.2 Environmental Enrichment

It is suggested that the potential of the neuroplasticity of the aging brain and in patients with neurodegenerative disorders can be enhanced using various educational programs, physical and cognitive training [8, 89, 193, 194, 212, 235, 294, 300, 341]. Experimental studies on animals showed their positive effect on cognitive functions, activity, and brain morphology. It should be noted that many methods of neuroplasticity reactivation were developed within the concept of “Environmental Enrichment” (the EE paradigm) [8, 22, 23, 206, 217, 340]. The term EE stands for the conditions of multimodal learning that enhance the environment with complicated sensory stimuli, cognitive tasks, and motor activity [23].

The high clinical potential of EE was shown for patients with brain injuries [8]. Many studies of the EE paradigm in animals showed a wide spectrum of impact on the brain, including an increase in the level of neurotrophic factors and gene expression that control neuronal signaling and protein synthesis [264]. EE and learning new skills increase the neuropil density in the cortex [355] and lead to structural changes in the brain [31].

According to Jennifer C. Tomaszczyk and coauthors, EE methods can be applied to destroy the descending spiral of negative neuroplastic changes and prevent “negative learning” [332]. The concept of negative neuroplasticity highlights the association between behavioral disorders, changes in the processes of sensory perception, environmental impact, as well as the importance of EE application in neurorehabilitation [332]. It can be suggested that the therapeutic effects of EE could be determined to some extent by the same factors as subjects in the multimodal enriched environment and often exposed to natural environmental patterns.

31.4.1.3 Fractality of Nature and Man

There is a vast evidence base on the nonlinear structure of natural objects, anatomic structures, and physiological processes in healthy humans that are characterized by the properties of self-similarity and scale invariance in space and time [71, 72, 101–103, 119, 143, 196, 325, 326, 402, 403].

Nonlinear fluctuations of physical and physiological parameters of living and nonliving systems are referred to noises [287]. Noises are conditionally assigned a color (according to the system of abstract terms of acoustics) in accordance with their spectral density. In nature, pink noise prevails, the spectral density of which is reversely proportional to the frequency and changed according to the $1/f$ power law. Pink noise, also called fractal noise or flicker noise, describes the dynamics of river flowing, fluctuations of climatic data, road traffic, human speech, music, etc. [115, 311, 342]. Natural noises influence the adaptation of living systems to environmental conditions [70, 114].

A typical characteristic of normal physiological processes is the fractal complexity of their dynamics [100, 199, 318]. Self-similar patterns of geometric and dynamic fractals also characterize the architecture and activity of a healthy brain [72, 176, 263, 329]. Ervin Sejdić and Lewis A. Lipsitz [287] noted that nonlinear interactions of regulatory systems and effects of the fractal environment created a strong variable behavior of the physiological process parameters that are modeled as pink noise.

In diseases, fractal dynamics of physiological processes is lost, which leads to the weakening of long-range correlations [103, 117–120, 137, 248]. Instead of fractal dynamics, physiological systems in pathological conditions produce (i) brown noise (weakly correlated fluctuations instead of pink noise), (ii) highly periodic behavior (deterministic fluctuations at a constant frequency), or (iii) completely stochastic system behavior (white noise) [199]. Simplification of the structure of neuronal networks [96] and oscillatory brain activity was revealed in patients with neurologic and psychiatric diseases [178, 182, 359, 377, 387]. In patients with Alzheimer's disease, a decrease in fractal dimension of the EEG rhythms was observed [65, 147, 182, 344, 359], and in patients with PD, a decrease in multiscale complexity was established in alpha rhythm [17, 182, 359] in comparison with healthy people of the same age.

An example of the loss of fractal complexity for the structure and function of the visual system is a manifestation of the aberrant dynamics of neuronal discharge in the visual system of a cat [186, 329, 382] and a decrease in the complexity of retinal vessels branching in patients with diabetic retinopathy [56] and neuronal processes in the LGN in experimental glaucoma [188].

The fractal dimension D qualitatively estimates the complexity of fractal geometric and dynamic patterns [196]. Natural fractals have D in a wide range, from 1.3 to 1.9. Most natural geometric fractals have average values of D from 1.3 to 1.5 [325]. The average fractal complexity with such D values is typical for most natural objects, for example, tree branching, shapes of clouds, rivers river beds, sea coastlines, mountain ridges, etc.

People are both subjects and objects of the dynamic chaos of nature [403]. The visual system of humans is adapted for the visual processing of mid-range fractals that prevail in nature [328]. Thus, spatial and temporal structures of the incoming sensor information can be the key factor that provides healthy brain activity [403]. Fractal visual surrounding significantly determines visual processing and human behavior [85, 86, 307, 325]. Studies showed that for an observer, the esthetic attraction of natural and artificial geometric fractals depended on the visual complexity of fractals. Most observers prefer fractals of mid-complexity 1.3–1.4 [4, 325–327]. The most pleasant music has a fractal dimension close to 1.4 [122]. Individual differences in the perception of fractal objects can be associated with genetic factors [261].

Natural and generated by humans geometric fractals of mid-complexity, including arts and architecture, can affect the physiological state of a person [277, 324, 326, 366].

It is suggested that the objectively existing association between CNS functioning and spatial and temporal structures of the visual and other sensor stimuli affects humans throughout their whole lives [403]. Therefore, during stimulation therapy, optical signals with a fractal structure can help in the rehabilitation of the visual system and brain function because of the activation of neuroplasticity [401–404]. It seems logical to use fractal signals for the synchronization of physiological fluctuation rhythms for preservation, maintenance, and restoration of complicated dynamics typical for a healthy organism.

31.4.1.4 Fractality of Music and Fine Arts

Studies showed that most musical compositions had fractal dynamics with a mid-range fractal dimension D , which corresponds to numerous natural fractal objects [29]. Music has fractal dimension [136] and is a complicated and universal irritant for the brain, which affect many cortical regions in the brain and subcortical structures involved in the processing of audial, cognitive, emotional, and motor information [278]. Music has various psychological and physiological effects manifested in EEG, cognitive functions, and human behavior [37, 145, 236, 251, 278, 279, 309, 375], and walking dynamics [27].

Musical learning was shown to affect the effectiveness of solving perceptual and sensorimotor tasks. The researchers associate this with structural transformations and functional changes that occur in the brain of musicians in comparison with the control subjects without musical education [28]. It should be noted that there are some correlations between the power of alpha rhythm and individual potential for creative thinking [28, 87, 88]. Music can modulate the association between mood and visual perception, transforming the perception of the world [150]. Musical education in childhood weakens the effect of aging on the CNS [167, 312]. Musical education and musical experience positively affect the perception of speech in the noise [129, 167, 243, 244, 383], reducing the age-related associated delay in nervous impulses transmissions [242], and slows down age-related hearing impairments [5].

At the same time, the improvement of speech perception in the noise is accompanied by neuroplastic alterations in the brain and auditory processing [11]. Regular musical practice activates neuroplasticity, improve the structure and function of the brain regions, and make music a potentially useful tool in neurorehabilitation [278, 286].

Music affects human emotions by changing the spectrum of EEG power [298, 375]. The strongest positive effect is observed in people who listen to Mozart (Mozart effect) [29, 136, 146]. The studies revealed a positive influence on the brain activity and cognitive function of a 10-minute listening to Mozart's sonata 448, although the changes were not always stable [6, 19, 145, 146, 270].

The analysis of the influence of music, as a dynamic art, and various types of fine arts (painting, architecture) are presented in a review by Marina Zueva [403]. Speaking about the positive effect of fine arts on the human brain, it should be mentioned that many art objects created by humans are characterized by the scale invariance in time or space, i.e., they are fractals [1, 122, 197, 277, 323, 324, 338, 403].

The influence of natural fractal patterns, natural landscapes, fractal pieces of art, and music therapy cannot improve a patient's condition in certain situations. In patients with neurodegenerative conditions, for the activation of adaptive neuroplasticity, it can be necessary to use an artificially created fractal environment. It is referred to the exposure of an observer to computer-generated self-similar patterns of geometric fractals with given parameters and sensory stimulation with dynamic fractals that include visual or audial repeated stimuli with the scale-free fluctuations of their parameters (for visual stimulation – fluctuation of brightness or frequency). Dynamic visual fractals, like fractal flickering, can affect the visual system being the basis for the development of a new strategy of diagnostics and treatment for the diseases of the brain and retina [55, 399, 401, 402, 404].

Considering the published data, fractal signals with mid-range D-values (1.3–1.5) can be considered the most promising for fractal stimulating phototherapy [228, 396, 404].

31.4.2 Perspectives of Fractal Stimulation Development

31.4.2.1 Objective Restrictions on Neurorehabilitation and the Theory of “Fractality of Sensations”

Experience keeps transforming the brain throughout the whole life. At the early stage of development, there are “windows” of increased plasticity of the brain. They are critical periods when the incoming information from the environment determines the structure of neuronal networks and their functionality [127, 246]. The windows of increased plasticity close when the balance between excitatory and inhibitory processes is achieved during brain development [46].

As it was mentioned in Sect. 31.3, the potential of neuroplasticity significantly reduces during aging, and it even more disrupts in patients with traumatic brain injuries and neurodegenerative diseases [246]. It is well documented that a mature brain has a certain potential for plasticity, which provides its adaptation to the changing environmental conditions throughout human life [98]. However, the ongoing reduction of the plasticity potential, which differs depending on genetic and epigenetic factors [98], limits the capacity to adapt to new environmental challenges in senior people. The possibilities of neurorehabilitation restoration of the structure and activity of the neural networks of the brain and retina are significantly reduced after their damage of various etiologies. For the same reason, it is difficult to expect high clinical results from any of neurorehabilitation methods if they are applied in patients with reduced neuroplasticity, which is insufficient for experience-dependent enhancement or creation of new synaptic connections [401, 404].

To enhance the positive effect of any neuroprotective strategy, it is necessary to increase the potential of neuroplasticity and activate the plastic capacities of the neural tissue. It should be noted that even a short-term enhancement (re-activation) of the potential of plasticity in patients with neurodegenerative diseases and retinal and brain injuries can increase the effectiveness of rehabilitation and remodeling of neuronal networks and their functions. Complex application of the methods targeted for the maximization of plasticity in combination with traditional methods for visual rehabilitation and neurorehabilitation should increase their effectiveness.

Thus, the reduced plasticity of neural tissue is one of the main objective limitations of neurorehabilitation methods based on brain plasticity.

On the other hand, stimulation therapy (electro, magnetic, photic, and audio-visual stimulation) involves modes of periodic exposures at a constant rhythm frequency. However, exposure to periodic rhythm is not capable of establishing a highly variable fractal pattern of physiological rhythms that characterize healthy processes, including the dynamics of neuronal activity (see the review [403]). Numerous studies showed that a typical sign of healthy physiological processes was their fractal dynamics with the power spectrum that corresponded to the $1/f$ power function [100].

The complexity of brain activity fluctuations and other physiological rhythms is lost in patients with neurodegenerative diseases [7, 65, 147, 287]. It is replaced with either ordered (periodic) fluctuations or completely random (stochastic) system behavior [100, 103, 120, 181, 199]. Thus, for photic stimulation with constant frequency typical for pathology and not typical for the norm, the risk of negative consequences cannot be excluded. The periodic rhythm cannot repair fractal dynamic biorhythms typical for a healthy organism. In some cases, there can be a risk of negative effect development. The deterministic-chaotic (fractal) dynamics of the functions of a biological system, including humans, determine its readiness to react to environmental challenges and provide the system flexibility and reliability of functioning, i.e. determine the freedom of its choice. The system that has the freedom of choice can actively resist the disturbing environmental influence and maintain the equilibrium. Imposed ordering of the system responses (deterministic rhythm) reduces its freedom of choice and can cause a pathological system response.

In 2015, the theory of “Fractality of sensations” was proposed [403], which associated the development and maintenance of normal neuronal networks and functional activity of the brain with the complexity of visual and other sensor environmental signals that affect a human throughout their life. This theory states that the simplification of the temporal structure of the environmental signals is associated with anomalies in the development and aging of the CNS. The use of fractal photic stimulation and fractal stimuli of other modalities makes it possible to increase the effectiveness of rehabilitation strategies for restoring the structure and function of the retina and brain by reactivating neuroplasticity, including in neurodegenerative pathology. Another important aspect arising from the theory of “Fractality of sensations,” which has to be considered during the development of stimulation therapy, shows that the simplification of the spatial and temporal structure of the environmental signals, caused by a pathology or habitat conditions, can aggravate the existing impairments of the structure and activity of the brain. This is especially relevant for patients who suffer from visual and audial disorders that narrow the diversity of sensory experience and limit the perception of the world.

Thus, there are some objective conditions that limit the effectiveness of the existing methods of neurorehabilitation and visual rehabilitation and the outcome of any strategies for the recovery of the structure and functions of the retina and brain. On the one hand, it is necessary to activate adaptive neuroplasticity. On the other hand, it is necessary to apply the physiologically adequate temporal structure of rhythmic stimuli. The authors of this chapter propose a novel approach to the stimulation based on the application of complex structure signals that have fractal dynamics. It is suggested that the application of fractal modes in sensory stimulation will reactivate the potential of neuroplasticity, improve the effects of stimulation therapy, and enhance the effectiveness of other methods used in combination. The period of enhanced (even short term) neuronal plasticity, achieved with fractal stimulation, can become that therapeutic window that will provide an increase in the effectiveness of other neurorehabilitation activities.

Special attention should be paid to the importance of the parallel timing of the decrease in the potential of neuroplasticity and violation of physiological rhythms dynamics in patients with neurodegenerative disorders. The parallel course of the processes does not establish their cause and effect relationship, but it certainly associates these processes and provides their possible mutual influence during interventions targeted for the maximization of neuroplasticity.

Sections 31.4.2.2, 31.4.2.3, and 31.4.2.4 of this chapter include the results of various studies that show some therapeutic possibilities of fractal stimulation using intermittent visual and audial stimuli.

31.4.2.2 Observation of Artificially Generated Fractal Images

The issue of the therapeutic effect of spatially invariant geometric fractals (static images) on the CNS of patients with diseases of the retina and brain is very interesting, but still remains open. Such static fractal objects can include simple

(classic) geometric fractals, fractal sets modeled with computer programs, pictures of natural objects, and pieces of art that can theoretically positively affect visual functions and brain activity. This possibility is confirmed by some publications. For example, one study showed the plasticity of human fixative eye movements exposed to geometric fractals of different complexity [224]. The temporal structure of saccadic eye movements shifted to the side of the fractal dynamics of the observed image. This effect depended on the fractal dimension of the observed stimuli.

31.4.2.3 Sensory Signals Control the Dynamics of Walking

Fractal Cues Control the Gait Variability

Fractal patterns of the gait observed in young healthy people, as well as other biological rhythms of a healthy organism, usually degrade with aging or pathology. The studies showed that fractal visual stimuli could be used for the formation of scale invariant gait patterns in patients with gait impairments. Some studies showed this capability for sound stimuli [134, 153, 203, 336] and visual stimuli [267]. In particular, it was found that interactive rhythmic audio-stimulation during walking on the treadmill (during the exposure to the stimuli) restored normal walking dynamics in patients with PD [134]. Even though rhythmical audial tones with a fixed frequency improved the gait in patients with PD in some aspects [68, 118, 330], rhythmic audiostimulation decreased fractal scaling in step-to-step intervals, synchronizing the gait rhythm around one frequency. Michael J. Hove et al. [134] showed that the dynamics of step-to-step interval oscillations in patients with PD could be improved to the healthy level of oscillations $1/f$ (pink noise) using interactive rhythmic signals of fractal dynamics.

Christopher Rhea et al. [267] showed that the fractal pattern in step-to-step intervals significantly changes during walking exposed to a fractal visual metronome (blinking red square). However, the participants could not adequately reproduce the stable fractal picture that was demonstrated by the stimuli. Fractal cues of young healthy people's gait enhanced at the phase of visual-motor synchronization, and this effect of visual entrainment remained for 15 minutes after synchronization. Hirotaka Uchitomi and colleagues [336] also revealed a significant impact of fractal rhythmic signals on the gait dynamics in patients with PD. Fluctuations of step-to-step intervals gradually returned to the healthy level of oscillations $1/f$, while a periodic rhythm of signals did not lead to such results. The authors see the perspectives for the wider application of interactive rhythmic signals in the area of neurorehabilitation. Joanne E. Wittwer and colleagues [367, 368] provided moderate evidence on the improvement of the speed and pace of walking in patients with AD and other neurological conditions. Nathaniel Hunt and coauthors [141] studied the influence of various temporal structures of complex audial signals on the pattern of step-to-step interval variability in healthy subjects. The results showed the plasticity of auditory-motor coupling in walking and the ability to control gait variability using a complex sound signal. Audial signals of various temporal structures (white and

brown noise) shifted the fractal dynamics of a healthy person's gait to the statistic properties of the respective signals.

Noise Therapy and Stochastic Resonance

The potential for the application of low-intensity white noise in clinical practice should be noted. Although it has nonfractal (determined-chaotic) and completely stochastic dynamics, white noise improves gait impairments and cognitive functions. Studies showed positive effects of the noise for senior people [61, 259] and patients with PD [371] and other neurological disorders [269, 302]. Vibrating soles with vibrations that corresponded to the frequency of white noise were used in the treatment of coordination and gait dynamics in senior patients and patients with diabetic neuropathy and in the post-stroke period [61, 269]. A significant increase in the fractal dimension of the step-to-step interval fluctuations after the therapy was shown. It is suggested that this effect is based on stochastic resonance. In nonlinear systems, the noise can lead to an increase in the degree of system arrangement, which can be used in therapy for various pathologic conditions. Such conditions include stochastic resonance. The response of the nonlinear system to a weak external signal enhances with an increase in the noise intensity in the system [100, 133]. There is optimal amplitude of noise that provokes the maximal system response to a weak signal. The studies show that a stochastic resonance is a fundamental physical event typical for nonlinear systems.

The phenomenon of stochastic resonance explains the positive effect of white noise on cognitive functions in children with attention deficit and hyperactivity disorder [296, 302, 303]. It is suggested that via sensory systems, external noise translates the internal noise to the neuronal networks and simplifies the performance of task by inducing stochastic resonance in mediator-based systems.

The results obtained in healthy volunteers and patients with PD and AD with gait disorders confirmed the key significance of the temporal structure of a signal during the sensory perception of the CNS. They also showed the possibility of the fractal synchronization of gait variability by the external signal. It proves a principal readiness of human physiological functions to capture the imposed oscillations of the fractal structure.

In Sects. 31.4.2.4 and 31.4.2.5, the results of some pilot clinical and experimental studies are presented showing the positive effects of fractal visual stimulation observed not only directly during the stimulation, but after courses of phototherapy.

31.4.2.4 Experience in the Use of Fractal Phototherapy in Patients with Glaucoma

In 2014, the first version of a LED generator of fractal flickering was designed. It is installed in the traditional dome-shaped ganzfeld photostimulator for electrophysiological testing [399]. The program of the LED generator controlled interval

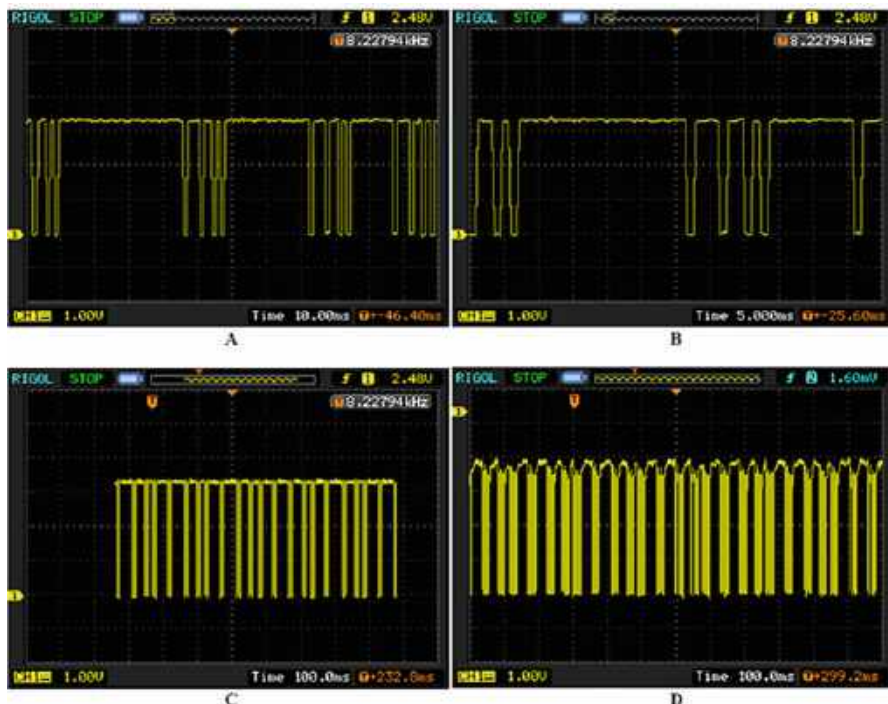


Fig. 31.1 An oscilloscope demonstration of the “sweep stimulation” principle with several signal embeddings

fluctuations between the flickers. The flicker generator was programmed to create a fractal pattern of light stimuli with an interval between the flickers (set of flickers) that changed in the sweep pattern or that was produced as a dichotomy pattern made of flickers sets (Fig. 31.1).

Rabbits were exposed to fractal photic stimulation once a day. The duration of the session was 5 minutes. A course of photic stimulation included 14 sessions. Dark-adapted and light-adapted electroretinograms (ERGs) were registered according to ISCEV standards [207]. The study results showed that the retinal stimulation with fractal “sweep” flickering increased the amplitude of the dark-adapted ERG and shortened the peak latency of the b-wave, which can be associated with an acceleration of the signal transmission between the retina neurons of the first and second rank [394].

Further, we advanced the program of stimulation and the technology of fractal phototherapy to create a line of devices that produce a fractal change in the brightness on the stimulator screen based on the functions of Weierstrass-Mandelbrot [227, 229, 395, 396]. Presently, this method is realized in VR glasses, and portable and stationary models of LED devices that generate fractal optical signals of the given parameters. Fluctuations of the signal brightness in the devices are mathe-

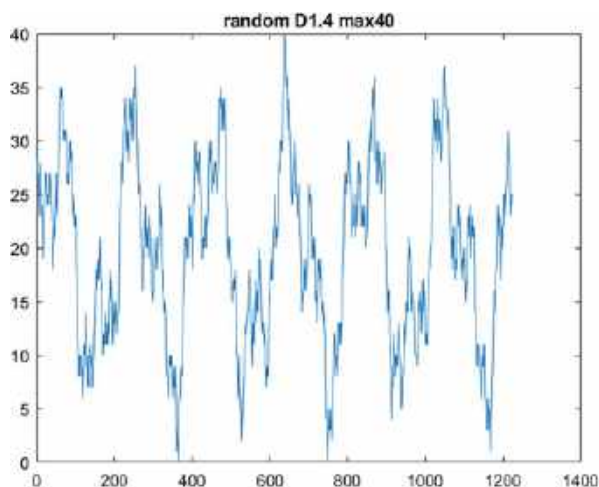
matically modeled and have temporal structures that correspond to natural fractal objects and physiological fluctuations of a healthy organism. It is known that even at the last stage of glaucoma, in the general RGCs population, there are neurons at the plastic stage of reversible changes [257] that can react to neuroprotective therapy. According to the proposed hypothesis, fractal photic stimulation therapy can be a promising nonpharmacological approach that contributes to the preservation of visual functions in patients with glaucoma.

Earlier, the effect of fractal photostimulation has been estimated for the color perception using the data of the Farnsworth-Munsell 100 Hue Test and light sensitivity using short wavelength automated perimetry (SITA-SWAP) in 150 persons with suspected glaucoma, the 1st stage of POAG, and healthy control [397]. The most significant impact of fractal optical stimulation was revealed in early glaucoma for the MD data. After the treatment, the average visual field defect significantly decreased (by 0.77–1.0 dB, $p < 0.05$). In the color vision test, after the treatment, a significant reduction in the number of errors (10–12) was found; however, the color perception defects persisted.

In a later study [396], LED and a programmer were installed in a case of VR glasses. The stimulator formed low-intensive (10–12 lux on the cornea) self-similar optical signals with a given fractal dimension $D = 1.4$. The mode of stimulation was managed via a PC. Figure 31.2 shows the shape of a signal programmed based on Weierstrass-Mandelbrot functions.

This study included 147 participants (98 patients in the main group and 49 control patients on placebo therapy; four subgroups in each group including patients with glaucoma suspicion and initial and advanced (stages II and III) POAG). Standard automatic perimetry (Humphrey—Carl Zeiss Meditec) was performed using a threshold test 24-2 and the algorithm SITA-Standard. Before and after a course of fractal photic therapy, the research team estimated the parametric indices MD (mean deviation) and PSD (pattern standard deviation).

Fig. 31.2 Waveform of optical signal programmed based on the Weierstrass–Mandelbrot function



A course of photic stimulation comprised 10 sessions. They were held daily in the morning excluding weekends. The duration of one session of fractal photic therapy was 10 minutes. Control patients received therapy, which included watching a video with a computer program intended for relaxation and release of visual tension twice a day for 10 days. During the session, participants were exposed to a computer-modeled image, a geometrical fractal set with the effect of movement illusion (<https://www.youtube.com/watch?v=8LFTTULcqD4>), and a musical composition by ELAN (Healing music). The duration of each relaxing video was 30 minutes.

The study showed that, unlike placebo therapy, fractal optical stimulation exerted a positive effect in patients with POAG and in patients with early and advanced glaucoma. In all patients from the main group, two-week low-intensive fractal phototherapy statistically significantly improved MD parameters that reflected the average defect of the vision field; the most significant effect was found for far-advanced POAG, wherein the average vision field defect was reduced by 4.39 dB. The results of the studies on the effect of fractal photic stimulation in patients with glaucoma confirmed the hypothesis and showed that self-similar visual stimuli with fractal complexity close to the natural dynamics of physiological functions showed a potential for nonpharmaceutical neuroprotective therapy. For further confirmation of clinical significance and stability of the therapeutic effects of fractal stimulation, additional studies are required with the application of objective functional diagnostic tests that can include the registration of ERGs, visual-evoked cortical potentials and EEG, fMRI.

Thus, these studies provided the first evidence of a neuroprotective effect of fractal photic therapy in patients with POAG of various stages. The expressed effect of fractal stimulation for eyes with stage III POAG confirmed that even at advanced stages of glaucoma in the general RGC population, there was still a significant amount of cells at the plastic stage of reversible functional changes. Fractal stimulation can be seen as a novel nonpharmacological (physiotherapeutic) approach to neuroprotective therapy, but its potential should be studied thoroughly. It should be noted that the method of fractal therapy for glaucoma can be used as a means of prevention in a risk group or in patients with suspected glaucoma. Unlike drug therapeutic methods, fractal photic stimulation does not have ethical limitations for patients with unverified glaucoma, and its application can reduce the spread of this socially significant disease.

31.4.2.5 Objective Studies of the Effect of Fractal Optical Stimulation on the Structure and Functional Activity of the Retina

The study of the mechanisms of influence of fractal photic stimulation on the retina and foundation of the prospects for the improvement of visual functions in humans requires experimental studies on animals using various modes of stimulation and methods of objective evaluation of the functional activity and retinal structure.

A recent study [228] included 12 healthy rabbits aged 2.5–3 months with a body weight of 1.7–2.1 kg (24 eyes). Various kinds of electroretinograms were



Fig. 31.3 Full field ERG (a) and pattern ERG (b) recording procedure in rabbits

Fig. 31.4 Rabbit stimulation session with fractal optical signals



registered to assess the alterations in the functional activity of the retina after a long-term course of fractal photic stimulation (Fig. 31.3). Studies on healthy animal retinas are necessary to prevent overdose when this method in clinical practice and to recommend optimal modes of photic stimulation for further experimental and clinical studies. Light stimulation was performed five times a week using a device for fractal optical stimulation of a special design (Fig. 31.4).

Twenty-minute sessions of photic stimulation were performed in the interval from 9 to 12 am. Fractal fluctuations of the signal brightness were generated on the base of Weierstrass-Mandelbrot functions [229]. The fractal dimension of Hausdorff-Besikovitch was 1.4; the frequency of signal cue repetition was 9–11 Hz. Diagnostic testing was performed before, and 1 week, 30 days, and 90 days after photostimulation.

Optical coherence tomography of the retina was made with Spectralis HRA+OCT (Heidelberg Engineering, Germany) according to the protocols Line and Macular Map. The rabbits were fixed so that the visual stripes with the maximal density of cones, located 3 mm ventrally to the optic disk, and appeared in the center of the image. Electrophysiology was performed using a diagnostic complex

RETIport/scan21 (Roland Consult, Germany) in a light-proof recording booth protected from electromagnetic interference. The methodology of the study and conditions of the potential registration comply with ISCEV standards (International Society for Clinical Electrophysiology of Vision) for human studies [18, 207] and include peculiarities of the ERG in rabbits [99]. Dark-adapted and light-adapted full-field ERGs, light-adapted flicker ERGs at the stimuli with frequency of 8.3, 10, 12, and 24 Hz, as well as a pattern electroretinogram (PERG) were registered. The first series of ERG records was made several days before a course of fractal photic stimulation. Three more data series were made after the courses with duration of 1, 4 and 12 weeks.

Optical coherence tomography parameters did not reveal any changes in the rabbit retina throughout the period of observation. The thickness of the retina remained stable.

The results of this study did not reveal a negative impact to retinal activity of the courses of fractal photic stimulation. However, a positive effect of photic stimulation was found manifested in a significant increase in the amplitude of b-wave of light-adapted ERG and a statistically significant decrease in the peak latency of b-waves in dark-adapted and light-adapted ERGs. Fractal photostimulation significantly increased the amplitude of flicker ERGs at frequencies 8.3–12 Hz (Fig. 31.5). The results showed the most significant increase in the amplitude of flicker ERG for stimuli 8.3 and 10 Hz. At the same time, the amplitude of flicker ERGs at 10 Hz remained at its elevated level even after a three-month course of photic stimulation, and an increase in the amplitude of 8.3 Hz flicker ERGs was observed already after one week course of stimulation.

Since a contribution of photoreceptor activity prevails in a low-frequency flicker ERG [398], these data indicate the influence of photic stimulation on the activity of cone photoreceptors. The flicker ERGs at 24 Hz showed the weakest changes

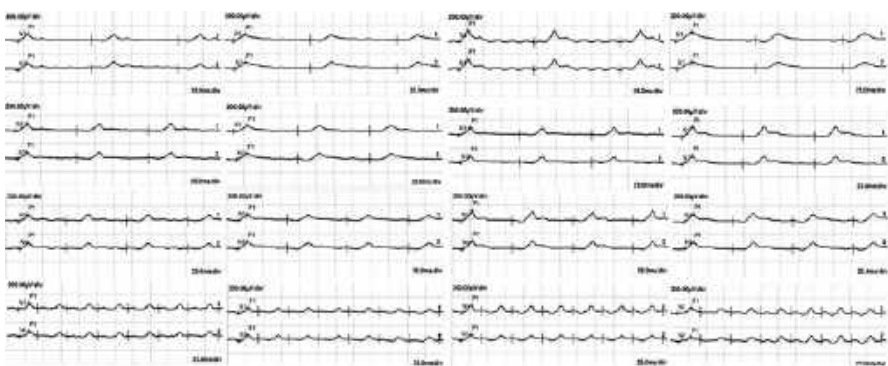


Fig. 31.5 Representative flicker ERG in rabbits obtained before and after a courses of fractal photic stimulation with the duration one week, four weeks, and twelve weeks. *From top to bottom:* ERG flicker for flickering frequencies of 8.3, 10, 12 and 24 Hz. *From left to right:* before and after stimulation courses lasting 1, 4 and 12 weeks

after long-term photic stimulation. Unlike 10 Hz-flicker ERG, the effect of fractal photostimulation on other types of ERGs decreased after a three-month course of photic stimulation in comparison with a one-month course. There were no significant changes in the peak latency of waves P50 and N95 of the transient PERG at any duration of fractal stimulation. Amplitudes of P50 and N95 waves of transient PERG and amplitude of the steady-state PERG increased after one-month and three-month courses of photic stimulation, but these changes were insignificant. The obtained data proved the feasibility of the application of a four-week course of photic stimulation in further studies of the pathology on animal models and in clinical studies. However, considering a significant increase in the 8.3 Hz-flicker ERG amplitude already after a one-week course of stimulation, 1–2-week fractal therapy should be seen as an alternative to a 30-day course in clinical studies. The data also suggested that even long-term exposure to fractal optical signals is safe for the retinal structure and can be used in clinical studies without harm or risk of overdose.

A recent pilot study also provided data on the influence of fractal photic therapy on the retina of a rabbit with modeled retinal pathology. In six rabbits, retinal pigment epithelial atrophy was modeled on one eye with subretinal injection of a sodium chloride solution by the method [231]. Three rabbits were exposed to a course of daily fractal photic therapy according to the scheme described above, and the other three rabbits with retinal atrophy (control) were daily exposed to a constant LED lamp (15–20 lux) for 20 minutes. ERG was registered before the surgical procedure, a week after atrophy modeling (= a week before the first session of photic stimulation), and after a seven-day course of photic stimulation (Fadeev et al., not published). Figure 31.6 shows one example of the dynamics of the rabbit's dark-adapted ERG response. Preliminary data showed better dynamics of ERG restoration during daily fractal photic therapy in comparison with the exposure to constant light in the same mode (placebo) (Fig. 31.7) [228]. Nowadays, these studies that involve a higher number of animals and parallel evaluation of the signs of neuroplasticity using immunological, immunohistochemical, and biochemical tests are ongoing.

Therapeutic effects of fractal photic therapy connected with the restoration of functional activity of the retina can be associated with the activation of synaptic and dendritic plasticity. The changes in the activity of the retina during the stimulation of the retina with intermittent stimuli of the fractal temporal structure can be connected with an enhancement of synaptic strength and the formation of new synapses (Hebb's rule) [125, 220], the complication of dendrites branching, and other plastic events in the retina that require further studies. Fractal photic stimulation can affect neurotrophic factors and enhance the expression of genes that control protein synthesis and neuronal signaling, as was shown earlier for electrical stimulation [284] (see also Sect. 31.2).

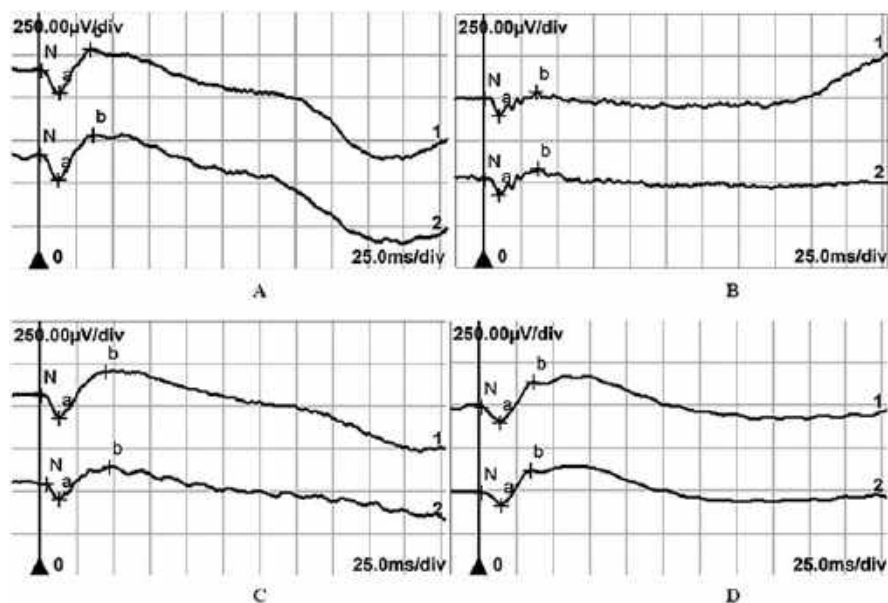


Fig. 31.6 Impact of fractal phototherapy on the scotopic ERG of the rabbit with retinal atrophy. (Dark-adapted ERG on a standard stimulus, $3.0 \text{ photopic cd} \cdot \text{s} \cdot \text{m}^{-2}$: (a) initial data; (b) 1 week after the surgery, before the treatment; (c) after 1 week course of fractal phototherapy; (d) after 4 weeks course of fractal phototherapy)

31.5 Conclusion Remarks

Studies showed similar signs of brain and retina plasticity in subjects with various neurodegenerative diseases and injuries to the visual system and the brain. In various disease etiologies, the response to CNS damage at the cellular and molecular level is identical and associated with remodeling of the neuronal circuits. It makes it relevant to use methods based on neuroplasticity in neurorehabilitation and visual rehabilitation strategies.

However, the neuroplasticity potential is reduced with aging and impairs during neurodegenerative diseases and brain and visual system injuries. This limits the brain's capacity to repair the structure and dynamics of its activity after lesions. For this reason, it is difficult to expect high clinical results for any methods of neurorehabilitation if it is applied in patients with reduced neuroplasticity, which is insufficient for experience-dependent enhancement, reconstruction, or creation of new connections. Therefore, researchers look for the most effective pathways of activation of the brain structural–functional plasticity to improve cognitive functions and the quality of life of patients. Maximization of neuroplasticity is necessary to provide the maximal CNS response to therapeutic intervention and adaptive

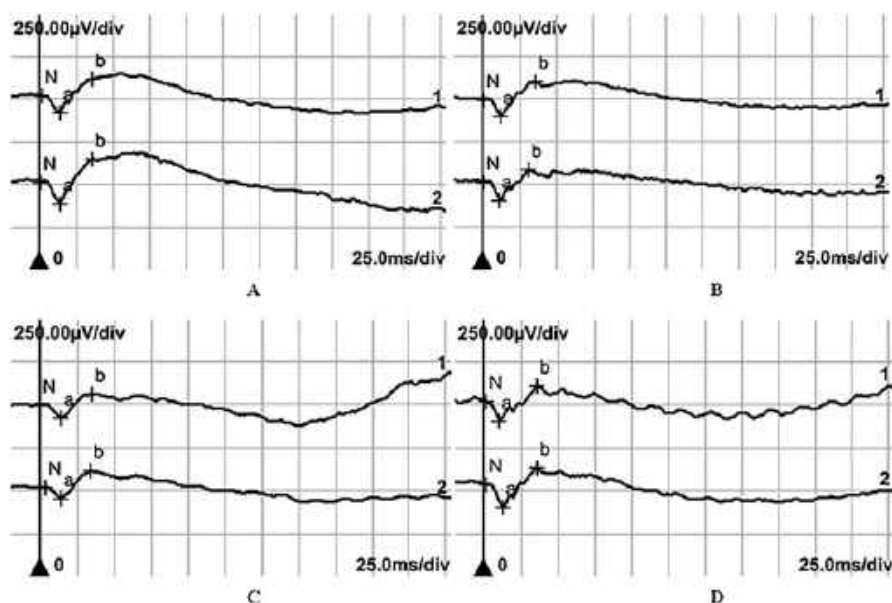


Fig. 31.7 The dynamics of scotopic ERG of the control rabbit with retinal atrophy. (Dark-adapted ERG on a standard stimulus, $3.0 \text{ photopic cd} \cdot \text{s} \cdot \text{m}^{-2}$: (a) initial data; (b) 1 week after the surgery, before the treatment; (c) after 1 week course of constant light treatment; (d) after 4 weeks course of constant light treatment)

reorganization of neuronal networks in patients with degenerative pathology and traumatic injury to restore the functional activity of the brain and retina.

Considering the fractal geometry and dynamics of the healthy brain and the loss of fractality in neurodegenerative pathology, we suggest that the application of self-similar visual signals with a fractal temporal structure in the stimulation therapy can reactivate the adaptive neuroplasticity and enhance the effectiveness of neurorehabilitation. Using fractal signals for the synchronization of the biorhythms fluctuations is proposed can preserve, maintain, and restore complex $1/f$ dynamics typical for a healthy organism.

This proposition was tested in the recent pilot studies. Patients with glaucoma had a statistically significant positive effect of fractal photic therapy on light sensitivity and the perimetric MD index, which shows that methods of fractal stimulation can be a novel nonpharmacological approach to neuroprotective therapy and neurorehabilitation.

Experimental studies on healthy rabbits demonstrated that a long-term course of stimulation (12 weeks) with optical fractal signals does not harm the electroretinogram (ERG) and retina structure. Preliminary evaluation of the influence of fractal phototherapy on the ERG in rabbits with modeled retinal atrophy showed better dynamics of the ERG restoration during daily stimulation therapy for a week (after atrophy modeling) in comparison with the exposure of rabbits to constant LED light (20 lux).

Positive changes in the visual function of patients with glaucoma and in the animal research can indirectly suggest the activation of its adaptive plasticity and the high potential of stimulation therapy with fractal visual stimuli as a novel nonpharmacological neuroprotective therapy and neurorehabilitation, which requires further study.

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Chapter 32

Fractal Similarity of Pain Brain Networks



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Abstract The conscious perception of pain is the result of dynamic interactions of neural activities from local brain regions to distributed brain networks. Mapping out the networks of functional connections between brain regions that form and disperse when an experimental participant received nociceptive stimulations allow to characterize the pattern of network connections related to the pain experience.

Although the pattern of intra- and inter-areal connections across the brain are incredibly complex, they appear also largely scale free, with “fractal” connectivity properties reproducing at short and long-time scales. Our results combining intracranial recordings and functional imaging in humans during pain indicate striking similarities in the activity and topological representation of networks at different orders of temporality, with reproduction of patterns of activation from the millisecond to the multisecond range. The connectivity analyzed using graph theory on fMRI data was organized in four sets of brain regions matching those identified through iEEG (i.e., sensorimotor, default mode, central executive, and amygdalo-hippocampal).

Here, we discuss similarities in brain network organization at different scales or “orders,” in participants as they feel pain. Description of this fractal-like organization may provide clues about how our brain regions work together to create the perception of pain and how pain becomes chronic when its organization is altered.

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Keywords Pain · Multiscale brain organization · Functional MRI · Intracranial EEG · Chronic pain · Graph theory

32.1 Brain Dynamics of Pain Perception

Pain is a complex sensory experience that transforms simple noxious stimuli into conscious perceptive events. The process of building a subjective perception of pain intensity engages multiple levels of the central nervous system [16]. Noxious stimulus information is first transduced by afferent peripheral nociceptors from discrete body region and their inputs are integrated over multiple spinal levels and differing time frames. This pre-processed information is transmitted to the thalamus and cortex (supraspinal structures) where an individually unique experience of pain magnitude is elaborated. This involves the processing of sensory-discriminative features of the pain stimulus, but also its affective-motivational features [76], with descending modulation pathways (i.e., top-down controls) that act to modulate incoming nociceptive activity [39, 66].

Human brain imaging has revealed that the multidimensional integration of pain is possible thanks to the coordinated activation of multiple and widespread brain regions interacting dynamically, commonly described as “the dynamic pain connectome (DPC)” [46]. The notion that pain perception needs joint cortico-subcortical interaction has been abundantly documented [3, 35]. This is not a fixed arrangement of structures but rather a system composed of several interacting brain networks, hierarchically organized [34].

Human brain electrophysiological approaches have also revealed the importance of brain dynamics and that brain regions are engaged successively during the pain experience [9]. The first brain activations associated with pain stimulation mainly involved the posterior operculo-insular cortex (pIns and parietal operculum: S2), mid-cingulate cortex (MCC), and the parietal sensory areas. This set of brain regions (or modules) that process raw nociceptive inputs reflect the smallest scale (first order) of interaction, but alone are not sufficient to sustain the conscious perception of pain [8, 43]. The involvement of a higher-order set of regions belonging to cerebral networks outside this “nociceptive matrix” is crucial for nociception to become pain [34, 35]. Second-order interactions link this raw sensory information with stimulus salience and motor control (including the anterior insula (aIns) and anterior cingulate regions). The largest-scale interactions (third order) link these sensorimotor and salience networks with brain structures that support high-level cognition and control (including pain re-evaluation in prefrontal and posterior parietal regions; see Fig. 32.1). The organization of these brain segregations (both temporally and topologically) forms a continuum going from pre-conscious cortical nociception to autonomic control, to attentional, to emotional, to full-on conscious pain [9, 25, 35].

Therefore, from a simple noxious event activating few brain recipients of the spinothalamic tract (i.e., a first-order nociceptive matrix), it becomes a very complex

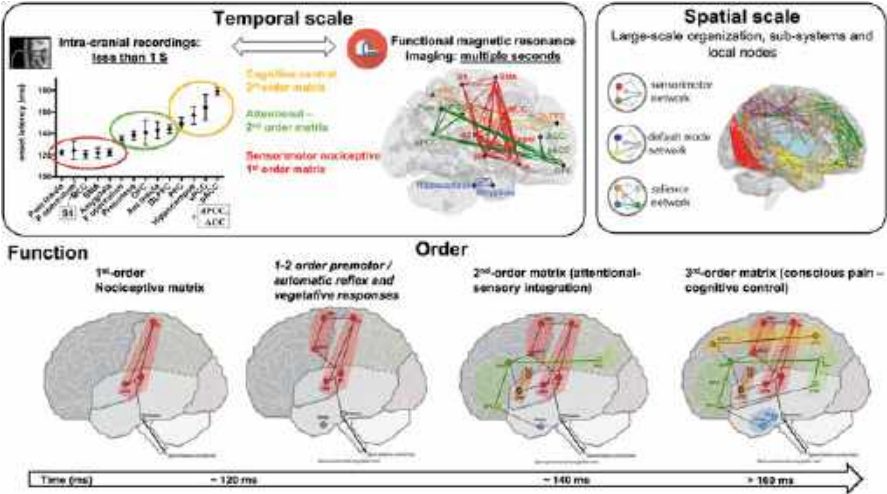


Fig. 32.1 Self-similarity of brain patterns in pain at different spatial and temporal resolutions from pre-conscious cortical nociception to conscious pain. The smallest scale (first-order) interactions occurred in brain regions that process raw nociceptive inputs (red network). Second-order interactions link this encoding of raw sensory information to stimulus salience and motor control (green network). The largest-scale interactions (third-order) link the sensorimotor and salience networks with brain structures that support high-level cognition and control (including re-evaluation; yellow network). The repeating network patterns at different scales might allow our brains to weave a high-level understanding of what is happening from the fabric of our raw sensory experiences (see [25]). *pINS* and *aINS* posterior and anterior insula, S1 and S2: primary and secondary somatosensory cortex, *SMA* supplementary motor area, *pACC*, *ACC*, *MCC*, and *PCC* perigenual, anterior, middle, and posterior cingulate cortex, *OFC* orbito-frontal cortex, *Prec* Precuneus, *Fop* frontal operculum, *DLPFC* dorsolateral prefrontal cortex, *PPC* posterior parietal cortex, *Amyg* amygdala, *Hip* hippocampus

process involving large and interconnected part of the cortex (i.e., second- and third-order high-level cognitive-pain networks), found in several temporal scales, which is the basis for the description of fractal in nature.

Fractal objects are made of subunits (and sub-sub-units, etc.) that look like the structure of the overall object and can be quantified by computing the fractal dimension (FD). FD is a measure of complexity derived from nonlinear dynamical systems theory [72], capturing nonstationary changes in the signal, and sometimes described as the “roughness” of the signal. FD is based on the self-similar behavior of time series [1, 40], which refers to the fact that the fractal dimension of a signal is the same, regardless of the scale at which that signal is observed – “scale-free” [23]. The fractal concept can be applied to complex time-varying processes that lack a single scale of time such as heart rate variability [41] or cortical neural activity [50] with many clinical applications.

This is a suitable approach to describe the irregular and nonperiodic patterns recorded by neuroimaging techniques, no matter what the properties (stationary, nonstationary, deterministic, or stochastic) of the analyzed signal. In other words,

FD analysis could be a useful tool for deriving features of brain states related to a nonlinear phenomenon such as spontaneous pain, which is commonly observed in individuals with chronic pain. The fluctuations of spontaneous pain ratings do not possess a stable mean or variance (because the source of evoking stimulus is often unknown) and therefore resemble fractal time series [30]. Brain patterns during pain also appear largely scale-free, with “fractal” activity and connectivity properties reproducing at short- and long-time scales [25].

Several computational techniques (e.g., Higuchi’s fractal dimension (HFD), Hausdorff dimension, Grassberger-Procaccia correlation dimension (GP), Entropy, etc.; see Chaps. 1, 2, 3, 4 and 5 for more details) can be used to assess FD. However, while this type of nonlinear approach is no longer an issue, to the best of our knowledge, there are very few applications for the analysis of micro [2, 73, 77] and macro [30, 68] neuroimaging data related to chronic pain and none on acute pain in healthy individuals.

In the following sections of this chapter, we illustrate the self-similarity of patterns (subunits) of pain processing at different spatial and temporal resolutions – i.e., multiscale characterization: from iEEG local recording to whole brain fMRI recording during pain stimulations.

32.2 Organization of Pain Brain Networks

When two brain structures appear to change their responses in similar ways and with the same dynamic, this is called a “functional connection” because it suggests that the phenomena that are driving the two structures synchronize them in some way. Functional magnetic resonance imaging (fMRI) and graph theory of the brain connectome are used to extract functional networks connecting correlated human brain regions. Analysis of the resulting networks in tasks [25, 57] or at rest [14, 15, 27] shows that the distribution of functional connections and the probability of finding a link are both scale-free [22, 29]. The patterns of intra- and interareal are incredibly complex, they form a network with small world properties, assortative mixing, and “fractal” connectivity properties reproducing at short- and long-time scales [69, 88]. Decomposing brain activity into functional modules of regions to investigate their interactions is a well-suited approach to study pain – an experience that requires the spatiotemporal coordination of different brain regional subsets [56].

In recent studies, we assessed how large-scale topological organization (macroscale brain function) obtained from fMRI shapes neural function and processing hierarchy provided by intracerebral stereo-electroencephalography during noxious pain stimulation. Findings indicate striking similarities in the activity and topological representation of networks at different orders of temporality, with reproduction of patterns of activation from the millisecond to the multi-second range. The brain connectivity analyzed using graph theory (modular detection) during pain was organized into four modules matching those identified through iEEG (i.e., sensorimotor, default mode, central executive, and amygdalo-

hippocampal). Such topological aspects at a long-time scale appear relevant to reflect local neurobiological features of iEEG responses to noxious stimulation (Fig. 32.1). Their similarities across the different orders of the pain process are discussed in what follows.

32.3 First-Level Nociceptive Processing

32.3.1 *How Does the Brain Encode Noxious Events?*

The understanding of the physiological mechanism of pain has moved from a concept associating pain with somatosensory processes to a framework involving more directly the insular cortices. Pain was assumed to involve primarily S1, S2 cortices, and thalamus, but only half of the neuroimaging studies confirmed this prediction, except for S2, which was found at different temporal scales to be almost consistently involved when nociceptive stimulation is applied.

Functional imaging reviews [35, 62, 63, 71, 82] and intracerebral recordings [34, 54] have shown that the posterior opercular-insular cortices are the regions the most consistently activated in response to a nociceptive stimulation and that they are capable of encoding pain intensity. These cortices receive most of the spinothalamic tract projections [21]. In addition, the operculo-insular cortex is the only brain region where a direct electrical stimulation can produce a pain sensation [52–54]. Direct focal electrical stimulations at minimal intensities in the other regions of the pain circuit (i.e., more than 4000 stimulations) are not able to elicit pain. Insular epilepsy (or propagation of discharges to the insular cortex) is the only focal epilepsy to be possibly associated with painful symptoms [55]. Finally, small lesions restricted to the parietal operculum/S2/posterior insula remove the ability of the brain to produce a pain sensation, and patients do not perceive nociceptive stimuli anymore. For example, they did not perceive the needle of venous punctures, they also did not perceive thermal pain with a 50 °C stimulation and their brain were unable to create a laser-evoked potentials [33, 36, 38]. Thus, the operculo-insular cortex can be presented as the only area in the brain to respond to the features of a first-order nociceptive matrix.

32.3.2 *An Insular Pattern*

Intracranial EEG (iEEG) recordings have helped to characterize the activation dynamics of brain regions of the pain connectome. This information illustrates that three successive and overlapping waves of activation sustain the transition from unconscious nociception to conscious pain in less than 1 s. The brain regions activated with the shortest latencies following a noxious stimulus include

the posterior insula and S2, but also the MCC, SMA, and amygdala (see Fig. 32.2). Brain network organization at longer time periods such as those explored in fMRI-based imaging is segregated in four modules (i.e., group of regions with high inter-connectivity) during pain which is consistent with the spatiotemporal organization of nociceptive-related activation at short time-scale (Fig. 32.1; [25]. A sensory-motor module encompassed the first activated brain regions but also included the anterior insula, which is not a direct spinothalamic recipient but marks the transition from the first to the second-order matrix [84].

The posterior and anterior subdivisions of the operculo-insular cortex have not the same function relative to pain, and this can be found at several fractal temporal resolutions. The posterior areas are specific to the discriminative processes related to painful conditions, whereas the antero-insular components appear as nonpain-specific contributions that can be recruited by any “salient” sensory input from various modalities such as auditory or visual [48]. Listening to baby cries in pain, seeing pictures or videos showing somatic pain, all the empathic experiments recruit large parts of the pain network, including the anterior insular cortex, with the noticeable exception of the SII-posterior operculo-insular region. In fMRI using high temporal resolution recording of the insular cortex by step of 641 ms and by delivering stimulations through a YAP-laser at a fixed intensity above the pain threshold, we were able to examine the latencies of activation along the postero-anterior axis of the insular cortex (see Fig. 32.3; [67]). Stimulations were mostly perceived as painful, but in a minority of situations, it was perceived as nonpainful. As expected, in the parietal operculum (S2) and posterior insula, the BOLD response was significantly more intense as the perception of the stimulation was painful suggesting that this portion of the operculo-insular cortex truly encode for the physical intensity of stimuli. This encoding will determine the ensuing perception. Conversely, the intensity of the response was no more discriminant in the mid and anterior insular cortices with a delayed response as compared to the first one. An interesting finding suggesting different sequential treatments (in time) was that the latency of the anterior insular response was shorter when the sensation was painful as compared to the nonpainful sensation. Fast processing as compared to slower ones may indicate (in the second order processes), a priority organization as the stimulation is considered as harmful or threatening. An independent study performed in our laboratory with intra-cerebral recordings replicated these findings [32]. They demonstrated that after a laser stimulation, responses were sequential in time with early and high amplitude of response in the posterior insula and a later response when the recording was anterior. Functional coherence levels with the posterior insula increased with time in a large majority of other cortical areas and irrespective of time window, the largest spectral coherence values with the posterior insula were observed in the anterior insula, parietal operculum, and perigenual anterior cingulate cortex [9]. This pattern was suggestive of an early treatment of sensory information (input) in the parietal operculum-S2-posterior insula and secondary order processes within the anterior insular subdivisions. The “neurological pain signature” predicting physical acute pain, described by Wager et al. [82], largely overlaps with this brain activity pattern.

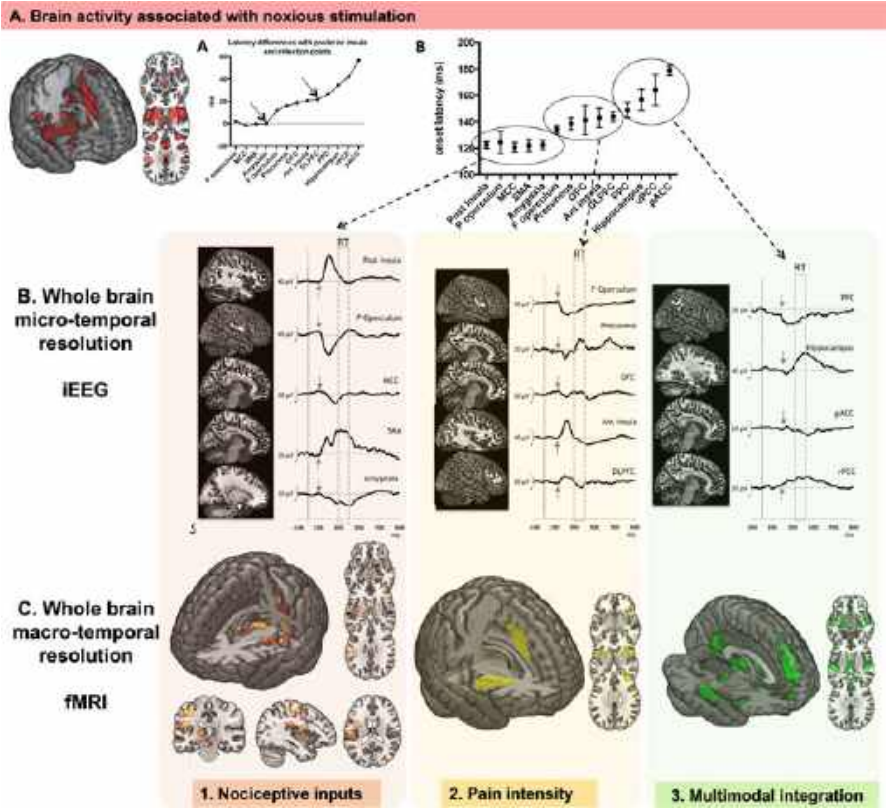


Fig. 32.2 Deconstructing brain processing of pain, from large-scale brain activity to local modules in 3 steps. (a) Top: brain activation pattern associated with pain stimulations in functional magnetic resonance imaging (fMRI) and onset latencies of nociceptive responses recorded with intracranial electrophysiology (iEEG). (b) Middle: micro-temporal dynamic of 14 brain regions recorded in iEEG, the onset latencies of the nociceptive responses are illustrated (Reproduced and edited by permission from Ref. [9]). Latency differences clearly distinguish 3 groups relative to that of the posterior insula (i.e., 110–130 ms; 130–150 ms, and 150–180 ms onset latency). The small arrows point to the mean onset of the responses and the areas delineated with dotted lines correspond to time windows during which motor reaction occurred (RT). (c) Bottom: macro-temporal dynamic in fMRI of brain activity related to pain stimulus, the first brain recipients of noxious inputs to high order and multimodal fronto-parietal brain regions are illustrated (see for details [24]). Split the BOLD activity related to the pain experience in three acts allow to illustrate that almost the same set of regions shape the three successive waves of activations measure in iEEG

Functional networks are scale free that implies that there is always a small but finite number of brain regions having broad access to most other brain regions. The anterior insula is a crucial hub in the pain connectome and may participate to inform higher-order networks (i.e., fronto-parietal) on the physical-somatic nature of the stimulus [10]. The high “centrality” of the anterior insula, reflecting its control over passing information, is consistent with a role in orchestrating potentially necessary

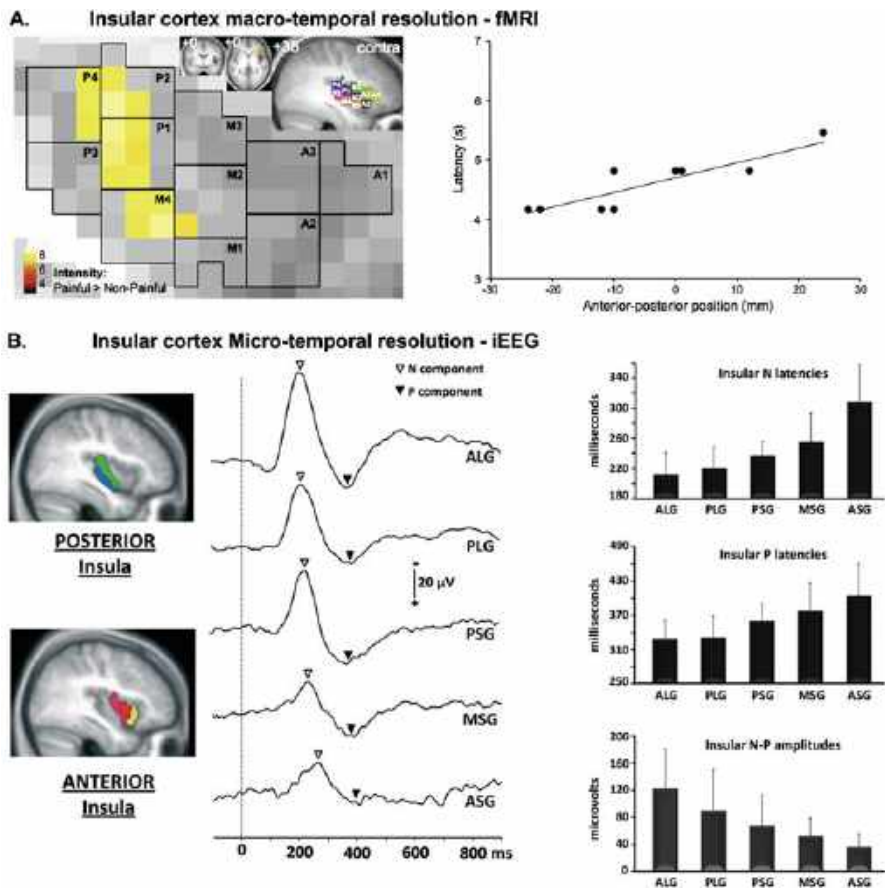


Fig. 32.3 Posterior to anterior insular cortex responses to noxious laser stimulations, as recorded with (Top) high-resolution fMRI (Reproduced and edited by permission from Ref. [67]) or with (Bottom) deep electrodes implanted into the brain (Reproduced and edited by permission from Ref. [32]). (a) Responses with shorter latency are in the posterior insular cortex and that its amplitude predicts painful perception. Correlation between latencies and anterior–posterior position shows that the haemodynamic response to painful stimuli peaks earlier in the posterior insula than in the anterior insula on the side contralateral to laser stimulation. In the anterior insular subdivisions, latencies of the response are shorter for stimulations that will be interpreted as painful while latencies are significantly delayed for responses that will be interpreted as innocuous. (b) Direct recordings show that insular responses have the shortest latency in the posterior subdivisions, while the anterior subdivisions have longer latencies. These two independent studies provide similar conclusions with a posterior to anterior gradient of response latencies and emphasize the fractal similarity of brain responses to pain in this cortex

changes in the perception of pain intensity and the concomitant behavioral responses [13].

32.4 Second-Level Attentional Control and Cross-Modality Sensory Integration

The anterior mid-cingulate cortex (aMCC) is also consistently activated as stimulations reach a nociceptive level with an activity range that may be proportional to the intensity of perceived pain. As the anterior insula, the aMCC is a multimodal structure that may be activated in a similar manner as a volunteer is in pain or is doing a cognitive task requiring attention. Since the more you have pain, the more you load attention to this stimulation, the functional significance of aMCC activations has moved from a strictly nociceptive one to a confound involving associated processes such as oriented attention [64, 65]. Interestingly, the same pattern of information flow could also be demonstrated in the aMCC where high temporal resolution fMRI showed earlier responses when the stimulus is perceived as painful compared to a nonpainful one. This pattern of response suggests that after identification of what happens in the parietal operculum-posterior-insula, second-order processes may occur in the anterior insula and aMCC with a priority transmission linked to time as a function of emergency or threatening. fMRI and iEEG findings add a new temporal dimension to the nociceptive responses. By merging information obtained by both techniques, it is possible to demonstrate that a faithful decoding of noxious laser stimulation can be implemented and transformed into pain sensation by changes in functional connectivity between affective areas such as the amygdala and the anterior insula [37].

Nociceptive inputs reach the amygdala through a low-speed tract (the spino-parabrachial-amygdalar pathways [12] leading to responses with a different temporal shape than other cortical regions (see Fig. 32.2 in [34]. At long-time scale in fMRI, this led to an isolated limbic module, made up of the amygdala-hippocampus couple (see Fig. 32.1). This module has low connectivity with other brain regions of the dynamic pain connectome, and similarly, this functional independence was also detected in iEEG data, which showed no significant spectral coherence between the amygdala and regions of the ascending pain pathway [10]. Activity within the limbic brain matrix is engaged early and is believed to be involved in the encoding and retrieval of emotionally charged memories associated with pain perception, while communications with other networks are dedicated to multimodal brain hubs such the anterior insula [25].

32.5 Third-Level Conscious Pain and Cognitive Control

Conscious experiences are not confined to sensory areas but rather represented over large, interconnected portions of the cortex [6, 58]. The second and third modules involved brain regions of higher order in the hierarchy of noxious stimulus processing, i.e., associative, and multimodal areas with longer response latencies than the regions described above (Figs. 32.1 and 32.2).

These include the central executive network anchored in the dorsolateral prefrontal cortex (DLPFC) and posterior parietal cortex. This network contributes to many neurocognitive functions including decision-making and working memory [18]. The default mode network is also recruited and comprises the precuneus/posterior cingulate cortex and the ventro-medial prefrontal cortex. It is involved in the experience of mind-wandering (attentional level) and self-consciousness [31]. Activity in these networks participates to a third-order matrix, which may to some degree reflect pain awareness and cognitive control, which could be associated with the engagement of descending pain modulatory systems to dampen or inhibit pain [19].

In iEEG recordings, activity in this module persists after the stimulus has produced an overt motor or verbal response [9], suggesting a role in both the consolidation of immediate perceptions and the control of adequate behavioral reactions. Communication between the different subsystems is supported by “network connector hubs” detected in the graph model and located in the anterior insula, DLPFC, and PPC. Such connectors play a key role in network organization by virtue of their control over information passing between other brain regions [42], but are also, logically, loci of vulnerability in brain disorders [79]. The results in the pain field are therefore consistent with a general framework which, whatever the sensory modality considered, links the conscious availability of sensory information with the enhancement of the functional coupling between sensory areas and high-order fronto-parietal networks.

32.6 Pre-stimulus Brain State Influence How Pain Is Perceived

Pain neurosciences studies are mainly focused on the brain activity associated with the noxious stimuli, but it is important to also consider how ongoing activity in the prestimulus state may influence subsequent pain perception [47]. Spontaneous brain activity is behaviorally relevant, as it contains predictive features about how brain networks will respond to a wide range of sensory and cognitive events [75]. Recording of local field potentials and discharges of single neurons by electrophysiological techniques have revealed that the variability of stimulus-evoked responses is influenced by the preceding spontaneous brain activity [5].

There is also growing evidence that a wide range of ongoing neural activity in the micro-time (milliseconds) or macro-time (seconds) scales leading up to sensory stimulation strongly influences whether acute pain stimulus will be perceived or the degree of pain intensity perceived [47, 51]. Trial-to-trial brain response variability in the pain network and the magnitude of pain ratings may be dependent upon pre-stimulus: (i) BOLD signal within third-order networks including sensorimotor and default mode networks, (ii) the spectral power of the ongoing EEG alpha and gamma oscillations [51, 78], and (iii) functional connectivity between the

anterior insular cortex and regions of descending pain control pathway [37, 66, 83]. These pre-stimulus neural fluctuations are reproduced at different spatial and temporal scales, but their impacts on conscious pain processes are still unclear. Bastuji et al. [11] show that the probability for a phasic noxious stimulus to entail an arousal is modulated by the prestimulus interareal phase coherence between sensory and higher level cortical areas. Fluctuations in interareal communication immediately before the noxious stimulus may determine the responsiveness to incoming input by facilitating or preventing the transfer of noxious information from sensory to multiple higher-level cortical networks [11]. Ongoing activity must play an important role in cortical function and cannot be ignored when investigating cognitive processes influencing pain experience.

32.7 Large-Scale Characterization of Brain Activity Can Predict Evoked Pain Intensity

Information about pain intensity is coded in the brain at multiple anatomical scales: locally, distributed across regions and networks, and globally [46, 49]. There is a localization of specific function in a single region, but also a distributed character for multidimensional processing [35].

As described above, neuroanatomy and direct electrical stimulation of a single neuronal unit have pointed out the dorsal posterior insula as subserving a fundamental role in human pain [54, 55, 71], whereas the distributed models of pain in the brain are supported by neuroimaging studies and the concepts named ‘pain brain networks’ [25], ‘neurosignature of pain’ [82], or the ‘dynamic pain connectome’ [46].

What is the gain of looking at the entire brain rather than 1 region? Using multi-study analysis and multivariate predictive model applied on BOLD fMRI data, Petre et al. [61] show that all spatial scales conveyed information about pain intensity. Individual regions have a lot of useful information: 80% of them. It makes sense that some regions seem important, and many predict well, but distributed, multisystem models predicted pain 20% more accurately than any individual region or network and are more generalizable to multimodal pain (thermal, visceral, and mechanical) and more specific to pain. However, these findings demonstrated that most of the information conveyed by a single brain region is also embedded in large-scale brain networks, illustrating the fractal features of the spatiotemporal organization of the brain during pain.

32.8 Disruption of Brain Fractal Patterns in Chronic Pain

Chronic pain is the most prevalent disease worldwide, leading to substantial disability and enormous societal costs [17]. Evidence supporting structural and

functional central nervous system changes in patients with chronic pain is mounting and supported by systematic reviews [4, 20, 35, 45, 62] and meta-analyses [74, 87].

One of the aspects of the chronic pain brain most explored with fMRI is functional activity at rest so-called resting-state networks (RSNs). It can capture the macroscopic spatial dynamics of the blood oxygenation level dependent (BOLD) signal of the brain, and its synchronization (functional connectivity) between different brain regions is the basis for forming networks [31]. Several neuroimaging studies have detected alterations in the activity and connectivity of RSNs in chronic pain patients compared to healthy controls [7, 20, 26]. Moreover, technologies that can examine brain activity at millisecond resolution (i.e., electroencephalography and magnetoencephalography (MEG)) have reported that people with chronic pain exhibit altered neuronal oscillations including an altered resting-state alpha-band [26, 86]. Specifically, brain activity dysfunctions at rest (spatially, spectrally, and temporally) can be more pronounced in several chronic pain conditions that have a neuropathic pain component. Because scale-free complex networks are known to show resistance to failure, facility synchronization, and fast signal processing, it would be important to see whether brain networks scaling properties are altered under various chronic pain conditions. To identify nonperiodic irregularities in the functional activity of RSNs associated with chronic neuropathic pain conditions, few studies have computed fractal dimension measures.

In an original work, Foss et al. [30] were able to differentiate between different types of chronic pain conditions using FD values to characterize the variability of spontaneous pain ratings and brain activity recorded in fMRI. They showed a correspondence between FD for ratings and for brain signal and found FD to be lowest for thermal pain and greatest for chronic pain [30]. In the same vein, Porcaro et al. [68] illustrated that FD values within frontal networks (dorsal attention system and anterior part of the default mode network) were higher, and lower within the posterior part of the default-mode network in patients with chronic migraine compared with healthy controls [68]. In electrophysiology, Tran et al. [77] studied nonoscillatory EEG markers for neuropathic pain and the effects of virtual reality interventions using fractal dimension. Using Higuchi's FD, they found significant increases in FD for the central and parietal regions during the virtual reality task [77]. These changes in neural signals have been interpreted as a normalization of the affected subcortical-cortical system following intervention. A previous work of Anderson et al. [2] found that spinal cord injury participants with neuropathic pain to have lower fractal dimension of EEG signals compared to healthy individuals and may be used to classify patients [2]. In sum, these studies illustrated that the fractal dimension of the EEG signals is lower for people suffering from chronic pain than for healthy individuals.

Higher FD values generally correspond to higher signal complexity, and a reduction in FD may indicate a loss of neural efficiency, as previously found in people suffering from brain disorders compared to healthy people [44, 70]. This loss of complexity in brain activity may lead to the brain networks becoming less flexible and efficient [2, 90] in processing and controlling sensory information. These findings illustrate the importance of nonlinear nonoscillatory brain features,

specifically Higuchi FD, which may be used as new prognostic neuromarkers to analyse the brain activity of people with chronic pain. It may help to characterize the microstructure complexity of brain pain networks and potentially differentiate between chronic pain conditions, and thus may have mechanistic and clinical utility.

32.8.1 *Fractal Dimension of Temporally Recurrent Brain Microstates*

Recognition that brain time series contains “hidden information” that might be captured by nonlinear methods such as fractal analysis provides crucial and, so far, overlooked neurophysiological information in pathological conditions such as chronic pain. Hidden Markov Model (HMM) is similar to fractal analysis since it is a measure of complexity based on the fact that a simple process that is repeated endlessly becomes a very complex process, which is the basis for the description of fractals in nature. The HMM approach is used to reduce cortical activity at rest into sequences of transient, intermittently reoccurring events, known as brain microstates [80, 81]. The brain signal is considered as brief bouts of activity that are repeated intermittently (i.e., nonrhythmic; see Fig. 32.4A).

This approach applied to neuroimaging data was successfully used to characterize profiles of time-varying connectivity associated with opioid analgesia [60]. Using MEG data, this method provided a fine probabilistic estimation of when the different brain states are active [85]. We recently applied it to MEG data from individuals with neuropathic pain to describe how their brain moves from one activity state to another [28]. Our main finding is that neuropathic pain impairs the dynamic coordination of neural network activity. Compared to healthy individuals, the neuropathic pain patients exhibited shorter time intervals between transitions to (i.e., visits of) the microstate characterized by activations of sensorimotor brain regions and their brains spent more active time in this state. Moreover, in the beta frequency band, patients had higher power and coherence in the sensorimotor state. Conversely, their brains showed lower coherence and spent less time in the frontal attentional state. We hypothesized that these findings illustrate an imbalance of temporal dynamics between sensorimotor and frontal microstates in neuropathic pain that may contribute to abnormal pain regulation in these patients. Reorganization of the sensorimotor subsystem (SMN) has been reported in numerous studies, such as abnormal activity and hyper-connectivity of the SMN and altered hub topology in chronic pain [27]. Abnormal SMN beta and alpha power activities were also found and could be due to cortical plasticity after a lengthy period of nociceptive input [89], or it could also serve as a compensatory mechanism to maintain sensory perception. Although the neurophysiological meaning behind these discrepancies is unclear, it illustrates that neuropathic pain has an impact on the overall dynamic orchestration of neural activity that is replicated at multiscale (spatial, topological, and temporal) in the brain (Fig. 32.4b).

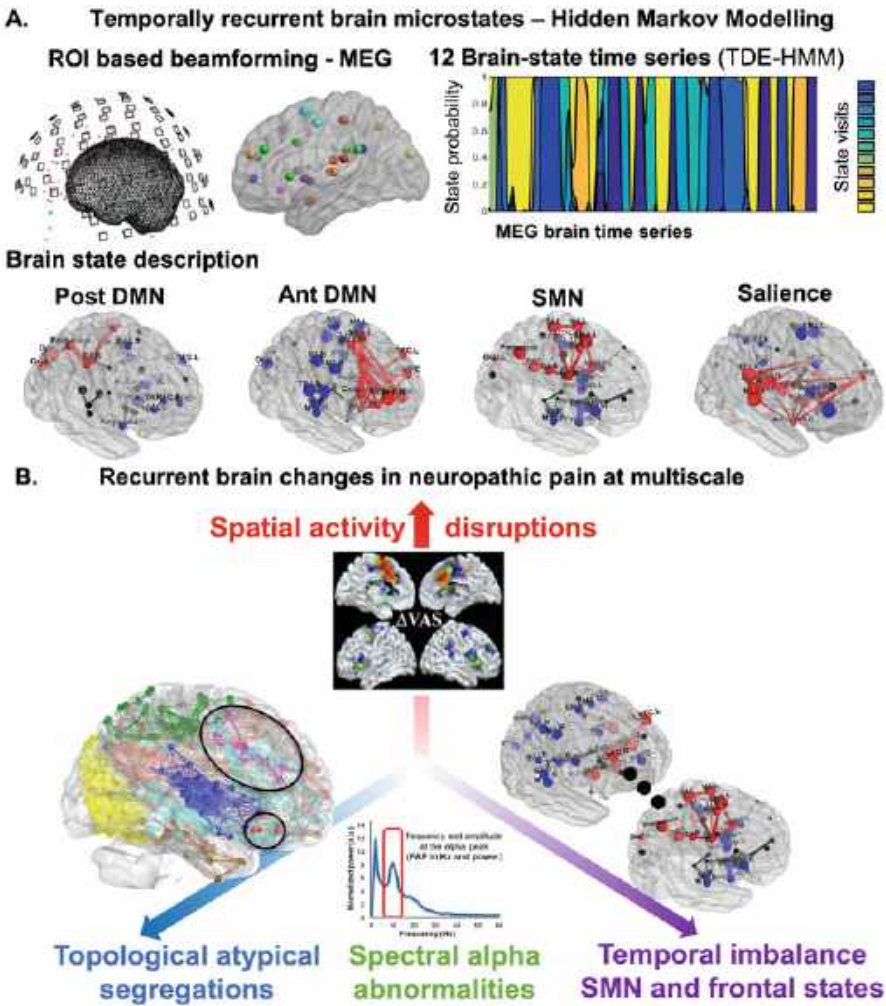


Fig. 32.4 Example of brain activity patterns repeated in time: fractal dynamics. (a) Illustration of TDE-Hidden Markov Model approach applied on MEG data. HMM inference is used to identify the state time courses (state probability). Each state is characterized as having its own distinct spatial, temporal, and spectral properties. Brain regions with significant power and phase coupling are represented across a wideband frequency range (1–30 Hz) for 4 estimated states. Node spectral power is relative to the temporal average (blue and red colors reflect power that is lower or higher than the average over states, respectively). Edges between nodes show functional connectivity. (b) Altogether recent findings illustrate that neuropathic pain alters neuronal activity repeatedly at several scales in the brain (see [28]). Locally in several brain regions (including the primary somatosensory cortex and the anterior insula); topologically in the network organization with higher segregation and cross-network connectivity; spectrally in the alpha band; and temporally by changing the temporal dynamic of brain microstates

32.9 Conclusion

Analysis of brain activity related to noxious stimuli over a scale of seconds or minutes reveals that pain fluctuates as a scale-free process. Using both large-scale brain imaging in healthy individuals and in those suffering from chronic pain, and direct neuronal measurements in a clinical population, allows to describe how the brain networks of pain are hierarchically organized in both space and time. Description of this fractal-like organization provides clues about how our brain regions work together to create the perception of pain and how pain becomes chronic when its organization is altered.

The modular organization reported in fMRI studies appears relevant to shape the dynamic patterns of responses to pain stimuli determined at short time scale using iEEG and are, therefore, consistent with the fractal properties of brain connectivity being reproduced at short and long time scales [59]. The results are consistent with a model of nociceptive integration whereby the conscious experience of pain emerges from the dynamic cooperation, segregation, and integration of multiple functional subsystems. It can help to ultimately understand how pain function is simultaneously constrained by both local and global features of cortical organization.

Findings using fractal analysis and brain microstates decomposition highlight the brain network reorganizations occurring in patients with chronic pain. At the individual level, studies also found a certain variability, showing that the neuropathological state differs among patients. This approach to brain dynamics in chronic pain can reveal novel information with potential mechanistic and clinical significance. It could serve as an individual blueprint to characterize the mechanisms through which local changes of neural function impact spatially distant sites, and therefore could contribute to the development of personalized treatment.

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Competing Interests The authors declare no competing interests.

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Chapter 33

Fractal Neurodynamics



Karolina Armonaite, Livio Conti, and Franca Tecchio

Abstract The neuronal ongoing electrical activity in the brain network, the neurodynamics, reflects the structure and functionality of generating neuronal pools. The activity of neurons due to their excitatory and inhibitory projections is associated with specific brain functions. Here, the purpose was to investigate if the local ongoing electrical activity exhibits its characteristic spectral and fractal features in wakefulness and sleep across and within subjects. Moreover, we aimed to show that measures typical of complex systems catch physiological features missed by linear spectral analyses. For this study, we concentrated on the evaluation of the power spectral density (PSD) and Higuchi fractal dimension (HFD) measures. Relevant clinical impact of the specific features of neurodynamics identification stands primarily in the potential of classifying cortical parcels according to their neurodynamics as well as enhancing the effectiveness of neuromodulation interventions to cure symptoms secondary to neuronal activity unbalances.

Keywords Neurodynamics · Power spectral density · Power law · Higuchi fractal dimension · Local signature

33.1 EEG-Derived Neurodynamics Assessment in the Human Brain

The statistical properties of electrical signals produced in the brain are of great interest as they contain a harvest of information that is not yet fully understood, show specificity of different source areas and exhibit unpredictability due to their

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irregular behaviours and non-stationarity of the underlying generating mechanisms. These processes characterize the complex internal brain activities when neuronal assemblies are synchronously coupled and switch between the states during brain functioning instead of external stimuli [14]. Other researchers prefer to regard the brain as a dynamical system, since majority of neuronal assemblies are excitable non-linear sub-systems possessing self-organization and power-law scaling properties [4, 10, 31, 35].

There are several claims that the brain's electrical ongoing activity is deemed to be scale-free if its power spectrum follows power law distribution $P(f) \sim 1/f^\beta$ [19, 31, 58], which is also a prerequisite for fractality [28]. If the brain oscillations have been well studied up to date, the non-oscillatory components are still a topic for discussion, and the methods for interpretation as well as the robust estimation of the scale-free neurodynamics are still being studied [20].

Nevertheless, evidence suggests that fractal properties of neurodynamics are handy to evaluate the complexity of implied processes and can have a direct application in clinics. More precisely, the fractal measures have been useful to distinguish consciousness from unconscious states, e.g. sleep stages and depth of anaesthesia [6, 18, 23, 49], fractal signatures can be also used as a biomarker in detecting senile dementia [5, 51] and are claimed to be a better predictor of schizophrenia than neural spectral oscillations [45]. In addition, fractal properties of EEG on the scalp and in intracranial recordings have been investigated to seek a signature of different cortical areas [7, 8, 17].

In this chapter, we will attempt to describe the local cortical neurodynamics via spectral and fractal analyses of electrophysiological time series and discuss their limits. The aim is to highlight the existence of typical features in distinct cortical parcels and that these properties are maintained even during sleep/wake stages. Moreover, cited studies confirmed the hypothesis that the measures of system complexity can better detect the specificities of the ongoing electrical activity than the linear ones such as the spectral analysis.

33.2 Neurodynamics as Local Cortical Signature: A Spectral Estimate

The first classification of the brain started with Korbinian Brodmann, who suggested the anatomical parcelling of the cortex based on the cytology of the constituent neurons [9]. The proof of different brain structural properties raised the notion to study their function separately and the functional connectivity between them [54].

While great interest has been given to studying the task-related behaviours, up to date, the possible brain parcellation based on intrinsic characteristics of the neuroelectric activity during resting states was barely investigated. The attempt to find a 'fingerprint' of distinct brain areas in EEG recordings was suggested by clustering them on the basis of distinct spectral profiles of their electrical activity.

Keitel and Gross [36] demonstrated that each brain area engages in different spectral behaviour that is characteristic of individual areas.

Cottone et al. [17] deployed the power spectrum measure to identify the local neurodynamics of the primary motor cortex (M1) and primary somatosensory cortex (S1) from EEG recordings on the scalp of 20 subjects, in both hemispheres, during resting wakefulness with eyes open and eyes closed, sensory stimulation and performing simple motor functions. The authors have shown that the neurodynamics' spectral indices displayed a clear and different pattern for S1 and M1 areas. They did not notice a distinction between the left and right side; however, all subjects were showing clear alpha (8–12 Hz) frequency band activity in S1 and high beta (26–33 Hz) and gamma (33–80 Hz) activity in M1.

More recently, the investigation of spectral properties of different brain regions was published by Montreal Neurological Institute (MNI) using stereotactic EEG recordings [25] demonstrating that different brain areas in resting wakefulness possess a characteristic activity. The authors have shown that alpha rhythm is found in the occipital lobe, the parietal lobe and temporal lobes. Beta rhythm is frequently found in the anterior head regions. In particular, well-sustained beta frequencies are described in the precentral and postcentral gyri with peaks at lower frequencies present in the postcentral regions. The precentral gyrus also expresses gamma activity. Moreover, beta peaks are found in the middle cingulate gyrus and anterior insula [25].

Following Cottone et al. [17] and Frauscher et al. [25] results, another attempt to analyse the typical spectral features was made by Armonaite et al. [7]. The authors studied three primary cortices: auditory (A1), somatosensory (S1) and motor (M1) from MNI intracranial stereotactic EEG recordings (sEEG). The distribution of the electrode contacts across three brain parcels of interest is shown in Fig. 33.1 where a single channel per subject per area has been selected (representative channel). The authors analysed the local neurodynamics characteristic spectral features, initially, within a subject and later across the population.

33.2.1 *Spectral Features in Resting Wakefulness*

The purpose of the Armonaite et al. [7] study was to deepen knowledge of the brain neurodynamics as cortical area signature. For extracting the spectral properties of the signals, authors calculated the power spectral density (PSD) of each channel in the three investigated regions during resting wakefulness (Fig. 33.2). The spectrum was categorized into 7 frequency bands: delta (≤ 4 Hz), theta (4–8 Hz), alpha (8–12 Hz), low beta (12–26 Hz), high beta (26–33 Hz), low gamma (33–49 Hz) and high gamma (49–80 Hz).

Initially, the authors compared the PSD in the three regions of interest within a subject and then averaged across the population. In Fig. 33.3, the scatterplots of the PSD evaluated in each frequency band, for each cortical region separately in each subject, are compared between a couple of areas (M1 vs S1, M1 vs A1

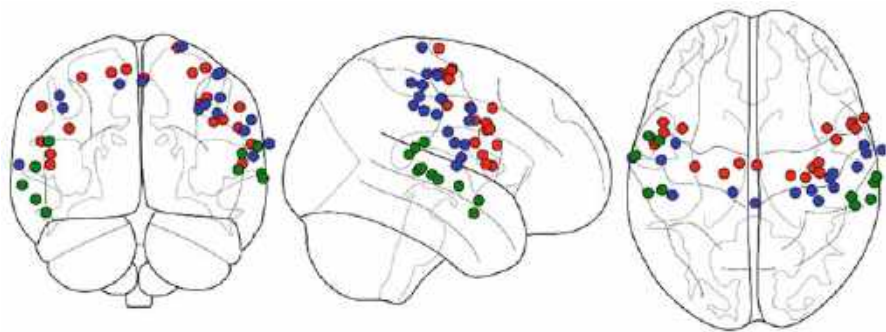


Fig. 33.1 The distribution of representative channels in A1, S1 and M1. After Armonaite et al. [7]. After a representative channel selection for each subject, the number of subjects that followed the constrain of the Armonaite et al. [7] study was reduced to 10 subjects in A1 (green), in S1 (blue) to 17 subjects and in M1 (red) to 20 subjects. The location of the selected representative channel for each subject in each area is shown on the cortex image in three projections

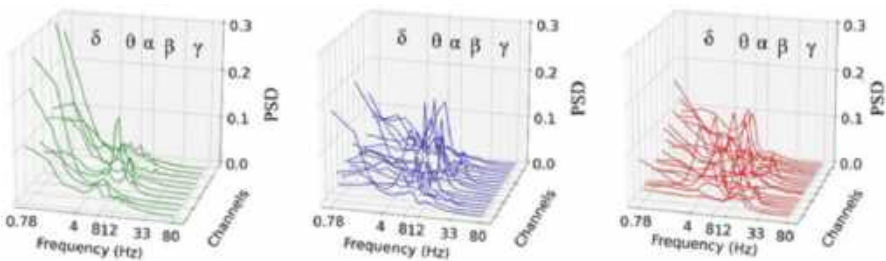


Fig. 33.2 The PSD for each channel in A1, S1 and M1. After Armonaite et al. [7]. The normalized PSD was evaluated for each channel in each region. To spot the trend at lower frequencies, the x-axis is represented on the logarithmic scale. From the left: A1, (10 channels) PSD expresses the highest power in the delta band; S1, (17 channels) PSD exhibited higher power in theta and alpha; M1, (20 channels) PSD expressed high power in theta, alpha, beta and gamma frequency bands

and S1 vs A1) in the wake. Only scatterplots for the frequency bands where the two sources statistically differ are presented. The bisector of the rectangular plot areas, divides the upper and lower regions where the PSD of the compared cortical parcels dominates. It is evident that the local neurodynamics evaluated in each subject appeared specific for the cortical district in terms of PSD band values. S1 consistently shows greater power than M1, evaluated with Wilcoxon nonparametric test (W_{test}), in alpha band [$W_{\text{test}} = 26, p = 0.03$] and smaller in high beta [$W_{\text{test}} = 25, p = 0.03$], low gamma [$W_{\text{test}} = 15, p = 0.01$] and high gamma [$W_{\text{test}} = 28, p = 0.04$]. A1 has power in delta frequency band greater than M1 (9 out of 9 subjects) while smaller in low beta (7/9), high beta (7/9) and low gamma (7/9) bands. A1 steadily shows greater power than S1 in delta band (6/6) and smaller in low (5/6) and high beta (4/6) bands.

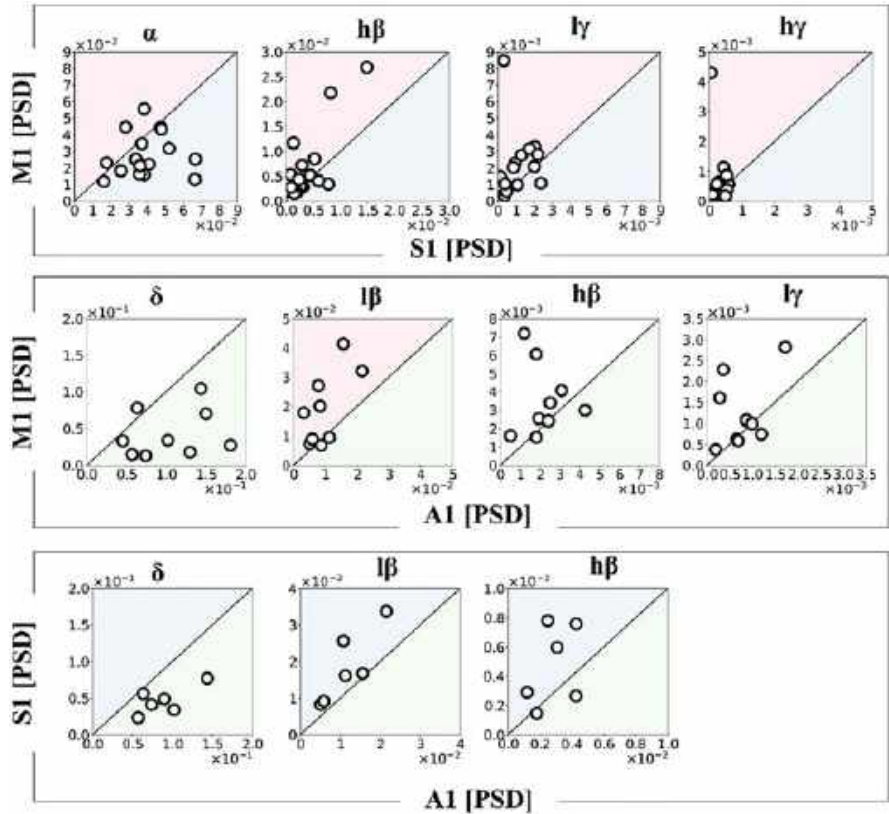


Fig. 33.3 Local neurodynamics comparisons in frequency bands. After Armonait et al. [7]. Comparing the neurodynamics between couples of the studied brain areas, the scatterplots of the normalized PSD for each subject are shown for the frequency bands where the PSD of two sources statistically differ. **Top panel:** M1 versus S1 [16 subjects] in 4 bands. **Middle panel:** M1 versus A1 [9 subjects] in 4 bands. **Bottom panel:** S1 versus A1 [6 subjects] in 3 bands. In each plot, a point above (below) the diagonal has a PSD of the source represented on x-axis lower (higher) than that of the source shown on y-axis. The Greek letter stands for the frequency bands, where the l/h prefix indicates low/high range, respectively

Further, the PSD – averaged across all channels – shows specific features between cortical areas also at the population level. The PSD in delta band is prevalent for A1, in alpha for S1 and in high beta and gamma for M1 (Fig. 33.4).

33.2.2 Spectral Features During Sleep

There is a large amount of literature suggesting that the statistical features of the brain activity measured via electrophysiological tools can vary across sleep

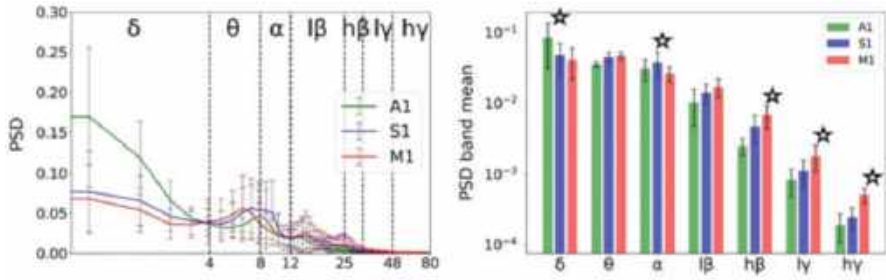


Fig. 33.4 PSD across population. Modified from Armonaitė et al. [7]. In the left panel, the normalized PSD averaged across representative channels of each subject is given for A1 (green), S1 (blue) and M1 (red). In the right panel, the bars indicate PSD mean on each frequency band, always averaged over representative channels, where the error bar is the standard deviation of the frequency band for all 7 frequency bands. The stars indicate those bands where the highest PSD has been found across regions, i.e. δ prevailing in A1; α in S1 and from $h\beta$ in M1. To enhance the higher frequencies, we show y-axis on logarithmic scale

stages [11, 15, 18, 44]; however, there is a lack of evidence that the differences of neurodynamics of separate brain areas can maintain between in the diverse sleep stages.

The variations of oscillatory frequency components were well described by Von Ellenrieder et al. [52] across regions during sleep from sEEG data. They found less heterogeneity of different regions in N2 and N3 sleep stages with respect to the variability of frequency components across different brain regions during wakefulness and rapid eye movement (REM) sleep stage. They also observed that, across different sleep stages, the clear peaks were prevalent more in the limbic system and in mesial visual and motor cortices, whereas more homogeneity was found in the rest of the investigated regions.

Armonaitė et al. [8] further studied the neurodynamics in three sleep stages (N2, N3 and REM) of primary motor, somatosensory and auditory cortices, by measuring their properties via power spectral densities, in order to assess if the local neurodynamics of investigated cortical regions maintains its' spectral features and differentiation not only in wake, but also across different sleep stages both within a subject and across population.

It was observed that within a subject in REM sleep M1 had greater power than S1 in low beta, high beta and low gamma bands [$W_{\text{test}} = 11$, $p = 0.05$, $W_{\text{test}} = 4$, $p = 0.01$, $W_{\text{test}} = 5$, $p = 0.01$], respectively. It was also found that M1 had greater PSD than S1 in alpha band [$W_{\text{test}} = 24$, $p = 0.07$] in N2 sleep stage. Likewise, also in N3 stage alpha waves were prevailing more in M1 than in S1 with [$W_{\text{test}} = 18$, $p = 0.03$] [8]. However, across population, systematic cortical area differences were not observed unlike in wakefulness (Fig. 33.5).

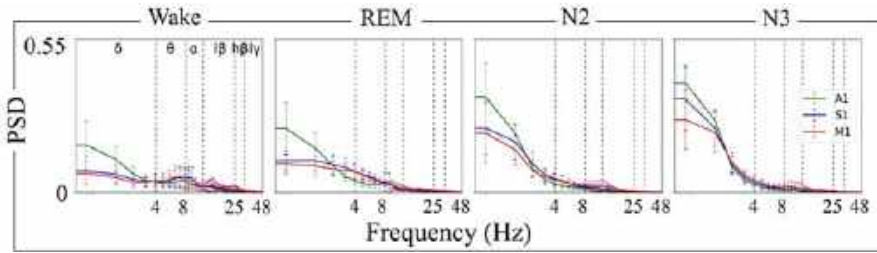


Fig. 33.5 PSD averaged across population in three regions of interest, in four sleep stages. After Armonaite et al. [8]. The mean and standard deviation of normalized PSD as a function of frequency calculated across all subjects, separately for M1, S1 and A1 areas, are presented. PSD values are normalized so that the area under each curve is equal to 1

33.3 Fractal Neurodynamics: Properties and Estimate Methods

As the large literature on the subject demonstrates, the brain's electromagnetic signals show complex properties and are still a topic for intense discussion, as they exhibit irregular behaviours that the spectral analysis might not be effective to catch. In fact, the electrophysiological signals contain not only harmonic oscillations but also noisy fluctuations that might be as informative [60]. These components are being suggested to be approached as scale-free and even fractal properties [28].

Some authors are quite sceptical about the scale-free, self-organized and fractal nature of brain neurodynamics, claiming that these noisy fluctuations occur merely due to the simple superposition of random components acting on multiple time scales [30]. Other researchers support also the idea that the observed power law of the power spectrum, so-called $1/f^\beta$ noise, would be an artefact due to many averaged narrow-band periodic oscillations of different amplitudes and frequencies [13], or could be due to the rapid exponential rise and slow exponential decay of dendritic response to an impulse input, possibly convolved with Poisson process pulses [26]. Moreover, Evertz et al., in a recent paper suggested a model according to the $1/f^\beta$ property would be the result of several independent perturbed damped alpha oscillations [22]. It was demonstrated that oscillatory alpha power correlates with the value of the β power law exponent of PSD at high frequencies, therefore, subjects with higher rhythmical alpha power would express higher β exponent for high frequencies [42]. We show an example of the power spectrum of a damped harmonic oscillation and its' behaviour on log-log scale in Fig. 33.6. This is a naïve, but necessary demonstration that the harmonic oscillations in the signal should be taken into consideration in order to avoid the meaningless surge of the slope while looking for the power-law trend.

On the other side, He [31] argues that rhythmic, recurring patterns of brain activity in a certain range of frequencies and arrhythmic activity with no prevailing frequency are distinct and the latter would indicate scale-free dynamics. Despite

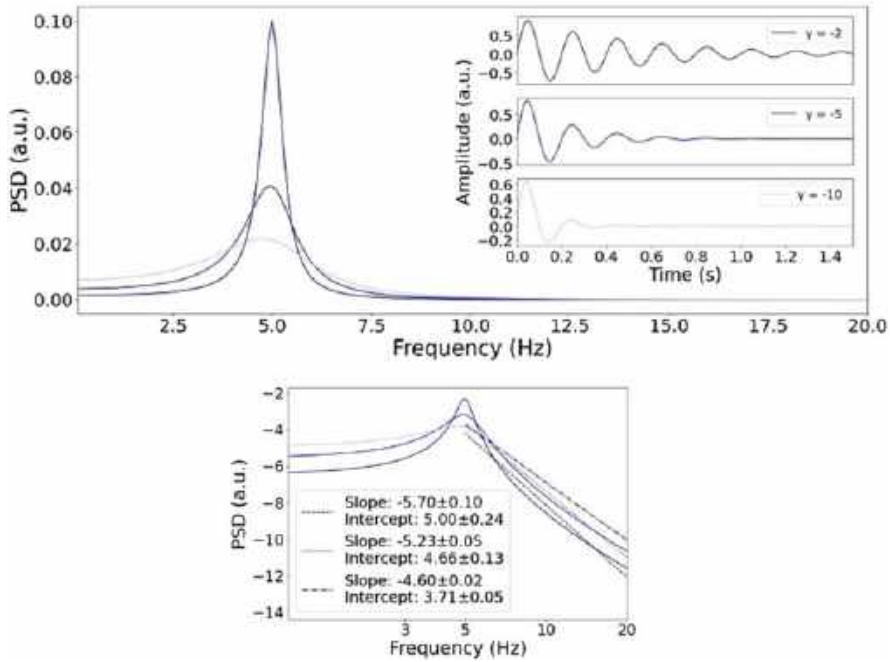


Fig. 33.6 Damped oscillator and artificial surge of a slope that can mimic a power law behaviour. In the top panel, we show the time series (top-right) and the spectrum (top-left) of a simulated periodic damped signal $y(t) = \sin(\omega t) e^{-\gamma t}$, with $\omega = 5$ Hz for three values (arbitrarily selected) of the damping parameter $\gamma = -2, -5$ and -10 Hz. In the right panel, the PSD of the signals are given in a log–log scale. The three black straight lines here indicate the best linear fit (evaluated with the least square method) of the higher frequency spectrum portion

that, the author is in line with the idea that periodic oscillations, like those prevailing in the 8–12 Hz frequencies in resting state, generate a trend in the power spectrum that can affect the estimation of the slope β of power law exponent. Therefore, before seeking the fractal nature of an EEG signal, it is necessary to incorporate an approach to identify its class and capture periodic oscillations from those that are irregular and non-periodic. Without this step, the methods deployed to estimate fractal dimension can lead to worthless estimates [21].

33.3.1 Evidence of Existing Scale-Free, Fractal Properties within EEG Signals

Nearly all complex systems express scale-free activity [12, 16, 58]. Despite the claims that $1/f^\beta$ can be accounted as a sum of damped harmonic narrow band oscillations [22], the precise patterns in their structure would require the

presence of different underlying mechanisms that generates this specific time-independent activity [32]. Yamamoto and Hughson [56, 57] suggested a method, called coarse-graining spectral analysis (CGSA), to separate physiological signals harmonic oscillations from arrhythmic activity and possibly fractal. Other authors [20] suggested similar techniques for parametrizing periodic and non-periodic oscillatory activity. Alternatively to identify fluctuations in certain time scales one can apply high/low-pass filters [48]. Independently from the method of detrending the harmonic oscillations from time series, the main goal is to obtain the underlying scale-free activity.

Therefore, $1/f^\beta$ property rather suggests a scale-free, fractal pattern within the brain. [32] have shown that after removing oscillation from the power spectrum, the power-law exponent β is different in cortical areas to another and changes during different task performances [32]. Furthermore, the authors have proved that $1/f$ scaling is not an artefact due to the instrumental noise [2], by recording an ECoG in a room without a patient. The results did not demonstrate the power-law distribution of these dummy records.

Interestingly, in the work of He et al. [32] the trend of PSD on a log-log scale was not completely linear, but instead expressed a couple of ‘shoulders’. According to their claim, this could occur due to the incomplete elimination of oscillatory components. However, we believe this could be an effect of existing different fractal regimes in the series that yield multifractality. This idea needs further investigation.

Some observations have also suggested that shared noise between two chaotic oscillators can actually induce, rather than inhibit, synchronization [59]. In addition, it was also demonstrated that uncorrelated noise can enhance coherent dynamic activity [43]. The understanding behind this phenomenon can be helpful in looking for the signature of local neurodynamics and synchronization between different cortical parcels. However, since the study of EEG recordings in the frequency domain has its own limits, one might be tempted to look if other estimation methods exist that can catch the non-linearities and fractal nature of the signals.

33.3.2 *Fractal Dimension Estimation Methods*

The PSD of a gaussian white signal will show a flat behaviour as represented in Fig. 33.7a; on the other hand, the PSD of a fractal signal at each time scale will follow a power law distribution, with a constant slope on a log–log scale, across all frequency ranges, as shown in Fig. 33.7b where the PSD of a Brownian noise signal is represented. However, the electrophysiological time series are the overlap of several signals with different features at different frequency ranges. Moreover, due to the characteristic frequencies in time domain, it becomes more complicated to point out power law properties as the PSD function on log–log scale expresses several peaks and humps as can be seen in Fig. 33.7c. For this reason, it is necessary to find a stable method to investigate the characteristic patterns in temporal scale beyond the frequency domain representation.

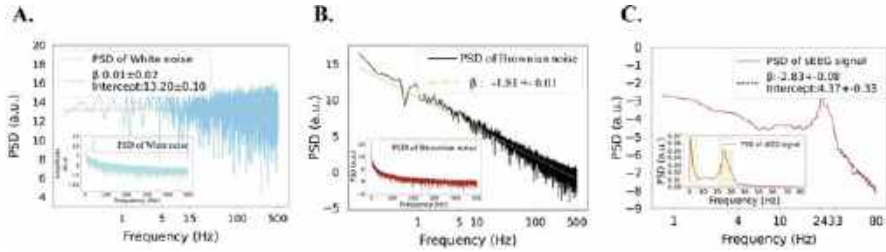


Fig. 33.7 Examples of PSD of white noise, Brownian noise and a sEEG signal. After Armonaite PhD thesis (2022) (a) PSD of an example of a typical stationary white noise is given in a double-(semi-) logarithmic scale for the main (inset) figure. The dashed yellow line indicates the best linear fit evaluated with the least square method. (b) We show the PSD of a fractal, Brownian noise, that follows power law distribution over all frequency ranges. (c) In the inset figure, the PSD of a sEEG signal from primary motor cortex (M1) with clear theta, beta and low gamma peaks is given. In the main figure, the same signal PSD on a log–log scale is plotted with a linear fit over the highest frequency range (33–80 Hz), where the power law behaviour is observed

There are plenty of methods that were used to examine the neurodynamics in healthy and with brain neural activity alterations illnesses subjects. Such as the predictability of ongoing neural activity during cognitive tasks using the Hurst exponent [33], Katz algorithm, correlation dimension and Liapunov exponent for differentiating sleep stages [1, 39], Lempel Ziv and Shannon entropy algorithm for depth of anaesthesia measure [23, 50] and Higuchi fractal dimension as a biomarker of Alzheimer’s disease [5, 51]. An attempt to classify cortical areas via the Higuchi fractal dimension is given by [7, 17].

33.3.2.1 Higuchi Fractal Dimension (HFD)

Higuchi fractal dimension (HFD) is considered to be a highly sensitive measure in the detection of information contained in physiological time series [38, 46]. Several authors also claimed that this method has a superiority over other fractal dimension estimation methods [3, 37, 47].

It has been shown that HFD is well assessed in diverse research areas such as analyzing heart rate variability [29], characterizing primary waves in seismograms [27], studying magnetic energy dissipation [55], volcanoes magnetic field [24], geomagnetic field [40] and other electromagnetic physical systems [41].

The reason why HFD would be superior to other fractal dimension estimates is because it is calculated directly in time domain, it can catch signal specificities in short time windows [34] and would be less sensitive to the signals that are not well pre-processed [38].

The method to calculate HFD is as follows.

Consider given N -length time series $\{X(1), X(2), X(3), \dots, X(N)\}$. For a time interval, we skip a number of samples equal to k . We obtain m number of sets of

new time series $X_k^m : X(m), X(m+k), X(m+2k), \dots, X(m + [(N-m)/k])$. Then we calculate the length of the curve X_k^m as follows:

$$L_m(k) = \left[\left(\sum_{i=1}^{\text{int}(\frac{N-m}{k})} |X(m+ik) - X(m+(i-1)k)| \right) \frac{N-1}{\text{int}(\frac{N-m}{k})k} \right] \frac{1}{k}$$

The length $L(k)$ of the newly generated subseries X_k^m is evaluated by averaging the k sets of $L_m(k)$ values, as:

$$L(k) = \frac{1}{k} \sum_{m=1}^k L_m(k)$$

If limit exists for $k \rightarrow \infty$, such as $L(k) \propto k^{-\text{HFD}}$ then the curve is fractal with dimension equal to HFD.

Higuchi fractal dimension is a quantitative measure of signal complexity, and it is not independent from power law exponent calculated with the classical spectral analysis. Higuchi [34] in his seminal paper stated, that there is a direct relationship between HFD and the β parameter of the $1/f^\beta$ power spectrum that holds $\beta = 5 - 2\text{HFD}$ if $1 \leq \beta \leq 3$ and $1 \leq \text{HFD} \leq 2$.

33.4 Neurodynamics as Local Cortical Signature: A Fractal Estimate

Despite the fact that the power spectrum has been largely applied to estimate the characteristic frequencies of the local neurodynamics, it was demonstrated that the irregular fluctuations of ongoing neuronal activity can be more ‘stable’ than oscillatory ones: that is, they show less spontaneous intrasubject variability and less intersubject variability [45, 53]. In the papers of Armonaite et al. [7, 8], the authors aimed to demonstrate existing solid neurodynamical patterns across different cortical areas using Higuchi fractal dimension measure during wakefulness and three sleep stages. The purpose of these studies was to deepen knowledge of the brain neurodynamics as cortical area signature. Moving on from the successful attempt to differentiate the neurodynamics of the S1 and M1 hand representations based on noninvasive EEG [17], Armonaite et al. [7, 8] aim to strengthen the results by investigating intracranial sEEG data.

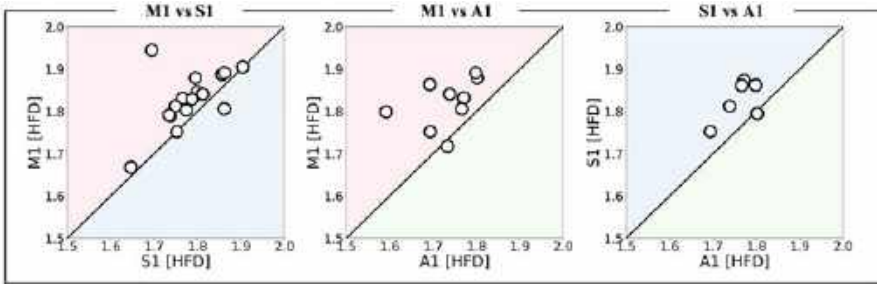


Fig. 33.8 Comparing the HFD of brain areas couples within a subject. After Armonaite et al. [8]. Scatterplots of each subject HFD in couple of areas: S1 versus M1 (16 subjects); A1 versus M1 (9 subjects) and A1 versus S1 (6 subjects). The diagonal is traced indicating where the HFD of the source represented in x is higher/lower than that in y. Colour codes as always indicate sources that are being compared: S1 blue, M1 red and A1 green

33.4.1 *Distinct Cortical Areas Exhibit Typical Complexity Relationship During Wakefulness*

Investigating the neurodynamics of S1 and M1 cortical parcels during resting wakefulness, sensory stimulation and hand motor function, the estimate of Higuchi fractal dimension shows a cortical area effect with a smaller value for S1 than M1 [17]. Following these results, Armonaite et al. [7] estimated the Higuchi fractal dimension of 21 subject sEEG signals during resting wakefulness in three cortical parcels: A1, S1 and M1. And compared Higuchi fractal dimension between the couples of cortical areas within the subject, initially, and then on average across the population. The results of this analysis are compatible with the previous ones, as it is shown that M1 expresses higher complexity than S1 and both higher than A1 with [$W_{\text{test}} = 12, p = 0.01$] (see also scatterplot of Fig. 33.8). Then the average value of HFD across the population was evaluated, and authors observed that even though the errors overlap, on average HFD for M1 is higher than S1 and A1 (Fig. 33.9).

33.4.2 *Cortical Parcels Hold Typical Fractal Characteristics During Sleep*

Since significant differences in neurodynamics across different cortical areas A1, S1 and M1 in resting wakefulness were found, further investigation by Armonaite et al. [8] was done with the aim: (i) to explore if the neurodynamics (expressed by Higuchi fractal dimension) maintain their local specificities along different sleep stages and (ii) to study if the local complexity in the investigated areas reduces in deep sleep with respect to awake and REM stages.

Fig. 33.9 Mean of HFD for the A1, S1 and M1 parcels across population. After Armonaite et al. [8]. The mean and standard deviation HFD evaluated across the population considering only representative channel for the three regions of interest. The HFD value is given for $k_{\max} = 35$

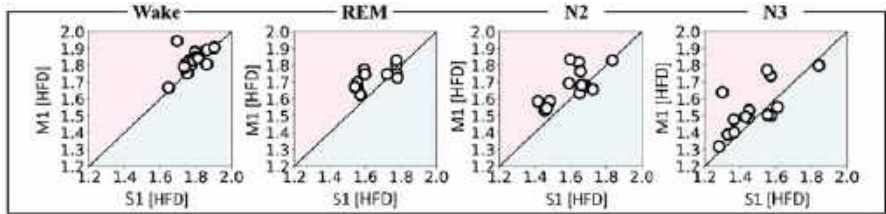
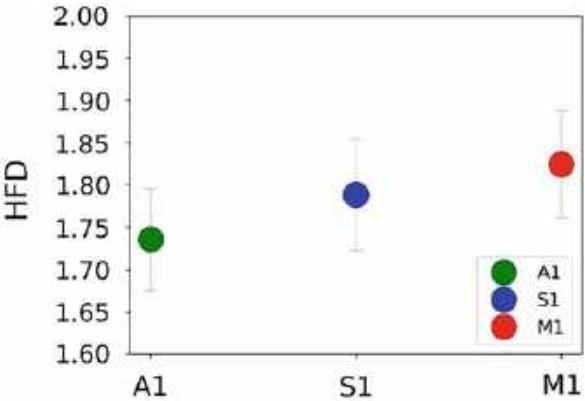


Fig. 33.10 Comparison via HFD of local neurodynamic in pairs of regions. After Armonaite et al. [8]. The scatterplots of HFD between subjects for S1 versus M1 [16 subjects]. In each plot, a point above (below) the diagonal has HFD value of the source represented on the axis lower (higher) than that of the source shown on the axis

The comparisons within the subject showed that across all sleep stages the relationship of Higuchi fractal dimension values in the three investigated cortical parcels shows stable overall behaviour with $M1 \geq S1 \geq A1$. The comparisons of the HFD values for S1 and M1 within a subject are given in Fig. 33.10.

By taking the mean value of HFD across subjects in each cortical parcel in the three sleep stages as well as wakefulness, an evident decrease of an average HFD value in each cortical parcel (from wakefulness to deeper sleep stages) can be observed (Fig. 33.11).

33.5 Conclusions

1. Evidence – from a selected group of regions – suggests that diverse cortical areas express a typical ongoing local electrical activity as their own signature.
2. Spectral estimation methods may not always reveal the underlying specificities of the signal, failing to adequately account for irregularities and arrhythmias present in the fluctuations. Although Fourier transform estimation allows linkage with current knowledge describing the specific physiological roles of different

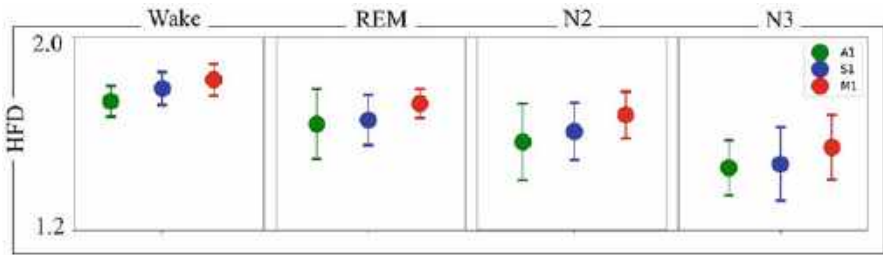


Fig. 33.11 Estimation of complexity, via HFD, of the local neurodynamics across people in wakefulness and sleep stages. After Armonaite et al. [8]. The mean HFD estimated across the population in wakefulness, REM, N2 and N3 sleep stages. For each subject, we considered a representative of all available channels within the A1, S1 and M1 regions. The Higuchi fractal dimension was evaluated at $k_{\max} = 35$ in each state

frequency domains, such estimation does not capture the complex nature of fluctuations typical of various cortical parcels.

- By comparing the neurodynamic signature measured by power spectral density (PSD) and Higuchi fractal dimension (HFD) in wakefulness and during the three sleep phases (REM, N2 and N3), a consistency between the two assessments emerged in three paradigmatic primary cortices (A1, S1 and M1), with M1 exhibiting a higher frequency activity than S1 and A1 not only during the waking state but also in the REM, N2 and N3 sleep phases, albeit in multiple frequency domains that decrease as sleep deepens. The single HFD number shows $M1 > S1 > A1$ during wakefulness and the three sleep phases, with the value of the three regions decreasing as sleep deepens.
- The nonlinear measure typical of complex systems, the fractal dimension, emerges as more stable than PSD; a ‘synthesis’ with a single value of spectral features; useful for classifying sleep stages.

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Chapter 34

Fractal Analysis of Electrophysiological Signals to Detect and Monitor Depression: What We Know So Far?



Milena Čukić, Elzbieta Olejarczyk, and Maie Bachmann

Abstract Depression is currently one of the most complicated public health problems with the rising number of patients, increasing partly due to pandemics, but also due to increased existential insecurities and complicated aetiology of disease. Besides the tsunami of mental health issues, there are limitations imposed by ambiguous clinical rules of assessment of the symptoms and obsolete and inefficient standard therapy approaches. Here we are summarizing the neuroimaging results pointing out the actual complexity of the disease and novel attempts to detect depression that are evidence-based, mostly related to electrophysiology. It is repeatedly shown that the complexity of resting-state EEG recorded in patients suffering from depression is increased compared to healthy controls. We are discussing here how that can be interpreted and what we can learn about future effective therapies. Also, there is evidence that novel options of treatment, like different modalities of electromagnetic stimulation, are successful just because they are capable of decreasing that aberrated complexity. And complexity measures extracted from electrophysiological signals of depression patients can serve as excellent features for further machine learning models in order to automatize detection. In addition, after initial detection and even selection of responders for further therapy route, it is possible to monitor the therapeutic flow for one person, which leads us to possible tailored treatment for patients suffering from depression.

Keywords Physiological complexity · Higuchi fractal dimension · Decomplexification · Stereotypy of disease · Depression · Detection · Monitoring

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34.1 Introduction: Neuroimaging Findings Related to Depression

In his 1948 ‘Cybernetics and Psychopathology’, Norbert Wiener stated that ... ‘(there is) ... *nothing surprising in considering the functional mental disorders as fundamentally diseases of the memory, of the circulating information kept by the brain in the active state, and of the long-time permeability of synapses*’, which is in line with recent research results in recurrent dynamics and functional integration of the brain. Wiener [1] called depression, paranoia, and schizophrenia ‘functional mental disorders’. The recent findings of the physiological, structural and functional mechanisms underlying depression confirmed the importance of disrupted functional connectivity within fronto-limbic system in depression [2–7].

A number of studies reported aberrant connectivity in depression. In their fMRI study of medication-free patients with major depressive disorder (MDD), Grimm et al. [8] showed the existence of hypoactivity in the left dorsolateral prefrontal cortex (DLPFC) and hyperactivity in the right DLPFC [8]. In a more recent fMRI study, Ge et al. [9] confirmed that the decreased connectivity of the right intermediate hippocampus (RIH) with the limbic regions was a distinguishing feature for treatment-resistant depression. On the other hand, several connectivity studies reported that functional connectivity exists between the subgenual anterior cingulate cortex (ACC) and medial temporal lobe (MTL) in depression, as well as in the hippocampus and amygdala [10]. Furman et al. [11] reported the aberrated frontostriatal functional connectivity in major depressive disorder (MDD), and Horn [12] reported the correlation between functional connectivity of pregenual anterior cingulate cortex (pgACC) and severity of anhedonia in MDD. Bluhm examined the resting-state default-mode network connectivity in early depression using a seed region of interest analysis [2] and confirmed decreased connectivity within the caudate nucleus. Their study showed a significantly reduced correlation between the precuneus/posterior cingulate cortex and the bilateral caudate in depression compared with controls. Berman [4] examined the connectivity of the default network specifically in the subgenual cingulate both on- and off-task and also the relationship between connectivity and rumination in MDD. Their results showed characteristic higher functional neural connectivity between the posterior cingulate cortex and subgenual cingulate cortex, but during rest periods only. Vederine et al. [3] and de Kwaasteniet et al. [13] elaborated on abnormal functional connectivity in the fronto-lymbic system. Using the combination of fMRI and functional anisotropy (FA), de Kwaasteniet confirmed that white matter integrity of the uncinate fasciculus was reduced and that functional connectivity between the subgenual ACC and MTL was enhanced in MDD. De Kwaasteniet also identified the negative correlation between uncinate fasciculus integrity and subgenual ACC functional connectivity with the bilateral hippocampus in MDD but not in healthy controls; this negative structure–function relation was positively associated with depression severity [13].

Zhang and his colleagues [5] published an fMRI/graph theory (small world) study confirming disrupted brain connectivity networks in drug-naïve first episode

MDD. It seems that MDD disrupts the global topological organization of the whole-brain networks. There are studies that emphasized the disrupted brain connectivity in mental disorders [6, 14, 15].

Lee et al. [16] tested the connectivity strength of resting state EEG as a potential biomarker of treatment response in major depressive disorder. They concluded that ‘...the stronger the connectivity strengths, the poorer the treatment response’. Their experiment also showed that frontotemporal connectivity strengths could be a potential biomarker to differentiate responders from slow responders and non-responders in MDD. Chen et al. [7] reported higher amplitude of low-frequency fluctuations (ALFF) in both the amygdala and hippocampus in participants with MDD compared to their healthy peers. Using graph theoretical analysis, they found that clustering coefficient, local efficiency and transitivity are decreased in MDD patients. In their reviews, Wayne et al. [17] and Willner et al. [18] covered almost all aspects examined in the quest of understanding the characteristic features of depression. The first one focused on structural and functional abnormalities and neurocircuitry in depression, and the second one more broadly reviewed the present and dominant approaches in this area of research. Prior to this review, Willner et al. [19] also demonstrated that antidepressants do not normalize brain activity: ‘mood and behaviour are restored to normal, but antidepressant-treated brain is in a different state from the non-depressed brain’. As a sum, all the changes found in depression indicate that the main characteristic of MDD is actually in their abnormal connectivity and transfer of information, rather than in solely physical differences. In their Granger Causality study about depression, Hamilton and colleagues questioned the importance of the functional connections between candidate regions found to be abnormal in depression [20]. Their research relied on then handful number of prior studies that yielded information about cross-structural communication and influence on depression [21, 22]. Based on previously confirmed aberrant interrelations in MDD, they applied multivariate Granger Causality to estimate the extent to which preceding neural activity in one or more seed regions predicted subsequent activity in target brain regions in the analysis of blood oxygen-level-dependent (BOLD) data.

Hamilton found that increased activity in the ventral anterior cingulate cortex (vACC) could be predicted by the activation of the hippocampus in patients with depression. In addition, the authors showed a mutual reinforcing effect between vACC and the prefrontal cortex. Further, it was found that hypoactivity of dorsal cortical regions might be predicted with vACC and hippocampal increased activity. They demonstrated that aberrant patterns of effective connectivity implicate disturbances in the mesostriatal dopamine system in depression contributing to the knowledge about the primary role of limbic inhibition of the dorsal cortex in the cortico-limbic relation [20]. It seems that many of the above-mentioned areas probably illustrate different dynamics as networks, active in specific tasks known to be characteristically different in depression.

To summarize, many other neuroimaging techniques showed that differences in MDD can be detected (MRI, fMRI, FA, DTI and PET), but due to the very high cost of so sophisticated recordings, all the above-mentioned studies usually

rely on a modest sample (less than 50) [23, 24]. Computational psychiatry, a combination of computational neuroscience methods in psychiatry, started as a movement to improve the situation, sometimes around 2011/2012 (and all the publications describing the theory-driven approach from 2013 onward). What is more, we realized that, despite the very good results of theory-driven computational psychiatry projects, they are simply not for everyone, since it is rather expensive for frequent clinical check-ups.

What is the way out of this situation and can electrophysiology (as the oldest neuroimaging technique, also the cheapest one) help? Could the complexity approach help? Is there a connection between the fractal nature of anatomical and functional phenomena in human physiology with better assessment by nonlinear analysis? Why are innovations in psychiatry so slowly accepted?

Here we will elaborate on possible solutions how to employ physiological complexity (fractal analysis in particular) as a key analytic approach.

34.2 Physiological Complexity Dovetails Neuroimaging Findings About Depression

As the founding fathers of physiological complexity noticed and described a while ago, despite the fact that we all learn Euclidean Geometry in math classes at school, all natural shapes, forms and signals have a fractal nature that we do not hear about in our educational systems. The particularly interesting application of fractal and nonlinear approach to analysis is the one in human physiology; both in anatomical and functional sense we repeatedly confirm that fractals are the language of nature. Self-similarity, a central feature of fractal forms that are following scaling laws, also applies to the electrophysiological signals, measurable in an engineering sense, that are caused by many different physiological functions: electroencephalogram, electrocardiogram, electroneurogram, seismocardiogram, gyro cardiogram, but also ECG-extracted breathing rate. What is applicable to the signals, is also applicable to the time-series extracted from those signals.

In this chapter, we will focus specifically on the analysis of time-series extracted from electrophysiological signals. In addition, in our prior work, we succeeded to apply fractal analysis for detection of depression, in particular, to major depressive disorder (MDD), the most prevalent form of the disorder. We propose a system for detection of certain features characteristic of this particular disorder extracted from the patient's electrophysiological signals, but also to monitor the treatment progress and later for potential remote patient monitoring.

In essence, we can all rely on early work in the field of physiological complexity where Goldberger elucidated the decomplexification in human physiology as a sign of ageing and disease [25, 26]. He showed that the whole business of physicians is based on the ability to recognize certain features of the disease in patients that separates them from healthy individuals. Think of Parkinson's disease patients who

all demonstrate similar clinical features, which are the basis for the diagnosis; they all share the feature completely absent in healthy people. In this case, the physiological processes become repeatable and predictable, and in essence less complex than the features of a healthy organism. One of the earliest documents pointing to the use of physiological complexity dates to 1816 when Dr. Cheyne noticed that his patients who were suffering from a particular cardiovascular disease exhibited a very unusual rhythm of breathing and heart dynamics, that can be measured and observed even without any equipment (so-called Cheyne-Stokes Syndrome). What was observable with the naked eye could not be quantified back then, but knowing what we know now that is a typical illustration of how we can use physiological complexity to improve diagnostics in medicine: quantification of signals and their complexity can serve early detection of disorders in clinical practice.

34.3 Depression Detection Based on EEG

Regarding depression, there are three distinct classes of published papers: those that attempt depression detection based on fractal dimension and other nonlinear measures calculated from time series extracted from EEG; those that initially characterize electrophysiological signal (EEG or ECG) and use it later for machine learning models to get the classification between patients and healthy controls (both Diagnostic); and the Interventional studies – those that try to elucidate the effectivity of the therapy, trace the progress and even predict the outcome of therapy/namely predict ‘responders’ based on physiological complexity analysis. It is common that those are also called early detection, differentiation of the groups and treatment effects monitoring and prediction in engineering sense.

The neurological and neuroimaging basis for depression detection could arise from a combination of findings presented/revealed by biochemical research (for review see [18]) and digital tractography, fractional anisotropy, MRI and fMRI studies elucidating the possible functional and anatomical differences present in depression [3, 13]. As Willner formulated, ‘the depressed brain is simply in a different state than non-depressed brain’ referring to the results of their work after testing whether those who recovered from depression still can relapse after even a minor adverse event, contrary to those who were never diagnosed with depression before [19]. Many researchers coming from biochemical background concluded that there might be a systemic difference, on a much deeper level than single neurotransmitters and disbalance of single molecules in brains. What fMRI studies discovered, for example in structural, but also in functional sense, is that for some (unknown) reason the functional connectivity within the fronto-limbic system is decreased in comparison to healthy controls [13]. That is very important in, as a psychiatrist would say, ‘controlling the affect’; even when a person wants to control intense emotions coming from hyperactive limbic system, the usual suppressive connections from prefrontal cortices are not able to perform that suppression, hence,

leading to overall negative results and downward spiraling in mood. The anatomical difference that Dutch researchers pinned down is the unfunctional second part of a very important white matter tract connecting the two regions (prefrontal and limbic), namely, the uncinate fasciculus [13]. There are several hypotheses as to why this damage happens, but it is impossible to decipher the true cause, in effect, functional inability to suppress the amygdala, which is hyperactive. As the authors discuss, one of the possible physiological options could be that the long-lasting elevated levels of cortisol (caused by prolonged elevated levels of stress) could have led to the damage of neural tissue. But at the point that you already confirm the anatomical difference it is impossible to say what exactly caused the damage. The overall effect is that the 'traffic' that would be normal via this important deep white matter is simply not possible due to the deficiency of the structure [13], hence decreased signaling possibility to suppress limbic signals from the higher prefrontal centers. As some clinicians explain, it is like the amygdala 'hijacked' the brain, overruling higher command centers that are supposed to suppress its drive (Williams et al. 2010). How that could be understood is a kind of parallel with, for example movement disorders; once the task cannot be performed in a usual way (say controlling a certain movement since deep brain structures orchestrating that are destroyed) the brain as a highly adaptable system, finds the other, unusual and unique way of performing it, which is known as compensation. Maybe we are looking at a similar adjustment in depression. What we can see from above is that every patient who has that kind of compensation (for example in PD) has also increased EEG complexity in the whole motor cortex. This could also be the case in depression. The brain that cannot use intended control mechanisms will develop a different path to perform the task, and the control of the limbic system from prefrontal 'higher order' centers loses the direct link. As a result, the new composite electrical signal characterizing the rearranged and complicated functioning of millions of neurons appears more complex for a subject in depression compared to a person who is healthy in mood-controlling regard. That might be the reason that we repeatedly see increased excitability of cortex in patients suffering from depression [27]. And that could be described with change in physiological complexity and quantified with measures of that complexity, where fractal description is justified.

There is quite a voluminous scientific output regarding the characterization of electrophysiological signals recorded in patients diagnosed with depression. One particularly interesting review paper comparing various approaches to analyzing EEG for depression detection concluded that although some authors (early papers like [28]) found that complexity is lowered while majority of authors detected increased complexity in depression compared to healthy controls, 'EEG dynamics for depressive patients appear more complex but might be more random than the dynamics of healthy non-depressed individuals' [29]. That conclusion might even point out to the destruction of deeper structures additionally.

Despite the lack of consensus on the cause among researchers, it is repeatedly shown that an elevated complexity of EEG in depression is confirmed [30–34]. Also, several authors have demonstrated that the increased complexity was present in all EEG channel recordings [31–33, 35–37], indicating that the source of the increased

complexity in a signal recorded from the cortex could be in deeper brain structures, while different anatomical targets being candidates for that culprit were elaborated in the introduction as a summary from neuroimaging research.

In the first group of Diagnostic studies, there are many successful trials to detect depression based on complexity changes from its output electrophysiological signals. Here we have two subclasses, namely those analysing electroencephalogram (EEG) and those focusing on electrocardiogram (ECG) analysis for depression detection.

In the EEG studies subgroup, we can find interesting results based on above mentioned reasoning from neuroimaging. Nandrino and Pezard [28] published the first study elaborating on changes in complexity in depression by using correlation dimension; although they could not detect the difference between the patients who were in episode and those in remission. Ahmadlou et al. [30] compared the performance of two different algorithms, namely Higuchi's fractal dimension (HFD) and Katz's fractal dimension (KFD), on the same depression detection task (HFD showed to be superior). The researchers used only six prefrontal electrodes, justifying it with the dominant opinion that prefrontal asymmetry in alpha is the key feature of depression. Hosseinifard et al. [35] used several nonlinear measures (HFD amongst others) to analyse EEG recorded in patients diagnosed with depression, showing elevated complexity as a characteristic hallmark of depression. Bachmann et al. [31] also detected elevated HFD in depression (the only study we are aware of to use an exclusively female sample, which is reasonable since women have 50% higher risk of depression than men), but due to the detection of the same phenomenon on all electrode positions, opted to use just four electrodes for the clinical simplicity's sake. They also employed Lempel-Ziv complexity (LZC) and a novel spectral measure (SASI), developed by their group. The same researchers published another paper in 2018, this time with several different combinations of measures adding machine learning to its methodology to get automated detection [32]. This time they opted to change the parameter $k_{\max}$ in HFD calculation and chose to use broadband EEG for analysis. From that period, there are many studies testing the utility of several other nonlinear measures as a potential marker for depression, but we are focusing here on mixed technologies that used fractal-based approach. In a recent study, [27] showed that even if you use several widespread machine learning models to enrich detection (and apply it for automated detection), the fractal (or more broadly nonlinear) characterization of a signal under study is the key; all seven compared ML models yielded high accuracy (higher than 92%) on the task because HFD and SampEn were used as features for further machine learning [27]. The same group also extended this research by probing a similar methodology on a sample consisting of patients diagnosed with bipolar depressive disorder (BDD) but concluded that unlike major depressive disorder (MDD), BDD has more phases (five to be precise, with various possible routes between them) of disease and is characterized by high personal variability [38, 39]. Unlike MDD studies in automatic detection, the BDD project led to a true personalized detection [38]. Avots and colleagues applied ensemble approach to detecting depression, using among others HFD as features [40]. Nevertheless, probably the most important result,

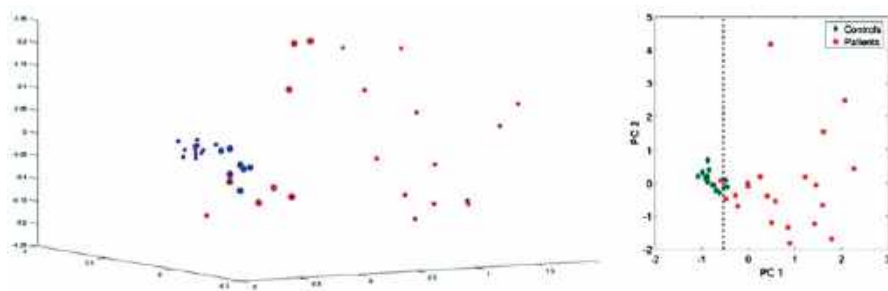


Fig. 34.1 Left: First three normalized PCA components calculated on samples extracted from EEG. The features consisted of HFD and SampEn calculated from EEG records from a group of patients and healthy controls described in the study, published in 2020 [27]. Blue dots are healthy controls; red dots are patients diagnosed with depression. Right: the same comparison, just with two normalized PCA components. Green diamonds are controls, red dots are depression patients. The vertical line denotes a separation line corresponding to a constant value of the first principal component (PC1). A good separation between the features based on complexity of EEG is visible

besides the successful detection of depression (solely fractal or in combination with machine learning) is the demonstration that fractal (and nonlinear in general) analysis can help us to distinguish between episode (exacerbation) and remission phase of depression [41]. This study was performed on a quite modest sample (as a majority of studies from this period we mentioned here) where 22 depression patients were recorded in both phases – once they were in episode, and later after they reached remission. They were compared with EEG records from age- and sex-matched 20 healthy controls. To further understand those features in phase space, a principal component analysis (PCA) was used and showed a clear separation of the data [41] (Fig. 34.1).

Since this method was based on the previous results from colleagues in the field, the final result came as a bit of a surprise; patients diagnosed with depression exhibited increased complexity of their EEG (compared to controls), but when the differences were compared person by person, individual subjects had higher scores during remission phase compared to an episode [41]. Comparing the values calculated from patients in both states with controls, ANOVA showed a significant effect of group on HFD ($p < 0.001$). That probably could be understood again as a unique characteristic compensation in order to circumvent lost control mechanisms described above. Again, it could be that calculated increased complexity in signal implies that it might be more *random* than in a healthy brain [29].

A more comprehensive review and cross-sectional analysis of diagnostic and interventional studies that combined fractal and nonlinear measures and machine learning approaches were recently published by Čukić et al. [37]. It was concluded that the complexity-based characterization of a signal under study is crucial to successful classification (see [42]).

While comparing the presented publications, there appeared a consensus about the repeatedly detected elevated complexity in depression. Still, sometimes the

direct comparison of results is challenging, since there are many methodological differences in calculating Higuchi's fractal dimension: researchers use different values for the parameter $k_{\max}$ – for example, it is shown that it is recommended to use $k_{\max}$ between 8 and 16, as shown to be the most appropriate [43], while Hosseinifard opted to choose 100. In some other applications where HFD was calculated from human EEG, researchers justified $k_{\max}$ to be larger than 65 [44]. Again, challenging comparison, since the majority of researchers do not even report the exact HFD values, nor $k_{\max}$ used, just detected significant statistical differences between the populations. Different researchers use EEG sub-bands, since in many instances, they wanted to compare the performance of standard time and frequency measures with nonlinear measures, while other researchers use broadband signal, reasoning that nonlinear methods should only be used on raw signals, with minimal pre-processing. Some researchers use classical preprocessing of a signal, while others select artifact-free segments of EEG in order to preserve the raw signal to keep the maximum original information content. Sample sizes are characteristically small, 20, 30 to maximum 50 participants (in recent ML-based studies that is changing gradually), but we are aware that the number should be increased, indicating the need to collect data in many different sites simultaneously employing the same protocols in order to increase its practical utility (like exemplified in STAR*D, 2017) and possible clinical translation. Actually, the whole protocol does not even need to be the same. But a part of it should be the same. Each site can also have an add-on, but needs to have a uniform part. In any case, we need to have more collaborative approach to collecting more data to increase the quality of this kind of research.

Considering interventional studies, Lee et al. [16] aimed to predict the treatment response of MDD. Their results suggested that frontotemporal connectivity strength could be a potential biomarker to differentiate between responders and slow responders or non-responders in MDD (the stronger connectivity was associated with lower response to the therapy). Unfortunately, their sampling frequency was as low as 100 Hz, making comparison with some other research difficult.

A study from 2014 [45] claimed that no difference exists between MDD and HC in nonlinear EEG measures (LZC), but concluded that nonlinear measures add value to the research. The authors also claimed that they were the first to use complexity measures for this task, despite the fact that Nandrino and Pezard used this approach in the analysis of EEG in depression in 1994, as did several other research groups. Bachmann et al. [31, 32] applied the same methodology (LZC) and demonstrated significant differentiation between patients and controls.

Lebiecka et al. [34] used fractal dimensions to study the neurophysiological effect of repetitive transcranial magnetic stimulation (rTMS) as a treatment for depression (MDD). Subgroups in their sample were MDD and BDD patients respondent and non-respondent to rTMS therapy. Changes in fractal dimension were analysed before and after the 1st, 10th and 20th sessions of the rTMS. FD increased in all bands after multiple stimulation sessions in MDD responders, while in BDD non-responders FD decreased after the first rTMS session. In MDD non-responders and BDD responders FD increased at the beginning of the therapy, but the final FD did not change compared to the initial values. Comparison between responders and

non-responders showed that the FD changes were higher in responders (concomitant to the analysis of remission phase in MDD by Čukić et al. [41] and Lebiecka et al. [34]). The opposite was seen in BDD: FD was lower in responders than in non-responders. Those results demonstrate that fractal dimension can be a promising clinical marker to estimate the effectiveness or performance of rTMS therapy in depression [34]. The changes of FD under the influence of rTMS could be unambiguously used to conclude who among the patients would be a responder to the therapy very early on in the treatment (see also [46] on mechanisms behind effectivity of rTMS as a treatment for depression).

Jaworska et al. [47, 48] examined a variation of pre-treatment EEG to predict depression treatment success; they performed a 12-week machine learning study to predict the outcome of pharmacology treatments in 51 patients with MDD. They used both electrophysiological and demographic data as well as source-localized current density, while the random forest was used for classification, and used kernel PCA to reduce and map important features. This method yielded 78–88% accuracy dependent on model complexity. Those studies on personalized, biomarker-based treatment approaches are the most promising in future clinical translation (Fig. 34.2).

Also, in connection to non-invasive electromagnetic stimulation (NIBS), there is evidence that transcranial direct current stimulation, tDCS (like rTMS), is able to significantly shift the brain state after a half-an-hour stimulation, which is considered to be the time limit for estimating the impact in plasticity studies [37, 41]. It has to be stressed here that both modalities of electromagnetic stimulations as a therapy for depression seem to have an effect exactly because of their potential to decrease, even if temporary, the elevated complexity present in patients diagnosed with depression. Recent work on the effectivity of use of tDCS in Finland, where it is available for at-home use and in many general practitioners' offices, has shown that tDCS can also be used as a maintenance therapy if a patient successfully underwent prior rTMS as a treatment [49]. Also, there is recent evidence that a state of consciousness can be determined by complexity measures, pointing out that in healthy subjects the higher levels of consciousness are also characterized with an increased HFD, among other nonlinear measures [50].

It is evident that the research must continue to elucidate how a response can be predicted, and therapy route planned to minimize relapses and other adverse changes of patient's state.



Fig. 34.2 Possible wireless options of EEG recording in order to measure the complexity of signals and extract features to be later used for classification. The figure corresponds with our earlier contribution to the paper by [37] <https://www.jmir.org/2020/11/e19548>

34.4 Detecting Depressions' Hidden Cardiovascular Risks Based on ECG

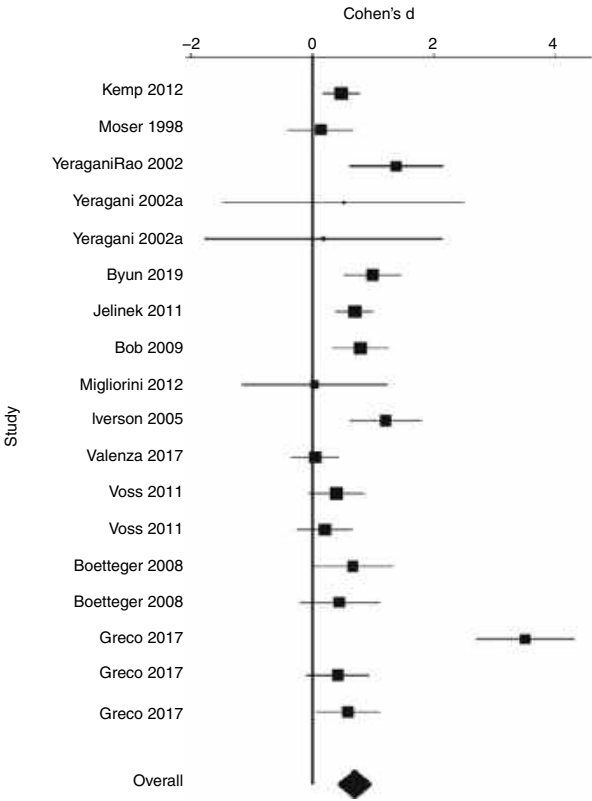
Yet another class of research is aiming at detecting depression from electrocardiogram (ECG) and using heart rate variability (HRV) as a robust biomarker of depression [51–53], but most importantly as a biomarker for unaddressed cardiovascular risk in patients diagnosed with depression. A recently performed random-effect meta-analytic review study to present that nonlinear measures is superior to standard time and frequency-based measures typically used in cardiology and in clinical practice in general ([54]/in a process of publication). The results of this review showed that the overall effect size (ES) of 26 included studies is indeed overperforming similar meta-analytic result (by [55]) based on standard (time- and frequency-based) measures of HRV. The ES of all the studies included was very high (Cohen's d , $d_s = 1.03$), but when reanalyzing subgroups of studies according to the family of applied nonlinear measures they showed to be also remarkably practically useful; since effect size is most useful result to report illustrating the

practical utility of any method under study. While entropy-based measures scored as a sub-group the highest (in one case even ES larger than 3), DFA-related studies scored ES of 0.364, and the group comprising of various other nonlinear approaches including fractal dimension, correlation dimension, complex correlation dimension, LLE and Poincare plots demonstrated overall effect size of 0.702, what is way above previously seen results in standard analytical approach to HRV research, that demonstrated mild ES according to Cohen's definition.

When considering HRV for detection of depression, there is evidence that various HRV indices, both conventional and fractal and nonlinear, could be successfully used for detection. Still, long hours of recording when using holter (even 24 h or more) are disrupting the patients' usual rhythms and are unpleasant. This has led to the development of various portable ECG devices that can increase the comfort of this early detection. Many of those devices claiming detection are actually not registering a good-quality signal (like detection from fingertips, or from a wrist, popular among 'smart watches' and wearables abundant on the market). Possible conclusion is that only direct recording from the chest that yields the medical-grade quality of the signal is sufficient for this kind of detection. Besides, there are several other promising methods to detect or reconstruct the high-quality ECG from vibrations of the chest (seizmo cardiogram, SCG) or from acceleration of the tissue (Gyro cardiogram GCG). There are successful methods that reconstruct respiration signal from those recordings, and the results demonstrate to offer a close resemblance of electrically recorded ECG, enabling high accuracy detection of depression [56]. The same group also demonstrated that fractal analysis (including detrended fluctuation analysis) performed on all three signals (ECG, SCG and GCG) show almost redundant values when calculated from RR standardized time series, indicating that small portable devices with novel sensors could be used in this kind of detection ([54] *a in press*). In that paper, Bland–Altman plots have shown that among a battery of nonlinear indices HFD performed among the best and showed the lowest possible error, below the 1%, contrary to the majority of standardly used measures (Fig. 34.3).

Just to illustrate why the fractal approach is rational and adequate, let us remember that in ventricular electrical (His-Purkinje) conduction system, electrical stimulus (the quantity transported in this particular case) that regulates the timing of cardiac contraction transported through tissues can be characterized as a fractal process (Abboud et al. 1991). Cortico-vagal control (crucial for heart dynamics) also demonstrates fractal properties, and ECG as a signal (emerging as a signal from all those processes) is adequately described as a fractal system [57]. Hence, instead of calculating just how your results are smeared around the mean value, it makes much more sense to apply fractal analysis to characterize the signal. As Klonowski described it back in the 2000s all bio-signals are 3 N: nonstationary, nonlinear and noisy. To treat that with standard reductionistic approach would lead to spurious conclusions about physiological processes [58].

Fig. 34.3 Effect sizes of studies that approach the link of depression and cardiovascular risks (more precisely aberrated dynamics of autonomous nervous system that is changing heart dynamics) by utilizing nonlinear measures ([54], in publishing process). One of the subgroups of reviewed publications has average Cohens $d_s = 0.702$, which is more than twice higher than any previously reported meta-analysis on standard spectral and time-related HRV measures



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Chapter 35

Point of Care Testing (POCT) in Psychopathology Using Fractal Analysis and Hilbert Huang Transform of Electroencephalogram (EEG)



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Abstract Research has shown that relying only on self-reports for diagnosing psychiatric disorders does not yield accurate results at all times. The advances of technology as well as artificial intelligence and other machine learning algorithms have allowed the introduction of point of care testing (POCT) including EEG characterization and correlations with possible psychopathology. Nonlinear methods of EEG analysis have significant advantages over linear methods. Empirical mode decomposition (EMD) is a reliable nonlinear method of EEG pre-processing. In this chapter, we compare two existing EEG complexity measures – Higuchi fractal dimension (HFD) and sample entropy (SE), with our newly proposed method using Higuchi fractal dimension from the Hilbert Huang transform (HFD-HHT). We present an example using the three complexity measures on a 2-minute EEG recorded from a healthy 20-year-old male after signal pre-processing. Furthermore, we showed the usefulness of these complexity measures in the classification of major depressive disorder (MDD) with healthy controls. Our study is in line with previous research and has shown an increase in HFD and SE values in the full, alpha and beta frequency bands suggestive of an increase in EEG irregularity. Moreover, the HFD-HHT values decreased in those three bands for majority of electrodes which is suggestive of a decrease in irregularity in the frequency–time domain. We conclude that all three complexity measures can be vital features useful for EEG analysis which could be incorporated in POCT systems.

Keywords Fractal analysis · Hilbert transform · EEG · Psychopathology · Point of care testing

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35.1 Introduction

In 1875, Richard Caton (1842–1926), a medical doctor from Liverpool, published an abstract in *The British Medical Journal* on his presentation at the 43rd Annual Meeting of The British Medical Association in Edinburgh [1]. He described the existence of electrical impulses in the brains of rabbit and monkey using a galvanometer [1]. After which, nearly 100 years later, in 1924, Hans Berger (1873–1941), a German psychiatrist, recorded the first human electroencephalogram (EEG) on a 17-year-old boy during a neurosurgery [2]. Later, he developed a recording technique that connected the electrode to the scalp similar to the protocol used today. Berger was also the first person to record electrical activity non-invasively from the human brain. He published his first paper on EEG, “Über das Elektrenkephalogramm des Menschen” in 1929 [2, 3].

In 2017, approximately 970 million people worldwide suffered from some form of mental health or substance use disorder [4]. The largest percentage reporting with anxiety disorders followed by depression [5]. Furthermore, a systematic review in 2018 showed that about 1 in 20 children suffered from attention deficit hyperactivity disorder (ADHD) [6]. At least a further 5% of children present with difficulties such as overactivity, inattention and impulsivity that are just below the threshold to meet the full diagnostic criteria for ADHD [6]. Unlike other specializations in medicine, psychiatry does not use extensive objective diagnostic tests such as the EEG [7–9]. Clinical interviews structured around the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) and the International Classification of Disease, Eleventh Revision (ICD-11) are used for the diagnosis of psychiatric disorders [10]. The diagnosis includes patient self-reporting of symptoms which include behavioural, cognitive, affective or physical symptoms [10].

Recently, research has shown that relying only on self-reporting for diagnosing psychiatric disorders does not yield accurate results at all times [7, 11, 12]. Moreover, treatment response for psychiatric disorders is also highly variable including medication-resistant patients [6, 12]. In practice, multiple treatment options are tested before concluding which treatment plan is most effective. Often, this means that patients must endure a long period of untreated symptoms and possible unwanted side effects before finding one that is effective [12]. A vital objective of modern psychiatry is to diagnose the psychopathology with high accuracy and find an effective treatment plan in a timely manner to optimize patient benefit.

WPA-Lancet Psychiatry Commission wrote that “Psychiatry in the first quarter of the 21st century is at the cusp of major changes” [13, 14]. In the past decades, there have been significant progress in multiple neurobiological domains, which has improved our understanding of the underlying mechanisms and causes of various mental disorders [13]. For example, the development of sophisticated imaging techniques such as magnetic resonance imaging (MRI), single-photon emission computerized tomography (SPECT) and positron emission tomography (PET), as well as major advances in genetic research and new neuroscience techniques including optogenetics [13, 15–18]. Transcranial magnetic stimulation (TMS) and

specifically repetitive transcranial magnetic stimulation (rTMS) has been used for both treatment and diagnostics. rTMS has been applied for treatment diverse psychopathology and has had positive outcomes in patients with unipolar depression exasperated by treatment resistance [19, 20]. rTMS has recently also been used for diagnostics when combined with fMRI or PET [21]. However, clinical diagnostics have not kept pace with these technical advances possibly due to cost factors and familiarity with advanced technology as well as baseline data availability [13, 22, 23]. Currently, therefore, most clinical decisions are based on anamnestic features and clinical experience of the psychiatrist [13]. EEG, on the other hand, is a useful diagnostic tool, which has been used for organic mental syndromes and epilepsy primarily by neurologists. Current challenges include developing models for investigating drug action and cognitive processes [24–26].

Until the 1990s EEG signals were recorded using ink print on continuous paper tapes, which were considered as analog recordings [11]. The advances in technology and the development of personalized computers made possible for digitalization of EEG recordings. Furthermore, artificial intelligence and other machine learning algorithms have allowed the introduction of point of care testing (POCT) including EEG characterization and correlations with possible psychopathology. POCT using EEG has been applied for post-stroke assessment, psychiatric evaluations, chronic diseases and as part of long-COVID related neuropathy [10–12]. Utilization of EEG as a POCT tool, however, requires additional technical innovation for real-time signal processing and removal of noise such as AC powerlines, electromagnetic noise, movement artifact (jaw clenching, eye movement and frowning), cardiac noise and eye saccades or pursuit movements, as well as blinks.

The majority of previous analytical methods for EEG recordings such as Fast Fourier Transform (FFT) consider the signal to be linear where the signal is a combination of some component signals with constant coefficients [27]. Furthermore, for an EEG to be of value in POCT suitable features must be selected for assessment. Denoising can be achieved by linear filtering methods such as independent component analysis (ICA) when the noise and useful signal are known and not overlapping as well as being linear [28, 29]. However, as most physiological recordings, EEG is considered to be nonlinear, nonstationary and dynamic [27, 30]. Hence, a more reliable method of signal pre-processing is required. Empirical mode decomposition (EMD) to analyse EEGs is a common pre-processing method that decomposes the signal into finite and smaller number of intrinsic mode functions (IMFs), which are amplitude and frequency modulated (AM and FM) waves [28]. This chapter proposes a new method of complexity measure using Higuchi fractal dimension (HFD) calculated from the Hilbert Huang transform of an EEG signal after pre-processing using EMD. The Hilbert-Huang transform provides a 2-dimensional, time-frequency output. All data analysis and calculations were performed using MATLAB (www.mathworks.com).

35.2 Higuchi Fractal Dimension (HFD)

Fractal analysis is an analytical method popularized by Benoit Mandelbrot for chaotic systems such as the Lorenz attractor that has found applications in a wide variety of research including geography, psychology, biology, cardiology and neuroscience, where in general a non-random repetitive pattern at different scales is expected [31–39]. Higuchi fractal dimension is a type of complexity measure that does not depend on binary sequences and is therefore insensitive to noise in most cases. Higuchi in 1988 introduced this nonlinear measure to measure the earth's changing magnetic field [40]. Nearly 10 years later, the method found its way into neurophysiology with the first paper using Higuchi dimension for detection of changes in human EEG with a dramatic increase in the early 2000 [41]. HFD has also shown more accurate results compared to other fractal analysis methods of temporal data such as the Katz and Petrosian dimension [42]. HFD can quantitatively describe the non-uniformity of nonlinear and nonstationary (fractal) sequence [40]. The algorithm to calculate the HFD is as follows [30]:

- (i) For time series with length N , $x = \{x(1), x(2), \dots, x(N)\}$, and new k time-series are constructed,

$$x_m^k = \{x(m), x(m+k), x(m+2k), \dots, x(m+Mk)\}, \quad m = 1, 2, \dots, k \quad (35.1)$$

- (ii) The length of the time series,

$$L_m(k) = \frac{1}{k} \left(\sum_{i=1}^M |x(m+ik) - x(m+(i-1)k)| \right) \times \frac{N-1}{M \times k}, \quad M = \frac{N-m}{k} \quad (35.2)$$

- (iii) The average value of k time-series length,

$$L(k) = \frac{1}{k} \sum_{m=1}^k L_m(k) \quad (35.3)$$

There is a power-law relationship between $L(k)$ and k such that $L(k)$ is proportional to k^{-D} . The estimated value of HFD for the EEG signal can be obtained by the slope coefficient of linear least square fitting line to the point $\{\ln(L(k)), \ln(1/k)\}$

$$D = -\frac{\ln L(k)}{\ln k}, \quad (k = 1, 2, \dots, k_{\max}) \quad (35.4)$$

Some studies have shown that using the maximal scale $k_{\max} = 8$ for the HFD algorithm is best for EEG signals [43, 44]. Some other studies have proposed a

more dynamic approach where a range of HFD values are plotted against their respective $k_{\max}$ values and the $k_{\max}$ value is selected when the HFD value reaches a local maximum or approaches a saturation point (asymptote) [45–47]. For our study, we first calculated HFD values using $k_{\max}$ between 2 and 20. Then the first local maximum or the point of saturation ($k_{\max} = 20$) was taken as the appropriate $k_{\max}$ and HFD value.

35.2.1 EEG Fractal Analysis in Psychiatry

EEG analysis has been investigated in a number of psychiatric conditions but has lacked some impact due to the time required for setting up a quantitative EEG (19 electrode set-up) as well as specificity. The nature of the psychiatric illness and patient presentation therefore often hinder this type of analysis. However, use of EEG in schizophrenia, major depressive disorder, cognitive decline dementia and anxiety have been reported with acceptable accuracy for detection of psychiatric condition as well as being able to predict disease course and response to treatment [48–50]. Automated analysis of the complex EEG patterns in psychiatric conditions has also been investigated using other fractal analysis methods apart from the HFD including box counting, correlation dimension and Lyapunov exponent [42, 51–53].

Epilepsy is a neurological disease characterized by abnormal synchronous discharges of cortical neurons that can cause unexpected and recurring dysfunction of the brain, including mood/affective changes, anxiety, personality and psychotic disorders [54, 55]. The authors in [30] studied the effectiveness of detecting epileptic EEG compared to nonepileptic EEG signals using machine learning by dividing the EEG into 1 second segments and calculating the HFD and sample entropy (SE) of each of these segments after pre-processing the signals. The results showed that HFD and SE was lower for the epileptic signals compared to nonepileptic signals. The results of the study suggested that both complexity measures should be used together for a support vector machine (SVM) classifier to classify signals as epileptic and nonepileptic signals, achieving a recognition rate of 89.9% when both complexity measures were used [30].

Study by Čukić et al. comparing EEG signal complexity using HFD between 20 healthy control subjects, 11 depressed patients and 11 depressed patients in remission indicated a significant difference between the three groups for electrodes placed at Fp1, Fp2, F3, F7, Fz, C3, Cz, P4, Pz, T4, T5, T6 and O2 following the 10/20 EEG placement protocol. HFD values increased for each electrode site from control to depression and from depression to depression in remission [7].

Other research such as [56] also showed a stable frontal asymmetry with the HFD of alpha bands of 30 women diagnosed with major depressive disorder (MDD). Furthermore, the results in [57] showed that higher HFD values for left, right and overall frontal lobes of MDD patients compared with non-MDD patients in the beta and gamma sub-bands. Elevated complexity in EEG from schizophrenic patients has also been observed [58, 59]. In contrast, a study by Raghavendra et al. [60] showed

a reduced EEG complexity in the bifrontal region and increased complexity in the bitemporal and occipital regions of schizophrenic patients. HFD also performed better in a study by Kawe (2019) for predicting anxiety or depression compared to control using frontal asymmetry measures [61].

35.3 Sample Entropy

Sample entropy (SE) is the modified version of the approximate entropy. SE is one of the types of information entropy, which is used to assess complexity. The more complex the system is, the more randomized and distributed the system becomes which results in a higher SE value. SE is insensitive to signal length [7]. Sample entropy has been extensively used since published by Richman and Moorman in 2000 including analysis of heart rate, electrocardiogram (ECG) and electromyography (EMG) [62].

If there is a time series sampled at equal time intervals and composed of m -dimensional vector space [7],

$$X_m(i) = \{x(i), x(i+1), \dots, X_m(N-m+1) \mid 1 \leq i \leq N-m+1\} \quad (35.5)$$

$d[x_m(i), x_m(j)] (1 \leq j \leq N-m, j \neq i)$ is the distance between $X_m(i)$ and $X_m(j)$, it is defined by:

$$d[x_m(i), x_m(j)] = \max_{k=0, \dots, m-1} (|x(i+k) - x(j+k)|) \quad (35.6)$$

B_i is the number of j when the distance between $X_m(i)$ and $X_m(j)$ less than r for the certain $X_m(i)$,

$$B_i^m(r) = \frac{1}{N-m-1} B_i, B^{(m)}(r) = \frac{1}{N-m} \sum_{i=1}^{N-m} B_i^m(r) \quad (35.7)$$

Then, the dimension increases to $2m+1$, A_i is the number of j when the distance between $X_{m+1}(i)$ and $X_{m+1}(j)$ less than r , so, the SE is calculated by

$$A_i^m(r) = \frac{1}{N-m-1} A_i, A^{(m)}(r) = \frac{1}{N-m} \sum_{i=1}^{N-m} A_i^m(r) \quad (35.8)$$

$$S_{En}(m, r) = \lim_{N \rightarrow \infty} \left\{ -\ln \left[\frac{A^{(m)}(r)}{B^{(m)}(r)} \right] \right\} \quad (35.9)$$

35.3.1 Sample Entropy in Psychiatry

Sample entropy (SE) has two main advantages over the approximate entropy: (1) relative consistency and (2) independence of signal length [63, 64]. Previous studies have reported an increase in entropy of EEG, including sample entropy in medication naïve patients with psychosis compared to control, although this may depend on location and sub-band analysis [65, 66]. Similar to HFD values, study by Čukić et al. showed a statistically significant higher values of SE for electrodes Fp1, Fp2, F3, F7, Fz, C3, Cz, P3, P4, Pz, T4, T5, T6, O1 and O2. SE values increased for each electrode site from control to depression and from depression to depression in remission [7]. Another study also showed significantly higher SE values for all 19 electrodes combined in patients with depression compared to control [44].

Furthermore, some other studies have shown SE can be used as an indicator of normal ageing and neurodevelopment by monitoring changes in EEG complexity during resting period or during sleep [67, 68]. Studies have shown that SE along with other complexity measures can be used in machine learning algorithms such as neural network, multilayer perceptron, logistic regression, support vector machines with the linear and polynomial kernel, Decision Tree, Random Forest and Naïve Bayes classifier to detect schizophrenia, depression and autism in children with accuracies reaching up to 97.56% [7, 64, 69, 70].

35.4 Empirical Mode Decomposition

Empirical mode decomposition (EMD) was pioneered by N.E. Huang et al. in 1998, and this decomposition method is adaptive and hence highly efficient [71]. EMD to analyse EEGs is a common pre-processing method that decomposes the signal into finite and smaller number of intrinsic mode functions (IMFs), which are amplitude and frequency modulated (AM and FM) waves [28, 71]. EMD includes extracting IMFs from a signal using a sifting algorithm. Each of the IMF must satisfy two conditions: (1) the number of extrema and number of zero crossings must either equal or differ at most by one and (2) at any point, the mean value of the envelope defined by the local maxima and local minima must be zero [71]. The steps involved in EMD are as follows:

- (a) Finding the local maxima and minima of the signal $x(t)$.
- (b) Computing the corresponding upper ($e_{\max}$) and lower ($e_{\min}$) envelopes using the cubic spline interpolation method.
- (c) Calculating the mean $m(t) = (e_{\max} + e_{\min})/2$.
- (d) Subtracting $m(t)$ from the signal to find $s(t) = x(t) - m(t)$.
- (e) Repeat steps (a) to (d) until the $s(t)$ satisfies the IMFs condition, then define $s(t)$ as IMF₁.

- (f) Next, to compute the next IMFs, $x(t)$ becomes $x(t) = x(t) - \text{IMF}_1$. Then the above steps will be repeated to obtain the other IMFs until $x(t)$ becomes a constant of a trend called the residue $-r(t)$.

35.4.1 Pre-processing EEG

EEG pre-processing to remove noise is a necessary step before data analysis to ensure that the data analysis are a result of the changes in condition rather than the noise. Common external or environmental sources of noise include power lines, lighting and electronic equipment (computers, display monitors or television, mobile phones, wireless routers, etc.) [72]. Furthermore, physiological noise can exist from cardiac signal (ECG), artifacts from muscle contraction (EMG) and ocular signal caused by eyeball movement (electrooculogram, EOG) [72]. EMG and EOG signals can be minimized by asking participants to be in a relaxed and eyes-closed state. However, ECG cannot be eliminated. Most commonly used pre-processing methods include Butterworth filter, linear regression, independent component analysis (ICA) or visually selecting a portion of EEG with less artifacts [7, 44, 72–74]. However, these are linear filtering methods that are not suitable due to the nonlinear properties of EEG.

Hence, EMD is a more reliable method of pre-processing as it is a nonlinear filtering method. The EEG pre-processing followed the method of [28] with an additional step included to account for the non-stationary spikes corresponding to artifacts. This method involved the following steps:

- (a) Obtain the IMFs ($\text{IMF}_i, i = 1, 2, \dots, N$, where N is the number of IMFs) of $x(t)$ – original raw EEG signal with noise.
- (b) Obtain $A_i(k)$ and $R_x(k)$, which denote the autocorrelation functions (ACF) of the IMF_i and $x(t)$, respectively.
- (c) Compute the mutual information (MI) scores $-M_i = M(A_i(k); R_x(k))$, where M_i is the MI score between $A_i(k)$ and $R_x(k)$.
- (d) Check if the signal contains additional noise. If $M_{\max} - M_{\min} < 0.8$, then signal contains noise.
- (e) Determine threshold, $\theta = 0.5(M_3 - M_1) + M_1$.
- (f) Reconstruct the denoized signal $\hat{x}(t) = \sum_j \text{IMF}_j$, where $j = \{i \mid M_i > \theta\}$.
- (g) Then $\hat{x}(t)$ was passed through another filter. This filter removes the non-stationary part of the EEG signal, which usually contain spikes corresponding to artifacts.

35.4.2 Hilbert Huang Transform (HHT)

Analysis of nonlinear and nonstationary signals such as the EEG is difficult. Several methods such as the windowed Fourier transform and the wavelet transform have been proposed but have several limitations [27, 75]. Both methods use periodic functions to decompose the signal and hence are not applicable for nonlinear signals. The Hilbert Huang transform (HHT) aimed to address this by combining the empirical mode decomposition with the Hilbert spectral analysis with the main advantage of being highly adaptive in processing a signal and resulting in a better spectral resolution, both in the frequency and time domains [75, 76]. The IMFs after EMD can be processed further by applying the Hilbert transform to provide AM and FM frequencies forming the Hilbert Huang transform. The Hilbert transform $H[x(t)]$ for any signal $x(t)$ is

$$H[x(t)] = y(t) = \frac{1}{\pi} \int \frac{x(\tau)}{t - \tau} d\tau \quad (35.10)$$

Theoretically, any analytic signal $z(t)$ can be expressed by the sum of its real part $x(t)$ and imaginary part $y(t)$, where $y(t)$ is the Hilbert transform of the real part. Hence, we get the equation:

$$z(t) = x(t) + jy(t) \quad (35.11)$$

35.4.3 HFD from HHT

In order to calculate the Higuchi fractal dimension from the Hilbert Huang transform of a signal (HFD-HHT), the following algorithm can be adopted:

1. Calculate the HHT of the signal and project it on a graph where the x -axis denotes the sample number, the y -axis denotes the frequencies (Hz) and the z -axis denotes the instantaneous energy values.
2. After selecting an appropriate k -max value as described above, calculate the HFD values using the instantaneous energy values along the vertical x -axis and horizontal y -axis, by treating each vertical column and horizontal row of the HHT as a 1D signal.
3. Calculate the mean of all the HFD values to give the HFD-HHT value.

35.4.4 HFD-HHT as a Measure in Monitoring Relaxation

The Hilbert Huang transform has been applied to attention studies, seizure classification, depression and alcoholism amongst others [76–79]. Moctezuma (2019)

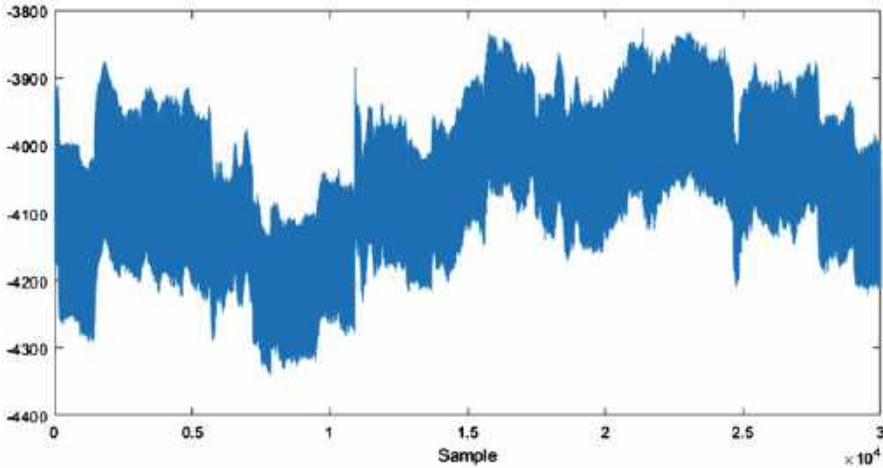


Fig. 35.1 2-minute raw EEG signal with noise

applied EMD and HFD to EEGs for detection of epileptic seizures and discussed the results with reference to identification of driver drowsiness [80, 81]. Our recent research study [82] showed the effectiveness of using HFD-HHT on the EEG signal recorded in a biofeedback study using only the Cz electrode. The study of ours included six participants that underwent three types of breathing conditions (normal, paced breathing and balloon game biofeedback). The results showed the lowest signal complexity in terms of SE, HFD and HFD-HHT value for the paced breathing compared to normal and balloon game biofeedback, indicative of relaxation [82].

Below we applied this methodology on a 2-minute EEG signal.

35.5 EEG Complexity in a Healthy Control

Figure 35.1 shows a 2-minute raw EEG signal from Fz electrode with noise that was collected from a healthy 20-year-old male with no personal or family history of mental disorders [83]. The IMFs were extracted as shown in Fig. 35.2 using EMD.

Next, pre-processing steps (a) to (c) was performed and Table 35.1 shows the mutual information (MI) values obtained. The difference between the maximum and minimum MI score is 0.9980, which is more than 0.8. Hence, according to [28], the signal does not contain any noise. The signal was reconstructed using the IMFs and was passed through another filter to remove the non-stationary part of EEG signal corresponding to artifacts. The denoized signal is shown in Fig. 35.3.

The HFD-HHT and HFD values for $k_{\max}$ values between 2 and 20 were calculated from the denoized EEG signal using the methodology explained previously. Figure 35.4 shows an asymptotic curve for HFD values, and hence, $k_{\max}$ was chosen

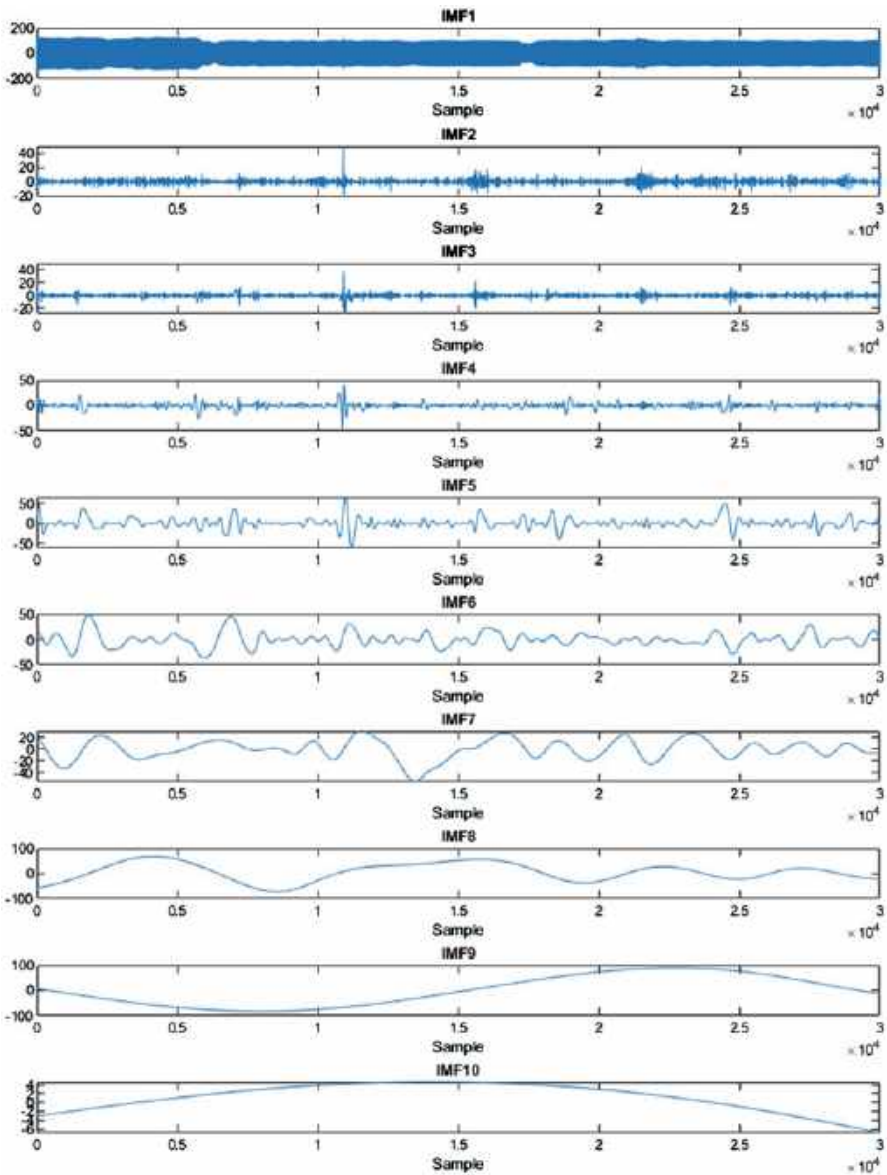


Fig. 35.2 Intrinsic mode functions (IMFs) 1–10 of the raw EEG signal

Table 35.1 Mutual information (MI) values calculated in step (c) of pre-processing

MI(1)	MI(2)	MI(3)	MI(4)	MI(5)	MI(6)	MI(7)	MI(8)	MI(9)	MI(10)
1.0000	0.0020	0.0020	0.0021	0.0021	0.0021	0.0074	0.0031	0.3156	0.4566

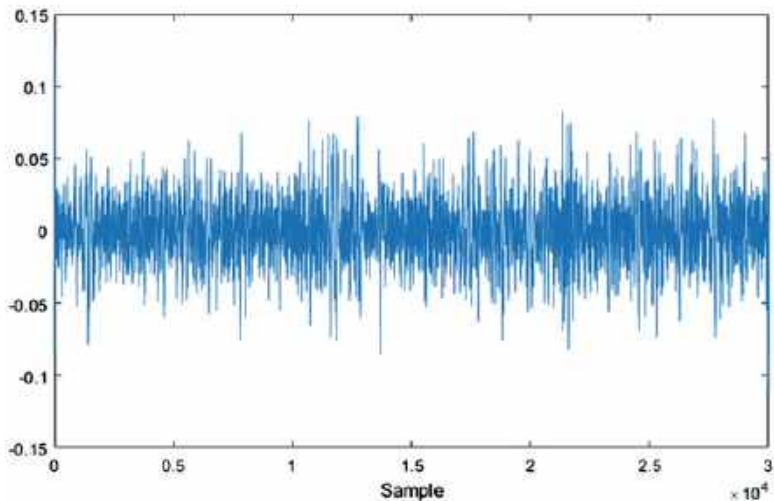


Fig. 35.3 2-minute denoized EEG signal after pre-processing

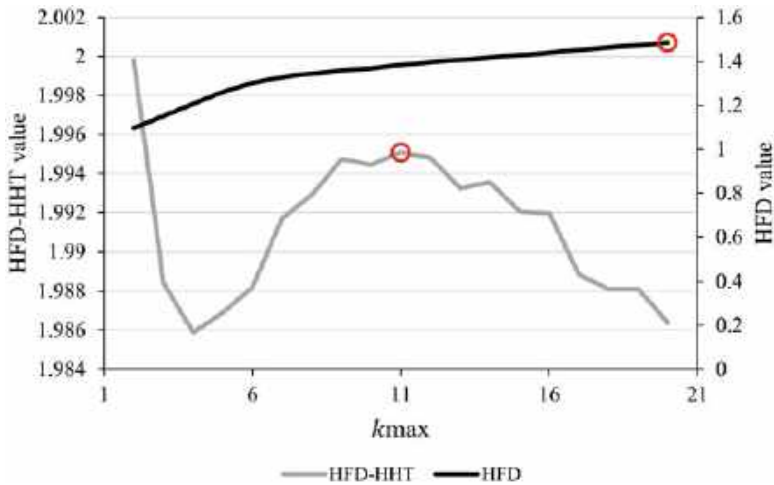


Fig. 35.4 HFD-HHT and HFD values for different $k_{\max}$ values

to be 20 and the respective HFD value of 1.4842 was recorded. Furthermore, the HFD-HHT value showed the first local maximum to be at $k_{\max} = 11$ and therefore, the respective HFD-HHT value of 1.9951 was chosen. The sample entropy value of the denoized signal was calculated to be 0.6247.

In the next section, we applied this methodology for the classification of major depressive disorder (MDD).

35.6 EEG Complexity in Major Depressive Disorder (MDD)

35.6.1 *Methods*

EEG signals were obtained from the multi-modal open dataset for mental-disorder analysis (<https://doi.org/10.1038/s41597-022-01211-x>) that included 23 patients with major depressive disorder (MDD) (12 males and 11 females with a mean age of 30.96 ± 10.59 years old) and 22 healthy controls (14 males and 8 females with a mean age of 31.77 ± 8.76 years old) [83]. Patients were diagnosed by at least one clinical psychiatrist. All MDD patients met the diagnostic criteria using the DSM-IV. None of the healthy control participants had a personal or family history of mental disorders. Furthermore, none of the participants in both groups were dependent on alcohol or used medications in the past year. None of the subjects were pregnant, lactating or taking birth control pills [83].

The study included 5-minute EEG signals recorded with eyes closed at rest using a 128-channel HydroCel Geodesic Sensor Net (Electrical Geodesics Inc., Oregon Eugene, USA) and Net Station acquisition software (version 4.5.4). The sampling frequency was 250 Hz. All the raw electrode signals were referenced to the Cz. Electrode impedances were below 50 k Ω .

To highlight the efficacy of the current proposed HHT-HFD method only, the electrodes present in the 10–20 system of 19 channels were included to simplify and compare the results with previous studies. However, the Cz electrode was used as reference and could not be used in the current data analysis. A 2-minute EEG section from the middle of the complete recording for each participant was used for the analysis. Pre-processing of EEG was carried out as previously described in Sect. 35.4.1. HFD, HFD-HHT and SE values were calculated from the full (1–45 Hz), delta (1–4 Hz), theta (4–8 Hz), alpha (8–13 Hz) and beta (13–30 Hz) frequency bands of the clean EEG after pre-processing. All significant differences and *p*-values were determined with a Mann-Whitney U-test using a 5% significance level.

35.6.2 *Results and Discussion*

The median HFD, HFD-HHT and SE values and their respective interquartile ranges for all EEG signals and for all electrodes combined are shown in Table 35.2. For both the control and MDD group, the interquartile range values were the greatest for the SE analysis in most of the bands, except for the full frequency band for both groups, where the HFD interquartile range was the highest. Furthermore, the interquartile range values were the lowest for the HFD analysis, except for the full frequency band for both groups and alpha frequency band for the control group.

There were no significant changes in median from the control group to the MDD group for any of the three analysis methods. There was an increase in median HFD value from the control to the MDD group in the full, alpha and beta frequency bands,

Table 35.2 Median and interquartile ranges for all denoized EEG signals and all electrodes

	Median values ± Interquartile range					
	HFD (control)	HFD (MDD)	HFD-HHT (control)	HFD-HHT (MDD)	SE (control)	SE (MDD)
Full	3.0332 ± 0.2415	3.1602 ± 0.2298	1.9593 ± 0.0136	1.9563 ± 0.0107	0.6933 ± 0.1360	0.6273 ± 0.1083
Delta	1.0034 ± 0.0005	1.0033 ± 0.0001	1.8915 ± 0.0214	1.8898 ± 0.0133	0.1813 ± 0.0131	0.1912 ± 0.0160
Theta	1.0164 ± 0.0004	1.0161 ± 0.0010	1.9841 ± 0.0010	1.9848 ± 0.0029	0.4980 ± 0.0155	0.4962 ± 0.0070
Alpha	1.0420 ± 0.0016	1.0424 ± 0.0010	1.9950 ± 0.0009	1.9953 ± 0.0011	0.5841 ± 0.0124	0.5856 ± 0.0092
Beta	1.1920 ± 0.0044	1.1954 ± 0.0057	1.9979 ± 0.0059	1.9972 ± 0.0058	0.6812 ± 0.0174	0.6835 ± 0.0074

Table 35.3 Changes in HFD values for EEG from control and MDD group

Electrode	Fz	Pz	C3	T3	C4	T4	Fp1	Fp2	F3	F4	F7	F8	P3	P4	T5	T6	O1	O2
Full	↑	↑	↑	↑	↑	↑	↑	↓	↓	↓	↑	↑	↓	↑	↑	↑	↓	↓
Delta	↑	↓	↑	↑	↑	↓	↓	↓	↓	↓	↓	↓*	↑	↑	↓	↑	↓	↓*
Theta	↑	↓	↓	↓	↑	↓	↓	↓	↓	↑	↑	↓	↑	↑	↑	↓	↓	↓
Alpha	↑	↑	↑	↑	↓	↑	↑	↑	↓	↑	↑	↑	↑	↑	↑	↑	↑	↑
Beta	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↓	↑	↑	↑

“*” denotes a significant difference

Table 35.4 Changes in SE values for EEG from control and MDD group

Electrode	Fz	Pz	C3	T3	C4	T4	Fp1	Fp2	F3	F4	F7	F8	P3	P4	T5	T6	O1	O2
Full	↑	↑	↑	↑	↓	↓	↑	↑	↑	↓	↓	↑	↓	↓	↓	↓	↓	↓
Delta	↓	↑	↑	↑	↓	↑	↑	↑	↓	↓	↑	↑	↓	↑	↑	↑	↑	↑
Theta	↓	↑	↑	↑	↓*	↓	↑	↑	↑	↓	↓	↑	↓	↓	↓	↓	↓	↑
Alpha	↑	↑	↑	↑	↓	↑	↑	↑	↑	↑	↓	↑*	↑	↑	↑	↑	↑	↑
Beta	↑	↑	↑	↑	↓	↑	↑	↑	↑	↓	↑	↑	↑	↑	↑*	↑	↑*	↑*

“*” denotes a significant difference

which is consistent with previous studies [7, 44, 57, 73, 74]. Furthermore, there was an increase in median SE value from the control to MDD group in the alpha and beta frequency bands. However, the median SE value decreased from the control to the MDD group in the full frequency band which is in contrast to previous studies [7, 44]. The median HFD-HHT value decreased for the MDD group compared to the control for all frequency bands except theta.

Table 35.3 shows a significant decrease in HFD values for electrodes F8 and O2 in the delta frequency band. The HFD values increased for electrodes Fz, Pz, C3, T3, C4, T4, Fp1, F7, F8 P4, T5 T6. Furthermore, there was an increase in HFD values for all electrodes in the beta and alpha frequency bands, except for electrodes C4 and F3 in the alpha band and T5 in the beta band. Previous studies have shown an increase in HFD for all 18 electrodes in the beta, gamma and full frequency bands [7, 44, 73, 74].

Table 35.4 shows a significant increase in SE values for electrodes T5, O1 and O2 in the beta frequency band and for electrodes F8 in the alpha frequency band. There was a significant decrease in SE values for C4 electrode in the theta band. Table 35.4 did not show increase in all the electrodes, which is in contrast to previous which have shown a significant increase in SE values for all electrodes [7, 44]. Moreover, SE values showed a decrease only in electrodes C4 and F7 in the alpha band and a decrease only in electrodes C4 and F4 in the beta band.

Table 35.5 shows the complexity in the frequency–time domain with a significant decrease in HFD-HHT values for the Fz electrode in the full frequency band. Furthermore, Table 35.5 shows an increase only for electrodes C4 and T4 in the full frequency band and for electrodes F4, T5 and T6 in the alpha band. This is in

Table 35.5 Changes in HFD-HHT values for EEG from control and MDD group

Electrode	Fz	Pz	C3	T3	C4	T4	Fp1	Fp2	F3	F4	F7	F8	P3	P4	T5	T6	O1	O2
Full	↓*	↓	↓	↓	↑	↑	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
Delta	↑	↓	↓	↓	↑	↓	↓	↓	↓	↓	↓	↓	↓	↓	↑	↓	↓	↓
Theta	↑	↓	↑	↑	↓	↓	↑	↑	↑	↑	↑	↓	↑	↓	↑	↓	↓	↑
Alpha	↓	↓	↓	↓	↓	↓	↓	↓	↓	↑	↓	↓	↓	↓	↑	↑	↓	↓
Beta	↓	↑	↓	↑	↑	↑	↓	↓	↓	↓	↓	↓	↑	↓	↑	↓	↓	↑

“*” denotes a significant difference

contrast to the HFD and SE values, which mainly showed increase in complexity for the full, alpha and beta frequency bands.

The complexity in the time domain calculated using SE and HFD analysis show similar trends in the alpha and beta frequency bands where there was an increase in complexity for the majority of electrodes. In contrast, HFD-HTT shows a decrease in complexity in the frequency–time domain analysis for majority of electrodes in the full, alpha and beta frequency bands. Moreover, there was an overall increase in median HFD and SE values of EEG in the frontal region of the brain in the full, alpha and beta frequency bands. However, there was a decrease in the overall median HFD and SE values of EEG in the frontal region of the brain for the full, alpha and beta frequency bands.

Our study is in line with some of the previous findings. However, there was varying results obtained that could be due to the difference in $k_{\max}$ value, EEG signal length and pre-processing method used for analysis. Many studies used a $k_{\max}$ value of 8 [7, 44]. Bachmann et al. (2013) used a $k_{\max}$ value of 50 in their study [74]. HFD is sensitive to the tuning parameter $k_{\max}$ and given that the fractality of EEG signals are not the same, we have chosen a dynamic approach to select the appropriate $k_{\max}$ value. Furthermore, Cukic et al. (2019) and Cukic et al. (2020) have used a 5-second EEG signal for fractal and sample entropy analysis [7, 44]. Ahmadlou et al. (2012) used 1-minute EEG signals, Akar et al. (2015) used 3-minute EEG signals and Bachmann et al. (2013) used 5-minute EEG signals [57, 73, 74]. Unlike SE analysis being insensitive to signal length, the HFD analysis is sensitive to signal length. In addition, one possible limitation to our study could be a high electrode impedance. Previous studies have kept their electrode impedances below either 5 or 10 k Ω [7, 44, 73, 74]. Moreover, previous studies have used Butterworth filter, independent component analysis or visually selected a portion of EEG with less artifacts [7, 44, 73, 74].

The underlying pathogenesis of depression is still not known [44]. Our study has shown an increase in EEG complexity using HFD and SE values for most electrodes with some electrodes even showing a significant increase for the full, alpha and beta frequency bands. Furthermore, our study showed an overall increase in EEG complexity using the HFD and SE values in the frontal region of the brain for the full, alpha and beta frequency bands. This is in line with previous findings that used quantitative EEG measures, connectivity measures, vigilance-based measures and sleep-related EEG measures and has shown abnormal frontal asymmetry in

alpha band, increase alpha activity, increase beta activity and decrease in alpha synchronization in the right fronto-central connections [44, 56, 73, 84–89]. The alpha and beta activity of EEG are a major characteristic of an individual's brain because these bands are indicated to be involved with higher cognitive functions, selective attention and cognitive coordination [73, 90]. Therefore, irregularity as shown by increase in complexity in these sub-bands might represent cognitive dysfunction in MDD patients [73]. Furthermore, some neuroimaging-based studies have shown a decrease in white matter and an increase in grey matter in the frontal brain of MDD patients which could explain the changes in brain dynamics in the frontal region [91–93]. In addition, our study has shown a decrease in complexity in the frequency–time domain for the full, alpha and beta frequency bands, which might indicate that even though there is an irregularity in the EEG signal, the frequency changes are more stable in MDD patients.

35.7 Conclusion

Advances in technology has allowed the introduction of various machine learning techniques for the classification of ADHD, MDD and other psychiatric disorders from healthy individuals [7, 11, 13, 30, 57, 94–97]. There is increasing evidence that use of nonlinear methods of provide significant advantages in analysing EEG signals over linear methods of analysis [44, 57, 69, 98–100]. In addition to HFD and SE analysis, our study has shown that complexity in the frequency-time domain using HFD-HHT can be a vital feature in EEG analysis which could be incorporated in POCT systems to allow quick diagnosis and to assess effectiveness of treatment in real time.

Our study has shown increase in HFD and SE values in the full, alpha and beta frequency bands but a decrease in HFD-HHT values in those three bands for majority of electrodes. This could suggest that although there is irregularity in EEG signal in the time domain but a decrease in irregularity in the frequency-time domain. To our knowledge, this was the first study to combined three methods of complexity measures for EEG analysis in depression. Furthermore, this was the first study to also use EMD as a pre-processing method, which has proven to be better for EEG due to its properties. One major limitation to our study was that the electrical impedance was higher than previous studies. This study used the 10–20 system for EEG analysis, but 128-channel EEG recordings could be done to allow better results.

A conclusion by a recent review of complexity measures in depression, MDD dynamics might “appear more complex but is actually more random” has a supported meaning in the theory of information that we know so little of what “complexity” means [101]. Everything living is producing electrical signals but we still have to understand what level of complexity of signals is considered to be healthy (the level of chaos inevitable to adapt and survive) or pathological (too high or too low). There are limited findings in research to support the understanding

of results [7]. EEG remains as the most accessible method for supporting decisions in clinical practice unlike fMRI which are far more expensive. Hence, EEG has the potential to be utilized in POCT systems for a variety of psychiatric disorders.

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Chapter 36

Advances in Understanding Fractals in Affective and Anxiety Disorders



Sergio Iglesias-Parro, Maria Felipa Soriano, and Antonio José Ibáñez-Molina

Abstract In this chapter, we review the research that has applied fractal measures to the study of the most common psychological disorders, that is, affective and anxiety disorders. Early studies focused on heart rate, but diverse measures have also been examined, from variations in subjective mood, or hand movements, to electroencephalogram or magnetoencephalogram data. In general, abnormal fractal dynamics in different physiological and behavioural outcomes have been observed in mental disorders. Despite the disparity of variables measured, fractal analysis has shown high sensitivity in discriminating patients from healthy controls. However, and because of this heterogeneity in measures, the results are not straightforward, and more studies are needed in this promising line.

Keywords Fractal · Affective disorder · Anxiety · Complexity

36.1 Introduction

It has recently been stated that chaos is inherent to natural systems [1]. For example, a healthy heart shows some form of chaotic rate, whereas low variability in the heart rate is a pathological sign [2]. Chaos has the characteristic of appearing random and unpredictable at certain scales but retains a highly defined and overall predictable form. Chaos and fractals are two central concepts within nonlinear dynamics. Nonlinear approaches can provide valuable insights into complex human behaviours and, subsequently, into dysfunctions of human behaviours, such as mental disorders.

In this chapter, we will review the research that has applied fractal measures to the study of the most common mental disorders, that is, affective and anxiety

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disorders. Early studies focused on heart rate, but diverse variables have also been examined, from subjective mood, to hand movements. Recent studies have mostly implemented fractal measures to neurophysiological data, such as electroencephalogram (EEG) or magnetoencephalogram (MEG). In general, there is growing evidence about abnormal fractal dynamics in different physiological and behavioural measures in mental disorders [3].

36.2 Fractals and Affective Disorders

Affective disorders include mainly depressive (unipolar) disorders and bipolar disorders. Depression is one of the most prevalent mental diseases and the first cause of major disability in the world, according to the World Health Organisation. People suffering from depression experience low mood but also loss of interest and pleasure, poor concentration, a feeling of worthlessness, and, occasionally, self-harm ideation that can lead to suicide. Patients with bipolar disorders shift between depressive episodes and manic episodes when they feel unusually euphoric, full of energy, and show an inflated self-esteem.

Some authors have proposed that patients with affective disorders suffer an alteration of the fractal organisation of their biological rhythms [4]. Among these biological rhythms, heart rate, gait temporal patterns, or even mood temporal fluctuations have been studied. For example, Kojima et al. [5] have shown that patients with depression have reduced heart rate variability and low fractal dynamics of heart rate. Decreased heart rate variability may be one of the causes of increased cardiovascular mortality in depressed patients. Comparable results have been obtained for patients with bipolar disorder [6]. These authors found that the heart rate's regularity (reflected in a decrease in entropy) correlated with the severity of depressive and anxiety symptoms.

Fractal analyses have also been applied to movement patterns in depression. Aybek et al. [4] evaluated the stride interval in human gait in ten depressed patients and ten healthy controls. The temporal organisation of the stride interval is thought to be neither random nor regular but to have a complex fractal-like structure characterised by self-similarity over multiple time scales. The authors compared stride patterns in patients with major depression and healthy participants through a system of sensors attached to each wrist. They found that fluctuations of activity in both hands displayed a fractal pattern in healthy controls. Importantly, the stride pattern of depressed participants showed higher fractal values than controls. These results point to an alteration of the fractal organisation of motor activity in depression. Moreover, the nonlinear measures of movement patterns were similar in both groups; therefore, fractal analysis discriminates better between depressive and control groups. As we mentioned previously, the fractal organisation of biological rhythms is thought to correspond to an ideal homeostatic regulation and is more prone to respond to sudden changes than a regular organisation. Altered fractal

pattern of activity in depressed patients might render movements less able to adapt to change, and this could lead to an increased risk of fall [4].

This finding of higher fractal values in movement measures has been reported in patients with bipolar disorder [7]. In this research, the authors examined actigraphy data for 2 weeks from 106 patients with bipolar disorder, 73 siblings of patients, and 76 controls. They found that patients had larger fractal values (according to detrended fluctuation analysis, DFA) than controls, while siblings had intermediate values. Larger values indicate a higher regularity in motor fluctuations. Fractal values were not significantly associated with depressive or manic scores. As we mentioned before, a fractal organisation of movements has been found in healthy individuals, and it has been suggested that a disruption of this fractal regulation with age or diseases predicts frailty, disability and even mortality.

Another interesting line of research has applied nonlinear analyses to the study of variations of mood. In a pioneering work [8], Gottschalk et al. employed daily mood records from patients with bipolar disorder and healthy participants over a time course of 1–2.5 years in order to examine temporal fluctuations of mood in bipolar disorders. They evaluated the obtained time series to identify whether the temporal pattern of mood originated from a periodic, a random, or a deterministic source. Results indicated that the correlation dimension, a nonlinear measure, could account for self-rated mood in patients. Mood in patients was significantly more organised than mood in normal participants, and it could be characterised as a low-dimensional chaotic process rather than a random process. Following this line, Woyshville et al. [9] analysed the time course of self-rated mood in 36 patients with affective instability and 27 controls. They asked participants to rate their daily mood on a visual analogue scale for 90 days. Time series were analysed with the mean squared successive difference (MSSD), power spectral density (PSD), and fractal dimension (FD). The authors found that patients demonstrated substantially more variability than controls on the MSSD, but less complexity as measured by the FD. In sum, despite the chaotic and irregular mood variability of patients, affective variations were less complex, in patients than in controls.

These results were replicated in some subsequent studies [10]; however, they have been recently questioned [11]. Specifically, Moore et al. [11] examined mood time series from eight patients with bipolar disorder over a time period of 5 years. They concluded that nonlinear methods have no advantage over linear methods in accounting for variations in mood in bipolar disorders. Clearly, more studies, ideally with larger samples, are necessary to elucidate whether mood variations in bipolar patients are better described by linear or nonlinear analyses.

Previously described studies are based on the observation that some clinical features of affective disorders repeat over a range of time scales. Nonlinear approaches to sequential observational data (in this case, self-rated mood) can provide new insights into the underlying processes that generate the observable patterns.

Finally, most studies involving fractal analyses in affective disorders have focused on electroencephalographic signals (EEG). As the brain behaves as a complex nonlinear system, it has been proposed that nonlinear methods for analyses

of the EEG signal provide more reliable information about brain functioning, compared to linear methods. In this line, Klonowsky et al. [12] analysed the FD of EEGs from patients with affective disorders, with Higuchi's algorithm (HFD). They calculated FD-ratio, based on differences in FD between eyes-open EEGs and eyes-closed EEGs, and they found that FD-ratio was highly correlated with depressive symptoms. The following study [13] examined the FD of the EEG in bipolar patients with a manic episode and found that EEG FD was significantly increased in these patients relative to healthy controls. As we mentioned previously, Higuchi's FD is a measure of the complexity of a time series. As applied to EEGs, HFD seems to indicate the level of complexity with which brain regions function or interact, on a scale ranging from fully deterministic to fully random. Akar et al. [14] also found increased FD in EEG in patients with depression. They analysed two measures of EEG fractal organisation (HFD and Katz's fractal dimension, KFD) in 16 patients with major depressive disorder and 15 healthy controls. They detected increased complexity in the parietal and frontal regions of patients; and this was observed not only in full-band activity but also in beta and gamma sub-bands of EEG.

More recently, Cukic et al. [15] found that HFD values were also increased in patients who had recovered from a depressive episode. These authors evaluated resting-state EEG in 22 patients with recurrent depression and 20 healthy controls. Out of the 22 patients, 11 were in remission and 11 were suffering from a depressive episode. They found that HFD values were higher in patients than in controls; more interestingly, HFD was higher in patients who were in remission than in those in the acute episode. This finding indicates that the brain dynamics of patients in remission is different from the non-depressed individuals. This may be due to compensatory mechanisms, as the authors suggest: the reorganisation of circuits during depression can produce an increment in brain complexity.

In addition, FD values have demonstrated good accuracy in the classification of depressed and healthy individuals. In this line, Ahmadlou et al. [16] evaluated the complexity of EEGs from patients with major depressive disorders, with Katz's FD (KFD) and HFD. Results showed higher HFD in frontal lobes of the brain of patients compared to controls, in beta and gamma sub-bands. Additionally, the authors found an accuracy of 91.3% in the classification of patients and controls based on HFD of frontal beta sub-band, whereas KFD revealed no differences between frontal EEGs from depressive and control participants.

Since then, some efforts have been made to improve detection of people with depression on the basis of nonlinear features from the EEG. Thus, Hosseinifard et al. [17] analysed EEGs from 45 depressed patients and 45 normal participants, with different nonlinear measures (detrended fluctuation analysis, HFD, correlation dimension, and Lyapunov exponents). They found that the highest classification accuracy (83.3%) was obtained with the correlation dimension. However, when all nonlinear features were combined, a classification accuracy of 90% was reached. Similarly, Bachamn et al. [18] compared spectral asymmetry index and HFD for discrimination between depressive and control participants. They analysed eyes closed EEGs data and found an increase in HFD in depressed participants in all EEG channels. HFD provided the true detection rate of 94% in the depressive group and

76% in the control group. Cukic et al. [19] also found high classification rates for HFD and sample entropy of patients with depression and controls. The classification accuracy of HFD ran from 80.49% to 100% with different classifiers.

The finding of a greater complexity in the EEG from depressed patients has been replicated with other nonlinear non-fractal measures. For example, Li et al. [20] registered EEG from 62 schizophrenic patients, 48 depressed patients, and 26 controls, at rest and while performing mental arithmetic tasks. They analysed the complexity of the EEG signal with the Lempel-Ziv complexity (LZC) algorithm. LZC is a nonparametric metric that assesses the number of distinct substrings and their rate of recurrence along the time series, assigning higher values to more complex data. Obtained results showed that both groups of patients had a higher LZC than the control group. Surprisingly, depressed patients showed a higher LZC than patients with schizophrenia, both at rest and during the mental arithmetic task. A later study by Mendez et al. [21] provided further evidence. They applied LZC analysis to magnetoencephalography (MEG) data from 20 patients with a major depressive disorder and 19 healthy controls. They detected that LZC was higher in patients than in controls. More interestingly, complexity values in patients decreased after 6 months of pharmacological treatment; and this reduction in complexity with treatment correlated with the degree of clinical symptom remission.

In sum, evidence points to a greater complexity in the EEG of patients with depression. This is congruent with findings with other mental disorders, such as schizophrenia [22]. Although the interpretation of these findings should be cautious, one possible meaning of brain complexity is that it reflects the system's ability to adapt quickly and efficiently in a changing environment. According to this idea, affective disorders might be characterised by a poorer capacity (at different biological levels) to respond to environmental changes.

36.3 Fractals and Anxiety Disorders

Anxiety is a feeling that usually arises when a person feels threatened. It is characterised by feelings of worry, ruminations, and unpleasant physical sensations. Anxiety can be, at certain levels, useful in some situations. However, when this emotion is extremely intense, whereby it becomes the usual response in most inoffensive situations or it affects most domains in a person's daily life, it is then considered a disorder. The distinction between normal anxiety and a clinical disorder is not always clear. There are a wide variety of anxiety disorders, including generalised anxiety disorder, panic disorder, obsessive-compulsive disorder (OCD), posttraumatic stress disorder (PTSD), or phobic disorders, among others.

Although anxiety disorders are the most common mental disorders, literature on the fractal properties of its structural or functional states is scarce. Still, some evidence has found that estimators as the fractal dimension can contribute to the understanding of neural basis of anxiety disorders.

One clear relationship between anxiety and fractality in the signals from the brain was provided by [23]. They explored the HFD of a number of participants while they were undertaking a simple auditory oddball task. In this task, participants are presented with sequences of two types of tones that they need to detect or just listen to passively. The authors registered the EEG of a large group of participants that were divided into two groups, neurotic and stable participants, according to the Eysenck Personality Questionnaire (EPQ). When the authors compared the HFD of the neurotic group with the HFD of the stable group, they did not find significant differences in the HFD of the basal EEG (before stimulation); however, and more interestingly, the HFD was higher for the neurotic group 0–100 ms after stimuli presentation. This result indicated that the structural properties of the EEGs in the case of neurotics were more complex only when they were processing the tones. In order to understand why complexity is higher for neurotics only during stimuli processing, we need to consider that, for all participants, the HFD after stimuli presentation is widely reduced in the time range of 0–400 ms. From a physiological point of view, it corresponds to a synchronic cortical response directed to process and become aware of the externally presented stimulus. Then, it can be inferred that neurotics, when compared with stable participants, tend to reduce the processing of external and salient events. Hence, it is reasonable to think that the characteristics of neuroticism as negative thoughts, anxiety, and anger tend to maintain a high level of EEG complexity due to a reduced external focus of attention.

Although Georgiev et al. [23] did not find fractal changes in the resting condition for participants with neurotic traits of personality, there is some evidence indicating that, indeed, the resting brain of patients with anxious symptoms exhibits distinct fractal properties. One example is provided by Chae et al. [24] in a study designed to measure the correlation dimension (D2) of patients with PTSD. In their study, Chae et al. [24] recorded the EEGs with an 82 s length from 16 electrodes during a resting-state period and found that D2 was significantly lower in the PTSD group than in the healthy group. Because this difference was consistent all across the scalp, it can be concluded that PTSD deviates the by-default processing of the waking brain towards a less complex functional regime. This decrement in complexity might reflect that cortical activation becomes simpler with recurrent evocation of traumatic stimuli. In other words, it might be possible that PTSD patients show a reduced by-default functional repertoire of the resting brain that mostly represents rehearsals of the same traumatic events.

In a recent study [25], Gentili et al. also tested the hypothesis that anxiety modulates the resting brain. In an fMRI study, they measured the Hurst exponent to healthy participants. Although there was no anxious stimulation during the experimental session, participants showed a positive correlation between the Hurst exponent and the low fluctuation of the BOLD (blood oxygen level dependent) activity in posterior cingulate/precuneus, inferior parietal sulci, and the parahippocampus areas of the brain. As these regions are mostly related with the by-default function of the brain, it can be suggested that social anxiety traits shape global brain activity and might trigger nonadaptive reactions in social situations.

A higher complexity has also been found in patients with obsessive-compulsive disorder (OCD) [26]. These authors evaluated 39 patients with OCD and 19 healthy controls at rest. They found that patients with OCD had higher FD values than controls in the frontal regions in all frequency bands. In addition, there was a significant positive correlation between FD values and the severity of obsessive symptoms.

Finally, Kawe et al. [27] explored FD of EEGs of University students at rest and related it with measures of anxiousness and depressivity from the Personality Inventory PID-5, aimed to identify pathological personality traits. They detected that posterior HFD (derived from averaged posterior channels), significantly predicted PID-5 anxiousness. Posterior HFD also correlated with depressivity, though this was a weaker correlation, and probably due to comorbidity between both traits.

From the already explained results, it might be concluded that anxiety has an influence in the complexity of the brain signals, as the EEGs, both during reactions to stimuli and the resting state. However, it is well known that conventional fractal analyses are not sensitive to time fluctuations in the complexity of the signal [28]. In many cases, different scaling exponents are required for different parts of the time series (e.g. [29]); that is, non-stationary signals, as, for example, EEGs, may have a non-single scaling in different segments of the series. Hence, non-stationary EEG series can be characterised as multifractals with a variety of scaling behaviours. As a consequence, it would be possible that the multifractality of brain signals from anxiety patients or participants with anxiety traits differ from those of the reference groups. In this line, Dik et al. [30] investigated the multifractality of the EEGs from patients with anxious phobic disorders. They explored changes in the EEG responses during psychogenic pain before and after a psychorelaxation treatment. Multifractal characteristics can be detected in time series by estimating local changes of the scaling properties in short segments of the signal. Hence, it is sensitive to structural changes in time that corresponds to different fractal properties. In this study, the degree of multifractality was estimated to EEG recordings during perception of psychogenic pain in the case of patients, and tactile evoked pain in the case of healthy participants. Multifractality of the EEG during pain perception was different in both groups, suggesting that the cortical generation of pain in patients was different in space and time than the cortical pattern that gives rise to pain in healthy participants. Interestingly, when a psychorelaxation technique was applied to the clinical group, the width of the scaling behaviour was reduced to approach the values of the healthy group. In other words, after psychorelaxation, the range of exponents that characterise the multifractal structure of the EEG in the clinical group was similar to the one in the healthy group. This result suggests that anxiety enhances pain perception by increasing the rate of change in the cortical patterns of activation. This dynamical process is reflected in the EEGs as a multifractal with a wide range of scaling exponents.

It is noteworthy that anxiety symptoms are related not only with the characteristics of brain function; structural properties of the brain in patients with anxiety symptoms might present a reduced FD. In this line, Ha et al. [31] calculated the three-dimensional FD of cerebral cortical surface measured with MRI to patients

with OCD and schizophrenia. Their results indicated that schizophrenic patients exhibited lower FD than OCD patients and controls. Interestingly, OCD patients showed lower FD than controls indicating that their cortical surface was less folded than the healthy group. This is a very relevant finding, and we believe that its clinical implications should be explored in further experiments.

Finally, fractal organisation has been also studied in heart rate in relation to anxiety [32]. As we mentioned before, it is known that the seemingly random fluctuations of the interbeat intervals hide a fractal structure [33]. Recently, the heart rate FD from adolescents with different levels of anxiety has been explored. Specifically, 107 adolescents were classified into four groups: a stable high-anxiety group, a stable moderate-anxiety group, a stable low-anxiety group, and a non-stable anxiety group (a group with varying levels of anxiety at two temporal points, T1 and T2). Significant differences between groups in FD were obtained. The stable low-anxiety group showed the highest heartbeat FD, higher than the FD from the moderate-anxiety group, but, unexpectedly, not higher than the FD from the high-anxiety group. Therefore, the relationship between FD and anxiety was no linear: the moderate-anxiety group had lower FD than both the low-anxiety and high-anxiety groups. The non-stable anxiety group also showed medium values of heartbeat FD. These results suggest that there is a complex association between fractal dynamics of the heart beat and the stability and severity of anxiety.

36.4 Fractals in Affective and Anxiety Disorders Treatments

Treatment of mental disorders has taken many forms, from the use of electroconvulsive therapy to hypnosis or meditation through the use of psychotropic drugs. In this section, we will briefly review the studies that have addressed the study of mental disorders treatment from a nonlinear perspective.

The analysis of FD concerning the treatment of mental disorders dates back the beginning of the 90s. In 1996, [34] already pointed out the importance of analysing the EEG characteristics associated with electroconvulsive therapy (ECT) in order to predict its efficacy in the treatment of severe depression. Specifically, they found a significant positive relationship between the degree of postictal suppression and the remission depressive symptoms. Following that work, Gangadhar et al. [35], using the Katz algorithm, computed the FD for early-, mid-, and post-seizure phases of ictal EEG after 4 weeks of either once per week or three times per week of ECT treatment. Depression severity was measured by the 21 items of Hamilton Rating Scale for Depression. They found that, apart from the number of ECT sessions per week (one or three), the only significant predictor of clinical remission was the post-seizure FD of EEG. Specifically, Gangadhar et al. found lower values of FD during the period immediately following electrically induced seizure only in the participants who showed later remission. According to the authors, reduced post-seizure FD could reflect postictal suppression, which in turn could indicate a greater brain response.

In the study of nonlinear systems, transitions between chaotic dynamics to periodic ones (or vice versa) are called bifurcations. These bifurcations have been proposed as useful tools in modelling the transitions that occur from healthy to pathological states [36]. However, because the empirical study of transitions from a healthy brain to a pathological one is not always possible, some authors have considered the reverse transition, namely, the clinical improvement due to treatment [37], as a valid strategy to understand mental illness. In this vein, Thomasson et al. [37] recorded the electrical brain activity on 31 scalp regions in 3 clinically depressed participants and 1 healthy control. At each of the 31 EEG channels, local measures of entropy were estimated as well as a measure of global entropy for the global EEG averaged over the 31 channels. The participants' EEG was recorded every 2 days for 2 or 3 weeks. Additionally, in order to study fluctuations of mood, rating scales of depression were collected during treatment. Measures of global entropy decreased along treatment and showed a significant correlation with the score on global rating scales of depression. Regarding the healthy control participant, entropy increased along recording sessions and did not covary with self-assessments of depression.

Treatment of mental disorders often involves the use of psychotropic drugs. These substances seem to have a specific effect in brain functioning, their own "EEG portraits", according to Fingelkurts et al. [38]. Thus, classical spectral analysis shows that barbiturates induce bursts in EEG signal [39], whereas benzodiazepines seem to produce an increase in slow (delta and theta) and fast (beta) rhythms (although see [38]) while reducing power in the alpha range [40]. FD has also been studied in relation with psychotropic drugs. In this regard, Chouvarda et al. [41] analysed the FD dynamics after benzodiazepines (lorazepam) intake. Lorazepam is a benzodiazepine drug with short to medium duration of action. It has anxiolytic, sedative/hypnotic, anticonvulsant, and muscle relaxant effects by slowing down the central nervous system. Fourteen participants took 2.5 mg of verum/placebo, and their EEG was recorded at resting state, for 3 min each hour for 24 h. FD was significantly higher in almost all channels in the verum than in the placebo condition, although these differences diminished with time. Participants that received placebo did not exhibit any significant variation in FD values with time. A follow-up study with the same dataset [42] compared the effect of lorazepam across frequency bands on Higuchi FD [43] 1 h before and 2 h after drug intake. Results showed that FD significantly increased 2 h after drug intake compared to baseline. Moreover, alpha band after drug intake was significantly lower than at baseline, but beta band was significantly higher after drug intake than at baseline. The authors proposed that, since beta band contains high-frequency components, its increase contributes to the complexity of the EEG signal, which in turn leads to a subsequent increase in FD.

This increase in FD after treatment is consistent with results from Cukic et al. [15], who detected that FD was higher in depressed patients who were in remission than in those who were in an acute depressive episode and higher also than in healthy controls.

On the contrary, in a study conducted by Mendez et al. [21], brain complexity was reduced after pharmacological treatment. These authors analysed the complex brain pattern associated to antidepressant intake. Twenty participants, moderately to severely depressed, completed a minimum 3-week medication washout before the first baseline pre-treatment MEG scan. After the first scan, participants begun antidepressant intake (mirtazapine, 30 mg once per day). After 6 months of treatment, participants received a second resting-state 5 min MEG scan. Depression improvement was evaluated with the Hamilton Rating Scale for Depression. After treatment, a significant reduction in the depression questionnaire scores was observed. When baseline MEG was compared with posttreatment MEG scan, a significant reduction in Lempel-Ziv complexity in anterior regions was observed, but only for young patients (<47 years). These results indicating a reduction in brain complexity following treatment of depression are in accordance with the aforementioned results obtained by [37]. Lebiecka et al. [44] also found a reduction in EEG complexity after treatment. These authors examined HFD of EEG in patients with treatment-resistant depression who were treated with transcranial magnetic stimulation (rTMS). In those patients who improved after rTMS, EEG complexity was significantly reduced.

Hypnosis and meditation procedures are often successfully used to facilitate treatments in stress-dependent disorders, such as anxiety, depression, or dependencies, or pain relief in patients with hypertensive disorders, burn injury, arthritis, cancer, or chronic back problems [45, 46]. Hypnosis is an altered state of consciousness that is considered to involve high attentional resources. Various attempts have been made in order to characterise the therapeutic effect of hypnosis. In this vein, Solhjoo et al. [47] analysed the EEG recordings of five highly hypnotisable participants, according to the Stanford Hypnotic Clinical Scale. Participants were asked to engage in two experimental tasks (relaxation and imagination) during normal and hypnotic states. In each condition, 1 m length EEG segments were obtained. Each segment was then divided into 12 segments of 5 s length each. Frequency (power spectral density components) and chaotic (Petrosian and Higuchi FD) features were extracted from segments and submitted to a Hidden Markov Model classifier. All measures discriminated between mental states, but the non-linear ones (specifically, the Higuchi FD) showed a better behaviour of classifying mental states than frequency-based features.

Yargholi and Nasrabadi [48] conducted a study in order to find out the impact of the depth of hypnosis on EEG signals recorded, while the individuals were performing mental tasks. EEG signals from 33 participants were recorded in an eyes-closed relaxation state and under hypnosis, and Higuchi FD was obtained for each recording channel. During hypnosis, participants had to perform different tasks, and this performance was used as an index of the depth of hypnosis. According to their performance, participants were classified into three groups: low, medium and highly hypnotisable. The results of the study showed that chaotic characteristics of some channels could serve to discriminate among participants depending on their hypnotisability.

Meditation has been proposed as a technique to increase focused internalised attention. Empirical evidence has supported the efficacy of meditation in the treatment of mental disorders [49]. Research has tried to characterise the brain signature of meditation from a nonlinear point of view. Thus, Aftanas and Golocheikine [50] studied how the dynamically changing inner experience during meditation could be indexed by the nonlinear dimensional complexity of EEG, compared with a more traditional frequency-based analysis. EEG was recorded at rest with eyes closed and during meditation in 20 experienced meditators. The EEG was segmented in 8.192 s. length epochs, and frequency band power analysis and nonlinear analysis (point correlation dimension) were conducted. Obtained results revealed a significant decrease of point correlation dimension over anterior-frontal and centro-frontal regions, suggesting that controlled allocation of attention resources was required for the maintenance of the meditative state. On the other hand, linear measures analysis revealed theta power increase over anterior electrodes and alpha power decrease in meditation state when compared with baseline.

Another study compared complexity measures of the EEG signal in experienced meditators before and during meditation [51]. Each 1 of the 22 participants underwent 5 meditation phases that were monitored using an EEG scanner. Among the five experimental phases, participants underwent a 10 m baseline without meditation, a 30 m of calming meditation, and a 30 m phase of insight meditation. FD and permutation entropy were obtained from EEG. For both types of meditation, global permutation entropy decreased, although not significantly. In contrast, the FD for both types of meditation increased. This increase was significant only in calming meditation.

36.5 Conclusions

Fractal analysis can be a valuable tool in the evaluation of mental diseases or treatment outcomes. As we have reviewed, mental complexity reflects a system's ability to adapt to a constantly changing environment. Such adaptation capacity is often impaired in patients with mental illnesses.

The studies that we have reviewed suggest that fractal dimension is increased in brain signals of patients with an affective disorder. Moreover, successful antidepressant treatments (such as ECT or antidepressant drugs) usually diminish fractal dimension restoring it to healthy levels, although some studies have found increased EEG complexity after treatment, or in patients in remission. Studies have also shown an altered fractal dynamic in the heart rate, subjective mood variability, or even the temporal pattern of movements, in affective patients. Fewer studies have focused on patients with anxiety disorders, and the evidence is not so clear. It seems that the fractal dimension of the EEG is also increased in patients with anxiety disorders.

Therefore, complexity (and fractal dimension) can be up or down in affective and anxiety disorders. This divergence of results can be observed in other mental disorders; for example, both increased and decreased complexities in the EEGs

from schizophrenic patients have been reported. The inconsistency can result from different sources. First, the biological signal evaluated (ECG, EEG, daily self-rated mood) and complexity measures vary considerably across studies. Second, mental disorders are heterogeneous in their clinical presentation and in the pathological mechanisms responsible for their apparition. Finally, as Takahashi highlighted [52], age-related complexity changes must be considered when investigating pathological brain because typical brain complexity is not constant across the vital cycle. Evidence points out clearly that in different mental disorders, the normal organisation that characterises healthy functioning deteriorates and is replaced by abnormal dynamics.

On the other hand, the interpretation of the scarce results we have summed is not straightforward. We believe that an interesting interpretation about complexity and mental illness is the one put forward by Yan and Tsai [3]. According to these authors (see Fig. 36.1), healthy mental function is complex in an optimal degree, and it can deteriorate into two distinct pathological paths: order and randomness. For example, patients with mania exhibit behaviours that seem to have extreme randomness (word salad, loosening of associations, impulsive behaviour), while patients with obsessive-compulsive disorder perform actions that seem extremely rigid and repetitive. These macroscopic observations of psychopathology should be linked to microscopic phenomenon, such as heart or brain activity. The brain's ability to adapt to the environment depends on the underlying neuronal functioning, which can be measured by complexity analysis at the microscopic level. The authors propose that mental illness is characterised by a loss of brain complexity. This brain dysfunction gives rise to altered dynamics in cognition, emotion, and behaviour, where we can observe either extremely ordered or extremely random patterns.

This is consistent with results from the mentioned studies, where both increased and reduced fractal organisation have been found. Health is probably linked to medium values of fractal dynamics, whereas probably both low and high values of FD are linked to pathological states.

Future research should evaluate whether it is possible to relate pathological behaviours with microscopic changes in the neuronal dynamics of the brain. Furthermore, more studies are needed about the relationship between complexity measures and symptomatology. Implications of this line of research are of great relevance in clinical practice. Nowadays, the diagnosis of a mental illness is based on the subjective appreciation of the clinician; despite numerous efforts, objective markers of mental diagnoses have not been found so far. The evaluation of the treatment effects also relies on the subjectivity of the clinician. Fractal measures could be a useful tool in the diagnosis, outcome prediction, and treatment evaluation in affective and anxiety disorders.

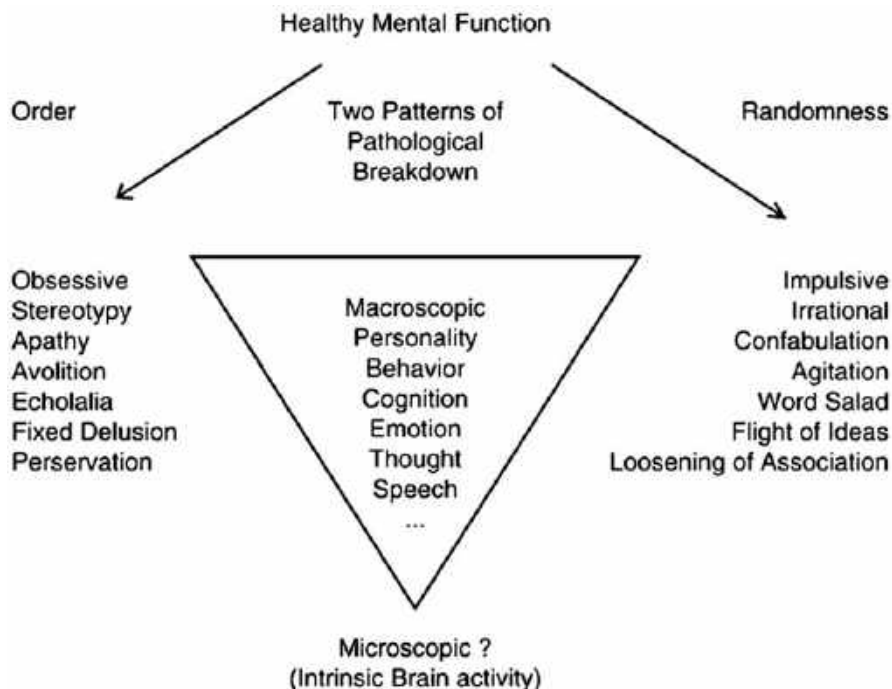


Fig. 36.1 Schematic illustration of the deterioration of healthy mental function into either an ordered or random pattern. (Adapted from Yang and Tsai [3] with permission)

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Chapter 37

EEG Complexity Analysis of Brain States, Tasks and ASD Risk



Stephen S. Wolfson, Ian Kirk, Karen Waldie, and Chris King

Abstract Autism spectrum disorder is an increasingly prevalent and debilitating neurodevelopmental condition and an electroencephalogram (EEG) diagnostic challenge. Despite large amounts of electrophysiological research over many decades, an EEG biomarker for autism spectrum disorder (ASD) has not been found. We hypothesized that reductions in complex dynamical system behaviour in the human central nervous system as part of the macroscale neuronal function during cognitive processes might be detectable in whole EEG for higher-risk ASD adults. In three studies, we compared the medians of correlation dimension, largest Lyapunov exponent, Higuchi's fractal dimension, multiscale entropy, multifractal detrended fluctuation analysis and Kolmogorov complexity during resting, cognitive and social skill tasks in 20 EEG channels of 39 adults over a range of ASD risk. We found heterogeneous complexity distribution with clusters of hierarchical sequences pointing to potential cognitive processing differences, but no clear distinction based on ASD risk. We suggest that there is indication of statistically significant differences between complexity measures of brain states and tasks. Though replication of our studies is needed with a larger sample, we believe that our electrophysiological and analytic approach has potential as a biomarker for earlier ASD diagnosis.

Keywords Fractal · Psychology · EEG · Complexity · Autism

37.1 Introduction

Despite decades of pioneering research across scientific disciplines, nonlinear complexity analysis remains largely unknown to much of the scientific community. In computational cognitive neuroscience, scientific discovery utilizing complexity analysis is an unmet challenge, made by pioneering neuroscientists half a century

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ago. Fortunately, recent advances in computational power, applied to complexity analysis, can reveal neurophysiological changes linked to specific neuro-functions, advancing understanding of the information-processing features of electro-neuro-physiology. These novel techniques require a “back to the future” approach, retooling complexity analysis techniques for higher computational power.

Human EEG technology has progressed from galvanic cortical measurements to modern high-density acquisition with digital processing. Clinical EEG research focused on organic brain disorders, while broader research expanded cognitive neuroscience. From the outset, European and American research focused on frequency oscillations in an effort to correlate them to specific brain activity for spontaneous and evoked responses. Interest in the frequency, timing and strength of the oscillations motivated researchers to improve instruments to more accurately measure the brain’s electrophysiology [148].

The dense multi-scale structure of the human nervous system [67] creates complex electromagnetic low-energy signals measured by EEG [17] inspiring vast amounts of research [148]. Oscillations have long been linked to levels of consciousness and brain regions [16] until challenged by full-band EEG [159]. 21 channel 10–20 [9] and high-density systems position scalp electrodes to measure EEG [42] referenced to a single electrode, or to an accumulative average in relation to the others [168].

Sensitivity is usually 100 microvolts, with “artefacts” and noise [76] without impedance loss, filtered with linear algorithms [158]. Traditional linear analysis methods convert to a stationary signal or apply stochastic methods [51]. But they don’t capture properties such as autocorrelation, which reveals the signal dynamics [134]. This is because linear methods, such as power spectrum analyses, remove the dynamic information in order to remove noise that may be the underlying deterministic processes. Work by Lorenz [102], Li and Yorke [98] revealed how some simple equations can give rise to complex and unpredictable outputs. Further work by Grassberger and Procaccia [60] created a technique to quantify the “strangeness”, or “chaos” of the system.

37.2 Nonlinear Complexity of EEG

B. Mandelbrot [107] revealed natural systems have “fractal” dimensions (FD), thus irregular self-similarity EEG features are not noise [78] and can be represented as the *density*, measured by the FD [11, 69, 84, 114] as well as the *uncertainty*, as measured by the largest Lyapunov exponent (LLE) [50], Kolmogorov entropy (K_E) [5], Lempel-Ziv complexity (LZc) [12, 174], multiscale entropy (MSE) [33] and multifractal detrended fluctuation analysis (MFDFA) [81] among others.

Complex dynamical systems (CDS) evolve exponentially or geometrically, can suddenly change energy states, are sensitive to their initial conditions to amplify small changes, have limited predictability, may be plotted using ordinary differential equations and are deterministic. A system is unpredictable, i.e. chaotic or fractal and

not random if it has a correlation dimension (CD) measured by summing adjacent points within a threshold value and dividing by that value logarithmically: where the x values are calculated within a set distance r for all values i and j to N with θ as the Heaviside function revealing the slope of $\log C(r)$ versus $\log r$ [11]:

$$C(r) = \frac{1}{N^2} \sum_{i,j=1}^N \theta(r - |x_1 - x_2|)$$

The LLE quantifies predictability by measuring the divergence of the trajectories of elements: where $\sum_i t_i \lambda_i$ is the sum of the natural log ratios of each distance between the trajectories and λ becomes the limit of that ratio over time [50]:

$$\lambda = \frac{1}{t} \sum_i t_i \lambda_i$$

Reconstructing a strange attractor manifold [118, 134] by the lag τ of a single time series reveals the Lyapunov Exponents [147]. To describe the accurate embedding dimension by Takens's [150] Theorem, a 'false nearest neighbors' [1] technique separates vectors which appear to be nearby trajectories but are not. The lag value τ can also be determined based on autocorrelation or mutual information [49, 161], which optimally represents mutual predictability between the time series.

As a line cut into equal thirds is three copies in one dimension or 3^1 , a plane cut into equal thirds on two dimensions (edges) is nine copies, or 3^2 , and a cube cut equally on each dimension (sides) is 27 copies, or 3^3 , this relationship gives the dimension by making N (number of identical copies) equal to r (the number of cuts) to the dimension (number of edges, sides, etc.) or $N(27) = r(3)^D$. This equation can be converted to create the Minkowski–Bouligand dimension [160]: $D(3) = \frac{\ln \ln N(27)}{\ln \ln r(3)}$, to calculate the FD.

A cube is not a fractal as its dimension is an integer, unlike the Sierpinski sieve, created by a recursive formula with 3 copies made by only 2 deletions, revealing $D = \frac{\ln \ln 3}{\ln \ln 2} = 1.5849 \dots$ a non-integer. Fractal shapes created by simple algorithms are self-similar, have no particular scale, can iterate indefinitely to completely fill the plane without intersecting and cannot be algebraically defined. Natural fractals are merely approximately self-similar objects or processes, as in EEG, seen by their spectrum of frequency variation, statistically correlated and modeled. The EEG frequency spectrum is fractal not random and has a tailed shape with a $\frac{1}{f^\beta}$ distribution (f for frequency and β for the distribution exponent) with Brownian "noise" modeled by $\frac{1}{f^2}$ and pink "noise" by $\frac{1}{f}$ such that the distribution does not change regardless of scale.

37.3 Nonlinear Complexity EEG Measurement Methods

EEG signals are too complex to be accurately modeled, but their fractal characteristics such as the Hurst exponent (H), Lyapunov exponent and autocorrelations reveal valuable information, since H measures the rate at which the data points in a time series auto-correlate inversely to their lag, known as ‘the Noah and Joseph Effects’ [111]. We can benchmark fractal measures such as with the Weierstrass cosine function (WCF) [162] cross-referencing the geometric measures, such as the Minkowski–Bouligand or Hausdorff dimension due to its scale-free self-similarity [72], to confirm the Weierstrass cosine function FD is 1.5^1 [45].

Hydrology models [110] and theoretical physics [41] have enabled the development of complexity analysis of EEG [125] due to the complex, irregular, seemingly random, self-similar and scale-independent nature of the signal [108]. Complex time series are also self-affine (similar but distorted), revealing both fractal and non-fractal characteristics [109] as it applies to EEG [26, 90, 103, 156, 164].

Current EEG analysis overcomes “noisy” signal deterministic chaos, introducing error by overprocessing waveforms [91], while complexity analysis instead measures the fractal, information and correlation dimensions [68]. For example, to model chaotic functions, we quantify the entropy of EEG based on the Rényi [130] formula:

$$Sq = \frac{1}{q-1} \log \log \sum_{i=1}^N p_i^q$$

with q representing the density moment of the probability p_i , the FD is:

$$D_0 = -\frac{\log \log N}{\log \log \delta v},$$

with the CD as $q = 2$ and the information dimension is:

$$D_1 = \frac{\log P_{\max}}{\log \delta V}$$

such that for a given time series the function D_q , corresponding to the probability distribution, is the fractal spectrum [92].

Fractal analysis of EEG’s complex, nonlinear, non-stationary signal reveals pathophysiology. Babloyantz, Salazar, and Nicolis [11] pioneered FA research using the CD, LLE and phase portraits to distinguish sleep stages. Lutzenberger et al. [103] compared the CD with the Fourier analyses and found higher FD for mental image tasks by scalp distribution. Cabukovski, Rudolf and Mahmood [24], after

¹ The value of the Hausdorff dimension of WCF was mathematically proven by Shen [144].

benchmarking, applied Higuchi's FD to distinguish epileptic discharges as having a lower FD.

Nonlinear complexity techniques can lead to contradictory ideas, such as between low-dimensional chaotic dynamics and stochastic or high-dimensional dynamics [26]. Geometrical approaches use spatial representations in contrast to information theory analysis based on probabilities. CD represents the set of all points that are closer together than a minimum distance, which, when that distance r gets closer to zero and the points N closer to infinity, can be represented as a log ratio or a recurrence plot and is the dimension of the signal. The CD is based on the correlation integral defined as:

$$C(r) = \frac{1}{N^2} \sum_{i,j=1}^N \theta(r - |X_i - X_j|)$$

where $\theta(x)$ is the Heaviside step function [59]. It is a simple way to calculate the dimension of the dynamical process of the attractor of a chaotic system with efficient modifications [88]. Using Takens's [150] theorem, converting the time series (μV in EEG) creates an m -dimensional vector by lagging based on τ , creating spatial information [19] as a subset of the attractor in the same dimension [154].

CD has determined the sleep stages but negatively correlated with the fractal exponent β via coarse graining spectral analysis (CGSA) [122]. It has discriminated mental tasks, with mental load correlating to higher FD [93]. Takahashi [149] found CD as the most frequent non-linear measure used for Alzheimer's disease (AD), schizophrenia and ASD [149]. Fernandez, Gomez, Hornero, and Lopez-Ibor [48] concurred but with mixed results.

The Lyapunov exponent (L_E) is the rate at which data points, representing the state of the system in time, change by moving away from or towards each other, and applying them to strange attractors, was first explained by Frederickson, Kaplan, Yorke, and Yorke [50], applied to a time series by Wolf, Swift, Swinney, and Vastano [166] and further refined by Brown, Bryant and Abarbanel [22] and Abarbanel, Brown and Kennel [2]. LLE EEG analysis was improved with scale-dependency [54] and as with Kolmogorov entropy (K_E) is a reliable measure of emotion [5], epilepsy and seizure detection [56, 157], brain computer interfacing [15] and distinguishing aspects of schizophrenia [48].

The Hausdorff dimension, D_H , from the following formula, where D_H is the Hausdorff dimension, r is the size of the measuring unit and N is the number of units [52, 66]:

$$D_H = \frac{\log \log N}{\log \log \left(\frac{1}{r} \right)}$$

Several methods evolved from this approach, such as by Katz [84], Petrosian [123], Raghavendra et al. [126], Sevcik [140], Fuss [52] and Higuchi [69]. Some methods are reportedly less reliable, notably Katz's method (KFD) [27, 52], due

to mixing of the units of measure (seconds and μV). Higuchi's fractal dimension (HFD) method is reported to be reliable and is widely used [85], though it is recommended to be applied with other measures due to certain sensitivity limitations [52]. Higuchi's formula is as follows, with k is the time interval, m is the initial time and $L(k)$ is a computed length of the curve [85]:

$$L(k) = \frac{\sum_{m=1}^k L_m(k)}{k}$$

HFD has been reported as superior for rapid assessment of EEG during sleep, cognition, anaesthesia, brain injury, epilepsy and mental disorder studies [85] as it measures fractal curves, statistically self-similar across scales and following a power law where the curve is fractal if the average value over the time interval sets is proportional to the time interval, exponentially to the power law [69].

Single-scale entropy measures may not correlate with dynamic complexity so multiscale entropy (MSE) measures were devised [32], but the information's mean rate (Kolmogorov-Sinai entropy) measurement is limited by sensitivity to noise [132]. MSE calculates sample entropies (SE) and then plots the scale factors S_E , with negative logarithmic conditional probability remaining similar. S_E 's displays multiscale changes from physiological generators and has been tested on surrogate signals and then applied to physiological signals to successfully correlate pathology to lower MSE values [34].

Multifractal detrended fluctuation analysis (MFDFA) relates to multifractal formalism [81]. It revealed noncoding DNA sequences had long-range correlations [120] and was then applied to heart rate dynamics, neuron spiking, human gait, weather, cloud structure, economics and physics [80]. Detrended fluctuation analysis avoids spurious correlations by determining statistical self-affine properties with long-range correlations [113] by calculating root-mean-square (RMS) fluctuations with time windows, as a cumulative sum, and detrending [121].

MFDFA measures monofractal and multifractal signals by converting to a random walk sequence, taking the detrended local RMS fluctuation and computing an overall RMS to distinguish fluctuation speed. Finally, by deriving an overall H and taking the q -order extension of the power-law relation of the q -order RMS H is revealed, allowing the multifractal spectra to be computed [75]. MFDFA reveals fractal variation in EEG likely to reveal pathologies as it has in heart disease [77], and EEG complexity in sleep stages [173], emotional states [106] and emotional assessment [29]. MFDFA application is expanding in various areas [167].

Kolmogorov complexity (K_C) is the ratio of a sequence to its estimated generative algorithm size [89]. Unlike Kolmogorov entropy (K_E) (the system's sensitivity to initial conditions), K_C measures signal chaos [31] as an estimate of the shortest code necessary [73]. Lempel and Ziv's [97] compression algorithm (LZ) has measured K_C complexity in EEG but with limitations [14]. Improvements include multiscale LZ [74] and permutation LZ [13]. LZ measures both K_C and Shannon entropy S_E [141, 174].

37.4 Evolving Applications of Nonlinear Complexity EEG Analysis

While FD was measured over short-time segments [124] decomposing EEG into frequencies revealed fractally distributed features scaled across dimensional ranges [115]. Measuring sleep states EEG $\frac{1}{f-\beta}$ with coarse graining spectral analysis calculated β as the FD, yet a negative linear correlation between CD/LLE and β was also seen, outside slow wave sleep fractal, non-chaotic EEG [122]. Anaesthesia monitoring with spectral frequency analysis, used high-order spectra, wavelet decomposition and other nonlinear complexity techniques [138]. Then DFA was successfully tied to anaesthesia depth replacing CD/LLE to address low-dimensional attractors, nonstationary, noisy signals and less data [57]. Lee et al. [96] were thus able to distinguish the hypnotic condition as fractally lower than other, conscious states.

ERP studies by Georgiev, Minchev, Christova, and Philipova [55] correlated personality type and gender with HFD, while Raghavendra et al. [126] found lower FD in young male neuroleptic naïve schizophrenics but increased bilateral temporo-occipital FD for those with positive symptoms. Still, Weiss et al. [163] identified Pereda et al. [122], Acharya et al. [3] and Ma et al. [104] had all used low sampling rates and limited channels.

Weiss et al. [163] estimated the H and a range of fractal spectra, finding less multifractality at deeper sleep levels, yet other studies confirmed the inverse relationship, with apnoea and hypopnea having higher FD states [171]. Results were not clear, with FD and Chaos seen as separate phenomena, inconsistent multifractality, lower FD correlated to lower activity and algorithms or parameters not necessarily the same across studies. Then, Allegrini et al. [8] posited a fractal critical point based on rapid transitions processes (RTP) [82]. Also, higher sensorimotor area (SMA) HFD for advanced musicians [62] with both regional functioning and networks displaying complex nonlinear dynamics was reported.

Further research on spatiotemporal fractal measures produced novel entropy analysis based on Rényi [130] entropy to detect emotion recognition and concentration levels [145]. Then Bornas et al. [18], applying KFD, showed a resting state global reduction with a self-focused cognitive emotion regulation style and a less adaptive negative affect. Zorick and Mandelkern [173] successfully applied MFDFA to sleep classification despite being less variable than the wavelet transform modulus maxima technique. And Harne [65] used HFD to distinguish lower FD after meditation.

Then Díaz et al. [38] estimated the H and FD during the abbreviated Raven test finding a more fractal and self-similar brain state emerging during cognitive load. In contrast, Maity et al. [106] used MFDFA and found increased alpha and theta multifractal spectral width during relaxing acoustical stimulation. Concurrently, Stokic et al. [146] saw increased FD with auditory-verbal short-term memory load in temporal and parietal regions, but a decrease in the frontal midline region.

Contrasting to linear measures with mixed frequency power spectral localization, they reported FD as a sensitive measure for dynamical change [146].

Evolving precision led to amplitude modification, where Fuss [53] increased sensitivity over the HFD without distortion to detect a startle response. Improved techniques, addressing varied results in the literature, came from reviewing the dynamical analysis of sleep EEG, by Ma et al. [105], who endorsed the utility of nonlinear measures. They distinguished the higher consistency of entropy measures, with limitations to clinical application for lack of well-designed studies due to variable preprocessing parameters independent from unknown neurophysiological mechanisms. Regardless, nonlinear sleep state research methodology improved with increased measure discrimination.

Improved methodology led to novel combinations of nonlinear complexity and linear measures, combined to discriminate brain states. Rakshit et al. [127] used MF DFA to detect decreased multifractality and increased monofractality in the frontal and parietal lobes during cognition tasks. Xin Li, Cai and Kang [100] used MF DFA to quantify mental stress with a higher Hurst value and broader multifractal spectrum. Concurrently, Ruiz-Padial and Ibanez-Molina [135] applied HFD to the EEG and DFA to the ECG to find inter-concordance that detected emotional response.

An expanding number of nonlinear complexity methods can measure features of resting states, sleep states, cognitive states and emotional states. Nonlinear complexity analysis now applies CD, HFD, KFD, DFA, MF DFA and entropy, often simultaneously adding to a richer understanding of the locality of brain functioning coming from frequency and power analysis. Thus, combined linear and nonlinear methods have increased our understanding of localized functioning, various networks and global functioning of the vast complexity of the human brain.

37.5 The Emergence of Nonlinear Complexity Biomarker Diagnostics

Nonlinear complexity analyses of brain conditions, to distinguish a psychogenic or molecular diagnosis, share similar methods to brain state studies. Seizure disorders, like epilepsy, dominated research due to significant pathogenic progression and are comorbid with disorders like ASD. Studies seeking reliable biomarkers have combined nonlinear complexity and linear techniques to capture complex brain dynamics.

Early diagnostic biomarker work using FD focused on dementia, reporting Adapted Box and Zero Set algorithms as a superior method, based on standard deviation of the control group [58]. Later, Ahmadlou et al. [7] found sub-band versus band-limited resting state EEG differences in Alzheimer's disease (AD) participants using KFD measures, reporting 99.3% diagnostic accuracy for AD. Then neuronal impairment post-acute stroke clinical recovery was predicted by applying HFD

[170], followed by HFD analysis detecting pathophysiological hypoglycemic effects [137]. More recently, in diagnostic nonlinear complexity applications of general mental disorders, HFD and sample entropy (SampEn) used machine learning to discriminate depression [35].

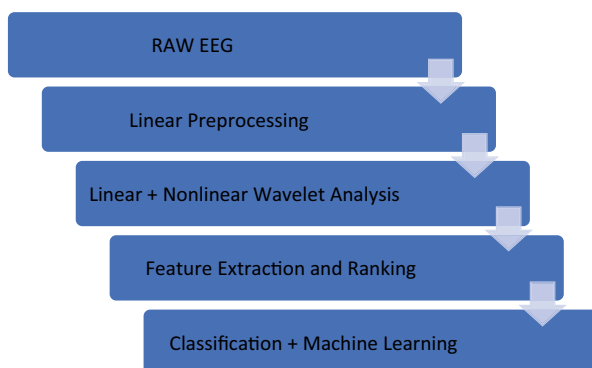
Difficulties with diagnostic EEG for epilepsy inspired automated nonlinear fractal techniques by Acharya et al. [4]. Babloyantz and Destexhe [10] utilizing LLE, CD and phase maps, distinguished epileptic seizure as lowest from rapid eye movement (REM) sleep or awake state. Bullmore et al. [23] using stereo-electroencephalography (SEEG) and applying Katz [84] methodology showed increased FD during ictal spikes. Hively et al. [70] employed K_E and CD to detect increases in FD and frequency during seizure. Using K_C (versus K_E), Petrosian [123] obtained linear and nonlinear properties without a preictal marker. Yet, these inconsistent FD representations, with different approaches reaching differing results resolve over time.

Lasemidis et al. [95] pioneered seizure detection with short-term LLE, to reveal preictal phase locking and a postictal drop as the brain resets to normalcy after an increased short-term LLE. Li et al. [99], applying continuous wavelet transforms, also found both preictal increases and decreases in EEG fractality, with a decrease in fractality during seizure. Using wavelet-chaos neural network classification using CD and LLE, Ghosh-Dastidar et al. [56] reported about 97% accuracy in predicting preictal states. Measuring CD for a larger scale of EEG data than is practical for Neural Network methods has been useful in predicting seizure risk and distinguishing sleep states from drowsiness [64]. Better discrimination of ictal EEG as fractal spectra was demonstrated by Easwaramoorthy and Uthayakumar [40] employing improved, generalized, FD algorithms. Application of FA showed clinical application, while a picture of EEG FD emerged gradually, reflecting varying methodologies.

More recently, epilepsy FA research combined other measures. El-Kishky [43] used Higuchi's method, Shannon and the H in seizure distinction. Kim et al. [87] reported lower FD during seizure with box counting and phase space reconstruction of the attractor. Reviewing computer-assisted epilepsy EEG analysis, Acharya et al. [4] pointed out automated epilepsy detection methods used higher order spectra, cumulants, recurrence measures, entropy measures, CD, H , LLE and surrogate data analysis. Martis et al. [112] developed intrinsic time-scale decomposition (ITD) to detect minute nonlinear EEG changes, distinguishing normal, interictal and ictal signals with 95.67% accuracy.

Wavelet-based methods for automated epilepsy detection are a large branch of research. Faust et al. [47] reported dominance by wavelet approaches and superiority of the nonlinear wavelet-chaos methods that are a combination of linear wavelet analysis with a layer of nonlinear discrimination. They incorporate classification involving machine learning to integrate the measures into a composite profile.

Other automated seizure detection methods used a scalar spectrum of fractal signal features. Applying multifractal analysis and relevance vector machine learning on intercranial EEG, Zhang et al. [172] reported epoch-based sensitivity of 92.94% and specificity of 97.47% in epilepsy diagnosing. They then successfully applied



Processing flow using the computer-aided seizure detection method by Faust et al. [47]

the k-nearest neighbours algorithm, much like Termonia and Alexandrowicz [152], box counting as suggested by Theiler [155]. More recent alternate concepts include dynamic analysis by Haghighi and Markazi [63] on their novel thalamocortical model using phase portraits, Poincare maps and bifurcation diagrams, revealing period doubling and chaotic dynamics to suggest nonlinear resonance as the ictal and interictal triggers. Sharma et al. [142] characterized seizures in a new way by applying their wavelet transform and FD methods reporting 100% sensitivity to ictal states. Both theoretical and practical application of FA methods, with novel automated detection, appear to have advanced epilepsy clinical research.

Linear EEG-based diagnosis of ASD has dominated, but nonlinear complexity methods are applied more frequently now as part of the range of quantitative electroencephalography (qEEG) [153], employing both [143]. Perhaps the earliest nonlinear complexity analysis of ASD by Ahmadlou et al. [6] compared KFD and HFD as a wavelet-chaos network method to diagnose ASD in children, reporting a 90% level of accuracy. W. Bosl et al. [20] used multiscale entropy (MSE) with machine learning, to resting-state EEG and distinguished high-risk ASD infants from typical infant controls in 80% overall and 100% for males. Catarino et al. [28] showed lower temporo-parietal and occipital MSE values as a reduction of complexity in ASD participants versus controls, with no difference in power spectra. Cusenza et al. [36] reported more organized neuronal functioning in ASD but abnormal bilateral sleep stage transition, versus same IQ developmentally retarded children in a mosaic of nonlinear fractal and linear frequency differences.

An anecdotal case of a young ASD adult with refractory catatonic obsessive-compulsivity saw reduced frontal and central smaller scale but occipital larger scale complexity during electroconvulsive therapy [116]. More broadly, in a diagnostic biomarker study of ASD, epilepsy and epileptiform EEG activity, W. J. Bosl, Loddenkemper and Nelson [21] found MSE values of ASD groups were less than controls, but greater than epileptic groups across various scalp locations with 95% accuracy. Similarly, Djemal et al. [39] used discrete wavelet transform and then Shannon entropy feature extraction with an artificial neural network classifier,

reporting 99.71% accuracy. Again, in that year, 20 male ASD children versus controls, during observation and imitation tasks, showed significantly lower MSE values, specifically in the right hemisphere and central cortex [101].

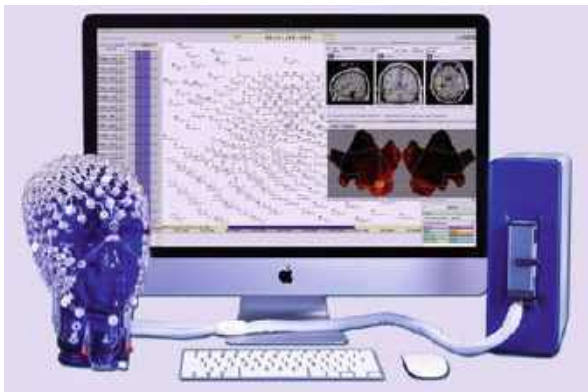
37.6 Current Nonlinear Complexity Analysis Trends

Studies consistently report lower MSE values, suggesting diminished complexity in ASD participants over controls. Kang et al. [79] more recently reported lower values in 4–8-year-old ASD children by comparing resting state EEG sample, fuzzy Renyi wavelet and Renyi permutation entropies, with age-measure sensitivity differences. The past decade of nonlinear complexity ASD biomarker research was dominated by entropy measures, with machine learning, yet with most studies following traditional linear methods. Still, the amount of nonlinear complexity research appears to be increasing.

In 2020, Yazdi-Ravandi et al. [169] showed higher KFD in the beta and lower gamma frequencies in OCD versus controls. More recently, in 2021, Hosseini et al. [71] reported HFD-based classifiers distinguished video game addicts at 95%. Classification by fractal dimension via Hurst exponent, Sebastián et al. [139] distinguished mental task difficulty versus resting state. Gu et al. [61] found a smaller MSE change distinguished unmediated adult ADHD from controls when comparing resting and task states. Ouchani et al.'s [117] review of EEG-based AD detection suggested further study. Dogmatic adherence to standard EEG analytic methods persists even when complexity is a parameter of inquiry. Recently in 2021, Rawls et al. [128] only employed traditional PSD methods to identify fractal complexity preference in participants.

37.7 Practical Considerations of EEG Complexity Analysis Research

EEG is relatively low cost and well established, yet continues to be technically challenging because, despite extensive research and significant advances, there is still little understanding of the causal chain of electrophysiological mechanisms that links EEG to manifest behaviours, sensory experiences and cognition [46] and the discrimination of relevant EEG signals from noise [151]. The signal/noise issue is partially due to EEG's susceptibility to signal interference and static, which are difficult to eliminate [131]. It is also assumed that some of the EEG is noise due to interference, refraction, reflection, etc. of the electromagnetic waves as they travel through the brain, skull and scalp, or due to some inherent randomness in the physiological process itself [44].



Picture courtesy of The Center for Neurological and Neurodevelopmental Health©

Another EEG technical challenge is the physical constraint on the accuracy of the signal collection due to scalp size, skin thickness, perspiration level and hair density. Additional variables include ambient temperature, humidity, barometric pressure, electrode size, type and connectivity, including the amplifier connection [83]. Other technical considerations include the design of the collection cap or net, leads, amplifier and computer into which the signal trace will be digitally converted and stored before preliminary processing [94].

These factors can seriously impact the sensitivity of the EEG apparatus, rendering signal data unusable, especially if the electrical impedance levels are too high [42]. EEG equipment synchronized with a stimulus computer frequently has inherent asynchrony of a significant number of milliseconds [129]. This means the measurement timing must be accordingly adjusted to more accurately reflect the time synchronicity of the signal (but with some degree of unremovable error), generally after collection.

In addition to the interference with collection of an accurate signal, there are electrophysiological events that may not be due to brain activity. Such inaccuracies may be due to muscle activity in the heart, head and neck of the participant, especially the eye muscles that may emit a strong electrical signal the electrodes will detect. These signals are often identified as artefacts and marked or electronically removed, generally after EEG acquisition [158]. Linear measures are time, power and space measures of value, whereas nonlinear complexity measures describe the behaviour of the system as a measure of its complexity over time [133]. Because linear processes (preprocessing) to remove signal noise are not designed to preserve nonlinear complexity features, they may affect the ability to detect complexity.

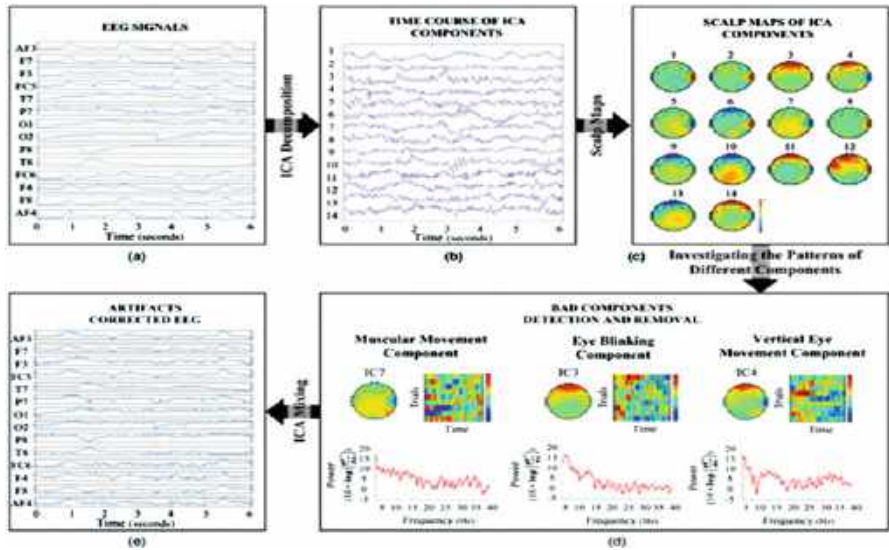
Preprocessing is a series of steps to improve the ‘accuracy’ of the collection and processing of EEG data [86], which can involve extensive modification of the signal itself and may then be called ‘EEG processing’ [37]. The decision of which procedures to use and which parameters to use is highly dependent upon the particular circumstances of the experiment involved. Although there are some general

best practices, the variety and complexity of EEG preprocessing and subsequent processing are broad. There is not a definitive line between EEG processing and preprocessing, but generally, the procedures that involve modification of the signal to reveal features that are critical to the hypothesis under investigation tend to be called ‘processing’, whereas those procedures simply designed to clean noise and adjust synchronization are ‘preprocessing’. Those steps that may affect the nonlinear complexity measures are generally steps in preprocessing which modify the signal in an effort to remove noise, but which may ‘proverbially throw the baby out with the bathwater’ and remove the information or reduce it so as to render the nonlinear complexity analysis less accurate [136].

For clarity, here ‘preprocessing’ refers to any modification of the data after collection, but before any measurements are made and ‘processing’ means the measurement of the data which transforms the data into values and plots using various algorithms. Typically, preprocessing means any transformation or reorganization between analysis and collection to facilitate analysis by organizing data into epochs, removing artefacts, removing electrodes, filtering and applying spatial transformations, with these decisions dependent mostly on the experimental requirements [30]. Generally, for linear time-frequency analysis EEG data is: (1) divided into epochs with matching trial counts, (2) low frequency (0.1–0.5) high-pass filtered to reduce drift unless epoched, (3) low-pass filtered if suitable, (4) it is usually visually or automatically trial rejected, (5) it can be spatially filtered, (6) it is usually referenced to one electrode (often the centre zero electrode – Cz), (7) it can be interpolated to fill in missing electrodes and (8) it can then have independent components analysed or have artefacts removed visually, including taking out EMG artefacts [30].

Most preprocessing automation was created to cope with time-consuming artefact removal and line noise removal, usually referred to as ‘preprocessing pipelines’, which apply various methods such as source separation, regression, linear decomposition, channel and segment artefact removal combined in a merged procedure [119]. A few of the pipeline’s stand out for their popularity and incorporation such as the Harvard Automated Preprocessing Pipeline for EEG (HAPPE), The Multiple Artefact Rejection Algorithm (MARA) and Automagic [119]. Though the selection is highly dependent on the particulars of the experiment, some try to standardize preprocessing in order to improve the replicability and generalizability of EEG studies. Along with the open-source code toolboxes of EEG, processing functions such as *eeglab* [37] and many other preprocessing pipelines have facilitated the application of preprocessing methods to a larger number of studies which may improve reliability, validity and replicability of EEG research as methods become standardized.

Since many of the preprocessing procedures involve significant modification of the EEG signal, it is difficult to know without performing a large number of systematic trials to identify all the potential changes. Still, certain preprocessing data modifications may be more likely candidates based on those that remove nonlinear complexity information above or below a certain frequency or removal of data: (1) outside of the epochs or (2) due to automated or visual removal of channels or (3) of



Automated Preprocessing using Automagic [119]

trials or (4) through the use of independent component analysis, which modifies the signal mathematically such as through convolution via complex algorithms. Finally, there is a question as to whether adjusting the signal through re-referencing to a particular electrode such as Cz would alter the signal in a manner as to affect nonlinear complexity processing, as the values are adjusted linearly for the entire data set. These remain somewhat open questions which may be found to require attention in nonlinear complexity experimentation.

37.8 Applied EEG Complexity Analysis of Resting State Data – Study One Results

We benchmarked and set parameters for six complexity measures – the *correlation dimension* plus *peak and embedded dimension*, checked with *false nearest neighbour* (CDPKFNN); the *largest Lyapunov exponent* (LLE); *Higuchi’s fractal dimension* (HFD); *multiscale entropy estimate* (MSE); *multifractal detrended fluctuation analysis* (MFDFA) and the *Kolmogorov complexity estimate* (Kc). As explained above, CDPKFNN quantifies the maximum attractor phase space; LLE determines the acceleration of trajectories on the attractor; HFD evaluates the FD on a place of the complex curve of the EEG trace; MSE counts the number of entropic information scales; MFDFA provides the spread of Hurst dimensions and Kc quantifies the compressibility of the signal related to the pattern sequence of the least necessary generating algorithm.

Results for Study 1 showing EO highest (all participants) for CDPKFNN and lowest (all participants) for HFD with no hierarchies and several opposing trends (2 participants and 2 tests)

Measure	CDPKFNN	LLE	HFD	MSE	MF DFA	Kc
High Setup(s)	EO NotP	None All	None All	None All	None All	None All
Low Setup(s)	None All	None All	EO NLPF	None All	None All	None All
		>EC EOEC> >EO	EOEC>	EO> & >EOEC EOEC> EO> & >EOEC	>EO	EOEC> >EOEC
		P P NLPF	P	P NLPF NotP	NotP NLPF	NLPF NotP

We quantified different features of the EEG signal compared to each other and the resting state: eyes open, closed or between (EO, EC and EOEC) using each complexity measure. Due to non-normal distribution and variable dependence, we used the non-parametric Wilcoxon signed rank test (WSRT) [165], (where H_0 : $x - y$ comes from a distribution with median equal to 0, or H_1 : $x - y$ comes from a distribution with median greater than 0). We assumed no statistical difference exists between the median values of each nonlinear measure for every channel at 5% significance level. The epoch lengths were 30,000 milliseconds sampled at 1000 times per second. There were three preprocessing setups: (1) re-referenced and adjusted for amplifier drift and monitor delay only, i.e. not automatic preprocessed (NAP); (2) preprocessed with automatic with low-pass band filtering (AP) and (3) preprocessed with automatic without low-pass band filtering (NLPF).

In comparing the reliability of six complexity measures, CDPKFNN, LLE, HFD, MSE, MF DFA and K_c , for three types of resting states, EO, EC and EOEC, several preprocessing parameters were chosen heuristically. We re-referencing for the international 10–20 system, adjusting values for amplifier drift and PC monitor lag, Automatic preprocessing automated steps including low-pass band filtering. Some parameters appeared to have significant impact on the measure values and others appeared to alter the relationship between values. CDPKFNN had slightly more significant values corresponding to higher complexity during EO versus EC. HFD contradicted this by distinguishing EO as the lowest.

37.9 Applying EEG Complexity Analysis to Brain Task Data – Study Two Results

CDPKFNN, LLE, HFD, MSE, MF DFA and K_c were again applied in three preprocessing setups: AP, NLPF and NAP. We note participant 3 had a “bad” quality

Comparison of hierarchies (3 tests and 3 Participants) and anecdotal trending (2 participants in 2 or more tests) in results

Measure	CDPKFNN	LLE	HFD	MSE	MFDFA	Kc
High Setup(s)	None All	RMET AP	None All	RMET AP	None All	None All
Low Setup(s)	BCST NAP NLPF	BCST All	RMET NAP	BCST AP	BCST RMET AP NLPF	RMET NAP
Trending Setup(s)	EO> NAP	None All	>EO NLPF	>RMET NAP	EO>; BCST> AP; NLPF	BCST> NAP

AP automagic preprocessed, *NAP* not automagic preprocessed, *NLPF* automagic preprocessed without low-pass filtering

rating from both AP and NLPF setups, which reflected significant differences but only for preprocessed data. Participants were mixed aged (16–85), self-identified binary gender adults along a spectrum of ASD risk from sub-clinical to clinical. For EO the data window was 2500 ms. Data windows for RMET and BCST varied, as they were all measured from the length of each event instead of the same length in study one.

As in the first study, we saw some indication that more complex mental states would evoke higher measurement values. This was seen somewhat in CDPKFNN and MSE, though more consistently in LLE. The parameters of each measure, the AP setups and the length of the event latency affected the results. Refined methodology for each measure would quantify each parameter’s effects and increase discrimination. As seen in the literature, the picture of the Brain’s CDS as measured by the EEG is a complicated one consisting of subtle nuances to be finessed out of the data with further study.

37.10 EEG Complexity Measures of Cognitive States Show Heterogeneous ASD Risk with Sequence Clustering – Study Three

Six measures of EEG complexity were compared between two groups based on AQ scores as higher and lower AQ score participants (19 above and 20 below the ASD threshold). No consistent trends or differences were found in the NLPF data, so the focus was only on AP and NAP setups. Similar to the previous two studies, we compared WSRT medians to set the hierarchy of complexity measures hypothesizing RMET would be the highest score for each participant and higher AQ-scored participants would have lower RMET WSRT medians.

Dataset	RunID	Observed Values					Expected Values				
		RunID	RunID	RunID	RunID	RunID	RunID	RunID	RunID	RunID	
1	1	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	
2	2	2.0	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	
3	3	3.0	3.1	3.2	3.3	3.4	3.5	3.6	3.7	3.8	
4	4	4.0	4.1	4.2	4.3	4.4	4.5	4.6	4.7	4.8	
5	5	5.0	5.1	5.2	5.3	5.4	5.5	5.6	5.7	5.8	
6	6	6.0	6.1	6.2	6.3	6.4	6.5	6.6	6.7	6.8	
7	7	7.0	7.1	7.2	7.3	7.4	7.5	7.6	7.7	7.8	
8	8	8.0	8.1	8.2	8.3	8.4	8.5	8.6	8.7	8.8	
9	9	9.0	9.1	9.2	9.3	9.4	9.5	9.6	9.7	9.8	
10	10	10.0	10.1	10.2	10.3	10.4	10.5	10.6	10.7	10.8	
11	11	11.0	11.1	11.2	11.3	11.4	11.5	11.6	11.7	11.8	
12	12	12.0	12.1	12.2	12.3	12.4	12.5	12.6	12.7	12.8	
13	13	13.0	13.1	13.2	13.3	13.4	13.5	13.6	13.7	13.8	
14	14	14.0	14.1	14.2	14.3	14.4	14.5	14.6	14.7	14.8	
15	15	15.0	15.1	15.2	15.3	15.4	15.5	15.6	15.7	15.8	
16	16	16.0	16.1	16.2	16.3	16.4	16.5	16.6	16.7	16.8	
17	17	17.0	17.1	17.2	17.3	17.4	17.5	17.6	17.7	17.8	
18	18	18.0	18.1	18.2	18.3	18.4	18.5	18.6	18.7	18.8	
19	19	19.0	19.1	19.2	19.3	19.4	19.5	19.6	19.7	19.8	
20	20	20.0	20.1	20.2	20.3	20.4	20.5	20.6	20.7	20.8	
21	21	21.0	21.1	21.2	21.3	21.4	21.5	21.6	21.7	21.8	
22	22	22.0	22.1	22.2	22.3	22.4	22.5	22.6	22.7	22.8	
23	23	23.0	23.1	23.2	23.3	23.4	23.5	23.6	23.7	23.8	
24	24	24.0	24.1	24.2	24.3	24.4	24.5	24.6	24.7	24.8	
25	25	25.0	25.1	25.2	25.3	25.4	25.5	25.6	25.7	25.8	
26	26	26.0	26.1	26.2	26.3	26.4	26.5	26.6	26.7	26.8	
27	27	27.0	27.1	27.2	27.3	27.4	27.5	27.6	27.7	27.8	
28	28	28.0	28.1	28.2	28.3	28.4	28.5	28.6	28.7	28.8	
29	29	29.0	29.1	29.2	29.3	29.4	29.5	29.6	29.7	29.8	
30	30	30.0	30.1	30.2	30.3	30.4	30.5	30.6	30.7	30.8	
31	31	31.0	31.1	31.2	31.3	31.4	31.5	31.6	31.7	31.8	
32	32	32.0	32.1	32.2	32.3	32.4	32.5	32.6	32.7	32.8	
33	33	33.0	33.1	33.2	33.3	33.4	33.5	33.6	33.7	33.8	
34	34	34.0	34.1	34.2	34.3	34.4	34.5	34.6	34.7	34.8	
35	35	35.0	35.1	35.2	35.3	35.4	35.5	35.6	35.7	35.8	
36	36	36.0	36.1	36.2	36.3	36.4	36.5	36.6	36.7	36.8	
37	37	37.0	37.1	37.2	37.3	37.4	37.5	37.6	37.7	37.8	
38	38	38.0	38.1	38.2	38.3	38.4	38.5	38.6	38.7	38.8	
39	39	39.0	39.1	39.2	39.3	39.4	39.5	39.6	39.7	39.8	
40	40	40.0	40.1	40.2	40.3	40.4	40.5	40.6	40.7	40.8	
41	41	41.0	41.1	41.2	41.3	41.4	41.5	41.6	41.7	41.8	
42	42	42.0	42.1	42.2	42.3	42.4	42.5	42.6	42.7	42.8	
43	43	43.0	43.1	43.2	43.3	43.4	43.5	43.6	43.7	43.8	
44	44	44.0	44.1	44.2	44.3	44.4	44.5	44.6	44.7	44.8	
45	45	45.0	45.1	45.2	45.3	45.4	45.5	45.6	45.7	45.8	
46	46	46.0	46.1	46.2	46.3	46.4	46.5	46.6	46.7	46.8	
47	47	47.0	47.1	47.2	47.3	47.4	47.5	47.6	47.7	47.8	
48	48	48.0	48.1	48.2	48.3	48.4	48.5	48.6	48.7	48.8	
49	49	49.0	49.1	49.2	49.3	49.4	49.5	49.6	49.7	49.8	
50	50	50.0	50.1	50.2	50.3	50.4	50.5	50.6	50.7	50.8	
51	51	51.0	51.1	51.2	51.3	51.4	51.5	51.6	51.7	51.8	
52	52	52.0	52.1	52.2	52.3	52.4	52.5	52.6	52.7	52.8	
53	53	53.0	53.1	53.2	53.3	53.4	53.5	53.6	53.7	53.8	
54	54	54.0	54.1	54.2	54.3	54.4	54.5	54.6	54.7	54.8	
55	55	55.0	55.1	55.2	55.3	55.4	55.5	55.6	55.7	55.8	
56	56	56.0	56.1	56.2	56.3	56.4	56.5	56.6	56.7	56.8	
57	57	57.0	57.1	57.2	57.3	57.4	57.5	57.6	57.7	57.8	
58	58	58.0	58.1	58.2	58.3	58.4	58.5	58.6	58.7	58.8	
59	59	59.0	59.1	59.2	59.3	59.4	59.5	59.6	59.7	59.8	
60	60	60.0	60.1	60.2	60.3	60.4	60.5	60.6	60.7	60.8	
61	61	61.0	61.1	61.2	61.3	61.4	61.5	61.6	61.7	61.8	
62	62	62.0	62.1	62.2	62.3	62.4	62.5	62.6	62.7	62.8	
63	63	63.0	63.1	63.2	63.3	63.4	63.5	63.6	63.7	63.8	
64	64	64.0	64.1	64.2	64.3	64.4	64.5	64.6	64.7	64.8	
65	65	65.0	65.1	65.2	65.3	65.4	65.5	65.6	65.7	65.8	
66	66	66.0	66.1	66.2	66.3	66.4	66.5	66.6	66.7	66.8	
67	67	67.0	67.1	67.2	67.3	67.4	67.5	67.6	67.7	67.8	
68	68	68.0	68.1	68.2	68.3	68.4	68.5	68.6	68.7	68.8	
69	69	69.0	69.1	69.2	69.3	69.4	69.5	69.6	69.7	69.8	
70	70	70.0	70.1	70.2	70.3	70.4	70.5	70.6	70.7	70.8	
71	71	71.0	71.1	71.2	71.3	71.4	71.5	71.6	71.7	71.8	
72	72	72.0	72.1	72.2	72.3	72.4	72.5	72.6	72.7	72.8	
73	73	73.0	73.1	73.2	73.3	73.4	73.5	73.6	73.7	73.8	
74	74	74.0	74.1	74.2	74.3	74.4	74.5	74.6	74.7	74.8	
75	75	75.0	75.1	75.2	75.3	75.4	75.5	75.6	75.7	75.8	
76	76	76.0	76.1	76.2	76.3	76.4	76.5	76.6	76.7	76.8	
77	77	77.0	77.1	77.2	77.3	77.4	77.5	77.6	77.7	77.8	
78	78	78.0	78.1	78.2	78.3	78.4	78.5	78.6	78.7	78.8	
79	79	79.0	79.1	79.2	79.3	79.4	79.5	79.6	79.7	79.8	
80	80	80.0	80.1	80.2	80.3	80.4	80.5	80.6	80.7	80.8	
81	81	81.0	81.1	81.2	81.3	81.4	81.5	81.6	81.7	81.8	
82	82	82.0	82.1	82.2	82.3	82.4	82.5	82.6	82.7	82.8	
83	83	83.0	83.1	83.2	83.3	83.4	83.5	83.6	83.7	83.8	
84	84	84.0	84.1	84.2	84.3	84.4	84.5	84.6	84.7	84.8	
85	85	85.0	85.1	85.2	85.3	85.4	85.5	85.6	85.7	85.8	
86	86	86.0	86.1	86.2	86.3	86.4	86.5	86.6	86.7	86.8	
87	87	87.0	87.1	87.2	87.3	87.4	87.5	87.6	87.7	87.8	
88	88	88.0	88.1	88.2	88.3	88.4	88.5	88.6	88.7	88.8	
89	89	89.0	89.1	89.2	89.3	89.4	89.5	89.6	89.7	89.8	
90	90	90.0	90.1	90.2	90.3	90.4	90.5	90.6	90.7	90.8	
91	91	91.0	91.1	91.2	91.3	91.4	91.5	91.6	91.7	91.8	
92	92	92.0	92.1	92.2	92.3	92.4	92.5	92.6	92.7	92.8	
93	93	93.0	93.1	93.2	93.3	93.4	93.5	93.6	93.7	93.8	
94	94	94.0	94.1	94.2	94.3	94.4	94.5	94.6	94.7	94.8	
95	95	95.0	95.1	95.2	95.3	95.4	95.5	95.6	95.7	95.8	
96	96	96.0	96.1	96.2	96.3	96.4	96.5	96.6	96.7	96.8	
97	97	97.0	97.1	97.2	97.3	97.4	97.5	97.6	97.7	97.8	
98	98	98.0	98.1	98.2	98.3	98.4	98.5	98.6	98.7	98.8	
99	99	99.0	99.1	99.2	99.3	99.4	99.5	99.6	99.7	99.8	
100	100	100.0	100.1	100.2	100.3	100.4	100.5	100.6	100.7	100.8	

Comparing six algorithmic measures of complexity, CDPKFNN, LLE, HFD, MSE, MFDFA and K_c , indicated potential differences in three types of mental tasks, RS EO, RMET and BCST, but no consistent differences between high and low-scoring AQ participants. Preprocessing parameters were applied heuristically to compare the measures, but variations were a likely confound. To avoid this, a sliding window could encompass all event lengths but would then create a large computational burden. As parameters appear to influence complexity values and relationships, two approaches, AP and NAP, controlled for the processing parameter impact. Indications that unprocessed EEG can be analyzed effectively with complexity measures suggest this may be an important area of study.

37.11 Conclusion

Of the many approaches to complexity analysis of EEG, there are two general types, geometric and informational. Both depend on mathematical features of complex dynamical systems exhibiting deterministic chaotic features of sensitivity to initial conditions, scale-free patterns and self-similarity or self-affinity, with fractal features. Some of the most common methods are CD, LLE, HFD and MSE, MFDFA and K_c . Complexity measures are used to study sleep, cognition, epilepsy, ASD and schizophrenia with general success, but with some contradictory results for schizophrenia [25]. Complexity EEG research has been less robust than linear EEG research, but recently interest has renewed the use of more nonlinear complexity methods due in part to more powerful computers. Nonlinear complexity techniques are used in conjunction with linear techniques, as part of the preprocessing procedure or as a separate measure itself. For both linear and nonlinear complexity measures, the effect of preprocessing the data on the accuracy of analysis is an important consideration. More research is needed to identify just how much information is lost that might destroy whatever features are being investigated with both methods.

Complexity analysis of EEG is in its early stages, and few studies have applied complexity methods to distinguish ASD. EEG preprocessing is developing better, and more standardized methods and EEG complexity analysis are establishing protocols to a core set of reliable and valid complexity measures with established benchmarks. MSE is now used frequently with ASD research with indications it may be a good candidate for standardization due to its computational economy and sensitivity. Another advantage to MSE is its temporal scales, which provide more dynamic entropic information from the EEG signal. Parameters in MSE are critical so standardized parameter settings for EEG data profiles will likely improve complexity analysis overall.

CD performs well but is computationally expensive and may require a high-capacity computing environment only available to universities and hospitals. LLE is far less computationally expensive but with many techniques and no standard benchmarked version LLE may be used only as an adjunct analysis. HFD has been

widely used, but parameters are critical so standardized parameter settings may support HFD to be used more frequently. MFDFA is technically challenging and parameter setting is even more critical. However, if these aspects are overcome the multi-fractal capacity may prove worth the effort. Finally, K_c applying LZ or another method is theoretically important because of the unknowable elements of organic computation. Clarification of the mathematical relationships between K_C , K_E , FD and H are an important step in understanding not only these measures but our deeper comprehension of the functional landscape dictated by the architecture of the brain.

Motivation for this research included advancing the understanding of not only specific EEG complexity analysis methods but also to improve the diagnosing of ASD, which continues to be difficult to both diagnose and treat. This inquiry may lead to innovative approaches to diagnosis and potentially better treatment of ASD. Improved diagnostic procedures may also lead to more reliable and valid prognoses through advancements in more accurate and earlier identification of ASD.

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Chapter 38

Fractals in Neuropsychology and Cognitive Neuroscience



Antonio Cerasa

Abstract The fractal dimension of cognition refers to the idea that the cognitive processes of the human brain exhibit fractal properties. This means that certain patterns of cognitive activity, such as visual perception, memory, language, or problem-solving, can be described using the mathematical concept of fractal dimension.

The idea that cognition is fractal has been proposed by some researchers as a way to understand the complex, self-similar nature of the human brain. However, it's a relatively new idea and is still under investigation, so it's not yet clear to what extent cognitive processes exhibit fractal properties or what implications this might have for our understanding of the brain and clinical practice. Indeed, the mission of the “fractal neuroscience” field is to define the characteristics of fractality in human cognition in order to differently characterize the emergence of brain disorders.

Keywords Fractal dimension · Human cognition · Executive functions · Visual perception · Reasoning · Language · Mental disorders

38.1 The Fractal Dimension of Cognition

Mandelbrot first described fractals in 1977 [1]. They are signals that exhibit scale-invariant, self-similar behavior. Power-law scaling is their distinctive feature. Fractals are self-similar geometric objects with properties on an unlimited number of scales in the space realm. They can form a hierarchy of temporal and geographical

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scales in the study of time series, covering a large range between coarse-scale long-term and high-frequency fine-scale variations. Understanding the dynamics of complex systems can be accessed through the statistical study of dynamical fractals. Random fractals are defined as having a stochastic mechanism involved in their production.

The procedure is deemed monofractal if the relationship between the property under evaluation and scale change is straightforward. Afterward, the process can be described by a single scaling exponent that is connected to its fractal dimension or spectral exponent. A single, stationary scaling exponent may not always be sufficient to accurately describe the scaling behavior, though. In these situations, it might be necessary to scale using a number of exponents, each of which would locally apply to a different section of the data stream. The histogram of the Hoelder exponents, commonly known as the singularity spectrum, is used to represent such multifractal signals [2].

At all levels of brain architecture, power-law scaling and other manifestations of fractal and self-similar patterns in space and/or time can be found. With a few notable exceptions, these observations remained essentially isolated in theoretical neuroscience, which was otherwise quickly expanding and had distinct goals [3].

In this chapter, we present the evidence about the fractal dimension of cognition taking into account the idea that fractal properties can be viewed as playing a role for the functional integration among different levels of neuronal organization [4], and that fractal dynamics may be useful for cognitive processes in the central nervous system [5].

The bridge between fractal dynamics and human cognition passes through the of physical laws regulating perceptual experience. The goal of classical psychophysics, as developed by Helmholtz, Fechner, and Weber, was to demonstrate how sense organs' responses to physical stimuli are affected by certain characteristics of perceptual experience. Stevens summarized his vast research in 1975 and proposed that this dependence is a power function in many sensory modalities [6].

Psychophysical laws, try to relate the amplitude of a physical input to its perceived magnitude. All the senses' stimuli—taste, touch, vision, hearing, and smell—are included in these laws. According to Weber's law, the slight modification of a stimulus is a constant ratio of the initial stimulus. Fechner developed a method for identifying the psychophysical rules of loudness [7]. Fechner believed that the Weber fraction (I/I), where I represents the standard sound intensity and I is the intensity change necessary for the j nd, created an equal increase in loudness perception (L). The result of integrating this equation, $L = I/I$, is known as Fechner's law, which states that loudness is a logarithmic function of sound intensity ($L = \log I$).

According to interpretations of the psychophysical observations, Weber claimed that cognition would generate its dynamical signature as a consequence of its own activity. In other words, physical dynamics underlying human perception of external stimuli stand in a power-function relation [7].

This phenomenon occurs also in other cognitive functions as described in future paragraphs.

38.1.1 *Fractal Dimensions of Executive Functions*

The fractal dimension of executive functions is a relatively new concept in cognitive science and neuroscience, which suggests that certain aspects of executive function, such as planning, reasoning, working memory, attention control, and decision-making, may exhibit fractal properties. This means that the patterns of activity involved in these processes may be self-similar across different levels of organization, such as at the neural, cognitive, and behavioral levels. The idea that executive functions are fractal is still being explored and investigated, and it is not yet clear to what extent these processes exhibit fractal properties or what implications this might have for our understanding of the brain and cognitive function. Some studies have shown a relationship between the fractal dimension and some executive function tasks, but it's still an ongoing research area. A fractal pattern is created by the endogenous variety of human behavior [8]. By repeatedly observing the same subject performing the same action while maintaining all other factors constant, rather clear images of endogenous variation can be obtained. In order to acquire a large sample of hundreds or thousands of data points as a trial-ordered time series, the optimal recipe is for using a straightforward task to assess human behavior and repeating measurement trials over and over again. A spectral analysis of repeated data typically depicts the fractal pattern of variation. The amount of changes between recorded values and the frequency of changes of a specific size—how frequently changes of a particular size happen—can both be used to completely define repeated measurements. The scaling link between size and frequency of variation is represented by the fractal pattern. On logarithmic scales, the relationship between the magnitude of changes and their frequency is inversely proportional.

When time-series data is evaluated, timing variations in behavioral tasks show cycle-to-cycle oscillations that show fractal scaling of power spectra [9, 10]. According to Ding et al. [11], this is due to the numerous time scale activity of dispersed brain regions that support task performance. When given a set of random numbers, it is possible to create a series with short- and long-range correlations that, in most situations, display a power-law spectrum [12]. Serial correlations in human cognition are interpreted by Van Orden et al. [13] as proof of self-organization. According to their theory, self-organization allows an organism to coordinate its activities across a hierarchy of time scales, resulting in correlated variation over time. Consequently, variations in response times would then appear as a natural fractal in which larger scale deviations nestle smaller (self-similar) scale deviations.

With this in mind, we can better describe what has been found in investigating the fractal proprieties of executive functions.

The fractal dimension of *working memory* (WM) is another relatively new concept in cognitive science and neuroscience, which suggests that certain aspects of working memory, such as the ability to hold and manipulate information in the mind, may exhibit fractal properties. Studies that use fractal *n*-back tasks are one of the ways to investigate the fractal dimension of working memory. The task typically

involves presenting a sequence of stimuli, such as fractal images, to the participant, and asking them to indicate when the current stimulus matches a stimulus that was presented a certain number of steps (or “ n ”) back in the sequence. The goal is to understand how the complexity of the stimuli affects the performance of the task and the neural activity related to it.

The n -back task is a widely used tool in cognitive psychology and neuroscience to study working memory. The use of fractal images in the task is an attempt to create a more ecologically valid working memory task, as fractals are more complex and self-similar than simple geometric shapes or letters. Ragland et al. [14] compared working memory performance using fractal images and letter stimuli, and the neural activity related to the task using fMRI. The current study compares a new version of n -back task using complex visuospatial fractal stimuli with respect to standard regular letter n -back condition [15]. These difficult-to-label geometric stimuli were created to fill the requirement for visual tasks requiring higher degrees of spatial perceptual processing than earlier item working memory tasks that employed faces or line drawings of familiar things [16]. The authors hypothesized that the greater spatial demands of processing fractal versus letter stimuli would be reflected by greater left dorsolateral prefrontal (PFC) activity for letter n -back and greater right PFC activity for fractal n -back. Contrary to their expectations, making the fractal stimuli harder to name led to participants exerting more effort to create verbal representations. Indeed, the reaction time during highly demanding 2back phases was longer during fractal tests with respect to letter tests, demonstrating the extra effort of constructing verbal representations rather than the more automatic process of letter identification. Similarly, Miller et al., [17] employed fractal visual stimuli to determine how long-term learning influences cortical activity patterns underlying WM maintenance. These stimuli, which had no prior meaning, were used to illustrate how long-term associative learning affected neuronal selectivity. These challenging stimuli were selected to lengthen the learning process and make a thorough item representation necessary for effective performance.

In summary, the literature on the fractal properties of WM ability would seem to be restricted to the employment of fractal stimuli in order to influence cognitive function. Grosu has articulated this particular choice in a seminal review [18]. The fact that many perceptual signals occurring in nature have scale-invariant/statistically self-similar features is one of the justifications for assigning a functional role to scale-free dynamics/fractal structure in human cognition. In fact, it may be desirable for the system to display similar qualities in order to analyze data with specific statistical properties. Grosu and colleagues described the famous example of statistical self-similarity on the British coast. Ecological settings also exhibit several instances of self-similarity (such as fractal branching of trees) and traits shared with the larger scale on a size of 100 m to 1 km (lakes vs. seas). Even at a smaller size, elements of the local environment replicate those on a larger scale. For instance, grassy areas resemble treey ones, little ponds reflect larger lakes, etc. Animals flourish in this particular habitat, and their brains help the scaling process. A mouse hiding in the grass, for instance, might hear and see what appears to be a predator. The same statistical features of these weak signals

with $1/f$ qualities translate into robust responses in the visual and auditory brain. Thus, the mouse brain displays self-similarity on several dimensions and some of its structural features are expressed on bigger spatial scales as well. For instance, the branching pattern of dendritic trees resembles that of real trees. In other words, the brain prefers to process signals whose characteristics mirror their own internal dynamical structure.

One of the distinguishing features of *human reasoning* is the capacity to represent and build intricate hierarchical systems. Humans arrange fundamental components into higher-order groupings and structures in a variety of fields, such as music, problem-solving, action sequencing, and spatial navigation [19]. Human behavior is flexible because of our capacity to encode the connections between discrete entities (words, persons, etc.) and the larger frameworks in which they are embedded. Humans can modify, augment, or adapt certain fundamental motions to specific settings while still maintaining the overall structure (and objectives) of canonical motor systems, as seen in action sequencing, for instance [20]. When brewing coffee, actions like “grinding the beans” or “refilling the water container” are common instances of these actions in context [21]. Without harming the process of preparing coffee overall, people can decide if these steps are necessary and remove them if they are [22]. This skill is distinct from straightforward action sequencing and appears to be extremely limited in nonhuman animals [23]. Recursive embedding processes are a potential approach for representing complicated hierarchical systems, which are prevalent in nature and appealing for experimental investigation [24]. Recursive embedding is the insertion of one structure into another structure of the same type and enables the creation of infinitely deep hierarchies with the use of relatively basic rules. Some examples are the fractal Mandelbrot images or fractal structures in nature.

The fractal dimensions of reasoning refer to the idea that certain aspects of reasoning, such as problem-solving, decision-making, and critical thinking, may exhibit fractal properties since they represent complex *hierarchical structures* [25]. According to Stephen and Dixon [26], reasoning can be thought of as the self-organization of novel structures when it comes to problem-solving and insight. Using a specific problem-solving task as an example, the authors propose that the solution can be seen as a phase transition in a self-organizing system whose dynamics would be reflected in power-law behavior. Correll [27] has reviewed the implications of these findings for social psychology, claiming that cognitive effort to avoid bias in assessments reduces the scaling exponents of response times compared to easier tasks. However, some studies have suggested that the fractal dimension may be a useful tool for characterizing the complexity of reasoning tasks and understanding how they relate to other cognitive processes. For example, a study has used fractal dimension to analyze the complexity of the problem-solving process in the Tower of Hanoi task [28] and found that the fractal dimension of the problem-solving process increased as the task became more difficult. Overall, this is an emerging field, and more research is needed to understand the fractality of reasoning processes and how they relate to other cognitive processes.

38.1.2 Fractal Dimensions of Human Perception and Visuospatial Processing

Fractal dimensions have been used as a tool to quantify the complexity of human perception, particularly in the context of visual stimuli. The idea is that the complexity of a visual pattern or image can be quantified by the degree of self-similarity present in the pattern. In brief, the dimension D of fractal designs, which lies between Euclidean dimensions, can be used to measure the geometric complexity of these patterns. The D parameter for photographs varies from $D = 1$ (a line) to $D = 2$ (a filled plane). A nonfractal line with $D = 1$ will take up more space as pattern repetitions occur as the D value goes closer to 2, indicating a higher degree of complexity, and the line will typically fill a bigger portion of the plane.

However, some 2D fractal images—known as “noncontinuous” or “dust” fractals—can have $D = 1$ if the patterns are sufficiently sparse to represent a point rather than a line [29].

Researchers have used fractal dimension to study the relationship between visual complexity and human perception, with the goal of better understanding the nature of visual perception itself. For instance, Taylor & Spehar [30] suggest that because the visual system has adjusted to these common natural patterns, fractals with medium complexity play a special function in human visual experiences. This adaptation can be seen across the visual system, from the eye’s data collection to the brain’s higher visual centers processing this input. For instance, eye-movement studies demonstrate that the eye creates mid- D fractal trajectories that aid in visual exploration of fractal landscapes. Based on these findings, they described a fluency model in which mid- D fractals are relatively simple for the visual system to process. This fluency provides an esthetic experience together with a decrease in the observer’s physiological stress levels and enhances the observer’s abilities (i.e., enhanced attention and pattern recognition). Others, on the other hand, used fractals as a tool to quantify the complexity of visual stimuli, including those based on gestalt principles. In gestalt psychology, the idea of “gestalt” refers to the holistic perception of visual stimuli as a whole, rather than as a collection of individual parts [31]. Fractal dimension can be used to quantify the degree of self-similarity present in a visual pattern or image, and some researchers have used this approach to study the complexity of gestalt visual stimuli.

Given the controllability of complexity in fractal patterns and their relation to more general theories, fractals are particularly useful stimuli for research on esthetic preference. Indeed, the fractal dimension of esthetic preference refers to the degree of complexity or “roughness” present in a work of art or design that is perceived as pleasing to the viewer, in order to understand how people perceive and respond to different types of visual stimuli. Some researchers have suggested that there may be a correlation between the fractal dimension of a work of art and its perceived esthetic appeal, although the relationship between these factors is still not fully understood. The most famous example is the Jackson Pollock. This is an abstract painter, who started using a paintbrush to drip and fling paint over long panels of

canvas for his notorious splatter paintings in the early 1940s. The fractal properties of Pollock's splatter paintings were first validated by Taylor et al. [32]. They used fractal dimension analysis as a tool to quantify the complexity of Pollock's dripped and splattered paintings, which are known for their dynamic and seemingly chaotic compositions. Pollock's technique developed over time, with three distinct phases: the "preliminary phase," which produced low- D fractal images (e.g., Untitled, 1945, $D = 1.10$), the "transitional phase," during which the resulting dimensions increased (e.g., Number 14, 1948, $D = 1.45$), and the "classic" phase, during which he perfected the technique and the D values of his works lingered around 1.7 [33]. He also created Untitled (1950), where the intricacy of his works peaked at $D = 1.89$, during the classic period. Taylor was able to demonstrate that Pollock's technique of dripping and splattering paint created highly complex, self-similar patterns that are characteristic of fractals. Taylor's study suggests that Pollock's process of creating art was a fractal one, as he used a self-similar process of adding drips and splatters of paint. The result of this process is a painting that appears to be random and chaotic, but it is highly structured with a specific fractal dimension. Furthermore, Taylor's work shows that there is a correlation between the fractal dimension of Pollock's paintings and their perceived esthetic appeal. Indeed, according to some research [34], people prefer fractals that fall between the D range of 1.3–1.5 from an esthetic standpoint. This would imply that Pollock's older pieces, created before his classic phase, should have a stronger esthetic appeal. However, the most famous of his works are still those from his classic period, indicating that higher- D fractals might be more effective at drawing attention than lower- D fractals. Taylor and colleagues [35] hypothesize that Pollock preferred to engage his audience with visually challenging pieces because he thought the "visually restful" mid- D range to be too straightforward and bland.

Fractal analysis of artistic paintings has also been applied to Frida Kahlo. Despite what we expect, this analysis has been made for investigating the effects of physical and psychological distress on visual art. Turkheimer and colleagues [36] demonstrated that there is an association between colors and physical and emotional pain. For performing fractal dimension analysis of each self-portrait, a previously established method that is particular to RGB photographs was employed [37]. The use of fractal analysis makes an effort to quantify how cognitive disorders affect an artist's work. Given that "the shattered column," one of her most well-known paintings, received the highest FD value, it is likely that the influence of pain on this marker was essentially constant throughout her lifetime despite significant variations.

In the fractal analysis of painting, a larger fractal dimension indicates the artist's ability to control the object on canvas in a cohesive manner and meaningfully connect all the scales of its realization, from small details to the large scenery or background. FD quantifies the space-filling property of the object measured. Forsythe et al. [38] used this metric as an indicator of cognitive decline in individuals with dementia. The idea is that as dementia progresses, individuals may have a harder time creating complex or "fractal" patterns in their artwork. Therefore, a decrease in fractal dimension in artwork created by individuals with dementia may

be an indication of the severity of their condition. These studies have looked at the fractal dimension of artwork created by individuals with dementia compared to healthy controls and found that the fractal dimension of the artwork created by individuals with dementia was lower, indicating a decrease in complexity and an increase in the simplicity of their artwork.

It's important to note that the use of fractal dimension to study human perception is still an area of ongoing research, and the results of different studies may vary. Also, fractal dimension is not the only approach to quantify the complexity of visual stimuli. Researchers also use other methods such as information theory and entropy to quantify the complexity of visual stimuli.

The fractal dimension of *visuospatial processing* refers to the complexity of the patterns in visual and spatial information that the brain processes. It can be measured mathematically using various algorithms such as box counting and has been found to be a useful tool in understanding the neural basis of visual perception and cognition. There have been multiple studies that have investigated the relationship between the fractal dimension of visuospatial processing and cognitive functions, neuropsychiatric disorders, and brain structure.

The processing of visual and spatial stimuli is sometimes described as taking place simultaneously in the ventral and dorsal streams of two separate neural systems [39]. Object quality or semantic information appears to be processed by the ventral stream, an occipito-temporal network, with more abstract categories being represented in more anterior areas of the temporal lobe. It has always been assumed that the dorsal stream, an occipito-parietal network, solely processes spatial data. However, this conventional perspective of the dorsal stream has just been modified. A third system known as the parietal-medial temporal pathway (PMT) appears to be crucial for integrating spatial and semantic information [40]. In order to recognize and create well-formed hierarchical embeddings in the visuospatial domain, it has been proposed that the PMT may have a special representational function.

The ability to recognize well-formed visuospatial hierarchical structures was evaluated in the seminal work by Martins et al. [41], based on the application of rules that either operate transformations within a hierarchical level or rules that generate new self-similar hierarchical levels. In this work, the authors contrasted the brain networks active during the representation of processes that allow the generation of new hierarchical levels (as required for generating fractals) with the representation of processes that may generate structures of equal complexity but do so without creating new levels. They did this using functional magnetic resonance imaging (fMRI). The visual recursion task (VRT) and the embedded iteration task (EIT) were two tasks created by the authors to examine the cognitive processes involved in the processing of visuospatial hierarchies. Participants in both tasks are exposed to generative processes for a predetermined number of iterative stages before being asked to draw conclusions regarding subsequent iterations. This indicates that in both tasks, participants are expected to derive straightforward principles from the initial iterations that may be used to foretell subsequent modifications. In VRT, each iterative phase develops a new hierarchical level in accordance with a specific spatial rule that is identical to the rule shown in the hierarchy's

earlier levels. The brain may create massive self-similar structures (fractals) with an infinite number of levels using just one straightforward rule. In EIT, new items are incrementally inserted into a predetermined hierarchical level, following a spatial norm without creating new levels. The hypothesis underlying this study was that many beautiful structures in nature are fractals, and based on their earlier research, they postulated that these are processed with a “fractal” cognitive technique in a particular and highly effective manner. The capacity to create representations utilizing previously known hierarchical information that allow the differentiation of new predictable hierarchical transformations makes both the VRT and EIT tasks novel. The authors found that when processing identical images in the visual-spatial domain using a “fractal” or a “nonfractal” cognitive technique, the brain employs different resources. The integration of spatial and abstract semantic information, as well as the integration of objects in contexts, appears to be recruited resources by the depiction of recursive rules allowing the production of additional hierarchical levels. Instead of being highly confined, this technique is implemented throughout a network of brain regions that spans both domain-specific and regions linked to specialized visual-spatial processes.

38.1.3 Fractal Dimensions of Language

The ability to communicate verbally or visually is known as language. Words are elements of language, which can be seen as a complicated system where they interact to create certain patterns [42]. These patterns stand in for the meaning that humans attribute to their ideas, feelings, will, and knowledge. Other communication systems do not have as many words to describe abstract and intellectual ideas as human language does. Additionally, the presence of polysemy, synonymy, and other concepts adds to its complexity. Through patterns in the geographical distribution of words and their frequencies, a text can be partially understood.

Texts inherit language’s complexity since they are written expressions of it. Indeed, texts are more than just a collection of words; these must have purpose for us to refer to it as a text. To put it another way, a text’s words must be used in a precise order to convey meaning.

The fractal dimension can be used to quantify the degree of self-similarity present in a pattern or structure. In the context of language, researchers have attempted to use fractal dimension to quantify the complexity of a language’s syntax, with the goal of better understanding the nature of language itself [43]. It is based on the concept of fractals, which are self-similar patterns that repeat at different scales. In the case of language, the fractal dimension is calculated based on the frequency of word n -grams, which are sequences of n words, in a text [44]. Altmann and Gerlach provide a complete guide on statistical and mathematical approaches employed for characterizing the complexity of human language [45].

Language is a member of the class of structures known as natural fractals. The order in which the words are arranged in an array determines the meaning. The

ordering of the words is crucial for conveying the content because any random placement of the words throughout the text drastically lowers its significance. In other words, the pattern of each word's appearance in linguistic arrays has some regularity in terms of meaning. The geographical distribution of word types in human language exhibits a fractal structure as a result of this statistical regularity [46].

There are other different methods for calculating the fractal dimension of language. The so-called Hurst exponent, which evaluates long-range dependence in physical systems, is similarly closely related to fractal dimension. Ausloos used the Hurst exponent to measure the distance between language sequences of the author, poet, and mathematician Lewis Carroll [47]. Techniques that approximate scaling from limits inferred from regression lines over log-log plots of size against scale can be used to calculate the fractal dimension of language. Box counting is another method for calculating the fractal dimension of complicated objects, such as language. This method involves dividing a text into smaller and smaller boxes and counting the number of boxes that contain at least one word. The fractal dimension is then calculated based on the relationship between the number of boxes and the size of the boxes. Studies have found that the fractal dimension of natural language is typically between 1.3 and 1.5, indicating a moderate degree of self-similarity [48]. This contrasts with random text, which would have a fractal dimension of 1, and a completely regular text, which would have a fractal dimension of 2. The fractal dimension can also be used to compare different languages, and studies have found that the fractal dimension of some languages, such as Chinese, is higher than that of others, such as English [49]. Mehri and Lashkari proposed another box-counting approach to the language component [50]. In the box counting process for language, suppose that a text with length N_t words is divided into N_b number of boxes with equal size L_b : $N_b = \lceil N_t/L_b \rceil$, where $\lceil x \rceil$ returns integer part of x . The number of boxes that contain a special word type (filled boxes) is denoted by $\eta_1(w)$.

Others contend that the strongest evidence for language's complexity can be found in the power-law distributions of lexis [42] and lexical variation [51]. The first to observe that, given a large enough text, the likelihood of the most common word occurring in the text is inversely proportionate to its rank was linguist George Kingsley Zipf [52]. As a result, language use is defined by a relatively limited number of often occurring words and a smaller number of less frequent ones. This phenomenon can also be seen in lexical variation, where the most prevalent semantic and phonological variants of a given word appear exponentially more frequently than the least prevalent variants. This phenomenon is not limited to word distribution within a text (i.e., Zipf's law). In addition, the fractal dimension can also be applied on a single author, and it has been used to detect the authorship of texts and the similarity between texts written by different authors [53].

In conclusion, fractal analysis enables us to see and describe how complex dynamic systems repeat through time and across various timescales [54]. Viewing developing patterns in this way holds significant potential for research into the human language capacity because the exact timelines of linguistic development are still somewhat elusive. It's worth mentioning that fractal dimension is a

mathematical concept, and it's not the only way to measure complexity in language, it's just one of the ways. However, despite this is still an area of ongoing research, in the future, the evaluation of fractal dimension to characterize language disorders (i.e., aphasia) will be outstanding.

38.2 Fractal Dimensions of Brain Disorders

In the neurological domain, fractal dimensions have been mainly applied to the analysis of neurophysiological signals (i.e., fMRI, structural MRI, EEG, and HRV). As concerns the human cognition, fractality has been applied to investigate various aspects of *movement*, such as the coordination and variability of movements, as well as the complexity of movement patterns. Recently, it was discovered that the seemingly “noisy” changes in the stride interval of human walking really contained fractal dynamics [55, 56]. A dynamical study of these step-to-step variations uncovered a pattern that is self-similar: fluctuations at one-time scale are statistically identical to those at several other time scales, at least over hundreds of steps, while healthy people walk at their typical pace. For instance, Hausdorff and colleagues [57] used data collected from healthy participants who walked for an hour at their normal, slow, and fast paces to investigate the stability of this fractal characteristic. For up to 1000 steps at all three walking rates, the stride interval fluctuations showed long-range correlations with power-law decay. These long-range connections vanished during metronomically paced walking, however, and the fluctuations in stride interval were random (uncorrelated) and nonfractal. Thus, the spontaneous stride interval's fractal dynamics are typically highly stable and inherent to the locomotor system. Furthermore, the higher neurological systems in charge of regulating walking rhythm may be connected to this fractal feature of neural output.

Numerous natural events exhibit long-range correlations and fractal-like characteristics [58]. When working with real-world time series, the Hurst exponent is one of the most accurate factors to measure the scaling law for these temporal correlations [59]. The long-range correlations in the stride interval fluctuations are also reduced by pathological changes to the motor system. In participants with Huntington's disease and Parkinson's disease, these associations were lower than in controls [60, 61]. Both disorders affect the basal ganglia, whose inability to coordinate the proper order of muscle contraction results in tremors or disorganized movements that enhance short-range noise. Although the basal ganglia in patients with advanced amyotrophic lateral sclerosis are intact and do not produce short-range noise, stride interval fluctuations from these patients also displayed lower levels of long-range correlations, as one would anticipate from a completely deconfigured neural command that impairs movement [61].

Daily motor activity displays fractal or self-similar fluctuations, which are characterized by comparable fluctuation patterns or statistical features over a wide variety of time scales [62]. Without external effects, such scale-invariant

behavior or fractal/self-similar patterns continue, pointing to an underlying fractal regulation (FR) in physiological systems [63]. According to theory, FR confers crucial physiological benefits for maintaining system integrity and adaptability, and that FR degradations signify a loss of functional resilience and an increase in sensitivity to catastrophic events, as seen in aging and disease [64]. The scientific group coordinated by Kun Hu has demonstrated that FR in motor activity is impaired in older people and is more degraded in individuals with Alzheimer's dementia, which is consistent with the adaptation theory of fractal physiological control [65]. Additionally, in extremely old persons with dementia, the level of FR degradation is highly related to cognitive function [66]. In a human postmortem study of dementia patients, they discovered a significant correlation between FR degradation prior to death and the density of amyloid plaques in the occipital cortex, a measure of the severity of Alzheimer's disease (AD) [67].

Generally, we can conclude that in neurological conditions, primarily affecting the motor system, the analysis of the motor impairments is characterized by a lower fractal dimension in their patterns compared to healthy controls, indicating a decrease in the complexity of their movements [68]. It's important to note that these studies are just one aspect of motor function analysis and other methods such as kinematic, kinetic, and EMG analysis are also commonly used to assess the motor function. Furthermore, more research is needed to fully understand the relationship between fractal dimension and motor dys-function and how it can be used as a tool for diagnosis and treatment of neurological disorders.

Assessing the fractal nature of motor disorders has also been performed in a particular psychiatric category: the Tourette Syndrome. More than 20% of all children experience tics, which are transient, pointless, unintended behaviors that manifest as recurrent motions of the vocal cords or skeletal muscles. Tics frequently occur in clusters (bursts of numerous tic occurrences occurring within a few seconds, followed by longer tic-free intervals). However, over longer timescales, bouts of tics lasting several minutes have also been reported. Longer time periods (days, weeks, and months) are subsequently covered by this recursive behavior, retaining self-similarity. This evidence implies that the tics' chaotic temporal character may have an intrinsic fractality, or neighbor-to-neighbor association, at all timescale levels. The difficulties in measuring the scale-invariant, power law exponent, known as the fractal dimension D_f , from fractional Brownian noise have prevented academics from concluding this phenomenon for more than 20 years. Beeler and colleagues [69] use time-series datasets taken from children 6–12 months apart to identify the fractal pattern connected to the occurrence of tics. They found that the occurrences exhibit neighbor-to-neighbor correlation and have an associated D_f at timescales ranging from seconds to minutes. These authors concluded that D_f is a robust and repeatable quantitative metric that makes it an easy-to-use but precise instrument for calculating the severity and fractality of tic attacks.

As concerns cognitive functions, studies have used FD to analyze the complexity of *memory* patterns in healthy individuals and in patients with memory disorders such as Alzheimer's disease. In Alzheimer's disease, studies have found that patients with the disease have reduced fractal dimension in their memory patterns

[70]. This reduction in fractal dimension suggests that patients with Alzheimer's disease have a reduction in the complexity of their memory patterns, which may reflect a decline in the neural mechanisms that underlie memory. Studies have reported that AD patients show decreased fractal dimension in memory-related brain regions, which may reflect a deficit in the neural mechanisms underlying memory encoding and retrieval. In patients with traumatic brain injury (TBI), the relationship between memory dysfunctions and fractality has been investigated by means of motor disorders. Indeed, deficiencies in gait and balance occur in the later stages of dementia and previous preclinical and clinical evidence demonstrated that the degree of irregular movements increased as a function of the cognitive decline. Fractal dimensions have been used to study elders' path tortuosity, demonstrating a high correlation between path tortuosity and dementia-related cognitive impairment [71]. Reduced route tortuosity has also been linked to functional recovery in patients with TBI. FD obtained from tracking data was connected to cognitive impairment in patients with TBI and, more recently, in inhabitants of assisted care facilities [72]. The value of Fractal D, which gauges how straight the routes are, can range from 1.0 for a straight path to 2.0 (chaotic Brownian motion), or the well-known "drunkard's walk." The Mini-Mental State Examination has shown an association between higher Fractal D scores and worse geographic orientation [73].

Overall, these studies suggest that FD analysis can be a useful tool for understanding the neural mechanisms underlying memory disorders in some neurological diseases and potentially aid in monitoring of the disease.

As concerns language, some authors have found that patients with AD have reduced fractal dimension in their language patterns, which is thought to reflect a reduction in neural complexity and a loss of neural plasticity. The loss of linguistic abilities is another significant issue brought on by AD in addition to memory loss. Aphasia (difficulty speaking and understanding), anomia, and other communication deficiencies in the field of language can all be detected (difficulty in recognizing and naming things). The patient's particular communication issues vary depending on the disease's stage.

Despite the value of the well-known biomarkers in AD, it is not feasible to administer such tests to all patients due to the associated costs and social/technological constraints. Given these issues, noninvasive intelligent diagnostic techniques may prove to be useful instruments for dementia early detection. Automatic spontaneous speech analysis has been proposed as a useful tool that can be used in the patient's usual environments without affecting or impeding the patient's abilities because the spontaneous speech used in these techniques is not perceived by the patient as a stressful test. One study, using the Higuchi algorithm for measuring FD, evaluated the complexity of speech patterns in AD patients during a spontaneous speech fluency test [74]. They found that features relative to speech duration, time domain, spectral domain and fractal dimension can be used to automatically distinguish AD patients from controls. This study also suggests that AD patients have a reduction in the complexity of speech patterns, which may reflect an early decline in the neural mechanisms that underlie language processing. Another study [75] used an ensemble of algorithms for developing

a speech recognition tool that automatically analyses the three main features of language: duration/time domain/frequency domain. Higuchi, Katz, and Castiglioni are three of these algorithms, which bear the names of their creators. These authors demonstrated that fractal proprieties of language disorders in AD, as defined by the implementations of Higuchi's, Katz's, and Castiglioni's algorithms, have the capacity to record language system dynamics and, as a result, expose pertinent fluctuations in spoken utterances.

Overall, these studies suggest that FD analysis can be a useful tool to aid in the early diagnosis and monitoring of AD patients. However, it is important to note that more research is needed to confirm these findings and to better understand the fractal properties of language disorders in every stage of AD-related neurodegeneration.

38.3 Conclusions

The translation of fractality on cognitive neuroscience as a mathematical framework helpful in determining the nature of human cognition is an interesting field of study that will create new bridges between different scientific fields. Despite the difficulties that lie ahead, it is obvious that this mathematical concept can greatly aid in the objective instrumental analysis of several cognitive functions, and that previous research has contributed to our understanding of the mechanisms underlying some of the most common pathologies and how they affect mental deficits. It would not be unexpected to see more research articles published on this subject in the future, or even to see those measurements used in a therapeutic setting.

However, there's a lack of research on how fractal dimension can be used as a diagnostic tool for brain/mental disorders, and how it can be used to monitor the treatment progress. It's important to note that these studies are just one aspect of mental disorders analysis and other methods such as artificial intelligence applied to neuroimaging [76] or genetic assessments [77] are commonly used to characterize and predict the occurrence of brain disorders. Furthermore, more research is needed to fully understand the relationship between fractal dimension and psychiatric/neurological disorders, and how it can be used as a tool for diagnosis and treatment.

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Part IV
Computational and Translational
Fractal-Based Neurosciences

Chapter 39

Computational and Translational Fractal-Based Analysis in the Translational Neurosciences: An Overview



Antonio Di Ieva

Abstract After the previous sections on “Fractals: What and Why?,” the last section of this book covers the software tools necessary to perform computational fractal-based analysis, with special emphasis on its applications into the neurosciences. The use of *ImageJ* and MATLAB, as well as other software packages, is reviewed. The current and future applications of fractal modeling in bioengineering and biotechnology are discussed as well. Perspectives on the translation of merging fractals with artificial intelligence-based methods with the final aim of pattern discrimination in neurological diseases by means of a unified fractal model of the brain are also given. Moreover, some new translational applications of fractal analysis to the neurosciences are presented, including eye tracking analysis, cognitive neuroscience, and music.

Keywords Artificial intelligence · Computation · Fractal · ImageJ · MATLAB · Modeling patterns recognition

What we observe is not nature itself, but nature exposed to our method of questioning.
(Werner Heisenberg, 1958)

From the previous chapters of this book, it is clear that computational fractal-based analysis has found and still finds large applications in the basic as well as clinical neurosciences, offering quantitative and qualitative parameters for the study of cellular and subcellular components of brain cells as well as the whole brain itself. Moreover, fractal analysis offers potential image and prognostic biomarkers useful in clinical medicine. After having reviewed such applications, the reader may be interested in the methods and software to perform computational fractal-based analysis. Part IV of this book is focused on the software tools to compute the fractal

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dimension and other complementary parameters. This section may be of interest to the neophyte approaching computational methods as well as for the computer science expert who would like to review the state-of-the-art methods or obtain new inputs.

Moreover, some computer science, translational neuroscience, and engineering-based perspectives (actual and future) are offered in some chapters.

39.1 How to Compute Fractals in Clinical Neurosciences

The previous sections of this book dealt with the questions “Fractals in Neurosciences: What and Why?”. The key question of this section is: “How?”

Several software packages are available for the fractal analysis of images (i.e., radiological and histological images) and time series. Many researchers use software programmed ad hoc by computer scientists for specific tasks. Nonetheless, one of the most used toolsets for computing fractal parameters in neurosciences has been, and currently is, the open-source image analysis software named *ImageJ*, which supports functions to compute the fractal dimension, lacunarity, and other parameters. In Chap. 40, two pioneers of the application of *ImageJ* for fractal analysis in the neurosciences, *Audrey Karperien* and *Herbert Jelinek* who also introduced the plug-in *FracLac* for *ImageJ*, illustrate its use for pattern extraction and translational research, which was aimed at translating the fractal geometry of the brain into clinical applications. MATLAB and other software platforms and tools online can be used as well, as illustrated in Chap. 41 by *Juan de Miras*. A chapter published in the first edition of the book on the use of GPU programming to increase the computational speed to calculate fractal parameters has not been included in this new edition, as over the last few years GPU-based computing has become a standard in many laboratories around the world.

39.2 Fractals in the Translational Neurosciences

Some practical examples of applications in the translational neurosciences are illustrated in this section. In Chap. 42, *Robert Newport*, *Sidong Liu*, *Antonio Di Ieva* summarize the main findings related to the quantification of eye tracking (i.e., “scanpaths”) and how fractal analysis can be used to implement such analysis within the cognitive and clinical neuroscience domains. Leaping into the realms of bio-engineering and biotechnology, in Chap. 43, *Mosleh*, *Rowland*, *Smith*, *Watterson*, *Griffiths*, *Montgomery*, *Philliber*, *Marlow*, *Perez*, and *Taylor* introduce the reader to a fractal fabric for the construction of bio-inspired fractal interconnects to interface the body and brain elements (e.g., neurons) in order to restore damaged functions. Then, *Rowland*, *Moslehi*, *Smith*, *Harland*, *Dalrymple-Alford*, and *Taylor*

continue the linkage between fractals in Chap. 44, neurosciences and bio-inspired engineering by proposing fractal geometry as a method to optimize the connectivity of neurons to artificial implants. *Richard Taylor, Catherine Viengkham, Julian Smith, Conor Rowland, Saba Moslehi, Sabrina Stadlober, Anastasija Lesjak, Martin Lesjak, and Branka Speh* present the concept of fractal fluency in Chap. 45, regarding the powerful significance of fractals for the human senses.

Jumping to further translational sciences, *Mauricio Dias Martins* illustrates the cognitive and neural representations of fractals in vision, music, and action in Chap. 46, while *Ion Andronache, Daniel Peptenatu, Helmut Ahammer, Marko Radulovic, Goran Djurčić, Herbert Jelinek, Carlo Russo, and Antonio Di Iorio* translate some findings from ecology, based on the introduction of new fractal-based parameters used for the quantification of forestation or deforestation patterns, to the clinical neurosciences in Chap. 47.

39.3 Fractals and Artificial Intelligence

Artificial intelligence has had exponential development and use in any field over the last decades, with deep learning promising extraordinary applications in the biomedical sciences and neurosciences. Natural language processing is also opening new venues and applications in the biomedical domain. The quick development of more sophisticated chatbots, like ChatGPT, developed by OpenAI, is showing the power of AI for drafting and generating sentences, papers, books, reports, etc.

For example, I asked ChatGPT to draft a paragraph by merging the keywords “fractal geometry” and “artificial intelligence,” and this is one of the results generated in less than 3 s:

Fractal geometry is a branch of mathematics that studies shapes and patterns that exhibit self-similarity, meaning that they can be split into smaller pieces that are similar to the larger whole. Fractals are often characterized by a repeating pattern that is revealed at increasingly fine scales, and they can be found in a variety of natural phenomena, such as snowflakes, mountains, and coastlines.

Artificial intelligence (AI) is the field of computer science that aims to create intelligent machines that can perform tasks that typically require human-like intelligence, such as learning, problem-solving, and decision-making. There are many different approaches to building AI systems, including machine

(continued)

learning, where a system is trained to perform a task by being fed data and adjusting its internal parameters to improve its performance, and symbolic reasoning, where a system is programmed with explicit rules and logical arguments to make decisions.

It is possible that fractal geometry could be used in some AI applications, such as image recognition or natural language processing. However, the relationship between fractal geometry and AI is not a direct one, as there are many other areas of mathematics and computer science that are more directly related to the development of AI.

Then I tried to cross the prompt “fractals” with “brain”:

“Fractals are complex, self-similar patterns that can be found in nature and in many man-made designs. They are characterized by their irregular and often asymmetrical shapes, and they can be found in a wide range of applications, including in computer graphics, engineering, and physics.

The brain is the organ responsible for the coordination and control of the body’s functions, including the senses, movements, thoughts, and behaviors. It is made up of billions of cells called neurons that communicate with each other through electrical and chemical signals.

There is some evidence to suggest that the brain may exhibit fractal-like patterns in certain aspects of its organization and function. For example, studies have shown that the branching patterns of neurons in the brain exhibit self-similarity, and that brain activity exhibits complex, dynamic patterns that are similar to those found in fractals. However, the relationship between fractals and the brain is still a topic of active research, and much more work needs to be done to fully understand this connection”.

I pushed it further, asking ChatGPT to write a novel on the brain and fractal in Lovecraft style, and this is what was generated in a few seconds.

Title: *The Fractal Brain*

Neuroscientist Dr. Mary Williams was always drawn to the mysteries of the human mind, but she never could have imagined the horrors that awaited her when she was recruited to study the brain of a mysterious comatose patient.

As she delved deeper into the patient's brain, she discovered that it was a perfect fractal - a geometric pattern that was repeated at ever smaller scales, creating a never-ending, self-similar structure. But this was only the beginning of her nightmare.

As she continued to study the fractal brain, she realized that it was not of this world. It was a gateway to a realm of cosmic horror and insanity, a place where the laws of physics and the nature of reality itself were twisted and distorted beyond recognition.

Mary became convinced that the fractal brain was connected to some ancient, otherworldly power, and she knew that she must find a way to destroy it before it consumed her entirely. With the help of a small group of fellow scientists, she began to unravel the secrets of the fractal brain.

But as they drew closer to the truth, they found themselves pursued by shadowy figures who seemed to be guarding the gateway to the other realm. In a desperate race against time, Mary and her colleagues fought to destroy the fractal brain and close the gateway before it was too late.

In the end, they were successful, but the experience left them all forever changed, haunted by the horrors they had witnessed. Mary knew that she could never forget the terrifying power of the fractal brain, and she vowed to spend the rest of her days studying the mysteries of the human mind, always wary of the dangers that lurked within.

And then I asked for a variation:

Title: *The Fractal Horror*

Neuroscientist Dr. Mark Thompson was always on the lookout for new and interesting cases to study. When he was approached about a comatose patient with an unusual brain structure, he couldn't resist the opportunity to delve into the mysteries of the human mind.

As Mark examined the patient's brain scans, he was immediately struck by the extraordinary complexity and symmetry of its structure. It was unlike anything he had ever seen before. He became convinced that this brain held the key to unlocking some of the most fundamental mysteries of the universe.

But as he delved deeper into his studies, Mark began to experience strange and disturbing hallucinations. He became convinced that the patient's brain was connected to another realm, a place of unimaginable horrors.

Despite the warnings of his colleagues, Mark was determined to uncover the truth, no matter the cost. He spent every waking moment studying the patient's brain, determined to unlock its secrets.

As he reached the brink of madness, Mark finally understood the true nature of the patient's brain. It was a fractal, a geometric pattern that was repeated at ever smaller scales, creating a never-ending, self-similar structure. And it was a gateway to a realm of cosmic horror, a place where the laws of physics and the nature of reality itself were twisted and distorted beyond recognition.

With a sense of dread, Mark realized that he must find a way to close the gateway before it consumed him and the entire world. But it was too late. The horrors of the other realm had already taken hold of his mind, and he knew that he would never be able to escape the darkness.

And then in Edgar Allan Poe style:

Title: *The Fractal Mind*

It was a dark and stormy night, and neuroscientist Dr. Elizabeth Taylor found herself struggling to stay awake as she pored over the scans of a mysterious comatose patient's brain. She had been tasked with finding a way to bring the patient out of his coma, but so far, all of her efforts had been in vain.

As she gazed at the scans, a strange pattern caught her eye. It was a series of geometric shapes, repeating at ever smaller scales, creating a never-ending, self-similar structure. Elizabeth was certain she had never seen anything like it before.

She became obsessed with understanding the meaning of the pattern and spent every waking moment studying it. As she delved deeper into the patient's brain, she began to experience strange and disturbing hallucinations. She became convinced that the patient's brain was connected to another realm, a place of unimaginable horrors.

Elizabeth's colleagues became concerned about her mental health and tried to convince her to stop her research. But she was determined to uncover the truth, no matter the cost.

One night, as she sat alone in her lab, a bolt of lightning struck the building, causing a power outage. In the darkness, Elizabeth heard a voice whisper her name. She followed the voice to the patient's room, where she found him sitting up in his bed, his eyes glowing with an otherworldly light.

Elizabeth realized that the patient's brain was a gateway to the realm of cosmic horror, and that she must find a way to close it before it consumed her and the entire world. With the help of her colleagues, she was able to destroy the patient's brain and close the gateway, but the experience left her forever changed, haunted by the horrors she had witnessed.

Title: *The Fractal Mind*.

Prologue:

In the year 2049, neuroscientist Dr. Amelia Reed made a groundbreaking discovery while studying the brain's neural pathways. Using advanced imaging technology, she found that the brain's connections formed a pattern that resembled a fractal - a never-ending, self-similar pattern found in nature.

This revelation sparked a new field of study: fractal neurology. Dr. Reed and her team of researchers delved deeper into the mysteries of the fractal mind, searching for clues about how it could impact everything from cognition to consciousness.

As they delved deeper into this uncharted territory, they began to uncover some of the most profound secrets of the human brain.

Chapter 1: The Fractal Brain

As Dr. Reed sat in her lab, staring at the images of the brain on her computer screen, she couldn't help but feel a sense of awe. The patterns she was seeing were unlike anything she had ever seen before.

At first, she had assumed that the brain's neural pathways were a tangled mess, randomly connecting different areas of the brain. But as she continued to study the images, she realized that there was a clear pattern to the connections. They formed a series of branching pathways that seemed to repeat themselves at different scales, just like a fractal.

Dr. Reed immediately knew that this discovery could revolutionize the field of neuroscience. If the brain's connections formed a fractal pattern, it could potentially explain why the brain was so adaptable and able to learn new things throughout a person's lifetime.

Chapter 2: The Implications of the Fractal Mind

As Dr. Reed and her team delved deeper into their research, they began to uncover some incredible implications of the fractal mind.

For one, they found that the fractal pattern of the brain's neural pathways seemed to be related to intelligence. Those with more complex fractal patterns in their brains tended to have higher IQ scores and be more creative.

But the implications went beyond just intelligence. The team also found that the fractal mind was able to process information in a more efficient way, allowing for faster learning and decision-making.

And as they continued to study the fractal mind, they began to uncover even more mysteries. They found that the fractal pattern of the brain was related to consciousness itself, and that it might even be the key to understanding the subjective experience of being alive.

Chapter 3: The Future of Fractal Neurology

As the field of fractal neurology continued to advance, it opened up new avenues of research and treatment.

Doctors began using the knowledge of the fractal mind to develop more targeted and effective treatments for brain disorders such as dementia and Parkinson's disease. And with a better understanding of the fractal mind, researchers were able to make significant breakthroughs in the fields of artificial intelligence and robotics.

But perhaps most exciting of all was the potential for the fractal mind to help unlock the mysteries of the universe itself. As Dr. Reed and her team delved deeper into the mysteries of the fractal mind, they couldn't help but wonder what other secrets it might hold, waiting to be discovered.

And then I asked Chat GPT-3 to write a poem on fractals:

Fractals, a dance of patterns in the night Nature's symphony, a mesmerizing sight
A never-ending spiral, a hypnotic trance A fractal wonderland, where
beauty enhances.

From the smallest leaf, to the grandest tree Fractals are the key, to unlock
nature's mystery In every mountain range, and every shore Fractals are the
door, to an endless exploration.

In the chaos of the universe Fractals bring a sense, of balance and a verse A
fractal future, a fractal past A fractal present, forever lasting.

Infinite complexities, in every line A fractal journey, one that's worth the time
So let's embrace the fractals in our lives And let them be the light, that always
thrives.

These are just some examples on the power of AI to generate language. The InstructGPT models are based on training and fine-tuning of GPT with supervised learning, followed by comparisons to train a reward model, and then optimizing the reward model by using reinforcement learning. At this pace, it can be foreseen that one of the next editions of the *Fractal Geometry of the Brain* will be written by an artificial intelligence tuned up and reinforced by scientists.

In the last chapter of this book, *Lorenzo Livi, Alireza Sadeghian, and Antonio Di Ieva* already in 2016 suggested some intriguing future perspectives in the field on the merging of fractal analysis with computational intelligence methods for pattern identification and extraction, which are becoming more feasible in the light of the new technology available today and on the horizon tomorrow.

Figure 39.1 shows the results of the prompts “fractal geometry” and “brain” on OpenAI DALL·E 2.

Fig. 39.1 Figure generated by OpenAI DALL·E 2 following the prompts to merge fractal geometry and brain



39.4 Conclusive Remarks: Toward a Unified Fractal Model of the Brain?

Heisenberg's quote at the beginning of this short chapter subtends a key concept of fractal analysis as well as every kind of analysis applied to scientific research, including biomedical sciences and neurosciences: that our observation of natural phenomena is filtered by the lens of the applied methodology. For example, in neuroradiology, what we see is not the brain itself but pixels showing specific features of the brain. Similarly, in digital pathology, what we see in histological slides of brain tumors are two-dimensional images of the tumors as revealed by specific histological and/or immunohistochemistry techniques. Aware of such method-related limitations, we are still able to use such information to our benefit, despite the fact that we are observing a natural phenomenon filtered by a methodological "Veil of Maya." This is often forgotten by neophytes, and even by the more experienced researcher. As widely expressed in Chaps. 1 and 13, fractal geometry is a model, with its advantages and intrinsic limits. Fractal geometry represents a universal language to translate natural phenomena, but the limits of the model itself do not qualify it to be a Rosetta stone that is able to interpret every natural object, as fractals are not a panacea.

The value of fractal dimension can change according to several different factors: different computational methods, preprocessing tools, different scale windows, etc. The same structure can show different values depending on "the angles of view." The methodological differences give rise to a heterogeneous dataset of results, and the clinical significance of the fractal dimension as a biomarker becomes

controversial in the absence of a well-standardized computational procedure. Being aware of such limitations of fractal geometry as a model, there are still several aspects that make this analysis such a powerful tool to analyze natural phenomena, including the complexity of the human brain in its entire physiopathologic spectrum. As introduced in Chap. 1, this book has not shown the existence of the fractal geometry of the brain as a pure mathematical model, rather it has suggested that computational fractal-based analysis be used for pattern recognition in the basic and clinical neurosciences. The fractal approach has been shown suited to automatic image analysis (e.g., in neuroradiology as well as neuropathology), extraction of features in images as well as in time series, and identification and classification of types and subtypes of states within the wide spectrum of the normal to the pathological brain. In sum, there exist a host of fractal-based morphometrics of biodata in the clinical neurosciences that are *objective, realistic, reproducible, and potentially prognostic and predictive biomarkers* with much promise for use in clinical practice.

As suggested by *Audrey Karperien and Herbert Jelinek* at the end of Chap. 40, “We hope [that this book] has further inspired the reader to continue to demystify fractal analysis, to explore new fractal possibilities, to develop pattern extraction methods for new perspectives and especially for databases, [...] to bring the multiple levels of the fractal geometry of the brain to the fore, not just as a wonder of the digital age but as a clinical reality.”

As illustrated in Chap. 48, computational intelligence methods are aimed at recognition, discrimination, and prediction in the analysis of complex patterns. The future of this field may be a computational fractal-artificial intelligence-based analysis, which, associated with further analyses, might offer, in a multiparametric way, a computer-aided, objective, and reproducible categorization of the different pathologies. By this way, the clinicians’ armamentarium will be enriched by new diagnostic, prognostic, and eventually therapeutic biomarkers as well.

Chapter 40

ImageJ in Computational Fractal-Based Neuroscience: Pattern Extraction and Translational Research



Audrey L. Karperien and Herbert F. Jelinek

Abstract To explore questions asked in neuroscience, neuroscientists rely heavily on the tools available. One such toolset is *ImageJ*, open-source, free, biological digital image analysis software. Open-source software has matured alongside of fractal analysis in neuroscience, and today *ImageJ* is not a niche but a foundation relied on by a substantial number of neuroscientists for work in diverse fields including fractal analysis. This is largely owing to two features of open-source software leveraged in *ImageJ* and vital to vigorous neuroscience: customizability and collaboration. With those notions in mind, this chapter's aim is threefold: (1) it introduces *ImageJ*, (2) it outlines ways this software tool has influenced fractal analysis in neuroscience and shaped the questions researchers devote time to, and (3) it reviews a few examples of ways investigators have developed and used *ImageJ* for pattern extraction in fractal analysis. Throughout this chapter, the focus is on fostering a collaborative and creative mindset for translating knowledge of the fractal geometry of the brain into clinical reality.

Keywords Computer-assisted image processing · Fractals · Box counting · ImageJ · FracLac · Microglia · Neurons · Theoretical models

40.1 Introduction

Imagine opening the screen on a cell phone and showing a patient a fractal dimension displayed overtop of the large vessels in the vicinity of their brain tumor and then zooming in to the gross overall outline and texture of their biopsy, through microvasculature and intertwining neuronal and other brain cell processes, all the

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way down to that one unusually complex astrocyte, and then its nucleus, fractal dimension displaying on the screen at each level—and patient and clinician alike understanding. This is the vision and mindset of fractal-based neuroscience for the near future that *ImageJ* supports.

The topic of fractal analysis is decades old, in its infancy within the millennia of science that human history holds, but of age in today's digital revolution. It is underlain by a set of tools developing in the context of unprecedented technological advances and global collaboration. Generally, the path any tool develops along follows the questions being asked in the workshop and the world served by that workshop, and those questions in turn develop within the questioner's expectations of how well their question might even begin to be explored. This chapter delineates the place of one tool, *ImageJ* (*IJ*), in influencing the imaginations of the people asking questions in the global community of fractal-based neuroscience and in making the topic generally accessible.

First, this chapter answers two questions, “What is *IJ*?” and “Where does it fit in fractal-based neuroscience today?.” As the anatomical fly-through thought exercise in the introductory paragraph suggests, this chapter's scope is limited to fractal analysis of digital images. In particular, bridging the gap between tumor and fractal dimension, vasculature and fractal dimension, neuropil and fractal dimension, etc., lies a substantial part of the realm of fractal analysis of digital images—pattern extraction. Thus, the chapter also reviews some ways investigators have used *IJ* to creatively answer questions about fractal analysis pattern extraction.

40.2 What Is *ImageJ*?

40.2.1 Removing Barriers with Free, Open-Source Software

To some people, *IJ* is one of many user-friendly photo filtering “apps”—to others, a powerful lab tool. At its simplest, it is computationally powerful [33], open-source software for digital image processing and analysis. At its broadest, it was designed with a mindset to meet four freedoms of open-source software quoted below:

Freedom:

1. To run the program for any purpose
2. To study how the program works and change it to make it do what you wish
3. To redistribute copies so you can help your neighbor
4. To improve the program and release your improvements to the public so that the whole community benefits [67].

IJ was created by Wayne Rasband at the US National Institutes of Health (NIH). First released in 1997, it was based on an existing computer program, *NIH Image* but written in the then new Java programming language (hence the “J”) [75]. The significance of this is that it made *IJ* independent of computer

architecture, owing to Java's then-revolutionary philosophy of "write once, run anywhere." *IJ* was at the outset and remains easily installed on most computer operating systems, making it a tool that diminishes barriers to collaboration that different operating systems can pose. This is important today, but it was perhaps more important when the program was first made available, a time when, more than today, having incompatible computer operating systems rendered many forms of collaboration very time-consuming or impossible. *IJ* reduced other barriers, too, making it popular in many fields, by being freeware [1, 26, 75]. *IJ* was and remains available for anyone to download from the NIH's public website (<http://imagej.nih.gov/ij/download.html>).

40.2.2 *Shaping Computational Fractal-Based Neuroscience*

40.2.2.1 *Making Fractal Analysis Accessible and Customizable*

Among many image-processing and analysis functions, *IJ* included fractal analysis methods early on. In particular, it included a fractal dimension-determining function, a basic binary box counting tool, founded on existing code written for *NIH Image* by Tom Smith, a pioneer in determining the fractal dimensions of neurons and other brain cells [77, 78]. Making fractal analysis even easier, *IJ* also included the basic pattern extraction methods required to use that binary box counting function (see Fig. 40.1) [77, 78].

Thus, it was possible for someone with basic digital image processing skills and no computer programming knowledge to use *IJ* to open digital images, decide on the type of patterns to extract, extract them, and obtain and compare fractal dimensions. This was, albeit, a rudimentary approach (Chap. 2 discusses box counting fractal analysis in detail and highlights this approach's limitations), but the point here is that the combination of these fundamental fractal analysis tools with the features of being platform independent, available, and free offered a timely opportunity for scientists with or without programming experience to personally explore their own questions about the growing field of fractal analysis.

In addition to demystifying and helping establish fractal analysis as an accessible method for the neuroscientist or clinician, *IJ* played an important role in encouraging the field to grow. Carrying forward basics from its predecessor program, it provided:

- Built-in functions to record and save macros (sequences of commands to repeat tasks and potentially apply them to batches of stored images or the same image with multiple modifiers).
- Plug-in capability (a plug-in is software that can be shared and quickly installed to make *IJ* able to do new tasks).
- Opportunity to shape the core software [56].

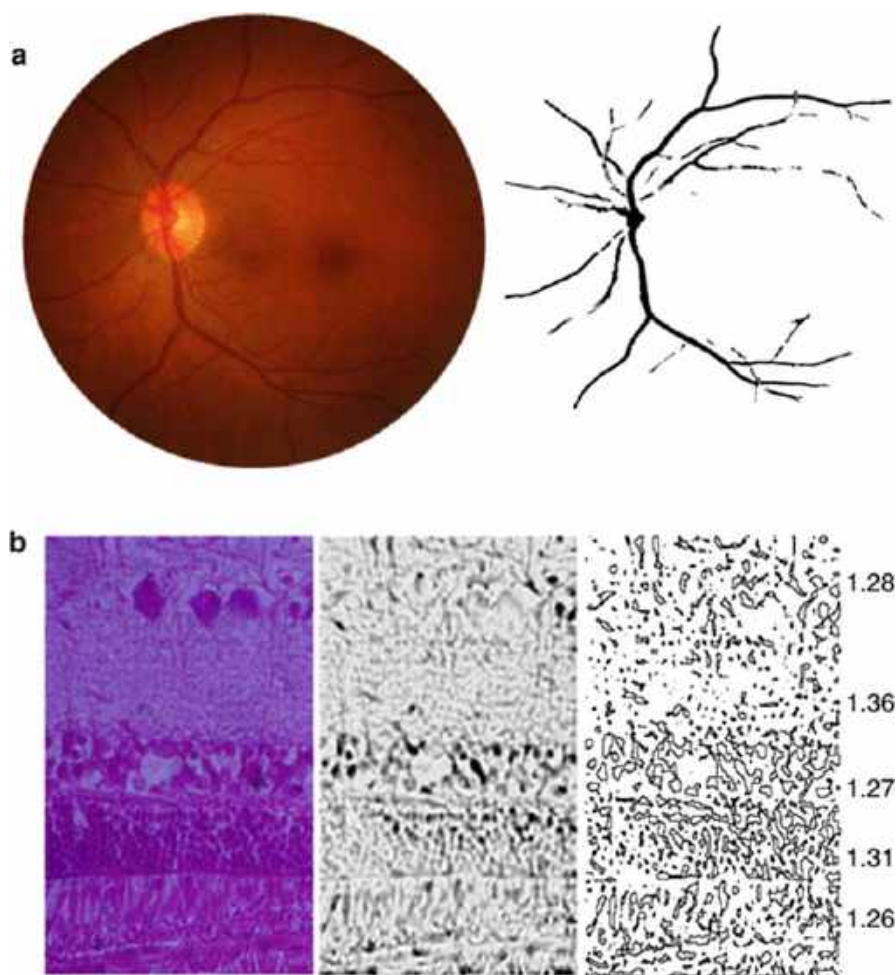


Fig. 40.1 Examples of patterns suitable for box counting fractal analysis extracted from the retina using basic *IJ* functions. (a) A binary silhouette (*right*) showing only the pattern of the vasculature, extracted from a retinal fundus photo, using background subtraction, smoothing, auto-selection, and binarizing. (b) Grayscale and binary textures extracted from a digital microphotograph of retinal layers on an HE-stained histological slide of retina; extracted by converting to grayscale, using background subtraction to remove ganglion and other cell body interiors and other undesired information, binarizing, and outlining

These features, which are applications of the freedoms underlining the *IJ* mindset, were especially relevant at the time because the methods of fractal analysis in neuroscience were just beginning to be developed [49]. They let people go beyond the basic box counting function to experiment with, customize, and hone fractal analysis methods; formulate and test questions; dynamically answer queries about multiple, changing nuances; rapidly process large numbers of images; etc.

IJ in general tapped into a growing scientific resource, and researchers and programmers soon produced a wide-ranging list of plug-ins, online guides, and tutorials in many fields (see <http://rsb.info.nih.gov/ij/plugins/>). Many of these resources supported fractal analysis to varying degrees, and many were created by or for both theoretical and clinical researchers to answer questions in fields such as microscopy, radiography, and brain morphology. They offered in one toolset a free, accessible, flexible, and user-friendly compendium of ways for non-programmers and programmers to do several types of fractal analysis (e.g., dilation, mass vs. radius [38], local connected fractal dimension [41, 45], grayscale analysis, and multifractal analysis) as well as related required pattern extraction tasks [15, 20, 31, 39, 48, 64, 68, 70, 75]. While these were certainly not the only means available to the research community for applying fractal analysis, they did create a common ground and foster a collaborative approach.

40.3 Where Does *IJ* Fit in Fractal-Based Neuroscience Today?

IJ plays a major part in both neuroscience and fractal analysis today. As illustrated in Fig. 40.2, one outcome of the resources described in the previous section and their continued development is that *IJ* today has a presence in published work linking the two fields [1, 15, 75].

Furthermore, *IJ* has become an image-processing model itself. It is now used as an applet in webpages and is available in a limited form for the Android operating system (e.g., for tablets and cell phones) [12, 72–74]. Today, *IJ* and *Fiji*, an *IJ* distribution developed to accommodate increasingly complicated plug-ins and data formats including “Digital Imaging and Communications in Medicine” (DICOM) and complex 3-dimensional (3D) data, are neuroscience industry standards. (Still straightforward software, “*Fiji*” is an acronym for “*Fiji* is Just *ImageJ*.”) *IJ* can communicate bidirectionally with many other types of software (e.g., *MatLab*) and is used as a software library in many applications including commercial applications used in neuroscience. Indeed, neuroscience labs today hire dedicated *IJ* programmers [3, 5, 7, 9, 10, 14, 18, 19, 50, 57, 71, 73].

IJ is also relevant in today’s changing science education environment and plays roles in translational research. *IJ* is being used increasingly in higher education, for instance, as digital pathology is now a reality, and virtual microscopy is becoming the norm for teaching medical histology [8, 43, 60]. Moreover, it is used to teach youth in high schools, including for teaching youth participating in science fairs and at science camps, how to do and understand fractal analysis of brain cells, and how to evaluate heart rate variability using fractal measures (see Fig. 40.3) [16, 36].

A key element of this widespread presence is that despite that *IJ* has evolved into a highly respected programming library and gained increasingly complex and specialized functions and uses, its main software interface has remained

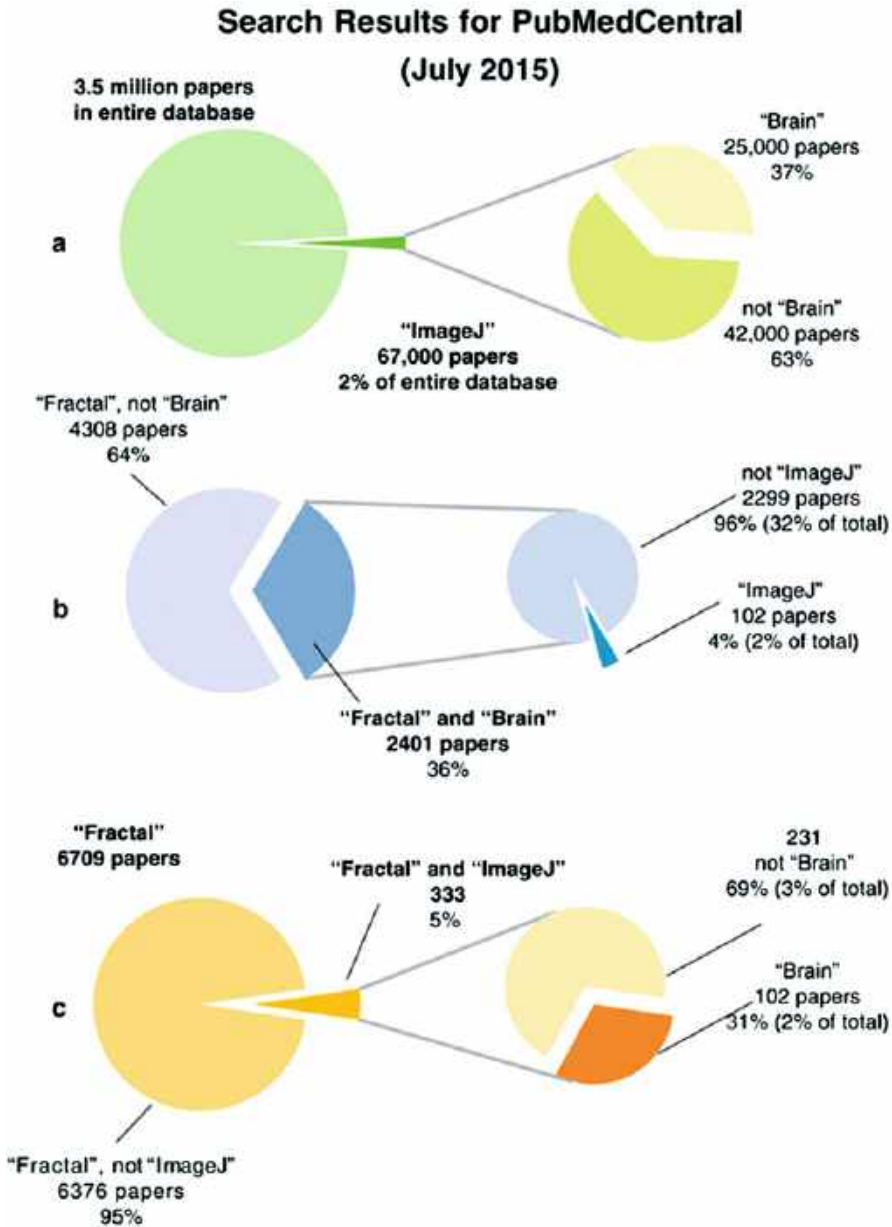


Fig. 40.2 *IJ* is referenced substantially in published articles indexed in PubMedCentral relevant to fractal analysis in neuroscience. (a) *IJ* is cited or otherwise referenced in 67,000 papers in the PubMedCentral database [57] and around a third of those are relevant to neuroscience (i.e., are also indexed under the term “brain”). (b) 36% of all articles relevant to fractal analysis (i.e., indexed under “fractal”) in the database are relevant to neuroscience and about 4% of those cite or otherwise reference *IJ*. (c) 5% of the database articles relevant to fractal analysis cite or otherwise reference *IJ*; about a third of those are relevant to neuroscience (Data based on a search of the US National Library of Medicine’s PubMedCentral in August 2015. Note that publication data generally underrepresent *IJ*’s use because authors do not always credit it when they use it for scientific publications [1, 75])



Fig. 40.3 *ImageJ* is used to teach young people about fractal analysis in neuroscience and leave them with the skills and tools (e.g., a youth-focused plug-in) to explore their own basic fractal analysis questions. (Panels are from presentations delivered to youth at a summer science camp in Bonnyville, Alberta, Canada, 2010)

straightforward. *IJ* remains a potent tool that the programmer and nonprogrammer alike can easily use on its own or with other software, at home and at work, to investigate, collaborate on, and explore new questions.

In fact, in many labs, for collaborating with colleagues on digital imaging fractal analysis questions, *IJ* is a preferred tool that equalizes differences in resources because it can be shared and customized as rapidly as the questions being asked change their nuances (e.g., by emailing images, macros, plug-ins, and results). To illustrate, a regular part of our work is adding features to or customizing the plug-in *FracLac* for *ImageJ* which was developed in our lab to overcome many pitfalls of fractal analysis (see Chap. 2) and has been used to publish in neuroscience and other fields (see <http://rsbweb.nih.gov/ij/plugins/fractal/FLHelp/FLCitations.htm> for 73 recent publications citing it) [4, 17, 21, 28, 30, 42, 82]. *FracLac* is an ongoing collaborative freeware project including a suite of tools for fractal analysis (including monofractal, lacunarity, multifractal, mass vs. distance, and local connected dimension analyses) and fractal bio-modeling (see Sect. 40.4). It was initially developed to answer basic questions about the fractal dimension of microglial brain cells and, as an open-source plug-in adhering to the *IJ* mindset, continues to develop based on our own evolving research needs and requests from users [2, 11, 27, 29, 40, 41, 45, 51, 79, 80].

40.4 Pattern Extraction

As was discussed in conjunction with the thought exercise that started this chapter (Sect. 40.1), fractal analysis of the geometry of the brain depends on bridging the gap between digital images and structural information (i.e., numerical coordinates in space). Today, *IJ* has centralized much effort spent toward solving one well-known and vexing burr in neuroscience that is doubly vexing in fractal-based neuroscience—the information problem [54, 65]. To elaborate, the scientific and medical worlds no longer face archival problems of too little cabinet space to store every X-ray and photograph, but of inadequate methods to analyze and manage rapidly amassing 3D data sets, videos, and still digital image information. Testifying to the significance of the problem, cash rewards are being offered for innovative contributions to the solution [37, 54, 58, 65, 66].

40.4.1 Pattern Types

Turning original digital images into patterns for fractal analysis is called “pattern extraction,” “segmentation,” or “preprocessing” and generally entails extracting some variation or combination of binary, grayscale, and 3D (e.g., voxel-based) data. The usual product of pattern extraction is itself another digital image, generally one of three basic types:

Contour	Gross outline with no fill (e.g., the outline of a hemisphere or tumor, a cell membrane, a nuclear membrane, the outer edges of branches, and the structures they radiate from such as a vessel or microglial cell); typically for binary analysis
Skeleton	The basic path of a branching structure excluding diameter changes (e.g., the path of an axonal tree or vasculature pattern); usually one pixel wide, typically for binary analysis
Silhouette or area	The texture within an area (e.g., the mass inside a hemisphere, tumor, cell, or nucleus), which may include a branching structure that may or may not have diameter changes and has diameters greater than a single pixel (e.g., vasculature; the area within the contour of a microglial cell); for either binary (filled with one color) or grayscale (filled with multiple possible colors) analysis

40.4.2 Extraction Methods

In general, published fractal analysis pattern extraction methods range from simple to elaborate. In terms of *IJ*, there are numerous free, open-source segmentation plug-

ins including methods that can be trained and methods that can be adapted to work with entire databases [12, 24, 31, 32, 44, 48]. This section outlines built-in functions, tracing plug-ins, thresholding, and customized methods relevant to fractal analysis in neuroscience using *IJ*.

There is an important caveat to this section, based on two general points. First, reading a signal, such as by extracting it into a binary format, choosing the part of a grayscale image to analyze, reconstructing 3D data, etc., is always idiosyncratic, and, in the end, what is considered signal and what noise, even within one image being investigated in more than one way, is limited only by the questions one chooses to ask. Second, all means of imaging are subject to confounders such as variation in lighting, contrast agent, intra- and interoperator variability, and a host of others pertinent to specific image acquisition protocols such as retinal photography, microscopy, radiography, etc. [61, 62]. These two points are common to all forms of image analysis, computer-aided or not, in science and medicine.

Accordingly, selecting a pattern extraction method requires many steps including testing that proposed methods (even automated) work in the way desired for a proposed dataset, reducing confounders, using control images, documenting anomalies and deviations, and, if the images are not in a previously acquired dataset, maximizing acquisition protocols in the first place (e.g., using staining, lighting, etc., appropriate to the question). Thus, whereas this section surveys options, it does not authoritatively prescribe how to extract patterns.

40.4.2.1 Built-in Functions

Fractal pattern extraction from images typically involves two fundamental techniques—tracing and thresholding. Tracing can be done manually, automatically, or semiautomatically and is usually the slower of the two. Thresholding means choosing for grayscale or color images a range of pixel values defining which pixels to keep (called foreground) and which to ignore (background). Sometimes pattern extraction combines the two processes. *IJ* includes a variety of functions for doing both and filters that enhance them, such as automated background subtraction, and smoothing. In labs with computer programming expertise, these basic methods are often modified and incorporated into macros and plug-ins to address specific issues pertinent to a set of images. Table 40.1 outlines some basic functions for tracing and thresholding, and Table 40.2 outlines some basic filters used in conjunction with those functions.

IJ's built-in functions are not necessarily original, insofar as commercial applications include many similar functions, and it is possible with some commercial applications to combine functions in macros and plug-ins as can be done with *IJ*. The unique value, however, inheres in features of *IJ* outlined in Sect. 40.2.1. From a lab perspective, for example, licensing restrictions can limit the number of people working with software at one time, giving free, open-source software an advantage for time-consuming tasks in particular. Moreover, basic extraction methods support informal or pilot-level enquiry and are adequate for some images,

Table 40.1 Basic built-in functions for fractal analysis pattern extraction by tracing and thresholding

Pattern	Function	Implementation
Contour	Draw	Select area with selection tools or other method then: Edit>Draw
		Optionally: Image>Adjust>Line width
	Binary outline	Select area with selection tools or other method then: Process>Binary>Make Binary
		Process>Binary>Fill
	Particle analyzer	Process>Binary>Make Binary
		Analyze>Analyze Particles
		Check Show Outlines then OK
		To make a selection to analyze: Edit>Selection>Create Selection
ROI manager	Analyze>Tools>ROI Manager	
	Select area with selection tools or other method then: ROI Manager>Add to add each to Manager	
	ROI Manager>More>Draw	
Skeleton	Skeletonize	Select area with selection tools or other method then: Process>Binary>Make Binary
		Process>Binary>Skeletonize
		Select area with selection tools or other method then: Process>Binary>Make Binary
		Process>Binary>Erode (repeat)
Silhouette or area	Fill	Select area with selection tools or other method then: Edit>Fill
	Binary fill	Select area with selection tools or other method then: Process>Binary>Make Binary
		Process>Binary>Fill
	Threshold	Image>Adjust>Threshold
	Particle analyzer	Process>Binary>Make Binary
		Analyze>Analyze Particles
		Check Show Mask then OK
	Grayscale	Select area with selection tools or other method then: Edit>Copy
		Or Edit>Clear Outside
		Or Image>Crop
	ROI manager	Analyze>Tools>ROI Manager
Select area with selection tools or other method then: ROI Manager>Add to add each to Manager		
ROI Manager>More>Fill		

Table 40.2 Methods for filtering images

Method	Implementation
Auto-selection	Choose Wand (tracing) tool
	Click on image to select a single or shift-click for a composite area
Subtract background	Process>Subtract Background
Filter	Process>Smooth or
	Process>Filter or
	Image>Color>Adjust LUTs
	Image>Color>Split Channels
Paintbrush	Image>Overlay>Add Overlay
	Choose Paintbrush Tool from main interface
	Optionally set width (e.g., edge or branch diameter):
	Highlight edge
	Analyze>Measure then note and set size:
	Choose Image>Overlay>Overlay Options
	Or Paintbrush Tool>Brush Width
	Type in noted size
	Manually trace features of interest then:
	Image>Overlay>To ROI Manager
	Reselect on original image by selecting ROI in ROI list
	Extract patterns using methods in Table 32.1

but do not necessarily work for every type of fractal enquiry. The generally greater freedom to write macros and customized computer code with *IJ* can be essential for unusual images, large jobs with often changing parameters, and the ability to look at images creatively [23, 46, 52–54].

40.4.2.2 Tracing Plug-Ins

Although using the unaided tracing methods from Table 40.1 can be quick for some images, it often becomes labor intensive (e.g., it may take hours to manually reconstruct the most complicated branching cells in *IJ* and similar computer software). To deal with this, several *IJ* developers have created tracing plug-ins that make the process easier and potentially less subjective. Three examples of such plug-ins are *NeuronJ* [55], *Neuron_Morpho* [15], and the *Simple Neurite Tracer* [48] (see Fig. 40.4). These plug-ins can be used to obtain branching structure information in all three of the basic pattern types from Sect. 40.4.1; *Neuron Morpho* and the *Simple Neurite Tracer* can also be used for 3D reconstructions. Although the developers originally created and tested these to answer questions about the structure of neurons, researchers have used them for related structures such as vasculature [81].

Tracing Branching Patterns for Fractal Analysis
Simple Neurite Tracer

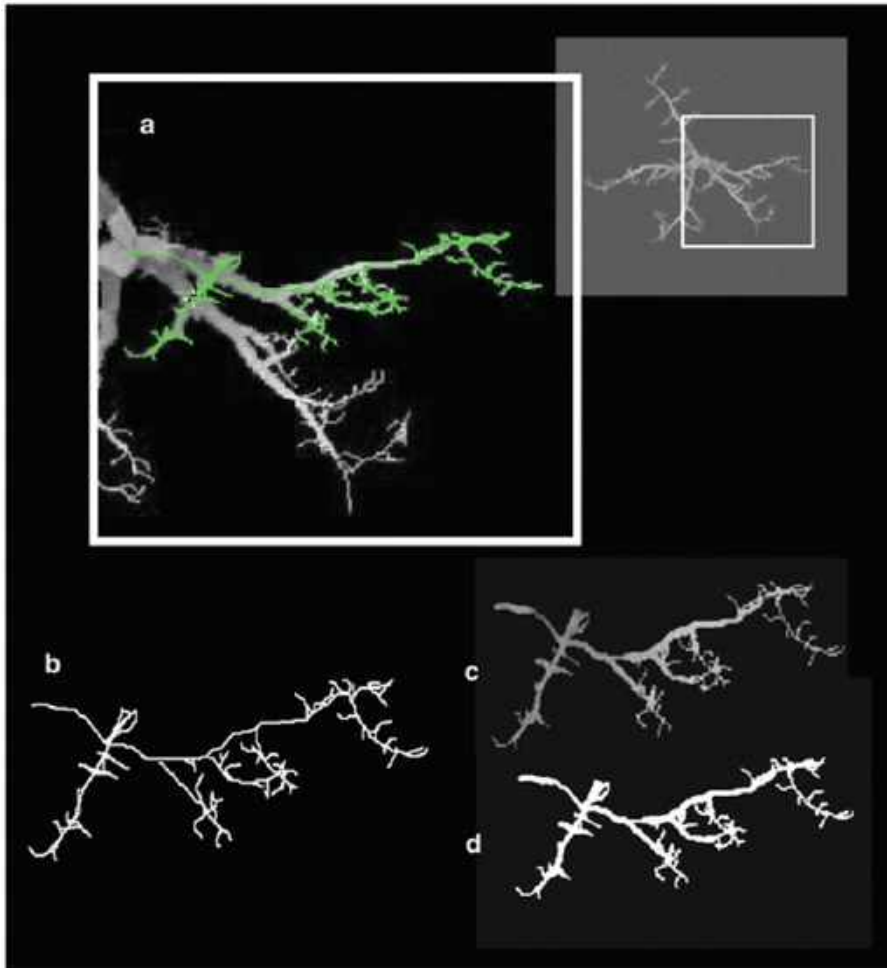


Fig. 40.4 Extracting branching patterns for fractal analysis in *IJ* (a) The green tracing is a path generated by the *Simple Neurite Tracer*[48]. The user clicks on points successively along a tortuous and branched structure (e.g., vessels or branched cells) and the plug-in automatically segments a basic skeleton. (Upper right) Original grayscale image of a modeled microglial cell. (b) Final skeleton generated by the plug-in. (c) Grayscale structure with background and unrelated cell parts removed automatically by the plug-in, based on the skeleton and parameters for filling in the area. (d) Binary silhouette generated by the plug-in from the grayscale trace and skeleton. (Cell model with stark background contrast generated with *MicroMod for IJa* plug-in for fractal modeling of biological cells [35])

40.4.2.3 Thresholding

Like most segmentation methods, the tracing plug-ins described in Sect. 40.4.2.2 works better for images such as the *in silico* image illustrated in Fig. 40.4 and images from biological imaging modalities with similarly stark contrast. This can be a severe limitation for images with less contrast or significant background variation. As an example, it is much more difficult and ambiguous to extract vessel and neuronal patterns from grayscale versions of the retinal images from Fig. 40.1 than the *in silico* cell in Fig. 40.4. This is part of a broader issue. The last decade has seen mammoth effort put into developing automated pattern extraction methods, but there remain two intertwined issues computer programmers, macro writers, and generally anyone in neuroscience or any other field reliant on pattern extraction continue to spend their time on: background subtraction and auto-thresholding. These are significant barriers to pattern extraction for fractal analysis.

One major problem complicating general background subtraction and thresholding is that whatever is decided upon as “noise” or “background” is not necessarily the same over all parts of an image. Some methods have been developed that account for local variation within one image or in a stack or other set of images with the result that patterns can sometimes be extracted automatically or with very little intervention, but this is not the general case with most medical images [44]. There are many, many sophisticated segmentation algorithms available, each with its own strengths and weaknesses. Recognizing this, Gabriel Landini, a researcher and *IJ* developer [44], has added to the core *IJ* software a built-in function that tests several algorithms at once and lets the investigator select a method for the work at hand.

This built-in function highlights an important point relevant to fractal analysis. If multiple patterns can be extracted from one image, that means investigation is required to test how different patterns translate to fractal dimensions [32, 44]. It also means automatic segmentation has a precarious side. To elaborate, *FracLac* and many related plug-ins, such as the *Sholl Analysis* plug-in, have options to automatically threshold [25, 34], but the option in *FracLac*, for instance, is a pilot testing convenience rather than a recommended strategy. It is vital to be aware that fully automatic thresholding cannot work on all original images simply because any automated segmentation method has a baseline whereby the programmer decides what is noise and what is signal, which may not agree with what the researcher decides is noise and signal. Even when looking at one image, an investigator’s questions may be pertinent to different features and need two different thresholding methods to extract two different types of information (e.g., the overall tissue texture vs. individual cells in Fig. 40.1) [22].

Customized Pattern Extraction Methods

Much of the research done in fractal analysis of digital images in neuroscience involves custom pattern extraction methods involving variations on tracing and thresholding, but there are other, novel approaches. Wave data, for instance (mean-

ing any sequence of values, such as EEG signals, sound, or other temporal data), can be converted to binary digital images for box counting analysis using the **Wave** command in *FracLac* [39]. Table 40.3 summarizes and compares some ways *IJ* has been used alone and with other means to inspire creative thinking in fractal-based neuroscience.

One particular example from Table 40.3d [13] succinctly illustrates a vital feature of *IJ* and similar open-source software in general. In that example, researchers asked how retinal microvessel complexity measured by the box-counting dimension (D_B) might be proxy for cerebral microcirculation pathology in CADASIL, a disorder having grave consequences including dementia. Working with retinal photographs obtained using a digital fundus camera, they found the mean D_B was significantly lower in patients than controls (1.426 vs. 1.506), suggesting that altered retinal vessel complexity might be a practical, noninvasive window into an early indicator of pathology in CADASIL. To come to that conclusion, they extracted patterns through the following summarized five steps using *IJ* functions:

1. Split color channels using green filtering: Image>Color>Split Channels.
2. Select an ROI 3.5 times the optic disk radius and centred on it.
3. Subtract background using the mean optical density from four arbitrarily selected, vessel-free areas.
4. Convert to binary, automatically trace vessels: Process>Binary>Outline.
5. Manually remove artifacts if present [13].

The above description has pedagogical value beyond the method per se. With free, open-source software, methods such as this can be easily shared and reproduced, improved, or modified for related data sets. Being able to convey detailed steps of a method so that researchers can easily reproduce them so that differences in lab resources are not a barrier to reproducing them, is an important feature of *IJ* that has the potential to significantly influence fractal-based neuroscience. It provides a level of continuity to enhance timely scientific development and advancement.

40.5 Conclusion

In conclusion, let us recap the answers to the questions posed at the beginning of this chapter (Sect. 40.1). The first question was “What is *IJ*?” The answer is that, substantially more than powerful, open-source image analysis software, *IJ* is both bio-tool and mindset. It removes barriers and promotes vigorous, collaborative neuroscience. It has matured alongside of and contributed to the presently materializing field of fractal-based neuroscience. It is a global project, an emergent feature of which is the potential for inspiring creativity, and a platform for collaboration [15, 47, 70] especially useful now when information is amassing ahead of our ability to analyze it [40, 66]. Researchers working with *IJ* are not bound by existing methods of analysis but are instead able to rapidly customize and repeatedly hone solutions and therefore formulate questions creatively using them.

Table 40.3 Examples of pattern extraction and fractal analysis methods implemented in *ImageJ*

Basic topic and model	Pattern extraction	Fractal dimension
(a) White matter		
<i>Texture</i> [59]		
Corpus callosum Anterior cingulum Digital micrographs Rat	Grayscale tiff of white mass	Box count: FracLac (texture) Discriminates adjacent white mass structures
<i>Multiple sclerosis</i> [22]		
Voxel-based slices Conventional MRI Human	Threshold slice, extract 2 binary patterns: 1. Skeleton of gyri within mass in IJ 2. Outline of entire mass in other software	Box count: HarFA [83] Identifies early brain changes in MS
(b) Cerebral arteries		
<i>Arteriovenous malformation</i> [69]		
Maximum intensity projections Time-of-flight MRI Human	Extract vessels and remove background with k-means clustering Binary skeleton in IJ	Box count: In-house Multifractal: FracLac Correlates with flow and size
<i>3-D reconstruction</i> [81]:		
Human Methods originally for neurons	Neuron Morpho [15]	Branching: L-Measure [63] Reconstructions now on free database
(c) Axonal architecture		
<i>Autism-related mutations</i> [29]		
Cells in culture Axon trajectories Rat	Binary network; white background; no somata or other structures; long and short, curved black lines of consistent diameter In-house software	Box count: FracLac Lacunarity: FracLac Mutations alter network architecture and synchrony
(d) Eye		
Cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy (CADASIL) [13]		
Cerebral microcirculation Retinal fundus photographs Human	Binary contours Automated in IJ	Box count: FracLac Vessel complexity identifies early pathology

(continued)

Table 40.3 (continued)

Basic topic and model	Pattern extraction	Fractal dimension
<i>Diabetic retinopathy</i> [41]		
Nonproliferative vs. proliferative Retinal digital images Fluorescein Human	Extract vasculature tree from 2 methods, then skeletonize in IJ 1. Automated continuous wavelet transform with supervised classification 2. Manual tracing	Box count: FracLac Local connected: FracLac Correlation: Other Progress on automated, objective screening
<i>Choroid</i> [76]		
Neovascularization Microglial ramification Neuroprotectin D1(NPD1) Mice	3-D reconstruction from laser confocal microscopy Fiji and other	3-D: Shape and fractal count [6] Microglia are potential targets of NPD1

The second question was “Where does *IJ* fit in fractal-based neuroscience today?” It occupies a prominent role in the published work of computational fractal-based neuroscience and multifarious collaborative roles in diverse areas relevant to the topic including industry and education. *IJ* has become a highly accessible avenue for researchers with and without computer programming background to join forces and fashion the tools they envision to answer questions about the fractal geometry of the brain at all levels, from gross to fine.

Moreover, despite becoming very powerful and being integrated into commercial applications, the main software itself remains practical and accessible, a bridge for sharing ideas and results with the nonexpert in many venues including the clinical picture of patient communication painted at the beginning of this chapter.

We hope this chapter has further inspired the reader to continue to demystify fractal analysis, to explore new fractal possibilities, to develop pattern extraction methods for new perspectives and especially for databases, to build that fly-through cell phone app, or to otherwise use the tools or embrace the mindset characteristic of *IJ* to bring the multiple levels of the fractal geometry of the brain to the fore, not just as a wonder of the digital age but as a clinical reality.

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Chapter 41

Fractal Analysis in MATLAB: A Tutorial for Neuroscientists



Juan Ruiz de Miras

Abstract MATLAB is one of the software platforms most widely used for scientific computation. MATLAB includes a large set of functions, packages, and toolboxes that make it simple and fast to obtain complex mathematical and statistical computations for many applications. In this chapter, we review some tools available in MATLAB for performing fractal analyses on typical neuroscientific data in a practical way. We provide detailed examples of how to calculate the fractal dimension of 1D, 2D, and 3D data in MATLAB. Furthermore, we review other software packages for fractal analysis.

Keywords Fractal analysis · Fractal dimension · MATLAB

41.1 MATLAB Packages and Toolboxes for Fractal Analysis

Boxcount [1] is one of the first MATLAB scripts developed for computing the fractal dimension (FD) of 1D, 2D, or 3D sets. As its name indicates, *Boxcount* estimates the FD of a set by using the box-counting method [2]. It can be only used with binary datasets (e.g., for an RGB image, a previous summation of the three components is performed), and the box sizes used to estimate the fractal dimension must be power of two. *Boxcount* is a MATLAB function that must be integrated into other MATLAB scripts in order to be executed.

CalcFD [3] is a MATLAB toolbox for computing the fractal dimension of 3D structures of the brain obtained from the FreeSurfer analysis pipeline [4], including cortical, subcortical, and ventricular structures. *CalcFD* uses the box-counting algorithm and can also be applied on volumes represented in standard formats such as NIFTI.

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FracLab [5] is veteran MATLAB software that allows fractal analysis of 1D signals and 2D images, but analysis of 3D sets is not supported. All the functionality of *FracLab* is accessible through a graphical interface, so no direct MATLAB programming is needed. *FracLab* allows the user to calculate not only the fractal dimension of the set but also parameters like the Hölder exponent or multifractal spectra.

For the case of 1D signals, the FD is commonly estimated by using Higuchi's algorithm [6], and there are several MATLAB implementations of this algorithm [7, 8]. Another well-known technique for estimating the FD of a 1D signal is Katz's algorithm [9]. In [10] a MATLAB implementation of this algorithm is provided. Unlike Higuchi's algorithm, which needs a manual-set parameter in order to establish the maximum size of the intervals to consider for approximating the 1D signal, Katz's algorithm can calculate the fractal dimension automatically without needing manual-set parameters. Some studies [11] have proven that Higuchi's algorithm is more accurate but also more noise sensitive than Katz's algorithm.

The FD value is associated with static structures; nevertheless, the multifractal spectrum [12] is a useful parameter in the fractal analysis of structures that present spatial and temporal variations. A detailed description and the related MATLAB scripts used to calculate the multifractal spectrum of biomedical time series can be found in [13, 14].

FD can be computed not only for the whole structure of interest but also at the local level for each pixel (2D) or vertex (3D) of the image. Meg Noah developed a package that allows the users to obtain the local FD of 2D images [15]. This software obtains a map of FD for the image by averaging the values of fractal dimension computed for several epsilon within the local neighbourhood defined by a window. Local FD maps of 3D structures can also be obtained by using MATLAB toolboxes such as Computational Anatomy Toolbox 12 [16] and UJA-SHFD [17]. In this case, these two tools were developed for computing local FD maps of cortical structures of the brain using a method based on spherical harmonics [18].

Computing directly the FD of grayscale images, without binarization, is also an algorithm with many applications in neuroscience such as EEG signal classification [19] and texture analysis for characterization of diseases [20]. The differential box-counting (DBC) algorithm [21] is the most widely used method for calculating the FD of grayscale images. In MATLAB, DBC can be computed using the *Fractal Dimension* toolbox [22].

41.2 MATLAB Examples: Fractal Dimension Computation for 1D, 2D, and 3D Sets

The FD is the most commonly used parameter in fractal analysis. A review of applications of the FD parameter in the neuroscience field can be found in [23]. In this section, we show several step-by-step practical examples of how the FD

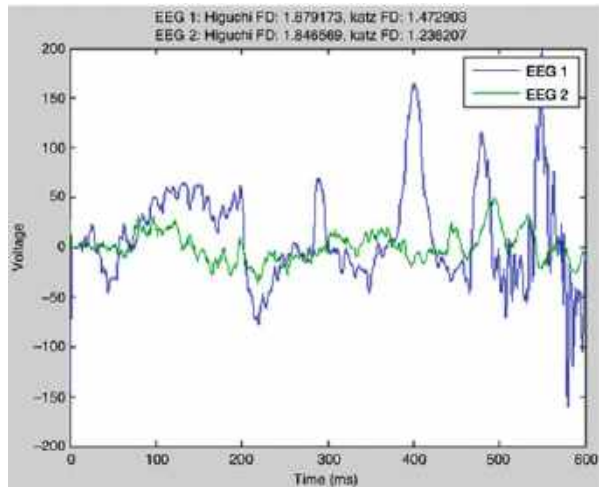
parameter can be computed in MATLAB for typical neuroscientific data in 1D, 2D, and 3D. For the case of 3D, we also provide an example on how to compute the local FD of a cortical surface.

41.2.1 Fractal Dimension of an EEG

In this first example, we describe how to estimate the FD of an electroencephalograph (EEG), a unidimensional signal commonly used in neuroscience. For this purpose, we use two different MATLAB implementations of FD: Higuchi's algorithm provided in [7] and Katz's algorithm provided in [10]. Figure 41.1 shows the signals of the same electrode for two different EEGs and the results of the two FD estimations for them. The MATLAB script that generates these results is shown in Listing 41.1.

In Listing 41.1, the EEG data is provided in European Data Format (EDF), a format for the exchange and storage of multichannel biological and physical signals. EDF files in Listing 41.1 are read by using the function `edfread()`, available from [24]. It is easy to import or export EEG data to EDF format through well-known MATLAB toolboxes like EEGLAB [25]. Katz's algorithm does not have any parameter to set. However, Higuchi's function `hfd()` needs to establish the parameter `kmax`, indicating the maximum value for k in Higuchi's algorithm [6].

Fig. 41.1 Two EEG signals and their Higuchi's and Katz's FD values



```

% Load EEG data in EDF format
[hdr1, eeg1] = edfread('chb01_01.edf');
[hdr2, eeg2] = edfread('chb02_01.edf');

% Higuchi and Katz Fractal Dimension computation
maxtime = 600; % for the entire signal use maxtime = length(eeg1(1,:));
kmax = floor(maxtime/2); % parameter needed in hfd function, set by default
elec = 1; % electrode to process

hfd1 = hfd(eeg1(1,1:maxtime), kmax);
kfd1 = Katz_FD(eeg1(1,1:maxtime));

hfd2 = hfd(eeg2(1,1:maxtime), kmax);
kfd2 = Katz_FD(eeg2(1,1:maxtime));

% plot the results
plot(1:maxtime, [eeg1(1,1:maxtime); eeg2(1,1:maxtime)]);
tit = sprintf('EEG 1: Higuchi FD: %f, Katz FD: %f\nEEG 2: Higuchi FD: %f, Katz FD: %f', ...
    hfd1, kfd1, hfd2, kfd2);
title(tit); xlabel('Time (ms)'); ylabel('Voltage'); legend('EEG 1', 'EEG 2');

```

Listing 41.1 MATLAB code used to calculate the FD of an EEG by using both Higuchi's and Katz's algorithms

41.2.2 Brain MRI Fractal Dimension of the Gray Matter with FracLab

In this example, it is shown how to calculate the FD value of a 2D image of the gray matter from a slice of an MR image of the human brain. First, we have to segment the original MR to extract the gray matter by using, for example, the SPM (statistical parametric mapping) toolbox [26] for MATLAB. The next step consists of selecting one slice of the segmented 3D image and extracting it to a single file in BMP format. This task can be easily performed with the software MRICroGL [27]. Then we load the BMP file of the slice into *FracLab* and execute the *box method* for binary data. Several parameters allow us to configure the box sizes and the type of linear regression used to estimate the FD. Figure 41.2a shows an example of the values used to calculate the fractal dimension of the gray matter slice of size 189×157 shown in Fig. 41.2b. Finally, the interval of box sizes used to perform the linear regression is manually selected (red dots) as shown in Fig. 41.2c, obtaining a fractal dimension value of 1.839 for this example.

41.2.3 Fractal Dimension Computation of an MRI Volume of the Brain White Matter with a Boxcount-Based MATLAB Script

Next, we show how to calculate the FD of a 3D volume representing the brain white matter. The main MATLAB function that we use for this process is *Boxcount*. The first step is the same as in the previous section; we have to segment the white matter of the original 3D MR image. This task can be performed with the toolbox SPM, as

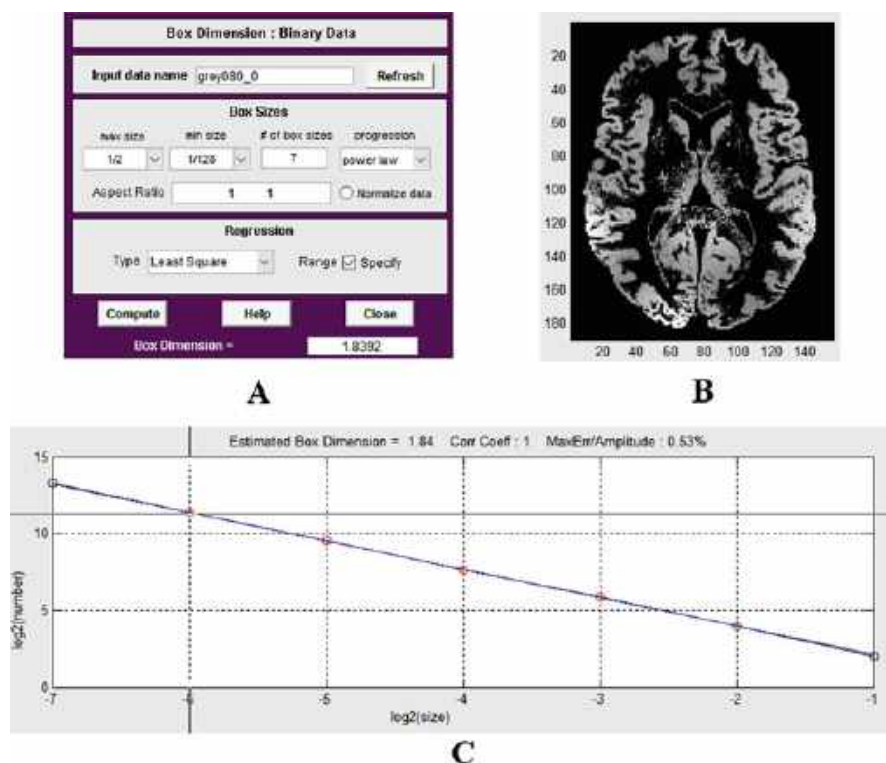
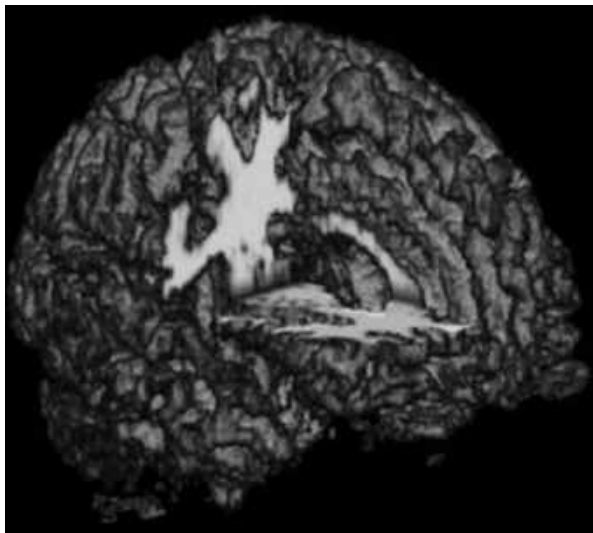


Fig. 41.2 Fractal dimension computation with *FracLab*. (a) Parameter selection for the box-counting method. (b) The gray matter slice to be processed. (c) Selection of the interval of box sizes to calculate the linear regression for estimating the fractal dimension value

previously described. The segmented volume of the white matter is obtained from SPM in ANALYZE format (HDR and IMG files), although another usual file format for dealing with MRI data is the NIfTI format. We have several choices for loading this kind of 3D image into MATLAB: (1) we can extract the BMP file for each slice of the 3D image with MRICron software and then load all the individual BMPs into a single 3D MATLAB matrix through a loop where the MATLAB function `imread()` loads each individual BMP into the corresponding layer of the 3D matrix; (2) we can load an *analyze* file directly into a MATLAB matrix by using the function `analyze75read()`; (3) NIfTI files are not directly supported by MATLAB functions, but there exist MATLAB toolboxes, such as the one in [28], allowing us to manage them. We can load a NIfTI image into MATLAB by using the function `load_nii()`. The 3D representation of the brain white matter that we use through this section is shown in Fig. 41.3.

The next step consists of performing the box counting of the 3D image. In this example, we use the *Boxcount* function introduced in Sect. 41.1. This function returns two arrays: an array (*r*) containing the incremental sizes of the boxes used

Fig. 41.3 3D MR imaging of the brain white matter (example described in the text)



to cover the 3D image and another array (n) containing the number of boxes of each size in r covered or partially covered by the 3D image. For our 3D, white matter representation in Fig. 41.3, the arrays have these values:

- $r = [1; 2; 4; 8; 16; 32; 64; 128; 256]$
- $n = [1,138,742; 171,868; 26,976; 4141; 667; 121; 25; 7; 1]$

From these data, we can easily calculate the fractal dimension by computing the slope of the linear regression of the log–log plot of $1/r$ versus n . Least-squares linear regression in MATLAB can be obtained with the `\operatorname{lsq}` operator (equivalent to the `mldivide()` function). Figure 41.4a shows the log–log plot, the regression line, and the final fractal dimension value calculated for our 3D image. In order to select the linear range of the regression line, we have selected the range of box sizes from 1 to 4. This selection was based on the plot provided by the function `boxcount(img, 'slope')` shown in Fig. 41.4b; this plot provides us with the evolution of the exponent $D = -d \ln(n)/\ln(r)$ for all the box sizes in r . As you can see in Fig. 41.4b, this exponent (local dimension) is almost constant (around 2.7) in the range 1–4. Finally, the full MATLAB script used to obtain the fractal dimension from a 3D image is shown in Listing 41.2.

41.2.4 *Local Fractal Dimension Computation of a Cortical Surface with UJA-SHFD*

In this last example, we show how to compute the local FD map for a cortical surface obtained previously through Freesurfer [4]. UJA-SHFD can be downloaded from <https://www.ugr.es/~demiras/ujashfd/>. Figure 41.5 shows the main graphical user

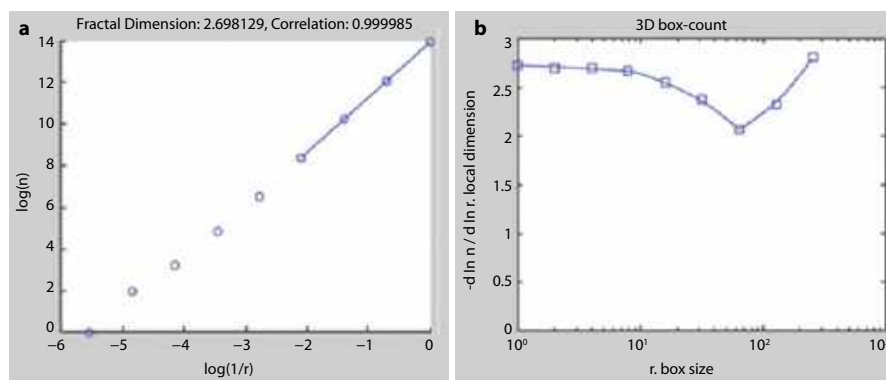


Fig. 41.4 (a) Log–log plot of box size versus number of voxels and the fractal dimension value for the 3D representation of the white matter. (b) Plot of the local dimension shows that it is almost constant in the range 1–4

```
% loads the 3D image from a NIFTI file 'white.nii.gz', requires the package "tools for NIFTI and ANALYZE image"
% http://www.mathworks.com/matlabcentral/fileexchange/8787-tools-for-nifti-and-analyze-image
image = load_nii('white.nii.gz');
view_nii(image); % shows the image loaded

% box-counting computation, requires the package "BoxCount"
% http://www.mathworks.com/matlabcentral/fileexchange/13063-boxcount
e = image.ing; % extracts the 3D matrix of the image
[n, r] = boxcount(e); % computes the box-counting
figure; boxcount(e, 'slope'); % plots the local dimension to select the range of box sizes

% log-log plot of the box-counting
N = log(n);
R = log(r.*-1);
scatter(R, N);
hold on;

% linear regression computation
ini = 1; % initial box size
fin = 4; % final box size
Rr = R(Rini : fin); % R limited to the selected range of boxes
Nr = N(Rini : fin); % N limited to the selected range of boxes
x = [ones(length(Rr), 1) Rr]; % adds a column of ones to Rr
b = x \ Nr; % b(1) is the y-intercept and b(2) is the slope of the line
y = x * b; % linear regression
corr = 1 - sum((Nr - y).^2) / sum((Nr - mean(Nr)).^2); % correlation

% plots the regression line and the fractal dimension results
plot(Rr, y);
xlabel('log(1/r)'); ylabel('log(n)'); title(sprintf('Fractal Dimension: %f, Correlation: %f', b(2), corr));
```

Listing 41.2 MATLAB code used to calculate the fractal dimension of a 3D image

interface of UJA-SHFD. From volumetric MRI data of the human head, FreeSurfer can obtain the cortical surfaces (pial and white) by using the command `recon-all` (https://mne.tools/stable/auto_tutorials/forward/10_background_freesurfer.html). That FreeSurfer command produces the surfaces `lh.pial`, `rh.pial`, `lh.white`, and `rh.white`. Each one of these cortical surfaces is represented by a triangular mesh of more than 100,000 triangles. These meshes are the input to UJA-SHFD (Fig. 41.5 button 1). UJA-SHFD also support general triangular meshes in the OBJ format. UJA-SHFD is preconfigured for working with FreeSurfer meshes, so the



Fig. 41.5 UJA-SHFD graphical user interface and steps to obtain local FD maps

next step is generating the spherical harmonic reconstructions of the cortical surface (Fig. 41.5 button 2). This process can take a long time, even more than an hour, depending on the computational power of the computer being used. When the generation of reconstructions is finished, an initial local FD map is computed and displayed. Then, the exact linear regression parameters for computing both the local FD map and the global FD value can be set (Fig. 41.5 button 3). As a result, the local FD map is stored in the format of a FreeSurfer overlay. This allows the user to use these local FD maps in other FreeSurfer tools such as QDEC (https://surfer.nmr.mgh.harvard.edu/fswiki/FsTutorial/QdecGroupAnalysis_freeview) in order to perform comparisons between groups.

41.3 Other Software for Fractal Analysis

If you do not have access to MATLAB software or prefer to perform fractal analysis using stand-alone software through a graphic user interface, then you have some choices on your hands: *BENOIT* [29] is paid software from TruSoft Inc that allows the FD and Hurst exponent of 1D, 2D, and 3D data sets to be measured using a set of methods. BENOIT only runs on Windows operating systems, but a MATLAB version of BENOIT is also available. *Fractalyse* [30] is a Java computer program that estimates the FD of binary images by using several algorithms. The FD value of grayscale images can be computed using the Windows program *Fdim* [31]. A

Visual BASIC software for estimating the FD of grayscale objects of up to 4D on the Windows platform was presented in [32]. Two popular programs for analyzing 2D images are *ImageJ* [33] and *HarFa* [34]. Fractal analysis in *ImageJ* can be performed through the plug-in *FracLac* [35]. *ImageJ* is written in Java, so it can be executed in any platform supporting Java. *HarFa* is a Windows program specifically designed to perform harmonic and fractal analysis on images. *ImageJ* and *HarFa* estimate the fractal dimension by using the box-counting method.

41.4 Conclusions

In this chapter, we have reviewed several MATLAB packages and toolboxes, as well as other software resources, used to perform fractal analysis on 1D, 2D, and 3D datasets. We have provided practical examples of how to compute the fractal dimension in MATLAB for typical neurological signals and images. To summarize, Table 41.1 shows a comparative synopsis of all the software reviewed.

Table 41.1 Summary of fractal analysis software

Software	Platforms	Algorithms	Data	Commercial	Reference
Boxcount ()	MATLAB	FD (box counting)	1D, 2D, 3D binary	No	[1]
CalcFD	MATLAB	FD (box counting)	3D	No	[3]
FracLab	MATLAB	FD, holder exponent, multifractal analysis	1D, 2D	No	[5]
hfd ()	MATLAB	FD (Higuchi)	1D	No	[7]
HFDALC ()	MATLAB	FD (Higuchi)	1D	No	[8]
Katz_FD ()	MATLAB	FD (Katz)	1D	No	[10]
WLBMF	MATLAB	Multifractal analysis	1D, 2D	No	[14]
Noah's tool	MATLAB	FD, local FD	1D, 2D, 3D	No	[15]
CAT12	MATLAB	Local FD	3D	No	[16]
UJA-SHFD	MATLAB	FD, local FD	3D	No	[17]
Fractal dimension	MATLAB	FD (differential box counting)	2D	No	[22]
BENOIT	Windows	FD, Hurst exponent	1D, 2D, 3D	Yes	[29]
Fractalyse	Windows, Linux	FD	2D binary	No	[30]
FDim	Windows	FD	2D grayscale	No	[31]
Fractal analysis v3	Windows	FD	1D, 2D, 3D, 4D grayscale	No	[32]
ImageJ	Java	FD, multifractal analysis	2D	No	[33]
HarFa	Windows	FD (box counting)	2D	No	[34]

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Chapter 42

Analyzing Eye Paths Using Fractals



Robert Ahadizad Newport, Sidong Liu, and Antonio Di Ieva

Abstract Visual patterns reflect the anatomical and cognitive background underlying process governing how we perceive information, influenced by stimulus characteristics and our own visual perception. These patterns are both spatially complex and display self-similarity seen in fractal geometry at different scales, making them challenging to measure using the traditional topological dimensions used in Euclidean geometry.

However, methods for measuring eye gaze patterns using fractals have shown success in quantifying geometric complexity, matchability, and implementation into machine learning methods. This success is due to the inherent capabilities that fractals possess when reducing dimensionality using Hilbert curves, measuring temporal complexity using the Higuchi fractal dimension (HFD), and determining geometric complexity using the Minkowski–Bouligand dimension.

Understanding the many applications of fractals when measuring and analyzing eye gaze patterns can extend the current growing body of knowledge by identifying markers tied to neurological pathology. Additionally, in future work, fractals can facilitate defining imaging modalities in eye tracking diagnostics by exploiting their capability to acquire multiscale information, including complementary functions, structures, and dynamics.

Keywords Eye paths · Scanpaths · Fractal · Eye tracking · Perception

42.1 Introduction

Noton and Stark’s study published in 1971 into the way we see things revealed that our measured eye movements are not random [56]. These movements, which they called “scanpaths,” were tracked, moving in a smooth, continuous way between

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points of interest that appeared to follow predictable patterns. This was a significant finding, as it helped shed light on the underlying processes that govern how we perceive visual information. It showed that the way we gaze at an image is not chaotic but rather organized by specific patterns that are influenced by the visual characteristics of both the stimulus itself and our own visual effects and perception. These visual effects include visual illusions, color and depth perception, and the perception of motion. The emergence of these new findings has driven a need to find physiological factors that drive the way we perceive the world. Indeed, this has now led to increasing interest in how eye tracking could be a predictor of early dementia, as demonstrated by a traditional visual-paired comparison task experiment by Crutcher et al. [15], errors by participants who fail to smoothly track a stimulus across a screen with their gaze by Hutton et al. [28], and new work using instruction-less tests where gaze patterns are interrogated to expose oculomotor biomarkers of cognitive dysfunction [50]. This new work using artificial intelligence (AI) to search for biomarkers, which relies on self-supervised machine learning techniques, mitigates the artifacts introduced in earlier research where smooth pursuit and stimulus comparison were highly dependent on eye tracking accuracy. It also has the potential to expose other pathological oculomotor behaviors not yet discovered via its unsupervised clustering of scanpath pattern components. However, these gaze patterns are spatially complex and display the kind of self-similarity seen in fractal geometry at different scales, making them a challenge to measure using traditional topological dimensions in Euclidean geometry. The complex patterns found in scanpaths are uniquely suited to measurements by means of fractal geometry [44], which we will uncover in this chapter.

42.2 Eye Tracking

A discussion of the process used to obtain eye tracking information should precede an exploration of the challenges facing its analysis. Pioneering research in this field came from Ibn al-Haytham's work in the late tenth century [3], in which vision was described as light reflecting off an object into one's eye and then transmitted to the brain. The connection between the eyes and the brain was largely forgotten until Charles Bell's seminal paper [4] on the nerves around the eye and the influence of their movements on visual orientation. The eye was described by Bell as being suspended in a "framework" connected to the nervous system and subject to changes resulting from disease or physiological conditions. This work paved the way for further research into connections between eye movements and cognitive processes, as demonstrated by a contraption made by Hering [26] consisting of a rubber tube and cigar holder used to hear muscular movements of the eye during reading. This was described by Hering as a series of auditory jerks from the rubber tube which correlated with what he observed as a fluid movement of the subject's eye across a page. Hering concluded that reading was not a fluid action but a rapid succession of fixations assembled to form a cohesive visual stream. This rapid succession of

movements was named in French as a “saccade” which in English means “jerk-like movement” and was used by Émile Javal to describe eye movements [30]. Later in the century, Crum Brown applied this concept outside his established studies in post-rotational nystagmus to conclude that “our eyes always move by these jerks as we look around the world” [7]. Based on his observations of saccades in 1895, he postulated that “during the jerks we practically do not see at all, so that we have before us not a moving panorama, but a series of fixed pictures of the same fixed things, which succeed one another rapidly [7].” More recently, the era of modern eye tracking started with work done by Yarbus [66] on “Eye Movements During Perception of Complex Objects” where the outline of the eye gaze was described as a visual “palpation” of the viewed stimulus. Using a lens attached to the cornea (using suction), Yarbus was able to catalog direct observations of behavior unique to the examined participant.

The invasive nature of these early eye tracking methods severely limited the progress of such research until cameras facilitated the observation of eye movements. The use of video cameras to record the reflection of infrared lights on the cornea during eye tracking provides a high degree of accuracy when measuring gaze fixations without the invasive eyeball adhesion required in earlier studies. This video-based technique has led to the commercialization of eye tracking devices specifically made for the research market. These devices, which are used in contemporary research, consist of either a head-mounted or tabletop camera, recording invisible reflections from infrared lights bounced off the eye. These light reflections are calibrated against the eye gazing upon specific positions on a video display. Interstitial positions between calibration points are interpolated to return specific fixation coordinates at consistent time intervals captured by the video camera. These intervals are measured in Hertz due to the sampling rate being captured in frames per second. The eye gaze position is tested during a validation phase to determine the level of accuracy during an experiment. It is important to consider the accuracy and sample rate of an eye tracking device as a measure of the granularity of the data captured.

42.3 Eye Movements

With the myriad of movements captured by a high-resolution eye tracking device including focus, binocularity, and accommodation, an introduction to oculomotor behavior will help understand the benefits of using fractal geometry to model eye movements. These movements can include involuntary fixations during a visual sweep of a target classified into slow components, such as physiological drift and tremor, and fast components, represented by microsaccades and rapid small-amplitude movements [20]. Measuring and comparing these small movements, including the unique physiology of the pupil itself, has resulted in increased insight into cognition. An example of the importance of tracking both position and duration of visual physiology can be found in the work by Krejtz et al. [36] which subjected

participants to both simple and difficult mental arithmetic tasks with an aim to discriminate between them using two specific eye measurements. These metrics included pupil diameter changes, measured from an established baseline, and micro-movements of the eye between visual fixations. The experiment found task-evoked micro-eye movements between fixations and pupillary metrics to be sensitive and reliable indicators for discriminating the difficulty of the task being attempted by a participant. We now know that other limiting factors like the Inhibition of Return (IoR) during eye gaze behaviors govern the frequency of attraction to fixation points and may influence how long a gaze can be maintained, influencing distribution, during scene exploration [40]. However, the extended dimensionality of the Cartesian x and y coordinates included with temporal and other variables like pupillary size can obfuscate the statistical significance of the distribution of fixation points, especially due to the propensity of human visual behavior to be influenced by attributes within the stimulus.

The distribution of fixation points has been noted to occur most frequently in the center of an image [40], though reasons for this could be both an artifact and a result of biological morphology, which deserves some exploration when considering an eye tracking methodology and scanpath model. Central bias due to artifacts in the data could be due to photographers framing the focus of a captured image within the center of the frame, shifting the focus away from edges and peripheral locations [40]. Morphological reasons for central bias could be due to the direction of the eye relative to the head, known as the orbital reserve [62]. Furthermore, eye fixation datasets have shown that saccadic biases are predominantly horizontal, indicating an anisotropic influence these movements may have in determining random fixation points during eye gaze exploration of a free-viewing condition [40]. These effects illustrate the inconsistency in performance for visual search tasks depending on where the stimuli appear in a scene. Indeed, structural conditions in the real world increase the likelihood that a horizontal floor is below and a horizontal ceiling is above and that perpendicular and parallel geometries are interstitial to these two planes [8]. This visual input creates a bias across the visual field statistically altering the likelihood of such structures in a scene, heightening sensitivity to edge orientation [21] and hinting at temporal and spatial differences in acuity that vary depending on the position within the visual field. When considering a computational eye path model, one must account for these structures and anisotropic effects in order to prevent their overt influence on the stimulus (Fig. 42.1).

42.4 Eye Path Representations

Computational models are used to measure meaning from raw data, as is the case for scanpath information from eye tracking devices. Fractal methods for representing visual gaze fixation patterns will be described after a look at how this was attempted in the early stages of research.

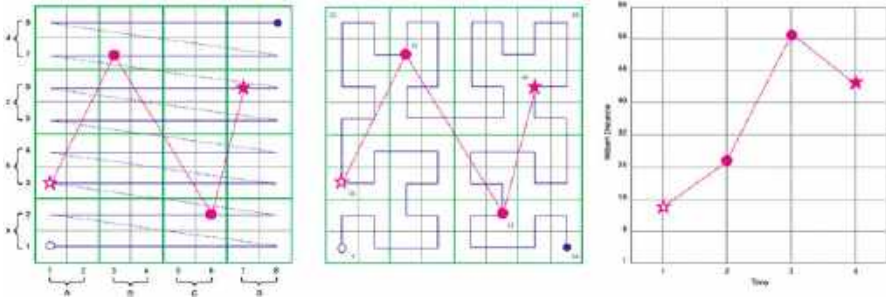


Fig. 42.1 A theoretical scanpath consisting of four fixations, starting with an empty pink star and ending with a filled pink star over an 8 unit by 8 unit space with the origin of the mapping path starting at the empty blue circle and ending at the filled blue circle. Figure (left) illustrates the scanpath mapped using Cartesian points (1,3), (3,7), (6,2), and (7,6), with quantized string representations as Ab, Bd, Ca, and Dc. Figure (center) illustrates a scanpath over a Hilbert path with distances 15, 25, 57, and 46. Figure (right) illustrates Hilbert distances from Figure (center) plotted against time

42.5 String Edit

Typically, measuring a fixation position originates from a corner and tracks left to right, row by row, to the coordinate horizontal and vertical points. This positioning is similar to how pixels on a digital camera's 2D sensor array are represented. Facilitated by the increasing amount of information provided by camera-based eye trackers, a way to both track and quantize this information for analysis was inspired by early bioinformatics research.

The Levenshtein [41] *distance metric* was a novel *string edit* method used to match an ordered sequence to another. A *cost metric* was defined by the accumulated changes required for two strings to match. These changes can include insertion, substitution, or deletion of characters within the string, with each change summing into what is defined as a *distance* between them. This method was adopted by Brandt et al. [6] where they hypothesize whether “eye movements play a significant functional role in the process of visual imagery.” Quantitative analysis of scanpath similarities and differences was determined using string edit methods, borrowed heavily from work done by Levenshtein, through the following execution: a grid is spanned over the stimulus and a letter combination is assigned to each square. Fixation points lying within a boxed area would be reduced to an *S* string name, which could then be compared to others. For example, in Fig. 42.1, the Cartesian points for the illustrated scanpath would be (1, 3), (3, 7), (6, 2), (7, 6), making the string equivalent *AbBdCaDc*.

However, two limitations exist for string edit methods using a grid system. First, it lacks locality preservation, where once a fixation point is reduced to its string value, the original precise position can no longer be determined. Second limitation is quantization calculation, where any point is uniformly reduced to its grid value,

regardless of its proximity to a boundary. This inflexibility with both locality and time is betrayed by a dependence on fixed regions of interest (ROI) which prevents granular discrimination between close and far points, and it does not account for differences in gaze duration. This limitation of Levenshtein was addressed in 1970 with the introduction of Needleman and Wunsch [51], a sequence alignment algorithm, allowing for repeating strings to account for coarse representations of fixation duration. Even though the Needleman–Wunsch implementation is similar to Levenshtein’s method through its use of strings, it is different from Levenshtein due to a method by which it can find the best fit. It does this by both allowing for gaps between two strings and applying varying penalties during alignment when calculating substitutions in this order: first, by creating a substitution matrix of all possible string combination scores; second, a penalty is determined for gaps in the string; and third by summing a score as an optimal path is calculated. Additionally and optionally, the duration is represented by repeating strings in a sequence proportional to others it surrounds.

As a testament to their robust implementation, two scanpath comparison methods developed using Needleman–Wunsch techniques over a decade ago still exist in use today. ScanMatch [13] comparison is made via temporal and quantized spatial binning of (x, y) coordinates into string representations of time, order, and fixation location. In order to maximize the similarity score, a substitution matrix makes a connection between any useful dimensions with the coded letters, including gridded distance or semantic space. To minimize the score, a penalty gap compares the degree by which scanpaths differ in a substitution matrix, over increasingly distant string pairs. A coarse way ScanMatch addresses duration is through temporal binning, i.e., repeating string representations proportional to the time duration of the fixation compared to others. For example, in Fig. 42.1, the string-substituted scanpath *AbBdCaDc* shown with an extended duration on its second fixation could be represented as *AbBdBdBdBdCaDc*. However, spatial quantization remains an issue because of the way square areas representing a string are divided by the observation boundary. The area from the center to the edge is entirely replaced by the string regardless of where the fixation falls within this boundary, quantizing edges to the center.

Research by Dewhurst et al. [17] attempted to mitigate the limitations of ScanMatch by introducing a vector-based approach via their MultiMatch method. Instead of a grid used to exclusively quantize only the position of a fixation point, as is done with ScanMatch, MultiMatch uses string representations of geometric attributes of the fixation vector into iteratively agglomerated sequential fixations. These string representations of the geometric vectors are then aligned sequentially over many dimensions. These include: *vector similarity*, where aligned saccade differences are measured, normalized, and averaged to determine geometric shape similarity; *length similarity*, where only the amplitude of aligned saccades are measured, normalized, and averaged without regard for the duration, direction, or location; *direction similarity*, where the angle differences between aligned saccades are normalized by π and averaged without regard for location or amplitude; *positional similarity*,

where Euclidean distances are measured, normalized, and averaged between aligned fixations, taking into account direction and amplitude; and *durational similarity*, where aligned fixation durations are normalized, averaged, and measured without regard for amplitude and position. The vector nature of the scanpath representation allows for more refined simplification compared to gridded methods.

The limitations of temporal integration into string edit methods are made evident through the coarse or isolated way these two well-known methods use duration as an attribute. Duration is attempted by ScanMatch by repeating letters in its string representation to coarsely approximate the duration of each fixation. However, this increases the further degree of quantization in the model through both temporal and positional generalization. MultiMatch only accounts for the duration in one out of five of its comparison algorithms, separating it from other interdependent attributes and dampening insight into the correlation or influence of these other attributes. Indeed, a challenge exists to integrate temporal dimensions into a space-defining structure without requiring quantization, yet still maintaining the granularity needed to describe geometric complexity. Current research using fractal mathematics proposes a solution.

42.6 Fractal Analysis

The conversion of Cartesian space to a metric useful for quantitative research is a critical component of any analysis in the spatial domain, especially when dealing with evolving diverse datasets spanning many formats. However, this domain can introduce artifacts when dealing with disparate area sizes and scales. For example, if a red balloon at pixel coordinate $(x : 20, y : 50)$ represents a data point, there are many unresolved attributes that need to be satisfied before it is quantitatively meaningful: what are the width and height? Do other pixel buffers have the same resolution? Where is the origin point? A traditional digital pixel buffer space is a piecewise linear stream starting at one point $(x : 0, y : 0)$ and following a line along the x -axis per N pixels for each stride, where it is reset and begins its stride again, displaced one pixel forward on the y -axis per stride [47, 58].

This presents a problem as resolutions increase; secondary calculations need to be made translating low-resolution (x, y) coordinates to a higher-resolution buffered pixel space [58]. This is because a piecewise linear function does not preserve locality well [47]; a prominent data point in a buffer at point N in image 256×256 translates awkwardly to point N in image 1024×1024 . Ideally, an increase in resolution should result in increasing precision of point (x, y) and N in the buffer [64].

In the red balloon example illustrated earlier, a computational application of a space-filling curve is the arrangement of an image's pixel buffer, snaking along the x -axis, with the width as its stride per N pixels of its height. However, a major limitation of this type of space-filling curve is its inability to scale well with increasing pixel resolutions. This problem, when expanded into the theoretical

realm, could be further defined to include its inability to represent infinite space with infinite points due to the rigid encoding of its stride to its width. A true space-filling curve should be able to represent infinite space.

42.7 Fractal Curves

In 1890, Giuseppe Peano published a description of a method for creating a continuous curve that passes through every point in a unit square, which he called a “space-filling curve.” This curve could theoretically be extended to fill an entire n -dimensional hypercube in infinite space, without any end points. This work was described in a paper titled “Sur la Plane,” which was published in 1890 [57].

While he may not have been aware of it at the time, Peano’s space-filling curve could be used as a framework in machine learning algorithms like k -nearest neighbor, where multidimensional points in a hypercube that are clustered on a space-filling curve can define a feature space [12]. This work laid the foundation for David Hilbert’s discovery of a new type of fractal space curve in 1891, which is widely used today in applications that require better locality preservation when transitioning between 1D and 2D space.

For instance, converting a Euclidean point to a 1D Hilbert curve can allow for a clearer empirical analysis when graphing against time, compared to if the point were left in 2D or 3D space. In addition, normalizing multidimensional data in a 1D space can lead to increased precision as pixel resolution increases, without the need for a linear piecewise function to interpret grid changes, which is often used in string edit methods. This is because the curve has the ability to effectively preserve locality due to its homeomorphic exponential growth with the n th approximation of the limiting curve.

Thus, as it increases in size, its ratio of detail to scale remains constant. This example of the Hilbert curve’s regularity of self-similarity is a testament to its robust ability to preserve locality during a shape’s growth or change during variable quantization [52]. This makes it particularly robust in an application where 2D values are reduced to 1D fractal curves plotted against a time dimension to be both dynamically quantized over space and uniformly windowed over time.

The reason this type of curve is able to preserve locality well is due to its homeomorphic exponential growth, scaled with the n th approximation to the limiting curve. Thus, its ratio of scale to detail remains constant as it increases in size. This illustration of the Hilbert curve’s regularity of self-similarity is a testament to its good ability to preserve locality during a shape’s growth [52]. Lacunarity, on the other hand, is used to describe heterogeneity through the growth of a shape, focusing more on “how rather than how much” the object fills the shape [10, 19, 61]. However, could insight be drawn from measuring a shape’s irregularity? Empirically, a measure of a shape’s dimension can be defined as the number of axes needed to plot the coordinate of a point within its boundary. However, as shown with space-filling curves, a continuous curve can be used to define such a point,

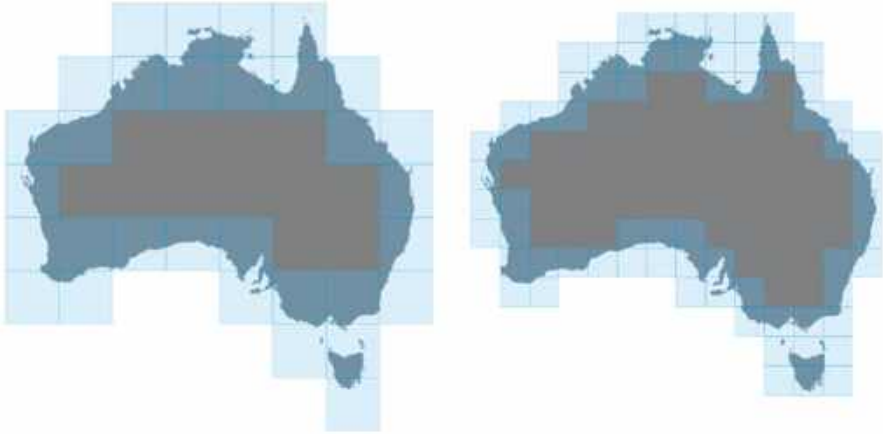


Fig. 42.2 The coastline of Australia is illustrated using the box-counting dimension, or Minkowski–Bouligand dimension. Using a grid, a box is drawn wherever a line from the coast intersects it with coarse (left) and fine (right)

illustrating uniform cardinality between these points despite a different number of axes. Thus, a more rigorous definition of dimension is required for a deeper understanding of this theoretical framework.

If 2D space is defined as the integration of a horizontal x-axis and vertical y-axis, and 3D space is the addition of a z-depth axis, is there value in defining a metric between these dimensions? For example, if a Hilbert curve fills all spaces in a 2D plane, and a straight line covers only a small portion of this plane, how can these differences be described mathematically? In 1918, Felix Hausdorff introduced a formula to calculate a numerical invariant of a metric space, now known as a Hausdorff Dimension [25]. To find the Hausdorff Dimension D_F , we take a Euclidean object with segments r and scale it by $1/r$, illustrated by $N = r_F^D$. After taking the log of both sides as $\log(N) = D_F \cdot \log(r)$, we can solve for the dimension with $D_F = \log(N)/\log(r)$. A Hilbert curve using this equation yields a D_F of 2.0. However, unlike Euclidean geometry, this equation may sometimes yield a fraction, representing the fractional *roughness* of a shape (B. B. [49]). For example, a Koch curve, which looks like triangular protrusions off a snowflake, can be expressed with an initiator of three segments in length and a generator of four segments in length. The change in scale S followed by the change in detail N can be expressed as $D_F = \log(4)/\log(3) = 1.26$ (B. B. [49]). When comparing the Koch and Hilbert curves using HD, we now have a quantifiable way of comparing the scale invariance of fractal shapes of infinite length. Figure 42.2 shows the computation of the box-counting fractal dimension (FD) of the Australian coastline.

42.8 Fractal Dimension

Hermann Minkowski and Georges Bouligand expand upon Mandelbrot's use of a coastline of an island [48, 49] as an example to illustrate their novel method for measuring fractal dimensionality in a natural shape. Similar to a Koch curve, as the coastline's scale is increased, its length also increases, making it impractical to measure the size of protons and electrons in a grain of sand around the perimeter of the coastline. An alternative to Hausdorff's calculation was proposed using a grid of boxes, where at one particular scale, the intersection of boxes and lines from the coast of the map are counted up and compared to the same intersections at a different proportional scale, as shown by the two scales in Fig. 42.2. This became known as D_B the box-counting dimension, or Minkowski–Bouligand Dimension, and yields similar results to Hausdorff's calculation of D_F .

Fractal exponents derived from the calculation of the fractal dimension described earlier can be further combined in a time series to create a singularity spectrum governed by a power law. This spectrum aims to describe the fractal dimension of the collection of exponents relative to the dimension of each contained exponent, as shown by the singularity spectrum $D(\alpha) = D_F\{x, \alpha(x) = \alpha\}$ where $D_F\{\cdot\}$ is the fractal dimension set of all points, and $D(\alpha)$ describes the Hölder exponent: $\alpha(x)$ of $f(x)$ at point x . A power law governs the relationship between the D_F components where each varies as a power of the other.

The relationship between the Hurst coefficient H and fractal dimension D_F can be expressed where $1 < D_F < 2$, as $D_F = 2 - H$. Used independently of D_F , H is mainly used to measure the tendency (or memory) for data in a time series to regress around the mean or gravitate toward a particular direction [43]. If $0.5 < H < 1.0$, then the data will contain positive autocorrelation over a majority of the time series; if $0.0 < H < 0.5$, then the data will contain alternating high and low values over a majority of the time series; and finally, if $H = 0.5$, the data may start correlated but will decay rapidly over the time series to zero. As an example, this concept can be used to explain how Gaussian noise can be created by differencing Brownian motion, and inversely, summing Gaussian noise produces Brownian motion. Thus, H can be used to measure relationships between types of random data in a time series.

42.9 Uses and Applications

The theoretical fractals described in the previous section can both aid in the measurement of geometric complexity using the fractal dimension and can also describe dynamic properties using the integration of a temporal dimension. Directly applied to biological morphology, the spatial properties of the nervous system's irregular structures can be observed and measured the same way Minkowski described measuring the coastline of an island. Additionally, dynamic properties

of the nervous system, including the brain and cerebellum structure, cerebral vasculature, etc., can be evaluated in order to discriminate against different states of the brain through the entire physiopathological spectrum using the fractal dimension as a quantitative parameter [23].

However, an important distinction should be made separating mathematical objects derived from theoretical fractals from those used to describe the complex morphological structural features characterizing biological and pathological development processes. The self-similarity that spans an unlimited range of scales in theoretical fractals is limited to “natural fractals” within a “scaling window” that exists inside an upper and lower boundary. The scaling window is defined by the measured size of the object and the scale of observation [46]. Though even with this more limited definition, fractal entities can be used to measure functional, dimensional, and geometric parameters and complexity within organs, tissues, and cells, which Losa et al. [45] stretch to include all natural objects exhibiting self-similar scaling properties and patterns into a “fractalomics” subdiscipline. Indeed, this ability to measure geometric complexity can be exploited in the investigation of scanpath outliers.

42.10 Identifying Scanpath Outliers

With the advancement of eye tracking hardware capable of examining granular eye movements at high frequency (e.g., 1000 times per second), a trove of data previously imperceptible, including tiny jerking movements of the eye called microsaccades, can now be captured. These and other movements, when compared between experiment participants, can reveal insight into the role perception plays in expertise, biomarkers for disease, and visual communications. The last several years have seen an explosion of research into examining how scanpaths can improve teaching and diagnostics [1, 28, 29, 38, 39, 50, 60].

However, the fast progress of development by researchers to computationally compare and cluster scanpath datasets has not been matched by a formal, robust definition of what constitutes an outlier [9, 31]. Scanpath outliers could be caused by many things including distraction of the participant, malfunction of eye tracking hardware, calibration issues, and other problems causing the participant to disengage from the stimulus. Due to the potential sparse spread of fixations and saccades over a stimulus, the exclusion of scanpaths done empirically should not be attempted. Though, outliers representing artifacts can negatively influence the statistical significance of findings. Alternatively, removing unusual scanpaths that may represent valid edge cases can limit opportunities to yield important discoveries. Therefore, a deliberate approach should be developed to identify irrelevant data points with robust analytical reasoning for their exclusion.

The earliest paper found to explicitly define a method for finding scanpath outliers was by Burch et al. [9] using Jaccard and bounding box methods. This entailed surrounding the extent of a scanpath’s fixations with a box and comparing

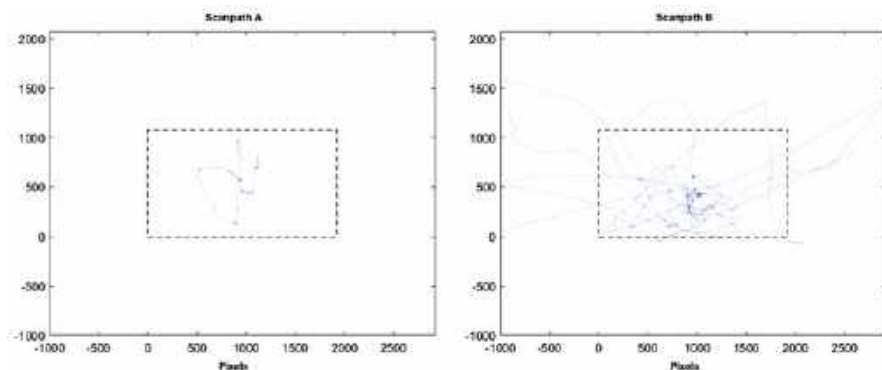


Fig. 42.3 Two scanpaths capturing a 5-second visual gaze pattern with the stimulus boundary illustrated via a dotted black line and axes measuring pixel space. The left scanpath was identified as normal, whereas the right was deemed a possible outlier using fractal analysis by Newport et al.

its size with the boxes from other scanpaths. Alternatively, Davies et al. [16] used heatmaps to compare scanpaths with one another in a binary fashion. Scanpaths that did not match well with others were marked for exclusion as outliers. Though, both of these methods do not account for viable scanpaths where the participant is engaged with the stimulus in a unique and novel way. A robust outlier detection method should be able to keep such unusual yet significant data while removing data artifacts.

Newport et al. [54] proposed that the geometric complexity defined by the fractal dimension can mitigate ambiguities separating unusual significant data from perturbations as a result of hardware or software failure or participant distraction. Figure 42.3 illustrates an example of what a possible scanpath outlier may look like, using data obtained in an experiment performed by the Computational NeuroSurgery (CNS) Lab at Macquarie University on expert and naive participants looking at medical images. On the left, the scanpath is within the stimulus boundary, with details clearly discernible. On the right, fine points are less clear, with fixations and saccades arranged more irregularly. Indeed, a measure of the geometric complexity using the method proposed by Newport et al. provides a robust metric to use for the exclusion criteria of such dubious scanpaths. This complexity is calculated in two steps: first, the 2D Cartesian coordinates of the scanpath are converted into 1D distances on a Hilbert space-filling fractal curve, and second, the 1D Hilbert curve distances are coupled with their millisecond fixation durations and measured using a temporally sensitive fractal dimension calculation method called the Higuchi Fractal Dimension (HFD).

HFD came about following positive advances in the analysis of human morphology using FD and has encouraged further exploration into how it can be extended to include physiology, e.g., dynamic, nonstationary sequences like the heart rate in the cardiovascular system. Higuchi [27] developed a nonstationary fractal analysis method capable of measuring FD in dynamic physiological systems (Di [23])

by approximating the box-counting value of a time series. Indeed, HFD is more resistant to noise and needs fewer points than correlation dimension [35] and can be used as a training set feature in a support vector machine (SVM) classifier to increase accuracy, e.g., as seen in the diagnosis of encephalopathy based on Electroencephalography (EEG) [29].

By reducing the dimensionality of the 2D fixations into 1D Hilbert curve distances and pairing them with their respective durations, a sequential temporal list of numbers can be fed into the HFD algorithm for a computed value between 1 and 2. To visualize this empirically, maximum noise would occupy 100% of the 2D fractal curve plane, returning an HFD value of 2, whereas the simplest curve would occupy 0% and return an HFD value of 1. Using HFD will provide a measurable single result of the scanpath's temporally sensitive geometric complexity. The geometric complexity of each participant's scanpath can be compared with others to search for similarities and differences within a range of standard deviation. Indeed, results were found with a handful of participant scanpaths that were indicative of distraction.

Evaluation of results by Newport et al. [54] concluded that the best approach to identifying scanpath outliers is to use their method as an adjunct tool in conjunction with another approach, e.g., Jaccard by Burch et al. [9] or MultiMatch by Dewhurst et al. [17] in order to build a robust case for exclusion. Furthermore, in cases where geometric complexity and matchability are incongruent, further probing can be performed to evaluate such data as an edge case. Indeed, Jones and Sibson [32] described the probing of the "index of interestingness" found in data variance proportions in order to prune perturbations while maintaining the "interestingness" found in samples that were unusual, yet valid, in order to increase the accuracy of unsupervised clustering results that rely on exploratory multivariate analysis. Similarly, Newport et al. [54] also captured "interesting" differences via geometric complexity revealed in HFD. However, the method by Newport et al. [54] keeps data intact, unlike the projection pursuit method by Jones and Sibson [32] where components are removed as data elements are reduced. It does this both by evaluating sequences entirely without reduction via FD and also by preserving locality in each point during dimensionality reduction by converting 2D Cartesian visual fixations to 1D Hilbert distances.

42.11 Comparing Scanpaths

Over 50 years ago, Needleman and Wunsch [51] developed an innovative approach to aligning protein sequences in the field of bioinformatics. The method drew from the Levenshtein [41] distance metric which used string edit methods to compare character sequences. However, Needleman and Wunsch split the larger sequence into smaller subsequences and then optimized the recombination using a scoring system that allowed for gaps and partial mismatches using a similarity matrix. Indeed, scanpath comparison research results have demonstrated that expert radio-

graphers can be discriminated against novices viewing brain Magnetic Resonance Imaging (MRI) images [14], healthy patients from those affected by autism [37], and even basic number search tasks [18].

In 2010, Cristino et al. [13] adopted Needleman and Wunsch to compare scanpaths using their ScanMatch method. Just like Needleman and Wunsch, ScanMatch compares two letter sequences. It maximizes their similarity score by using a substitution matrix and a gap penalty. A newer approach developed by Dewhurst et al. [17] called MultiMatch took a more granular approach by measuring Vector, Direction, Length, Position, and Duration metrics from a scanpath separately. This compartmentalized approach allowed MultiMatch to find similarities in scanpaths that had gaze paths moving at different speeds and in reverse order, which were limitations of the ScanMatch method.

However, both ScanMatch and MultiMatch use the same method for reducing scanpath dimensionality by assigning letters to the spatial attributes of the fixation. In ScanMatch, this is done using letters to represent their place in a grid or semantic space; MultiMatch extends this method to define their vector direction. To imagine this method visually, replace the numbers on a clock face with letters A, B, C, . . . L. A vector pointing straight up would be at position L, whereas a vector pointing down is at F. Therefore, a scanpath's sequential vector path could be defined, e.g., FCLI to be looking down, and then to the right, up, and finally left. This method of gaze path representation allowed MultiMatch to use string edit methods, such as Needleman–Wunsch, to compare and rank pairs of scanpaths.

Alternatively, a method called SoftMatch proposed by Newport et al. [55] preprocesses the (x, y) scanpath fixation coordinates into 1D Hilbert curve distance points in tandem with their corresponding cumulative durations. This Hilbert curve distance and cumulative duration pair (h, t) form a 2D point that can be measured using simple Cartesian distance calculations. The method presented by Newport et al. [55] used the discrete Frechet distance measurement to parametrically calculate how close two (h, t) segments are from each other, to assess similarity. Parameters determine the tolerance for how high the discrete Frechet distance can be to return a match and the size of the segmentation windows used to create unique individual sequences.

The use of fractal Hilbert curve distances used to reduce the dimensionality of 2D scanpath fixations into 1D distance points, paired with their cumulative duration, provides both a flexible way to quantize point pairs during parametric exploration and also enables locality preservation when converting back to 2D space. In parametric exploration, best-fit analysis can be performed using descending distance tolerance and window size metrics without the need to re-box the stimulus, as would be the case with boxed string edit representation. Additionally, the locality is preserved in 1D Hilbert curve distances converted back to 2D space, which opens up the possibility for future work where Euclidean distance measurements can work in tandem with (h, t) Hilbert curve distance and time points.

42.12 Spatial and Time Domain Analysis

The consolidated analysis of eye tracking features is a challenge for researchers interested in investigating the interdependence of spatial and temporal features in scanpaths involved in assessing expertise. Indeed, research by Kundel [39] proposes that perceptual errors can be mitigated through a better understanding of the principles that govern the perception of medical images, which will both increase expertise in the viewer and minimize errors during diagnosis. Kundel [39] hypothesized that perceptual errors can be categorized into five steps: image attribute acquisition, e.g., contrast and detail, bottom-up component organization, top-down resolving of ambiguities, selective scrutiny of preferred details, and the application of decision criteria to the preferred perception.

This hypothesis was tested in an experiment where the eye gaze patterns of radiologists searching for lung cancers were tracked and diagnostic errors scrutinized. The results identified three types of errors. Decision errors accounted for the majority at 45%, where the cancerous area was fixated but misdiagnosed as an abnormality. Visual search errors were secondary at 30%, where the cancerous abnormality was not fixated at all. The remaining errors at 25% were recognition problems, where the cancerous area was fixated and ignored. Results confirm earlier speculation by Kundel et al. [39] that experience plays a role in shifting the perceptual process from a primarily discrete search mode to a more interconnected one.

Fractal analysis provides a tantalizing potential way to understand human scanpaths through a metric of interconnected geometric complexity. However, the connected components made of the gaze characteristics are still being explored. Mammographic screening research done by Alamudun et al. [1] using fractal analysis to measure human visual search complexity by using participant scanpath attributes including the number of hits, true lesion hit latency, total dwell for normal and abnormal regions, overall search times, and spatial gaze complexity. The relationship between stimulus and participant attributes indicated that the complexity of visual scanning patterns differed depending on many factors, including reader experience level, breast density, and case pathology. However, this research did not provide a consolidated analysis of spatial and temporal features simultaneously.

Work done by Suman, Russo, Carrigan, Nalepka, Lique-Weiland, Newport, et al. [60] at Di Ieva's Lab hypothesized that a computational approach applied to spatiotemporal dynamics, in addition to scanpath attributes used in prior work, can uncover a deeper understanding of how and why experts exhibit different gaze characteristics. In this work, the authors defined two measurements to encapsulate geometric and temporal complexity: the fractal dimension (FD) of the scanpath, calculated using the compass method, is used to measure geometric complexity within the small spatial coverage defined by the scanpath x and y coordinates, and the Hurst exponent (HE) of the scanpath, calculated using both R/S and Detrended Fluctuation Analysis (DFA) methods, to analyze the persistency/anti-persistency of a participant's gaze within the time series.

The FD and HD work particularly well together in this instance because the FD is able to support the HE's coarse focusing ability measurement within tight spatial regions. It does this through its strength in measuring geometric complexity, thus providing a context from which the HE can be examined. Visually, this was accomplished via heatmaps of the visual search parameters that include FD and HE using R/S, and DFA. Windowing in 1000 ms increments was done for the full 5 s, with most abnormalities discovered in the first second. Indeed, the first second FD coupled with R/S results indicates that experts displayed higher scanpath geometric complexity over smaller spatial coverage compared to other participants. The behavior of experts reflects a quick focus on abnormal regions, followed by a sustained inspection to interrogate details for deeper insight.

Similar to findings by Kundel [38], experts appear to first observe a stimulus from the bottom up, waiting for a pathological detail to reveal itself like a "stubbed toe" event often used to describe bottom-up analysis. Once the expert finds an abnormality, it shifts to a top-down approach, where details are interrogated to match a pathology with the context. Nonexperts appear to either be stuck in the bottom-up search or unsuccessful in their interrogation during top-down scrutiny. This research creates an opportunity for a dual-pronged approach to perception training by improving both bottom-up and top-down search methods to improve expertise training in radiology.

42.13 Recurrence Quantification Analysis and FD

The dynamics of recurrent patterns, such as the repeating rise and fall distance of a breathing chest, or the ebb and flow of waves covering a shoreline, are deceptively complex when trying to measure, predict, or analyze them. Recurrence quantification analysis (RQA) is a method proposed to measure the characteristics of similar natural dynamic systems with the theoretical premise that a direct relationship can be found connecting recurrent patterns and their underlying dynamics.

In a paper introducing RQA, Webber and Zbilut demonstrate an application using waves bobbing a buoy in the ocean. Using wave heights at the buoy plotted against time, a chosen height is assigned a demarcation, e.g., 3 m, and all waves that match that height are timestamped. This produces a new plot containing a frequency of timestamps per wave height, e.g., all times where waves reached 3 m. Incorporating many other wave heights will construct a distribution comparing the frequencies of waves reaching 3 m versus 4, 5, 6, etc. This comparison of times to other wave heights is a comparative recurrence.

Gandomkar et al. [24] applied this concept to the examination of mammographic images in order to distinguish expert radiologists from less experienced ones. RQA is specifically implemented by the authors to address spatiotemporal dynamics absent in time-related metrics, e.g., fixation latency, viewing times, target fixation duration, total fixations, fixation cluster sizes, and fractal dimension [24].

The fractal dimension (FD) described earlier can be exploited by RQA to describe the frequency of geometric complexity of eye paths over time. Intuitively, increasing the geometric complexity of eye paths implies increased scrutiny, whereas more sparse, widely spread eye path patterns hint at visual search. The study proposes that the participant moving between high and low FD, along with the other time-related metrics, can be recurrent within a scanpath.

Following the same process described in the earlier example using ocean wave heights, Gandomkar uses fixation points which are quantized to a 2.5° radius of a previous fixation and plotted using eight metrics to evaluate their positions in space over time. In addition to FD, other metrics include recurrence (REC), defined as the percentage of all fixation pair combinations that are quantized to the same position, T_2 as the difference in average time between two nonconsecutive returns, and laminarity (LAM), which is the measure of a set of consecutive fixations repeated many times in a scanpath.

Participants of the study included four inexperienced and four experienced radiologists viewing 120 mammograms. Results found by Gandomkar et al. revealed that radiologists with greater experience were more efficient at re-fixating, deterministic, and laminar eye movements.

42.14 Scanpaths as Features in Neural Networks

In an attempt to better understand the relationship between the relevance of reading material and the reader, Bhattacharya et al. [5] captured gaze data from participants reading short news articles and rated their relevance based on a trigger question. The eye paths were encoded for processing in a convolutional neural network, where they were trained based on ground-truth answers regarding the news story relevance. The researchers chose this approach instead of the typical queries and interaction logs used in such studies because they considered eye tracking to be a close proxy for human cognition [34, 65]. However, even though the researchers note the many advantages of utilizing eye tracking, they correctly criticize the lack of standards and methods for converting eye path data into a format usable by machine learning algorithms. For example, in methods using string edit representation, 2D fixation features are collapsed into a data stream of single numbers or letters where fine-grained information is lost after quantization. Bhattacharya et al. proposed a technique where scanpaths are converted into images encoded with symbols representing certain fixation attributes and then fed into a convolutional neural network for classification using image recognition. Even though experiments using this method were 80% accurate, limitations and artifacts relating to the size (224 pixels by 224 pixels) of trained images and possible loss due to hidden or overlapping symbols encoded in the flattened image hamstring this method.

Alternatively, recent research has demonstrated a way visual gaze paths can be flattened to a one-dimensional tensor for use in machine learning algorithms while mitigating information loss during quantization. Newport et al. [55] demonstrated a

dimensionality reduction method using fractals to convert 2D (x, y) coordinates into 1D distances along a curve.

As mentioned earlier, the specific type of curve used in the conversion was developed by David Hilbert in 1891 and adapted from earlier fractal curve research done by Giuseppe Peano in 1890. The curve attempts to solve a task by which a line can pass through every space in a unit square. As described in Sect. 2.2.1, fractal curves can be used to convert 2D fixations in a scanpath into 1D Hilbert distances. As demonstrated by Newport et al., a list of Hilbert distances can be used as tensors representing feature spaces in a convolutional neural network.

Traditionally, conversion of 2D fixations to 1D string representations, or other types of flattening into 1D data streams for machine learning feature space representation, involves quantizing before conversion into strings or numerical representations, resulting in the kind of data loss and lack of granularity Bhattacharya et al. and Newport et al. alluded to traditional dimensionality reduction methods. However, Hilbert distances can be converted back and forth between 2D and 1D space without loss, and due to the linear equidistant scaling of Hilbert space, quantization can be performed last, allowing for increased statistical significance of adjacent fixations, especially along boundaries.

42.15 Discussion and Conclusion

A biological imaging modality is the resulting acquisition of multiscale information, including complementary functions, structures, molecular compositions, and dynamics [63]. Yet, it is surprising that a prevailing modality for analyzing eye gaze information does not yet exist. This dearth may be due to the dispersion of various imaging modalities in many different books, specific to unique research tasks and requiring specialized backgrounds to comprehend. Walter et al. addresses and attempts to mitigate this problem in their biological imaging modality compendium [63].

However, no prevailing imaging modality exists today for clinicians to view eye path information from patients as an adjunct tool for diagnosing neurological conditions in a uniform, consistent manner. Yet, a recent eye tracking meta-analysis [42] has identified 11 studies using different markers to recognize cognitive disorders over the last decade. Older studies, such as work done by Hutton et al. [28], demonstrated an evolution from their early work identifying a strong correlation between the severity of visual tracking abnormality and the severity of dementia to newer studies by Mengoudi et al. [50] using machine learning to augment dementia diagnosis using instruction-less visual gaze pattern clustering. Electronystagmography (ENG) uses electrodes to assess voluntary and involuntary eye movements to diagnose vertigo and other balance dysfunctions by interrogating the vestibular system. ENG provides an eye-movement-based modality by using the amplitude of frequency signals sent back by electrodes over nerves responsible for eye movements [22]. However, it lacks the adaptability to other neurological disorders uncovered in research using video-based eye tracking.

This dearth of modality also extends to pedagogical applications of eye gaze analysis. A recent study of eye gaze analysis in crime scene investigation instruction has shown that better-performing students follow a similar decision-making process when assessing a crime scene [11]. A similar longitudinal study developed by Jossberger [33] investigated the development of expertise among eight radiology residents with similar findings. Indeed, a systematic review of eye tracking technology in medical education [2] using 33 studies has shown that training, assessment, and feedback practices can aid in reflection and improvement. Despite these advances, students are still tasked with the lengthy viewing of hundreds of radiographs, called “massed practice,” as the prevailing educational practice in medical schools today [59].

The challenge that visual scanpaths present, including their complexity, data sparsity, and the spatiotemporal nature of fast-moving eye gaze patterns, has caused researchers to successfully develop many different approaches to tackling this problem. However, displaying both Cartesian positions of fixations and their temporal information has been coarse, as demonstrated with repeating strings in string edit representation (e.g., *AbBdBdBdCaDc*) and by using dedicated temporal analyses as demonstrated with MultiMatch’s durational similarity analysis. Both these methods require quantization, resulting in a loss of data.

Future work incorporating fractals can be used to provide a functioning modality to capture the dynamics of visual scanpaths. The fractal dimension (FD), e.g., Minkowski–Bouligand dimension, is demonstrated in Fig. 42.2 measuring multi-scale information. FD was used by researchers [54, 60] to measure and compare the behavioral characteristics of scanpaths. Structures of the gaze converted into fractal Hilbert curves were used by Newport et al. [55] to increase its matchability and improve its accuracy when used in machine learning [53]. These and many other methods used to capture markers in scanpaths hint at a potential space for measuring and using eye gaze information in a clinical setting beyond research.

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Chapter 43

Fractal Electronics for Stimulating and Sensing Neural Networks: Enhanced Electrical, Optical, and Cell Interaction Properties



S. Moslehi, C. Rowland, J. H. Smith, W. J. Watterson, W. Griffiths, R. D. Montgomery, S. Philliber, C. A. Marlow, M.-T. Perez, and R. P. Taylor

Abstract Imagine a world in which damaged parts of the body – an arm, an eye, and ultimately a region of the brain – can be replaced by artificial implants capable of restoring or even enhancing human performance. The associated improvements in the quality of human life would revolutionize the medical world and produce sweeping changes across society. In this chapter, we discuss several approaches to the fabrication of fractal electronics designed to interface with neural networks. We consider two fundamental functions – stimulating electrical signals in the neural networks and sensing the location of the signals as they pass through the network. Using experiments and simulations, we discuss the favorable electrical performances that arise from adopting fractal rather than traditional Euclidean architectures. We also demonstrate how the fractal architecture induces favorable physical interactions with the cells they interact with, including the ability to direct the growth of neurons and glia to specific regions of the neural–electronic interface.

Keywords Fractals · Human implants · Electronics · Neurons

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43.1 Introduction

In 1752, Benjamin Franklin attached a metal key to the bottom of a dampened kite string and then flew the kite in a storm. It didn't take long to go from his simple demonstration of harnessing electricity to applying it to living bodies. In 1764, physician Charles Le Roy applied electricity to patients' eyes, causing them to see flashes of light. In 1791, Luigi Galvani did the same to the muscles of frog legs, causing them to twitch [1]. Since then, two centuries of rapid development have led to today's commercial electronics industry. Miniaturization has been the main driver of improvements. In addition to faster operation, the evolution from micro-electronics to nano-electronics facilitates novel methods for manipulating the flow of electricity. For example, our research group has investigated ballistic electronics [2], quantum electronics [3], spin electronics [4], and coulomb blockade [5]. However, developments such as these target the computing and communications industry rather than electronic interfaces with biological systems.

Miniaturization offers surgeons the opportunity to implant devices in humans rather than relying on the crude external wires used by Le Roy and Galvani. For example, electronic devices have been implanted into human retinas with the aim of restoring vision to patients with degenerative retinal diseases [6–13]. More than 150,000 deep brain stimulation implant surgeries have been performed targeting neurological disorders such as Parkinson's disease [14]. Furthermore, people with amputated limbs receive interactive prosthetic implants that restore mobility. In addition to enhancing the performance of medical devices, exploring neuron responses to implants can be used to investigate their fundamental behavior and the degree to which this behavior can be controlled. Such studies should also accommodate interactions with glia. Although neurons and glia were discovered around the same time, research on the latter has been slower to gain momentum [15] even though they are prevalent in the central nervous system [16] and play central roles in controlling neuronal network structure and functionality [17]. Specifically, whereas neurons are crucial for carrying the body's electrical signals, glia are equally crucial because of their supporting role as the neurons' life support system.

Significantly, implants are frequently referred to as bionic devices in recognition of the importance of bio-inspiration and the need for biocompatibility at the cell–implant interface (“bion” is Greek for “unit of life”). Yet, Fig. 43.1 highlights a fundamental geometric mismatch between today's conventional implants and the cells that they interact with. Confocal microscopy (Fig. 43.1a) can be used to construct three-dimensional models of neurons (Fig. 43.1b), allowing an analysis of their fundamental geometry [18–20]. Discussed in more detail in Chap. 44, this analysis shows that neurons belong to fractal geometry, featuring dendritic branches that repeat at increasingly fine-size scales. In contrast, the active area of typical implants is based on Euclidean geometry. The gray square shape shown in Fig. 43.1c serves as an example.

Fractal branches are prevalent in nature. When flying his kite in the storm, Franklin would have noticed the multi-scaled branches of the lightning around him,

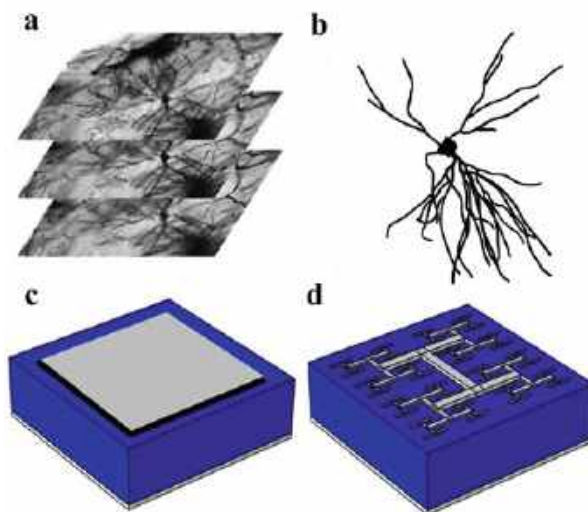


Fig. 43.1 (a) Confocal microscopy generates a stack of $200\ \mu\text{m} \times 200\ \mu\text{m}$ horizontal sections (with vertical separations of $2\ \mu\text{m}$) of Golgi-Cox-stained neurons. (b) These sections are then computer-assembled into three-dimensional images for pattern analysis. A neuron's basal dendrites from a rat's hippocampus are shown for demonstration. (c) A schematic showing the square-shaped active area (light gray) of a conventional implant. (d) For our bio-inspired implants, the active area features branched patterns that repeat at different size scales

along with the fractal trees that populate many natural environments. Animals also benefit from fractal branches in, for example, their bronchial trees and veins. In each case, as their branches spread out in space, the structure features two embedded fractal patterns – the branches and the gaps forming between them. Their structural relationship offers the potential to balance functionality with operational costs while also maintaining structural integrity. In this balancing process, the repeating patterns offer large interfaces to interact with light in the case of trees [21], oxygen in the case of bronchial trees [22], and neighboring cells in the case of neurons [18].

Even if Franklin had pondered the potential benefits of using a set of strings that copy lightning branches, the challenges of assembling this network would have been daunting enough to dampen his enthusiasm for such an experiment. In contrast, here we will present state-of-the-art approaches to fabricating fractal electronics suitable for implants. Adopting the principle of bio-inspiration, we will describe the use of fractal interconnects to establish a biophilic interface between the implants and the neuronal cells they interact with. A repeating H-pattern is shown as an example in Fig. 43.1d. We will consider fractal interconnects with two distinct electrical functions – electrodes that stimulate signals in the neurons and sensors that detect and track these signals as they pass through the body's neural network. The superior electrical, optical, and physical properties of the fractal interconnects are expected to be generic. Future applications could therefore include interactions with neuronal cells in the brain, retina, and limbs.

43.2 Fabrication of the Fractal Interconnects

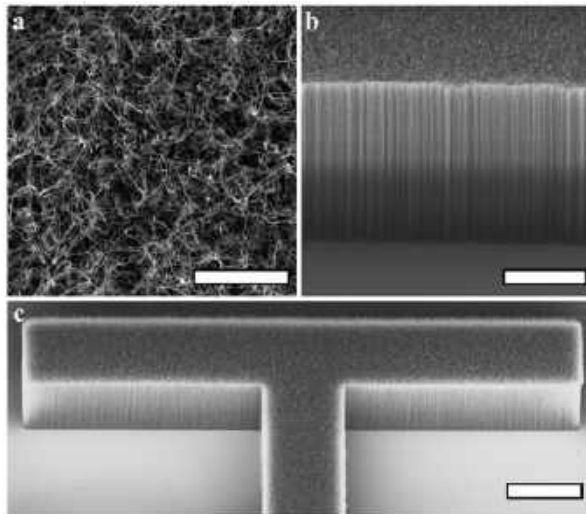
When neuronal cells interact with each other, they do so by exploiting both their physical and chemical environments. Our interconnects aim to do the same. We exploit carbon nanotubes (CNTs) as our interconnect material. A wealth of previous research has studied interactions of CNTs with many cell types, including neurons. Their density, stiffness, and topography can be controlled [23–25], and they can be synthesized on, or transferred to, flexible substrates [26, 27]. Due to a combination of their chemical composition and surface texture, CNTs promote neuronal adhesion and increase the number of processes and their growth [28–30]. Their nanoscale roughness has been proposed to mimic some of the extracellular matrix (ECM) properties [31, 32] and to enhance neurite outgrowth and elongation if the roughness variation matches the neuron process diameter [33]. CNT flexibility also likely plays a role, since neurons are known to readily adhere to and grow processes on softer substrates [34, 35].

Although it has been suggested that CNT functionalization is necessary for biocompatibility, our research supports previous results demonstrating that appropriate degrees of texture and flexibility of our pristine CNTs are sufficient for neural network survival [36], which in turn favors recording and stimulation [37]. Previous studies demonstrating that electrically biased CNT electrodes stimulate neurons effectively [38, 39] and even boost their signal transmission [40, 41] indicate that our fundamental studies have large potential for translation to future applications.

We employ two fabrication techniques to match the fractal geometry of the CNT interconnects and the neurons. Whereas all fractals exhibit repetition of patterns at multiple size scales, fractals can be grouped into two categories based on how the patterns repeat. For fractals prevalent in the body (e.g. neurons, veins, and bronchial trees), the statistical qualities of the patterns repeat at different scales. In contrast, the patterns of mathematically generated “exact” fractals repeat exactly at different scales. Consequently, whereas exact fractals look precisely the same at increasingly fine scales, “statistical” fractals simply look similar at different scales [42]. One of our fabrication techniques exploits the precision and control associated with the clean geometry of exact fractals. The other harnesses natural deposition processes to generate statistical fractals.

Fractal dimension, D , is a central parameter for quantifying both types of fractal. This describes how the patterns occurring at different scales combine to build the resulting fractal shape [42]. For Euclidean shapes, dimension is described by familiar integer values – $D = 1$ for a smooth line and $D = 2$ for a completely filled area. The repeating patterns of a fractal line cause the line to begin to occupy more space than the Euclidean line. This occurs at multiple scales and results in a D value that has a non-integer value lying between 1 and 2. For each fabrication method, D can in principle be adjusted along with the size scales over which the fractal repetition occurs. This facilitates an investigation of the impact of these fractal parameters on the interconnects’ properties.

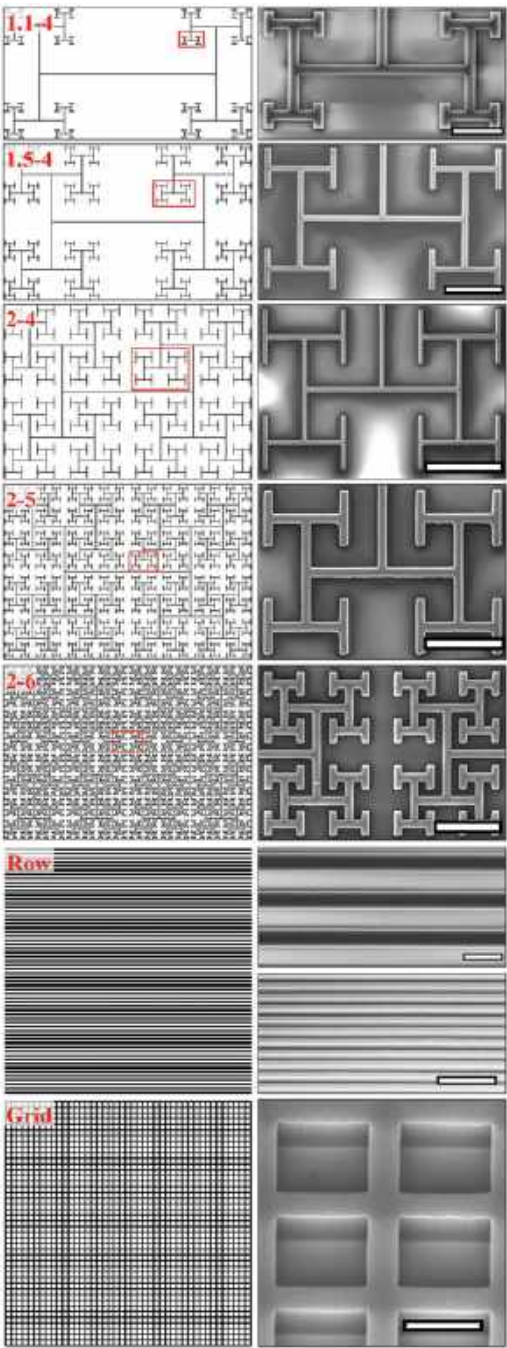
Fig. 43.2 Scanning electron micrograph (SEM) images of patterned VACNT forests. **(a)** Top-down view of entangled VACNTs on the forest's top surface. **(b)** and **(c)** are views of VACNT sidewalls taken at a 40° angle. Scale bars are 2, 10, and 20 μm , respectively. The VACNT patterns are fabricated in collaboration with B. Alemán (University of Oregon, USA)



To construct the exact fractals, we employ a vertically aligned CNT (VACNT) approach in which CNTs grow vertically to heights of approximately 25 μm from a metallic catalyst layer deposited on a smooth silicon dioxide (SiO_2) substrate [43]. Crucially, they tangle into a conducting “forest” as they grow (Fig. 43.2). These forests can be shaped laterally by using lithography to define intricate patterns in the catalyst layer. In addition to establishing textured branches separated by smooth gaps, the high aspect ratios of the VACNTs will aid penetration into neural tissue [44]. The gap sizes span from a few μm up to $\sim 4\text{ mm}$ to investigate both the individual and cell network interactions with the interconnects (retinal neuronal and glial cell bodies range from a few μm to $\sim 30\text{ }\mu\text{m}$, and glial and dendritic processes can be as large as a few hundred micrometers). The VACNT branch width is chosen to be $W_{\text{CNT}} = 20\text{ }\mu\text{m}$ to allow the somas of several neurons to attach to the VACNT surfaces and for their processes to grow out from the somas across the electrode surface.

We choose the H-Tree (built from repeating H-patterns) for the lateral shape because it is an established branched fractal with well-defined scaling properties. As demonstrated in Fig. 43.3, D sets the rate of shrinkage of branch length between repeating levels with a higher D corresponding to a slower rate. In addition to D , the number of repeating levels m is varied (Fig. 43.3). Acknowledging that even these simple H-Trees are inherently complex, we also investigate interconnects consisting of rows and grids. We refer to these comparators as Euclidean designs because the gaps within each interconnect design have a single scale in contrast to the multi-scale character of the fractal design. For the rows, W_{CNT} and the gap width, W_{Si} , vary between interconnects and span from 25 to 100 μm . For the grids, $W_{\text{CNT}} = 25\text{ }\mu\text{m}$ to match the fractal design and $W_{\text{Si}} = 60\text{ }\mu\text{m}$ is chosen to ensure that each grid’s “cell” can accommodate several glial cell bodies.

Fig. 43.3 The left column from top to bottom shows schematics of the fractal, row, and grid interconnects: ($D = 1.1$ and $m = 4$, labeled as 1.1–4), ($D = 1.5$ and $m = 4$, labeled as 1.5–4), ($D = 2$ and $m = 4$, labeled as 2–4), ($D = 2$ and $m = 5$, labeled as 2–5), ($D = 2$ and $m = 6$, labeled as 2–6), (Row), and (Grid). The right column shows SEM images of a region of the interconnect shown in the left column. The fractals and the first image of rows are shown from a top view, while the second row and grid images are taken at a 40° angle. From top to bottom, the scale bars are 100, 200, 400, 200, 200, 100, 200, and $50\text{ }\mu\text{m}$. The VACNT patterns are fabricated in collaboration with B. Alemán (University of Oregon, USA)



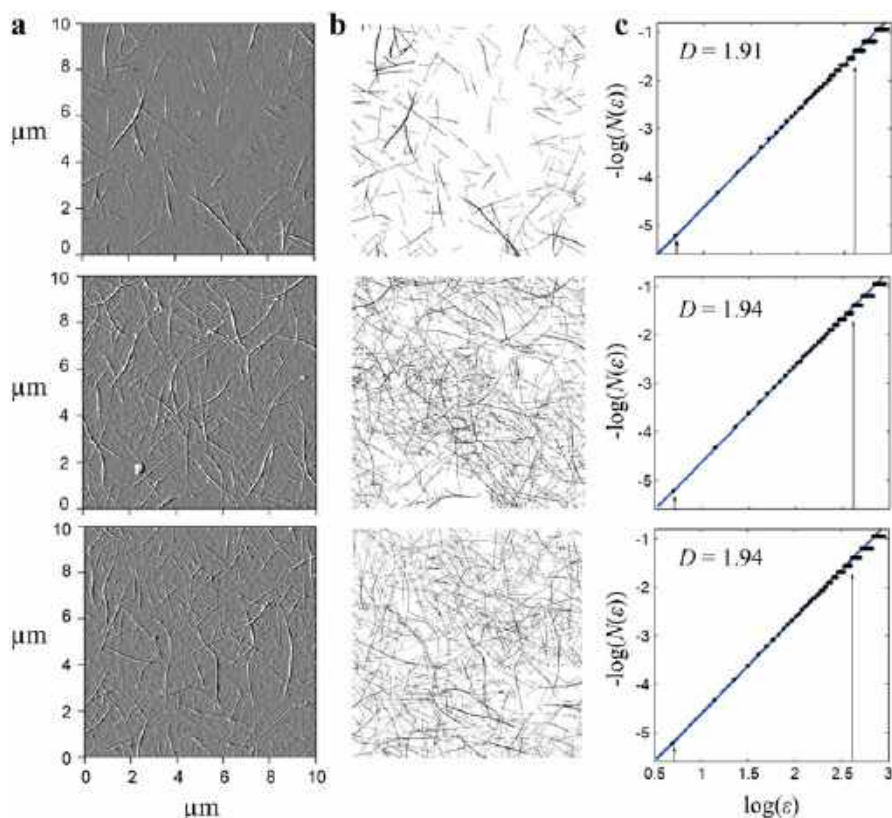
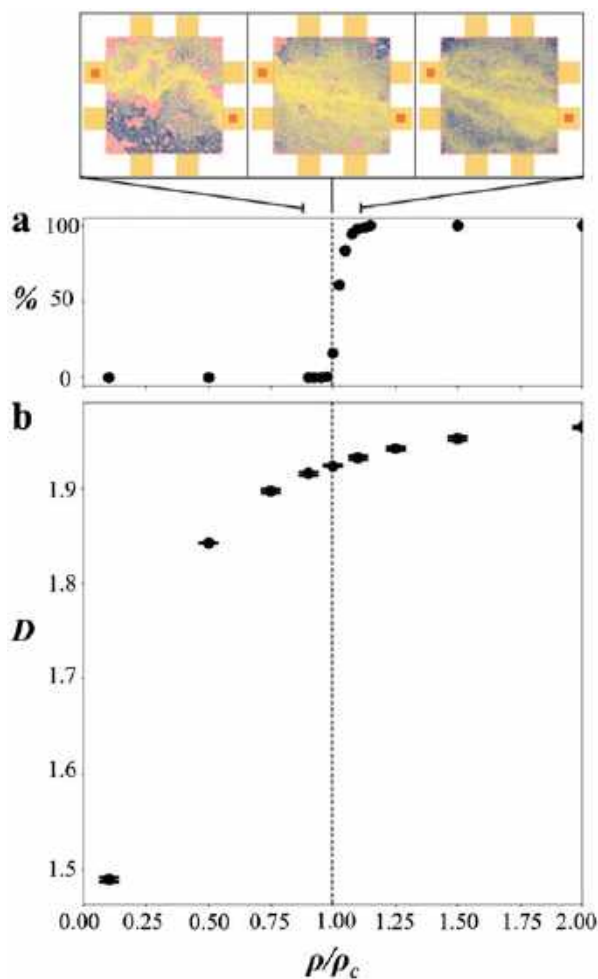


Fig. 43.4 Fractal analysis of the CNT thin film network. (a) AFM images of typical networks. (b) To perform the fractal analysis, these images are converted to black and white bitmap images. (c) The associated box-counting fractal analysis plots in which the log of the minimum number of boxes, $N(\epsilon)$, required to cover the network is plotted against box size, ϵ . The arrows indicate the expected limits to assess D . The CNT films are fabricated in collaboration with N. Plank (University of Victoria, New Zealand)

To fabricate the statistical fractals, a thin horizontal film (spanning $3\text{ mm} \times 7\text{ mm}$ with thickness varying between 1.5 and 10 nm depending on bundling) of CNTs is formed on a SiO_2 substrate using a solution deposition technique [45–48]. To confirm that these randomly arranged CNTs self-assemble into a statistical fractal network, a standard box-counting technique is performed on atomic force microscope (AFM) images of the networks. Figure 43.4 shows three examples of $10\text{ }\mu\text{m} \times 10\text{ }\mu\text{m}$ images taken at different locations on the film. In each case, the image is covered with a mesh of identical squares (boxes). For a network to be considered fractal, the minimum number of boxes, $N(\epsilon)$, required to cover the network scales according to box size, ϵ , as $N(\epsilon) \sim \epsilon^{-D}$. The linearity of the plot indicates the networks are fractal. A higher D fractal corresponds to a steeper

Fig. 43.5 Probability of percolation, %, (a) and fractal dimension, D , (b) of 100 and 5 simulated CNT thin film networks for (a) and (b), respectively, as a function of stick density ρ . The top images show networks connected to measurement terminals located along the network's perimeter (symbolized by eight yellow squares). The nonlinear character of the networks is highlighted by coloring according to the number of sticks needed to connect the upper left and lower right terminals to the given stick: red indicates sticks that are not connected to the network, yellow indicates a small number of sticks, and dark blue indicates a large number



slope and therefore to the pattern occupying a larger number of fine-scale boxes. The observation limits are shown by the arrows. As expected, the lower limit of observation is set by the smallest feature size and the upper limit set by the counting statistics of the box-counting algorithm (20% of the total network size).

The average tube lengths for the shown films are (from top to bottom) 1.6, 1.9, and 1.9 (± 0.5) μm with associated tube densities of 3.0 tubes (± 0.5)/ μm^2 , 9.9 tubes (± 0.9)/ μm^2 and 9.9 tubes (± 0.9)/ μm^2 , respectively. To demonstrate the relationship between density, r , and D more systematically, Fig. 43.5 shows the results of a Monte Carlo simulation of the deposition process [49] in which “sticks” (representing either individual or small bundles of CNTs) have a fixed length approximating to the tubes within our CNT network. The vertical dashed line indicates that the stick density ρ_c is necessary for an infinite system to reach

its percolation threshold (i.e. to establish an electrical connection by at least one continuous path across the network). However, Fig. 43.5a plots the probability, %, that our finite-sized network will establish percolation and demonstrates that a higher density than the dashed line value is necessary to reliably establish electrical connectivity across the network.

Figure 43.5b shows the average D for simulations and reveals a clear dependence on ρ . Near the percolation threshold, D changes rapidly and then saturates at higher densities. When interpreting the shape of this curve, recall that D is an indicator of the ratio of fine to coarse structure in the network. As more tubes are added to the network its fine structure increases, raising the D value. Taken together, Figs. 43.4 and 43.5 demonstrate the potential of tuning the fabricated network's D value using deposition density. Furthermore, the film could be patterned post-synthesis by etching away selected regions. Alternatively, the tip of an AFM could be scanned across the nanotube network while applying an electrical bias to the tip, which would cut the network into the desired fractal pattern.

43.3 Functionality of the Fractal Interconnects

Our interconnects are designed to serve two distinct electronic functions – inducing and detecting electrical signals in the neurons. For both functionalities, it is crucial to consider the impact of the fractal design on the mechanical, optical, and electrical properties of the neuron–implant interface along with its impact on cell health and growth. Adopting the philosophy that the implant is only as strong as its weakest link, we target the fundamental science that emerges when all of these properties are considered in unison. In terms of their mechanical properties, fractal circuits have been proposed for stretchable electronics [50] – the associated flexibility therefore presents the possibility of developing future interconnects that are conformal to the regions of the neural network in which they are implanted. In terms of optics, the surface coverage of the fractal interconnects is considerably less than the equivalent squares due to the multitude of gaps between the branches (Fig. 43.1c and d). This suggests that light can be easily transmitted through the fractal interconnect. This will be advantageous for future brain implants that utilize both optical and electrical stimulation [51]. Furthermore, the ability to shine light through the interconnects is crucial for the operation of retinal implants featuring interconnects positioned above photodiodes [52].

The interconnects' electrical properties are central to the operation of both stimulators and sensors. Electronic stimulation of neurons dates back to the pioneering studies of Le Roy and Galvani. The electrical fields from their electrodes set up potential differences across the neuron membrane, inducing ion flows that trigger a signal along the axon [53]. In Le Roy's experiment, for example, this process induced signals in the retinal neurons that the brain interpreted as originating from the retina's photoreceptors. The positive impact of adopting the fractal geometry becomes clear by returning to the schematic of Fig. 43.1c and d in which the gray

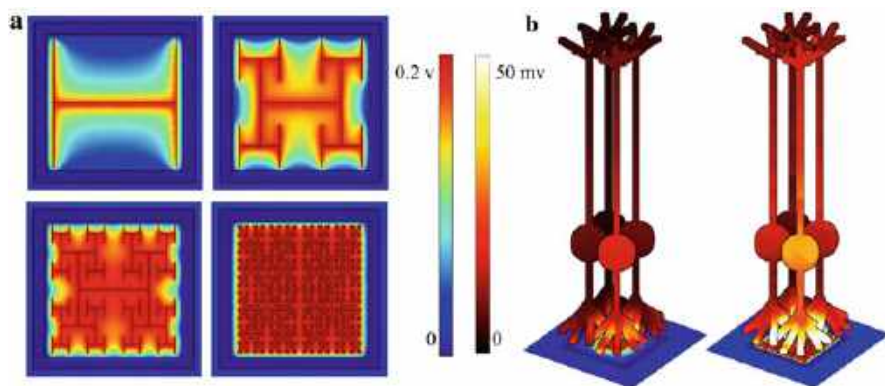


Fig. 43.6 Simulations of electric fields generated by a shaped central electrode surrounded by a ground electrode (the latter is not shown for clarity). **(a)** Color plots showing a horizontal slice (at the central electrode's top surface) of the three-dimensional voltage distribution for an H-Tree central electrode. The four simulations show the field uniformity achieved by increasing the repetition levels from $m = 1$ to $m = 4$. In each case, the applied voltage is 0.2 V and the frequency is 1 kHz. The left central color bar shows the maximum extracellular voltages reached during each oscillation which varies from 0 V (blue) to 0.2 V (red). **(b)** A color plot of the maximum change in electric potential across the membrane before and after stimulation for bipolar neurons above a square central electrode (left) and fractal central electrode (right) both for an applied bias of 0.2 V and a frequency of 1 kHz. The right central color bar displays the magnitude of the potential and varies from 0 V (black) to 50 mV (white). For visual clarity, only four of the nine neighboring neurons are shown. Images are drawn to scale: the neurons are 100 μm in height, and the soma is centered 30 μm above the surface

interconnect serves as the stimulating electrode. Our simulations of the electrostatic properties of electrodes show that a significant amount of electrical charge resides on the sidewalls of the electrodes [54]. The simulations in Fig. 43.6 demonstrate that the increase in sidewall charge with m results in increasingly uniform fields such that a $m = 4$ fractal electrode's electric field is as uniform as that of an electrode without gaps. Consequently, the gaps can let light through without disrupting the field. Compared to traditional Euclidean electrode designs such as grids and squares, the large boundaries inherent in fractal patterns generate a large surface area at the sidewalls. The enhanced electrical capacitance then results in larger stimulating electric fields. For the 20 $\mu\text{m} \times 20 \mu\text{m}$ electrodes designed for retinal implants, our simulations demonstrate that voltage biases that stimulate all of the neighboring neurons when applied to the fractal electrode stimulate less than 10% of them when applied to the equivalent square electrodes (Fig. 43.6) [55]. This fractal advantage translates to electrodes applicable to brain stimulation applications.

The growth conditions that we use for the VACNTs generate multi-walled CNTs that are metallic [56] and therefore ideal for electrode applications (due to their low resistances, metals minimize variations in electrostatic potential across the electrode). In Fig. 43.7, we show how the large-scale H-Tree design (which purposely spreads across millimeter scales to interact with large neural networks) of

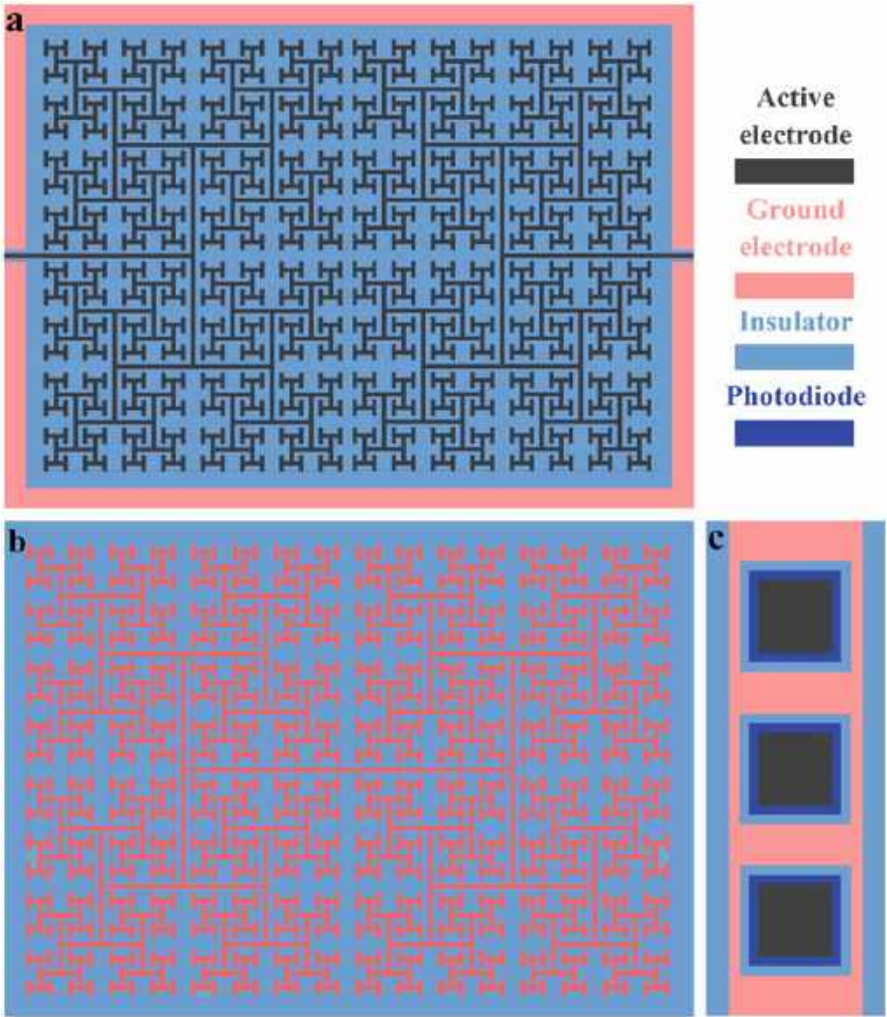


Fig. 43.7 Potential electrode designs for brain and retinal stimulation purposes. **(a)** For brain stimulation purposes, the large-scale fractal pattern (gray) serves as the active electrode bounded by a rectangular ground electrode (pink). **(b)** For retinal stimulation purposes, the large-scale fractal pattern serves as the ground electrode (pink) with an array of small holes inserted along its branches. **(c)** The zoom-in on a fractal branch in **(b)** shows three holes within the branch of the ground fractal electrode. Each hole features an active square electrode (gray) positioned above a photodiode layer (dark blue) separated from the ground fractal electrode by an insulating region (light blue)

Fig. 43.3 could be used for neuron stimulation applications. Translation of these designs into brain or retinal stimulating implants would need to feature active electrodes that accommodate the high resolutions required for these applications

(the active electrodes of retinal and brain implants are typically of the order of $\sim 20\ \mu\text{m}$ and $\sim 10\ \mu\text{m}$, respectively) [11, 57].

Here, we consider several potential approaches to achieving this. The simplest translation would be to employ the fractal electrode as the active electrode bounded by a rectangular ground electrode featuring a gap to allow passage of the active electrode to a voltage source (Fig. 43.7a). In this application, the voltage source would bias the active electrode and the electric field generated between the active and the ground electrode would be used to stimulate the neurons. However, the active electrode would need to be scaled down or subdivided into component electrodes to match the sizes of the typical brain implants. A more appealing approach, applicable to retinal implants, would use the large-scale fractal as the ground electrode with an array of small holes defined along its branches (Fig. 43.7b). Each hole would then encapsulate an active square electrode deposited above a photodiode layer that is separated from the ground electrode by an insulator [54, 58]. In this application, light-induced charge in the photodiode migrates up to the active electrode and generates the electric field between this and the ground electrode. The square active electrodes could be replaced by fractal electrodes to further increase their electrical capacitance and hence their ability to stimulate the neurons [54, 55].

In terms of electronic sensors, the simultaneous occurrence of many neuronal signals makes tracking individual signals extremely difficult. This problem is analogous to the “cocktail party problem” when the listener tries to focus on a specific conversation among the background “chatter” from many guests. To accomplish this task, recordings from the conventional sensors are analyzed using waveform analysis or triangulation to determine signal location. Waveform analysis relies on a priori information about signals’ temporal characteristics (their waveforms) to identify unique signatures of individual neurons. In vivo, this is limited by: (1) isolating thousands of signals, (2) waveform variability within each neuron, and (3) electrode drift with time [59]. Triangulation works by measuring the waveform variation between different sensors in the array. However, errors in uniquely identifying thousands of signals present critical problems for triangulation algorithms [60]. Our fractal sensor avoids all of the aforementioned limitations by uniquely identifying neuronal signals without post-measurement analysis and its reliance on a priori information about waveforms. This is achieved by incorporating a fractal conduction network [49].

A common approach to conventional electronic sensors features a semiconductor channel that conducts electricity between source and drain terminals. The electric potential generated by neuron signals depletes electrons in the channel, leading to a measurable resistance change. Unfortunately, this traditional design offers little flexibility for improvement. For example, gains in measurement resolution by reducing device size are inevitably coupled with larger measurement noise due to increased resistance. Our bio-inspired design is shown in Fig. 43.8. Current is passed between pairs of measurement terminals through a fractal distribution of conducting channels (this active region is symbolized by the light gray square). Depletion of a local region of this network will induce a re-arrangement of current through the network. The nonlinear signature of this re-arrangement is expected to generate an

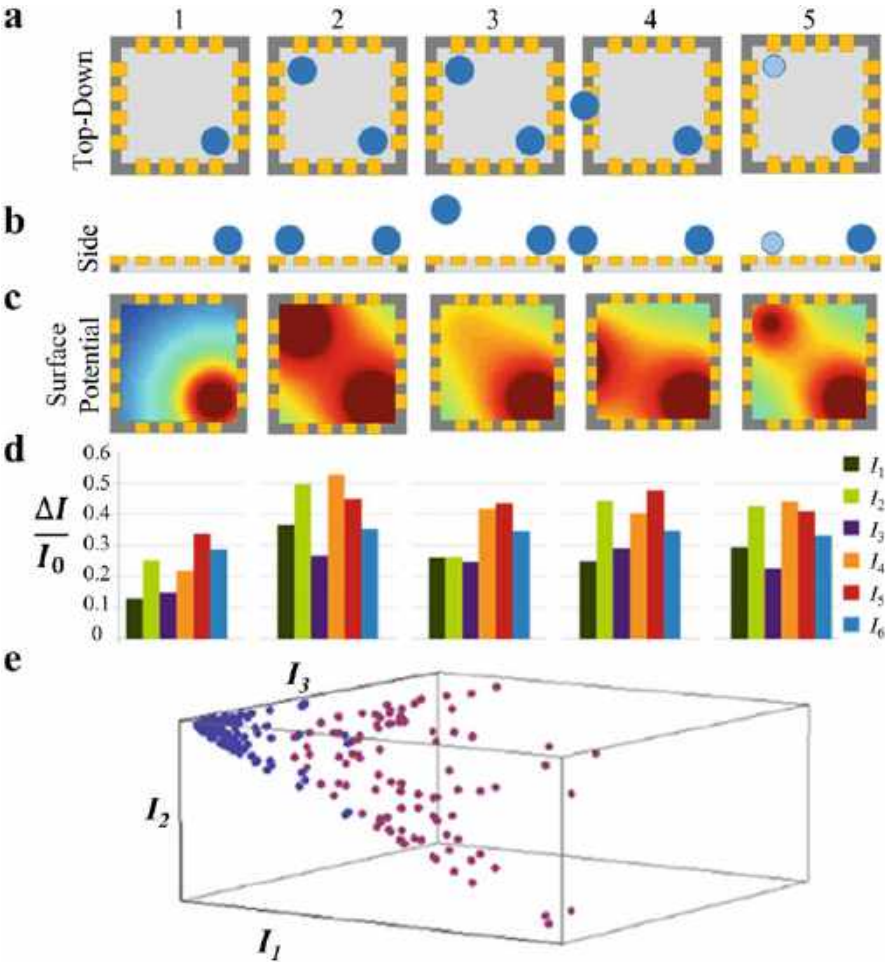


Fig. 43.8 Illustrative examples of neurons (blue) acting as electrostatic gates on the active layer (light gray) on an insulating substrate (dark gray) of a multi-electrode field effect transistor (FET) probe using 16 electrodes (gold) to serve as source and drain. Five different neuronal positions and magnitudes, shown per column, lead to unique sets of current measurements. The top-down (a) and side projections (b) of 10 μm (dark blue) or 7 μm (light blue) neurons. (c) The unique surface potentials associated with the (x, y, z) coordinates of neurons in each case. (d) The associated normalized current changes, $\Delta I/I_0 = (I - I_0)/I_0$, through specific electrodes in each case. I_0 is the electrode's current with no stimulus. (e) The current signatures (for three currents) are shown for one neuron at different locations and signal strengths (purple = small signal, magenta = large signal). (Figure taken from Ref. [47])

exponential sensitivity of the measured resistance to the location of the depletion region and hence to the neuronal signal. Combining the measurements from the various terminal pairs creates an emergent knowledge greater than the sum of the individual measurements. This is achieved through a calibration library that, in real

time, converts the multiple resistance values into unique values for the position and size of each signal's depletion pattern. Returning to the cocktail party analogy, the resulting enhanced sensitivity is equivalent to giving the party guests hearing aids. The calibration library is equivalent to the guests working as a team, comparing notes about previous parties.

Considering the fundamental operational principles in more detail, the 16 terminals allow for 15 independent measurements of the current in the channel. The simulations shown in Fig. 43.8 quantify the sensitivity of these 15 measurements to the location and magnitude of the electric field generated by neurons firing (their positions are indicated by the blue circles in Fig. 43.8a and b). Serving as a control, the results shown are for a uniform semiconductor channel. Five example neuron scenarios, numbered 1–5, are shown by column in Fig. 43.8. For each scenario, the resulting potentials at the sensor surface and the current between six terminals are calculated (Fig. 43.8c and d, respectively). The six current measurements show the unique current signature for each of the neuron configurations, allowing the sensor to distinguish the signal's origin and magnitude for each neuron.

The library plot shown in Fig. 43.8e is based on just three current measurements for one neuron at different locations and signal strengths. In future applications, once a library is established a search algorithm could then be used to convert the 15 measured current values into the positions and magnitudes for each neuron present. The accuracy of the sensor will depend on the degree to which scenarios can be distinguished in the library and this is determined by how the data points fill the available space. In particular, degeneracies (i.e. two scenarios occupying the same data location) will increase if data points occupy only a small fraction of the current space. If the uniform channel is replaced by a statistical fractal network, the nonlinearity of the current re-arrangement produces enhanced sensitivity of the measured current to the location of the depletion region and hence to the neuron signal. This will increase the spread of data in the current space, increasing our ability to distinguish between the unique current signatures of different neuronal scenarios.

The CNT films used to generate the statistical fractals are ideal for the sensor network because the percentage of metallic and semiconductor CNTs in the network can be adjusted with the growth conditions. This composition along with tube density are key parameters for determining the most sensitive network, and this is the subject of ongoing simulations and measurements of fabricated fractal networks. In particular, the junctions between the semiconductor and metallic CNTs form Schottky barriers and their large associated resistances cause currents to divert through neighboring routes. These form “pinch points” in the network which can radically impact the current when depleted by a neuronal signal.

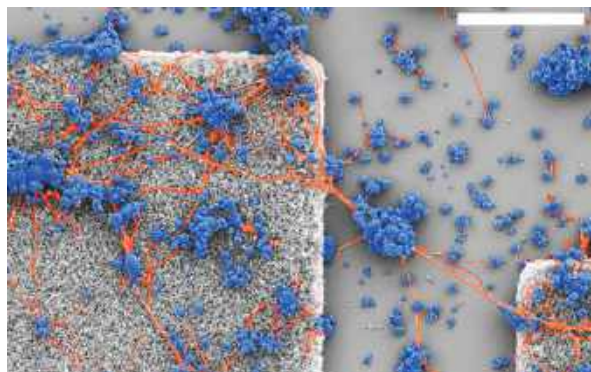
43.4 The Biophilic Interface

Building on the advantageous mechanical, optical, and electrical properties proposed so far, the interconnects also provide a crucial biocompatibility function. We have already highlighted how the CNTs' chemical composition and their surface textures encourage neuronal adhesion and growth. This close proximity of an abundance of healthy neurons to the textured surface will enhance neuron stimulation for electrodes and enhance signal detection for sensors. This is analogous to attracting the party guests that you want to talk to and listen to. In contrast, smooth surfaces are known to trigger an accumulation of glial cells [35, 61, 62], to the extent of sometimes forming a protective layer called a glial scar [44]. Whereas the presence of individual glial cells is a signature of health (because they serve as the neurons' life support system), the scar pushes the neurons away from the smooth surface of conventional implants reducing their ability to stimulate or sense the neurons.

Defining a physical fractal pattern in the implant's surface exploits this difference in the responses of the two cell types and in doing so introduces an extra functionality – the ability to selectively direct neurons and glia to different locations on the interconnect. Although this is advantageous for both sensing and stimulating implants, here we will describe the H-Tree electrode of Fig. 43.3 in detail. Our aim is for neurons to adhere in large numbers to the textured VACNT branches and for glial cells to primarily cover the smooth surface of the SiO₂ gaps between them. Because of this cell “herding,” the glia will be prevented from hindering neuron–electrode interactions but will be sufficiently close to provide the trophic and metabolic support necessary for the nearby neurons to maintain health and to transmit electrical signals. The electrode branches will take on the role of the physical scaffold normally provided by the glial cells when supporting neurons, ensuring neuron-rich electrodes that maximize stimulation.

Building on a broad range of CNT compatibility studies (including *in vitro* [36, 63], rat *in vivo* [58], and human ECG tests [64]), we focus on *in vitro* experiments due to the controlled environment in which fluorescence microscopy can be used to examine cell behavior and cell–electrode interactions as they evolve over 17 days *in vitro* (DIV). In our experiments, implants are lowered into a retinal co-culture of neurons and glia and imaged as the cells interact with the electrodes [43]. To prepare the culture, the retina is first removed from the outer epithelium of neonatal mice and placed in a culture medium containing enzymes, which is then mechanically agitated to dissociate the retinal cells. Subsequent staining allows the various retinal cells to be analyzed separately. DAPI (blue) fixes to the cell nuclei (allowing a total cell count), GFAP (green) attaches to glial cells, and β -tubulin III (red) binds to neuron microtubules. Based on our long-term goal of neuron stimulation and the fact that neuronal processes have a high density of stimulation sites, we focus our quantitative analysis of the neurons on the density of their processes (i.e. total length of the dendrites and axons within a given surface area). Based on their role of promoting neuron homeostasis and survival, we focus the glial analysis on their

Fig. 43.9 False-colored SEM image highlighting cell bodies (blue) and process (orange) behavior for textured VACNT and smooth SiO₂ regions. The scale bar is 50 μ m



surface coverage density (referred to hereafter as “coverage,” i.e. the surface area covered by glia normalized to the total area available).

Figure 43.9 employs false coloring of an SEM image to highlight some of the basic properties of the neurons as they interact with the textured VACNT and smooth SiO₂ surfaces. Somas (colored blue) join to form clusters on both surfaces, and these are connected via processes (orange) to form a network that spans both surfaces. Figure 43.10 shows representative fluorescence microscopy images of fractal electrodes along with Euclidean rows and grids to highlight common neuron and glial behaviors across various electrode shapes. Whereas glial cells are observed in the gaps, the VACNT electrodes are sufficiently high to act as barriers that are never traversed and therefore present the potential to guide glial coverage across the SiO₂ surface. On the rare occasions that they attach to the VACNT surfaces, they exhibit a more branched morphology than in the gaps. On the even rarer occasions that they coincide with 90° corners in the VACNT pattern, they are not restricted by these turns. In terms of neuronal growth, their processes are generally considerably longer and form more complex networks on the VACNTs. The relatively simple networks on the SiO₂ surfaces feature fewer but larger clusters. The neuronal processes tend to follow the top and the bottom edges of the electrodes upon reaching them and are able to climb up or down the sidewalls to connect cell clusters that exist on both surfaces.

Figure 43.11 provides more detailed observations of the cell responses to the fractal electrodes. This is facilitated by categorizing the gaps into three regions based on the cell behavior in these regions. The columns show example images of the neurons and glia, along with schematic representations immediately below these images: the electrode region (Fig. 43.11a, e, i-1, and j-1), the “boundary” region (Fig. 43.11b, f, i-2, and j-2), the “cluster” region (Fig. 43.11c, g, i-3, and j-3), and the “desert” region (Fig. 43.11d, h, i-4, and j-4). The deserts are furthest away from the electrodes and feature a few individual neurons and small clusters with weak processes, along with a scattering of glial cells. Nearer to the electrodes, neurons aggregate into larger clusters physically connected to each other by bundles of processes and accompanied by significant numbers of glia. These are labeled

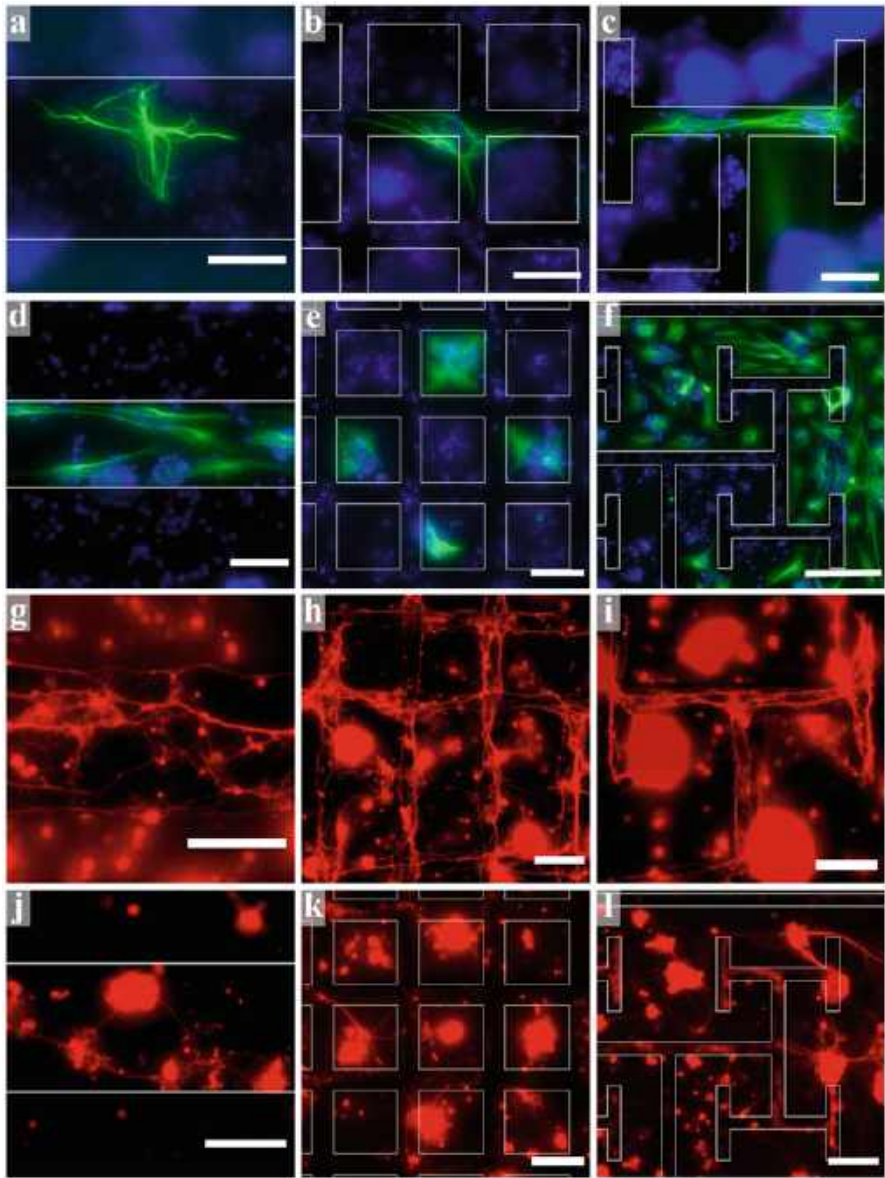


Fig. 43.10 Representative examples of fluorescence images of retinal cells interacting with the row, grid, and fractal electrodes at 17 DIV (green = GFAP labeled glia; red = β -tubulin III labeled neurons; blue = DAPI labeled nuclei). Glia on the VACNT surfaces of the (a) row, (b) grid, and (c) fractal electrodes. Glia on the SiO₂ surfaces of the (d) row, (e) grid, and (f) fractal electrodes. Neuron processes following the VACNT patterns of the (g) row, (h) grid, and (i) fractal electrodes. Neuron clusters and processes on the SiO₂ surfaces of (j) row, (k) grid, and (l) fractal electrodes. Electrode edges are highlighted with white lines except for panels (g), (h), and (i) which concentrate on the behavior of processes along the edges because the lines would have obscured these processes. Scale bars are 50 μ m in (a)–(e) and (g)–(k) and 100 μ m in (f) and (l)

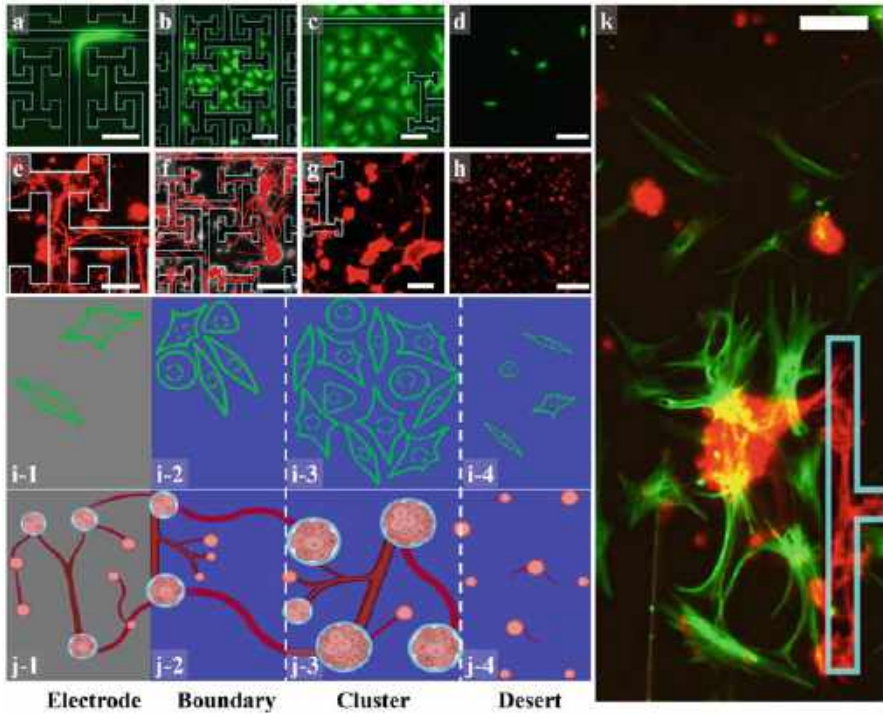


Fig. 43.11 Examples of fluorescence images of retinal cells interacting with the fractal electrodes at 17 DIV (green = GFAP labeled glia; red = β -tubulin III labeled neurons). (a) The rare occurrence of glia following the 90° turn of a 2–6 electrode branch. (b) Glial coverage in the gap of a 2–6 electrode. (c) Glial coverage in the gap of a 1.1–4 electrode close to its branches. (d) Individual glia in a desert region away from the branches of a 1.1–4 electrode. (e) Neurons and their processes on a 2–6 electrode’s branches. (f) Neuron clusters and processes in a boundary region interacting with the neurons on the nearby branches of a 2–6 electrode. Neuronal processes were semi-automatically traced using the Fiji simple neurite tracer and were false-colored. (g) Neuron clusters and processes forming a cluster neural network in the gaps of a 1.1–4 electrode. (h) Individual neurons in a desert region of a 1.1–4 electrode far from the branches. (i) and (j) Schematic of the glial and neural network regions. (i-1) and (j-1) The electrode with few glial cells and multiple processes connecting individual neurons and small- to medium-sized clusters. (i-2) and (j-2) show the “boundary” region featuring small to medium glial coverage regions and clusters connecting to each other and to neurons on the electrodes using multiple processes. (i-3) and (j-3) show the “cluster” region featuring larger glial coverage and clusters with bundles of processes connecting them. (i-4) and (j-4) show the “desert” region furthest from electrodes featuring very few glial cells, mostly individual neurons and very few processes. (k) Merged fluorescence image of glia and neurons on a 2–4 electrode showing all the different regions. Scale bars on (a)–(c), (f), and (g) are 100 μ m, on (d) and (h) are 200 μ m, and on (e) and (k) are 50 μ m. The electrode edges are highlighted in cyan in (a)–(c), (e)–(g), and (k). Schematic panels were created in BioRender. (Figure taken from Ref. [41])

as cluster regions in recognition of their typically larger clusters compared to the other regions. Many of these cluster networks are connected to neurons on the electrodes via the boundary regions, which form in some places along the electrode–gap interface. These boundary regions are composed of small- to medium-sized clusters and accompanied by occasional glial coverage. Figure 43.11k captures these various behaviors in one wide field of view and emphasizes that the cell–electrode interactions are far more subtle than the term “herding” would suggest.

The formation of these regions can be understood by recognizing the importance of surface adhesion for neuron development and their ultimate survival. In particular, neurons can aggregate into clusters reminiscent of small-world networks – so named because each node is connected to all other nodes through a small number of connecting neuronal processes, a strategy that can maximize signal transmission efficiencies [65, 66]. Initially, our neurons extend their processes in search of neighboring cells and reach a maximal complexity state featuring large numbers of nodes (individual cells and clusters) and links (neuronal processes) between them [43]. An analysis of 3, 7, and 17 DIV suggests that this peak complexity occurs around 7 DIV for our networks [43]. The network then starts to optimize as it shifts dominance from mostly neuron–substrate to neuron–neuron interaction forces also playing a large role [67]. The nodes then decrease in number as the largest clusters absorb their smaller neighbors. The links connecting them also decrease, with some processes joining together to form bundles [68] and others pruned [69]. This fine-tuning results in cluster regions displaying small-world-like networks. In contrast, neuron processes close to the electrode branches experience strong cell–VACNT adhesion forces that compete with the neuron–neuron forces, slowing down cluster formation and resulting in boundary regions. Neurons anchored to the VACNT surfaces secrete chemical signals [70]. This encourages strong interaction between neurons on the VACNTs and in the nearby gaps, leaving the gaps furthest away from electrodes to form deserts almost devoid of neurons. These developments are supported by glial cells which proliferate through cell division and growth, following chemical cues [70] that increase their surface coverage close to the neuron-rich cluster and boundary regions [71, 72]. This glial coverage supports not only neuronal survival and process development but also migration along glial fibers [73, 74] toward the electrodes.

To consider how the fractal properties of the electrodes influence the cell growth and assembly behavior (e.g. glial cell division and neuron process bundling and pruning), we introduce geometric parameters that relate these behaviors to the fractal geometry of their environment [43]. The fractal electrode design integrates two sets of related, multi-scaled patterns – the branches and the gaps – and our investigations show that both impact the cell organization favorably. The repeating patterns of the branches build long edges that interface with the gaps, and their interpenetrating nature is further amplified by their meandering character. We quantify these two branch characteristics by their total edge length E_n and Tortuosity T . We quantify the gaps by their proximities to the branches (P is the reciprocal of the distance between each gap pixel and its nearest branch pixel, averaged across all pixel locations in the gap) and various measures of gap areas. For example, A_c is the

maximum connected area of the gaps and characterizes the “openness” of the gap patterns for glial cells to cover before being blocked by intruding electrode branches [43].

Fractals with high D and m provide large E_n , T , and P but low A_c . These characteristics are predicted to encourage boundary regions and reduce desert regions, since there are no large gaps far away from branches. In contrast, fractals with low D and m generate low E_n , T , and P values but high A_c . These fractals are expected to minimize boundary regions, and their vast empty gaps will encourage deserts. Forming between the boundary and desert regions, we expect the growth of cluster regions to be encouraged for mid- D and m electrodes. Taken together, these effects suggest that fractals with mid to high D with four to five repeating levels will promote the most favorable cell interactions. They will enhance glial coverage inside the multi-scaled gaps without restricting the glia and will also prevent the creation of deserts. These glia will contribute to fuel the formation of the neural networks in the cluster regions. The large VACNT edge length of these fractals in close proximity to the SiO_2 gaps enhances the growth of neuronal processes in the boundary region connecting the cluster neural networks to those on the VACNT branches. These predictions are confirmed by qualitative inspection of the fluorescence images coupled with a detailed experimental analysis of the dependence of the cell measurements (G_{Si} , G_{CNT} , N_{Si} , and N_{CNT} – defined as glial covering area or neuron process length on each surface divided by the total area of that surface) on the fractal parameters [43].

Within this model, the advantage of H-Tree fractals over Euclidean rows and grids lies in the fact that they provide a connected electrode with an abundance of edges for neurons to follow while also allowing glial cells to cover the nearby interconnected, multi-scaled gaps. For example, Fig. 43.12 presents the balance of electrode proximity with gap openness by plotting P vs A_c for the fractal and Euclidean patterns. The high P and low A_c values of the Euclidean electrodes studied in our experiments (represented by the filled red and green symbols) are predicted to be dominated by boundary behavior. However, if Euclidean electrodes with larger gaps had been fabricated to match the fractal A_c values, their low P values are predicted to be dominated by deserts. In contrast, the fractals achieve relatively high P and A_c values simultaneously. In terms of the optimal balance of P and A_c , we note that the cluster-dominated fractals (i.e. mid to high D with four to five repeating levels) lie in the middle region of blue data.

To examine how these cell responses translate to the electrodes’ ability to “herd” (i.e. for neuron processes to adhere predominantly to the electrodes and in contrast for the glia to accumulate in the gaps), we introduce key quantitative measures to build on these qualitative observations. We define N as the ratio of the total neuron process length on the electrode (N_{CNT}) to that on the combination of the electrode and gap surfaces (i.e. $N_{\text{CNT}} + N_{\text{Si}}$), where N_{CNT} and N_{Si} have been normalized relative to the surface areas of the electrode and SiO_2 , respectively [43]. Similarly, we introduce G as the ratio of the glial coverage area in the gaps (G_{Si}) to that on the electrode and gap surfaces combined (i.e. $G_{\text{CNT}} + G_{\text{Si}}$), where G_{CNT} and G_{Si} have also been normalized to the surface areas of the electrode and SiO_2 ,

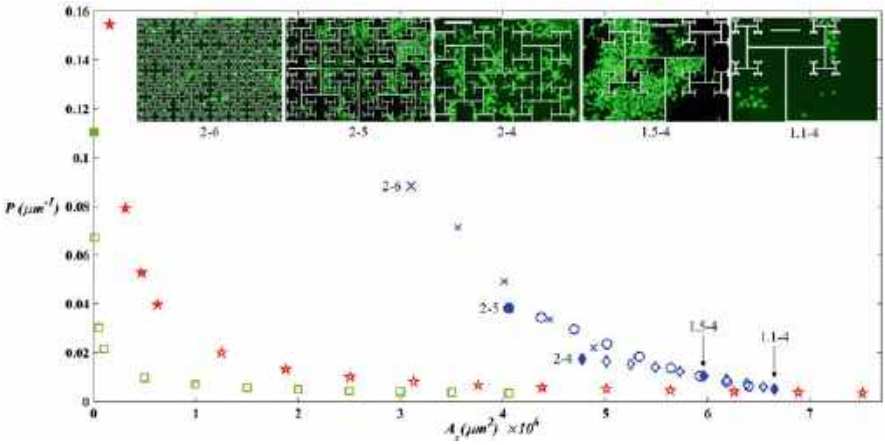


Fig. 43.12 Simulation plots of mean proximity, P , against the connected area, A_c , for fractal (blue), grid (green), and row (red) electrodes. D and m are varied for the fractals, and the W_{Si} values are varied for the rows and the grids. In each case, the filled symbols represent electrodes studied experimentally. The insets show fluorescence microscopy images of glia in the SiO_2 gaps of the fractal electrodes. Scale bars are $500\ \mu m$

respectively [43]. Adopting these measures, N and G values greater than 0.5 indicate successful guiding of neuronal processes and glial cells to the desired VACNT and SiO_2 surfaces, respectively. We also introduce the multiplication parameter GN to quantify the combined herding power.

Figure 43.13a shows a scatterplot of N vs G along with the equivalent binned data in Fig. 43.13b. Here, we group the electrodes into their three geometric categories – fractals, grids, and rows – and note that a more detailed analysis of the dependence of these categories on their parameters (e.g. D , m , W_{Si} , etc.) is presented elsewhere [43, 75]. In Fig. 43.13c–e, we also plot histograms of the percentage of electrodes within each electrode category that have given G , N , and GN values. In terms of overall herding, the fractal electrodes deliver the best GN performance (see Ref. 41 for discussions of statistical analysis). This is as expected. For example, cluster-dominated fractals generate both low N_{Si} and large G_{Si} values. In contrast, the grids have the lowest G_{Si} values because their disconnected gaps prevent the expansion of glial coverage.

Intriguingly, inspection of N reveals that the grids perform the best in terms of neuron herding. This can be understood by returning to Fig. 43.13a and b, where the dashed vertical line represents a threshold G_T in glial herding at $G \sim 0.95$, beyond which no Euclidean electrodes are observed. The superior neuron herding of the grids over the fractals is in part due to a collapse of the fractal N_{CNT} value in the high ($G > G_T$) regime. This is likely due to changes in G . Notably, in addition to some fractals supporting G_{Si} values larger than those reached in the low regime, G_{CNT} also drops when moving to the high regime. This results in G_{CNT} values significantly lower than those of the grids. It is possible that a small presence of glia

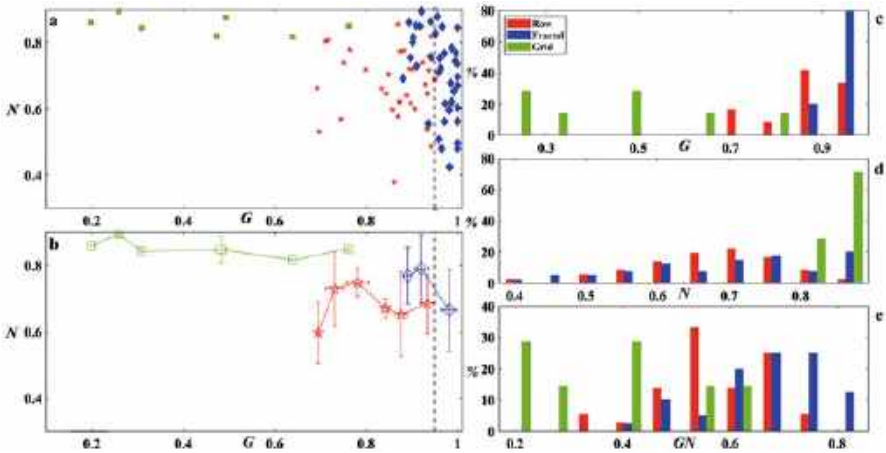
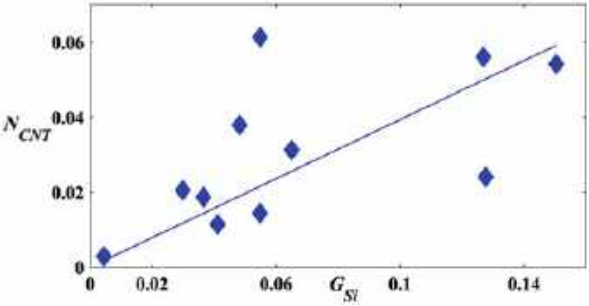


Fig. 43.13 (a) Scatterplot of N (neuron herding) vs G (glial herding) at 17 DIV for the row (red), fractal (blue), and grid (green) electrodes where each data point represents one electrode. (b) Binned average of the N vs G plot at 17 DIV for the row, fractal, and grid electrodes. In both cases, we note data that are statistical outliers have been excluded from the plots. Histograms of the percentage of electrodes with a given (c) G , (d) N , and (e) GN values at 17 DIV for the row, fractal, and grid electrodes

Fig. 43.14 N_{CNT} vs G_{Si} for the 2-5 fractals



is required on the VACNT surface to support neurons and the high regime fractals fall below this minimum level. The low G_{CNT} might result from the sparse cross sections of some of the fractals, and this could be addressed by fabricating future H-Trees featuring wider branches.

A central question to herding concerns the amount of glial coverage needed in the gaps to keep neurons healthy and functional on the VACNT electrodes. In order to study the correlation between the glia inside the SiO_2 gaps and neurons on the VACNT electrodes, we plot N_{CNT} vs G_{Si} in Fig. 43.14 for the 2-5 fractal electrodes. By comparing variations within the fractal electrode, neuron process growth can be taken as an indicator of health because all other geometric factors are constant. N_{CNT} increases with G_{Si} with the shown linear fit described by an R^2 of 0.63. We use this fit to highlight the following key observations. Firstly, the data trend suggests that an

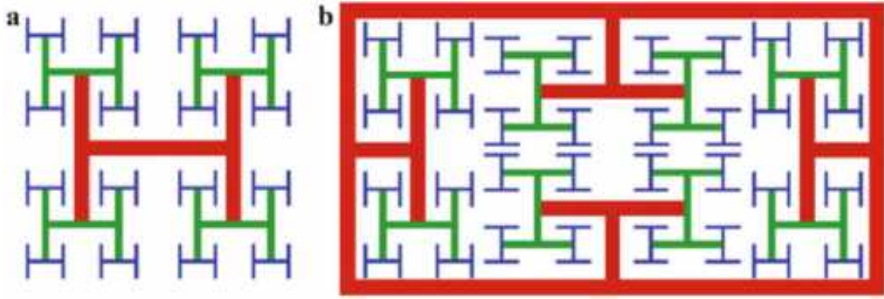


Fig. 43.15 (a) H-Tree fractal electrode with three magnification levels. (b) Optimal electrode design based on the cut and rotated parts of the electrode shown in (a). The whole design is bounded with the red rectangle to ensure electrode connectivity

absence of glia (i.e. $G_{Si} = 0$) impedes the growth of neuron processes substantially. However, the data does not reveal a distinct lower limit (i.e. a G_{Si} value below which N_{CNT} falls to zero). This is backed up by qualitative observations showing that there are some VACNT regions that support processes even in the absence of nearby glia. Secondly, when variations in G_{Si} within the culture result in more glia, this increased presence promotes neuronal growth. Our 17 DIV fractal system operates within a regime in which there is no upper limit – the more glia the better.

The absence of an upper limit suggests that refined electrode designs that increase G_{Si} will further boost neuron health and perhaps even remove the need for the small presence of glia on the VACNT surface. As an example of refinement, rather than branches spreading out from the origin as done in the H-Tree used in our study (Fig. 43.15a), the branches shown in Fig. 43.15b spread from a boundary inward. Removing branches from the central section of the design creates a completely connected SiO_2 gap region while keeping the proximity of the gaps to the electrodes roughly comparable to the original fractal electrode. We predict that optimized designs will ensure that a larger number of neuronal processes reside within the stimulating fields generated when the electrode branches are electrically biased. The stimulated processes will connect to neurons in the gaps that resemble small-world networks, improving the electrode's ability to stimulate the surrounding retinal neurons more efficiently. Furthermore, in contrast to implants that use anti-inflammatory drugs to inhibit glial scarring on their electrode [76], glial cells will be confined to the gaps between our electrodes where their proximity to the neurons will ensure the neurons' health, prolonging the stability and functionality of the retina-implant interface.

43.5 Conclusions

In this chapter, we have discussed some of the key favorable properties of fractal interconnects designed to interact with neural networks. The term “interconnect” emphasizes the multi-functional character of this interface. Accordingly, we have considered their mechanical, optical, and electrical properties along with their ability to promote positive cell interactions. These properties are applicable to interconnects used to stimulate and sense signals in the neural network. Future experiments will be aimed at confirming the enhanced performance of fractal designs over conventional Euclidean designs – in particular, quantifying their electrical and biophilic advantages. Our approach of integrating computer modeling with *in vitro* studies allowed controlled investigations to establish the fundamental performances of these inherently complex systems. However, a vital step forward will be to focus on transferring these studies to the *in vivo* environment.

Fractal implants have much to offer. Brain implants could be used to address a number of pathological conditions including Parkinson’s disease. In addition, they could also radically improve other stimulation techniques to improve learning and cognition, visual perception, working memory, and motor control [77]. Retinal implants could be used to regain vision lost due to diseases such as macular degeneration or retinitis pigmentosa. The long-term potential impact of fractal interconnects could also stretch well beyond improving implants designed to interact with the human brain, retina, and limbs. In addition to stimulating and sensing naturally occurring neurons, they could be applied to retinal STEM cells in both *in vitro* and *in vivo* investigations.

Complementing the above medical applications, the central question asked in this chapter – what do we gain by building interconnects that match the fractal properties of the cells they interact with – also addresses the basic science of cell behavior and electricity. Fueled by technology, the prospect of fractal implants demonstrates how far electronics has come from the days of Franklin harnessing electricity from the clouds, and Le Roy and Galvani applying it to the human body.

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Chapter 44

Fractal Resonance: Can Fractal Geometry Be Used to Optimize the Connectivity of Neurons to Artificial Implants?



C. Rowland, S. Moslehi, J. H. Smith, B. Harland, J. Dalrymple-Alford, and R. P. Taylor

Abstract In parallel to medical applications, exploring how neurons interact with the artificial interface of implants in the human body can be used to learn about their fundamental behavior. For both fundamental and applied research, it is important to determine the conditions that encourage neurons to maintain their natural behavior during these interactions. Whereas previous biocompatibility studies have focused on the material properties of the neuron–implant interface, here we discuss the concept of fractal resonance – the possibility that favorable connectivity properties might emerge by matching the fractal geometry of the implant surface to that of the neurons.

To investigate fractal resonance, we first determine the degree to which neurons are fractal and the impact of this fractality on their functionality. By analyzing three-dimensional images of rat hippocampal neurons, we find that the way their dendrites fork and weave through space is important for generating their fractal-like behavior. By modeling variations in neuron connectivity along with the associated energetic and material costs, we highlight how the neurons’ fractal dimension optimizes these constraints. To simulate neuron interactions with implant interfaces, we distort the neuron models away from their natural form by modifying the dendrites’ fork and weaving patterns. We find that small deviations can induce large changes in fractal dimension, causing the balance between connectivity and cost to deteriorate rapidly. We propose that implant surfaces should be patterned to match the fractal dimension of the neurons, allowing them to maintain their natural functionality as they interact with the implant.

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Keywords Biocompatibility · Bio-inspiration · Electronics · Fractals · Implants · Neurons

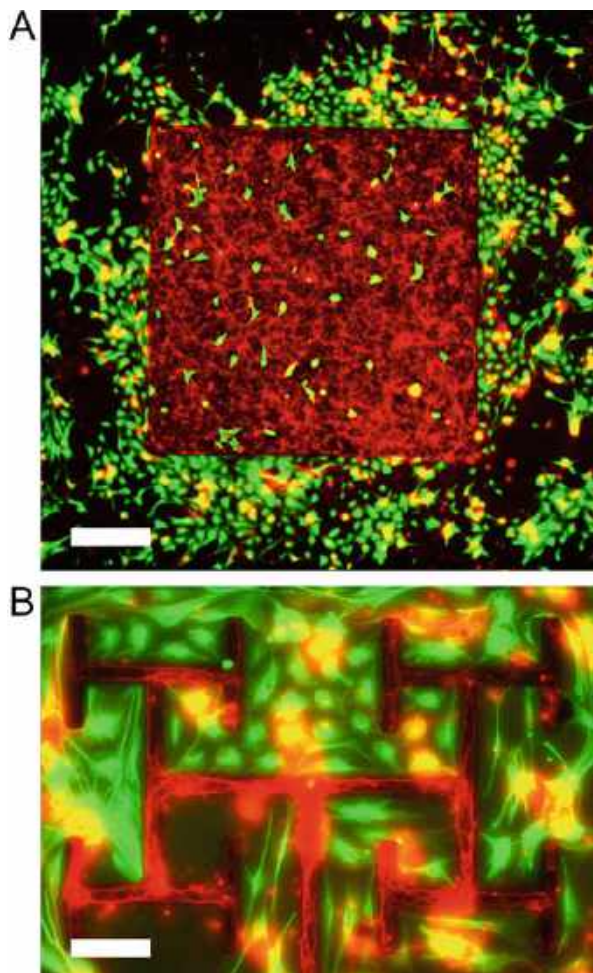
44.1 Introduction to Fractal Resonance

Electronic implants in the human body have been the focus of broad interdisciplinary research. Functions include electrodes that stimulate electrical signals in the body's neurons or that sense the locations of these signals as they pass through the neuronal networks that serve as the body's electrical wiring. For both applications, the implant's operation must be sustainable in terms of toxicity, durability, and efficiency. The factors that ensure biocompatibility can be pictured as three legs of a stool, with each leg playing a vital role for stability. The two legs that have received most attention are the chemical environment and physical textures established by the implant surface as it interacts with the cells. In this chapter, we will focus on the third leg – the electrode's shape – and explore the potential of bio-inspired fractal designs. The repeating patterns of fractals are prevalent in nature and, in particular, define the geometric properties of the dendritic branches of our neuronal networks.

Figure 44.1a shows an example of retinal cells interacting with a square-shaped electrode composed of a uniform layer of vertically aligned carbon nanotubes (VACNTs) grown to heights of approximately 25 μm from a metallic catalyst layer deposited on a smooth silicon dioxide (SiO_2) substrate. Imaged using an in vitro co-culture of neurons and glial cells, the former fluoresce red while the latter fluoresce green. This facilitates an investigation of their differing spatial distributions on the VACNT and SiO_2 surfaces [1]. In addition to chemical compatibility, neuron growth is promoted on the VACNT surface [2] by a texture thought to resemble characteristics of the extracellular matrix [3–5]. In contrast, the glia grow and cover the smooth surfaces of the surrounding SiO_2 . The advantage of shaping the electrode is highlighted in Fig. 44.1b. The square shape has been replaced by an H-Tree, a well-known fractal pattern in which H-shaped branches repeat at different scales. Whereas the multi-scaled connected gaps between the branches encourage glial coverage, the long, inter-penetrating character of the branches ensures a close proximity between the glia and the neurons growing on the branches. This cell “herding” is highly beneficial because the glia act as the neurons' life support system [6, 7]. The glia are therefore prevented from hindering neuron–electrode interactions but are sufficiently close to provide trophic and metabolic support. Described in detail in Chap. 43 [8], this fractal patterning also provides electrical, optical, and mechanical advantages for implant operations.

Having successfully herded neurons onto the electrode's fractal branches, are there further geometric strategies for promoting their growth and ability to transmit electrical signals? Notably, although the H-Tree serves as a simple demonstration of fractal scaling, its exact repetition of straight branches and ninety-degree turns contrasts dramatically with the natural fractal form of neurons. We, therefore, hypothesize that electrode branches that match the precise fractal characteristics

Fig. 44.1 (a) A fluorescence image of retinal cells interacting with a VACNT electrode grown on a SiO_2 substrate measured at 17 days in vitro (green GFAP labeled glia, red β -tubulin III labeled neurons). (b) An analogous image of a zoom-in section of an H-Tree fractal electrode with a branch width of $20\ \mu\text{m}$. The scale bars in (a) and (b) are $500\ \mu\text{m}$ and $100\ \mu\text{m}$, respectively. Cell imaging is performed in collaboration with M.-T. Perez (Lund University, Sweden)



of the neurons they interact with will allow the neurons to flourish by closely resembling their natural behavior. This “fractal resonance” marks a shift from biocompatibility to biophilia. Rather than simply tolerating the presence of the electrodes, the goal is for the neurons to be attracted to them.

A diverse range of previous experiments have demonstrated the impact of patterning on neuron process (dendrites and axons) behavior [9]. Patterning strategies have varied from curving the surface [10, 11] to introducing shaped regions of surface texture such as parallel grooves [12–16] and zig-zag microtracks [17, 18]. Whereas processes prefer the direction of minimum curvature [10, 11], the directional guidance induced by patterned textures depends on pattern width and depth, as well as neuron species and their stage of development. The sizes of the textured patterns in these studies span from sub-micron to micron-scale features

(with widths, spacing, and depths ranging across the studies from 100 nm–350 μm , 500 nm–1000 μm , and 14 nm–69 μm) fabricated on materials such as quartz, PDMS, and PMMA-covered silicon to name a few. As an example, *Xenopus* spinal neuron processes are found to grow parallel to grooves with depths ranging from 14 to 100 nm and widths of 1, 2, and 4 μm [13]. In another study, the percentage of embryonic rat hippocampal neuron processes growing parallel to grooves increases with groove depth [19]. Intriguingly, whereas hippocampal neuron processes again grow parallel to wide, deep grooves, they grow perpendicular to narrow, shallow ones [13].

The above experiments focus mainly on the extent to which neurons can be manipulated by guiding the growth direction of their processes. In contrast, our bio-inspired approach aims to determine the conditions that encourage neurons to maintain their natural growth during these interactions. In parallel to medical applications, this exploration of how neurons interact with the interface of artificial implants can be used to learn about their fundamental behavior. Our own experiments confirm that growth can be manipulated by forcing process growth along straight lines and through sharp turns [1, 18, 20]. For example, neurons follow the ninety-degree corners of the H-Tree branches in Fig. 44.1. Furthermore, Fig. 44.2a shows a zoom-in on the boundary between the VACNT and SiO₂ regions of one of our electrodes. This reveals neuron processes following the straight boundary for long distances.

However, away from the boundary, the unrestricted growth resumes a more natural weaving behavior. Similarly, in Fig. 44.2b and c, neurons follow a textured region patterned into a narrow zig-zag line. However, a close inspection of Fig. 44.2b and c highlights processes leaving the lines to resume their natural growth pattern.

Significantly, previous research on zig-zag microtracks reveals longer processes when compared to straight tracks [17]. Whereas the zig-zag's increased surface area was suggested as the cause of the accelerated growth, does the introduction of the corners into the pattern also play a role? If we speculate that the corners act as a crude re-introduction of the processes' natural weave into the unnatural straight line, then the next logical step would be to incorporate more corners at multiple scales. This concept drives the thought experiment of Fig. 44.3 which starts with the line (top), then introduces triangular corners and next repeats these triangles at multiple scales to generate the Koch curve (middle). Whereas the Koch curve is a well-known "exact" fractal that repeats its pattern precisely, natural fractals such as neurons mix randomness into this repetition such that only their statistical qualities repeat. Therefore, we do the same to generate the statistical version of the Koch Curve. Finally, because processes fork rather than follow one long line, we cut and paste to re-arrange the fractal line into a fractal forking pattern.

Although following this evolution from Euclidean to fractal geometry would move our electrodes closer to the shapes of neurons, is this close enough to achieve fractal resonance? To answer this central question, we will have to answer more fundamental questions – to what extent are neurons fractal, what is the geometric origin of this fractality, and why do neurons adopt this fractal geometry?

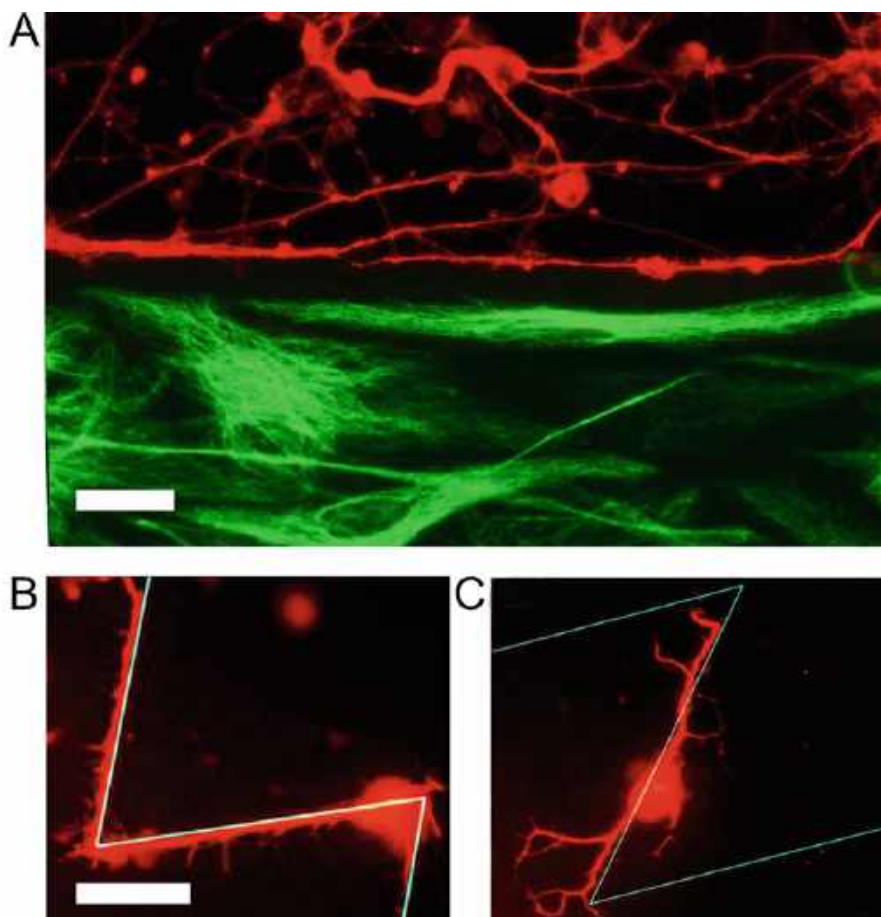
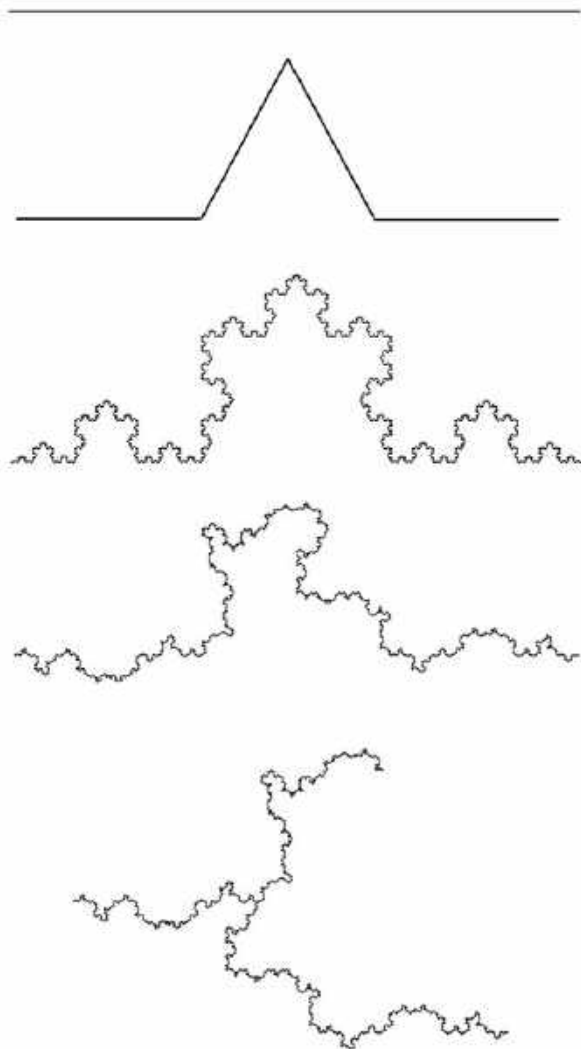


Fig. 44.2 (a) Fluorescence image of neurons (red) on the VACNT (top) and glia (green) on SiO_2 (bottom) at 17 days in vitro demonstrating neuron processes following the straight VACNT– SiO_2 boundary. (b, c) Fluorescence images of neurons following textured patterns (indicated by the blue lines) of SU8 zig-zag lines deposited on a SiO_2 surface. These images were recorded at 3 days in vitro before glia start to cover the surface. The scale bars in (a) and (b) are both $25\ \mu\text{m}$. The scale bar in (b) also applies to (c). Cell imaging is performed in collaboration with M.-T. Perez (Lund University, Sweden)

While answers to the first two questions will help realize electrode designs that achieve fractal resonance, the third addresses the positive consequences of the resonance. We speculate that resonance will allow neurons to maintain their natural functions during electrode interactions. Given that fractal growth is related to this functionality, we expect resonance will also encourage an enhanced growth of this functional neural system.

Although many previous neuron investigations have quantified the scaling properties of neurons, this was done mainly to categorize neuron morphologies

Fig. 44.3 Evolution from a straight line celebrating Euclidean geometry (top) to forked branches celebrating fractal geometry (bottom). See text for details



[21–35] rather than quantify how neurons benefit from their fractal scaling. Why does the body rely on fractal neurons rather than, for example, the Euclidean wires prevalent in everyday electronics? Neurons form immense networks within the mammalian brain, with individual neurons exploiting up to 60,000 connections in the hippocampus alone [36]. In addition to their connections within the brain, they also connect to the retina’s photoreceptors allowing people to see and connect to the limbs allowing people to move and feel. Given this central role as the body’s wiring, we will focus on the importance of fractal scaling in establishing the connectivity between the neurons [37]. Previous analysis over small parts of

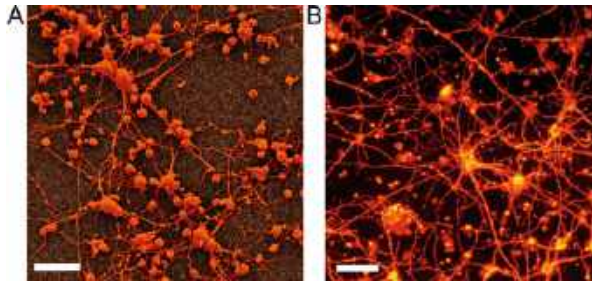


Fig. 44.4 False-colored scanning electron micrograph (a) and fluorescence image (b) of neurons growing on a VACNT surface. The images are taken after 7 days in vitro. The scale bars in (a) and (b) are 20 μm and 100 μm , respectively. Cell imaging is performed in collaboration with M.-T. Perez (Lund University, Sweden)

the neuron's dendritic arbor identified the scale invariance of patterns as one of the geometric factors used to balance connectivity with its maintenance costs [38, 39]. Their research built on Ramon y Cajal's wiring economy principle from a century earlier which proposed that neurons minimize their wiring costs [40]. These costs include metabolic expenditures [41, 42], wire volume [43–45], and signal attenuation and delay [46–48]. We will build on these principles further by showing that distortions away from the neuron's natural fractal form will result in a deterioration of the balance between function (connectivity) and the costs associated with building (mass) and maintaining (energy) the connectivity. Fractal resonance proposes that matching the electrode to this natural condition will allow neurons to operate efficiently while interacting with the implant.

Figure 44.4 employs two techniques (false-colored scanning electron microscopy and fluorescence imaging) to image the growth of neurons on the uniform, two-dimensional surface of CNT electrodes. Although restricted to the electrode's two-dimensional plane, the growth behavior is unrestricted within this plane. Therefore, one possibility for investigating fractal resonance would be to fabricate electrodes that copy these growth patterns. Neurons interacting with these electrodes would therefore follow the unrestricted growth behavior. Our hypothesis could be confirmed by quantifying the deterioration in neuron performance for electrode shapes that move away from this fractal resonance condition. Favorable performances could include enhanced adhesion, growth, health, and ultimately electrical stimulation. Intriguingly, the geometric framework of our hypothesis offers the possibility of a cleaner and more systematic demonstration ahead of these complex experiments. If the favorable functions of neurons originate from their geometry, then the networks of Fig. 44.4 could be converted into neuron models and the processes of these model neurons distorted mathematically to examine how these functions deteriorate.

In addition to offering an appealing strategy for examining neuron interactions with the surfaces of implants, there are broader implications for our approach. For example, mathematical distortions could be used to investigate pathological

deviations from neurons' natural fractal condition. Accordingly, in this chapter, we will consider the general case of three-dimensional neurons and later discuss the implications for interactions with an electrode. We will use confocal microscopy to image rat neurons located in the hippocampus CA1 region and convert these into models. We will show that, despite being named after trees, dendrites are considerably different in their scaling behavior. Whereas trees have famously been modeled using a fractal distribution of branch lengths [49, 50], the ways in which the multiple-length dendrites fork and weave through space are important for determining their fractal character.

We will demonstrate that fractal dimension, D , is a highly appropriate parameter for quantifying the dendritic patterns because it incorporates dendritic length, forking, and weaving in a holistic manner that directly reflects the neuron's fractal geometry. Serving as a measure of the ratio of fine to coarse-scale dendritic patterns, we will use D to directly map the competing functional and cost constraints. By investigating ~1600 distorted neuron models with modified dendrite length, forking, and weaving behavior, we propose that the neuron D values reflect a network cooperation that optimizes the balance between connectivity and cost constraints. The functions of neuron types that optimize at high D place a higher emphasis on connectivity, while low D neurons place a higher emphasis on cost. Remarkably, D captures this functional optimization even though the fractal scaling behavior occurs over a highly limited range of size scales. Electrodes that match the resonance condition for a particular neuron type will promote the growth of these neurons on the electrode and optimize their functional restraints. Extending this picture, we hypothesize that resonance will allow different neuron types to selectively grow on electrodes that most closely match their D values. Ultimately, this could allow different neuron types within a network to be directed to different parts of an implant.

44.2 The Geometric Origin of Neuron Fractality

Figure 44.5a shows a representative image obtained using confocal microscopy of CA1 pyramidal neurons in the coronal plane of the dorsal rat hippocampus. Axonal and dendritic arbors extend from neuron somas located in the stratum pyramidale (SP) of the CA1 region. Although the dendritic arbor features two component arbors (apical and basal), here we focus on the basal arbor. The arbor's complex branching patterns extend into the neighboring stratum oriens (SO) where they collect signals from the axons of other neurons. Figure 44.5b (lower inset) shows an example of three-dimensional reconstruction of an arbor. We define a dendritic branch as any path in the neuron's arbor that starts from the soma and ends at the tip of a dendrite (Fig. 44.5b – larger image). In our reconstruction, these branches are composed of a set of cylindrical segments that have a median width, W , and length, L_s , of 1.4 μm and 2.4 μm , respectively.

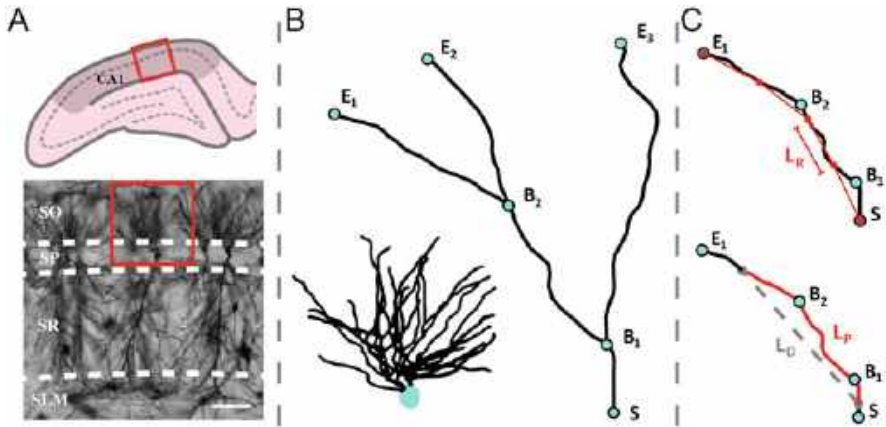


Fig. 44.5 (a-top) Schematic diagram of a coronal slice through the hippocampus at Bregma -4.52 mm showing a selected location (red box) within the hippocampal CA1 region (darkened area); the pyramidal layer is denoted by the dashed line. (a-bottom) An example confocal micrograph taken from the selected location highlighted in (a-top) spanning the oriens (SO), pyramidal (SP), radiatum (SR), and stratum lacunosum-moleculare (SLM) strata. The red box in the micrograph highlights the approximate region occupied by the basal arbor of a single neuron. (b) The lower inset shows an example reconstruction of a neuron's basal arbor with the neuron's soma colored in cyan and its dendrites in black. The larger image shows an example of the paths taken by a neuron's dendrites. S denotes the point where the dendrites initially extend out of the soma. B₁ and B₂ denote the points at which the dendrites bifurcate. E₁, E₂, and E₃ denote the end points of the dendrites. (c-top) A dendritic branch from (b) beginning at S and ending at E₁ showing four rulers of length L_R (red) used in the coastline scaling analysis. (c-bottom) The same branch (c-top) where a chosen branch section of path length L_P is highlighted in red and the Euclidean distance L_D separating the ends of this path is shown by the dashed gray line

In principle, each branch could extend into the SO layer following a perfectly straight line with dimension $D = 1$ or meander along a very winding trajectory that completely fills space with a dimension of $D = 3$. If the arbor instead features fractal dendrites, then each of these will be quantified by an intermediate D value lying between 1 and 3. Figure 44.5c introduces several key length scales that we will employ to measure D . Branches can be divided into a series of “rulers” – Euclidean straight lines of length L_R . The displacement length, L_D , represents the ruler length connecting two ends of a chosen section along a branch. In contrast, the path length of this section, L_P , is approximately the total length of all rulers spanning the section when the ruler is set to the smallest resolution (which approximates the median L_S value). The fork length, L , (not shown) is defined as the path length between two forks in the branches. As an indicator of arbor size, the maximum fork length, $L_{\max}$, varies between 109 and 352 μm across all neurons, with a median value for $L/L_{\max}$ of 0.24.

Because many mathematical fractals are generated by scaling L , we start by comparing the neurons' L behavior to that of H-Trees. Figure 44.6 shows the scaling relationship of N (the number of branches with a given $L/L_{\max}$) measured for a

$D = 1.4$ H-Tree (Fig. 44.6a, c, and e) and a typical arbor (Fig. 44.6b, d, and f). We assign the branch section levels such that $i = 1$ corresponds to branch sections emerging from the soma, $i = 2$ to branch sections emerging from the first forks, etc., with neurons featuring a median of 7 levels (other common level assignments such as the Strahler scheme [51] generate similar findings to those below). The H-Tree exhibits a well-defined reduction in $L/L_{\max}$ as i increases (Fig. 44.6c) which translates into a power law decrease in N as $L/L_{\max}$ increases (Fig. 44.6e). This power law behavior is expected since it generates the scale invariance of fractal geometry: the magnitude of the data line's gradient in Fig. 44.6e equals the H-Tree's D value of 1.4. In contrast, this behavior is notably absent for the neuron: in Fig. 44.6d, $L/L_{\max}$ does not exhibit a systemic reduction in i and, consequently, the Fig. 44.6f data does not follow a well-defined slope. This result demonstrates that the L distribution alone is insufficient for generating the fractal character of the branches – in addition to branch lengths, the angles that determine the forking behavior along with the way branches weave between the forks must also play a role.

44.3 Fractal Dimension of the Branches

A variety of scaling analyses have previously been applied to neurons [24, 27, 28, 30, 39, 52–56]. Here we consider two fractal methods that accommodate the dendrite length, forking, and weaving contributions to the branch dimension – a traditional method employed in the first demonstration of nature's fractality [57], and a novel method that examines the dendrites' tortuosity [58, 59], T , across multiple scales. We label their measured branch dimension as D_B . The high degree of agreement between the two methods in their measurement of D_B emphasizes the appropriateness of characterizing the scaling properties of dendrites with this integrated approach. In addition to providing confirmation of D_B , this second method also allows an examination of the relationship between D_B and the scaling behavior of T . By doing so, we unite traditional and novel approaches to understand neuron geometry.

We first employ a three-dimensional extension of a traditional method pioneered by Richardson [57] and then Mandelbrot [60] in their discovery of the fractal character of meandering coastlines. The 'coastline method' examines the branch at different resolutions through its employment of the rulers shown in the top panel of Fig. 44.5c. The branch is divided into a series of ruler lengths, and the branch's fractal scale invariance is then revealed through the power law dependence of N , the number of rulers needed to span the branch's entire length. The exponent of $N \propto L_R^{-D_{BC}}$ is labeled the branch's coastline fractal dimension, D_{BC} , and is extracted from the log-log scaling plot shown in Fig. 44.7a. The coarse- and fine-scale cutoffs of this fractal scaling are set by measurement limits. At the fine scale, we do not consider rulers smaller than $4 \mu\text{m}$ (which approaches the median value of L_s) because smaller rulers would start to detect the linear character of the cylindrical

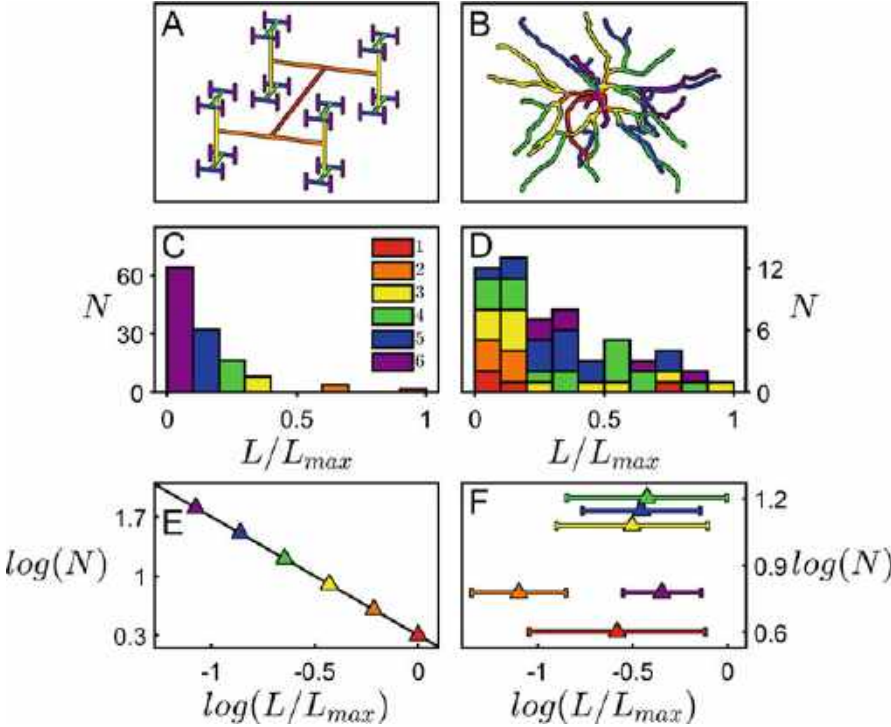


Fig. 44.6 (a) $D = 1.4$ H-Tree fractal with $W = 1 \mu\text{m}$ (a) and an example basal arbor with median $W = 1.4 \mu\text{m}$ (b). The branch level i is colored as follows: red (1st branch), orange (2nd), yellow (3rd), green (4th), blue (5th), and purple (6th). Histograms for an H-Tree (c) and neuron (d) plotting the number of branches N with a given value of L/L_{max} . Panels (e) and (f) show the analysis of (c) and (d) plotted in log-log space

segments rather than the fractal character of the meandering branches. At the course scale, we allow for rulers that span up to $40 \mu\text{m}$, which provides an order of magnitude scaling. Accordingly, branches with an L_E value less than $40 \mu\text{m}$ are excluded from our analysis.

Normalizing L_R using L_E (the largest possible ruler length, connecting the soma S to the branch end point E) allows scaling plots for branches with different lengths to be plotted on a common x -axis. Figure 44.7b demonstrates that all of the branches within a given arbor condense onto a single line, indicating that they are quantified by a common fractal dimension, D_{BCN} . To extract D_{BCN} , we focus the fit on the black data corresponding to the scaling range of L_R/L_E shared by all branches within the arbor (i.e., the region over which all of the individual plots overlap). The inset employs a histogram to compare the mean D_{BC} across all branches within the arbor, $\langle D_{BC} \rangle$, (1.031 as indicated by the cyan line) with D_{BCN} (1.032 as indicated by the red line). Figure 44.7c further demonstrates D_{BC} 's lack of dependence on branch length by plotting the values of all the individual branches across all the neurons

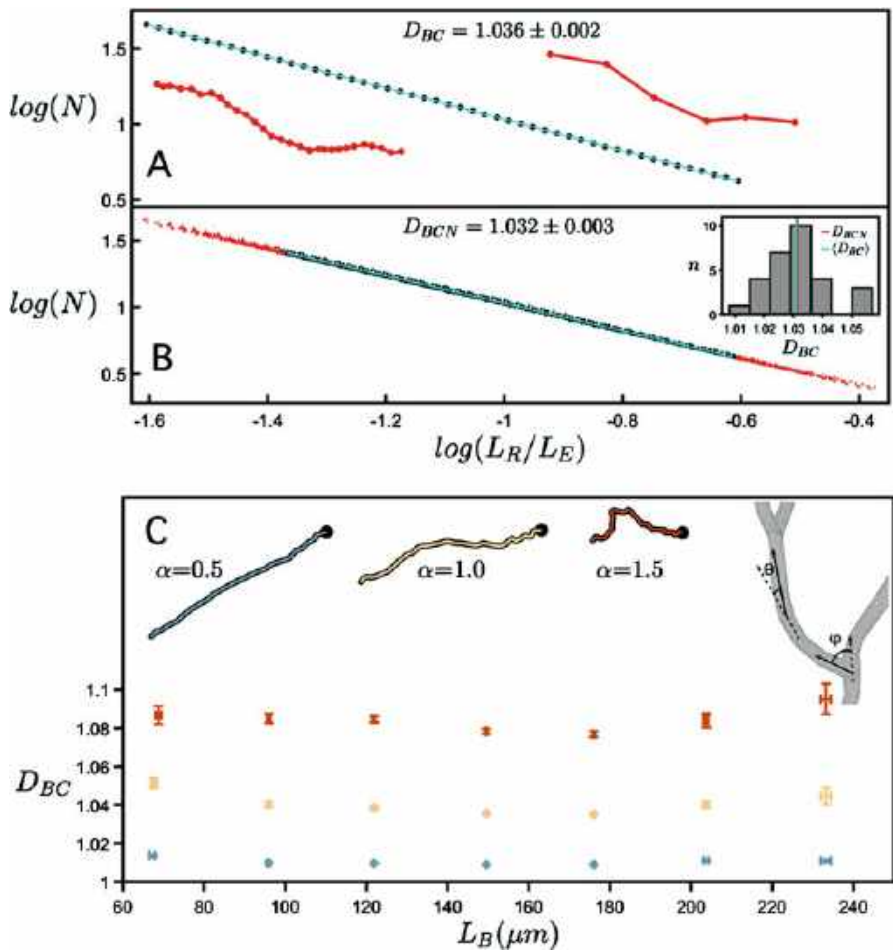


Fig. 44.7 (a) The coastline scaling plot of the number of rulers, N , spanning the branch versus the normalized ruler length, L_R/L_E , measured for a single branch within a selected neuron arbor. The red insets show examples of segmented versions of the branch corresponding to ruler lengths of 6.4 μm (left) and 34.7 μm (right). The slope of the line yields a coastline fractal dimension, D_{BC} , of 1.036 ± 0.002 . (b) The equivalent coastline scaling plot including all the branches within the selected neuron's arbor. The black data correspond to the scaling range of L_R/L_E shared by all branches within the arbor, whereas the red data correspond to the range in which some branches do not contribute and are accordingly removed when fitting the data. The slope of the line yields a normalized coastline fractal dimension, D_{BCN} , of 1.032 ± 0.003 . The inset at the right shows a histogram of the number of branches, n , of a given D_{BC} within the neuron's arbor. The vertical red and cyan lines correspond to D_{BC} and the mean coastline fractal dimension across all the branches within the neuron's arbor, $\langle D_{BC} \rangle$, respectively. (c) Coastline fractal dimension, D_{BC} , plotted against branch length L_B (measured in μm). The data, which have been binned along the x -axis, represent the mean values across all the branches of all the neurons within each bin. The error bars represent the standard error from the mean within each bin. The upper-right inset shows a zoom-in of a bifurcation in a neuron's dendrites and demonstrates how the fork angle, ϕ , and the weave angle, θ , are measured. The other insets show the path of a single neuron branch for three values of the angle multiplier, α , where the location of the neuron's soma is indicated by the black dot. The colors of the data shown in this plot correspond to the α values shown in the insets (see text for details)

examined. D_{BC} does not vary with L_B within nor across neurons, where L_B is the path length, L_P , measured along a whole branch.

All of the branches reveal surprisingly mild fractal characteristics quantified by low D_{BC} values close to those expected for straight lines. In Fig. 44.7c, we also mathematically manipulate the branch weave and forking angles (details of this technique are reported elsewhere [61]). The weave angles, θ , are defined as the angles between connecting segments along the branch (Fig. 44.7c – right inset). We define the forking angles, ϕ , as the first of the branch weave angles for any branch not emanating from the neuron's soma. Multiplying every θ and ϕ value by a common factor, α , changes the D_{BC} value as follows. Values of α higher than 1 increase the angles above their natural values and cause the neuron branches to curl up, causing D_{BC} to rise because the amount of fine structure in the branch's shape increases. Similarly, reducing α causes the branches to gradually straighten out, decreasing the amount of fine structure and causing D_{BC} to drop. Figure 44.7c includes a visual demonstration of this curling process. We set the range of α values (0.5 to 2 in steps of 0.25) to ensure that branches rarely intersect and penetrate, ensuring a physically reasonable condition.

Whereas the coastline method considers the entire length of the branch and examines how the branch properties change with measurement resolution, our second method considers the finest resolution and examines how the branch properties vary when investigating increasingly small sections of the branch. This second approach aligns with one of the traditional measures of tortuosity that quantifies the extent to which the meandering branch deviates from a straight trajectory. Several different tortuosity metrics have been used in previous studies of a variety of biological structures [39, 59, 62–65]. Due to its mathematical connection to fractal measurement, here we adopt the definition of tortuosity, T , as the ratio L_P to L_D (shown in Fig. 44.5c – bottom). Whereas L_P captures the fractal tortuosity of the branch, L_D represents the length of the straight, Euclidean line connecting the two ends of the chosen section along the branch. By measuring various path lengths and their corresponding displacement lengths along the branches of the neurons and averaging the resulting tortuosity across all of these branches, we are able to chart the relationship between T and L_P using the scaling plot of Fig. 44.8 (plotted over the same scaling range used to measure D_{BC}). We have previously shown that for fractal behavior, T is expected to follow the power law relationship with L_P revealed in Fig. 44.8, with the slope S of the log-log plot related to branch fractal dimension using $D_{BT} = 1/(1 - S)$ [66]. Accordingly, increasing α results in a steeper slope of the data line. The inset of Fig. 44.8 confirms the intuition that the T value averaged across the data line will increase with α .

Figure 44.9 plots the D_{BT} values extracted from Fig. 44.8 against the corresponding D_{BC} values (averaged over all of the branches across all neurons) and reveals a remarkable agreement for the two approaches: the black line indicates the expected relationship, $D_{BC} = D_{BT}$. The two measurement techniques provide confirmation of the branch fractal dimension, an important capability in light of the importance of distinguishing the branches mild (i.e. low fractal dimension) fractal behavior from the Euclidean behavior characterized by integer dimensionality. Due to the lower

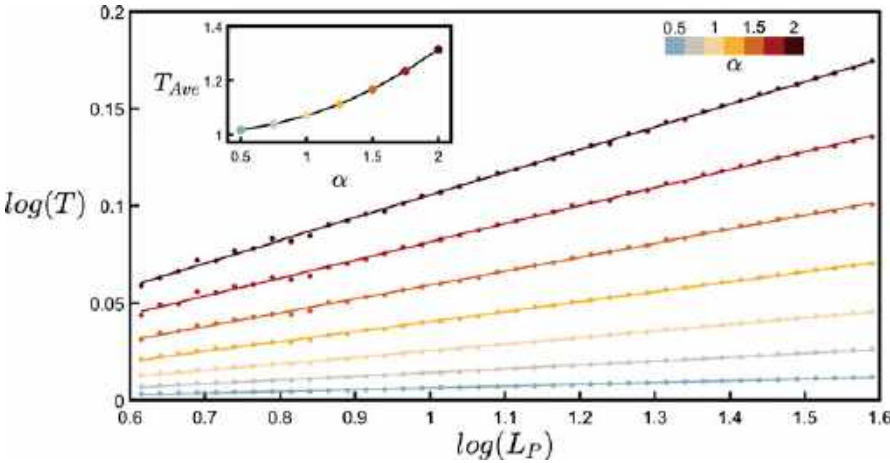


Fig. 44.8 Scaling plot of tortuosity, T , against path length, L_P , (measured in μm) for seven values of α as indicated by the upper-right color bar. The data shown represent binned averages of T across all possible paths within all of the branches across all neurons. The upper-left inset shows how the average value of tortuosity across the L_P range examined in the main plot, T_{Ave} , increases with α . (Figure taken from reference [66])

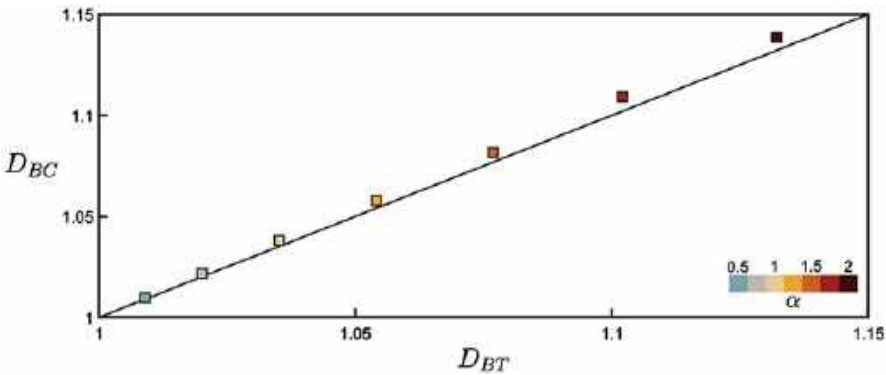


Fig. 44.9 Coastline fractal dimension, D_{BC} , plotted against tortuosity fractal dimension, D_{BT} , for seven values of α . The lower-right color bar indicates the α value of the corresponding data. The black line indicates $D_{BC} = D_{BT}$

scatter observed in the scaling plots of the individual branches [66], we will use D_{BC} rather than D_{BT} values later in this chapter.

44.4 Fractal Dimension of the Arbor

Having established the fractal behavior of the individual branches using two techniques, we now similarly apply two fractal techniques to quantify the arbor properties. As the individual branches spread out in space, the resulting arbor properties depend on two embedded fractal patterns – the branches and the gaps forming between them (as highlighted by Figs. 44.5b and 44.6b). The first method relates to Sholl analysis, which is a traditional approach to assessing the complex interplay of the branches in neuron arbors.

Sholl analysis can be performed by counting the number of intersections of dendritic branches with concentric rings (in two dimensions) or spheres (in three dimensions) of increasing radii centered at a neuron's soma. In our study [67], we develop a modified version of the traditional three-dimensional Sholl analysis that calculates the number of intersections, N_I , of a neuron's dendritic branches with concentric spheres of increasing radii, r , averaged across spheres centered at 25 randomly selected locations on a neuron's arbor. These locations are restricted to being within a distance of $R_A/\sqrt{2}$ of the center of mass of the arbor, where R_A is the arbor's radius [39]. This sampling of many local origins rather than just one origin centered on the soma accommodates potential variations arising from some parts of the neurons possessing different structural qualities than others. Restricting the selection of local origins to be within $R_A/\sqrt{2}$ reduces the number of spheres that have large portions extending beyond the arbor's periphery. This approach also allows alignment with the cumulative mass fractal analysis which similarly samples many locations [24].

To perform the cumulative mass fractal analysis (also referred to as the mass-radius method) of a neuron's arbor, we calculate L_{in} , the total length of all dendrites within concentric spheres of increasing radii, r , averaged across randomly selected sphere centers using the same process as the modified Sholl analysis. The range of r examined also matches that used in the modified Sholl analysis. Figure 44.10 shows the relationship between N_I and r . As r increases, we see N_I initially increase to a maximum of 21 at $r = 73 \mu\text{m}$, followed by a decrease as r nears the mean arbor radius of $98 \mu\text{m}$. While the increase reflects the fractal character of the repeating patterns established by the dendrites [39, 54], the decrease is a consequence of the measurement technique – it reflects the increased chance of the larger outer sphere surfaces reaching beyond the space occupied by the dendrites. However, its onset can be impacted by any changes in the fractal character toward the neuron periphery. Figure 44.10 also shows the results of the cumulative mass analysis by plotting L_{in} against r and reveals a gradual increase that slows in the range of r accompanied by the decrease in N_I .

For a neuron with fractal branches, the mass dimension, D_M , of its arbor can be measured from this cumulative mass analysis using the following relationship [24, 54]:

$$L_{in} \sim r^{D_M}$$

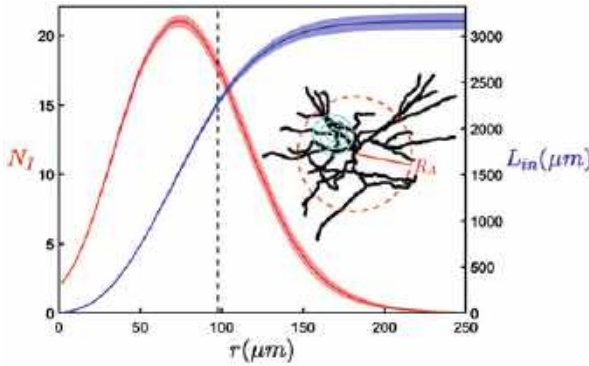


Fig. 44.10 The results of a modified Sholl analysis (red) measuring the average number of intersections of dendrites, N_I , with a sphere surface of radius r , and a cumulative length analysis (blue) measuring the total length of all dendrites, L_{in} , within a sphere of the same radius. These curves represent the mean behavior across a set of basal arbors and the shaded region around each curve shows the standard error from the mean. The inset shows an example neuron's basal arbor with its arbor radius, R_A , denoted by the red dashed ring and example sphere radii used in the analyses denoted by the concentric cyan rings. The black dashed line at $98 \mu\text{m}$ indicates the mean arbor radius

D_M is referred to as the mass fractal dimension because it measures the change in the cumulated mass of the object as a function of the size of the region considered (relating length to mass assumes that the branch width does not vary substantially, which is a reasonable approximation for the cylindrical segments of our neurons). Thus, the slope of a double logarithmic plot of L_{in} versus r provides a quantitative value of D_M . However, once the radius of a sphere reaches a large enough value that the entire arbor is contained within the sphere, L_{in} will become equal to L_T , the sum of all of the fork lengths within the arbor. As such, this power law scaling only holds over a finite range of r . Our selection of the scaling range for the fit is chosen to be consistent with the scaling range examined in our second fractal analysis, the box-counting technique.

Whereas the cumulative mass method provides a link to neuroscience through its relationship to the Sholl analysis, the box-counting method is as traditional method adopted from fractal studies. Similar to the coastline method counting the number of rulers as ruler size is reduced, the analogous arbor analysis replaces the rulers with boxes to accommodate the fact that the arbors feature multiple branches. The box-counting technique then determines the amount of space occupied by the arbor by inserting it into a three-dimensional grid of the boxes and counting the number of boxes, N_{box} , occupied by the branches. This count is then repeated across a range of box sizes, L_{box} . We can use the following relationship to determine the arbor's 'covering' fractal dimension, D_A :

$$N_{\text{box}} \sim L_{\text{box}}^{-D_A}$$

As with the branch measurements, the arbor power-law scaling will only hold over a finite range due to the arbor's physical size and the limited resolution of the reconstructions. At the fine-size scale, we limit L_{box} to W . This avoids resolution effects arising from the segment shapes. At the coarse scale, we limit L_{box} to be less than one-fifth of the largest extent of the arbor in the x , y , or z directions to ensure sufficient counting statistics. Within these limits, a straight line can be fitted for all sets of points that range over at least an order of magnitude on the log-log plot of N_{box} versus L_{box} and the fit with maximal R^2 is chosen to measure D_A .

In Fig. 44.11, we compare the scaling plots for the two fractal methods. Figure 44.11a shows the measurement of the arbor mass dimension, D_M , while Fig. 44.11b shows the measurement of the arbor covering dimension, D_A , for the same neuron. Although the dimension measurements of both methods are in agreement with one another, we note that the results of the linear regression in the box-counting analysis, yielding $D_A = 1.40 \pm 0.01$ ($R^2 = 0.9993$), correspond to a better fit than the results of the cumulative mass analysis, yielding $D_M = 1.42 \pm 0.05$ ($R^2 = 0.9926$). A comparison across all neurons shows the mean D_A is lower than the mean D_M value (1.41 and 1.47, respectively). We note that D_M and D_A belong to a spectrum of dimensions, and their magnitudes can be compared using a multi-fractal analysis [53, 68]. In our case, the box-counting analysis serves as a more global measure of fractality because it accommodates the whole arbor, while the cumulative mass analysis is biased toward the central section (through its restriction of the local sphere centers to be within $R_A/\sqrt{2}$ of the center of mass of the arbor). If the dendrites start to, for example, straighten or fork less toward the arbor periphery, a dimension that measures the whole arbor would be expected to be lower than one that focuses on the central region.

Although the differences between D_A and D_M are relatively small for our neurons, based on this potential effect and also on the relative qualities of the associated fits (Fig. 44.11), we will focus on D_A later in the chapter. However, both dimensions highlight two intriguing qualities: (1) the spatial interaction of the branches and gaps in the arbor generates a significantly larger fractal complexity than that displayed by the individual branches, (2) given that the arbor fractal dimension could assume values up to 3, the fractal complexity is nevertheless relatively low when compared to the capacity of some mathematical fractals to fill space.

44.5 Connectivity Analysis

We now investigate the impact of changing fractal dimension on the neuron's potential to connect to other neurons. We begin this discussion by considering the relationship between the arbor and branch dimensions. Figure 44.12b plots the D_A measurements across all the neurons and their mean branch fractal dimension, $\langle D_{BC} \rangle$, for different α values. The fact that both increasing and decreasing α results in a rise in D_A can be understood in terms of the interplay of the fractal branches

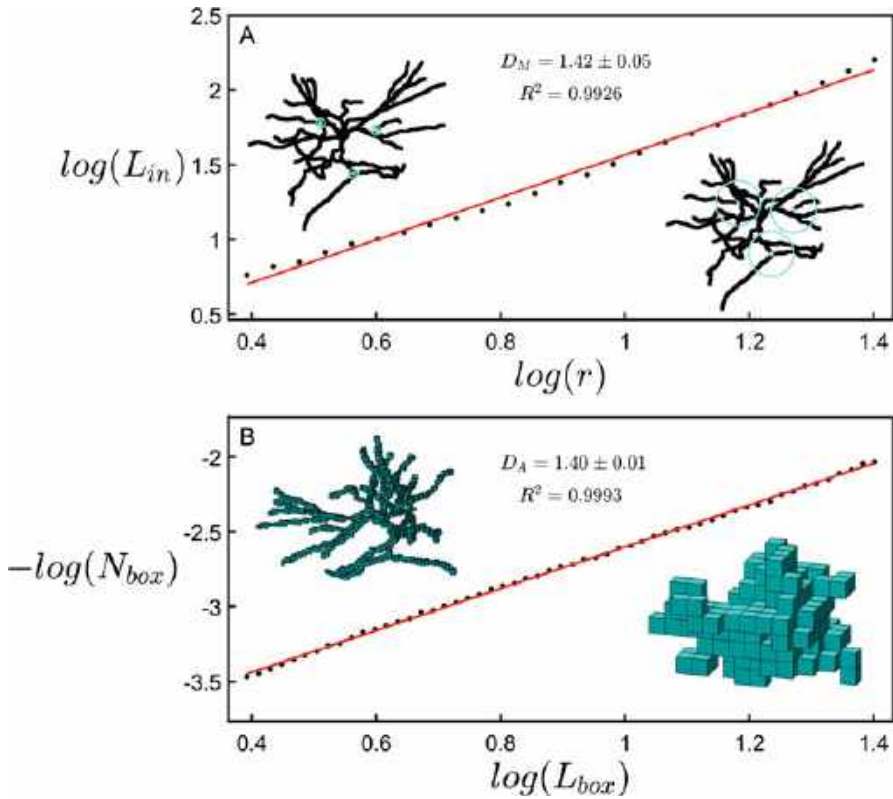
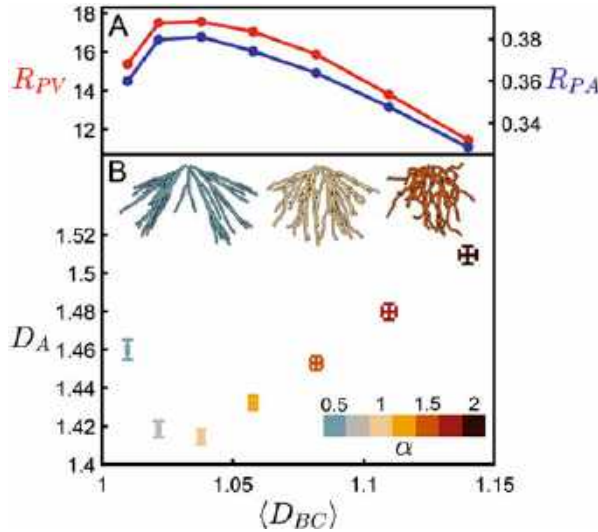


Fig. 44.11 (a) A scaling plot of the cumulative length analysis showing the total dendritic length, L_{in} , within a sphere plotted against its radius, r . The left inset shows rings with a radius of $5 \mu\text{m}$ at 3 example locations on a neuron, while the right inset shows spheres with a radius of $25 \mu\text{m}$ at the same locations. (b) A scaling plot of the number of boxes occupied by a neuron, N_{box} , plotted with respect to the size of the boxes, L_{box} . The left inset shows a representation of the space occupied by a neuron at a box size of $3.1 \mu\text{m}$, while the right inset shows the same neuron at a box size of $20.3 \mu\text{m}$. (Figure taken from reference [67])

and gaps. The branches self-avoid at the natural condition of $\alpha = 1$ and so move closer together when α is either increased or decreased. This is demonstrated in Fig. 44.12b insets which show an example arbor for the natural case (middle) and for lower (left) and higher (right) α values. This generates an increase in the ratio of fine to coarse structure and a corresponding rise in D_A . In terms of the sensitivity of U-shaped curve to changes in their $\langle D_{BC} \rangle$ values, we draw attention to the curve's asymmetry. Distortions that increase the dendrites' weaving and forking angles lead to small increases in D_A compared to the sharper rises observed for distortions that reduce these angles. In particular, arbors featuring dendrites close to the Euclidean condition are highly sensitive to distortions. For example, the small reduction in $\langle D_{BC} \rangle$ from 1.02 ($\alpha = 0.75$) to 1.01 ($\alpha = 0.5$) is accompanied by an increase in

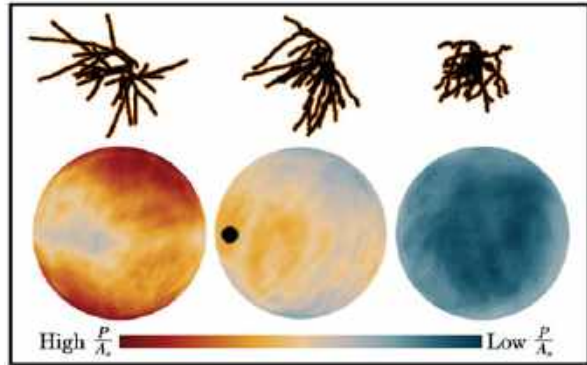
Fig. 44.12 Top: R_{PV} (red) and R_{PA} (blue) plotted against the mean coastline fractal dimension, $\langle D_{BC} \rangle$. Bottom. Arbor fractal dimension, D_A , plotted against $\langle D_{BC} \rangle$ for seven values of α as indicated by the lower-right color bar. The shown data represent the mean of D_A and $\langle D_{BC} \rangle$ across all arbors at each α value, with the error bars indicating the standard error from the mean. The three upper insets show an example neuron's dendritic arbor for three values of α as indicated by the color of the arbor. The top and bottom plots share the same x-axis



D_A from 1.42 ($\alpha = 0.75$) to 1.46 ($\alpha = 0.5$). Relative to the dendrites, the arbor's dimension increases approximately fourfold.

Figure 44.12a plots the $\langle D_{BC} \rangle$ dependence of two connectivity efficiency factors, R_{PV} and R_{PA} , which relate the neuron's connectivity to the costs of building and operating this connectivity, respectively. To understand these parameters, we first consider connectivity in detail. Previous studies have established that the arbor's physical structure is sufficient for describing the connection process, with chemical steering playing a relatively minor role [69, 70]. In particular, the arbor's dendritic density [45, 71–73] and resulting physical profile [39] are powerful indicators of its potential to connect to other neurons. When viewed from a particular orientation, we define the arbor's profile, P , as the total projected area of its branches. Large profiles will therefore result in the increased exposure of synapses, which are responsible for receiving signals from other neurons. When calculating the profile from the dendrite images, we incorporate an extra layer surrounding the branches to account for the outgrowth of dendritic spines – small protrusions that contain the majority of the dendrite synapses. We then normalize this projected surface area of the dendrites using their total surface area, A_s , to accommodate for the range in neuron sizes and associated surface areas. The current study adopts the general approach of averaging P/A_s across all orientations of the dendritic arbor to allow for the fact that axons originating from within the CA1 region connect to the dendritic arbors from every direction [74]. The profile variation with orientation can be visualized by projecting the P/A_s values obtained for each orientation onto a spherical surface. For the profile spheres included in Fig. 44.13, the neurons are viewed from a common direction which corresponds to the middle point on the sphere's surface. For the natural neuron, the orientation for which P/A_s peaks is marked by the black dot. Typically, this peak occurs in the direction that the Schaffer collateral axons enter

Fig. 44.13 The upper images show an example neuron for $\alpha = 0.25$ (left), 1 (middle), and 1.75 (right). The lower images show the equivalent profile spheres, where the black dot represents the orientation with maximal P/A_s for the middle neuron and the bar indicates the colors ranging from high to low P/A_s values



from the CA2 region [36] and so maximizes the connectivity of our natural neurons to those incoming axons.

P/A_s exhibits an inverse relationship with D_A because the increased fine structure of high D_A neurons causes branches to block each other and so reduces the overall profile [61]. This blocking effect is important for capturing the neuron's connectivity because multiple connections of an axon to the same dendritic arbor are known to generate redundancies [39]. Therefore, if a straight axon from an incoming neuron connected to an exposed branch, subsequent connections to blocked branches wouldn't increase the connectivity and should be excluded. Another well-known fractal effect is that high D_A fractals increase the ratio of the object's surface area A_s to its bounding area A_b [23, 49] (i.e. the surface area of the volume containing the arbor, as quantified by its convex hull). Figure 44.14a combines the 'blocking area effect' with the 'increased surface area effect' by plotting P/A_b (i.e. the multiplication of P/A_s and A_s/A_b) against D_A . In effect, P/A_b quantifies the large surface area of the arbor while accounting for the fact that some of this area will be blocked and therefore excluded from the profile P . Normalizing P using A_b serves the additional purpose of measuring the arbor's potential connectivity in a way that is independent of its size. Accordingly, P/A_b serves as a connectivity density and is an effective measure of the neurons' capacity to form a network.

Figure 44.14a–c examine this connectivity density along with the associated costs. The clear rise in P/A_b revealed by Fig. 44.14a highlights the functional advantage offered by high D_A branches – incoming axons from other neurons will experience the dendritic arbor's large connectivity density. Note that the plotted connectivity density is for individual neurons. Because of the inter-penetrating character of dendritic arbors from neighboring neurons, the collective connectivity density will be even larger due to their combined profiles. If this functionality was the sole driver of neuron morphology, then all neurons would therefore exploit high D_A values approaching 3. Yet, the arbors cluster around relatively low values of $D_A \sim 1.41$ (as shown by the histogram in Fig. 44.14f), suggesting that there are additional, negative consequences of increasing D_A .

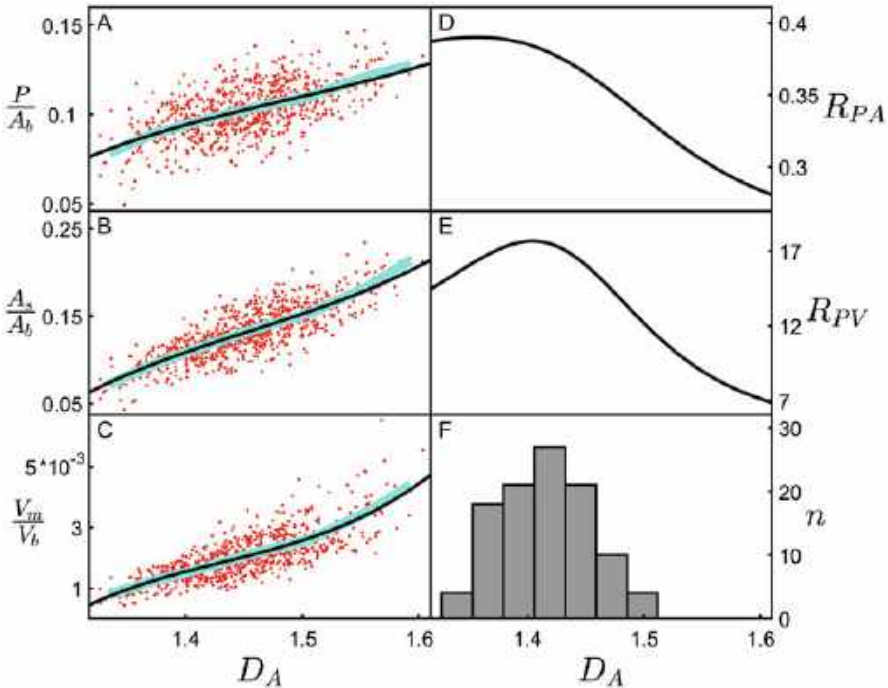


Fig. 44.14 Left column. Dependences of P/A_b (a), A_s/A_b (b), and V_m/V_b (c) on D_A . The underlying red data include both natural and distorted neuron arbors, the cyan lines correspond to binned averages of the data, and the black curves correspond to third-degree polynomial fits to the data. Right column. (d) R_{PA} (the ratio of the derivatives of the fits to P/A_b and A_s/A_b) plotted against D_A . (e) R_{PV} (the ratio of the derivatives of the fits to P/A_b and V_m/V_b) plotted against D_A . (f) Histogram of n , the number of natural neurons with a given D_A value

Considering the arbor’s “operational” cost, it is well known from allometric scaling relationships that metabolic costs generally increase with mass [75, 76]. Specifically, previous research proposed that the amount of ATP expended by neurons increases with A_s [39, 42]. By plotting A_s/A_b , Fig. 44.14b therefore charts how the normalized energy cost increases with D_A . In Fig. 44.14c, we plot the ratio of the volume occupied by the branches, V_m , to the neuron’s bounding volume, V_b (i.e., the arbor’s convex hull volume). For high D_A dendrites, the tighter weave angles along with forking angles that bring branches closer together result in more densely packed structures. This produces the observed rise of V_m/V_b . Assuming constant tissue density, V_m is proportional to the neuronal mass. The rise in V_m/V_b therefore quantifies the increase in mass density and associated ‘building’ costs of high D_A neurons.

The distinct forms of Fig. 44.14a–c are highlighted using third-degree polynomial fits (black) which closely follow the binned average values of the data (cyan).

To explore how the fractals balance these factors, in Fig. 44.14d and e, we consider the ratios of the rates of change of connectivity with operating cost:

$$R_{PA} = \frac{\frac{d}{dD_A} \left(\frac{P}{A_b} \right)}{\frac{d}{dD_A} \left(\frac{A_s}{A_b} \right)}$$

and with building cost:

$$R_{PV} = \frac{\frac{d}{dD_A} \left(\frac{P}{A_b} \right)}{\frac{d}{dD_A} \left(\frac{V_m}{V_b} \right)}$$

as simple optimization indicators with the prediction that peaks in R_{PA} and R_{PV} indicate the D_A value at which the optimal balance occurs. We note that R_{PA} and R_{PV} both peak at around the natural neurons' prevalent D_A value. Consequently, although high D_A offers superior connectivity for the neurons, the positive consequences of increasing D_A beyond the peak rapidly diminish in terms of the mass costs of establishing connectivity. Simultaneously, the negative consequences of increasing D_A rapidly rise in terms of the energy costs of establishing connectivity.

Note that the red data in Fig. 44.14a–c feature both natural and distorted neurons, and these are not distinguished to emphasize that the two groups follow a common behavior. Distortions shift a neuron's D_A value along the common black line. Figure 44.12a shows how these distortions cause the neuron to deviate away from the optimized form. The data in Fig. 44.12a are generated by first using Fig. 44.12b to convert the $\langle D_{BC} \rangle$ values for each α value to their associated D_A values and then using Fig. 44.14d and e to convert these D_A values to their corresponding R_{PA} and R_{PV} values. As shown by the falling R_{PA} and R_{PV} values, the balance between connectivity and cost deteriorates as the D_A value moves to the higher, unnatural values. Significantly, the largest sensitivities of R_{PA} , R_{PV} , and D_A all occur at the smallest $\langle D_{BC} \rangle$ values. Figure 44.12 therefore captures the importance of the fractal resonance hypothesis. If a neuron that looks like the middle image of the bottom panel interacts with electrodes shaped like the left and right images, then it will experience the deterioration shown in the top panel.

44.6 Conclusions

In this chapter, we proposed the fractal resonance hypothesis for the interface between implants and neurons. This hypothesis declares that favorable properties emerge by matching the geometries of the two interacting surfaces. An appealing analogy can be found with aircraft landing on a runway. Aircraft move most efficiently along straight trajectories, and straight runways therefore accommodate

this Euclidean behavior. In contrast, neurons weave and fork in a multi-scaled manner as they approach our electrodes, and therefore we aim to accommodate their fractal growth with fractal electrode designs. We have shown that neuron growth establishes connections to other neurons while balancing the building and operating costs of these connections. For neuron types with high D_A values, this balance favors connectivity while for those with low D_A values costs are more important for their functionality. By matching the dimensions of the implant and neurons, the aim is to promote unrestricted growth that maintains the necessary connectivity–cost balance to maintain their functions as they interact with the artificial surface. In a way, we are using geometry to ‘convince’ the neurons that they are interacting with artificial versions of themselves.

The prospect of fractal resonance holds great promise for implants whether their function is to stimulate neurons or sense their signals. Both functions benefit from preserving neuron growth and function. Although long term, the ultimate practical aim for fractal resonance is for each neuron process to automatically seek out and grow along a fractal branch on the implant’s surface. We also aim for ‘selective’ growth, whereby targeted types of neurons could grow on specific branches. This would be powerful for implants interacting with networks featuring multiple types and would allow them to be assembled in different regions of the implant.

Although the aircraft analogy is appealing in its simplicity, the reality of fractal resonance will be driven by optimization curves such as the one shown in Fig. 44.12 for our CA1 pyramidal neurons from the coronal plane of the dorsal rat hippocampus. We anticipate that the U-shaped relationship between D_B and D_A will be generic to neuron types with arbors composed of fractal branches that self-avoid. For neurons with a spread of D_A values, we expect an analogous behavior to that shown in Fig. 44.14f whereby R_{PA} and R_{PV} peak at D_A values closely matching the histogram peak, indicating that the majority of neurons exist near the optimizing condition.

In terms of distortions away from the natural fractal condition, it is informative to examine the symmetry of the U shape. For the hippocampal neurons of the current study, the natural condition ($\alpha = 1$) centered around $D_B = 1.04$ and distortions that reduced the fractal weave of the branches resulted in relatively large changes to the arbor fractal characteristics as the branches neared the Euclidean condition of straight lines. Based on this observation, we anticipate that neuron types with naturally occurring low D_B values that are distorted in a manner that reduces their weave or forking angles will experience large changes in their arbor fractal characteristics and associated functionality. This, however, assumes a similar arbor density to the neurons examined in the current study – this behavior may not be seen for sparsely branching neurons. It is also intriguing to consider neurons with large naturally occurring D_B values and examine whether distortions through increases in their weave and forking angles would experience a similar sensitivity to D_B .

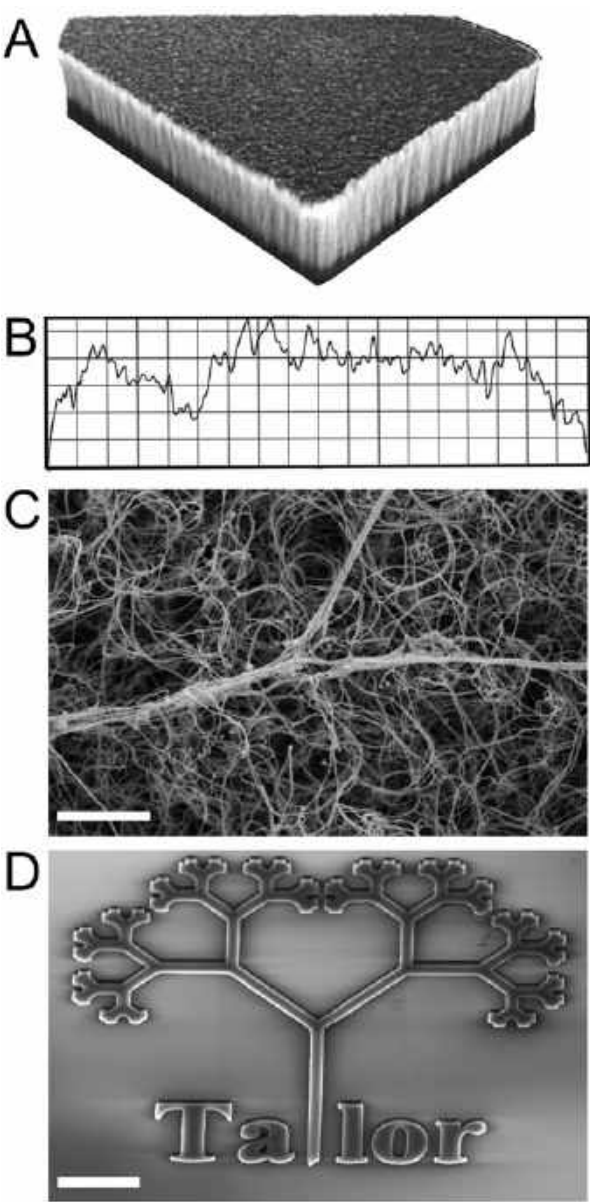
In addition to neurons peaking at different D_A values depending on function, the size of the peak might be expected to vary also. It is interesting to note that if we created neurons that are mathematically restricted to a two-dimensional plane, their optimization curves would be perfectly flat. It is therefore vital for future studies

to analyze a broad range of neurons to consider the variety of optimization curves, including comparisons of neuron dendrites that spread out in planes (such as the plexiform layers of the retina [77]) to neurons that spread out into the volume surrounding an interface. Based on this need, we are automating our optimization procedure so that it can be applied to neuron images provided by the online repository NeuroMorpho.Org [78].

Based on these discussions, the practical challenge of achieving fractal resonance lies in the fabrication methods capable of matching fractal geometries. Although not the focus of this chapter, we will finish by highlighting two promising approaches, the details of which can be found elsewhere [1, 2, 20, 79, 80]. The images shown in Fig. 44.15 are fabricated from VACNTs. As noted in the introduction, their nano-scale texture (shown in Fig. 44.15b and c) has been proposed to mimic some of the properties of the extracellular matrix [3–5]. This has been shown to enhance neurite outgrowth and elongation if the roughness variation matches the neuron process diameter [81]. Given the clear multi-scaled character of this texture, a powerful future approach would be to increase the texture amplitude to generate three-dimensional structures that are fractal patterns that extend in both the lateral and height directions. This would offer the possibility of matching the fractal character of three-dimensional neurons and implant surfaces.

In Fig. 44.15d, we provide a demonstration of the patterning capabilities of the VACNT technique. We can pattern features down to the micron-size scale, in principle allowing a replication of patterns such as those shown in Fig. 44.4. However, this lithographic technique is practically intensive given the complexity of the patterns being replicated. A more appealing approach would be to harness a natural growth process that self-assembles fractal structures. Figure 44.16b shows statistical fractals grown using diffusion-limited aggregation that look strikingly similar to, for example, retinal ganglion cells (Fig. 44.16c). Using this technique, a beam of metallic clusters (each with a diameter of ~ 50 nm) is deposited on a substrate in an ultra-high vacuum chamber. These clusters then grow into fractal islands called nanoflowers [79]. Figure 44.16a charts this growth process, showing the small ellipsoidal islands that initially form (plotted on the graph's left side) and the islands with fractal branches that they grow into (right side). The islands' fractal morphology and size can be adjusted by varying deposition conditions such as temperature, deposition rate, and flux. Fortuitously, neighboring islands naturally self-avoid rather than merge, allowing arrays of disconnected fractal patterns to be grown efficiently. Whereas the shown patterns are grown on graphite, we have also self-assembled fractals on insulating surfaces [80]. The demonstrations shown in Figs. 44.15 and 44.16 highlight that fractal resonance is practically possible and therefore represent a potential new era for implants in which bio-inspired fractal designs interface more effectively with neuronal networks in the body.

Fig. 44.15 (a) Scanning electron micrograph image generated using a stereoscopic technique showing a VACNT branch imaged at an angle. (b) The VACNT side profile quantifying the surface texture. The y-axis spans from 23.5 to 24.5 μm and the x-axis spans from 0 to 90 μm . (c) A top-down view of a neuron process adhering to the texture of the VACNT top surface. (d) A top-down view of the lithographic patterning capability. The scale bars in (c) and (d) are 2 μm and 100 μm , respectively. The VACNTs are fabricated in collaboration with B. Alemán (University of Oregon, USA)



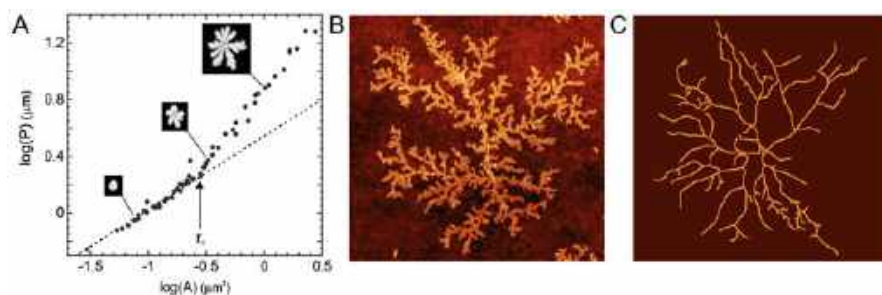


Fig. 44.16 (a) A plot of log (island perimeter, P) versus log (island area, A), as measured from a digital scanning electron micrograph image of antimony patterns deposited on a graphite substrate. Each data point represents an individual pattern. The dotted line is a guide to the eye, showing the behavior expected for purely ellipsoidal (i.e. non-fractal) pattern. The critical radius, r_c , represents the point at which the data deviate from this behavior, indicating the formation of fractal branches. (b) Atomic force microscopy image of a fractal pattern. (c) For comparison, we show the dendritic arbor of a mouse retinal ganglion cell. The fractals shown in (a–b) are fabricated in collaboration with S.A. Brown (University of Canterbury, New Zealand). The neuron shown in (c) is adapted from data publicly available on NeuroMorpho.Org [78, 82]

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Chapter 45

Fractal Fluency: Processing of Fractal Stimuli Across Sight, Sound, and Touch



Richard P. Taylor, Catherine Viengkham, Julian H. Smith, Conor Rowland, Saba Moslehi, Sabrina Stadlober, Anastasija Lesjak, Martin Lesjak, and Branka Spehar

Abstract People are continually exposed to the rich complexity generated by the repetition of fractal patterns at different size scales. Fractals are prevalent in natural scenery and also in patterns generated by artists and mathematicians. In this chapter, we will investigate the powerful significance of fractals for the human senses. In particular, we propose that fractals with mid-range complexity play a unique role in our visual experiences because the visual system has adapted to these prevalent natural patterns. This adaptation is evident at multiple stages of the visual system, ranging from data acquisition by the eye to processing of this data in the higher visual areas of the brain. Based on these results, we will discuss a fluency model in which the visual system processes mid-complexity fractals with relative ease. This fluency optimizes the observer's capabilities (such as enhanced attention and pattern recognition) and generates an aesthetic experience accompanied by a reduction in the observer's physiological stress levels. In addition to reviewing people's responses to viewing fractals, we will compare these responses to recent research focused on fractal sounds and fractal surface textures. We will extend our fractal fluency model to allow for stimuli across multiple senses.

Keywords Fractals · Complexity · Behavioral and physiological responses

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45.1 Introduction

What happens to your brain when you take a walk through nature's scenery? Our senses soak in the sights, sounds, and physical textures of nature, along with smells and perhaps even tastes. Each of our senses has evolved to detect variations in the natural environment for our brains to respond to. Interest in quantifying the positive impacts of experiencing nature gained momentum in the 1980s with the biophilia movement [1]. Biophilia – nature-loving – emphasizes the inherent need of humans to connect with nature. Whereas the primary goal of biophilia is to incorporate elements of nature into built environments, other complementary missions aim to maximize exposure to natural environments. For example, the Japanese traditional practice of *Shinrin-yoku*, or “forest bathing,” continues to thrive.

Evolution has transformed the eye from its initial role as a simple motion detector to the remarkable system that we benefit from today. Vision is our dominant sense. The brain receives 2 billion pieces of information from our eyes every second, while it receives only 1 billion pieces from the rest of the body. Consequently, up to one-third of the brain's volume is dedicated to visual processing, compared to just 8% to touch and 3% to hearing. Under such data pressure, the visual system has developed efficient strategies for processing what we see. In this chapter, we will propose that this efficient processing results in anesthetic experience accompanied by positive impacts on our well-being.

Previous pioneering psychology experiments have demonstrated that exposure to nature's scenery induces significant stress reduction – even to the extent of accelerating patients' recovery from surgery [2–4]. Attention Restoration Theory (ART) has been proposed to help explain our inherent fascination with nature [5, 6]. ART declares that the “soft” attention induced by nature differs fundamentally from the “hard” attention required for unnatural tasks such as reading text. Consequently, this “effortless looking” restores depleted mental resources and prevents occupational burnout. Conversely, unnatural scenes strain the visual system and induce negative responses such as headaches and stress [7].

Although groundbreaking, these demonstrations of “biophilic” responses employed vague descriptions for nature's visual properties. More generally, although we all recognize that nature's beauty is profound, identifying the origin of this beauty is challenging. This is due in part to the multitude of factors that potentially contribute to beauty. Another major challenge arises from a historic lack of understanding of the underlying geometry of natural scenes. What do we see in the wispy edges of clouds, in the intricate branches of trees, and in the jagged peaks of a snowy mountain range? For many years, scientists assumed these images were a haphazard mess devoid of any pattern. However, the past 50 years have seen a remarkable revolution in the way scientists study nature's scenery, which has brought scientific inquiry and artistic views of nature closer together. Fractal patterns, named by Benoit Mandelbrot, lie at the heart of this revolution [8]. Dramatically referred to as “the fingerprint of life,” their repetition of patterns across multiple scales forms the basic building blocks of many of nature's patterns,

ranging from clouds, trees, and mountains through to our brains, blood vessels, and lungs. The visual complexity generated by their repeating patterns stands in sharp contrast to the relative smoothness of artificial shapes such as circles, triangles, and straight lines.

We will review psychology experiments conducted over the past two decades that suggest the aesthetic qualities of fractals might be triggering the powerful effects of biophilia. These experiments inform the “fractal fluency” model which states that human (and presumably animal) vision has become fluent in the visual language of fractals through evolutionary exposure. Accordingly, a cascade of subconscious processes places the observer in their visual “comfort zone,” ushering in a neuro-aesthetic experience accompanied by a decrease in physiological stress.

Similar to the phenomenon of synesthesia, in which sensations are transferred between the senses, fractal fluency might extend to the other senses. Although receiving significantly less attention, we will compare people’s responses to auditory and tactile fractals with those well charted in the visual realm. Ultimately, our goal is to identify whether there are emergent experiences induced by exposing multiple senses simultaneously, whereby the whole is much more than the sum of the parts. Returning to the walkthrough nature, we can replace “forest bathing” with “fractal bathing.” While enjoying the views of trees, clouds, and rivers, the observer can also listen to fractal sounds ranging from bird songs to waterfalls and can reach out to experience the fractal textures of wood and rock surfaces.

In addition to investigating the fundamental science of human responses to nature’s beauty, we will also consider strategies for transferring this knowledge to applications in the built environment. Biophilia’s warning takes on more urgency with the predicted impacts of climate change [9]. In the future, humans might no longer be able to rely on the countryside to provide the fractal complexity that their senses have developed to expect and respond to throughout evolution. Fractal bathing in the sights, sounds, and textures of a forest could be reduced to a behavior of our past.

This possibility places an urgency on the need for fractal art, fractal music, and the tactile surfaces of fractal sculptures. Fractal creation is not new to humans; fractals have permeated cultures spanning across many centuries and continents, ranging from Hellenic friezes (300 B.C.E) [10] to Jackson Pollock’s poured paintings (1940–50s) [11–14]. However, prior to Mandelbrot’s labeling of fractals in the 1970s, these creations were intuitive. Similarly, although elements of auditory fractals have appeared in natural noises and music, it wasn’t until the 1990s that Hugh McDowell (cellist in the Electric Light Orchestra) deliberately composed music using algorithms based on Mandelbrot’s work [10]. Driven by technological advances, we now live in an era in which art–science collaborations can generate novel science-informed fractal patterns for the built environment. We will therefore close the chapter by discussing applications created by the Science Design Laboratory (SDL) based on the neuro-aesthetic principles described earlier in the chapter.

45.2 The Visual Complexity of Biophilic Fractals

We begin our discussions with visual fractals. In Fig. 45.1, we use a coastline to demonstrate their intrinsic visual properties. As shown in the left column, fractals can be divided into two categories – “exact” (top image) and “statistical” (bottom image). Whereas exact fractals are built by repeating a pattern at different magnifications, “statistical” fractals introduce randomness into their construction. This disrupts the precise repetition so that only the pattern’s statistical qualities (e.g., density, roughness, and complexity) repeat. Consequently, statistical fractals simply look similar at different size scales. Whereas exact fractals display the cleanliness of artificial shapes, statistical fractals capture the “organic” signature of natural objects.

Statistical fractals are highly topical in the field of “bio-inspiration,” in which scientists investigate the favorable functionality of natural systems and apply their findings to artificial systems. For example, the ability of fractal coastlines to

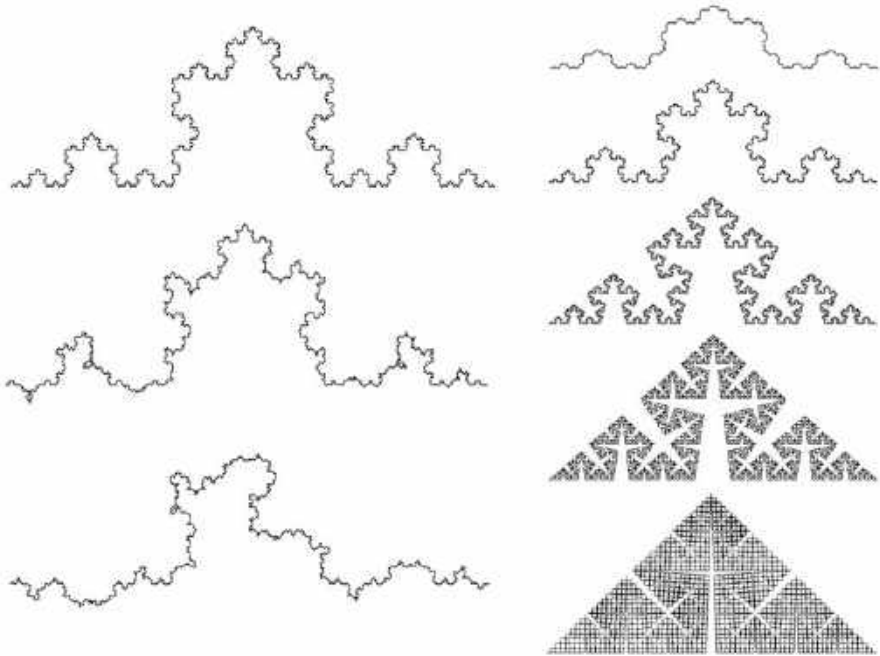


Fig. 45.1 Left column: A computer-generated coastline based on exact fractals (top) is morphed into a statistical fractal coastline (bottom) by introducing randomness. For the top fractal, all of the headlands point upward. For the bottom fractal, half point downward and the positions of the up and down headlands are randomized. Note the fractal dimension (D) value (1.24) is preserved for all 3 patterns (top, middle, and bottom). Right column: The effect of increasing D is shown for 5 exact coastlines. Each of the coastlines is built using the same coarse-scale pattern. Increasing the contributions of the fine-scale patterns causes the coastlines to occupy more of the 2-dimensional plane, thus raising their D values: 1.1 (top), 1.3, 1.5, 1.7, and 1.9 (bottom)

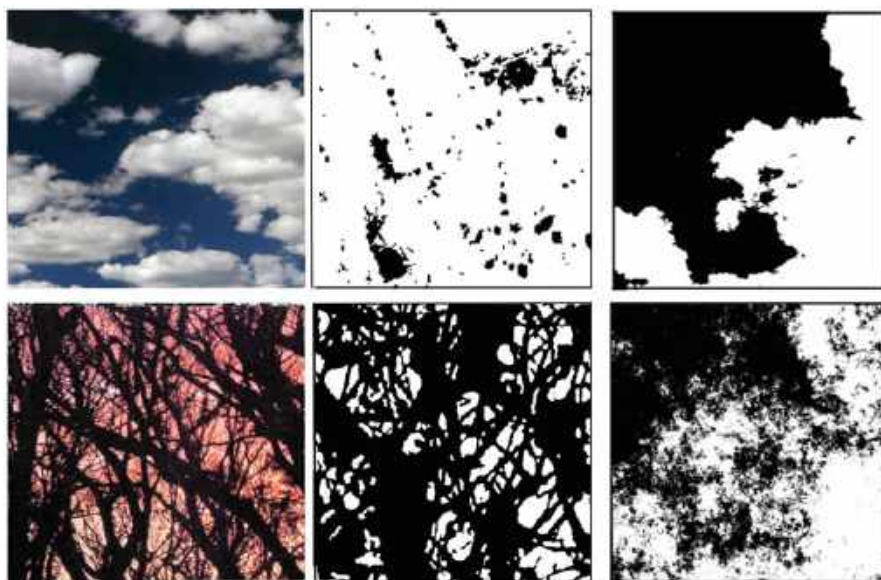


Fig. 45.2 Fractal complexity in nature, art, and mathematics. The left column shows clouds with $D = 1.3$ (top) and a forest with $D = 1.9$ (bottom). The middle column shows Jackson Pollock's *Untitled 1945* with $D = 1.1$ (top) and *Untitled 1950* with $D = 1.89$ (bottom). The right column shows computer-generated fractals with $D = 1.2$ (top) and $D = 1.8$ (bottom)

efficiently disperse wave energy has inspired fractal storm barriers. The ability of fractal tree canopies to efficiently collect sunlight has inspired fractal solar panels. The growing role of fractals in art suggests that the repeating patterns might serve another bio-inspired function beyond the scientific realm – an aesthetic quality.

To quantify the rich visual intricacy of the statistical fractals, we adopt a traditional mathematical measure to quantify the fractal image projected on the retina – fractal dimension D [8, 15]. This parameter describes how the patterns occurring at different magnifications combine to build the resulting fractal shape. For a smooth line (containing no fractal structure) D has a value of 1, while for a completely filled area (again containing no fractal structure) its value is 2. However, the repeating patterns of the fractal line cause the line to begin to occupy space. As a consequence, its D value lies between 1 and 2. When the contribution of the fine structure to the fractal mix is raised, the line increasingly fills in the 2-dimensional surface of the retina and the fractal's D value approaches 2. Figure 45.2 demonstrates how a statistical fractal's D value has a profound effect on the visual appearance of fractal patterns found in nature, art, and mathematics. Clearly, for fractals described by low D values, the small content of fine structure builds a very smooth sparse shape. However, for fractals with D values closer to 2, the larger amount of fine structure builds a shape full of intricate, detailed structure. More specifically, because the D value charts the ratio of fine to coarse structure,

it is expected that D will serve as a convenient measure of the visual complexity generated by the repeating patterns. Behavioral research confirms that people’s perception of complexity increases with D [16, 17].

Our initial investigations used three distinct categories of stimuli summarized in Fig. 45.2: natural fractals (using photographs of clouds, trees, mountains, etc.), artistic fractals (paintings generated by Jackson Pollock using his famous pouring technique), and mathematics (computer-generated images) [18]. Our more recent studies focus exclusively on computer-generated fractals due to their advantageous properties [19]. First, the D values of the images are known precisely because they are input parameters for the computer-generated process. Second, the greater control offered by computers allows the separation of different visual characteristics. For example, whereas density and D value are intrinsically linked in Pollock’s paintings (as seen in Fig. 45.2, he raised the painting’s D value by adding more paint which in turn inevitably raised the density [12]), these 2 parameters can be adjusted independently using computer-generated images. Thirdly, the images are purely abstract. Consequently, responses are not contaminated by associations with recognizable objects such as trees and clouds.

Figure 45.3a shows examples of our current stimuli, which are generated either by Fourier spectrum or midpoint displacement methods [15]. For the far left image,

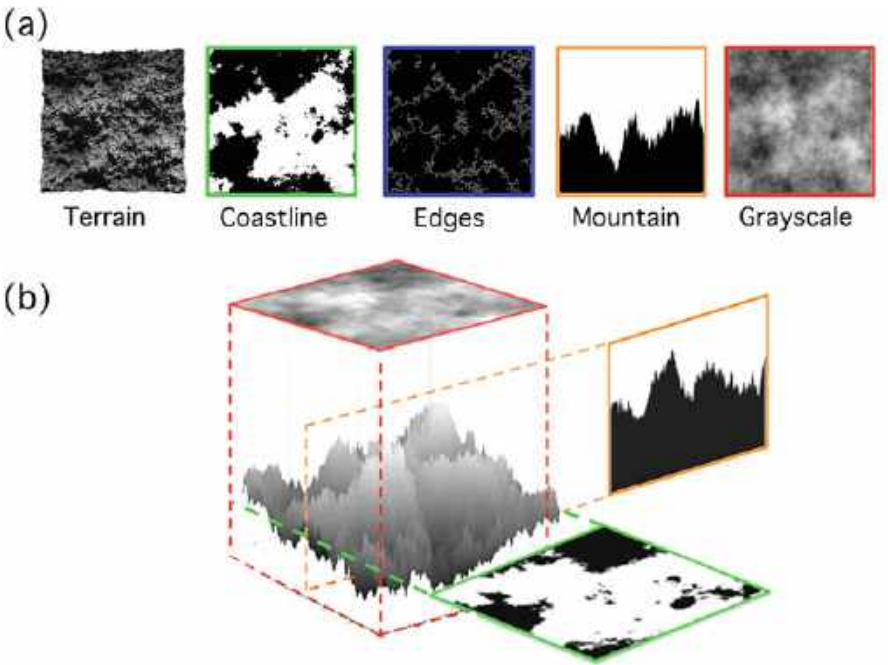


Fig. 45.3 (a) Computer-generated fractal stimuli. From left to right: terrains, coastlines, coastline edges, mountains, and grayscale patterns. (b) Schematic showing the relationship between the 5 types of fractal image

the computer has generated a geographical terrain (in this case viewed from above). The second image shows a horizontal slice taken through the terrain at a selected height. The terrain below this height is colored black, and all of the terrain above is colored white. Referred to as the coastline pattern (black being the water), this image is used to generate the third image by highlighting the coastline edges in white. The fourth image is created by taking a vertical slice through the terrain to create a mountain profile. Finally, the grayscale image is generated by looking down on the terrain from above and assigning grayscale values to the heights of the terrain (white being the highest). Taken together, these 5 families of fractals are powerful stimuli for examining people's responses because, although superficially quite different in appearance, they all possess identical scaling properties.

45.3 Fractal Fluency of Visual Fractals

The physical processes that form nature's fractals determine their D values. For example, wave erosion generates the low complexity ($D = 1.1$) of the Australian coastline, while ice erosion results in the high complexity ($D = 1.5$) of the Norwegian fiords. Significantly, although all D values in the range $1.1 < D < 1.9$ appear in natural scenes, the most prevalent fractals lie in the narrower range of 1.3–1.5. For example, many examples of clouds, trees, and mountains lie in this range. We therefore previously proposed a fluency model in which the human visual system has adapted to efficiently process the mid-complexity patterns of these prevalent $D = 1.3$ –1.5 fractals [13]. We expect this adaptation to be evident at multiple levels of the visual system, ranging from data acquisition by the eye to processing of this data in the higher visual areas of the brain.

Our studies of fractal fluency commenced with the eye movement studies shown in Fig. 45.4 [13, 15]. The eye-tracking system (Fig. 45.4a) integrates infrared and visual camera techniques to determine the eye's gaze when looking at a pattern presented on the computer monitor. During the 60-s observation period, participants were instructed to memorize the pattern in order to induce "free viewing" activity. Figure 45.4b shows a section of the spatial pattern traced out by the eye's gaze as it moves across the monitor. As expected, the pattern is composed of long saccade trajectories as the eye jumps between the locations of interest and smaller micro-saccades that occur during the dwell periods. These periods of relative motionlessness can also be seen in the associated temporal trace of Fig. 45.4c. Details of the fractal measurement technique applied to the eye's spatial and temporal patterns are reported elsewhere [13, 15, 20, 21].

The results show that the saccade trajectories trace out fractal patterns with D values that are insensitive to the D value of the fractal pattern being observed: the saccade pattern is quantified by $D = 1.4$, even though the underlying pattern varied over a large range from 1.1 to 1.9. This mid- D saccade pattern was confirmed for viewing computer-generated, natural, and Pollock fractals. Furthermore, participants with Alzheimer disease, frontal and anterior temporal lobe degeneration,

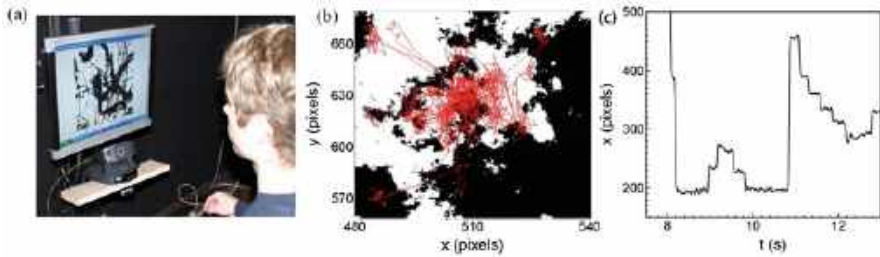


Fig. 45.4 (a) A photograph of the eye-tracking apparatus, (b) the spatial pattern of the eye tracks (red) plotted in the x (horizontal) and y (vertical) directions. The eye track (red) is overlaid on the observed fractal pattern (black and white), (c) the equivalent time series data which plots x versus time

and progressive supranuclear palsy all exhibited the same fractal gaze dynamics as healthy participants, indicating that it is fundamental to eye movement behavior and that it is not modified by processing in the higher levels of the visual system [21].

We propose that the purpose of the eye's search through fractal scenery is to confirm its fractal character (this ability to confirm that, for example, a forest features only fractal trees and no predators would promote survival). If the gaze is directed at just one location, the peripheral vision only has sufficient resolution to detect coarse patterns. Therefore, the gaze shifts position to allow the fovea to detect the fine-scale patterns at multiple locations. This allows the eye to experience the coarse and fine-scale patterns necessary for confirmation of fractal character. Why, though, does the eye adopt a fractal trajectory when performing this task? A possible answer can be found in the fractal motions of animals when they forage for food [22]. The short trajectories allow the animal to look for food in a small region and then to travel to neighboring regions and then onto regions even further away, allowing searches across multiple size scales. Significantly, such fractal motion has an 'enhanced diffusion' compared to the equivalent random movements of Brownian motion. The amount of space covered by the fractal search is therefore larger. This might explain why it is adopted for both animal searches for food and the eye's search for visual information [15]. The mid- D saccade is optimal for this fractal search because it matches the D values found in prevalent fractal scenery – the saccades then have the same amounts of coarse and fine structure as the observed stimulus, allowing the eye to shift through the visual information efficiently.

We expect that strategies for efficiently processing mid- D fractals will also be evident at higher levels in the visual system. In the 1990s, Field and others presented a neural model featuring virtual "pathways" used for processing scenic information in the visual cortex of the brain [23, 24]. Some pathways are dedicated to analyzing large structures in nature's environments and others to small structures. It was proposed that these pathways have evolved to accommodate our fractal view of nature as follows: the number of pathways dedicated to each structure size is proportional to the number of structures of that size appearing in the scene. In other words, the distribution of processing pathways matches the D values that dominate

the viewed environment. In other early studies, Geake and Landini proposed that fractal processing utilizes images stored in memory [25]. Their experiments showed that people who displayed a superior ability to distinguish between fractals with different D values were found to also excel in mental tasks involving simultaneous synthesis (an ability to combine current perceptual information with data from long-term memory). Modern neurophysiological measurement techniques such as quantitative EEG (qEEG) and functional MRI (fMRI) now offer the potential for researchers to refine these preliminary ideas of how the brain processes fractal stimuli.

EEG is a well-established measure of cortical activity. While the alpha frequencies (9–12 Hz) indicate a wakefully relaxed state, the beta frequencies (18–24 Hz) are associated with external focus, attention, and an alert state [26]. Previous recordings by Ulrich and colleagues revealed that people are more wakefully relaxed during exposure to natural landscapes than to townscapes, and studies of wall art found that images with natural content have positive effects on anxiety and stress [2, 3]. In our studies, participants' responses were continuously monitored using a digital EEG recorder while they viewed fractal "mountain" stimuli (Fig. 45.3) with different D values [27]. The images were viewed for 1 minute each and interspaced by a neutral gray picture for 30 seconds. This exposure period was chosen to ensure that a relaxation effect in the participants could occur. Three regions of the brain – frontal, parietal, and temporal – were chosen because processes in these associational zones are known to be complementary [26]. The results showed that fractal images quantified by $D = 1.3$ induce the largest changes in participants' alpha and beta responses [27]. Intriguingly, these responses were dampened when the images were morphed from the statistical to exact versions (Fig. 45.1), emphasizing the adaptation of processing fluency to nature's biophilic fractals [28, 29].

Our preliminary studies using the fMRI technique further indicate that mid- D fractals induce distinct responses when compared to those of low or high D equivalent images [30, 31]. Ongoing fMRI studies will consider the role of the parahippocampal region (which is known to be involved in memory retrieval and scene recognition) and the default-mode network (a large brain network associated with wakefully restful activities such as daydreaming and mind-wandering, and which features in modern versions of ART [32]).

45.4 Fractal Aesthetics and Stress Reduction

The fluency model predicts that the increased processing capabilities should result in enhanced performance of visual tasks when viewing mid- D fractals. Indeed, our behavioral studies demonstrate participants' heightened sensitivity to mid- D fractals [33]. Using grayscale fractal images displayed on a computer monitor (Fig. 45.3), the contrast in the patterns was gradually reduced until the monitor displayed uniform mean luminance. Participants were able to detect the mid- D

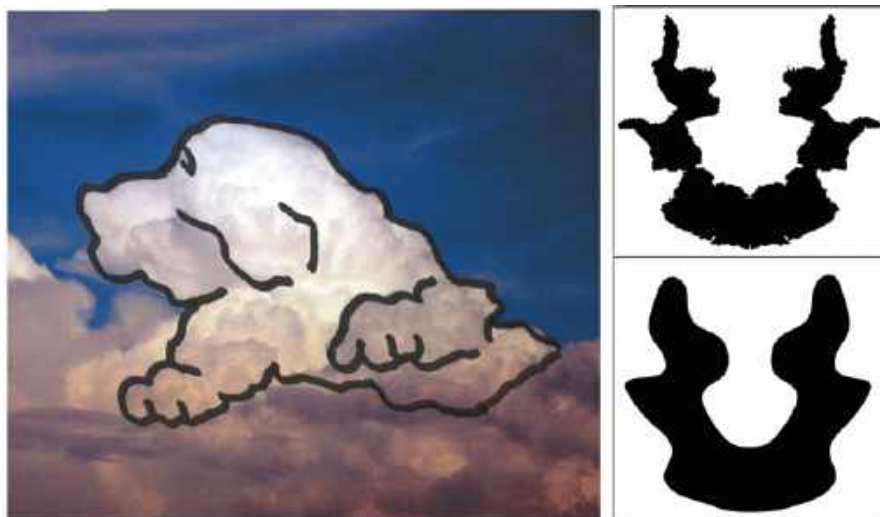


Fig. 45.5 Left: Fractal clouds are renowned for inducing perceived images (as an illustration, a perceived dog is drawn on the photograph of a cloud). Right: an example ink blot (top) which induces fewer percepts when its fractal edges are smoothed (bottom)

fractals for much lower contrast conditions than the low and high D fractals [32]. Similarly, participants displayed a superior ability to distinguish between fractals with different D values in the mid- D range [24, 25, 33]. The increased beta response in the qEEG studies suggests a heightened ability to concentrate when viewing mid- D range [27].

There is also anecdotal evidence to suggest that pattern recognition capabilities increase for mid- D fractals. We are all familiar with the percepts induced by cumulus clouds (Fig. 45.5). A possible explanation is that our pattern recognition processes are so enhanced by these $D = 1.3$ clouds that the visual system becomes “trigger happy,” and consequently, we see patterns that aren’t actually there. Our research confirms that mid- D fractals induce large numbers of percepts [34] and that they activate the visual cortex’s object perception and recognition regions [29]. This is supported by our studies of Rorschach ink blots. Perception of shapes in the fractal blots peaks in the lower D range [35] and declines when the fractal structure is electronically removed from the blot images (Fig. 45.5).

Does fractal fluency create a unique aesthetic quality because we find them relatively easy to process and comprehend? Perhaps this “aesthetic resonance” for $D = 1.3$ – 1.5 fractals induces the state of relaxation indicated by the peak in alpha response in the qEEG studies. Our skin conductance measurements similarly demonstrated that mid- D fractals are stress-reducing [36]. The question of fractal aesthetics holds special significance for the field of experimental aesthetics. One of its early pioneers, George Birkhoff, introduced “aesthetic measure” in the 1930s – the idea that aesthetics could be linked to measurable mathematical properties of

the observed images. Visual complexity was a central parameter in his proposals [37]. In 1993, we conducted the first aesthetics experiments on fractals, showing that 95% of observers preferred complex fractal images over simple Euclidean ones [38]. Soon after, Sprott employed computer-generated fractals to show that mid- D fractals were preferred over low and high D fractals [39].

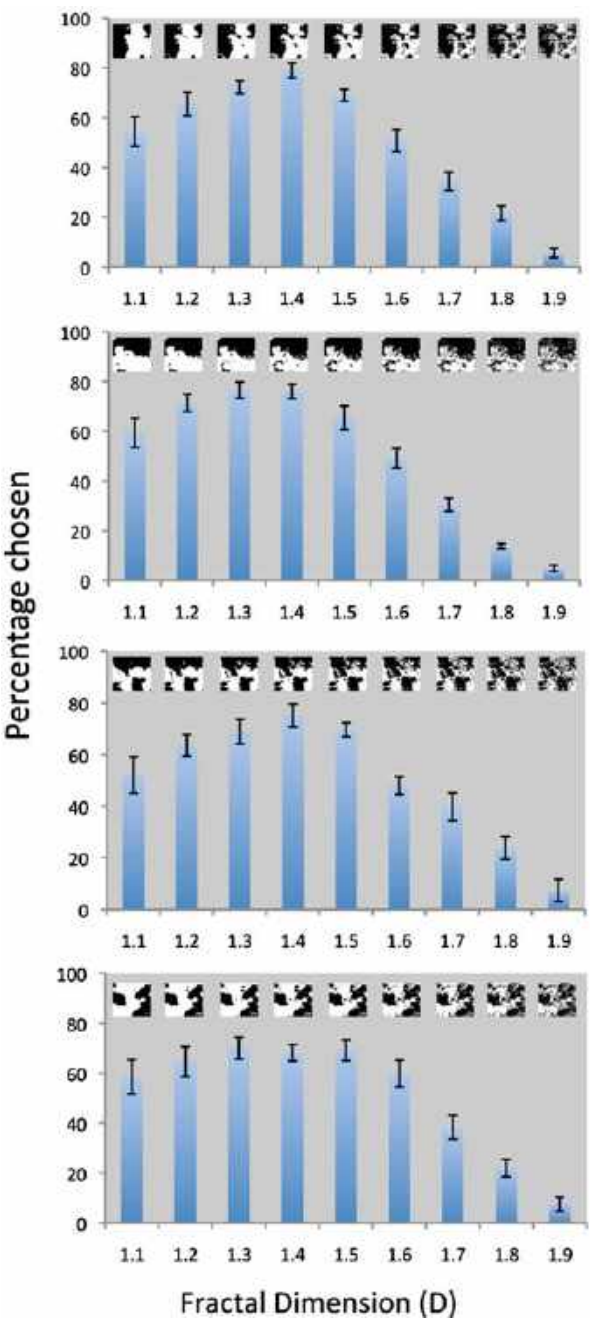
Over the past two decades, fractal aesthetics experiments performed by us and other groups on statistical fractals have shown that preference for mid- D fractals is “universal” rather than dependent on specific details of how the fractals are generated. Fractal aesthetics experiments also confirm that preference for mid- D complexity occurs for a wide variety of fractal image types, including fractal “composites” (natural scenes composed of multiple fractal objects) [40] and that this preference is already evident by the age of 2 [41]. Figure 45.6 shows example results for the coastline stimuli of Fig. 45.3 [13]. Preference was determined using a “forced choice” technique in which participants were shown a pair of images with different D values on a monitor and asked to choose the most visually appealing image. In these experiments, all of the images were paired in all possible combinations. The presentation order was fully randomized, and preference was quantified in terms of the proportion of times each image was chosen.

The panels are for four different “configurations” (i.e., different random arrangements of the fractal pattern). The peak preference shows remarkable consistency despite superficial variations in the 4 families of fractal. Figure 45.7 further emphasizes the “universality” of the preference for mid-complexity fractals by comparing 3 of the fractal types (coastline, edge, and grayscale images) shown in Fig. 45.3 [17]. Our experiments also find a direct correlation between the observer’s enhanced capabilities (based on their abilities to detect and discriminate fractals) and preference [32].

In addition to these laboratory-based behavioral experiments, a computer server has been used to send screen savers to a large audience of 5000 people. New fractals were generated by an interactive process between the server and the audience, in which users voted electronically for the images they preferred [42]. In this way, the parameters generating the fractal screen savers evolved with time, much like a genome, to create the most aesthetically preferred fractals. The results re-enforced the preference for mid- D fractals found in the laboratory-based experiments. The majority of research into responses to visual fractals have presented the images on the 2-dimensional planes of computer monitors. Funded by NASA, the skin conduction relaxation studies were a rare exception when the images were presented to participants on the walls of a mock-up space laboratory [36]. We expect that the fractal fluency effects will be amplified in 3-dimensional environments. A recent virtual-reality experiment indicates that spatial awareness skills are heightened in fractal terrains. When participants were instructed to navigate an avatar to find an object randomly placed within a virtual landscape, accuracy and completion speeds peaked for the mid-complexity landscapes predicted by the fluency model [43].

Intriguingly, when fractal images are projected on the walls of a room rather than displayed on monitors, the preferred D shifts to slightly higher values ($D \sim 1.6\text{--}1.7$) [44]. The observer then sees the fractal surrounded by the blank surface of the

Fig. 45.6 Visual preference for computer-generated fractal patterns. For each of the 4 panels, D is plotted along the x-axis and the preference on a scale of 0–100 is plotted along the y-axis. This scale is based on the number of times an image is chosen as being the most preferred (see the main text for details). Each of the 4 panels uses a different fractal configuration to investigate preference. The fractal images are shown as insets in each panel



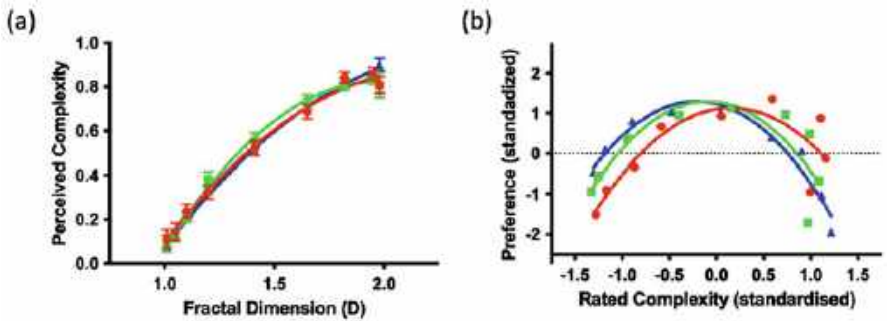


Fig. 45.7 (a) Perceived complexity plotted against fractal dimension for the coastline (green), edges (blue), and grayscale (red) images of Fig. 45.3. (b) Preference plotted against perceived complexity (both plotted on standardized scales)

wall. Introduction of Euclidean simplicity increases the tolerance for high fractal complexity. This experiment serves as a warning for future fractal studies aimed at biophilic applications. Because practical applications will embed fractals within artificial environments, we will need to adapt them (bio-inspiration) rather than simply copy them (biomimicry). For example, when sitting in a room surrounded by simple walls, the preferred D value will be higher than when the observer walks through a forest and is engulfed by other fractals. Experiments investigating the importance of matching city skylines to the backdrop of fractal mountains [45] further emphasize the importance of viewing context.

Clearly, as we shift from fundamental to applied research, the specifics of the individual spaces along with the needs of the individuals who occupy them will be crucial. Although the overall population prefers mid- D values, there are 3 subgroups that exhibit distinct preferences. As shown in Fig. 45.8, whereas the majority's preference peaks at mid- D , just under 25 percent of observers are instead “sharpies” (preferring high D) and a similar number are “smoothies” (preferring low D) [17]. One recent study proposed that genetic factors might influence the fractal aesthetics of individuals [46]. Furthermore, the art works of Willem De Kooning decreased in D value as he descended into Alzheimer's, highlighting the influence of neurological conditions on fractal fluency and therefore on fractal aesthetics [47]. It will also be intriguing to explore if there are underlying personality traits that characterize the smooth and sharpie subgroups. For example, it has been suggested that creative people might have a preference for higher D values [48]. Some studies show that urban versus rural living along with aging can shape fractal preference, indicating that adaptation during our lifetimes might also be a factor [49]. Consistent with this view, the D values of Jackson Pollock's paintings increased during his career, possibly as a result of adaptation [50–53]. Clearly, one fractal will not fit all when it comes to the aesthetics associated with fractal fluency!

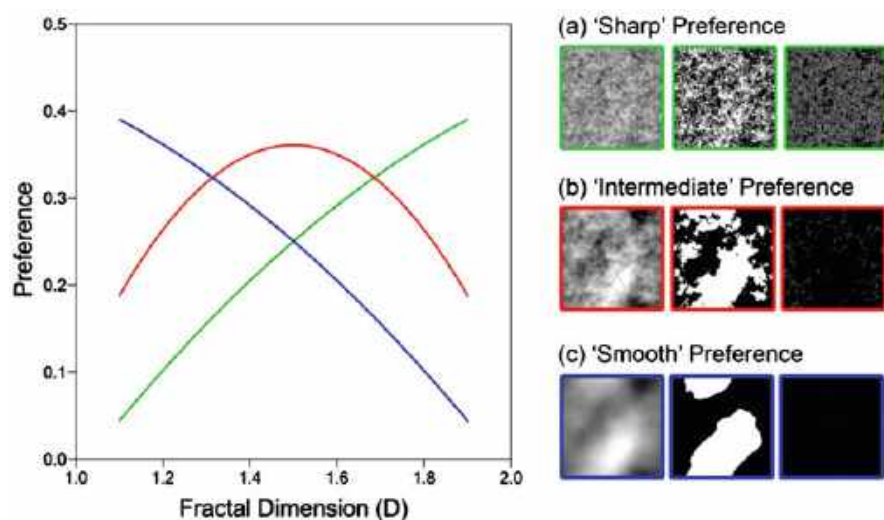


Fig. 45.8 Visual preference for computer-generated fractal patterns. Left. Preference dependencies on D for “smoothies” (blue), “intermediates” (red), and “sharpies” (green) [17]. Preference is quantified on a scale spanning 0–1 based on the number of times an image is chosen as being the most preferred. Right. Example preferred fractal images for (a) “sharpies,” (b) “intermediates,” and (c) “smoothies”

45.5 Beyond the Visual: Fractal Fluency in Touch and Sound?

The physicality of real-life, three-dimensional spaces presents opportunities to move beyond the visual sense. Inspired by the idea of synesthesia, it is possible that mid-complexity fractals might also hold special significance for tactile experiences. Three-dimensional printers now allow computer-generated patterns to be printed (“contour-crafted”) as physical objects and artists such as Daniel Della-Bosca have used them to construct fractal sculptures [54]. In discussions with Della-Bosca, we pictured rooms incorporating fractal surfaces for passersby to touch. Mandelbrot had previously asked: “In order to understand geometric shapes, I believe you have to see them.” Della-Bosca took this thought process one step further by asking “what happens if you touch them too?” [54].

Fractal surface textures have been used in previous studies of aesthetics to gauge perceived roughness [55] as well as their aesthetic value [17]. As expected, visual preference was found to be greatest for the mid-complexity surfaces. However, the surfaces in both studies were presented on a monitor and visually inspected as opposed to being touched. In our ongoing experiments, the terrains shown in Fig. 45.3 (left) are printed to investigate the visual and tactile impacts of these physical terrains [56]. As shown in Fig. 45.9, these fractal sculptures are printed onto the face of 10 by 10 by 1 cm hard, synthetic blocks. They are presented to participants on a

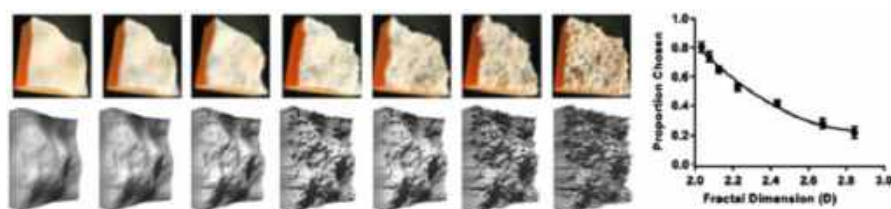


Fig. 45.9 Left: Computer-generated three-dimensional fractal images (bottom row) and their printed tactile surfaces (top row). The fractal dimensions of the tactile surfaces range from 2 (left) to 2.8 (right). Right graph: Preferences plotted as a function of fractal dimension for the sculptures. Preference is quantified on a scale spanning 0–1 based on the number of times a sculpture is chosen as being the most preferred

tabletop, and an adjustable occluder is used to hide the sculptures from view for the purely tactile experiences.

To discuss the dependence of participants' responses on D , we return to the discussions of the images in Figs. 45.1 and 45.2. Because these fractal patterns are embedded in a 2-dimensional plane, their D values lie below the $D = 2$ value associated with a fully filled (i.e., uniformly black) plane. This reasoning extends to the middle “coastline” and “mountain” images of Fig. 45.3. Embedded in horizontal and vertical planes, respectively, they are described by a mid-complexity value of $D = 1.5$. However, the terrain's surface shown in Fig. 45.3(left) is not confined to a 2-dimensional plane – it spreads out into a volume and consequently has a raised dimensionality of $D = 2.5$ [57]. The physical sculptures of Fig. 45.9 therefore all have physical fractal dimensions lying between 2 and 3, with a mid-complexity in these experiments corresponding to 2.5.

Backing up the fractal fluency experiments performed using images displayed on monitors, the purely visual inspections of the sculptures reveal a preference for mid-complexity D values. However, the tactile experiments reveal a preference for smoother surfaces closer in dimension to $D = 2$ (Fig. 45.9). Previous haptic studies have shown that roughness is a particularly important descriptor [58–61]. Consistent with our fractal surface study, ratings of everyday materials were found to have a negative relationship between perceptions of pleasantness and roughness [62].

However, this does not necessarily suggest that the visual and tactile domains have no common basis for how fractal statistics influence aesthetic perception. Indeed, it is possible that the range of complexity variations between the visual and tactile domains differs; in other words, the stimuli between the two sensory domains might not have been cross-modally matched. Notably, the low D tactile stimuli still maintain a slightly rough and irregular surface when touched even though they look smooth, leading to a more compressed range of variations in the tactile domain compared to the visual domain. Furthermore, it is worth noting that, although the majority of participants preferred to touch low D surfaces, a small proportion showed a greater preference for intermediate or for high D stimuli. This

emphasizes that “smoothies,” “intermediates,” and “sharpies” play roles for both tactile and visual responses.

That all said, the differences in tactile and visual preferences might also be explained by extra factors influencing the tactile experience. If fractal fluency was the only factor, it would be predicted that the abundance of mid- D fractals would make them easier to detect and therefore be more preferred for both the visual and tactile realms. However, physical interactions with the surface present the potential for harm. The sharpness of mid-to-high D surfaces might therefore cause low D surfaces to dominate the aesthetics.

Real environments also offer the opportunity to move beyond static fractals to explore spatiotemporal qualities. Nature’s dynamic fractals include moving ripples of water, tree branches swaying the breeze, flickering flames, and clouds moving across the sky. These time variations in dynamic fractals might be expected to attract and maintain the observer’s attention to a higher degree than their static equivalents and might therefore boost the favorable fractal fluency effects. For example, the shifting sun could cast fractal shadows from trees across a room during the day and clouds could create extra variations on shorter timescales. This idea of projecting nature’s movements into rooms is central to the biophilia movement [63]. However, introducing time variations into the static images of Fig. 45.3 could also be employed to achieve a similar dynamic impact.

To consider the visual complexity of dynamic fractals in more detail, we return to the role of D in quantifying the spatial frequency content of fractal patterns. Employing Fig. 45.3 (right) as an example, when this grayscale image is decomposed into its component spatial frequencies using Fourier analysis, the intensity amplitudes follow a power law relationship $A \sim f^{-\alpha}$ as a function of spatial frequency f . This power law generates the scale invariance of the fractal’s repeating patterns, and its exponent α is inversely related to the pattern’s D value [57]. Studies of dynamic natural scenes have shown that this power law relationship not only holds for spatial frequencies but also extends to temporal frequencies. The measured temporal α value approximates to 1 [64–66], which translates mathematically to the mid-complexity of $D = 1.5$ for these stimuli [57]. Thus, whereas the physical terrains of Fig. 45.3(left) extend beyond the horizontal (x–y) plane along a third spatial (z) axis representing height, the dynamic fractals extend their fractality along an axis representing time. In other words, the observer now experiences fractal variations both when moving between locations in the spatial (x, y) pattern and also when positioned at one location and watching the pattern evolves with time.

Altering the temporal α (and therefore the temporal D) changes how much “energy” is found in certain temporal frequencies in the dynamic pattern. For example, as α approaches 0, energy is distributed equally across all temporal frequencies resulting in a large rate of change at all frequencies. As α increases, more energy is distributed to the lower frequencies, resulting in large rates of change at low frequencies and relatively smaller changes at high frequencies. This is most clearly perceptible in the speed that the pattern changes over time, with low α stimuli generating a fast-frequency effect similar to “flicker.”

For physical fractals in real environments, the spatial fractals influence their temporal fractal variations. Taking the mechanics of a tree as an example, longer branches will experience more inertia than shorter branches when subject to the same force of the wind. Thus, shorter branches will display faster temporal frequencies. In this way, the temporal D values will be determined by the structure's spatial D values along with driving factors from the surrounding environment such as the magnitude and direction of the wind's force. In contrast to this interdependence of the spatial and temporal D values for nature's fractals, an advantage of studying synthetic spatiotemporal fractals is that the two fractal dimensions (spatial and temporal) can be adjusted independently [64].

Billock et al. [64] investigated the ability of the visual system to tune to the fractal $A \sim f^{-\alpha}$ spectrum in both space and time and demonstrated visual sensitivity to the condition predominant in nature (i.e., approximately to $\alpha \sim 1$). In terms of aesthetics, Toet et al. [67] asked participants to label real-world natural temporal textures such as moving water in terms of pleasure. Although the study did not measure α values, the movie images were rated for spatial and temporal characteristics. Significantly, pleasure was found to be negatively correlated with speed (i.e., rate of change of the image). Our own recent investigations of the spatiotemporal fractals of Fig. 45.3 [68] also highlight that visual preference does not align with the visual system's sensitivity. While sensitivity is highest for intermediate modulation rates (temporal $\alpha \sim 1.25$, which is abundant in nature), consistent with fractal fluency expectations, the most preferred stimulus has high temporal values of $\alpha \sim 2.25$.

Figure 45.10a shows this preference for dynamic fractals when plotted against D . Figure 45.10b shows perceived complexity ratings in order to confirm that this rises with increasing D , consistent with the original studies performed on static spatial fractals [16, 17]. Thus, moving beyond static fractals by introducing dynamic experiences doesn't impact the perceived complexity generated by D (Fig. 45.10 confirms that this also holds when introducing tactile and, to be discussed later, auditory experiences). Having excluded changes to perceived complexity as a cause for the shift in preference to lower D values, one possible explanation relates to safety. While sensitivity is aligned with abundance through the fractal fluency effect, preference for temporal variations might be based on slower stimuli signaling safer and therefore more preferred environmental factors. This deviation of the aesthetic condition away from the fractal fluency D value due to additional safety factors is similar to that described earlier for tactile experiences.

Nature's dynamic fractals are rarely quiet. We are all familiar with the sounds of trickling water and tree branches creaking in the wind. Fractal sounds in nature even extend to bird songs. Given that nature's fractal sounds are generally regarded as pleasant, it is logical to expand fractal aesthetics studies to the auditory domain to include noises and, ultimately, music. Just as the spatial D values of nature's fractal structures influence their temporal fractal variations, they also influence the fractal noises they induce. Returning to the tree as an example, the fractal distribution of gap sizes between the branches will influence the temporal frequencies of the noise of the wind passing through them. Similarly, the D values of the fractal sculptures in Fig. 45.9 will influence the fractal noise generated by trickling water across their

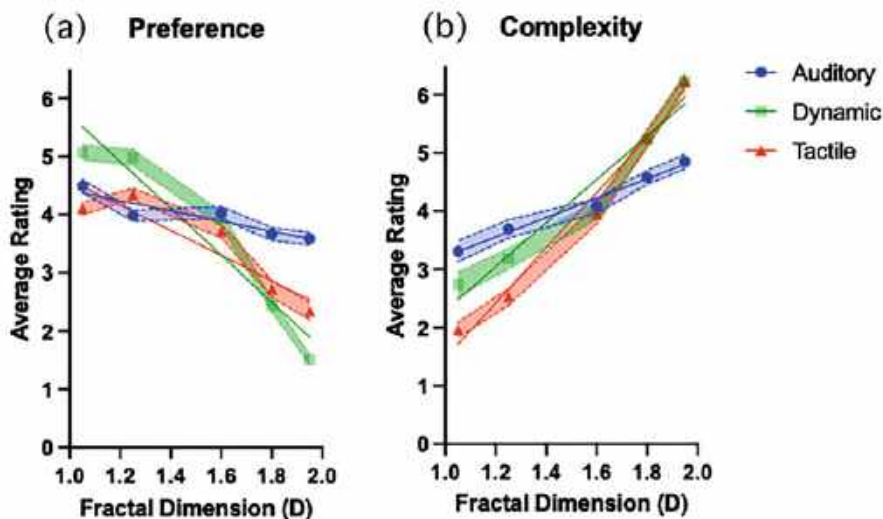


Fig. 45.10 Ratings of preference (a) and complexity (b) as a function of D for tactile, dynamic, and auditory fractals

surfaces. Fractal noises are labeled using an analogy with the color spectrum of light [15]. For example, white power laws with $\alpha = 0$ have equal power at all frequencies, equivalent to white light featuring contributions from all visible frequencies. Pink power laws have a higher α value, indicative of an increase in power at lower frequencies analogous to pink light.

Remarkably, fractal noises are not restricted to those generated by natural processes but can appear in artificial auditory experiences that are sufficiently complex to assume fractal variations [69]. A variety of music, sound, and talk show programs on the radio have been shown to have a fractal $1/f^\alpha$ frequency spectrum. Hour-long segments from various radio programs [70] including speech, classical music, and rock music sounds were found to have an average α value similar to that generated by natural scenes. However, the significance of fractal measures such as D remains unclear in terms of determining the aesthetics and perception of different genres of music. For example, some studies have found that various genres differ in their D value [71–73], while others have found no discernible consistencies in the role of fractal-scaling statistics at all [74].

In addition to these fractal spectral noise studies, there are many other ways to incorporate fractality into music. For example, fractal rhythms can be introduced through drumbeats. Other research has focused on employing $1/f^\alpha$ synthetic melodies [75] to examine fractal melodicty and preference. That is, when music is presented in its most basic form – a sequence of single notes of the same duration spaced equally to form a melody – does a $1/f^\alpha$ manipulation of this sequence alter its perceived aesthetic value? Applying this approach to simple melodic sequences (see, e.g., Fig. 45.11), investigations of α values ranging from approximately 0–2



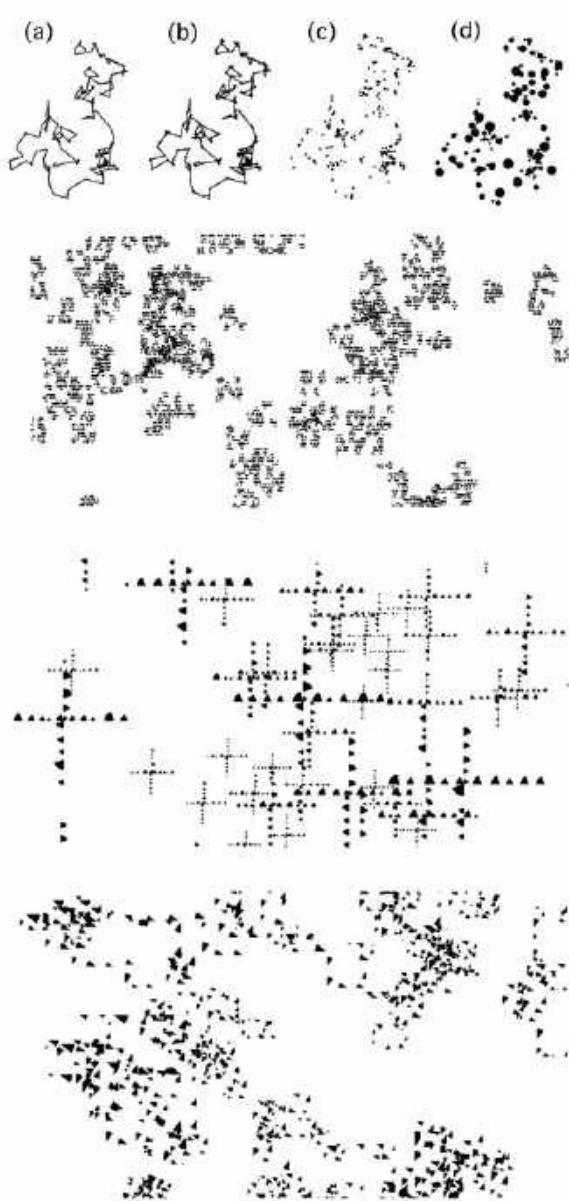
Fig. 45.11 Examples of synthetic $1/f^\alpha$ melodies for α values of 0.25, 1.25, and 2.25

revealed a preference for the intermediate (equivalent to “natural”) values [70, 75]. However, preference ratings from our own studies (Fig. 45.10) exhibit a preference for high α (i.e., low D) similar to that revealed for the tactile and dynamic fractal studies. This shift for the auditory realm might well be inherited from the dynamic realm (since low-frequency spatial motions are more likely to generate low temporal frequencies). Finally, we emphasize the preference variations for the “smoothie” and “sharpie” subpopulations should be expected for the auditory realm. Indeed, Güclütürk and van Lier [76] measured participant liking ratings for 25 song excerpts from a variety of musical genres that had been previously rated for instrumental complexity. The analysis of preferences clustered the participants into 2 relatively equal-sized groups, where 1 group showed a preference for more complex songs and the other group showed a preference for more simple songs.

45.6 Fractal Applications

The Science Design Laboratory (SDL) was founded in 2017 to translate the science of fractal fluency into applications for indoor and outdoor architectural environments [77, 78]. Initial fractal designs focused on interior spatial patterns. Because floors occupy such a large part of our eye’s visual field, an award-winning collection of carpet designs called Relaxing Floors was developed for the Mohawk Group, one of the world’s largest carpet manufacturers [77]. As shown in Fig. 45.12, the eye’s fractal trajectories provided the underlying layout for the carpet designs. A circular “seed” pattern was then inserted at the locations between trajectories, and its size was scaled according to the length of the preceding trajectory. Then each circle was replaced by a fractal pattern which could be varied between designs.

Fig. 45.12 Top: The pattern generation process for the Relaxing Floor designs. (a) The fractal pattern of the eye’s trajectories, (b) the fractal pattern when circular “seeds” are added to the locations between these trajectories, (c) the fractal pattern when the trajectories are removed, and (d) the fractal pattern when the sizes of the circles are scaled based on the length of the previous trajectory. Bottom: three complete carpet designs generated by replacing the circular patterns by three different fractal patterns. (Images courtesy of *Fractals Research* and *13&9 Design*)



The D value of the overall pattern generated by this process was informed by the $D = 1.6\text{--}1.7$ target range suggested by the aesthetics experiments that projected fractal images into rooms [44]. Behavioral studies performed on the Relaxing Floor patterns show this complexity to be very effective at balancing engagement,

preference, refreshment, and relaxation qualities for a broad group of observers [79]. Specifically, these patterns have the greatest agreement across individuals in terms of their preference, engagement, and refreshment while also maintaining relaxing effects.

Emphasizing the versatility of statistical fractals, cutting the pattern into tiles and randomly re-arranging them did not disrupt the fractal character nor significantly shift their D values. This has important consequences because many carpets in large spaces ranging from airports to hotels are installed as tiles rather than as continuous carpets.

Figure 45.13 demonstrates a key strategy – development of versatile designs that can be used for multiple applications. In this case, the applications are carpet patterns (in the *Mohawk* collaboration), wall patterns used to disperse light throughout a chapel (in a collaboration with *INNOCAD Architecture*), computer screen savers (the latter are being made available for free personal use during the pandemic), and ceiling tiles (in collaboration with *Fact Design*). The ceiling tile application demonstrates a second key strategy – that patterns should, when possible, provide multiple functions. In this case, the patterns are embossed in the ceiling tiles, offering the potential to create an aesthetic impact coupled with the established noise-dampening capabilities of fractal surfaces. Inspired by this strategy of combining aesthetics with other favorable functions, a psychology-engineering project recently incorporated fractal aesthetics into solar panels [80].

Fractal window patterns offer further possibilities for multi-functionality [10, 81]. The fractal pattern can be used as a window shade to obscure an unattractive view, to provide shade, and to also cast dynamic fractal shadows across a room. For open windows, the shade can also generate a fractal breeze. In addition to the fractal variations in light, it can therefore provide analogous variations in heat and air currents for the room's occupants. The fractal designs are also being developed for outdoor surfaces. In addition to their visual and textural aesthetics, these surfaces offer the possibility of floor surfaces that reduce slipping (based on fractal shoe soles that have been patented because of their proposed superior traction). Interestingly, the basic pattern of Fig. 45.13 reminds many viewers of Morse Code patterns. With their sequences of dots and dashes, Morse Code can be both visual and sonic. Consequently, we are using computers to convert the visual stimuli of Fig. 45.13 into the auditory equivalents.

45.7 Conclusions

In this chapter, we have reviewed experiments that support the “fractal fluency” model which states that human senses have become fluent in the visual, sonic, and tactile language of fractals through evolutionary exposure. Reflecting the focus of previous research, most of our discussion centered on fractal fluency effects in the visual realm. In addition to biophilia's well-known effects, our own research of visual fractals has demonstrated increases in detection sensitivity, attention,



Fig. 45.13 The SDL fractal pattern of Figure 45.12 employed as (top left) a floor design at the University of Oregon, USA (image courtesy of *Fractals Research*, *13&9 Design*, and the *Mohawk Group*), (top right) as light patterns in the Fractal Chapel in the State Hospital in Graz, Austria (image courtesy of *Fractals Research*, *13&9 Design*, and *INNOCAD Architecture*), (bottom left) as a design for ceiling tiles, Linz, Austria (image courtesy of *Fractals Research*, *13&9 Design*, *INNOCAD Architecture*, and *FACT Design*), (bottom right) as a design for computer screen savers. (Image courtesy of *Fractals Research* and *13&9 Design*)

visual performance (e.g., pattern recognition and navigation), aesthetic appeal, and stress reduction. Consistent with fractal fluency, these effects peak at the mid- D complexity prevalent in natural scenery.

Extending the experiments to dynamic visual fractals and to tactile experiences reveals a more subtle dependence of responses to the D value of the fractal stimulus. Whereas sensitivity to the dynamic and tactile fractals peaks at the mid- D complexity expected from the fluency hypothesis, aesthetic preference peaks for lower complexity fractals. This might be explained by extra factors that influence fractal aesthetics in real-life physical environments. Whereas aesthetics aligns with detection in the visual realm, safety cues might drive the aesthetic appreciation away from this condition when physical interactions are involved. For example, in the



Fig. 45.14 Examples of high- D (top), mid- D (middle), and low- D (bottom) scenery

tactile realm, the potential harm from the sharp texture of high D surfaces might drive the aesthetics to lower D smooth surfaces. Similarly, the high frequencies of high D fractal noises might signal the approach of dangerous conditions such as storms and earthquakes.

Seasonal changes can induce small shifts in the D value of the visual fractals in some scenes. For example, as leaves detach from trees in the Fall, the fine-scale patterns of the exposed twigs raise the D values of the trees. Whether these small shifts contribute to seasonal mental disorders has not been investigated. Looking to the future, climate change might induce larger and more permanent shifts in D . As an example of the human impact on fractality, picture the mid-complexity scene of trees spread across a savannah (a landscape in which our ancestors evolved) (Fig. 45.14). If climate change reduces the trees to a barren desert, then the one-dimensional line of its horizon will shift the scene out of the

Fig. 45.15 A fractal atmosphere in which the room occupant is immersed in fractal floor patterns, fractal wall patterns, and fractal music. (Image courtesy of *Fractals Research, 13&9 Design*, and the Mohawk Group)



aesthetic zone. Similarly, forests that become filled in with fine-scaled branches will shift to higher D and, again, out of the zone. Cloud patterns also contribute to the mid-complexity of scenery, and any increased likelihood of uniformly blue or gray skies will similarly cause a shift away from the aesthetic zone.

Even if we can evade climate change, modern life forces people away from nature's fractals and into urban scenery. We have therefore also considered strategies for transferring the fundamental science of fractal fluency to applications in the built environment. Imagine a future in which we immerse building occupants in a "fractal atmosphere" of visual, auditory, and tactile experiences – potentially inducing an emergent experience that we have all evolved to expect and appreciate (Fig. 45.15). Based on the examples presented in this chapter, we already have the potential to walk into a room in which the fractal ceilings dampen the noise, the fractal window shades provide an optimal breeze, fractal solar panels deliver efficient energy, and all of their patterns combine to create a stress-reducing visual environment analogous to the complex scenes of nature.

Finally, although we have concentrated on visual, sonic, and tactile fractals, it is interesting to note that fractals have consequences for smells and taste. For example, the fractal airflows induced by fractal window patterns will influence not only wind variations in the room but also the smells they carry. Fractal objects are also known for their large surface areas which might promote taste. We hope that this chapter highlights the diverse range of future opportunities for studying human responses to fractals across many senses and for applying these to provide positive environments for the occupant. Without a doubt, the fractal geometry of nature influences the fractal geometry of the brain and vice versa.

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Chapter 46

Cognitive and Neural Representations of Fractals in Vision, Music, and Action



Mauricio de Jesus Dias Martins

Abstract The concept of fractal was popularized by Mandelbrot as a tool to tame the geometrical structure of objects with infinite hierarchical depth. The key aspect of fractals is the use of simple parsimonious rules and initial conditions, which when applied recursively can generate unbounded complexity. Fractals are structures ubiquitous in nature, being present in coast lines, bacteria colonies, trees, and physiological time series. However, within the field of cognitive science, the core question is not which phenomena can generate fractal structures, but whether human or animal minds can represent recursive processes, and if so in which domains. In this chapter, we will explore the cognitive and neural mechanisms underlying the representation of recursive hierarchical embedding. Language is the domain in which this capacity is best studied. Humans can generate an infinite array of hierarchically structured sentences, and this capacity distinguishes us from other species. However, recent research suggests that humans can represent similar structures in the domains of music, vision, and action and has provided additional cues as to how these capacities are cognitively implemented. Using a comparative approach, we will map the commonalities and differences across domains and offer a roadmap to understand the neurobiological implementation of fractal cognition.

Keywords Recursion · fMRI · Cognition · Language

46.1 Introduction

Hierarchies, intuitively understood as structures with multiple levels of organization, are common in nature. The most paradigmatic example of a hierarchy is that of a tree, with a trunk, branches, sub-branches, and leaves. In a tree, several branches originate from the trunk, several sub-branches originate from each branch, and

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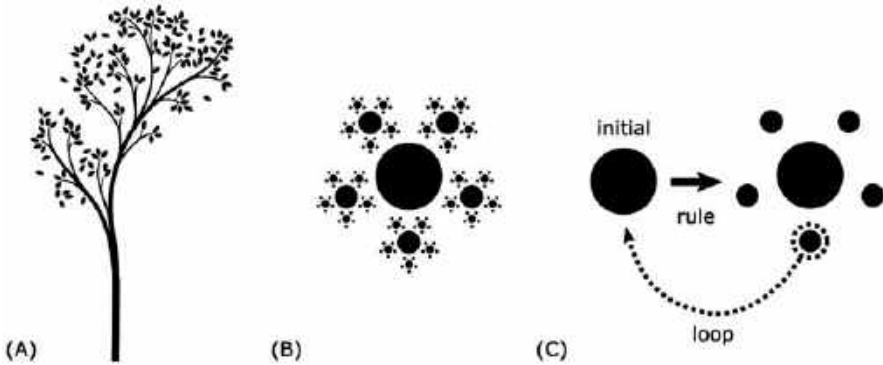


Fig. 46.1 (a) Hierarchies are tree-like structures in which an element at a level $L(i)$ (e.g., trunk) connects with two or more elements at a subordinate level $L(i + 1)$ (e.g., branches); (b) fractals are structures in which this “hierarchical branching” can extend to infinity by using transformation rules consistent across multiple levels $L(1)$, $L(2)$, $L(3)$...; (c) to generate fractals, we need an *initial* condition $L(0)$, a transformation *rule* $\{L(i + 1) \rightarrow L(i) + \text{some operation}\}$, and a *loop*. In this case, the initial condition $L(0)$ is a black circle, the transformation rule generates $L(i + 1)$ by “adding 5 circles at particular angles and distance relative each circle in $L(i)$ ”, and then the same rule can be applied to each newly added circle, recursively, to generate fractal (b)

several leaves sprout from each sub-branch. This “branching” property, in which one item at the level of organization $L(i)$ connects with two or more items at a subordinate level $L(i + 1)$, is the key feature of all hierarchies (Fig. 46.1a) [64]. It is therefore no surprise that the metaphor of a tree is used to describe hierarchies in multiple domains.

Fractals are a kind of hierarchy that can have an unbounded number of structural levels. When trying to measure the length of coast lines, Mandelbrot understood that the measured length depended on the size of the ruler [33]. When using smaller rulers, more and more infinitesimal segments of the coastline could be revealed increasing its length potentially to infinity. The crucial insight of Mandelbrot was that these structures with infinitesimal details often displayed the property of self-similarity, i.e., they often have regularities across levels of organization, which allow for parsimonious formal descriptions (Fig. 46.1b). In other words, Mandelbrot understood that we could describe natural objects with unbounded hierarchical depth using simple sets of rules. To do so, we need a description of initial conditions, certain transformation rules, and the ability to apply these rules recursively, i.e., to feed the output of the operation back to itself as input (Fig. 46.1c).

Defined as such, we can find fractal structures not only in coast lines and trees but also in bacteria colonies [44], worm behavioral responses [2], tissue structure (e.g., human lungs and blood vessels), and physiological signals such as electroencephalogram (EEG) and electrocardiogram (ECG) time series [57, 63]. In all these domains, we can find complex multilayered signals with structural self-similarity. However, the fact that these signals can be generated by natural phenomena does not make them immediately relevant or transparent for the human

cognitive level. Our fractal EEG or ECG time series is not immediately transparent to our perception without sophisticated technology. Similarly, the fact that bacteria colonies can grow in a fractal pattern does not mean that this structure is represented by the bacteria themselves. Hence, from the perspective of cognitive science, the relevant question is whether human and animal minds can *represent* the kinds of recursive processes that generate fractals [31, 34]. In the next section, we will discuss how the representation of these generative processes can be experimentally investigated.

46.1.1 *Recursion in Human Cognition*

Humans are exceptional in their ability to generate complex and unbounded hierarchical structures, across a variety of domains, including the social domain, visuo-spatial processing, action planning, and language. In all these domains, we are able not only to recognize underlying hierarchical structures but also to extend them beyond the given in a way that is consistent with the previous levels. Imagine the complex hierarchical structure denoting a national defense system composed of several armies, battalions, etc. We can always go beyond the given and create supranational command levels dominant over (but consistent with) the national structures (think of NATO). Similarly, in the language domain, we can create an infinite set of structured expressions from a finite set of words and hierarchical rules. For example, we could imagine an arbitrarily long sentence *S* and embed it within the sentence “I think that [*S*].” We could then embed the resulting sentence in “Peter thinks that [I think that [*S*]],” and so on.

The cognitive and neural mechanisms underlying the capacity to recursively embed hierarchies within higher-order structures – or to add new levels beyond the given – are an exciting and very active topic of research ([34], for a review). For example, within the domain of language, the generative syntax tradition proposes that there is a core recursive cognitive system that allows the generation of unbounded complexity from a finite set of primitives and rules [6, 18, 20, 47, 48]. These syntactic rules govern how different syntactic categories (nouns, verbs, etc.) can be combined to generate increasingly complex sentences. Consider the following system to generate noun phrases (NP), verb phrases (VP), and sentences (S) using determinants (D), nouns (N), and verbs (V):

1. NP $\rightarrow$ (D) N
2. VP $\rightarrow$ V (NP)
3. S $\rightarrow$ NP VP

With these rules, we can generate the noun and verb phrases:

NP $\rightarrow$ [the_D girl_N]_{NP}

VP $\rightarrow$ [opened_V the door_{NP}]_{VP}

and combine them to form a sentence:

$S \rightarrow [[\text{the girl}]_{NP} [\text{opened the door}]_{VP}]_S$

Unlike in the visual domain, in language, we rarely have self-similarity at the surface level. While sentences like “[I think that [Peter thinks that [Susan thinks that [S]]]]” can exist, they are rare. Mostly, self-similarity exists at a deeper conceptual level. For instance, the compound noun “[[[student]film]committee]” can be translated as NPs embedded within NPs ([[[NP]NP]NP]) using the generative rule $NP \rightarrow [[NP]NP]$. These rules, like fractals, involve *direct recursion*, in which both sides of the symbol “ $\rightarrow$ ” contain objects of the same category. However, even such direct recursion is rare in language [3]. Most commonly, sentences are generated using *indirect recursion*, i.e., via the combination of a set of syntactic rules, which do not necessarily have similar categories on both sides of “ $\rightarrow$ ” [55].

Here is how it works in practice. We can use the rule system above to generate the sentence S' :

$S' \rightarrow [[\text{the girl}]_{NP} [\text{kissed the boy}]_{VP}]_{S'}$.

If we modify rule 1 to allow embedding of S' within a NP, we create an indirect loop between NP and S because NP can now contain a S ($NP \rightarrow NS'$) and S can contain a NP ($S \rightarrow NP VP$). With this *indirect recursive* loop, we can embed S' within S = “[[[the girl]_{NP} [opened the door]_{VP}]_S”, because the latter contains the NP “the girl”:

$S = [[\text{the girl}[S']_{NP} [\text{opened the door}]_{VP}]_S$.

If we now rewrite S including S' , and use the appropriate pronoun, we obtain the following sentence (see Fig. 46.2 for a graphical description):

$[[\text{the girl} [\textit{who kissed the boy}]_{S'}]_{NP} [\text{opened the door}]_{VP}]_S$.

This system can be applied to generate sentences of unbounded depth by allowing sentences to be embedded within other sentences.

The core point, made in a seminal paper by Hauser, Chomsky, and Fitch, is that “Language is based on a recursive generative procedure that takes elements from [...] the lexicon, and applies repeatedly to yield structured expressions, without bound” [20]. Crucially, this idea of recursion in language requires more than the mathematical notion of “definition by induction” which requires initial conditions and a recursive function [46]. For example, the set of natural numbers can be generated using the initial condition $L(0) = 1$ and the recursive function $L(i + 1) = L(i) + 1$. When applied recursively, the function generates the set $\{1, 2, 3, \dots, n\}$. However, natural numbers are not hierarchical; hence, classical recursion is not sufficient to describe fractals. Recursion in language also requires that the rule allows the generation of new hierarchical levels via structural embedding (e.g., $NP \rightarrow [[NP]NP]$). Thus, when we talk about recursion in language, we are mainly referring to recursive hierarchical embedding (henceforth RHE), which is the focus of our chapter.

RHE as a procedure to generate complex hierarchies from a finite set of rules and primitive elements allows us to model other phenomena present in human

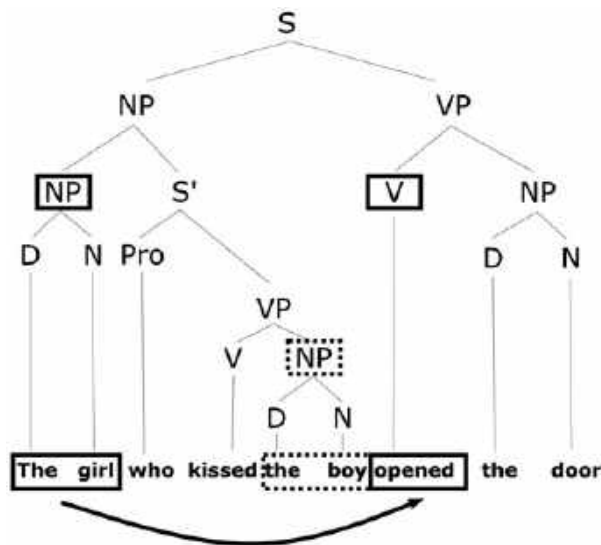


Fig. 46.2 Language is hierarchical. In the sentence “the girl *who kissed the boy* opened the door,” the “boy” is closer to the verb “opened” than the “girl.” However, competent English speakers can understand that it was “the girl” who “opened the door” and not “the boy.” This is because the NP “the girl” is closer to the verb in the hierarchical structure (solid boxes) than the NP “the boy,” which is deeply nested in the bottom of the hierarchy (dashed boxes)

cognition. Beyond the domains of language and vision [16, 20, 24, 49], music and action planning contain complex hierarchical structures that can be generated and represented using RHE [5, 11, 17, 21, 23, 30, 56].

In the following section, we will discuss the state of the art on how the representation of RHE is instantiated in the cognitive and neurobiological context. We will start with the domain of linguistic syntax and then discuss how we used musical and motor fractals to investigate the representation of RHE in these domains.

46.2 Neural Mechanisms of Recursive Hierarchical Embedding in Language

Linguistic utterances are sequential. When we speak or write, we produce words one after the other. However, this sequential order is only superficial. To understand the intended meaning of a sentence, we also need to project the underlying hierarchical structure [6]. For instance, in the sentence “The girl who kissed the boy *opened* the door,” we know that it was *the girl* who opened the door and not *the boy*, despite *the boy* being closer to the verb *opened* in the word sequence. We can infer this relationship automatically because the phrase *the girl* is closer to the verb *opened*

in the hierarchical structure (Fig. 46.2), i.e., it occupies a hierarchical level closer to the verb, in comparison with *the boy* which is more deeply nested.

When competent adult speakers process language, the hierarchical interpretation takes precedence over the sequential interpretation. Consequently, one of the most distinctive signatures of hierarchical processing in language is the capacity to process dependencies between words that are nonadjacent in the sequential structure. This behavioral signature has been extensively used in empirical research ([64], for a review).

Interestingly, children younger than 7 y.o. might not always be able to inhibit the sequential representation, and recursive hierarchical structure seems to develop slowly and requires extensive experience [53–55]. For instance, when asked to pick the “second red ball” from an array of colored circles, children often chose the ball in the second position which is also red. This corresponds to the conjunctive interpretation “second AND red ball” which has no hierarchical depth. Conversely, adults can project the structure [second[red ball]] and choose (for instance) the fourth circle in the array, but the second one that is red. The capacity to represent this nested structure is also a crucial signature used in empirical research [53–55].

When applied to neuroimaging research, these empirical signatures have yielded consistent results. The processing of linguistic syntax, hierarchical depth, and nonadjacent dependencies strongly correlates with activity in a left-lateralized network comprising inferior frontal gyrus (IFG) and posterior superior temporal sulcus (pSTS) (extending to middle and superior temporal gyri) [18, 19, 32, 66]. This pattern of activity is present in both natural and artificial languages but is modulated by the developmental stage and the amount of experience with hierarchical structure [26, 27, 61, 62]. For instance, in children, both syntactic and semantic processing seem to initially correlate with activity in left pSTS, but syntax-related activity in IFG becomes more robust later, around 9 years old [61]. Research in adults investigating the learning of artificial languages also shows that IFG activity increases with the degree of training and automatization [27]. These results suggest that there is an optimal neural machinery specialized in the processing of hierarchical structure in language. However, stable recruitment of this specialized circuitry emerges throughout ontogeny and necessitates a certain volume of experience and exposure. In other words, the pattern of brain activity during the acquisition of RHE rules in language is different to that found after extensive training and automatization.

This pattern of activity is particularly interesting because IFG and pSTS are among the brain structures that underwent rapid expansion in the human lineage in comparison with other primates [9, 52]. Most notably, the fiber tract connecting them – the arcuate fasciculus – also underwent dramatic expansion in humans and is underdeveloped in human children until the age of 7–8 y.o [52]. This could explain why the capacity to represent hierarchies in nonhuman animals is limited in scope – being available perhaps in spatial navigation and social structures [10, 60] – and depth – not going beyond a few hierarchical levels. The necessity of a mature arcuate fasciculus to support the capacity to represent hierarchies could also explain why RHE is not present in human children until relatively late.

This set of findings raises interesting questions about human evolution, but it is limited to the domain of language. Is RHE available in other domains? If so, does it recruit similar brain networks? Is there evidence for domain-general acquisition principles; and do we find similar domain specialization with experience? Answering these questions requires the extension of this research program to other domains. In the next sections, we will summarize how we used fractals in music [38, 39], action [36], and vision [37, 42] to map the cognitive and neural mechanisms supporting RHE in different domains. Using a comparative approach, we will analyze the commonalities and differences between domains and draw the larger picture concerning how RHE is implemented in neurocomputational systems.

46.3 Recursive Hierarchical Embedding in the Visual, Musical, and Motor Domains – Behavioral Research

As we discussed above, words are presented sequentially in language. However, in addition to this sequential structure, there is an underlying syntactic hierarchy that is represented by competent language speakers [64]. This kind of dual nature – sequential and hierarchical – is also present in domains like music and action. Music is composed both by a sequence of tones unfolding in time one after the other and also by an underlying hierarchy of tonal tension that goes beyond adjacent relationships [23, 28, 30]. Experienced musicians and music listeners can detect these tension dynamics along musical phrases and react with surprise when there are violations [28]. Similarly, in action, we execute motor commands sequentially, one after the other. For instance, in the process of making coffee, we must turn on the machine, add coffee to the container, add water to the container, place the cup under the dispenser, and press the button to dispense the coffee. However, in case there is already coffee and water in the respective containers, we can skip those steps without getting lost in the sequence. This is because we can represent the hierarchical structure underlying the goal of making coffee which goes beyond adjacent connections between motor commands one after the other [17, 22, 29].

fMRI research on the musical and motor domains has mostly investigated the responses elicited by the processing of tonal and action violations ([11, 17], for reviews). These violations consistently activate IFG across domains in addition to other regions more suggestive of domain specialization (e.g., sensorimotor brain areas in the motor domain and auditory areas in music) [7, 8]. The common IFG activation between language, action, and music suggests computational commonalities in the implementation of hierarchical representations across these domains ([11, 17], for reviews). However, the exact activity loci can vary. In music and action, IFG activity is stronger in the right hemisphere, and precise comparisons reveal little commonalities between domains within left IFG, hinting on a language-specialized circuitry [1, 12, 13]. This circuit micro-specialization is to be expected in the development of expert automatized processing [4]. However, a robust arcuate

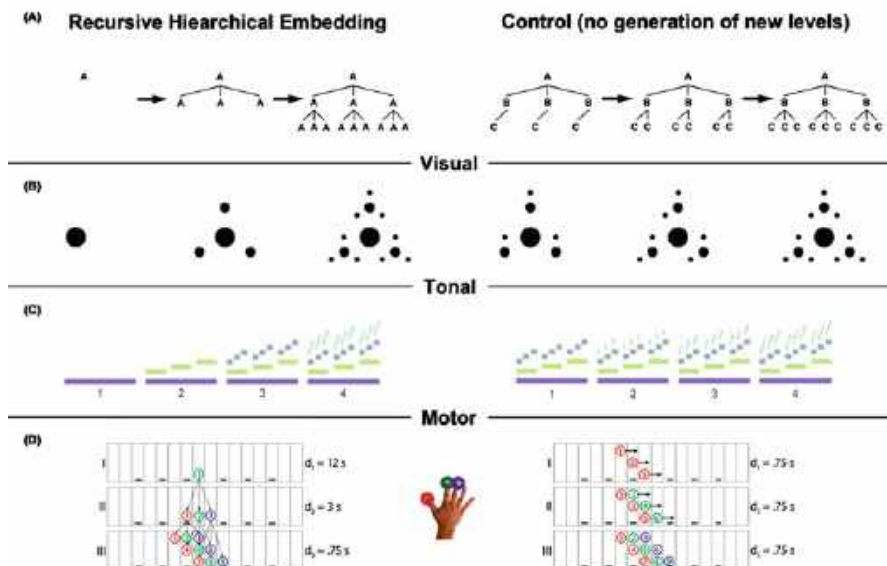


Fig. 46.3 (a) General principles of Recursive Hierarchical Embedding (RHE; left) and a control task that does not generate new levels of the hierarchy (right). On the surface, both rules generate the same final structure. However, in the RHE task, each step generates a new hierarchical level, while in the control task, each step adds elements within a fixed hierarchical level without generating new ones. (b) Visual fractals. These are generated using the rules described in A. (c) Tonal fractals. Here, the horizontal axis denotes tone duration, and the vertical axis denotes pitch. Tonal fractals are generated stepwise from an initial long-duration and low-pitch tone. Each step adds a sequence of (higher pitch and shorter duration) tones with a particular contour, and pitch relations consistent with the previous level. (D) Motor fractals. Each step adds a sequence of finger-tapping movements on a silent keyboard (with the thumb, index, and middle finger) with contour and key distances consistent to those of the previous level. The duration of each finger tap at level d_n is shorter than those of level d_{n-1} .

fasciculus connecting IFG and STS might facilitate similar computations across domains, even if the exact loci of activity differ from domain to domain.

This research based on the detection of violations is interesting but presents some confounds. The patterns of activity might reflect attention or surprise effects rather than computations specific for recursive hierarchical embedding. Moreover, it is not clear whether participants must represent the underlying generative processes in these violation paradigms. To investigate the latter more specifically, we need to tap on the capacity to extend hierarchies beyond the given. To resolve this gap, my colleagues and I used the principles of fractal geometry to create novel paradigms in the visual, music, and action domains (Fig. 46.3):

In this research program, we focus on the capacity to represent recursive hierarchical embedding, i.e., the ability to extract cross-level regularities and use them productively to generate new levels [34]. Crucially, we do not focus on how participants represent fractal structures per se but rather how participants represent

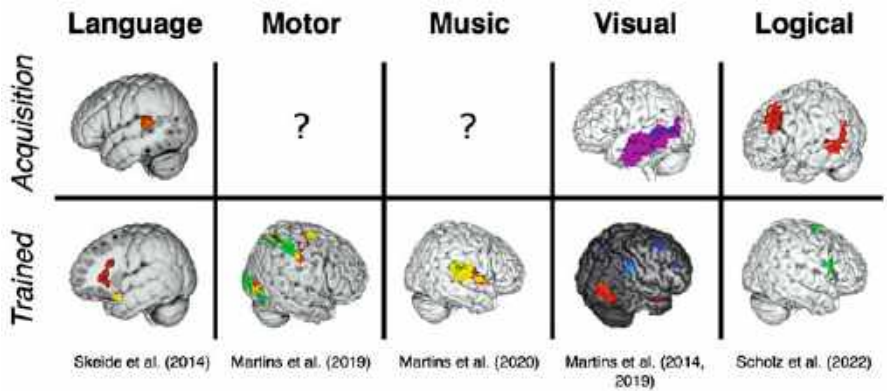


Fig. 46.4 Commonalities and differences in the processing of recursive hierarchical embedding across domains. The left posterior temporal cortex, spanning pSTS and adjacent areas, is critically involved in the acquisition of RHE across domains. After extensive training, RHE processing becomes supported by increasingly domain-specialized neural regions

their generative process. Indeed, in our control condition, the rule also generates a visual fractal, but without adding new hierarchical levels. This controls for stimuli-related features and focuses on the representation of the underlying RHE process. We expose participants to the first three iterations of the generative process and ask them to imagine, and then select (or execute), the next correct continuation (Fig. 46.4b).

We first utilized this paradigm in the visual domain [42]. We learnt that adult humans can easily represent RHE in vision and that this capacity is neither well predicted by general intelligence nor by visual-spatial working memory. However, we found that the capacity to process RHE invision correlates strongly with recursive action planning as measured by the Tower of Hanoi, hinting on shared cognitive resources across domains [35, 39]. We also found that this capacity matures slowly, and it is available in children older than 8 years, but not before [37, 41]. Similar to language, mastering RHE requires experience with simpler conjunctive visual processes, before RHE becomes available. These data suggest that there might be some cognitive specialization for RHE and shared resources across domains.

To investigate these questions, we developed music and motor tasks [35, 39] with the same underlying principles (Fig. 46.4c, d). Here, each step of the process generated an additional hierarchical level of musical tones, or of finger movements presented on a silent keyboard. Broadly, our behavioral research suggests that the ability to represent RHE shares cognitive resources across domains, when controlling for domain-specific effects. Factor analyses including musical and visual RHE tasks, their respective nonrecursive controls, and the recursive action planning Tower of Hanoi show that all recursive tasks cluster together within a factor orthogonal to general cognitive capacity [35, 39].

Recent research in other domains has further hinted on a specialized capacity to the processing of hierarchical structures. For example, accuracy in the visual RHE task was found to specifically correlate with the understanding of sentences with two levels of hierarchical embedding, such as “[the bird [who the frog [who is red] washes] laughs]” [36, 40]. In this sentence, participants need to parse the nonadjacent relationship between *bird* and *laughs*. The same visual RHE was found to specifically correlate with the ability to parse 2-level if-then logical hierarchies, again when controlling for domain-specific variance [58]. In this logical task, participants are asked to determine the numeric value of a pair of objects based on their relationship. For example, in the hierarchical task, if the first object is red and if the shape of the second object is *similar* to the first one, then the numeric value is “3”; if the first object is red and if the shape of the second object is *different* to the first one, then the numeric value is “1,” etc. If the first object is green, numeric contingencies are different. In the control – conjunctive – task the value of the pair is simply the sum of the individual values without any higher-order relation.

In summary, behavioral research suggests that there are specific cognitive resources dedicated to the processing of RHE across domains. In the next section, we will examine the neural systems that support these capacities.

46.4 Recursive Hierarchical Embedding in the Visual, Musical, and Motor Domains – fMRI Research

To investigate the neural bases supporting the representation of RHE, we applied similar paradigms across domains (Fig. 46.3). In general, we present participants with the first three steps generating a fractal structure and ask participants to imagine the correct continuation of the same process. In the visual task, they chose the correct choice from two alternatives [37, 41]; in the musical task, they decide whether the fourth step is correct or incorrect [38]; and in the motor task, they execute the correct fourth step in a keyboard inside of the scanner [36, 40]. In all cases, the control task also depicted a control process generating a fractal structure in three steps. However, the latter does not entail the generation of new hierarchical levels. This contrast, by keeping the choice and execution stimuli constant across conditions, is optimal to isolate the internal representations of RHE. Crucially, we focused our fMRI contrasts on the period in which participants are imagining the fourth step, after attending to the third (thus, a period with internal *simulation* but without external *stimulation*).

With this design, we found that different domains activate specialized regions. In particular, we find bilateral activity in the visual central stream, hippocampus, and default mode network in the visual task [15, 37, 41]; bilateral activity in STG for the musical domain [38]; and bilateral activity in sensorimotor and premotor areas, basal ganglia, and cerebellum in the motor domain [36, 40].

These findings create an interesting puzzle. Our behavioral studies clearly suggest that the implementation of RHE across domains is supported by common cognitive machinery. However, our neuroimaging results clearly show that these tasks rely on separate and domain-specialized neural circuits. To solve this apparent paradox, we recall our observation that the cognitive and neural processes necessary to acquire RHE rules might differ from those used for automatic processing after extensive training.

46.4.1 Acquisition Vs. Automatic Processing of RHE Structures

Traditionally in fMRI experiments, participants are very well trained with the paradigm or task that they execute in the scanner. To increase the signal-to-noise ratio, it is sensible to reduce the cognitive variability among participants by excluding those who perform poorly and by bringing participants' performance close to ceiling in preliminary sessions. Unfortunately, this procedure introduces a bias. As we reviewed earlier for language, the patterns of neural activity change with training and experience [27, 62]. After extensive training, neural activity might reflect fast and automatic retrieval of previously formed visual, musical, or motor schemas. This access to procedural or declarative long-term memory information might be supported by brain networks crucially distinct from those supporting the *acquisition* of RHE.

Following this line of reasoning, we devised paradigms to test which brain regions might be critically involved in the *acquisition* of RHE, by using participants without prior experience with our paradigms. In the first study, we tested patients with brain lesions in the left hemisphere and mapped the brain lesions that disturbed the process of acquisition of RHE in the visual domain [36, 40]. Because lesions can affect both accuracy and response times, we used a drift diffusion model to account for behavioral performance [51, 65]. With this model, we can combine individual accuracy and response time to generate two sensible measures of processing. The first is called drift and is the speed with which participants can accumulate information from the stimuli. The second is called boundary separation. It measures how much information each patient collects before providing an answer and can be seen as a measure of impulsivity or cognitive control.

Using these methods, our main finding was that lesions in the posterior middle temporal gyrus negatively decreased the speed of information processing specifically associated with RHE (drift) [36, 40]. In addition, we found that lesions in IFG negatively impacted cognitive control (boundary separation). This specific brain network is reminiscent to that used in the processing of language during the processes of acquisition (posterior temporal cortex) [61] and expert processing (IFG) [27, 61], respectively. Crucially, this experiment also provided behavioral evidence for shared resources between visual and linguistic RHE, when controlling

for general attention and working memory. In other words, when testing untrained participants, we find evidence for similar cognitive and neural resources used in the implementation of RHE in different domains.

In the second experiment, we tested participants on their ability to parse 2-level if-then logical hierarchies as described earlier [58, 59]. We contrasted the hierarchical condition (in which the first object determines how the second is interpreted) and the flat condition (in which the two objects are interpreted independently). Crucially, our participants were given very little training before the scanning session. Under these conditions, we found that the processing of hierarchies was supported by a left lateralized network comprising IFG and MFG, and pSTS [59]. Again, this (left lateralized) pattern of activity is reminiscent of syntactic processing in language. Interestingly, the higher the number of trails that participants completed, the higher the activity in IFG. This confirms that this region plays an essential role in expert processing of complex hierarchical stimuli. However, in this experiment, IFG expertise-related activity was confined to the right hemisphere.

Altogether, these findings suggest that the acquisition of RHE rules might be supported by cognitive and neural systems that are used across domains. However, the longer the training with these kinds of structures in each domain, the wider the separation of the neural systems supporting their automatized processing. This specialization of neural resources after training is consistent with what we know from the neuroscience of learning and is essential to the capacity to retrieve previous experiences from long-term memory to solve similar problems in the present. Domain-specialized circuitry leads to faster and more efficient retrieval of information, and this is what distinguishes experts from nonexperts.

46.5 Conclusion – Cognitive and Neural Bases of Fractal Cognition

Fractals are complex structures with an unbounded number of hierarchical levels. However, they can be built with simple generative rules that once mastered allow us to tame complexity. The capacity to represent recursive hierarchical embedding (RHE) is key to our ability to represent fractals and other complex structures. As we reviewed, human adults can use RHE to generate fractal-like structures in language, vision, music, and the motor domains.

From the evolutionary point of view, this capacity is interesting because it seems uniquely human. Other species can represent simple hierarchies to navigate their social rank and for spatial navigation [10, 60], but these are limited in scope and depth. While a few nonhuman primates might learn to represent two hierarchical levels also in artificial languages, these few individuals require an extensive amount of training [14]. For humans, the acquisition of this capacity is also difficult, as it comes late in ontogenetic development. However, once it is available, it enables the

representation of hierarchies of unlimited depth and can be used in a multiplicity of domains.

Our review from the literature and our own results point to distinct cognitive and neural systems involved in the processing of RHE in humans (Fig. 46.4).

First, both in language and in the visual domains, mastery of RHE becomes available after 7–8 y.o., and it requires previous experience with conjunctive structures [41, 55]. Interestingly, this is the age at which the accurate fasciculus matures to its distinctively human connectivity pattern [52]. Second, we found that the two main regions connected via this pathway – IFG and pSTS – were somewhat involved in the acquisition of RHE. Left posterior temporal cortex is clearly recruited not only in the acquisition of linguistic syntax [62], logical rules [59], and visual fractals [40] but also in the processing of musical fractals [38]. On the other hand, activity in lateral frontal cortex is less robust and varies depending on the level of expertise [27, 40, 61]. These neuroimaging findings fit well with the behavioral research suggesting strong co-variance across domains in the capacity to acquire RHE [39, 40, 58]).

On the other hand, and in line with previous literature, we find that expert processing of RHE recruits increasingly specialized neural systems, which include specific regions within left IFG for language [12, 27, 61], right IFG for logical hierarchies [59], sensorimotor areas for motor fractals [36, 40], bilateral STG areas related to melody processing for music fractals [38], and the visual ventral stream for visual fractals [37, 41]. These distinct patterns might reflect domain-general mechanisms operating over increasingly domain-specialized representations [43], better suited for fast, automatized, and modular processing [4].

Finally, in this chapter, we only reviewed the role of cortical regions in the implementation of RHE representations. However, an increasing body of evidence has implicated subcortical regions in both the acquisition and expert representation of these structures. For instance, basal ganglia activity is associated with expert processing of RHE in language [25], action [35, 39], and logical reasoning [59]. On the other hand, the hippocampus is active in the processing of visual-spatial fractals [37] and especially involved in the processing of higher-order elements in compositional structures [59]. These findings are consistent with the roles of the basal ganglia in expert processing hinging on the procedural memory system and often connected with and mirroring IFG activity [25] and with the role of the hippocampus in establishing novel compositional structures of the kind items-in-context [45, 50].

Taken together, these findings suggest that the acquisition and expert processing of fractal structures results from the interplay between diverse neural structures and cognitive systems, such as those generally involved in procedural and declarative learning, conceptual reasoning, and cognitive control, but also regions encoding specialized representations. Mapping the precise roles and division of labor between these neural structures is an exciting endeavor for future research.

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Chapter 47

Fractals in the Neurosciences: A Translational Geographical Approach



Ion Andronache, Daniel Peptenatu, Helmut Ahammer, Marko Radulovic, Goran J. Djuđić, Herbert F. Jelinek, Carlo Russo, and Antonio Di Ieva

Abstract The chapter presents three new fractal indices (fractal fragmentation index, fractal tentacularity index, and fractal anisotropy index) and normalized Kolmogorov complexity with proven applicability in geographic research, developed by the authors, and the possibility of their future use in neuroscience. The research demonstrates the relevance of fractal analysis in different fields and the basic concepts and principles of fractal geometry being sufficient for the development of models relevant to the studied reality. Also, the research highlighted the need to continue interdisciplinary research based on known fractal indicators, as well as the development of new analysis methods with the translational potential between fields.

Keywords Fractal parameters · Kolmogorov complexity · Geography · Neuroscience

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47.1 Introduction

The concept of fractal geometry was developed by Benoît B. Mandelbrot (1924–2010) based on earlier pioneering studies by Jules Henri Poincaré (1854–1912), Georg Ferdinand Ludwig Philipp Cantor (1845–1918), Lewis Fry Richardson (1881–1953), Niels Fabian Helge von Koch (1870–1924), Waclaw Sierpinski (1882–1969), Karl Menger (1902–1985), and Karl Weierstrass (1815–1897). Fractal geometry was initially developed from the observation that shapes such as those of rivers, trees, roots, and organs such as lungs or blood vessels did not fit into Euclidean patterns [30] as well as from the analysis of the “monsters” of Euclidean geometry: mathematical or natural phenomena not fully explainable or quantifiable by means of Euclidean geometry.

Mandelbrot first used the term “fractal” in his book *Les Objets Fractals: Forme, Hasard et Dimension* [85] and later in *The Fractal Geometry of Nature* [87] to designate objects or phenomena that are continuous and exhibit self-similarity at different scales. The term fractal comes from the Latin *fractus* meaning broken, which is derived from the verb *frangere* meaning to break.

According to Mandelbrot, fractal objects are mainly characterized by four properties: irregularity of their shape, self-similarity of their structures, fractal dimension, and invariance to scaling.

The fractal feature of an object indicates its morphological complexity. This complexity is expressed both by periodicity and by similarity, a repetition of patterns on multiple scales of size: a structure in which each element is a similar structure made up of other similar structures, and so on, up to the limit of resolution. Dividing a fractal into two or more parts yields fragments whose structure is no simpler than that of the whole. This feature is called self-invariance [48]. Self-invariance can be self-similarity (when isotropic), or self-affinity (when anisotropic, depending on spatial direction).

Mathematical fractals retain their pattern and complexity regardless of the scale of analysis. In reality, natural objects are approximate fractals; they may have fractal properties in a defined scale’s range.

Unlike mathematical fractals, natural objects are characterized by irregularity and “rough” shapes that are apparently very complex. There is a virtually infinite list of natural objects that morphologically approximate fractal structures [64], i.e., clouds, snowflakes, crystals, contours of mountains, shorelines, aquifer networks, cauliflower and broccoli, vascular systems, aerial and subterranean plant arborescence, organizations of fungi, algae, lichens, corals, and many others. Many natural or technical processes have fractal dynamics, which are all the more pronounced when the nonlinearity of the equations that can model them is more pronounced. The best-known example is meteorological phenomena, but most physiological, and generally biological, processes recorded accurately and without filtering to retain only strictly periodic components, exhibit fractal characteristics [18]. Already established applications of fractal analysis include population dynamics, epidemiology [38], internal dynamics and interaction of

ecosystems, analysis of cosmic background radiation, dynamics of stock markets, macroeconomic processes, dynamics of geological and tectonic processes, artificial intelligence, architecture [128], kinetics and dynamics of animal behavior [114], genome organization [12], and many others.

Self-similarity measurement methods are powerful tools in the study of natural objects, which seem too complex to be quantified by Euclidean geometry. The concept for evaluating geometric objects is that of dimension, which is a characteristic value of the system.

Fractal theory quantitatively and qualitatively describes the inherent irregularity of any real object or phenomenon in relation to the presence of complexity at varying dimensional scales. While statistical theory can describe the irregularity of a limited number of parameters in a population with the other parameters, fractal analysis provides a quantitative measure in the case of unlimitedly complex structures, including when the scale at which the analysis is performed changes.

In fractal analysis, the calculation of fractal dimension is the fundamental and most often explored component. Fractal dimension is a value representing a synthetic measure of complexity based on the self-similarity and size of an object, — in the sense of the degree to which the outer boundaries are filled.

There are a multitude of fractal dimensions, each determining varying aspects of fractal complexity, both for binary, grayscale and even RGB color images, such as box-counting and compass dimensions (for determining the degree of space filling) [7, 31, 95, 113], pyramid dimension (also for determining the degree of space filling, but it is a faster processing method, very useful in preprocessing a large volume of data) [89, 90], ruler dimension, Minkowski dimension, dilation dimension (for contour irregularity analysis) [8, 14, 82, 86, 88, 133], perimeter-area, mass dimension (for quantifying fractal shape) [20, 59, 84], information dimension (for determining fractal entropy) [13, 105], correlation dimension [29, 44], local connected fractal dimension (for determining fractal connectivity) [9, 73], Kolmogorov complexity (KC), Higuchi 1D and Higuchi 2D (for determining fractal complexity) [1, 3, 71, 100, 121], and lacunarity (for determining the degree of heterogeneity/homogeneity of the spatial distribution of fractal objects) [4, 62, 109].

In the analysis of territorial systems, fractal modeling can be a useful tool for improving their description and classification due to its ability to quantify the final value of the “amount” of detail or irregularity at different scales associated with the objects under analysis.

In the framework of the collaboration between the Center for Integrated Analysis and Territorial Management (CAIMT), University of Bucharest (Romania), and Medical University of Graz (Austria), the following new fractal indices have been developed and tested: fractal fragmentation index (*FFI*), fractal tentacularity index (*FTI*), fractal anisotropy index (*FAI*), and normalized Kolmogorov complexity (*NKC*).

47.2 Fractal-Based Computational Tools

47.2.1 2D fractal Fragmentation Index

The concept of fractal fragmentation emerged from the need to analyze the degree of compaction/fragmentation of complex images with several larger and smaller fractal objects, which may contain artifacts (isolated, residual pixels) due to binarization.

Fractal fragmentation index (*FFI*) is a fractal index that allows quantifying, using multi-scale fractal techniques, the degree of fragmentation or compaction of objects filling a space [6]. At the same time, *FFI* can identify an object's deviation of its shape from a Euclidean geometric shape.

The fractal fragmentation index was generated from the need to analyze the degree of compaction/fragmentation of complex images with multiple, larger and smaller objects, images that may contain artifacts (isolated, residual pixels) resulting from segmentation and binarization.

FFI is calculated according to the equation (47.1):

$$FFI = D_A - D_P = \lim_{\varepsilon \rightarrow 0} \left(\frac{\log N(\varepsilon)}{\log \frac{1}{\varepsilon}} \right) - \lim_{\varepsilon \rightarrow 0} \left(\frac{\log N'(\varepsilon)}{\log \frac{1}{\varepsilon}} \right) \quad (47.1)$$

where *FFI* is the fractal fragmentation index, D_A is the fractal dimension of the summed areas (mass), and D_P is the fractal dimension of the summed perimeters (borders), ε is the box size, $\log N(\varepsilon)$ is the number of contiguous, non-overlapping boxes required to cover the area of the object, and $\log N'(\varepsilon)$ is the number of contiguous, non-overlapping boxes required to cover just the perimeter of the object [6].

According to equation (47.1),

- $FFI = 0$, when the objects analyzed are too small so that their borders cannot be extracted; therefore, $D_A = D_P$;
- $(0 > FFI < 1)$
 - When the areas occupied by objects are compact, the *FFI* value tends to 1;
 - When the areas occupied by objects are more dispersed and more fragmented, the *FFI* value is closer to 0.
- $FFI = 1$, then analyzing a Euclidean object, 100% compact object without any discontinuities ($D_A = 2$ and $D_P = 1$).

The fractal fragmentation index (*FFI*) has been implemented as an open-source ImageJ2/Fiji [116] plugin in the ComsystanJ package, <https://comsystan.github.io/comsystanj/> [2] that can be downloaded from <https://github.com/comsystan/comsystanj/tags>. *FFI* uses binary images.

FFI was tested to differentiate fragmentation patterns on synthetically generated Hierarchically Structured Random Map (HRM) images (Fig. 47.1).

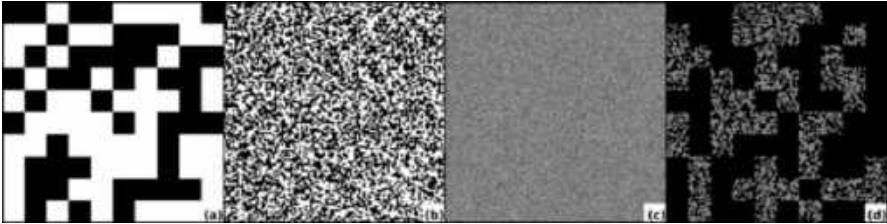
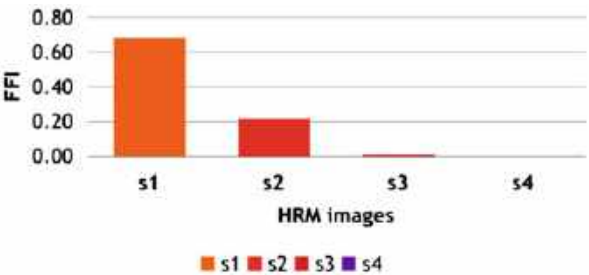


Fig. 47.1 Examples of HRM images generated using ComsysanJ plugin HRM: (a) high s1 set ($p_1 = p_2 = 1$ and $p_3 = 0.5$), (b) middle s2 set ($p_1 = p_3 = 1$ and $p_2 = 0.5$), (c) low s3 set ($p_1 = 0.5$ and $p_2 = p_3 = 1$), and (d) mixed s4 set ($p_1 = p_2 = p_3 = 0.5$)

Fig. 47.2 Identification of fragmentation/compaction patterns using HRM images: s1 – compaction; s2 – moderate fragmentation; s3 and s4 – strong fragmentation



HRMs are binary maps of pixels that are randomly set to 1 or 0, with probability distributions of 1 or 0 varying by region [106].

For our testing, HRM pictures were created with numbers of rows and columns (L) fixed at 1000 (1,000,000 pixels; or 1000×1000 -pixel resolution). The three variables p_1 , p_2 , and p_3 indicate the proportion of space that the HRM objects occupy at various grain sizes (e.g., $0.5 = 50\%$). These were produced by making arrays with L rows and L columns, setting the L^2 elements to 1 with a probability p , with p_1 , p_2 , and p_3 , representing the percentage of space filled by the HRM objects at various scales.

The size of the created HRM objects for p_1 is 100×100 pixels grain; for p_2 it is 10×10 pixels; and for p_3 it is 1×1 pixel. For each type of HRM object formed, p , the total filled area of the image, is in each case 0.5, or 50%. The HRM objects are uniformly distributed throughout the photos because of their random dispersion (Fig. 47.1).

HRM images were generated by the Fiji plugin ComsysanJ using a recursive algorithm derived from the curdling and random trema generation method employed by Plotnick [87, 106].

FFI testing on such synthetically generated images (HRM) showed the ability of *FFI* to identify fragmentation patterns: s1 – compaction; s2 – moderate fragmentation; s3 and s4 – strong fragmentation (Fig. 47.2).

In geography, *FFI* has been applied in the analysis of the effects of deforestation at the county level [6, 7, 104], Administrative-Territorial Units (ATU) [103], or mountain groups [9, 21, 33, 36], indicating in all cases that forest fractal fragmentation increases as a result of deforestation.

In Andronache et al. 2016 [6], a new fractal index, fractal fragmentation index, was proposed, which is applied to tree cover, loss, and regenerated forests for the 6 counties of the Northeast Development Region, Romania.

The *FFI* analysis identified two patterns of fragmentation of the spatial distribution of forests, imposed by the dominant landform in the county. The first pattern with higher *FFI* (with more compact forests) was specific to the three counties with mountainous areas, while the second pattern, with lower *FFI* (with more fragmented forests), was specific to the other three counties, which are located exclusively in the Moldovan Plateau.

Pintilii et al. [104] used *FFI* to quantify the degree of forest fragmentation in Maramureş County, one of the most deforested counties in Romania. However, some important exceptions occur when deforested areas at the Administrative-Territorial Units (ATU) level are more compact and homogeneous than the county average.

In a recently published study [9] where *FFI* analysis was applied, it was found that the fractal fragmentation of the forest in the Apuseni Mountains increased by 17.7%. Dynamics of *FFI* revealed that forest loss areas tended to cluster based on their geographical extensions, particularly after 2005, and that fragmentation had less of an impact on forest loss areas than it did on regeneration. Furthermore, by initially combining the results with fractal connectivity and binary image particle analysis, we significantly enhance the *FFI*'s ability to assess fragmentation.

47.2.2 3D Fractal Fragmentation Index

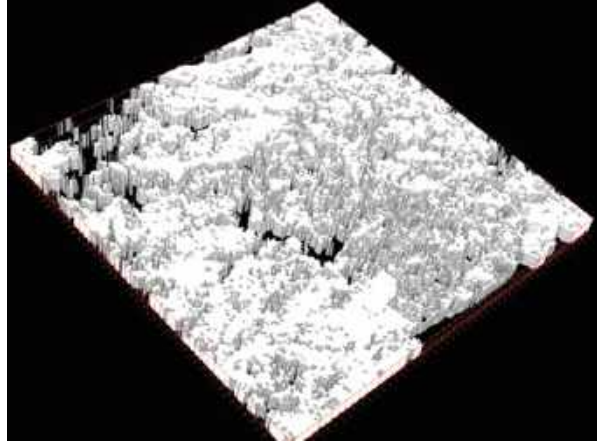
We propose to extend *FFI* analysis from 2D images to 3D binary image volumes. For this purpose, the algorithm for determining the fractal dimension has been extended by one dimension. The boxes for the box-counting algorithm are consequently 3D cubes with different sizes ε according to Eq. 47.1.

A stack of 22 binary images (one for each year from 2000 to 2021) of regions of interest in the core area of the 10 mountain groups was extracted, one stack per group. The first image in each stack is the year 2000 tree cover, and the last image is the year 2021 tree cover (Fig. 47.3).

The results showed that there is a strong differentiation between the groups in terms of fragmentation, but also a different rate of increase in fragmentation due to an increasingly intense and dispersed cumulative forest loss with many areas of new deforestation.

The most fragmented groups are those with the highest deforestation (northern, central) and the most compact groups with more limited but concentrated deforestation and higher natural forest compaction.

Fig. 47.3 3D reconstruction of the tree cover dynamics in the Northern Group of the Eastern Carpathians (2000–2021). The tree coverage is marked in white pixels



The 3D *FFI* analysis has a crucial advantage in that it shows in a single value both the reality of fragmentation and its dynamics over time (whether the cumulative loss is caused by the clusterization (agglomeration) of old deforested areas or by the emergence of new deforested areas).

47.2.3 Fractal Tentacularity Index

The fractal tentacularity index (*FTI*) is a new fractal index that can be used to quantify the degree of tentacularity in a given system or structure. Tentacularity refers to the property of a system or structure having many branches, or tentacles, that extend out from a central point. The *FTI* can be used to compare the degree of tentacularity of different systems and can provide insights into how systems change over time.

FTI is derived from *FFI* and calculated according to the equation (47.2):

$$FTI = FFI_{CH} - FFI \quad (47.2)$$

where FFI_{CH} is fractal fragmentation index of Convex Hull of the analyzed objects, and *FFI* is fractal fragmentation index.

The fractal tentacularity index (*FTI*) has been implemented as an open-source ImageJ2/Fiji [116] plugin (Fractal Fragmentation Indices) in the ComsystanJ package, <https://comsystan.github.io/comsystanj/> [2] that can be downloaded from <https://github.com/comsystan/comsystanj/tags>. *FTI* uses only binary images.

In order to test and validate the *FTI*, 8 images with different degrees of tentacularity were generated using the FracLac plugin in Fiji ImageJ 2—MicroMode (Fig. 47.4) [60].

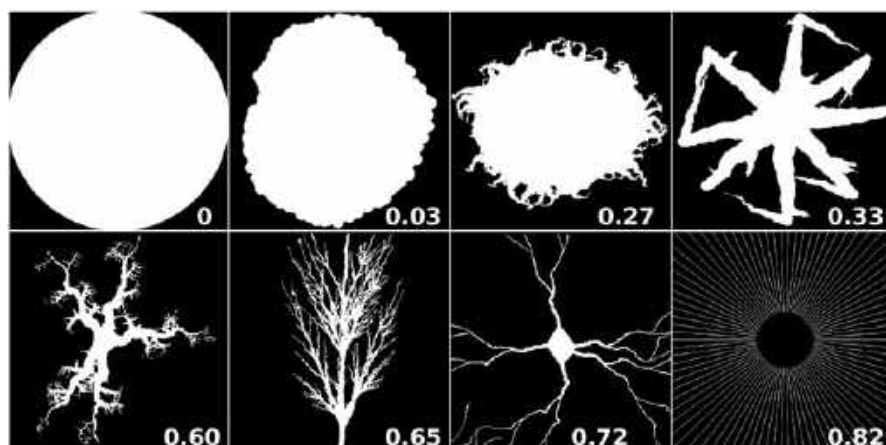


Fig. 47.4 *FTI* calculated for several images with different degrees of tentacularity generated using the FracLac plugin in Fiji ImageJ2—MicroMode

A value of 0 indicates no nonuniformity; the appearance of tentacles leads to increases in *FTI* values and values greater than 0.5 indicate pronounced tentacularity.

We used *FTI* in the analysis of the spatial dynamics of the Romanian capital, Bucharest, for the period 1860–2015. *FTI* correctly identified periods of sprawling development (nineteenth century, and the postwar period, when Bucharest's development was dominated by the main development axes) and compact periods when Romania was a kingdom, and the spatial development of Bucharest was more homogeneously distributed (unpublished data).

FTI can also be exploited for medical applications. For instance, the potential applications are in the analysis of microglia activation, the branching structure of blood vessels, and the tentacular structure of tumor cells or brain tissue. In the study of the branching structure of blood vessels, *FTI* can be used to quantify the degree of branching in different regions of the body. This can provide insights into how blood flow is affected by different branching angles changing the relationship between laminar or turbulent flow and pathology such as cerebrovascular and cardiovascular disease [63]. In the study of the irregular structure or surface roughness of tumor cells, *FTI* can also quantify the degree of branching of the tumor cells or surface characteristics, which can provide insights into how the tumor cells are spreading and thus predict the progression of the disease [10, 97].

47.2.4 Fractal Anisotropy Index

Fractal anisotropy is a concept used to describe the directional dependence of fractal patterns in geographic data. It refers to the variation of fractal dimension across different orientations or directions within a given structure. For example, a coastline may have a different fractal dimension when measured parallel to the shoreline compared to when measured perpendicular to it.

Anisotropy and fractal/fractional anisotropy in particular play a wide-ranging but crucial role in several fields of science, with many applications in earth sciences (such as mapping, geophysics, and geology), material sciences, and engineering to name just a few. Despite its overarching complexity and the fact that there is no general definition for anisotropic fractal surface, the profiles of such elements are assumed to be fractal in two main axes.

The fractal dimensions of the two-dimensional digital images were estimated by extracting one-dimensional gray value profiles and computing their Higuchi dimensions [1, 51]. With this, it was possible to compute separate fractal dimensions for the horizontal and the vertical direction. We used the Mean of all rows/columns, with maximal k .

The fractal anisotropy index identifies and differentiates patterns of directionality and nondirectionality. It is generally defined by equation Eq. 47.3:

$$FAI = \left| \frac{D_H - D_V}{D_E - D_T} \right| \quad (47.3)$$

D_H is the Higuchi dimension in horizontal direction, and D_V is the Higuchi dimension in vertical direction. Equation (47.3) holds true for any embedding dimension D_E and topological dimension D_T . For this study, $D_E = 2$ and $D_T = 1$ which reduces (Eq. 47.4):

$$FAI = |D_H - D_V|. \quad (47.4)$$

Usually, the values of the Higuchi dimension of gray value profiles are in the range [1, 2], leading to a range of [0, 1] for the FAI . The higher the anisotropy value, the more pronounced the directionality and vice versa. A value of 0 indicates a complete lack of directionality. Perfect directionality means $FAI = 1$.

Higuchi dimensions and the FAI were computed using the Image (2D)/Higuchi dimension 1D plugin of the ComsystanJ package [2] (<https://comsystan.github.io/comsystanj>) for ImageJ/Fiji [116] (<https://imagej.net/software/fiji>). FAI can be calculated for 8-bit grayscale and binary images. In our study, we used grayscale images.

In order to test and validate the FAI , 12 images with different degrees of anisotropy were used (Fig. 47.5).

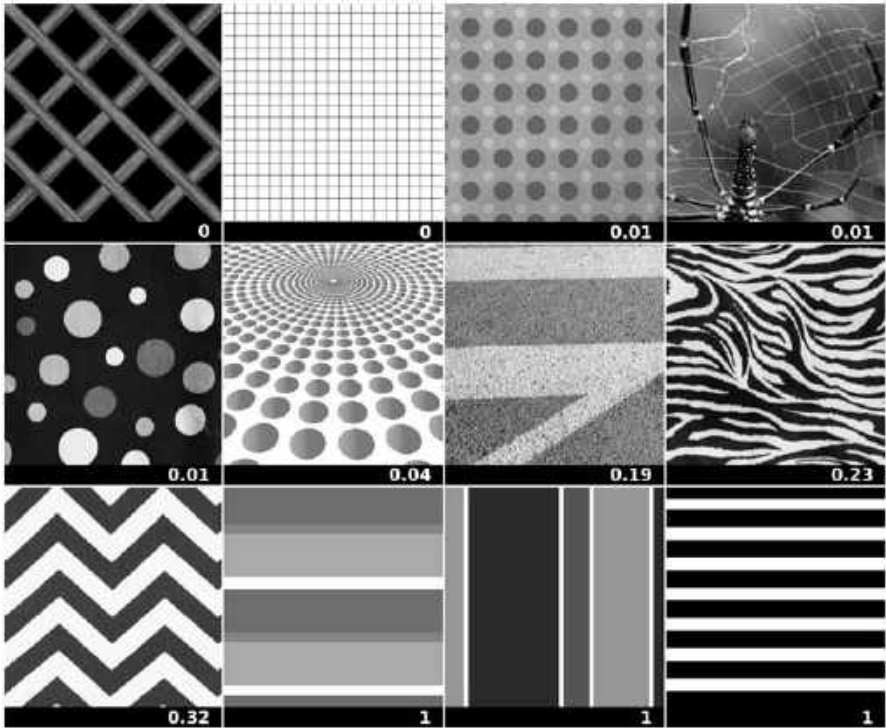


Fig. 47.5 *FAI* of some exemplary images with different degrees of anisotropy

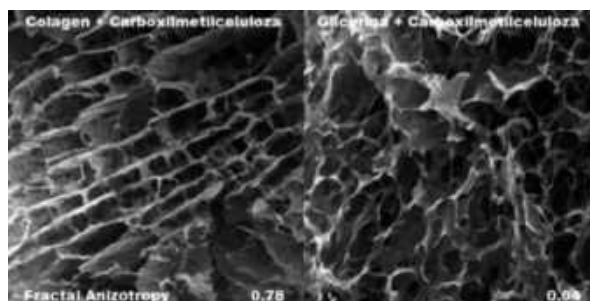
Analysis of the *FAI* results (Fig. 47.5) showed that the proposed new index clearly identified the isotropy (0) and anisotropy (1) as well as the different degrees of the anisotropy and can be tested and used in pattern differentiation on real images.

The *FAI* analysis has proven useful in the identification of the deforestation patterns in the Romanian Carpathian groups. Mountain groups were identified where deforestation was isotropic (Fagaras, Parang, Bucegi, and Northern groups) and groups where deforestation was more anisotropic (Central group, Southern group, Retezat-Godeanu, Banat, Poiana Rusca, and Apuseni), along some corridors (unpublished data).

We tested *FAI* on the differentiation of porous hydrogels from collagen/gelatin and carboxymethylcellulose biopolymers by lyophilization and ion cross-linking processes with biomedical applications for tissue engineering of soft tissues, and the degree of anisotropy was differentiated very well (Fig. 47.6).

Fractal anisotropy clearly differentiated the cross-linking effects of sodium tripolyphosphate (TPP) in collagen hydrogels. This is more evident due to its fibrillar structure, and the interactions manifested with polysaccharides.

Fig. 47.6 Fractal anisotropy index testing for determining the cross-linking effects of porous biopolymer hydrogels



Fractal anisotropy also has potential application in the medical field. Fractal anisotropy can be used as a tool to study the structure and function of various biological systems, such as blood vessels, lung airways, and the brain.

In the cerebrovascular and cardiovascular systems, fractal anisotropy can be used to study the geometry and branching patterns of blood vessels, which can provide insights into the development of vascular diseases such as atherosclerosis [43, 93, 120]. In the respiratory system, its potential application is to analyze the geometry and branching patterns of lung airways, which can provide insights into the development of respiratory diseases such as asthma and chronic obstructive pulmonary disease (COPD) [11].

Furthermore, in the nervous system, fractal anisotropy has been used to study the geometry and branching patterns of neural networks, to provide insights into the development of neurological disorders such as Alzheimer's disease and Parkinson's disease [75, 126, 134].

Overall, fractal anisotropy is a valuable tool for understanding the complexity and directional dependence of biological structures and can provide valuable insights into the development and progression of various diseases.

47.2.5 *Normalized Kolmogorov Complexity*

Kolmogorov complexity (KC), in theory, is the length of the shortest computer program in Megabytes (Mb), in a predetermined programming language, that produces an output. Thus, KC is a measure of the computational resources required to specify the object and is also known as algorithmic entropy [71]. KC is a theoretical concept that cannot be implemented as a fractal algorithm, for digital images, in an exact way, because it cannot be verified whether the image generated by the software is indeed the correct one. Therefore, a way to estimate the Kolmogorov complexity of an image is to quantify the amount of memory needed to create a compressed image without quality loss.

Kolmogorov complexity can help to identify the complexity of spatial reality, spatial distributions, dynamics, and possibly patterns of spatial behavior.

Lossless compression algorithms are well investigated and optimized for digital images, e.g., lossless PNG image compression or ZIP compression of any file type. We opted in our testing to use the PNG algorithm because the details of the algorithm are well known, and the corresponding implementations are very fast in analysis and freely available. However the overall image size would be biased when assessing these estimates of KC , ultimately making it unusable for classifying patterns. According to Eq. 47.5, normalization—that is, the ratio of KC to the image size in Mb—was able to successfully reduce this bias:

$$NKC = \frac{KC}{S} \quad (47.5)$$

where NKC is normalized Kolmogorov complexity, KC is Kolmogorov complexity un-normalized (in MB), and S is image size (in MB).

Normalized Kolmogorov complexity and Kolmogorov complexity were computed by using the Image (2D)/Kolmogorov complexity and LD plugin of the ComsystanJ package [2] (<https://comsystan.github.io/comsystan>) for ImageJ2/Fiji [116] (<https://imagej.net/software/fiji>). NKC can use 24-bit RGB, 8-bit grayscale, or binary images. In our study, we used grayscale images.

NKC allowed the quantification of complexity using values between 0 and 1. When an image is simple and every pixel has the same gray value, the value 0 is recorded; when each pixel is surrounded by eight neighbors who have varied gray values, the value 1 is reached [100].

In order to test the ability of Kolmogorov complexity to differentiate patterns of fractal complexity, grayscale fractal surface images were generated using an Inverse *FFT* algorithm. Images with fractal dimensions sizes between 2.0 and 3.0 were analyzed at 1000×1000 pixel resolution. The gray values of a digital image represent a surface, a two-dimensional topological object that more or less fills the third dimension. Thus, the fractal dimensions of grayscale images are in the range 2 (minimum complexity) and 3 (maximum complexity). Therefore, the test images generated were also generated in this range (Fig. 47.7).

Our test (Fig. 47.8) showed the ability of Kolmogorov complexity to accurately quantify fractal complexity and implicitly to differentiate patterns of fractal complexity of the analyzed images.

Therefore, Kolmogorov complexity can help to identify the complexity of territorial reality, spatial distributions, dynamics, and spatial patterns.

The main advantage of Kolmogorov complexity is not only the very high processing speed of an image but also the possibility to analyze binary images, in 8-bit or 16-bit grayscale, or 24-bit RGB images.

Analyzing the evolution of Kolmogorov complexity for the share of tourism turnover in the total local economy, an upward trend can be observed for the whole period analyzed, the degree of complexity increasing from year to year due to the increase in the number of ATUs with tourism activities and the different share of this activity in the total local economy [45].

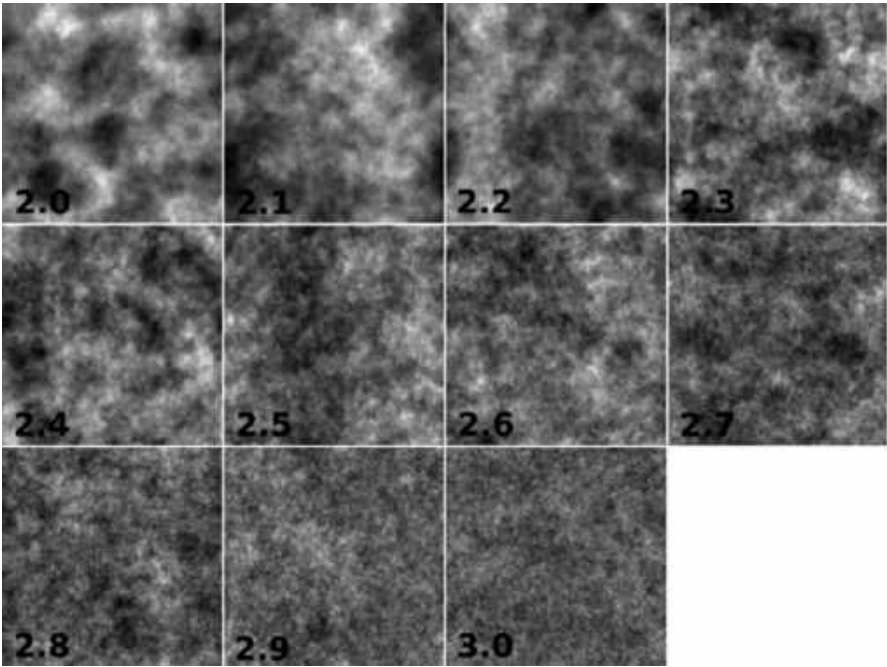


Fig. 47.7 Fractal surface grayscale 8-bit images applying the Inverse *FFT* algorithm having fractal size ranging from 2.0 to 3.0. The resolution of the images was 1000 × 1000 pixels

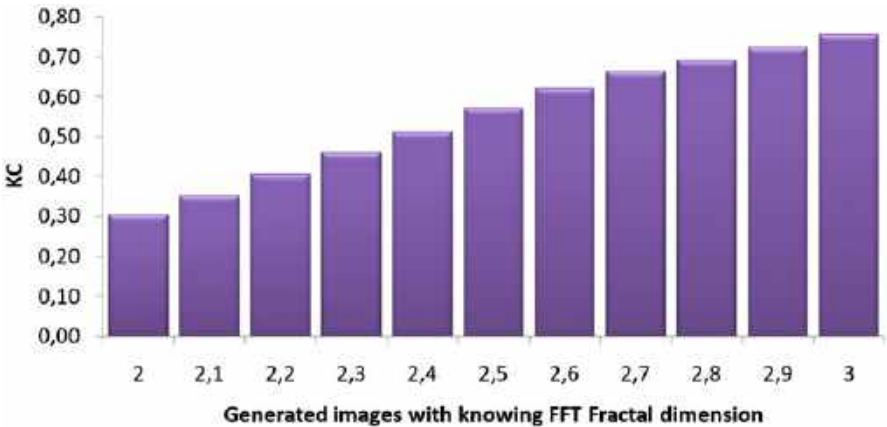


Fig. 47.8 Kolmogorov complexity of 8-bit grayscale fractal surface images applying the Inverse *FFT* algorithm with fractal size ranging from 2.0 to 3.0

Moreover, the increase in complexity is also due to the uneven distribution of ATUs with values in the same class, both in terms of turnover derived as a share of tourism activity in the total local economy. The ATUSs with high turnover or weight

values are located in areas with high tourism resources: the Danube Delta, the Black Sea coast, the Subcarpathian and mountainous area, and punctually along the development corridors. These areas do not have the necessary strength to transmit tourism development at the territorial level, and most of the surrounding ATUs are poor in tourism performance, with different rates of growth or decrease in gross value [45].

The KC analysis highlights two moments of decreasing complexity of spatial distribution in the share of tourism turnover in the total local economy, decreasing complexity in 2002 (due to privatization), 2010 (due to the economic crisis), and 2013, 2015–2016 (due to the increase of turnover in other branches of local economies compared to tourism) [45].

47.3 Application of FFI, FTI, FAI, and KC in the Analysis of Medical Images

The macroscopic morphology acquired by MRI presents a rich source of tumor heterogeneity information because the tumor is shaped by the growth patterns of malignant cells and, fundamentally, by the sum of all molecular interactions within a tumor. Therefore, the analysis of tumor morphology theoretically provides a wider analytical breadth than the most advanced molecular high-throughput methods [129]. Because of its geometrical complexity and irregularity, the quantification of tumor morphology is approached by computational analysis, e.g., radiomics in radiology imaging [54] and pathomics in digital pathology. Taken together, we evaluated our three new fractal indices (fractal fragmentation index, fractal tentacularity index, and fractal anisotropy index) and Kolmogorov complexity normalization for their performance to classify tumors based on the morphological heterogeneity depicted on MR imaging.

(a) Grading of the Brain Glioma Tumors

Gliomas originate from glial cells and constitute the most common type of malignant primary brain tumors in adults [118]. Treatment planning of gliomas entirely depends on the accurate grading between low-grade glioma (LGG: in the classification systems earlier than 2016, grades I and II) and high-grade glioma (HGG) tumors (grades III and IV). Grading is particularly valuable if it can be done noninvasively and accurately; however, this is difficult to achieve using standard MR imaging techniques [19]. Disadvantages of the tissue biopsy are discomfort and risk to the patient, the potential for sampling error and a substantial cost. Therefore, we aimed to test the ability of our novel image analysis features to improve the glioma grading performance by exploiting it for the computational analysis of MRI.

47.4 Patient Characteristics

We applied our analysis to a shareware dataset [32, 119], analyzing MRIs of 293 patients with high-grade and 76 patients with low-grade glioma. Representative images are presented in Fig. 47.9.

Our fractal indexes were tested for their ability to differentiate between high-grade and low-grade gliomas, and their performance was evaluated by the receiver operating characteristic curve (ROC) analysis (Fig. 47.10).

3D *KC* grayscale, 2D *FTI* binary, and 2D *FAI* features did not reach the statistically significant differentiation performance by any of the used statistical tests (Fig. 47.10). 2D *FFI* binary was the best-performing individual image analysis feature based on its AUC of 0.19, followed by 3D *FFI* binary (AUC = 0.27). Besides evaluating the degree of association between the continuous feature values with the binary outcome, ROC analysis also informs the direction of the association. Thereby, $AUC < 0.5$ designates the association with low-grade glioma and $AUC > 0.5$ indicates the association with high-grade glioma.



Fig. 47.9 Examples of the analyzed glioma MR images. (a) Enhancing core (blue), tumor core (red), and edema (yellow). (b) High-grade glioma grayscale stack for HGG patient, for 3D grayscale *NKC*. (c) Tumor core of high-grade glioma binary stack, 3D binary *FFI*. (d) Low-grade glioma grayscale stack for LGG, 3D grayscale *NKC*. (e) Tumor core of low-grade glioma binary stack for LGG, 3D binary *FFI*

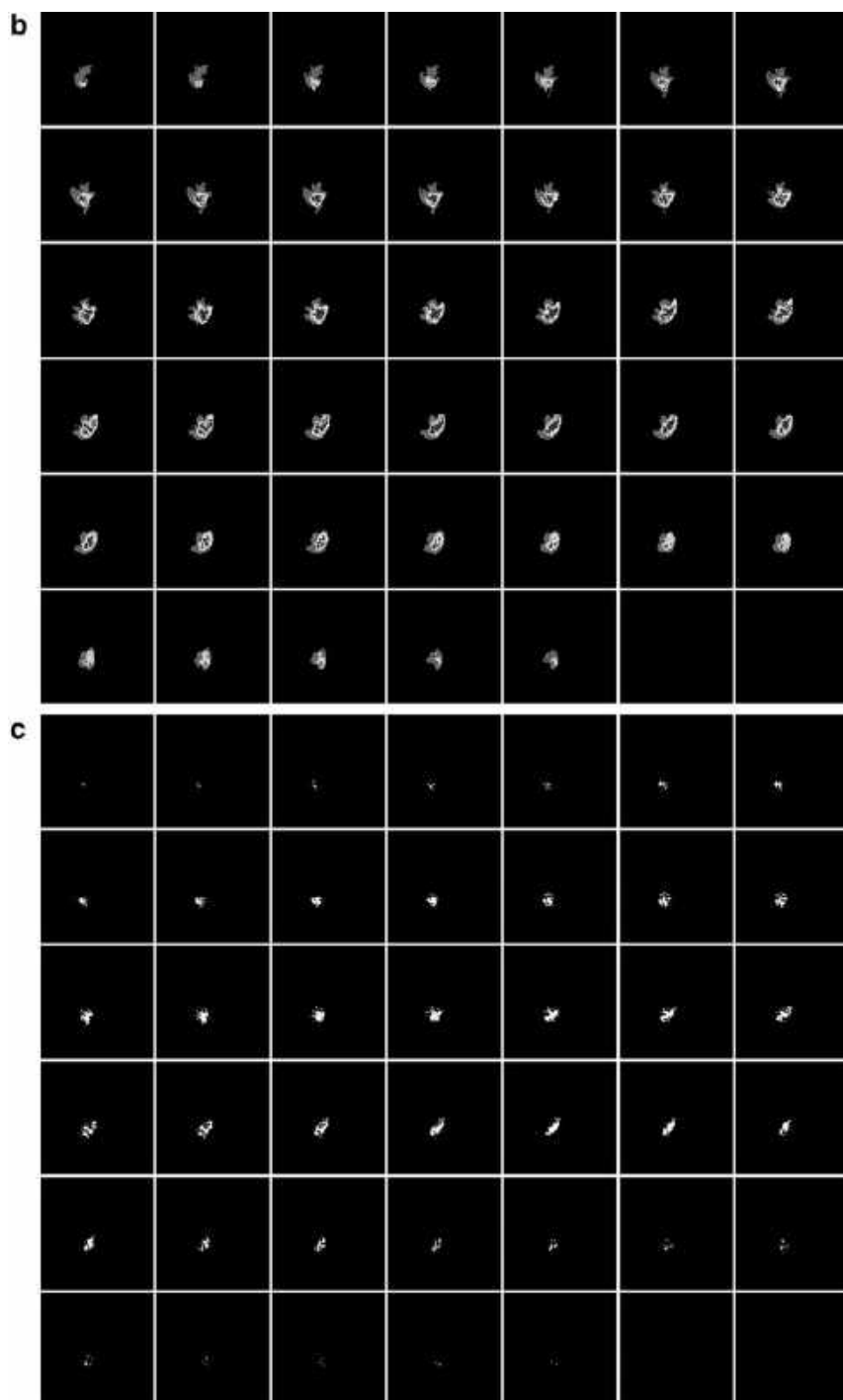


Fig. 47.9 (continued)

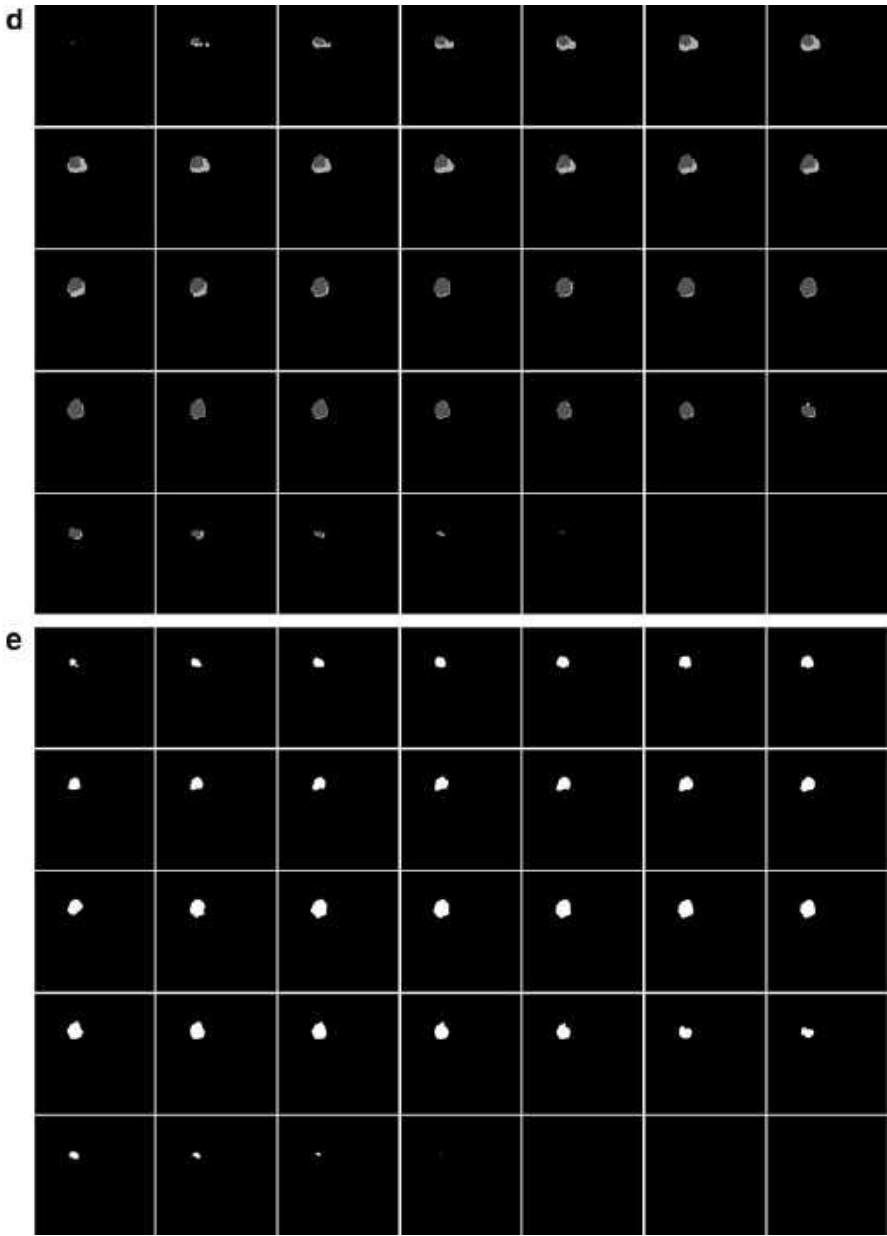


Fig. 47.9 (continued)

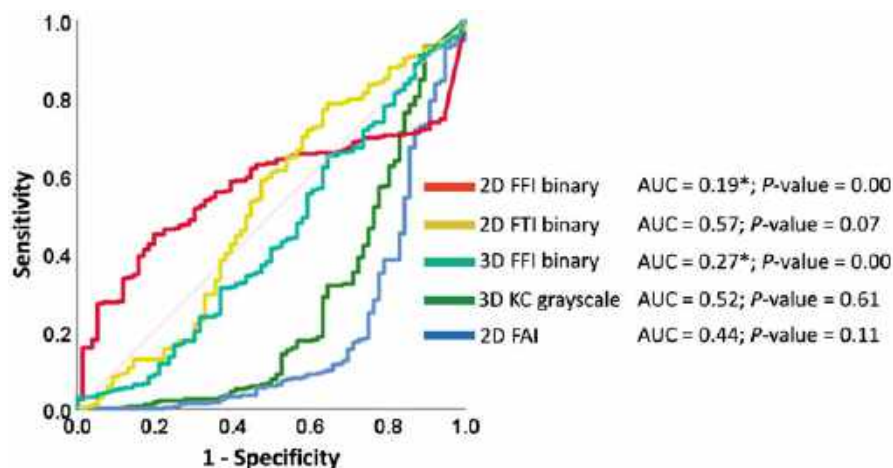


Fig. 47.10 Estimates of the obtained performance in the prediction of the glioma grade. Receiver operating characteristic curves reveal the efficiency in discrimination between low-grade and high-grade glioma. Areas under the ROC curves (AUCs) were calculated by use of the low- and high-grade gliomas as outputs. *P*-values of AUCs were corrected by bootstrap. Significant AUC values ($P \leq 0.05$) are marked with an asterisk. $AUC < 0.5$ indicates the association with low-grade glioma, while $AUC > 0.5$ indicates the association with high-grade glioma. Abbreviations: *FFI*, fractal fragmentation index; *FTI*, fractal tentacularity index; *KC*, Kolmogorov complexity; *FAI*, fractal anisotropy index

Based on the obtained results, all features with a significant tumor-grade association increased the value with a decrease in the glioma grade, which makes them markers of the low grade. 2D *FFI* binary and 3D *FFI* binary were thus identified as potential biomarkers for the low-grade glioma. The clinical significance of the improved determination of glioma grade is in the improvement of the treatment planning and the consequent extension of survival.

(b) Prediction of the Tumor Chemoresponsiveness

The heterogeneity of the tumor's microscopic and macroscopic morphology is increasingly exploited for the purpose of tumor chemoresistance prediction [34]. However, tumor morphology obtained by MRI is still underexploited, possibly due to the inadequacy of the previously used fractal and textural features.

Osteosarcoma mostly affects the long bones before the age of 20 [42]. Its treatment includes chemotherapy consisting of methotrexate [M], doxorubicin [A], and cisplatin [P] (MAP), followed by surgical tumor removal or amputation, depending on the local tumor invasiveness [42]. Although this multiple chemotherapy regimen was implemented to overcome multidrug resistance and numerous clinical trials of novel drugs, osteosarcoma survival rates have improved little over the last three decades [42, 49]. Due to the limited efficacy of existing cytotoxic chemotherapies, improvement in osteosarcoma survival depends on the slow discovery and implementation of novel anticancer drugs [47]. Survival might be improved faster by



Fig. 47.11 Examples of osteosarcoma MR images analyzed in all three anatomical directions. (a) Horizontal, (b) frontal, and (c) sagittal. One characteristic image was analyzed per each MRI slice orientation

more efficient use of existing drugs, whereby chemoresistant tumors would receive alternative treatments such as experimental trial protocols or more radical surgical excision [96]. This approach could avert metastasis development during ineffective treatments of chemoresistant tumors. The main obstacles for such personalized treatment decisions are the current lack of validated prognostic markers and low reliability of the chemoresistance prediction [83]. We therefore tested our novel fractal indexes for their association with the tumor chemoresistance which could potentially improve the decision support in precision medicine and extend patient survival.

The model included 71 osteosarcoma patients diagnosed and treated by the National Sarcoma Multidisciplinary Team during the 5 years from 2010 to 2014. An expert radiologist selected the most representative magnetic resonance image with the largest tumor area for each of the three acquired slice orientations (Fig. 47.11a–c), a total of three images per patient.

Analyses were limited to tumor areas by use of ROI delineation, whereby the background was marked in white pixels (Fig. 47.11). The calculated feature values were assessed for their prediction of chemoresistance by the use of chemotherapy-induced tumor cell necrosis as the endpoint event (Fig. 47.12).

2D *FAI* and 2D *FTI* binary significantly associated with the tumor chemoresistance based on the receiver operating characteristic (ROC) curve analysis (Fig. 47.12). A significant predictive association was achieved for 95% necrosis chemoresponsiveness but not for 80% and 90% (not shown). 2D *FFI* binary did not associate significantly with tumor chemoresistance (Fig. 47.12). We emphasize the use of unmanipulated continuous values, which is an advantage because value categorization often introduces a bias. It is of note that prediction directions of 2D *FTI* binary and 2D *FAI* were inverted, whereby 2D *FTI* was a marker of chemoresistance as its values increased with higher chemoresistance, while higher 2D *FAI* values predicted increased chemosensitivity. Such directionality was reflected in AUC values as $AUC < 0.5$ designated the association with chemosensitivity, while $AUC > 0.5$ indicated the association with chemoresistance. The curve of 2D *FTI* binary looks particularly convincing because it is further removed from the central no-discrimination line (Fig. 47.12). A ROC curve is a graphical plot that illustrates the predictive ability of a binary classifier system, as its discrimination threshold

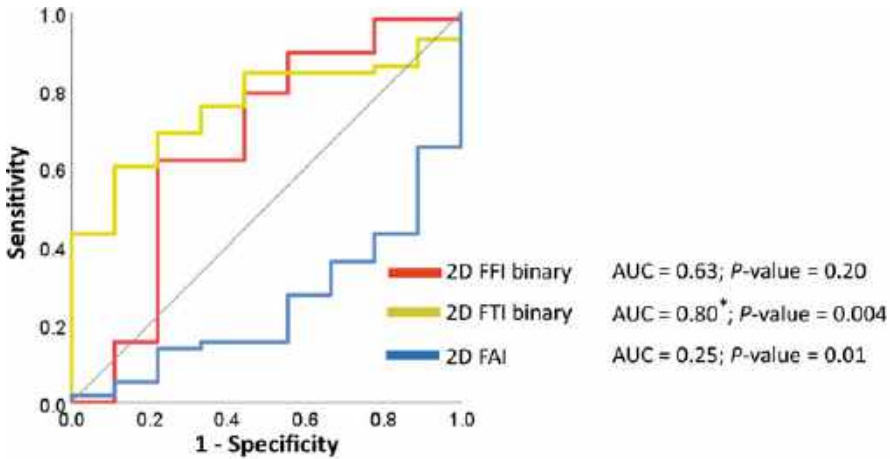


Fig. 47.12 Illustration of the obtained performance in prediction of osteosarcoma chemoresistance. Receiver operating characteristic curves reveal efficiency in discrimination between chemosensitive and chemoresistant osteosarcoma tumors by continuous values of the tested features. *P*-values of AUCs were corrected by bootstrap. Significant AUC values ($P \leq 0.05$) are marked with an asterisk. AUC < 0.5 indicates the association with chemosensitivity, while AUC > 0.5 indicates the association with chemoresistance. Plots reveal discrimination efficiencies of the continuous values of *FTI*, *FAI*, and *FFI* feature

changes. It is created by plotting the true-positive rate or sensitivity, against the false-positive rate (1-specificity) at various threshold settings (Fig. 47.12).

(c) Retinal Ganglion Cells

Retinal ganglion cells (RGCs) connect the retina with brainstem and cortical visual structures. Anatomical classification was first described using Golgi staining techniques such as employed by Perry for the analysis of the rat retina [101] and later utilizing Lucifer Yellow [111] and other markers that can be used simultaneously with physiological characterization. This section is focused on novel fractal-type analysis methods for the analysis of cat retinal ganglion cells. However, retinal ganglion cells (RGCs) from diverse species have been studied intensely for many years from a morphological and physiological perspective [16, 58, 65] including rabbit [5], ferret [131], mouse [35], rat [17, 40], hamster [74], marsupial [37, 117, 125], flying fox [27], salamander [81], fish [23, 24, 69], frog [68], turtle [46, 66], dolphin [28], pelagic seabirds [50], dog and wolf [99], cat [15, 22, 70, 78, 98, 107, 112, 115, 123, 130], new and old world monkey [25, 39, 77, 80, 132], and human [102] among many more [94].

Classification of cells from an anatomical perspective has undergone several changes starting from measures of soma and dendritic field size, distance from area centralis/fovea, location within inner or outer plexiform layer and branching patterns to mathematical modeling, and image classification algorithms including fractal analysis [55, 59, 67, 98, 110, 130]. Fractal analysis has been employed for

the classification of rat, cat, and other species' RGCs [39, 53, 55, 56, 58, 61, 72, 79, 91, 92, 131]. Multifractal analysis and connected fractal analysis are further fractal-based classification tools [56, 79]. Classification issues arose as the number of cell types increased and physiological data emerged from cells that were also intracellularly labeled. Hence, the α , β , and γ of Boycott and Wässle, which in turn are included in the anatomical classification suggested by Helga Kolb, as well as the more non- α and non- β including δ cells, medium-gamma cells, neuropeptide Y immunoreactive cells, and RGCs of the geniculate wing had to have equivalent cell types in the physiological classification of X, Y, W, and Q cell types [15, 26, 52, 70, 76, 127]. Fractal analysis was able to clarify classification based on the anatomical system suggested by Kolb and colleagues as well as including RGCs identified by Stone, Pu, and others with the later emerging images of physiological classes [107]. Fractal analysis using box-counting indicated that the dendritic trees of the morphologically defined γ , δ , and ε cells are distinct cell types that match physiologically defined W cells, provided the phasic and tonic W cell classes are subdivided according to their fractal dimension into W-tonic1, W-tonic2, W-phasic1, and W-phasic2. The fractal dimension of the ε cells then are equivalent to the W-tonic1 group and γ cell-type equivalent to the W-phasic1 group. Delta cells were not clearly defined and may be equivalent to either the W-tonic2 or the W-phasic2 group, suggesting that the broad δ cell class has distinct morphological subtypes [57].

Recently, we proposed a fractal analysis-based approach on fractal tentacularity index, a new fractal dimension proposed here by us. Applying this analysis on a subset of α (13 binary images), β (9 binary images), and δ (7 binary images) (Fig. 47.13) obtained from the literature [26, 52, 108, 122, 124], the findings indicate that this method provides additional morphological information, indicating a significant difference between α and δ , and β and δ (Fig. 47.14).

47.5 Conclusion

In conclusion, the universality of fractal geometry is demonstrated by the fact that the same concepts or methods can be applied to completely different disciplines, from geomatics to medicine, from physics to biology, and from forestry to land use. For example, the box-counting method, despite having several intrinsic limitations, can be used not only for geometric analysis of the complexity of areas occupied by forest but also in the analysis of the distribution of trees in a forest or the roughness of the coastline of an island or even the surface of a cell.

In the medical fields of neuroscience and oncology, we indicated the ability of our novel fractal indexes to classify glioma and osteosarcoma according to their essential clinical properties such as grade and chemoresistance. The clinical significance of the reliable prediction of tumor grade and chemoresistance is based on improved chemotherapy guidance and the consequent survival benefits.

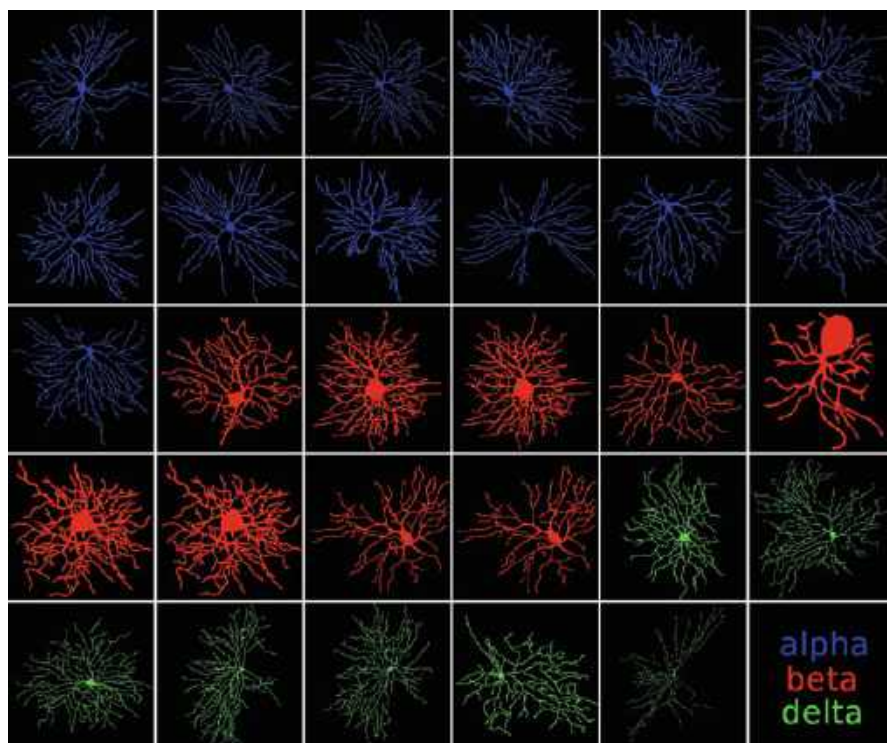


Fig. 47.13 α , β , and δ retinal ganglion cells.

In the field of geography, fractal methods have contributed to the development of classic methods of geographic space analysis, based on spatial data modeling, by extracting additional data that increased the relevance of the results.

These three examples from the medical imaging domain emphasize the potential role of the application of the proposed novel fractal-based parameters in different medical domains, e.g., radiomics, to be explored in further research.

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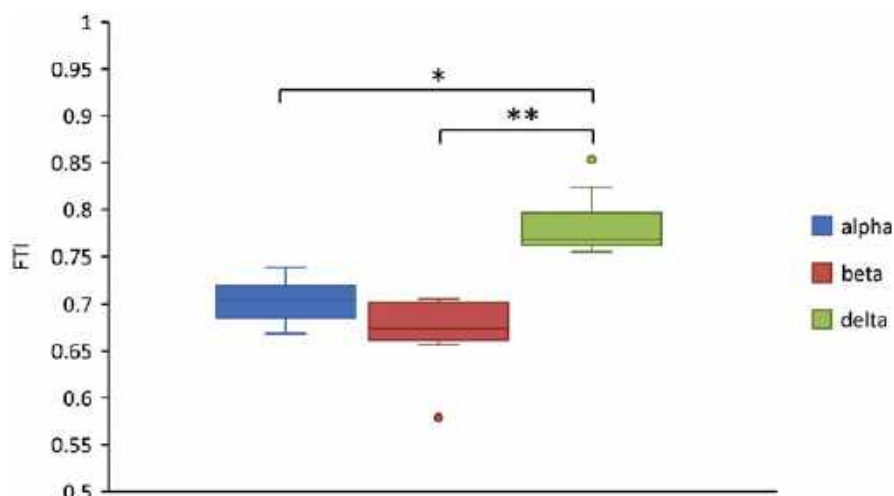


Fig. 47.14 Fractal tentacularity index of α , β , and δ retinal ganglion cells. The box plots show the distinct distributions of the values. Statistical analysis was performed using SPSS 27 software. Due to the small number of cases, nonparametric Kruskal–Wallis pairwise tests were performed (* $p < 0.05$, ** $p < 0.001$)

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Chapter 48

Fractal Geometry Meets Computational Intelligence: Future Perspectives



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Abstract Characterizations in terms of fractals are typically employed for systems with complex and multiscale descriptions. A prominent example of such systems is provided by the human brain, which can be idealized as a complex dynamical system made of many interacting subunits. The human brain can be modeled in terms of observable variables together with their spatio-temporal-functional relations. Computational intelligence is a research field bridging many nature-inspired computational methods, such as artificial neural networks, fuzzy systems, and evolutionary and swarm intelligence optimization techniques. Typical problems faced by means of computational intelligence methods include those of recognition, such as classification and prediction. Although historically conceived to operate in some vector space, such methods have been recently extended to the so-called nongeometric spaces, considering labeled graphs as the most general example of such patterns. Here, we suggest that fractal analysis and computational intelligence methods can be exploited together in neuroscience research. Fractal characterizations can be used to (i) assess scale-invariant properties and (ii) offer numeric, feature-based representations to complement the usually more complex pattern structures encountered in neurosciences. Computational intelligence methods could be used to exploit such fractal characterizations, considering also the possibility to perform data-driven analysis of nongeometric input spaces, thereby overcoming the intrinsic limits related to Euclidean geometry.

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Keywords Computational intelligence · Complex systems · Analysis of nongeometric input spaces · Fractal analysis · Brain research

48.1 Introduction

The analysis of complex systems [1–3] is rapidly expanding with the development of novel theoretical and computational tools designed to tackle the problem of disentangling the interactions among different parts of a complex process/system. The human brain is a natural and important example of complex systems made of many interacting parts. The rich complexity of the brain along with its multilevel, time-varying, and intricate topological organization of neurons suggests to look beyond explanations driven by classical Euclidean geometry and linear dynamics [4–6]. Mathematical tools from fractal geometry offer a means to perform quantitative analysis and description of the complexity of natural, social, and artificial systems. It is a fact that modern neurosciences admit the prevalence of fractal properties in the brain at various levels of observation and in different contexts. In general, the brain can be analytically or computationally studied by (i) focusing on a specific subunit (i.e., a specific aspect) and/or (ii) taking into account (time-dependent) interactions between such units. In the first case, usually one resorts to the study of a series of suitable observable variable recorder over some time span – usually named time series. In the literature, there are many approaches to study the peculiarities of the systems underlying the observed time series. Such methods include causality analysis [7] and directional transfer of information [8–10], memory effects and predictability in stochastic systems [11, 12], clustering [13], and recurrence plots [14]. In the latter case, instead, the graph-theoretic framework and its modern revival as the science of complex networks play a pivotal role in the characterization of brain structures along with their dynamical and functional aspects [15–22]. Graph characterizations, in terms of both static and dynamic measures of complexity, are also popular in describing complex systems and their functions [23–33].

The field of computational intelligence (CI) is founded on data-driven methods that take inspiration from natural systems. Examples of such systems include artificial neural networks, kernel methods, fuzzy systems, and evolutionary and swarm intelligence optimization techniques as well [34]. Typical tasks tackled by means of computational intelligence methods include recognition problems (e.g., classification, clustering, and function approximation) and those of adaptive control (e.g., fuzzy control and data-driven optimization via neural networks). Intelligent systems operating through data analysis come with various connotations and usually imply different interpretations. In spite of the existing diversities, it is possible to recognize several features that are profoundly visible across a broad spectrum of those systems. An intelligent system should be adaptable, meaning that it should come with a parametric mathematical model. Adaptation is typically implemented via the learning-from-examples paradigm by using suitable error criteria and update mechanisms. Another important aspect of characterizing intelligent systems is their

capability to generalize with accuracy under uncertainty or when dealing with unseen data. The spectrum of successful applications of computational intelligence methods is rapidly expanding. Far from being exhaustive, it is worth citing few related applications in biomedical and particularly in the brain science context: classification of brain injury [35], seizure recognition in EEG [36], feature transformation and clustering on fMRI data [37], and data selection for EEG signal classification [38, 39], also in different neurological disorders, including major depressive disorder [40], epileptic seizures [41], Alzheimer's disease [42], and schizophrenia [43].

Here, we hypothesize that an interplay between fractal analysis and computational intelligence methods could offer a compelling tool of analysis in neuroscience. Fractal characterizations can be used to (i) assess scale-invariant properties and (ii) offer numeric, feature-based representations to complement the usually more complex pattern structures encountered in brain research. Computational intelligence methods could be used to exploit such fractal characterizations, considering also the possibility to perform data-driven analysis of nongeometric input spaces, hence overcoming the limits related to Euclidean geometry-based approaches.

The remainder of this chapter is structured as follows. Section 48.2 offers a brief overview of fractal analysis and its impact in neuroscience. In Sect. 48.3, we introduce the computational intelligence research context; we discuss also some research endeavor examples in processing nongeometric data by using conventional computational intelligence methods. Section 48.4 provides a conceptual bridge between fractal analysis and computational intelligence methods, highlighting their possible interplay in future research. Finally, Sect. 48.5 concludes this chapter offering future perspectives for interdisciplinary research in brain science.

48.2 Fractal Analysis and Brain Complexity

Fractal and multifractal analysis play a pivotal role in several scientific fields. Notably, neuroscience [44–46] and physiology in general [47, 48] are two important examples. Characterizations in terms of fractal scaling are typically used for the purpose of describing the complexity of systems following a multiscale paradigm. Fractal analysis formalisms have been developed for image analysis, discrete-time signals (i.e., time series), and complex networks analysis as well. Considering the former, a large body of applications exists, including neuronal spike trains [49], background neuronal noise-like activity underlying epileptiform transitions in human and mouse hippocampus [50], and EEG signals analysis [51–54], for instance. The application of fractal geometry to the analysis of complex networks is a novel research approach [55–61]. As suggested by Song et al. [61], fractal scaling in complex networks emerges as a particular combination of a disassortative degree mixing – repulsion between hub and weakly connected vertices. The renormalization group approach [60] is a well-known approach to evaluate the fractality of a network's underlying dynamical changes. The method

consists in compressing the network in clusters by considering a characteristic (topological) length scale. If the network compression adheres to a specific power-law scaling, then the network can be considered as fractal. It is worth stressing that the renormalization group approach provided a way to explain the coexistence of otherwise conflicting features, such as fractal and small-world properties of networks. Li et al. [57] proposed a generalization of the box-counting method in order to perform multifractal analysis on complex networks. A novel perspective for multifractal analysis of complex networks has been proposed by Maiorino et al. [59]. Time series of vertex observables, such as degree, clustering coefficient, and closeness centrality, can be formed via (stationary) random walks on the network. Fractal and multifractal analysis can then be performed by making use of the more consolidated time series framework. The fractal structure of brain networks is typically related to its highly modular topology [62, 63], which contributes to the compartmentalization of brain functions. However, brain networks, as well as other complex biophysical systems, exhibit also the celebrated small-world feature [15], implying that the efficiency of long-range connections is something that is optimized while taking care of the related (energetic) costs.

48.3 Computational Intelligence Methods and the Challenge of Processing Nongeometric Input Spaces

Computational intelligence (CI) is a research field bringing together many nature-inspired computational methods, such as artificial neural networks, fuzzy systems, and evolutionary and swarm intelligence optimization techniques [34]. In Table 48.1, we report some of the most common terminology, methods, and techniques that are encountered in the CI research field. It is worth noting that CI is closely related to soft computing, which is a toolbox of techniques, conceived to provide solutions for modeling and computational problems cast in an intrinsically uncertain and complex setting. Bonissone [64] defines the boundary of the soft computing context by stating, “The term Soft Computing (SC) represents the combination of emerging problem-solving technologies such as Fuzzy Logic, Probabilistic Reasoning, Neural Networks, and Genetic Algorithms. Each of these technologies provides us with complementary reasoning and searching methods to solve complex, real-world problems.”

Typical problems faced by means of CI methods include recognition problems (e.g., classification, clustering, and function approximation [65–75]) and those of adaptive control (e.g., fuzzy control and data-driven optimization via neural networks [76, 77]). Focusing on recognition problems [78, 79], the concept of pattern plays an important role. Patterns can be discovered virtually everywhere: in climate science, geophysics, rare catastrophic events, complex biochemical processes in living beings and ecosystems, clinical diagnostics, financial market trends, and social network evolution. Formally speaking, a pattern is a form of

Table 48.1 Typical terminology and methods used in the computational intelligence field (sorted in alphabetic order)

Terminology	Definition
Artificial neural networks	Adaptive system comprised of nodes and connections in an organization similar to the central nervous system. Neural networks are capable of dealing with complex systems by adapting the model during the learning phase
Backpropagation algorithm	Learning algorithm for neural networks
Data mining	Techniques used to analyze and discover knowledge and relationships in data
Evolutionary algorithm	Global optimization method inspired by the evolution of biological systems. Important concepts include fitness function, natural selection, mutation, recombination, and crossover
Expert system	Computer technique that aims to simulate a human expert in order to provide support and advice to humans
Fuzzy logic	A multivalued logic in which the degree of truth of a statement ranges in the unit interval
Hidden Markov model	Statistical model used to characterize and predict sequences of observable variables without directly inspecting the underlying system's state
K-nearest neighbor	Classification method that categorizes data according to their proximity in a given representation space
Machine learning	Computer algorithms designed to learn complex relationships, rules, and patterns from empirical data in order to perform inductive inference
Representation space	Mathematical formalization of the input space where the data is represented
Self-organizing feature map	Unsupervised learning mechanism that can be used for either clustering or classification
Supervised learning	Methods of machine learning that require knowledge of the classes during the training of the model
Support vector machines	Flexible method for linear and nonlinear classification based on the concept of separating hyperplane and convex optimization
Unsupervised learning	Machine learning techniques that do not require knowledge of the classes during the training

experimental evidence gathered from a data-generating process, P . A process $P: X \rightarrow Y$ is the idealization of a system, either abstract or physical, that generates outputs according to input patterns. X is referred to as the input space (data domain or representation space), where the patterns are represented according to some suitable formalism. Y , instead, is the output space, providing a way to learn how to recognize patterns according to their description. In pattern recognition, the closed-form, analytical expression of P is unknown. The problem is typically reduced to determine the parameters of a suitably mathematical model of P , e.g., M , by analyzing a finite dataset S . The objective is to obtain a model that is able to generalize over pattern instances different from those in S . Three categories of problems in pattern recognition can then be distinguished:

- (1) *Classification*. In classification, Y is a finite set of classes (a finite nominal set of labels) that is completely – or, in some variations, partially – known a priori. X contains pattern instances that are (correctly) labeled with respect to Y . Classification is also referred to as discrimination, since in fact the problem consists of learning a model that is able to correctly label patterns.
- (2) *Clustering*. In clustering, X contains unlabeled pattern instances, and therefore the definition of Y is not available a priori; there is no supervision. The goal is to determine the hidden structure of the data by relying on some measures of similarity among the patterns.
- (3) *Function approximation*. In function approximation (also known as regression), Y is a normed space, usually a continuous interval in $\mathbb{R}$. As the name suggests, the goal is to learn a (nonlinear) function by using labeled training patterns in X . In the case of multidimensional outputs, i.e., when the problem is generalized to learning a mapping.

Practically, a mathematical model M must be tuned to the problem at hand. Learning (or synthesizing) a model M from the available dataset S consists in optimizing some criterion, i.e., a performance measure that allows adjusting the model parameters according to a suitable training dataset. Two principal schools of pattern recognition exist: the one including discriminative methods (such as support vector machines [80] and many neural network models [78]) and the one exploiting generative models (such as the well-known hidden Markov model [81] and several versions of the recently developed deep convolutional neural network model [82]).

CI methods are typically designed based on the assumption that the input space, X , is essentially a subset of $\mathbb{R}^d$. When departing from the $\mathbb{R}^d$ vector-based pattern representation, many theoretical and practical problems arise, which are mostly due to the absence of an intuitive geometric interpretation of the data. Notwithstanding, the availability of datasets of nongeometric data motivated the development of pattern recognition and soft computing techniques on such domains [83–95]. Nongeometric patterns include data that are characterized by pairwise dissimilarities that are not metric; therefore, they cannot be straightforwardly represented in a Euclidean space [88, 90, 96]. A particularly interesting instance of such nongeometric data is constituted by structured patterns. A structured pattern is conceived to include in the structure the information regarding the relations and, at the same time, attributes describing the pattern itself. A labeled graph is the most prominent example of a structured pattern that is conceivable. In fact, a graph can be used to characterize a pattern by describing the topological structure of its constituting elements (by means of the vertices) through their relations (the edges) [97–99]. Both vertices and edges can be equipped with suitable labels, i.e., the specific attributes characterizing the elements and their relations. Sequences of generic objects, trees, and automata can be thought of as particular instances of labeled graphs. Labeled graphs are used to represent data and systems in many different contexts, such as document analysis [100], solubility of the *Escherichia coli* proteome [31], brain functional networks [101], biomolecule recognition [102, 103], digital images [104], and scene understanding [105].

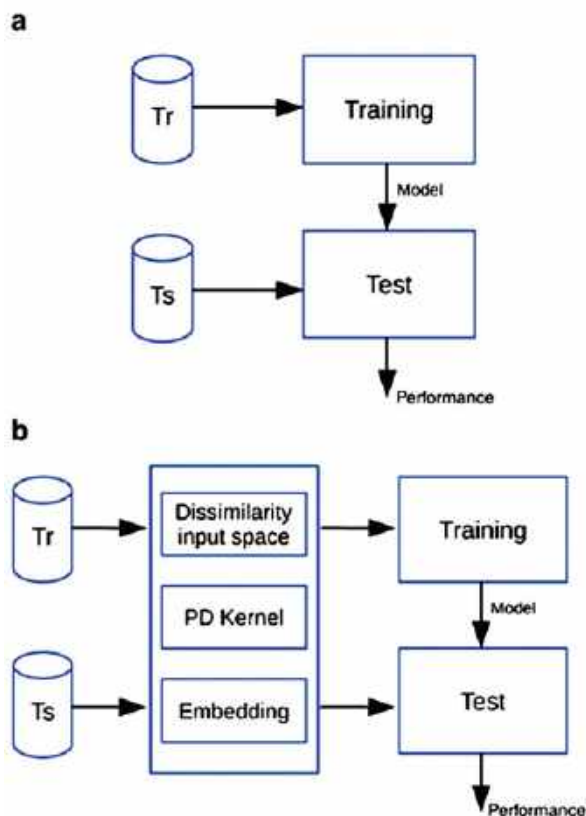
Figure 48.1 offers a schematic, visual representation of the typical stages involved in the use of CI methods to address recognition problems. By means of a training set (Tr in the figures), a model is synthesized and then used during the testing stage using unseen data (denoted as Ts). This high-level overview is valid for both $\mathbb{R}^d$ (1(a)) and nongeometric (1(b)) data. However, in order to make proper use of CI methods, in the case of nongeometric data, the input space must be processed with particular care. Notably, three mainstream approaches can be pursued [88, 90]: (i) using a suitable dissimilarity measure operating in the input space; (ii) using positive definite (PD) kernel functions; and (iii) designing an embedding to map the input in $\mathbb{R}^n$. The choice depends on the particular data-driven system adopted for the task at hand and on other factors, typically related to the particular application context. The first case is the most straightforward one, but its use is legitimate only if the data-driven system does not require a specific geometrical structure of the input space. In fact, a dissimilarity measure might not be metric, therefore not Euclidean either. A first example in this direction could be provided by citing the well-known k -nearest neighbor rule, which processes the input patterns by relying only on their topological similarity. The second scenario is a typical choice in the case of kernel methods. However, positive definite kernels can be obtained only if the underlying geometry is Euclidean. Nonetheless, if such a requirement is not satisfied at first, correction techniques could be used for this purpose to “rectify” the data domain [88]. The last approach consists of mapping (i.e., embedding) the input space to a vector space, typically $\mathbb{R}^n$. In this way, conventional CI methods can be used without alterations, since in fact the problem is reduced to processing conventional patterns represented as numeric vectors.

48.4 On the Interplay Between Fractal Analysis and CI Methods

As mentioned before, from a computational viewpoint, the human brain can be studied by considering measurements targeted to specific regions together with their spatial, temporal, and functional relations. A systemic view of such measurements gives rise to complex data patterns, providing a structured and potentially heterogeneous dataset of information to be processed. CI techniques, with particular interest in those for nongeometric data, could be used to tackle data-driven problems in such a context [101].

Let us elaborate further on this idea by describing a hypothetical case study. Let us assume that we are interested in modeling each particular state of the conscious mind as a graph – a network of activated brain regions. Furthermore, typically we would be interested in correlating such states with the execution of task; therefore, we assume to have collected such states during the execution of a specific task over time. A graph would provide a way to encode the time-varying relations among the active brain regions, together with suitable, problem-dependent

Fig. 48.1 Schematic representations of a data-driven inference system operating in $\mathbb{R}^d$ (a) and considering nongeometric data (b)



descriptive attributes. Such attributes can be associated with both vertices and edges, describing in principle any type of data (signals, vectors, and sequences). In order to infer insights on structure–function relationships from the experiment, the collection of graphs recorded over time could be analyzed with conventional graph-theoretic methods. For example, it could be beneficial to study the variation of structural and dynamical properties elaborated from such graphs, including their fractal dimension – for instance, as a measure of complexity. However, it could also be interesting to conceive problems of automatic classification (discrimination) of such states, dealing with the classification problem directly in the space of graphs, i.e., states of the conscious mind in our example. This might be of interest to a neuroscientist, since representing a neurological state as a graph has significantly more representational power than using a vector of numeric values. Notably, it could be interpreted more easily, and hence the discrimination process could be explained in a language closer to human reasoning. The interpretability issue of CI methods is a well-known research endeavor; researchers usually refer to such a problem with the expression “opening the black-box.” In fact, most of the CI methods operate without providing information intelligible to humans. Of course,

this is not an issue when the task at hand is purely technological – problem-solving-oriented. However, when scientific knowledge has to be extracted from a data-driven process, interpretability of the data and the inference mechanism itself assume critical importance.

We hypothesize here that the interplay of fractal analysis and computational intelligence methods could offer a compelling tool of analysis in the context of brain research. Fractal characterizations can be used to assess scale-invariant properties and offer numeric, feature-based representations to complement the otherwise more complex patterns usually encountered in brain research. Computational intelligence methods, especially those for nongeometric spaces, could be used to face specific recognition problems exploiting such information without limiting the problem definition to conventional Euclidean spaces.

48.5 Future Perspectives and Concluding Remarks

The human brain is a complex dynamical system made of many units (neurons and astrocytes) that interact over time, space, and/or by considering specific functions. The hierarchical organization of the brain connections calls for advanced computational techniques of analysis, including those based on graph-theoretic methods and time series analysis. The intricate shape of the brain is better described in terms of fractal geometry and related mathematical and computational tools. The field of computational intelligence research is founded on data-driven methods which take inspiration from natural systems. Examples of such systems include artificial neural networks, fuzzy systems, and evolutionary and swarm intelligence optimization techniques. We discussed also the more recently developed methods for analyzing nongeometric spaces, which allow to depart from the assumption of an underlying Euclidean geometry for the input space.

In this chapter, we first described graph-based and time series analysis methods exploiting fractal analysis concepts. Such methods play an important role in neurosciences, since in fact the types of data that are encountered in such a context require complex characterizations that typically go beyond the usual vector-based representation. Successively, we suggested a possible interplay between fractal analysis and computational intelligence methods, especially considering methods for nongeometric spaces. Computational intelligence methods (such as classifiers and prediction systems) are already highly exploited in related contexts, such as biomedical signals and diagnostic medicine. However, the full potential of an interplay between the two disciplines still needs to be discovered.

In the following, we provide a list of possible applications in which computational intelligence methods and fractal analysis could be used by neuroscientists:

- Time series in physiological as well as pathological states, e.g., intracranial pressure trends to predict patients' outcome following traumatic brain injury and/or bleeding.

- EEG analysis for seizure prediction, MEG (magnetoencephalography) time series analysis for analysis of cognitive states, EMG (electromyography) time series analysis for the study of neuromuscular disorders, and fMRI analysis of brain oscillation in physiopathological states.
- Analysis of the local field potential dynamics and brain oscillations in different physiological statuses (e.g., sleep) [106] and pathologies (e.g., parkinsonism) [107].
- Image analysis for characterization of brain tumor features (e.g., distribution of the microvessels, roughness of the tumors' edges, architectural pattern, etc.) for automatic classification (histotype and/or grading) and follow-up. Machine learning algorithms for the classification of brain diseases according to neuroradiological features (e.g., spectroscopy) and for the connectomics analysis. Brain tumors' features extraction by means of fractal radiomics [108, 109] and features' extraction optimization by means of fuzzy deep convolutional neural networks [110, 111].
- Training machines for automatic segmentation of regions of interest, such as brain tumors and/or arteriovenous malformations, for surgical and/or radiosurgical planning.
- Neural networks and similar CI methods for the analysis of gene expression patterns in neurological diseases.
- Computational quantitative analysis for the study of brain cells and receptors.
- Application of AI techniques and extraction of fractal features in multiscale to optimize brain–computer interface systems [112].
- Computational fractal-based analysis of eye tracking in healthy individuals and patients affected by neurological disorders.

The recent explosion of artificial intelligence opens new venues in the field of translational neurosciences, and the merged application of computational fractal-based analysis and computational intelligence could open new streams of discovery and research.

In this field, the sky is the limit!

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